fib Model Code for Concrete Structures 2020

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Commission 9 – Dissemination of Knowledge

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Preface

Throughout its long history, the Fédération Internationale du Béton (*fib*), International Federation for Structural Concrete, and its predecessor associations Comité euro-international du béton (CEB), Euro-International Committee for Concrete, and Fédération Internationale de la Précontrainte (FIP), International Federation for Prestressing, has been engaged in the advancement of concrete structures, with its first model code in 1978, followed by codes in 1990 and 2010 – MC 2020 is the next generation in the series of international model codes.

The inception of MC2020 was the outcome of several significant events, including the *fib* international workshop held in The Hague in June 2015 and the *fib* MC2020 Core Group meeting conducted in Madrid in December 2015. As a result of these discussions and associated planning events, Task Group 10.1 was formed, under the guidance of Commission 10, which was newly established at that time. The task group's inaugural meeting was held in October 2016, and was led with tremendous drive and ambition by Stuart Matthews (Convenor), Agnieszka Bigaj-van Vliet (Deputy Convenor), and Gerrie Dieteren (Technical Secretary), until its completion in 2023.

The MC2020 project was supported and further developed through a series of fourteen follow-up international consultations and workshops on the latest advancements in structural codes held between September 2016 and March 2023, which included special sessions on Model Code held during the *fib* Symposium in Maastricht in June 2017 and during the *fib* Congress in Oslo in June 2022.

From its inception in 2015, through to its completion in 2023, the *fib* MC2020 project was an ambitious and challenging undertaking that sought to build upon the achievements of MC2010, particularly its treatment of activities for the conservation of existing concrete structures. The vision for MC2020 was to go beyond the state-of-the-art at its inception, extending the code's application to existing structures.

MC2020 is a comprehensive code that integrates provisions for designing new concrete structures with those related to existing concrete structures. The code is designed to effectively address the design of structures as well as all activities associated with the management of existing concrete structures, including in-service assessment and interventions. To achieve this, sustainability is prioritized in MC2020, taking a holistic approach that considers societal needs and impacts, life-cycle cost, and environmental impacts.

Concrete is a remarkable material, providing strength while allowing itself to be freely moulded into almost any shape, a protector of humanity while allowing architects and engineering to bring life to modern societies, across regions and vast continents. And, while through its modern history it has driven economies to provide a quality of life to the citizens it defends, it is also a non-inconsequential contributor to climate change in what is described by the United Nations as "*the defining issue of our time*".

The important global movement of nations declaring not just their ambition for, but strong commitments to carbon neutrality by 2050 served as a compelling catalyst for the increased emphasis on sustainability in MC2020, distinguishing it from its preceding editions. Although not directly controlling, this movement played a vital role as a strong motivator. Its profound influence resonated with *fib*, prompting us to add new measures to address the sustainability of structural concrete and its CO_2-eq footprint as a crucial matter that demanded an immediate response.

Thus, MC2020 emphasizes sustainability as a fundamental requirement for design, based on an all-inclusive treatment of societal needs and impacts, life-cycle cost, and environmental impacts. The code adopts an integrated life-cycle perspective, which means that the requirements for social performance will have defining implications for subsidiary performance requirements that are crucial in designing new structures and assessing existing ones. These performance requirements include aspects such as human and environmental safety, serviceability, durability, robustness of the structure, resilience of its functionality, aesthetics, adaptability, maintenance, and reducing CO_2-eq emissions.

MC2020 takes into account factors such as structural safety, serviceability, durability, and sustainability, which are linked to broader issues such as the life-cycle management, cost, environmental, and societal impacts of concrete structures.

Given the aspiring nature of the project, all *fib* commissions, together with other bodies able to contribute on relevant topics, was drawn upon to assemble the breadth of knowledge and expertise required for the successful completion of the MC2020 project.

Derived from the collective efforts and expertise of more than 1400 contributors, from 67 countries, this cutting-edge code represents the best and most current state-of-thinking in tested and reliable models' development. It draws from extraordinary advancements in the science of concrete structures and materials, across design and assessment, brought together through consensus of its contributors and is supported by its 40 national member nations, making it a truly global effort.

We acknowledge the tremendous efforts of all involved and congratulate them on the production of this excellent code.

Stephen Foster	Akio Kasuga	Tor Ole Olsen
President of *fib*	Immediate Past President	Chair, Commission 10

The ***fib*** Model Code for Concrete Structures was an initiative taken by ***fib***'s predecessors CEB (Comité Euro-International du Béton) and FIP (Fédération Internationale de la Précontrainte) at a time when there were hardly any international codes. Since CEB and FIP were both organisations aiming to synthesize international research and experience, it was regarded as an important step forward to convert this knowledge and experience into practical documents for design, so that national code commissions could take advantage of it. The first code-like recommendations in 1964 and 1970 were used in this way. The Model Code 1978 [1-1] also contributed to international harmonization. The Model Code 1990 [1-2] and Model Code 2010 [1-3] provided confirmation of that intention, among others, by serving as an important basis for the Eurocode 2 [1-4] and its recent revisions. The vision for ***fib*** MC2020 is to go beyond the point reached by *fib* Model Code 2010 [1-3], recent ISO codes, such as ISO 16311 [1-5], and the current activities to extend the application of the Eurocodes to existing structures.

1 Scope

1.1 Aim of the *fib* Model Code 2020

The ***fib*** Model Code for Concrete Structures 2020 (***fib*** MC2020) is intended to serve as the basis for future codes for concrete structures. Whereas existing operational codes are legal documents based on mature knowledge, the ***fib*** Model Code also considers the newest developments with respect to concrete structures, concrete as a structural material, reinforcing materials for concrete and a broad range of new techniques for construction and interventions. It also reflects the newest ideas with respect to formulating requirements and recognises the diverse needs of engineering communities across different regions of the world. By doing so, it ensures that structures achieve optimum behaviour according in accordance with contemporary insights and the evolving demands of today's world.

In this ***fib*** Model Code, those requirements refer not only to traditional demands with regards to safety, serviceability and durability, but take sustainability as a fundamental requirement to establish links to wider issues such as social responsibility, environmental quality and economic efficiency. The main intention of ***fib*** MC2020 is to deal effectively with both the design of new structures and all the activities associated with the life-cycle management of existing concrete structures, including matters such as in-service assessment and interventions to extend the life / improve the performance of these structures.

For those who will be involved in updating existing codes or developing new codes for concrete structures, the ***fib*** Model Code should be a source of comprehensive and accurate knowledge. Whereas a normal operational code predominantly gives sets of application rules that should be transparent enough to be applied by professional designers, while also being accurate enough to be economical, the ***fib*** Model Code goes further by supplying its users with essential background information and reference materials to support their work.

Nonetheless, while serving also a reference for the research and development work, and a resource for professional education, the ***fib*** Model Code is meant also to be an operational document for everyday design situations and structures.

Part I, Scope and terminology addresses in chapters 1-2 subjects such the structure and format of the Model Code, as well as providing definitions of the terminology, notations and acronyms employed.

Part II, Basic principles comprises chapters 3-10 which take a through-life project perspective upon the creation and life-cycle management of concrete structures. In chapters 3-5 the topics addressed include the general implementation of the overarching principles of sustainability and the performance-based approach, along with those for the life-cycle management, of both new and existing structures. Chapters 5-8 are concerned with the principles of structural design and assessment, execution and the conservation of existing structures. In chapter 9 consideration is given to the principles to be adopted in respect of circularity and dismantlement, including the subsequent reuse of construction materials in new structures. Finally, chapter 10 describes the principles of quality and information management which are to be employed.

Part III, Principles of structural performance evaluation comprises chapters 11-12, with chapter 11 setting out the basis of structural performance evaluation (verification), namely the principles of the limit-state approach, service life design and the reliability-based methodology. Chapter 12 is concerned with the principles of structural design and assessment, primarily concentrating upon the various verification methods and formats which may be employed.

Part IV, Actions on structures gives in chapter 13 definitions of actions and the guiding principles for dealing with actions on structures.

Part V, Input data for materials covers in chapters 14–19 the properties of the main structural materials, including concretes, reinforcing and prestressing steel, non-metallic reinforcement and fibre reinforced concrete and UHPC, along with those of protective materials and systems.

Part VI, Input data for interfaces provides in chapters 20-25 characteristics for interfaces between concrete and steel or non-metallic reinforcement, and between new concretes and existing concretes. Consideration is given to both embedded and surface-or near-surface mounted reinforcement. Moreover, the properties of the interface between concrete and structural-steel-sections and the characteristics of anchorages in concrete are discussed

Part VII, Design and assessment is the major technical component of ***fib*** MC2020 and comprises chapters 26-34, which take a through-life project perspective upon the creation and life-cycle management of concrete structures.

1.2 Structure of the *fib* Model Code 2020

The ***fib*** MC2020 is subdivided into ten parts. The sequence of the parts reflects the basis of life cycle thinking:

Part I: Scope and Terminology
Part II Basic Principles
Part III Principles of Structural Performance Evaluation
Part IV: Actions on Structures
Part V: Input Data for Materials
Part VI: Input Data for Interfaces
Part VII: Design and Assessment
Part VIII: Execution
Part IX: Conservation
Part X: Circularity and Dismantlement

Chapter 26 is concerned with the processes of conceptual design, and chapters 27 & 28 address the approaches taken to design and assessment, respectively. Chapter 29 is concerned with various aspects of structural analysis and, especially, the structural effects of time-dependent behaviours. Chapter 30 is a very substantial technical component as it contains the provisions for evaluation (verification) of the various aspects of structural performance, for a wide range of loads and environmental conditions. Evaluation of other non-structural aspects of social performance are contained in chapter 31 and environmental quality and economic efficiency are considered in chapters 32-33, respectively. Chapter 34 is concerned with sustainable decision making.

Part VIII, Execution provides in chapters 35-37 execution rules for concrete, steel, formwork and intervention works, along with the management of the associated execution processes.

Part IX, Conservation comprises chapter 38 which deals with the activities associated with conservation and the life-cycle management (LCM) of existing structures including the different strategies employed, condition survey and evaluation, assessment, monitoring, selecting and making appropriate interventions, LCM after interventions have been undertaken and recording of LCM information.

Part X, Circularity and dismantlement comprises chapter 39 which completes the life-cycle approach to the use and re-use of construction material by presenting information upon circularity and dismantlement including the reuse of materials, with the goal of improving the uptake of circularity principles.

fib MC2020 presents an illustrative conceptual framework comprising a four-tier layout representing a continuum of overlapping performance- or outcomes-based requirements, specifications and testing regimes. The performance focus shifts from broad higher-level operational goals aligned with the asset owner objectives and the desires of other stakeholders (Tier 1), through engineering technical definitions for a scheme or project (Tier 2) and then focuses upon the increasingly detailed contractual and technical performance aspects (Tiers 3 and 4 in this example).

As ***fib*** MC2020 adopts an integrated life-cycle perspective, which means that the requirements for social performance should have defining implications for subsidiary performance requirements that are crucial in designing new structures

1.3 New developments in *fib* Model Code 2020

Amongst the many new advances and innovations incorporated in ***fib*** MC2020, perhaps the most significant philosophical development is the further evolution in the approach to sustainability associated with adoption of a “sustainability-orientated performance- or outcome-based approach”, to provide a framework for the design and life-cycle management of concrete structures using sustainability-orientated performance requirements and related verification processes.

In taking this step ***fib*** MC2020 seeks to make sustainability a fundamental requirement for design and all the activities associated with the management of existing concrete structures, including in-service assessment and interventions. This is based on the concept of an all-inclusive treatment of societal needs and impacts, life-cycle cost, and environmental impacts. ***fib*** MC2020 recognizes the importance of

and in the life-cycle management of existing ones. These performance requirements include aspects such as human and environmental safety, serviceability, durability, robustness of the structure, resilience of its functionality, aesthetics, adaptability, maintenance, upgrading, and reducing CO2-eq emissions.

sustainability approach and provides guidelines for consistent treatment of sustainability demands to new and existing concrete structures.

Other technical advances and innovations incorporated into ***fib*** MC2020 include, but are not limited to:

- Implementing an integrated life-cycle perspective which promotes a holistic treatment of structural safety, serviceability and durability that is linked to wider issues such as the societal, environmental and economic impacts of concrete structures.
- Incorporating, in a consistent manner, reliability concepts aimed at taking better account of uncertainties and risk.
- Making use of the performance-based concept to close the gap between material engineers and structural engineers and to remove specific constraints upon novel types of concrete and reinforcing materials.
- Fully integrating provisions based on generalised models and the implementation of the level of approximation approach applicable for both the design of new structures and all the activities associated with the assessment, interventions and the life-cycle management of existing concrete structures.
- Taking account of the effects of material degradation, especially in respect of the assessment of reinforcement corrosion initiation and propagation phases and the effects upon structural, performance.
- Dealing with structural arrangements which are not recognised in contemporary design codes (non-compliant detailing).
- Improving the application of non-linear finite element methods (NLFEM) to support verifications of structural performance.
- Taking advantage of information acquired by testing of structural materials and (in situ) testing and monitoring of concrete structures to better assess the properties of materials and structures, their real condition and behaviour of structures, potentially reducing uncertainties in the design and assessment including the prediction of the performance of the structures over time.
- Considering end-of-service-life issues to better address matters such as circularity and the re-use of construction materials in the context of

The reference sources include ***fib*** Bulletins, especially those specifically written as support documents for ***fib*** MC2020, and earlier CEB-FIB Bulletins providing background information expanding upon the main provisions presented within ***fib*** MC2020, other codes and consensus documents or specifications (especially those prepared by international organisations such as ISO & RILEM), as well as technical papers in scientific journals.

dismantlement (both deconstruction and demolition) , including consideration of the associated safety and structural aspects.

1.4 Format

The format of this ***fib*** Model Code follows the earlier CEB-FIP tradition, namely:

– the main technical provisions are presented on the right-hand side of the page in a logical sequence of topics. Structural requirements are stated, followed by the relevant design or assessment criteria, appropriate engineering models and/or design or assessment rules: their application is intended to satisfy the relevant structural requirements;

– explanations are given on the left-hand side of the page, with specific diagrams, alternative (simplified) rules, short justifications of the options presented on the right-hand side of the page, along with references to other sources that were used to derive the design and / or assessment recommendations.

References

[1-1] CEB-FIP Bulletin 172. CEB-FIP Model Code 1970.

[1-2] CEB-FIP Bulletin 213/214.. CEB-FIP Model Code 90. 1993

[1-3] *fib* Model Code 2010. *fib* Model Code for Concrete Structures 2010, Fédération Internationale du Béton, Ernst & Sohn, 2013.

[1-4] EN 1992-1-1:2004. Eurocode 2: Design of concrete structures – Part 1-1: General rules and rules for buildings.

[1-5] ISO 16311-1:2014. Maintenance and repair of concrete structures – Part 1: General principles.

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Examples of the *action effects* are stresses, stress resultants, reactions, deformations, displacements, as well as other effects, depending on the type of structure.[1]

Aesthetics of structures is usually associated with the visual sense and, to some extent, the senses of sound and texture, as well as with the perception of the recognized associations and the context.

Although any person's response to the aesthetics of a structure will be unique to that individual, many aesthetic principles can be identified and used by the creator

2 Terminology

2.1 Definitions

This section defines the various technical terms that appear in the *fib* Model Code 2020. Definitions are based on the sources listed in the References of this chapter.

Acceptance:
Agreement of the stakeholders (i.e. owners, users, contractors, society) [2-1] to take over the structure or a part of it as its own property.

Accidental action:
Design situation involving exceptional conditions of the structure or its exposure, including fire, explosion, impact or local failure.[2]

Accidental design situation:
Design situation taking into account accidental conditions for the structure or its components under consideration.[1]

Accompanying action:
Action accompanying the leading action considered.[1]

Action effect:
Effect of action(s) on structural members (e.g. internal force, moment, stress, strain) or on the whole structure (e.g. deflection, rotation).

Actions:

a) set of forces (loads) applied to the structure (direct action);
b) set of imposed deformations or accelerations caused, for example, by temperature changes, moisture variation, uneven settlement or earthquake (indirect action).[2]

Adverse state:
State in which the performance criterion is not met.

Aesthetics of structures:
Aspects of the appearance of a structure perceived in terms of visual aesthetic considerations.

of the structure to achieve specific aesthetic effects. Effects relevant for structures include for instance repetition, symmetry/asymmetry, rhythm, perspective, proportion, harmony, contrast, pattern, ornamentation, texture, colour, granularity and interaction of sunlight and shadows.

In order to derive an *analytical model*, use is made of basic relationships such as equilibrium conditions, constitutive relationships and kinematic conditions.

Availability refers to the probability that a structure is actually available for use during the period of time when it is supposed to be available.[3]

The *birth certificate* should provide specific details on parameters that are important to the durability and service life of the structure concerned (e.g. cover to reinforcement, concrete permeability, environmental conditions, quality of workmanship achieved) and the basis on which future knowledge of through-life performance should be recorded.[5]

The framework laid down in the *birth certificate* should provide a means of comparing actual behaviour/performance with that anticipated at the time of design of the structure.[5]

The *birth certificate* should offer reference to facilitate ongoing (through-life) evaluation of the service life which is likely to be achieved by the structure.[5]

Analytical model:
Mathematical relationship between the forces and imposed deformations exerted on the structure or a structural element and its response to those forces (e.g. deformations, displacements or internal forces).

Assessment:
see *Condition assessment*.

Availability:
The ability of a structure to operate satisfactorily at any point in time, excluding times when the structure is under repair.[3]

Basic variable:
Part of a specified set of variables representing physical quantities, which characterize actions and environmental influences, geometrical quantities and material properties.[4]

Basis of design:
Technical description of the implementation of the service criteria agreement.[1]

Bearing:
Device to transfer a mainly compressive force for supporting an element.

Biological actions:
The aggression of biological organisms (bacteria, insects, fungi, algae) affecting and influencing the structure or its components.

Birth certificate:
A document, report or technical file (depending on the size and complexity of the structure concerned) containing engineering information formally defining the form and the condition of the structure after construction.[5]

Buildability:

Capacity design:
Method of seismic design with appropriately defined areas of plastic deformations exhibiting adequate ductility, together with other areas of the structure that are provided with increased yielding resistance to ensure elastic behaviour.[1]

Characteristic value of a material property:
The value of a material property (e.g. structural material or soil) having an a priori specified probability of not being attained in the supply produced within the scope of the relevant material standard.[6] The characteristic value generally corresponds to a specified fractile of the assumed statistical distribution of the particular property of the material or product. A nominal value is used as the characteristic value in some circumstances.[2]

Characteristic value of a geometrical property:
Value usually corresponding to the dimensions specified in the design.[6] Where relevant, characteristic values of geometrical quantities may correspond to some prescribed fractiles of the statistical distribution.[2]

Characteristic value of an action:
Principal representative value of an action.[6]

Chemical actions:
The reactive transport of chemicals (e.g. salts, acids, alkaline substances and organic compounds) affecting and influencing the structure or its components.

Circularity of concrete:

Approach to minimize the need for new materials either by extending the lifetime of existing or new structures or by the application of elements/ components and/or subsystems of structures in a new life-circle.

Collapse:
Catastrophic physical disruption, giving-way or breakdown of elements or components of a structure, to such an extent that the structure is unable to perform its intended loadbearing function.[5]

Commissioning:
Start of planned use.[1]

Composite element:
Element consisting of at least two different structural materials which cooperate in satisfying the requirements for ULS and/or SLS.

Conception: Identifying, developing and assessing different design alternatives.

Collapse may be a sudden occurrence, giving limited warning of the impending calamity.[5]

Composite elements can consist of basically different materials but also of variants of similar materials, such as concretes cast at different times.

Conceptual design: All activities and developments leading from the design criteria to a suitable structural solution.

Condition assessment: A process of reviewing information gathered about the current condition of a structure or its components, its service environment and general circumstances, allowing a prognosis to be made of current and future performance, taking account of active deterioration mechanisms and, if appropriate, predictions of potential future damage.

Condition control: The overall through-life process for conserving the condition of a structure, involving condition survey, condition assessment, condition evaluation, decision-making and the execution of any necessary interventions, performed as a part of the conservation process.

Condition evaluation: Similar to *condition assessment*, but is concerned with establishing the adequacy of the structure for future service, judged by its ability to comply with specified performance requirements comprising a defined set of loadings and environmental circumstances.

Condition evaluation would generally consider whether any subsequent intervention is required to meet the specified performance requirements (original or revised), or the implementation of structure management measures to allow the structure to remain in service, such as a reduction of the permitted imposed loading.

The term *condition assessment* may be used more commonly in connection with damaged or deteriorated structures.[5]

A wide range of parameters might be included in *condition survey*, with data being obtained by activities such as visual inspection and various ways of testing. *Condition survey* would also seek to gain an understanding of the (previous) circumstances which have led to the development of that state, together with the associated mechanisms causing damage or deterioration.

Condition survey: The process of acquiring information relating to the current condition of the structure with regard to its appearance, functionality and/or ability to meet specified performance requirements with the aim of recognizing important limitations, defects and deterioration.

Configuration: Creation of an aesthetic expression by means of spatial arrangement, shaping and choice of structural materials.[5]

Connection: Transition between structural elements able to transmit forces and/or moments.

Conservation activities may involve restoring the current condition of a structure to a satisfactory state, or include preventive measures which aim to ensure that the future condition of a structure remains within satisfactory bounds, or improvements to meet revised performance requirements. For this, the effects of potential future deterioration should be considered.

Conservation: Activities and measures taken which seek to ensure that the condition of a structure remains within satisfactory bounds to meet the performance requirements for a defined period of time, with respect to structural safety, serviceability and sustainability requirements, which may include considerations such as aesthetics.

Conservation plan: The overall plan for controlling and conserving the condition of a structure; that is, condition survey, condition assessment, condition evaluation, decision-making and the execution of any necessary intervention.

For comparison, see definition of *Structural materials*.

The *construction* is deemed to include any necessary preparatory works (e.g. excavation, landfill) and finishing works required to be carried out at a particular site or location to facilitate the creation of the desired entity (e.g. bridge).[5]

Construction products are either construction materials or various components, elements and assemblies made of construction materials, which are used during construction.

Cumulative knowledge of through-life performance concerns the evolution of certain properties or parameters relevant to safety, serviceability and/or durability of the structure, the type of loading (especially if fatigue effects are of potential concern), data on the characteristics of the environment(s) affecting the structure, and so on.[5]

Construction: see *Construction process*

Construction documents: Contract documents, construction programmes, minutes of meetings and records of construction inspections, together with the daily record of work carried out.[1]

Construction inspection plan: Specifying the type, extent, execution and timing of construction inspections, including information on quality requirements and admissible deviations as well as resolving questions of responsibilities and information flow.[1]

Construction inspections: Checking whether the design specifications are implemented correctly during execution.[1]

Construction materials: Structural and non-structural materials used in a construction process.

Construction process: The overall process of assembling construction elements or products to create a structure.

Construction product: Any product that is manufactured for erecting a building or infrastructural facility.

Construction work:
Carrying out the construction according to contract.[1]

Construction works documents:
Documents specific to construction works.[1]

Control measurement:
Measurement to monitor selected physical quantities (e.g. geometrical characteristics or structural deformations).[1]

Cumulative knowledge of through-life performance:
Information on the performance of a structure, based on systematic gathering and evaluation of data during the service life.[5]

Damage:
Physical disruption or change in the condition of a structure or its components, caused by external actions, such that some aspect of either the current or future performance of the structure or its components will be impaired.[5]

Defects may be in-built or may be the result of deterioration or damage.[7]

Design of structures (process) may be subdivided into *conceptual design*, *structural analysis* and *dimensioning*.

In the context of performance-based design, sets of *performance requirements* are used as input for the *design of structures*. Therefore performance-based *design of structures* will be preceded by the *conceptual design* including a *requirements development phase* (which may be preceded by a *feasibility study* of the project).

Decommissioning:
Discontinuation or interruption of use.[1]

Deconstruction:
Process of controlled dismantling by systematic selective removal of parts of the structure with planned recycling. In this case separation of material fractions is strived after.

Degradation: Worsening of condition with time; see also *Deterioration*.

Defect: A specific *deficiency* or inadequacy in the structure or its components which affects their ability to perform according to their intended function at the required level, either now or at some future time.[5]

Deficiency: Imperfection, possibly arising as a result of an error in design or construction, which affects the ability of the structure to perform according to its intended function, either now or in the future.[5]

Deformation capacity: (Elastic and/or plastic) deformation of a structure or a structural component reached at failure or at any other defined state of loading.

Demolition:
The process of conventional dismantling and removal of existing structures without previous demounting of structural members or structural materials.[5]

Design: Developing a suitable solution, taking due account of functional, environmental and economical requirements.

Design alternatives: Feasible alternatives to solve the design assignment.

Design boundary conditions: Space, time, legal, financial, structural, material-related, execution-related and service-related conditions for design.[1]

Design criteria: see *Performance criteria*.

Design of structures: Process of developing a suitable solution, taking due account of safety, functionality and sustainability of a structure during its intended service life.

Design service life: see *Specified (design) service life*

The *design situations* considered will include all foreseeable conditions that can occur during construction and use. The design will demonstrate that the relevant limit states are not exceeded for the identified design situation.

The *design value of a geometrical property* is generally a *nominal value*.

Where relevant, the *design value of a geometrical property* may be equal to the characteristic value, and correspond to some prescribed fractile of the statistical distribution. However, it may be treated differently in cases where the limit state under consideration is very sensitive to the value of the geometrical property.[8]

Alternatively, the *design value of a geometrical property* can be established on a statistical basis, with a value corresponding to a more appropriate fractile (e.g. rarer value) than applies to the characteristic value.[4]

Typically, *deterioration* of a structure or its components will be driven by chemical, mechanical or physical processes or agents, or combinations of those actions.

The term *diagnosis* is typically applied to forms of deterioration and degradation or other mechanisms causing an alteration in the expected or desired behaviour of the structure or its components.[5]

Dimensioning is usually performed in combination with numerical verifications by design equations.[1]

Design situations: Sets of defined actions and physical conditions representing the real situation expected during a specified time interval, for which the design is performed.

Design value of a geometrical property: Specified minimum or maximum value of geometrical dimension, which should not be exceeded.

Design value of an action: Value obtained by multiplying the representative value by the partial safety factor, corresponding to the design situation considered.

Design value of material or product property: Value obtained by dividing the characteristic value of the material or product property considered by a partial safety factor or, in particular circumstances, by direct determination.[2]

Desired state: State in which the performance criteria should be met.

Destruction: Loss of reliability, serviceability or durability due to damage to a structure that is of such severity that repair is not a practical or viable option.

Detailing: Determining the dimensions of structural components and reinforcement layout and geometry in local areas of the structure and specifying the structural details.

Deterioration: Worsening of condition with time, or a progressive reduction in the ability of a structure or its components to perform according to their intended functional specifications.[5]

Deterioration mechanism:
(Scientifically describable) process of the cause and development of deterioration.[1]

Diagnosis:
Identification of the cause or explanation of the mechanism(s) by which a phenomenon affects the behaviour or the condition of a structure or its components, based on an investigation of signs and indications exhibited.[5]

Dimensioning:
Determining the dimensions, the structural materials (including their properties) and the detailing of a structure on the basis of structural and execution-related considerations.[1]

In the context of performance-based design of structures, *durability* refers to the fulfilment of the performance requirements within the framework of the planned use and the foreseeable actions, without unforeseen expenditure on maintenance and repair.[1]

Dimensioning criteria:
see *Design criteria*

Dimensioning situations:
see *Design situations*

Dimensioning value:
see *Design value*

Disintegration:
Severe physical damage and disruption of a structure or its components which results in its (localized) break-up into fragments, with the possibility of gross impairment of their functional capability.[5]

Dismantlement:
Demolition of a structure with or without separation of the structural members and structural materials, fulfilling disposal requirements.[1]

Ductility:
Plastic deformation capacity characterized by irreversible deformations and energy dissipation, usually referred to quantitatively as the ratio between plastic deformation and the limit of the elastic behaviour.

Durability:
The capability of structures, products or materials to fulfil the requirements defined, determined after a specified period of time and usage.[3]

Durability assessment report (DAR) or durability plan

These are some of the alternative names that may be given to a Durability Deliverable Report for concrete structures. This is a formal stand-alone document which is authored by the durability consultant.

Durability checklist

Tabular/matrix form summarising durability issues relating to concrete structures, their members and other components of the project. Durability checklists are authored by the durability consultant.

Durability consultant

Person or group who complete the expert durability assessment and is / are the author of the durability assessment report and durability checklists. Refer *fib* Bulletin XY [2-2] and other contemporary guidance in CIA Z7-01 [2-3], including for durability assessment report and durability checklist.

Environmental influences need to be taken into account during planning of service life, design and construction of a particular structure or asset.[5]

Environmental influences may need to be considered at different scales ranging from macro level (affecting the overall structure), meso level (affecting an individual element or component) down to micro level (localized influences).[5]

The execution specification is not one document but signifies the total sum of documents required for the execution of the work as provided by the designer to the constructor and includes the project specification prepared to supplement and qualify the requirements of ISO 22966 [2-4], as well as referring to the national provisions relevant in the place of use.

In the context of limit state design, *failure* is reached when the criteria of the limit state under consideration are not met.

Associated Note: This would be a person or group who can apply materials deterioration knowledge to construction materials and construction processes, additional to more common structural, civil, geotechnical and other engineering knowledge of design, construction and maintenance works.

The durability consultant shall have relevant technical qualifications and experience (e.g. qualified materials scientist, engineer, etc), appointed / approved by the concrete structure owner (or authorised representative responsible to approve the durability consultant for a project).

Economy:
Moderate use of financial means and natural resources in relation to the whole period of design, execution and service.[1]

Economic pillar:

Environmental influences:
Physical, chemical and biological actions resulting from the atmospheric conditions or characteristics of the surroundings to the structure. (Loads associated with wind or wave effects are classified as mechanical loads.)

Environmental pillar:

Estimate:
Estimated mean value of a quantity.[1]

Examination: Condition survey and evaluation, including recommendation of remedial measures occasioned by special circumstances.[1]

Execution: All the activities and measures involved in the physical creation of a structure, including preparation for construction.[1]

Execution specification: Documents covering all drawings, technical data and requirements necessary for the execution of a particular project.

Failure: The state where the performance level of a structure or a structural element is inadequate.

In the context of performance-based design, a *feasibility study* may be carried out before starting the *requirements development phase* and the design of structure.

Ground can be built on (e.g. foundations to structures), built in (e.g. tunnels, culverts, basements), built with (e.g. roads, runways, embankments, dams) or supported (e.g. retaining walls, quays).

Often the term *ingress* is associated with the entry of substances which cause deterioration (e.g. chlorides into reinforced or prestressed concrete, sulfates and carbon dioxide (CO_2) into concretes).[5]

Interventions may be preventative (applying some form of treatment or taking action to ensure that the condition of a structure remains within satisfactory bounds or that an unsatisfactory performance condition is not reached) or reactive (taking action after damage has become visible, e.g. cracking or spalling of concrete).[5]

Interventions may be planned or unplanned. Planned interventions tend to be classified as maintenance. Unplanned interventions tend to be classified as repairs.

Interventions might be instigated for the purposes of, for example, repair, rehabilitation or remediation of the structure concerned.[5]

Fatigue resistance: Ultimate resistance under frequently repeated actions.[1]

Feasibility study: Preliminary analysis of all possible solutions to a problem and a recommendation on the best solution. A *feasibility study* is undertaken to ascertain the likelihood of the project's success.

Fixed action: Action with fixed distribution over the structure or structural member; everywhere the magnitude and the direction follow clearly from the information at a point.[1]

Free action: An action whose distribution over the structure is not fixed.[1]

Geometrical properties: Planned dimensions and unwanted imperfections of a structure.[1]

Ground: Subsurface material (e.g. sand, silt, clay, gravel, boulders or rock) in the area under or adjacent to a structure.

Hazard: An occurrence which has the potential to cause deterioration, damage, harm or loss.[5]

Hazard scenario: Critical situation characterized by a leading hazard and defined circumstances.

Ingress: The entry of substances into structural and/or non-structural components of a structure.[5]

Inspection: A primarily visual examination, often at close range, of a structure or its components with the objective of gathering information about their form, current condition, service environment and general circumstances.[5]

Integration: Adaptation of a structure to the natural and manmade environment.[1]

Intervention: A general term relating to an action or series of activities taken to modify or preserve the future performance of a structure or its components.

Inventory may be established to assist in the management of the structures.[5]

The process of inquiry might employ sampling, testing and various other means of gathering information about the structure, as well as theoretical studies to evaluate the importance of the findings in terms of the performance of the structure.[5]

Limit state represents the transition between the *desired state* and the *adverse state* (*failure*).

Maintenance activities involve recurrent or continuous measures which enable the structure to fulfil the requirements for reliability.[7]

The term *maintenance* is commonly applied in the context of building fabric components with a limited life, components associated with water management and rainwater run-off, items where regular intervention is required to maintain their effective operation and so on. The term *maintenance* is commonly applied to ancillary items such as gutters, drains, sealants, movement joints and bearings.

Maintainability refers to the probability that an item will be restored to specified conditions within a given period of time when maintenance action is performed in accordance with prescribed procedures and resources.[3]

Management of structures often involves conflicting requirements and objectives, which invariably requires compromise and judgement about the action to be taken or not taken, due to limitations in the available resources.[5]

Inventory: Detailed list or register of items or elements, possibly classified by type, function or some other principal attributes.[5]

Investigation: The process of inquiry into the cause or mechanism associated with some form of deterioration or degradation of the structure and the evaluation of its significance in terms of its current and future performance. The term may also be employed during the assessment of defects and deficiencies.[5]

Irreversible serviceability limit states: Serviceability limit states where some consequences of actions exceeding the specified service requirements will remain when the actions are removed.[2]

Leading action: Main action in a load case.[1]

Leading hazard: Main hazard in a hazard scenario.[1]

Limit state: State beyond which the structure no longer satisfies the relevant performance criteria.[2]

Load: see *Mechanical loading*

Load case: Compatible load arrangements, sets of deformations and imperfections considered simultaneously with fixed variable actions for a particular verification.[2]

Maintainability:

Maintenance: A set of planned (usually periodic) activities performed during the service life of the structure, intended to either prevent or correct the effects of minor deterioration, degradation or mechanical wear of the structure or its components in order to keep their future serviceability at the level anticipated by the designer.[5]

Maintenance plan: Instructions for maintenance specific to the structure considered.[1]

Maintainability: The ability of a structure to meet service objectives with a minimum expenditure of maintenance effort under service conditions in which maintenance and repair are performed.[3]

Management (of structures): Processes and procedures adopted for the operation, maintenance, inspection, testing, assessment and repair or other remedial action of structures in order to provide effective control against

(predetermined) criteria to ensure the continued safe service of individual structures or wider groupings of structures and related assets.[5]

Material: Metal, non-metallic inorganic or organic material with useful technical properties.[1]

Mechanical loading: (External) pressure, force or imposed displacement to which the structure or its components are subjected.

Method of construction: Manner in which the construction is carried out.[1]

Method statement: A description how construction will be performed. The Method Statement includes (but is not limited to) sequencing, planning (including the impact of phases on formwork/ falsework and reinforcement details) , temporary provisions, working methodology (including equipment required and prestressing procedures), applicable specifications and tests/verifications, safety/ health precautions and interference with, and care for the environment, logistics, required permits and approvals

Pending the stage of development/ construction stage of the scheme, the Method Statement is more global or fully detailed.

Modification: Making changes to a structure for the purpose of adapting it to new requirements.[1]

Monitoring: To keep watch over, recording progress and changes in materials and/or structural properties with time; possibly also controlling the functioning or working of an associated entity or process (e.g. by using warning alarms based upon parameters such as applied load, element deflection or some other aspect of structural response).[5]

Structural *monitoring* typically involves gathering information by a range of possible techniques and procedures to aid the management of an individual structure or class of structures. It often involves the automatic recording of performance data for the structure and possibly some degree of associated data processing.[5]

Monitoring involves similar activities to surveying, but with measurements being undertaken on an ongoing and possibly quasi-continuous basis. Monitoring could involve installed instrumentation. If so, this will introduce ways of measurement and data gathering different from those used for a survey. Under some circumstances, these activities might possibly include various forms of local/global response measurement or testing.

Monitoring plan: Instructions for monitoring specific to the structure.[1]

Nominal value: Value fixed on a non-statistical basis, for instance on acquired experience or on physical conditions, or a planned prescribed value.[2]

The *nominal value* of a material or a product property is normally used as a characteristic value and established from an appropriate document such as a standard.[4]

Objective of protection: Qualitative and quantitative specification of the requirements of a structure for the case of accidental occurrences and conditions.[1]

Observation: Examining the serviceability by simple and regular checks.[1]

Observational method: Possible procedure in the case of insufficiently reliable basic information for the design, execution and use of a structure, involving certain acceptable risks, a prediction of behaviour and the specification of associated limit values, together with corresponding monitoring and safety measures.[1]

Operational instruction: Instructions for the owners and users on the handling and operation of the technical equipment.[1]

Overall stability: State of stable equilibrium for the whole structure as a rigid body.[1]

Partial safety factor: A factor employed to deal with the uncertainties in the model variable.

The uncertainties in material properties are dealt with by the *partial safety factor* for a material property. The uncertainties of the (resistance) models (including geometric deviations associated with them, if these are not modelled explicitly) are dealt with by the *partial safety factor* for the (resistance) model. The uncertainties in the actions are dealt with by the *partial safety factors* for loads and environmental actions.

Passive state/passivity: The state in which, by virtue of a protective oxide film, steel does not spontaneously corrode.[7]

Penetration: The entry of substances into structural and/or non-structural components of the fabric of a building or structure.[5]

In many instances the term *penetration* is used interchangeably with the term *ingress*, but it may also be used in the context of evaluating the depth to which a deleterious agent has penetrated the component concerned (e.g. chlorides have penetrated to the depth of the reinforcing steel).[5]

The term *penetration* may also be associated with the introduction of agents which will help to extend the service life of the structure (e.g. the introduction of resins or corrosion inhibitors into concrete).[5]

Performance: The behaviour of a structure or a structural element as a consequence of actions to which it is subjected or which it generates.

Performance aspect: Aspect of the behaviour of a structure or a structural element for a specific action to which it is subjected or which it generates.

Performance based design:

In the context of limit state design, *performance criteria* are the limit values that describe for each limit state the conditions to be fulfilled.[4]

Performance criteria: Quantitative limits, associated to a *performance indicator*, defining the border between desired and adverse behaviour.

A *performance indicator* is associated with and gives meaning to the *performance criteria* used to define the *performance requirements* for a design, an actual, a potential or an intended intervention option.[5]

Performance indicator: A measurable/testable parameter (i.e. characteristic of materials and structures) that quantitatively describes a performance aspect.

Performance requirements are established by means of *performance criteria* and associated *performance indicators* and constraints related to *service life* and *reliability*.

Performance requirements refer to the fulfilment of the essential demands of the stakeholders (i.e. owners, users, contractors, society) during the intended lifetime of structures or structural elements.[5]

Sets of *performance requirements* are used as input into the performance-based design of structures.

Performance requirements are established by means of *performance criteria* and associated *performance indicators* and constraints related to *service life* and *reliability*.

Performance requirements refer to the fulfilment of the essential demands of the stakeholders (i.e. owners, users, contractors, society) during the intended life time of structures or structural elements.[5]

Sets of *performance requirements* are used as input into the performance-based design of structures.

Physical actions are usually caused by change of humidity or temperature (e.g. shrinkage, creep, fire exposure, heating and cooling, freeze–thaw, salt weathering) or movement of agents of wind, water, solid, ice (e.g. water erosion, wind erosion).

Products that are commonly fabricated by precasting, include beams and joists, slab units, wall panels, columns and utility items such as pipes and ducts.[3]

Performance level: Qualification of a structure or a structural element, which is established by verifying its behaviour against the performance requirements. A satisfactory performance level is reached when a structure or a structural element has demonstrated a sufficient behaviour to meet the performance requirements. In the opposite case, the performance level of a structure or a structural element is considered to be unsatisfactory.

Performance requirement: A condition for design, or an actual, potential or intended option for intervention, aiming at meeting a specified performance criterion during the service life with appropriate reliability and in a sustainable way.

Performance requirement: A condition for a design, an actual, potential or intended intervention option that the performance criterion must be met during the service life with appropriate reliability and in a sustainable way.

Permanent action: Action almost constant or monotonically approaching a limiting value during a reference period.[1]

Persistent design situation: Design situation which is relevant during a period of the same order of magnitude as the design service life.[2]

Physical actions: Physical phenomena other than mechanical loads (e.g. hydro-thermal processes, weathering, erosion processes) affecting and influencing the behaviour of the structure or its components.

Precast concrete: Concrete that is produced by wet-casting or extruding and cured at a location other than its final position in a structure.[3]

Precast element: element manufactured in compliance with a specific product standard in a factory, or in a location other than its final position in the structure.

Precast structure: a structure made of precast elements.

The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended. The treatment or action is taken before deterioration and/or damage become apparent/visible on the structure, for example due to cracking or spalling of concrete.

In the context of the Model Code, the *(owner's) professional team* means those engaged or commissioned by the *stakeholders* to advise and assist through the appropriate provision of technical and related services. Some, possibly all, of the individuals may reside within the entity or organization owning the facility concerned.[5]

Protection involves an action or series of actions undertaken to seek to defend a structure from the effects of further or future deterioration by providing a physical or chemical barrier to aggressive species (e.g. chloride ions) or other deleterious environmental agents and loadings upon the in-service performance and durability of a structure. Typically this will often be provided by surface coatings, impregnation treatments, overlays, membranes, electrochemical treatments, enclosure or surface wrappings applied to the concrete structure, elements or parts thereof.[5]

Typically, the *prudent estimate* is concerned with soil properties.

Preparation for construction: Invitation to tender, tendering, evaluation of tenders, conclusion of contract for materials and work, as well as preparation of construction work.[1]

Preventive intervention: A proactive conservation activity concerned with applying some form of treatment or taking action that anticipates a change in a material property (e.g. carbonation or chloride ingress causing deterioration) adversely affecting the ability of a structure, or parts of it, to meet the required performance levels.

(Owner's) professional team: A group of persons, generally from one or more organizations, who together are skilled in the various technical aspects and processes required for the design, construction and maintenance of buildings, works and other facilities of public or commercial utility.[5]

Protection: A measure which prevents or reduces the development of defects.[7]

Prudent estimate: A value which, compared to the estimate, is provided with an adequate margin to meet the required reliability.[1]

Quality management plan: DEFINITION TO BE ADDED

Reactive intervention: A reactive conservation activity, undertaken after deterioration and/or damage has become apparent/visible (e.g. cracking or spalling of concrete) such that, because of the deterioration, it has adversely affected the ability of the structure, or parts thereof, to meet the required performance levels (which may include consideration of issues such as aesthetics).

Re-birth certificate: A document, report or technical file similar to the birth certificate for a structure, but related to the information and circumstances associated with a project for the repair/remediation/ refurbishment of the structure, or a part thereof, to extend its anticipated service life.[5]

Rebuild: To create a new structure or structural component to replace the original damaged, defective or deteriorated entity after its destruction or demolition, without restriction upon the materials or methods employed.[5]

Typically, *recalculation* is concerned with in-service performance assessment and structural load capacity in particular. The process may utilize similar steps and procedures to design but fundamentally differs from this by seeking to take into account the actual form and condition of the structure as found, including deterioration. This will often include a more realistic consideration of the actual loading regimes, rather than the idealized values used in design. The recalculation process may be used to predict future structural performance, taking into account the influence of ongoing deterioration processes and any remediation actions.[5]

Generally, *reconstruction* is concerned with meeting specific objectives such as strength or future durability requirements.[5]

This includes in particular the use for the production of new concrete of coarse fractions (e.g. recycled concrete aggregates), fine fractions (e.g. sieve sand) and/or binders extracted from construction and demolition waste flow.

COMMENT: Ask AG10 whether they agree with this definition which is given in chapter 3.

The aim of *rehabilitation* is in principle similar to the aim of reconstruction, but possibly with greater emphasis upon the serviceability requirements associated with the existing or proposed revised usage of the structure.[5]

Recalculation: A process of analytical examination using mathematical models or simplified representations of an existing structure or structural elements in order to make an estimate of the performance, taking into account the actual form and condition of the structure as found, including deterioration.

Reconstruction:
Restore or reinstating all or part of a structure or component that is in a changed, defective or deteriorated state compared to its original or higher level of performance, without restriction upon the methods or materials employed.[5]

Record of construction:
Collection of construction works documents updated according to the state of the execution.[1]

Recycling:
Is the (repetitive) use of construction materials and/or constituents of construction materials after extracting them from residual waste flows.

Redundancy:
Ability of a structure to redistribute internal forces in the event of failure of key elements, which can be achieved by means of load-path redundancy (i.e. alternative and sufficient load path is existing), structural redundancy (i.e. the boundary conditions or supports are such that failure of the member merely changes the boundary or support conditions but does not result in the collapse of the superstructure), and internal redundancy (i.e. alternative load paths exist in the same member)

Reference period:
Chosen period of time used as a basis for assessing statistically variable actions, and possibly for accidental actions.[2]

Refurbishement and repurposing:
Service life extension of existing structure for new functional or technical requirements, which can be achieved by functional or structural modifications.

Rehabilitation:
Intervention to restore the performance of a structure or its components that are in a changed, defective, degraded or deteriorated state to the original level of

In some instances, the *rehabilitation* may not be intended to bring the structure or its components back to the original level of serviceability or durability. The work may sometimes be intended simply to reduce the rate of deterioration or degradation, without significantly enhancing the current level of serviceability.[5]

In the context of performance-based design of structures, *reliability* refers to the ability of a structure or a structural member to fulfil the performance requirements during the service life for which it has been designed[4] at a required failure probability level corresponding to a specified reference period.

Possible *remedial interventions* are wide ranging and may involve structural strengthening through to preventative measures, such as applying surface coatings to provide a barrier to the ingress of deleterious environmental agents (e.g. chloride ions). The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended.

The term *remodelling* is often employed where changes principally involve appearance, rather than alteration of the structural components.[5]

In some instances, the *repair* may not be intended to bring the structure or its components back to its original level of serviceability or durability. The work may sometimes be intended simply to reduce the rate of deterioration or degradation, without significantly enhancing the current level of performance.[7]

Repairability is related to level of damage/degradation in a structure, which is to be repaired. Decision-making for intervention is conducted based on the level of damage/deterioration. The definition is provided in accordance with ISO 19338:2014.

performance, generally without restriction upon the materials or methods employed.[5]

Reliability:
Ability of a structure or a structural member to perform its intended function satisfactorily (from the viewpoint of the stakeholder) for its intended life under specified environmental and operating conditions.[3] Reliability is usually expressed in probabilistic terms.[4]

Reliability differentiation: Measures intended for socio-economic optimization of the resources to be used to build structures, taking into account all expected consequences of failures and the cost of the structures.[2]

Remediation: see *Remedial intervention*

Remedial intervention: A conservation activity undertaken after a change in a material property (e.g. that caused by the influence of carbonation or chlorides) has adversely affected the ability of the structure, or parts thereof, to meet the required performance levels because of deterioration.

Remodelling: Changes or alterations to a structure to meet revised functions, performance requirements, usage or occupancy.[5]

Removal: Removing parts from a structure.[5]

Renewal: To reinstate the performance of a damaged or deteriorated component or structure using original methods and materials.[5]

Repair: Intervention taken to reinstate to an acceptable level the current and future performance of a structure or its components which are either defective, deteriorated, degraded or damaged in some way so that their performance level is below that anticipated by the designer; generally without restriction upon the materials or methods employed.

Repairability (restorability): ability of a structure or structural element to be repaired physically and economically when deteriorated/damaged under the effects of considered actions.

Representative value of an action: The value of an action used for the verification of a limit state. A representative value may be the characteristic value,

Replacement may include improvements and strengthening, but does not usually involve a change in function.[5]

The *required service life* is the basis for determining the *specified (design) service life* (for new structures) and the *specified (design) residual service life* (for existing structures).

The *requirements development phase* may be subdivided into gathering the requirements from *stakeholders*, checking for consistency and completeness, definition (writing down descriptive requirements) and specification (creating an initial bridge between requirements and design). The *requirements development phase* may have been preceded by a *feasibility study* of the project.

While *stakeholders* usually believe that they know which performance of a structure they are requesting, it may require skill and experience in structural engineering to recognize incomplete, ambiguous or contradictory requirements.

The *residual service life* is related to the *required service life*, as given by the stakeholders (i.e. owners, users, contractors, society) of the structure and to the other implications of service criteria agreement, for example with regard to structural analysis, maintenance and quality management.

Reuse options may include use in new structures of e.g. structural members or structural components design for multi-life-cycle or reclaimed from construction and demolition waste

the combination value, the frequent value and the quasi-permanent value, but it may also be another value of an action.[2,6]

Replacement: Action to provide substitute new components for ones which have experienced deterioration, damage, degradation or mechanical wear.[5]

Required service life: The demand stated by the stakeholders (i.e. owners, users, contractors, society) for the period in which the required performance has to be achieved.

Requirements development phase: Phase of extracting and describing performance requirements for a structure.

Resilience

The ability of a system, community or society exposed to hazards to resist, absorb, accommodate and recover from the effects of a hazard in a timely and efficient manner, involving the preservation and restoration of its essential basic structures and functions, through the use of risk management techniques. *[Adapted from 11]*

Resistance: Capacity of a member or component, or a cross-section of a member or component of a structure, to withstand actions.[4]

Residual service life: The demand for the remaining period in which the required performance has to be achieved, used in the re-design of existing structures.

Restoration: Intervention to bring the structure or its components back to their original condition, not only with regard to function and performance level anticipated by the designer, but also with regard to aesthetic appearance and possibly other (historical) considerations.[5]

Reuse:
The (repetitive) use in new structures of structural members of structural components reclaimed from existing structures.

Robustness indicates the ability of a structural system to mobilize alternative load paths around an area of local damage. It is related to the strength and form of the structural system, particularly the degree of redundancy (number of potential alternative load paths) within the structural system.[5]

In the context of performance-based design of structures, *safety* is one of the basic performance requirements. For comparison, see the definition of *structural safety*.

For comparison see the definition of *required service life*, *specified (design) service life*, *residual service life*.

CEN documents use the term *working life* where this Model Code uses the term *service life.*

Serviceability may be evaluated under various headings, and consideration would normally be given to a number of issues affecting either the whole structure or parts thereof. The issues would typically include various limit state cases (e.g. deflection, vibration, thermal movements, appearance).[5] In the context of performance-based design of structures, *serviceability* is one of the basic performance requirements.

Risk: The combination of the likelihood of occurrence of a particular hazard and its consequences.[5]

Robustness:
The ability of a structure to withstand adverse and unforeseen events (like fire, explosion, impact) or consequences of human errors without being damaged to an extent disproportionate to the original cause[6]

Safety: Ability of a structure or structural element to ensure that no harm would come to the users and the people in the vicinity of the structure under any (combination of) expected actions.[10]

Safety criterion: Performance criterion for the ultimate limit state (ULS).

Service life: The period for which the required performance of a structure or structural element is achieved, when it is used for its intended purpose and under the expected conditions of use.[4,5]

Serviceability:
Ability of a structure or structural element to perform adequately for normal use under all (combinations of) actions expected.[6]

Serviceability limit: Specified limit of serviceability.[1]

Serviceability limit state (*SLS*): State that corresponds to conditions beyond which specified service requirements for a structure or structural member are no longer met.[2]

Serviceability criterion: Performance criterion for a serviceability limit state (SLS).[2]

Service criteria: Requirements for the behaviour of a structure resulting from the planned use.[1]

Service criteria agreement: Description of the utilization and protection aims of the stakeholders (i.e. owners, users, contractors, society) as well as the basic conditions and regulations for the design, execution and use of the structure.[1]

Service instructions: Instructions for the owner and the operator on the use of the construction works.[1]

Service situations: Physical circumstances and conditions during the design service life.[1]

Social pillar:

The s*pecified (design) service life* is the service life, as required by the stakeholders (i.e. owners, users, contractors, society) and to the other implications of service criteria agreement, such as with regard to structural analysis, maintenance and quality management.

Specified (design) service life: The period during which the required performance must be achieved, used in the design of new structures.

As a rule, the key *stakeholders* would be the founders, the owners, the users, the neighbours (if construction interferes with them), the contractor, the design and constructing team, maintenance and demolition team, facility management, society in general and other which can vary in each case. Other *stakeholders* may be the government and society.

Stakeholder: Person or organization that has a legitimate participation in a project.

Strengthening: An intervention made to increase the strength (load resistance/load capacity) and/or possibly the stiffness of a structure or its components, and/or to improve overall structural stability and/or the overall robustness of the structure to a performance level above that adopted by the designer.

Structural analysis:
Determination of action effects by means of a structural model, if necessary in steps, using different analytical models for the structures as a whole, individual members and local effects.[1]

Structural design concept: The basic idea underlying the structural design.[1]

Structural integrity: The ability of structural components to act together as a competent single entity.[5]

Construction materials that are used primarily for decoration, insulation or other than structural purposes are not included in group of *structural materials*.[3]

Structural materials: Construction materials which, because of their ability to withstand actions, are considered in the design of a structure.[3]

Structural member: Physically distinguishable part of a structure, such as a column, a beam, a slab or a foundation pile.[1]

Structural model: Result of delimiting and idealizing the structural system.[1]

Structural resilience:
Ability of a structure to respond, absorb, and adapt to, as well as recover from a disruptive event, whereby a structure is expected to be able to resist to an extreme event with minimal damages and functionality disruptions during the event and whereby after the event, the structure should be able to rapidly recovery its functionality similar to the pre-event level.

Structural safety is usually expressed by the ratio (safety factor) between the actions that would cause collapse or other similar forms of structural failure and the actions that are imposed upon it in service.[3]

Survey is taken to mean the range of activities used to evaluate conformity with the design data for actions and/or material and/or product properties used in the service life design (SLD) on a periodic basis during the service life of the structure. Survey activities would be expected to include a visual inspection undertaken in conjunction with various forms of localized condition testing and measurement (e.g. measurement of depth of cover to reinforcement).

The term *survey* may be applied to the inspection of a number of similar structures/components to obtain an overview. The term *survey* is also used to describe the formal record of inspections, measurements and other related information which describes the form and current condition of a structure and its components.[5]

Various types of *testing* are recognized, their classification being primarily based on the amount of damage or interference caused to the structure. The main divisions are:

– non-destructive testing, which does not cause damage to the structure by the test procedure (e.g. testing with cover meter, radar, acoustic emission, load testing in the elastic range);

Structural safety: Ability of a structure and its members to guarantee the overall stability as well as an adequate ultimate bearing resistance, corresponding to the assumed actions and the required reliability for the specified reference period.[1]

Structural system: Arrangement of interacting structural members offering a potential solution to provide bearing resistance to a specified combination of actions.

Structure: Product of human design, intended to fulfil societal functions with adequate reliability with regard to safety, serviceability and sustainability, for a defined period of time.

Substrate: The surface layer in which a protection or repair material has been applied or is to be applied.[5]

Survey: The process, often involving visual examination or utilizing various forms of sampling and testing, aimed at collecting information about the shape and current condition of a structure or its components.[5]

Sustainability: Ability of a structure or structural element to contribute positively to the fulfilment of the present needs of humankind with respect to nature, society, economy and well-being, without compromising the ability of future generations to meet their needs in a similar manner.

Technical report: Explanatory report on design work.[1]

Tender documents: Text of the planned contract for materials and work, special conditions, bill of quantities or work description, plans and general conditions.[1]

Testing: Procedure aimed at obtaining information about the current condition or performance of a structure or its components.[5]

– destructive testing, which may cause damage to the structure or marking of the surface finishes (e.g. pull-out tests, material sampling, load testing beyond the elastic range).[5]

Generally, the *ultimate limit state* corresponds to the maximum load-carrying resistance of a structure or structural member.[2]

Upgrading (*retrofitting*) relates particularly to the strengthening of structures as a means of minimizing damage during specified loading events.

SOURCE ISO 9001

SOURCE ISO 9001

Tie: Tensile continuous element acting across the structure, horizontally and/or vertically.

Transient design situation: Design situation that is relevant during a period much shorter than the design working life of the structure and which has a high probability of occurrence.[2]

Ultimate limit state (*ULS*): State associated with collapse or with other similar forms of structural failure.[2]

Ultimate resistance: Limit of resistance.[1]

Upgrading (*retrofitting*): Intervention to enhance the functionality or form of a structure or its components so as to improve some aspect of future performance above that defined/achieved during design and construction; typically undertaken to achieve an improved (higher) load resistance against specified loads/actions.

Use: Utilization of a structure as described in the service criteria agreement and in the basis of design.[1]

Validation: In general, validation is the confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled.

Variable action: Action which is not permanently acting, not constant or not monotonically changing during a reference period.[1]

Verification: Activities conducted to ensure that the design and development outputs meet the input requirements;

Vulnerability

The possibility that a hazard or threat can cause damage, disruption or some form of harm to the entity or system affected.

Associated Note: A critical consideration in respect of vulnerability is the magnitude of the harm or disruption of the functionality of the entity or system which may be caused by a hazard or threat incident / shock event.

The degree of harm or disruption of the functionality caused to the entity or system will commonly depend upon the magnitude or severity of the hazard or threat incident / shock event experienced by that entity or system.

The level of the vulnerability of the entity or system may be judged by the degree of harm or disruption caused relative to the magnitude / severity of the hazard / threat incident or shock event. For example, the relative vulnerability of the entity or system to the hazard / threat might be described in terms of it being "*Low*", "*Medium*", "*High*", "*Extremely high*".

2.2 Notations

Meaning of roman capital letters

A area
C torsional moment of inertia; serviceability constraints
D fatigue damage factor; diffusion coefficient
E modulus of elasticity; earthquake action; load (action) effect
F action in general; local loading
G permanent action; shear modulus
H horizontal component of a force
I second moment of a plane area
J creep function
K (permeability) coefficient
M bending moment; coefficient of water absorption; safety margin
N axial force
P force
Q variable action
R resistance; strength (resisting load effect); reaction at a support; resultant
S static moment of a plane area
T torsional moment; temperature
V shear force, volume
W modulus of inertia
X material or soil properties in general; reaction or force in general, parallel to x axis
Y reaction or force in general, parallel to y axis Z reaction or force in general, parallel to z axis

NOTE: Roman capital letters can be used to denote types of material, e.g. C for concrete, LC for lightweight concrete, S for steel, Z for cement.

Meaning of roman lower case letters

a deflection; distance; acceleration
b width
c concrete cover
d effective height; diameter (see also h)
e eccentricity; sets of loads (actions)
f strength
g distributed permanent load; acceleration due to gravity; limit state function
p prestressing
q distributed variable load
r radius; resistance variables; resistance function
s spacing; standard deviation of a sample
t time; torsional moment per unit length or width; thickness of thin elements
h total height or diameter of a section; thickness
i radius of gyration
j number of days
k all coefficients with dimension
l span; length of an element
m bending moment per unit length or width; mass; average value of a sample
n normal (longitudinal, axial) force per unit length or width
u perimeter
v velocity; shear force per unit length or width
w width of a crack
x coordinate; height of compression zone
y coordinate; height of rectangular diagram coordinate; lever arm

Use of Greek lower case letters

alpha	α	angle; ratio; coefficient
beta	β	angle; ratio; coefficient
gamma	γ	safety factor; density; shear strain (angular strain)
delta	δ	coefficient
epsilon	ε	strain
zeta	ζ	coefficient
eta	η	coefficient
theta	θ	rotation
lambda	λ	slenderness ratio; coefficient
mu	μ	relative bending moment; coefficient of friction; mean value of a whole population
nu	ν	relative axial force; Poisson's ratio
xi	ξ	coefficient; ratio
pi	π	mathematical use only
rho	ρ	geometrical ratio of reinforcement; bulk density
sigma	σ	axial stress; standard deviation of a whole population
tau	τ	shear stress
phi	φ	coefficient
chi	χ	coefficient
psi	ψ	coefficient; ratio
omega	ω	mechanical ratio of reinforcement

Mathematical symbols and special symbols

S	sum
Δ	difference; increment (enlargement)
Ø	nominal diameter of a reinforcing bar or of a cable
′	(single prime) compression (only in a geometrical or locational sense)
e	base of Naperian logarithms
exp	power of the number e
π	ratio of the circumference of a circle to its diameter
n	number of ...
w/c	water/cement ratio
<	smaller than
>	greater than

General subscripts

a	support settlement; additional; accidental load
b	bond; bar; beam
c	concrete; compression; column
d	design value
e	elastic limit of a material
f	forces and other actions; beam flange; bending; friction
g	permanent load
h	horizontal; hook
i	initial
j	number of days
k	characteristic value
l	longitudinal
q	variable load
r	cracking
s	ordinary steel; snow; slab
t	tension;* torsion;* transverse
u	ultimate (limit state)
v	shear; vertical
w	wind; web; wire; wall
x	linear coordinate
y	linear coordinate; yield
z	linear coordinate
1, 2, 3	particular values of quantities
cc	conventional asymptotic value

m	mean value; material; bending moment
n	axial force
o	zero
p	prestressing steel

Subscripts for actions and action effects

a(A)	support settlement; accidental action
cc	creep of concrete
cs	shrinkage of concrete
ep	earth pressure
ex	explosion; blast
g(G)	permanent load
im	impact
lp	liquid pressure
m(M)	bending moment
n(N)	axial force
p(P)	prestress
q(Q)	variable load
s(S)	snow load
t(T)	torsion; temperature
v(V)	shear
w(W)	wind load

Subscripts obtained by abbreviation

abs	absolute
act	acting
adm	admissible, permissible
cal	calculated, design
crit (or cr)	critical
ef	effective
el (or e)	elastic
est	estimated
exc	exceptional
ext	external
fat	fatigue
inf	inferior
int	internal
lat	lateral
lim	limit
max	maximum
min	minimum
nec	necessary
net	net
nom	nominal
obs	observed
pl	plastic
prov (or pr)	provisional (stage of construction); provided
red	reduced
rel	relative; relaxation
rep	representative
req	required
res	resisting, resistant
ser	serviceability
tot	total
var	variable

Notation list

Roman lower case letters

* When confusion is possible between tension and torsion, the subscripts tn (tension) and tr (torsion) should be used.

2.3 Acronyms

AAEM	age adjusted effective modulus (for creep calculations)
AAR	alkali aggregate reaction
ACI	American Concrete Institute
AFRP	aramide fibre reinforced plastic
ASR	alkali silica reaction
ASTM	American Society for Testing and Materials
BCD	birth certificate document
CCL	condition control level
CCP	condition control plan
CEB	Comité Euro-Internationale du Béton / Euro-International Committee for Concrete
CEN	European Commission for Normalization
CEM	indication for cement type
CFRP	carbon fibre reinforced plastic
CMOD	crack mode opening displacement
CTE	coefficient of thermal expansion
DIN	German Institution for Normalization
ECE	electrochemical chloride extraction
EDC	equivalent durability concept
EE	embodied energy
EIC	environmental impact calculation
EN	European Norm
ETA	European Technical Approval
ETAG	European Technology Assessment Group
fib	fédération internationale du béton / International Federation for Structural Concrete (created from the merger of CEB and FIP)
SIA	social impact assessment, or Swiss Union of Engineers and Architects
SFRC	steel fibre reinforced concrete
SLD	service life design
FIP	Fédération Internationale de la Précontrainte / International Federation for Prestressing
FRC	fibre reinforced concrete
FRP	fibre reinforced plastic
GFRP	glass fibre reinforced plastic
GHG	green house gas
GWP	global warming potential
Hz	hertz
IABSE	International Association for Bridges and Shell Structures
ISO	International Organization for Standardization
JCSS	Joint Commission on Structural Safety
JSCE	Japanese Society of Civil Engineers
JSSC	Japanese Society of Steel Construction
LC	indication for lightweight concrete strength class
LCC	life cycle cost
LCF	life cycle file
LCM	life cycle management
LoA	level of approximation
LWAC	light weight aggregate concrete
MC	Model Code
MPa	megapascal
PC	prestressed concrete
PL	protection level
PQP	project quality plan
QM	quality management
RC	reinforced concrete
SLS	serviceability limit state
RILEM	International Union of Laboratories and Experts in Construction Materials, Systems and Structures

RH	relative humidity
SETRA	French Road and Motorway Technical Studies Department
SCA	service criteria agreement
SCC	self compacting (consolidating) concrete
UFC	Unified Facilities Criteria (code for military structures)
UHPFRC	ultra high performance fibre reinforced concrete
ULS	ultimate limit state
UTS	ultimate tensile strength

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References

[2-1] *fib* Bulletin 65/66. *fib* Model Code for Concrete Structures 2010.

[2-2] fib Bulletin XY Durability planning for concrete structures [IN PREPARATION, COM8]

[2-3] CIA Z7/01 Durability Planning, which is a Recommended Practice Report forming part of the CIA Concrete Durability Series, Concrete Institute of Australia (CIA), Sydney, 2014.

[2-4] ISO 22966 ISO 22966:2009, Execution of concrete structures, ISO, 2009

The definitions given in section 2.1 are partly based on the following sources:

1 SN 505 260 (SIA 260:2003), Basis of Structural Design, 2003

2 CEN, EN 1990:2002, Eurocode – Basis of Structural Design, 2002

3 McGraw-Hill Encyclopedia of Science and Technology Online, in http://www.accessscience.com/, last modified Sept. 2003

4 ***fib*** Bulletin 34, Model Code for Service Life Design. Fédération Internationale du Béton, 2006

5 ***fib*** Bulletin 17, Management, maintenance and strengthening of concrete structures. Fédération Internationale du Béton, 2002

6 ISO 2394:2015, General principles on reliability for structures, 2015

7 CEN, ENV 1504:1997: Part 9, Products and systems for the protection and repair of concrete structures – Definitions, requirements, quality control and evaluation of conformity – Part 9: General principles for the use of products and systems, 1997

8 “Probabilistic Model Code”, Joint Committee on Structural Safety (JCSS PMC), 2000

9 SN 505 262 (SIA 262:2003), Concrete Structures, 2003

10 Asian Concrete Model Code, ACMC 2006

11 United Nations Office for Disaster Risk Reduction publication, UNISDR Terminology and Disaster Risk Reduction, Geneva, 2009

In 1987, the Report of the World Commission on Environment and Development: Our Common Future (Brundtland Report) defined the principle of sustainable development as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs." In 2015, United Nations (UN) adopted in 2015 Resolution 70/1: Transforming our world: the 2030 Agenda for Sustainable Development. This Agenda of Sustainable Development Goals (SDG) is a plan of action for people, planet and prosperity and should stimulate action up to 2030 in areas of critical importance for humanity and the planet. The principal goal is to end poverty, protect the planet and ensure prosperity for all. One of the SDG Agenda priorities is sustainable urban development, which is a crucial aspect influencing the quality of life of people. Construction industry as a main stakeholder responsible for the use of material and energy resources has a key role in the implementation of SDG actions in the process of design, development and operation of buildings, roads, bridges and other infrastructure creating urban built environment.

For the practical achievement of SDGs, consideration of ESG (Environment, Social and Governance) concept is also important.

Crucial importance of concrete structures in enabling sustainable development follows from the amount of produced concrete, which is the most widely used construction material in the world with large impacts on the environment.

Enhancing the performance of concrete elements and structures can be achieved for instance by:

(i) enhancement of design and assessment methods,

(ii) developments of alternative concrete mixtures, composite materials and ways of reinforcement,

(iii) upgrade of technology for concrete elements production,

3 Sustainability perspective

3.1 Principles of design and assessment with respect to sustainable development

Sustainability is a global goal of sustainable development focusing to assurance of quality life for the future generations.

Concrete products and structures should, through their performance within the entire life cycle, contribute to sustainable growth of society and sustainable development of life by reducing negative impacts and increasing positive affect on the society, environment and the economy.

Sustainability approach is a key conceptual principle to be considered through the life cycle of concrete structures in all activities, including design, production, construction, operation, maintenance, repair and dismantlement or reuse of any building or civil engineering work forming the built environment.

(iv) development of integrated design procedures for the different systems and technologies (including structural systems) in buildings and infrastructures,

(v) innovations in construction techniques,

(vi) assessment and re-design of existing structures and associated life-cycle management processes and

(vii) innovations in maintenance, repair, demolition and recycling processes.

The three pillars of sustainability are also referred to as people, planet and profits.

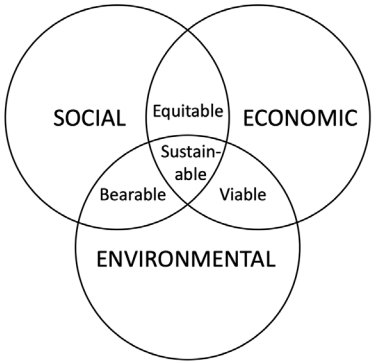


Figure 3.1-1: *Three pillars of sustainability and their interconnections*

Further guidance on sustainability assessment can be found in ISO 21931-2 [3-3] and EN 17472 [3-4].

A considerable gain in sustainability can also be achieved by focusing on certain parts of the life cycle e.g. on design, on design and execution or on design, execution and conservation. Particular aspects of sustainability could be assessed by LCA, LCC and other methods of assessment of structures using existing standardized approaches in generally accepted standards (ISO, ACI, EN, etc.). By changing the extent of the life cycle stages considered, a different level of ambition is aimed at.

According to many organizations, including the United Nations (e.g. UN Resolution 70/1), ISO (e.g. ISO 13315 [3-1]) or CEN (e.g. EN 15643 [3-2]), sustainability can be described by means of three interdependent, interconnectedly linked and mutually reinforcing pillars of sustainability on which sustainable development depends:

- Social pillar – covering essential social aspects
- Environmental pillar – covering essential environmental aspects
- Economic pillar – covering essential economic aspects

In the design of new structures and in the assessment of existing structures, evaluation of performance should be performed for all three pillars of sustainability, described in section 3.2 (social performance), section 3.3 (environmental performance) and section 3.4 (economic performance).

Sustainability approach requires application of performance based design and assessment of concrete structures which is described in chapter 4. Hence, the relevant aspects of social, environmental and economic issues are to be dealt with as performances with the associated necessary requirements to be considered in order to provide the functions satisfactorily throughout the working life of the structure.

It is essential to consider changes and development of performance within the entire life of a structure. Therefore a life cycle approach is important in the sustainability evaluation. This consideration corresponds to the perspective of life-cycle management given in chapter 5.

In order to insure sustainability approach to be complete, it is essential that all stakeholders that are related to the structure are considered. The main stakeholders are: founders, owners, users, neighbours, contractor, design and construction team, maintenance and demolition team, facility management, society in general and other which can vary in each case.

The many aspects of sustainability are to be considered when making choices in design of concrete structures, with the objective to obtain the optimum structural solution under the given circumstances. This process is in practice managed by the lead engineer and in many cases this is the civil/ structural engineer with overall responsibility for the design. The lead engineer should cooperate already in an early stage with the other responsible persons for architectural design, construction and production of relevant structural or non-structural elements. The level of knowledge and experience of the lead engineer and senior team members is the decisive factor in this process. To deal with the context in which structural design is carried out, it shall be coordinated with the design of any other system (e.g. architectural components, MEP systems, ICT systems) of buildings and infrastructures, in order to optimize and integrate the mutual contribution of the different systems to the

The relevant aspects of social, environmental and economic performance related to fulfilling performance requirements and contributing to achievement of sustainable development goals during the entire life of a structure shall be defined based on society and stakeholder needs. They should be specified during briefing phase in the client brief and during the subsequent phases of the design and assessment process they should be dealt with in a way that is compatible with the decision process employed by the owner. In particular:

- during the briefing phase, the client's brief should specify the legal framework and the client specific requirements. As sustainability comprises many aspects which may interfere each other, strengthen/ weaken each other or can even be conflicting, the brief should contain an order of significance or any other hierarchy / valuation to allow engineers to make choices (see subsection 10.2.3),
- during scouting phase (see subsection 10.1.4);
- during basis of design phase (see subsection 10.1.5);
- during project specification phase (see subsection 10.1.6);
- during final design phase (see subsection 10.1.7);
- during detailed design phase (see subsection 10.1.8);
- during operation and use phase (see subsection 10.3);
- during demolition phase (see subsection 10.4).

General principles of considering sustainability in the conceptual design and during the process of the assessment of concrete structures are described respectively in chapters 26 to 28.

sustainability targets. To improve the integrated design procedures, digital design tools and processes, e.g. BIM systems, may be adopted.

Performance requirements with respect to structural safety, serviceability and durability of a structure are expressed in terms of reliability. Reliability is defined as ability of a structure or structural member to fulfil the specified requirements, during the working life, for which it has been designed. Treatment of reliability in verification of structural performance is discussed in section 4.

Other appropriate methods could be applied based on regional specifics.

Consistent with the scope definitions in sections 3.2, 3.3. and 3.4, the provisions for the evaluations for the three pillars of sustainability are described in chapter 27 and chapter 31 for social performance, in chapter 32 for environmental performance and in chapter 33 for economic performance. Basis of the overall evaluation of sustainability for the three pillars as a whole is described in a chapter 30.

3.2 Social performance

3.2.1 General

A structure should be designed and constructed in order to provide high quality and safe life for society. It means, that the performance of a structure should provide operation of structure within its entire life which should not cause any adverse influence to society. On the other hand, the structure should indorse beneficial changes in the society for current and future generations. Social performance of concrete structure concerns:

– Structural performance (see subsection 3.2.2),

– Other aspects of social performance (see subsection 3.2.3).

3.2.2 Structural performance

Structural performance is the basic aspect of social performance of any concrete structure. The basis of performance-based approach to structural performance is described in section 4. Assurance of high-quality structural performance is one of the most important design and engineering tasks and covers performance aspects such as:

- structural safety,
  defined as ability of a structure or structural member to avoid exceedance of ultimate limit states, including the effects of specified accidental phenomena, with a specified level of reliability, during a specified period of time;

- serviceability
  defined as ability of a structure or structural member to perform adequately for normal use under all (combinations of) actions expected;
- durability
  defined as capability of a structure or any structural member to satisfy with planned maintenance the design performance requirements over a specified period of time under the influence of the environmental actions;
- repairability
  ability of a structure or structural element to be repaired physically and economically when deteriorated/damaged under the effects of considered actions;
- robustness
  defined as ability of a structure to withstand adverse and unforeseen events (like fire, explosion, impact) or consequences of human errors without being damaged to an extent disproportionate to the original cause;

Robustness can be achieved e.g. by means of reducing the possibility of occurrence of accidental actions, preventing the propagation of a possible initial failure, by designing the structure to withstand accidental and unidentified actions or by increasing redundancy (e.g. by the indirect method and the alternative load path method). Redundancy is defined as ability of a structure to redistribute internal forces in the event of failure of key elements, which can be achieved by means of load-path redundancy (i.e. alternative and sufficient load path exists), structural redundancy (i.e. the boundary conditions or supports are such that failure of the member merely changes the boundary or support conditions but does not result in the collapse of the superstructure), and internal redundancy (i.e. alternative load paths exist in the same member)

Considering an increasing number of natural and man-made disasters and increasing economic and social problems, it is necessary to modify existing principles and methods of structural design, corresponding techniques of construction and operation of structures, in order to be more resilient. The principal goal is to enhance structures and the entire built environment to be more resilient in the situation of changing natural and socio-economic conditions in the world. Generally, resilience should be pursued in a systemic approach at an urban or regional scale, looking at the social, economic and physical components of cities and territories.

However, resilience should be taken into account in the design of single structures as well. In this case, a resilient structure is expected to be robust (capable to absorb external shocks and incur moderate damages) and easily repairable/recoverable (so that any inspection, repair or retrofit intervention is not particularly complicated, expensive or time consuming). Thus, resilience of a

- structural resilience,
  defined as ability of a structure to respond, absorb, and adapt to, as well as recover from a disruptive event, whereby a structure is expected to be able to resist to an extreme event with minimal damages and functionality disruptions during the event and whereby after the event, the structure should be able to rapidly recovery its functionality similar to the pre-event level, immediately or after minor structural repairs (see also section 27.7).

structure is related to structural performance that can be achieved with proper structural design.

Although structural performance is the first objective of any structural design, contemporary society is demanding more from structural designers. They are asked to integrate the design of structural performance with the design of environmental, economic and other social performances, in close cooperation with designers from other disciplines (e.g. energy, ICT, acoustics, environmental science).

Some of these aspects are more important for buildings (e.g. thermal and acoustic comfort) than for infrastructure. Moreover, different length of the design life for buildings and infrastructures should be considered.

Buildability, maintainability, demountability and circular use of structures and recycle-ability of elements and materials can have a significant importance for economic efficiency and environmental quality.

Flexibility can be achieved by applying large floor spans and design of floors with residual capacity.

Basis of evaluation of structural performance is described in Part III (chapter 11 and chapter 12) and chapter 30.

3.2.3 Other aspects of social performance

The objective of considering other aspects of social performance than structural performance is taking into account the satisfaction perceived by all stakeholders involved with or impacted by the structure during its entire life cycle. The awareness of the impact of structural design on other aspects of social performances is crucial for structural designers and must be fostered by modern structural codes. While structural performance is highlighted by structural engineers as the basic aspect of social performance that structures are required to exhibit, all other aspects of social performances are to be treated as additional requirements that structural design must pursue. These aspects include but are not limited to:

- buildability;
- maintainability;
- demountability of structures and recycle-ability of elements;
- flexibility of the structure to fulfil other functions in future;
- suitability of the spatial layout for the intended use;
- aesthetics;
- safety of workers and external persons;
- hindrance to neighbouring stockholders;
- disturbances to traffic, communication and other operations;
- acoustic comfort;
- thermal comfort;
- operation security.

Basis of approach to evaluation of other aspects of structural performance is described in chapter 31.

In some cases, advanced design and implementation in concrete industry can adopt processes to be beneficial to the environment.

3.3 Environmental performance

A structure should be designed, constructed, managed and taken care of during the life cycle in order to minimize harmful impacts to environment and providing environmental benefits.

Key aspects that should be considered within environmental performance are:

- use of resources;
- use of energy;
- use of land;
- emissions of harmful substances to air, water and soil;
- noise and vibration;
- waste production;
- biodiversity loss (species and entire ecosystem).

Basis of evaluation of environmental performance is described in chapter 32.

3.4 Economic performance

Economic performance concerns cost and benefits of concrete structure covers:

- construction cost;
- operation cost;
- maintenance cost;
- refurbishment and repurposing cost;
- dismantlement cost;
- reuse and recycling cost,
- cost of externalities;
- cost of capital;
- support of local economy;
- benefits provided by the structure.

An essential aspect of economic performance is the tender process and acceptance of alternatives (for instance in case of Construct type contracts) or award of contract based upon most economically attractive tender (MEAT), where the many aspects are in the end expressed in an economic unit, generally the monetary unit. This process is supporting an optimum in economic performance.

Both direct and indirect costs (including overhead and risks) need to be considered and specified in the Client's brief.

Binding requirements will have cost as consequence and it is the duty of the engineer to optimize the solution in order to meet the requirements of the client. Each individual client might have additional value aspects. These might include different ambitions for different sustainability aspects. As long as value can be created against costs which are lower than the value as specified by the client, it will form part of the design.

Basis of evaluation of economic performance is described in chapter 33.

References

[3-1] ISO 13315-1:2012. Environmental management for concrete and concrete structures – Part 1: General principles.

[3-2] EN 15643:2021. Sustainability of construction works - Framework for assessment of buildings and civil engineering works

[3-3] ISO 21931-2 Sustainability in buildings and civil engineering works — Framework for methods of assessment of the environmental, social and economic performance of construction works as a basis for sustainability assessment — Part 2: Civil engineering works

[3-4] EN 17472 Sustainability of construction works. Sustainability assessment of civil engineering works. Calculation methods

As a rule, the key stakeholders would be the founders, the owners, the residents, the users, the neighbours (if construction interferes with them), the contractor, the design and construction team, the tenancy management and maintenance team. Other stakeholders may be the government and the society.

While stakeholders usually believe that they know which performance criteria they should define for a structure, it may require skill and experience in structural engineering to recognize incomplete, ambiguous or contradictory demands.

The objective of this code is to disseminate the concept of performance-based approach and thereby encourage rational design and technological progress so as to build superior infrastructure for future generations.

For concrete structures, the frequently formulated prescriptive requirements are materials specifications and construction specifications.

4. Principles of performance-based approaches

4.1 Performance specifications supporting sustainability

The stakeholders have to give demands for the objectives of a structure or a structural component during its life cycle and the functions to achieve the objectives. These demands reflect the role(s) that a structure or a structural element should have under the intended conditions of construction, service and dismantlement.

Consistent with the sustainability framework described in chapter 3, the demands of the stakeholders are to be dealt with from the sustainability perspective and shall be formulated considering all three pillars of sustainability and performance aspects described in section 3.2 (social performance), section 3.3 (environmental performance) and section 3.4 (economic performance).

Structures and structural members must be designed, constructed and maintained in such a way that they perform adequately and in a social, environmental and economically reasonable way during the whole life cycle. The life cycle perspective shall be treated in accordance with the provisions of chapter 5.

Performance-based design or assessment is defined as a concept that requires structures to achieve a desired level of satisfaction of the stakeholders with the performance of structures, while complying with statutory responsibilities and requirements, however without specifying conditions for how this is to be achieved by specific materials, detailing, or construction method.

MC2020 should, whenever possible, facilitate the use of performance-based design or assessment to remove inappropriate constraints on the use of novel types of concrete and reinforcing materials, as well as those materials and processes used to make adaptations and interventions to extend the useful life or enhance the performance of existing concrete structures.

As an alternative, when there are not fully developed specific performance requirements for the different pillars of sustainability, prescriptive requirements may be used.

4.2 Performance requirements

A simplified model of such an approach is illustrated in Figure 4.2-1. This presents an overview of a performance-based approach (PBA) to the design of new structures, focussing upon the 'Requirements' and the 'Verification' aspects involved and portraying the decomposition of the owner's & stakeholders overall sustainability performance requirements for structure into appropriate technical performance requirements through the associated members of the construction supply chain.

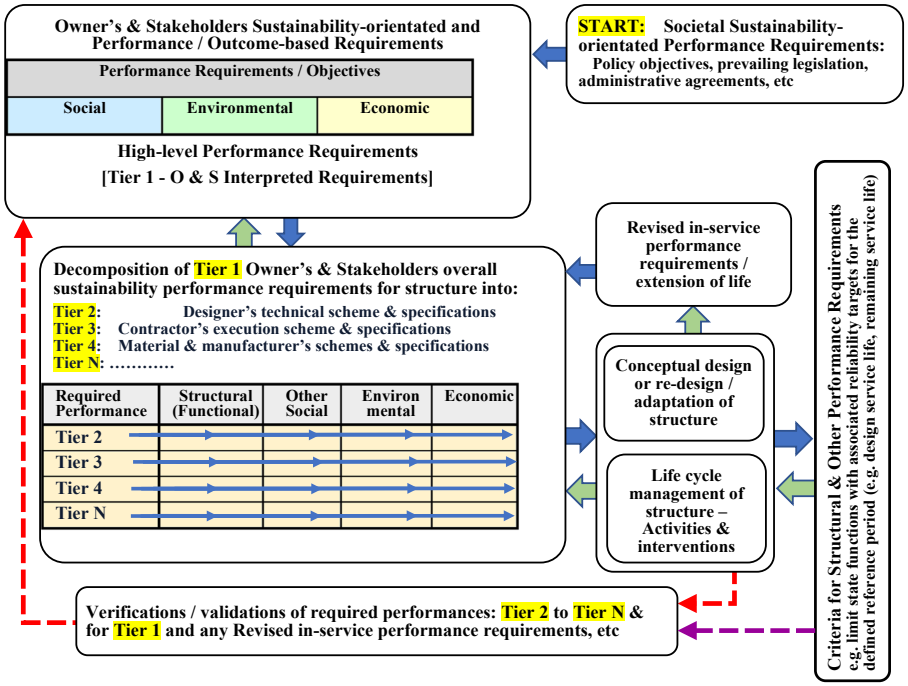


Figure 4.2-1*: A (simplified) overview of the performance-based approach – Requirements and Verifications*

A simplified model of such an approach is illustrated in Figure 4.2-2. This presents an overview of a performance-based approach (PBA) to the design of new structures and the life-cycle management (LCM) processes and activities for existing structures.

The demands of the stakeholders and statutory responsibilities and requirements are the basis for specifying the performance requirements. For each aspect of the demands that is relevant for a structure or structural component, the performance

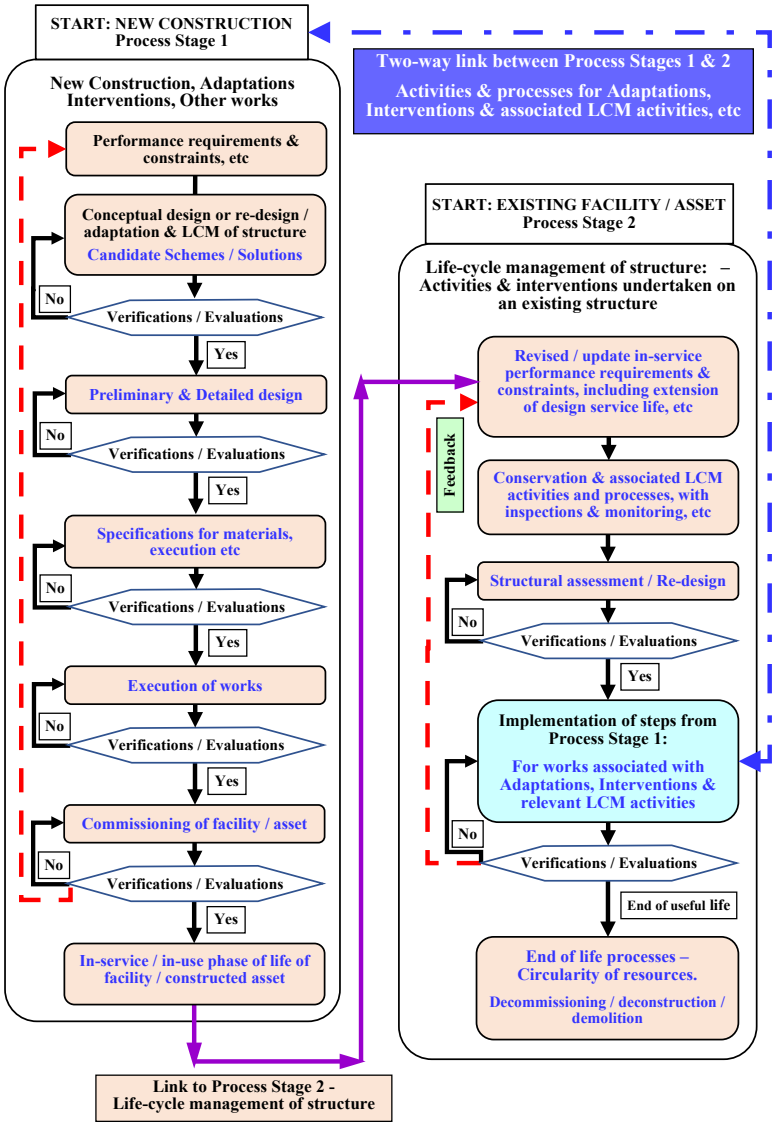


Figure 4.2-2: *A (simplified) overview of the performance-based approach to the design of new structures and the LCM processes & activities for existing structures.*

requirements must be specified. Accordingly, the performance requirements refer to the fulfilment of the essential demands of the stakeholders.

Figure 4.2-3 presents a representation of the sustainability-orientated performance-based approach for design and LCM of structures, portrayed with a focus upon the aspects relating to the performance requirements.

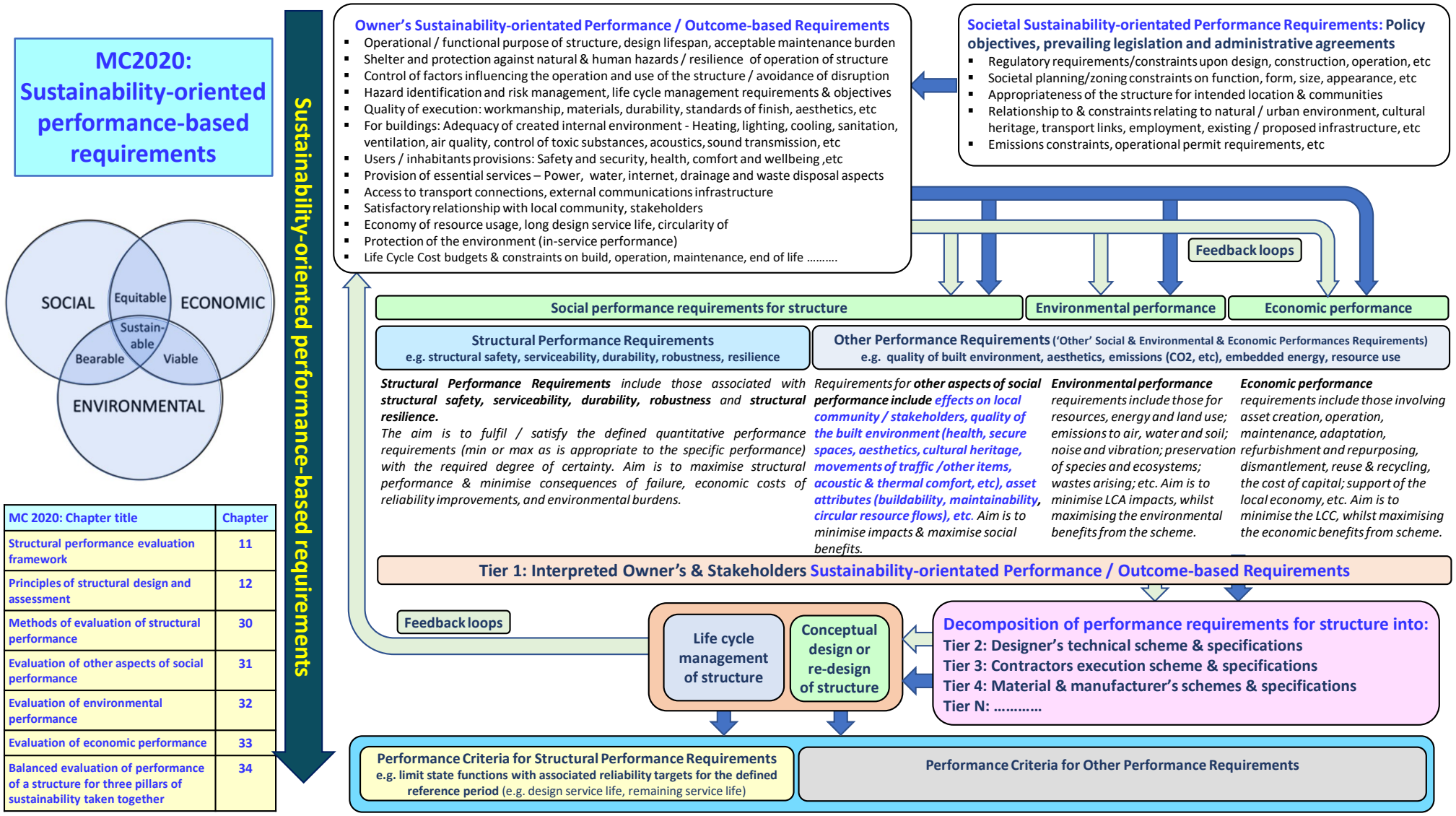
Performance requirements are established by means of the performance criteria. Performance criteria can be qualitative or quantitative limits set in balance with each other, defining the border between the desired and the adverse behaviour, relevant for the specific aspect of performance. Performance criteria depend on the particular needs of the stakeholders for a certain structure.

Developing performance requirements, as an alternative to the current prescriptive requirements, requires consideration of four main tiers of linked performance- or outcomes-based requirements, specifications and testing regimes, namely:

Tier 1: Owner's and user's / stakeholder's higher-level performance- or outcomes-based requirements : Owners and users / stakeholders need to be able to define and specify clear social, environmental and economic performance requirements for the structure / constructed asset / functional system which is being created / adapted / renovated, etc;

Tier 2. Technical consultant's, engineer's and other professional's design, assessment and interventions strategies and structural performance specifications : Technical consultants, engineers and other professionals need to be able to turn the owner's / user's / stakeholder's higher-level performance requirements into strategies and performance specifications, employing associated metrics which are able to assess whether the performance requirements/ specifications have been met and to what degree;

Tier 3. Contractor's execution performance specifications: Contractors need to be able to employ appropriate performance-based specifications so they can select and procure suitable materials and/or manufactured systems from their suppliers and employ appropriate construction (execution) processes in the creation of the required structure / constructed asset / functional system which they are tasked with creating / adapting / renovating, etc. These performance specifications need to be allied to appropriate testing procedures so that they can verify that the required performance specifications have been achieved with the desired degree of certainty (reliability);



MC 2020: Chapter title	Chapter
Structural performance evaluation framework	11
Principles of structural design and assessment	12
Methods of evaluation of structural performance	30
Evaluation of other aspects of social performance	31
Evaluation of environmental performance	32
Evaluation of economic performance	33
Balanced evaluation of performance of a structure for three pillars of sustainability taken together	34

Figure 4.2-3: *The sustainability-orientated PBA for design and LCM of structures, portrayed with a focus upon the Performance Requirements.*

Figure 4.2-4 presents a representation of the sustainability-orientated performance-based approach for the design and LCM of structures, portrayed with a focus upon the design processes and activity flows.

Tier 4. Suppliers and manufacturers material's and manufactured system's performance specifications : Material suppliers and manufacturers need appropriate performance-based specifications so they can provide suitable materials and manufactured systems. The performance specifications need to be allied with appropriate testing procedures and regimes so that it can be demonstrated that the performance specifications have been achieved with an appropriate degree of certainty (reliability).

The four-tier layout represents a continuum of overlapping performance- or outcomes-based requirements and specifications, whose focus shifts from the broad higher-level operational goals aligned with owner objectives and the desires of the users (Tier 1), though engineering technical definition for the scheme or project (Tier 2) and focusing upon increasingly detailed contractual and technical performance aspects (Tiers 3 and 4).

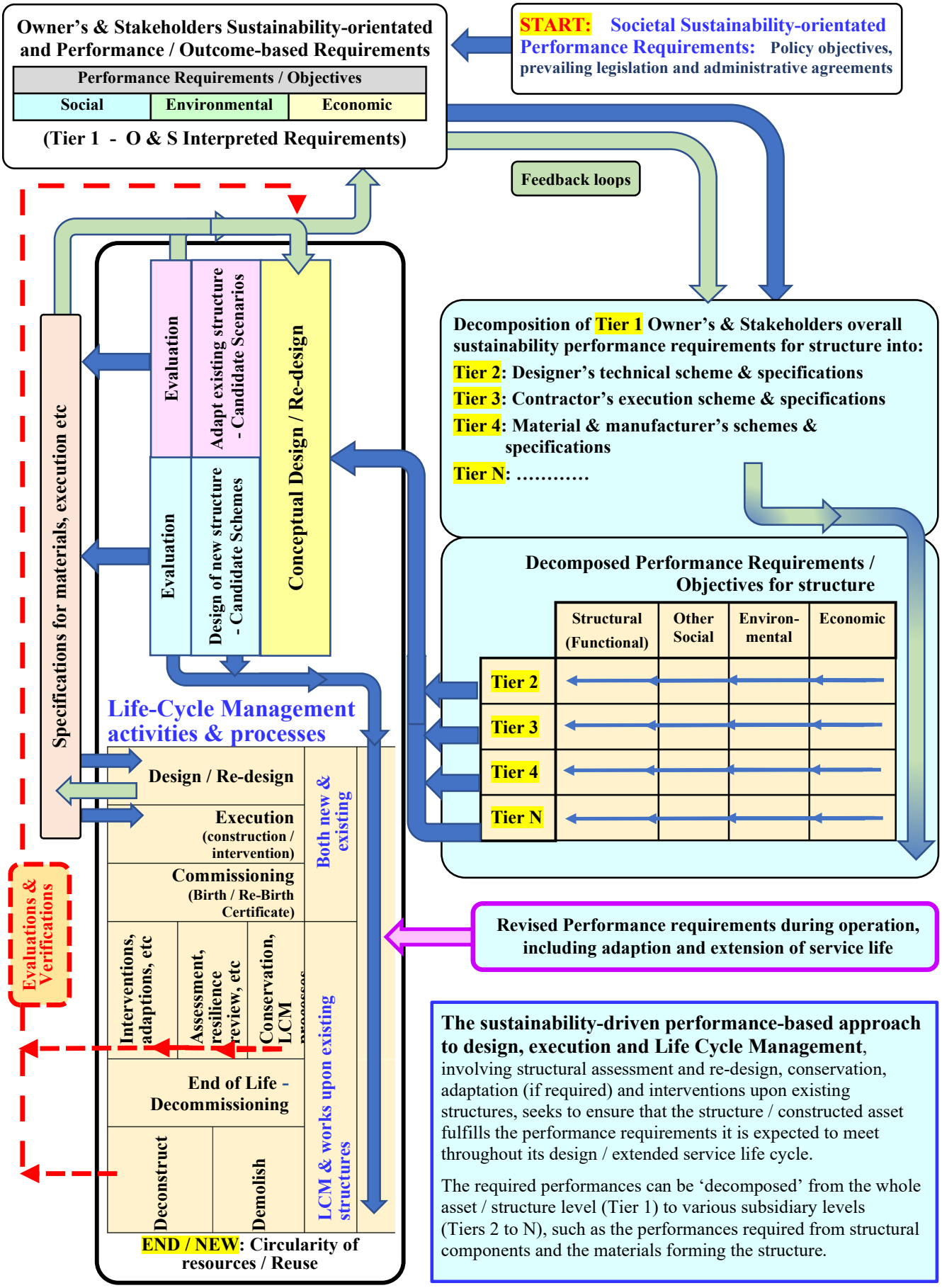


Figure 4.2-4: *Sustainability-orientated PBA for the design and LCM of structures, with a focus upon design processes and activity flows.*

Methodologies for evaluation of performance may involve verification and validation. Verification procedures involve performing modelling or simulating performance of element, component, or system within the structures, followed by a review or analysis of the modelling results. Validation procedures involve establishing evidence that provides a high degree of assurance that an element, component, or system within the structure accomplishes its intended requirements. In case of structural performance, evaluation of ??? is usually done by verification procedures. For other aspects of performance of structures (e.g. hindrance to neighbouring stakeholders) validation procedures may rather be used.

An integral methodology to assure compliance of design, construction and maintenance with requirements is given by ISO 15288 [4-2].

It should be noted that the requirements for existing structures may be different from those for new structures.

Clarify in informative difference between new and existing (deign and assessment)

4.3 Performance evaluation

Performance is evaluated using appropriate methodologies by verifying the behaviour, quality or impact of a structure or a structural component against the specified performance requirements with the associated performance criteria. An adequate performance is reached when a structure or a structural component has demonstrated to satisfy in a desired balance the needs and the demands of the stakeholders throughout its life cycle. In the opposite case, the performance of a structure or a structural component is considered to be inadequate.

The performance-based design of a new structure or a structural component is completed when it has been shown that the relevant performance requirements are adequately satisfied for all relevant aspects of performance related to social, environmental and economic performance.

In case of existing structures, by using a performance-based approach it is assessed whether the actual performance of an existing structure or structural members and their performance during the residual life satisfies the demands of the stakeholders. The performance-based assessment of an existing structure or a structural component is completed when it has been identified whether all relevant performance requirements are satisfied or not. In the latter case the performance of a structure or a structural component is qualified as inadequate.

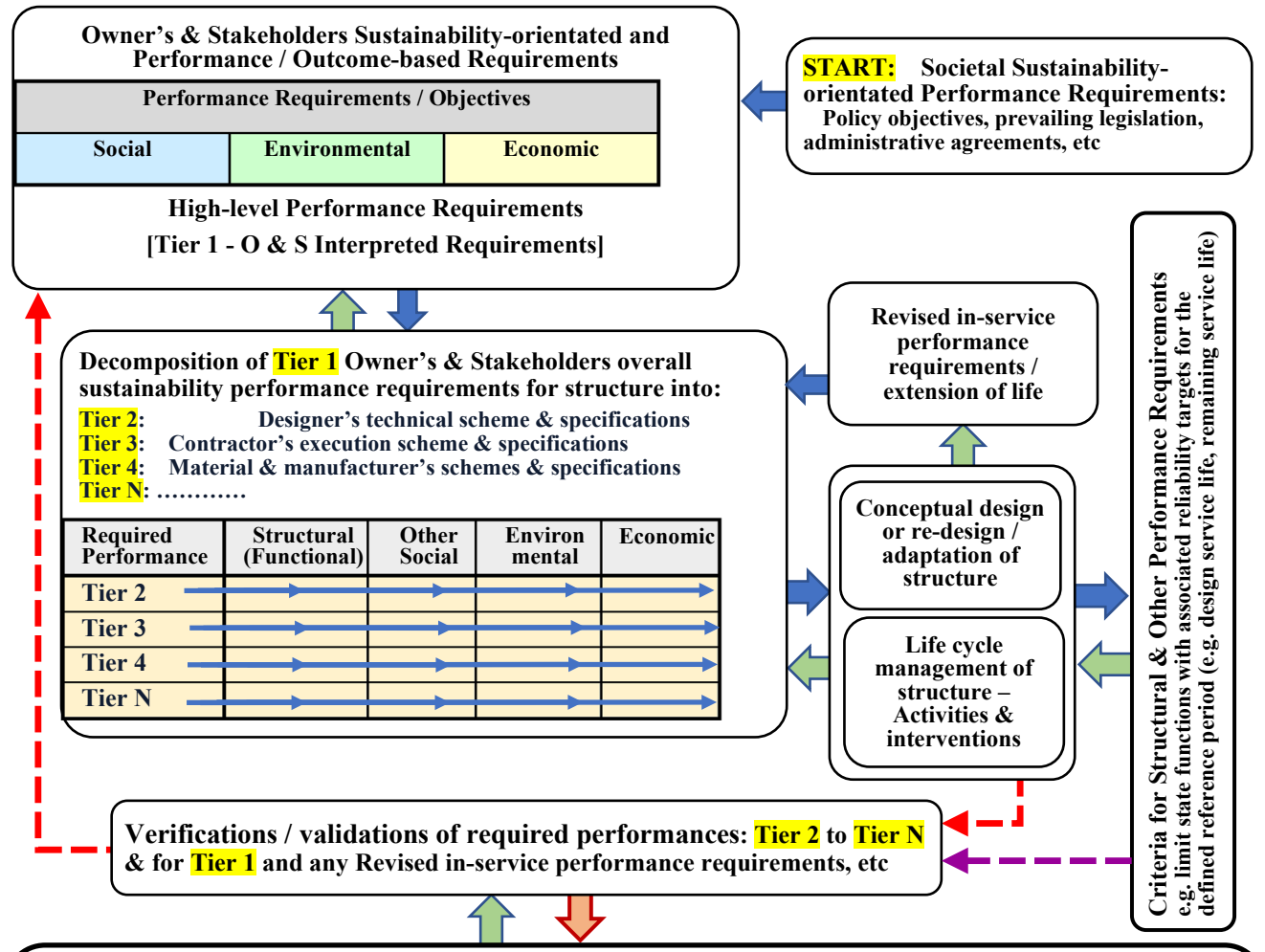
In chapter 26 and in chapter 27, the process of transforming the stakeholders demands into the performance requirements and process of selecting performance criteria is addressed, respectively for the design of new and assessment of existing concrete structures.

Basic principles for evaluation of structural performance are given in Part III (chapter 11 with regard to structural performance evaluation framework and in chapter 12 with regard to principles of structural design and assessment) and the methods of evaluation of structural performance are given in chapter 30. Provisions for evaluation of other aspects of social performance are given in chapter 31, for

Figure 4.3-1 presents a representation of the sustainability-orientated performance-based approach for the design and LCM of structures, showing a summary representation of the Performance Requirements, but with a focus upon the available performance verification / evaluation options.

evaluation of environmental performance in chapter 32 and for evaluation of economic performance in chapter 33.

Chapter 34 discusses sustainable decision-making. This seeks to achieve an appropriate performance balance between the three-pillars of sustainability; namely between social, environmental and economic considerations.



Verifications / evaluations of sustainability-oriented performances seek to demonstrate that a satisfactory balance between the requirements is achieved throughout the life-cycle of a structure.
Performance is evaluated using appropriate methodologies to verify or validate the behaviour, quality or impact of a structure or a structural component relative to the specified performance requirements by means of the associated performance criteria (which provide appropriate metrics).

Verification / validation methods		New structures	Existing structures
Risk-informed method	**– see Section 12.5.2**	✓	**LCM activities**
Reliability-based method	**– see Section 12.5.3**	✓	✓
Partial factor method	**– see Section 12.5.4**	✓	✓
Global resistance factor method (Applies to specific performances)	**– see Section 12.5.5**	✓	✓
Deemed-to-satisfy approach (May only apply to specific performances)	**– see Section 12.5.6**	✓	✓
Avoidance approach (May only apply to specific performances)	**– see Section 12.5.7**	✓	**LCM activities**
Assisted by testing and / or monitoring (May only apply to specific performances)	**– see Section 12.5.8**	✓	✓

Figure 4.3-1: *Sustainability-orientated PBA for the design and LCM of structures showing a summary representation of the Performance Requirements, but focussing upon the Performance Verification / Evaluation options.*

References

[4-1] CONREPNET Matthews et al: A Performance Based Approach: Concepts and application for the use of performance concepts to achieve durable remediation of concrete structures, CONREPNET Project final publishable report, IHS-BRE Press, Report EP77, 2007

[4-2] ISO 15288:2015. Systems and software engineering – System life cycle processes.

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There are a number of phases in the development and use of a concrete structure. Typically, the sequence of events through the life of such a concrete structure progresses as follows:

- Definition phase - where the owner's basic requirements and needs are established with reference to the drivers arising from consideration of the overarching 'three-pillars' of sustainability.
- Concept phase – where the identified requirements are examined, potential solutions satisfying these are defined, evaluated and a suitable structural concept chosen for further detailed development. Many iterations of the process are commonly required to refine the solution concepts to accord with the functional requirements and comply with associated sustainability/financial/other constraints.
- Design - usually involving preliminary and detailed design phases.
- Construction / Execution - the process whereby the structure is built.
- Operation and use – with various management activities / interventions being undertaken in a timely manner to maintain its functionality and ensure the adequacy of its life-cycle performance; with these activities / interventions being recorded (eg. Rebirth Certificate and associated documents) so the knowledge is utilised in development of subsequent LCM scenarios.
- Extended or alternative use – this is the process by which the useful life of the concrete structure can be extended or the concrete structure brought into a new use by being adapted, sold or via some other means.
- Disposal or re-use - the process by which the concrete structure is decommissioned and removed; being either dismantled, with the components being re-used as far as possible, or demolished and the materials recycled.

5 Life-cycle management

5.1 Life-cycle management objectives

Life cycle management (LCM) is the overall philosophy / strategy to be used in managing a concrete structure through its life-cycle, particularly during its operation and use phase. Accordingly, the LCM strategy adopted shall be developed from the performance requirements for the new or altered concrete structure, as well as influencing the development of the associated candidate design solutions / selection of relevant interventions as appropriate. To do this, the owner will usually need the support of a team of technical advisers, depending upon the expertise held by the owner.

Critical aspects of these considerations will be derived from the sustainability objectives defined for the concrete structure under the three pillars of sustainability (social responsibility, environmental quality and economic efficiency) discussed previously in Chapter 3. The LCM strategy shall contribute to achieving an acceptable balance between these factors.

Thus, the LCM process shall respond to those factors in a coherent and integrated way providing input to the processes of design, construction, usage, and conservation of concrete structures, as well as, if appropriate, their adaptation to new uses / changed performance requirements.

Accordingly, successful implementation of LCM principles for a new concrete structure will involve all stages of planning, design, execution and use of concrete structures, including those associated with extending the useful life, undertaking adaptation works or upgrading the performance of structures in-service by undertaking interventions, with the preparation and subsequent use of relevant documentation (eg. Birth / Rebirth Certificates), together with associated end of life aspects and possibilities for the reuse of elements.

Chapter 34 addresses the balanced evaluation of performance of a structure that embraces the three pillars of sustainability: namely, the social, environmental and economic aspects as a whole.

In contemporary engineering practice the goal has often been to minimise the costs associated with achieving the required performance (i.e. to meet relevant performance criteria during the service life at the required reliability level) while achieving an appropriate (minimum) quality requirement.

Optimisation involves making trade-offs between competing objectives. Interactions and interdependencies between a wide range of factors, including cost, risk and quality, need to be considered. Accordingly, the process of making LCM evaluations shall be approached with caution. As an optimisation problem, previously the goal of LCM has been to minimise the expected costs on a net present value basis. However, increasingly the expectation is that this shall be done in conjunction with minimising adverse environmental and social impacts, whilst seeking to maximise beneficial social, environmental and economic outcomes.

ISO 22040 [5-1] describes the general principles for implementing LCM of concrete structures throughout the stages of planning, design, execution, use and end-of-life, as well as the framework and procedures involved in delivering LCM.

The available conservation strategies and respective tactics are classified on the basis of their proactive versus reactive characteristics as follows:

– Category A: Proactive conservation activities;
– Category B: Reactive conservation activities;
– Category C: Situations where conservation activities are not feasible.

An increase in the level of the performance requirements during the use stage may necessitate an intervention being made to increase the performance or possibly the durability of the concrete structure.

5.2 Life-cycle management framework

5.2.1 Introduction

A fully integrated approach to LCM is complex and requires realistic life-cycle cost (LCC) calculations and life-cycle analysis (LCA) of the environmental impacts and benefits associated with the concrete structure through its life-cycle, assuming appropriate service lives for the various members and components making up the structure.

Accordingly, LCM concepts can be applied:

- to the construction of new concrete structures, as well as to existing or adapted structures, and
- to both the entire life-cycle or to one or more stages of the life-cycle of a concrete structure.

“Durability Planning” makes an important contribution to life-cycle management activities both in respect of the Conceptual Design process (in Section 26.1.3) and also in the overall Approach to Design (in Section 27.6).

LCM facilitates informed choices being made between various design, construction, use, conservation, adaptation and end of life options on the basis of formulating appropriate life cycle management scenarios (LCM scenarios) at the planning stage, as a means of ensuring the performance requirements are attained.

Figure 5.2-1 presents an indicative LCM procedure flow for new and existing structures. This illustrates how the LCM scenario(s) can be modified during the design /assessment stage, the execution stage, and the use stage of a structure, if necessary. Figure 5.2-1 indicates that when the scenario is modified, the management process reverts to the previous stage.

The risk of performance degradation of the structure during its use, as well as its impact on the environment, shall be considered during the formulation of the LCM

When undertaking LCC evaluations the high degree of uncertainty in the assumptions made about prospective behaviours and circumstances in (say) the latter half of the design life shall be recognised and taken into account.

scenarios. The scenarios shall be appropriately updated to take account of the influence of such changes.

LCM decisions shall be made so that the work at each stage of planning, design, execution, use, etc can be appropriately linked and managed based on a consistent concept.

5.2.2 LCM in the planning and design / assessment stages

In the planning and design / assessment stages, the performance requirements of the structure shall be defined, along with the structural forms, materials and dimensional details so that the structure satisfies the performance requirements for the design working life at the chosen reliability level. The envisaged methods of maintenance / intervention shall be defined at this time. It is necessary to ensure that appropriate maintenance / intervention activities can be undertaken during the use stage. Further details are given in Section 26.1.3, in *fib* Bulletin XY [5-2] and in CIA Publication Z7-01 [5-3].

5.2.3 LCM in the execution stage

A structure shall be constructed so as to fully achieve the performance level(s) set at the design stage. Exceptionally, if subsequent post-design changes are proposed, the implications of these for the design, the LCM and the resulting performance level(s) for the structure shall be fully evaluated prior to the implementation of the changes to avoid adverse or inappropriate impacts upon the performance parameters.

Information acquired during the execution stage, including materials, construction methods, and quality control records, which are necessary for future LCM scenarios, shall be recorded and transferred to the use stage for as-built information.

5.2.4 LCM during the in-use stage

In regard to LCM in the execution stage, Appendix G of ***fib*** Bulletin 44 [5-4] sets out steps for the development of supplementary documents to support the preparation of a project specification for the execution works for concrete structures, linking with the requirements of the execution standard ISO 22966 [5-1].

Prior to entering the use stage, confirmation shall be made as to whether or not the design intentions have been achieved and that the execution was appropriately carried out (Birth Certificate). Defects and deficiencies shall be repaired as required. Subsequent LCM scenarios shall also be revised at this time.

In the use stage of a structure, periodic inspection / monitoring / diagnosis shall be conducted to confirm the retained performance of the structure. A prognosis of subsequent performance shall be made as required. Not only physical measures including repair / upgrading, but also operational measures including use restrictions, or the possibility of termination of use, shall be considered as elements of the LCM of the concrete structure.

An increase in the level of the performance requirements during the use stage may necessitate an intervention being made to increase the performance and/or durability of the concrete structure. When an intervention is to be made on an existing structure, the thought processes and procedures employed for conceptual design (as for new construction) shall be used – refer Chapter 26.

Information acquired from inspection / monitoring / diagnosis during the use stage, information related to applied measures, and information on performance recovery shall be transferred to the party responsible for maintenance. Information acquired during the maintenance / intervention stage shall be recorded (eg. preparation of Rebirth Certificate and associated reports) and utilised in development of subsequent LCM scenarios. Further details on LCM activities during the in-use stage are given in Chapter 8

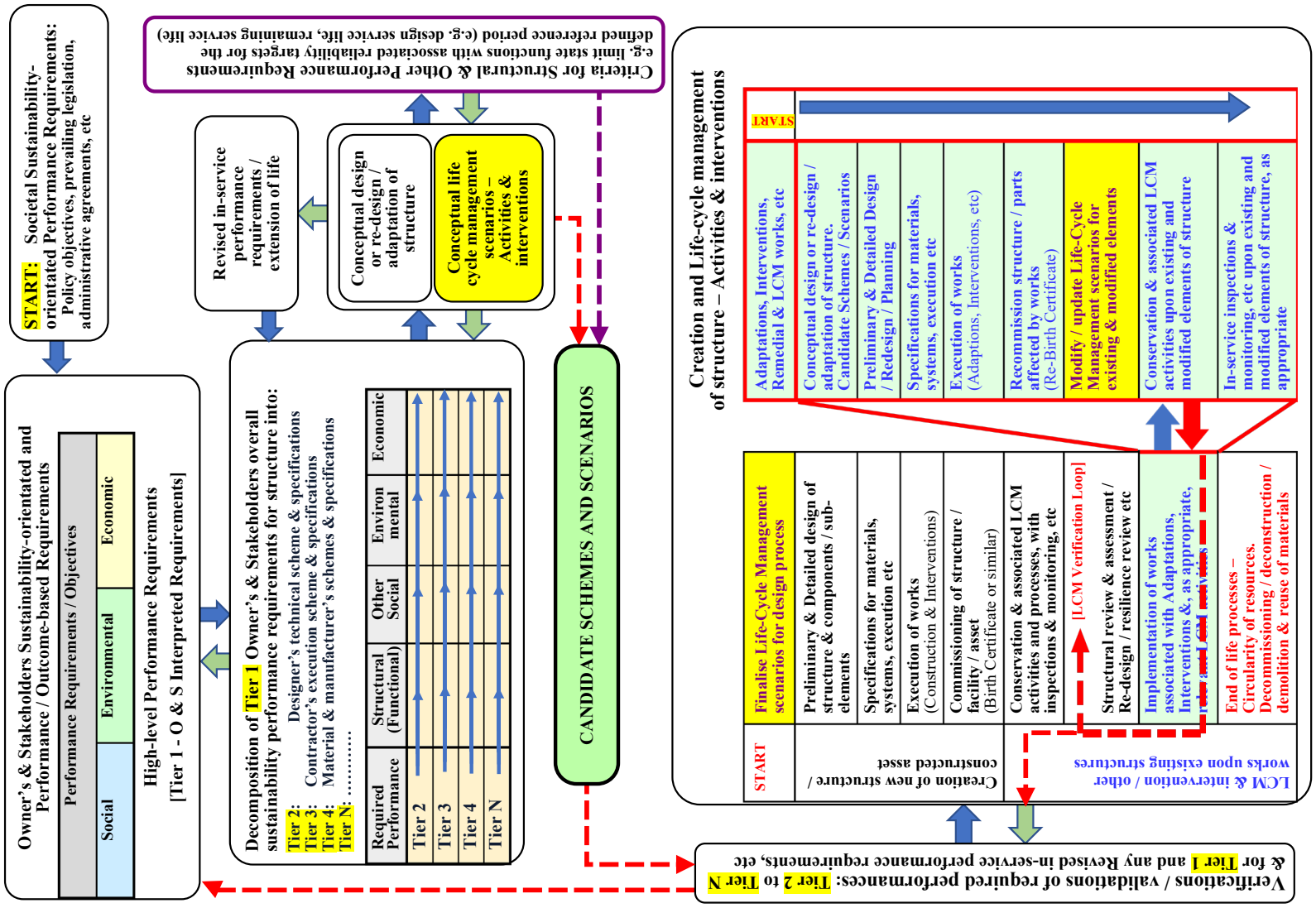

Figure 5.2-1 Indicative LCM procedure flow for new and existing structures INCLUDE REUSE AND OTHER END OF LIFE ASPECTS

The principles of circularity / the circular economy require that planning be undertaken at the time of design to facilitate 'higher-value' future uses of construction members and materials, particularly the reuse of structural members whenever possible, in preference to simple recycling of materials and a focus upon 'down-cycling' processes.

In general, the life-cycle management process for an existing concrete structure involves the types and sequence of activities detailed below.

- **Condition survey:** This involves gathering information on the current condition and performance, and factors relevant to the development of that condition (ie. the technical status quo of the structure). This might involve activities such as inspections, surveys, testing and monitoring. An initial inspection shall be conducted as the first step in collecting information on the performance of the structure throughout its life. Such information may be available in the Birth Certificate for the concrete structure – refer *fib* Bulletin 93[5-6].
- **Performance evaluation:** This involves analysing the available data to establish the current performance of the concrete structure, and to make a prognosis of future performance, including identification of deterioration mechanisms and prediction of future damage, as well as the need for, timing of and nature of any interventions required.
- **Decision-making for LCM activities and interventions:** This is concerned with establishing the implications of (say) deterioration upon the life-cycle management and conservation of an concrete structure, evaluating potential management and conservation options, selecting

5.2.5 LCM during the end-of-life stage

This concerns the process by which the concrete structure is decommissioned and removed; being either dismantled, with the components being re-used as far as possible, or demolished and the materials recycled. The objective shall be to reduce the amount of waste and promoting the reuse members / recycling of construction materials.

To ensure that the reuse of structural members is promoted and maximized, the structure shall be designed so that it can be dismantled at end-of-life to facilitate the reuse of the existing components in another structural application.

5.2.6 LCM activities

To achieve effective and efficient life-cycle management, the process needs to be adequately managed and documented. In particular, the planning shall be prepared in a manner consistent with the maintenance works / intervention strategy for the structure or its component parts defined during the service life design stage. The general flow of the conservation process procedures is shown in Figure 8.3-2.

In this Model Code, principles of quality and information management are given in Chapter 10. Specific methods of achieving required performance of structures at different phases of the life cycle are given in MC2020 Part VII for design and assessment, in Part VIII for execution, Part IX for conservation and in Part X in respect of the circular economy, dismantlement and reuse processes.

appropriate management and/or intervention options and the timing of such activities and works.

- Execution of life-cycle management activities and/or preventative interventions or remedial works.
- Undertaking on-going condition survey and/or monitoring activities, including recording the information required for LCM.

Further details on the principles of implementing life-cycle management of concrete structures have been developed by ISO – refer ISO 22040 [5-1].

5.3 Associated considerations

5.3.1 Learning from life-cycle monitoring, forensic engineering and building pathology investigations

Forensic engineering and building pathology investigations, which are defined below, are able to provide insight into the reasons why a concrete structure may not have achieved the performance requirements specified at the time of design or following an intervention being undertaken. These disciplines provide a basis for learning from the in-service performance of concrete structures and for providing feedback to improve professional practice

Forensic engineering: The application of engineering knowledge to the investigation and assessment of the nature, condition and performance of an engineered system.

Building pathology: The systematic treatment of building defects, their causes, their consequences and their remedies.

These approaches use scientific and engineering principles to study:

- Deterioration, damage or abnormal changes, inadequate performance or failure of constructed facilities or members thereof;
- their causes, symptoms and means of remediation; and

Life-cycle monitoring procedures including structural health monitoring and associated techniques, together with forensic engineering and building pathology investigations, provide important means of driving learning and improving the standard of professional practice by providing feedback into the development of analytical / design / assessment techniques and procedures used for the design and life-cycle management of concrete structures.

These procedures shall be a component of the techniques and procedures employed for the life cycle management of concrete structures, with their ability to add to the body of knowledge available for the design of future structures, and to contribute to the subsequent life-cycle management of existing concrete structures.

- the methods of investigation and assessment which can be employed.

Under the building pathology approach, it is a fundamental principle that there shall be no apportionment of liability associated with the events or circumstances investigated.

Regulatory / governmental organisations preparing formal reports upon structural safety and fire safety issues associated with buildings and other structures in the built environment include:

- The HSE (*Health and Safety Executive*) [EN 5-1]
- NIST (*National Institute of Standards and Technology* [EN 5-2]
- CROSS (*Collaborative Reporting for Safer Structures*) [EN 5-3]

Almost all countries have some form of regulatory / governmental organisation concerned with structural and fire safety issues arising in the built environment.

5.3.2 Learning via other feedback routes

Use shall also be made of other routes to learning from past performance and professional / industry practices. This includes sources such as formal reports upon structural safety and fire safety issues relating to buildings and other structures in the built environment prepared by regulatory / government organisations. Such reports help to create positive technical and cultural change to improve safety in buildings and other structures.

Confidential reporting schemes provide another means for professionals to learn and also to report matters of concern; such as a structural / fire related technical issue / event or a safety near-miss or a professional / industry practice of concern. These provide learning events which otherwise might not be made known / formally published.

References

[5-1] ISO 22040 *Life cycle management of concrete structures*, Geneva, International Organisation for Standardisation (ISO, 2021).

[5-2] *fib* Bulletin XY *Durability planning for concrete structures, fib* Technical report, [IN PREPARATION, COM8]

[5-3] CIA Publication Z7-01 *Durability Planning,* which is a Recommended Practice Report forming part of the CIA Concrete Durability Series, Concrete Institute of Australia (CIA), Sydney, 2014.

[5-4] *fib* Bulletin 44 *Concrete structure management – Guide to ownership and good practice, fib* Guide to Good Practice report, (*fib*, 2008)

[5-5] ISO 22966 *Execution of concrete structures*, Geneva, International Organisation for Standardisation (ISO, 2009)

[5-6] *fib* Bulletin 93 *Birth Certificate and through-life management documentation*, *fib* Technical report, (*fib,* June 2020)

Endnotes

[EN 5-1] The HSE (*Health and Safety Executive*) which is the UK's Building Safety Regulator, along with being the national regulator for workplace health and safety (www.hse.gov.uk/building-safety/index.htm).

[EN 5-2] NIST (National Institute of Standards and Technology), which has a Disaster and Failure Studies Program (www.nist.gov/disaster-failure-studies). This undertakes research into building and infrastructure performance / industry practice, which has led to significant changes in practices, standards and codes.

[EN 5-3] CROSS (Collaborative Reporting for Safer Structures) operates a secure and (importantly) confidential safety reporting system allowing professionals to share their experiences to help others. CROSS (www.cross-safety.org/uk) currently operates in the United Kingdom, Australasia and the United States, but it has a wider international network sharing safety information between countries.

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6 Principles of Quality and Information Management

6.1 Principles of Quality Management

6.1.1 General

Quality management (QM) is a life cycle process to ensure that concrete structures will achieve the required quality and performance. Its purpose is to ensure that the assumptions on which the design is based on are fulfilled and that the structure achieves the required performance with the required level of confidence.

QM enables quality improvement through quality planning, that comprises quality assurance and quality control issues, at all stages of the life cycle: QM involves both organisational measures and controls, as explained in Chapter 10 for design phase (section 10.1), for execution of construction works and inteventions phase (section 10.2), for operation and conservation phase (section 10.3) and dismantlement phase (section 10.4).

The provisions for design and assessments of new and existing structures in MC2020 comply with the principles given in ISO 2394 [6-1] (see section 6.4.3).

ISO 2394 [6-1] specifies the general requirements for a Quality Management system. The philosophy in ISO 2394 [6-1] is in line with that of JCSS Probabilistic Model Code [6-2] and e.g. the European EN 1990 [6-3].

ISO 2394 [6-1] assumes that the execution of the concrete work on the construction site is complying with a certain quality level (QL).

The provisions for execution of concrete structures in MC2020 comply with the principles given in ISO 22966 [6-4]. ISO 22966 [6-4] applies to the execution of concrete structures to achieve the intended levels of reliability and serviceability that are given in ISO 2394 [6-1].

ISO 22966 [6-4] share significant similarities in its requirements and guidelines with e.g. the European EN 13670 Execution of concrete structures [6-5].

ISO 22966 [6-4] assumes that the concrete production complies with the requirements in ISO 22965 Part 1 [6-7] and ISO 22965 Part 1 [6-8].

ISO 22965 [6-6] is based on the European EN 206 [6-9].

As such, CC's can directly be related to a differentiation in structural applications.

The level of QL is then directly related to the calibration of the corresponding partial factor for material

It is assumed that QM includes the full "Plan-Do-Check-Act" cycle as described in ISO 9001 [6-10].

Plan: *establish the objectives and processes necessaryto deliver results in accordance with customer requirements and the organization's policies*
Do: *implement the processes.*
Check: *monitor and measure processes and product against policies, objectives and requirements for the product and report the results*
Act: *take actions to continually improve process performance*

ISO 22965-Part 2 [6-9] gives provisions for factory production control, but does not offer a 3-level differentiation like its mother standards.

Communication between parties involved in the development of the project is vital. It is important that the client remains engaged in the process even after his basic needs have been defined. It is essential to monitor progress and to communicate with the client throughout the whole project development.

Communication needs to take place throughout the whole process from project inception to its life-end. The iterative nature of the design process needs to be

6.1.2 Quality Levels

MC2020 applies a reliability framework where the target reliability level (β_{target}) for the structure depends on the consequences of failing to meet the performance criteria. Consequences related to exceedance of an ultimate limit state state are classified in Consequence Classes, according to section 11.3.3.

Section 11.3.2 defines recommended target reliability levels correponding to the 3 Consequence Classes (CC) considered. Section 12.5 recommends the use of partial factors γ_M for materials for the 3 CCs based on the target reliability levels given in section 11.3.

QM is an important tool to ensure that the design assumptions take into account the actual statistical distribution of the design variables and for the avoidance of gross human errors.

ISO 2394 [6-1] and MC2020 distinguish 3 Quality Levels (QL) for both the design and the execution, see section 10.1.3.

The QLs can be directly related to the Consequence Classes.

The specified QL will have a direct influence on the partial factor for materials, γ_M See section 12.5.4.2

ISO 22966 [6-4] and MC2020, like EN 13670 [6-4], describes 3 Execution Classes (EC) corresponding to the 3 QLs, see section 6.1.3.

To make QM effective, there must be a clear and unambiguous understanding between the owner and the designer about the performance requirements and criteria, along with the strategies to be applied in the design, construction, conservation and dismantlement/demolition phases (including the maintenance strategy of the project).

The QL to be used in design and execution shall relate to the actual CC and shall be specified in the basis of design as well as in the execution specification.

recognized. For most of the individual phases of the project communication procedures are generally formalized. But at interfaces, communication should get special attention. This is especially the case at the start of the design phase where realistic, feasible and clear requirements and criteria need to be agreed between the client and the designer. This is often an iterative process where the designer should support the client by providing feedback on how various starting points may affect economic and technical feasibility of the scheme and its sustainability and to advise upon alternatives.

Similar to the ISO standards representing 165 national standardization bodies, the main European standards relating to concrete structures are also written as framework standards and have to be implemented in the 30 CEN Member States by supplementing national determined parameters (NDPs) and/or national annexes.

In particular this is the case for QM where the level of safety, durability and economy remains within the competence of the Member States.

6.1.3 Quality Level differentiation

ISO 2394 [6-1] suggests a QL differentiation based on the elements given in Table 10.1-1 (extract from ISO 2394 [6-1])

	Description	*Personnel and organizations involved*
QL1	Basic QL	Self-control by the person carrying out the work
QL2	Increased QL	As for QL1 plus systematic QM by the organization carrying out the work
QL3	Extensive QL	As for QL2 plus checking by a party independent of the organization carrying out the work

6.1.4 Local implementation

QM and required level of safety for structures are in general dealt with in national legislation. The building traditions and boundary conditions for an effective implementation of the general requirements given in ISO standards result in a wide variety of solutions throughout the world, all supposed to ensure the required level of safety.

The ISO standards are written as framework standards where the practical implementation has to be given in a national supplementing annex or on a project

basis. MC2020 is intended for use world-wide. A similar national or regional flexibility concerning the implementation of QM measures is therefore also intended in this document.

6.1.5 Action in the event of non-conformity

ISO 22966, clause 4.4 [6-5] states:

(1) Where inspection reveals a non-conformity, appropriate action shall be taken to ensure that the structure remains able to perform as designed.

(2) The following aspects shall be investigated in the listed order:

- *the implications of the non-conformity on further execution and fitness for intended design purpose;*
- *the measures necessary to make the component acceptable;*
- *the necessity of rejection and replacement of the non-repairable component.*

(3) If required in the execution specification, the rectification of non-conformity shall be in accordance with a procedure stated in the execution specification or agreed.

6.2 Principles of Information Management

6.2.1 General

The life cycle file is a living document, which continues to be developed throughout the entire life cycle of the project. The life cycle file should be initiated during the design phase and populated with the first set of the relevant information/documents. Later phases further complete the life cycle file. The life cycle file also serves as an interface document managing the collection and transfer of information from one phase of the project into the next.

Activity	*Resulting document*	*User of document*
Conceptual design	Basis for design	Designer of the structure
Design	Execution specification	Constructor of the structure
Design and execution	Concrete specification	Concrete producer
Execution	Inspection report As-built geometry Birth certificate	Owner and/or Operator of the structure

	Maintenance plan Condition Assessment plan	
Operation	Re-birth certificate Service life file	Owner and/or Operator of the structure
Dismantlement planning	Dismantlement document Waste management document	Constructor

To allow effective and efficient QM, the project quality status/progress should be documented. Therefore, development documents should be integrated with QM activities.

In the documents, information to manage the project throughout the service life should be available. Therefore, they should contain all relevant data, such as relevant engineering documents, engineering instructions, specifications, test results and certificates, such as built documentation, maintenance strategy, factual maintenance data and the decommissioning strategy of the scheme.

The contents of an:

- Execution specification is given in ISO 22966, annex A [6-5];
- Concrete specification is given in ISO 22965-Part 1 [6-6];
- Inspection report is given in ISO 22966 [6-4] and ISO 16204 [6-11];
- Birth and Re-birth certificate are given in *fib* Bulletin 93 [6-12];
- Maintenance plan is given in ISO 16204 [6-11];
- Condition (Inspection) assessment plan is given in ISO 16204 [6-11].

These documents may either be on traditional paper format or the information may be integrated in Building Information Modelling (BIM).

The documents continue to be developed throughout the entire life cycle of the project. Thus data on the quality metrics are collected throughout the life cycle, through comprehensive verification and validation processes, including process audits, peer reviews, analysis and testing, as appropriate.

MC2020 and ISO documents describe a number of documents to ensure a proper interface and communication between the different parties involved in a project.

6.3 Project quality plan

Appropriate planning allows the parties involved to ensure alignment between project and quality goals. For proper quality planning, it is necessary to determine quality goals and quality metrics, and to use an agreed set of criteria and a standard methodology for defining the desired levels of quality.

Quality planning is required to give structure to the measures, to assure coherence between the various disciplines and stages of development and to allow quantitative management of quality. For quality planning, a project quality plan (PQP) is widely used and often required.

ISO 10005 [6-13] gives further advice on the development, acceptance, application and revision of quality plans.

Requirements for quality assurance and quality control may be defined in terms of parameters such as design supervision levels, execution classes and condition control levels. A systematic approach using these concepts is given in *fib* Bulletin 34 [6-14]. Minimum levels for the quality assurance and quality control may be defined in national legislation of each country.

Reviews are an important aspect of quality assurance and quality control, and therefore of the general management of the overall design and construction process. Reviews should be planned in advance and their timing should be linked to decisive milestones within the overall schedule of activity. It is desirable that the first review is undertaken shortly after completion of the basis of design phase or at the start of the design, in order to have the basis of the design reviewed and, as such, confirmed.

A typical contents list of a PQP is as follows:

- general: description of the project, description of the assignment, quality objectives in general, distribution and revisions of the PQP, abbreviations;
- financial: contract data, change procedure, cost control, invoicing, project evaluation;
- risk management: risk inventory, risk mitigation and management, safety and health plan;
- organizational: project organization, sub-consultants/contractors, interface management, communication procedures (reporting, meetings);
- time schedule: planning schedule, milestones, document planning, review and audit planning schedule;
- information management: document control, acceptance procedures, change management, filing, as-built documentation, confidentiality agreements;
- process quality: overview of applicable procedures, progress reporting, non-conformities, audits, customer satisfaction, project evaluation;
- product quality: functional requirements, boundary conditions, basic data and criteria, codes and practices, verification plan, design validation plan, design and drafting tools.

Checklists may be useful for the implementation of a project quality plan. Examples are given in CEB Bulletin 194 [6-15].

For standard schemes handled by a single source company with a certified company quality plan, a simple reference can be made to such a plan for most of the

The PQP should define the tasks and responsibilities of all parties involved and provide adequate control and checking procedures and the organization and filing of adequate documentation of the building process and its results.

The PQP should cover quality assurance and quality control issues.

The PQP should address or refer to:

- objectives and criteria applicable to the project;
- organizational structure;
- technical and organizational working methods and procedures;
- lines of communication;
- tasks and responsibilities;
- QM measures applicable to the outsourcing/subcontracting of activities;
- key personnel involved;
- handling of non-conformities.

The extent of a PQP may differ depending on the nature and size of the project, type of contract and parties involved, each development phase should have a plan, or the

items to be addressed in the PQP. For more complicated schemes and/or schemes handled by a combination of partners, the PQP will generally be project-specific. In such cases the ISO 9000 series of codes may be a useful support.

There is a crucial interaction with the skills of the individuals involved. Although subjective, requirements for skills and qualifications need to be assessed. Where these are deficient, training and education measures should be instigated or more appropriate staff assigned to the project, or a combination of these measures implemented.

While the ISO 9000 [6-16] series of standards is accepted worldwide as the model approach for QM, with the focus in contemporary standards upon the concept of the "continuous improvement" of an organization's management system in order to improve overall performance and customer satisfaction, sole reliance on this concept can present various difficulties in respect of the construction of concrete structures on site. In this context there is a need to prevent the occurrence of nonconformities in the "one-off" circumstances associated with the site placement of concrete in a particular structure or component, especially where these may impact upon the structural capacity, performance or durability of the finished entity. Thus there needs to be a focus – within the practices and procedures for assuring quality – upon preventive measures that minimize the risk of nonconformities occurring. This is compatible with a risk-based approach and related methodology to QM.

For more information upon pre-construction planning, the role of the project specification and of QM during execution of concrete structures, see Annexes F and G of *fib* Bulletin 44 [6-17]. These Annexes are especially relevant for design and construct type contracts, where the designer may be required to prepare tender documents for an owner with a view to anticipating potential nonconformities in construction (including those caused by the Contractor's design detailing). Appropriate PQP requirements need to be specified in the tender documents.

plan may cover a number of phases. Coherence and transfer of information and/or instructions between phases is critical. For non-standard and/or complicated projects, a project specific risk analysis should be conducted to define the issues to be addressed specifically in the PQP.

Quality cannot be assured by procedures and an organizational structure only, so the methods of improving quality practices need to be introduced into the process for potential benefits to be realized.

References

[6-1] ISO 2394:2015. General principles on reliability for structures.

[6-2] JCSS. Probabilistic Model Code. 2001.

[6-3] EN 1990:2002. Eurocode – Basis of structural design.

[6-4] ISO 22966:2009. Execution of concrete structures.

[6-5] EN 13670:2009. Execution of concrete structures

[6-6] ISO 22965 Concrete

[6-7] ISO 22965-1 Concrete – Part 1: Methods of specifying and guidance fort he specifier.

[6-8] ISO 22965-2 Concrete – Part 2: Specification of consistuent materials, production of concrete and compliance of concrete.

[6-9] EN 206:2014. Concrete – Specification, performance, production and conformity.

[6-10] ISO 9001 Quality management systems — Requirements

[6-11] ISO 16204:2012. Durability – Service life design of concrete structures.

[6-12] *fib* Bulletin 93. Birth Certificate and Through-Life Management Documentation

[6-13] ISO 10005:2005. Quality management – Guidelines for quality plans.

[6-14] *fib* Bulletin 34. Model Code for Service Life, 2006.

[6-15] CEB Bulletin 194. Modelling of Structural Reinforced and Prestressed Concrete in Computer, 1990.

[6-16] ISO 9000:2015. Quality management systems – Fundamental and vocabulary

[6-17] *fib* Bulletin 44. Concrete structure management – Guide to ownership and good practice.

Guidance for the development of health and safety systems can be found in: ISO 45001 [7-2].

Guidance for the development of systems for environmental protection can be found in ISO 13315-1 [7-3] , and in chapter 3 of this Model Code

Guidance for the development of systems for quality assurance and control can be found in ISO 9001 [7-4].

Guidance for the development of Risk Management Principles and Guidelines can be found in ISO 45001 [7-2] and in chapter 31 of this Model Code.

7 Principles of execution

7.1 General principles of execution

The execution of concrete works at the construction site must comply with the requirements given in ISO 22966 [7-1] or a technical specification that fulfils the minimum requirements of this standard.

The execution of concrete works at the construction site must comply with governing codes and regulations regarding health and safety. Additionally agreed requirements may apply.

The execution of concrete works at the construction site must comply with governing codes and regulations regarding protection of the environment. Additionally agreed requirements may apply

Details of the General Principles of Execution are presented in chapter 35 and chapter 36.

7.2 Principles of execution management

Execution management at the construction site must assure and control compliance of quality of the works with the contract documents, governing codes and regulations, and the execution specification.

Before commencement of execution of any part of the works, the execution specification, relevant to that part of the works must be complete and available.

Prior to the development of the quality management plan, a risk analysis and evaluation must be performed to identify risk, required assurance measures and/ or mitigation measures. These measures must be included in the quality management plan. Upon completion of the works, an interface document must be prepared to support life cycle management during the service life of the works

Before commencement of execution of any part of the works, management must ensure that available workforce is appropriately trained and has adequate experience to perform the tasks. If needed, additional assurance measures shall be taken to adequately counterbalance the potential risk from limited experience and/ or training.

The BIM technology allows a systematic and integral, collection and storage of relevant information.

Guidance for the development of an Emergency Preparedness plan can be found in ISO 14001 [7-5].

To demonstrate compliance with requirements, verifications must be performed. Execution management at the construction site must have documented procedures that can confirm whether verifications are compliant or non- compliant (see also chapter 35.4).

7.3 Principles of construction works

Before commencement of execution of any part of the works, the execution specification, as referred to in ISO 22966 [7-1], relevant to that part of the works must be complete and available.

Before commencement of execution of any part of the works, a workplan must be available which includes the method statement, the inspections and tests to be performed, and the protocol for the collection of data for the birth certificate. The method statement must include site preparation works related to safe access by equipment and personnel to the construction site.

Before commencement of execution of any part of the works, workforce must be briefed the specific risk and assurance measures relevant for the part of the works concerned. This includes site instructions by design, governing considerations underlaying the method statement, and safety issues.

Before commencement of execution of the works, an Emergency Preparedness plan must be available and briefed to all involved in the execution of the works.

Temporary structures must allow safe construction of the permanent structure and must enable construction within the applicable specifications.

7.4 Principles of execution of interventions

Before commencement of execution of any part of the works, the assessment plan and the repair plan, as referred to in ISO 16311 [7-6], relevant to that part of the works must be complete and available. These must include the category and level of the assessment, the investigation plan and the selected method of inspection.

Before commencement of execution of any part of the intervention works, an execution plan according to ISO 16311 [7-6] (or ISO 16711 [7-7] for seismic retrofitting) must be available, which includes the method statement, the inspections and tests to be performed for quality assurance and the method of recording. The

Guidance for the development of an Emergency Preparedness plan can be found in ISO 14001 [7-5].

method statement must include site preparation works related to safe access by equipment and personnel to the construction site.

Before commencement of execution of any part of the works, workforce must be briefed the specific risk and assurance measures relevant for the part of the works concerned. This includes site instructions by design, governing considerations underlaying the method statement, and safety issues.

Before commencement of execution of the works, an Emergency Preparedness plan must be available and briefed to all involved in the execution of the works.

Non-standard and/or unfamiliar materials or techniques employed must ensure safety and minimum disturbance to the environment and existing structure.

References

[7-1] ISO 22966:2009 Execution of concrete structures.

[7-2] ISO 45001:2018 Occupational health and safety management systems – Requirements with guidance for use.

[7-3] ISO 13315-1:2012 Environmental management for concrete and concrete structures – Part 1: General principles.

[7-4] ISO 9001:2015 Quality management systems – Requirements.

[7-5] ISO 14001:2015 Environmental management systems – Requirements with guidance and use.

[7-6] ISO 16311-1:2014 Maintenance and repair of concrete structures – Part 1: General principles.

[7-7] ISO 16711-1:2021 Requirements for seismic assessment and retrofit of concrete structures

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Special performance requirements may be defined for special structures, such as heritage structures. Special performance requirements, such as preservation of structural and architectural features in the structures, may restrict the types of intervention which can be undertaken.

Information obtained via conservation activities can provide feedback to the design employed (in particular to the verification of condition limit states associated with durability – see section 30.6), facilitating assessment of compliance with the original performance requirements or revised performance requirements, as appropriate.

An increase in the level of the performance requirements during the use stage may necessitate an intervention being made to increase the performance or possibly the durability of the structure / constructed asset.

Further background information on conservation of concrete structures, as treated in this section, is given by Matthews et. al [8-1].

These are activities to achieve the design service life of a structure or of the respective component parts. Different elements of a structure may have different lengths of design service life. For example, considering a bridge structure; the waterproofing, surfacing, jointing and sealant components might have the shortest design service lives; whereas the bearings, drainage and parapets would have longer design service lives; and the main deck structure, piers and foundations would have the longest design service life.

These are activities to achieve an extended service life of a structure / its respective component parts or revised performance requirements.

8 Principles of conservation

8.1 Introduction / General

In this Model Code, conservation is defined to mean all activities aimed at maintaining or returning a structure to a state which satisfies the defined performance requirements. The goal of conservation is to help protect owners, users and society more generally from various factors, such as injury and the possibility of death (social impacts), economic losses and impairment of the environment, which can result from the in-service deterioration of the structure.

This chapter provides a general basis for the conservation of concrete structures and the components thereof, during their planned and/or extended service life, in order to ensure that the performance level of the structure/its components remain above that required for structural safety, serviceability and repairability.

Typically, three different conservation objectives are distinguished:

– conservation activities concerned with enabling a structure to meet its design service life, as envisaged at the time of design;

– conservation activities concerned with extending the planned service life of a structure; or

– enabling a structure to meet revised performance requirements (e.g. enhanced loading or changed functionality).

The maintenance works / intervention strategy required for the conservation of a concrete structure must be considered when designing the structure. These maintenance works plans shall be developed on the basis of the design assumptions and a prognosis for the behaviour under the envisaged environmental and loading conditions over its design service life, considering the performance requirements that need to be met.

Similarly, the maintenance works / intervention strategy required for the conservation of a concrete structure to extend its design service life or accommodate revised performance requirements shall be developed when re-designing the structure on the basis of knowledge of the current condition of the structure, a prognosis for future behaviour under the envisaged environmental and

While one conservation strategy might be chosen for the management of an entire structure, in other circumstances different conservation strategies could rationally be adopted for the management of different groups of components or parts of a structure / asset.

To achieve the specified design service life objectives, it is necessary for durability to be considered in the design, construction / execution processes and activities, along with those concerned with Life-Cycle Management LCM) / operational maintenance aspects, to avoid premature deterioration of concrete structures.

This process is commonly referred to as *Durability planning* – refer Sections 27.6, 35.6 and 38.1.2 for further details upon the overall durability planning approach, its various facets and the associated component parts.

Previous engineering practice for conservation management for new structures has often been based on minimising the cost of achieving and maintaining a minimum level of performance / quality requirements (e.g. minimising the costs of design, construction and conservation while satisfying appropriate performance / quality requirements).

For existing structures, optimisation of conservation schemes has often been similarly based on minimising of the costs of conservation, while satisfying minimum performance / quality requirements.

loading conditions and recognising the implications of the revised performance requirements.

8.2 Conservation strategies and tactics

8.2.1 General

The necessity for conservation works varies widely and is influenced by many factors including the design and detailing of the structure, the environmental and loading conditions, along with the performance requirements.

Appropriate conservation strategies (one or possibly more) must be chosen for the management of the structure, with suitable conservation tactics being chosen for implementation.

For new structures, the conservation strategy and tactics must be defined during design stage. Both the chosen strategy and tactics must be confirmed/revised on the basis of the outcome of the condition evaluation, plus any associated intervention works, performed after construction of the structure.

For existing structures without a maintenance works / intervention strategy in place, the preliminary conservation strategy and tactics must be set out during the re-design stage. Both the chosen strategy and tactic must be confirmed/revised on the basis of the outcome of the current condition evaluation and current performance requirements.

In the case of revised performance requirements, the conservation strategy and tactics must be revised to meet the new performance requirements. Both the chosen strategy and tactics must be set out/confirmed/revised during the re-design stage on the basis of the outcome of the current condition evaluation.

The goal has usually been to minimise life-cycle costs for the asset / structure, as discussed in section 5.2.3. However, it is also appropriate to similarly seek to minimise life-cycle environmental impacts for the asset / structure, as discussed in section 5.2.2, in order to enhance the sustainability of the asset / structure, as discussed in Chapter 30.

The choice of conservation strategy and tactic to be used depends on various factors including:

Within the overarching framework for sustainable design and construction adopted for this Model Code, it would be appropriate to seek to minimise life-cycle environmental impacts for the asset / structure concerned.

Rational optimisation requires an approach which weights the direct and associated indirect life-cycle costs of conservation against the benefits such as a longer service life, fewer faults or greater safety. Increasingly, attempts are being made to deal with conservation management on the basis of rational risk analysis - see discussion on dealing with risk-based approaches in section 8.2.2.3. For more details, see also CUR Report 190 [8-2].

Table 8.2-1: *Basis for selecting "proactive" or "reactive conservation"*

Reasons for the adoption of a Proactive approach may include:
• When the client / stakeholders require proactive life-cycle management • The conservation strategy requires planned interventions / maintenance • Where changes / deterioration needs to be kept within predetermined bounds, perhaps for particular parts of the structure (e.g. safety / aesthetic issues) • For monitoring aggressive / severe loading and environmental conditions • When early identification of possible problems and risk issues is critical • When a rapid / significant change or loss of performance could occur • When damage / deterioration has occurred to the structure • The design service life is long (e.g. monumental / important structures)
Reasons for the adoption of a Reactive approach may include:
• When the client / stakeholders stipulate the use of a reactive approach • When regular inspections and/or maintenance works are not practical (e.g. because they are not feasible or are too expensive) • When deterioration rates are expected to be very slow • When the deterioration does not /will not compromise compliance with critical performance requirements • When it may be acceptable to wait for a specified failure / non-compliant condition to develop /will not compromise performance requirements • It is acceptable to employ an ad-hoc inspection and/or testing regime

Conservation activities can be categorised on the basis of whether they:

- adopt proactive or reactive activities;
- adopt planned or unplanned activities;
- amend the reference performance level of the structure or component part; whether the activities are to maintain the intended performance level (i.e. a
- consequences of a potential failure, which are related to the social, economic and environmental importance of the structure, the impact on third parties, the function of the structure or components thereof, the design and so on;
- feasibility of evaluating the condition of the structure; that is, how easy it will be to undertake a condition survey (which is related to the ease of inspections, measurements, the nature of the design etc.) and make an associated performance evaluation;
- the uncertainty associated with predictions of the service life of the structure or its components, which is related to factors such as the design service life, the environmental conditions and knowledge of the asset;
- records or information noting the nature and magnitude of the past loading and/or environmental actions that were experienced during the service life of the structure (e.g. mechanical loadings, physical, chemical and biological actions to which the structure was subjected or exposed to);
- feasibility of preventative or remedial interventions (e.g. repairs, replacements etc.);
- cost of conservation activities, which are related to matters such as the direct cost of the works; plus the frequency of inspections, repairs, replacements, indirect costs and so on, when a life-cycle perspective is being taken.

The available conservation strategies and respective tactics are classified on the basis of their proactive versus reactive characteristics as follows:

- Category A: Proactive conservation activities;
- Category B: Reactive conservation activities;
- Category C: Situations where conservation activities are not feasible.

maintenance or repair activity) or are to enhance the current performance level (i.e. strengthening activity).

The reasons for adopting a proactive approach might include consideration of factors relating to technical, functional, aesthetic or economic issues. For example, this might be where interventions will be expensive and/or technically difficult if undertaken once appreciable deterioration has occurred. A proactive approach is likely to be particularly appropriate for structures having a long design service life (e.g. monumental, important or sensitive structures).

A proactive conservation strategy assumes use of planned maintenance works and interventions (as appropriate). For a new structure these requirements would be defined during the service life design (see section ???).

A proactive conservation strategy would be appropriate for structures, or parts thereof, where deterioration needs to be kept within acceptable and predetermined bounds. For example, a proactive conservation strategy would be appropriate if deterioration rates are expected to be high, and would result in a rapid and relatively large loss of structural capacity within a short period of time. In this situation, where the time to respond to possible degradation would be very limited, a proactive conservation approach would typically bring significant benefits.

This approach requires appropriate condition limit states to be defined beforehand. As the assessed condition of the structure draws near to the selected condition limit state, this may lead to an increase in the frequency of evaluating the condition of the structure before a decision is made to implement structure management actions or to proceed with intervention(s). Condition-based conservation may also be referred to as state-dependent conservation.

8.2.2 Strategy using proactive conservation measures

8.2.2.1 Introduction

A proactive conservation strategy is based on preventative (or protective) measures and interventions aimed at avoiding and minimising future deterioration or loss by applying some form of treatment or taking action prior to damage becoming visible.

A proactive approach to conservation of concrete structures is desirable as it should enable early identification of problems and possible risk issues affecting the condition of the structure, potentially enabling early preventative action being taken to minimise the overall cost of ownership. A proactive conservation strategy shall be applied to structures or parts thereof where it is necessary or desirable that the performance be kept above a specified minimum performance requirement.

Tactics for delivering a proactive conservation strategy may involve various approaches including the following, which are illustrated below:

– condition-based conservation;

– risk-based conservation; and

- time-based conservation.

8.2.2.2 Condition-based conservation

With this approach the condition of the structure or its components is determined regularly and the evaluated condition must be registered, together with changes over time. The frequency at which the data is gathered and the associated evaluations are made may vary with time, potentially being related to the perceived criticality of the condition of the structure.

This approach requires that an appropriate parameter (or parameters) exist which can be evaluated and used to reflect the condition of the structure or its

Condition-based conservation may be applied when it is difficult to make a prognosis of the future condition of a structure in the absence of information about the actions on the structure. Alternatively, condition-based conservation may be adopted when regular determination of the condition of the structure by direct inspection is simple and cheap.

An example of this approach is the monitoring of the number of load or action cycles applied to a structure. Conservation works are undertaken after the registration of a certain number of load units (i.e. after registering an extreme action or after registering a predetermined measure of cumulative actions). Thus, the application of a load-dependent conservation tactic involves a load unit norm. The criteria and threshold values (limit states) are to be established beforehand during the service life design stage. Similar criteria might be applied for other types of actions.

components. The condition-based tactic involves norms related to limit states. Exceeding the predetermined limit states invokes a preventative intervention to extend the service life of the structure.

Condition-based conservation might involve consideration of factors such as:

- the number of load or action cycles applied;
- the development of cracking and/or spalling;
- loss of structural strength or stiffness;
- aesthetic effects (i.e. unacceptable deterioration in appearance).

8.2.2.3 Risk-based conservation

Depending on the specific nature of the constructed asset / structure and its context, requirements other than those concerning life safety may apply, e.g. in relation to risk of damage to the environment, loss of reputation and finances. Such additional requirements shall be identified as part of the hazard identification and risk screening processes.

Section 12.5.2 addresses the general use of rational risk-based analysis principles to evaluate hazards to which a constructed asset / structure is exposed and to assess the implications thereof.

The various hazards which might affect the structure or component in question need to be identified and estimates made of their likely frequency / probability of occurrence. Consideration must also be given to the potential consequences of the hazards. Thus, these factors provide a basis for estimating the relative importance of the potential risks arising from the various types of hazards affecting the structure or component in question.

There are numerous aspects to be considered including a need to:

- Establish the nature of the hazards involved.
- Define the risks to be carried and the criteria used in this process.

Risk assessment and management is a systematic approach which is used to avoid, reduce or control hazards and the risks arising from them. The course of action followed is to assess uncertainty by identifying and assessing hazards, understanding, acting on and communicating the risk issues involved. The goal of risk management is to help protect owners, users and society more generally from various factors such as injury and the possibility of death (social impacts), economic losses and impairment of the environment. There needs to be a balance between the cost and burdens of managing the risks arising from a hazard and the benefits expected to arise from accepting the risks.

Systematic risk-based structural performance assessment procedures shall be employed to categorise structural system failure modes in accordance with their total consequences and the probabilities of occurrence of these outcomes.

In this context hazard identification, risk assessment and management procedures would be expected to form part of the overall life-cycle management activities adopted by the bodies owning and / or operating a constructed asset / structure.

Risk-based review of the hazard scenarios shall be employed where it helps to identify:

- hazards which can be eliminated (if possible) or their consequences reduced;

– Consider whether there are particular classes of risk, such as those imposed by statutory obligations under law or any associated with specific business / organisation objectives.
– Consider the potential outcomes.
– Clarify whether the risks change with time.
– Identify the internal and external stakeholders involved.
– Establish what hazard scenarios shall be considered.
– Clarify when a risk shall be controlled or reduced and how to do this.

Once the relevant hazards have been identified and evaluated by some procedure, the decision has to be made whether the associated risks can be accepted or not. If risks are too large for direct acceptance, the standard approach is to investigate and plan appropriate counter measures. The aim is to detect those events or processes where a significant benefit can be obtained from a proportionally small input effort.

It is also relevant to note that sources of hazards can extend beyond the physical, they can also arise also from 'soft' issues such as the people involved (owners, designers, contractors, etc.) and the processes used. These issues have been termed the '3Ps' – People, Processes and Products.

A common approach to managing risks are the ALARP principles, that is seeking to ensure that risks are '*as low as reasonably practicable'*. However, an alternative term which may be used is SFARP: '*so far as is reasonably practicable*'.

The social acceptance of risk to human life is commonly presented in a so-called F–N curve. The F–N curve is a tool used to establish whether incidents are likely to be considered to be "acceptable" or "unacceptable" in a graphical form, with the annual frequency (probability) of occurrence presented on the (logarithmic) vertical axis and the potential number of fatalities on the (logarithmic) horizontal axis.

There are various specialised system analysis techniques which are used for failure analysis, particularly of operational systems. Examples of these techniques includes *Failure mode and effects analysis* (FMEA) and *Failure mode, effects and criticality analysis* (FMECA), which is an extension of FMEA. FMEA may also be used in conjunction with Fault Tree Analysis and other techniques.

ISO 31000 [8-3] sets out the principles, a framework and a process for the management of risks that are applicable to any type of organisation in the public or private sector. As this guidance is intended for organisations and the interactions between them, it may not be directly relevant to the management of a constructed asset.

– what risk-reduction measures may be appropriate; and
– if the remaining (residual) risks are tolerable or if other appropriate risk-reduction measures need to be identified and implemented.

Design failure mode scenarios shall be evaluated with respect to the efficiency of implementing further risk reducing measures for life safety and / or protection of the environment.

The assessment of the necessity for further risk reducing measures for life safety and / or protection of the environment shall be undertaken based on the ALARP concept.

Two approaches commonly used to undertake risk analysis and assessment of constructed assets / structures are:

– Qualitative risk assessment.
– Quantitative risk assessment.

Qualitative risk assessment: When it is not practical to make an explicit quantitative (numerical) evaluation of all the significant factors contributing to the risk environment, a common approach is to use some form of risk matrix (frequency versus severity) concept. This provides a simpler basis to support aspects of the decision making needed for assessing or managing the structure concerned. The approach typically involves allocating either high, medium and low ratings against the factors under consideration.

Quantitative risk assessment approaches:

IEC 31010 [8-4] sets out the principles of risk management and provides extensive information upon associated risk assessment techniques.

This approach to risk reduction and management is sometimes referred by the acronym ‘ERIC’ which stands for actions to Eliminate, Reduce, Inform about and Control hazards and associated risks.

Eliminating hazards is the prime objective and then, if this has not proved possible, to invoke steps to reduce risks from the remaining hazards.

An approach to performance-based asset management, which is used by the Rijkswaterstaat – the Ministry of Infrastructure & Water Management in the Netherlands to provide a framework for infrastructure construction and maintenance activities, is RAMSSHE€P. This is an acronym for Reliability, Availability, Maintainability, Safety, Security, Health, Environment, economics (€) and Politics. Taken together these form criteria, referred to as RAMSSHE€P aspects, which are employed in ‘performance-based risk analyses’ to evaluate the performance of the infrastructure system under consideration.

The outcomes define performance and maintenance requirements, along with associated maintenance plans. Both qualitative and quantitative risk analysis procedures are employed.

The qualitative maintenance plans are derived using qualitative risk analysis, namely *Failure Mode, Effect and Criticality Analysis* (FMECA); with the estimates of the probability and severity of the different RAMSSHE€P aspects being based on expert judgement.

The performance-based maintenance plans for major assets are derived using quantitative risk analysis, which utilises mathematical modelling to estimate the probability of system failure. This focuses exclusively on ‘reliability’ and ‘availability’ aspects and not on the other RAMSSHE€P aspects.

The performance-based risk analyses and the resulting maintenance plans are reported to assist optimal planning of availability of constructed assets and in reducing their unplanned outages (periods where they are unavailability).

The risk to human life can be evaluated on the basis of the estimated probability of death in respect of:

a. Individuals (i.e. single fatalities), and

b. Groups (i.e. multiple fatalities)

The group criterion is more commonly adopted for constructed assets.

Risk-based assessment procedures can also help guide the life-cycle management of constructed assets / structures, with the goal of:

– Eliminating hazards where practicable.

– Reducing hazards and controlling risks to the structure of these buildings, which are generally at very low levels / acceptable levels, as far as is possible.

Performance-based risk analysis procedures can be employed for the life-cycle management of constructed assets / infrastructure, enabling asset management activities and responsibilities to be clearly focused upon the performance of the ‘system’ under consideration.

The performance-based risk approach facilitates the adoption of performance-based service level agreements (SLA's) between the owner of the constructed asset / infrastructure and those responsible for its operational management, maintenance and undertaking remedial intervention or other forms of life-cycle management activities.

For further details on the RAMSSHE€P methodology see [8-5].

8.2.2.4 Time-based conservation

Registration of the actions which cause deterioration is often difficult. In such situations, a conservation tactic that involves a time-based norm may be more appropriate and practical to implement.

Time-based conservation may be adopted when regular determination of the condition of the structure is not feasible or is expensive compared with the anticipated intervention costs. Different preventative interventions could be undertaken at different times / frequencies.

Time-based conservation is likely to be employed when the condition- and risk-based conservation approaches are not applicable.

This approach involves the use of a time-based norm to define when a preventative intervention shall be undertaken to extend the service life of a structure.

This involves estimating an appropriate timing of preventative interventions or the time period between such interventions. The goal is to estimate a time period (life span) corresponding to a sufficiently low probability of failure/non-compliance for the actions being considered, typically taking them as a random variable over the period of time in question.

8.2.3 Strategy using reactive conservation measures

Reactive conservation strategy may be applied to structures or components thereof when apparent deterioration does not compromise meeting critical performance requirements. A reactive approach may be appropriate where regular inspections and maintenance works are not practical (e.g. because they are not feasible or too expensive).

A reactive conservation approach would be appropriate if deterioration rates are expected to be very slow. For example, if the corrosion rate after depassivation of the reinforcement was very slow, there should be a long period before there was an appreciable loss of structural capacity or the development of significant concrete cracking. Accordingly, in this situation there should be adequate time to make an intervention to address potential degradation impacting upon the SLS or the ULS conditions.

A reactive conservation strategy is based on remedial (or corrective) measures and interventions, aimed at arresting currently active processes which are causing deterioration or damage. Typically, remedial interventions involve some form of treatment or the taking of measures after damage has become apparent, presumably involving visual indications (e.g. cracking or spalling of concrete).

Tactics for delivering a reactive conservation strategy may include waiting for a specified failure/non-compliant condition to develop.

A reactive conservation strategy could employ the following condition survey / inspection regimes. For a new structure these requirements shall be defined during the service life design process (see Sections ???):

– A reactive approach utilising an ad-hoc inspection and testing regime;

– A reactive approach that utilises an initial planned visual inspection.

The failure dependent conservation tactic involves a predetermined failure norm / non-compliant performance condition. Thus, on this basis a remedial intervention, perhaps involving a repair or replacement, to a structural component is not undertaken until it no longer meets the specified performance requirements. The failure dependent conservation tactic might potentially be applied to those parts of the structure that do not contribute to the probability of failure of the structural system as a whole. However, appropriate review and judgment of the consequences of failure is required because use of a failure dependent conservation tactic would often not be acceptable because of the potentially large consequences of failure.

8.2.4 Situations where conservation measures are not feasible

There are parts of structures, such as foundations, where it may not be feasible to undertake inspections or conservation measures.

For a new structure these requirements shall be defined during the service life design process (refer Section ???).

In such circumstances it may be possible to gather indirect indications of structural performance through ancillary behaviours. For example, the performance of foundations might be inferred from related indirect parameters, such as differential displacements and/or relative movements within the various parts of the structure.

Should significant damage or deterioration occur (and be detected), a potentially difficult and expensive reactive / remedial intervention could be required. In addition to cost, such major works are also likely to have significant implications in terms of environmental impact.

There are parts of structures where inspection and the application of conservation measures is very difficult or not practical, or where it would be difficult economically and/or technically for preventative or remedial measures to be undertaken.

However, it may be feasible to obtain indirect indications of condition or performance of these elements via activities such as land surveying, structure or ground movement measurement, or other indirect but related activities such as leakage detection. Before this approach is adopted an appropriate review and judgment shall be made of the consequences of failure and the feasibility, cost and environmental impacts of undertaking maintenance works / performing interventions in this manner.

8.3 Conservation management

For the chosen conservation strategy, it is necessary to develop an appropriate life-cycle management plan, with the associated maintenance works / intervention activities.

Condition survey and monitoring (including SHM), performance evaluation and decision-making for LCM activities and interventions are processes which are performed as a part of the overall conservation process.

Figure 8.3-1 shows an idealised representation of the life-cycle performance of a structure requiring a remedial intervention to meet its design service life.

After completing the construction, the first life-cycle management activities are performed and the Birth Certificate document is recorded. During the post-construction

8.3.1 Life-cycle management and conservation process

To allow effective and efficient life-cycle management, the conservation process must be adequately managed and documented. In particular, the conservation planning shall be prepared in accordance with the selected conservation categories for the structure or its component parts (see below).

In general, the life-cycle conservation process for a structure involves the following types and sequence of activities:

- *Condition survey and monitoring:* This involves gathering information on the current condition and performance, including undertaking on-going condition surveys and/or structural health monitoring activities, including recording of the information required for future life-cycle management of the structure – see section 38.2;
- *Identification of deterioration and damage mechanisms:* This involves identification of deterioration and damage mechanisms, plus prediction of the degree and rate of deterioration and damage – see section 38.3;
- *Performance evaluation:* This involves analysing the available data to establish current performance and to make a prognosis of future performance for management of the asset– see section 38.4;
- *Decision-making for LCM activities and interventions:* This is concerned with establishing the implications of (say) deterioration upon the life-cycle management and conservation of an asset, evaluating potential management and conservation options, the selection of appropriate management and intervention option(s) and the timing of such works to remedy non-conformity with the performance requirements – see sections 38.5 and 38.6;
- *Execution of LCM activities and intervention works* – see chapters 37 and section 38.6;
- *Documenting LCM activities, maintenance works and interventions* – see section 38.7.

The general flow of conservation process procedures for new structures and for existing structures is shown in Figure 8.3-2. In parallel, it is indicated which information shall be recorded to meet the requirements of life-cycle quality management, as discussed in chapter 10 with regard to development of the life cycle file.

lifetime, knowledge of the performance of the structure is accumulated by means of on-going condition surveys and/or structural health monitoring.

Further details on these topics are provided in the following publications:

a. *fib* Bulletin 93 [8-6];

b. *fib*, Structural Concrete 2020 [8-7].

Evaluation of the performance records against the performance criteria supports decision-making as to whether there is a need for an intervention. After carrying out the (required) intervention, a condition survey is performed / data is gathered by testing and/or monitoring procedures and a re-Birth Certificate for a structure is prepared. During the post-intervention lifetime, the knowledge of the performance of the (possibly altered) structure is accumulated by means of on-going condition surveys and/or structural health monitoring procedures.

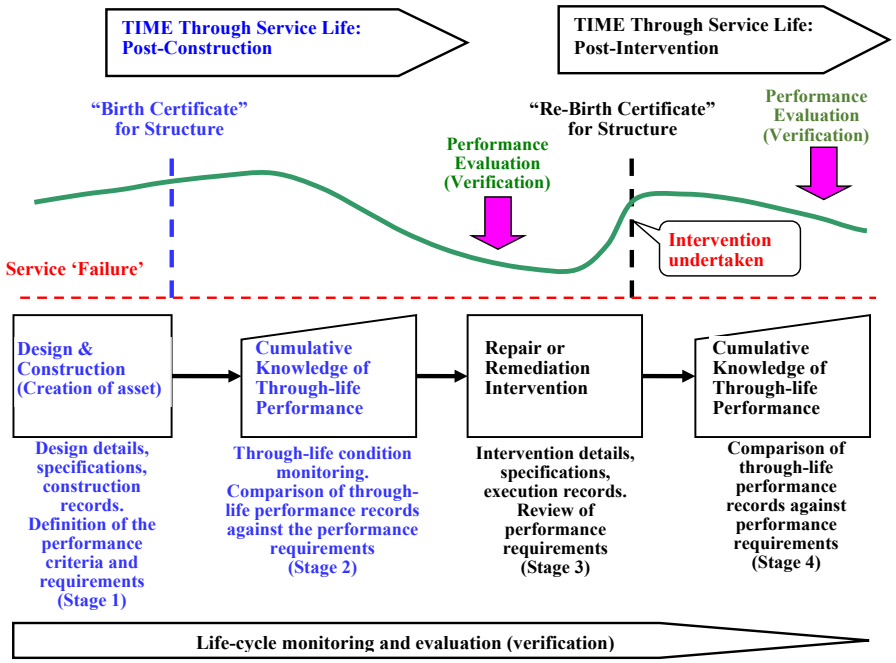

Figure 8.3-1: *Idealised representation of the life-cycle performance of a structure requiring a remedial intervention to meet its design service life*

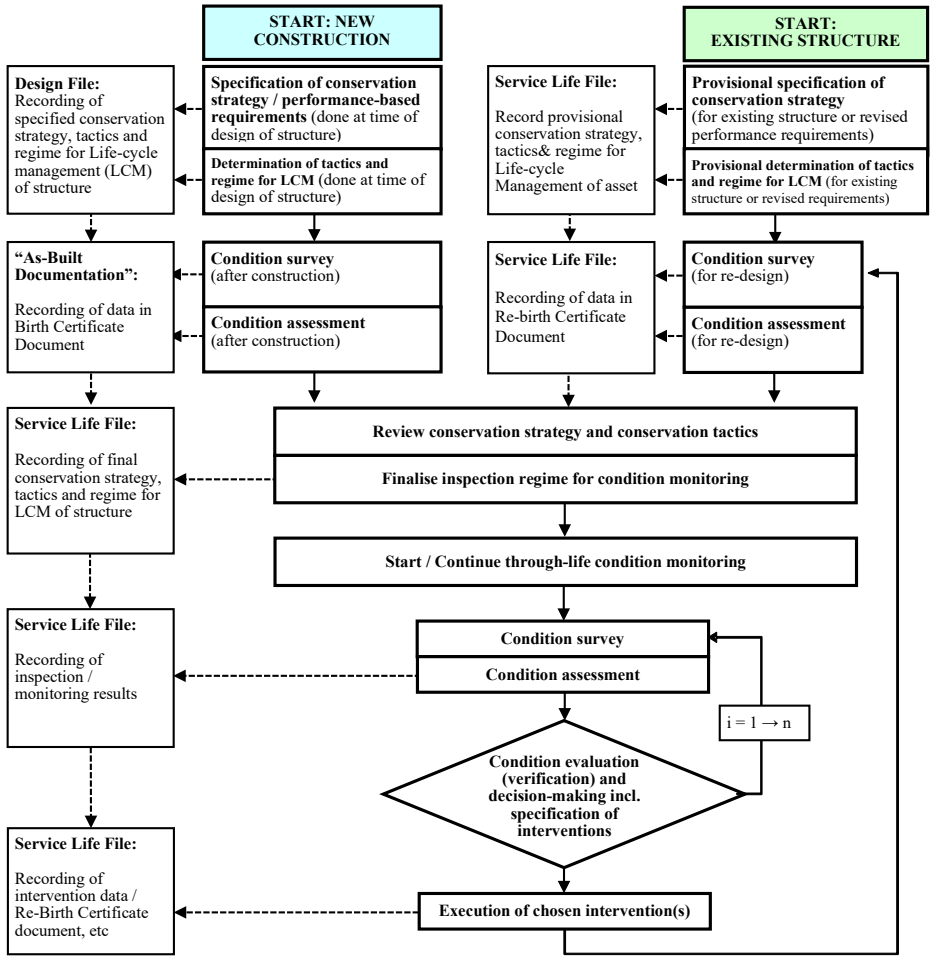

Figure 8.3-2: *Conservation management: Life-cycle management and conservation process and associated recording of information.*

Notes to Figure 8.3-1:

The main stages and factors to be considered are:

Stage 1. Details about the design concepts and construction of the original structure along with the expectations for its performance, ideally including material specifications and information upon the quality on execution.

Stage 2. Information upon / monitoring of through-life performance post construction and prior to the repair or remediation intervention.

Stage 3. *As item1* but for repair or remediation intervention(s), plus the definition of appropriate success criteria upon which to evaluate the subsequent performance of the intervention works.

Stage 4. Monitoring of through-life performance post repair or remediation intervention, using defined performance indicators and environmental parameters and related factors establishing the context within which this data should be reviewed and evaluated for success or failure.

Stage 5. Use of a Performance Evaluation / Verification methodology to assess the overall condition / structure performance of the structure. This could take the form of a Performance Index.

The necessity for maintenance and intervention works varies widely, depending on factors such as the social, economic and environmental importance of the structure, the nature of its design and that of its components, potential threats posed to third parties and the design (planned) service life of the structure. Account needs to be taken of the potential difficulties of performing inspections and associated condition survey activities, the required maintenance works, along with predictions of the potential severity and rate of deterioration and the difficulty of undertaking intervention works.

Examples of structures within the three conservation categories are:

Conservation category A:

– structures whose deterioration must not be apparent or would be technically unacceptable from a client requirement or technical performance perspective;

– monumental, important or sensitive buildings and structures.

Conservation category B:

– structures for which remedial measures can be taken once a small degree of deterioration becomes apparent (presumably visually, but potentially with the assistance of NDT / SHM techniques);

– buildings and other common structures.

Conservation category C:

The conservation planning adopted in Chapter 38 must accord with the selected conservation categories for the structure and/or its component parts, as defined during the service life design process (see below).

Different categories are defined depending on factors such as the importance of the structure, its function, design service life, impact on third parties, environmental conditions, ease of maintenance and cost. Requirements for conservation shall be defined at the time of design. The categories are as follows:

Category A: structures or parts thereof which are to be managed by planned proactive maintenance works and interventions.

Category B: structures or parts thereof which are managed by reactive interventions.

Category C: structures or parts thereof for which maintenance works and interventions would be extremely difficult or are not practical.

– structures where it would be difficult economically and/or technically for preventative or remedial measures to be taken, such as foundations, for which direct inspection is so difficult and/or costly that assessment and judgment may need to be made based on information gathered on indirect performance indicators.

The Conservation Plan shall state the types of inspection, testing and condition monitoring that have to take place, what components of the structure are to be inspected / monitored, what the frequency of the inspections shall be and so on.

The provision for planned activities within the conservation plan must address the significant deterioration and damaging processes that are relevant to a structure during its service life.

This process leads to the final conservation plan to be adopted for the life-cycle management of the structure.

The RAMSSHE€P analysis approach (see [8-5]) can be used during the development of the required strategy for maintenance and repair interventions upon a constructed asset / existing structure.

The above mentioned RAMSSHE€P processes can be employed to develop a conservation plan which is to be adopted for the future life-cycle management of the structure / constructed asset.

8.3.2 Conservation plan

The conservation plan shall state the type and frequency of the foreseen conservation activities and may include:

– specification of the regime for inspection, testing and condition / structural health monitoring activities – see section 38.2;

– identification of deterioration and damage mechanisms, plus methods of estimating the degree and rate of progression – see section 38.3;

– methods of evaluating structural and other performances against the performance requirements (criteria to be met) – see section 38.4;

– decision-making, including selecting and implementing the LCM activities required to manage the structure, as well as undertaking any interventions which may be required to remedy non-conformity with the performance requirements – see section 38.5;

– LCM activities and the execution of intervention works – see chapter 37 and section 38.6;

– methods of documenting LCM activities, maintenance works and interventions – see section 38.7.

For new structures, the conservation strategy and conservation tactics must be defined and the conservation plan must be developed during the design stage. Both strategy and tactics shall be reviewed on the basis of the outcome of the condition evaluation performed after construction of the structure (Birth Certificate – see *fib* Bulletin 93 [8-6]).

For existing structures without a conservation plan in place, a conservation plan shall be developed during the re-design stage on the basis of the current condition evaluation and consideration of the current performance requirements.

In case of the revised performance requirements, the conservation strategy and tactics, along with the conservation plan, shall be revised to meet new performance requirements/criteria to be met. These shall be reviewed during the

re-design stage, on the basis of the outcome of the current condition evaluation and consideration of the revised performance requirements.

References

[8-1] Matthews, S., Bigaj-van Vliet, A. Conservation of Concrete Structures according to *fib* Model Code 2010. *Structural Concrete*, 14(4): 362-377.
To be REPLACED WITH A NEW PAPER TO BE PUBLISHED FOR MC2020

[8-2] CUR Report 190. Probability in Civil Engineering, 1997.

[8-3] ISO 31000:2018. Risk management – Guidelines.

[8-4] IEC 31010:2019. Risk management – Risk assessment techniques.

[8-5] RAMSSHE€P. Guidelines on Performance-based Risk Analyses (PRA) – Enabling asset management based on system performance, 1997. Rijkswaterstaat – Ministry of Infrastructure & Water Management, the Netherlands.

[8-6] *fib* Bulletin 93 Birth Certificate and Through-Life Management Documentation, *fib* Technical Report, *fib* 2020

[8-7] *fib* Structural Concrete 2020 Existing concrete structures: Structural Health monitoring and testing for condition assessment, *Special Theme (15 papers) in Structural Concrete journal,* 2020, 21(4): 1464 – 4177.

9 Principles of circularity and dismantlement

9.1 General principles of circularity

Circular economy principles applied to concrete structures shall contribute towards creating a closed-loop system for built environment, minimising the use of natural resources and the creation of waste, pollution and carbon emissions.

Circularity of concrete structures contributes to sustainable design by decreasing dependency on natural resources through a variety of structural solutions: structures, structural components and construction materials may be:

– kept in prolonged use

– refurbished and repurposed

– deconstructed and reused

– demolished and recycled.

The circular economy aims to keep products, materials, equipment and infrastructure in use for longer, thus improving the productivity of these resources. Waste materials and energy should become input for other processes through waste valorisation: either as a component or recovered resource for another industrial process or as regenerative resources for nature.

Circularity of concrete structures is an approach with the aim to minimize the need for new materials if structures are subjected to an extended lifespan or for new structures. This could be achieved by the application of elements/ components and/or subsystems of structures in a new life-circle.

Prolonged use is service life extension of existing structure within the original functional or technical requirements, which can be achieved by maintenance and interventions. Principles of prolonged use of structures are further addressed and explained in chapter 8 and other relevant chapters.

Refurbishment and repurposing refers to service life extension of existing structure for new functional or technical requirements, which can be achieved by functional or structural modifications. Principles of refurbishment and repurposing of structures are further addressed and explained in chapter 8 and other relevant chapters.

Reuse is the (repetitive) use in new structures of structural members of structural components reclaimed from existing structures. Reuse options may include use in new structures of e.g. structural members or structural components design for multi-life-cycle or reclaimed from construction and demolition waste. Reuse options may vary in the level of decomposition i.e. system, sub-system, component, element (structural member). In most cases larger benefit for the environment is to be reached if reuse option is implanted at the higher level (e.g. reuse of bridge decks, floor systems, load bearing frames vs. reuse of beams, columns or slabs).

Recycling is the (repetitive) use of construction materials and/or constituents of construction materials after extracting them from residual waste flows. This includes in particular the use for the production of new concrete of coarse fractions (e.g. recycled concrete aggregates), fine fractions (e.g. sieve sand) and/or binders

extracted from construction and demolition waste flow. Waste flows other than construction and demolition waste can also be mobilised e.g. for replacing cement with alternative binders and fillers from new residual waste flows, which additionally contributes to reducing demand for cement for new concrete structures.

It should be noted that already in the design phase of structures, the possibility of using the structure for another function than the original one should be regarded. As an example, buildings with large span floors with residual bearing capacity, and facades allowing the entrance of ample daylight are often more suitable for a change of function than buildings with small rooms and heavy separation walls. A good conceptual design of structures can therefore prevent later dismantlement, since the structure can be re-used for another function.

To optimize the potential of circularity for new structures it is advised to include the interaction between the structure and installations for consequences regarding circularity of the structure.

In the past, structures have rarely been designed taking into account the possibility of adaptation or re-use in future for other aims. The increasing need to demolish structures has therefore led to an accelerated development of all types of demolition techniques. However, those new technologies often go along with other non-satisfactory phenomena resulting from insufficient analysis of their influence on the stability of the structure and with other negative effects for environment.

In case of new concrete structures design decisions should be taken with due consideration of sustainability of various scenarios, inducing the design options involving reuse and/or recycling of construction materials and/or structural elements recovered from construction and demolition waste. Next, in design for service life, due consideration should be given to a potential future need for refurbishment or adaptation to changing needs. Finally, during design the demands of the end-of life dismantlement enabling the re-use of parts of materials of the structure involved should be taken into account in order to enable circularity of the structures via unproblematic execution of the future deconstruction or demolition.

In case of existing structures for which decisions with regard to terminating their service life are still to be made, a path of optimal use incl. service life extension may be more sustainable from the overall life cycle perspective. Therefore, for existing structures the end-of-life decisions should be taken with due consideration of sustainability of various scenarios, involving service life extension (with or without refurbishment and/or repurposing) as well as (partial) deconstruction and/or demolition.

In case of resolved dismantlement of existing structures, enabling and implementing circular solutions such as deconstructed & reused and demolished & recycled is the encouraged path leading towards the improvement of the suitability of concrete structures.

To optimize the potential of circularity for new structures, the opportunities should be addressed and investigated right from the start of the design process and continued through the construction phase. Such considerations should include:

- the optimization of whole-life scenarios, including considerations for adaptation to changing needs.
- the design/ method statement for deconstruction, including matters of the execution of the future deconstruction or demolition.

For existing structures the availability of reliable data with regard to the quality and the condition of the structure and construction materials is a decisive factor to indicate whether circularity is feasible and if so, which of the above listed applications would be appropriate. To enable the anticipated circularity in future, it is essential to collect, process and archive the required data from design, construction and maintenance properly and in a accessible format for future use. BIM models could serve as an effective tool.

In case of an eventual reuse of materials and/or recycling of reclaimed elements and components it should be considered that the design consideration, execution conditions and conservation needs might differ from those for structures made with new materials.

Specific provisions for implementation of circularity principles in life cycle of the structures are still under development.

Dismantlement of structures is a process of disassembly of structures by deconstruction or demolition with or without separation of the structural members and structural materials, fulfilling disposal requirements.

Deconstruction is process of controlled dismantling by systematic selective removal of parts of the structure with planned recycling. In this case separation of material fractions is strived after. Deconstruction aims to maximize the reuse potential of dismantled structures. In this case separation of reclaimed parts of the structures is strived after. Deconstruction techniques remove components of the

– the collection and management of data from existing structures for evaluation of reuse and recycling options in case of dismantlement.

Optimization of decisions during life-cycle of the structure within the context of circularity shall follow the rules of the overall evaluation of sustainability as described in chapter 30, and comprises the investigation to reach the balance between investment, value creation and financial feasibility. At all time, compliance with regulations and project-specific requirements shall be met.

For general provisions for implementation of circularity principles in life cycle of the structures, reference is made to Chapter 39.1

9.2 General principles of dismantlement

The dismantlement of large structures requires careful preparation. The aim of dismantlement should be defined with regard to its extent (partial or full), requirements for eventual re-use of the material or components considering circularity principles given in section 9.1, and the time available for demolition.

The structure to be dismantled should be investigated with regard to its bearing structure and capacity, the materials used, its stability in the various stages of dismantlement, and the effects of eventual contamination.

An appropriate type and method of dismantlement should be chosen, taking into account the aim of dismantlement, the safety aspects of the deconstruction operation (safe working conditions), the economy and the boundary conditions of its eventual continued use during the (partial) dismantlement process.

In general, dismantlement can be subdivided into two types:

– controlled dismantlement by deconstruction;

structure piece by piece rather than dismantling it while causing their damage by destructive demolition techniques.

Demolition is process of conventional dismantling of a structure without previous demounting of structural members or structural materials. Demolition aims to maximize the recycling potential of (construction) materials and/or constituents of construction materials after extracting them from dismantled structures. In this case separation of material fractions is strived after.

REFERENCES

- **No references -**

– conventional dismantlement by demolition.

During dismantlement all principles as presented in this Model Code for the construction of new structures with respect to hazards and risks, health, safety, environment provisions, quality management and execution management do apply.

For general provisions specifically related to dismantlement, reference is made to chapter 39.2.

The aim of the design process is to translate the performance requirements of an asset owner into detailed information to enable the contractor to actually build the asset. Generally, an iterative procedure is employed to develop the starting points formulated by the owner into sufficiently detailed specifications and drawings. Through a series of cycles, the QM-plan takes shape, its contents gradually become better defined and more refined. The cycles form recurrent actions in the design process, which eventually will result into specific outputs that support the decision-making process. Without an iterative design process that engages effectively with the decision-making process, there may be a risk that substantial re-working of the design may be required at a later stage.

, 27 an d28Although there are various ways to organize the consecutive design phases, decision models are generally based on GO/NO-GO milestones, with a requirement for an associated increase in the accuracy of the prediction of the required project budget. Generally, both the amount and quality of the engineering input will gradually increase in the consecutive phases of the design process. The greater the confidence that the objectives and requirements are correctly formulated and the project is likely to proceed, the greater will be the justification for more detailed design efforts. Accuracy levels (plus and minus) of the budget will typically be about 30% in the scouting phase, about 20% at the start of design phase, about 10% at project specification phase, about 5% at the final design/detailed design phase. While these estimates are typically related to project costs, they could also be applied to aspects such as quantified levels of environmental impact and of sustainability parameters.

10 Implementation of Q&IM during LCM

10.1 Q&IM during design phase

10.1.1 Objectives

In a project the design process is generally split into a number of consecutive phases to enhance its effectiveness and efficiency. Each phase in the process should be formulated in a way that is compatible with the decision process.

During design phase, a through-life project perspective upon the creation and life-cycle management of sustainable concrete structures should to be adopted. Chapters 26, 27 and 28 address these objectives in more details, with particular focus on the processes of conceptual design, and the approaches taken to design and assessment, respectively.

Generally, the following consecutive design phases can be distinguished:

- briefing phase, see section 10.1.3;
- scouting phase, see section 10.1.4;
- basis of design phase, see section 10.1.5;
- project specification phase, see section 10.1.6;
- final design phase, see section 10.1.7;
- detailed design phase, see section 10.1.8.

10.1.2 Design File

The design file of the project should be addressed in the briefing phase. Upon completion of the detailed design phase, all relevant documents from the design should be included in the design file. The design file must contain the following documents:

- stakeholder's brief, see section 10.1.3;
- scouting report, see section 10.1.4;
- serviceability criteria agreement, see section 10.1.5;
- project specification document, see section 10.1.6;
- final design report, see section 10.1.7;
- calculations report, technical report and design drawings, see section 10.1.8.

Upon completion of the design phase, the design files should be included in the life cycle file and handed over to the asset owner for evaluation and if needed for further development in the next phase of the project. *What is the next phase; why not name it?*

10.1.3 Briefing phase

Objectives

In case a performance-based approach is adopted, general performance requirements should be developed during the initial stage of design of new structures or during the condition assessment of existing structures. General, objectively formulated, statements must be used to define the structural and functional performance requirements, according to section 10.4.1.

The stakeholders should define the required performance of the structure during its design service life. Minimum performance requirements, such as those specified in applicable national standards, should not be violated.

The clearer the performance requirements are defined, the better the effectiveness and the efficiency of the process will be. Performance requirements that are insufficiently clear and are not measurable may remain open to variable interpretations, which is likely to impair the process.

Setting realistic performance requirements involves a sound engineering and economic analysis to determine what is achievable and at which cost.

Preferably, the performance requirements of the structure are to be defined by the asset owner together with the other stakeholders and in conjunction with the project team and the (owner's) professional team.

The project team, including the (owner's) professional team and the designer, is a group of persons who have experience on the various technical aspects and processes relevant for the design, construction and maintenance of structures.

The stakeholder must not refrain from interaction/communication with other parties involved once their basic needs have been established. It is important to monitor progress and communicate with the stakeholders during all stages of the project. Communication needs to take place throughout the whole project process, from project inception to its life-end.

In many instances, the brief is a continually evolving document. In the briefing phase the brief does not provide all the information, but it should pose questions and challenges for the designers. The discussion on and clarification of the owner's final requirements comes during the scouting phase (section 3.5.3.4).

Key issues to address when developing an initial owner's brief include:

- type(s) of structure and its location (decided after examination of other means of achieving the general objectives – a process which is undertaken before deciding to build);
- planned function(s) of the structure and its components;
- requirements for appearance/aesthetics (initially and throughout the operational life of the structure);
- requirements for usable space, dimensions, services and fittings;
- the period of (operational) service which constitutes the end of service life and the requirements for the structure at the end of this period;
- need for future changes of use (e.g. to increase flexibility and minimize the risk of obsolescence);
- time, budget and/or quality constraints.

It is common practice to limit the expenses of the design effort because the feasibility of the project will usually be uncertain at this stage. The objective of making an initial estimate of the overall project costs with limited staff input (and hence incurred costs) will normally require sufficiently experienced personnel to develop an outline project concept and to make judgements about potential costs, sustainability impacts and so on. At this stage the target accuracy for the estimate of overall project costs may typically be ± 30%. However, this target level could also be applied to other performance requirements such as environmental impact and the evaluation of sustainability parameters.

Stakeholder's brief

The stakeholder's requirements must be written down in a formal document called the (initial) stakeholder's brief.

The owner's brief addresses the relevant needs and aims of the project, resources to be provided by the owner, the details of the project and any appropriate design requirements. It sets a framework within which all subsequent briefing (when needed) and design can take place.

Goals in the initial brief need to be prioritized into "must haves" and "desirables" in order to guide the project team and help them make compromises when the need arises (e.g. prioritizing of time, cost and quality).

10.1.4 Scouting phase

Objectives

The scouting phase is an initial (basic/simplified) feasibility evaluation of the project/scheme. Generally, it will be based on an outline project concept established from the global performance requirements defined in the briefing phase.

One approach which is commonly adopted is to review relevant former schemes, adapting them to the specific circumstances and requirements of the current project. In order to do so in an effective and efficient way, with limited staff effort, the designer needs to be well-experienced and to understand the general cost drivers associated with the new and previous project concepts. The goal is to identify project specific, decisive points of attention and cost drivers that need to be considered in detail during the next phase of the development of the design.

At this stage the target accuracy for the estimate of the overall project costs may typically be ± 20%. However, this indicative value could also be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

Examples of relevant basic data include:

- geotechnical data;
- meteorological and oceanic data;
- topographical and bathymetrical data;
- climatological data;
- environmental data (earthquake, hurricanes, the aggressiveness of the service environment);
- material properties.

Operational and maintenance requirements may comprise:

- the use of chloride-based de-icing salts;

It will usually be necessary to prepare an indicative budget to support the owner's decision-making process.

Document

The output will be in the form of a scouting phase evaluation report on the feasibility of the project and the project scheme, with global functional requirements, outline concept and budget estimate.

10.1.5 Basis of design phase

Objectives

During this phase the functional requirements, basic data and design criteria will be developed and the service criteria will be agreed upon with the stakeholders. A conceptual design (see also section 7.1) will also be developed to support a more accurate budget estimate. Quite some effort is required at this stage as the basis of design should be agreed upon, and fixed upon completion of this stage. An essential part of this phase is the service criteria agreement.

Service criteria agreement

The service criteria must be clearly specified *by which party or person?* in the service criteria agreement, which must comprise:

- general aims for the use of the construction works; *or "structure"?*
- basic relevant data, including third-party interactions;
- operational and maintenance requirements;
- special requirements of the stakeholders;
- objectives for consideration of protection against and treatment of special risks;
- loadings and loading combinations;
- codes and regulatory requirements.

- replacement strategy of components subjected to wear;
- flexibility in terms of space requirements, future extension or loadbearing capacity.

The objectives and the degree of protection of the structure against deterioration should be based on a risk evaluation.

Fixing the performance criteria for serviceability and structural safety should consider:

- the hazards, together with measures by which these hazards can be avoided, reduced, mitigated, controlled, managed or resisted;
- the nature and consequences of deterioration and failure;
- the resistance and mitigation mechanisms.

The specification of the (design) service life or the residual service life of a structure or of individual structural components should consider aspects such as:

- the required service life of a structure, as given by the asset owner and/or stakeholders;
- which events or condition constitutes the end of service life of individual components of the structure;
- a need for differentiation of service life for individual components of the structure (e.g. depending on aspects such as maintainability and replaceability);
- the implications of other service criteria, for example with regard to structural analysis, maintenance and QM.

Fixing the target reliability levels for condition, serviceability and ultimate limit states should consider aspects such as:

- type and consequences of failure;
- amount of acceptable damage;
- importance of the structure with respect to dealing with a hazard following an accidental event;
- expenditure to reduce the risk;
- possibilities for monitoring, maintenance and repair during operation of a structure as well as the corresponding expenditure;

In particular, the specification of the service criteria must address:

- performance criteria for serviceability and structural safety, see section 3.3.1;
- service life constraints, see section 3.3.2;
- reliability constraints, see section 3.3.3;
- performance requirements for sustainability, see section 3.4.

Progress toward the performance requirements should be traceable

- need for differentiation of the target reliability levels depending on the limit state and reference period, either for the whole structure or for individual structural components;
- possible hazard scenarios, which should be considered and evaluated, and suitable measures specified in order to keep the hazards under control or to limit them to an acceptable extent.

The following principles may be applied to mitigate hazards considered to be relevant:

- elimination, prevention or reduction of a hazard;
- control or alarm systems;
- choice of structural systems that are less susceptible to the hazards under consideration;
- choice of structural systems that can tolerate local damage as well as the loss of one or more structural components of the structure without total failure;
- choice of structural systems that do not fail without prior warning;
- limiting the spread of fire by the provision of fire compartments;
- choice of suitable structural materials that, if well maintained, will not substantially deteriorate during the required service life;
- accepting a shorter service life for structural components, which may be replaced one or more times during the specified service life;
- appropriate structural analysis and dimensioning;
- careful detailing;
- dimensioning the structure in a manner that allows for or compensates for deterioration during the specified service life;
- choice of an appropriate execution method;
- execution carried out as planned and with the necessary care;
- planning and applying suitable protective and mitigating systems;

appropriate monitoring and conservation, including inspections at fixed or condition-dependent time intervals and necessary preventive intervention or remedial activities.

Fixing the performance criteria for sustainability should follow consideration of factors such as:

- the importance of the structure to the global, regional and local environments;
- the required achievements with respect to sustainability, as given by the asset owner;

- the type and consequences of not meeting the required achievements with respect to sustainability;
- the flexibility to allow future extensions and/or modifications of the functional requirements;
- the expenditure to reduce the risk of not meeting the required achievements with respect to sustainability;
- the need to differentiate the required achievements with respect to sustainability for individual parts of the structure (e.g. depending on factors such as their replaceability).

Well-defined performance requirements allow evaluation of the achievement of performance goals throughout the design, execution, operation and dismantlement/demolition of the structure.

At this stage the target accuracy for the estimate of overall project cost might typically be ± 10%. However, this target level could also be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

The output of this stage can serve as the technical part of the invitation to bid for a design/construct contract.

10.1.6 Project specification phase

Objectives

With the basis of design as the starting point, the design will be developed first into a preliminary design. Specifications for workmanship, materials and detailed design will then be developed. Significant effort is generally required at this stage.

At this stage, alternative structural concepts will generally be developed and evaluated against each other (see also section 7.1). Numerous aspects should be included in this judgement, potentially including the following:

- robustness of the concept;
- constructability of the concept;
- planning schedule for the concept;
- economy of the project/overall life cycle cost and its achievements with respect to sustainability parameters;
- feasibility of future extensions;
- reliability of the concept as a whole and critical components especially;
- maintenance and repair considerations;
- dismantling of the structure/demolition aspects.

In order to develop the structural concept issues such as the following should be taken into account:

- the service criteria agreement;

Factors influencing the constructability/economic feasibility of the project may include:

- accessibility of the construction site;
- bearing capacity of the subsoil at the site with respect to anticipated construction equipment loads;
- lifting capacity at the site;
- minimum/maximum size of structural components;
- clearance between energy units necessary for construction;
- quality, availability and reusability of construction materials;
- restrictions regarding the design and construction times, and the budget limitations;
- legal aspects (laws, ordinances, directives);
- construction methods, transport and assembly techniques;
- transport distance of deliverables like concrete, precast elements and installations to the construction site
- equipment and facilities for monitoring and maintenance measures;
- maintaining the use of infrastructure facilities (pipelines etc.);
- demolition approach at the end of service life;
- life cycle cost considerations.

The following deviations should be considered:

- deviation from the assumed values of the actions;
- deviations from the planned values of the ultimate resistances of the structure or the soil;
- eccentricities due to construction tolerances, imperfections in the dimensions of structural components.

A structure can be designed for flexibility, anticipating on possible future changes of its function.

A clear statement must be given, indicating which data are fixed, which data need further development, which data have been assumed and which assumptions have been made.

- constructability/economic feasibility of the scheme;

- the critical actions and action effects, as well as the sensitivity of the concept to deviations from the anticipated values;

- the foreseeable service situations, which should be considered and evaluated and appropriate measures taken to ensure serviceability;
- aspects of sustainability in agreement with the requirements of the owner, stakeholders and/or governing authorities.
-

Project specification document

The project specification document should include information such as the following:

- the chosen structural system;
- the specified (design) service life;
- the service conditions considered;

In the context of partial safety factor verification, ensuring the required reliability level requires adequate consideration of the uncertainties regarding actions, structural modelling and the determination of action effects. The differentiation of the partial safety factors depending on the uncertainties in actions, material properties and applied models is addressed in chapter 4.

At this stage, the target accuracy for the estimate of overall project cost may typically be ± 5%. However, this target accuracy could also be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

- the hazard scenarios considered;
- the requirements for structural safety, serviceability, robustness and sustainability, together with the measures needed to achieve them, including attribution of responsibilities, processes, controls and corrective mechanisms;
- a reliability qualification statement for the data used for design;
- the most important dimensions, construction material properties and construction details;
- the assumed soil conditions;
- the important assumptions in the structural and analytical models;
- the accepted risks;
- advised/required additional investigations;
- other conditions relevant to the design;
- comments on the envisaged methods of construction;
- specifications for detailed design, materials and workmanship.

The extent and content of the project specification document should be adapted to the importance of the structure and the associated hazards and environmental risks.

10.1.7 Final design phase

At this stage, all primary structural components will be specified and typical details will be designed.

Structural analysis and calculations report

The structural analysis should consider the behavior of the structure in relation to the envisaged dimensioning situations, taking into account the relevant factors that significantly influence the potential performance of the structure or the structural components concerned.

The methods of structural analysis must be based on well-established theories – experimentally confirmed if necessary – and engineering practice.

The results of the structural analysis must be checked for reliability; for example, the results should be subject to a critical review utilizing general engineering judgement.

Final design report

The final design report must contain all data used for design, all phases considered, including construction phases, applied static schemes, structural analysis, applied criteria and material properties, construction method considered and a traceable demonstration of compliance with the project specification.

The final design report should also contain a risk file. The risk file must present the identified risks, how they have been managed and, if any, instructions for the next phases of design and construction.

Drawings must present the overall layout of the project, as well as the geometry, shape and dimensions of primary structural components and typical details.

10.1.8 Detailed design phase

Objectives

The output of this phase should allow construction at the start of the project and demountable at the end of the project. All calculations that are needed to demonstrate compliance with codes and requirements/ specifications of the project are prepared during this stage. The level of detail of drawings and specifications/site instructions should allow unambiguous understanding by the contractor of what is required and how the structure has to be constructed, as well as how compliance with the serviceability and design files should be demonstrated. A risk file must be prepared to inform the contractor on the risks involved, how these risks have been managed in the previous stages of design and how the remaining risks should be managed. Issues which require special attention must be clearly noted on the construction drawings.

Dimensioning

Dimensioning concerns the determination of the dimensions, the structural materials and the detailing of a structure based on structural and execution-related considerations or numerical verifications.

Detailing, limit measures and special provisions supplement the use of analytical models for various purposes, such as:

- to avoid superfluous calculations;
- to satisfy the minimum performance requirement or comply with deemed-to-satisfy provisions with regard to unidentified or poorly quantified hazards. These measures include provision of a minimum resistance to lateral forces, multiple load paths and ties between structural components (see sub-sections 2.1 and 3.2.3 of CEB Bulletin 191 [10-1];
- to ensure the validity of calculation models, for example by minimum ratios of reinforcement;

– to ensure a good standard of execution and/or durability, for example by rules for bar spacing and concrete cover depth.

The Execution Management of structures is covered in chapter 35 of this MC2020 and specific details of Concrete Construction Works are addressed in chapter 36. In addition, there are several documents that draw attention to relevant construction processes, including potential problems which must be resolved during the execution of concrete structures. These documents include ISO 22966 [10-2], EN 13670 [10-3] and 13369 [10-4] and Guides and Reports by ACI Committees 304, 305, 306, 308, 309, 311, 315. *fib* Bulletin 53 [10-5], constitutes an essential reading for achieving safe and durable concrete structures (sections 5.5, 5.10, 5.13, 5.14 and 5.15 are particularly applicable to this chapter).

ISO 22966 [10-2] (similar to EN 13670 [10-3]): "Execution of concrete structures" is the generally accepted starting point. However, it is a 'framework standard' and not a complete specification. To be used in a contract situation, the items in the standard's checklists have to be "filled in", to convert them into a set of specified requirements

The dimensioning may be assisted by testing, for example, if:

– actions, structural materials or soil properties are not adequately known;
– no appropriate analytical models are available;
– the structure contains components for which there is limited experience and which have a critical influence on the reliability of the structure.
–

Calculations report

The basis and the results of the detailed design phase should be well- documented.

Technical report and design drawings

The constituent materials, the structural dimensions and the detailing of a structure as determined during design should be documented in the technical report and design drawings.

10.2 Q&IM during execution

10.2.1 Q&IM during execution of construction works

10.2.1.1 Objective

The objective of Quality Management (QM) during the execution of construction works is to ensure that the resulting built structure satisfies the assumptions adopted in the design (geometrical aspects, materials characteristics and materials performance, etc.), expressed in the drawings and specifications, in compliance with the applicable execution specifications and standards.

For the real performance of the structure to resemble that predicted by the models applied during the design phase, strict controls during its execution are required, following a detailed quality and inspection plan that should be prepared at the end of the design phase and included in the tender (mandatory for QL2 and QL3).

In particular, the location and orientation of the structural elements as well as their geometric dimensions shall be within the stipulated tolerances. This also applies to the positioning of the steel reinforcement and prestressing. The quality of the steel, concrete and other products shall conform to the specifications and, especially

which, together, form the project execution specification which then becomes the centrepiece of the tender and contract documents.

The design process involves the application of idealized models which predict the response of the structure to the applied loads (mechanical, environmental, etc.). In the case of structural design, the response to the applied mechanical loads (static and dynamic) is expressed in terms of visco-elastic and plastic deformations, cracking and, ultimately, failure. In the case of durability design, the response to the environmental loads is expressed in terms of service life or the time to reach some pre-defined limit state (e.g. depassivation of steel reinforcement).

The following subjects of inspection are typical of concrete construction but not exclusive of other that may be required by the specific characteristics of the project:

Preplacement inspection:

- Scaffolding, formwork and falsework
- Quality and positioning of the steel
- Prestressing components and operations
- Concrete mixture proportions and supporting test data
- Positioning and zero reading of sensors and reference points

Placement inspection:

- Concrete quality through checks of delivery tickets and sampling and testing for conformity (for the case of performance specifications see section 12.4)
- Concrete processing operations (conveying, placement, consolidation, finishing, treatments, initial protection)
- Precast erection

Postplacement inspection:

- Moist curing and temperature control
- In-place strength before removal of shoring or loads application
- Post-tensioning operations
- Eventual damages and remedial treatments
- In-place concrete construction quality by non-destructive and/or cores' testing (e.g. thickness and permeability of the cover zone, as per section 2.3.6 of *fib* Bulletin 93 [10-6])

regarding concrete, its processing on site must ensure that the intrinsic potential of the material is achieved during the construction phase.

For mechanical performance, the cross-section dimensions, quality (especially strength and stiffness) and uniformity of the concrete, as well as quantity, quality and positioning of the reinforcement and prestressing steel are key factors. Regarding durability, besides avoiding deleterious internal reactions, the focus of the QM activities shall be on achieving a sufficiently thick and tight cover concrete.

10.2.1.2 QM Activities

The Quality Management (QM) activities in the Execution phase of Construction works cover three stages:

- Preplacement inspection
- Placement inspection
- Postplacement inspection

The result of these activities shall be compiled in an Inspection Report, which is mandatory for QL2 and QL3.

Correct positioning and firm fixation of the reinforcing and prestressing steel, as well as other reinforcing materials, shall precede concreting. Adequate and sufficiently closely located spacers are essential to achieve cover thicknesses conforming to the specification and tolerances, as well as to ensure sufficient room to allow the free flow of concrete between the bars and in the cover zone. A performing mix design, providing adequate workability of the concrete to achieve adequate compaction, especially between the reinforcing bars and the formwork surface, is key to achieve a tight cover concrete, to be complemented by proper curing conditions of the just placed concrete. Control and monitoring of these aspects shall be carried out in accordance with the Execution Specifications and in full conformity with the approved procedures of the Project Quality Plan.

This section refers to the documentation to be supplied to the owner and/or operator of the structure at handover/commissioning. It shall provide detailed information enabling the engineers (sometimes several generations of them) in charge of monitoring and maintaining the structure to make decisions on what, where and when monitoring and maintenance activities shall be conducted.

A comprehensive description of a Birth Certificate, including examples, can be found in "Birth Certificate and Through-Life Management Documentation", *fib* Bulletin 93 [10-6].

The birth certificate should facilitate ongoing (through-life) evaluation of the original service life which is likely to be achieved by the structure and/or individual components and shall contain:

- Structure Identification
- Durability Specifications and Standards
- Environmental Exposure Conditions
- Deterioration Mechanism and Models
- Construction Materials Type and Quality
- Durability Test Methods
- Structure Component Durability Data
- In-Service Conservation Plan (see also section 10.4)
- Dismantling Plan (see also section 10.5)

10.2.1.3 Documentation at Handover/Commissioning

The objective is to provide the owner/operator with an as-detailed-as-possible documentation to facilitate monitoring and maintenance of the structure during its whole operational service life.

The supplied documentation shall contain:

- Design File (see section 10.1.2) incl.:
  - Basis for design
  - Execution specification
  - Concrete specifications
- Execution File (see sections 10.2.1 and 10.2.2) incl.:
  - Birth certificate (see section 10.2.1.4) and/or Re-birth certificate (see section 10.2.2.3)
  - As-built geometry
  - Inspection report(s)
- Maintenance and Condition assessment plan (see section 10.3 and chapter 38)
- Dismantlement and reuse plan (see section 10.4 and chapter 39)

10.2.1.4 Birth Certificate

The Birth Certificate is a document, report or technical file (depending on the size and complexity of the structure concerned) containing engineering information formally defining the form and the condition of the structure on completion of construction.

The birth certificate document (BCD) should provide and quantify specific details on parameters important to the durability and service life of the structure concerned (e.g. cover to reinforcement, concrete permeability, environmental exposure conditions, quality of workmanship achieved, etc.), and the basis upon which future knowledge of through-life performance should be recorded. These parameters should be categorized as either assumed, estimated, observed/measured, or inferred/deduced.

The framework laid down in the birth certificate should provide a means of comparing actual in-service behavior / performance with that anticipated at the time of design and/or the time of construction of the structure.

The BCD might serve as the basis for monitoring the condition of the structure and for planning conservation activities during its service life. Recommendations on

conservation procedures, which depend on the specifics of the project, are given in chapter 38.

10.2.2 Q&IM during execution of interventions

10.2.2.1 Objective

Under certain circumstances, preventive or remedial interventions are required with the intention of keeping or restoring the performance of a new or existing structure to an acceptable level. The objective of Quality Management (QM) with respect to the execution of interventions is to ensure that the principles and methods applied result in a performing structure during the specified or expected design or residual life.

10.2.2.2 QM Activities

The QM activities in the execution of an intervention cover three instances:

- Preparation
- Execution
- Inspection

As a possible outcome of Life-Cycle Management activities (see section 34.5), the decision of conducting a preventative or remedial intervention on the structure may be adopted. That decision may be a consequence of the detection of defects during the QM during Construction Execution of new structures (e.g. non-conforming concrete quality, honeycombing, lack of thickness or tightness of the concrete cover) or of the detection, during the QM during Operation and Conservation of existing structures, of signs of distress (e.g. excessive deflections or vibrations, stains, cracks) or of insufficient performance (e.g. too deep carbonation or chloride penetration in relation to cover thickness).

The type and purpose of the intervention are described in section 34.6, in general agreement with ISO 16311-4 [10-7].

The execution of interventions will be conducted in accordance with an Execution Plan, as described in chapter 33, which should include a quality management plan. Generally, interventions, when properly conducted, improve the current performance of the structure.

Often, interventions consist in complementing, strengthening or replacing the existing concrete with sealing compounds, special products or new concrete; sometimes the steel reinforcement is the focus of the intervention. Almost invariably, interventions require a thorough preparation of the treated surfaces to remove loose or low-quality material, improve bond, etc. so as to generate a firm stratus for the application of different materials and products. This is a crucial step that needs special QM attention for the success of the intervention. The preparation phase includes also the selection of the materials or products to be applied, the quality of which shall be certified by technical documentation and/or adequate testing.

During execution, key QM activities are: verification that the ambient conditions are suitable for the type of products used and that the modality and rate of application

meet the specifications (e.g. lts/m²) and that full coverage of the affected area is achieved. Protection/curing of the freshly placed products is another matter of attention.

Once the products have achieved the service maturity, inspection shall focus on the performance of the intervention, measured through site tests, both destructive and non-destructive (bond, strength, hardness, permeability, etc.).

10.2.2.3 Re-Birth Certificate

The goal of Birth and Re-Birth Certificates is the same – recording the information related to structure durability in order to assist in the prediction of the time that a structure is able to remain in service and define what ongoing condition assessment and maintenance is required.

Birth and Re-Birth Certificates are organized in the same manner, see *fib* Bulletin 93 [10-6].

The Re-Birth Certificate is a document, report or technical file (depending on the size and complexity of the structure concerned) containing engineering information defining the condition of the structure after completion of a major preventative or remedial intervention.

It should contain a post-intervention conservation plan, made according to the estimated long-term performance of the applied intervention method.

10.3 Q&IM during Operation and Conservation

10.3.1 Objectives

A proper inspection regime during the service life of a structure, as well as the documentation of inspection results, will give the owner or responsible party (or who is established to ensure the structure performance requirements) the possibility to perform condition control during the service life and to apply protective measures when service life design expectations are not met.

The suggestions provided in the present chapter are applicable to the majority of structures. Complementarily, the reader may refer to other detailed documents for specific information concerning the analyzed structure type.

The objective of Quality Management (QM) in conservation is to control and manage the activities and measures taken, which seek to ensure that the structure condition remains within satisfactory limits in order to meet the performance requirements for a defined period of time; this applies to structural and functional performance requirements, which may include considerations of aspects such as reliability, availability, user's safety, economy and environmental impact.

This is achieved through activities that may involve condition survey, monitoring the structure performance through its service life, condition assessment, condition evaluation, decision-making and the execution of any necessary intervention; the corresponding conservation activities and measures undertaken shall be properly documented.

10.3.2 Service life file

The service life file shall document all conservation activities carried out during the operational life of a structure. This file shall also include results from inspections performed on the structure or its components, during its service life. The service life file shall include:

- Structure classification and conservation strategy;
- Reference to relevant drawings, and details of immediate and surrounding environment;
- Details concerning inspection and evaluation procedures, including inspection and monitoring results, data on observed deterioration, estimated deterioration rate and condition rating of the structure;
- Details of the plan and actual design and execution of preventative and corrective interventions carried out.

An extract of the service life file, called the re-birth certificate document (RCD), includes results of in-service inspection of an existing structure after preventative or corrective interventions have been undertaken – see Figure 8.3-2. The content of the RCD usually corresponds to the information included in the birth-certificate document (BCD).

The in-service inspection level of detail must be stipulated in accordance with the quality levels, defined in chapter 6.1, and with national standards. The information collected over the structure life-cycle should, however, be as relevant as possible to enable standardization to be achieved.

The standardization of information facilitates the estimation of functional performance requirements for the definition of life-cycle management (LCM) activities, quality control plans (QCP), and physical intervention strategies for the implementation of efficient performance-based management approaches.

The medium or long-term actions can be defined based on the development of different scenarios, allowing the choice between different intervention strategies – see

For new structures, recording during conservation would be expected to draw upon information obtained for and detailed in the birth-certificate document (BCD) containing the as-built drawings. For existing structures, there is the expectation that recording during conservation would draw upon/contribute to the preparation of a re-birth certificate document (RCD), depending on whether a previous version had been prepared and was to be updated.

The RCD would provide a record of at least the following:

- Verification of the structure condition after an intervention (preventative or corrective) has been made, and a record of the standard of execution/variability achieved in that process and previously;
- Updated (in-service) benchmark for reference on service life design matters;

Updated data as required for revision of verification of the limit states, and in particular, limit states associated to durability.

The data gathered would also allow

- A review of service life predictions, based on updated (in-service) measured data and a revised prognosis on future performance;
- Assessment of compliance/non-compliance to design requirements and planning for any future preventative and corrective interventions required.

An adequate categorization of information facilitates the integration of building information modeling (BIM) with structure management systems. The detailed description of structural components enables a more reliable estimate of the performance of the structure, being inputs for the design and assessment processes of similar structures types.

The integration of BIM throughout the life-cycle of the structure, including operation and conservation, complemented with condition survey and monitoring activities, provides a wider understanding, not only of the structure as a whole, but also of its components. This understanding enables better quality and information

management and thus to better decision-making on the definition of QCP and maintenance activities.

There may be a range of additional activities associated with the dismantlement/demolition works, such as those involved in the cleaning up and/or treatment of the site to decontaminate it and/or make it suitable for future use or redevelopment.

For dismantlement, a plan should be made covering at least the following:

- the adequate stability and structural integrity and personnel safety in all stages of dismantlement;
- minimization of societal hindrance by dust, dirt and noise;
- minimization of contamination of soil, respecting at least the local regulations;
- conditioning and removal of wastes in such a way that the principles of sustainability, as formulated in section 3.4, are satisfied;
- recycling the appropriate parts of the dismantled material;
- cleaning the site and reintegration in the environment after dismantlement.

The document needs to be adjusted to the type of concrete structure to be dismantled, and should address: site and structure description, survey data (presence of hazardous substances in the structure and assess the condition of individual components, using the classification described in section 35.2.2); technical data; time constraints; selective dismantling/demolition planning and phases (for a maximization

section 34.6. The pre-determination of needs, costs and the best timing of these interventions allows for an efficient planning of activities, as well as a long-term cost reduction.

The service life file must be preserved as long as a structure remains in service. It may also be desirable to keep such records for an indefinite period for reference purposes, namely for the design, construction and conservation of other similar structures. Such records must be kept in a format which can easily be understood.

10.4 Q&IM during Dismantlement and Circular Use

10.4.1 Objectives

The objective of QM in dismantlement is to control and manage the activities and measures taken to allow the safe removal of an existing structure and the clearance of the site as appropriate through:

- preparing dismantlement;
- dismantling the structure into its components;
- demolishing the structure by physically breaking it up;
- or a combination of such measures, facilitating the reuse and/or recycling of the original components parts and materials for new uses in a manner that minimizes the associated environmental and social impacts;
- waste management.

10.4.2 Dismantlement document

The dismantlement document sets down the activities, measures and procedures which will allow the safe removal of an existing structure and the clearance of the site in a manner that minimizes the associated environmental and social impact.

of the materials/wastes to be re-use); equipment's removal planning, safety equipment planning; labor safety; emergency action plan; pollution control (noise, dust and erosion control); cleaning planning; and environmental protection. next to a dismantlement document the quality of the elements for reuse should also be determined.

From a circular economy point of view, the re-use of all components of the entire structure is the best option. However, factors such as space, integrity, aesthetics, refurbishment costs and client satisfaction need to consider the potential of structures to be re-used or recycled (for further details refer to chapter 9). This could be established by a material passport, which states all properties of a structure, per element enabling assessment of the potential for reuse or recycling. A re-use, and recycling plan (on-site or regional re-use or regional disposal); transportation planning for regional re-use or/and disposal should form part of the re-use and recycling documents.

The waste management document should address the waste volume produced during the dismantling, the waste identification (nature, use, hazardous or non-hazardous) and quality, the disposal plan (on-site or regional re-use or regional disposal) and the transportation planning for regional re-use or/and disposal.

10.4.3 Re-use and recycling documents

The re-use and recycling document sets down a system for documenting essential information about materials and elements used in structures and intende for re-use and recycling, including details such as material type, composition, dimensions, manufacturing date, and any relevant certifications. It should therefore include criteria for assessing the quality of materials and elements for re-use and recycling, consider factors such as durability, structural integrity, contamination levels, and compatibility with recycling processes, and include or refer to the acceptable limits or thresholds for each criterion based on industry standards and best practice.

10.4.4 Waste management document

The waste management document sets down the activities, measures and procedures which will allow to re-use some wastes generated during the dismantling phase in a manner that minimizes the environmental and economic impact and increases the circular economy of the concrete sector.

References

[10-1] CEB Bulletin 191. General Principles on Reliability for Structures.

[10-2] ISO 22966:2009. Execution of concrete structures

[10-3] EN 13670:2009. Execution of concrete structures.

[10-4] EN 13369:2018. Common rules for precast concrete products.

[10-5] *fib* Bulletin 53. Structural Concrete Textbook, second edition, volume 3. 2009.

[10-6] *fib* Bulletin 93

[10-7] ISO 16311-4:2014. Maintenance and repair of concrete structures – Part 4: Execution of repairs and prevention.

The limit states refer to the entire structure, to structural members or to local regions of the members.

In engineering practice, most of the limit states refer to simplified models describing the exposure and the structural response. However, a limit state may also account for several actual limit states simultaneously.

The durability criteria are implicitly involved in the requirement that structures are designed for structural safety and serviceability for a design service life or assessed for a remaining service life (section 11.2.1).

11 Structural performance evaluation framework

11.1 Principles of limit state approach

11.1.1 General

In this Model Code, the concept of limit state verification is applied to evaluation of structural performance during design of new or assessment of existing concrete structures.

In the context of the limit state verification, the structural performance of the whole structure or part of it should be described with reference to a specified set of limit states, which separate desired states of the structure from adverse states.

Limit states are states beyond which the performance requirements are no longer satisfied.

Conceptually, limit states correspond to a discrete representation of the structural response under specified exposure to which specific losses and/or damages can be associated.

Limit states are related to design or assessment situations. They may relate to persistent situations during the service life of the works, transient situations during the execution of the construction works (stage of construction and/or assembling or repair), extreme actions and environmental influences, unintended use or accidents.

In the context of the limit state verification, performance criteria for serviceability and structural safety are specified by:

- serviceability limit states criteria (section 11.1.2);
- ultimate limit states criteria (section 11.1.3).

In some cases, limits between the serviceability limit states and the ultimate limit states may be defined based on the condition of a structure. In such case performance criteria for serviceability and structural safety can be specified by condition limit states, distinguishing:

- condition limit states often associated with durability (section 11.1.4);

– partial damage limit states (section 11.1.5).

In the context of the limit state verification, special consideration shall be given to :

– robustness (general principles are given section 11.1.6.1 and procedures for achieving robustness are given in section 30.9);
– damage tolerance (section 11.1.6.2);
– temporary stages during constructions, interventions and dismantlement (section 11.1.6.3)

11.1.2 Serviceability limit states (SLS)

Serviceability limit states correspond to the states beyond which specified demands for a structure or a structural component related to its normal use or function are no longer met.

In the cases of irreversible local damage or irreversible unacceptable deformations, the exceedance of a serviceability limit state causes inadequate serviceability of the structure, that is, failure. Some repair may be necessary for the structure to be fit-for-use.

In other cases (such as temporary local damage by, for instance, wide cracks, temporary large deformations or vibrations) the exceedance of a serviceability limit state may be reversible. In those cases, failure occurs:

– the first time that the serviceability limit state is exceeded, if exceedance is considered unacceptable;
– if exceedance is acceptable but the time during which the structure is in the undesired state is longer than specified;
– if exceedance is acceptable but the number of times that the serviceability limit state is exceeded is larger than specified.

Frequently exceeding the serviceability limit states may affect the efficient use of a structure, its components (tanks, pipes, canals) or its appearance. In many cases, the risk of damage is indirectly excluded by ultimate limit state verifications or by detailing.

The serviceability limit states address fitness-for-use of a structure.

The corresponding serviceability limit state criteria are related to:

– functionality of the structure related to its normal use;

The serviceability limit state criteria may refer to, for example:

– unacceptable deformations or deflections which impair the functionality of the construction works or their contents, cause damage to non-structural

components, cause discomfort to people, affect the appearance of structural or non-structural components or the functioning of equipment (the conditions to be fulfilled with regard to limiting the deformation are associated with the type of building or the civil engineering structure, and are often, for the sake of simplification, substituted by rough approximations);

– excessive vibrations which limit the functional effectiveness of the construction works, affect non-structural components, impair the comfort of users or the functioning of equipment. (Although such limit states may be characterized by the magnitude of the vibrations, they are commonly indirectly covered by limiting the fundamental period of vibrations of the structure or some of its structural components, in comparison to the expected period of the excitation vibrations);
– local damage (e.g. cracking, slip in connections) which does not affect structural safety but may affect the efficiency or appearance of structural or non-structural components;
– local or global degradation due to environmental actions (e.g. depassivation of reinforcement, weathering) which may affect the efficiency or appearance of structural or non-structural components;
– lack of tightness or defective sealing that restrict the functionality or impair the user's comfort.

The effect of exceeding an ultimate limit state is almost always irreversible and causes failure the first time it occurs. The ultimate limit state can be the result of a single extreme action event or of a degradation process over time followed by a (less) extreme action event.

– the comfort of users of the structure;
– the appearance of the construction works.

The limit values that define serviceability limit state criteria differ depending on consequences associated with exceeding the limit state and possible repair or upgrade cost, structural malfunction cost etc.

Design principles regarding the formulation of performance criteria for the analysis of the serviceability limit states are given in chapter 12.

The procedures for the verification of the serviceability limit states are given in section 30.5 (for RC and PC structures) and in section 30.7 (for FRC structures).

11.1.3 Ultimate limit states (ULS)

Ultimate limit states are limit states associated with the various modes of structural failure or stages close to structural failure which, for practical purposes, are also considered as ultimate limit states. Ultimate limit states can refer to structural elements, as well as to structural systems; the latter including the effects

The ultimate limit states which may require consideration include (non-exhaustive):

– attainment of the maximum resistance of structures, structural members and sections (regions), for example resulting from:
  - material failure, excessive deformations or settlement;
  - loss of capacity caused by fire;
  - loss of capacity caused by degradation of structural components due to environmental actions (e.g. corrosion of reinforcement, corrosion induced cracking and spalling, alkali silica reaction);
  - impact or explosion;
  - earthquake;
  - permanent deformations exceeding a certain limit after an earthquake;
– rupture of structural members caused by fatigue under essentially repetitive loading or other time-dependent effects;
– loss of stability of the structure or any part of it, including supports and foundations, for example:
  - sudden change of the assumed structural system to a new system (e.g. transformation into a kinematic mechanism, snap through or large crack formation);
  - buckling of slender structures or structural members, in which second order effects play a role;
  - loss of equilibrium of the structure or of a part of the structure, considered as a rigid body (e.g. sliding or overturning) including that caused by impact, explosion or an earthquake;
  - loss of equilibrium caused by impact or explosion;
  - sliding beyond a certain limit or overturning due to an earthquake;
– failure of the ground by hydraulic heave, internal erosion, or piping.

of system behaviour and robustness; see section 30.9. Consequences related to exceedance of an ultimate limit state are classified in section 11.3.1 .

The corresponding ultimate limit states criteria are related to:

– resistance of critical regions;
– fatigue;
– stability;
– loss of equilibrium of the structure or any part of it, considered as a rigid body;
– soil failure.

The limit values that define the ultimate limit state criteria vary depending on failure consequences.

Design principles regarding the formulation of performance criteria for ultimate limit state analysis are given in chapter 12.

The procedures for verification of the ultimate limit states are given in section 30.1 (for predominantly static loading of RC and PC structures), section 30.2 (for non-static loading of RC and PC structures,) and in section 30.7 (for FRC structures).

11.1.4 Condition limit states (CLS)

CLS is commonly a well-defined and controllable limit state.

In principle, the requirement for sufficient durability is already covered by the fact that safety and serviceability are required for a certain period of time. However, for practical reasons it might be helpful to add specific durability related limit states (as in ISO 16204 [11-1], ISO 13823 [11-2] and ISO 13822 [11-3]) or limit states referring to certain (noncritical) conditions.

Material properties, such as strength, stiffness, permeability, diffusivity or bond strength between concrete and reinforcement can be degraded in such a way that the material property assumed in design should not be considered anymore.

Normally it is assumed that a sudden loss occurs immediately when the limit state is exceeded. If however the losses occur gradually, a solution might be to introduce a subdivision of the undesired outcome space (a sequence of conditional limit states) corresponding to several loss levels.

The condition limit states may correspond to the following situations:

- Limit state of damage/deterioration beyond which the assumption made to evaluation of structural performances, such as structural safety and serviceability, no longer apply.
- An approximation to the real limit state that is either difficult to define or difficult to calculate. Examples are the use of the elastic limit as ultimate limit state and the use of depassivation as a limit state for durability (often also referred to as the initiation limit state).
- Local damage (including cracking) which can reduce the durability of the structure or affect the efficiency or appearance of structural or non-structural elements.
- Additional limit state thresholds in the case of a continuously increasing loss function due to deterioration processes.

The reliability requirement for the condition limit state should be consistent with the ultimate or serviceability limit state under consideration.

11.1.5 Partial damage limit states (PDLS)

In particular cases a limit between the serviceability limit states and the ultimate limit states may be defined, a so-called “partial damage limit state”.

The “partial damage limit state” can be for example identified in the case of earthquake damage of structures, be associated with the maintenance and repair criteria or with the safe shutdown of a structure, infrastructure or facility. For more details, see the JCSS Probabilistic Model Code, Part 1 [11-4].

These engineering analyses must be accurate enough to provide realistic results but not too sophisticated to be beyond the best engineering practice.

The main assumption of this approach relies on the principle that possible partial damage limit states can be identified via engineering analyses.

A structure satisfies the minor damage limit state when the damage is limited and no repair works are deemed necessary. This limit state can be used e.g. to ensure an early warning and provide sufficient time to plan maintenance and repairs.

Generally, a structure satisfies the operational limit state criteria if all the following conditions are met: the facility has suffered practically no damage and can continue serving its original intention with little disruption of use for repairs, supported either by undamaged lifelines or by back-up systems, and any repair that is necessary can be deferred to some future time without disruption of normal use.

Generally, a structure satisfies the immediate use limit state criteria if all of the following conditions apply:

- the structure itself is very lightly damaged (i.e. localized yielding of reinforcement, cracking or local spalling of concrete, without residual drifts or other permanent structural deformations);
- the normal use of the facility is temporarily but safely interrupted (in the case of an industrial plant, after a safe shutdown) and can be restored as soon as utility systems are back in operation;
- risk to life (referring to persons at risk due to a structural failure, including also persons who may be outside the construction work but affected by structural failure; see section 11.3.1) is negligible;
- the structure fully retains its necessary strength and stiffness and its ability to withstand loading;
- the (minor) damage of non-structural components and systems can be easily and economically repaired at a later stage.

Generally, a life-safety limit state is reached if any of the following conditions are met (but not exceeded):

- the structure is significantly damaged, but does not collapse, not even partly, retaining its integrity;
- the structure does not provide sufficient safety for normal use, although it is safe enough for temporary use;
- secondary or non-structural components are seriously damaged, but do not obstruct emergency use or cause life-threatening injuries by falling down;

Accordingly, the partial damage limit states that could be considered can be described as:

- minor damage limit state;
- maintenance, repair and operational limit states;
- immediate use limit states;
- life-safety limit states;

– the structure is on the verge of losing capacity, although it retains sufficient loadbearing capacity and sufficient residual strength and stiffness to protect life for the period until repair is completed;

repair is economically questionable and demolition may be preferable.

Generally, a structure has reached the near-collapse limit state if any of the following conditions are met:

– the structure is heavily damaged and is at the verge of collapse;
– most non-structural components (e.g. partition walls in buildings) have collapsed;
– although life safety is mostly ensured during the loading event, it is not fully guaranteed because there may be life-threatening injury situations due to falling debris;
– the structure is unsafe even for emergency and would probably not survive additional loading;
– the structure presents low residual strength and stiffness but is still able to support the quasi-permanent loads.

By virtue of its robustness, the structural system should be able to continue to fulfil the function for which it was created, modified or preserved, without being damaged to an extent disproportional to the cause of the damage.

The limit states which may require consideration are related to:

– near-collapse limit states.

11.1.6 **Special considerations**

11.1.6.1 Robustness

Robustness is important for maintaining the ability of the structural system to fulfil its function during events such as accidental loading or due to consequences of human error.

Robustness of the structural system addresses:

– life safety;
– property and environment protection;
– protection of operations.

Accordingly, the robustness criteria are related to:

– disproportional failure of a large part of the structure or the whole structure caused by an accidental load or failure of a structural component (e.g. due to explosion, loads by extremely high water table, flooding, loads due to extreme circumstances such as fire, impact, explosion or earthquake), resulting in:
  – system collapse;
  – life-threatening component collapse.
  – resistance of the structural system;
  – special functions (e.g. shelter from climatic phenomena, containment of substances, providing fortification, security, shade etc.).

The general principles and the procedures for the verification of robustness are given in section 30.9.

11.1.6.2 Damage tolerance

[Not used at this time, to be updated in future edition]

11.1.6.3 Temporary stages during constructions, interventions and dismantlement

[Not used at this time, to be updated in future edition]

11.2 Service life

11.2.1 Design service life and remaining service life

Service life is the period during which the structure has to satisfy the agreed performance criteria. Service life has to be specified.

Design service life refers to new structures and remaining service life to existing ones.

Depending on structure-specific conditions, the remaining service life of an existing structure may be shorter than the design service life intended for a structure in the original structural design. If needed, the service life can be extended by construction (repair or upgrade) and/ or operational (e.g. monitoring, change of use or limits on imposed or traffic loads) interventions (CEN/TS 17440 [11-5]). In many practical cases, it is sufficient to check the first and last years of the required service life.

The design service life or the remaining service life follow from the required service life as given by the stakeholders and from other implications, for example specified on the basis of structural analysis, maintenance and quality management.

A differentiation between replaceable and non-replaceable components of the structure may be considered when choosing the design or remaining service life for the structure and its components.

Limit states associated with the time-dependent material degradation are, for example, initiation of reinforcement corrosion, cover concrete cracking or spalling due to corrosion.

Due consideration is needed to decide whether limit states related to a change of performance due to material degradation should be regarded as:

- serviceability limit states (which may be a failure to achieve some performance, such as good appearance of the structure) and/ or;
- ultimate limit states (which may be a failure such as falling of spalled concrete that may diminish the resistance or be harmful to people around the structure).

For practical reasons it might be helpful to add specific durability related limit states or limit states referring to specified conditions – see section 11.1.4.

The required service life should be given by the owner in consideration of the interests of other stakeholders (i.e. users, contractors, society).

The nominal/formal end of the service life is reached when the performance criteria are no longer met for a required reliability level.

11.2.2 Verification of service life

The performance verification must be conducted with proper consideration of the change of performance over time, for instance due to degradation or time-dependent effects. Effects of creep and shrinkage of concrete on the structural performance over time must be evaluated according to the guidelines of section 29.2.3.

Service life verification demonstrates that during the service life degradation does not result in violation of the performance criteria. In the case of deteriorating structures, reliability assessment should be conducted in each year of the remaining service life or taking the year with the most unfavourable conditions, taking into account degradation and defects.

Principles of reliability verification and the procedures for service life design are given in chapter 12 and section 30.6, respectively.

Accordingly, the service life verification is performed according to the procedures described in section 30.1 (verification of structural safety for predominantly static loading of RC and PC structures), section 30.2 (verification of structural safety for non-static loading), section 30.5 (verification of the serviceability for RC and PC structures), section 30.6 (evaluation of condition limit states (CLS) associated with durability), section 30.7 (verification of safety and serviceability for FRC structures) and section 30.8 (verification of safety and serviceability for structures with non-metallic reinforcement).

11.3 Principles of reliability-based approach

11.3.1 Consequence Classes

11.3.1.1 General

For the purpose of reliability differentiation, consequence classes (CC) should be established by considering all the potential consequences of a structural failure depending on the type of a structure, its use, and structural robustness. Failure consequences may include human and economic losses, environmental impacts, social and political consequences, loss of cultural heritage value etc. Direct and follow-up (e.g. in the transportation network or in the society) consequences should be considered. A higher CC should be considered for structures with high levels of occupancy as well as for structures with essential function for the communities (hospitals, fire stations, important network bridges, escape routes in buildings with high occupancy, etc.; see also section 11.3.1.2.2). Failure consequences should be estimated considering all relevant scenarios (with relevant likelihoods) that may occur within a service life of the structure.

Depending on the type and properties of the structural system, a particular structural member may be permitted to be classified independently in a higher or lower CC than assigned to the entire structure; this could be particularly useful in the assessment of existing structures or in special situations such as hand rails close to large differences in levels where many people can gather, large foundation pits whose failures may result in a large number of fatalities during construction phases, etc..

11.3.1.2 Recommendations for buildings and bridges

This Model Code recommends applying the classification according to Table 11.3-1 (based on prEN 1990 [11-6]) unless a more detailed approach is adopted; see for instance Table 11.3-2 and Table 11.3-3.

Foundations and geotechnical structures that support a building or bridge should be classified with the same or higher consequence class as the building or the bridge.

Persons (lives) at risk due to a structural failure include persons who may be outside the structure but affected in case of failure of the structure. When considering the population at risk, attention should also be given to long-term risks and not only to those immediately created during a structural failure.

Table 11.3-2 for buildings and Table 11.3-3 for bridges are based on a critical review of categorisation of structures (Diamantidis & Sykora [11-7]). Although the classes reflect potential consequences of failure, they are defined at the exposure level similar to ASCE 7-16 [11-8] aiming to facilitate the use in practice. They are related to the whole structure and reflect the system level.

Table 11.3-1: *Qualitative definition of consequence classes and examples for buildings and bridges.*

CC	Qualitative description	Examples
CC1	Small number of persons at risk, and negligible economic or environmental losses.	Buildings where people do not normally enter (e.g. storage buildings, greenhouses). Low rise buildings with only few people. Bridges - short-span structures such as culverts, short river crossings.
CC2	All buildings not classified as CC1 or CC3	Residential and office buildings. Most buildings up to 15 stories. Bridges not in other consequence classes.
CC3	Large number of persons at risk, or substantial economic or environmental losses.	Buildings where people may congregate (e.g. grandstands, concert halls), buildings designated as essential facilities (e.g. hospitals, schools) or buildings whose failure could pose a substantial hazard to a community (e.g. facilities for manufacture or storage of hazardous substances). High-rise buildings. Railway bridges, bridges over railways. Bridges over or under major roads.

Table 11.3-2: *Indicative quantitative criteria for consequence classes for buildings based on a number of persons at risk.*

CC	Number of persons at risk N	Description
CC1	$N < 10$	single occupancy houses, agricultural structures
CC2	$10 \leq N < 1000$	all other buildings/structures
CC3	$N \geq 1000$	structures with $N \geq 1000$ (permanent use), grandstands with more than 5000 persons, structures utilised in important industries, resilience critical structures serving communities with more than 5000 persons

For buildings, an independent reliability differentiation of structural members can be performed based on the potential consequences to persons associated with a particular member collapse scenario, which will depend on (Hingorani et al. [11-9]):

- the damage extensions of the collapse scenario, quantified through the affected building net room area, A_{col} – see section 11.3.2.2.2, and
- the occupancy ratio of this area, Ocu_{col}/A_{col} (number of persons at risk Ocu_{col} per unity of area A_{col}).

The occupancy ratio Ocu_{col}/A_{col} (hereafter in persons/m^2) related to a specific member collapse scenario, may also be used to determine a CC – an example of such classification is as follows:

– CC1: small number of persons at risk N within the collapsed area, $Ocu_{col}/A_{col} \leq 0.01$ (typical examples are areas in buildings where people do not normally enter such as agricultural facilities);

– CC2: medium N within A_{col}, $0.01 < Ocu_{col}/A_{col} \leq 0.1$ (examples – residential and office areas, excluding congregation areas);

– CC3: large N within A_{col}, $Ocu_{col}/A_{col} > 0.1$ (examples – congregation areas including lobbies, entrance halls, waiting rooms, convention centres, assembly halls, sport and swimming facilities, grandstands, cinemas, theatres, museums, auditoriums, class-rooms and similar).

When a collapsed area can be associated with different occupancy ratios, a conservative value or weighted average may be adopted to establish a CC. The occupancy ratio Ocu_{col}/A_{col} may depend on the nature of the considered failure mode. Although ductile failure might give individuals an early enough warning to evacuate the building in time, such a scenario may be conservatively ruled out when estimating occupancy ratios unless the feasibility of escape is analysed in detail.

Table 11.3-3 indicates quantitative criteria for consequence classes for bridges.

Table 11.3-3: *Indicative quantitative criteria for consequence classes for bridges.*

CC	Description
CC1	- bridges for pedestrian or cycle use only with traffic less than 50 persons per hour where people are unlikely to gather - road bridges in rural areas with AADT≤ 5000* vehicles per day **or** with a span length L < 10 m - temporary bridges whose failure will lead to no fatalities **and** small economic and environmental impact
CC2	- all other bridges
CC3	- important pedestrian bridges, e.g. with long span(s) and/or where people are likely to gather - road bridges with either AADT ≥50,000* vehicles per day **or** no possibility of detour (detour length greater than 25 km in rural areas and greater than 10 km in urban areas) **or** span length greater than 100 m - railway bridges on transnational, national and main regional lines and special railway bridges (metro and tram bridges, bridges in important industrial plants) - bridges essential for emergency situations

*Annual average daily traffic; to be defined based on national traffic characteristics.

With respect to environmental losses, Consequence Classes can analogously be defined on the basis of the extent of damage expressed through e.g. volume of contaminated soil, surface or underground water, or air pollution (see for instance StFV Verordnung über den Schutz vor Störfällen (Störfallverordnung) [11-10]) or through the recovery time.

In particular for Ultimate Limit States, the target reliability levels, related acceptable probabilities of failure and possibly acceptable levels of risk are determined by the relevant national authorities. The criteria related to serviceability and durability can be specified for a project and agreed with the client; these criteria then have to comply with the target levels required by the relevant authorities. In these regards all the values provided in this section are indicative.

ISO 2394 [11-11] defines a target reliability as an average acceptable reliability level that is to be reached as close as possible. Failure consequences may include human and economic losses, environmental impacts, social and political consequences, loss of cultural heritage value, etc. Further considerations for the choice of the level of reliability are provided in JCSS PMC [11-4], ISO 2394 [11-11], and *fib* Bulletin 80 [11-12].

The costs involved when upgrading the performance of existing structures (e.g. increasing their safety) are usually high compared to the costs of improving the same performance by a structural design in the case of a new structure. Upgrading existing structures may entail relocation of occupants and disruption of activities or influencing heritage values, which does not play a role in the design of most new structures.

The relationship between P_f and β-values is illustrated in Table 11.3-4.

Table 11.3-4: *β-values related to the failure probability P_f.*

P_f	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-6}
β	1.28	2.32	3.09	3.72	4.75

Reliability management has to be supported by suitable databases of different types of structures and their performance over time, taking into account various degradation processes. Therefore, data have to be collected in order to quantify risk, and hence decide on the target reliability values.

In many reinforced concrete structures there is more than one dominant failure mode for example due to more than one critical member. It may be appropriate to

11.3.2 Target reliability levels

11.3.2.1 General

In accordance with ISO 2394 [11-11], the choice of the target level of reliability should take into account all possible consequences of failure. The choice of the target level of reliability also takes into account the amount of expense and effort required to reduce the risk of failure. The target reliability levels may be based on explicit risk analysis, economic optimisation, and/or on calibration with regard to current best practice.

Due to large differences in the outcome of such considerations, due attention should be given to differentiating the required reliability level of structures to be designed and that of existing structures.

Reliability requirements for structures to be designed and for existing structures may adequately be expressed in terms of the reliability index β:

$$\beta = -\Phi^{-1}\left(P_f\right) \tag{11.3-1}$$

where $\Phi^{-1}(\cdot)$ is the inverse of standard normal cumulative probability distribution function and P_f is the failure probability corresponding to a specified reference period.

In order to select appropriate target β_t-values, the reference period has to be specified and consequences of failure and cost of safety measures have to be analysed for the specific case under consideration.

The target reliability indices given in section 11.3.2.2 and section 11.3.2.3 for ultimate limit states relate to the dominant failure mode at member or local level in

require an increase of 0.5 per structural member in terms of reliability index for structures with multiple, equally important failure modes (e.g. for a structure with many similar members). Such an increase may correspond to the requirement of using a one-class higher consequence class.

Experienced reliabilities are often higher than the target values. This may be a result of hidden safeties in resistance and load effects (such as neglecting the contribution of longitudinal reinforcement to shear resistance or conservative simplifications in zoning in climatic action maps), ULS overdesign due to dominant SLS criteria, additional reliability due to constructional arrangements etc. These aspects can be taken into account for the assessment of existing structures on the basis of a careful analysis.

Design or remaining service life and a reference period for the target value can be selected independently. The target β_t -value and the extreme value distribution used to represent the time-variable actions (or the leading action in a load combination) are based on the same reference period. When a reliability verification is based on an annual reference period, the annual target reliability index is considered along with actual resistance and action effects estimated for the year of verification; appropriate annual extremes of the leading time-variable action should be thereby taken into account. In this way, each year of a service life of the structure needs to be verified. A target reliability $\beta_{t,fref}$ related to a given reference period may be recalculated to a different reference period using the methodology in (Holicky et al. [11-13]), considering dependency between annual failure events.

The semi-probabilistic format for ultimate limit states in this Model Code is calibrated to the target reliability levels in Table 11.3-5 and Table 11.3-6; see section 12.5.4.

The principles of probabilistic structural limit state design with a possibility for differentiating the reliability level are described in JCSS PMC [11-4] and in ISO 2394 [11-11]. In most design situations, the design values obtained considering relation to the most unfavourable load combination. The requirements for the reliability of the components of the system will depend on the system characteristics. Therefore, structures with multiple, equally important failure modes should comply with a higher level of reliability per component than recommended in this Model Code; see section 11.3.3 for details.

The target reliability indices given in section 11.3.2.2 and section 11.3.2.3 are recommended for ductile structural components whose failure allows redistribution of internal forces and thus it does not immediately lead to a global or partial structural failure, allowing taking measures to avoid severe consequences. Therefore, by explicit requirements or by appropriate detailing it should be assured that brittle failure does not occur. A structural component that would be likely to collapse suddenly without warning should comply with a higher level of reliability than those recommended in this Model Code.

The annual target reliabilities should be fulfilled in every year of a service life of the structure.

To satisfy performance requirements at the target reliability levels as recommended in section 11.3.2.2 for ultimate limit states, it is normally sufficient to follow the safety concepts, as explained in chapter 12.

11.3.2.2 Target reliability levels for structural design (ultimate limit states)

11.3.2.2.1 Recommended levels

For structural design in persistent, transient or accidental design situations, this Model Code recommends applying either the annual target reliability indices in Table 11.3-5 or the 50-year target reliability indices in Table 11.3-6. All

either $\beta_{t,eco,1}$ or $\beta_{t,eco,50}$ values in Table 11.3-2 and Table 11.3-3, respectively, differ insignificantly provided that the appropriate values of the sensitivity factors α_R and α_E are taken into account; see section 12.5.4.1.4. The small differences do not affect the average reliability level achieved by the different sets of partial factors or probabilistic approach. For more information see Meinen and Steenbergen [11-14] and Steenbergen et al. [11].

In Table 11.3-5 the default case for structural design is CC2 and Normal (B) relative cost of safety measure. Normal (B) relative cost of safety measure is associated with medium variabilities of the yearly extreme of the total loads and resistances and normal service life duration of 30-70 years.

Deviation from Normal (B) relative cost of safety measure can be considered based on the following characteristics:

- Degree of uncertainty; large uncertainty in either loading or resistance may increase the relative cost of safety measure to Large (A). This may be relevant for structures with low quality execution or structures dominated by highly variable loads such as snow or wind.
- Small uncertainty in both loading and resistance may decrease to Small (C). However, this is unrealistic for concrete structures.
- Service life and/ or obsolescence; for new structures with a short design service life or otherwise rapid obsolescence in the order of less than 10 years the relative cost of safety measure may be classified as Large (A).

See Köhler [11-15] for further details.

Due to a significant influence of permanent actions and structural resistance, on average the annual failure events are moderately correlated for concrete structures. The annual value $\beta_{t,eco,1}$ = 4.2 in Table 11.3-5 hence on average corresponds to 50-year $\beta_{t,eco,50}$ = 3.8 in Table 11.3-6 for concrete structures in persistent design situations, Meinen and Steenbergen [11-14] and Baravalle and Köhler [11-16]. Further, the $\beta_{t,eco}$-values provided in Table 11.3-5 and Table 11.3-6 were derived for non-degrading structural members. When degradation (possibly controlled by inspections and repairs) is a mechanism contributing to failure, the annual target of 4.2 seems to be a reasonable requirement for the most unfavourable year as structural reliability in other years will exceed this target (thus an annual target of 4.7 would often be overly conservative), see also JRC 2022 [11-17].

$\beta_{t,eco}$-values depend on many parameters affected by the economy of a country including failure consequences (e.g. societal consequences and impact to the economy of a region affected by the failure), relative costs of safety measure, consequences related to structural collapse, failure or malfunction should be considered including for instance persons affected indirectly as can be the case for hospitals, fire stations or buildings providing shelter in the case of emergency situations.

Table 11.3-5: *Annual target reliability indices, $\beta_{t,eco,1}$, for Ultimate Limit States based on economic optimisation, adopted from JCSS PMC [11-4] and ISO 2394 [11-11].*

Relative cost of safety measure	**Consequence Class**		
	CC1	CC2	CC3
Large (A)	3.1	3.3	3.7
Normal (B)	3.7	**4.2**	4.4
Small (C)	4.2	4.4	4.7

Table 11.3-6: *Informative target reliability indices for structures to be designed, related to a 50-year reference period, $\beta_{t,eco,50}$ (assumed to be based on economic optimisation)*

Consequence Class		
CC1	CC2	CC3
3.3	**3.8**	4.3

obsolescence and interest rates. $\beta_{t,eco}$-levels are thus commonly adjusted considering the level of development of a country.

The target reliability indices $\beta_{t,eco}$ provided in Table 11.3-5 and Table 11.3-6 are indicative for developed countries. For countries under development, lower β_t-values may be justified considering relevant socio-economic parameters, using for instance the Life Quality Index approach provided in ISO 2394 [11-11].

In exceptional cases, target reliability levels may need to be determined by probabilistic risk optimisation; see section 12.5.2.

No specific guidance for the target levels for fatigue verifications is provided here. In the absence of detailed justification, it is assumed that the $\beta_{t,eco}$- values in Table 11.3-5 or in Table 11.3-6 may be applied.

Lower target levels are often applied in seismic design situations; see section 11.3.2.2.5.

Human safety verifications may be particularly relevant for key members (see also section 30.9.4.5), where a local member failure might induce a partial or total collapse of the structural system. The LQI approach provided in ISO 2394 [11-11] may be applied. It requires estimations of the expected number of fatalities N in a specific collapse scenario. Guidance on the modelling of N for various collapse scenarios can be found in Hingorani et al. [11-9].

The application of the requirements according to the LQI approach may yield higher risk levels compared to current practice (e.g. design according to Eurocodes). To achieve a level of human safety similar to that associated with current practice, the following indicative annual target reliability indices depending on constant C representing the Consequence Class (CC) and the area affected by the collapse, A_{col}, can be considered for design of members in buildings exposed to persistent situations:

$$\beta_{t,cp,hs,1} = C - 2A_{col}^{-0.1} \qquad (11.3\text{-}3)$$

The use of Eq. (11.3-3) is recommended along with the CC according to the occupancy ratio related to a specific member collapse scenario; see section 11.3.1.2. Indicative values for the corresponding constant C in Eq. (11.3-3) are 4.75, 5.25 and 5.75, for CC1, CC2 and CC3, respectively. This approach is based on the procedure proposed by Tanner & Hingorani [11-18].

11.3.2.2.2 Indicative criteria for human safety

If structural failure is expected to result in human losses, it may be necessary to ensure that human safety risks comply with societal preferences. The resulting target reliability level is then:

$$\beta_t = \max(\beta_{t,eco}; \beta_{t,hs}) \qquad (11.3\text{-}2)$$

where $\beta_{t,eco}$ and $\beta_{t,hs}$ denote the annual target reliability levels based on economic optimisation and human safety criteria, respectively.

The reliability levels in Eq. (11.3-3) apply to failure modes of individual structural members. However, given the individual member failure, the estimation of the associated collapse area (A_{col}) should account for a possible (sub-) system failure depending on the type and the characteristics of the structural system (see also Section 30.9 on *Robustness*).

fib Bulletin 80 [11-12] provides indicative reliability levels for bridges.

The definition of rational safety levels for temporary structures depends on many variables, most of which may not be as important as for permanent structures, such as structural costs, failure consequences, or variability of permanent loads. Economic optimisation and risk assessment are suitable tools to specify appropriate target levels. In light of economic considerations and depending on the specific design situation (single or possible multiple use of a temporary structure), different levels of reliability, relative to the permanent structures, might be optimal for the temporary structures.

For temporary repeatedly used structures such as scaffoldings or bridge falseworks, economic optimisation studies typically lead to a range of $\beta_{t,eco,1}$-values from 3.2 to 4.4. For short-term erection phases such as during a bridge launching, an indicative value is $\beta_{t,eco,tref} \approx 3$, related to the duration of the erection phase t_{ref}; the compliance with human safety criteria may need to be verified when relevant.

Such larger uncertainty increases costs of safety measures and may justify to imply lower target reliability when failure cost is relatively small. However, when all failure consequences (e.g. damage to temporary and permanent structures, personnel, equipment, delays in construction, other indirect costs) are taken into account, the additional uncertainty in most cases does not result in reducing the target reliability since the cost of reducing the uncertainty level becomes negligible in comparison to failure consequences.

11.3.2.2.3 Indicative criteria for temporary structures

For temporary structures, dismantlement of structures and other temporary situations, the same principles as for permanent situations, including Eq. 11.3-2, apply. According to JRC 2022 [11-17], the total time of use should be considered for temporary structures that are repeatedly re-used.

Design of the temporary structures used in construction should take into account that the continuously changing structural system and dominant action may lead to load effects associated with larger uncertainty than those associated with the permanent structure.

11.3.2.2.4 Indicative criteria for accidental situations

The design of a structure for accidental situations may be necessary in order to verify compliance with requirements on robustness (see section 30.9.4). Such a

For accidental actions, the large relative cost of safety measure can be commonly assumed , see JRC 2022 [11-17].

In Eq. (11.3-4), $\beta_{t|acc}$ represents the *conditional* target reliability index associated with the admissible probability of failure given the occurrence of an accidental situation, $p_{f|acc}$. Indicative $\beta_{t|acc}$-values corresponding to different accidental situations are given below. The same approach was adopted in [16] where it is considered that P(failure given fire) × P(fire) ≤ target probability. In this way the target level for the accidental situation (given its occurrence) can be specified.

When applying Eq. (11.3-4), the relative cost of safety measures and failure consequences relate to the accidental situation under consideration. This may result in a different classification and thus in a different β_t-value in comparison to normal design situations.

The investigation of the reliability levels associated with current design practice leads to the following indicative $\beta_{t|acc}$-values for selected accidental situations (see Levendecker & Ellingwood [11-20]):

– As the basis for design values of forces due to impacts of lorries to bridge piers, ISO 10252 [11-19] specifies the target reliability index $\beta_{50} = 4.0$, and thus $\beta_1 \approx 4.8$ for independent annual failure events. This leads for example to $\beta_{t|acc} \approx 1.5$ for $\lambda = 10^{-5}$ / year (the hazard occurrence rate being dependent on many parameters including pier geometry, distance to pier and truck traffic intensity). Note that FprEN 1992-1-1 [11-21] adopts the target reliability index $\beta_{t|acc} = 2.0$ for accidental situations generally.

– Reliability of RC structural members designed according to Eurocodes and exposed to the ISO 834-11 [11-22] standard fire is approximately $\beta_{t|acc} \approx 1.35$ given a developed structurally significant fire (the lifetime-cost economic optimisation by Hopkin et al. [11-23] revealed that optimum target levels for fire safety depend on design should comply with reliability requirements for both persistent and accidental situations.

In accordance with ISO 10252 [11-19] for bases for design of structures exposed to- accidental actions, the appropriate degree of reliability should, in accordance with ISO 2394 [11-11], be selected with due regard to the possible consequences of failure, the associated expense and the level of efforts and procedures required to reduce the risk of failure and damage.

This can be achieved by considering appropriate target reliability index β for ULS and hazard occurrence rate λ (for $\Phi(-\beta) / \lambda < 1$, normally considering annual reference period for β and λ):

$$\beta_{t|acc} = -\Phi^{(-1)}\left[\frac{\Phi(-\beta)}{\lambda}\right] \qquad (11.3\text{-}4)$$

where $\Phi(\cdot)$ is the standard normal cumulative probability distribution function ($\Phi^{-1}(\cdot)$ being its inverse); $\beta_{t|acc}$ is the target reliability given the occurrence of the accidental situation.

the expected fire load, safety measure cost and failure consequences with values mostly in the range $\beta_{t|acc} \approx 1.5\text{-}2.5$).

Hingorani et al. [11-24] suggested that key members in CC2 buildings (see section 11.3.1), where gasses are burned, regulated, transported or stored, should comply with a life safety risk-based requirement, given a gas explosion and the subsequent member exposure to the generated pressure wave.

$$\beta_{t|acc} = 3 - 7.75 A_{col}^{-0.4} \qquad (11.3\text{-}5)$$

where A_{col} is the area affected by the member collapse in m^2 (see also section 30.9 on Robustness). When A_{col} is less than a minimum value $A_{col,min}$, no explosion-specific reliability requirements for member design are required. $A_{col,min} = 20\ m^2$ may be considered.

Structural design for seismic situations can refer to one or more Limit States (LS), such as Fully Operational (OP), Damage Limitation (DL), Significant Damage (SD), Near Collapse (NC).

Each LS is verified using a different seismic intensity having a return period $T_{LS,CC}$ dependent on the Consequence Class (see section 11.3.1).

Inherent acceptable (target) annual probability of exceedance of a LS, associated with the structures designed to seismic criteria and assuming the linear relationship between intensity measure of the ground motion and structural demand, can be estimated as (ASCE/SEI 43-05 [11-25], FEMA P695 [11-26], Žižmond and Dolšek [11-27], Monti et.al. [11-28]):

$$P_{f,LS} = \frac{1}{\left(\gamma_{R,LS}\right)^{k}} \cdot \frac{1}{T_{LS,CC}} \cdot exp\left(\frac{1}{2} k^2 \cdot \beta_{LS}^2\right) \qquad (11.3\text{-}5)$$

where:

- $\gamma_{R,LS}$ is the global overstrength factor of the designed structure, which multiplies the design capacity at the LS under consideration; when

11.3.2.2.5 Indicative criteria for seismic situations

In accordance with ISO 2394 [11-11], the appropriate degree of reliability should be selected with due regard to the possible consequences of failure, the associated expense and the level of efforts and procedures required to reduce the risk of failure and damage. Where relevant, human safety criteria should be fulfilled.

Verification of structural reliability in seismic situations may be based on annual target probability of exceedance of Near Collapse Limit State, $P_{f,t,cp,seismic,NC,CC}$. For structures of consequence class CC2, this target probability is $P_{f,t,cp,seismic,NC,2} = 2 \times 10^{-4}$ / year, corresponding to a mean return period of NC-exceedance of 5000 years. Return periods may be specified by the competent authority or the owner for specific projects.

$\gamma_{R,NC}$ is in the order of 2.5, the target $P_{f,t,cp,seismic,NC,2}$ is attained under the seismic action with return period $T_{NC,2}$,

- $T_{LS,CC}$ is the return period of the design seismic action causing the design capacity at the LS considered for a particular Consequence Class,
- k is the slope of the seismic hazard (in the log-log plane), assumed as linear, which in Europe varies in the range 1.4-3; the minimum of $P_{\mathrm{f,LS}}$ in Eq. (11.3-5) over the European territory is found at sites with $k = 2.5$,
- β_{LS} is the log-standard deviation of the safety margin $\ln\frac{D}{C}$ between demand D and capacity C, in the order of 0.6 for NC LS.

Therefore, Eq. (11.3-5) provides the reliability levels associated with seismic verifications based on the aforementioned assumptions.

As an example for NC LS, $T_{NC,CC}$ can be taken equal to 800, 1600, and 2500 years for CC1, CC2, and CC3, respectively. Using Eq. (11.3-5), the resulting annual reliability levels are listed in Table 11.3-7.

Table 11.3-7: Annual target reliability levels for structures designed to the seismic NC LS, for different Consequence Classes.

	Consequence Class		
	CC1	CC2	CC3
$P_{f,t,cp,seismic,NC,CC,1}$	4.0×10^{-4}	**2.0×10^{-4}**	1.3×10^{-4}
$\beta_{t,cp,seismic,NC,CC,1}$	3.3	**3.5**	3.7

According to (CEN/TS 17440 [11-5]) the target reliability levels of existing structures may be different from those assumed for new structures. The following

11.3.2.3 Target reliability levels for assessment of existing structures (ultimate limit states)

11.3.2.3.1 Recommended levels

The assessment of existing structures essentially requires specifying a minimum target reliability $\beta_{t,0}$ below which the structure should be upgraded (or another safety

aspects motivate the differentiation of target reliability levels between new and existing structures:

– economic considerations: the relative cost of interventions in existing structures with the aim to increase the reliability level can be very high, whereas the incremental cost of increasing the reliability at the design stage of new structures is generally low;

– societal considerations: possible resettlement of inhabitants, relocation of activities or loss of heritage values must be considered in relation with existing structures, whereas such aspects normally do not affect the design of most of new structures;

– sustainability considerations: repair or retrofitting of existing structures normally enables the choice of the most appropriate materials, compared to building new structures.

For more details, see also *fib* Bulletin 80 [11-12] and ISO 2394 [11-11].

When reliability of the existing structure is insufficient, a safety measure to be taken can be structural intervention (repair or upgrade) and/ or operational measure (e.g. monitoring, change of use or limits on imposed or traffic loads).

For existing structures with multiple, equally important failure modes, an increase of a $\beta_{t,0}$-level by 0.5 per structural member could be appropriate.

Human safety verifications may be particularly relevant for key members (see also section 30.9.4.5) where a local member failure might induce a partial or total collapse of the structural system (see also section 30.9 on Robustness).

measure should be taken) and the target level for structural upgrade $\beta_{t,up}$; see section 11.3.2.3.5 for the latter.

$\beta_{t,eco,0}$-values may be based on the informative annual target reliability indices given in Table 11.3-5 for Large (A) relative cost of safety measure:

– CC1: $\beta_{t,eco,0,1} = 3.1$
– CC2: $\beta_{t,eco,0,1} = 3.3$
– CC3: $\beta_{t,eco,0,1} = 3.7$.

For existing structures with multiple, equally important failure modes, a higher level of reliability per structural member should be considered.

11.3.2.3.2 Indicative criteria for human safety

If structural failure is expected to result in human losses, it may be necessary to ensure that human safety risks comply with societal preferences. The resulting target reliability level is then:

$$\beta_t = max(\beta_{t,eco,0}; \beta_{t,hs}) \quad (11.3\text{-}6)$$

For new and existing structures, similar human safety criteria should be used.

The assessment of human safety associated with such members in existing buildings may be based on Eq. (11.3-3). For that purpose, constant C in Eq. (11.3-3) may be reduced by 0.5.

No specific guidance is available for existing bridges; see *fib* Bulletin 80 [11-12] for indicative levels.

In exceptional cases with very high failure consequences and/ or very high cost of safety measure (e.g. upgrade including losses due to temporary structural malfunction), an appropriate $\beta_{t,0}$-level should be determined by probabilistic risk optimisation.

For the assessment of key members in parts of existing buildings corresponding to consequence class CC2 (see section 11.3.1), where gasses are burned, regulated, transported or stored, the conditional target reliability index $\beta_{t|acc}$ according to Eq. 11.3-6 may be reduced by 1.0. For the minimum area affected by the collapse, below which no explosion-specific reliability requirements are required, $A_{col,min}$ = 100 m^2 can be considered.

No specific guidance is available for other types of accidental actions in the assessment of existing structures.

Following the principles provided in section 11.3.2.2.5, Eq. (11.3-5) can be used to determine the inherent acceptable (target) annual probability of exceedance of a LS for the existing structures assessed considering $\gamma_{R,LS}$ = 1; the minimum of $P_{f,LS}$ over the European territory is found at the sites with $k = 1.4$.

11.3.2.3.3 Indicative criteria for accidental situations

The assessment of an existing structure for accidental situations may be necessary in order to verify compliance with the requirements on robustness (see section 30.9.4).

The appropriate degree of reliability should, in accordance with ISO 2394 [11-11], be selected with due regard to the possible consequences of failure, the associated expense and the level of efforts and procedures required to reduce the risk of failure and damage.

A conditional target reliability index may be obtained according to Eq. (11.3-4) by considering an appropriate target reliability index β_t for ULS and hazard occurrence rate λ (for $\Phi(-\beta_t) / \lambda < 1$, normally considering annual reference period for β_t and λ).

11.3.2.3.4 Indicative criteria for seismic situations

The assessment of an existing structure for seismic situations may be based on target failure probabilities that can be higher than for design situations.

As an example for NC LS, $T_{NC,CC}$ can be taken equal to 800, 1600, 2500 years for CC1, CC2, and CC3, respectively. Using Eq. (11.3-5), the resulting annual reliability levels are listed in Table 11.3-8.

Table 11.3-8: Annual reliability levels for structures assessed to the seismic NC LS, for different Consequence Classes.

	Consequence Class		
	CC1	CC2	CC3
$P_{f,t,cp,seismic,NC,CC,1}$	1.8×10^{-3}	**8.9×10^{-4}**	5.7×10^{-4}
$\beta_{t,cp,seismic,NC,CC,1}$	2.9	**3.1**	3.3

While slightly lower values can be normally justified for $\beta_{t,up}$-levels in comparison to the design target levels, it is common and reasonable to require the compliance with the design levels when upgrading the structure.

The principles for human safety verification provided for structural design are applicable in the same way for specifying the $\beta_{t,up}$-levels.

In exceptional cases, an appropriate $\beta_{t,up}$-level and the selection of safety measure(s) may need to be based on probabilistic risk optimisation; see section 12.5.2.

Following the principles provided in section 11.3.2.2.5, Eq. (11.3-5) can be used to determine the inherent acceptable (target) annual probability of exceedance of a LS for the structures upgraded to seismic criteria considering values of $\gamma_{R,LS}$ related to the specified upgrade level, defined as the ratio between upgraded global mean capacity ($\gamma_{R,LS} \leq 1$) and the corresponding mean demand for the LS of interest; lower values of $\gamma_{R,LS}$ are related to higher relative cost of safety measures.

11.3.2.3.5 Target reliability for upgrades

For ULS, $\beta_{t,up}$-values may be based on the target reliability indices given in Table 11.3-5 or in Table 11.3-6 for structural design.

The reliability level for an upgrade of the existing structure for seismic situations depends on the selected upgrade level. This is given as the ratio between the upgraded global mean capacity and the corresponding mean demand for the LS of interest. Lower upgrade levels are related to higher relative cost of safety measures.

The target reliability indices provided in Table 11.3-9 are intended to provide the basis for probabilistic verifications and calibration of the semi-probabilistic verification formats. JCSS PMC [11-4] indicates that a variation of around 0.3 from the target reliability indices given in Table 11.3-9 can be considered.

Condition limit states are defined in ISO 2394 [11-11]. Examples of limit states with direct (negative) consequences are ULS and SLS.

ISO 13823 [11-2] includes an Initiation Limit State as one of the possible Durability Limit States that can be considered, which is defined by the initiation of deterioration of a component that precedes the occurrence of the serviceability or ultimate limit states.

Guidance on establishing and validating the requirements for concrete structures associated with Durability Limit States can be found in ISO 16204 [11-29].

In case of the design of new structures, the conventional way to verify the design service life is to comply with deemed-to-satisfy requirements based on classes for compressive strength, concrete composition in terms of maximum w/b and minimum cement content, minimum cover to steel reinforcement and crack width limitations, all as a function of the exposure class. However, LoA has been introduced to encourage more complex design solutions where practical, in order to allow for more cost efficient and/or sustainable solutions. Probabilistic methods may provide suitable approaches, particularly for structures of major importance. Some codes (e.g. AS 3600 [11-30]) also note that the code may not always provide a safe

11.3.2.3.6 Target reliability levels for serviceability limit states

Target values for serviceability limit states can be derived by cost optimisation. For irreversible Serviceability Limit State verifications, this Model Code recommends the target reliability indices provided in Table 11.3-9.

Table 11.3-9: *Annual target reliability indices for irreversible Serviceability Limit States based on economic optimisation, $\beta_{t,eco,SLS,1}$, (examples) according to JCSS PMC [11-4].*

Relative cost of safety measure	$\boldsymbol{\beta_{t,eco,SLS,1}}$
Large	1.3
Normal	1.7
Small	2.3

11.3.2.3.7 Target reliability levels for condition limit states

General

Condition limit states are limit states acting as a proxy for a limit state which has direct negative consequences. In applications related to durability aspects, a condition limit state is often referred to as a durability limit state

For a specific project the target reliability level and relevant condition limit states related to durability as well as the associated design criteria shall be specified and agreed with the client and the relevant stakeholders.

The design strategy regarding condition limit states shall be specified and agreed with the client and the relevant stakeholders. This strategy can – besides design by avoidance of degradation – relate to a Level of Approximation for the verification of the design service life, e.g. (1) verification by a deemed-to-satisfy approach, (2) verification by durability indicators (3) verification by a partial factor method, (4) verification by a full-probabilistic approach. In case the consequences are expected to play a significant role, also a risk-based approach can be considered.

solution. Probabilistic methods which consider a higher number of model parameters may also be appropriate where there is doubt about deemed-to-satisfy provisions, provided that for each model parameter the statistical distribution to be used is based on representative data. Guidance on assessing target values for the reliability index may be found in RILEM Report 14 [11-31], DuraCrete Final Technical Report [11-32] and *fib* Bulletins 34 [11-33], 76 [11-34] and 80 [30-12].

For deterioration mechanisms for which there is no general international consensus on time-dependent models, e.g. for freeze-thaw, acid and sulphate attack, alkali-aggregate reaction and delayed ettringite formation, the avoidance-of-deterioration verification format should be applied.

An undesirable event in the case of durability performance can for example relate to:

a) the carbonation front exceeding the cover depth;

b) the chloride content at the depth of the reinforcing steel exceeding a critical level (i.e. corresponding to a limit state of depassivation);

the occurrence of active corrosion of the reinforcement over a specified reference period (commonly the design service life).

In view of the absence of broad international consensus target β-values are merely provided for condition limit states related to reinforcement corrosion.

The target value is generally based on expert opinion and reflects current deemed-to-satisfy rules. For calibration of the durability provisions in EN 1992-1-1:2023 [11-21], a target value $\beta = 1.5$ was chosen, corresponding to a probability of approximately 7% for the undesirable event of depassivation of the steel reinforcement including a portion of the propagation period characterised by a uniform corrosion attack of 50 µm (carbonation-induced corrosion) or a local corrosion attack of 500 µm (chloride-induced corrosion). This is expected to be relevant for the majority of concrete structures designed according to this standard.

These levels of corrosion attack are based on a corrosion-induced concrete strain that, when combined with common allowance for load induced strain, may just induce cracking.

For most situations these levels of corrosion attack are assumed not to compromise the serviceability limit state (SLS) and the ultimate limit state (ULS), however verification is recommended, particularly for specific situations considering the potential negative consequences and applicable target β-value for

Performance-based design of new structures

In case of a performance-based approach in which the verification of the design service life is based on a full-probabilistic method, the performance of the concrete structure or of an individual component with respect to durability can be expressed in statistical probabilistic terms by a reliability index β, corresponding to the probability of occurrence of an undesirable event.

The relevant condition limit states and the associated target reliability values shall be agreed with the client and relevant stakeholders of a specific project.

The selection of appropriate target β-values should be based on a risk-based approach, taking into account structural, serviceability, safety, maintenance and economic consequences of subsequent deterioration, the implications of the rate of structural deterioration and the life-cycle management regime which is to be implemented for the structure.

either design or assessment purpose. Guidance may be found in section 27.6 & 27.11 and *fib* Bul. TG 3.2 (in preparation) [11-35].

These levels of corrosion attack are considered to serve as a proxy for direct consequences. The target β-value = 1.5 is considered appropriate for most types of concrete structures and concrete components.

A reliability differentiation can be considered for the specific project based on aspects like:

- accessibility of the structure (or specific components) for inspection, maintenance and repair;
- consequences of deterioration for user safety and availability of the structure;
- life-cycle cost;
- any associated change in life-cycle environmental impacts / benefits.

However, it should be emphasized that the target β-value is based on assumptions about the statistical distribution of model parameters, and hence care should be exercised in a full-probabilistic calculation when comparing the outcome to such a β-value.

Regarding the limit state of solely depassivation of steel reinforcement literature provides a wide range of suggested target β–values. Generally, no entirely sound and objective reasoning is currently available to substantiate a specified β-value. However, a value of $\beta = 1.3$ is often used assuming that there will be no major repairs required during the design life, although this will also depend on the actual structure and reinforcement detailing.

The use of β = 1,5 for the depassivation period in combination with a well-specified part of the subsequent propagation period is assumed to result into a broadly similar safety level than considering $\beta = 1,3$ with depassivation only.

In contrast to verification of ultimate and serviceability limit states in most design codes, durability provisions are only given for one reliability class. However, depending on the consequences (economic, societal and/or environmental), a differentiation in reliability classes may be considered.

The target reliability levels to be used for the assessment of existing structures with respect to durability may differ from those applied to new structures.

The same considerations listed in the commentary to 11.3.2.2.1 are also valid for CLS.

In the case of a full-probabilistic durability-based performance of existing undamaged structures, the same target reliability level as for new structures can be applied unless the following justify a different β-level:

a) ULS and/ or SLS verifications or

b) the application of risk-based methods or

c) the cost of intervention.

The observation of clear signs of reinforcement corrosion should be taken as a warning that assessment of the structural capacity at the time of assessment should be carried out. The target reliability levels for such assessments are given in 11.3.2.2.1.

Specific additional investigations should be carried out taking into account the spatial variability of damage, carbonation depth and ingress of chlorides.

Performance-based assessment of existing structures differs from the design approach adopted for new concrete structures. The choice of appropriate alternative values for target reliability levels of relevant condition limit states for existing structures will be influenced by various factors e.g. altered service life, altered requirements on structural behaviour over the remaining service life (i.e. cracking, spalling etc.) and the actual progress of the deterioration.

In case of existing structures, a performance-based assessment of the remaining service life may be based on an ultimate limit state or a serviceability limit state verification, e.g. in case damage has already occurred. Such a SLS- and ULS-verification serves as an addition to the verification of condition limit states. In case of SLS and ULS, the verification should be performed using annual target reliability values, corresponding to the target reliability levels as specified in Section 11.3.2. Such verifications are based on a coupled modelling of the initiation period and propagation period and the actual condition of the structure should take into account:

- the uncertainties in relation to the initiation phase modelling;
- the uncertainties in relation to the propagation phase modelling;
- the uncertainties typically considered in structural reliability calculations for ULS/SLS (as done for non-degrading structures);
- the uncertainties associated with monitoring, inspections and the interpretation of such data, if applied.

These uncertainties can be reduced by updating the model parameters based on measurements and inspections performed on-site.

A possible target reliability β = 1.5 with respect to a uniform corrosion attack of 50µm (carbonation-induced corrosion) or to a local penetration depth of 500µm (chloride-induced corrosion) can be considered.

Component reliability is the reliability of one single structural component which has one dominating failure mode.

If several failure modes are of similar importance, structure-related risks (implicitly accepted according to current best practice) are significantly higher compared to cases with a single dominant mode and the structure can be less redundant.

System reliability is the reliability of a structural system composed of a number of components or the reliability of a single component which has several failure modes of nearly equal importance.

A probabilistic approach provides an appropriate tool to investigate system behaviour. A reliability and robustness assessment should be carried out to establish the probability of scenarios which may lead to structural system failure including identification and consideration of:

- the sequences of failures of individual constituent failure modes leading to system failure modes;
- the ability of the structure to allow for redistribution of internal forces in given failure states;
- possible means of ensuring that reliable redistribution of internal forces is facilitated (tying);
- possible means of ensuring that unreliable redistribution of internal forces is avoided (segmenting);

In the case of full-probabilistic durability-based performance assessments of existing structures, the same target reliability level as for new structures can be applied unless ULS and/ or SLS verifications, or the application of risk-based methods or the cost of intervention justify a different β-level.

11.3.3 Component reliability and system reliability

Structural analysis methods, as described in this Model Code, are primarily concerned with component behaviour. Each limit state equation is, in most cases, related to a single mode of failure of a single component.

However, individual components may also be susceptible to a number of possible failure modes. Therefore, in reliability verification, the susceptibility of the individual components to a number of possible failure modes must be checked where relevant, by checking a number of limit state equations.

Furthermore, most structures are an assembly of structural components. System failure is usually the most serious consequence of component failure. Therefore, the likelihood of system failure following an initial component failure should be assessed in relation to robustness with respect to accidental events, redundancy (alternative load paths) and complexity of the structure (multiple failure modes; see section 30.9). Accordingly, the system analysis must be carried out as a part of the reliability verification.

The system analysis requires considerable inventiveness and initiative from the engineer. In general, the system behaviour of structures can be quantified in terms of the limit state design by deterministic approach (e.g. progressive collapse analysis) or by a probabilistic approach.

– dependencies between failures of individual constituent failure modes and system failure modes.

It is noted that besides various failure modes, reliability verification may need to account for different load and degradation scenarios. For more details see JCSS PMC [11-4] and ISO 2394 [11-11].

References

[11-1] ISO 16204:2012 Durability — Service life design of concrete structures

[11-2] ISO 13823:2008. Genral principles on the design of structures for durability.

[11-3] ISO 13822:2010. Bases for design of structures.

[11-4] JCSS PMC Probabilistic model code – Part 1: Basis of design, 2001. Joint Committee on Structural Safety.

[11-5] CEN/TS 17440. Technical Specification – Assessment and Retrofitting of Existing Structures, 2020.

[11-6] prEN 1990:2019. Eurocode – Basis of structural and geotechnical design. ### 2023?.

[11-7] Diamantidis, D., Sykora, M. Reliability Differentiation and Uniform Risk in Standards: a Critical Review and a Practical Appraisal. *Future Trends in Civil Engineering*, 2019.

[11-8] ASCE 7-16. Minimum Design Loads and Associated Criteria for Buildings and Other Structures. American Society of Civil Engineers, 2017.

[11-9] Hingorani, R., Tanner, P., Prieto, M., Lara, C. Consequence classes and associated models for predicting loss of life in collapse of building structures. *Structural Safety*, 85, 2020.

[11-10] StFV. Verordnung über den Schutz vor Störfällen (Störfallverordnung). Switzerland, 814.012, 2019.

[11-11] ISO 2394:2015. General principles on reliability of structures.

[11-12] *fib* Bulletin 80. Partial factor methods for existing concrete structures. 2016.

[11-13] Holicky, M., Diamantidis, D., Sykora, M Reliability levels related to different reference periods and consequence classes. 2018. *Supplement: 16th International Probabilistic Workshop* 113(S2): 22 – 26.

[11-14] Meinen, N., Steenbergen, R. D. J. M. Reliability levels obtained by Eurocode partial factor design – A discussion on current and future reliability levels. 2018. *Heron* 63(3): 243 – 301.

[11-15] Kohler, J. Reliability targets for semi-probabilistic design standards. Life-Cycle Analysis and Assessment in Civil Engineering: Towards an Integrated Vision, *Proc. IALCCE 2018* p. 387 – 395.

[11-16] Baravalle, M, Köhler, J. , A risk-based approach for calibration of design codes, Structural Safety 78, 2019, p. 63-75.

[11-17] JRC 2022 JRC Technical Report for Reliability Background Eurocodes – report under preparation by CEN TC250 / SC10/ AHG Reliability background.

[11-18] Tanner, P., Hingorani, R. Acceptable risks to persons associated with building structures. *Structural Concrete*, 2015, 16(3): 314 – 22.

[11-19] ISO 10252:2020. Bases for design of structures – Accidental actions.

[11-20] Levendecker, E. V., Ellingwood, B. Design Methods for Reducing the Risk of Progressive Collapse in Buildings, 1977. U.S. Departement of Commerce, National Bureau of Standards.

[11-21] EN1992-1-1:2023 Design of concrete structures – Part 1-1 General rules, rules for buildings, bridges and civil engineering structures, ### 2023

[11-22] ISO 834-11: 2014. Fire resistance tests.

[11-23] Hopkin, D., Fu, I., Van Coile, R. Adequate fire safety for structural steel elements based upon life-time cost optimization. *Fire Safety Journal*, 2021, 120.

[11-24] Hingorani, R., Tanner, P., Zanuy, C. Life safety risk-based requirements for concrete structures in accidental situations caused by gas explosions. *Structural Safety*, 2019, 76: 184 – 196.

[11-25] ASCE/SEI 43-05 ASCE, Seismic design criteria for structures, systems, and components in nuclear facilities, ASCE/SEI 43-05, Reston, VA. (2005)

[11-26] FEMA P695 FEMA, FEMA P695: Quantification of building seismic performance factors, Federal Emergency Management Agency (FEMA) - Washington, DC, USA (2009)

[11-27] Žižmond, J., Dolšek, M. Formulation of risk-targeted seismic action for the force-based seismic design of structures, Earthquake Engineering & Structural Dynamics 48 (2019) 1406–1428.

[11-28] Monti, G., Demartino, C. Gardoni P. Towards risk-targeted seismic maps for Europe, Structural Safety, 2021 (under review)

[11-29] ISO 16204:2012. Durability – Service life design of concrete structures.

[11-30] AS 3600 AS 3600 – Concrete structures, Standards Australia, 2018

[11-31] RILEM Report 14 ??????????????????????????????????????

[11-32] DuraCrete Brite EuRam III - Modelling of degradation : DuraCrete, probabilistic performance based durability design of concrete structures, CUR-Gouda, 1998

[11-33] *fib* Bulletin 34 Model Code for Service Life Design. Model Code, *fib*, 2006

[11-34] *fib* Bulletin 76 Benchmarking of deemed-to-satisfy provisions in standards, fib, 2015

[11-35] *fib* Bulletin ## State of the art report: Modelling of structural performance of existing concrete structures, fib, report under preparation by TG3.2

For complex structures and structures with a sequential change of the structural system during construction, or in use, which are sensitive to time dependent behaviour, the consideration of load- or deformation history may be necessary. In such a case it may be required to carry out both an initial and a long term reliability assessment.

The various types of design situations are defined in section 3.2.2 of CEB Bulletin 191 [12-1].

Accidental action is defined as action of usually short duration that is unlikely to occur with a significant magnitude on a given structure during the design service life, but its consequences might be catastrophic, for example fire, explosions or impact from vehicles.

Unlike accidental actions, which cannot be associated with a statistical probability of being exceeded, seismic actions can be classified in terms of probability of occurrence and severity.

Construction states can be considered as persistent or transient design situations. Accidental design situations involve either the accidental situation itself or they refer to the situation immediately after the accidental event.

12 Principles of structural design and assessment

12.1 Design and assessment situations

Actions, environmental influences and structural properties may vary with time. Such variations, which occur throughout the lifetime of the structure, should be considered by selecting design situations (assessment situations in the case of existing structures), each one representing a certain time interval with associated load cases and other hazards, conditions and relevant structural limit states. The design or assessment situations considered must include all foreseeable conditions that can occur during execution and use.

In the design or assessment procedures, various design or assessment situations should be identified as relevant, by distinguishing:

- persistent situations, which refer to conditions of normal use of the structure and are generally related to the structure's design or remaining service life, respectively for new and existing structures;
- transient situations, which refer to temporary conditions of the structure, in terms of its use or its exposure;
- accidental situations, which refer to exceptional conditions of the structure or its exposure;
- seismic situations, which refer to conditions of the structure under an earthquake event.

In many cases judgement is necessary to supplement codified provisions, in order to identify those design or assessment situations that are to be taken into account for a particular structure.

For persistent situations a reference period t_R is commonly considered equal to the design service life for new structures or the residual service life for existing ones. Usually, for persistent situations in case of new structures a reference period t_R of 50 years is adopted for buildings and 100 years for bridges and tunnels. Even when

reliability verification is based on a shorter reference period such as 1 year, structural performance during the whole design service life should be verified.

Assessment of existing structures may be based on different reference periods than the design of new structures. These periods may be shorter than the anticipated remaining service life. In the case of deteriorating structures, the reliability assessment should be conducted by considering structural performance at the end of the remaining service life, based on a reference period of, for example, 1 year and taking into account degradation and defects. In any case, structural performance during the whole remaining service life should be verified.

For transient situations a reference period t_R of 1 year is normally taken.

Reference is made as well to EN 1991-1-6 [12-2] where the method for determining characteristic values of the climatic actions is specified for different nominal durations.

In the design for accidental actions the failure probability depends on the occurrence of the particular event considered.

In seismic design the failure probability is found by convoluting the probabilities of occurrence of seismic actions greater or smaller than the design event during the design service life for new structures or the residual one for existing structures.

Accidental situations are considered to be instantaneous and the corresponding reference period t_R is defined as the duration of the design event.

In the context of seismic situations, a reference period t_R is normally taken equal to 1 year.

12.2 Design and assessment strategies

Structures must be designed for all relevant design situations (i.e. persistent, transient, accidental and seismic, if relevant). Analogously, in the assessment of existing structures, all relevant assessment situations must be taken into account.

Depending on the type of action or damage state, the following strategies must be applied in design for different categories of the design situations:

- strategies applied in persistent and transient design situations for limiting the consequences of identified permanent and variable actions, which are:
  - design the structure to sustain the action;
  - design the structure or introduce protective measures to avoid or reduce the action or its effects;
  - design the structure for damage limitation;
- strategies applied in accidental or seismic design situations for limiting the consequences of identified accidental or seismic actions are:

Failure of the structural components and failure of the system must be analysed for all possible damage states and exposure events relevant for the design or assessment situation under consideration.

Section 3.2.3 of CEB Bulletin 191 [12-1] gives similar guidance on the choice of a design procedure appropriate to limit damage due to identified or unidentified hazards.

The general principles and the procedures for the verification of robustness are given in section 30.9. Besides the design for identified accidental actions, the structure can also be designed for robustness against unidentified actions, see section 30.9.

- design the structure to sustain the action;

  or

- design the structure to avoid the action or its effects;

  And

- design the structure for sufficient robustness.

Depending on the results and conclusions from the assessment with regard to the performance objectives for an existing structure, appropriate recommendations should be formulated to the client, taking into account the conditions for possible interventions and future operation of the construction work, established in the utilization plan.

The recommendations can include intervention measures belonging to different categories (operational interventions: monitoring, change in use; construction interventions: repair, upgrading). In the planning of such interventions, equivalent strategies can be applied as those mentioned for the design of new structures.

12.3 Level of approximation approach

Level I is reserved for structures where high accuracy is not required. It can also be used for pre-design of structures in a more general sense. Higher level methods can be used in cases where higher accuracy is required. An example of this is the assessment of an existing structure for its bearing capacity, supporting the decision of whether repair is necessary or not.

Various levels of approximation are possible for the design and assessment of concrete structures. Therefore in a number of chapters methods are offered with different levels of accuracy. Level I methods generally represent the most simple and straightforward approach, valid for standard cases. Higher levels are presented, which generally require more effort but may lead to more economic solutions.

All analyses performed for the design and assessment of structural members are approximations of reality. These approximations have different levels of accuracy.

The LoA approach is based on the use of rational theories that are based on physical models. The behaviour and strength of structural members are characterized through a series of parameters and a set of limit state functions. The parameters may involve physical variables (such as crack widths), mechanical properties (such as concrete compressive strength) or geometrical parameters (such as the width of a member).

A levels-of-approximation (LoA) approach is a design (assessment) strategy where the accuracy of the estimate of a structural member's response (behaviour or strength) can be, if necessary, progressively refined through a better estimate of the physical parameters involved in the limit state functions.

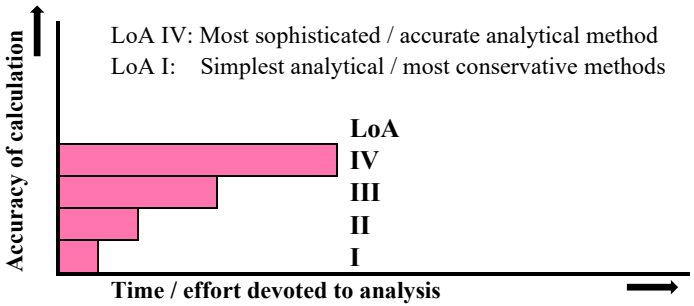

Figure 12.3-1: Accuracy on the estimate of the actual behaviour as a function of time devoted to the analysis for various levels-of-approximation.

In the LoA approach, the accuracy in the estimate of the various physical parameters is refined in each new LoA by devoting more time to the analyses, so that the accuracy in the behaviour and strength provided by the limit state functions is also improved.

Building projects typically involve a number of design phases, such as preliminary design, tender design and execution design. The required accuracy of the estimate of the structural behaviour and strength (and the available time to do it) increases as a project evolves. A suitable design strategy consists of using low-order LoAs for the first design phases and higher LoAs for the last design phases. In design, a potential for further increasing the LoA is often limited by the time available to perform the design and the relative cost of doing the design compared to the reduction in construction materials to be achieved.

This strategy also applies to the assessment of existing structures where increasing the LoA is commonly stopped when reliability is shown to be sufficient.

Numerical procedures such as non-linear finite element analysis typically allow the best estimates of the physical parameters of limit state functions to be obtained. They are normally used on the highest-order LoAs. The use of such LoAs can however be very time consuming and is only advised for the final design of very complex structures or for the assessment of critical existing structures. This is justified when a

The choice of a suitable LoA depends on the type of analysis performed, on the context of the analysis (preliminary or detailed calculations) and on the potential savings that can be provided if a higher-order LoA is performed.

The first LoA has to provide simple and safe hypotheses for evaluating the physical parameters of limit state functions. It leads to safe (yet realistic) values of the behaviour and strength of the structural member. This LoA is simple and low time consuming and usually sufficient for preliminary design and preliminary assessment purposes. Also, the first LoA can be used to check whether a given failure mode cannot be governing (in case a structure shows sufficient strength under the safe assumptions of the first LoA). In such a case, performing further analyses by using higher-order LoAs is not necessary.

The estimate of the first LoA can be refined progressively in successive LoAs by devoting more time to the estimate of the physical parameters involved. This can be done by using analytical or numerical procedures.

For higher LoA (second or third levels), the physical parameters of the limit state functions are typically evaluated through simplified analytical formulas accounting for the internal forces and other geometrical and mechanical parameters. These LoAs are still low time consuming and are usually sufficient to cover most design or assessment situations. Their use is advised e.g. for the tender and final design of new structures as well as for the assessment of existing structures.

more accurate estimate of the physical parameters can lead to significant savings by avoiding or limiting strengthening of the structures.

The limit states either refer to the entire structure, to structural elements or to local regions of elements.

12.4 Verification methods of structural performance

12.4.1 Verification principles

The structural performance of a whole structure or part of it should be described with reference to a specified set of limit states which distinguish desired states of the structure from adverse states. Limit states relevant for verification of structural performance are described in section 11.1.

In general terms, attainment of a limit state can be expressed as:

$$g(\mathbf{e}, \mathbf{r}) = 0 \tag{12.4-1}$$

where:

$g(\mathbf{e}, \mathbf{r})$ is the limit state function, $\mathbf{e}$ represents sets of loads (actions) and $\mathbf{r}$ represents resistance variables.

Conventionally, failure (i.e. an adverse state) is represented as:

$$g(\mathbf{e}, \mathbf{r}) \leq 0 \tag{12.4-2}$$

Although limit state equations representing different limit state conditions are various, the limit state function $g(\mathbf{e}, \mathbf{r})$ can often be subdivided into a resistance function $r(\mathbf{r})$ and a loading (or action effect) function $e(\mathbf{e})$. In such a case, Eq. (12.4-1) can be expressed as:

$$r(\mathbf{r}) - e(\mathbf{e}) = 0 \tag{12.4-3}$$

Consequently, Eq. (12.4-3) lends itself to the following representation of failure:

$$r(\mathbf{r}) \leq e(\mathbf{e}) \tag{12.4-4}$$

12.4.2 Verification methods for new structures

Verification of a structure with respect to a particular limit state is carried out via a model describing the limit state in terms of a function (called the limit state function) whose value depends on all relevant design parameters.

Verification of the limit states must be realised by a probability-based method. This Model Code recommends for verification of the limit states to use one of the methods:

– risk-informed method – see section 12.5.2;

– reliability-based method – see section 12.5.3;

The probabilistic safety format (sometimes referred to as full-probabilistic design method) allows to explicitly include the reliability requirements in terms of the reliability index β and the reference period. This may be used for structures to be designed and for existing structures in cases where such an increased effort is economically justified. However, it will seldom be used for the design of new structures due to lack of statistical data. The reliability-based format is more suited for the assessment of existing structures, in particular for the calculation of residual service life. The same holds for the risk-informed method.

– partial factor method – see section 12.5.4;

The partial safety factor format is the usual way of verifying structural design. It is a simplified verification concept, which is based on past experience and calibrated in such a way that the general reliability requirements are satisfied with a sufficient margin during a defined period of time. In the future, this safety format might also be applicable for the verification of service life, provided that sufficiently long term experience is gained or a sufficient amount of data becomes available for a calibration by the probabilistic method.

– global resistance factor method – see section 12.5.5;

In the global resistance format the resistance is considered on a global structural level, as compared to local verification of sections with partial safety factors. It is especially suitable for design based on non-linear analysis, where verification of limit states is performed by numerical simulations (see section 30.10).

– deemed-to-satisfy approach – see section 12.5.6;

The deemed-to-satisfy approach includes a set of appropriate values from a set of predetermined alternatives given in a standard. This method is the normal way of verifying service life design of new structures.

– design by avoidance approach– see section 12.5.7;

Design by avoidance is applicable both for the verification of traditional structural design and design for service life.

– design assisted by testing – see section 12.5.8.

For each specific limit state the relevant basic variables should be identified, that is the variables which characterize actions and environmental influences, properties of materials and soils, geometrical parameters, etc.

The variables pertaining to the various limit states may be time-dependent.

The variability of basic variables must be analysed based on the available information. In the case of the probabilistic format the basic variables are treated as random variables, or random fields. In the case of the partial factor format, the basic

In a component analysis with one dominating failure mode, the limit state condition can normally be described by a single limit state equation. In a system analysis, where more than one failure mode may be governing, several equations may apply.

The partial factor method represents the default approach to be used for the assessment of common existing structures.

After the partial factor method with the recommended values has been utilized, adjusting the partial factors, probabilistic methods, and the risk assessment approach can be appropriate for:

– overcoming the conservatism of partial factor methods;
– cases of structural failures with serious consequences;
– evaluating the efficiency of monitoring and maintenance strategies;
– making fundamental decisions concerning a whole group of structures.

When using the risk-informed assessment method, the reliability of an existing structure with significant failure consequences (economic, societal, environmental, loss of cultural heritage value, etc.) shall be verified by comparing the expected risk to the corresponding target risk level.

When using the probabilistic method, reliability of a structure shall be verified in terms of failure probability P_f.

As in design of new structures, the deemed-to-satisfy approach might be useful in the assessment. For instance, the rules for span-to-depth ratio can be applied in the assessment of deflections, in respect of deflection. Also, in-service performance of the

variables are treated as deterministic quantities. In the case of the global safety format, the global resistance is treated as a random variable.

For each limit state, models should be established that describe the behaviour of a structure. These models include mechanical models, which describe the structural behaviour, as well as other physical or chemical models, which describe the effects of environmental influences on the material properties. The parameters of such models should in principle be treated in the same way as the basic variables and model uncertainties must be regarded.

Models for the verification of the limit states can be either analytical (sections 30.1–30.6) or numerical (section 30.10), possibly supported by testing (section 30.11).

12.4.3 Verification methods for existing structures

The actual reliability of an existing structure shall be verified using one of the following types of approaches:

– risk-informed method – see section 12.5.2;
– reliability-based method – see section 12.5.3;
– semi-probabilistic methods:
  - partial factor method– see section 12.5.4;
  - global resistance factor method – see section 12.5.5;
– deemed-to-satisfy approach – see section 12.5.6;

existing structure to date is commonly a good indication of future performance when the future conditions will not be significantly different to those experienced so far in-service. Thus, the structural element could potentially be acceptable on the basis of satisfactory past performance; see section 28.9.

Assessment by avoidance approach can be utilised in the framework of LCM of the structure, as restrictions on use and/ or loading, controlling exposure (through implementation of protection measures), etc.

Examples of the cases when the recommended values of partial factors should be updated include:

– changes of the target reliability level and/ or of the sensitivity factors,

– decreased or increased variability of a basic variable such as a material property, geometry, load effect or model uncertainty,

– a basic variable having another probabilistic distribution than that assumed when deriving the γ-values in section 12.5.4.2.

Risk- informed structural performance verification normally involves the evaluation of

i) the probabilities of all relevant hazard events,

ii) the conditional probability of relevant structural element damage states given the relevant individual hazard events,

– assessment by avoidance approach– see section 12.5.7;

– assessment assisted by monitoring and testing – see section 12.5.8.

12.5 Verification formats

12.5.1 Level of refinement in verification formats

The basic method for reliability verification in the Model Code is the partial factor method provided in section 12.5.4. The recommended values of partial factors may be applied given that conditions of the structure under assessment comply with the assumptions detailed in a respective sub-section. Alternatively, a more advanced approach may be adopted. More advanced approaches include adjusting partial factors according to section 12.5.4.2.3, probabilistic approach in section 12.5.3.2, and risk analysis in section 12.5.2.

The higher-level methods should be applied when structure-specific conditions deviate from those commonly acceptable for structural design. It is recommended to start with the verification based on the recommended values of partial factors (if the assumptions on its application are fulfilled) and gradually proceed with higher levels of verification if needed.

12.5.2 Risk-informed format

The use of risk-informed structural performance verification is commonly motivated by:

- very high failure consequences (possibly also in relation with insufficient robustness),
- extremely high costs of increasing reliability (including the case of making fundamental decisions concerning a whole group of structures),

iii) the conditional probability of relevant structural system damage states given all relevant element damage states and given all relevant individual hazard events.

Depending on the specific structure and its context other requirements than those concerning life safety may apply, e.g. with respect to risk to the qualities of the environment, loss of reputation and finances. Such additional requirements should be identified as part of the risk screening process.

- lack of target reliability for the structure,
- unusually large uncertainties (in resistances and/ or load effects).

Design failure mode scenarios should be evaluated with respect to the efficiency of further life safety risk reducing measures.

The assessment of the necessity for further life safety risk reductions may be based for instance on the LQI approach according to ISO 2394 or the ALARP concept.

The analysis and results of risk- informed performance verification, including all assumptions that affect the conclusions, should be thoroughly documented such that

- an independent check can be carried out or
- the performance verification may be updated, if and when required during the design service life of the structure.

The risk- informed verification of structural design and assessments shall follow the flow illustrated in Figure 12.5-1.

It may be necessary to adapt the quantitative risk assessment by re-calculations during its implementation to account for the effects of measures introduced for reducing risks and improving life safety

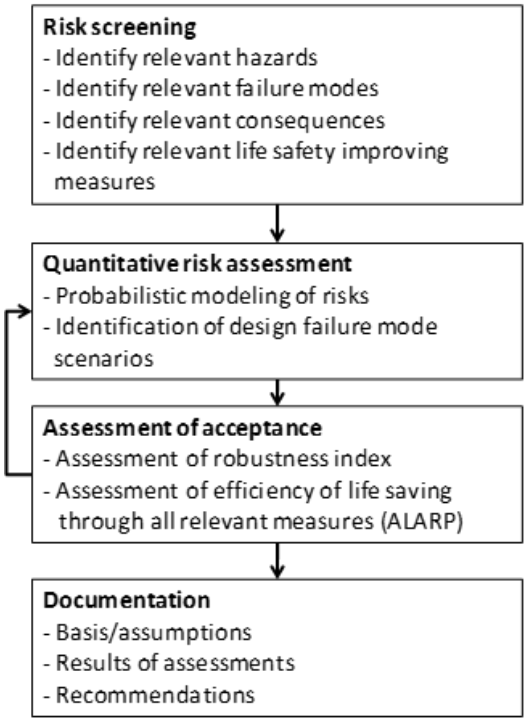

Figure 12.5-1*: Risk-informed verification of structural design and assessments*

Relevant information for the risk-informed performance verification includes information available from both the conceptual and the detailed design of the structure.

The risk-informed verification of structural design and assessments should take into account all relevant information concerning the structure, its intended use and envisaged measures for its life cycle integrity management.

Assigning hazards to a specific category can be based on the categorization provided in Table 12.5-1.

Table 12.5-1**:** *Categorization of potentially relevant hazards.*

Hazard	Category
Internal gas explosion	1
Internal dust explosion	1
Bomb explosion	2
Fire (natural or unintentional anthropogenic)	1
Fire (malevolence and vandalism)	2
Impact by vehicle, aircraft, ship, …	1

A structural risk-informed verification shall address the following three categories of hazards:

- Category 1: natural hazards and unintentional anthropogenic hazards;
- Category 2: malevolence and vandalism;
- Category 3: negligence and human errors.

Overloading	1
Earthquakes	1
Landslide	1
Mining subsidence	1
Tornado and Typhoons/Hurricanes/Cyclones	1
Avalanche	1
Rock fall	1
High groundwater	1
Flood	1
Storm surge	1
Volcanic eruption	1
Environmental attack	1
Tsunami	1
Vandalism	2
Public disorder effects	2
Design or assessment error	3
Material error	3
Construction error	3
User error	3
Lack of maintenance (deterioration)	3
Errors in communication.	3

Concurrent and causally dependent hazard events are often more severe than hazards acting individually. Structural risk assessments shall include these. Some examples include:

- Gas explosion and/or fire following an earthquake;
- Tsunami following an earthquake;
- Fire following either a gas explosion or bomb blast;
- Fire following a tornado or other wind storm; Structural deterioration, following damage from an accidental action.

In the assessment of consequences to life safety the quantification of risks to personnel may be supported by occupancy statistics.

The representation of hazards should encompass a probabilistic representation and analysis of the temporal and spatial extent of their occurrences and intensities as well as their causal and stochastic dependencies over time and space.

Depending on the situation, the non-exhaustive list of consequence types given in Table 12.5-2 should be considered.

Table 12.5-2 *Types of consequences associated with structural damage states.*

Consequence Type	Consequence

Health	Fatalities
	Injuries
	By damaging vital facilities (e.g. hospitals), spread of diseases
	Delayed long term health effects
	Psychological
Economic/Property	Damage to the building/structure
	Damage to surrounding properties
	Damage to contents
Business Continuity	Loss of income
	Loss of customers
	Inability provide vital services and/or activities
	Costs of detours and delays
	Costs to the economy of a region
Environmental	Reversible environmental damage.
	Irreversible environmental damage
Social and Political	Loss of reputation
	Increase of public fears
	Loss of political support/enforcement of stringent new measures.
	"Blight"/long-term evacuation

12.5.3 Reliability-based format

12.5.3.1 General

A probabilistic safety format must be applied in accordance with the principles and recommendations laid down in the JCSS PMC [12-3] and in the RILEM report 032 [12-4]. Note that according to ISO 2394 [12-8], failure is insufficient load-bearing capacity or inadequate serviceability of a structure or structural member, or rupture or

The main objective of a reliability analysis by the probabilistic approach is a probabilistic assessment of the safety of the structure by estimating the failure probability (or the reliability index β).

excessive deformation of the ground, in which the strengths of soil or rock are significant in providing resistance. It is emphasized that according to prEN 1990:2023, specified reliability requirements do not account for human errors and failure probabilities are thus not directly related to the observed failure rates, which are highly influenced by failures involving some effects of human errors.

Examples of cases characteristic of existing structures, where reliability of existing structures may need to be assessed, are the following:

- doubts about the performance of the structure;
- the expiration of (design or residual) service life (e.g. granted on the basis of design or an earlier assessment of the structure);
- detection of design or construction errors;
- occurrence of unusual incidents during use, which could have damaged the structure;
- a planned change of the use of the structure.

Examples of design situations that are out of the range of application of this Model Code and therefore must be analysed according to a probabilistic safety format are the following:

- actions and hazards outside the range covered by this Model Code;
- use of structural materials and combinations of structural materials outside the usual range of experience;
- new structural materials with properties outside the range covered by this Model Code;
- service life requirements outside the range covered by this Model Code;
- reliability level not covered by this Model Code;
- extraordinary structural systems;
- cases where failure would lead to exceptionally serious consequences such as major containments and storages of toxic materials, major offshore facilities, major dams and dikes, etc.
-

The probabilistic safety format is a suitable approach for the assessment of the performance of existing structures.

The probabilistic approach may support the design and assessment according to the partial factor format or deemed-to-satisfy approach, for example to ensure an appropriate robustness of structures or to account for specific requirements out of the range of application of this Model Code.

A proper choice of the distribution of the basic random variables is of importance, since the results of the reliability analysis can be very sensitive to the type of distribution adopted. For actions information is given in Section 13.

When specific information on the structure under investigation is available, partial factors for materials and actions can be updated. In the assessment of an existing structure, uncertainties in resistances and load effects can often be reduced on the basis of inspections, measurements, and tests. However, it is also possible that uncertainties to be considered in the reliability verification of the existing structure exceed those

12.5.3.2 Basic rules for probabilistic approach

The verification of a structure with respect to a particular limit state is carried out via estimation of the probability of occurrence of failure in a specified reference period and its verification against reliability requirements.

With the failure criteria formulated according to Eq. (12.4-2), the probability of occurrence of failure can be generally expressed as:

$$P_f = \text{Prob}\{g(\mathbf{e}, \mathbf{r}) \le 0\} = \text{Prob}\ \{M \le 0\} \qquad (12.5\text{-}1)$$

where:

$M = g(\mathbf{e}, \mathbf{r}) = 0$ represents the limit state

If the limit state function is expressed in the form of Eq. (12.4.-4) and parameters characterising actions, environmental influences, material and geometry are represented by the random variables **E** and **R**, the probability of occurrence of failure can be expressed as:

$$P_f = \text{Prob}\{r(\mathbf{R}) \le e(\mathbf{E})\} = \text{Prob}\ \{R \le E\} \qquad (12.5\text{-}2)$$

where

$E = e(\mathbf{E})$ and $R = r(\mathbf{R})$ are the basic random variables associated with loading and resistance, respectively.

Deriving representative values of environmental actions should be made with caution as they may be affected over time by the effects of climate change.

12.5.4 Partial factor format

12.5.4.1 General principles of partial factor method

12.5.4.1.1 Introduction

This section describes derivation and application of the partial factor method for the reliability assessment of new or existing structures conforming to ISO 2394 and EN 1990.

considered for a relevant design situation and the partial factors adopted for design might be insufficient.

Case-specific updating for model uncertainties is uncommon in engineering practice. The model uncertainty characteristics provided herein for resistance and load effects are consistent with the recommendations of Muttoni [12-5]; FprEN1992-1-1:2022 [12-6]; *fib* bulletin 80 [12-7] and JCSS Probabilistic Model Code [12-3]. They are based on the conventional probabilistic description of model uncertainties that is considered to be applicable for common reinforced concrete structures with a reasonable level of approximation.

In this section, the following types of distributions are assumed to provide appropriate models:

- Lognormal distribution for resistance variables, model uncertainties, and time-invariant components of variable load effects;
- Normal distribution for permanent loads;
- Gumbel distribution for extremes of time-variant components of variable loads.

The coefficients of asymmetry and kurtosis can provide valuable information for determining the appropriate theoretical model (probability distribution function) and can indicate that other than the aforementioned type of distribution is appropriate for a particular variable (e.g. normal distribution for a material property or lognormal distribution for annual maxima of ground snow loads).

If the actual distribution shows a multimodal character, the choice of one single distribution can lead to considerable error.

For non-dominant permanent actions, either $\xi\,\gamma_G(\alpha_E$ for a leading action) or $\gamma_G(\alpha_E$ for an accompanying action) can be applied, with α_E the sensitivity factor for the load effect, see section 12.5.4.1.4.

In accordance with this method, the partial factors γ_X should be derived from the actual distribution of the variable X under consideration (based on prior information, or results of tests or the combination of both).

The choice of probability distribution functions should be made with caution, considering possible skewness.

The partial factors provided in section 12.5.4 shall be applied in conjunction with appropriate load combination rules.

This section refers to the analysis of basic structural elements such as beams, slabs and columns investigated by simplified (analytical) models. The safety formats to be applied in conjunction with advanced numerical models are provided in section 30.10.

This separation is a practical simplification that may provide crude estimates when basic variables are statistically mutually dependent. Hence, constant values given in partial factors should be considered as approximations having a limited range of validity. Using fixed values for partial factors may fail for instance in the following cases:

– strongly non-linear limit state equations;
– mutually correlated variables;
– design by testing.

The general method of deriving the updated design values to be used in the partial factor method in the case of existing structures is given in ISO 2394 [12-8] and ISO 13822 [12-9].

These reliability margins seem to cover the whole set of uncertainties, but a part of the model uncertainties is commonly directly covered by the codified models themselves.

This does not exclude the fact that some actions (e.g. shrinkage) can be negligible in particular cases. What is to be considered as one individual action is defined in the

The partial factor format separates the treatment of uncertainties and variabilities originating from various causes by means of design or assessment values assigned to variables. In this Model Code the representative values of the variables and the partial safety factors are chosen in such a way that the reliability requirements, which are expressed in terms of β related to the reference period, are met.

In the case of existing structures, the same principles of the partial factor format can be applied as for new structures. However, the assessment values of the variables (i.e. the characteristic values and the partial factors) for existing structures need to be updated in order to guarantee that the reliability requirements for the assessment of existing structures are satisfied at the level discussed in section 11.3.1.3.

It shall be verified that in all relevant design or assessment situations, no limit state is exceeded when design or assessment values for actions, material properties and other relevant parameters are used. In common cases, the reliability margin can be expressed as $g = R_i - E_i$. A structure can be considered to comply with the reliability requirements when:

$$g = R_i - E_i \geq 0 \qquad (12.5\text{-}3)$$

where:

g is the reliability margin;
E_i is the design or assessment value of the effect of actions;
R_i is the design or assessment value of the corresponding resistance.

12.5.4.1.2 Basic variables

For basic variables, design values include reliability margins. For other variables, whose dispersion may be neglected or is covered by a set of partial factors, they are normally taken as equal to their most likely values.

In this Model Code the following variables are considered as basic:

– actions (F), unless specified otherwise in particular sections;

corresponding standard and explained in section 4.2.1 of CEB Bulletin 191 [12-1]. For prestress, see section 12.5.4.1.5.2 of this Model Code.

For these basic geometrical quantities, tolerances should be carefully fixed (see section 12.5.4.1.5.4) and controlled. For the other geometrical quantities, tolerances generally reflect usual practice. For all geometrical quantities it would not be realistic to specify tolerances less than twice the mean deviation expected or minimum attainable. As a consequence, tolerances may, according to the case considered, be either the basis for the design or necessary complements to the design.

Identifying and selecting the other relevant basic variables is one of the major responsibilities of a designer who faces a problem involving some unusual aspects.

To simplify the text, the term “design value” is applied in this section – the same principles apply for “assessment value” to be used in reliability assessment of existing structures.

– material or product properties (X), unless specified otherwise in particular sections (e.g. strengths (f), creep (φ) and friction coefficients (μ));
– some geometrical quantities (a);

– variables which account for the model uncertainties (θ).

Occasionally other variables should be considered as basic variables. This may be the case for the numbers of repetitions of loads in fatigue verifications.

12.5.4.1.3 Design or assessment condition

With reference to the representation of failure given in Eq. (12.4-2), the design condition can be expressed in terms of design values of basic variables as:

$$g\,(F_d, X_d, a_d, \theta_d, C) \geq 0 \qquad (12.5\text{-}4)$$

where:

F_d are design values of actions;

X_d are design values of material and soil properties;

a_d are design values of geometrical quantities;

θ_d are design values of the variables which account for model uncertainties;

C are serviceability constraints.

The relationship given in Eq. (12.4-4) lends itself to the following representation of the partial factor checking format:

$$e(F_d, \ldots) \leq r(X_d, \ldots) \qquad (12.5\text{-}5)$$

Typically, the design value x_{di} of any particular variable x_{ki} is given by:

$x_{di} = \gamma_i \, x_{ki}$ in case of loading variables; (12.5-6a)

Or

$x_{di} = x_{ki}/\gamma_i$ in case of resistance variables; (12.5-6b)

where:

x_{ki} is a characteristic value strictly defined as the value of a random variable which has a prescribed probability of not being exceeded (or of being attained); for material properties, a specified or nominal value is often used as a specified characteristic value;

γ_i is a partial safety factor with a value commonly greater than unity.

Some actions (e.g. non-closely bounded hydraulic actions) should be expressed in another way. Furthermore, for verifications relating to fatigue and vibrations, the format is generally different (see subsection xxx for verifications relating to fatigue and subsection xxx regarding limitation of vibrations).

For material properties other than strengths (e.g. modulus of elasticity, creep, friction coefficients) see the relevant parts of chapters xx and xx.

Numerical values of γ_M may be different in various parts of the limit state equation given by Eq. (12.4-4), especially for the calculations of $e(\mathbf{e})$ and $r(\mathbf{r})$.

12.5.4.1.4 Design values of basic variables

In this Model Code the design values of the basic variables are expressed as follows:

(a) *Design values of actions*:

$F_d = \gamma_F \, F_{rep}$ (12.5-7)

where:

F_{rep} is the representative value of the action, defined in section 12.5.4.1.5.1;

$\gamma_F = \gamma_f \gamma_{Sd}$ is a partial safety factor for the action effect;

γ_f is a partial safety factor for the action accounting for unfavourable deviations of action values from the representative value;

γ_{Sd} is a partial safety factor taking account of model uncertainty in estimation of the load effect.

Alternatively, the model uncertainty may be treated as a contributing variable of the load effect model and included in the quantification of γ_F.

It may be necessary to account for the effect of climate change and related uncertainties when deriving design values of climatic action effects.

(b) *Design values of material or product property*:

$f_d = f_k/\gamma_m$ (12.5-8a)

or in case uncertainty in the model is taken into account separately:

$f_d = f_k/\gamma_M = f_k/(\gamma_m \cdot \gamma_{Rd})$ (12.5-8b)

where:

For concrete and steel, γ_M usually covers the deviations of structural dimensions not considered as basic variables and includes a conversion factor η converting the strength obtained from test specimens to the strength in the actual structure. For practical applications, see the provisions regarding γ_M in section xxx).

Other factors – applied to f_d or implicitly included in design formulas – take into account the variations of strength due to non-standardized loading conditions.

γ_M may in some cases be substituted by one or two partial factors γ_{Rd}, applicable to the resistance, and a partial factor γ_m applicable to f_k.

Liquid levels representing hydraulic actions should in some cases be expressed as $a_k + \Delta a$, where a_k is a characteristic level and Δa an additive or reducing reliability margin.

For existing structures model uncertainties are commonly considered in the same way as in the design of new structures, unless previous structural behaviour (especially damage) indicates otherwise. In some cases model factors, coefficients and other design assumptions may be established from measurements on the existing structures (e.g. wind pressure coefficient or effective width values). For more information, see ISO 2394 [12-8], ISO 13822[12-9].

For a resistance parameter X, the updated design value x_d can be obtained from the following procedure according to ISO 13822 [12-9]:

$x_d = \mu(1 - \alpha\,\beta\,V)$ for a normal random variable (12.5-9a)

or

$x_d = \mu \exp(-\alpha\,\beta\,\sigma - 0.5\sigma^2)$ for a lognormal random variable (12.5-9b)

where:

x_d is the updated design value of X;

μ is the mean value of the resistance parameter X;

α is a sensitivity factor;

f_k is the characteristic value of a material property;

γ_m is a partial safety factor for a material property accounting for unfavourable deviation of the material or product properties from their characteristic values and the random part of conversion factors, if these are not modelled explicitly;

γ_{Rd} is a partial safety factor associated with the uncertainty of the (resistance) model and geometric deviations, if these are not modelled explicitly;

$\gamma_M = \gamma_m \cdot \gamma_{Rd}$ is a partial safety factor for a material property accounting for the uncertainties covered by γ_m and γ_{Rd}.

Alternatively, the model uncertainty may be treated as a contributing variable of the resistance model and included in the quantification of γ_M.

(c) *Design values of geometrical quantities* to be considered as basic variables are generally directly expressed by their design values a_d.

In the design of new structures the design values of the basic variables should be determined using representative values of the basic variables and partial safety factors given in section 12.5.4.1.5.1 (representation of actions), 12.5.4.1.5.2 (representation of prestress), 12.5.4.1.5.3 (representation of material properties), 12.5.4.1.5.4 (representation of geometrical quantities).

When assessing existing structures, reconsideration of the design values of the basic variables may be required. Guidance is given in section 12.5.4.1.5.1 to 12.5.4.1.5.4, where relevant.

β is the target reliability index;

V is the updated coefficient of variation;

$\sigma^2 = \ln(1 + V^2)$.

The values of α can be taken equal to those commonly used for new structures ($\alpha_{E50} = -0.7$ for the dominating parameter at the action side, $\alpha_{R50} = 0.8$ for the dominating parameter at the resistance side, and $0.4 \times -0.7 = -0.28$ for non-dominating parameters at the action side and $0.4 \times 0.8 = 0.32$ for non-dominating parameters at the resistance side, according to ISO 2394 [12-8]) when considering a reference period of 50 years.

The change of reference period from 50 years to 1 year requires adjusting the commonly applied sensitivity factors that are predominantly proportionally dependent on coefficients of variation of basic variables, V_X. According to the extreme value theory, the coefficient of variation of annual extremes of the variable action is larger than that of 50-year maxima, $V_{Q1} > V_{Q50}$, and thus $|\alpha_{Q1}| > |\alpha_{Q50}|$. In simplification for standardisation purposes, it might be considered for load effects and resistances that $|\alpha_{E1}| > |\alpha_{E50}|$ and $\alpha_{R1} < \alpha_{R50}$. Following Meinen and Steenbergen [12-10] and the recommendations of JCSS [12-11] it is recommended to consider $\alpha_{R1} = 0.7$ and $\alpha_{E1} = -0.8$ for the dominating resistance and load effect parameters, respectively, when considering a reference period of one year.

As an alternative procedure, one might also determine first a characteristic value x_k and calculate the design value by applying the appropriate partial factor γ_m.

Here:

$$x_d = x_k/\gamma_m \qquad (12.5\text{-}10)$$

And

$$x_k = \mu(1 - kV) \text{ for a normal random variable} \qquad (12.5\text{-}11a)$$

or

$$x_k = \mu \exp(-k\,\sigma - 0.5\sigma^2) \text{ for a lognormal random variable} \qquad (12.5\text{-}11b)$$

where:

k = 1.64 is generally used

For loads and geomechanical properties, a similar procedure may be applied, but usually other distribution types will be more appropriate. For more information, see ISO 2394 [12-8] and ISO 13822 [12-9].

For practical classifications of the most common actions, see the relevant Appendices to ISO 2394 [12-8] .

Seismic action is considered to be an accidental action or as a variable one, depending on the geographic location of the structure (see for instance EN 1998-1 [12-12]). In general, seismic action is considered as a variable action wherever the available information is sufficient to quantify the representative values of the seismic action with a prescribed probability of not being exceeded during a reference period t_{Ref}. If there is not sufficient information to this end (for instance in regions of very low seismicity), the seismic action is considered as accidental.

Permanent actions, self-weight included, although usually classified as fixed, may have to be considered as partially free where the effects are very sensitive to their variation in space, for example for static equilibrium and analogous verifications.

Soil reactions, for example soil pressure underneath foundation slabs or footings, are strongly influenced by soil–structure interaction. They should be determined by analysis, but the result should commonly be considered widely uncertain, especially the distribution in space.

Load arrangements are sometimes defined in the load standards. If several actions are free, the load cases (fixing the arrangements of all actions by taking into account their compatibility) are sometimes defined in the same documents. More information on load arrangements is given in section 4.2.3 of CEB Bulletin 191: [12-1] See also EN 1991-2 [12-13] for the load arrangements due to traffic actions.

When overloading has been observed in the past, it may be appropriate to increase representative values. When some loads have been reduced or removed completely, the representative values of the load magnitudes can be appropriately reduced and/or the partial factors can be adjusted. Guidelines are given in ISO 2394 [12-8] and ISO 13822 [12-9].

12.5.4.1.5 Representative values of basic variables

12.5.4.1.5.1 Representation of actions

Actions should be classified as:

- direct or indirect;
- permanent, variable or accidental;
- static, quasi-static or dynamic;
- closely bounded or not-closely bounded;
- fixed or free.

Reactions, mainly on supports, should also be distinguished from directly imposed actions. Although they are taken into account like actions for some verifications, they are in reality effects of actions and may need specific reliability measures in design.

For each free action, different load arrangements should be defined.

The representative values of actions to be applied in design of new structures are given below.

When assessing existing structures, the load characteristics should be introduced with values corresponding to the actual situation.

In the first two cases, G is considered as a mean value and should be calculated from nominal dimensions.

In the third case, it is defined as G_{sup} or G_{inf}.

The difference between G_{sup} or G_{inf} and the mean value G_m should not exceed 0.1 G_m. For some types of prestressed structures this maximum acceptable difference may have to be reduced to 0.05 G_m.

This case is mainly applicable to finishes and equipment. G_{sup} and G_{inf} may normally be defined as corresponding to 0.95 and 0.05 fractiles plus (or minus) the expected variation in time of G_m.

For structures to be designed for the most common variable actions these values are given in standards or codes associated with the same γ_F values as in this Model Code.

Ψ values depend on the model of the action – see ISO 2394 [12-8].

An example of the choice of the coefficients ψ_i according to EN 1990 [12-14], is given in Table 12.5-3.

Table 12.5-3: *The coefficients ψ_i according to EN 1990 [12-14].*

Action	ψ_0	ψ_1	ψ_2
Variable loads in buildings:			
Category A: domestic, residential	0.7	0.5	0.3

Representative values of permanent actions

Each permanent action is represented by a single representative value G if at least one of the following conditions is satisfied:

– the variability of the action in time and with regard to the design is small;

– the influence of the action on the total effect of the actions is small;

– it is evident that one of the two representative values (the upper or the lower) governs for all parts of the structure.

In the other cases, two representative values (upper and lower, G_{sup} and G_{inf}) should be defined, taking into account variations which can be foreseen.

Nominal numerical values of densities are given in subsection 13.2 for plain, reinforced and prestressed concrete. For future possible permanent equipment an upper value should be specified.

The representative values of the prestress are defined in section 12.5.4.1.5.2.

Representative values of variable actions

Each variable action may be represented by

– characteristic value Q_k;

– combination value $\Psi_0 Q_k$;

– frequent value $\Psi_1 Q_k$;

– quasi-permanent value $\Psi_2 Q_k$;

where:

Ψ_0 is the coefficient for the combination value of a variable action, taking into account the reduced probability of simultaneous occurrence of the most unfavourable values of several independent actions;

Ψ_1 is the coefficient for the frequent value of a variable action, generally representing the value that is exceeded 5% of the reference period;

Ψ_2 is the coefficient for the quasi-permanent value of a variable action, generally representing the value that is exceeded 50% of the reference period.

Category B: office areas	0.7	0.5	0.3
Category C: congregation areas	0.7	0.7	0.6
Category D: shopping areas	0.7	0.7	0.6
Category E: storage areas	1.0	0.9	0.8
Category F: traffic area, ≤ 30 kN	0.7	0.7	0.6
Category G: traffic area, 30–160 kN	0.7	0.5	0.3
Category H: roofs	0	0	0
Snow load: H ≤ 1000 m a.s.l	0.5	0.2	0
Wind loads on buildings	0.6	0.2	0

These values are associated with the methods of verification defined in section 12.5.4.2.

For structures to be designed, these values are normally defined by the competent public authority or by the client and correspond to the values beyond which a high probability of integrity of the structure can no longer be assured.

For ordinary facilities appropriate multiple representative seismic actions are the following:

– for the serviceability limit states as defined in section 11.1.2:
  - for the operational limit state: a “frequent” seismic action, expected to be exceeded at least once during the design service life of the structure (i.e. having a mean return period much shorter than the design service life);
  - for the immediate use limit state: an “occasional” earthquake, not expected to be exceeded during the design service life of the structure (e.g. with a mean return period of about twice the design service life);
– for the two ultimate limit states defined in section 11.1.3:
  - for the life safety limit state: a “rare” seismic action, with a low probability of being exceeded during the design service life of the structure;
  - for the near collapse limit state: a “very rare” seismic action, with very low probability of being exceeded in the design service life of the structure.

Besides, for some variable actions, specific representative values are defined for fatigue verifications.

Representative values of accidental actions

Each accidental action can be given by a single representative value, which is usually the design value A_d.

Representative values of seismic actions

A representative seismic action, with a prescribed probability of not being exceeded during a reference period t_{ref}, is defined for each limit state considered.

Depending on the use and importance of the facility, competent authorities will choose how many and which limit states should be verified as a minimum and to which representative seismic action they will be paired off.

For facilities whose consequences of failure are very high, the “very rare” seismic action may be appropriate for the life safety limit state. For those which are essential for the immediate post-earthquake period a “rare” seismic action may be appropriate for the immediate use limit state or even the operational limit state.

It is not sufficient to define a representative seismic action by scaling standard spectral shapes to a single ground motion parameter, notably the effective or the peak ground acceleration. Instead, the seismic action should be defined in terms of its full spectrum, throughout the full range of structural periods of relevance.

The basic definition of each representative seismic action is through its elastic response spectrum for a single-degree-of-freedom oscillator, as a function of viscous damping (the default value being 5% of critical damping). The spectrum applies to the top of the ground under free-field conditions and should be specified taking into account the site’s subsoil conditions and the local topography and geology, if relevant.

The elastic response spectrum is the same for the horizontal components of the ground motion, but should be specified separately for the vertical direction.

Normally it is sufficient to consider only the two horizontal translational components of the ground motion.

For buildings or similar structures, in general the vertical component may be neglected, with the possible exception (depending on seismicity) of:

– horizontal members with significant concentrated masses along the span;
– long horizontal spans (e.g. over 20 m) or cantilevers (e.g. over 5 m);
– prestressed horizontal members.

In bridges, the vertical component should always be taken into account for the design of prestressed decks or bearings.

The components of the seismic action should be taken to act simultaneously.

Simulated records are produced from mathematical models of the seismic source which dominates the seismic hazard, including the rupture event, the wave propagation through the bedrock to the site and through the subsoil to the ground surface.

Historic records should come from seismic events, with magnitude, fault distance and mechanism of rupture at the source that are consistent with those dominating the seismic hazard for the representative seismic action in question. Their travel path and the subsoil conditions of the recording station should preferably resemble those applying at the site.

Artificial (or “synthetic”) records, mathematically derived from the target elastic response spectrum, are not realistic if they are rich in all frequencies in the same way

Time-histories of the relevant components of the ground motion are needed for response-history analyses of the structure.

Preference should be given to historic or simulated records over artificial ones.

as the target spectrum. Therefore, perfect matching of the elastic response spectrum should be avoided.

The period range of interest may be taken to extend from twice to 20% of the fundamental period of the structure in the direction of the seismic action component in question.

For the estimation of peak response quantities, a minimum of seven such events is needed if the corresponding results of the analyses are averaged, or a minimum of three, if the most adverse peak response from the analyses is used. Many more seismic events than these minimum numbers are necessary for the estimation of residual deformations through non-linear response-history analyses.

To conform to the basic definition of the representative seismic action, each individual component time-history should be scaled so that the values of its elastic response spectrum for the default damping are at least 90% of the specified spectrum throughout the period range of interest.

A sufficient number of independent seismic events (in terms of component time-histories) should be used for the derivation of meaningful and robust statistics of the action effects.

12.5.4.1.5.2 Representation of prestress

Generally, during prestressing, the external forces are imposed and the associated elongations of the tendons are controlled. The prestressing load is determined at the time of its application

Prestressing forces are regarded as actions on the structure.

Even where prestress has to be considered as an action, a prestrain $\varepsilon_p(x, t)$ has commonly also to be considered in some parts of the calculations, especially in verifications with regard to the ULS. Where only immediate losses are considered $\varepsilon_p(x, t)$ is deduced from $P(x, t)$ by dividing it by the product E_pA_p. Where also long term losses are considered, this simple division may have to be supplemented by a correction transforming the relaxation of the tendon into a variation of strain.

Length and angular deviation may be considered small if the ratio $\Delta P_m(x, t)/P(0, 0)$ is not, at any time t, higher than 0.30. The partial factors for forces in prestressing tendons are discussed in section 12.5.4.2.4.

Representative values

Losses are numerically defined as mean values $\Delta P_m(x, t)$ in the subsections 16.2.5 and 16.2.6, assuming that the structure is submitted to the quasi-permanent combination of actions defined in section 12.5.4.5.

For a given set of tendons, considered in the same calculation of losses, the mean value of the prestressing force is defined as:

$P_m(x, t) = P(0, 0) - \Delta P_m(x, t)$ (ΔP in absolute value)

Two characteristic values of the prestressing force are also defined.

In the cases where the length and angular deviation of the tendons are not exceptionally large, the following formulas, although conservative if the angular deviation is small, may be used as acceptable approximations.

(a) Bonded tendons

$P_{k\,\mathrm{sup}}(x, t) = 1.1\, P_m(x, t)$

$P_{k\,\mathrm{inf}}(x, t) = 0.9\, P_m(x, t)$

(b) Unbonded tendons

$P_{k\,\mathrm{sup}}(x, t) = 1.05\, P_m(x, t)$

$$P_{k\,\inf}(x, t) = 0.95\, P_m(x, t)$$

12.5.4.1.5.3 Representation of material properties

Representative values

Strengths and other material properties to be considered as basic variables are represented by their characteristic values f_k (strength) or generally X_k or by their mean values.

When assessing existing structures, the material properties must be considered according to the actual state of the structure.

The significance of these values is shown in section 6.3 of CEB Bulletin 191 [12-1]. In exceptional cases, where an increase of the strength results in a decrease in reliability, upper characteristic values and specific γ_m values (smaller than 1) should be used.

When the original design documents are available and no serious deterioration, design errors or construction errors are observed or suspected, the characteristic value in accordance with the original design should be used. If appropriate, destructive or non-destructive inspections should be performed and evaluated using statistical methods. For more information, see ISO 2394 [12-8] and ISO 13822 [12-9].

Mean and characteristic values for strength properties of concrete and steel are given in section 14 and section 15. Time-dependent concrete hardening often has a positive effect while long-term permanent loading might have a negative one.

12.5.4.1.5.4 Representation of geometrical quantities

Representative values

The representative values of geometrical quantities to be applied in design of new structures are defined below.

When assessing existing structures the dimensions of the structural elements must be taken according to the actual state of the structure.

When the original design documents are available and no change in dimensions has occurred or other evidence of deviations is known, the nominal dimensions in accordance with the original design documents should be used in the analysis. These dimensions must be verified by inspection to an adequate extent. For more information, see ISO 2394 [12-8] and ISO 13822 [12-9].

In this section, only geometrical quantities representing the structure are considered. For most of the quantities, their deviations within the specified tolerances should be considered as statistically covered by γ_F and γ_M factors. Only those quantities, which might in some verification be one of the main variables, should, in those verifications only, be taken as basic.

Unintentional eccentricities, inclinations and parameters defining curvatures affecting slender columns, walls and shells, are unique geometrical quantities defined in this Model Code to be taken into account as basic variables, if not specified

The depths of reinforcement in thin members are taken into account by modifying their nominal values by additive reliability margins.

Dimensions in slabs larger than intended may significantly increase the self-weight, whereas smaller dimensions and/or lever arms of steel bars may significantly reduce the resistance. Similarly, a concrete cover smaller than the nominal value may endanger the durability or the anchorage resistance of steel bars. An unintended inclination of columns may disproportionately increase their action effects.

Due to the complicated nature of the related phenomena, no explicit figure of general validity can be given on the amount of such performance reduction, but it is considerably less than 4%.

In the absence of a more justified set of tolerances, the following limitations may apply:

(a) Table 12.5-4: *Tolerances for concrete sectional dimensions, according to ISO 22966 [12-15] (for Tolerance Class 1 and 2).*

Elements and dimension (mm)	**Permitted deviation** Δ (mm)	
Beams slabs and columns	Class 1	Class 2
a < 150 mm	± 10 mm	± 5 mm
a = 400 mm	± 15 mm	± 10 mm
a ≥ 2500 mm	± 30 mm	± 30 mm

with linear interpolation for intermediate values.

(b) Table 12.5-5: *Tolerances for the location of ordinary and prestressing reinforcement, according to ISO 22966 [12-15] (for Tolerance Class 1 and 2).*

Height of cross-section h (mm)	**Permitted deviation** Δ (mm)	
Ordinary reinforcement	Class 1	Class 2
a < 150 mm	± 10 mm	± 5 mm
a = 400 mm	± 15 mm	± 10 mm

otherwise. The other geometrical quantities are as specified in the drawings of the design.

The basic geometrical variables are directly fixed as design values in the chapters where the relevant limit states are treated.

Tolerances

The possible deviations in the geometry of the concrete elements, of the cover or of the position of steel, must not alter significantly either the SLS or the ULS performance of the relevant elements.

As a general rule for these geometrical basic variables, the corresponding specified tolerances may be taken equal to their design values of the deviations divided by 1.2 and should be controlled accordingly.

For the other geometrical variables, the values of the material partial safety factors included in this Model Code are meant to cover small reductions of performance (resistances, mainly) which may result from their deviations.

Depending on the quality assurance scheme applicable, relevant tolerance values should be respected for each category of possible deviations under well-specified conditions of measurements and evaluations. Possible foreseen higher deviations should lead to additional design steps, taking into account all the consequences of deviations that exceed the specified tolerances.

a ≥ 2500 mm	± 20 mm	± 20 mm
Prestressing reinforcement		
h ≤ 200 mm	± 0.03 *h*	
h > 200 mm	the smaller of ± 0.03 *h* or ± 30 mm	

(c) Tolerance of cover: $c_{nom} - c_{act} < 10$ mm.

(d) Table 12.5-6: *Tolerances of unintentional deviations of columns, walls, beams and slabs according to ISO 22966 [12-15] (for Tolerance Class 1).*

Elements and type of deviation	**Permitted deviation** Δ (mm)
Columns and walls	
Inclination of a column or wall at any level in a single- or multi-storey building	The larger of h/300 or 15 mm where h is free height
deviation between centre	the larger of t/30 or 15 mm but not more than 30 mm where t = (t1 + t2)/2
Lateral deviation of a column between adjacent storey levels	The larger of t/30 or 15 mm but not more than 30 mm where t = (t1 + t2)/2
location of a column or a wall at any storey level, from a vertical line through its intended centre at base level in a multistorey structure	the smaller of 50 mm or Σh/(200 n 1/2), where h is free height and n is the number of storeys and n > 1
Beams and slabs	
location of a beam-to-column connection measured relative to the column	the larger of ± b/30 or ± 20 mm, where b is dimension of column in the same direction as Δ
position of bearing axis of support when structural bearings are used	the larger of ± l/20 or ± 15 mm, where l is intended distance from edge

The tolerance values apply to the structure, after compaction and hardening of the concrete.

The basic design rules differ according to the limit state under consideration.

In some cases, defined in other chapters, some limit state calculations may be substituted by detailing rules or special provisions.

These numerical values are considered to be appropriate in the design of new structures in countries with the socioeconomic conditions similar to those in most European countries. In countries where different conditions prevail (and possibly depending on the type of building or civil engineering works), γ factors for design may be adjusted.

12.5.4.2 Basic rules for partial factor approach

12.5.4.2.1 General

In design and assessment by the partial factor method it should be proven that the structure, given the design (assessment) values for the basic variables, does not exceed the relevant limit states. The basic design rules given in this section are applicable to the limit states as defined in chapter 11.

In principle, all relevant limit states should be considered, as well as all relevant design situations, load arrangements and load cases and combinations of actions.

The numerical values of γ factors given in the following sections are applicable to the design of new structures and assessment of existing structures.

In the following sections explanations are given with regard to updating the design (assessment) values of the variables. After the evaluation of the updated design (assessment) values, one may check the structural reliability of existing structures using the standard procedures for new structures.

The numerical values of γ factors given in the following sections are applicable to the design (assessment) of buildings and civil engineering works not subject to variable actions having an exceptional variability.

At the action side, at least the following variables should be differentiated:

- self-weight of the structure;
- other permanent loads;
- variable actions;
- prestressing;
- other actions (earthquake, fire, accidental actions etc.).

At the resistance side, at least the following parameters should be differentiated:

- concrete strength;
- steel strength;
- model uncertainty.

Relationships Eq. (12.5-12) and (12.5-13) assume that:

- Resistance is obtained as a linear product of resistance model uncertainty, geometrical property, and of material property (which may include a conversion factor);
- Characteristic value of the material property (commonly strength) corresponds to a 5% fractile of the respective lognormal distribution;
- Unity characteristic values are considered for resistance model uncertainty and conversion factor, respectively;
- Nominal value of a geometrical property corresponds to its mean;
- Statistical uncertainty is reflected in the estimate of characteristic resistance.

Factor η is commonly not considered for reinforcement.

Representative probabilistic models for concrete compressive strength, conversion factor, and yield strength of steel reinforcement are specified in section 12.5.4.2.4.2 for design and in section 12.5.4.2.4.3 for assessment. The models for resistance model uncertainties and geometrical properties that can be common to various design and assessment situations are presented in this section.

For non-deteriorated structures, Table 12.5-7 provides the indicative values of resistance model uncertainty characteristics for the governing failure modes in reinforced concrete structures as given by (Muttoni [12-5]). Reinforcement properties

12.5.4.2.2 Derivation of partial factors

12.5.4.2.2.1 Material factor γ_M

Following the definition of the design (assessment) value R_d of the resistance variable R provided in section 12.5.4.1.4, the partial factor γ_M should be obtained as:

design situation:

$$\gamma_M = R_k / R_d \approx [\exp(-1.645 V_f)] / [\mu_{\theta R}\, \mu_a\, \mu_\eta \exp(-\alpha_R\, \beta_t \sqrt{(V_{\theta R}^2 + V_a^2 + V_\eta^2 + V_f^2)})] \quad (12.5\text{-}12)$$

assessment situation:

$$\gamma_M = R_k / R_d \approx [\exp(-1.645 V_{fis})] / [\mu_{\theta R}\, \mu_a \exp(-\alpha_R\, \beta_t \sqrt{(V_{\theta R}^2 + V_a^2 + V_{fis}^2)})] \quad (12.5\text{-}13)$$

where

a	geometrical property;
f	material property (based on control specimens for concrete);
f_{is}	material property based on specimens taken from or on the structure;
V	coefficient of variation;
α_R	sensitivity factor provided in section 12.5.4.1.4;
β_t	target reliability index provided in section 11.3.2;
η	conversion factor – commonly ratio between in-situ strength and strength based on control specimens; for concrete, $\eta = f_{c,is} / f_c$ primarily accounts for the effects of casting, vibration, curing, consolidation of fresh concrete, and possibly differences in strength of horizontal and vertical members;
θ_R	resistance model uncertainty;
μ	bias in a variable – systematic deviation of random values of the variable from its characteristic (nominal) value, expressed as the ratio of mean to characteristic (nominal) value.

The subscripts “k” and “d” denote characteristic and design (assessment) value, respectively.

The distinction should be made between resistance model uncertainties for the failure modes dominated by reinforcement or concrete properties. In the absence of

are mostly dominating in bending whereas concrete compressive strength dominates in compressive crushing of columns without buckling. Resistance model uncertainties are assumed to have a lognormal distribution (JCSS PMC [12-3]); for further information see also Annex A of *fib* Bulletin 80 [12-7].

more detailed information, the resistance model uncertainty characteristics provided in Table 12.5-7 may be considered.

Table 12.5-7: *Indicative values of resistance model uncertainty characteristics.*

Failure mode	$\mu_{\theta R}$	$V_{\theta R}$	*Note*
Bending	1.09	0.045	Considered in 12.5.4.2.4 to derive γ_S.
Crushing of columns	1.02	0.07	Considered in 12.5.4.2.4 to derive γ_C.

Examples of the geometrical property a are an effective depth of the beam exposed to bending and/ or shear or a concrete section area of the column in compression.

Based on (Muttoni [12-5]), common levels of uncertainty in design may be associated with:

- $\mu_a \approx 1$ and $V_a \approx 0.04$ for concrete section area – for a square column with fully correlated dimensions $b \times b$, the coefficient of variation can be obtained as $V_a = 2V_b = 0.04 \times (300 \text{ mm} / b)^{2/3}$; when the two widths are statistically independent, the factor of 2 should be replaced by $\sqrt{2}$ and thus $V_a = 0.0283 \times (300 \text{ mm} / b)^{2/3}$
- $\mu_a \approx 0.95$ and $V_a \approx 0.05$ for effective depth $d = 200$ mm; for other effective depths $\mu_a = 1 - 0.05(200 \text{ mm} / d)^{2/3}$ and $V_a = 0.05(200 \text{ mm} / d)^{2/3}$ may be considered

Reduced uncertainty in the geometrical property is expected for instance in the following cases:

- when a decisive geometrical property is measured in the finished structure – $\mu_a = 1$ and $V_a = 0.015$ may then be considered for section area, and $\mu_a = 1$ and $V_a = 0.01$ for effective depth;

In design, common levels of uncertainty in a geometrical property a on which resistance is based may be taken into account, considering respective quality of execution. In the absence of structure-specific information, normal uncertainty in geometry may be considered, characterised by $\mu_a = 1$ and $V_a = 0.04$ for concrete section area (considered to derive the recommended γ_C-values for design) and $\mu_a = 0.95$ and $V_a = 0.05$ for effective depth (considered for γ_S for design).

In assessment of the existing structure, uncertainty in the geometrical property a should be verified by in-situ measurements. The value of V_a should include all uncertainties related to measurements including measurement error. When a decisive geometrical property is measured in the existing structure, uncertainty in geometry may be characterised by $\mu_a = 1$ and $V_a = 0.015$ for concrete section area (considered for γ_C in assessment) and $\mu_a = 1$ and $V_a = 0.01$ for effective depth (considered for γ_S in assessment).

– an increased execution quality and quality control are required in design – for instance when higher execution quality is reached (ensuring that the geometrical deviations of Tolerance Class 2 according to EN 13670 [12-16] are fulfilled), $\mu_a = 1$ and $V_a = 0.02$ may be considered for section area. For effective depth, $\mu_a \approx 0.975$ and $V_a \approx 0.025$ may be considered for $d = 200$ mm; for other effective depths $\mu_a = 1 - 0.025(200 \text{ mm} / d)^{2/3}$ and $V_a = 0.025(200 \text{ mm} / d)^{2/3}$ may be considered.

Note that the variability of effective depth may be ignored ($\mu_a = 1$ and $V_a = 0$) if design resistance is based on the design value of effective depth obtained as $a_d = a_{nom} - \Delta a$ where a_{nom} is the nominal value of the effective depth determined on the basis of the nominal cover; and Δd is the deviation value of the effective depth:

– $\Delta d = 15$ mm for reinforcing and post-tensioning steel;
– $\Delta d = 5$ mm for pre-tensioning steel.

The partial factor γ_M should be applied to resistance as follows:

$$R_d = R(f_{ck}/\gamma_C, f_{yk}/\gamma_S, \ldots) \qquad (12.5\text{-}14)$$

where R denotes the resistance function; and γ_C and γ_S are the partial factors for concrete and reinforcement, respectively.

12.5.4.2.2.2 Permanent action factor γ_G

Following the definition of the design (assessment) value G_d of the load effect of a permanent action G provided in section 12.5.4.1.4, the partial factor γ_G should be obtained as:

$$\gamma_G = G_d / G_k \approx 1 - \alpha_E \beta_t \sqrt{(V_{\vartheta E}^2 + V_g^2)} \qquad (12.5\text{-}15)$$

where:

g permanent load;

α_E sensitivity factor provided in section 12.5.4.1.4;

β_t target reliability index provided in section 11.3;

θ_E load effect model uncertainty.

The subscripts “k” and “d” denote characteristic and design (assessment) value, respectively.

Relationship Eq. (12.5-15) assumes that:

- Characteristic value of the permanent load effect corresponds to its mean value.
- The permanent load effect is normally distributed.
- Unity characteristic value is considered for unbiased load effect model uncertainty.
- Nominal value of a geometrical property corresponds to its mean.

When a favourable load effect is considered (e.g. in the case of stabilising forces), Eq. (12.5-15) can be used along with an appropriate value of α_R.

See chapter 13 for further details.

These conventional values yield design (assessment) values comparable to those reported in Annex A of *fib* bulletin 80 [12-7] and the references provided therein; see also chapter 13.

It is noted that $V_{\theta E} > 0.075$ may be associated with the load effects on complex systems. In particular for permanent load effects, $V_{\theta E} < 0.075$ may apply for simple static systems with well-defined boundary conditions where load effects can be determined without uncertainties and permanent actions are based on measurements of actual volume densities and dimensions.

See chapter 13 for further details.

In general, maxima of the variable load effect over a given reference period, Q_{tref}, depend on the time-variant component $q_0(t)$, the time-invariant component C_0 (e.g shape, gust, roughness and other factors affecting wind pressure), and on uncertainties in the load effect θ_E. Probabilistic model for Q_{tref} can often be based on:

$$Q_{tref} = \vartheta_E \times C_0 \times \max_{tref}[q_0(t)] = \vartheta_E \times C_0 \times q_{0,tref} \qquad (12.5\text{-}17)$$

Probabilistic approach (see section 12.5.3.2) is normally used to estimate the partial factor from relationships (12.5-16) and (12.5-17).

Where a Gumbel distributed time-variant component is a dominating source of variability (commonly for climatic and imposed loads), the partial factor γ_Q can be estimated as:

$$\gamma_Q = \delta \times \mu_{Q,tref} \times \{1 - V_{Q,tref}[0.45 + 0.78\ln(-\ln \Phi(-\alpha_E \beta_t))]\} \qquad (12.5\text{-}18)$$

Model uncertainty in load effect is affected by many factors that should be considered in reliability analysis. For complex or redundant systems, the sensitivity of load effect on the assumptions adopted in reliability analysis should be verified and conservative choices may need to be taken.

The probabilistic model of θ_E should reflect all possible biases and uncertainties in the load modelling. In the absence of further information, $\mu_{\theta E} = 1$ and $V_{\theta E} = 0.075$ may be considered (these values are considered to derive the recommended values of partial factors for unfavourable load effects for design and assessment).

The probabilistic model of permanent load g should reflect all possible biases and uncertainties in modelling of the permanent load.

For design, the indicative values for V_g provided in chapter 13 may be used in the absence of detailed statistical data.

In the case of assessment of the existing structure, a V_g-value should be based on measurements of actual dimensions and volume densities specified from cores. When a permanent load effect influences structural reliability insignificantly, the indicative values for V_g provided in chapter 13 may be used.

12.5.4.2.2.3 Variable action factor γ_Q

Following the definition of the design (assessment) value Q_d of an unfavourable load effect of the variable action Q provided in section 12.5.4.1.4, the partial factor γ_Q should be obtained as:

$$\gamma_Q = Q_d / Q_k = F_{Q,tref}^{-1}[\Phi(-\alpha_E \beta), t_{ref}] / Q_k \qquad (12.5\text{-}16)$$

where:

Q_d design (assessment) value of variable action effect;

Q_k characteristic value of variable action effect;

$F_{Q,tref}^{-1}$ inverse cumulative distribution function of maxima of the variable load Q_{tref} during a reference period t_{ref}, for which a target reliability index β_t (see section 11.3) is specified;

where

δ correction factor to account for imprecision of the approximation (for numerical values see section 12.5.4.2.4.2 for design and in section 12.5.4.2.4.3 for assessment)

$\mu_{Q,t\text{ref}}$ mean of maxima of variable load effect, related to t_{ref} (relatively to its characteristic value)

$V_{Q,t\text{ref}}$ coefficient of variation of variable load effect, related to t_{ref}

The characteristics of $Q_{t\text{ref}}$ can be estimated as:

$$\mu_{Q,t\text{ref}} \approx \mu_{\theta E} \times \mu_{C0} \times \mu_{q,t\text{ref}} \quad (12.5\text{-}19)$$

$$V_{Q,t\text{ref}} \approx \sqrt{(V_{\theta E}^2 + V_{C0}^2 + V_{q,t\text{ref}}^2)} \quad (12.5\text{-}20)$$

where

θ_E load effect model uncertainty (see 13.4)

C_0 time-invariant component of variable load (see 13.3)

$q_{t\text{ref}}$ time-variant component of variable load (13.3, note that the symbol of "$Q_{t\text{ref}}$" is used in 13.3)

Where the lognormally distributed time-invariant factors, θ_E and C_0, are a dominating source of variability (commonly for road traffic loads), the partial factor γ_Q can be estimated as:

$$\gamma_Q = \delta \times \mu_{Qt\text{ref}} \times \exp(-\alpha_E \beta_t V_{Q,t\text{ref}}) \quad (12.5\text{-}21)$$

The numerical values of δ are provided in 12.5.4.2.4.2 and 12.5.4.2.4.3. The characteristics of $Q_{t\text{ref}}$ can be estimated according to Eq. (12.5-19) and (12.5-20).

Assuming the Gumbel distribution and independent annual extremes of $q_{t\text{ref}}$, the characteristics of extremes of q related to t_{ref2} can be obtained from mean and standard deviation of q related to t_{ref1}, $\mu_{q,t\text{ref1}}$ and $\sigma_{q,t\text{ref1}}$ respectively:

$$\mu_{q,t\text{ref2}} = \mu_{q,t\text{ref1}} + 0.78\, \sigma_{q,t\text{ref1}} \ln (t_{\text{ref2}} / t_{\text{ref1}}); \sigma_{q,t\text{ref2}} = \sigma_{q,t\text{ref1}} \quad (12.5\text{-}22)$$

α_E sensitivity factor provided in section 12.5.4.1.4.

Favourable effects of variable actions are not considered in reliability verifications based on the partial factor method.

Model uncertainty in a variable load effect should be considered according to section 12.5.4.1.2.

Information on probabilistic modelling of C_0 and $q_{0,tref}$ is provided in chapter 13.

The probabilistic model of the time-invariant and time-variant components, C_0 and $q_{0,tref}$ respectively, in Eq. (12.5-16) should reflect all possible biases and uncertainties in modelling of the variable load.

12.5.4.2.2.4 Combination of actions

These rules must be amended for accidental situations (see the section regarding general rules for combinations of actions in the sequel) and if possible simplifications or refinements regarding combinations of actions are applied – see Eq. (12.5-17).

In most cases, γ_F factors should be applied globally as follows:

$$E_d = E\left\{\gamma_G G + \gamma_P P + \gamma_Q\left(Q_{1k} + \sum_{i>1}\Psi_{oi}Q_{ik}\right)\right\} \qquad (12.5\text{-}23)$$

Eq. (12.5.4-24) is the more general. Particular cases are mainly those where:

– E_d is an under-proportional function of the actions (or the principal of them); in these cases Eq. (12.5.4-23) may be unsafe, or:

– the effects of some actions have a sense opposite to the effects of the other actions and are of the same order of magnitude; in these cases Eq. (12.5-27) may be too conservative (this may be the case for the isostatic effects of prestress).

In particular cases, defined in the relevant sections of other chapters or to be identified by judgement, for persistent or transient situations, this formula may be substituted by:

$$E_d = \gamma_{Sd} E\left\{\gamma_g G + \gamma_P P + \gamma_q\left(Q_{1k} + \sum_{i>1}\Psi_{oi}Q_{ik}\right)\right\} \qquad (12.5\text{-}24)$$

These two formulas are partially symbolic and should be applied by following in detail the combination rules given in the sequel.

The use of a sum of permanent actions $\gamma_{G,i}G_{k,i}$ instead of a single permanent load G is allowed.

This rule (not splitting γ_M into γ_m and γ_{Rd}) is not applicable in design by testing.

γ_M factors should generally be applied globally.

For the definition of individual actions, see subsections 1.2.1 and 6.2.1 of CEB Bulletin 191 [12-1].

For the Ψ factors, see the information regarding representative values of variable actions in section 12.5.4.1.5.1.

The combinations of design (assessment) values to be taken into account for applying Eqs. (12.5-23) and (12.5-24) are as follows, in symbolic presentation:

– fundamental combinations applicable for persistent and transient situations

$$E_d = E\left\{\gamma_{G\,sup}G_{sup} + \gamma_{G\,inf}G_{inf} + \gamma_P P + \gamma_{Q,1}Q_{k,1} + \sum_{i>1}\gamma_{Q,i}\Psi_{0,i}Q_{k,i}\right\} \qquad (12.5\text{-}25)$$

Ψ factors take account of the reduced probability of simultaneous occurrence of actions. The choice between $\Psi_{1,1}Q_{k,1}$ or $\Psi_{2,1}Q_{k,1}$ depends on the type of accidental design situation, such as impact, fire or survival after an accidental event or situation.

In seismic situations masses are consistent with the gravity loads corresponding to the combination $G_{\text{sup}} + G_{\text{inf}} + \sum_{i \geq 1} \Psi_{2,i} Q_{k,i}$.

Prestressing P should be added, if relevant.

In most cases some variable actions, which obviously are not the leading ones for a given verification, need not be considered as $Q_{k,1}$.

For fire situations, apart from the temperature effect on the material properties, A_d should represent the design value of the indirect thermal action due to fire.

In general, there will be two different levels of A_{Ed}, one for each ultimate limit state introduced in section 3.3.1.2.

The cases of incompatibility or negligible compatibility are very numerous. They are given in the codes or standards on actions or identified by judgement (e.g. snow and maximum climatic temperature).

If justified to be relevant for the structure under investigation, prior information can be combined with new information obtained e.g. by measurements and/ or tests through Bayesian updating [12-3] + [12-4] in any level of a verification format.

– accidental combinations, applicable for accidental situations

$$E_d = E\left\{G_{sup} + G_{inf} + P + \left(A_d \text{ or } 0\right) + \left(\Psi_{1,1} \text{ or } \Psi_{2,1}\right)Q_{k,1} + \sum_{i>1} \Psi_{2,i} Q_{k,i}\right\} \quad (12.5\text{-}26)$$

– seismic combinations, applicable for seismic situations

$$E_d = E\left\{G_{sup} + G_{inf} + P + A_{Ed} + \sum_{i \geq 1} \Psi_{2,i} Q_{k,i}\right\} \quad (12.5\text{-}27)$$

In these combinations:

– G_{sup} and G_{inf} refer to the unfavourable and favourable parts of the permanent actions when the variation of the permanent , respectively;

– P refers to prestressing;

– $Q_{k,i}$ refers to any variable action, in succession;

– A_d denotes the unique accidental action associated with the accidental situation, if this situation is due to this action. If it is due to another event or to a past action, A_d is substituted by 0;

– A_{Ed} denotes the design seismic action having a prescribed probability of not being exceeded during the reference period t_d and associated with the ultimate limit state of interest in this specific seismic situation;

The actions to be included in any combination are only those that are mutually compatible or are considered as such, as an acceptable approximation. Non-simultaneous actions should be considered in the same combination if their effects are simultaneous.

12.5.4.2.3 Recommended values of partial factors for ULS

12.5.4.2.3.1 General

This section provides recommended values of partial factors for ULS – section 12.5.4.2.1.1 for design situations and section 12.5.4.2.1.2 for assessment situations, given that conditions of the structure under assessment comply with the assumptions detailed in the left-hand column in a respective sub-section. Alternatively, a more

Examples of the cases when the γ-values should be updated include:

– changes of the target reliability level and/ or of the sensitivity factors,

– decreased or increased variability of a basic variable such as a material property, geometry, load effect or model uncertainty,

– a basic variable having other probabilistic distribution than that assumed when deriving the γ-values in section 12.5.4.2.

The partial factors in Table 12.5-8, Table 12.5-9 and Table 12.5-10 are derived using the approach presented in section 12.5.4.2.3, considering:

- the informative target reliability indices β related to a 50-year reference period as provided in Table 11.2-3;
- sensitivity factors $\alpha_R = 0.8$ and $\alpha_E = -0.7$;
- the model uncertainty characteristics provided in Table 12.5-7 for resistances and in section 12.5.4.1.2 for load effects;

Similar values of the partial factors are obtained when the informative annual target reliability indices β_t given in Table 11.3-2 (normal relative cost of safety measure) along with $\alpha_R = 0.7$ and $\alpha_E = -0.8$ are considered.

Further, the partial factors in Table 12.5-8 are based on:

- representative characteristics for concrete:
  - $V_{fc} = 0.1$ ($V_{fc} = 0.06$ could be considered for precast concrete);
  - $\mu_\eta = 0.95$ and $V_\eta = 0.12$ (conversion factor for in-situ cast concrete – ratio between in situ and control specimen concrete compressive strength; for precast concrete bias close to unity and lower V_η may apply (Moccia et.al [12-17] ; Muttoni, May 2021 [12-##])
- $V_{fy} = 0.045$ for reinforcement.

advanced approach to specify partial factors can be adopted. More advanced approaches include adjusting the partial factors according to section 12.5.4.2.3, probabilistic approach in section 12.5.3.2, and risk analysis in section 12.5.2.

The higher-level methods should be applied when structure-specific conditions deviate from those commonly acceptable for structural design. Special care in updating should be taken when characteristic values are defined in a different way than those assumed in section 12.5.4.2.3. It is recommended to start with the verification based on the recommended values of partial factors (if the assumptions on its application are fulfilled) and gradually proceed with higher levels of verification if needed.

In particular for existing structures, the partial factors should be updated considering structure-specific information.

12.5.4.2.3.2 Recommended values of partial factors for structural design

For design of new structures, this Model Code recommends applying the partial factors provided in Table 12.5-8 for materials, Table 12.5-9 for permanent actions and Table 12.5.4-10 for variable actions.

Table 12.5-8: *Recommended values of the partial factors γ_M for materials (design).*

CC	γ_C	γ_S		
		normal unc. in geo*	higher exec. qual.**	finished str.***
CC1	1.4	1.1	1.05	1.0
CC2	**1.5**	**1.15**	1.075	1.05
CC3	1.6	1.175	1.1	1.05

In Table 12.5-8 uncertainty in geometry is described for three cases—normal, higher execution quality and measurements in the finished structure—according to section 12.5.4.1.1. A reference level of effective depth of 200 mm is considered; for deeper structural members lower variability in geometry may be considered according to section 12.5.4.1.1, resulting in lower values of γ_S.

Note that the creep and shrinkage coefficients given in chapter ??? represent mean values. A characteristic value or a design value may need to be defined for such coefficients; for instance in design of joints or when second order effects need to be considered. These coefficients may be associated with large uncertainty, expressed through coefficients of variation exceeding 0.25.

The value of μ_η represents concretes in tops of columns; for other governing regions the bias may be increased to 1.0 or to 1.05 for bottoms of columns. Similarly, for smaller members ($h < 450$ mm) lower biases $\mu_\eta = 0.95$ may be considered while for larger members ($h \geq 450$ mm) more favourable values may be applied, $\mu_\eta = 1.03$.

Further guidance may be provided to reflect the effects of non-standard quality of execution, various types of binding materials, seasonal aspects, size of the structure etc.

Further to the above assumptions common to Table 12.5-8 to Table 12.5-10, the partial factors in Table 12.5-9 are based on $V_g = 0.04$ (representative for self-weight) and $V_g = 0.1$ (most of other permanent actions), see section 13.2. For favourable load effects, $\gamma_{G,\text{fav}} = 0.9$ can be considered where $V_g = 0.04$ and $\gamma_{G,\text{fav}} = 0.85$ can be considered where $V_g = 0.1$.

Prestressing is in most situations intended to be favourable and a general value of $\gamma_P = 1.0$ is mostly appropriate. This also applies to tendons in cross-sections which might be considered to act unfavourably as a single element but favourably in combination with other tendons.

In particular cases like the verification of the ultimate limit state for stability with external prestress where an increase of the prestressing force can be unfavourable, $\gamma_{P,\text{unfav}} = 1.2$.

[*]Normal uncertainty in effective depth characterised by $\mu_a = 0.95$ and $V_a = 0.05$.
[**]Higher execution quality or effective depth considerably exceeding 200 mm: $\mu_a = 0.975$ and $V_a = 0.025$.
[***]Effective depth measured in the finished structure: $\mu_a = 1$ and $V_a = 0.01$.

For FRC in compression or tension (limit of linearity) or tension (residual strength) γ_F is as for plain concrete.

Table 12.5-9: *Recommended values of the partial factor γ_G for unfavourable effects of permanent actions (design).*

CC	γ_G	
V_g	0.04	0.1
CC1	1.2	1.3
CC2	**1.25**	**1.35**
CC3	1.25	1.4

$\gamma_{P,\text{fav}} = 1$ is recommended for the prestress force for bonded and unbonded tendons in verifications where an increase in prestress is favourable. $\gamma_{P,\text{unfav}} > 1.0$ is recommended for verifications where an increase in prestress is unfavourable.

Further to the above assumptions common to Table 12.5-8 to Table 12.5-10, the partial factors in Table 12.5-10 are based on:

- the indicative statistical characteristics of the time-variant components of selected variable loads for a reference period of 50 years provided in Table 13.3-1 (assuming a Gumbel distribution),

- the indicative statistical characteristics of the time-invariant components of the selected variable loads provided in Table 13.3-2 (assuming a lognormal distribution).

It is emphasised that in particular characteristics of climatic load effects can be significantly region- or location-dependent and the respective γ_Q-factors provided in Table 12.5.4-10 may need to be updated for structure-specific conditions.

The values of the partial factors given in Table 12.5-10 are based on (12.5-18) and obtained by numerical integration. The approximations based on Gumbel (for imposed, snow, and wind) and lognormal (road traffic) distributions, given in (6.4.3-7) and (6.4.3-10) respectively, negligibly underestimate γ-values (errors around 1%) and the correction factor can be considered as unity, $\delta = 1$.

Table 12.5-10: *Recommended values of the partial factor γ_Q for unfavourable effects of variable actions (design)*

CC	γ_Q			
Action	Imposed*	Snow	Wind	Road traffic
CC1	1.3	1.75	1.6	1.25
CC2	**1.5**	**1.95**	**1.85**	**1.3**
CC3	1.7	2.2	2.1	1.4

*Imposed loads in office buildings with a dominating sustained component of the load, see section 13.3.

12.5.4.2.3.3 Recommended values of partial factors for assessment of existing structures

12.5.4.2.3.3.1. Verification of need of safety measure(s)

In order to identify the need for a safety measure to achieve a required reliability level for the existing structure, the use of the partial factors provided in Table 12.5-11 to Table 12.5-13 is recommended.

The partial factors in Table 12.5-11 to Table 12.5-13 are based on:

- the informative target reliability indices $\beta_{t,eco,0,1}$ related to a one-year reference period as provided in section 11.3.2.3.1.
- sensitivity factors for the one-year reference period $\alpha_R = 0.7$ and $\alpha_E = -0.8$;

the model uncertainty characteristics provided in Table 12.5-7 for resistances and as explained further in this section for load effects. When using lower partial factors for assessment of ULS requirements according to this section, this can imply that high stresses occur under service load situations. This is why stresses in the serviceability limit state should be limited to avoid yielding of the reinforcement and development of micro-cracks and cracks in concrete.

According to 11.3.2.2.2, human safety may require higher reliability levels than those provided in 11.3.2.3.1 if structural failure is expected to result in human losses. In particular for special cases e.g. with many persons at risk or with large failure

consequences, target levels higher than the $\beta_{t,eco,0,1}$-values in 11.3.2.3.1 might be more appropriate.

The conditions that may imply the need for updating the partial factors may be related e.g. to structural damage including deterioration, differences in material properties, detailing provisions or execution tolerances. These may for instance result in the need to utilise resistance models associated with model uncertainty different than that assumed in 12.5.4, revise the probabilistic models of basic variables, or change assumptions concerning load redistribution.

Further examples of common deviations from the assumptions include failure to meet the minimum requirements for using the design equations, governing failure modes different from those in design situations e.g. as a consequence of spatial variability.

Note that the partial factors for resistance may need to be increased to cover epistemic uncertainties related to the lack of knowledge about important details that may have significant influence on structural behaviour such as actual geometry or amount of reinforcement in joints of frames. In these cases, a sensitivity analysis may help to explore the effect of such uncertainties on structural behaviour. As an example, assumptions for representative favourable and unfavourable situations might be made and structural analysis can then reveal the effect of this uncertainty on structural resistance.

Further, the partial factors in Table 12.5-11 are based on (Muttoni [12-5]):

- coefficient of variation of in-situ concrete compressive strength $V_{fc,is} \geq 0.08$ estimated from core tests – considered in Eq. (12.5-2) instead of V_{fc};
- $V_{fy} = 0.045$ for reinforcement;

Note that the conversion factor η is implicitly covered by test results for in-situ compressive strength, $f_{c,is}$; $\mu_\eta = 1$ and $V_\eta = 0$ are considered in Eq.(12.5-2).

Uncertainty in geometry is described for two cases—measurements in the finished structure and normal—according to section 12.5.4.1.1. The case with normal uncertainty assumes that reliability verification adopts information on geometry from documentation that is checked by in-situ measurements, but the geometry of the decisive section is not directly verified.

It should be ensured that all the assumptions adopted for deriving the partial factors in 12.5.4 apply and critical checks should always be made before applying the γ-values in 12.5.4.

In addition, the partial factors in Table 12.5-11 should only be applied when:

- concrete compressive strength is verified by adequate destructive tests (following standards on evaluation of concrete in-situ strength such as EN 13791 [12-18]);
- type and amount of reinforcement is verified;
- degradation does not affect structural behaviour significantly.

Table 12.5-11: *Recommended values of the partial factors γ_M for materials for assessment to verify the need of safety measure(s).*

CC	**γ_M**			
	γ_C		γ_S	
$V_{fc,is}$	0.08	0.15	measurements ex. str.*	normal unc. in geo.**
CC1	1.1	1.1	0.975	1.075
CC2	**1.125**	**1.15**	**1.0**	1.075
CC3	1.15	1.2	1.0	1.1

*Effective depth measured in the structure: $\mu_a = 1$ and $V_a = 0.01$.**Normal uncertainty in effective depth: $\mu_a = 0.95$ and $V_a = 0.05$.

Further to the above assumptions common to Table 12.5-11 to Table 12.5-13, the partial factors in Table 12.5-12 are based on:

- Common load effect model uncertainty ($V_{\theta E}$ = 0.075), or reduced load effect model uncertainty ($V_{\theta E}$ = 0.025) – see section 12.5.4.2.3.2;
- V_g = 0.04 (for self-weight based on dimensions, reinforcement ratios and volume densities measured on the finished structure) and V_g = 0.1 (most of other permanent actions).

For favourable load effects, $\gamma_{G,\text{fav}}$ = 0.95 can be considered in the cases with $V_{\theta E}$ = 0.025 and V_g = 0.04 and $\gamma_{G,\text{fav}}$ = 0.9 in other cases.

For further guidance see 12.5.4.2.4.2.

Further to the above assumptions common to Table 12.5-11 to Table 12.5-13, the partial factors in Table 12.5-13 are based on the probabilistic models for variable action effects given in section 13.3. In particular, characteristics of climatic load effects can be significantly region- or location-dependent and the respective γ_Q-factors provided in Table 12.5-13 might not be generally applicable and might need to be adjusted on the basis of region- or location-specific data.

The values of the partial factors given in Table 12.5-13 are based on (12.5-18) and obtained by numerical integration. The approximations based on Gumbel (for imposed, snow, and wind) and lognormal (road traffic) distributions, given in (6.4.3-7) and (6.4.3-10) respectively, underestimate γ-values (errors around 5%) and the correction factor can be considered as δ = 1.05.

Table 12.5-12: *Recommended values of the partial factor γ_G for unfavourable effects of permanent actions for assessment to verify the need of safety measure(s)*

CC	**γ_G**			
V_g	0.04		0.1	
$V_{\theta E}$	0.025	0.075	0.025	0.075
CC1	1.125	1.2	1.25	1.3
CC2	**1.125**	**1.225**	**1.275**	**1.325**
CC3	1.15	1.25	1.3	1.375

The assumption on reduced load effect model uncertainty may only be adopted when properly justified; e.g. when sufficient information on dimensions and volume densities should be available and should confirm modelling assumptions. Further, there should be no uncertainties in the load effect of the permanent action e.g. due to a system behaviour or the effect of joints.

$\gamma_{P,\text{fav}}$ = 1 is recommended for the prestress force for bonded and unbonded tendons in assessments where an increase in prestress is favourable. $\gamma_{P,\text{unfav}}$ > 1.0 is recommended for assessments where an increase in prestress is unfavourable.

Table 12.5-13: *Recommended values of the partial factor γ_Q for unfavourable effects of variable actions for assessment to verify the need of safety measure(s)*

CC	**γ_Q**			
Action	Imposed*	Snow	Wind	Road traffic
CC1	1.0	1.05	1.05	1.05
CC2	**1.075**	**1.15**	**1.15**	**1.075**
CC3	1.25	1.35	1.3	1.15

*Imposed loads in office buildings with a dominating sustained component of the load, see section 13.3.

12.5.4.2.3.3.2. Assessment of existing parts of upgraded structure

The partial factors in Table 12.5-14 to Table 12.5-16 are based on the informative target reliability indices β_t related to a one-year reference period as provided in Table 11.3-2 (normal relative cost of safety measure). Unless stated otherwise, the other assumptions adopted for Table 12.5-11 to Table 12.5-13 apply also for Table 12.5-14 to Table 12.5-16.

In contrast to the assumptions adopted for Table 12.5-11, uncertainty in geometry related to the case with in-situ measurements is only considered for upgraded structures. Uncertainties in geometry are thus characterised by $\mu_a = 1$ and $V_a = 0.015$ for section area, and by $\mu_a = 1$ and $V_a = 0.01$ for effective depth.

Besides considering different target reliability levels, the partial factors in Table 12.5-15 are based on the assumptions as adopted for Table 12.5-12.

For favourable load effects, the following $\gamma_{G,fav}$-values can be considered:

- $\gamma_{G,fav} = 0.95$ when $V_{\theta E} = 0.025$ and $V_g = 0.04$ (variation related to self-weight of concrete);
- $\gamma_{G,fav} = 0.9$ when $V_{\theta E} = 0.025$ and $V_g = 0.1$ (variation of permanent actions in general), or when $V_{\theta E} = 0.075$ and $V_g = 0.04$;
- $\gamma_{G,fav} = 0.85$ when $V_{\theta E} = 0.075$ and $V_g = 0.1$.

In the reliability assessment of existing parts of an upgraded structure, the use of the partial factors provided in Table 12.5-14 to Table 12.5-16 is recommended.

For new parts of an upgraded structure, the approach in section 12.5.4.2.4.2 can be used.

In addition, the partial factors in Table 12.5-14 should only be applied when:

- concrete compressive strength is verified by adequate destructive tests (following standards on evaluation of concrete in-situ strength such as EN 13791 [12-18]);
- type and amount of reinforcement is verified;
- decisive geometrical properties are verified by measurements in the structure;
- degradation does not affect structural behaviour significantly.

Table 12.5-14: *Recommended values of the partial factors γ_M for materials for assessment of an upgraded structure.*

CC	γ_M		
	γ_C		γ_S
$V_{fc,is}$	0.08	0.15	V_{fy}= 0.045
CC1	1.15	1.175	1.0
CC2	**1.175**	**1.25**	**1.025**
CC3	1.20	1.275	1.05

Table 12.5-15: *Recommended values of the partial factor γ_G for unfavourable effects of permanent actions for assessment of an upgraded structure.*

CC	γ_G			
V_g	0.04		0.1	
$V_{\theta E}$	0.025	0.075	0.025	0.075
CC1	1.15	1.25	1.3	1.35
CC2	**1.15**	**1.3**	**1.35**	**1.4**

For further guidance see 12.5.4.2.4.2.

Besides considering different target reliability levels, the partial factors in Table 12.5-16 are based on the assumptions as adopted for Table 12.5-13. In particular characteristics of climatic load effects can be significantly region- or location-dependent and the respective γ_Q-factors provided in Table 12.5-16 might not be generally applicable and might need to be adjusted on the basis of region- or location-specific data.

The values of the partial factors given in Table 12.5-16 are based on (12.5-18) and obtained by numerical integration. The approximations based on Gumbel (for imposed, snow, and wind) and lognormal (road traffic) distributions, given in (6.4.3-7) and (6.4.3-10) respectively, underestimate γ-values (errors around 10%) and the correction factor can be considered as $\delta = 1.1$.

These partial factors are derived using the approach presented in 12.5.4.2.3, considering:

- the informative target reliability indices $\beta_{t|acc} = 2.0$ as provided in 11.3.2.2.4;
- sensitivity factors $\alpha_R = 0.8$ and $\alpha_E = -0.7$;
- resistance characteristics according to 12.5.4.2.4.2.

Design or assessment values for concrete tensile strength are specified according to ???.

CC3	1.175	1.3	1.35	1.45

$\gamma_{P,fav} = 1$ is recommended for the prestress force for bonded and unbonded tendons in assessments where an increase in prestress is favourable. $\gamma_{P,unfav} > 1.0$ is recommended for assessments where an increase in prestress is unfavourable.

Table 12.5-16: *Recommended values of the partial factor γ_Q for unfavourable effects of variable actions for assessment of an upgraded structure.*

CC	**γ_Q**			
Action	Imposed*	Snow	Wind	Road traffic
CC1	1.25	1.35	1.3	1.15
CC2	**1.5**	**1.6**	**1.55**	**1.2**
CC3	1.65	1.75	1.7	1.25

*Imposed loads in office buildings with a dominating sustained component of the load, see section 13.3.

12.5.4.2.3.4 Indicative values of partial factors for materials in accidental design and assessment situations

This section provides indicative values of partial factors for materials, to be used in accidental design and assessment situations for structures. Where costs of safety measures are significant or high failure consequences may be caused by the accidental situation, the use of the probabilistic approach in 12.5.3 or of risk analysis in 12.5.2 is recommended.

For design of new structures or assessment of existing structures, this Model Code provides the following indicative values of partial factors for materials:

- $\gamma_C = 1.15$ for concrete compressive strength;
- $\gamma_S = 1.0$ for reinforcement tensile and compressive strength.

Static actions not repeated more than 10^4 times or for which $\psi_1 = 0$ are considered unable to produce fatigue failure. Examples of actions able to cause fatigue are loads due to vehicles, moving machinery, wind (gusts, turbulence, vortices etc.) and wave action.

This is an indirect verification that the loss of strength will not be significant.

In assessing the stress range, stress variations in opposite senses (due for example to successive arrangements of a moveable load) must be, if relevant, taken into account.

Other design properties associated with tensile stress of concrete (e.g. a formal shear stress) may also have to be considered.

12.5.4.3 Fatigue verification

12.5.4.3.1 Design principles

Fatigue design must ensure that in any fatigue-endangered cross-section, the expected damage D will not exceed a limiting damage D_{lim}. The verifications of this requirement can be performed according to four methods of increasing refinement.

12.5.4.3.2 Level I approximation

This is a qualitative verification that no variable action is able to produce fatigue. If the conclusion of this verification is not positive, a verification according to one of the higher levels must be made.

12.5.4.3.3 Level II approximation

This is a verification by a simplified procedure. It is verified whether the following stresses or stress ranges:

- the maximum design stress range in the steel

  $\gamma_{Sd}\ \Delta\sigma_s(G, P, \Psi_1 Q_k)$;
- the maximum design concrete compressive stress

  $\gamma_{Sd}\ \sigma_{c,max}\ (G, P, \Psi_1 Q_k)$;
- the maximum design tensile stress in plain concrete

  $\gamma_{Sd}\ \sigma_{ct,max}\ (G, P, \Psi_1 Q_k)$;

do not exceed the limit values given in section 30.2.2.1.3.

The load factor γ_{Sd} is assumed to be 1.1, or 1.0 if accurate stress analysis is possible. In the resistance model, the strength of the materials is divided by $\gamma_{s,fat} = 1.15$ for the steel and $\gamma_{c,fat} = 1.5$ for concrete.

In Eq. (12.5-28) the term between the brackets is the static part and the term Q_{fat} is the dynamic part.

For Q_{fat}, in many cases the frequent value $\Psi_{1,1}Q_k$ may be used as an equivalent or conservative approach.

12.5.4.3.4 Level III approximation

This verification refers to a representation of the variable load dominant for fatigue by a single load level Q associated with a number of repetitions n during the required lifetime.

The stresses in the structural materials, or the stress range, are calculated under the following combination of actions:

$$\left(G_{sup}+G_{inf}+P+\psi_{1,1}\cdot Q_{k,1}+\sum_{i>1}\left(\psi_{2,i}\cdot Q_{k,i}\right)\right)+Q_{fat} \qquad (12.5\text{-}28)$$

where:

Q_{fat} is the relevant fatigue load (e.g. traffic load or other cyclic load).

The stresses found under the load according to Eq. (12.5-28) are multiplied by a factor γ_{Sd} = 1.1, or 1.0 if accurate stress analysis is possible. At the resistance side the strength of the materials is divided by $\gamma_{s,fat}$ = 1.15 for the steel and $\gamma_{c,fat}$ = 1.5 for the concrete.

12.5.4.3.5 Level IV approximation

This is a verification based on an assessment of the fatigue damage resulting from various magnitudes of loads. According to this method, the load history during the required life is represented by a spectrum in a discretized form. The accumulation of fatigue damage is calculated on the basis of the Palmgren–Miner summation.

The load factor γ_{Sd} is assumed to be 1.1, or 1.0 if accurate stress analysis is possible. In the resistance model, the strength of the materials is divided by $\gamma_{s,fat}$ = 1.15 for the steel and $\gamma_{c,fat}$ = 1.5 for concrete.

12.5.4.4 Verification of structures subjected to impact and explosion

Impact and explosions are regarded as accidental loads, so Eq. (12.5-26) applies.

As mentioned in subsections 30.5.2.4.6. and 30.5.2.5.2.4. some of these rules may be substituted by stress limitations, detailing rules or other indirect verifications.

The α-factor (e.g. 0.6 for excessive compression) describes the limit state and is not a reliability factor.

In such equations f_d generally is not to be considered as a basic variable.

This rule may in some cases be substituted by a maximum slenderness ratio.

If not fixed by the Code, C_d should be fixed by the contract or chosen by the designer, possibly depending on non-structural parts.

See subsection 30.5.2.6.

Pragmatic values smaller than 1 may be envisaged for indirect actions.

12.5.4.5 Serviceability

Design principle

(a) Limit state of cracking and excessive compression

It should be verified that in any cross-section:

$\sigma(F_d) < \alpha f_d$ for crack formation and excessive creep effects;

$w_d(F_d, f_d) < w_{lim}$ for design crack width;

$\sigma(F_d) \leq 0$ for crack re-opening;

where:

σ is a defined stress;

F_d is the design value of action;

f_d is a tensile, shear or compressive design strength;

w_d is a defined crack width;

α is a reduction factor for the case considered, with $0 \leq \alpha \leq 1$.

(b) Limit state of deformations

It should be verified that:

$$a(F_d, f_d) \leq C_d \qquad (12.5\text{-}29)$$

where:

a is a defined deformation (generally a deflection or a rotation at a member end);

F_d and f_d are values as defined under (a);

C_d is the limit value for the deformation considered.

(c) Limitation of vibrations

In the most common cases, the limitation is ensured by indirect measures, such as limiting the deformations or the periods of vibration of the structure in order to avoid the risk of resonance. In the other cases, a dynamic analysis is necessary.

Values of partial factors

(a) γ_F factors are taken equal to 1;

(b) γ_M factors are taken equal to 1.

Combinations of actions

In general, there will be two different levels of A_{Ek} – one for each serviceability limit state introduced in subsection 11.1.2.

In the seismic situations masses are consistent with the gravity loads corresponding to the combination $G_{sup} + G_{inf} + \sum_{i \geq 1} \Psi_{2,i} Q_{k,i}$.

General rules

The combinations which should be considered depend on the particular limit state under consideration and are identified in the corresponding chapters.

They are defined as follows, in a symbolic presentation:

characteristic: $$G + P + Q_{k,1} + \sum_{i>1} (\Psi Q_{k,i}) \quad (12.5\text{-}30)$$

frequent: $$G + P + \Psi_{1,1} Q_{k,1} + \sum_{i>1} (\Psi_{2,i} Q_{k,i}) \quad (12.5\text{-}31)$$

quasi-permanent: $$G + P + \sum_{i \geq 1} (\Psi_{2,i} Q_{k,i}) \quad (12.5\text{-}32)$$

seismic: $$G + P + A_{Ek} + \sum_{i \geq 1} (\Psi_{2,i} Q_{k,i}) \quad (12.5\text{-}33)$$

where:

G is taken according to subsection 12.5.4.1.5.1;

P is the mean value of the prestressing load, as defined in subsection 12.5.4.1.5.2, where the most unfavourable value (with or without losses) should be applied;

$Q_{k,i}$ refers to any variable action, successively;

A_{Ek} is the representative seismic action prescribed for the serviceability limit state of interest.

12.5.4.6 Recommended values of partial factors for CLS

[Not used at this time, to be updated in future edition]

12.5.4.7 Recommended values of partial factors for PDLS

[Not used at this time, to be updated in future edition]

12.5.5 Global resistance format

The global resistance approach was initiated by the introduction of non-linear analysis, which is based on a global structural model and offers tools for the safety assessment. It is a general approach, which follows the probabilistic safety concept more rationally than the partial factor method. It is applicable to the safety check on structural level. However, it can be applied also to members or sections as well.

The global safety factor reflects the variability of the structural response due to random properties of basic variables. The effect of random variation of basic variables, such as strength f on resistance R is dependent on the type of limit state function $r(f,..)$. The limit state function is represented by non-linear numerical analysis. Thus, for dominating concrete failure the resistance variability is much higher than for steel failure. This also means that the variability of resistance is, in general, not constant for a given set of material parameters and their random variations and depends on the structural model considered.

The global resistance has a general meaning and usually describes the response of an element or a structure to given load actions. The resistance can be described by a scalar, vector or a function, depending on design and limit state formulation. A significant feature of the structural resistance is the integration of various random effects of material properties, dimensions and so on, and their interactions. Unlike in the partial factor design method, the uncertainties are evaluated on a global structural level and not in local material points.

The meaning of global resistance can be illustrated by an example of a simple beam under the action of a force. The global resistance is expressed by the ultimate force, which can be resisted by the beam. This resistance covers all material properties, geometry, reinforcement, boundary conditions and modes of failure. Typically, the beam can fail in bending or in shear and both of these failure modes are described by the same variable – maximum force resisted by the beam. The same calculation model, for example a finite element analysis, can be used and the failure mode is detected automatically in the analysis. However, it is emphasised that various failure modes

12.5.5.1 General

The global resistance format treats the uncertainties of the structural behaviour as described by the limit state condition according to Eq. (12.4-3) on the level of structural resistance. The effects of various uncertainties (of material properties, geometrical dimensions etc.) are integrated in a global design resistance and can also be expressed by a global safety factor. The representative values of the global resistance variables and the global safety factors should be chosen in such a way that the reliability requirements for the design or assessment of new or existing structures, respectively.

12.5.5.2 Basic rules for global resistance approach

12.5.5.2.1 Representative variables

The representative variable for the global resistance is the structural resistance R.

The uncertainty of resistance is expressed by the following values of resistance:

R_m mean value of resistance;

R_k characteristic value of resistance (corresponding to a 5% fractile);

R_d design value of resistance (or assessment value for reliability assessment of existing structures).

The basic variables, defined for the partial factors in subsection 12.5.4.2.1, are used for calculating the resistance values. The values of these variables (f, a,…) should be chosen in accordance with the safety formats described further in this chapter. The value of action F is considered in the same way as in the partial factor method.

may need to be related to different target reliability levels depending on consequences associated with exceeding a particular limit state.

The uncertainty of resistance R is described by its random distribution function with its parameters: function type, mean, standard deviation (and possibly others). The parameters of scatter for a given random distribution of resistance can be used to derive the mean, characteristic and design values of resistance R_m, R_k, R_d. The global safety can be expressed either by a global safety factor or by a reliability index.

In contrast, if the same beam is verified by the partial safety factor method, a specific section is considered and local checks are made for specific actions in a cross-section. Two separate verifications are performed in the section, one for the bending failure and another one for shear failure. The global safety is not evaluated, but it is guaranteed by the formulation of partial safety factors.

In many cases it is possible to estimate the mean and the characteristic values of resistance by the values of resistance derived from mean and characteristic values of the basic variables, respectively. When the mean value obtained in this way differs from the mean value obtained by other means (e.g. experiments) special care is advised.

It is important to recognize that in the present formulation the global safety factor γ_R^* is related to the mean variable. To distinguish this from the partial safety factors, which are referring to characteristic values a notation with asterisk superscript is used.

Furthermore, it is useful to introduce a scaling factor for a loading pattern. In general, action F_d and resistance R_d, may include many components (e.g. vertical and horizontal forces, body forces and temperature) and can be described by a point in a multidimensional space. The resistance scaling factor k_R describes the relation between resistance and action and has the same meaning as a safety factor. In a symbolic form, considering a pair of corresponding components it can be defined as:

$$k_R = \frac{R_m}{F_d} \tag{12.5-37}$$

Then, the design condition formulated in Eq. (12.5-34) can be rewritten as:

$$k_R \geq \gamma_R^* \tag{12.5-38}$$

12.5.5.2.2 Design condition

The design condition derived from Eq. (12.4-4) for the global format takes the following form:

$$e(F_{d,..}) \leq r(R_{d,..}) \tag{12.5-34}$$

In a simplified force representation, it can take the form:

$$F_d \leq R_d \tag{12.5-35}$$

The design and mean values of resistance are related as:

$$R_d = R_m/\gamma_R^* \tag{12.5-36}$$

where γ_R^* is the global safety factor for mean resistance.

The global safety factor $\gamma^*{}_R$ accounts for random uncertainties of model parameters, namely of material properties. An uncertainty due to model formulation, must be treated by a separate safety factor for model uncertainty γ_{Rd}. This can be applied either to the action or to the resistance. In the latter case, the design resistance takes the form:

$$R_d = \frac{R_m}{\gamma_R^* \gamma_{Rd}} \tag{12.5-39}$$

where γ^*_R is a required global safety factor for resistance. If relevant, the global safety factor can include the model uncertainty. The factor k_R can be used to calculate the relative safety margin m_R for resistance:

$$m_R = k_R - \gamma_R^* \qquad (12.5\text{-}40)$$

The model uncertainty factor γ_{Rd} should be chosen based on the knowledge of the design conditions of the structure during its service life. The value $\gamma_{Rd} = 1.0$ should be used only in exceptional cases, when an evidence of the model validation in the design conditions is available. An example of such a condition is the case of assessment of an existing structure.

The value $\gamma_{Rd} = 1.06$ should be used for models based on a refined numerical analysis, such as non-linear finite element analysis. The model should be objective (low mesh sensitivity) and validated. The factor 1.06 does not cover the errors due to approximations in the numerical model. It covers the other effects not included in the numerical model, such as time effects and environmental effects. An example of such a case is the usual design according to the partial safety factor method.

The value $\gamma_{Rd} = 1.1$ should be used for models sufficiently validated as in the case above, but with a higher uncertainty of structural conditions due to an unknown design situation. An example of such a case is a design under uncertain loading history due to actions.

The value of the model uncertainty factor depends on the quality of formulation of the resistance model. The recommended values are:

$\gamma_{Rd} = 1.0$ for no uncertainties;

$\gamma_{Rd} = 1.06$ for models with low uncertainties;

$\gamma_{Rd} = 1.1$ for models with high uncertainties.

12.5.6 Deemed-to-satisfy approach

12.5.6.1 General

The deemed-to-satisfy approach is applicable both for the traditional structural design and for the design associated with durability. The method may comprise sets of predetermined alternatives given in a standard. In most operational standards the design associated with durability is based on the deemed-to-satisfy approach.

The deemed-to-satisfy approach is a set of rules for

- dimensioning;
- material and product selection; and
- execution procedures

that ensures that the target reliability for not violating the relevant limit state during the design service life is not exceeded when the concrete structure or component is exposed to the design situations.

Traditionally, durability related deemed-to-satisfy provisions include requirements to the workmanship, concrete composition, possible air entrainment, cover thickness

The specific requirements for design, materials selection and execution for the deemed-to-satisfy approach must be determined in either of two ways:

to the reinforcement, crack width limitations and curing of the concrete. However, other provisions may also be relevant.

Examples of the calibration of deemed-to-satisfy criteria based on a probabilistic safety format and data derived from 10–15 years old structures are given in *fib* Bulletin 34 [12-19].

– on the basis of statistical evaluation of experimental data and field observations according to requirements of section xx regarding the probabilistic safety format;
– on the basis of calibration to a long term experience of building tradition.

The limitations to the validity of the provisions – such as the range of cement types covered by the calibration – must be clearly stated.

12.5.6.2 Durability related exposure categories

In the absence of a more specific study, the durability related exposure categories related to environmental conditions may be classified for concrete with reinforcement or embedded metal as given in Section 13.5.1.

12.5.7 Design by avoidance

Traditional structural design involving the avoidance method includes a concept based on avoiding or reducing the detrimental effect, for example sheltering the structure from certain loads such as environmental loads, wind, wave loads impact by vehicles or missiles.

In design for durability the avoidance-of-deterioration method implies that the deterioration process will not occur, due to for instance:

– separation of the environmental action from the structure or component, for example by cladding or membranes;
– using non-reactive materials, for example certain stainless steels or alkali-non-reactive aggregates;
– separation of reactants, for example keeping the structure or component below a critical degree of moisture;
– suppressing the harmful reaction, for example by electrochemical methods.

In seismic design, seismic isolation may be introduced at certain horizontal levels:

– between the superstructure of buildings or similar structures and the foundation or the ground;

The assumed effectiveness of the actual concept must be documented, for instance for products by complying with relevant minimum requirements in product standards.

– between a bridge deck and the tops of the piers and abutments;
– between sensitive equipment, containers of hazardous materials, important artefacts and such like, and the supporting structure or foundation.

The specific requirements for design, materials selection and execution for the avoidance-of-deterioration method can in principle be determined in the same way as for the deemed-to-satisfy approach.

The limitations to the validity of the provisions must be clearly stated.

12.5.8 Design and assessment by testing and monitoring

12.5.8.1 General principles of design by testing

[Not used at this time, to be updated in future edition]

12.5.8.2 General principles of assessment by testing and monitoring

[Not used at this time, to be updated in future edition]

References

[12-1] CEB Bulletin 191. General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS. 1988.

[12-2] EN 1991-1-6. Eurocode 1: Actions on structures.

[12-3] JCSS PMC Probabilistic model code – Part 1: Basis of design, 2001. Joint Committee on Structural Safety.

[12-4] RILEM Report 032. Probabilistic Assessment of Existing Structures – JCSS Report. 2001.

[12-5] Muttoni, A. Background document for prEN1992-1-1:2021, Part on 4.3.3 and Annex A. Partial safety factors for materials (CEN/TC250/SC2/WG1/TG6: prEN 1992-1-1:2020(D7) or CEN/TC250/SC2 N1939), CEN 2021

[12-6] FprEN1992-1-1 FprEN 1992-1-1 Design of concrete structures – Part 1-1 General rules, rules for buildings, bridges and civil engineering structures, Sept 2022

[12-7] *fib* Bulletin 80. Partial factor methods for existing concrete structures. 2016.

[12-8] ISO 2394:2015. General principles on reliability of structures.

[12-9] ISO 13822:2010. Bases for design of structures.

[12-10] Meinen, N., Steenbergen, R. D. J. M. Reliability levels obtained by Eurocode partial factor design – A discussion on current and future reliability levels. 2018. *Heron* 63(3): 243 – 301.

[12-11] JCSS Comments JCSS (2018). JCSS Comment on draft prEN1990:2017, Reliability Requirements in Annex C. February 2018

[12-12] EN 1998-1:2004. Eurocode 8: Design of structures for earthquake resistance – Part 1: General rules, seismic actions and rules for buildings.

[12-13] EN 1991-2 Eurocode 1: Actions on structures – Part 2

[12-14] EN 1990. Eurocode – Basis of structural design.

[12-15] ISO 22966:2009. Execution of concrete structures.

[12-16] EN 13670:2009. Execution of concrete structures.

[12-17] Moccia F, Kubski X, Fernández Ruiz M, Muttoni A., The influence of casting position and disturbance induced by reinforcement on the structural concrete strength, Structural Concrete. 2020; 1–28

[12-18] EN13791 EN13791:2019: Assessment of in-situ compressive strength in structures and precast concrete components, Aug 2019

[12-19] *fib* Bulletin 34. Model code for Service Life Design. 2006.

[12-20] ISO 22965-1:2007. Concrete – Part 1: Methods for specifying and guidance for the specifier.

[12-##] Muttoni, May 2021; ??????????????????????????????????

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Beyond scope of this chapter is, for instance, the guidance on:

- Prestressing and its time-dependent losses
- Thermal loads, railway traffic loads, and various specific types of imposed loads (rooms in hotels and hospitals, labs, libraries, schools, shopping areas, storages, congregation halls etc.)
- Settlements
- Accidental loads
- Duration of load effects, dynamic effects, fatigue, seasonal effects, climate change effects, interactions between load effects or spatial randomness

etc.

Further information is provided in JRC 2022 [13-1] and JCSS PMC [13-2].

Detailed guidance for specification of environmental actions is provided by *fib* bulletin 34 [13-3] and *fib* bulletin 76 [13-4].

In this section loads are considered as random variables with a certain distribution type. However, modelling of variable and accidental actions as a stochastic process or field may be needed, for instance when combining several variable actions with similar magnitudes of effects or when combining effects of variable actions with degradation effects.

13 Actions

13.1 Introduction

This chapter provides the guidance for specification of actions and action effects on concrete structures. It is primarily focused on permanent actions (self-weight of concrete structures and other permanent actions), on selected types of variable actions (imposed loads in residential and office buildings, road traffic loads, and climatic—wind and snow—loads) , and on the load effect model uncertainties. Further, the principles of specification of environmental actions are provided.

13.2 Classification of actions

An action is:

- a group of forces acting on a structure (direct action);
- a group of imposed deformations or accelerations e.g. caused by variation in temperature or humidity, differential settlements or earthquakes (indirect action).

Based on their variation in time, actions are classified as:

1) permanent actions *G*: actions that are present during the entire duration of the reference period. Their change in function of time is negligible with respect to their mean value or the change is constant until a limiting value is reached (e.g. the water level at a spillway). Examples are self-weight,

permanent equipment, floor or road finishing, shrinkage, differential settlements, etc.;

2) variable actions Q: actions that are normally not present during the entire reference period. Their changes in time are not negligible with respect to their mean value. Examples are imposed, traffic loads or climatic (wind, snow, thermal, etc.) loads;
3) accidental actions A: actions that have a low probability of occurrence during the reference period, but can have an important influence on structural reliability due to their magnitude (e.g. impact, explosion or fire);
4) seismic action A_E: action that arises due to earthquake.

Actions that act together and reach their maximum at the same time are most often considered as one action in practical calculations. Actions that are only weakly correlated may be considered to act independently.

Actions may also be classified with respect to the structural response:

1) Static loads: these do not produce significant accelerations in the structural system or its components;
2) Dynamic loads: these can produce significant accelerations in the structural system or its components.

The distinction is also made between fixed loads (e.g. snow or wind loads with prescribed patterns on the structure) and free (or mobile) loads such as traffic loads.

13.3 Permanent actions

Permanent actions may be subdivided into self-weight and other permanent actions. Self-weight covers the weight of structural and non-structural, permanently fixed components. The main characteristics of permanent actions are:

- The probability of occurrence at an arbitrary point-in-time is close to one.
- The variability with time is normally negligible.
- The uncertainties of the magnitude are normally small in comparison with other types of loads such as variable or accidental actions.

The weight density and dimensions of a structural part are assumed to have a normal distribution. To simplify the calculations of the self-weight, it is approximated to also have a normal distribution. Mean values and coefficients of

The values provided in Table 13.3-1 refer to large populations and are based on data from various sources. The values for ordinary concrete are valid for concrete without reinforcement and with stable moisture content. In the case of continuous drying under elevated temperature, the stable volume weight after 50 days is 1.0-1.5 kN/m³ lower. The mean values for concrete which are not provided depend on the mix, composition and treatment.

Further information e.g. on permanent actions due to ballast is provided in JRC 2022 [13-1].

variation for the total variability of the weight density of steel and concrete are given in Table 13.3-1.

Table 13.3-1: Mean value and coefficient of variation for weight density

Material	**Mean value [kN/m³]**	**Coefficient of variation**
Steel	77	< 0.01
Concrete		
- Ordinary concrete	24	0.04
- High strength concrete	24-26	0.03
- Lightweight aggregate concrete	*	0.04-0.08
- Cellular concrete	*	0.05-0.10
- Heavy concrete for special purposes	*	0.01-0.02

*Refer to section 14.2.1.2 for the definition of classes and corresponding values.

Table 13.3-2 gives the probabilistic models for basic variables affecting permanent action effects (adapted from *fib* Bulletin 80 [13-5], in accordance with JRC 2022 [13-1].

Table 13.3-2: Probabilistic models for selected basic variables affecting permanent action effects.

Variable	**X**	**Distr.**	**μ_X/X_k**	**V_x**
Self-weight of in situ concrete	g_{SW}	normal	1	0.04
Other permanent actions	g_{pa}	normal	1	0.10

Model uncertainty in relation to the permanent load effect may be incorporated according to section 13.5.

The variable loads include imposed, traffic and climatic (such as wind, snow or thermal) actions.

Given the mean value μ_Q and the coefficient of variation V_Q, the Gumbel distribution parameters are calculated as:

$$\beta_{\text{scale}} = (\sqrt{6})V_Q\mu_Q/\pi \qquad \text{(scale parameter)} \qquad (13.4\text{-}2a)$$

$$\mu_{\text{location}} = \mu_Q - 0.577\,\beta_{\text{scale}} \qquad \text{(location parameter)} \qquad (13.4\text{-}2b)$$

Note that annual maxima of the time-variant components of the variable loads are considered as statistically independent in time.

In particular wind and snow effects may be strongly region and location dependent. Representative values of the coefficient of variation of annual maxima of the ground snow loads in the Czech Republic are $V_{S,1} \approx 0.6\text{-}0.7$ for lowlands and $V_{S,1} \approx 0.4\text{-}0.6$ for mountains; see JRC 2022 [13-1] for further information. The model for basic wind speed should always be based on local meteorological data; the value of the coefficient of variation of annual maxima of basic wind speed $V_{vb,1} = 0.15$ is representative for coastal areas in The Netherlands.

As an approximation, both maximum wind speed and maximum wind pressure may be represented by a Gumbel distribution although this is not theoretically sound. Available measurements suggest that the Gumbel distribution fits the maximum wind pressure well for many locations. The coefficient of variation of the wind pressure (V_W) can be obtained directly from data. For another approximation see *fib* Bulletin 80 [13-5].

The model for 5-year maxima of an imposed load (offices) Q_{imp} assumes that a sustained component of the imposed load is dominating over an intermittent component. For the sustained component of the imposed load in office buildings, the basic reference period of 5 years can be assumed, CEN ENV 1991-1 [13-6]. For the reference period of 1 year, the values of the 5-year period should be used since the load is assumed to be fully correlated over this period of time. Note that the imposed load is mostly the sum of sustained and intermittent parts. As annual maxima of the intermittent part are lower than its 5-year maxima, the consideration

13.4 Variable actions

In general, a variable load effect depends on the time-variant component $q_0(t)$ and on the time-invariant component C_0.

Extremes of the time-variant component of the variable load over a specified reference period are commonly described using a Gumbel distribution. Suggested values for the properties of the distribution, depending on the reference period considered, are provided in Table 13.4-1. The mean value of annual maxima of the load may be obtained by multiplying a characteristic value (Q_k) by the bias obtained as:

$$\mu_{Q,1} / Q_k \approx 1 / [1 - V_{Q,1}[0.45 + 0.78\ln(-\ln(P_{Qk})]] \qquad (13.4\text{-}1)$$

where $V_{Q,1}$ is coefficient of variation of annual maxima and $P_{Qk} = \text{Prob}(Q_1 > Q_k)$ is the exceedance probability of the fractile corresponding to the characteristic value.

Table 13.4-1: Indicative statistical characteristics of the time-variant component of selected variable loads for various reference periods (assuming Gumbel distribution)

Variable load, load parameter	Ref. period	Probab. P_{Qk} [-]	CoV (V_Q) [-]	Bias (μ_Q / Q_k) [-]
Imposed (Q_{imp})	5 years	0.995	1.1	0.20
	50 years	0.95	0.37	0.59
Basic wind speed (v_b)	1 year	0.98	0.15	0.72
	5 years	0.904	0.13	0.86
	50 years	0.364	0.10	1.05
Basic wind pressure (W = constant × v_b^2)	1 year	0.960	0.32	0.60
	5 years	0.817	0.23	0.85
	50 years	0.132	0.16	1.19
Road traffic (T)	1 year	0.999	0.075	0.73
	5 years	0.995	0.069	0.80
	50 years	0.95	0.061	0.90
Ground snow load	1 year	0.98	0.65	0.37

of a 5-year reference period for the total imposed load is a conservative approximation. In detailed analysis, the duration of the load pulse (taken here as deterministic 5 years) can be described by a probabilistic distribution.

Note that the statistical characteristics suggested in Table 13.4-1 for the imposed load are based on the JCSS PMC [13-2] and on the background studies that supported the development of EN 1990:2002 [13-7]. The 2022 background document for the Eurocodes [13-1] considered a slightly different model with higher (more unfavourable) biases and lower (more favourable) coefficients of variation; this model would lead to slightly higher γ_Q-values in section 12.5.4.2.4 (for design by about 5% and for assessment by about 9%).

Bias – mean of annual maxima of the traffic load to its characteristic value $\mu_{T,1}$ / T_k = 0.55-0.85 and coefficient of variation of annual maxima $V_{T,1}$ = 0.03-0.15 are reported in ARCHES D08 [13-8].

The statistical characteristics of the time-invariant components are mainly based on JCSS PMC [13-2]. The time-invariant components are related to the load models commonly provided in standards such as Eurocodes. Different statistical characteristics can be considered when more detailed load models are applied (e.g. models based on in-situ measurements, wind tunnel tests, numerical modelling by FEM or computational fluid dynamics etc.); in such cases biases are typically close to unity and CoVs are lower than those indicated in Table 13.4-2.

The time-invariant component of the imposed load effect, $C_{0Q\text{imp}}$, primarily describes uncertainties related to the spatial variability of the load and to the combination of intermittent and sustained load components. Due to the lack of data, the model for $C_{0Q\text{imp}}$ is based on experts judgement, JRC 2022 [13-1].

The time-invariant component of the wind action effect, C_{0W}, describes uncertainties related to the wind pressure model including the factors for roughness, gust, shape (pressure coefficient), friction, force, or dynamic effects.

The model for the time-invariant component of the traffic load on road bridges, C_{0T}, describes uncertainties related e.g. to a random lateral position of heavy vehicles on the bridge, dynamic amplification factor, changes in axle loads due to acceleration or braking, and assumptions on combining loads in several traffic lanes.

The time-invariant component of the snow load effect, C_{0s}, describes uncertainties related primarily to conversion from ground to roof snow loads, covering the effects of shape and size of the roof, wind exposure, heat flux through the roof, etc. The model for V_{C0S} for the snow load effect is based on JRC 2022 [13-1].

(S)	5 years	0.904	0.36	0.68
	50 years	0.364	0.22	1.11

Coefficient of variation (V_Q) for all types of loads should be specified in accordance with structure-specific conditions. The biases may also need to be adjusted.

The time-invariant component of the variable load is commonly described by a normal or lognormal distribution. Indicative statistical characteristics are provided in Table 13.4-2.

Table 13.4-2: *Indicative statistical characteristics of the time-invariant component of selected variable loads (normal or lognormal distribution).*

Variable load	**Probab. (P_{C0}) [-]**	**CoV (V_{C0Q}) [-]**	**Bias (μ_{C0Q} / C_{0Qk}) [-]**
Imposed ($C_{0Q\text{imp}}$)	-	0.1	1
Wind (C_{0W})	0.95	0.3	0.65
Road traffic (C_{0T})	-	0.1	1
Snow (C_{0S})	0.9	0.2	0.8

These factors can include:

- The complexity of a static system, ability of load re-distribution and its modelling (e.g. assumptions on stiffness of structural members and joints, effect of diaphragms, interaction of slabs with beams, effect of cracking, etc.)
- Construction method (system changes during construction)
- Simplifications of boundary conditions (supports, joints)
- Failure mode and behaviour (in case of ductile failures with sufficient deformation capacity, the uncertainties related to action effects are limited, whereas they can be significant in case of brittle failures of highly redundant structures)
- Material and geometrical uncertainties affecting action effects
- Simplifications in numerical models

etc.

The values given in Eq. (13.5-1) are adapted from CEB Bulletins 127 and 128 with an additional distinction between brittle and ductile failures according to Yu et al [13-9].

It is common that the practitioners adopt rather conservative choices about the parameters for which detailed information is missing, and that may more or less affect the load effect. These choices can be related for instance to the assumptions about span length (considering axial distances of columns) or about supports, stiffness of joints or adjacent structural members, or re-distribution through various layers of different materials. This is why a conservative bias in the load effect can be expected in many cases, $\mu_X / X_k < 1.0$. As quantification of these effects at a general level is difficult, the unity bias can be considered.

Coefficient of variation $V_{Sd,1}$ can be affected by the complexity of the load combination. The default value can be used in the absence of more detailed information.

Model uncertainty in relation to the variable load effect may be incorporated according to section 13.5.

13.5 Model uncertainties for load effects

The model uncertainty in relation to the action effect is affected by many factors and should be considered in the reliability analysis. In specific cases such as for:

- complex or redundant systems,
- statically indeterminate structures with significant system changes during construction and
- structures interacting with the soil,

the sensitivity of the load effect to the assumptions adopted in the reliability analysis should be verified and conservative choices may need to be taken.

Unless a more detailed investigation is conducted or case-specific information is available, the following probabilistic models and indicative statistical characteristics for model uncertainties for load effects may be assumed:

- distribution: lognormal
- $\mu_X / X_k = 1.0$
- $V_{Sd} = \sqrt{V_{Sd,1}^2 + k_{Sd}^2 \cdot V_{Sd,2}^2}$ (13.5-1)

where:

$V_{Sd,1} = 0.04$ accounts for the effects of load combinations;

$k_{Sd} = 1.0$ for brittle failures and $k_{Sd} = 0.5$ for ductile failures;

$V_{Sd,2} = 0.0$ for statically determinate structures;

Regarding specification of $V_{Sd,2}$ simple statically indeterminate structures are typically continuous beams and similar structures composed of linear elements where the modelling assumptions (such as stiffness distribution or shear deformations) affect insignificantly the action effects. Otherwise, the structures should be considered as complex.

More detailed information with respect to the influence of structural modelling on the action effects can be gained by a sensitivity analysis where the main assumptions with respect to stiffness (also depending on the load level due to cracking and other nonlinearities) are varied.

System changes (such as span-by-span construction of a continuous bridge girder or upper floors in a building executed when the supporting elements have been already deformed by the actions of the lower floors), can have an influence on the action effects related to self-weight. In these cases, creep, and the related uncertainties, can have a significant impact on the action effects. In case of complex structural systems or in case of significant system changes, a value $V_{Sd,2} > 0.13$ may need to be considered.

$V_{Sd,2} = 0.08$ for simple statically indeterminate structures without system change during construction;

$V_{Sd,2} = 0.11$ for simple statically indeterminate structures with system change during construction and

$V_{Sd,2} = 0.13$ for complex statically indeterminate structures.

13.6 Environmental actions

13.6.1 Durability-related exposure classes

13.6.1.1 Principles

Exposure classifications have been defined in current Codes with the aim of defining concrete properties in relation to environmental aggressiveness. Most Codes differ in the description of the exposure classes and related requirements for concrete properties in order to resist environmental conditions, however the requirements for concrete production are in general very similar worldwide. In the application of LoA 1 and 2 (refer section14.3 and 30.6), the generic classifications given in most present Codes are adequate in principle. In contrast, LoA 3 and 4 are based on time-explicit models, require more in-depth analysis related to the specific project under consideration, and will be applicable mostly to special structures, or for owners of large infrastructures (road and rail agencies, port agencies, etc.).

In the DtS prescriptive approach, design criteria are defined and verified based on pre-accepted solutions, which generally include requirements to the workmanship, concrete composition, possible air entrainment, cover thickness to the reinforcement, crack width limitations and curing of the concrete.

The design of concrete structures for durability requires the identification of exposure conditions related to the expected mechanisms of deterioration. The practical application of a classification of exposure conditions in durability design are divided into the following four Levels of Approximation (LoA), with an increasing level of complexity:

1. Standard exposure classes in combination with prescriptive design requirements (Deemed-to-satisfy (DtS)).
2. Standard exposure classes in combination with performance-based design, which includes experimental or analytical evaluation of concrete properties (Performance Indicators).
3. Quantification of the environmental load in combination with performance-based design; application of partial safety factors.

Performance Indicators (PIs) relevant to durability design are transport properties such as permeation, diffusion, and capillary suction, and properties related to resistance against carbonation, chloride ingress, freeze-thaw, ASR, DEF, and chemical attack. Performance-based design approaches for durability are typically based on using experimentally obtained PIs in deterioration models. In principle, performance approaches can be used in combination with standard exposure classes, requiring limiting performance parameters (e.g. diffusion coefficient, carbonation resistance, penetrability parameters, etc.) for certain standard exposure conditions (LoA2). For both LoA 1 and 2, prescriptive requirements or PIs must be justified either by the full probabilistic approach or by calibration to long-term experience. In a more refined approach, quantifiable environmental loads can be used in the deterioration time-explicit models (LoA 3 and 4).

The verification of limit states associated with durability can relate to various aspects, including full probabilistic modelling, modelling using partial factors derived from full probabilistic modelling, and avoidance by adoption of a detail or material that reduces the assessed likelihood of failure in the design service life to an acceptably low level of occurrence.

Durability-related exposure categories in design situations are classified as given in Table 13.6-2. The standard exposure classes presented here may differ from related definitions found in the European CEN or ISO standards.

Standard exposure classes aim at presenting conservative approximations of the aggressiveness of the environment and will be applicable to the majority of common structures. Engineers can classify the environment with respect to individual exposure categories. In some cases, a combination of categories can be used where required. Engineers need to consider that the standard exposure classes define exposure in general and various items can result in a more aggressive exposure than the prevailing atmosphere would suggest. For example, evaporative concentration, particularly where associated with wick action, can create much more severe exposure than the prevailing atmosphere would suggest. Other examples are environments with large daily temperature fluctuations, low or high atmospheric pressure, or high wind speeds.

Exposure to environmental actions causing abrasion and erosion are not included in the general exposure categories. These aspects are considered separately in Sections 30.6.7 and 14.8.8.

4. Quantification of the environmental load in combination with Service Life Modelling and performance-based design; limit-state design using full probabilistic approaches.

13.6.1.2 Standard exposure categories

In the absence of a more specific study, the durability-related exposure categories associated with environmental conditions may be classified for concrete with steel reinforcement or embedded metal as given in Table 13.6-1.

Table 13.6-1: *Durability-related exposure categories related to environmental conditions*

Exposure categories	Environmental conditions
XO No risk of corrosion or attack	Exposure to very dry environments
XC Corrosion induced by carbonation	Exposure to air and moisture
XD Corrosion induced by chlorides other than from seawater	Exposure to de-icing agents, capillary rise, in ground chlorides or airborne chlorides
XS Corrosion induced by chlorides from seawater	Exposure to seawater
XF Freezing and thawing attack	Exposure to moisture and freeze-thaw-cycles
XA Chemical attack	Exposure to aggressive chemicals, for example components exposed to aggressive chemical environments (gas, liquid or solid) from natural exposure (soil, water) or industrial exposure

Table 13.6-2: *Exposure classes related to environmental conditions*

Class	Description of environment
No risk of corrosion or attack	
X0	Concrete without reinforcement or embedded metal: all exposures except where there is freeze/thaw, abrasion or chemical attack
	Concrete with reinforcement or embedded metal; very dry (generally RH<60%) [1], for example concrete inside buildings with low air humidity, concrete in dry environments, concrete in arid environments (annual rainfall 250 mm or less)
Corrosion induced by carbonation	
XC1	Dry or permanently wet, for example concrete inside buildings with low humidity, concrete permanently submerged in water [2]
XC2	Wet, rarely dry, for example concrete subjected to long-term water contact
XC3	Moderate humidity (RH 60 – 90%) [3], for example concrete inside buildings with moderate or high air humidity, external concrete sheltered from rain
XC4	Cyclic wet and dry, for example concrete surfaces subject to water contact, not within exposure class XC2
XC5	Increased or high CO_2 concentrations, for example concrete surfaces subject to increased or high CO_2 concentrations and water contact. Industrial environments, tunnels, parking structures [4]
Corrosion induced by chlorides other than from sea water [5]	
XD1	Moderate humidity, for example concrete surfaces exposed to airborne chlorides
XD2	Wet, rarely dry, for example swimming pools, concrete exposed to industrial waste containing chlorides
XD3	Cyclic wet and dry, for example parts of bridges exposed to spray containing chlorides, pavements, car park slabs [6, 7]
Corrosion induced by chloride from sea water [5, 8]	

XS1	Exposed to minor contamination with airborne salt but not in direct contact with sea water, for example structures near the coast, for which measurements, modelling or experience indicate a low risk of chloride ingress [9]
XS2	Permanently submerged, for example parts of marine structures below the low tide mark
XS3a	Exposed to significant contamination with airborne salt but not in direct contact with sea water, for example structures affected by chloride-laden winds, fog, mist, and other sources of chloride contamination, beachfront and marine above the splash zone [10]
XS3b	Tidal, splash and spray zones, for example parts of marine structures above the low tide mark to design life maximum wave height. [11, 12]
Freeze/thaw attack with or without de-icing salts	
XF1	Moderate water saturation, without de-icing agents, for example vertical concrete surfaces exposed to rain and freezing
XF2	Moderate water saturation, with de-icing agents, for example vertical concrete surfaces of road structures exposed to freezing and airborne de-icing agents
XF3	High water saturation, without de-icing agents, for example horizontal concrete surfaces exposed to rain and freezing
XF4	High water saturation, with de-icing agents or sea water, for example road and bridge decks exposed to de-icing agents, concrete surfaces exposed to direct spray containing de-icing agents and freezing, splash zones of marine structures exposed to freezing
Chemical attack [5, 7, 13]	
XA1	Slightly aggressive chemical environment
XA2	Moderately aggressive chemical environment
XA3	Highly aggressive environment

[1] This applies if the ambient RH is generally below 60%. In locations with seasonal increases in RH above 60%, the associated risk of deterioration needs to be evaluated.

[2] The risk of carbonation-induced reinforcement corrosion in structures exposed to XC1 is insignificant. In the design for durability, it is recommended to treat class XC1 the same as class X0.

[3] Even if the annual average RH is moderate, only periods with high RH are able to support significant reinforcement corrosion. XC3 therefore needs to be interpreted locally depending on the assumed variation of RH over the year.

[4] The following levels are recommended for classification of the exposure (adopted from CIA Publication Z7-02 [13-10]):

- Normal atmospheric CO_2 < 500 ppm
- Increased CO_2 500 – 1000 ppm (e.g., parking structures)
- High CO_2 1000 – 5000 ppm (e.g., tunnels)

It should be noted that high CO_2 concentrations alone do not necessarily result in an increased risk of carbonation or carbonation-induced steel reinforcement corrosion. When applying this class, ambient moisture conditions and exposure to water need to be considered.

[5] The deterioration mechanisms associated with this exposure class are more significant at elevated temperatures. In the design for durability, the associated risk of deterioration in structures exposed to hot environments (e.g. dry or humid) needs to be evaluated accordingly.

[6] More severe conditions may occur in car parks on surfaces subjected to frequent exposure to ice and snow slush followed by drying. Such situations require special consideration.

[7] Evaporation of water with even relatively low sulfate and chlorides concentrations can lead to high surface concentration, and result in premature reinforcement corrosion and sulfate attack. This is apparent on plinths where water ponds at their base and immediately above water level in tanks.

[8] These classes refer to reinforced concrete elements and concern the risk of chloride-induced reinforcement corrosion, which for these structures is of more concern than possible sulfate attack from sea water. Marine exposure for unreinforced concrete elements, e.g. those used for shoreline protection, has to be considered separately under exposure category XA (in most cases, this would be class XA1).

[9] Depending on the prevailing conditions, mild air-borne exposure to chlorides, especially in environmental conditions that also favour carbonation, can result in an increased risk of reinforcement corrosion. In the design for durability, it is recommended to treat class XS1 the same as class XS2, unless practical experience on representative structures suggests otherwise.

[10] The aggressiveness of airborne marine chloride exposure depends strongly on the prevailing environmental and geographical conditions. This needs to be assessed on a case-by-case basis unless suitable experience exists or standard national or local guidelines are available.

[11] Exposure should be specifically considered for elements with frequent long wet or long dry exposures, e.g. dry docks.

[12] Some industrial exposure may lead to unusually high chloride concentrations (e.g. desalination plants, waste containers). Such conditions need to be considered separately.

[13] Standard definitions of the aggressiveness of chemical exposure according to XA1, XA2 and XA3 are provided in EN 1992-1-1 [13-11] (and for example also in ACI 318:2014 [13-12]). For industrial environments with non-standard chemical exposure (e.g., certain production facilities, mines, sewage treatment plants and pipes, etc.), the associated risk of deterioration has to be evaluated separately. Similarly, chemical attack in rural areas, related to farming activities, may need special consideration.

In specific cases, the standard exposure categories given in Table 13.6-1 may not apply. Non-standard exposure classes can be identified based on experimental measurements of the environmental load, experience with the performance of existing structures in the specific environment, or relevant deterioration models.

In some cases, it may be possible to relate non-standard exposure conditions to one of the existing standard exposure classes, if the same design principles would result in durable

When the performance testing approach is used, the exposure conditions are defined by the particular test method used (e.g., permeability, resistivity, absorption, diffusion, etc.). In some cases, it is possible to adjust the test exposure conditions to those expected in the actual environmental exposure. In other cases, factors of equivalence should be developed between the test exposure and the real one.

concrete. In other cases, it will be necessary to quantify the environmental load so that the quantified values can be used as input parameters in deterioration and service life prediction models.

Taking frost as an example, the testing relates to a given number of standardized freeze/thaw cycles. This can be adapted to a different number of cycles or different maximum and minimum temperatures. The threshold values of the specific conditions should be established with respect to the values obtained with standardized testing conditions. Currently, it is not clear in all cases how the result of a performance test relates to the external environment. As stated in ISO 16204 [13-13], if test methods not referenced in ISO 22965 [13-14] are to be applied, the sampling, these test methods, and the statistical interpretation of their results, shall be stated in the execution specification.

In the parameters time-explicit models of Section 30.6 some represent the environmental loads. For example, they may relate to the external concentration of CO_2 or the chloride surface concentration.

For the load actions, specified limits are typically stipulated as characteristic values, partial safety factors for design are considered by the codes. Partial factors for these parameters can be found in Section 30.6. However, there is a lack of a general framework for the characterization of partial factor evolution with time and with regards to the geographic particularities.

In the case of carbonation, for example, the environmental characterization will depend on the model selected. Typical environmental exposure characteristics related to carbonation of concrete include CO_2 concentrations (kg/m^3), ambient RH (%), and availability of moisture (e.g. duration of wet periods).

In present case of Section 30.6.2, the environmental parameters in the model are:

C_S the CO_2-concentration in the air in kg/m^3;
$W(t)$ weather function [–];

Typical environmental exposure characteristics related to chloride ingress modelling include salinity (%) and temperature, refer to Section 30.6.3. Chloride ingress models based on Fick's laws of diffusion require the quantification of the chloride surface concentration ($C_{s,0}$) as an environment-related model input parameter. The value of $C_{s,0}$ depends on factors such as exposure time, marine exposure conditions, salinity of the seawater, exposure to wetting/drying cycles, ambient RH and temperature. It is further dependnt on concrete material properties such as absorption and desorption characteristics, penetrability, and cementitious binder chemistry. The quantification of $C_{s,0}$ therefore relates to the quantification of both environmental exposure and material response. Suitable values for $C_{s,0}$

13.6.1.3 Quantification of the environmental load

When considering the use of time-explicit models, relevant input parameters need to be defined, describing, amongst other aspects, environmental conditions for each concerned exposure condition.

Currently, there exists no general agreement on partial safety factors applicable for all the environmental inputs for modelling.

When the full probabilistic approach is used, the exposure conditions are defined by model input parameters (e.g. temperature CO_2 concentration, surface Chloride concentration, etc.). Universally applicable correlations between test environments and actual environmental load are usually lacking.

For the design of a new structure, the environmental factors may be derived from literature or existing structures where the concrete composition, execution and exposure conditions have been similar to those expected for the new structure. The variation of environmental factors may be found in the literature or obtained in the field. If the quantification of environmental factors is obtained from field data, the variation is determined by the number of samples taken.

When assessing the remaining service life of an existing structure, the environmental actions may be quantified from measurements made upon the structure.

can be obtained from experience or experimental investigations. Detailed information on the quantification of $C_{S,0}$ as well as other model parameters for chloride ingress prediction are presented in *fib* bulletin 34 [13-3] and *fib* bulletin 76 [13-4].

For exposure to aggressive chemicals from natural exposure or exposure to industrial environments, typical environmental exposure characteristics related to the expected rates of deterioration include the concentration of the chemical component under consideration, as well as environmental factors such as ambient temperature and humidity, and exposure to flowing or standing water. The quantification of these parameters can be based on relevant experience or obtained with suitable test methods. Currently, approved time-dependent models for most deterioration mechanisms other than chloride- and carbonation-induced reinforcement corrosion are still not available.

Avoidance-of-deterioration can, for example, be established with the use of surface protection systems (coatings, sealants, membranes, etc.), which prevent ingress of aggressive substances such as CO2, chlorides, dissolved chemicals, and water. The protective systems need to be selected based on the resistance they provide against the prevailing exposure conditions / the ingress of contaminants.

13.6.1.4 Separation of environmental actions

If the avoidance-of-deterioration approach is used, adequate protective measures need to be selected based on the prevailing environmental exposure condition and related deterioration mechanisms. In this case, detailed characterisation of the environmental exposure is often not required.

References

[13-1] JRC 2022 — JRC Technical Report for Reliability Background Eurocodes – report under preparation by CEN TC250 / SC10/ AHG Reliability background.

[13-2] JCCS PMC — Probabilistic model code – Part 1: Basis of design, 2001. Joint Committee on Structural Safety.

[13-3] *fib* Bulletin 34 — Model Code for Service Life Design. Model Code, *fib*, 2006

[13-4] *fib* Bulletin 76 — Benchmarking of deemed-to-satisfy provisions in standards, *fib*, 2015

[13-5] fib Bulletin 80. — Partial factor method for existing concrete structures. 2016.

[13-6] CEN ENV 1991-1. — Basis of design (2nd draft). Report by the Eurocode 1.1 project team. 1996.

[13-7] EN 1990:2002 — ??

[13-8] ARCHES D08. — Recommendation on the use of results of monitoring on bridge safety assessment and maintenance – Annex A: Recommendations on bridge traffic load monitoring. Final report. 2009.

[13-9] Yu et al — Q. Yu, P. Valeri, M. Fernández Ruiz, A. Muttoni, A consistent safety format and design approach for brittle systems and application to textile reinforced concrete structures, Engineering Structures 249, 2021.

[13-10] CIA Publication Z7-02 — CIA Z7/02-2018, Exposure Classification, Concrete Institute of Australia (CIA), Sydney, 2018

[13-11] EN 1992-1-1 — EN 1992-1-1:2004: Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, CEN, 2004

[13-12] ACI 318:2014 — ACI CODE-318-14: Building Code Requirements for Structural Concrete and Commentary, American concrete Institute (ACI), 2014

[13-13] ISO 16204 — ISO 16204:2012: Durability — Service life design of concrete structures, ISO 2012

[13-14] ISO 22965 — ISO 22965-1:2007: Concrete — Part 1: Methods of specifying and guidance for the specifier, ISO, 2007

This chapter provides the designer with the best possible code-type characterization of the material properties of concrete to be used in their specific design models concerning both structural and durability design. Naturally this is best obtained from full-scale testing of in-field exposed structures. As this normally cannot be realized, the alternative is direct testing, while the last option should be to derive material properties from other material characteristics (e.g. tensile strength based on compressive strength or permeability based on strength or water/cement ratio).

All models and relations given in chapter 14 are physically sound within the specified range of application and are based on the evaluation of experimental data as well as available field data.

The constitutive relations given in these sections are applicable for the entire range of concrete grades dealt with in this Model Code.

In the *fib* Bulletin 70 [14-1], the background of the models and relations subsequently presented will be given, together with fundamental data as well as relevant references.

Add the moment of publication of this Model Code only insufficient information on the influence of other binders, such as Calcined Clays, Belite Cements or calciumsulfoaluminate Cements etc. onto the performance of concrete was available. In case such binders shall be used for the production of a concrete, performance-based design schemes might allow for integrating such new types of binders and resulting concretes into the existing design scheme of the Model Code.

Green concrete (also known as sustainable or ecological concrete) may be characterized by having a significantly improved sustainability compared to ordinary structural concrete. This holds particularly true, if the CO2 emission associated with a concrete is significantly reduced and/or the energy necessary to produce the concrete and its constituent materials is considerably lower than for ordinary concrete. So far, no generally accepted limiting values and benchmarks exist.

Green concretes based on Portland cement binders as defined in chapter 14.1, first paragraph generally contain higher amounts of secondary cementitious materials and may thus deviate in their behaviour especially regarding to durability parameters. In case of

14 Concretes

14.1 Scope, definitions and range of applicability

The relations given in this Model Code apply to concretes, where the primary binder is Portland cement, eventually containing secondary cementitious materials (SCM) such as blast furnace slag (as in blast furnace slag cements), fly ash, silica fume, natural pozzolans and/or rock powders. Here it is irrelevant, whether the above named SCMs are added as part of the cement (i. e. as a composite cement) or as cement replacement material in the concrete production process.

doubt such as in deformation sensible structures or structures with pronounced durability loadings, a performance based design approach (see chapter ???) is recommended.

Throughout chapter 14 the following sign conventions are maintained which may differ from those used in other parts of this Model Code:

- all material properties are used is absolute terms ;
- tensile stresses and tensile strains (elongations) are positive;
- compressive stresses and compressive strains (contractions) are negative;
- where multiaxial stress states are considered, $\sigma_1 > \sigma_2 > \sigma_3$ is valid for the principal stresses.

It is assumed that the concrete complies with ISO 22965-1 [14-2]and ISO 22965-2 [14-3], with the amendments and alterations given in this Model Code.

The typical example of the special concrete or mortar as materials for intervention are as follows:

- polymer cementitious mortar or concrete (polymer hydraulic cement mortar or concrete);
- mortar or concrete with high early strength cement, such as ultrarapid hardening cement or high early strength Portland cement
- plasticized concrete or high-fluidity concrete, in some cases, together with an expansive material

When conventional concrete is used as material for intervention, the relations given in the subsequent sections should be used. The additional aspects as material for intervention can be found in chapter 19.

The subsequent sections apply to structural concrete with normal and lightweight aggregates, composed and compacted so as to retain no appreciable amount of entrapped air other than intentionally entrained air.

Though the relations in principle also apply for heavyweight concrete, special consideration may be necessary for such concretes (Performance Based Design).

Concerning compressive strength, *fib* Model Code for Concrete Structures 2020 covers concretes up to a characteristic strength of 120 MPa, that is normal strength concrete (NSC, $f_{ck} \leq 50$ MPa) and high strength concrete (HSC, $f_{ck} > 50$ MPa) are dealt with; see subsection 14.6.1.

The subsequent relations also apply for self-compacting concrete, unless additional information is provided. Many comparative tests have been carried out comparing self-compacting concrete and conventional concrete with the same strength. Those tests were related to properties like E-modulus, tensile strength, bond strength, interface shear and chloride penetration. Those tests showed that the properties considered were in the same range of scatter. Therefore in this chapter no difference will be made between self-compacting and conventional concrete.

The relations given also roughly apply for green concrete, as far as the composition of such concrete deviates from the composition of ordinary structural concrete only by the replacement of a certain amount of Portland cement by fly ash, silica fume, blast furnace slag and natural pozzolans – that is, chemically reactive substitutes.

This chapter of the ModelCode and the models given therein are primarily focused on predicting and describing the concrete properties based on its characteristic strength at the age of 28 days. Here it is assumed that the chosen characteristic strength is representative of the properties of the concrete in the structure. This parameter thus is a key parameters for compliance control. This representativeness can only be realistically assumed if the structure has been built applying sound placing, compaction and curing concreting practices, such as those prescribed in codes and standards (e.g. EN 13670 [14-4] and/or ISO 22966 [14-5]).

Deriving complex concrete properties from the characteristic strength only however is a substantial simplification and influences from the mixture composition on certain individual concrete properties eventually cannot be

precisely represented. In order to obtain a more accurate prediction, additional tests can be carried out, for instance to obtain characteristics like those related to permeability or diffusion, required for a more advanced durability analysis.

The models in this chapter also include equations to describe the aging of the material. When evaluating existing concrete structures it however must me noted, that the aging models given in this chapter cannot cover the broad bandwith of loadings and environmental actions the concrete has been exposed to during its service life. For the assessment of existing structures therefore either non-destructive or tests on specimens taken from the structure itself should to be carried out for a better calibration. Material characterizations and testing methods for the assessment of existing structures are dealt with in the chapters 30.11.

Special concrete as material for interventions may have the relations different from those given in the subsequent sections for conventional concrete. However, the necessary properties are basically the same as those for conventional concrete. The properties of special concrete for intervention should be obtained by test, which can be with the same way as that for conventional concrete unless otherwise specified. The relevant information is provided in section 19.2.1 and 19.3.

The information given in sections 14.6.1.1, 14.6.1.2, 14.6.1.4 and 14.6.2 is valid for monotonically increasing compressive stresses or strains at a constant range of approximately 0.2 MPa/s $< |\dot{\sigma}_c| <$ 10^7 MPa/s and $10 \cdot 10^{-6}$ $s^{-1} < |\dot{\varepsilon}_c| < 3 \cdot 10^2$ s^{-1}, respectively.

For tensile stresses or strains, this information is valid approximately for 0.02 MPa/s $< \dot{\sigma}_{ct} <$ 10^7 MPa/s and $1 \cdot 10^{-6}$ $s^{-1} < \dot{\varepsilon}_{ct} < 3 \cdot 10^2$ s^{-1}, respectively.

14.2 Concrete classification

14.2.1 Classification of conventional concrete

14.2.1.1 Introduction

The concept of concrete grades or strength classes in this Model Code is primarily used for facilitating the design process. All equations and constitutive relations in this Model Code do not rely on the strength class but on the actual mean compressive strength value.

In this Model Code concrete is classified on the basis of its compressive cylinder strength (see section 14.6.1.1). Design is based on a grade of concrete

If not known or experimentally determined, the characteristic strength may be calculated from $f_{ck} = f_{cm}$ - 8 MPa.

Production control and attestation of conformity of concrete must be in accordance with ISO 22965-2 [14-3] or similar applicable local standard.

The specification of concrete given to the concrete producer must include all assumptions made during the design as well as those properties needed to ensure that the needs during transportation and execution are considered. It is to be considered that durability and compressive strength are not directly correlated.

This Model Code uses the designations Cxx and LCxx, while ISO 22965 [14-2] uses Bxx and LBxx, respectively. The dual designation of concrete grades (e.g. C30/37) has been abandoned.

However, as in many countries 150 mm cubes are used as standard specimens for measuring strength, characteristic compressive strength values of normal weight concrete, measured on 150 mm cubes ($f_{ck,cube}$), can be converted into equivalent values of Ø150x300 mm cylinders (f_{ck}), applying as an example Eq. (14.2-1) (strength values in MPa).

$$f_{ck} = f_{ck,cube} \cdot \left[0.76 + 0.2 log_{10}\left(f_{ck,cube} / 19.6\right)\right] \qquad (14.2\text{-}1)$$

The standard reference age at which concrete properties are defined is 28 days. For specific cases, to be evaluated on a case by case basis, other ages may be used, subject to verification concerning structural design and durability e.g. for concretes made with binders containing active supplementary cementitious materials.

Nevertheless, keep in mind that some specifications – for example the requirements defined for the different exposure classes – are based on the 28-day compressive strength.

which corresponds to a specific value of its characteristic compressive strength f_{ck} as defined in section 14.6.1.1.

Concrete grades for normal weight concrete (C) can be selected from the following series:

C12, C16, C20, C25, C30, C35, C40, C45, C50, C60, C70, C80, C90, C100, C110, C120.

Concrete grades for lightweight aggregate concrete (LC) can be selected from the following series:

LC8, LC12, LC16, LC20, LC25, LC30, LC35, LC40, LC45, LC50, LC55, LC60, LC70, LC80.

The numbers following the symbols C and LC denote the specified characteristic strength f_{ck} in MPa.

Unless specified otherwise, the compressive strength of concrete and the tensile strength of concrete is understood as the strength value obtained at a concrete age of 28 days. The indicated compressive strength values correspond to testing of Ø150x300 mm cylinders.

Characteristic compressive strength values for normal weight and lightweight concrete are given in section 14.6.1.1 (Tables 14.6-1 and 14.6-2).

14.2.1.2 Classification by density

This classification corresponds to ISO 22965 [14-2] and is complemented in Model Code 2020 by the density class D0.8, as these types of concrete have been successfully applied in applications where both load carrying capacity and thermal insulation are required. Note that substantially more experience is available for concretes with dry densities between 700 and 800 kg/m³ compared to concretes with densities between 600 and 700 kg/m³. As the mechanical properties of the lightweight concrete pronouncedly depend on the type of aggregate used and considering the influence of lightweight

Concrete is classified in three categories of oven-dry density:

- lightweight aggregate concrete (600–2000 kg/m^3);
- normal weight concrete (>2000–2600 kg/m^3);
- heavyweight concrete (>2600 kg/m^3).

aggregates on the fresh concrete properties, it is highly recommended to carry out a performance based design for lightweight concretes.

Lightweight aggregate concrete with a density <600 kg/m^3 can usually not be used for structural applications, as it is not covered by design codes.

With increasing compressive strength, concrete generally contains more cement and less water, resulting in a higher density of HSC compared to NSC. Also, HSC members may contain more reinforcement than NSC members. Nevertheless, the relevant density values may vary within relatively wide limits, depending on mix composition and density of aggregate materials (both may vary between countries), reinforcement ratio and air content.

The values given in Table 14.2-1 assume an air content of 2%. A change of air content by 1% results in a density change of 1%. The values may be used for design purposes in calculating self-weight or imposed permanent loading. Where a higher accuracy is required than is provided by Table 14.2-1, the concrete density may be determined experimentally, for example according to ISO 1920-5 [14-6].

The values given in Table 14.2-2 are valid for plain and reinforced lightweight aggregate concrete with usual percentages of reinforcement. The values for in-situ density may be used for design purposes in calculating self-weight or imposed permanent loading.

Where a higher accuracy is required than is provided by Table 14.2-2 the concrete density may be determined experimentally, for example according to ISO 1920-5 [14-6].

In addition to the density class specifications, a further option is the definition of the so-called “target-density” – see for example ISO 22965-1 [14-2].

Lightweight aggregate concretes possess a very favourable relationship between strength and specific weight and have successfully been applied with the strength class as high as LC70.

For ordinary normal weight concrete, both, normal strength (NSC) and high strength concrete (HSC), the in-situ density may be estimated from Table 14.2-1.

Table 14.2-1: In-situ density [kg/m3] of NSC and HSC, plain and with different steel reinforcement ratios.

Reinforcement ratio	**C30** (w/c ≈ 0.65)	**C80** (w/c ≈ 0.35)	**C120** (w/c ≈ 0.25)
0.0%	2350	2450	2500
1.0%	2400	2500	2550
2.0%	2450	2550	2600

The classification of lightweight aggregate concrete according to its oven-dry density is given in Table 14.2-2.

Table 14.2-2: Density classes and corresponding design densities of lightweight aggregate concrete.

Density classes		D0.8	D1.0	D1.2	D1.4	D1.6	D1.8	D2.0
Oven-dry density ρ [kg/m^3]		601-800	801–1000	1001–1200	1201–1400	1401–1600	1601–1800	1801–2000
In-situ density [kg/m^3]	**Plain concrete**	850	1050	1250	1450	1650	1850	2050
	Rein-forced concrete	950	1150	1350	1550	1750	1950	2150

Detailed durability classes of concrete, which are going to be introduced in the next generation of Codes, are not specified in present version of fib MC2020. However, these classes can be made considering Level of Approximation (see section 14.3). They can be based on performance tests which can provide "Durability Indicators" (LoA2) thresholds which are ranked according to the experience to their link with durability, or can be a concrete property introduced in a time-explicit model (LoA3 or LoA4 depending on the type of model and probabilistic treatment). Sections 14.7 and 14.8 outline the types or performance tests taking into account several deterioration processes described in section 14.8.1.

There is no general definition for non-conventional concrete. Moreover, what is non-conventional at the time of release of MC2020, might be conventional at a later stage. In this document non-conventional concrete is understood as concrete which does not meet the definitions and prescriptions given in section 14.2.1; prominent examples for currently non-conventional concretes are green concretes or concretes made with unusual binders or concretes having a very low cement content.

For other types of binder than clinker-based binders, e.g. geopolymers etc., the relationships detailed might be different and are not covered in this ModelCode. These materials can alternatively be validated by testing related to both structural and durability related performance (e.g. as per European Technical Assessment or similar).

The listed concrete properties to be determined by tests are considered as mandatory in order to classify a non-conventional concrete and in particular to enable structural design and the application of this concrete. The volume and individual number of the tests must be fixed by an expert. The test results shall enable the expert to derive adapted constitutive relations and concrete models for the non-conventional concrete considered on the basis of the models given in this design code.

14.2.1.3 Classification by other properties

Concrete is specified by the user not only by strength, but also by other characteristics as well, typically:

- Maximum size of the aggregate;
- Fresh consistency (e.g. slump) at delivery;
- Resistance to pre-defined exposure conditions.

14.2.2 Classification and use of non-conventional concrete

The classification of non-conventional concrete should correspond to section 14.2.1. The application of non-conventional concrete needs experimental investigations in order to specify the properties of the considered concrete. By means of tests the subsequent mechanical properties have to be determined:

- Compressive and tensile strength;
- Modulus of elasiticity;
- Time development of strength and stiffness;
- Creep and shrinkage;
- Behaviour under high sustained loads.

Related to durability, at least the following characteristics generally have to be determined:

- Carbonation behaviour;
- Chloride diffusion characteristics.

Depending on the particular environment, additional test are mandatory, such as e.g. related to freeze-thaw resistance with and without de-icing agents, resistance to chemical attacks and wear.

Such approaches are given in most concrete standards (ISO 22965-1 [14-2], ASTM C94 [14-7], EN 206 [14-8]).

MC 2010 already introduced the concept of design by performance in opposition to design by prescriptions of the proportion of raw materials of the concrete. This was so because the so-named “prescriptive design approach” has been identified as insufficient in some aggressive or specific media and as not being applicable to new concrete constituents, thus becoming a barrier for innovation. It is the philosophy of MC 2020 to extend the performance concept in design and assessment of concrete structures, including ensuring and specifying their durability.

The performance approach can be divided into two categories:

- the method of performance-based design based on the use of Indicators (threshold values in the tested specimens without any further modelling);
- the method based on the use of time-dependent models whose input parameters indicate the performance at the design service life.

In both types of performance approaches, the concept of “equivalent performance” of the concrete properties by comparison of a new concrete type with a known one, can be applied.

14.3 Specification approach

14.3.1 Introduction

Regarding concrete specifications, two approaches - or a combination thereof - are adopted:

a) Prescriptive approach

b) Performance approach

In the prescriptive approach, the purchaser specifies the type and characteristics of the constituents to be used to produce the concrete, providing a full mix design or a partial mix design (e.g. specifying the desired consistency instead of water content).

In the performance approach, a set of requirements (including limiting values) is established for certain characteristics of concrete that are essential for or bear a significant role on the capability of the structure to perform under the applied loads (mechanical, thermal, environmental, etc.).

The characteristics or properties to be specified shall:

- be meaningful, i.e. as closely connected as possible to the actual mechanical, physical and/or chemical mechanisms that govern the performance of the structure;
- be expressed as limiting values of the characteristic or property, that can be verified directly through tests;
- be measurable in a reasonably short time, through well-defined, standardized methods of test or analysis conducted at suitable, well defined Points of Performance (laboratory, production plant, delivery point, “in situ”, in service, etc.);
- allow assessment of conformity with the specified requirements through straightforward pass/fail compliance rules;
- be directly or indirectly (e.g. via robust correlations) included as parameters of existing performance prediction models.
-

14.3.2 Durability Specification

MC2010 already introduced different verification methods to be used in design (called safety formats in MC2010). For durability design or assessment the verification methods have the hierarchy of design by avoidance, deemed to satisfy, partial factor method and full probabilistic method (The last two are reliability based) with an increasing level of complexity. There are verification combinations that could constitute intermediate levels.

In the "prescriptive approach" (concrete composition prescriptions or rules by tests and measurements I and II), rules shall be calibrated as stated in ISO 16204 [14-9], *fib* Bulletin 34 [14-10]. The technical responsibility for the mix design, i.e. for achieving the required performance of the concrete (strength, durability, etc.), lies on the specifier who must correctly apply the standard set up by relevant code committees.

The specific requirements for design, materials selection and execution for the Deemed to satisfy I and II, shall be determined in either of two ways:

- on the basis of statistical evaluation of experimental data and field observations;
- on the basis of calibration to a long-term experience of building tradition.

For "concrete composition prescriptive requirement" (Deemed to satisfy I), the producer's responsibility is limited to using the specified raw materials and to batching them within the tolerances stated in the standards. The best-known examples of prescriptive specifications typically refer to durability by limiting the (maximum) w/b ratio, and often also the (minimum) binder content. These prescriptive limitations are intended to be valid for conventional service lives (typically 50 years) but must be extrapolated to longer/shorter service lives or possibly updated when specific concrete properties are needed (e.g. thermal conductivity, low rate heat release, etc.).

For "rules by tests and measurements (Performance indicators)" (Deemed to satisfy II) approach, the technical responsibility for the mix design, i.e. for achieving the required performance of the concrete (strength, durability, etc.), is on the producer, who establishes the mix design according to experience or testing. The best-known example of performance approach is compressive strength, included invariably in all codes and standards, that fulfills all the indicated list of requisites. The performance approach allows the purchaser to check compliance of the delivered concrete by sampling and testing.

Typically, the more complex reliability based method (used for calibration of the other level of approximations) should increase the accuracy introduced, although this is dependent on accuracy and representativeness of the values for the variables that are critical for the design situation. In other words, increased accuracy of results by considering more sophisticated methods is mainly linked to the accuracy of input parameters used for modelling and not just by use of the increased sophistication of assessment approach.

As indicated in the Figure 14.3-1 or in Table 14.3-1, several approaches to durability performance specification are available, corresponding to one of the different verification methods as described in 12.4.2.

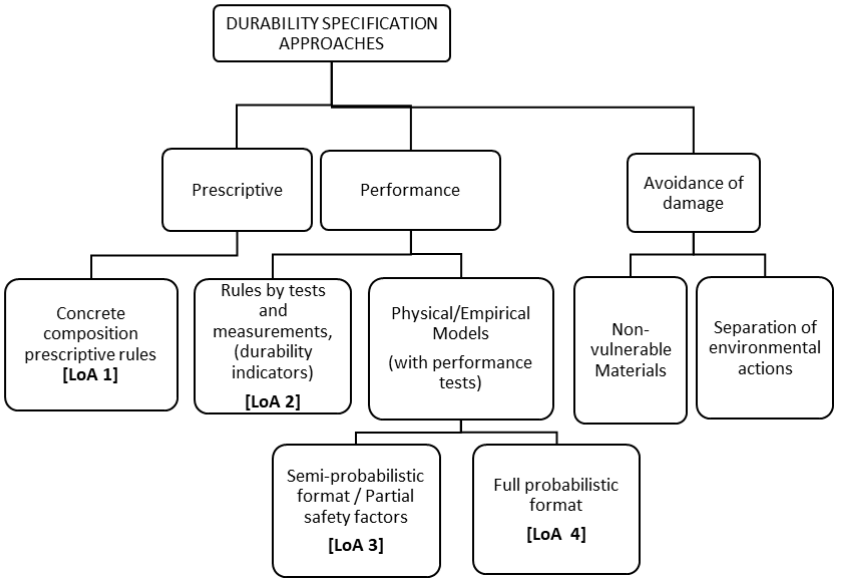


Figure 14.3-1: Framework of durability specification in the context of Verification methods for design.

- Deemed to satisfy I- Concrete composition prescriptive rules: Concrete is prescribed in relevant Codes by prescriptive requirements (obligation of means for concrete composition).
- Deemed to satisfy II – Performance indicators: Concrete is prescribed in relevant Codes by performance requirements (i.e. by measuring the concrete performance on specimens and/or directly on site is compared to pre-established threshold values, according to the classes of exposure faced by the structural element. These thresholds should be part of the project's specification or contained in specific standards).

The limitations to the validity of the provisions, e.g. the range of cement types or threshold value for a given durability indicator covered by the calibration, shall be clearly stated.

All proposed approaches remain "performance-based" as they are supposed to ensure the minimum performance of the structure over the whole required service life.

The full probabilistic approach is mostly used in the evaluation of existing structures. In this case, the relevant material properties are determined on the structure itself, thus implying the influence of the environment and the quality of construction. In this way, a representative forecast of the remaining service life can be made.

- Partial factor method and full probabilistic method: In these approaches, the threshold values are not obtained from tables in function of exposure classes, but by calculation using predictive models having time as an explicit parameter. Partial factors (semi-probabilistic models) are based on full probabilistic models and are developed for certain reliability indices. In partial factor method the characteristic values of the model parameters are applied, whilst in full probabilistic method the assumed or ascertained statistical distribution functions of the variables are considered and given in the limit state equation for the reliability-based analysis.

One specific approach is the "Avoidance of Deterioration" strategy of durability design, by which a set of materials, with characteristics known to make them non susceptible to damage during the design service life, are specified. Examples are the exclusion or neutralization of alkali-reactive aggregates or the use of "stainless" steel to avoid or slow down reinforcement corrosion.

None of the four verification methods includes the use of models by the contractor/concrete producer. They only have to relate to the results of modelling by the designer / code committee who is the responsible part for writing the execution specification.

It is to be noted that for Deemed to satisfy II, two approaches are alternatively applicable:

- The use of the "performance-related design method" is for example mentioned in section 5.3.3 and annex J of EN 206 [14-8]. The durability properties of the concrete to be tested/qualified are compared to durability indicator threshold values which are defined using the experience obtained from local practices in local environments and the use of proven predictive models;
- The use of the "equivalent performance concept" is for example mentioned in section 5.2.5.3 and annex E of EN 206 [14-8]. This concept is based on the definition of a "reference" concrete for a dedicated exposure class. The durability properties of a concrete to be tested/qualified are compared to those of the "reference" concrete. This "reference" concrete should be a concrete composition complying with the standard's prescriptive and/or performance requirements, according to the classes of exposure to which the structural element is subjected. The concrete to be tested/qualified typically corresponds to concrete compositions not fully complying

Another durability specification approach is the design by avoidance approach, the avoidance of damage is considered either by using non-vulnerable materials or by the separation of environmental actions from the structures.

For the design and assessment of the durability of concrete structures, the previous approaches are classified as indicated in Table 14.3-1.

Table 14.3-1: Concept of Level of Approximation (LoA) for durability specification.

	MODEL	TEST	PARAMETERS INVOLVED	QUALITY CONTROL
1	No model, concrete composition prescriptive rules	As per codes, usually only strength	Strength classes and prescriptive requirements (cement type and min. content, w/c_{max},,…), curing classes	By prescriptive specifications with conformity criteria provided by the codes
2	No model, concrete performance prescriptive rules	Performance tests (durability Indicators)	Performance tests for: gas and water permeability, chloride diffusion, chloride migration, Electrical resistivity, carbonation rate.	By performance with conformity criteria related to concrete properties: For initial testing

with the prescriptive specifications of the Codes.

To apply the performance-based design approach, it is important to define the responsibilities and commitments of the various players and to clarify the contractual framework for its implementation. Currently, international, or even approved detailed national guidelines, defining a contractually applicable framework to implement the design by performance concept for a project are missing. A signed agreement by all stakeholders (owner, designer, contractor, laboratory, etc.) on the whole adopted process is advisable.

			freeze-thaw, alkali silica reaction, delayed ettringite formation, leaching, abrasion, etc... Passed/failed rules (on mean, max/min or characteristic threshold values)	During production
3	Physical model with analytical or numerical solution (Partial-safety factor format)	Performance tests (characteristic values with partial safety factors)	Time, cover thickness, carbonation rate, chloride diffusion coefficient, chloride migration coefficient, conductivity coefficient, ageing factor, gas permeability coefficient, corrosion rate, and other parameters related to deterioration rate. (inputs possibly in function of constitutive parameters / laws)	By performance and modelling with conformity criteria for: Selected Models/Laws Model's input parameters For Initial testing (concrete properties) During production (concrete properties)
4	Physical model with analytical or numerical solution (full-probabilistic)	Performance tests with associated statistical distribution of parameters		By performance and modelling with conformity criteria for: Selected Models Model's input parameters Probability of failure or reliability index as Limit State functions For initial testing of model's input parameters

				During production (concrete properties)

When designing for durability by the performance approach, it has to be taken into account that experience by calibration to long-term experience is still scarce internationally with the prescriptive approach, and for the time-dependent models and of the threshold values of the controlling parameters. This lack of long-term calibration introduces a certain degree of uncertainty that, however, should not discourage their use as gathering feedbacks and setting up database is the only manner to get them calibrated in the future.

Irrespectively of the considered approach, relevant durability indicators must be selected considering approved testing standards or protocols (international or national). Of utmost importance is that compliance shall be established by clear rules considering the expected variability of the measured property.

14.4 Concrete material quality control

The quality control regime should distinguish on whether the specification is prescriptive or by performance (prescribed or designed concrete as per ISO 22965-1 [14-2]).

In the case of prescriptive specification approach (LoA1 in Figure 14.3-1 and Table 14.3-1), the type of constituent materials and their proportions are defined by the owner (see sections 14.3.1 and 14.3.2). Conformity control is performed by checking the supplier's delivery records, verifying that the materials have been batched within the tolerances specified for the project or established in the applicable standard. In addition, delivery certificates by the concrete constituents' supplier should show conformity with the specified materials types (ISO 22965-1).

In the case of Performance specification approach (LoA2, 3 and 4 in Figure 14.3-1 and Table 14.3-1), for regular quality control, two parallel systems may operate simultaneously (ISO 22965-2 [14-3]):

- The Production Control, exercised by the concrete supplier to make sure that the production process is under control (e.g. central values and scatter parameters are within tolerances with respect to the specified quality). Typically, it follows a preventive approach including, e.g.: checking the constituents' quality, inspecting the batching records, calibrating the scales, recording and compensating sand moisture, measuring temperature, consistency, air content, unit weight and early compressive strength (3 to 7 days of age). The test at 28 days serves just to verify that the preventive measures successfully achieved the target quality.

This section refers exclusively to actions intended to control the quality of concrete as a material, either by testing cast specimens, drilled cores or by non-destructive testing of the as-built structure at the jobsite (see chapter 30.11). As such, it is one part of a more comprehensive Quality Management system (see chapter 10) concerned with the overall quality of the structure, in particular during Execution (see chapter 10.5), for which there are many other supplementing means to ensure the correct and predictable quality of the product.

Quality control is set up following the concept of Levels of Approximation defined in section 14.3, as presented in Table 14.3-1.

A Quality Control system should include actions to be undertaken by both the supplier (producer) and the purchaser (client). The actions undertaken by the producer are defined as Production Control, whilst those undertaken by the client are defined as Conformity Control. In some parts of the world (like Europe), the conformity control is normally taken care of by the producer, and for materials of some importance, also supported by a third-party surveillance/certification. It is noted that some of the performance tests last very long, and in some cases accelerated methods can be used for this verification, subject to calibration, and/or use of historical data.

The common goal of both Producer and Client is to ensure that concrete, of quality complying with the specified threshold value P_s, of property P, is poured in the structure. Usually, the threshold value is associated to a certain (low)

- The Conformity Control, exercised by the owner/purchaser by which samples of the delivered concrete are taken at predefined rates from which specimens are cast, cured and tested (usually at 28 days), all according to the applicable standards. The test results are evaluated according to certain statistical rules (specified for the project or established in the applicable standards), leading to a decision of accepting or rejecting the corresponding lots.
-

Principles and framework for determination of the environmental impact are defined in ISO 14040 [14-11]. ISO 13315 [14-##] series provides general procedure for environmental evaluation for each specific lifecycle stage of concrete and concrete structure and ISO 13315-2 [14-##] defines system boundaries for measuring the environmental impact of concrete and raw materials, which is an essential information to perform life cycle assessment.

The production of concrete and its primary constituent, cement, have a pronounced environmental impact. This is why so-called eco-concretes or green concretes with reduced amount of Portland cement clinker, new types of binders and recycled aggregates are increasingly gaining importance as sustainable alternatives. Whereas the reduced environmental impact resulting from the production of such alternatives to ordinary concrete is clearly beneficial, limited information is available on the (long-term) performance of such concretes and on their durability.

In case no information on the concrete environmental impact is available during the design process, the designer can refer to databases such as GaBi [14-12]. EcoInvent [14-13], or Oekobaudat [14-14].

probability of accepting “defective” concrete, i.e. concrete violating the threshold value P_s.

14.5 Environmental performance of concrete

The evaluation of the environmental impact of a concrete structure is highly complex and comprises a great bandwidth of aspects reaching from emissions and resources consumption resulting from the production of the concrete and other building materials, the impact resulting from the building process itself, impacts resulting from the use of the structure (such as heating, cooling etc.) as well as impacts from the demolition of the structure. Principles of evaluation of environmental performance of concrete structures are provided in section 32.3.

Concrete, however, by definition, cannot be sustainable or non-sustainable in itself. It is rather in the responsibility of the designer to use the given material properties in the most sustainable manner during design, execution, and in-service operation of the structure throughout its entire life cycle.

The production of concrete (including provision of raw materials such as cement) is afflicted with pronounced environmental impacts. These impacts comprise the emission of detrimental substances such as e.g. carbon-dioxide (CO_2), as well as the depletion of natural resources such as sand, gravel or water. A reduction of both emissions and resources consumption during concrete production is mandatory, however must be carried out in a sustainable manner, respecting both the requirements on structural performance as well as on other social, economic and environmental performance aspects posed onto a structure during its life cycle. Evaluation of environmental performance of concrete structures may be supported by verification of the environmental impact of concrete material itself and impact resulting from the concrete production process, with respect to reference concrete.

General format for verification of concrete environmental performance limit state is proposed with Eq. (14.5-1):

In order to provide the designer with a indicator to evaluate whether the concrete under consideration has a high potential to build a sustainable structure, the so-called Concrete Environmental Performance Potential (*CEPP*) can be considered. Concrete Environmental Performance Potential (*CEPP*) is defined in Eq. (14.5-2):

$$CEPP = \frac{f_{ck}\, t_{SL}}{EIC} \tag{14.5-2}$$

with:

f_{ck} the characteristic strength of the concrete according to section 14.2 in [MPa] representing the possible performance of the material;

t_{SL} the potential service life of the concrete under the specific environmental actions to be expected in the lifetime of the building member in years

EIC the environmental impact calculation, expressed as the CO_2-eq. for 1 m³ of concrete.

The *CEPP*-index recognizes the fact that high strength concretes may have high emissions for producing 1 m³ of concrete, but that their strongly improved structural behaviour and durability allow to produce highly sustainable structures. It also

$$ELS_{cal} = \frac{\left[\frac{\Sigma EI}{SL}\right]_{eco}}{\left[\frac{\Sigma EI}{SL}\right]_{ref}} \leq ELS_{predefined} \leq 1.0 \tag{14.5-1}$$

with:

ELS_{cal} calculated concrete environmental performance limit state;

$ELS_{predefined}$ limit value that defines the *ELS* criteria;

EI environmental impact of concrete and concrete production;

SL service lifetime.

The index *ref* indicates the value calculated for a reference concrete. The index *eco* indicates the value calculated for a concrete for which a optimization has been carried out in such a way that the predefined limit state criteria ($ELS_{predefined}$) is fulfilled.

For practical application, Equation 14.5-1 can also be simplified, for example by focusing the limit state consideration exclusively on the of CO_2-eq emission. In such case, ΣEI = CO_2-eq mass per cubic metre [kg/m³] of concrete and SL = 1,0. For more details, see Haist et al. [14-xx], *fib* Bulletin XX [14-25].

Verification of the environmental impact of concrete material and concrete production process may also be supported by the calculation of indicators such as for instance the Concrete Environmental Performance Potential (*CEPP*), see Haist et al. [14-yy].

acknowledges that eco-concretes will only be deemed sustainable if they meet the required structural performance and durability criteria. Note that the dimension of the *CEPP*-index (i.e. [MPa · year/(kg CO_2-eq)] does not hold a physical interpretation, but rather represents the impact of crucial input values on the concrete's environmental performance.

Minimizing the environmental impact by choosing a low CO2-concrete might potentially affect the properties of the concrete. Because in this Model Code, all concrete properties related to structural performance are expressed as a function of the characteristic compressive strength f_{ck}, which is a key parameter for quantifying structural performance and well-known to the designer, f_{ck} is incorporated in the *CEPP* definition. Reducing the environmental impact by choosing a low CO_2-concrete should neither lead to a reduction in durability. In recognition of the significance of durability, the potential service life t_{sl} of the concrete under a given environmental impact is also introduced in the *CEPP* definition. For concrete production, the most dominant environmental impact results from the greenhouse gas (GHG) emission, including CO_2 For this reason the CO_2-equivalent is here chosen as the lead environmental impact parameter in the *CEPP* definition. The CO_2-equivalent can be calculated by summing up the CO_2-equivalent contribution of the different concrete constituents. The individual CO_2-equivalent of each constituent material must be obtained from the specific producer. Further, emissions e.g. resulting from the transport of the raw materials to the concrete plant and emissions resulting from the mixing process etc. should be considered. For more details see Haist et al. [14-yy].

14.6 Mechanical properties

14.6.1 Properties related to static loading

14.6.1.1 Compressive strength

The reference compressive strength of the concrete according to this Model Code is measured on cylinders 150/300 mm in accordance with ISO 1920-3 [14-15]; for classification, see section 14.2.1.

In analysis and design of concrete structures, the characteristic compressive strength f_{ck} [MPa] is applied. This value may be derived from strength tests by the criterion that 5% of all possible strength measurements for the specified concrete

For special requirements or in national codes, test specimens other than cylinders 150/300 mm and stored in other environments than those specified in ISO 1920-3 [14-15] may be used to specify the concrete compressive strength. In such cases conversion factors should either be determined experimentally or, when given in national codes, used accordingly for a given category of testing equipment.

In the case where concrete cubes of 150 mm size are used, the characteristic strength values given in Table 14.6-1 must be obtained for the various concrete grades of normal

weight concrete, whereas Table 14.6-2 gives the corresponding characteristic strength values for lightweight aggregate concrete.

The characteristic strength f_{ck} is the strength value for the reference age t_{ref} which should be taken as 28 days in general or may be taken between 28 and 91 days when specified for a project. For some verifications in design, or for an estimate of other concrete properties, it is necessary to refer to a mean value of compressive strength f_{cm} (or f_{lcm} for lightweight aggregate concrete) associated with a specific characteristic compressive strength f_{ck} (or f_{lck} for lightweight aggregate concrete). In this case f_{cm} and f_{lcm} may be estimated from Eqs. (14.6-1) and (14.6-2), respectively:

$$f_{cm} = f_{ck} + \Delta f \qquad (14.6\text{-}1)$$

$$f_{lcm} = f_{lck} + \Delta f \qquad (14.6\text{-}2)$$

where:

$\Delta f = 8$ MPa

Background information on the strength independence of Δf may be found in Muller et al. [14-16].

are expected to fall below the value f_{ck}; see also section 14.2.1.1 and Tables 14.6-1 and 14.6-2.

Table 14.6-1: Characteristic strength values of normal weight concrete [MPa].

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50
f_{ck}	12	16	20	25	30	35	40	45	50
$f_{ck,cube}$	15	20	25	30	37	45	50	55	60

Concrete grade	C55	C60	C70	C80	C90	C100	C110	C120
f_{ck}	55	60	70	80	90	100	110	120
$f_{ck,cube}$	67	75	85	95	105	115	130	140

Table 14.6-2: Characteristic strength values of lightweight aggregate concrete [MPa].

Concrete grade	LC8	LC12	LC16	LC20	LC25	LC30	LC35
f_{lck}	8	12	16	20	25	30	35
$f_{lck,cube}$	9	13	18	22	28	33	38

Concrete grade	LC40	LC45	LC50	LC55	LC60	LC70	LC80
f_{lck}	40	45	50	55	60	70	80
$f_{lck,cube}$	44	50	55	60	66	77	88

Although the uniaxial tensile testing is the most appropriate method for determining the tensile strength of concrete, it is rarely used anywhere other than in research because of the experimental difficulties in performing such experiments. Therefore, in many instances the splitting tensile strength or the flexural tensile strength are determined.

When testing tensile strength, special attention should be paid to possible effects of moisture gradients.

Table 14.6-3 gives tensile strength values for normal weight concrete, estimated from the characteristic compressive strength f_{ck} according to Eqs. (14.6-3) to (14.6-5).

Table 14.6-3: Tensile strength in MPa for different concrete grades.

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50
f_{ctm}	1.4	1.9	2.3	2.7	3.0	3.3	3.5	3.8	3.9
$f_{ctk,min}$	1.0	1.3	1.6	1.9	2.1	2.3	2.5	2.6	2.8
$f_{ctk,max}$	1.8	2.5	3.0	3.5	3.9	4.3	4.6	4.9	5.1

Concrete grade	C55	C60	C70	C80	C90	C100	C110	C120
f_{ctm}	4.1	4.3	4.5	4.8	5.0	5.2	5.4	5.5
$f_{ctk,min}$	2.9	3.0	3.2	3.4	3.5	3.6	3.8	3.9
$f_{ctk,max}$	5.3	5.6	5.9	6.2	6.5	6.7	7.0	7.2

If there is no test procedure agreed or given in national guidelines, tests may be performed according to RILEM CPC 7 [14-17].

Eq. (14.6-3) was derived by evaluating available data from axial tension and compression tests. The data from splitting and flexural tests were not considered, in order to avoid evident uncertainties resulting from indirect testing – see *fib* Bulletin 42 [14-18].

14.6.1.2 Tensile strength and fracture properties

14.6.1.2.1 Tensile strength

The tensile strength of the concrete and the term “tensile strength”, unless stated otherwise in this Model Code, refer to the uniaxial tensile strength f_{ct} determined in related experiments.

In the absence of experimental data, the mean value of tensile strength f_{ctm} in MPa may be estimated for normal weight concrete from the characteristic compressive strength f_{ck}:

$$f_{ctm} = 1.8 ln(f_{ck}) - 3.1 \quad (14.6\text{-}3)$$

where:

f_{ck} is the characteristic compressive strength in MPa according to Table 14.6-1;

The lower and upper bound values of the characteristic tensile strength $f_{ctk,min}$ and $f_{ctk,max}$ may be estimated using, respectively:

$$f_{ctk,min} = 0.7 f_{ctm} \quad (14.6\text{-}4)$$

$$f_{ctk,max} = 1.3 f_{ctm} \quad (14.6\text{-}5)$$

To estimate a mean value of the tensile strength f_{lctm} for lightweight aggregate concrete, f_{ctm} according to Eq. (14.6-3) must be multiplied by a reduction factor η_l:

$$f_{lctm} = \eta_l \cdot f_{ctm} \quad (14.6\text{-}6a)$$

$$\eta_l = (0.4 + 0.6 \cdot \rho / 2'200) \quad (14.6\text{-}6b)$$

where:

ρ is the oven-dry density of the lightweight aggregate concrete in kg/m^3.

The lower and upper bound values of the characteristic tensile strength $f_{lctk,min}$ and $f_{lctk,max}$ may be estimated for lightweight aggregate concrete using Eqs. (14.6-4) and (14.6-5), respectively, replacing f_{ctm} by f_{lctm}.

If the tensile strength is measured as splitting tensile strength $f_{ct,sp}$ or as flexural tensile strength $f_{ct,fl}$ a conversion factor α should be determined by means of uniaxial tension tests.

If such conversion factors are not available, the mean uniaxial tensile strength f_{ctm} may be estimated from the mean splitting tensile strength $f_{ctm,sp}$ as:

$$f_{ctm} = \alpha_{sp} \cdot f_{ctm,sp} \qquad (14.6\text{-}7)$$

where:

$f_{ctm,sp}$ is the mean value of splitting tensile strength determined according to ISO 1920-4 [14-20];

α_{sp}= 1.0.

The same conversion factor α_{sp} = 1.0 may be used for lightweight aggregate concrete.

In order to estimate the mean uniaxial tensile strength f_{ctm} from the mean flexural tensile strength $f_{ctm,fl}$ we can use:

$$f_{ctm} = \alpha_{fl} \cdot f_{ctm,fl} \qquad (14.6\text{-}8a)$$

where:

$f_{ctm,fl}$ is the mean flexural tensile strength;

$$\alpha_{fl} = \frac{0.06 h_b^{0.7}}{1 + 0.06 h_b^{0.7}} \qquad (14.6\text{-}8b)$$

where:

h_b is the beam depth [mm].

14.6.1.2.2 Fracture energy

In existing national and international codes and standards values of the conversion factor α_{sp} may be found, which vary from 0.67 to 0.95. However, comprehensive new research results show that this factor is beyond 1.0; see Malárics & Müller [14-19]. The data indicates that for cast specimens, $\alpha_{sp} = 2.08 \cdot (f_{cm})^{-0.16}$ with f_{cm} in MPa. For crushed aggregates the value for α_{sp} may be increased up to 20%.

As a compromise, α_{sp} = 1.0 has been chosen.

Eqs. (14.6-8a) and (14.6-8b) were deduced from fracture mechanics considerations. In CEB-FIP MC 1990 [14-21] the relation given by Eq. (14.6-8b) was given for normal strength concrete. Since the ratio of flexural strength to uniaxial tensile strength of concrete $f_{ctm,fl}/f_{ctm}$ should decrease for a given beam depth if the concrete becomes more brittle, α_{fl} should depend on the brittleness of the concrete and decrease as brittleness increases. This means that in Eq. (14.6-8b) the number 0.06 for high strength concrete and for lightweight aggregate concrete should be replaced by a value lower than 0.06, to be determined by experiments.

The fracture mode of concrete subjected to tension allows the application of fracture mechanics concepts, that is, energy considerations. In those concepts, the fracture energy of concrete G_F is often used as a material characteristic to describe the resistance of concrete subjected to tensile stresses.

G_F should best be determined from uniaxial tension tests. Most frequently, however, indirect tests, first of all three-point bend tests on notched beams are used, which are easier to perform.

For normal weight concrete the fracture energy depends primarily on the water-to-cement ratio, the maximum aggregate size and the age of concrete. Curing conditions also have a significant effect on experimentally determined G_F values. Further, G_F is affected by the size of a structural member and in particular by the depth of the ligament above a crack or notch. The fracture energy of high strength normal weight concrete is also influenced by the above-mentioned parameters, but not to the same extent as in the case of normal strength concrete. The aggregate type and content seem to affect the fracture energy of concrete much more than the size of aggregate. This phenomenon is caused by the transition from interfacial fracture to trans-aggregate fracture. For high strength concrete, the effect of curing conditions on G_F is somewhat less pronounced than for normal strength concrete, but it is still significant. For further information, see *fib* Bulletin 42 [14-18].

The knowledge of fracture mechanisms of lightweight aggregate concrete (LWAC) is still insufficient, and the dependence of fracture energy of LWAC on different parameters (density, types of aggregates, strength etc.) must be addressed by future research. LWAC is notch sensitive (most important for this sensitivity are eigenstresses, because of moisture gradients). The maximum crack opening depends on the kind of matrix and the kind of aggregates, respectively. Thus, tests to determine fracture energy and softening behaviour are mandatory if relevant information on LWAC needs to be used for analysis and design.

This failure criterion is one among several acceptable formulations. It has been chosen since it is not too difficult to use and agrees well with test data. For further details and the range of applicability of Eq. (14.6-11), refer to the CEB Bulletin 156 [14-22], to Ottosen [14-23] and to Speck & Curbach [14-24].

The fracture energy of concrete G_F [N/m], defined as the energy required to propagate a tensile crack of unit area, should be determined by related tests.

In the absence of experimental data G_F in N/m for ordinary normal weight concrete may be estimated as:

$$G_F = 85 f_{ck}^{0.15} \qquad (14.6\text{-}9)$$

where:

f_{ck} is the characteristic compressive strength in MPa.

As an approximation for estimating the fracture energy of lightweight aggregate concrete, we can use:

$$G_{F,l} = G_{FoA} + 16 f_{lctm} \qquad (14.6\text{-}10)$$

where:

$G_{F,l}$ is obtained in N/m;

G_{FoA} = 24 N/m for lightweight aggregate concrete with normal weight sand;

= 0 for lightweight aggregate concrete with lightweight sand;

f_{lctm} is the mean value of tensile strength in MPa.

14.6.1.3 Strength under multiaxial states of stress

The mean value of strength under multiaxial states of stress may be estimated from the failure criterion given by Eq. (14.6-11).

The criterion applies for monotonic stress increase until failure. The load capacity remains as long as the stress exists. At unloading or modification of the stress ratio there remains only a low load capacity. This effect occurs especially with lightweight concrete, but also with normal concrete at stress ratios with high hydrostatic stresses. In this case, introducing a cap function may be useful, which closes the open top failure curve. For normal concrete, concerning the intersection point of the cap function with the hydrostatic axis, different information ranging from 1.6 to 2.3 times the uniaxial strength can be found.

The invariants of the stress tensor (I_1) and the stress deviators (J_2 and J_3) used in Eqs. (14.6-11) to (14.6-13) may be calculated as follows:

$$I_1 = \sigma_1 + \sigma_2 + \sigma_3$$

$$J_2 = \frac{1}{6}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]$$

$$J_3 = (\sigma_1 - \sigma_m)\cdot(\sigma_2 - \sigma_m)\cdot(\sigma_3 - \sigma_m)$$

$$\sigma_m = (\sigma_1 + \sigma_2 + \sigma_3)/3$$

The stress coordinates σ_o and τ_o (octahedron stresses) may be calculated as:

$$\sigma_o = \frac{I_1}{3} \qquad \text{and} \qquad \tau_o = \sqrt{\frac{2}{3}\cdot J_2}$$

Note that f_c and f_{c2c} are defined as positive values; all other compressive stresses and strengths are negative values.

No standardized test method is available for determining the multiaxial strength. During the test the load has to be applied by special test devices, which follow the deformation of the specimen and prevent parts of the load being introduced through friction into the lateral load application system. Detailed information is available in [Background document [14-25]].

In the absence of experimental data the biaxial compressive strength and the triaxial compressive strength at one point on the compressive meridian may be estimated from the uniaxial compressive strength. Note that no consolidated experience exists for a stress level above σ_{com} = −240 MPa for normal weight concrete and σ_{lcom} = −60 MPa for

$$\alpha\frac{J_2}{f_{cm}^2} + \lambda\frac{\sqrt{J_2}}{f_{cm}} + \beta\frac{I_1}{f_{cm}} - 1 = 0 \tag{14.6-11}$$

where:

$$\lambda = c_1 \cdot cos\left[\frac{1}{3}\cdot arccos\left(c_2 \cdot cos\,3\theta\right)\right] \tag{14.6-12}$$

$$cos\,3\theta = \frac{3\sqrt{3}}{2}\cdot\frac{J_3}{J_2^{\frac{3}{2}}} \tag{14.6-13}$$

The parameters J_2, J_3 and I_1 in Eqs. (14.6-11) to (14.6-13) represent the invariants of the two stress deviators and the stress tensor, respectively, characterizing the state of stress considered.

For lightweight aggregate concrete f_{cm} in Eq. (14.6-11) has to be replaced by f_{lcm}.

The coefficients α, β, c_1 and c_2 are material parameters which depend on the uniaxial compressive strength f_{cm} (or f_{lcm} for lightweight aggregate concrete), the uniaxial tensile strength f_{ctm} (or f_{lctm}), the biaxial compressive strength f_{c2cm} (or f_{lc2cm}) and the triaxial compressive strength at one point on the compressive meridian ($\sigma_1 = \sigma_2 > \sigma_3$) described by σ_{com} and τ_{com} (or σ_{lcom} and τ_{lcom}). To determine these coefficients, the parameters have to be calculated as:

$$k = \frac{f_{ctm}}{f_{cm}};\ f_{2c} = \frac{f_{c2cm}}{f_{cm}};\ x = \frac{\sigma_{com}}{f_{cm}};\ y = \frac{\tau_{com}}{f_{cm}};\ h = -\frac{\sqrt{2}\cdot x + y}{\frac{y}{\sqrt{2}} - \frac{1}{3}} \tag{14.6-14}$$

$$\alpha = \frac{h\cdot\beta - \sqrt{2}}{y}; \qquad \beta = \frac{\sqrt{2} - \frac{3\cdot y}{k\cdot f_{2c}}}{h - \frac{9\cdot y}{f_{2c} - k}} \tag{14.6-15}$$

$$\lambda_c = \lambda(\theta = 60°) = \left(1 - \frac{h}{3\cdot y}\right)\cdot\sqrt{3}\cdot\beta + \sqrt{3} + \frac{\sqrt{2}}{\sqrt{3}\cdot y} \tag{14.6-16}$$

lightweight aggregate concrete. No sufficient experimental data exist for self-compacting concrete, steel fibre concrete or other types of concrete.

$$f_{c2c} = \left(1.2 - \frac{f_c}{1000}\right) \cdot f_c$$

where:

$f_c = f_{cm}$ for $f_{c2c} = f_{c2cm}$;
$f_c = f_{ck}$ for $f_{c2c} = f_{c2ck}$;
$f_c = f_{lcm}$ for $f_{c2c} = f_{lc2cm}$;
$f_c = f_{lck}$ for $f_{c2c} = f_{lc2ck}$.

$$\tau_{com} = \left[185 - 180 \cdot \frac{f_{cm}}{100} + 260 \cdot \left(\frac{f_{cm}}{100}\right)^2 - 84 \cdot \left(\frac{f_{cm}}{100}\right)^3\right]$$

$$\tau_{lcom} = \left[250 \cdot \frac{f_{lcm}}{100} - 460 \cdot \left(\frac{f_{lcm}}{100}\right)^2 + 310 \cdot \left(\frac{f_{lcm}}{100}\right)^3\right]$$

$$\tau_{cok} = \left(0.8 + \frac{f_{ck}}{1000}\right) \cdot \tau_{com}$$

for $\sigma_{com} = \sigma_{cok} = -240$ MPa and $\sigma_{lcom} = \sigma_{lcok} = -60$ MPa

with f_{cm}, f_{ck}, f_{lcm} and f_{lck} in MPa.

The coefficients for normal weight concrete given in Figure 12.6-1 are the results of Eqs. (14.6-14) to (14.6-19b).

$$\lambda_t = \lambda(\theta = 0°) = \left(2 \cdot \sqrt{3} - \frac{f_{2c} \cdot h}{\sqrt{3} \cdot y}\right) \cdot \beta + \frac{\sqrt{3}}{f_{2c}} + \frac{\sqrt{2} \cdot f_{2c}}{\sqrt{3} \cdot y} \tag{14.6-17}$$

$$c_1 = [2 \cdot \cos\theta - 1] \cdot \lambda_t + 4 \cdot [1 - \cos\theta] \cdot \lambda_c \quad \text{for } \frac{\lambda_c}{\lambda_t} \le \frac{1}{2} \tag{14.6-18a}$$

$$c_1 = \frac{\lambda_c}{\cos\left(\frac{\pi}{3} - \frac{1}{3} \cdot \arccos(c_2)\right)} \quad \text{for } \frac{\lambda_c}{\lambda_t} \ge \frac{1}{2} \tag{14.6-18b}$$

$$c_2 = 1 \quad \text{for } \frac{\lambda_c}{\lambda_t} \le \frac{1}{2} \tag{14.6-19a}$$

$$c_2 = \cos\left\{3 \cdot \arctan\left[\frac{\left(2 \cdot \frac{\lambda_c}{\lambda_t} - 1\right)}{\sqrt{3}}\right]\right\} \quad \text{for } \frac{\lambda_c}{\lambda_t} \ge \frac{1}{2}. \tag{14.6-19b}$$

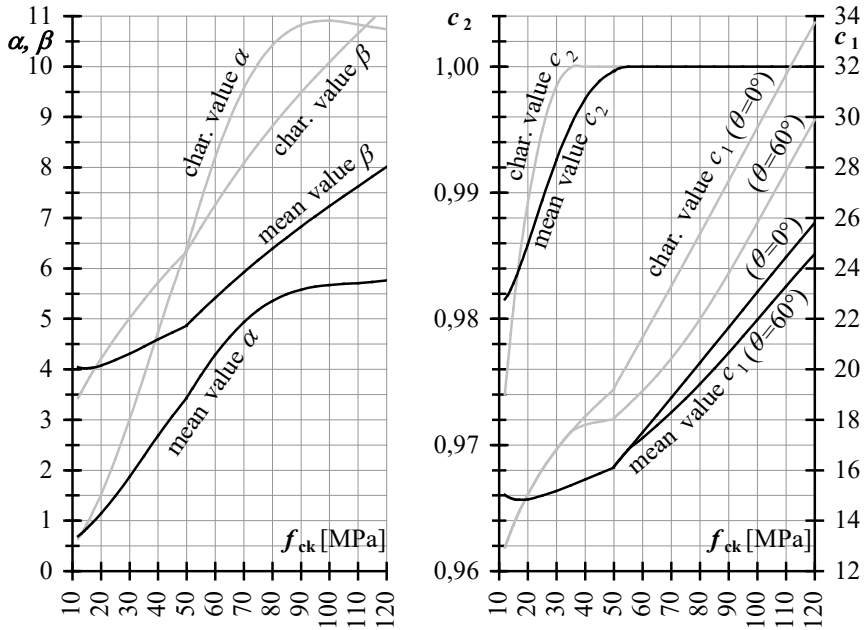


Figure 14.6-1: Coefficients for Eq.(14.6-11), normal weight concrete.

To estimate a characteristic multiaxial strength, in Eqs. (14.6-11) and (14.6-14) the mean values of uniaxial compressive and tensile strength, biaxial and triaxial compressive strength have to be substituted by the characteristic values of these strengths.

The strength of concrete under biaxial states of stress ($\sigma_3 = 0$) may be estimated from the same criteria as given in Eqs. (14.6-11) to (14.6-19).

14.6.1.4 Modulus of elasticity and Poisson's ratio

14.6.1.4.1 Range of application

The information given in this section is valid for monotonically increasing compressive stresses or strains at a rate of $\dot{\sigma} \approx 0.6 \pm 0.4$ MPa/s or $\dot{\varepsilon} \approx 15 \cdot 10^{-6}$ s^{-1}, respectively. For tensile stresses or strains, it is valid for $\dot{\sigma} \approx 0.06$ MPa/s or $\dot{\varepsilon} \approx 1.5 \cdot 10^{-6}$ s^{-1}, respectively.

14.6.1.4.2 Modulus of elasticity

The modulus of elasticity E_{ci}, as obtained from Eqs. (12.6-20) and (12.6-21), is defined as the tangent modulus of elasticity at the origin of the stress–strain diagram. It is approximately equal to the slope of the secant of the unloading branch for rapid unloading and does not include initial plastic deformations. It has to be used for the description of the stress–strain diagrams for uniaxial compression and uniaxial tension according to sections 5.1.8.1 and 5.1.8.2, respectively, as well as for an estimate of creep according to Eq. (14.6-53), subsection 5.1.9.4.3. The reduced modulus of elasticity E_c according to Eq. (14.6-23) includes some irreversible strains.

The elastic deformations of concrete largely depend on its composition (especially the type of natural aggregate). The values given in this Model Code (Table 14.6-5) should be regarded as indicative for general applications. However, the modulus of elasticity should be specifically assessed or experimentally determined if the structure is likely to be sensitive to deviations from these general values. This particularly applies to concretes made with recycled aggregates and concretes that significantly differ from common compositions (e.g. concretes with cements not covered by EN 197-1 [14-26] and sand-rich concretes). In this context, see RILEM CPC 8 [14-27]; a similar test procedure is published in ISO 1920-10 [14-28], ASTM C 469 [14-29] and EN 12390-13 [14-30]. However, the latter standard describes the determination of the secant modulus of elasticity in compression The secant modulus are only convertible into the tangent modulus in the above-mentioned unloading branch (see Figure 14.6-2).

Compared to the use of quartzite aggregates the modulus of elasticity can be increased by 20% or decreased by 30% only by changing the type of aggregate. Eqs. (14.6-20), (14.6-21) and Table 14.6-4 give the qualitative changes α_E in the modulus of elasticity for different types of aggregate.

Table 14.6-4: Effect of types of aggregate on the modulus of elasticity.

Types of aggregate	α_E	$E_{c0} \cdot \alpha_E$ [MPa]
Stiff aggregates, e.g. Basalt*)	1.2	25800

Values for the modulus of elasticity for normal weight concrete with natural sand and gravel can be estimated from the specified characteristic strength using:

$$E_{ci} = E_{c0} \cdot \alpha_E \left(\frac{f_{ck} + \Delta f}{10} \right)^{\frac{1}{3}} \qquad (14.6\text{-}20)$$

where:

E_{ci} is the modulus of elasticity in MPa at the concrete age of 28 days;

f_{ck} is the characteristic strength in MPa according to section 14.6.1.1;

Δf = 8 MPa;

E_{c0} = $21.5 \cdot 10^3$ MPa;

α_E is 1.0 for quartzite aggregates. For different types of aggregate qualitative values for α_E can vary between 0.7 and 1.2; see Table 14.6-4.

Where the actual compressive strength of concrete at an age of 28 days f_{cm} is known, E_{ci} may be estimated from:

$$E_{ci} = E_{c0} \cdot \alpha_E \left(\frac{f_{cm}}{10} \right)^{\frac{1}{3}} \qquad (14.6\text{-}21)$$

The modulus of elasticity for lightweight aggregate concrete E_{lci} can be estimated from:

$$E_{lci} = \eta_E \cdot E_{ci} \qquad (14.6\text{-}22)$$

where:

$\eta_E = \left(\frac{\rho}{2200} \right)^2$ is reduction factor;

ρ is the oven-dry density of the lightweight aggregate concrete in kg/m^3;

E_{ci} is the modulus of elasticity in MPa according to Eq. (14.6-20) or Eq. (14.6-21); here α_E = 1.0 for all types of lightweight aggregates.

Ordinary aggregates, e.g. Quartzite*)	1.0	21500
Soft aggregates, e.g. Sandstone*)	0.7	15100

*)Properties can strongly vary depending on local geology

The modulus of elasticity E_{ci} does not include the initial plastic strain due to its definition. While the limit for the stress σ_c reached in the serviceability limit state (SLS) is set to $\sigma_c = -0.4 \cdot f_{cm}$ this stress level gives an upper limit for the reduction factor α_i (Figure 14.6-2, Eq. (14.6-23)). This factor $\alpha_i = E_c/E_{ci}$ is increasing with increasing concrete strength. For concrete grades higher than C80 the difference between first loading up to $\sigma_c = -0.4 \cdot f_{cm}$ and the unloading branch is smaller than 3% and could be neglected.

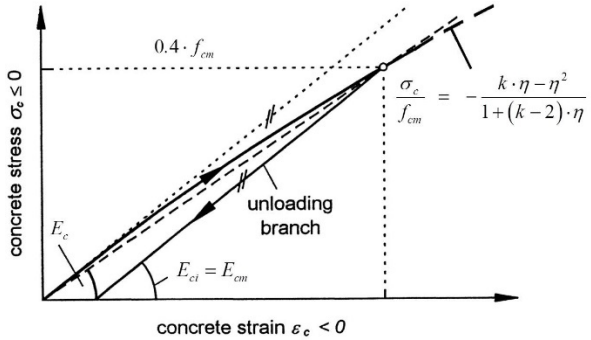


Figure 14.6-2: Definition of different moduli of elasticity (according to fib Bulletin 42 [14-18]).

Note that E_{ci} is considered as the mean value of the tangent modulus of elasticity; hence $E_{ci} = E_{cm}$.

Self-compacting concrete (SCC), being produced with an increased binder content (powder type SCC), may have a reduced value of E_{ci} up to approximately 20% at maximum compared to conventional concrete of equal strength. However, the E_{ci} values are within the scatter band for ordinary structural concrete.

Where only an elastic analysis of a concrete structure is carried out, a reduced modulus of elasticity E_c according to Eq. (14.6-23) should be used in order to account for initial plastic strain, causing some irreversible deformations.

$$E_c = \alpha_i \cdot E_{ci} \tag{14.6-23}$$

where:

$$\alpha_i = 0.8 + 0.2 \frac{f_{cm}}{88} \leq 1.0 \tag{14.6-24}$$

Values of the tangent modulus E_{ci} and the reduced modulus E_c for different concrete grades are given in Table 14.6-5.

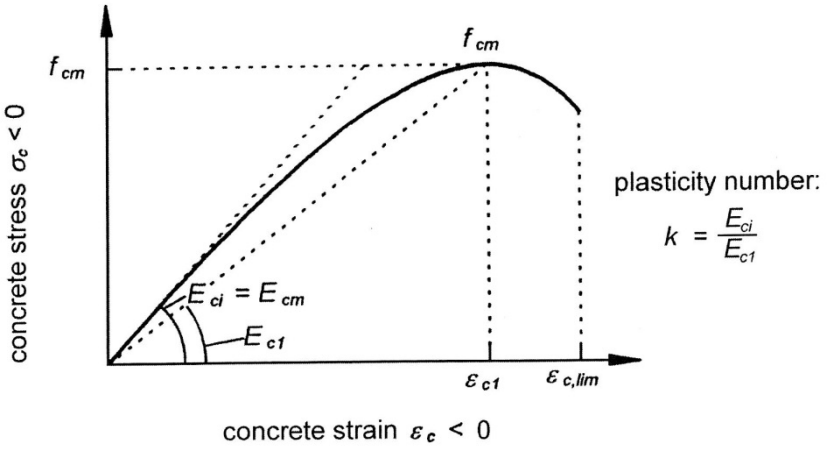
Table 14.6-5: Tangent modulus and reduced modulus of elasticity.

Concrete grade	**C12**	**C16**	**C20**	**C25**	**C30**	**C35**	**C40**	**C45**	**C50**
E_{ci} [GPa]	27.1	28.8	30.3	32.0	33.6	35.0	36.3	37.5	38.6
E_c [GPa]	22.9	24.6	26.2	28.0	29.7	31.4	33.0	34.5	36.0
α_i	0.845	0.855	0.864	0.875	0.886	0.898	0.909	0.920	0.932

Concrete grade	**C55**	**C60**	**C70**	**C80**	**C90**	**C100**	**C110**	**C120**
E_{ci} [GPa]	39.7	40.7	42.6	44.4	46.0	47.5	48.9	50.3
E_c [GPa]	37.5	38.9	41.7	44.4	46.0	47.5	48.9	50.3
α_i	0.943	0.955	0.977	1.0	1.0	1.0	1.0	1.0

E_c for normal weight concrete and E_{lc} for lightweight aggregate concrete are defined as the reduced or secant value of the modulus of elasticity.

The value of $\nu_c = 0.20$ is also applicable for all other types concrete covered by this Model Code (such as Lightweight Concrete, Self-Compacting Concrete and High Strength Concrete). Poisson's ratio can be assumed to be constant for $\sigma_c \leq 0.6 \cdot f_{ck}$. ν_c should be specifically assessed or experimentally determined for concretes made with recycled aggregates and concretes which significantly differ from common compositions (e.g. concretes with cements not covered by EN 197-1 [14-26] and sand-rich concretes).



The modulus of elasticity for lightweight aggregate concrete E_{lc} can be estimated by multiplying E_c from Eq. (14.6-23) with the reduction factor η_E given in Eq. (14.6-22):

$$E_{lc} = \eta_E \cdot E_c \tag{14.6-25}$$

14.6.1.4.3 Poisson's ratio

For a range of stresses $-0.6 \cdot f_{ck} < \sigma_c < 0.8\ f_{ctk}$ the Poisson's ratio of concrete ν_c ranges between 0.14 and 0.26. Regarding the significance of ν_c, for the design of members, especially the influence of crack formation at the ultimate limit state (ULS), the estimation of $\nu_c = 0.20$ meets the required accuracy.

14.6.1.5 Stress–strain relations for short term loading

14.6.1.5.1 Compression

The relation between σ_c and ε_c for short term uniaxial compression shown in Figure 14.6-3 is described by Eq. (14.6-26):

$$\frac{\sigma_c}{f_{cm}} = -\left(\frac{k \cdot \eta - \eta^2}{1 + (k-2)\eta}\right) for \left|\varepsilon_c\right| < \left|\varepsilon_{c,lim}\right| \tag{14.6-26}$$

where:

$\eta = \varepsilon_c / \varepsilon_{c1}$;

$k = E_{ci} / E_{c1}$;

ε_{c1} is the strain at maximum compressive stress (Table 14.6-6);

E_{c1} is the secant modulus from the origin to the peak compressive stress (Table 14.6-6);

k is the plasticity number (Table 14.6-6).

Figure 14.6-3: Schematic representation of the stress–strain relation for short term loading in uniaxial compression (according to fib Bulletin 42 [14-18]).

[Checking whether relation in EN 1992-1-1 is more appropriate.]

The stress–strain relations for concrete in compression generally comply with the schematic representation shown in Figure 14.6-3. The descending branch of the stress–strain relations should be considered as the envelope to all possible stress–strain relations of concrete, which tends to soften as a consequence of concrete micro-cracking.

The descending part of the stress–strain curve in compression is strongly dependent on the specimen or member geometry, the boundary conditions and the possibilities for load redistribution in the structure. In tests, a strong influence of the rigidity of the testing device used can be observed. During the softening process micro-cracking occurs in a fracture zone of a limited length and width (strain localization). One single fracture zone is supposed to be decisive for the failure of a given member. The stress in the fracture zone drops down with a shear displacement in local shear bands of $w_c \approx 0.5$ mm.

The ultimate strain $\varepsilon_{c,lim}$ is caused by the displacement w_c related to a certain length which is given in Figure 4-4 in *fib* Bulletin 42 [14-18]). The descending portion of the stress–strain relation is size-dependent and therefore not only a material property (see Figure 4-5 in *fib* Bulletin 42 [14-18]).

The stress–strain relation may be best determined by corresponding tests. If only the modulus of elasticity is available from experiments, this value may be used for estimating the stress–strain diagram. However, an accurate stress–strain diagram can only be found if the plasticity number k was investigated.

Eqs. (14.6-25) and (14.6-27) should be used with care when lightweight aggregate concretes with oven-dry densities < 1600 kg/m^3 are considered.

Table 14.6-6: *Moduli E_{ci}, E_{c1}, strains ε_{c1}, $\varepsilon_{c,lim}$ and plasticity number k for normal weight concrete.*

Concrete grade	**C12**	**C16**	**C20**	**C25**	**C30**	**C35**	**C40**	**C45**	**C50**
E_{ci} [GPa]	27.1	28.8	30.3	32.0	33.6	35.0	36.3	37.5	38.6
E_{c1} [GPa]	11.1	12.2	13.3	14.9	16.5	18.2	20.0	21.6	23.2
ε_{c1} [‰]	−1.9	−2.0	−2.1	−2.2	−2.3	−2.3	−2.4	−2.5	−2.6
$\varepsilon_{c,lim}$ [‰]	−3.5	−3.5	−3.5	−3.5	−3.5	−3.5	−3.5	−3.5	−3.4
k	2.44	2.36	2.28	2.15	2.04	1.92	1.82	1.74	1.66

Concrete grade	**C55**	**C60**	**C70**	**C80**	**C90**	**C100**	**C110**	**C120**
E_{ci} [GPa]	39.7	40.7	42.6	44.4	46.0	47.5	48.9	50.3
E_{c1} [GPa]	24.7	26.2	28.9	31.4	33.8	36.0	39.3	42.7
ε_{c1} [‰]	−2.6	−2.7	−2.7	−2.8	−2.9	−3.0	−3.0	−3.0
$\varepsilon_{c,lim}$ [‰]	−3.4	−3.3	−3.2	−3.1	−3.0	−3.0	−3.0	−3.0
k	1.61	1.55	1.47	1.41	1.36	1.32	1.24	1.18

For the calculation of ε_{lc1} for lightweight aggregate concrete a factor κ_{lc} is introduced, taking into account different types of sand:

$$\varepsilon_{lc1} = -\kappa_{lc} \frac{f_{lck} + 8}{E_{lc}} \qquad (14.6\text{-}27)$$

where:

f_{lck} is the characteristic strength value for lightweight aggregate concrete in MPa according to Table 14.6-2;

E_{lc} is the modulus of elasticity in MPa for lightweight aggregate concrete according to Eq. (14.6-25);

κ_{lc} 1.1 for lightweight aggregate concrete with light sand;

1.3 for lightweight aggregate concrete with natural sand.

The stress–strain relation for unloading of the uncracked concrete may described as:

$$\Delta\sigma_c = E_{ci} \cdot \Delta\varepsilon_c \qquad (14.6\text{-}28)$$

where:

$\Delta\sigma_c$ is the stress reduction;

$\Delta\varepsilon_c$ is the strain reduction.

14.6.1.5.2 Tension

Tensile failure of concrete is always a discrete phenomenon. Thus, to describe the tensile behaviour a stress–strain relation should be used for the uncracked concrete, and a stress-crack opening relation, as shown in Figure 14.6-4, should be used for the cracked section.

Since the post-cracking curve, as shown in Figure 14.6-4, is size-dependent it is recommended to carefully use this approach when constitutive relations for concrete need to be derived.

For uncracked normal weight concrete subjected to tension, the following bilinear stress–strain relation may be used (Figure 14.6-4):

$$\sigma_{ct} = E_{ci} \cdot \varepsilon_{ct} \text{ for } \sigma_{ct} \le 0.9 \cdot f_{ctm} \qquad (14.6\text{-}29$$

$$\sigma_{ct} = f_{ctm}\left(1 - 0.1\frac{0.00015 - \varepsilon_{ct}}{0.00015 - 0.9 \cdot \frac{f_{ctm}}{E_{ci}}}\right) \text{ for } 0.9 f_{ctm} < \sigma_{ct} \le f_{ctm} \qquad (14.6\text{-}30$$

where:

E_{ci} is the tangent modulus of elasticity in MPa according to Eq. (14.6-20);

ε_{ct} is the tensile strain;

σ_{ct} is the tensile stress in MPa;

f_{ctm} is the tensile strength in MPa from Eq. (14.6-3).

For a cracked section a bilinear approach for the stress-crack opening relation according to Figure 14.6-4 can be estimated by:

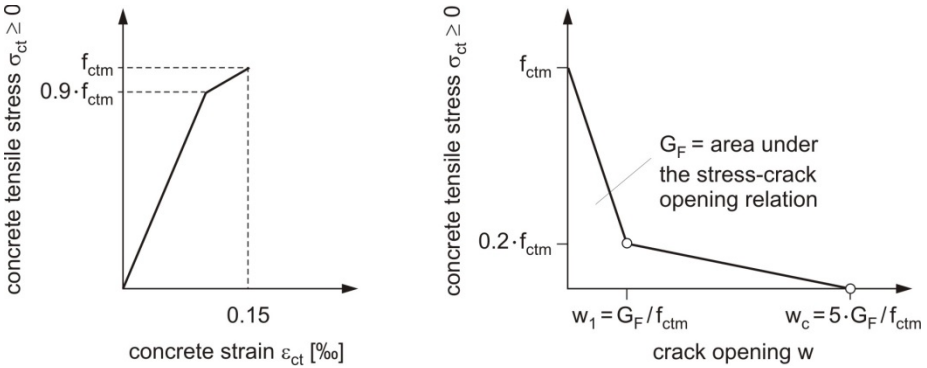


Figure 14.6-4: Schematic representation of the stress–strain and stress-crack opening relation for uniaxial tension (according to fib Bulletin 42 [14-18]).

At tensile stresses of about 90% of the tensile strength f_{ct}, micro-cracking starts to reduce the stiffness in a small failure zone (Eqs. (14.6-29) and (14.6-30). The micro-cracks grow and form a discrete crack at stresses close to the tensile strength. All stresses and deformations in the fracture process zone can be related to a fictitious crack opening w (according to *fib* Bulletin 42 [14-18]).

Neglecting the small energy consumed by a complete loading cycle in the stress–strain relation, the maximum strain $\varepsilon_{ct,max}$ can be estimated as $\varepsilon_{ct,max} \approx f_{ctm}/E_{ci}$. For the analysis of the fracture zone a strain $\varepsilon_{ct,max} = 0.15$ ‰ can be assumed. Due to the localization of micro-cracking in the fracture zone and the large uncracked areas outside the damage zone, this strain is only valid inside the fracture zone.

Regarding the fracture energy in general, see section 5.1.5.2. To describe the stress–strain relation for uniaxial tension for lightweight aggregate concrete, see Faust [14-31].

$$\sigma_{ct} = f_{ctm}\left(1.0 - 0.8\frac{w}{w_1}\right) \quad \text{for} \quad w \le w_1 \tag{14.6-31}$$

$$\sigma_{ct} = f_{ctm}\left(0.25 - 0.05\frac{w}{w_1}\right) \quad \text{for} \quad w_1 < w \le w_c \tag{14.6-32}$$

where:

w is the crack opening in mm;

w_1 $= G_F/f_{ctm}$ in mm when $\sigma_{ct} = 0.20 \cdot f_{ctm}$;

w_c $= 5 \cdot G_F/f_{ctm}$ in mm when $\sigma_{ct} = 0$;

G_F is the fracture energy in N/mm from Eq. (14.6-9);

f_{ctm} is the tensile strength in MPa from Eq. (14.6-3).

14.6.1.5.3 Multiaxial states of stress

Homogeneous macroscopic approaches are mandatory to describe behaviour of structures with spatial dimensions of [m]. Mesoscopic approaches with distinction of cement matrix, aggregates and interfacial transition zones are used for specimen with dimensions of [mm]. Microscopic approaches treat the crystalline structure of the cement matrix with the dimension of [µm].

Different spatial – microscopic, mesoscopic, macroscopic – and temporal – high strain rate, short term, long term – scales have to be distinguished to analyse multiaxial concrete behaviour. Homogenized macroscopic stress-strain behaviour in short terms neglecting influence of time is treated in the following.

Two different fundamental approaches are available with tensor based for stress and strain or vector based for stress and strain. The latter is known as micro-plane

Types of macroscopic behaviour become distinguishable with unloading from a phenomenological point of view where concrete proves to be a combination of damage (→ degradation) and plasticity (→ permanent strains).

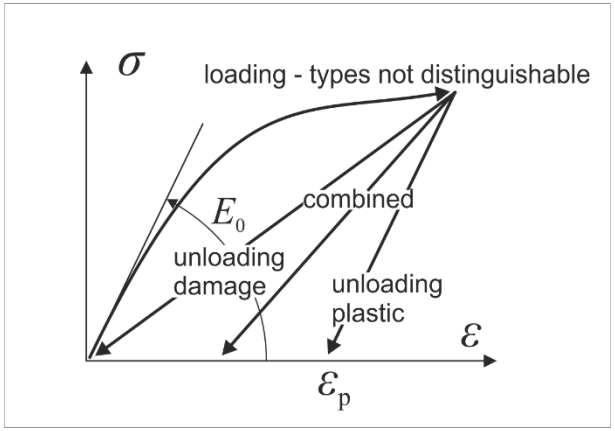


Figure 14.6-5: Approaches for material behaviour description.

Strain-based isotropic damage formulations with scalar internal variables which distinguish tensile as well as compressive isotropic damage can be found for example in:

- Mazars & Pijaudier-Cabot [14-32];
- Mazars et al. [14-33].

Isotropic damage states the same material degradation in all directions. This cannot model load induced anisotropy where a material point sustains compression in one direction while failing to tension in an orthogonal direction.

Thus, an extension is given with orthotropic damage based on a second-order integrity tensor as internal variable as is described, e.g., in:

- Carol et al. [14-34].

For a further extension with an anisotropic formulation with a higher order tensorial damage variable refer, e.g., to:

- Govindjee et al. [14-35].

approach. Both can be formulated in terms of nonlinear elasticity, elasto-plasticity, damage or combinations of them. The application is generally restricted to small deformations. Multiaxial strength conditions have to be applied as a constraint or naturally arise from stress-strain relations. In the initial unloaded state concrete is assumed to be an isotropic material with initially elastic behaviour with an elasticity tensor $\boldsymbol{E}_0$. This generally evolves to load induced anisotropy during a loading history from a phenomenological point of view due to concrete cracking mechanisms.

Symmetric 2nd order stress and strain tensors $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$ are arranged in 6 component vectors for convenience. They may be connected by 4th order material tensors arranged in 6 x 6 component matrices. A tensor based popular formulation combining elasto-plasticity and isotropic damage is given by:

$$\sigma = (1-D)\, E_0 \cdot (\varepsilon - \varepsilon_p) \tag{14.6-33}$$

with the scalar damage $0 \le D \le 1$ and plastic strain $\boldsymbol{\varepsilon_p}$. This yields pure damage and pure elasto-plasticity as special cases. A strain based evolution of scalar damage is ruled by a damage limit function $F(\boldsymbol{\varepsilon}, \beta)$ with an equivalent strain measure $\beta \ge 0$. It is connected to D by a damage evolution equation, e.g. with a simple example:

$$D(\beta) = \begin{cases} 0 & \beta \le \beta_0 \\ 1 - e^{-\left(\frac{\beta-\beta_0}{\beta_d}\right)^g} & \beta > \beta_0 \end{cases} \tag{14.6-34}$$

whereby the material parameters β_0, β_d and g may be derived from uniaxial material behaviour. The evolution of the damage variable D is ruled by so-called Kuhn-Tucker conditions:

$$\dot{D} \ge 0\,,\ F\dot{D} = 0,\ F \le 0 \tag{14.6-35}$$

The jacked dot $(\bullet)$ generally marks an increment with respect to a measure of time. The Kuhn-Tucker conditions imply $\dot{D} = 0$ with $F < 0$ but $\dot{D} \ge 0$ in combination with:

$$\dot{F} = \frac{\partial F}{\partial \boldsymbol{\varepsilon}}\,\dot{\boldsymbol{\varepsilon}} + \frac{\partial F}{\partial \beta}\,\dot{\beta} = 0 \tag{14.6-36}$$

A given $\dot{\boldsymbol{\varepsilon}}$ yields the evolution $\dot{\beta}$ and further a scalar damage D through integration of $\dot{\beta}$ in time. With D increasing during loading and $\boldsymbol{\sigma} = \mathbf{0}$ in case of

In the case of coinciding functions g and f the flow rule Eq. (14.6-37) is of the associated type, otherwise it is of the non-associated. Non-associated rules should be used in concrete plasticity models to describe the inelastic volume change under compression, which is characteristic of frictional materials.

Yield and flow functions f and g are chosen based on multiaxial failure criteria for concrete. They should depend not only on shear stresses, but also on the first invariant I_1 of the stress tensor to consider the influence of the hydrostatic pressure on the material ductility and strength. Thus, formulations such as:

- the Rankine criterion, where tensile failure occurs when the maximum principal stress reaches the uniaxial tensile strength f_{ct};
- the Drucker-Prager criterion, which is the modification of von Mises criterion including the influence of hydrostatic pressure on yielding;
- the Mohr–Coulomb criterion, where the maximum shear stress is the decisive measure of yielding, and the critical shear stress value depends on hydrostatic pressure;

and modifications or combinations of them can be used in concrete plasticity models. For further criteria and detailed basic information, see:

- Chen & Saleeb [14-36];

  Jirásek & Bažant [14-37].

A number of examples for elaborated plasticity models are available in the literature.

See, e.g., for coupling of damage with plasticity:

- Häussler-Combe & Hartig, J. [14-38];

total damage a limit condition for material strength is implied. The functions $F(\boldsymbol{\varepsilon}, \beta)$, $D(\beta)$ are material functions which have to be determined on the basis of experimental data.

In contrast to damage plasticity is generally stress based. Occurrence of permanent plastic strain increments is determined by a flow rule:

$$\dot{\boldsymbol{\varepsilon}}_p = \lambda \frac{\partial g}{\partial \boldsymbol{\sigma}} \tag{14.6-37}$$

with the plastic potential g and the plastic multiplier λ. The plastic potential g is a function of stress $\boldsymbol{\sigma}$ and state variables $\boldsymbol{\alpha}$, representing the load history. The multiplier λ is again determined by respective Kuhn-Tucker conditions:

$$\lambda \geq 0\,,\ f\lambda = 0,\ f \leq 0 \tag{14.6-38}$$

with a yield function f. The yield function f is also a function of stress $\boldsymbol{\sigma}$ and state variables $\boldsymbol{\alpha}$ and should include a limit condition for the material strength. The Kuhn-Tucker conditions imply $\dot{\boldsymbol{\varepsilon}}_p = 0$ associated with $f < 0$ but $\dot{\boldsymbol{\varepsilon}}_p \neq 0$ in combination with:

$$\dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \boldsymbol{\alpha}} \dot{\boldsymbol{\alpha}} = 0 \tag{14.6-39}$$

This condition and an evolution law for the plastic state variables

$$\dot{\boldsymbol{\alpha}} = \lambda\, \boldsymbol{h}(\boldsymbol{\sigma}, \boldsymbol{\alpha}) \tag{14.6-40}$$

determine the plastic multiplier λ, the plastic strain increments $\dot{\boldsymbol{\varepsilon}}_p$ and further plastic strains $\boldsymbol{\varepsilon}_p$ through integration in time. The functions g, f and $\mathbf{h}$ are material functions which again have to be determined on the basis of experimental data. The elasto-plastic formulation may be extended by multiple yield surfaces and plastic potentials.

Damage limit functions and evolution relations on the one hand and flow function, yield function and evolution functions for plasticity on the other hand have to adjusted to each other in case of combining damage with plasticity. Generally they cannot be treated independently when combined.

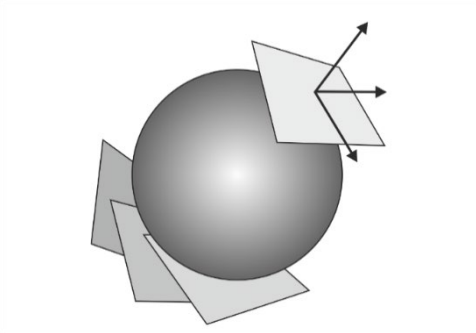


Figure 14.6-6: Micro-planes in material point unit sphere (Leukart & Ramm [14-39]).

Microplane has a number of versions or splits regarding derivations of strain vectors from tensors and the implementation of material types, see, e.g.;

- Ožbolt et al. [14-40];
- Leukart & Ramm [14-39].

Increments $\dot{\sigma},\dot{\varepsilon}$ of stresses and strains have to be discretized with respect to loading time in case of quasi-static problems or physical time in case of dynamic problems. As $(\dot{\bullet})$ stands for time derivations this is approximated by finite difference schemes in time and implemented into an incrementally iterative approach to solve nonlinear problems, see, e.g.

Bathe [14-41];

- Belytschko et al. [14-42].

Consistent derivations of the tangential material stiffness is generally given together with the description of the particular material models and their specific forms.

For methods and implementation of regularization see, e.g.:

- Bažant & Planas [14-43];
- Jirásek [14-44].

Tensorial strains, e.g., are projected into a number of planes placed in a material point under consideration leading to a 1st order tensor or 3 component strain vector **e** for each micro-plane. This is split in volumetric and deviatoric parts and yields a 1st order tensor or vector **t** of stresses. They may be connected by 2nd order material tensors arranged in 3 x 3 component matrices. A vector based formulation combining elasto-plasticity and isotropic damage is given by:

$$\boldsymbol{t} = \left(1-D\right)\boldsymbol{E}_0^{3x3}\left(\boldsymbol{e}-\boldsymbol{e}_p\right) \tag{14.6-41}$$

in analogy to the above tensor based approach. This is generally further simplified with 3 scalar equations for each vector component and done separately for each micro-plane. The set of derived micro-plane stresses is transformed into an equivalent 2nd order stress tensor under the condition of integral equilibrium.

The approaches for damage and plasticity described above basically may be translated to vector or scalar formats. In case of scalar relations for micro-planes they can be formulated in a more simple way. A distinct feature of the micro-plane approach is its inherent ability to model load-induced anisotropy due to the different response of micro-planes.

Apart from trivial cases the use of numerical methods, preferably the Finite Element Method is obligatory to apply multiaxial stress-strain relations. The quality of the involved stress-strain relations generally is decisive for the quality of numerical computations. Due to nonlinear concrete behaviour corresponding incremental stress-strain relations:

$$\dot{\boldsymbol{\sigma}} = \boldsymbol{C}_T \cdot \dot{\varepsilon} \tag{14.6-42}$$

with a tangential material stiffness $\mathbf{C}_T$ are used for a linearization within solution procedures. This is generally updated depending on a loading history whereby the specific forms of $\mathbf{C}_T$ have to be derived from a consistent derivation using Eq. (14.6-33) relating finite strains and stresses or their micro-plane counterparts with their specific formulations of limit-, yield- and flow-functions and corresponding evolution laws.

Concrete behaviour furthermore is characterized by a softening behaviour, i.e. increasing strains with decreasing stresses after having reached a stress limit or strength. This leads to a fundamental mesh sensitivity of standard finite elements whereby results with respect to fracture energy items will differ for all different discretizations. Regularization methods like crack-band approaches, gradient approaches or discrete modelling of cracks are required to overcome this.

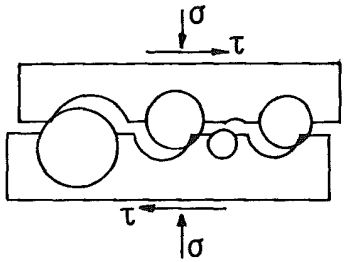


Figure 14.6-7: Principle of shear friction in concrete crack (Walraven [14-##]).

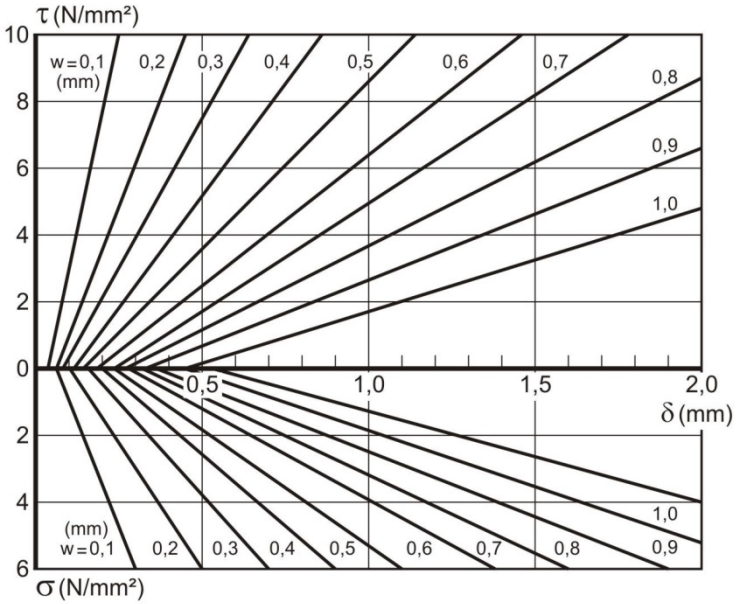


Figure 14.6-8: Relations according to Eqs. (14.6-43) and (14.6-44) for f_{cm} = 30 MPa (Walraven [14-##]).

The relations shown in Fig. 14.6-8 are based on experiments on push-off specimens. The aggregate grading in the mixture followed the Fuller curve, whereas glacial river

14.6.1.5.4 Shear friction behaviour in cracks

If in an open crack the crack faces are subjected to shear displacements with opposite signs, resisting shear stresses and normal (wedging) stresses develop as a result of the roughness of the crack faces.

The mean shear stress τ [MPa] and the mean normal stress σ [MPa] may be calculated from the following general relations:

$$\tau = C_f \left\{ -0.04 f_{cm} + \left[1.8 w^{-0.8} + \left(0.292 w^{-0.7} - 0.25 \right) f_{cm} \right] \delta \right\} \quad (14.6\text{-}43)$$

$$\sigma = C_f \left\{ -0.06 f_{cm} + \left[1.35 w^{-0.63} + \left(0.242 w^{-0.55} - 0.19 \right) f_{cm} \right] \delta \right\} \quad (14.6\text{-}44)$$

where:

δ is the shear displacement in mm;

w is the crack width in mm;

f_{cm} is the mean compressive strength in MPa at an age of 28 days.

C_f is an aggregate effectiveness factor, which is 1.0 if the aggregate does not fracture upon cracking of the concrete. For concrete with weak aggregates, like used for lightweight concrete, or high strength concrete (with strong cement paste), in which most of the particles are broken, for C_f a value of about 0.35 applies. More accurate values for C_f can be found by carrying out a push-off tests.

The crack opening path (development of shear displacement in relation to crack opening) can be constructed from diagrams as shown in Figure 14.6-8. If the relation between normal stress σ and crack opening w is given (analogous to spring stiffness), the corresponding values δ and τ can be read.

For the assessment of existing structures it should be noted that the concrete may have substantially increased compared with the 28-days strength. For some types of concrete this may imply that in the actual stage the aggregate particles fracture at crack formation, whereas they remained uncracked in earlier stages. Therefore the aggregate effectivity factor C_f should be chosen appropriately.

For the assessment of existing structures the concrete strength is often determined on drilled cores. If those cores are subjected to splitting to determine

aggregate was used. The Eqs. (4.6-43) and (4.6-44) were derived from tests on push-off specimens with concrete strengths up to f_{cm} = 47 MPa. The maximum particle diameter was 32 mm.

The compressive strength of concrete at an age t depends on the type and strength class of the cement, the type and amount of admixtures and additions, the water/cement ratio and environmental conditions, such as temperature and humidity.

The tensile strength of concrete primarily depends on those parameters which also influence the compressive strength of the concrete. However, tensile and compressive strength are not proportional to each other, and particularly for higher strength grades, an increase of the compressive strength leads only to a small increase of the tensile strength.

The development of tensile strength with time is strongly influenced by curing and drying conditions (internal stresses, surface cracking) as well as by the dimensions of the structural member. For a concrete age $t < 28$ days residual stresses may cause a temporary decrease of the tensile strength.

If no experimental data are available, the subsequent equation may be used to estimate crudely the tensile strength $f_{ctm}(t)$:

$$f_{ctm}(t) = [\beta_{cc}(t)]^{0.6} \cdot f_{ctm}$$

For high strength concrete the decrease of the tensile strength due to shrinkage cracks seems to be more pronounced than for normal strength concrete.

Where the development of tensile strength with time is important it is recommended to carry out experiments taking into account exposure conditions and dimensions of the structural member.

the concrete tensile strength, they can be used as well for a visual inspection of the crack faces, in order to ascertain whether particle cracking has occurred, and in this way enable an appropriate choice for C_f.

14.6.1.6 Time effects

14.6.1.6.1 Development of strength with time

For a mean temperature of 20°C and curing in accordance with ISO 1920-3 [14-15] the relevant compressive strength of concrete at various ages $f_{cm}(t)$ may be estimated from:

$$f_{cm}(t) = \beta_{cc}(t) \cdot f_{cm} \qquad (14.6\text{-}45)$$

with:

$$\beta_{cc}(t) = exp\left\{ s_C \left[1 - \left(\frac{t_{ref}}{t} \right)^{0.5} \right] \left(\frac{28}{t_{ref}} \right)^{0.5} \right\} \qquad (14.6\text{-}46)$$

where:

$f_{cm}(t)$ is the mean compressive strength in MPa at an age t in days;

f_{cm} is the mean compressive strength in MPa at an specified reference age t_{ref} in days;

$\beta_{cc}(t)$ is a function to describe the strength development with time;

t is the concrete age in days adjusted according to Eq. (14.6-80) (taking into account the temperature during curing);

t_{ref} age of concrete at which the concrete strength is determined in days;

s_C is a coefficient which depends on the strength development class of concrete as given in the following Table 14.6-7 and Table 14.6-8.

Table 14.6-7: Coefficient s_C to be used in Eq. (14.6-46) for different classes of concrete (see Table 14.6-8).

Concrete strength, f_{ck} [MPa]	s_C		
	Class CR	Class CN	Class CS
$f_{ck} \leq 35$	0.3	0.5	0.6
$35 < f_{ck} < 60$	0.2	0.4	0.5
$f_{ck} \geq 60$	0.1	0.3	0.4

Table 14.6-8: Strength development classes of concrete.

Class	Composition and properties of the cement or the binder of the concrete		
	Concrete containing only cement		Concrete containing binder composition
	Type of cement[a]	Cement strength class[b], e.g.	Composition of the binder (Portland cement clinker[c] und SCM[d])
Class CS	CEM III, CEM II/B	32.5 N, 42.5 N	Portland cement clinker and more than 65 M.-% of ground granulated blast furnace slag (ggbs) or more than 35 M.-% of fly ash (fa)
Class CN	CEM II/A, CEM I	32.5 R, 42.5 N	Portland cement clinker and more than 35 M.-% but less than 65 M.-% of ground granulated blast furnace slag (ggbs) or more than 20 M.-% but less than 35 M.-% of fly ash (fa)
Class CR	CEM II/A, CEM I	42.5 R, 52.5 N, 52.5 R	Portland cement clinker and less than 35 M.-% of ground granulated blast furnace slag (ggbs) or less than 20 M.-% of SCM other than ggbs

[a] CEM IV, CEM V and special cements may be classed considering Portland cement clinker content and strength class

[b] according to EN 197-1

[c] observe that the content of Portland cement clinker in CEM I may be 95 M.-%

[d] considered as binder for the water-binder-ratio according to EN 206 [14-8].

The compressive strength of the concrete and may either be taken from Table 14.6-7 or may be experimentally determined according to ISO 1920-4 [14-20].

The compressive strength of lightweight aggregate concrete mainly depends on the strength of the cement paste. Lightweight aggregate concrete has a relatively rapid early strength development and a relatively slow long term strength development because of the low strength of the lightweight aggregates. Therefore hardly any strength gain may be observed after a certain concrete age when the cement paste approaches the strength of the lightweight aggregates with ongoing hydration. The age at which this situation is reached depends on the strength of the lightweight aggregates. An age in the range 1–4 weeks is a realistic estimate in most cases.

For lightweight aggregate concrete Eq. (14.6-49) has to be applied with caution. For structures sensitive to deformations, related tests have to be carried out.

The hydration of the cement in self-compacting concrete is basically controlled by the same mechanisms as that of vibrated concrete. Thus no particular difference in the time-development of properties is expected.

A high sustained compressive stress can be defined as a stress of a magnitude of above 75% of the short time strength $f_{cm,}$, applied at an age of $t_0 = t_{ref}$. The value of 75% depends

For lightweight aggregate concrete the compressive strength in MPa at various ages may be estimated from:

$$f_{lcm}(t) = \beta_{lcc}(t) \cdot f_{lcm} \qquad (14.6\text{-}47)$$

where:

$\beta_{lcc}(t)$ is a function describing the development with time;

$\beta_{lcc}(t) = \beta_{cc}(t)$ according to Eq. (14.6-46) where s_C has to be replaced by s_{lc};

s_{lc} 0.05 for lightweight aggregates of high strength;

0.25 for lightweight aggregates of low strength;

f_{lcm} is the mean compressive strength in MPa at an age of 28 days.

14.6.1.6.2 Development of modulus of elasticity with time

The modulus of elasticity of concrete at an age t ≠ 28 days may be estimated from:

$$E_{ci}(t) = \beta_E(t) E_{ci} \qquad (14.6\text{-}48)$$

with:

$$\beta_E(t) = \left[\beta_{cc}(t)\right]^{0.33} \qquad (14.6\text{-}49)$$

where:

$E_{ci}(t)$ is the modulus of elasticity in MPa at an age t in days;

E_{ci} is the modulus of elasticity in MPa at the reference age t_{ref}, according to Eq. (14.6-20);

$\beta_E(t)$ is a coefficient which depends on the age of concrete t in days;

$\beta_{cc}(t)$ is the coefficient according to Eq. (14.6-46).

14.6.1.6.3 Strength under sustained loads

14.6.1.6.3.1 Sustained compressive strength

When subjected to sustained high compressive stresses the compressive strength of concrete decreases with time under load due to the formation and

on the concrete strength class, and literature results show that this value may vary from 70% for lightweight aggregate concrete to 80% for high strength concrete.

Due to the counteracting effects of the parameters influencing the strength under sustained loads, $f_{cm,sus}(t,t_0)$ passes through a minimum. The duration of loading for which this minimum occurs is referred to as the critical period. The length of the critical period depends significantly on the age at loading and on the strength class of concrete (refer to Table 14.6-7). It is typically higher for higher ages at loading and for lower values of the parameter s_C (rapid early strength development) and it may range between some days and some years. Detailed values for the assessment of the critical period can be found in Tasevski et al. [14-45]. Research has shown a slight increase of the sustained load strength with increasing compressive strength of the concrete. However, due to the limited number of tests on high strength concrete the sustained load strength behaviour of normal strength concrete should be applied also for high strength concrete.

Resulting from the reduced strength gain of the lightweight aggregate concrete, as soon as the strength of the cement paste approaches the strength of the aggregates, the critical period is extended and the strength under sustained loads equals 70–75% of the short time strength at the age of 28 days.

The concrete strength reduction and potential failure under high sustained loads is associated to the development of nonlinear creep strain, which is considered as a detrimental effect. As a consequence, internal stress redistributions can potentially occur, resulting in a beneficial influence in terms of structural strength. Such redistributions can be observed in the compression zone of members in bending or in reinforced columns under high compression stresses .

propagation of micro-cracks. This strength reduction is counteracted by the strength increase with time due to continued cement hydration. The combined effect of high sustained stresses and of continued cement hydration is given by:

$$f_{cm,sus}\left(t,t_0\right) = f_{cm} \cdot \beta_{cc}\left(t\right) \cdot \beta_{c,sus}\left(t,t_0\right) \tag{14.6-50}$$

with:

$$\beta_{c,sus}\left(t,t_0\right) = \beta_{t0}\left(t_0\right) + \left[1-\beta_{t0}\left(t_0\right)\right]\left[1+10^4\,\frac{\left(t-t_0\right)}{t_0}\right]^{-0.1} \tag{14.6-51}$$

where:

$f_{cm,sus}(t,t_0)$ is the mean compressive strength of concrete in MPa at time t when subjected to a high sustained compressive stress at an age at loading $t_0 < t$;

$\beta_{cc}(t)$ is the time strength development function according to Eq. (14.6-46));

$\beta_{c,sus}(t,t_0)$ is a function which describes the decrease of strength with time under high sustained load. It depends on the time under high sustained loads t-t_0 in days, as well as on the age of the concrete at loading in days. The function is defined for $(t\text{-}t_0) > 0.015$ days (= 20 min);

$\beta_{t0}(t_0)$ is a parameter which considers the maturity of the concrete at loading. It can be calculated as $\beta_{t0}(t_0) = 0.64 + 0.01 \cdot \ln(t_0)$, with t_0 in days;

t_0 is the age of concrete at loading in days. Eq. (14.6-51) is valid for $t_0 \geq 7$ days;

t-t_0 is the time under high sustained loads in days. For $t\text{-}t_0 > 10$ years, it shall be taken $t\text{-}t_0 = 3650$ days.

Eq. (14.6-52) has been taken from *fib* Bulletin 42 [14-18].

The distinction between creep and shrinkage is defined by convention. For further background information, see *fib* Bulletin 70 [14-1].

Also, separation of initial strain and creep strain is a matter of convention. In structural analysis, the total load-dependent strain as given by the creep function (section 14.6.1.6.4.3) is of importance. The initial and creep strain components have to be defined consistently, so that their sum results in the correct load-dependent strain.

The initial elastic strain $\varepsilon_{ci}(t_0)$ is directly connected with the tangent modulus of elasticityby Hooke's law:

$$\varepsilon_{ci}(t_0) = \frac{\sigma_c(t_0)}{E_{ci}(t_0)} \qquad \text{(14.6-##)}$$

The initial plastic strain occurring at first loading and being observed upon rapid unloading (see Figure 14.6-2) is considered to be part of the creep strain.

14.6.1.6.3.2 Sustained tensile strength

The tensile strength under sustained loading $f_{ctk,sus}$ in MPa can be estimated from:

$$f_{ctk,sus} = \alpha \cdot f_{ctk} \qquad (14.6\text{-}52)$$

where:

f_{ctk} is the short term strength in MPa;

α = 0.60 for normal strength concrete and 0.75 for high strength concrete.

14.6.1.6.4 Creep and shrinkage

14.6.1.6.4.1 Definition of strains

The total strain at time t, $\varepsilon_c(t)$, of a concrete member uniaxially loaded at time t_0 with a constant stress $\sigma_c(t_0)$ may be expressed as:

$$\varepsilon_c(t) = \varepsilon_{ci}(t_0) + \varepsilon_{cc}(t) + \varepsilon_{cs}(t) + \varepsilon_{cT}(t) \qquad (14.6\text{-}53)$$

or

$$\varepsilon_c(t) = \varepsilon_{c\sigma}(t) + \varepsilon_{cn}(t) \qquad (14.6\text{-}54)$$

where:

$\varepsilon_{ci}(t_0)$ is the initial elastic strain at loading;

$\varepsilon_{cc}(t)$ is the creep strain at time $t > t_0$;

$\varepsilon_{cs}(t)$ is the shrinkage strain;

$\varepsilon_{cT}(t)$ is the thermal strain;

$\varepsilon_{c\sigma}(t)$ is the stress-dependent strain: $\varepsilon_{c\sigma}(t) = \varepsilon_{ci}(t_0) + \varepsilon_{cc}(t)$;

$\varepsilon_{cn}(t)$ is the stress-independent strain: $\varepsilon_{cn}(t) = \varepsilon_{cs}(t) + \varepsilon_{cT}(t)$.

The formulations as presented for creep and shrinkage do not predict local rheological properties within the cross-section of a concrete member such as variations due to internal stresses, moisture states or the effects of local cracking.

The predictions are not applicable to

– concrete subjected to extreme temperatures, high (e.g. nuclear reactors) or low (e.g. LNG tanks);

– very dry climatic conditions (average relative humidity RH < 20%).

The effect of temperature variations during hardening can be taken into account in accordance with Eq. (14.6-80). The effect of temperature in the range 0°C < T < 80°C is dealt with in section 14.6.1.7.

Here, concrete is considered as an ageing linear viscoelastic material. In reality, creep is a non-linear phenomenon. The non-linearity with respect to creep inducing stress may be observed in creep experiments at a constant stress, particularly if the stress exceeds $0{,}4f_{cm}(t_0)$, see section 14.6.1.6.4.3 (d), as well as in experiments with a variable stress history even below stresses of $0{,}4f_{cm}(t_0)$.

In this Model Code, creep after a given duration of loading is described by means of the creep coefficient, which is a descriptive figure for the magnitude of creep effects. The creep coefficient is defined as the ratio of the creep strain to the elastic strain of concrete at an age of 28 days (reference elastic deformation) under the same stress.

In Müller et al. [14-16] background information of the creep coefficient $\varphi(t,t_0)$ in combination with the modulus of elasticity E_{ci} is given.

Creep and shrinkage prediction are generally associated with considerable uncertainties. Their main source is the effect of the constituents of structural concrete. This would also hold true, even if the parameters of the concrete composition would be considered in the models. As these parameters are mostly unknow at the stage of design, the compressive strength, which is known to the designer, is used as a substitute parameter to consider the effect of the concrete composition (see also below). The uncertainty in the estimation of creep and shrinkage can be considerably reduced if the given models are

14.6.1.6.4.2 Range of applicability

The relations for creep and shrinkage given below predict the time-dependent mean cross-section behaviour of a concrete member moist cured at normal temperatures for not longer than 14 days.

Unless special provisions are given, the relations are valid for ordinary structural concrete (C20 to C120)) subjected to a compressive stress $|\sigma_c| \le 0.4 f_{cm}(t_0)$ at an age at loading t_0 and exposed to a mean relative humidity in the range of 20 to 100% at a mean temperature in the range of 5°C to 30°C. The age at loading should be at least 1 day.

It is accepted that the relations apply as well to concrete in tension, though the relations given in the following are directed towards the prediction of creep of concrete subjected to compressive stresses.

14.6.1.6.4.3 Creep

(a) Assumptions and related basic equations

Within the range of service stresses $|\sigma_c| \le 0.4 \cdot f_{cm}(t_0)$, creep is assumed to be linearly related to stress.

For a constant stress $\sigma_c(t_0)$ in MPa applied at time t_0 this leads to the subsequent creep strain $\varepsilon_{cc}(t,t_0)$:

$$\varepsilon_{cc}(t,t_0) = \frac{\sigma_c(t_0)}{E_{ci}} \varphi(t,t_0) \qquad (14.6\text{-}55)$$

where:

$\varphi(t,t_0)$ *is the creep coefficient;*

E_{ci} is the modulus of elasticity at the age of 28 days according to Eqs. (14.6-21) or (14.6-22) in MPa.

adjusted according to short-time test results carried out on the concrete used in practice. It can be assumed that the coefficient of variation drops from roughly about V = 30% to V = 10%. Details on the procedure to be followed and further background information are given in Bulletin XX.

The application of the principle of superposition is consistent with the assumption of linearity. However, due to the actual non-linear behaviour of concrete, some prediction errors are inevitable when linear superposition is applied to creep of concrete under variable stress, particularly for unloading or decreasing strains, respectively. These deviations are mainly caused by the neglect of hygral effects, interaction of stresses and ageing and material damage, including cracking and fracture. For linear creep prediction models, the error depends also on the type of model underlying the creep prediction; see CEB Bulletin 177 [14-46].

The structural effects of time-dependent behaviour of concrete are dealt with in detail in section 29.2.3 of this Model Code, in CEB Bulletin 215 [14-47] and section 4.1.6 of *fib* Bulletin 52 [14-48].

The relations to calculate the creep coefficient are empirical. They were calibrated on the basis of laboratory tests (creep in compression) on structural concretes. Total creep is separated into the components of basic creep and drying creep, reflecting the associated different physical mechanisms.

In this prediction model only those parameters are taken into account that are normally known to the designer, that is characteristic compressive strength, dimensions of the member, mean relative humidity to which the member is exposed, age at loading, duration of loading and type of concrete. It should be pointed out, however, that creep of concrete does not depend on its compressive strength or age at loading per se, but rather on its composition and degree of hydration; creep of concrete decreases with decreasing water/cement ratio, decreasing cement paste content, increasing stiffness of the aggregates and increasing degree of hydration at the age of loading.

The stress dependent strain $\varepsilon_{c\sigma}(t,t_0)$ at time t may be expressed as:

$$\varepsilon_{c\sigma}(t,t_0)=\sigma_c(t_0)\left[\frac{1}{E_{ci}(t_0)}+\frac{\varphi(t,t_0)}{E_{ci}}\right]=\sigma_c(t_0)J(t,t_0) \qquad (14.6\text{-}56)$$

where:

$J(t,t_0)$ is the creep function or creep compliance, representing the total stress-dependent strain per unit stress;

$E_{ci}(t_0)$ is the modulus of elasticity at the time of loading t_0 according to Eq. (14.6-51); hence $1/E_{ci}(t_0)$ represents the initial elastic strain per unit stress at loading.

For practical applications concrete may be considered as an ageing linear viscoelastic material, and for variable stresses and strains, the principle of superposition is assumed to be valid. On the basis of these assumptions and definitions given above, the constitutive equation for concrete may be written as:

$$\varepsilon_c(t)=\sigma_c(t_0)J(t,t_0)+\int_{t_0}^{t}J(t,\tau)\frac{\partial\sigma_c(\tau)}{\partial\tau}d\tau+\varepsilon_{cn}(t) \qquad (14.6\text{-}57)$$

(b) Creep coefficient

The creep coefficient $\varphi(t,t_0)$ may be calculated from:

$$\varphi(t,t_0)=\varphi_{bc}(t,t_0)+\varphi_{dc}(t,t_0) \qquad (14.6\text{-}58)$$

where:

$\varphi_{bc}(t,t_0)$ is the basic creep coefficient according to Eq. (14.6-59);

$\varphi_{dc}(t,t_0)$ is the drying creep coefficient according to Eq. (14.6-62);

t is the age of concrete in days at the moment considered;

t_0 is the age of concrete at loading in days adjusted according to Eqs. (14.6-68) and (14.6-80).

Due to the inherent scatter of creep and shrinkage deformations, the errors of the model and the general uncertainty caused by randomness of material properties and environment, a prediction of the deformation may result in a considerable error. After short durations of loading or drying, the prediction error is higher than after long durations of loading and drying.

Based on a computerized database of laboratory test results a mean coefficient of variation for the predicted creep function $V_c = 30\%$ has been found.

Assuming a normal distribution, this corresponds to a 10 and 5 per cent cut-off, respectively, on the lower and the upper side of the mean value of:

$$\varphi_{0.10} = 0.6\varphi; \quad \varphi_{0.05} = 0.5\varphi$$

$$\varphi_{0.90} = 1.4\varphi; \quad \varphi_{0.95} = 1.5\varphi$$

The prediction error should be taken into account in a probabilistic approach where appropriate.

It is not known whether creep approaches a finite value or not. However, in this constitutive approach the development of basic creep is predicted by a logarithmic function having no finite value, whereas a hyperbolic function was chosen for the description of drying creep which approaches an asymptotic value for $t \to \infty$.

Evaluations on the basis of test results indicate that these equations give a reasonably good approximation for the time development of creep up to 50 years of loading under the conditions indicated in Tables 14.6-9. From laboratory observations of creep up to 30 years, we can conclude that the increase of creep from 50 years up to 150 years of duration of loading will not exceed 10% of the creep after 50 years. Some observations in practice primarily on bridges show after 30-50 years in service considerable higher deformations than expected from creep predictions. The reasons are not yet clarified. There might be excessive creep strains but also different constructional reasons.

In cases where a lower level of accuracy is sufficient, the values given in Table 14.6-9 can be accepted as representative values for the creep coefficient after 50 years of loading of a normal weight structural concrete.

Table 14.6-9: Creep coefficient $\varphi(50y,t_0)$ of an ordinary structural concrete after 50 years of loading (service life according to Table 11.2-1).

Age at loading t_0 [d]	**Dry atmospheric conditions (RH = 50%, indoors)**	**Humid atmospheric conditions (RH = 80%, outdoors)**

The basic creep coefficient $\varphi_{bc}(t,t_0)$ may be estimated from:

$$\varphi_{bc}(t,t_0) = \beta_{bc}(f_{cm}) \cdot \beta_{bc}(t,t_0) \tag{14.6-59}$$

with:

$$\beta_{bc}(f_{cm}) = \frac{1.8}{(f_{cm})^{0.7}} \tag{14.6-60}$$

and the time development function

$$\beta_{bc}(t,t_0) = ln\left\{\left(\frac{30}{t_{0,adj}} + 0.035\right)^2 (t - t_0) + 1\right\} \tag{14.6-61}$$

where:

f_{cm} is the mean compressive strength at an age of 28 days in MPa according to Eq. (14.6-1);

$t_{0,adj}$ is the adjusted age at loading in days according to Eq. (14.6-68).

The drying creep coefficient $\varphi_{dc}(t,t_0)$ may be estimated from:

$$\varphi_{dc}(t,t_0) = \beta_{dc}(f_{cm}) \cdot \beta(RH) \cdot \beta_{dc}(t_0) \cdot \beta_{dc}(t,t_0) \tag{14.6-62}$$

with:

$$\beta_{dc}(f_{cm}) = \frac{412}{(f_{cm})^{1.4}} \tag{14.6-63}$$

$$\beta(RH) = \frac{1 - \frac{RH}{100}}{\sqrt[3]{0.1 \cdot \frac{h}{100}}} \tag{14.6-64}$$

$$\beta_{dc}(t_0) = \frac{1}{0.1 + t_{0,adj}^{\,0.2}} \tag{14.6-65}$$

The development of drying creep with time is described by:

$$\beta_{dc}(t,t_0) = \left[\frac{(t - t_0)}{\beta_h + (t - t_0)}\right]^{\gamma(t_0)} \tag{14.6-66a}$$

with:

for strength development classes of concrete			$h = 2A_c/u$ [mm]				$h = 2A_c/u$ [mm]			
CS	**CN**	**CR**	**100**	**200**	**500**	**1 000**	**100**	**200**	**500**	**1 000**
3	**1**	**1**	4,2	3,8	3,4	3,1	3,0	2,8	2,6	2,5
10	**7**	**3**	3,1	2,8	2,5	2,3	2,2	2,1	2,0	1,9
32	**28**	**23**	2,4	2,2	1,9	1,8	1,7	1,6	1,6	1,5
91	**91**	**91**	1,9	1,7	1,5	1,4	1,4	1,3	1,2	1,2
365	**365**	**365**	1,4	1,3	1,1	1,0	1,0	0,9	0,9	0,8
Corr. exponent A			0,82	0,79	0,75	0,72	0,71	0,68	0,66	0,64

The creep coefficients are mean values with a coefficient of variation of about 30%. The values of the creep coefficient apply to $f_{ck,28}$ = 35 MPa. For other strength in the range of 12 MPa ≤ $f_{ck,28}$ ≤ 100 MPa, the values should be multiplied by the factor $35/[f_{ck,28}]^A$, where A is the correction exponent in the table.

[a] For Classes CS, CN and CR, see Table 14.6-8.

The data given in Tables 14.6-9 apply for a mean temperature of the concrete between 10 °C and 20 °C. Seasonal variations of temperature between −20 °C and +40 °C can be accepted. The same holds true for variations in relative humidity around the mean values given in the same table.

Creep of SCC is affected when a high paste content is applied. In general, the creep deformation is approximately 10–20% higher than that of conventional concrete of equal strength. However, the deformations are within the scatter band for ordinary structural concrete, which is defined to be ±30%. If the structural response is sensitive to variations in creep behaviour tests are highly recommended.

The higher creep tendency of lightweight aggregate concrete due to the reduced stiffness of the aggregates is partially compensated by the lower creep capability of the stiffer cement paste matrix.

A more sophisticated and comprehensive model for calculating the creep deformations of normal and high strength lightweight aggregate concrete with expanded clay aggregates was published by Kvitsel [14-49].

$$\gamma(t_0) = \frac{1}{2.3 + \frac{3.5}{\sqrt{t_{0,adj}}}} \tag{14.6-66b}$$

$$\beta_h = 1.5 \cdot h + 250 \cdot \alpha_{f_{cm}} \leq 1500 \cdot \alpha_{f_{cm}} \tag{14.6-66c}$$

with:

$$\alpha_{f_{cm}} = \left(\frac{35}{f_{cm}}\right)^{0.5} \tag{14.6-66d}$$

where:

f_{cm} is the mean compressive strength at an age of 28 days in MPa according to Eq. (14.6-1);

RH is the relative humidity of the ambient environment in %;

h = $2A_c/u$, is the notional size of the member in mm, where A_c is the cross-section in mm² and u is the perimeter of the member in contact with the atmosphere in mm;

$t_{0,adj}$ is the adjusted age at loading in days according to Eq. (14.6-68).

For lightweight aggregate concrete the relevant creep coefficient φ_l may be calculated according to Eq. (14.6-67):

$$\varphi_l = \eta_E \cdot \varphi(t, t_0) \tag{14.6-67}$$

where:

$\eta_E = \left(\frac{\rho}{2200}\right)^2$, with oven-dry density ρ in kg/m^3;

$\varphi(t,t_0)$ is the creep coefficient according to Eq. (14.6-58).

For concrete grades LC12 and LC16, the creep coefficient φ_l has to be additionally multiplied with a factor 1.3.

(c) Effect of type of concrete and curing temperature

The effect of the type of concrete on the creep coefficient of concrete may be taken into account by modifying the age at loading t_0 to $t_{0,adj}$:

$$t_{0,adj} = t_{0,T}\left[\frac{9}{2+t_{0,T}^{1.2}}+1\right]^{\alpha} \geq 0.5 \text{ days} \qquad (14.6\text{-}68)$$

where:

$t_{0,T}$ is the age of concrete at loading in days adjusted according to Eq. (14.6-80);

α is a coefficient which depends on the strength development class of the concrete (see Table 14.6-8)

$\alpha = -1$ for class CS;

$\alpha = 0$ for class CN;

$\alpha = 1$ for class CR.

(d) Effect of high stresses

For stress levels in the range $0.4 f_{cm}(t_0) < |\sigma_c| \leq 0.6 f_{cm}(t_0)$ the non-linearity of creep may be taken into account using:

$$\varphi_\sigma(t,t_0) = \varphi(t,t_0)\cdot exp\left[1.5\left(k_\sigma - 0.4\right)\right] \qquad \text{for } 0.4 < k_\sigma \leq 0.6 \qquad (14.6\text{-}69)$$

where:

$\varphi_\sigma(t,t_0)$ is the non-linear creep coefficient;

$\varphi(t,t_0)$ is the creep coefficient according to Eq. (14.6-58);

Different types of binders used for concrete result in different degrees of hydration at the same age. Creep of concrete depends on the degree of hydration reached at a given age rather than on the age of the concrete. Therefore, the effect of the type of binder, resp. concrete is taken into account by modifying the age at loading so that, for a given modified age, the degree of hydration is approximately independent of the type of concrete. Note, that the duration of loading $(t-t_0)$ used in Eqs. (14.6-61) and (14.6-66a) is the actual time under load.

Eq. (14.6-68) was developed based on experimental results primarily with CEM I and CEM III cements and first presented in CEB-FIP ModelCode 1990 [14-21]. The transition from "type of cement effect" to type of concrete effect (see Table 14.6-8) has been carefully checked. Nevertheless if other cement types or large amounts of pozzolans are and the development of the creep deformations has high relevance for the design creep tests should be carried out.

The non-linear behaviour of concrete under high stresses mainly results from micro-cracking. Eq. (14.6-69) represents a simplification in so far as it just describes the increase of the magnitude of creep but does not take into account the observation that non-linearity decreases with increasing duration of loading. Further this approach neglects the differences in non-linear behaviour to be observed between basic creep and drying creep.

It should be noted that delayed elastic strains upon total unloading may be assumed as linear functions of stress up to stress levels of $|\sigma_c| = 0.6 f_{cm}(t_0)$ though some experiments indicate some over-proportionality.

$k_\sigma = \dfrac{|\sigma_c|}{f_{cm}(t_0)}$, which is the stress-strength ratio.

14.6.1.6.4.4 Shrinkage

Due to microstructural mechanisms becoming dominant for high strength concrete, the shrinkage has to be separated into basic shrinkage and drying shrinkage.

For curing periods of concrete members t_s < 14 days at normal ambient temperatures, the duration of moist curing does not significantly affect shrinkage. Hence, this parameter as well as the effect of curing temperature is not taken into account.

In Eqs. (14.6-72) and (14.6-77) the actual duration of drying (t-t_s) has to be used.

Similar to creep, total shrinkage does not depend on concrete compressive strength per se. Drying shrinkage decreases with decreasing water/cement ratio and decreasing cement paste content, whereas basic shrinkage increases with decreasing water/cement ratio and decreases with decreasing cement paste content. The compressive strength serves as a convenient substitute parameter, always known at the design stage.

If the composition of concrete deviates considerably from ordinary structural concrete (e.g. green concrete) it is recommended to run tests. This also holds true for ordinary concrete in case the structural response is sensitive to shrinkage deformations or in case cement types are used other than CEM I, CEM II and CEM III, or if high amounts of pozzolans are used in partial replacement of CEM I.

Tests should be performed according to:

- RILEM TC 107-CSP [14-50];
- ISO 1920-8 [14-51].

In some countries outside Europe, higher cement contents than recommended by EN 206 [14-8] are applied for structural concrete. This leads to an increased shrinkage up to 20% compared to the prediction of this model.

The shrinkage or swelling strains $\varepsilon_{cs}(t,t_s)$ may be calculated as:

$$\varepsilon_{cs}(t,t_s) = \varepsilon_{cbs}(t) + \varepsilon_{cds}(t,t_s) \tag{14.6-70}$$

where shrinkage is subdivided into the basic shrinkage $\varepsilon_{cbs}(t)$ which occurs even if no moisture loss is possible:

$$\varepsilon_{cbs}(t) = \varepsilon_{cbs0}(f_{cm}) \cdot \beta_{bs}(t) \tag{14.6-71}$$

and the drying shrinkage $\varepsilon_{cds}(t,t_s)$ giving the additional shrinkage if moisture loss occurs:

$$\varepsilon_{cds}(t,t_s) = \varepsilon_{cds0}(f_{cm}) \cdot \beta_{RH}(RH) \cdot \beta_{ds}(t-t_s) \tag{14.6-72}$$

where:

t is the concrete age in days;

t_s is the concrete age at the beginning of drying in days;

(t-t_s) is the duration of drying in days.

The basic shrinkage component $\varepsilon_{cbs}(t)$ may be estimated by means of the basic notional shrinkage coefficient $\varepsilon_{cbs0}(f_{cm})$ and the time function $\beta_{bs}(t)$:

$$\varepsilon_{cbs0}(f_{cm}) = -\alpha_{bs}\left(\frac{0.1 f_{cm}}{6 + 0.1 f_{cm}}\right)^{2.5} 10^{-6} \tag{14.6-73}$$

$$\beta_{bs}(t) = 1 - exp\left(-0.2\sqrt{t}\right) \tag{14.6-74}$$

where:

f_{cm} is the mean compressive strength at the age of 28 days in MPa according to Eq. (14.6-1);

A mean coefficient of variation of predicted shrinkage has been estimated on the basis of a computerized database, resulting in V_s = 30%. The corresponding 10 and 5% cut-off values are:

$$\varepsilon_{cs0.10} = 0.6\varepsilon_{cs}\,;\varepsilon_{cs0.05} = 0.5\varepsilon_{cs}$$

$$\varepsilon_{cs0.90} = 1.4\varepsilon_{cs}\,;\varepsilon_{cs0.95} = 1.5\varepsilon_{cs}$$

when a normal distribution is assumed.

In cases where a lower level of accuracy is sufficient, the values given in Table 14.6-11 can be accepted as representative values for shrinkage after 50 years of drying of a normal weight structural concrete. Usually these values may be taken as final shrinkage values.

Though shrinkage reaches a final value, little information exists on the shrinkage strains of large members after long durations of drying. Therefore, the values calculated using Eq. (14.6-77) for $2A_c/u$ = 1000 mm, and the values given in Table 14.6-11 for shrinkage of members with a notional size of $2A_c/u$ = 1000 mm, respectively, are uncertain and may overestimate the actual shrinkage strains after 50 years of drying.

α_{bs} is a coefficient, dependent on the strength development class of concrete (see Table 14.6-10).

Table 14.6-10: Coefficients α_i used in Eqs. (14.6-75) and (14.6-75).

Strength development class of concrete	α_{bs}	α_{ds1}	α_{ds2}
CS	800	3	0.013
CN	700	4	0.012
CR	600	6	0.012

The drying shrinkage $\varepsilon_{cds}(t,t_s)$ is calculated by means of the notional drying shrinkage coefficient $\varepsilon_{cds0}(f_{cm})$, the coefficient $\beta_{RH}(RH)$, taking into account the effect of the ambient relative humidity, and the function $\beta_{ds}(t\text{-}t_s)$ describing the time-development:

$$\varepsilon_{cds0}\left(f_{cm}\right) = \left[\left(220 + 110\alpha_{ds1}\right)\cdot exp\left(-\alpha_{ds2}\cdot f_{cm}\right)\right] 10^{-6} \qquad (14.6\text{-}75)$$

$$\beta_{RH} = \begin{cases} -1.55\left[1-\left(\dfrac{RH}{RH_{eq}}\right)^3\right] & for\ 20\% \le RH \le RH_{eq} \\ -1.55\left[1-\left(\dfrac{RH}{RH_{eq}}\right)^2\right] & for\ RH_{eq} < RH < 100\% \\ -1.55\left[1-\left(\dfrac{RH}{RH_{eq}}\right)^2\right] + 0,25 & for\ RH = 100\% \end{cases} \qquad (14.6\text{-}76)$$

$$\beta_{ds}\left(t-t_s\right) = \left(\frac{\left(t-t_s\right)}{0.035\,h^2 + \left(t-t_s\right)}\right)^{0.5} \qquad (14.6\text{-}77)$$

$$RH_{eq} = 99\left(35 / f_{cm}\right)^{0.1} \le 99 \qquad (14.6\text{-}78)$$

Table 14.6-11: Total shrinkage values $\varepsilon_{cs,50y} \cdot 10^3$ of an ordinary structural concrete after a duration of drying of 50 years (service life according to Table 11.2-1).

Strength development of concrete	$f_{ck,28}$	Dry atmospheric conditions (RH = 50%, indoors) h = 2Ac/u [mm]				Humid atmospheric conditions (RH = 80%, outdoors) h = 2Ac/u [mm]			
		100	200	500	1 000	100	200	500	1 000
Class CS	20	0,57	0,56	0,48	0,36	0,33	0,32	0,28	0,21
	35	0,53	0,51	0,45	0,35	0,31	0,31	0,27	0,22
	50	0,49	0,48	0,43	0,35	0,30	0,29	0,27	0,23
Class CN	20	0,67	0,65	0,56	0,41	0,38	0,37	0,32	0,24
	35	0,60	0,59	0,51	0,39	0,34	0,34	0,30	0,24
	50	0,55	0,54	0,48	0,37	0,31	0,31	0,28	0,23
	80	0,48	0,48	0,43	0,36	0,30	0,30	0,28	0,25
Class CR	35	0,76	0,74	0,65	0,48	0,42	0,41	0,36	0,28
	50	0,67	0,66	0,58	0,44	0,36	0,35	0,32	0,26
	80	0,55	0,54	0,49	0,39	0,31	0,30	0,28	0,25

The values are mean values with a coefficient of variation of about 30 %.

[a] For Classes CS, CN and CR, see Table 14.6-8.

The shrinkage of powder type SCC is affected by its high paste content. The ultimate shrinkage deformation is approximately 20% higher than that of conventional concrete of equal strength. However, the deformations are within the scatter band, which is defined to be ±30%. If the structural response is sensitive to variations in the shrinkage behaviour, tests are highly recommended.

where:

α_{ds1}, α_{ds2} are coefficients, dependent on the strength development class of concrete (see Table 14.6-10);

f_{cm} is the mean compressive strength at the age of 28 days in MPa according to Eq. (14.6-1);

RH is the relative humidity of the ambient atmosphere in %;

h $= 2A_c/u$, is the notional size of the member in mm, where A_c is the cross-section in mm^2 and u is the perimeter of the member in contact with the atmosphere in mm;

t is the concrete age in days;

t_s is the concrete age at the beginning of drying in days;

$(t-t_s)$ is the duration of drying in days.

Higher fly ash contents in concrete (e.g. green concrete) tend to decrease the total shrinkage deformations, which may result from the reduced cement content. As the given model considers slowly hardening cements which would correctly describe the delayed hydration of fly ash concretes but not the reduced cement content, shrinkage experiments are recommended when shrinkage deformations are decisive in the design of green concrete structures.

In contrast to normal weight concrete the shrinkage behaviour of lightweight aggregate concrete (LAWC) is characterized by swelling deformations in the young concrete age. This results from water stored in the porous aggregates which is only slowly released into the cement paste matrix. This shrinkage characteristic of LWAC is not taken into account in Eq. (14.6-79). The observed swelling deformations are turning into shrinkage deformations only after a longer duration of drying. The final value of drying shrinkage depends on the moisture content of the aggregates.

For details on shrinkage characteristics of LWAC and the related modelling, see information on left column referring to Eq. (14.6-67) and Müller et al. [14-16].

For structures sensitive to shrinkage deformations tests are recommended.

The shrinkage of lightweight aggregate concrete $\varepsilon_{lcs}(t,t_s)$ may be roughly estimated as:

$$\varepsilon_{lcs}(t,t_s) = \eta \cdot \varepsilon_{cs}(t,t_s) \qquad (14.6\text{-}79)$$

where:

$\varepsilon_{cs}(t,t_s)$ is calculated according to Eq. (14.6-70);

η = 1.5 for LC8, LC12, LC16;

= 1.2 for LC20 and higher.

14.6.1.7 Temperature effects

14.6.1.7.1 Maturity

Eq. (14.6-80), originally developed for normal strength concrete, is based on an activation energy for cement hydration of 33 kJ/mol. Research has shown that the activation energy not only depends on the type and strength class of cement, but also on the water/cement ratio, additions and admixtures. Nevertheless, there is no data basis available which would enable a modification of Eq. (14.6-80) regarding the use of additions and admixtures in common normal strength and high strength concretes.

Note that the number 13.65 in Eq. (14.6-80) results from the ratio 4000/(273+20).

Note that it is useful to limit the temperature range to 65°C during the hydration process (very young concrete) in order to avoid the risk of delayed formation of ettringite in hardened concrete.

The effect of elevated or reduced temperatures on maturity is prominent only until the compressive strength reaches about 50% of the 28-day value.

The effect of elevated or reduced temperatures on the maturity of concrete may be taken into account by adjusting the concrete age:

$$t_T = \sum_{i=1}^{n} \Delta t_i \, exp\left[13.65 - \frac{4000}{273 + T(\Delta t_i)}\right] \qquad (14.6\text{-}80)$$

where:

t_T is the temperature-adjusted concrete age which replaces t in the corresponding equations in days;

Δt_i is the number of days where a temperature T prevails;

$T(\Delta t_i)$ is the mean temperature in °C during the time period Δt_i.

The coefficient of thermal expansion depends on the type of aggregates and on the moisture state of the concrete. Thus it may vary between approximately $6 \cdot 10^{-6}$ K^{-1} and $15 \cdot 10^{-6}$ K^{-1}. The design value of $10 \cdot 10^{-6}$ K^{-1} is valid for normal strength and high strength concrete, as well as for self-compacting concrete and concrete containing high amounts of fly ash (e.g. green concrete).

Dependent on the stiffness and the coefficient of thermal expansion of the aggregates, the coefficient of thermal expansion of lightweight aggregate concrete ranges between $5 \cdot 10^{-6}$ K^{-1} and $11 \cdot 10^{-6}$ K^{-1}.

Where the structural response is sensitive to thermal strains, tests should be performed according to RILEM TC 129-MHT I [14-52].

Eqs. (14.6-82) and (14.6-83) are valid for sealed concrete tested in the hot state shortly after completion of the heating. In the past, these equations have also been accepted for drying ambient conditions. Considering all experimental data, a large scatter of the compressive strength values can be observed; see Müller et al. [14-16]. If a higher accuracy is required tests must be performed, for example according to RILEM TC 129-MHT II [14-53].

14.6.1.7.2 Thermal expansion

Thermal expansion of concrete may be calculated as:

$$\varepsilon_{cT} = \alpha_T \Delta T \tag{14.6-81}$$

where:

ε_{cT} is the thermal strain;

ΔT is the change of temperature in K;

α_T is the coefficient of thermal expansion in K^{-1}.

For the purpose of structural analysis, the coefficient of thermal expansion may be taken as $\alpha_T = 10 \cdot 10^{-6}$ K^{-1} for normal weight concrete, and $\alpha_T = 8 \cdot 10^{-6}$ K^{-1} for lightweight aggregate concrete.

14.6.1.7.3 Effect of temperature on compressive strength

14.6.1.7.3.1 Concrete under sealed conditions or submerged in water

The effect of temperature in the range $0°C \le T \le 80°C$ on the compressive strength of sealed (i.e. non-drying or under water) normal strength and high strength normal weight and lightweight aggregate concrete, $f_{cm}(T)$ and $f_{lcm}(T)$, respectively, may be calculated as:

$$f_{cm}(T) = f_{cm}(1.06 - 0.003T) \tag{14.6-82}$$

$$f_{lcm}(T) = f_{lcm}(1.04 - 0.002T) \tag{14.6-83}$$

where:

$f_{cm}(T), f_{lcm}(T)$ is the compressive strength in MPa at the temperature T in °C;

f_{cm}, f_{lcm} is the compressive strength in MPa at $T = 20$ °C from Eqs. (14.6-1) and (14.6-2);

T is the temperature in °C.

When concrete is subjected to mean temperatures substantially greater than 20 °C and in parallel to drying, the mechanical properties of the concrete will be affected in different ways. Generally, Eq. (14.6-82) represents the lower bound for the compressive strength. Drying will normally increase strength. Further, at elevated temperatures, an interaction between the cement paste and the aggregates might occur, which should also be considered. Both effects can be calculated according to the given Eq. (14.6-84).

Sustained moderately elevated temperatures or low relative humidity of the environment after sufficient curing may slightly increase the compressive strength compared to strength development at normal ambient environmental conditions if drying of the member is possible. A more precise approach, which includes the effect of moisture content (drying) and thermal incompatibilities is given by Eq. (14.6-84). For further details see *fib* Bulletin XX [14-67] and Acosta [14-54]. This approach is only verified for concrete with siliceous aggregates.

This physically based approach considers as a decisive parameter the local relative humidity h_T in the concrete pores. It can be determined by measuring the humidity with moisture sensors using pre-tests or by calculating the drying process using diffusion equations according to Bazant & Najjar [14-55]. The subsequent Table 14.6-12 gives representative values for h_T.

Table 14.6-12: .Representative values for h_T[TITLE TO BE INSERTED]

Temperature [°C]	Duration of drying [d]	Dry atmospheric conditions (RH = 50 %, indoors) $h = 2A_c/u$ [mm]				Humid atmospheric conditions (RH = 80 %, indoors) $h = 2A_c/u$ [mm]			
		100	200	500	1000	100	200	500	1000
40	7	0.88	0.94	0.97	0.99	0.95	0.97	0.99	0.99
	30	0.76	0.87	0.95	0.97	0.91	0.95	0.98	0.99
	90	0.66	0.79	0.91	0.95	0.86	0.92	0.96	0.98
	365	0.56	0.66	0.83	0.91	0.82	0.86	0.93	0.96
	18250	0.50	0.51	0.53	0.60	0.80	0.80	0.81	0.84
60	7	0.79	0.89	0.95	0.98	0.91	0.95	0.98	0.99
	30	0.65	0.78	0.90	0.95	0.86	0.91	0.96	0.98
	90	0.57	0.68	0.84	0.92	0.83	0.87	0.94	0.97
	365	0.52	0.57	0.72	0.84	0.81	0.83	0.89	0.94

14.6.1.7.3.2 Concrete subjected to drying conditions

The effect of temperature in the range of 20°C ≤ T ≤ 80°C on the **c**ompressive strength of normal strength and high strength normal weight concrete with siliceous aggregates, $f_{cm}(h_T,T)$, subjected to drying, may be calculated as:

$$f_{cm}(h_T,T) = f_{cm}(h_{ref},T_{ref}) \cdot (1+D_{f_c}+S_{f_c}) \qquad (14.6\text{-}84)$$

where:

$f_{cm}(h_T,T)$ is the compressive strength in MPa at the temperature T in °C and a relative humidity h_T in the concrete pores;

$f_{cm}(h_{ref},T_{ref})$ is the compressive strength in MPa at T_{ref} = 20 °C at reference storage conditions;

h_T is the relative humidity of the concrete pores [-] at temperature T to be determined by tests or predicted by diffusion equations, approximated values may be taken from Table 14.6-12;

h_{ref} is the relative humidity of the concrete pores [-] at reference temperature T_{ref} = 20 °C; h_{ref} = 100% = 1.0;

T is the temperature in °C;

T_{ref} is the reference temperature: 20°C.

	18250	0.50	0.50	0.51	0.54	0.80	0.80	0.80	0.82
80	7	0.67	0.80	0.92	0.96	0.87	0.92	0.97	0.98
	30	0.56	0.67	0.83	0.91	0.83	0.87	0.93	0.97
	90	0.52	0.58	0.74	0.85	0.81	0.83	0.90	0.94
	365	0.51	0.52	0.61	0.74	0.80	0.81	0.84	0.89
	18250	0.50	0.50	0.50	0.51	0.80	0.80	0.80	0.81

If the tensile strength is a major input parameter in the design of a structure, tests are recommended and should be performed according to RILEM TC 129-MHT III [14-56].As a first approximation alternatively the values calculated by Eq. (14.6-90) may be reduced or increased by 20%, due to the large scatter of data.

The effect of drying is taken into account in Eq. (14.6-84) by the parameter D_{fc}.

$$D_{f_c} = -0.32(h_{T_{ref}})^{26} + 0.32 \quad (14.6\text{-}85)$$

with:

$$h_{T_{ref}} = h_T - K(h_T)\cdot(T - T_{ref}) \quad (14.6\text{-}86)$$

with:

$$K(h_T) = 0.0135 h_T \frac{1-h_T}{1.25-h_T} \quad (14.6\text{-}87)$$

The effect of thermal dilation incompatibilities between the hardened cement paste and the aggregates is taken into account by introducing the factor S_{fc}.

$$S_{f_c} = 0.35\left(\frac{T-20}{100}\right)^2 + a_{f_c}\left(\frac{T-20}{100}\right) \quad (14.6\text{-}88)$$

with:

$$a_{f_c} = 2.3\cdot 10^{-9}\exp(19.1 h_{T_{ref}}) - 0.55\exp(0.64 h_{T_{ref}}) \quad (14.6\text{-}89)$$

14.6.1.7.4 Effect of temperature on tensile strength and fracture properties

14.6.1.7.4.1 Concrete under sealed conditions or submerged in water

In the range 0 °C ≤ T ≤ 80 °C, the uniaxial tensile strength f_{ctm} of normal strength concrete is significantly affected by temperature according to:

$$f_{ctm}(T) = f_{ctm}(1.16 - 0.008T) \quad (14.6\text{-}90)$$

where:

$f_{ctm}(T)$ is the uniaxial tensile strength in MPa at the temperature T in °C;

f_{ctm} is the uniaxial tensile strength in MPa at T = 20 °C from Eq. (14.6-3);

T is the temperature in °C.

In the range 0 °C ≤ T ≤ 80°C, the dependency of the splitting tensile strength $f_{ct,sp}$ on the temperature of normal strength normal weight concrete at the time of testing is described by:

$$f_{ct,sp}(T) = f_{ct,sp}(1.06 - 0.003T) \qquad (14.6\text{-}91)$$

where:

$f_{ct,sp}(T)$ is the splitting tensile strength in MPa at the temperature T in °C;

$f_{ct,sp}$ is the splitting tensile strength in MPa at T = 20 °C;

T is the temperature in °C.

To estimate the effect of elevated or reduced temperatures on flexural strength $f_{ct,fl}$ of normal strength normal weight concrete, we can use:

$$f_{ct,fl}(T) = f_{ct,fl}(1.1 - 0.005T) \qquad (14.6\text{-}92)$$

where:

$f_{ct,fl}(T)$ is the flexural strength in MPa at the temperature T in °C;

$f_{ct,fl}$ is the flexural strength in MPa at T = 20 °C;

T is the temperature in °C.

If moisture gradients may occur, the flexural tensile strength may be lower by up to 20%.

Eqs. (14.6-93a) and (14.6-93b) might give in some cases a somewhat more pronounced effect than observed. Available experimental data show a considerably broad scatter band.

Fracture energy G_F is strongly affected by temperature and moisture content of concrete at the time of testing. The effect of temperature on G_F of normal strength normal weight concrete may be estimated as:

dry concrete: $$G_F(T) = G_F(1.06 - 0.003T) \qquad (14.6\text{-}93a)$$

mass concrete: $$G_F(T) = G_F(1.12 - 0.006T) \qquad (14.6\text{-}93b)$$

where:

$G_F(T)$ is the fracture energy in N/m at a temperature T in °C;

Sustained moderately elevated temperatures or low relative humidity of the environment after sufficient curing may slightly increase the tensile strength compared to strength development at normal ambient environmental conditions if drying of the member is possible. A more precise approach, which includes the effect of moisture content (drying) and thermal incompatibilities is given by Eq. (14.6-94). For further details see *fib* Bulletin XX [14-67] and Acosta [14-54]. This approach is only valid for concrete with siliceous aggregates.

G_F is the fracture energy in N/m at $T = 20$ °C from Eq. (14.6-9);
T is the temperature in °C.

14.6.1.7.4.2 Concrete subjected to drying conditions

The combined effect of temperature and drying on the tensile strength of normal strength and high strength normal weight concrete with siliceous aggregates, $f_{ctm}(h_T,T)$ in the temperature range of $20°C \le T \le 80°C$ may be calculated as:

$$f_{ctm}\left(h_T,T\right) = f_{ctm}\left(h_{ref},T_{ref}\right)\left(1+D_{f_{ct}}+S_{f_{ct}}\right) \quad (14.6\text{-}94)$$

where:

$f_{ctm}(h_T,T)$ is the uniaxial tensile strength in MPa at the temperature T in °C and a relative humidity h_T in the concrete pores;

$f_{ctm}(h_{ref},T_{ref})$ is the uniaxial tensile strength in MPa at $T_{ref} = 20$ °C from Eqs. (14.6-3);

h_T is the relative humidity of the concrete pores [-] at temperature T, to be determined by tests or predicted by diffusion equations, approximated values may be taken from Table 14.6-12;

h_{ref} is the relative humidity of the concrete pores [-] at reference temperature T_{ref} ; $h_{ref} = 1.0$;

T is the temperature in °C;

T_{ref} is the reference temperature: 20 °C.

The effect of drying is taken into account in Eq. (14.6-94) by the parameter D_{fct}.

$$D_{f_{ct}} = -0.44\left(h_{T_{ref}}\right)^4 + 0.44 \quad (14.6\text{-}95)$$

With $h_{T_{ref}}$ according to Eq. (14.6-86).

The effect of thermal dilation incompatibilities between hardened cement paste and aggregates is taken into account with S_{fct} and may be calculated as:

$$S_{f_{ct}} = 0.45\left(\frac{T-20}{100}\right)^2 + a_{f_{ct}}\left(\frac{T-20}{100}\right) \quad (14.6\text{-}96)$$

with:

$$a_{f_{ct}} = 7.6\cdot 10^{-5}\, exp(8.5 h_{T_{ref}}) - 0.72\, exp(0.48\ h_{T_{ref}}) \quad (14.6\text{-}97)$$

The temperature range is limited to 20°C ≤ T ≤ 80°C.

14.6.1.7.5 Effect of temperature on modulus of elasticity

14.6.1.7.5.1 Concrete under sealed conditions or submerged in water

The effect of elevated or reduced temperatures at the time of testing on the modulus of elasticity of normal strength and high strength normal weight concrete and lightweight aggregate concrete at an age of 28 days may be estimated from:

$$E_{ci}(T) = E_{ci}(1.06 - 0.003T) \quad (14.6\text{-}98)$$

$$E_{lci}(T) = E_{lci}(1.04 - 0.002T) \quad (14.6\text{-}99)$$

where:

$E_{ci}(T)$, $E_{lci}(T)$	is the modulus of elasticity in MPa at the temperature T in °C;
E_{ci}, E_{lci}	is the modulus of elasticity in MPa at T = 20 °C from Eq. (14.6-20) and (14.6-22);
T	is the temperature in °C.

Eqs. (14.6-98) and (14.6-99) are valid for sealed concrete. In the past, these equations have also been accepted for unsealed conditions. The observed scatter of data is indicated in Müller et al. [14-16]).

If the structural response is sensitive to concrete stiffness, tests are recommended according to RILEM TC 129-MHT IV [14-57]).

14.6.1.7.5.2 Concrete under drying conditions

The combined effect of temperature and drying on the modulus of elasticity of normal strength and high strength normal weight concrete with siliceous aggregates, $E_{cm}(h_T,T)$ in the temperature range of 20°C ≤ T ≤ 80°C may be calculated as:

$$E_{cm}(h_T,T) = E_{cm}(h_{ref},T_{ref})(1 + D_{E_c} + S_{E_c}) \quad (14.6\text{-}100)$$

where:

$E_{cm}(h_T,T)$ is the modulus of elasticity in MPa at the temperature T in °C and a relative humidity h_T in the concrete pores;

$E_{cm}(h_{ref},T_{ref})$ is the modulus of elasticity in MPa at the temperature T_{ref} = 20°C at reference storage conditions h_{ref} =100% =1.0.

The effect of drying is taken into account in Eq. (14.6-100) by the parameter D_{Ec}.

$$D_{Ec} = 0 \qquad \text{for } h_{Tref} = 1.0$$
$$D_{Ec} = -0.12\left(h_{Tref}\right)^{19} - 0.14 \qquad \text{for } h_{Tref} < 1.0 \qquad (14.6\text{-}101)$$

with

$h_{T_{ref}}$ according to Eq. (14.6-86)

The effect of thermal dilation incompatibilities between hardened cement paste and the aggregates is taken into account with the factor S_{Ec} and may be calculated as:

$$S_{E_c} = a_{E_c}\left(\frac{T-20}{100}\right) \qquad (14.6\text{-}102)$$

With

$$a_{E_c} = 2.3\cdot 10^{-5}\cdot\exp\left(9.0h_{T_{ref}}\right) - 0.2\cdot\exp\left(0.7h \right. \qquad (14.6\text{-}103)$$

14.6.1.7.6 Effect of temperature on creep and shrinkage

14.6.1.7.6.1 Creep

The relations to predict the effect of temperature up to 80 °C on creep given in this section are only rough estimates. For a more accurate prediction considerably more sophisticated models are required which take into account the moisture state of the concrete at the time of loading. Neglecting this parameter the relations given in this section are generally more accurate for thick concrete members with little change in moisture content than for thin members where significant changes in moisture content occur, particularly at elevated temperatures.

If the structural response is sensitive to concrete creep, tests are urgently recommended, for example according to RILEM TC 129-MHT V [14-58].

The effect of temperature prior to loading may be taken into account using Eq. (14.6-80).

Eqs. (14.6-104) to (14.6-108) below describe the effect of a constant temperature differing from 20°C while a normal weight concrete is under load.

The effect of temperature on the time-development of creep is taken into account using $\beta_{h,T}$:

$$\beta_{h,T} = \beta_h \cdot \beta_T \tag{14.6-104}$$

with:

$$\beta_T = exp\left[\frac{1'500}{(273+T)-5.12}\right] \tag{14.6-105}$$

where:

$\beta_{h,T}$ is a temperature dependent coefficient replacing β_h in Eq. (14.6-66a);

β_h is the coefficient according to Eq. (14.6-66c);

T is the temperature in °C.

The effect of temperature on the creep coefficient is taken into account using:

$$\varphi_{bc,T} = \varphi_{bc} \cdot \varphi_T \tag{14.6-106}$$

$$\varphi_{dc,T} = \varphi_{dc} \cdot \varphi_T^{1.2} \tag{14.6-107}$$

with:

$$\phi_T = exp\left[0.015(T-20)\right] \tag{14.6-108}$$

where:

$\varphi_{bc,T}$ is a temperature dependent coefficient which replaces φ_{bc} in Eq. (14.6-58);

$\varphi_{dc,T}$ is a temperature dependent coefficient which replaces φ_{dc} in Eq. (14.6-58);

φ_{bc} is the basic creep coefficient according to Eq. (14.6-59);

φ_{dc} is the drying creep coefficient according to Eq. (14.6-62);

Here the additional creep (transient creep) is addressed which is observed when a temperature increase occurs while concrete is under constant load (temperature increase after loading).

If the structural response is sensitive to concrete creep, tests are recommended according to RILEM TC 129-MHT VI [14-59].

Some experiments indicate not only an acceleration of shrinkage but also an increased basic shrinkage deformation if the concrete is subjected to ongoing elevated curing temperatures. This effect decreases with increasing concrete strength and is neglected in this simplified approach.

If shrinkage at temperatures deviating from 20 °C is a major input parameter tests may be performed according to RILEM TC 129-MHT VI [14-59].

T is the temperature in °C.

For an increase of temperature while the structural member is under load, creep may be estimated from Eq. (14.6-106):

$$\varphi\left(t,t_0,T\right)=\varphi\left(t,t_0\right)+\Delta\varphi_{T,trans} \tag{14.6-109}$$

with:

$$\Delta\phi_{T,trans}=0.0004\left(T-20\right)^2 \tag{14.6-110}$$

where:

$\varphi(t,t_0)$ is the creep coefficient according to Eq. (14.6-58) and temperature-adjusted according to Eqs. (14.6-106) to (14.6-108);

$\Delta\varphi_{T,trans}$ is the transient thermal creep coefficient which occurs at the time of the temperature increase;

T is the temperature in °C.

14.6.1.7.6.2 Shrinkage

Temperatures between 0 °C and 80 °C affect both components of shrinkage being defined by Eq. (14.6-70).

For basic shrinkage mainly the time-development is influenced. To take this effect into consideration, the concrete age t in Eq. (14.6-74) has to be replaced by the effective concrete age t_T according to Eq. (14.6-80).

The effect of a constant temperature differing from 20 °C while the concrete is drying is described by means of Eqs. (14.6-111) to (14.6-117) below.

The effect of temperature on the time-development of drying shrinkage is taken into account using $\alpha_{sT}(T)$:

$$\alpha_{sT}\left(T\right)=0.035\cdot h^2\,exp[-0.06\left(T-20\right)] \tag{14.6-111}$$

where:

The effect of elevated temperatures on shrinkage is influenced considerably by the moisture content of the concrete prior to heating and the moisture loss after an increase of temperature.

Whether a concrete specimen is shrinking or swelling under particular ambient climate conditions is determined by its internal relative humidity and the temperature-dependent water sorption capacity. The transition point at the relative humidity between shrinkage and swelling (RH_T) is therefore dependent on the concrete temperature and the concrete compressive strength (substitute parameter for the concrete composition).

Note, Eq. (14.6-114) is a simplified version of Eq. (14.6-76).

$\alpha_{sT}(T)$ is a temperature-dependent coefficient replacing the product 0.035 h^2 in Eq. (14.6-77);

T is the temperature in °C.

The effect of temperature on the magnitude of drying shrinkage is taken into account using:

$$\beta_{RH,T} = \beta_{RH} \cdot \beta_{sT} \tag{14.6-112}$$

$\beta_{RH,T}$ is a temperature-dependent coefficient, which replaces β_{RH} in Eq. (14.6-72). It is calculated using:

$$\beta_{sT} = 1 + \left(\frac{4}{103 - RH}\right) \cdot \left(\frac{T - 20}{40}\right) \tag{14.6-113}$$

$$\beta_{RH} = \begin{cases} -1.55 \cdot \left[1 - \left(\frac{RH}{100}\right)^3\right] & \text{for } 40 \le \text{RH} < RH_T \\ 0.25 & \text{for } RH \ge RH_T \end{cases} \tag{14.6-114}$$

$$R_{H_T} = 99\beta_{s1} + \beta_{s1,T} \le 100\% \tag{14.6-115}$$

$$\beta_{s1} = \left(\frac{35}{f_{cm}}\right)^{0.1} \le 1.0 \tag{14.6-116}$$

$$\beta_{s1,T} = \left(\frac{T - 20}{25}\right)^3 \tag{14.6-117}$$

where:

RH is the relative humidity of the ambient environment in %;

T is the temperature in °C;

f_{cm} is the mean compressive strength at the age of 28 days in MPa according to Eq. (14.6-1).

14.6.2 Properties related to non-static loading

14.6.2.1 Properties related to fatigue

In this chapter, the S-N relations describing the material's fatigue behaviour in terms of load cycles to failure are given. They are determined as mean value curves for concretes with concrete ages higher than 28 days. Fatigue tests exhibit large scatter of the numbers of cycles to failure. The necessary safety is covered by a safety factor and the reduction of the fatigue reference strength in the fatigue design. Therefore, appropriate assessment of design fatigue resistance is only possible using the S-N relations given in section 30.2.2.1.

S-N relations are usually given using maximum levels $S_{c,max}$ or rather $S_{ct,max}$ and minimum stress levels $S_{c,min}$, which are calculated using acting fatigue stresses divided by a reference strength of the concrete. In this chapter, the material's fatigue behaviour is described. Therefore, the mean values of concrete strengths f_{cm} or f_{ctm}, tested for the same concrete, with the same concrete age and the same specimen geometry, in the same environment and with the same concrete moisture content, are used as reference strengths for the determination of the maximum and minimum stress levels. The concrete strengths f_{cm} and f_{ctm} should be taken from experiments.

For further general information, see CEB Bulletin 188 [14-60] and *fib* Bulletin 42 [14-18].

The S-N relations in Eqs. (14.6-118) to (14.6-121) for pure compressive fatigue loading in "dry" condition are valid for concrete specimens stored in standard climate of approximately 20 °C, 65% relative humidity and tested in air. The curves were developed based on experiments on ultra high strength concrete (up to C180) and were validated for high strength and normal strength concrete as well. The curves have been verified with experiments up to 10^7 load cycles to failure ($log\ N = 7$). For $log\ N > 8$, the curves asymptotically approach the minimum stress level of the respective curve. Further information can be found in Lohaus et al. [14-61].

Concrete which is water saturated and/or submerged in water has a lower fatigue resistance. The S-N relations for "wet" condition were developed using concrete specimens submerged in water immediately after casting and stored under water until fatigue testing, which was executed submerged in water. Further information can be found in Oneschkow et al. [14-62].

The S-N relations Eqs. (14.6-118) – (14.6-121) for "dry" condition ($\omega_w = 1.00$) and "wet" condition ($\omega_w = 0.75$) are applicable for frequencies $f > 0.1$ Hz.

14.6.2.1.1 Fatigue resistance

For constant stress amplitude, the number N of cycles causing fatigue failure of plain ordinary concrete may be estimated from Eqs. (14.6-118) to (14.6-123) below. They are valid for pure compression, compression-tension and pure tension, respectively.

The applicability of the S-N curves for lightweight concrete and concrete with recycled aggregate has to be proven.

(a) Pure compression

For $S_{c,max} < 0.90$ and $0 \le S_{c,min} \le 0.80$, the following equations apply:

$$log\ N_1 = \left[\frac{8}{Y-1}\cdot\left(S_{c,max}-1\right)\right] \qquad (14.6\text{-}118)$$

$$logN_2 = \left[8+\frac{8ln(10)}{Y-1}\left(Y-S_{c,min}\right)\cdot log\left(\frac{S_{c,max}-S_{c,min}}{Y-S_{c,min}}\right)\right] \qquad (14.6\text{-}119)$$

with:

$$Y = \frac{0.45+1.8\cdot S_{c,min}}{1+1.8\cdot S_{c,min}-0.3\cdot S_{c,min}^2}$$

where:

(a) if $log\ N_1 \le 8$, then $log\ N = \omega_w \cdot log\ N_1$ (14.6-120)

(b) if $log\ N_1 > 8$, then $log\ N = \omega_w \cdot log\ N_2$ (14.6-121)

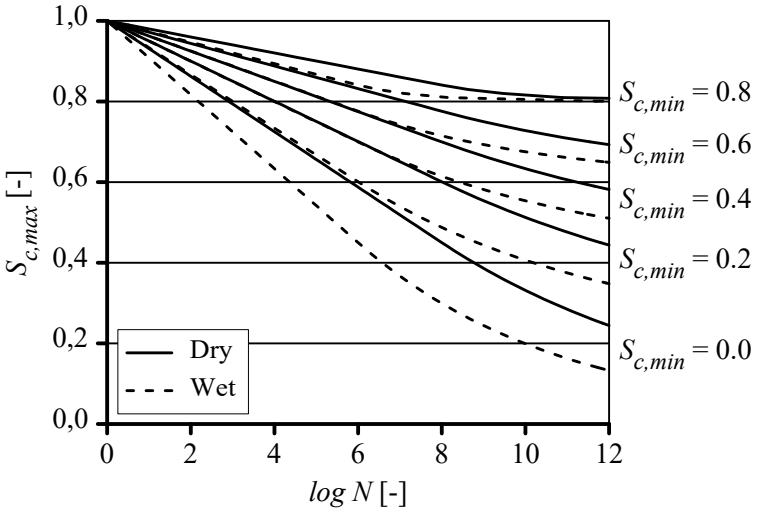


Figure 14.6-9: S-N relations according to Eqs. (14.6-118) to (14.6-121) for "dry" condition (ω_w = 1.00) and "wet" condition (ω_w = 0.75).

Though experimental evidence has still to be given, the S-N relations Eq. (14.6-118) – (14.6-121) may be assumed to apply also for self-compacting concrete.

Eq. (14.6-122) and (14.6-123) are applicable for dry condition and frequencies f > 0.1 Hz. If Eq. (14.6-122) is applied, it may be assumed that the concrete always fails in compression.

For concrete in tension, the crack propagation can be different for different types of concrete due to the differences in the internal material structure. However, test results have shown that the fatigue life seems to be relatively equal for the various concrete types; see *fib* Bulletin 42 [14-18].

with:

$$S_{c,max} = |\sigma_{c,max}| / f_{cm}$$

$$S_{c,min} = |\sigma_{c,min}| / f_{cm}$$

where:

$\omega_w = 1.00$	for concrete stored in standard climate and tested in dry environment
$\omega_w = 0.75$	for water saturated concrete and/or concrete submerged in water
f_{cm}	mean value of concrete compressive strength (taken from experiments)

For $S_{c,min} > 0.8$, the S-N relations for $S_{c,min} = 0.8$ are valid.

(b) Compression-tension with $S_{c,max}$ and $S_{ct,max} < 0.90$ and $\sigma_{ct,max} \le 0.026|\sigma_{c,max}|$

$$log\, N = 9\left(1 - S_{c,max}\right) \qquad (14.6\text{-}121)$$

with:

$$S_{c,max} = \frac{|\sigma_{c,max}|}{f_{cm}}$$

(c) Pure tension and tension-compression with $S_{c,max}$ *and* $S_{ct,max} < 0.90$ and $\sigma_{ct,max} > 0.026|\sigma_{c,max}|$

$$log\, N = 12\left(1 - S_{ct,max}\right) \qquad (14.6\text{-}122)$$

with:

$$S_{ct,max} = \frac{|\sigma_{ct,max}|}{f_{ctm}}$$

For higher stress levels and lower frequencies, lower values of log N than predicted by Eqs. (14.6-118) to (14.6-123) may be expected. For further details, see CEB Bulletin 188 [14-60] and *fib* Bulletin 42 [14-18].

In Eq. (14.6-124) it is assumed that creep due to repeated loading is equal to creep under a constant stress $(|\sigma_{c,max}| + |\sigma_{c,min}|)/2$ acting for a time $(t - t_0) = (1/1440)\cdot(n/f)$ = duration of repeated loading [days], where:

n is the number of cycles applied at a frequency f;

f is the frequency of repeated loading [min^{-1}].

Therefore, Eq. (14.6-124) gives only a rough estimate of the strains due to repeated loads. It does not take into account variations of E_c due to repeated loads or tertiary creep, which develops prior to fatigue failure. For further details, see CEB Bulletin 188 [14-60]. New research results indicate that the strain developments due to fatigue loading might be underestimated by Eq. (14.6-124).

where:

N is the number of cycles to failure;

$S_{c,max}$ is the maximum compressive stress level;

$S_{c,min}$ is the minimum compressive stress level;

$S_{ct,max}$ is the maximum tensile stress level;

$\sigma_{c,max}$ is the maximum compressive stress in MPa;

$\sigma_{c,min}$ is the minimum compressive stress in MPa;

$\sigma_{ct,max}$ is the maximum tensile stress in MPa;

f_{cm} is the mean value of concrete compressive strength in MPa (taken from experiments);

f_{ctm} is the mean value of concrete tensile strength in MPa (taken from experiments);

ω_w is a coefficient taking into account the moisture content and environment of concrete.

14.6.2.1.2 Fatigue strains

For maximum compressive stresses $|\sigma_{c,max}| < 0.6f_{ck}$ and a mean stress $(|\sigma_{c,max}| + |\sigma_{c,min}|)/2 < 0.5f_{ck}$ the strain at maximum stress due to repeated loads of a given frequency f may be estimated as:

$$\varepsilon_{cf}(n) = \frac{\sigma_{c,max}}{E_{ci}(t_0)} + \frac{\sigma_{c,max} + \sigma_{c,min}}{2E_{ci}}\varphi(t,t_0) \qquad (14.6\text{-}124)$$

where:

ε_{cf} is the strain at maximum stress due to repeated loads;

$\sigma_{c,max}$ is the maximum compressive stress in MPa;

$\sigma_{c,min}$ is the minimum compressive stress in MPa;

E_{ci} is the modulus of elasticity of concrete in MPa at a concrete age of 28 days according to Eq. (14.6-20);

$E_{ci}(t_0)$ is the modulus of elasticity of concrete in MPa at a concrete age t_0 according to Eq. (14.6-48);

The strain rate effect on mechanical properties of concrete as it was expressed in MC2010 was based on test results. Meanwhile, computer simulations have revealed that two effects in the experiments cannot be separated: the true material effect and the inertia effect. The material effect is due to the rate process which can be expressed by a power function. The inertia effect is inherent in the testing equipment and specimen. What is measured can be called apparent strength.

The formulae in this code imply the rate effect, i.e. the true material behaviour which is the input in fully dynamic analyses. The apparent strength as given in MC2010 can be used in hand calculations. However, one has to be aware of the fact that inertia depends on size, shape, failure mode and boundary conditions of the structural element to be analysed.

The given constitutive relations are valid also for lightweight aggregate concrete.

No information is available for self-compacting concrete. However, it is supposed that the behaviour of self-compacting concrete does not deviate significantly from that of ordinary structural concrete; that is, data on self-compacting concrete are supposed to lie within the scatter band of ordinary concrete.

$\varphi(t,t_0)$ is the creep coefficient according to Eq. (14.6-58);
t_0 is the age of concrete at the beginning of repeated loading in days;
t is the age of concrete at the moment considered in days.

14.6.2.2 Properties related to impact and explosion

14.6.2.2.1 Range of applicability

The information given below is valid for monotonically increasing compressive stresses or strains at a constant range of approximately 1 MPa/s < $|\dot{\sigma}_c|$ < 10^7 MPa/s and $30\cdot10^{-6}$ s^{-1} < $|\dot{\varepsilon}_c|$ < $3\cdot10^2$ s^{-1}, respectively. In the corresponding equations all strain and stress values have to be used as absolute values.

For tensile stresses or strains the information is valid approximately for 0.03 MPa/s < $\dot{\sigma}_{ct}$ < 10^7 MPa/s and $1\cdot10^{-6}$ s^{-1} < $\dot{\varepsilon}_{ct}$ < $3\cdot10^2$ s^{-1}, respectively.

The values should be used in fully dynamic analyses of concrete structures.

14.6.2.2.2 Compressive strength

For a given strain and stress rate, respectively, the compressive strength under high rates of loading may be estimated as:

$$\frac{f_{c,imp,k}}{f_{cm}} = \left(\frac{\dot{\varepsilon}_c}{\dot{\varepsilon}_{c0}}\right)^{0.014} \qquad (14.6\text{-}125)$$

with $\dot{\varepsilon}_{c0}$ = $30\cdot10^{-6}$ s^{-1};

and:

$$\frac{f_{c,imp,k}}{f_{cm}} = \left(\frac{\dot{\sigma}_c}{\dot{\sigma}_{c0}} \right)^{0.014} \quad (14.6\text{-}126)$$

with $\dot{\sigma}_{c0}$ = 1 MPa s^{-1},

where

$f_{c\ imp,k,}$ is the impact compressive strength;

is the strain rate in s^{-1};

f_{cm} is the mean compressive strength in MPa;

is the stress rate in MPa/s.

14.6.2.2.3 Tensile strength and fracture properties

(a) Tensile strength

For a given strain and stress rate, respectively, the tensile strength under high rates of loading $f_{ct\ imp,k}$ may be estimated as:

$$\frac{f_{ct,imp,k}}{f_{ctm}} = \left(\frac{\dot{\varepsilon}_{ct}}{\dot{\varepsilon}_{ct0}} \right)^{0.018} \quad (14.6\text{-}127)$$

with $\dot{\varepsilon}_{ct0} = 1 \cdot 10^{-6}\ s^{-1}$

and:

$$\frac{f_{ct,imp,k}}{f_{ctm}} = \left(\frac{\dot{\sigma}_{ct}}{\dot{\sigma}_{ct0}} \right)^{0.018} \quad (14.6\text{-}128)$$

with $\dot{\sigma}_{ct0}$ = 0.03 MPa s^{-1}.

(b) Fracture energy

The information available regarding the effect of stress or strain rate on the fracture energy is too incomplete to be included in this Model Code.

There is little information regarding the effect of high stress or strain rates on the shape of the stress–strain diagram.

No information is available for the strain-softening region.

14.6.2.2.4 Modulus of elasticity

The effect of stress and strain rate on the modulus of elasticity $E_{c\ imp}$ may be estimated as:

$$\frac{E_{c,imp}}{E_{ci}} = \left(\frac{\dot{\sigma}_c}{\dot{\sigma}_{c0}} \right)^{0.025} \tag{14.6-129}$$

$$\frac{E_{c,imp}}{E_{ci}} = \left(\frac{\dot{\varepsilon}_c}{\dot{\varepsilon}_{c0}} \right)^{0.026} \tag{14.6-130}$$

with $\dot{\sigma}_{c0}$ = 1 MPa s^{-1} and $\dot{\varepsilon}_{c0}$ = 30·10^{-6} s^{-1} for compression;

with $\dot{\sigma}_{ct0}$ = 0.03 MPa s^{-1} and $\dot{\varepsilon}_{ct0}$ = 1·10^{-6} s^{-1} for tension.

14.6.2.2.5 Stress–strain relations

For monotonically increasing compressive stresses or strains up to the peak stress, as an approximation Eq. (14.6-26) may be used together with Eqs. (14.6-125) - (14.6-128) for the peak stress $f_{c,imp}$, Eqs. (14.6-129) and (14.6-130) for the modulus of elasticity $E_{c,imp}$ and Eq. (14.6-131) for the strain at maximum stress $\varepsilon_{c1,imp}$.

The effects of high stress and strain rates on the strains at maximum stress in tension and compression may be estimated as:

$$\frac{\varepsilon_{c1,imp}}{\varepsilon_{c1}} = \left(\frac{\dot{\sigma}_c}{\dot{\sigma}_{c0}} \right)^{0.02} = \left(\frac{\dot{\varepsilon}_c}{\dot{\varepsilon}_{c0}} \right)^{0.02} \tag{14.6-131}$$

with $\dot{\sigma}_{c0}$ = 1 MPa s^{-1} and $\dot{\varepsilon}_{c0}$ = 30·10^{-6} s^{-1} for compression;

with $\dot{\sigma}_{ct0}$ = 0.03 MPa s^{-1} and $\dot{\varepsilon}_{ct0}$ = 1·10^{-6} s^{-1} for tension;

where:

$\varepsilon_{c1,imp}$ is the impact strain at maximum load for compression and tension, respectively;

ε_{c1} is the strain at maximum load for static loading from sections 14.6.1.5.1 and 14.6.1.5.2 for compression and tension, respectively.

14.6.3 Properties related to extreme thermal conditions

14.6.3.1 Properties related to fire

For the material properties of concrete at high temperatures, see *fib* Bulletin 38 [14-63] and *fib* Bulletin 46 [14-64] as well as chapter 3 "Material properties" in EN 1992-1-2:2004 [14-65] and EN 1992-1-2:2004/AC:2008 [14-66].

For the material properties of concrete at high temperatures, see *fib* Bulletin 38 [14-63] and *fib* Bulletin 46 [14-64] as well as chapter 3 "Material properties" in EN 1992-1-2:2004 [14-65] and EN 1992-1-2:2004/AC:2008 [14-66].

14.6.3.2 Properties related to other high temperature conditions

The material properties are negatively affected by the influence of high temperatures. Changes to the microstructure of concrete during a high temperature exposure result in corresponding changes in mechanical and physical properties which are mainly caused by the thermo-hydraulic and thermo-mechanical behaviour of the material.

For a structural fire design point of view see section 30.4.2 of this Model Code.

14.6.3.3 Properties related to cryogenic condition

The compressive strength increases at low temperatures as a function of temperature and moisture content of the concrete. The strength gain can be estimated as:

$$\Delta f_{cm} = 12m\left[1-\left(\frac{T+170}{170}\right)^2\right] \qquad (14.6\text{-}132)$$

where:

Approximate moisture content m of structural concrete [% by mass]:

- general indoor: 2%;
- general outdoor: 4%;
- exposed to rain: 6%.

Eq. (14.6-132) is also valid for lightweight aggregate concrete.

No information is available for self-compacting concrete, high strength concrete and for concretes that differ significantly from common compositions. However, it is supposed that the behaviour of self-compacting concrete does not deviate significantly from that of ordinary structural concrete; that is, data on self-compacting concrete are supposed to lie within the scatter band of ordinary concrete.

The ingress of aggressive agents from the environment (CO_2, Cl^-, SO_4^{2-}) into concrete is determined by the pore structure of the surface layers of a concrete member and the resulting transport properties. Additionally, the water content of pores as well as wetting/drying cycles influence the uptake of harmful substances. Compressive strength, on the contrary, is a bulk property. As a result, transport properties and compressive strength are poorly correlated.

The parameters governing these transport mechanisms can be measured by appropriate test methods (see section 14.8.2 and 14.8.3), some are described in this chapter and the resulting values may be used for compliance with performance specifications and/or for predictive modelling.

Cracks effects on the transport properties is described in *fib* Bulletin XX [14-67].

More details and background information on the transport of liquids, gases, and ions are given in *fib* Bulletin XX [14-67].

Note that migration is not a real transport process as it is caused by an accelerated performance test method (more information is given in *fib* Bulletin XX [14-67]).

Δf_{cm} is the gain in compressive strength in MPa;

m is the moisture content in % by mass;

T is the temperature in °C.

Eq. (14.6-132) is valid for T between $0° \geq T \geq -170°C$, for a single temperature drop.

Tensile strength and modulus of elasticity increase at low temperature. They can be estimated by means of Eqs. (14.6-3) and (14.6-20) inserting the respective compressive strength.

Also look at text of Section 30.4.3.4.

14.7 Transport of liquids, gases and ions in hardened concrete

14.7.1 General

Liquids, gases, or ions may be transported through hardened concrete by one or a combination of different mechanisms: permeation, diffusion, and migration.

The flow of matter (gases, liquids, ions) through concrete is frequently a dominant factor in durability, whenever the transported substances may lead to physical, chemical and/or electrochemical processes resulting in the deterioration of the concrete itself or of the embedded reinforcing steel.

Although not strictly aggressive in itself, water also contains ions and is involved in almost all degradation processes. Frost damage, sulfate attack, alkali-aggregate reaction and steel corrosion do not progress in the absence of water. Shrinkage and creep are also dependent on water content. Water penetrates and moves within concrete by different mechanisms discussed below (permeation, capillary suction, and diffusion).

Water-permeability of concrete is relevant to structures that contain or transport water (or other liquids), in particular dams, tanks containing water or other liquids, retaining walls, canals, culverts, pipes, etc.

Similarly, gas-permeability of concrete is relevant to structures that contain or transport gases, in particular tanks and pipes, underground gas reservoirs to store/release energy or to sequester CO_2, evacuated tunnels for high-speed trains, etc. It plays an important role in the release of water vapour under fire, thus decreasing the risk of explosive spalling in the event of fire.

Under certain conditions, aggressive substances can penetrate the concrete by permeation. The initial water content affects the pore space available for the transport of the gases and liquids, thus affecting water and gas permeability and capillary suction.

The water permeability of concrete is not constant because, as water penetrates, it may lead to further cement hydration, swelling of mineral phases and precipitation of products from the paste that block the pores and reduce the permeability. This process also helps in sealing cracks with widths around approx. 0.1 – 0.2 mm (possibly larger in some cases) and is known as “self-healing”.

There are no standard test methods to measure the coefficient of water-permeability of concrete by direct application of Eq. (14.7-1) or Eq. (14.7-2).

As a substitute, the Water Penetration under Pressure test is alternatively used, standardized in European Standard EN 12390-8 [14-68] and in Chinese Standard GBJ 82-85 [14-69]. The test result is the maximum penetration of the waterfront x_d (mm).

The specific permeation of water is described by Darcy’s law, Eq. (14.7-2):

$$Q = -K_w A \frac{\Delta h}{\Delta L} \qquad (14.7\text{-}2)$$

where:

K_w Darcy's coefficient of water permeability (m/s)

14.7.2 Permeation

14.7.2.1 Introduction

Permeation describes the mechanism by which fluids, that can be liquids or gases, flow through the concrete porous network driven by a pressure gradient.

14.7.2.2 Liquids permeability

The more general permeation law is based on a generalization of Darcy’s law (strictly applicable to water flow), extending it to other liquids. It assumes a laminar viscous flow and reads:

$$Q = K_l \frac{A}{\mu} \frac{dP}{dL} \qquad (14.7\text{-}1)$$

where:

Q	flow rate (m³/s);
K_l	intrinsic coefficient of permeability to liquids (m²);
A	cross section area of the body traversed by the liquid (m²);
μ	dynamic viscosity of the liquid (Pa.s);
dP/dL	pressure gradient across the body (Pa/m).

$\Delta h/\Delta L$ pressure gradient, with pressure expressed as water head (m/m).

fib Bulletin XX [14-67] gives the equations enabling the calculation of the penetration time and the conversion between the coefficient of water permeability and the intrinsic coefficient of liquids permeability.

The liquid permeability could also be assessed by means of the modelling of drying and an experiment on mass loss (see for instance Auroy et al [14-70]).

14.7.2.3 Gas permeability

Eq.(14.7-3) assumes that the flow of gas through concrete corresponds to a laminar viscous flow. In fact, this flow coexists with a molecular flow, making K_g dependant on the applied pressure (Klinkenberg effect), resulting in higher K_g values for lower applied pressures.

Amongst the standardized test methods to measure the permeability of concrete to gases, we cite the following (all based on Eq. (14.7-3)):

- RILEM-Cembureau test (kO), covered by Recommendation RILEM TC 116-PCD [14-71] and Spanish Standard UNE 83966 [14-72].

All gas-permeability tests need to take into consideration the important effect of moisture content of the concrete at the time of testing.

For the interpretation of test data, Table 14.7-1 provides a rating of the permeability of concrete, measured at 28 days, according to various standardized permeation tests.

Table 14.7-1: Classification of Concrete Penetrability.

Concrete Penetrability	**Permeation** WP_{max}
Class	**mm**
High	> 80
Moderate	50-80
Low	30-50
Very Low	< 30
Negligible	---

Note: gas permeation is measured on specimens dried 3 - 7 days at 50°C

WP_{max}: maximum water penetration after EN 12390-8 [14-68].

More detailed information is provided in fib Bulletin XX [14-67].

The gas permeability law, assuming laminar viscous flow, is slightly different from Eq. (14.7-1), to take gas compressibility into account:

$$Q = K_g \frac{A}{\mu L} \frac{P_2^2 - P_1^2}{2P_1} \qquad (14.7\text{-}3)$$

K_g intrinsic coefficient of permeability to gases (m²);
A cross section area of the body, traversed by the gas (m²);
P_2 gas inlet pressure (Pa);
P_1 pressure at which Q is measured (Pa), usually outlet pressure;
L thickness of the body (m);
μ dynamic viscosity of the gas (Pa.s).

Capillary suction takes place when unsaturated concrete is put in contact with water.

Exponent n is usually assumed as 0.5, which is a fair approximation for highly porous and/or dry concrete. But for denser and less dried concrete reported values for n are in the range of 0.20 to 0.35 for example due to self-desiccation, ongoing hydration, capillary imbibition, gravity.

In standardized test conditions, usually the initial coefficient of adsorption is determined where a linear relationship between capillary water uptake and the square root of time is determined (ASTM C1585-13 [14-73], EN ISO 15148:2002 [14-74]). Detailed information can be found in *fib* Bulletin XX [14-67].

Diffusion is a relevant transport mechanism in concrete, closely associated with durability. Gases, typically CO_2 (that provokes carbonation, leading to steel depassivation) and O_2 (also needed for steel corrosion), may penetrate concrete from the surrounding atmosphere, mainly by diffusion.

The main material property that governs the phenomenon is the diffusion coefficient D, that depends on the pore structure of the concrete and on the nature of the transported substance, including its possible reactions with reactive phases in the cement paste.

The transport of water vapour phase under steady state conditions can be described by Fick's first law of diffusion, with the gradient of the relative pore humidity as the driving force. The diffusion coefficient D is a non-linear function of the local relative pore humidity H. The volume of water flowing is given by:

$$V = D(H)\frac{dH}{dx}A \cdot t \qquad (14.7\text{-}6)$$

with:

V	is the volume of transported water in (m^3);
$D(H)$	is the diffusion coefficient (m^2/s) at relative pore humidity H *(-);*
dH/dx	is the gradient in pore relative humidity (m^{-1});
A	is the penetrated area (m^2);

14.7.2.4 Capillary suction

Liquids, particularly water and solutions, may be transported into concrete by capillary suction or absorption. Water absorption may be expressed by Eq. (14.7-4):

$$w = M_w \cdot t^n \qquad (14.7\text{-}4)$$

with:

w	is the water absorbed per unit area at time t (m^3/m^2);
t	is the duration of water absorption (s);
n	≤ 0.5;
M_w	is the coefficient of water absorption ($m/s^{0.5}$).

14.7.3 Diffusion

Diffusion is a transport mechanism by which gases, liquids or dissolved ions flow across a porous body, driven by a concentration gradient.

When gases, liquids, and dissolved substances are transported under a constant concentration gradient (steady state) Fick's first law of diffusion applies, as given in Eq. (14.7-5):

$$Q = -D \cdot A\frac{dc}{dx} \qquad (14.7\text{-}5)$$

Q	is the flow rate of substance transported ($mol.s^{-1}$)
dc/dx	is the concentration gradient ($mol.m^{-4}$)
A	is the penetrated area (m^2)
D	is the diffusion coefficient (m^2/s)

The negative sign in this equation arises because diffusion occurs in the direction opposite to the direction of increasing concentration.

t is the time (s).

The term $\frac{\partial c}{\partial t}$ in Eq. (14.7-7) can be neglected if no enrichment/depletion is required.

In standards water vapour permeability coefficient in mg/(m².h.Pa) is determined by dividing the mass gain per unit time G in mg/h through the sample area A in m and the water vapour partial pressure Dp in Pa according to EN ISO 12572 [14-75] or EN 12086 [14-76]. Sample precondition, moisture and temperature difference strongly affect the water vapour permeability coefficient.

Eq. (14.7-8) shows the error function solution for unidirectional diffusion of substances in general. Models for chloride ion diffusion or leaching are referred to section 30.6.3 and 30.6.8.

Table 14.7-1 provides classification of concrete penetrability based on standardized chloride diffusion test. D_{nss}: coefficient of chloride diffusion after ASTM C1556 [14-77] or NT Build 443 [14-78]. It is to be mentioned that more recent ISO 1920-11 [14-79] and EN 12390-11 [14-80] tests use 90 days exposure that will lead to different values of recorded D_{nss}.

Deterioration potentially leads to material, structural, safety and environmental consequences, as a result of various effects including, the:

- loss of strength and modulus of elasticity due to cracking or material softening.
- loss of bearing area and volume due to cracking and spalling;
- loss in steel/concrete bond;
- failure of tensioning cables;
- falling pieces of concrete creating safety hazards;
- aggressivity / pollution of ground or surface waters;

Fick's second law relates the unsteady diffusive flux to the concentration gradient. $\frac{\partial c}{\partial t}$ applies if enrichment or depletion of diffusing substance is considered.

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} \pm \frac{\partial c}{\partial t} \qquad (14.7\text{-}7)$$

Assuming that D and the surface concentration of the substance are constant, Eq. (14.7-7) accepts an explicit solution (semi-infinite element, unidirectional diffusion):

$$c(x,t) = c_0 + (c_s - c_0).\left[1 - \mathrm{erf}\left(\frac{x}{\sqrt{4.D.t}}\right)\right] \qquad (14.7\text{-}8)$$

with:

c_0 initial concentration of the substance (g/m³);
c_s surface concentration of the substance (g/m³);
erf error function.

Eq. (14.7-8) is at the root of test methods to measure chloride diffusion and of models to predict the penetration of chlorides into concrete with time (see section 30.6.3).

14.8 Properties describing / defining durability

14.8.1 General

The durability of structural concrete is related to its resistance to physical, chemical, and biological attack processes associated either with aggressive substances already present in the concrete (e.g., alkali aggregate reactions and delayed Ettringite formation) or with the penetration into the concrete of CO_2, chlorides, sulfates, and other aggressive substances. Deterioration processes affecting concrete structures are usually considered on the following basis:

– physical deterioration and damage processes in concrete.
– chemical deterioration processes in concrete.

The above listed deterioration mechanisms lead to the following main consequences:

reduction in service life leading to greater consumption of concrete and other resources due to need to undertake remedial interventions; and

increase in direct and consequential costs, plus social and environmental impacts arising from remedial works (eg. due to traffic congestion, greater natural resource use, higher CO_2 emissions, social disruption).

In the case of "external" deterioration processes, freeze thaw degradation occurs when the concrete has a critical degree of water saturation and is subjected to a suitably low external temperature. In this case the durability depends on the proportion of the critical controlling parameters (chemical compositions, air-voids network, …), as well as the external exposure (e.g. temperature and water availability).

There are also the so-called "internal" deterioration processes, which are associated with the intrinsic raw materials (e.g. alkali-aggregate reaction) or arise from the hydration processes (e.g. delayed ettringite formation)

Concrete mix design, production, placing, finishing and curing all have a role to play in determining the characteristics of the in-place concrete, and hence, its durability.

- biological deterioration processes in concrete; and
- corrosion of steel reinforcement / prestressing components.

The different deterioration mechanisms are considered in detail in the following subsections.

Concrete interacts with the surrounding environment and changes occur with time. Whether this aging is deleterious or not will depend on the exposure aggressivity, which is specific to each type of concrete. Exposure aggressivity classification is usually ranked on the basis of the risk of concrete deterioration.

The exposure environments can be different on each face of a concrete member, and consequently, the requirements relating to concrete, cover and cracking could potentially be different on each of these faces.

The potential deterioration processes experienced by the concrete structure can be classified in different manners. Table 14.8-1 presents one classification.

Table 14.8-1: Examples of concrete deterioration mechanisms.

Deterioration processes		Caused by
Physical damage		Temperature effects (including freeze/thaw) Early thermal cracking Abrasion and erosion Plastic settlement and shrinkage Water and water vapour transport, drying shrinkage, autogenous shrinkage Physical salt crystallization and scaling Weathering, discoloration and soiling of surfaces
Chemical deterioration	External exposure	Acid attack, sulfates (soil/groundwater), nitrates, soft water (leaching) Thaumasite formation Moisture and its transport mechanism
	Internal effects	Alkali-aggregate reaction (AAR) Delayed ettringite formation (DEF) Deleterious aggregates (e.g. Pyrites)
Biological deterioration		Biological growth on concrete Bacterial attacks through the production of organic or mineral acids that could attack the cement paste and/or the exposed steel
Corrosion (physico-chemical process)		Chloride-induced steel reinforcement corrosion due to chlorides present in the mix or chloride penetration. Carbonation-induced steel reinforcement corrosion from CO_2 (natural air or exhausts) diffusion in the porous network

According to chapter XXX a typical value of an action is the value of an action used for the verification of a limit state.

The definition of initiation and propagation periods is given in section 30.6.2.

14.8.2 Carbonation induced corrosion

14.8.2.1 General

In the present section all input parameters to the deterioration models related to concrete materials of both phases, the initiation period and the corrosion propagation period, are consecutively treated for the several Level of Approximation approaches (section 14.3).

All input parameters related to concrete materials of both phases, the initiation period and the propagation period, are given as characteristic or mean values; the design values are described in section 30.6.2.

14.8.2.2 Physical models for carbonation-induced corrosion

Eq. (14.8-1) was developed in the European research project DuraCrete and slightly revised in the research project DARTS.

$$x_c(t) = \sqrt{2 \cdot k_e \cdot k_c \cdot (k_t \cdot R_{ACC,0}^{-1} + \varepsilon_{t,d}) \cdot C_{s,d}} \cdot W(t) \cdot \sqrt{t} \qquad (14.8\text{-}1)$$

k_e environmental function [-];
k_c execution transfer parameter [-];
k_t is a regression parameter [-];
$R^{-1}_{ACC,0}$ is the inverse carbonation resistance of concrete [$mm^2.year^{-1}.kg^{-1}.m^3$];
$\varepsilon_{t,d}$ error term for inaccuracies which occur conditionally when using the accelerated test method in [$mm^2.year^{-1}.kg^{-1}.m^3$];
$C_{s,d}$ is the CO_2 concentration of the ambient air [kg/m^3];
$W(t)$ weather function [-].

The physical models are further described in section 30.6.2, the input parameters will be given in the following parts.

The effect of parameters of the mix design on carbonation have been reviewed/studied by RILEM TC 281-CCC WG1&2 and 5 [14-81].

Designers should consider the most severe exposure conditions including potential future usage of a structure to determine wetting/drying events frequency that would make the exposure more severe than expected.

Consideration should also be given to maintenance works, their effects on the exposure, and the provisions that should be included to avoid unforeseen corrosion damage.

The influencing factors that can lead to considerable corrosion damage in carbonated concrete have been reviewed/studied by RILEM TC 281-CCC WG5 (2020) [14-82].

In dry/semi dry conditions (favourable for carbonation initiation progress but leading to very low propagation process), most codes consider implicitly the propagation period by setting requirements for both the cover depth and the concrete composition.

The simplest method to measure the carbonation front advance is the pH colour change.

The carbonation rate is also named V_{CO2}, A, k_{NAC} or k_N depending on the standards and literature.

14.8.2.3 Concrete composition prescriptive rules

14.8.2.3.1 Concrete composition prescriptive rules – initiation period

Current standards give limiting values and rules for the composition and the constituent materials of concrete by some or all of following:

- maximum water/binder ratio,
- minimum binder content,
- maximum permissible fines content,
- permitted binder type,
- minimum compressive strength.

The rules are in dependence on the exposure conditions for the structures. These parameters describe the resistance of the concrete and the concrete structural members to possible depassivation of the rebars. They vary depending on local tradition and experience feedback.

It is assumed that when fulfilling such limiting values, combined with some construction parameters (e.g. cover thickness) (see section 30.6.2.3) the concrete structure is functional without unpredicted maintenance works and safe during the design service life period considered.

14.8.2.3.2 Concrete composition prescriptive rules -propagation period

Present codes do not consider explicitly the corrosion propagation.

14.8.2.4 Performance indicators

14.8.2.4.1 Performance indicators considering the initiation period

The relevant material property needed as input parameter to describe the carbonation progress over time with simplified engineering models is the carbonation rate k provided in some testing standards.

The carbonation depth X_c is determined by splitting the specimen into two parts, spraying thymolphthalein or phenolphthalein solution on the freshly split faces and reading the length of the colorless part

The reduction of the cross-sectional area of a reinforcing bar can be determined in different ways: s Rilem recommendation (2004) [14-83], gravimetric method, calliper and electrochemical measurements… more details are given in *fib* bulletin XX [14-67].

The concrete resistivity, and consequently the corrosion rate (see Eq. 30.6-8) can be obtained as indicated in Rilem Recommendation (2000) [14-84] and in ASTM C1876 -19 [14-85]. Both are based on the use of a probe with four electrodes equally separated or by a single electrode that are made to enter in contact to the concrete surface. It gives the "surface resistivity".

The corrosion potential can be measured following Rilem Recommendation (2003) [14-86] and ASTM C 876-15 [14-87]. It consists of measuring the difference in electrical potential between the bar and a reference electrode placed on the concrete surface. This indicator, although providing information on the probability of corrosion being active, does not give indication about the corrosion rate.

The carbonation rate includes both material property and environmental load. The carbonation rate (or alternatively the carbonation resistance) is determined under natural or accelerated test conditions. The carbonation rates derived from theses test methods are generally not identical since the k will depend on the CO2 concentration. Thus, it is highly important to describe the test conditions under which the carbonation test is carried out. Further details are given in *fib* Bulletin XX [14-67].

The carbonation rate (or alternatively the carbonation resistance) is determined under natural or accelerated test conditions.

- Natural carbonation is usually referred to tests under natural CO_2 concentration, 400ppm or slightly higher, but usually not exceeding 500-600ppm.

- Accelerated carbonation is usually referred to tests where the CO_2 concentration is increased to speed up the carbonation depth progress.

14.8.2.4.2 Performance indicators considering the propagation period

The relevant material property used to design for the propagation period over time with simplified engineering models is the corrosion rate which measures the velocity of the reduction of the cross-sectional area of a reinforcing bar along time. In cases where the corrosion rate is difficult to obtain, an alternative performance indicator can be the concrete resistivity.

14.8.2.5 **Semi probabilistic approach**

14.8.2.5.1 Input parameters for the initiation period

The material input parameter relevant for the carbonation-induced corrosion model is described in 14.8.2.4.1. Other input parameters are described in 30.6.2.2. The characteristic value of the parameters is assessed based on the obtained variability in the test data.

14.8.2.5.2 Input parameters for the propagation period

On how to measure the corrosion rate see section 14.8.2.4.2. The characteristic value of the parameters is assessed based on the obtained variability in the observations (see Bulletin XX [14-67]).

14.8.2.6 **Full probabilistic approach**

14.8.2.6.1 Input parameters for the initiation period

In the full probabilistic approach, all uncertain material parameters are described by their distribution function and distribution parameters (*fib* Bulletin XX [14-67]).

The carbonation rate k (also named V_{CO2}, A, k_{NAC} or k_N depending on the standards and literature) and the inverse carbonation resistance $R_{ACC,0}^{-1}$ can be described with a normal distribution (Gehlen [14-88]).

Refer to 14.8.2.4.1. The statistical distribution of the parameters is assessed based on the obtained variability in the tests.

The standard deviation of carbonation rate applied in durability design is a consequence of the variability of raw materials, mix proportions, production conditions, execution, climatic conditions in the concrete production, climatic conditions, binder type, execution ….

14.8.2.6.2 Input parameters for the propagation period

The measurement method of the corrosion rate can be found in 14.8.2.4.2.

The corrosion rate V_{corr} is highly dependent on the particular climatic conditions at meso and macro level and it varies from negligible level in dry conditions to high values when the concrete is wet.

However, the accumulated corrosion P_{corr} (the total corrosion depth attack after certain period) is reasonably constant if averaged annually due to the repetition of the weather in the consecutive seasons. This spatial variability has not been characterized yet.

On how to measure the corrosion rate see section 14.8.2.4.2. The characteristic value of the parameters is assessed based on the obtained variability in the tests.

The critical chloride content is dependent both on material properties and the environmental factors as well as the steel-concrete interface condition, which in turn influences the corrosion potential of the steel bar. The critical chloride content and some other not material related input parameters for the modelling of the chloride ingress in concrete are also presented in this chapter.

14.8.2.7 Avoidance of carbonation induced corrosion

Refer to section 30.6.2.

14.8.3 Chloride ingress

14.8.3.1 General

The present chapter deals with all material parameters of concrete which are relevant for assessment of the durability of concrete structures in both initiation and propagation of the reinforcement corrosion due to the chloride exposure. This is undertaken considering the Level of Approximation approaches for durability specification which are defined in section 14.3.

The material related descriptions and specifications in the following chapters can be used in conjunction with the corresponding specifications in section 30.6.3 in order to assess the durability of reinforced concrete structures with regard to chloride-induced corrosion.

14.8.3.2 Physical models for chloride-induced corrosion

In the European joint research projects DuraCrete and DARTS a model for the prediction of time and depth dependent chloride content has been developed and validated. The variation in chloride content of concrete exposed to chloride ingress is given as follows:

$$C(x,t) = C_0 + \left(C_{S,\Delta x} - C_0\right) \cdot erfc \frac{x - \Delta x}{2 \cdot \sqrt{D_{app}(t) \cdot t}} \qquad (14.8\text{-}2$$

$C(x, t)$ is the chloride concentration at depth x at time t [% weight of binder];

C_0 is the initial chloride content of the concrete [% weight of binder];

$C_{s,\Delta x}$	is the chloride concentration at depth Δx (equivalent surface chloride concentration) [% weight of binder].
Δx	skin zone; is the depth range in which the chloride penetration behaviour may deviate from the behaviour according to Fick's law of diffusion owing to intermittent chloride attack [m];
x	is the depth with a corresponding chloride content $C(x, t)$ [m];
t	is the concrete exposure duration [s];
$erfc$	is the complementary Gauss error function (= 1-erf);
$D_{app}(t)$	is the apparent chloride diffusion coefficient of the concrete [m²/s]. For each t, the variable is assumed constant over entire exposure period.

The physical models are further described in chapter section 30.6.3, the input parameters will be given in the following parts.

14.8.3.3 Prescriptive rules on concrete composition

14.8.3.3.1 Initiation of reinforcement corrosion

Current standards give limiting values and rules for the composition and the constituent materials of concrete by some or all following:

- maximum water/binder ratio,
- minimum binder content,
- minimum/maximum binder content,
- permitted binder type,
- minimum compressive strength,
- maximum initial chloride content of the concrete.

The rules are in dependence on the exposure conditions for the structures. These parameters describe the resistance of the concrete and the concrete structural members to possible depassivation of the rebars (and maybe a minor corrosion propagation) due to chloride ingress from the environment. They vary depending on local tradition and experience feedback. It is assumed that when fulfilling such limiting values, combined with some construction parameters (see section

There is no approved performance test for the critical chloride content of concrete to consider this as an indicator. Furthermore, critical chloride content is not only a material property and depends on the environmental factors such as temperature and humidity, and on the steel/concrete interface condition which in turn, influences the corrosion potential.

The corrosion rate is the visual or gravimetric homogeneous corrosion attack measured on site divided by the duration (years) of the corrosion period. In the case of measuring pit depth with the caliper, this depth is also divided by the duration for its pit are determination. The homogenous value of the corrosion rate so measured can be multiplied by the pitting factor α=10 in order to obtain an approximation the maximum pit depth.

30.6.3.3) the concrete structure is functional without unpredicted maintenance works and safe during the design service life period considered.

14.8.3.3.2 Propagation of reinforcement corrosion

Refer to section 14.8.2.3.1.

14.8.3.4 **Performance indicators**

14.8.3.4.1 Performance indicators considering the initiation period

Some material properties can be considered as predominant parameters for concrete resistance (ageing factor, critical chloride content, apparent chloride diffusion coefficient…) to chloride ingress which can be used as a direct or an indirect indicator to assess the concrete performance. These parameters can be measured by defined test methods. The values of the parameters can be ranked to form resistance classes. Some examples of such indicators are shown in *fib* Bulletin XX [14-67] with the corresponding test methods and suggested resistance classes.

14.8.3.4.2 Performance indicators considering the propagation period

See section 14.8.2.4.2.

14.8.3.5 **Partial safety factor format (semi-probabilistic approach)**

14.8.3.5.1 Partial safety factor format – initiation period

The characteristic value of the parameters is assessed based on the obtained variability in the tests.

Other alternative models to describe the chloride transport in concrete may also be used, provided they are sufficiently validated against in-field performance. In the fully probabilistic durability design each model variable is considered as a statistically distributed quantity. The current chapter deals with the description of the model parameters. For the statistical quantification of the parameters please see *fib* Bulletin 34 [14-10], *fib* Bulletin 76 [14-89] and Concrete Society Technical Report 61 [14-90] comprehensively.

This ageing effect is contributed by several phenomena as pore blocking effect due to chloride ingress etc. (DARTS 2004 [14-91]) but it is still not comprehended thoroughly.

The general formulation of the apparent chloride diffusion coefficient can be depicted as follows:

$$D_{app}(t) = D_{app}(t_0)\left(\frac{t_0}{t}\right)^{\alpha} \qquad (14.8\text{-}3)$$

where:

$D_{app}(t)$ is the apparent chloride diffusion coefficient [m²/s] at time t [s];
$D_{app}(t_0)$ is the apparent chloride diffusion coefficient [m²/s] at the reference point of time t_0 [s];
α is the ageing exponent [–].

In the case that the profile shows the maximum chloride content inside of the concrete, the fitting should be made on the internal part of the decreasing profile by shifting the x

The same model and the same laboratory tests as in the full probabilistic format (section 14.8.3.6.1) are to be applied.

14.8.3.5.2 Partial safety factor format - Propagation period

The measurement of the corrosion rate is depicted in section 14.8.3.4.2.

14.8.3.6 **Probabilistic safety format**

14.8.3.6.1 Probabilistic safety format – Initiation period

The mathematical model for the transport of chloride in concrete is shown in Eq. (30.6-3). The model parameter related to the material properties of concrete is the so-called apparent chloride diffusion coefficient $D_{app}(t)$ which is described hereinafter.

The other input parameters required to undertake modelling are detailed partly in this section as well even if they are not related to material properties.

14.8.3.6.1.1 Apparent chloride diffusion coefficient $D_{app}(t)$

The apparent chloride diffusion coefficient $D_{app}(t)$ which represents the material resistance as a constant over a defined time period, is subject to considerable scatter and reduces with increasing exposure time. This implies that generally $D_{app}(t_2) < D_{app}(t_1)$ with $t_2 > t_1$.

axis to be zero on the position of the maximum. This will be the "substitute surface chloride concentration" and not the extrapolated value.

For modelling of the chloride ingress, the Δx is the distance from the surface to the maximum of chloride concentration. This distance must be added to the value obtained from the maximum, assuming that its position is not time-evolving after a certain exposure period [x]. The other frequently used method to obtain the Apparent diffusion coefficient is to extrapolate the chloride concentration to the concrete surface, which gives a "fictious" chloride surface concentration. The ageing exponent α can take values between 0 and 1.0 (in theory). The higher α the stronger is the decrease of resistance of the concrete against chloride ingress over time. It is recommended to limit the ageing exponent to a value between 0.20 and 0.80.

14.8.3.6.1.2 Critical chloride content

The critical chloride content does not depend only on the concrete "quality" but also on the environmental conditions and of the corrosion potential of the steel bar which reflects the impact of the steel/concrete interface conditions.

The threshold concentration is usually expressed as acid soluble (total) chloride content by mass of cement, or more practically, by mass of concrete. There are however some standards where the threshold is expressed as water soluble (free) chloride content in concentration or percentage in the mixing water or by mass of cement or concrete.

The threshold value of acid soluble chloride is considered between 0.4 and 0.6% by cement weight in numerous Codes/Standards. However, in submerged conditions values from 1 to 1.5% in cement content may not produce depassivation.

If limited in the mixing water, the chloride threshold contents vary from 0.06% by cement weight in prestressed elements to 0.25- 0.30 % in reinforced concrete exposed to chlorides.

The critical chloride content is considered here as the overall chloride content which causes depassivation of the surface of the rebars and initiation of corrosion of the reinforcement (provided the other corrosion conditions are satisfied).

14.8.3.6.2 Probabilistic safety format – Propagation period

The measurement method of the corrosion rate can be found in section 14.8.3.4.2.

14.8.3.7 Avoidance of chloride ingress

See section 30.6.3.7.

14.8.4 Freeze-thaw degradation – Internal structural damage and scaling

14.8.4.1 General

Freeze-thaw durability for structural concrete is defined by its capacity to resist repetitive freeze-thaw cycles throughout a defined service life with acceptable degree of deterioration due to internal cracking or scaling.

Besides moisture content, environmental factors such as the minimum freezing temperature, the rate of freezing, number of freeze-thaw cycles and the cation types in the de-icing agent are important.

The degree and type of damage caused by freeze-thaw attack depends on:

– material properties determined by concrete composition including porosity, pores size distribution and strength;

– the actual service environment, i.e. the conditions at the concrete surface and their variation with time covering relative humidity, surface contact with water and temperature;

– the degree of saturation which varies with time and location in the concrete due to moisture transport by capillary suction, water vapour diffusion together with capillary condensation and water vapour sorption

The limit state functions presented in 30.6.4.2.3 (internal structural damage) and 30.6.4.3.3 (scaling) are conceptual, therefore no modelling input is presented for both the partial safety format and the full probabilistic format.

14.8.4.2 Prescriptive rules on concrete composition

Properties related to concrete composition prescriptive rules parameters that are specific to freeze-thaw action are:

– the environmental exposure conditions (wetness/saturation, chlorides, frost etc.) on which to base the design on (see section 30.6.4).

– the material requirements (air-void requirements, w/b-ratio, binder type and minimum content, composition, strength, aggregate types, etc.) for the actual environmental exposure conditions

Examples of the concrete composition prescriptive approach are given in the EN 206 [14-8], ACI 201.2R-16 [14-92], ACI 318-14 [14-93], and also in many national application documents.

Freeze-thaw resistant aggregates: concrete consists of 60% to 75% (in volume) aggregate. Therefore, aggregate freeze-thaw performance should be taken into consideration;

Air entrainment: the most important factor affecting freeze-thaw performance is the quality of the air-void system (volume, size, and spacing of the air-voids). The air-void system parameters typically used to assess the quality of an air-void system include the air

content (volume %), spacing factor (mm), and specific surface (mm-1). These are determined in hardened concrete.

Field experience/research have shown that internal damage is almost non-existent if the spacing of air-voids is low enough (< 0.20 – 0.25 mm).

Detailed information is given in *fib* Bulletin XX [14-67]. The intention is to simulate more severe freezing and thawing environmental load by accelerated testing ("torture performance tests"). Due to the accelerate nature of these tests, correlation to the actual in situ deterioration is not well known.

When two aggressive components are present simultaneously; for example, in H_2SO_4 or solutions containing Mg and SO_4 ions, interactions arise and one of the processes a) or b) is dominant. There are standards presenting limiting values for many of the named chemical substances, for example EN 206 [14-8].

– the test methods (air-voids, freeze-thaw performance tests, spacing factor of air voids, porosity, strength etc.) to either confirm compliance of the end product or determine performance characteristics of the mix design.

14.8.4.3 Performance indicators

In freeze-thaw test methods samples are subjected to alternating freezing and thawing. Salt solution is used if freeze-thaw durability in the presence of deicing salt or sea water is of interest.

14.8.5 Chemical attack

14.8.5.1 General

This chapter describes the input parameters pertaining to the models on chemical attack described in 30.6.5.

Aggressive substances resulting in a chemical attack on concrete can be divided into two groups:

a) Components resulting in a loss of strength or mass: the most important among these are acids, solutions containing aggressive carbon dioxide, and solutions containing ammonium or magnesium ions;

b) Components resulting in expansions: the most important of these are sulfates (excess sulfate in the concrete or sulfate ions in solution), free lime or periclase in the cement and reactive aggregates (alkali aggregate reaction).

Description of mechanisms and available physical models are detailed in *fib* Bulletin XX [14-67].

14.8.5.2 Description of inputs for available physical models

So far, no prediction formula for the constant k_c (given in *fib* Bulletin XX [14-67]) has been given. This constant should be determined by appropriate experiments (Beddoe et al [14-94], Pavlik [14-95]).

The input parameters introduced in *fib* Bulletin XX [14-67] can be determined by appropriate experiments.

14.8.5.2.1 Sulfate and magnesium ions attack

No physical approved models are presently available.

14.8.5.2.2 Inorganic and organic acid attack

The deterioration depth induced by acid attack may be estimated from the general comprehensive formula:

$$d = k_c \cdot c^{\alpha} \cdot t^{\beta} \qquad (14.8\text{-}4)$$

where

d is the depth of deteriorated cementitious material in mm

k_c is a coefficient that translates the effects of the properties of concrete, and the properties of the acid-affected layer formed at the surface. It is to be noted that the porosity, the chemical composition of the matrix (cement type, addition), and the nature of aggregates can all affect the value of k_c.

c is the concentration of acid in mol/L

a and b are positive real numbers. They depend on the type of acid, its chemical properties (dissociation constant) and the physico-chemical properties of the salts (solubility, physical properties). $0 < a < 1$, $0 < b < 1$

14.8.5.2.3 Microbial induced deterioration in sewage and wastewater

There are no finalized integrated models for the prediction of the deterioration kinetics of concrete in sewers, that consider all the influencing parameters. Some empirical models were established, for example, based on Pomeroy model.

14.8.5.2.4 Aggressive CO2 and Ammonium Attack

For estimations of k_c (in *fib* Bulletin XX [14-67]) by laboratory experiments, see for example Yu and Zhang (2017). Most laboratory work has been carried out with c = 6 M. In this case, using the values obtained from Yu and Zhang, k_d can be taken equal to 1.

Chloride diffusion coefficient can be used as a concrete performance property for soft water attack and leaching since it is highly dependent on the pore network tortuosity and connectivity.

Several models are used to determine the effective diffusion coefficient of a species in cementitious materials (i.e. the power law, the exponential law). Here the experimentally derived model from Tognazzi [14-96] is given as applied in De Larrard et al. [14-97]:

$$D_e(\phi) = D_0 e^{k\phi} \qquad (14.8\text{-}6)$$

where:

- $D_e(\phi)$ is the effective diffusion coefficient in the cementitious material in m^2/s;
- D_0 is the effective diffusion coefficient of Ca in water in m^2/s;
- ϕ is the porosity [-];
- k is a regression parameter dependent on the material (for cement pastes, k=9.95).

Chloride concentration at reinforcement depth is modelled using the diffusion property of the concrete, as for marine exposure. (section 14.8.3)

The deteriorated depth *d* that describes the degree of degradation of concrete caused by ammonium attack is proportional to the square root of time as follows:

$$d = k_c \cdot k_d \cdot t^{0.5} \qquad (14.8\text{-}5)$$

where

- d is the depth of deteriorated cementitious material in mm
- k_c is a coefficient that translates the effect the properties of concrete driven by its porosity, the chemical composition of the matrix (cement type, addition), and the nature of aggregates,
- k_d is a coefficient that translates the concentration of ammonium nitrate of the attacking solution.

14.8.5.2.5 Soft water attack and leaching

Currently, no physical approved models are presently available. *fib* Bulletin XX [14-67] provides a detailed description of a simplified equation to assess the leaching progress.

14.8.5.2.6 Attack by chlorides in groundwater and de-icing salts

Chloride concentration at reinforcement depth is modelled using the diffusion property of the concrete, as for marine exposure. (Section 14.8.3).

Chloride in groundwater or deicing salts, in contact with a concrete surface will diffuse progressively into the concrete, in the same way as with other forms of chloride exposure. However, for these situations there has been relatively little discussion or consensus concerning appropriate values for modelling the diffusion process.

The limiting values of the prescriptions can be found in currents standards on concrete. They vary depending on local tradition and experience.

It is assumed that fulfilling such limiting values, the concrete structure is functional within any unpredicted maintenance works and safe in the life period considered.

While not shown explicitly, the specifications provided in some relevant international standards as developed within *fib* Bulletin XX [14-67] cover situations where aggressive CO2 may be present.

An increased cement content provides more calcium to buffer the pore solution. Therefore, with respect to the effect of leaching on concrete integrity (strength, porosity, creep, corrosion, durability) the progress of leaching depth is slowed down with increasing amount of cement.

14.8.5.2.7 Seawater attack to plain concrete

No physical approved model presently available

14.8.5.2.8 Food production and processing

No physical approved model presently available

14.8.5.3 **Prescriptive rules on concrete composition**

The prescriptive rules related to material properties are not clearly available and these are generally covered in prescriptive requirements on concrete mix design (30.6.5).

In the case of leaching, concrete composition prescriptive rules can be provided by considering the following parameters:

- the amount of soluble species i.e. calcium per unit area
- the water to cementitious ratio
- the type of binder, and the amount of clinker
- the cement content
- the compressive strength

14.8.5.4 **Performance indicators**

14.8.5.4.1 Sulfate and magnesium ions attack

For specific cases of concrete subject to highly aggressive environment. It is generally advisable to carry out performance tests with the binder combinations to be used in the project with respect to their sulfate resistance. Although there are a number of variants of the test in terms of the procedure followed in different countries, and the aspect of testing itself is still under discussion.

While the concentration of hydrogen ions is defined by the measure “pH”, the effect is not uniquely linked to this property, even if other prevailing conditions such as the renewal rate of acid at the surface are unchanged. With “weak” acids, having a low level of dissociation, there is a reserve of hydrogen, which is freed when those initially present are neutralized by the cement.

To date, there is no international consensus on relevant applicable standardized performance test.

14.8.5.4.2 Inorganic and organic acid attack

To date, there is no international consensus on relevant applicable standardized performance test.

14.8.5.4.3 Soft water attack and Leaching

As leaching is a coupled dissolution diffusion process, performance indicators are either indirect by determining the pore structure or transport properties that can be correlated with the diffusion of the species:

- Porosity (and tortuosity)
- Hydraulic conductivity
- Water permeability

Direct indicators are dependent on the model applied and can be for example:

- Leaching rate of Ca

Diffusion coefficient

14.8.6 Internal Expansion Mechanisms

Adverse reactions leading to volumetric expansion of the concrete can occur from sources derived from the constituents of concrete. The description of the process and models for alkali aggregate reaction and delayed ettringite formation are addressed in 30.6.6.

Preventive measures have been developed in most countries where ASR is a problem, and these seem to have been effective. (See for instance Institution of Structural Engineers 1992 [14-98], LCPC Recommendations 1994 [14-99], Concrete Society 1999 [14-100], French standard 2014 [14-101], RILEM 2015 [14-102], CSA A23.1 2019 [14-103].

Some heuristic approaches, however, tend to relate the expansion potential (in reference conditions) to quantitative features related to the constituents. Moreover, some models relate the ASR development (actual expansion, degree of advancement) to degradation of the design parameters (Young's modulus, tensile and compressive strength), especially for reassessment of affected structures. Models describing the expansion kinetics are also available. Such models generally need a certain degree of experimental calibration.

Table 14.8-2: ASR influencing parameters and mitigation measures.

Influencing parameters	Mitigation measures
Equivalent alkalis content	Limit the equivalent alkali content of the cement and concrete Use supplementary cementitious materials
Aggregates reactivity	Avoid the use of Alkali-reactive aggregates
Moisture content	Prevent contact with water and reduce concrete moisture content
Temperature (impact on reaction kinetics)	None

Measures effective in the prevention of damage due to ASR, such as controlling the alkali content of the concrete or using SCM, are not generally effective in controlling

14.8.6.1 Alkali aggregate reaction

14.8.6.1.1 General

There are no approved models that link the expansion produced both to the reactive constituents and to the possible degradation in design material characteristics (30.6.6).

There are three necessary requirements for ASR to occur: an alkali source, reactive aggregate, and sufficient moisture to sustain the reaction and induce expansion. Mitigation and preventive measures must address one or more of these factors.

14.8.6.1.2 Prescriptive rules on concrete composition

The measures summarized in Table 14.8-2 have been found to be effective in minimising or reducing the risk of ASR. These are applicable for concrete composition prescriptive approach and may correspond to precise locally mandatory requirements and Standards.

14.8.6.1.3 Alkali Carbonate reaction

There are no known models for damage produced by alkali carbonate reaction (30.6.6).

expansion due to ACR. The use of alkali-carbonate reactive rocks should, therefore, be avoided in concrete construction.

Some heuristic approaches however tend to relate the expansion potential (in reference conditions) to quantitative features related to the constituents. Models are being developed to quantify the effect of the temperature history at early age on the DEF-induced expansion potential.

The measures summarized in Table 14.8-3 have been found to be effective in minimising or reducing the risk of DEF. These are applicable for a prescriptive approach and may correspond to precise locally mandatory sets of requirements.

Table 14.8-3:DEF influencing parameters and mitigation measures.

Influencing parameters	Mitigation measures
Temperature during hardening process	Select low heat of hydration concrete mix design Reduce concrete temperature at placing Control heat-treatment for heat-cured precast elements Avoid concreting during high temperatures Consider the thermal conductivity of formworks Consider sequencing to maximise heat loss – e.g. thinner concrete layers cast at suitable time intervals Provide cooling systems within the mass concrete elements
Cement composition	Use low C_3A and SO_3 content Portland cement Use supplementary cementitious materials
Alkali content	Limit the equivalent alkali content of the concrete Use supplementary cementitious materials
Moisture content	Prevent contact with water and reduce concrete moisture content

14.8.6.2 **Delayed ettringite formation**

14.8.6.2.1 General

There are no approved models that predict the expansion produced by DEF and the possible consequent degradation of the engineering properties of the concrete (30.6.6).

14.8.6.2.2 Prescriptive rules on concrete composition

In mass concrete structures, or when concrete members are subjected to heat curing (or steam curing), the internal temperatures during hydration can be sufficiently high (typically above 65 to 70°C) to lead to the formation of unstable ettringite or to the incorporation of the sulfates in other cement phases. Consequently, the sulfates and aluminates are kept latent within the structure of the CSH. With the availability of moisture in hardened concrete, these phases may lead to the formation of secondary ettringite within the CSH or at the interface between paste and aggregates, leading to the generation of expansive pressures, causing cracking. Once cracks form, the ettringite tends to deposit in the cracks.

14.8.7 Leaching

[Not used at this time, to be updated in future edition]

14.8.8 Abrasion, erosion and cavitation

14.8.8.1 General

Abrasion, erosion, and cavitation represent a special durability-related exposure for concrete surfaces. In certain situations, exposure of these phenomena may govern the specification of the concrete and dictate lifespan. The terms abrasion and erosion are often used interchangeably in the literature, for the purpose of this report, these are specifically defined as follows:

- Abrasion is typically understood as the deterioration due to wear on dry surfaces.
- Erosion occurs when the concrete surface is in contact with hydrodynamic flows (and solid particles) as well as with wind-borne particles.
- Cavitation is caused by pressure changes in high velocity fluid flows.

14.8.8.2 Abrasion

Abrasion wear is characterised by sliding/scraping, rolling, and percussive impact forces that lead to scratches, hutches, cavities or cracks in the concrete surface.

The abrasion resistance of concrete is strongly affected by the hardness of the surface zone. Therefore, the compressive strength (which can be related to the hardness of aggregate and cement matrix) of concrete is often the main resistance parameter if mainly sliding/scraping and rolling dominates the exposure.

The resistance of concrete against abrasion, erosion and cavitation is difficult to test and to estimate because the type of damage varies depending on the special exposure on the structure.

If there is pure impact exposure, the concrete must have a certain ductility in the surface zone. Also, the bond behaviour between the aggregate and the cement matrix is important.

The test-related determination of the resistance of concrete to abrasion depends on the exact wear conditions (see also tribological analyses). For example, for the different exposures exist rubbing/sliding tests, sandblasting tests, and pure impact tests are all available.

In the case of impact loads caused by bulk freight or liquids (with solids), it has to be distinguished between predominantly tangential and normal loads. A predominant tangential load leads to a fast wear of the cement matrix. Therefore, mainly the resistance (e.g. hardness) of the aggregate is relevant. If the normal load is predominant the behavior of the cement matrix and the packing density of the aggregate are important.

When present erosion damages to concrete hydraulic structures range from a few centimetres to several decimetres, (Jacobs et al. [14-104], Spörel [14-105]). Rates of erosion depth have been recorded in excess of 10 mm/year in hydraulic structures incorporating concretes regarded as having good erosion resistance, thus demonstrating the importance of this phenomenon in certain exposure conditions (Cunningham et al. [14-106]).

Material properties are of minor importance for cavitation, as even strongest materials cannot resist a cavitation exposure sufficiently. Therefore, construction principles must be considered in order to avoid cavitation damage. Approved laboratory testing procedures are not available.

14.8.8.3 Erosion

Erosion results from water-borne sand, gravel, rocks, and other debris, which circulate over concrete surfaces of hydraulic structures. This exposure leads to a progressive destruction of concrete by an alternating removal of cement matrix and aggregate.

The rate of erosion wear depends on the kind of exposure and on the durability of concrete. The main factors regarding the exposure are quantity, hardness, shape and size of the particles, and their velocity. The hardness, shape, size and packing density of the aggregates, the bond between the aggregates and cement matrix, and the hardness and elasticity of the cement matrix influence the concrete resistance. To investigate the abrasion resistance of concrete an adequate test method is required.

14.8.8.4 Cavitation

Cavitation is the formation of bubbles in a liquid caused by a local drop of pressure to a value that causes a vaporization of the water. The collapse of these vapor bubbles results in a high immediate pressure impacting on the surface resulting in pitting.

At hydraulic structures in open channels, cavitation damages on concrete surfaces can occur at high flow velocities of water and irregularities at the concrete surface (ACI 201.1R [14-107]).

14.9 Evaluation of equivalent performance for non-conventional concrete

14.9.1 Mechanical behaviour

14.9.1.1 Performance under static loading

All the mechanical properties should be measured by means of normalized tests in order to assess the following relations:

- Relation between mean compressive strength and characteristic one (update of D_f in Eq. 14.6-2)
- Relation between tensile strength and characteristic strength (update of Eq. 14.6-3)
- Relations between lower and upper bound values of the characteristic tensile strength and the tensile strength update of Eq. 14.6-4 and 14.6-5)
- Relation giving the fracture energy (Eq. 14.6-9)
- Relation giving the modulus of elasticity and the reduced modulus (Eqs. 14.6-20 and 14.6-24)
- Stress strain relationship (Eq. 14.6-26)
- Relation giving the strength development with time (Eq. 14.6-45) and coefficient s_C in Eq. 14.6-46
- Relation giving the development of modulus of elasticity (Eq. 14.6-48)
- Relations giving the development of creep (Eqs. 14.6-60, 14.6-61 and 14.6-63 to 14.6-66d)
- Relations giving the development of shrinkage (Eqs. 14.6-73, 14.6-74 and 14.6-75 to 14.6-78)

14.9.1.2 Performance under non-static loading

Under a non-static loading, the properties presented in 14.6.2 should be evaluated.

14.9.1.3 Performance under extreme thermal conditions

Under extreme thermal conditions, the properties presented in 14.6.3 should be evaluated.

14.9.2 Transport of liquids and gases in hardened concrete

[Not used at this time, to be updated in future edition]

14.9.3 Durability

The prescriptive approach could not be applied. The rules by tests and measurements (performance indicators) presented in 14.8 are applied.

14.10 Concretes with recycled concrete aggregates

14.10.1 Definitions

τ_{RCA} is the substitution rate of recycled coarse concrete aggregates (d > 4mm) = quantity of recycled coarse aggregates/total quantity of coarse aggregates

τ_{TRA} is the total substitution rate of recycled aggregates = quantity of fine+coarse recycled aggregates/total quantity of aggregates

14.10.2 Principles

For reinforced concrete, when only recycled coarse aggregates are used, in order to facilitate recycling, when $\tau_{RCA} \le 0,25$ there is no change in the mechanical and structural properties. When $0,25 < \tau_{RCA} \le 1$ properties could be measured or relations given below could be used. When fine recycled aggregates are used the properties should be measured.

For prestressed concrete, when only recycled coarse aggregates are used and $0 < \tau_{RCA} \le 0,4$ properties could be measured or relations given below could be used. When fine recycled aggregates are used or $\tau_{RCA} > 0,4$ the properties should be measured.

Proposal of the guidelines of the French project RECYBETON [14-108].

In the Spanish code: for total replacement reduction of 5-15% of the density

No discernible effect of full RCA incorporation on the standard deviation" (Pacheco et al. 2019-1 [14-109]).

Full RCA incorporations had no effect on the variability of the splitting tensile strength" [Pacheco et al. 2019-2 [14-110]].

The equations and factors given are identical to those given in FprEN 1992-1-1:2023 [14-%%%].

Relations adapted from the project RECYBETON, papers from Tosic et al ([14-111], [14-112], [14-113]) and Silva et al [14-114].

Note that there is a high dispersion in the experimental results concerning creep and shrinkage of concretes made with recycled concrete aggregates. In case of a sensitive structure, measurements of creep and shrinkage is highly recommended.

From tests by Wardeh et al [14-115].

The effect of the use of adjustments of E_{cm} and creep and shrinkage are not enough to explain the increase of the deflection (Tosic et al. 2019 [14-113]).

14.10.3 Physical properties

The density is expressed by the following relation:

$$\rho = 2'500 - 220\,\tau_{TRA} \ [\text{kg/m}^3] \qquad (14.10\text{-}1)$$

Unless more accurate information is available, the linear coefficient of thermal expansion may be taken equal to 10 10^{-6} K^{-1}.

14.10.4 Mechanical properties

The relation between the mean value of the compressive strength and the characteristic strength is unchanged.

The variability of the tensile strength is unchanged.

$$E_{cm} = k_{ERCA} \cdot f_{cm}^{1/3} \text{ with } k_{ERCA} = k_E\,(1 - 0,25\tau_{TRA}) \qquad (14.10\text{-}3)$$

Basic and drying shrinkages should be multiplied by a factor:

$$\eta_{shRAC} = 1 + 0,8\,\tau_{TRA} \qquad (14.10\text{-}4)$$

The creep coefficient for basic and drying creep should be multiplied by a factor:

$$\eta_{cRAC} = 1 + 0,6\,\tau_{TRA} \qquad (14.10\text{-}5)$$

14.10.5 Design properties

Rotation capacity: multiply ε_{cu} by:

$$1 + 0,2\,\tau_{TRA} \qquad (14.10\text{-}6)$$

Deflection control:

$$\zeta = 1 - \beta_{tRAC}\left(\frac{\sigma_{sr}}{\sigma_s}\right)^2 \geq 0 \qquad (14.10\text{-}7)$$

Adapted from RECYBETON [14-108] proposal.

Rules for ensuring durability of structures where recycled aggregate are used are given in standards like FprEN 1992-1-1: 2023 [100-%%%] based on exposure class, types of recycled aggregates and percentage of aggregate replacement."

The mechanical characteristics given in Table 14.11-1 represent average values on the basis of the concrete compressive strength for a wide range of compositions.

General rules for design by testing are given in section 30.11.

with

$\beta_{tRAC} = 1$ for a single short term loading and

$\beta_{tRAC} = 0,25$ for sustained loads or many cycles of repeated loading;

14.10.6 Durability

The minimum clear cover $c_{min,dur}$ due to durability requirement shall be increased by 5mm in the case of risk of corrosion due to carbonation (except in conditions corresponding to XC1 in EN206 where no increase is necessary) and by 10mm in the case of risk of corrosion due to chlorides. These values could be not considered if the measured durability performances are equivalent to a concrete with natural aggregates in the same exposition.

14.11 Assumptions used for design

14.11.1 Dimensioning values for evaluation of structural performance

14.11.1.1 Strength and strain characteristics

The compressive strength of concrete is denoted by concrete strength classes which are related to the characteristic (5%) cylinder strength f_{ck} or the cube strength $f_{ck,cube}$ at an age of 28 days.

The characteristic strengths for f_{ck} and the corresponding mechanical characteristics necessary for design are given in Table 14.11-1.

Concrete can be tailored for particular applications. In that case the relation between the compressive strength of the concrete and relevant mechanical properties may deviate from the relations found in Table 14.11-1. In such case, different relations can be obtained by testing.

In certain situations (e.g. prestressing) it may be appropriate to assess the compressive strength of the concrete before or after 28 days. General relations

Axial tensile tests are very sensitive to the way the test is carried out. For further information on axial tensile testing of concrete, see van Mier van & Vliet van [14-116].

See also section 14.6.1.2.

between the compressive and tensile strengths of the concrete at other ages than 28 days are given in section 5.1.

The tensile strength of concrete can be determined directly by a uniaxial tensile test or by a splitting tensile test. Where the tensile strength is determined as the splitting tensile strength $f_{ct,sp}$ an approximate value of the axial tensile strength f_{ct} is found from:

$$f_{ct} = 1.0 f_{ct,sp} \qquad (14.11\text{-}1)$$

Table 14.11-1: Strength classes for concrete.

Concrete grade	**C12**	**C16**	**C20**	**C25**	**C30**	**C35**	**C40**	**C45**	**C50**	**C55**	**C60**	**C70**	**C80**	**C90**	**C100**
f_{ck} (MPa)	12	16	20	25	30	35	40	45	50	55	60	70	80	90	100
$f_{ck,cube}$ (MPa)	15	20	25	30	37	45	50	55	60	67	75	85	95	105	115
f_{cm} (MPa)	20	24	28	33	38	43	48	53	58	63	68	78	88	98	108
f_{ctm} (MPa)	1.6	1.9	2.2	2.6	2.9	3.2	3.5	3.8	4.1	4.2	4.4	4.6	4.8	5.0	5.2
$f_{ctk,0.05}$ (MPa)	1.1	1.3	1.5	1.8	2.0	2.2	2.5	2.7	2.9	3.0	3.1	3.2	3.4	3.5	3.7
$f_{ctk,0.95}$ (MPa)	2.0	2.5	2.9	3.3	3.8	4.2	4.6	4.9	5.3	5.5	5.7	6.0	6.3	6.6	6,8
E_{cm} (GPa)	27.1	28.8	30.3	32.0	33.6	35.0	36.3	37.5	38.6	39.7	40.7	42.6	44.4	46.0	47.5

The values in Table 12.11-1 are based on the following analytical relations (in all formulas the strengths f_{cm}, f_{ck}, f_{ctm} and f_{ctk} are in MPa and the strains ε in ‰).

$f_{cm} = f_{ck} + 8$ with f_{cm} and with f_{ck} in MPa

$f_{ctm} = 0.30\, f_{ck}^{2/3}$ for concrete classes ≤ C50

$f_{ctm} = 2.12 \ln (1 + f_{cm}/10)$ for concrete classes > C50

$f_{ctk;0.05} = 0.7\, f_{ctm}$

$f_{ctk;0.95} = 1.3\, f_{ctm}$

$E_{cm} = 21.5(f_{cm}/10)^{1/3}$ with f_{cm} in MPa

Further see section 14.6.1.1 (compressive strength), section 14.6.1.2.1 (tensile strength), section 14.6.1.4.2 (modulus of elasticity) and section 14.6.1.5.1 (stress–strain relations in compression for short term loading).

The constant 0.06 in Eq. (12.11-2) is appropriate for normal strength concrete. With increasing brittleness of the concrete the coefficient decreases. This means that for high strength concrete and lightweight concrete lower values for the constant in Eq. (12.11-2) apply – see section 14.6.1.2.1 and the comment about Eqs. (14.6-8a,b).

Table 12.11-1 shows a survey of design values for concrete strength classes up to C100. Although in section 5.1 mechanical properties of concrete are given up to a strength class C120, in this table the strength classes are limited to C100. The reason is that for strength classes > C100 not all areas of application have sufficient data yet for all aspects of behaviour.

The flexural tensile strength can be formulated as a function of the axial tensile strength. A general relation is:

$$f_{ctm,fl} = f_{ctm} \frac{1 + 0.06 h_b^{\,0.7}}{0.06 h_b^{\,0.7}} \qquad (14.11\text{-}2)$$

where:

f_{ctm} is the mean axial tensile strength [MPa];

$f_{ctm,fl}$ is the mean flexural strength [MPa];

h_b is the overall depth of beam [mm].

This relation applies as well to the characteristic values of the concrete strength.

14.11.1.2 Elastic deformations

The elastic deformations of concrete largely depend on its composition (especially the aggregates). The values given in Table 14.11-1 should be regarded as indicative for general applications. The values given in Table 14.11-1 are approximate values for the E modulus E_{cm}, being the secant value between $\sigma c = 0$ and $0.4\ f_{cm}$ for concrete with quartzite aggregate, subjected to short term loading. More detailed information for concrete with other aggregates is given in section 5.1. The elastic deformations should be specially assessed if the structure is likely to be sensitive to deviations from the indicative values. More detailed information on the E modulus is given in section 5.1.

The Poisson's ratio may be taken equal to 0.2 for uncracked concrete and 0 for cracked concrete.

Unless more accurate information is available, the linear coefficient of thermal expansion may be taken equal to $10 \cdot 10^{-6}\ K^{-1}$

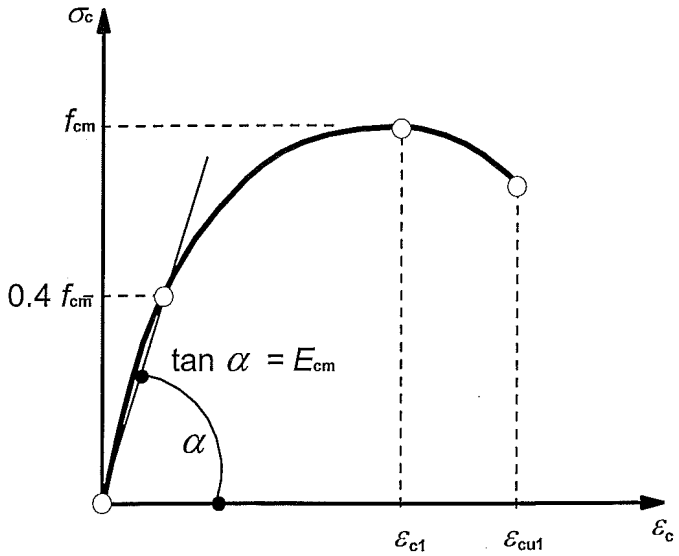


Figure 14.11-1: Schematic representation of the stress–strain relation for structural concrete.

14.11.1.3 Stress-strain relationship

The relation between σ_c and ε_c shown in Figure 14.11-1 (compressive stress and shortening strain shown as absolute values) for short term uniaxial loading is described by the expression:

$$\frac{\sigma_c}{f_{cm}} = \frac{k \cdot \eta - \eta^2}{1 + (k-2)\eta} \qquad (14.11\text{-}3)$$

where:

$\eta = \varepsilon_c / \varepsilon_{c1}$;

ε_{c1} is the strain at peak stress according to Table 14.11-1;

$k = 1.05 E_{cm} |\varepsilon_{c1}| / f_{cm}$ (f_{cm} according to Table 14.11-1).

Eq. (14.11-3) is valid for $0 < |\varepsilon_c| < |\varepsilon_{cu1}|$ where ε_{cu1} is the nominal ultimate strain.

Other idealized stress–strain relations may be applied, if they adequately represent the behaviour of the concrete considered.

Generally the first variable loads on a structure are applied months after the determination of the 28-days strength. Since then the strength of the concrete has increased by continued hydration of the cement. This increase approximately balances the unfavourable effect of sustained loading, so that $\alpha_{cc} = \alpha_{ct} = 1.0$ is appropriate. For concrete strength determined at an age greater than 28 days, the effect of hydration may not be able any more to compensate the effect of sustained loading, so that $\alpha_{cc} = \alpha_{ct} = 0.85$ is more suitable, unless tests show otherwise.

The strength reduction factor η_{fc} was defined in MC2010 for the case of diagonal compression in shear. It applies also to other cases as compression in columns, compression zone due to bending, struts in strut-and-tie models, partially loaded areas and other cases. Coefficient 40 is a design value accounting for the partial safety factor γ_C. For comparisons to test results, it should be replaced by 30 (see detailed explanation in Moccia et. al. [14-117].

14.11.1.4 Design compressive and tensile strengths

The value of the design compressive strength is defined as:

$$f_{cd} = \alpha_{cc}\eta_{fc}f_{ck} / \gamma_C \qquad (14.11\text{-}4)$$

where:

γ_C is the partial safety factor for concrete, being 1.5 for transient and persistent situations and 1.2 for incidental situations;

α_{cc} is a coefficient taking account of long term effects on the compressive strength and of unfavourable effects from the way the load is applied. For normal design situations it may be assumed that the increase of the compressive strength after 28 days compensates the effect of sustained loading, so that $\alpha_{cc} = 1.0$ for new structures.

η_{fc} is a strength reduction factor accounting for the increased brittleness for higher strength concrete and may be calculated as:

$$\eta_{fc} = \left(\frac{40}{f_{ck}}\right)^{1/3} \leq 1.0 \quad (f_{ck} \text{ in MPa}) \qquad (14.11\text{-}4a)$$

The value of the tensile strength f_{ctd} is defined as:

$$f_{ctd} = \alpha_{ct}f_{ctk} / \gamma_c \qquad (14.11\text{-}5)$$

For the same reason as mentioned for αcc, for new structures $\alpha_{ct} = 1.0$

References

[14-1] *fib* Bulletin 70. Code type models for structural behaviour of concrete – Background of the constitutive relations and material models in *fib* MC 2010.

[14-2] ISO 22965-1:2007. Concrete – Part 1: Methods of specifying and guidance to the specifier.

[14-3] ISO 22965-2:2007. Concrete – Part 2: Specification of constituent materials, production of concrete and conformity of concrete.

[14-4] EN 13670 Execution of concrete structures, CEN, 2009

[14-5] ISO 22966 ISO 22966:2009, Execution of concrete structures, ISO 2009

[14-6] ISO 1920-5:2018. Testing of concrete – Part 5: Density and water penetration depth.

[14-7] ASTM C94/C94M – 14. Standard Specification for Ready-Mixed Concrete. 2014.

[14-8] EN 206:2014. Concrete – Specification, performance, production and conformity.

[14-9] ISO 16204:2012. Durability – Service life design of concrete studies.

[14-10] *fib* Bulletin 34. Model Code for Service Life Design. 2006.

[9] ACI 329R-14. Report on Performance-Based Requirements for Concrete. 2015.

[14-11] ISO 14040:2006. Environmental management – Life cycle.

[14-##]] ISO 13315 Environmental management for concrete and concrete structures

[14-##] ISO 13315-2:2014 Environmental management for concrete and concrete structures — Part 2: System boundary and inventory data

[14-12] GaBi Software. GaBi Solutions, https://gabi.sphera.com/ (obtained on 26-11-2021).

[14-13] ecoinvent. The ecoinvent Association, Zurich, Switzerland, https://ecoinvent.org (obtained on 26-11-2021).

[14-14] Oekobaudat. Sustainable Construction Information Portal, Federal Ministry of the Interior, Building and Community, Germany, https://oekobaudat.de (obtained on 26-11-2021).

[14-xx] M. Haist, K. Bergmeister, M. Curbach, P. Mark, C. Müller, H.S. Müller et al:, Climate limit state (CLS) for building structures – A possible companion of ULS and SLS limit states. Proceedings of the fib International Congress, Oslo, Norway, June 12-16, 2022.

[14-25] *fib* Bulletin $$ Background document

[14-yy] M. Haist, K. Bergmeister, M. Curbach, P. Mark, C. Müller, H.S. Müller et al:, Nachhaltig konstruieren und bauen mit Beton. In BetonKalender (eds K. Bergmeister, F. Fingerloos and J.-D. Wörner), 421-531, 2022.

[14-15] ISO 1920-3:2019. Testing of concrete – Part 3: Making and curing test specimens.

[14-16] Muller et al Muller, H. S., Anders, L., Breiner, R., Vogel, M, .Concrete treatment of types and properties in *fib* Model Code 2010, *Structural Concrete*, 14(4), 2013.

[14-17] RILEM CPC 7. Direct tension of concrete specimens, 1975.

[14-18] *fib* Bulletin 42. Constitutive modelling for high strength/high performance concrete, 2008.

[14-19] Malarics & Muller Malarics, V., Muller, H., Evaluation of the splitting tension test for concrete from a fracture mechanical point of view, *Proceedings of the 7th International Conference on Fracture Mechanics of Concrete and Concrete Structures*, CD: 05-06, 709 – 716, 2010.

[14-20] ISO 1920-4:2020. Testing of concrete – Part 4: Strength of hardened concrete.

[14-21] CEB-FIP N 213/214. CEB-FIP Model Code 90, 1993.

[14-22] CEB Bulletin 156. Concrete under multiaxial states of stress – constitutive equations for practical design, 1983.

[14-23] Ottosen. Ottosen, N., A Failure Criterion for Concrete, *Journal Engineering Mechanics Division, ASCE*, 103, EM4, 1977.

[14-24] Speck & Curbach. Speck, K., Curbach, M., Fracture criterion for all concretes – Normal, lightweight, high- and ultrahigh -performance concrete, *3rd International fib Congress, Washington, D.C.*, 2010.

[14-26] EN 197-1:2018. Cement – Part 1: Composition, specifications and conformity criteria for common cements.

[14-27] RILEM CPC 8. Modulus of elasticity of concrete in compression, 1975.

[14-28] ISO 1920-10:2010. Testing of concrete – Part 10: Determination of static modulus of elasticity in compression.

[14-29] ASTM C469/C469M-14. Standard Test Method for Static Modulus of Elasticity and Poisson's Ratio of Concrete in Comparison, 2010.

[14-30] EN 12390-13:2019. Testing hardened concrete – Part 13: Determination of secant modulus of elasticity in compression.

[14-31] Faust Faust, T., Lightweight concrete in structural engineering, 2002.

[14-32] Mazars & Pijaudier-Cabot. Mazars, J., Pijaudier-Cabot, G., Continuum Damage Theory – Application to Concrete, *Journal of Engineering Mechanics*, 1989.

[14-33] Mazars et.al Mazars, J., Hamon, F., Grange, S,.A new 3D damage model for concrete under monotonic, cyclic and dynamic loading, *Materials and Structures, 2015.*

[14-34] Carol et al Carol, L., Rizzi, E., Willam, K., On the formulation of anisotropic elastic degradation. I. Theory based on pseudo-logarithmic damage tensor rate. II. Generalized pseudo-Rankine model for tensile damage, *International Journal of Solids and Structures*, 38, 2001.

[14-35] Govindjee et al Govindjee, S., Kay, G. J., Simo, J. C., Anisotropic modelling and numerical simulation of brittle damage in concrete, *International Journal for Numerical Methods in Engineering*, 38, 1995.

[14-36] Chen & Saleeb Chen, W. F., Saleeb, A. F., Constitutive Equations for Engineering Materials, 1994.

[14-37] Jirasek & Bazant. Jirasek, M., Bazant, Z. P., Inelastic Analysis of Structures, 2002.

[14-38] Häussler-Combe & Hartig Häussler-Combe, U., Hartig, J., Formulation and numerical implementation of a constitutive law for concrete with strain-based damage and plasticity, *Int. Journal of Non-Linear Mechanics*, 43, 2008.

[14-39] Leukart & Ramm Leukart, M., Ramm, E., Identification and Interpretation of Microplane Material Laws, *Journal of Engineering Mehcanics,* 132:3, 295.

[14-40] Ozbolt et al Ozbolt, J., Li, Y.-J., Kozar, I., Microplane model for concrete with relaxed kinematic constraint, *International Journal of Solids and Structures*, 38, 2001.

[14-41] Bathe Bathe, K. J., Finite Element Procedures, 1996.

[14-42] Belytschko et al Belytschko, T., Liu, W. K., Moran, B., Nonlinear Finite Elements for Continua and Structures, 2000.

[14-43] Bazant & Planas Bazant, Z. P., Planas, J., Fracture and Size Effect in Concrete and Other Quasi-brittle Materials, 1998.

[14-44] Jirasek Jirasek, M., Nonlocal models for damage and fracture: comparison of approaches, *International Journal of Solids and Structures*, 35, 1998.

[14-##] Walraven — Walraven, J.C., Fundamental analysis of aggregate interlock, Journal of the Structural Division, 1981, ASCE, SP-149, pp. 311-330

[14-45] Tasevski et al — Tasevski, D., Fernandez-Ruiz, M., Muttoni, A., Assessing the compressive strength of concrete under sustained actions: from refined models to simple design expressions, *Structural Concrete*, 20, 971 – 985, 2019.

[14-46] CEB Bulletin 177. — Summary and Analysis of Observations concerning the Revision of the CEB-FIP Model Code 1978 with discussion documents on new or revised clauses, 1987.

[14-47] CEB Bulletin 215. — Structural Effects of Time-Dependent Behaviour of Concrete, 1993.

[14-48] *fib* Bulletin 52. — Structural Concrete Textbook on behaviour, design and performance, Second edition Volume 2: Basis of design, 2010.

[14-49] Kvitsel, V. — Kvitsel, V., Prediction of shrinkage and creep of normal strength and high-strength structural lightweight concrete made with expanded clay aggregates, 2011.

[14-50] RILEM TC 107 CSP. — Creep and shrinkage prediction models: Principles of their formation. Recommendation for measurement of time-dependent strains of concrete, *Materials and Structures*, 31, 507 – 512, 1998.

[14-51] ISO 1920-8:2009. — Testing of Concrete – Part 9: Determination of drying shrinkage for samples prepared in the field or in the laboratory.

[14-52] RILEM TC 129-MHT I — Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 6: Thermal strain, *Materials and Structures,* 17 – 21, 1997.

[14-53] RILEM TC 129-MHT II — Test methods for mechanical properties of concrete at high temperatures. Recommendations: Compressive strength for service and accident condition, *Materials and Structures*, 28, 410 – 414, 1995.

[14-54] Acosta — Acosta Urrea, F., Influence of Elevated Temperatures up to 100 °C on the Mechanical Properties of Concrete, *KIT Scientific Publishing,* 84, 2017.

[14-55] Bazant & Najjar — Bazant, Z.P. and Najjar, L.J., “Drying of concrete as a non-linear diffusion problem” (Cement and Concrete Research, Vol.1, pp. 461-473, 1971)

[14-56] RILEM TC 129-MHT III — Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 4: Tensile strength for service and accident conditions, *Materials and Structures*, 33, 219 – 223, 2000.

[14-57] RILEM TC 129-MHT IV — Test methods for mechanical properties of concrete at high temperatures. Recommendations: Modulus of elasticity for service and accident conditions, *Materials and Structures*, 37, 139 – 144, 2004.

[14-58] RILEM TC 129-MHT V — Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 8: Steady state creep and creep recovery for service and accident conditions, *Materials and Structures*, 37, 6 – 13, 2000.

[14-59] RILEM TC 129-MHT VI — Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 7: Transient creep for service and accident conditions, *Materials and Structures*, 31, 290 – 295, 2000.

[14-60] CEB Bulletin 188. — Fatigue of concrete structures, 1988.

[14-61] Lohaus et al — Lohaus, L., Oneschkow, N., Wefer, M., Design model for the fatigue behaviour of normal-strength high-strength and ultra-high-strength concrete, *Structural Concrete*, 13: 182 – 192, 2012.

[14-62] Oneschkow et al — Oneschkow, N., Hümme, J., Lohaus, L., Compressive fatigue behaviour of high-strength concrete in a dry and wet environment, *Construction and Building Materials*, 262, 2020.

[14-63] *fib* Bulletin 38. — Fire design of concrete structures – materials, structures and modelling, 2007.

[14-64] *fib* Bulletin 46. — Fire design of concrete structure – structural behaviour and assessment, 2008.

[14-65] EN 1992-1-2:2004. — Eurocode 2: Design of concrete structures – Part 1-2: General rules – Structural fire design.

[14-66] EN 1992-1-2:2004/AC:2008 — Eurocode 2: Design of concrete structures – Part 1-2: General rules – Structural fire design.

[14-67] *fib* Bulletin XX — ??????????????????????

[14-68] EN 12390-8:2019. — Testing hardened concrete – Part 8: Depth of penetration of water under pressure.

[14-69] GBJ 82-1985. — Testing methods for long term and long-lasting performance of ordinary concrete.

[14-70] Auroy et al. 2015 — Auroy et al., Impact of carbonation on unsaturated water transport properties of cement-based materials, Cement and Concrete Research 74 (2015) 44–58

[14-71] RILEM TC 116-PCD. — Permeability of Concrete as a Criterion of its Durability, *Materials and Structure*, 32, 163 – 173.

[14-72] UNE 83966:2008. — Concrete durability. Test methods. Conditioning of concrete test pieces for the purpose of gas permeability and capilar suction tests.

[14-73] ASTM C1585-13. — Standard Test Method for Measurement of Rate of Absorption of Water by Hydraulic-Cement Concretes.

[14-74] ISO 15148:2002. — Hygrothermal performance of building materials and products – Determination of water absorption coefficient by partial immersion.

[14-75] ISO 12572:2016. — Hygrothermal performance of building materials and products – Determination of water vapour transmission properties – Cup method.

[14-76] EN 12086:2013. — Thermal insulating products for building applications – Determination of water vapour transmission properties.

[14-77] ASTM C1556. — Standard Test Method for Determining the Apparent Chloride Difussion Coefficient of Cementitious Mixture by Bulk Difussion.

[14-78] NT Build 443. — Accelerated Chloride Penetration.

[14-79] ISO 1920-11:2013. — Testing of concrete – Part 11: Determination of the chloride resistance of concrete, unidirectional diffusion.

[14-80] EN 12390-11:2015. — Testing hardened concrete – Part 11: Determination of the chloride resistance of concrete, unidirectional diffusion.

[14-81] RILEM TC 281-CCC-WG1&2 — ??

[14-82] RILEM TC-281-CCC-WG5 — ??

[14-83] Rilem recommendation (2004) — ??

[14-84] Rilem recommendation (2000). — ??

[14-85] ASTM C1876 - 19. Standard Test Method for Bulk Electrical Resistivity or Bulk Conductivity of Concrete.

[14-86] Rilem recommendation (2003) ??????????????????????????????????????

[14-87] ASTM C 876-15. Standard Test Method for Corrosion Potentials of Uncoated Reinforcing Steel in Concrete.

[14-88] Gehlen. Probabilistische Lebendsdauerbemessung von Stahlbauwerken - Zuverlässigkeitsbetrachtungen zur wirksamen Vermeidung von Bewehrungskorrosion, *Heft 510 des Schriftenreihe des DAfStb*, 2000.

[14-89] *fib* Bulletin 76. Benchmarking of deemed-to-satisfy provisions in standards, 2015.

[14-90] TR 61. Part 1: Enhancing reinforced concrete durability: Guidance on selecting measures for minimizing the risk of corrosion of reinforcement in concrete*, Concrete Society*, 2004.

[14-91] DARTS 2004. ??????????????????????????????????????

[14-92] ACI 201.2R-16:2016. Guide to Durable Concrete.

[14-93] ACI 318-14:2014. Building Code Requirements for Structural Concrete and Commentary.

[14-94] Bedoe et al. 2009. ??????????????????????????????????????

[14-95] Pavlik Pavlik, V., Corrosion of hardened cement paste by acetic and nitric acids. Part I: Calculation of corrosion depth, *Cem. Concr. Res.*, 24(3): 551 – 562, 1994. https://doi.org/10.1016/0008-8846(94)90144-9

[14-96] Tognazzi Tognazzi, C. , Cracking and leaching of cement-based materials, experimentations. PhD Thesis INSA Toulouse, 1998.

[14-97] De Larrard et al De Larrard, T., Benboudjema, F., Colliat, J.B., Torrenti, J.M., Deleruyelle, F., A probabilistic approach for modelling calcium leaching in concrete structures` Concrete under Severe Conditions: Environment and Loading. Proceedings of the 6th International Conference on Concrete Under Severe Conditions (CONSEC'10), Merida, Yucatan, Mexico, 7-9 June 2010, pp.625--632.

[14-98] Institution of Structural Engineers 1992. ??????????????????????????????????????

[14-99] LCPC Recommendations 1994 ??????????????????????????????????????

[14-100] Concrete Society 1999 ??????????????????????????????????????

[14-101] French standard 2014 ??????????????????????????????????????

[14-102] RILEM 2015 ??????????????????????????????????????

[14-103] CSA A23.1 2019. Concrete Materials and Methods of Concrete Construction/Test Methods and Standard Practices for Concrete.

[14-104] Jacobs et al — Jacobs, F., Winkler, K., Hunkeler, F., Volkart, P., Minor, E.-H. (Hrsg.)., Concrete abrasion in hydraulic engineering, basics – field tests – recommendations, 2001.

[14-105] Spörel — Spörel, F., Hydroabrasive exposure and concrete resistance against abrasion erosion. ACI SP-326 Durability and Sustainability of Concrete Structures (DSCS 2018) – Proceedings 2nd International Workshop, 75.1-75.10.

[14-106] Cunningham et al — Cunningham, L.S., Farrington, B., Doherty, A., Briefing: Abrasion Performance of Concrete in Coastal Structures. Proceedings of the Institution of Civil Engineers – Maritime Engineering, Volume 168, Issue 4, 2015, 157-151.

[14-107] ACI 210.1R-94. — Compendium of Case Histories on Repair of Erosion-Damaged Concrete in Hydraulic Structures, 1999.

[14-%%%] FprEBN1992-1-1:2023 — CEN, "Eurocode 2 - Design of Concrete Structures - Part 1-1: General Rules - Rules for Buildings, bridges and civil engineering structures (Final Draft FprEN 1992-1-1:2023)". CEN European Committee for Standardization Brussels, Belgium, 2023, 405 p.

[14-108] RECYBETON. — Projet national RECYBETON, Comment recycler le béton dans le béton – Recommandations du projet national RECYBETON, IREX 2018.

[14-109] Pacheco et al 2019-1 — Pacheco, J., de Brito, J., Chastre, C., Evangelista, L., Experimental investigation on the variability of the main mechanical properties of concrete produced with coarse recycled concrete aggregates, *Construction and Building Materials* 201 (2019) 110–120.

[14-110] Pacheco et al 2019-2 — Pacheco, J., de Brito, J., Chastre, C., Evangelista, L., Reliability of the bond strength of recycled coarse aggregate concrete, Proceedings of the Symposium 2019, Concrete - Innovations in Materials, Design and Structures, Krakow, 2019.

[14-111] Tosic et al 2018-1 — Tosic N., Marinković S., de Brito, J., Deflection control for reinforced recycled aggregate concrete beams, Experimental database and exentsion of the *fib* Model Code 2010, *Mater. Struct. Constr.*, 2018.

[14-112] Tošić et al 2018-2 — Tošić, N., de la Fuente, A., Marinković, S., Shrinkage of recycled aggregate concrete experimental database and application of the *fib* Model Code 2010, *Mater. Struct. Constr.*, 2018.

[14-113] Tošić et al 2019 — Tošić, N., de la Fuente, A., Marinković, S., Creep of recycled aggregate concrete: Experimental database and creep prediction model according to the *fib* Model Code 2010, *Constr. Build. Mater.*, 195, 590 – 599, 2019.

[14-114] Silva et al. — Silva, R.V., de Brito, J., Dhir, R.K,.Establishing a relationship between the modulus of elasticity and compressive strength of recycled aggregate concrete, *J. Clean. Prod.*, 112, 2171 – 2186.

[14-115] Wardeh et al. — ??

[14-116] Mier van & Vliet van — Mier van , J. G. M., Vliet van, M. R., Uniaxial tension test for determination of fracture parameters of concrete: state of the art, *Engineering Fracture Mechanics*, 69(2): 235 – 247, 2002.

[14-117] Moccia F., Yu Q., Fernández Ruiz M., Muttoni A. — Concrete compressive strength: From material characterization to a structural value, Structural Concrete, (2020)): Vol 22, Issue S1, P 634-654, https://doi.org/10.1002/suco.202000211

15 Reinforcing steel

15.1 Scope definition and range of applicability

Products used as reinforcing steel may be:

– bars;
– wires;
– welded fabric.

Reinforcing steel is characterized by:

– chemical properties:
  • elements in carbon equivalent;
  • carbon equivalent;
– geometrical properties:
  • diameter (size);
  • surface characteristics;
– mechanical properties:
  • yield strength and tensile strength;
  • bond characteristics;
  • ductility;
  • fatigue behaviour;
  • thermal expansion;
  • behaviour under extreme thermal conditions;
– technological properties:
  • bendability;
  • weldability;
  • durability.

Reinforcing steel bars can be either directly produced in straight lengths or by straightening reinforcing steel from coils (Decoiling).

Reinforcing steel wires can be plain, ribbed or indented cold-reduced product generally of constant cross-section throughout its length.

Surface characteristics include the characteristics of the ribs or indentations by means of which bond with the concrete is achieved, and the characteristics of the coating if applicable.

Bond aspects are covered in chapter 20.

Durability mainly covers the corrosion protection properties of “special” reinforcing steel such as (either metallic or organic) coated steel or corrosion resistant steel. Behaviour at extreme temperatures may include either the behaviour at low temperature (e.g. for cryogenic applications) or high temperature (e.g. behaviour in case of fire).

Examples of relevant international product standards are ISO 6935-1 [15-1], ISO 6935-2 [15-2], and ISO 6935-3 [15-3].

Types of reinforcement not covered by product standards may be used after it has been shown that they meet the specified minimum requirements.

Requirements for couplers for mechanical spicing of reinforcing steel bars are specified in product specifications (i.e. European Assessment Document EAD 160129-00-0301 [15-4], ISO 15835-1 [15-5).

The simultaneous use of steels of various types on the same site is allowed only on condition that no confusion between the types is possible during construction. The strength and ductility class can be identified by rib pattern and/or special marking (e.g 50B – "50" for 500MPa, "B" for ductility class B). It is considered good practice to mark reinforcing steel bars for the identification of source.

It should be possible to distinguish clearly between:

– plain bars of various grades and/or of various ductility classes;
– high bond bars of various grades and/or of various ductility classes;
– reinforcement that is weldable and that which is not.

Reinforcing steels shall comply with international or national product standards applicable at the location of the structure. The standards specify geometrical, mechanical and technological properties.

Couplers for splicing of reinforcing bars shall comply with international or national product standards or product specifications. Couplers should be capable of developing the static strength, fatigue strength (if required) and ductility of the spliced reinforcement.

15.2 Classification

The designation of reinforcing steels normally includes:

– the relevant product standard;
– the nominal diameter (size);
– the steel grade related to the characteristic yield strength, the ductility properties and the weldability.

Each product should be clearly identifiable with respect to this designation.

The properties of reinforcing steels are generally confirmed by certification programmes and certificates of compliance. Requirements to certification of conformity should follow relevant international or national standards. A factory production control system that complies with ISO 9001 [15-6] and addresses the requirements of the relevant product standard is considered as appropriate.

Fabrication may include decoiling, straightening, cutting, bending, mechanical splicing and welding of steel reinforcement.

Cutting and bending of steel reinforcement may be controlled via a standard such as BS 8666 [15-7].

Welding of steel reinforcement may be controlled via a standard such as ISO 17660-1 [15-8], and ISO 17660-2 [15-9].

Installation of steel reinforcement may be controlled via a standard for execution of concrete structures such as ISO 22966 [15-10] and EN 13670 [15-11].

The nominal diameter is a conventionally fixed value in product standards which serves as a basis for the calculation of the nominal cross-sectional area taken as the area of a circle with a diameter equal to the nominal diameter. For ribbed and indented reinforcing steel, the actual cross-sectional area may be less than the nominal cross-sectional area.

The actual cross-sectional area is determined by weighing a given length of reinforcing steel, assuming a steel density of 7850 kg/m^3.

For welded fabric the following applies:

- twin bars or wires are allowed in one direction only;
- adequate stiffness of the welded fabric should be ensured either by a limitation of the maximum spacing of the bars or wires, or by introducing a minimum ratio between the diameter of the transverse bars or wires; and the diameter of the longitudinal bars.

15.3 Provisions for quality control

The manufacturing and fabrication of reinforcing steel shall be subject to factory production control by the manufacturer and fabricator, and continuous external control by an independent certification body, which includes certification and regular audits.

15.4 Geometrical properties of reinforcing steel

15.4.1 Diameter (Size)

The diameter of reinforcing steel is defined by a nominal diameter for bars and wires, and a set of nominal diameters for welded fabric.

Maximum diameter should be considered:

- Ribbed reinforcing steel $Ø \leq 50$ mm;
- Indented reinforcing steel $Ø \leq 16$ mm.

The difference between actual and nominal cross-sectional area shall not exceed the limiting values specified in relevant product standards.

–

Plain wires and bars should only be used for reinforced concrete in non-structural applications such as spacers, except in the form of welded fabric.

The rib parameters may be specified by either the relative rib area f_R, or by a combination of rib spacing, rib height and rib inclination of the transverse ribs.

The indentation parameters may be specified by either the relative indentation area f_P, or by a combination of indentation spacing, indentation depth and inclination of the indentations.

The straightening of ribbed or indented bars and wires from coils can alter the properties of the bar. This can lead to reductions in the relative rib or indentation area and thus the bond properties of the straightened products. Additionally, the straightening can result in changes to the ductility properties of the bar.

The standardised requirements are, however, specified for the straightened product.

The common coatings applied to reinforcing steel are either metallic (e.g. Zinc or Zinc alloy) or organic (e.g. Epoxy).

Examples of relevant international product standards are ISO 14654 [15-12] and ISO 14657 [15-13].

15.4.2 Surface characteristics

Bond characteristics of reinforcing steel should be classified according to its surface characteristics. The surface of reinforcing steel may be:

– ribbed;

– plain;

– indented.

Ribbed bars, wires and some indented products should be considered as high bond steel reinforcement if they satisfy the conditions and requirements imposed by the relevant product standards.

Products not satisfying these requirements should be treated as plain bars with respect to bond. Indented products, which cannot be considered as high bond reinforcement, shall be treated according to relevant standards or technical specifications.

The characteristics of the coatings are in general:

– finish and appearance;

– adherence;

– mass of the coating (e.g. Zinc or Zinc alloy) deposited per unit area;

– thickness of the coating (e.g. Epoxy);

– continuity.

These characteristics shall conform to the requirements specified in the relevant product standards.

For quality control purposes and design calculations, the mechanical properties of a product are referred to the nominal cross-sectional area.

Reference is made to the test methods for reinforcing steel given in ISO 15630 Part 1 (Bars, rods and wire) [15-14], Part 2 (Welded fabric and lattice girders) [15-15] and Part 3 (Prestressing steel) [15-16].

The requirements apply to the product in the condition in which it is delivered. In the case of reinforcing steel delivered in coils (wire or rods), the requirements apply to the product after straightening.

The value of f_{yk} corresponds to the 0.2% offset in the characteristic stress–strain diagram.

For steels totally or partially cold-worked by means of axial tension, it is generally the case that:

$$f_{yc} \neq f_{yt}$$

where f_{yc} and f_{yt} are the actual yield strengths, for compression and tension respectively. The value of f_{yc} to be used in a calculation should therefore be specified in the relevant standards.

Note: For design purposes f_{yc} is often assumed to be identical to f_{yt}.

There are a number of situations where the designer expects reinforcement to yield before failure of a member or structure. If in fact the reinforcement is appreciably stronger than assumed in the design, it is possible that, in a flexural situation, the concrete may reach its ultimate compression strain before the yielding of the tensile reinforcement. This may lead to brittle, rather than ductile failure. It is therefore important to ensure that the characteristic yield strength of the reinforcement is not significantly higher than that assumed in the design. An over-strength in the order of 30% may be appropriate.

See also CEB Bulletin 242 [15-17].

Four ductility classes are defined for design purposes.

15.5 Mechanical properties of reinforcing steel

The mechanical properties are defined on the basis of standardised tests.

15.5.1 Tensile strength and ductility properties

The characteristic values of

- the tensile strength (f_t);
- the yield strength (f_y);
- the strain at maximum force (ε_u).

are respectively denoted as f_{tk}, f_{yk} and ε_{uk}.

These classes are defined by minimum specified values for the characteristic value of the ratio f_t/f_y and the characteristic strain at maximum stress ε_{uk} as follows:

- Class A: $(f_t/f_y)_k \geq 1.05$ and $\varepsilon_{uk} \geq 2.5\%$;
- Class B: $(f_t/f_y)_k \geq 1.08$ and $\varepsilon_{uk} \geq 5\%$;
- Class C: $(f_t/f_y)_k \geq 1.15$ and ≤ 1.35 and $\varepsilon_{uk} \geq 7.5\%$;
- Class D: $(f_t/f_y)_k \geq 1.25$ and ≤ 1.45 and $\varepsilon_{uk} \geq 8\%$.

Adequate ductility is necessary, whether or not moment redistribution is taken into account in design.

The characteristic value of the ratio (f_t/f_y), *that is* $(f_t/f_y)_k$, corresponds to the 5% fractile of the relation between actual tensile strength and actual yield strength.

Ductility class definitions A, B, C and D are only valid for steel grades with a characteristic yield strength ≤ 700 MPa.

Classes C and D should be used where high ductility of the structure is required (e.g. in seismic regions). In seismic design an additional requirement for $f_{y,act}/f_{yk}$ for classes C and D (e.g. $f_{y,act}/f_{yk} \leq 1.3$) can be introduced.

Should it be required to quantify a level of ductility in relation to the deformation capacity of a concrete member, it may be misleading to focus on isolated physical characteristics of the steel. In practice, quantification of ductility may be done by considering the "equivalent steel concept", where the overall steel ductility parameter p may be regarded as equivalent to:

for cold-worked steel

$$p = \varepsilon_u^{0.75}\left[\frac{f_t}{f_y} - 1\right]^{0.8} \qquad (15.5\text{-}1a)$$

for hot-rolled steel

$$p = \left((\varepsilon_u - \varepsilon_y) + 3 \cdot \varepsilon_y\right)^{0.75}\left[\frac{f_t}{f_y} - 1\right]^{0.8} \qquad (15.5\text{-}1b)$$

Different types of reinforcing steels can show comparable ductility performance in a structure if they have the same p values.

For more information, see CEB Bulletin 218 [15-18] and CEB Bulletin 242 [15-17].

Couplers for splicing of reinforcing steel bars shall be capable of developing the maximum tensile strength f_t, the maximum compressive force with yield strength f_y, and the strain at maximum force ε_u required for the spliced reinforcement.

(2) The permanent slip measured after unloading the specimen shall not exceed 0,10 mm.

15.5.2 Stress–strain relation for short-term loading

Due to the diversity and evolution of the manufacturing processes for bars and wires, various stress–strain relations may be encountered.

Indicative stress–strain diagrams of reinforcing steel in tension are shown in Figure 15.5-1.

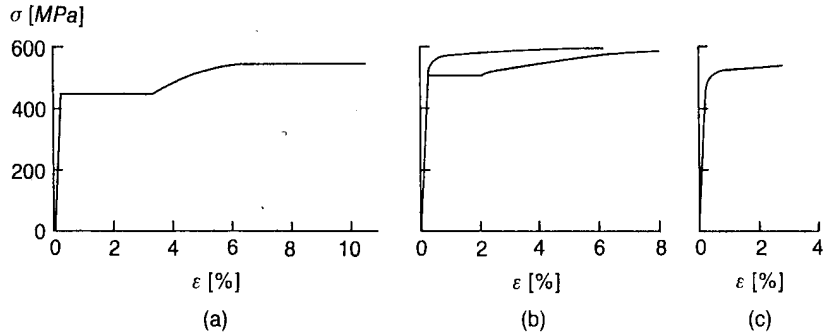


Figure 15.5-1: *Indicative stress–strain relationships of reinforcing steel: (a) hot-rolled bars; heat-treated bars; micro-alloyed bars; (b) low carbon, heat-treated bars (lower curve): cold-worked bars (upper curve); (c) old-worked wires.*

15.5.3 Fatigue behaviour

Reinforcing steel bar and straightened products should be subject to fatigue testing for each steel grade and process route if it is required by international and national standards. The fatigue strength of the reinforcing steel for acceptance testing of the product should be assessed with testing to international and national standards such as ISO 15630-1 [15-14].The test is generally performed in room temperature using stress range which is dependent on the diameter of the product

The fatigue strength of reinforcing steel is tested as part of initial assessment and under the factory production control in accordance with relevant test standards.

and product standard. Typically, reinforcing steel should endure minimum 2 million cycles up to maximum 5 million cycles. All diameters should be tested over a specified period (e.g. 5 years).

The fatigue performance of reinforcing steel placed in concrete considered in design is commonly given in S-N curves (see 30.2.2, Table 30.2-1) and depends on factors such as diameter, rib geometry, bending of bars and welded joints.

See also CEB Bulletin 188 [15-19].

If couplers are used in fatigue design the fatigue strength of these should also be tested as part of initial assessment and under the factory production control in accordance with relevant test standards.

15.5.4 Coefficient of thermal expansion

Within the temperature range between –20°C and 180°C, the coefficient of thermal expansion of reinforcing steel may be taken as $10 \cdot 10^{-6}$ $degC^{-1}$.

15.5.5 Behaviour under extreme thermal conditions

When specifying the use of reinforcing steel that is expected to operate in concrete under extremely high or extremely low temperatures – that is, outside of what may be considered as the "normal" temperature range (−40°C to +100°C) – consideration may be given to using steel manufactured specifically for such purposes. This recognises that factors such as the chemical composition and processing conditions related to its manufacture can affect the steel's performance.

Detailed information on the behaviour of steel reinforced concrete at extreme temperatures may be found in FIP Special Report SR 88/2 [15-20]. Information on the behaviour of ribbed steels at very low temperatures may be found in G. Hartl, 1985 [15-21] or *fib* Bulletin 98 [15-22].

The properties of reinforcing steel are valid for "normal" temperatures between −40°C and + 100°C.

Tensile strength and yield strength decrease and the strain at maximum stress increases if the temperature increases above the "normal" temperature range.

Tensile strength, yield strength and modulus of elasticity increase if the temperature decreases below the "normal" temperature range. The percentage total elongation at maximum stress increases initially but decreases rapidly below a temperature of about −150°C.

15.5.6 Effect of strain rate

For details on the strain rate effect on the reinforcing steel properties, see CEB Bulletin 187 [15-23].

The tensile properties of steel increase with the speed or the rate of strain application.

Typical bond tests provide good bond conditions to the anchorage zone of the reinforcing steel. Attention should be paid to verify bond performance of reinforcing steel also under poor bond conditions.

General regulations for bond of reinforcing steel are given in section 20.

Anchorage length of high bond reinforcing steel is given in section 30.13.3.5.

For non-conventional concretes and/or rebars a standard test procedure is given in section 20.7.

Alloying elements can make the corrosion layer denser and adherent. These types of reinforcing steels can still rust, but the rust can become protective under highly alkaline conditions so that corrosion can become less rapid than with carbon reinforcing steels.

The extent of change of stress/strain relation depends on particular conditions of the reinforcing steel placed in the member and hence, has to be assessed for the specific member. See also section 15.11.2.1. Reference is made to research at Institute of Structural Engineering (IBK), at ETH Zurich, Prof. Dr. W. Kaufmann.

Ductility and elongation at maximum force of corroded reinforcing steel reduces when compared to non-corroded steel. The extent of reduction depends on particular conditions of the reinforcing steel placed in the member and hence, has to be assessed for the specific member.

15.5.7 Bond characteristics

Bond performance should be tested in accordance with procedures specified in product and test standards.

Relative rib and indentation area may be used as quality control measure for products previously confirmed in bond testing.

15.6 Properties describing / defining durability

15.6.1 Properties related to corrosion of reinforcement

Alloying elements such as Cu, Cr and Ni can be added to improve corrosion resistance of reinforcing steels.

15.6.1.1 Stress-strain relation properties of corroded reinforcing steel

The stress-strain relation may be affected by corrosion, with a reduction of the ultimate strain and the corresponding tensile strength.

15.6.1.2 Ductility properties of corroded reinforcing steel

There are no standard tests for assessment of the ductility properties of corroded reinforcing steel.

Reference:

- Structural Engineering (IBK), at ETH Zurich, Prof. Dr. W. Kaufmann. Other research Prof. Dr. G. Mancini

- Cairns J et al. [15-24].

15.6.1.3 Properties related to other degradation mechanisms

In addition to stress-strain relation also the fatigue and bond properties are affected by corrosion.

15.7 Technological properties

15.7.1 Bendability

Reinforcing steels should not be bent to a radius less than that used in the relevant bend and/or rebend test specified in the product standards by some tolerance. A tolerance of one bar diameter against the bend test has been specified for the minimum mandrel diameter in some national standards.

Bending of steel at temperatures below −5°C is permitted only if allowed by the project specification and additional precautions should be taken as mentioned in EN 13670 [15-25].

The requirements concerning bendability are specified in the relevant standards.

15.7.2 Weldability

Inappropriate welding procedures can adversely affect tensile or other properties of reinforcing steel.

The requirements concerning weldability are generally given by requirements to the chemical composition of the reinforcing steel, specified in the relevant product standards.

Depending on the type of reinforcement used, the methods for welding may be restricted.

15.7.3 Welded joints in welded fabric

The welded joints of the welded fabric for acceptance testing of the product should be assessed with testing to international and national standards such as ISO 15630-2 [15-15].

Provision of cross wires with properly welded joints may significantly reduce the bond length of longitudinal wires. This may under certain conditions induce strain localisation in the longitudinal wires.

Where welded joints are taken into account for the calculation of the anchorage or lap length, each welded joint shall be capable of withstanding a shear force not less than 0.25 $A_s f_{yk}$, where A_s denotes the nominal cross-sectional area of the anchored wire.

15.8 Old types of reinforcing steel

If relevant properties of old types of reinforcing steel are not available from international or national standards or product compliance documentation of the time when these types were produced, testing of a relevant number of samples should be performed.

15.9 Special types of steels

The following special types of reinforcing steel that show enhanced corrosion protection properties can be used, subject to verification of all properties relevant for design:

- galvanised steels;
- epoxy-coated steels;
- stainless steels.

Further information on these three special types of steels may be found in *fib* Bulletin 49 [15-26].

Some specific effects may be considered in design and/or detailing for special types of steels: for example, possible sensitivity to fretting fatigue of some stainless steels, see Concrete Society Technical Report 51 [15-27].

Coating of reinforcing steel may affect in particular the bond performance.

Stainless steel reinforcement such as ferritic and austenitic stainless steel grades can improve service life especially at high corrosive applications such as marine structures.

Other special types of reinforcing steel that are not mentioned, such as low-carbon chromium steel reinforcement, with stainless steel cladding or similar new developments are available.

Combination of non-coated and stainless reinforcing steel does not cause problems – see Concrete Society Technical Report 51 [15-27].

Stainless steels may be classified according to their corrosion resistance.

15.10 Assumptions used for design

15.10.1 Dimensioning values for evaluation of structural performance

The parameters of reinforcing steel to be used for design are as follows:

- modulus of elasticity, E_s;
- characteristic yield strength, f_{yk} (or characteristic value of 0.2% proof strength, $f_{0.2k}$);
- ductility parameters, that is characteristic strain at maximum force ε_{uk} and characteristic ratio tensile strength/yield strength $(f_t/f_y)_k$.

As a simplification, actual stress–strain diagrams of reinforcing steel in calculations may be replaced by an idealised stress-strain diagram according to Figure 15.10-1, assuming a modulus of elasticity E_s equal to 200 GPa.

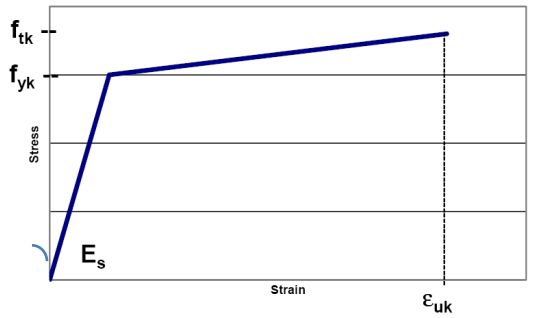


Figure 15.10-1: *Idealised stress–strain diagram.*

The actual stress-strain diagram for a particular steel may be used if it is duly verified. See also commentary relating to section 15.5.1 on over-strength of reinforcement.

For high strength steels, the stress–strain diagram is non-symmetrical in compression and in tension.

Some cold-worked steels have a lower modulus of elasticity in compression than in tension. The difference is not important in practice.

The Menegotto–Pinto idealisation (see section 15.5.2) may also be used for design purposes. In that case, the parameters of the idealisation should be adjusted to the relevant characteristic values of the stress–strain diagram.

Some types of stainless reinforcing steels have lower moduli of elasticity than carbon reinforcing steel. As a simplification, the stress–strain diagram of Figure 15.10-1 may, in general, be used for ultimate strength verification also.

The main parameters of reinforcing steel to be used for fire design are as follows:

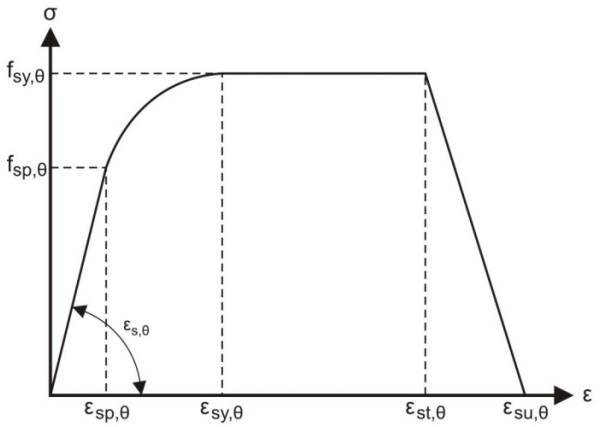
- modulus of elasticity at temperature θ, $E_{s,\theta}$;
- proportional limit at temperature θ, $f_{sp,\theta}$;
- maximum stress at temperature θ, $f_{sy,\theta}$.

The values for the parameters $E_{s,\theta}$, $f_{sp,\theta}$ and $f_{sy,\theta}$ given in Table 15.10-1 may be used.

Table 15.10-1: *Values for the parameters $E_{s,\theta}$, $f_{sp,\theta}$ and $f_{sy,\theta}$.*

Steel temperature θ (°C)	$E_{s,\theta}/E_s$		$f_{sp,\theta}/f_{yk}$		$f_{sy,\theta}/f_{yk}$	
	Hot-rolled	Cold-worked	Hot-rolled	Cold-worked	Hot-rolled	Cold-worked
20	1.00	1.00	1.00	1.00	1.00	1.00
100	1.00	1.00	1.00	0.96	1.00	1.00
200	0.90	0.87	0.81	0.92	1.00	1.00
300	0.80	0.72	0.61	0.81	1.00	1.00
400	0.70	0.56	0.42	0.63	1.00	0.94
500	0.60	0.40	0.36	0.44	0.78	0.67
600	0.31	0.24	0.18	0.26	0.47	0.40
700	0.13	0.08	0.07	0.08	0.23	0.12
800	0.09	0.06	0.05	0.06	0.11	0.11
900	0.07	0.05	0.04	0.05	0.06	0.08
1000	0.04	0.03	0.02	0.03	0.04	0.05
1100	0.02	0.02	0.01	0.02	0.02	0.03
1200	0	0	0	0	0	0

For fire design, the idealised diagram according to Figure 15.10-2 can be used.



Strain range	**Stress σ_θ**	**Tangent modulus**
$\varepsilon_{sp,\theta}$	$\varepsilon E_{s,\theta}$	$E_{s,\theta}$
$\varepsilon_{sp,\theta} \le \varepsilon \le \varepsilon_{sy,\theta}$	$f_{sp,\theta} - c + (b/a)[a^2 - (\varepsilon_{sy,\theta} - \varepsilon)^2]^{0,5}$	$\frac{b(\varepsilon_{sy,\theta} - \varepsilon)}{a\left[a^2 - \left(\varepsilon - \varepsilon_{sy,\theta}\right)^2\right]^{0,5}}$
$\varepsilon_{sy,\theta} \le \varepsilon \le \varepsilon_{st,\theta}$	$f_{sy,\theta}$	0
$\varepsilon_{st,\theta} \le \varepsilon \le \varepsilon_{su,\theta}$	$f_{sy,\theta}[1-(\varepsilon - \varepsilon_{st,\theta})/(\varepsilon_{su,\theta} - \varepsilon_{st,\theta})]$	-
$\varepsilon = \varepsilon_{su,\theta}$	0	-
Parameters	$\varepsilon_{sp,\theta} = f_{sp,\theta} / E_{s,\theta}$ $\varepsilon_{sy,\theta} = 0{,}02$ $\varepsilon_{st,\theta} = 0{,}15$ $\varepsilon_{su,\theta} = 0{,}20$ Class A reinforcement: $\varepsilon_{st,\theta} = 0{,}05$ $\varepsilon_{su,\theta} = 0{,}10$	
Functions	$a^2 = (\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta})(\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta} + c/E_{s,\theta})$ $b^2 = c\,(\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta})\,E_{s,\theta} + c^2$ $c = \frac{(f_{sy,\theta} - f_{sp,\theta})^2}{(\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta})E_{s,\theta} - 2(f_{sy,\theta} - f_{sp,\theta})}$	

Figure 15.10-2: Idealised stress–strain diagram for fire design and corresponding mathematical model.

The stress-strain relations of Figure 15.10-3 apply mainly to carbon reinforcing steel but may be used for stainless reinforcing steel also. Attention should be placed to cases where the modulus of elasticity could have a significant effect on the performance since some stainless steels have lower moduli than 200 GPa.

For stainless reinforcing steel, it is suggested to limit the design ultimate strain ε_{ud} to the same value as used for normal reinforcing steel.

15.10.2 Design tensile strength

The design tensile strength of reinforcing steel f_{yd} follows from:

$$f_{yd} = f_{yk} / \gamma_S = f_{0.2k} / \gamma_S \qquad (15.10\text{-}1)$$

where:

f_{yk} is the characteristic yield stress of the steel;

γ_S = 1.15 for persistent and transient situations and 1.0 for accidental situations.

The behaviour of reinforcing steel in tension and compression is idealised in accordance with the stress–strain relations given in Figure 15.10-3.

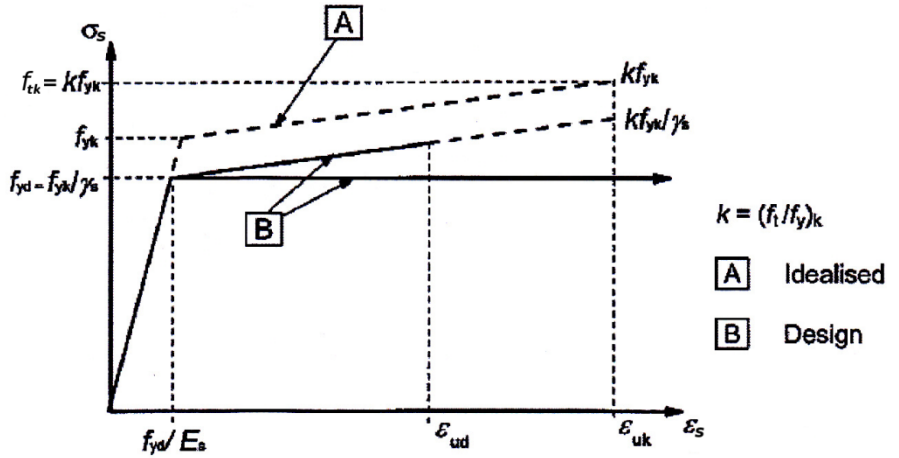


Figure 15.10-3: *Idealised and design stress–strain relations for reinforcing steel in tension and compression.*

For situations in which the plastic structural deformations are of importance, it is suitable to assume a linear strain-hardening behaviour of the reinforcing steel. In this case, the maximum stress $k f_{yk} / \gamma_s$ is reached at a strain of $\varepsilon_{ud} = 0.9 \varepsilon_{uk}$. The value k follows from $k = (f_t / f_y)_k$. For the values of the ultimate strength f_t and the yield strength f_y – see section 15.10.1.

In general, the structural analysis and dimensioning may be based on the mean value of the modulus of elasticity E_s given in section 15.10.1.

For refined verifications, the actual modulus of elasticity and the Menegotto-Pinto idealisation or Ramberg-Osgood formulation may be used.

The Menegotto–Pinto formulation gives an idealisation of the actual stress–strain relation of the reinforcing steel. The strength and strain values should correspond to the actual material values.

The Menegotto–Pinto function is:

$$\frac{\sigma}{\sigma_0} = b\left(\frac{\varepsilon}{\varepsilon_0}\right) + d = b\left(\frac{\varepsilon}{\varepsilon_0}\right) + \frac{(1-b)\left(\frac{\varepsilon}{\varepsilon_0}\right)}{\left[1+\left(\frac{\varepsilon}{\varepsilon_0}\right)^n\right]^{\frac{1}{n}}} \qquad (15.10\text{-}2)$$

where b is the ratio of the final to the initial stiffness and d is a value that is graphically defined in Figure 15.5-2. In the normalized space of stress and strain, the initial stiffness has a slope 1, the slope of the final tangent stiffness is b, and d varies from 0 to $(1 - b)$ as $\varepsilon/\varepsilon_0$ progressively increases from 0 to a maximum value at the last data point. The parameter n defines the transition between elastic and post-yield slopes. The transition tends to a pure bilinear curve with sharp yield value when $n = \infty$ (in practice n > 15), and to a smooth curve if n is low.

The Menegotto–Pinto idealisation also applies well to special types of steel such as stainless steel and to prestressing steel – see section 15.9.

The Menegotto–Pinto idealisation may also be applied to cyclic straining to varying amplitudes by restarting a branch at each strain reversal while adjusting the value of n (see reference below). In this way, it is commonly used in modelling seismic behaviour.

Relevant references are: Menegotto & Pinto [15-28], Menegotto & Pinto [15-29].

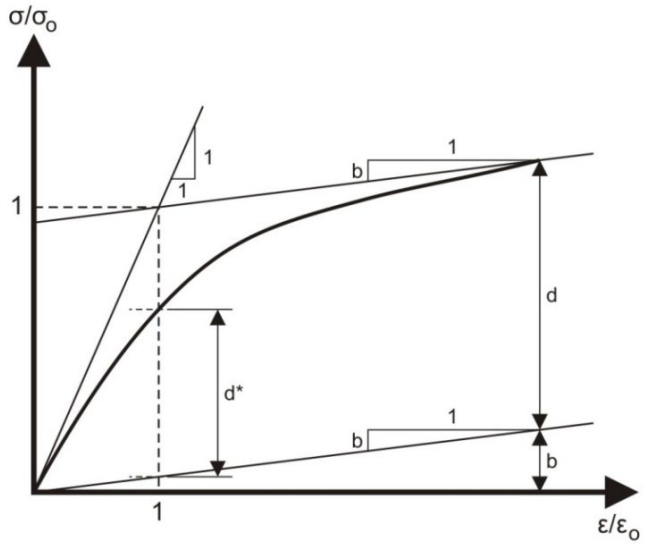


Figure 15.10-4: Menegotto–Pinto expression for the stress–strain relation of reinforcing steel.

15.11 Assumptions used for assessment

15.11.1 Reinforcing steel without deterioration or defects

Design assumptions according to section 15.10.2 may be used for assessment of structures with reinforcing steel without deterioration or defects.

15.11.2 Reinforcing steel with deterioration or defects

Testing of samples of the reinforcing steel with deterioration or defects should be performed when an effect of deterioration or defect is considered relevant for the performance of a member or structure.

15.11.2.1 General

Guidance on the effect of corrosion on anchorage of reinforcement steel is given in section 20.3.5.

Corrosion of reinforcing steel can occur due to different mechanisms:

- Carbonation: CO_2 penetrates through the concrete cover and reacts with the $Ca(OH)_2$, due to which the passive film around the reinforcing steel may be destroyed locally or over a larger area. If sufficient water and oxygen reaches the steel, corrosion may occur.
- Chloride penetration: if chlorides reach the reinforcement in a sufficient concentration, a reaction of chloride, water and oxygen may lead to corrosion. Chlorides can reach the reinforcement either by diffusion through the concrete cover or through the cracks. In case of chloride corrosion so-called pitting corrosion occurs, which can lead to substantial local reductions of the cross-sectional areas of the steel bars.

The reduction of the cross-sectional area of a reinforcing bar can be determined in different ways:

a) Gravimetric method: if samples can be taken from a structure, parts of the bars can be cut and subsequently be weighed. The relative weight loss is determined after removing the rusted parts from the corroded bar, without affecting the sound metal, and comparing the remaining weight with the weight of the undamaged bar.
b) Calliper measurements of bar diameters at different locations along the bar, on the basis of which the cross-section can be estimated. A methodology taking into account the geometry of a deformed bar and pit locations is proposed in ACI 364.14T-17 [15-30].
c) Electrochemical measurement of the True Polarization Resistance technique (measured at the site by modulated confinement of the auxiliary electrode or by using the critical length value for appraising correctly the area polarized). This technique enables the calculation of the corrosion rates l_{cor} which in turn gives the loss in cross-section if it is integrated during the corrosion propagation period and it is an average annual value. A description is given in the corresponding Rilem Recommendation Pro048 [15-31] and Contecvet Manual [15-32].

Corrosion of reinforcing steel may have two effects on the properties of the reinforcement:

- Reduction of the cross-sectional area of the reinforcing steel;
- Reduction of the ductility of the reinforcing steel.

Two main types of corrosion are considered: uniform and local corrosion. Uniform corrosion occurs in case of carbonation. When chloride penetration is the main reason for deterioration, the corrosion can be first localised, to develop subsequently to approximately uniform. In case of corrosion by chloride penetration the most severe type of localised corrosion is pitting corrosion. In case of symmetric (uniform) corrosion, the loss of cross-section can be determined as shown in Figure 15.11-1 (a). However, in most cases the attack is not symmetric with respect to the bar diameter, because the aggressive reaction is driven from one side. In that case, the remaining bar cross-sectional area can be taken as $\pi \cdot x_b \cdot y_b/4$ where x_b and y_b are measured in perpendicular directions and x_b and y_b are the minimum widths (Figure 15.11-1 (b)). For local corrosion (e.g. pitting corrosion), the width x_c and depth y_c of the pitting corrosion are measured, Figure 15.11-1 (c) and (d), and the total cross-sectional area is reduced by $x_c \cdot y_c$. For more irregular boundaries of damage, as shown in Figure 15.11-1 (e), an approximation of the uncorroded cross-sectional area can be made, based on the net cross-sectional area.

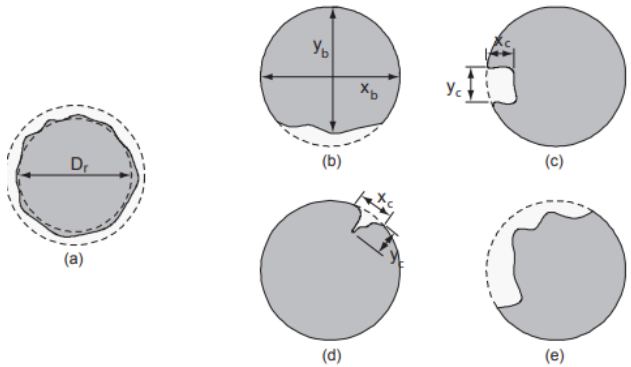


Figure 15.11-1. Residual steel cross-section according to BA38/98 [15-33].

Further to the loss of tensile resistance of reinforcing bars the loss of ductility is a point of concern. In many behavioral models for the determination of the structural resistance redistribution of forces is a prerequisite and if the ultimate strain of the steel is reduced; this may require restrictions to the application limits of the model considered.

It should be noted that method (a) is most appropriate for the case of uniform corrosion. In case of pitting corrosion, the method (b) is more appropriate. In that case not only the local reduction of the cross-sectional area should be measured, but also the distance between the pits. The combination of those data might be used to derive general stress strain relations of the bars. Method (c) is suitable to be able to measure and predict the progress of corrosion.

Figure 15.11-2 shows a single pit with a reduced rebar cross-section $A_{s,red}$. In this case the loss of cross-sectional area $A_s - A_{s,red}$ can be compensated by an increase of the steel stress in the remaining cross-sectional area after passing ε_y in case of hardening. This local increase of stress goes along with a local increase of the strain, which is restricted to the length Δl, so that the strain ε_u will never be reached over the whole length of the bar.

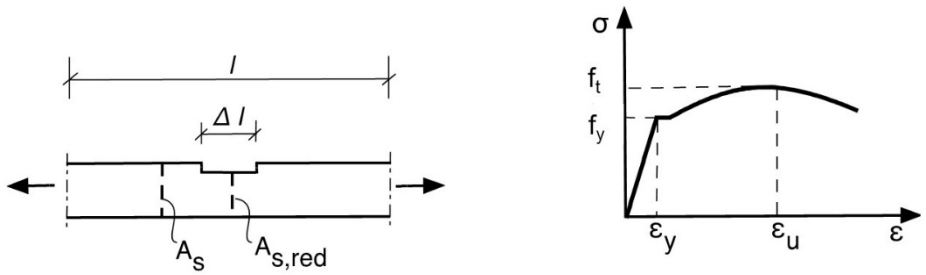


Figure 15.11-2 Relation between pit geometry and strain properties

So the elongation of the bar is largely determined by a number of pits with different geometries and distances, which govern the overall behavior. Nevertheless experiments have shown that there is a relation between the overall properties of the bar after corrosion and the relative loss of weight M_{corr} , see Eq. (15.11-1).

For determining the effect of corrosion on the strength characteristics and ductility of the reinforcing steel the loss of mass of a steel bar due to corrosion can be used. The mass loss ratio is defined as

$$M_{corr} = \left(m_0 - m_1\right) / m_0 \qquad (15.11\text{-}1)$$

where:
m_0 is the weight of the reinforcing bar before corrosion
m_1 is the weight of a bar with the same diameter and length after corrosion

The loss of strength is determined assuming that the bar diameter remains the same and that the yield stress, the tensile strength and the ultimate strain are reduced. For the description of the loss of yield stress and tensile strength linear descending relations have been derived. For the reduction of the ultimate strain an exponentially descending relation appeared to be more suitable.

For uniform corrosion the following decay relations can be used for determining the losses of yield stress, tensile strength and ultimate strain:

$$\frac{f_{y,corr}}{f_y} = 1 - 1.43 M_{corr} \qquad (15.11\text{-}2)$$

$$\frac{f_{t,corr}}{f_t} = 1 - 1.25 M_{corr} \qquad (15.11\text{-}3)$$

$$\frac{\varepsilon_{t,corr}}{\varepsilon_t} = e^{-2.0 M_{corr}} \qquad (15.11\text{-}4)$$

For pitting corrosion the following degradation equations apply:

$$\frac{f_{y,corr}}{f_y} = 1 - 2.0 M_{corr} \qquad (15.11\text{-}5)$$

$$\frac{f_{t,corr}}{f_t} = 1 - 1.86 M_{corr} \qquad (15.11\text{-}6)$$

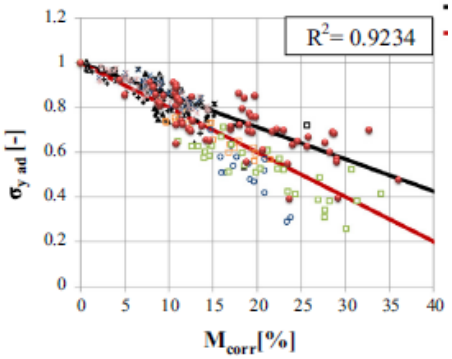


Figure 15.11-3a Reduction of yield stress f_{sy} as a function of relative loss of weight M_{corr} [%] for uniform corrosion (black line, Eq. (15.11-2) and for pitting corrosion (red line, Eq. (15.11-5) (Imperatore et al. [15-34])

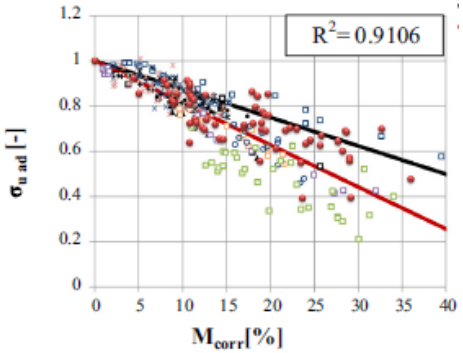


Figure 15.11-3b Reduction of tensile strength f_{su} as a function of relative loss of weight M_{corr} [%] for uniform corrosion (black line, Eq. 15.11-3) and for pitting corrosion (red line, Eq.15.11-6) (Imperatore et al. [15-34])

$$\frac{\varepsilon_{t,corr}}{\varepsilon_t} = e^{-5.5M_{corr}} \qquad (15.11\text{-}7)$$

The equations (15.11-3) to (15.11-7) are the result of a statistical evaluation of 80 tests from 12 sources (Imperatore et al. [15-34]). The various tests were carried out on bars with diameters ϕ8 - ϕ20

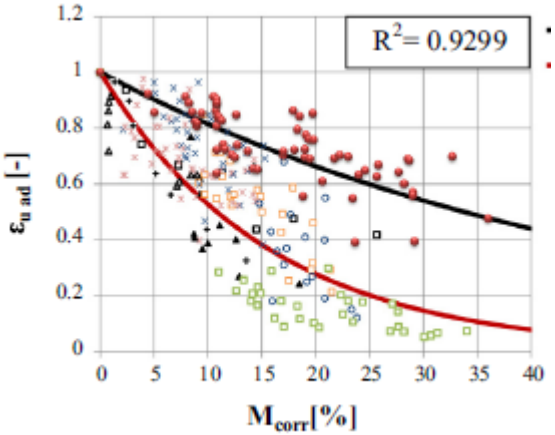


Figure 15.11-3c Reduction of ultimate strain ε_{su} as a function of relative loss of weight M_{corr} [%] for uniform corrosion (black line, Eq. 15.11-4) and pitting corrosion (red line, Eq.15.11-7) (Imperatore et al. [15-34])

Further to statistical evaluations of tests also behavioural models have been developed taking into account the shapes of the pits (measured by X-ray tomography), their distance (assuming a lognormal distribution based on measurements in realistic circumstances) allowing to derive the behaviour of corroded rebars by a probabilistic analysis. Also this analysis confirmed the suitability of linear decay curves for yield stress and ultimate tensile strength and an exponential decay curve for the ultimate strain (Zeng et al. [15-35]).

15.11.3 Reinforcing steel with noncompliant properties

Testing of samples of the reinforcing steel with noncompliant properties should be performed when the noncompliance is considered relevant for the performance of a member or structure.

References

[15-1] ISO 6935-1. Steel for the reinforcement of concrete - Part 1: Plain bars.

[15-2] ISO 6935-2. Steel for the reinforcement of concrete - Part 2: Ribbed bars.

[15-3] ISO 6935-3. Steel for the reinforcement of concrete - Part 3: Welded fabric.

[15-4] EAD 160129-00-0301 Couplers for mechanical splices of reinforcing steel bars, EOTA January 2020

[15-5] ISO 15835-1 Steels for the reinforcement of concrete - Reinforcement couplers for mechanical splices of bars - Part 1: Requirements

[15-6] ISO 9001. Quality Management Systems

[15-7] BS 8666. Scheduling, dimensioning, bending and cutting of steel reinforcement for concrete – Specification.

[15-8] ISO 17660-1. Welding of reinforcing steel – Part 1: Load-bearing welded joints

[15-9] ISO 17660-2. Welding of reinforcing steel – Part 2: Non-load bearing welded joints.

[15-10] ISO 22966. Execution of concrete structures.

[15-11] EN 13670. Execution of concrete structures.

[15-12] ISO 14654. Epoxy-coated steel for the reinforcement of concrete.

[15-13] ISO 14657. Zinc-coated steel for the reinforcement of concrete.

[15-14] ISO 15630-1. Steel for the reinforcement and prestressing of concrete - Test methods - Part 1.

[15-15] ISO 15630-2. Steel for the reinforcement and prestressing of concrete - Test methods - Part 2.

[15-16] ISO 15630-3 Steel for the reinforcement and prestressing of concrete - Test methods - Part 3.

[15-17] CEB Bulletin 242. Ductility of Reinforced Concrete Structures. 1998.

[15-18] CEB Bulletin 218. Ductility – Reinforcement – Progress Report. 1993.

[15-19] CEB Bulletin 188 Fatigue of concrete structures, 1988

[15-20] FIP Special Report SR 88/2.. Information on the behaviour at very low temperatures of ribbed steels. 1988.

[15-21] Hartl, G. Beton- und Stahlbetonbau 10/1985, pp. 271–274, and 11/1985, pp. 293–299

[15-22] *fib* Bulletin 98. Acceptance of post-tensioning systems for cryogenic applications, 2021

[15-23] CEB Bulletin 187. Concrete Structures under Impact and Impulsive Loading – Synthesis Report. 1988.

[15-24] Cairns J, Plizzari A. G.,.Du Y., Law W. D., Franzoni, C. Mechanical properties of corrosion damaged reinforcement. ACI Materials Journal, Vol 102. No. 4 pp. 256-264, 2015.

[15-25] EN 13670. Execution of concrete structures.

[15-26] fib Bulletin 49. Corrosion protection of reinforcing steels. 2009.

[15-27] Concrete Society Technical Report 51. Guidance on the use of stainless steel reinforcement. 1988.

[15-28] Menegotto M. and Pinto P. E. Method of analysis for cyclically loaded RC plane frames including changes in geometry and non-elastic behaviour of elements under combined normal force and bending. Preliminary Report, *IABSE Symposium: Resistance and ultimate deformability of structures acted on by well-defined repeated loads* – IABSE Vol. 13, 1973.

[15-29] Menegotto M. and Pinto P. E. Strength of reinforced or prestressed concrete columns under biaxial load. Preliminary Report, IABSE Symposium: Design and safety of reinforced concrete compression members – IABSE Vol. 16, 1974.

[15-30] ACI 364.14T-17. Section loss determination of damaged or corroded reinforcing steel bar. 2017.

[15-31] Rilem Recommendation Pro048. Calculation of initiation and propagation periods of service life of reinforcements by using the electrical resistivity. 2004.

[15-32] Contecvet Manual. A validated user's manual for assessing the residual life of concrete structures. 2001.

[15-33] BA38/98 Assessment of the fatigue life of corroded or damaged reinforcing bars. 1998.

[15-34 Imperatore, S., Rinaldi, A., Drago, C. Degradation relationships for the mechanical properties of corroded steel rebars, Construction and Building Materials 148 (2017) 219-230, https://doi.org/10.1016/j.conbuildmat.2017.04.209

[15-35] Chaoqun Zeng, Ji-Jua Zhu, Cheng Xiong, Yanru Li, Dawang Li, Joost Walraven Analytical model for the prediction of the tensile behaviour of corroded steel bars, Construction and Building Materials, Vol. 258, 2020, https://doi.org/10.1016/j.conbuildmat.2020.120290

16 Prestressing steel & prestressing systems

16.1 Prestressing steel

16.1.1 Scope definition and range of applicability

Steels for prestressing are delivered as:

– wire;
– 2-wire strands, 3-wire strands, 7-wire strands, 19-wire strands;
– bars.

Prestressing steel is characterized by:

– geometrical properties:
  • surface characteristics: plain, indented or ribbed;
  • nominal diameter;
  • nominal cross – sectional area;
  • nominal mass per metre;
  • nominal dimensions of indentations (wire and strand);
  • pitch length (strand);
  • left or right lay (strand);
  • nominal pitch and dimensions of ribs or threads (bars);
  • left or right thread (bars);
– mechanical properties:
  • tensile properties: ultimate tensile strength (UTS), 0.1% proof stress and total elongation at ultimate tensile strength;
  • modulus of elasticity;
  • fatigue behaviour;

The standard tests are defined in ISO 15630-3 [16-1].

The 0.1% proof stress is sometimes called “yield stress”.

Behaviour at extreme temperature may cover either the behaviour at low temperature (e.g. for cryogenic applications) or high temperature (e.g. behaviour in case of fire).

Coatings mainly serve as supplementary corrosion protection (e.g. zinc coating) but can also serve as main corrosion protection (e.g. epoxy coating). Some coatings modify the bond characteristics of prestressing steel.

Sheathings can either serve as corrosion protection and/or for containment of a lubricating filler inside the sheathing. For soft fillers inside the sheathing, like grease or wax, the prestressing steel remains permanently unbonded.

Resins have also been used as filler. Resins with delayed hardening can provide bond. [COMMENT 16-3]

Examples of product standards for prestressing steels are: prEN10138 [16-2], ASTM A416 [16-3], A421 [16-4], and A722 [16-5], JIS G3536 [16-6] and JIS G3109 [16-7].

- behaviour under extreme thermal conditions;
- bond characteristics;
- relaxation behaviour;
- ductility properties corresponding to the product type such as the percentage reduction of area, resistance to reverse bending and resistance to bending.

Prestressing steel can be provided with the following coatings or sheathing:

– metallic coating;

– organic coating,

– exterior sheathing with a filling product.

Prestressing steels shall comply with national or international product standards that specify their geometrical, mechanical and technological properties. Testing of prestressing steel should be carried out in accordance with national or international standards such as ISO 15630-3 [16-1].

16.1.2 Classification

The classification of the product should consist of:

– reference of the relevant product standard;

– steel designation consisting of:

- type of the prestressing steel (wire, strand, bar);
- nominal ultimate tensile strength (UTS) in MPa;
- nominal diameter;
- relaxation class;

- fatigue class;
- stress corrosion resistance class.

Information on complementary characteristics specific to a product may include:

– for wire:
  - type of surface: plain or indented (with the type of indentation);
  - type of coating (if any);
– for strand:
  - number of wires (2, 3, 7 or 19);
  - type of coating (if any);
  - type of sheathing and filler (if any);
  - type of surface: plain, indented or compacted;
  - performance level of deflected tensile test behaviour;
– for bars:
  - type of surface: plain or ribbed;
  - type of ribs: hot-rolled or cold-rolled.

Each product should be clearly identifiable with respect to this designation.

16.1.3 Provisions for quality control

The properties of prestressing steels are generally ensured by certification schemes and certification compliance. The methods of testing and of certification of conformity are defined in the relevant national or international standards or recommendations by RILEM.

The fabrication of prestressing steels shall be subject to factory production control by the manufacturer, and continuous external control by an independent qualified body, including certification and regular audits.

16.1.4 Geometrical properties

Geometrical properties are mainly nominal diameter and nominal cross-sectional area.

Products should comply with the geometrical properties specified in national or international product standards.

The difference between actual and nominal geometrical properties shall not exceed the tolerances specified in the relevant product standards.

16.1.5 Mechanical properties

16.1.5.1 General

The standard tests are defined in ISO 15630-3 [16-1].

16.1.5.2 Tensile properties

The characteristic values of

- UTS (f_{pt});
- 0.1% proof stress ($f_{p0.1}$);
- strain at maximum stress (ε_{pu});

are respectively denoted as f_{ptk}, $f_{p0.1k}$ and ε_{puk}.

16.1.5.3 Stress–strain diagram

Indicative stress–strain relations for prestressing steel in tension are represented in figure 16.1-1.

The process of manufacture of prestressing steel may influence several properties. This is particularly true for bars and the following properties:

- modulus of elasticity;
- bending;
- ratio 0.1% proof stress to UTS;
- ductility.

The UTS value expressed in MPa is often denoted as the grade of prestressing steel.

The ratio of $f_{p0,1k}$ to f_{ptk} according to prEN10138 [16-2] is:

- ≥ 85% for wires and strands;
- ≥ 80% for bars.

The ratio for bars should be declared by the manufacturer.

Some standards specify an upper limit of the tensile strength (UTS) which is about 15% above the characteristic strength. [COMMENT16-7] Such an upper limit of the strength is given to ensure a certain homogeneity between lots of prestressing steel (small variations indicating good quality control during fabrication).

Due to specific details of the manufacturing process the stress–strain relation may differ between manufacturers.

As an idealization of the actual stress–strain relation the Menegotto–Pinto formulation presented in section 15.5.2 for reinforcing steel may be used for prestressing steel as well.

The nominal value of the modulus of elasticity of the prestressing steel E_p may be taken equal to:

– 205,000 MPa for wires;

– 195,000 MPa for strands (approximately).

The value for bars should be declared by the manufacturer. This since for bars the modulus of elasticity very much depends on the manufacturing process applied.

For design purposes a nominal value of the modulus of elasticity of the prestressing steel may be assumed. For verification of tendon elongations during stressing, the value declared by the manufacturer should be used.

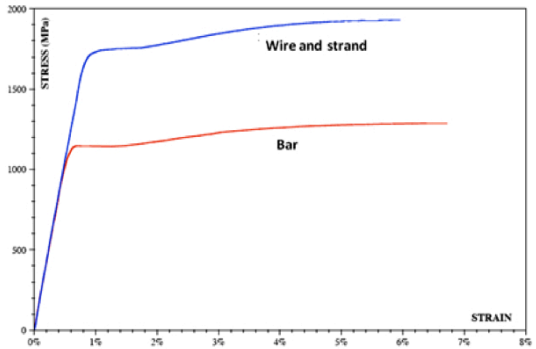


Figure 16.1-1: *Typical stress–strain diagrams for prestressing steel.*

16.1.5.4 Fatigue behaviour

The fatigue behaviour of prestressing steel is tested on straight samples in air under standard test procedures.

Other fatigue behaviour may be required if the steel is in an aggressive environment (e.g. near the sea) or at temperatures other than room temperature.

Recommendations for fatigue testing:

The frequency of load cycles should not exceed:

– 120 Hz for wires and bars;

– 20 Hz for strands.

The temperature during the test should not exceed 40°C.

The minimum fatigue stress ranges indicated in table 16.1-1 are valid for prestressing steels tested in air. Once installed as tendons, the prestressing steel is in contact with concrete/grout or metal surfaces, transverse stresses due to tendon curvature may occur, and anchorages may be provided. These conditions reduce the fatigue strength of prestressing steel (e.g. due to fretting fatigue). Therefore, this Model

Prestressing steel must comply with fatigue stress range requirements determined at $2 \cdot 10^6$ load cycles performed at an upper stress level of 70% or 80% of the UTS. The minimum fatigue stress range must comply with Table 16.1-1.

Table 16.1-1: *Minimum fatigue stress range of prestressing steels at upper stress level of 70% or 80% of the UTS [10% quantile].*

Type of steel		Stress range [MPa]
Plain wire		200
Indented wire		180
Plain strand		190
Indented strand		170
Plain bars	$d \leq 40$	200
	$d > 40$	150
Ribbed bars	$d \leq 40$	180
	$d > 40$	120

Code provides S–N curves only for tendons, see section 30.2.2, Table 30.2-2, but not for the prestressing steel in air.

See COMMENT [16-13]

It may be assumed that prestressing steels are typically suitable for use under cryogenic conditions showing a 0.1% proof stress and an ultimate tensile strength higher than at room temperature and a strain at maximum load over 2% when tested at –196°C.

Stress–strain relations are suitable for the assessment of the behaviour at cryogenic temperatures.

Reference is made to the FIP State of the Art Report [16-8] and FIP SR88/2 [16-9], *fib* Bulletin 98 [16-10].

The threshold value is in the temperature range 150°C to 250°C.

Figure 16.1-2 illustrates the actual ultimate tensile strength of prestressing steel when exposed to and tested at a given temperature. Figure 16.1-3 illustrates the residual ultimate tensile strength of prestressing steel tested at 20°C after previous exposure over 3 hours to a given temperature. Refer to Atienza et al. [16-11]. It should be noted

16.1.5.5 Coefficient of thermal expansion

Within the temperature range from −40°C to 180°C the coefficient of thermal expansion of prestressing steel may be taken as $10 \cdot 10^{-6} \mathrm{degC}^{-1}$.

16.1.5.6 Behaviour under extreme thermal conditions

The above values are valid for “normal” temperatures between −40°C and +100°C.

16.1.5.6.1 Cryogenic conditions

Prestressing steels for cryogenic conditions should be specifically ordered for this purpose. The temperature for which they are tested shall be specified.

The producer should provide test results of the prestressing steel for the specified cryogenic temperature.

16.1.5.6.1.1 High temperature

The maximum stress and the 0.1% proof stress decrease and the strain increases if the temperature increases beyond a threshold value limiting the normal range.

that exposure to temperatures higher than 20°C increases the losses due to relaxation of prestressing steel – see section 16.1.7.1.

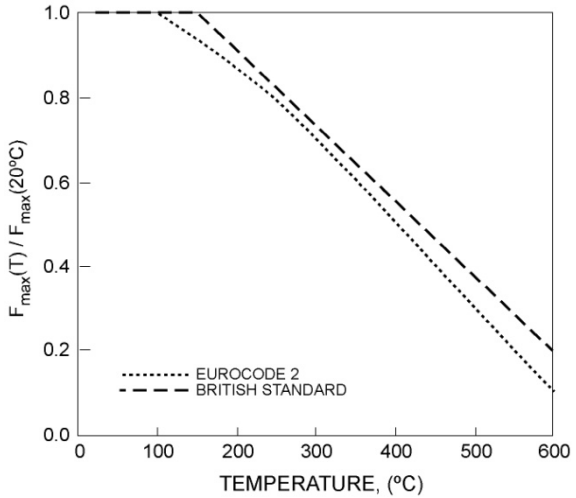


Figure 16.1-2: *Effect of temperature on the ultimate tensile strength of prestressing steel when tested at a given temperature.*

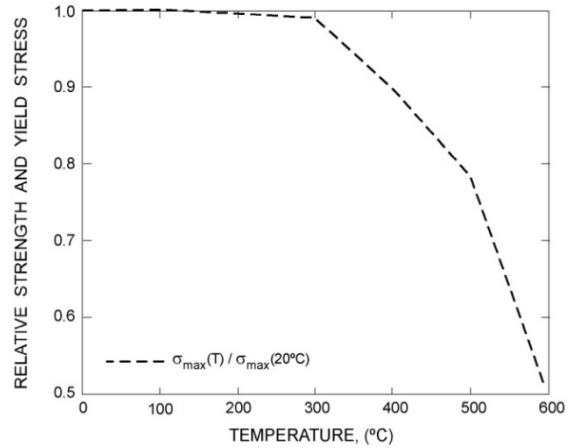


Figure 16.1-3: *Effect of temperature on the ultimate tensile strength of prestressing steel when tested at 20°C after 3 hours exposure to a given temperature.*

Tests on prestressing steel show marginal effect of strain rate on the yield and ultimate tensile strength. Reference is made to: Galvez et al. [16-12], and CEB Bulletin 187 [16-13].

Quality control tests for the bond characteristics of prestressing steel should be performed in accordance with national or international standards. Alternatively, the ASTM A1081 [16-14] may be used. Minimum bond strength should comply with relevant standards or should be specified by the engineer.

Bond behaviour and strength for design purposes are covered in section 20.2.2. Provisions for transmission and anchorage length of prestressing steel wire and strand are given in 30.13.4.2.

In prestressed concrete structures the high stress level in the wires or tendons adds to the possibility of inducing a much more rapid propagation of the cracks and lowering of the steel toughness leading to brittle failures. Stress corrosion is characterized by the coupling between the conventional localized corrosion (pitting attack in a chloride environment or local attack in carbonated concrete) and the high stress level which in some particular media induces the progress of microcracking at the steel surface, in general associated with hydrogen diffusion in the crack tip ending in metal embrittlement. Microcracking at the wires or tendons may lead to brittle failure of the prestressing steel for a level of corrosion which is hardly recognized from outside. Since this type of deterioration goes along with reduced ultimate strains, there may be no visible warning announcing tendon failure, which can strongly affect structural safety. Because pits due to chloride attack can also be considered as surface defects or notches, the evolution of SCC from pits may turn into prestressing wire failure for a low level of local corrosion. The susceptibility of prestressing steel to SCC and hydrogen embrittlement (HE) is highly dependent on the metal microstructure. Where, on the one hand, quenched and tempered steel may show poor performance, on the

16.1.5.7 Effect of strain rate

Any increase in yield and ultimate tensile strength of prestressing steel due to high strain rate should only be taken into account when tests on the particular prestressing steel confirm such improved values.

16.1.5.8 Bond characteristics

The bond characteristics of prestressing steel vary depending on the surface characteristics and the manufacturing process. For prestressing steel intended to be used in bonded conditions, the producer shall document the bond characteristics through representative quality control testing performed on each production batch.

16.1.6 Properties describing / defining durability

16.1.6.1 General

Further to the types of deterioration observed in reinforcing steel (section 15.6), where electrochemical corrosion processes can lead to homogeneous or local reductions of the cross sectional area of bars, in prestressing steel as well “stress corrosion cracking” (SCC) can occur. So, for prestressing steel the following modes of failure are possible:

- Mechanical brittle failure by surpassing the residual strength of corroded steel wires due to the loss of cross sectional area;
- Stress corrosion fracturing because of the combination of a localized attack with a high rate of stress or deformation at the defect tip;
- Fatigue failure of the corroded reinforcement (by the effects of fatigue and/or fretting) [COMMENT 16-22].

other hand cold drawn prestressing stress-relieved steels are not typically susceptible to this type of corrosion.

At the current state of research, deterioration functions to model the processes described in this section are not yet available.

The NH4SCN test (solution A of ISO 15630-3 [16-1]) is considered a suitable test for quality control of prestressing steels. It is a legitimate test when the dominant failure mechanism under service conditions is by hydrogen embrittlement.

Any new type of prestressing steel should be subject to stress corrosion testing using solution A of ISO 15630-3 [16-1] to establish minimum individual and median values of lifetime to failure for such new types of prestressing steels. Where regulations require the application of solution B for stress corrosion, the additional test solution B shall be used. In general prestressing steels should pass 2000 hours in solution B specified in ISO 15630-3 [16-1].

New types of prestressing steel should include but not necessarily be limited to different chemical composition, different (in particular higher) ultimate tensile strength, different manufacturing process and so on.

The process of manufacture may influence several properties such as:
- stress relaxation;
- stress corrosion resistance (environmental cracking).

Relaxation tests at an initial stress of 80% of the actual ultimate tensile strength and at higher temperatures than 20°C over 1000 hours may be agreed between producers and purchasers.

16.1.6.2 Stress corrosion resistance

Prestressing steels shall be subject to ongoing quality control testing for stress corrosion. The minimum individual and median values of lifetime to failure shall be determined using solution A of ammonium thiocyanate specified in ISO 15630-3 [16-1]. The values shall be in accordance with the values given in the relevant product standards.

New types of prestressing steel shall be subject to initial approval testing for stress corrosion. Stress corrosion tests are based on ISO 15630-3 [16-1] standard (solution A) and additional tests specific to national certification bodies when required.

16.1.7 Technological properties

16.1.7.1 General

The standard tests are defined in ISO 15630-3 [16-1].

16.1.7.2 Isothermal stress relaxation

The loss of stress by relaxation should be established by testing at a nominal temperature of 20°C for a period of 1000 h from an initial stress of 70% of the actual ultimate tensile strength of the prestressing steel.

In the past, mostly two classes of relaxation were used for wire and strands (very low and normal relaxation). At one stage even three classes were used (very low, low and normal relaxation).

The trend worldwide is to use mostly very low relaxation wire and strand. In Japan, wire and strand with a relaxation of 8% is also manufactured (ordinary products) which corresponds to the above-mentioned normal relaxation.

The CEB-FIP Model Code 1990 differentiated between three classes of relaxation:

– Class 1: normal relaxation characteristics for wire and strand;

– Class 2: improved relaxation characteristics for wire and strand;

– Class 3: relaxation characteristics for bars.

This Model Code considers only wire and strand with very low relaxation (MC 90 Class 2), and bars (MC 90 Class 3). For information on MC 90 Class 1 (normal relaxation), see MC 90.

The loss by relaxation increases if the temperature of the prestressing steel increases above 20°C. Figure 16.1-4 illustrates typical relaxation losses of very low relaxation wire and strand as a function of time when exposed to constant temperature.

Relaxation losses at 20°C may be conservatively assumed to be valid at lower temperatures than 20°C.

Testing at 80% initial stress should be considered if tendons in structures are intended to be stressed to initial stresses higher than 70% of the actual ultimate tensile strength.

Prestressing steels are available in different classes of relaxation. Table 16.1-2 gives values for very low relaxation wire and strands, and for prestressing bars.

Table 16.1-2: *Maximum specified values of stress loss of prestressing wire and strand with very low relaxation and for prestressing bars after 1000 h.*

Type of steel	**Initial stress**	**Specified maximum loss ρ_{1000}**
Wire/strand	70% UTS	2.5%
Wire/strand	80% UTS	4.5%
Bars ≤ 15mm	70% UTS	6%
Bars > 15mm	70% UTS	4%

Temperatures higher than 20°C accelerate the relaxation losses and increase the magnitude of relaxation loss of prestressing steel. When relevant, the relaxation losses of prestressing steel shall be determined in relaxation tests performed at the relevant temperature to which the prestressing steel will be exposed for a significant period of time.

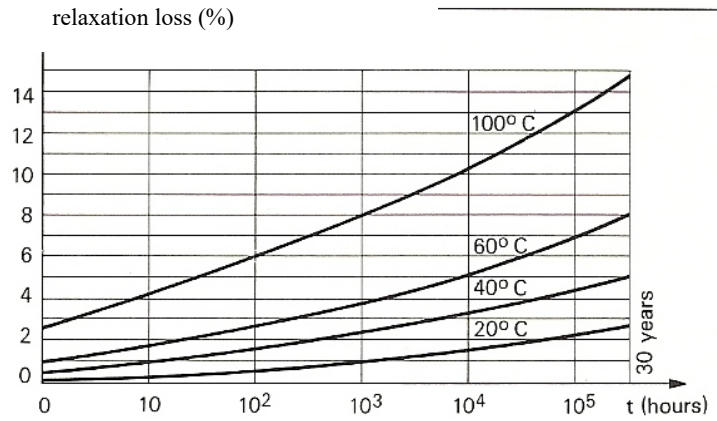


Figure 16.1-4: *Typical relaxation losses of very low relaxation wire and strand as a function of time up to 30 years when exposed to constant temperature (information from prestressing steel manufacturer)* [COMMENT 16-27].

See ISO 15630 [16-1] for the test procedure.

This test simulates the behaviour of strand in anchorages and curved ducts of post-tensioning tendons.

For details, see FIP recommendations “Deflected tensile test” (FIP, 1996).

For specific applications (stay cables or specific post-tensioning), a lower maximum reduction of 20% may be specified to the manufacturer.

Residual stresses, particularly the stresses created during cold-drawing of wires, have an influence on the shape of the stress–strain curve (i.e. the ratio of the 0.1% proof stress to UTS), on the stress relaxation losses and on cracking due to environmental influences.

Usually it is assumed that prestressing wires of very low relaxation have surface residual tensile stresses lower than 50 MPa.

16.1.7.3 Deflected tensile behaviour

The reduction of tensile strength of strands with nominal diameter ≥ 12.5 mm is verified in a test where the strand is deflected on a standardised mandrel.

For normal prestressing applications the maximum permitted reduction of the tensile strength of prestressing steel in a deflected tensile test is 28%.

16.1.7.4 Residual stresses

Residual stresses influence the technological properties of prestressing steel.

For more information see: Elices [16-15].

Prestressing steels are available with coatings and sheathing either for improved corrosion protection and/or to maintain the prestressing steel permanently unbonded inside the concrete or grout.

Metallic coating may be applied to wire, strand and bar.

Organic coating may be applied to wire, strand and bar.

See ISO 14655 [16-16] for epoxy coated strands.

The sheathing containing the filler is mostly applied to strand.

Properties of plastics and wax or grease are specified in XP A 35-037 [16-17] , NBNI10-008 [16-18], EAD-16-0004 [16-19] or EAD-16-0027 [16-20].

16.1.8 Special types of prestressing steel

16.1.8.1 General

Prestressing steels are available with coatings and with sheathing containing different fillers.

16.1.8.2 Metallic coating

This coating process is made by hot-dip immersion in a zinc bath or zinc alloy bath. It should be made before the final thermo-mechanical treatment.

Specific quality controls are specified in the relevant standards (or in approval documents) and include:

- mass per square metre of coating;
- appearance of the coating;
- continuity of the coating;
- adherence of the coating on the steel.

16.1.8.3 Organic coating

Organic coating is applied after the final thermo-mechanical treatment. Specific quality controls on the organic coating are specified in the relevant product standards.

16.1.8.4 Exterior sheathing with a filling product

Sheathing should be made of high-density polyethylene or polypropylene. Fillers may be either grease or wax.

Application of sheathing and filler is made after the final thermo-mechanical treatment of the prestressing steel.

In Japan, see "Recommendation for design and construction of partially prestressed concrete (class III of prestressed concrete) structures" by the Architectural Institute of Japan [16-21].

The strand with low-friction sheathing is typically used for internal unbonded tendons (tendons cast inside concrete) and for external tendons with ducts filled with grout. This type of strand is filled with grease.

The strand with high-friction sheathing is typically used outside of concrete for non-injected tendons such as stay cables. The sheathing in this type of applications is filled either with wax or grease. If such strand is cast into concrete or inside grouted ducts, stressing is difficult or impossible due to the adherence.

All specific tests are defined in XP A 35-037 [16-17], NBN I10-008 [16-18] or in JIS K2220 [16-22], JIS K 2246 [16-23], JIS K 2265 [16-24], and JIS K 6922-2 [16-25] and EAD-16-0004 [16-19].

Typically, a minimum thickness of the sheathing is specified.

The sliding test measures the actual adherence of the sheathing on the strand:

– for post-tensioning (low-friction sheathing);
– for stay cable (high-friction sheathing).

Static and dynamic leak tightness tests have been specified.

In some cases also an splitting (cracking) test on the sheathing is performed.

Two basic types of sheathed products are available:

– strand with a low-friction sheathing (sliding strand) filled with grease;
– strand with high-friction sheathing (adherent strand) filled with either grease or wax.

Specific controls on the sheathing and on the filling products are specified in the relevant standards (or in assessment documents) and include:

– mass of the sheathing;
– mass of the filling product;
– type of the sheathing and its geometrical characteristics;
– properties of the finished product based on the following tests:
  • sliding test;
  • bonding test under thermal variation;
  • leak tightness test;
  • impact resistance test.

16.1.9 Assumptions used for design

16.1.9.1 Dimensioning values for evaluation of structural performance

The main parameters of prestressing steel to be used for design are as follows:

– characteristic UTS, f_{ptk};
– characteristic 0,1% proof stress, $f_{p0,1k}$;
– characteristic strain at maximum stress, ε_{puk} ;
– modulus of elasticity, $E_{p;}$
– relaxation loss at 1000 hours at 70% of actual tensile strength and 20°C.

For design purposes the Menegotto–Pinto formulation given in section 15.5.2 (Figure 15.5-2) may be used. The strength and strain values must be limited to the corresponding characteristic material values.

For design a stress–strain relation of prestressing steel similar to the real behaviour, see Figure 16.1-1, but limited to a maximum stress of f_{ptk}, may be used. However, an idealized bilinear stress–strain relation, as shown in Figure 16.1-5, may also be used assuming a nominal value for the modulus of elasticity of the particular type of prestressing steel up to the characteristic yield stress, and a second straight line up to characteristic ultimate tensile strength and characteristic strain at maximum stress.

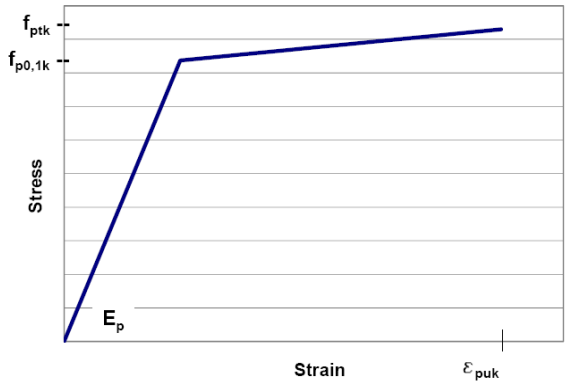


Figure 16.1-5: *Idealized stress–strain relation for prestressing steel.*

Alternatively to tests performed on the actual batch of prestressing steel, values based on documented experience of the producer for the same type of prestressing steel (size and grade) may be considered.

The relaxation loss at 1000 hours should be taken either as the maximum value specified in the relevant standard or may be taken as the average of three tests performed on the prestressing steel production batch intended to be used in the particular application. Tests should be done for a duration of 1000 hours minimum, at an initial stress of 70% of the actual tensile strength or the percentage relevant for the application, and at 20°C or at a temperature relevant for the particular application.

16.1.9.2 Design tensile strength

The design value of the yield strength of the prestressing steel follows from:

$$f_{pd} = \frac{f_{p0,1k}}{\gamma_S}$$

The behaviour of prestressing steel under tensile stresses is idealized in accordance with the stress–strain diagram given in Figure 16.1-6.

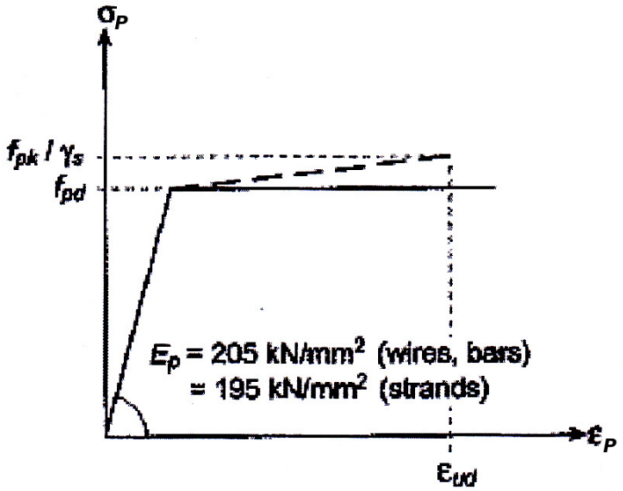


Figure 16.1-6: *Design stress strain diagram for prestressing steel.*

The dimensioning is based on the nominal cross-sectional areas of the prestressing steel.

In general, a perfectly plastic behaviour may be assumed.

For situations in which the plastic structural deformations are of importance, a linear strain hardening behaviour of the prestressing steel may be assumed. The ultimate strain must be limited to $\varepsilon_{ud} = \varepsilon_{uk}$ as specified in the relevant product standard.

In general, the structural analysis and dimensioning may be based on the mean value of the modulus of elasticity E_p according to this section.

The design value for the modulus of elasticity E_p may be assumed to be 200 000 MPa or the nominal modulus of elasticity. If more accurate values are required (e.g. for verification of tendon elongations), values from test certificates from supplier or independent laboratory should be used.

The actual value of the modulus of elasticity for wires can range from 195 to 205 MPa. For strands it can range from 190 000 MPa to 200 000 MPa. Depending on the manufacturing process and the geometrical configuration of the prestressing steel. For

bars, same range as wires for the modulus exists however, the secant modulus between 0 and 0,7f_{pk} may be as low as 170 000 MPa depending on the manufacturing process.

16.1.10 Assumptions used for assessment

16.1.10.1 Prestressing steel without deterioration or defects

Design assumptions according to section 16.1.9.2 may be used for assessment of structures with prestressing steel without deterioration or defects.

16.1.10.2 Prestressing steel with deterioration or defects

Members may be checked for the sensitivity of their behaviour / performance to assumed changes in the properties due to deterioration or defects.

Testing of samples of the prestressing steel with deterioration or defects should be performed when an effect of deterioration or defect is considered relevant for the performance of a member or structure.

16.1.10.3 Prestressing steel with noncompliant properties

Members may be checked for the sensitivity of their behaviour / performance with reinforcing steel with noncompliant properties.

Testing of samples of the prestressing steel with noncompliant properties should be performed when the noncompliance is considered relevant for the performance of a member or structure.

16.2 Prestressing systems

16.2.1 Scope definition and range of applicability

Prestressing steels are provided as wires, strands or bars.

FRP materials are provided as wires, strands, bars or plates / laminates.

The prestressing tendons considered in this Model Code are made either of prestressing steel in accordance with section 16.1, or of FRP materials in accordance with chapter 17. These tendons may be:

- internal to the concrete, and:
  - pretensioned, or

Extradosed tendons and stay cables which are partially or totally outside the outline of the structure, and prestressed ground anchors are not covered by this Model Code. See: *fib* Bulletin 89 [16-26], "SETRA Recommendations" [16-27], and FIP Recommendations for Prestressed Ground Anchorages [16-28].

Optional features of prestressing tendons may be provided such as to obtain:

- re-stressable tendons;
- exchangeable tendons (*fib* Bulletin 97 [16-29]);
- tendons for cryogenic applications (*fib* Bulletin 98[16-10]);
- fully encapsulated tendons;
- monitorable tendons.

In the absence of applicable national or international approval procedures, EAD 16-0004 [16-19] is recommended to be used. It includes detailed test procedures and acceptance criteria for verification of the system performance.

The CEN Workshop Agreement CWA 14646 [16-30] is recommended as reference for the qualification of specialist companies and for the training of supervisors and personnel. This Workshop Agreement is currently converted under CEN into a European standard.

Information relating to anchorage arrangements is given in the approval documents. When the assumptions or service conditions differ from those envisaged by these approval documents, additional experimental checks may be necessary.

There are three basic types of anchorages:

- stressing anchorages which permit stressing of the tendon;
- fixed anchorages which do not permit stressing of the tendon;

  - post-tensioned – in this case they may be bonded by grouting, or temporarily or permanently unbonded;
- external to the concrete but totally within the external outline of the structure.

Prestressing may be used for any type of structure for:

- new construction;
- repair and strengthening of existing structures.

Post-tensioning systems should comply with national or international approval procedures.

Prestressing tendons with all necessary components should be installed by qualified specialist companies with sufficiently experienced supervisors and suitably trained personnel.

16.2.2 Post-tensioning system components and materials

16.2.2.1 Anchorages and coupling devices

16.2.2.1.1 General

After hardening of the concrete, the tensile elements of tendons are tensioned, and their extremities are fixed within anchorages, which transfer the prestressing forces to the concrete.

– coupling devices to connect the end of a tendon that has been tensioned first, to a second tendon placed as an extension of the first, and which will be tensioned in a second stage.

Fixed anchorages can be mechanical devices or a tendon anchored by bond.

It may be necessary to place intermediate anchorages, functioning in both directions, or additional non-prestressed reinforcement, to reduce the risk of progressive collapse, when the strength of the structure is achieved by one set of tendons extending over many spans.

When tendons are bonded by cement grouting to the concrete structure, the transfer of the prestressing force may be ensured by bond in sections adjacent to the failure.

The deviators need to transmit both transverse (deviating) and longitudinal (friction) effects and the corresponding displacements of the tendons.

It is recommended to make special provisions at the time of design and construction for access and anchorage attachment in the diaphragms, and for deviation devices to permit future addition of external tendons. These provisions should be made for an additional prestressing force specified by the engineer.

The future addition of external tendons is intended for additional future load or deflection adjustment, if ever needed. A minimum provision for 10% of the primary prestressing force or moment capacity has been recommended by some specifications.

Relevant test procedures are specified in the FIP "Recommendations for the acceptance of post-tensioning systems" [16-31] or in EAD 16-0004 [16-19].

The performance of tendons made of prestressing steel is based on either the actual ultimate tensile strength or the strength specified in standards. A minimum elongation of 2% is specified to ensure yielding and ductility of the tendon.

With unbonded tendons (internal and external), special attention should be given in design to the potential consequences of an accidental tendon failure, because the tendon force is lost over the entire tendon length.

With external prestressing, deviating devices are placed between the tendons and the structure, to deflect the tendon as needed. These devices and their fixing zones, should be designed to transfer the corresponding design actions, taking the permissible installation tolerances into account.

With external prestressing, provision should be made for the future replacement of the prestressing tendons.

16.2.2.1.2 Anchorage and coupling device performance

Post-tensioning tendon anchorages and coupling devices should have the following minimum performance in the specified tests:

(a) Tendons made of prestressing steel:

– in the tensile test, achieve not less than 95% of the actual tensile strength of the prestressing steel with an elongation under maximum load of not less than 2%;

– in the fatigue test, have not more than 5% of the tendon cross-section fail over $2 \cdot 10^6$ load cycles, for a stress range of 80 MPa at an upper stress of 65% of the tensile strength of the prestressing steel;

Since the material properties for FRP tendons are not yet standardized, the anchorage performance of tendons made of FRP materials is based on the tensile strength declared by the supplier, which takes into account:

- the short term tensile strength of the tendon for the given tendon/anchorage assembly;
- the level of strength retention during service life, whereas the residual strength of the tendon for the permissible permanent prestress and the given tendon/anchorage assembly is at least 95% of the short term tensile strength.

Since FRP materials do not yield, the minimum elongation of the tendon at maximum load must be declared by the supplier and be used as reference for acceptance.

Design of structures with FRP tendons must be based on a permissible permanent prestress, depending on the characteristic long term strength of the FRP tendon. Any applicable detrimental environment, like exposure to temperature humidity and alkalinity, must be taken into consideration. In addition the permissible permanent prestress has to fulfil the strength retention criterion mentioned above.

Fracture toughness properties of new materials used for anchor heads and coupling devices without prior experience for this application should be assessed. Susceptibility to environmentally assisted cracking for the particular exposure conditions of these materials should be ascertained.

- in the load transfer test, anchorage zone resist to not less than 110% of the specified tendon strength at the specified minimum concrete strength for stressing to the maximum force.

(b) Tendons made of FRP materials:

- in the tensile test, should achieve not less than 95% of the declared tensile strength of the FRP tendon and a minimum elongation at maximum load which is equal to or exceeds the value declared in the system documentation;
- in the fatigue test, shall have not more than 5% of the tendon cross-section fail over $2 \cdot 10^6$ load cycles, for a stress range of 80 MPa (or the value specified by the designer) at an upper stress equal to the permissible permanent prestress of the FRP tendon;
- in the load transfer test, the anchorage zone should resist to not less than 110% of the declared tendon strength at the specified minimum concrete strength for stressing to the maximum force.

Anchorages must be made of materials suitable for this purpose.

16.2.2.1.3 Technological aspects

Friction loss characteristics of the tendons inside the anchorages and coupling devices, and tendon seating loss characteristics in the anchorages and coupling devices, should be declared in the post-tensioning system documentation, and in technical approval documents where these exist.

16.2.2.2 Ducts

16.2.2.2.1 General

Depending on the intended use, ducts for prestressing systems have to provide one or several of the following features:

- forming a cavity in the structure for the installation of the tensile elements and defining the tendon path in the structure for internal tendons;
- forming a conduit for the installation of the tensile elements and deviating the tendon at specified locations in the structure for external tendons;
- providing an interface suitable for the transfer of bond stresses from the tensile elements to the structure for bonded tendons;
- providing a low friction interface suitable for unbonded tendons;
- providing corrosion protection to the tensile elements if made from a non-corrodible material and forming a leak tight envelope.

When FRP tendons are used as external tendons, they are often applied without ducts.

Other types of ducts may be considered if they provide the required features and if they offer equivalent performance to corrugated metal and polymer ducts or to smooth steel and plastic pipe, as may be applicable. The formation of deleterious galvanic cells between the tensile elements and the duct material should be avoided.

Ducts for prestressing tendons typically are either:

- corrugated metal ducts;
- corrugated polymer ducts;
- smooth steel pipes;
- smooth polymer pipes.

16.2.2.2.2 Performance of ducts

Ducts should be designed, fabricated and installed such that they are fit for the intended purpose and that they have the expected durability.

Performance requirements for ducts, testing procedures and corresponding acceptance criteria may be found for the different types of ducts in:

- corrugated metal ducts: EN 523 [16-32] and EN 524 [16-33];
- corrugated polymer ducts: *fib* Bulletin 75 [16-34];
- smooth steel pipe: ISO 4200 [16-35];
- smooth plastic pipe: EAD 16-0004 [16-19].

The cross-sectional area of the ducts should normally be in the range of 2.0 to 2.5 times that of the actual area of the tensile elements, depending on the length and geometry of the tendon as well as on the installation method used (pushing strand by strand versus pulling the complete bundle). The above recommendations correspond to duct filling ratios between 0.4 and 0.5. Smaller values of degree of filling, 0.35 to 0.40, are recommended for long tendons or if the tensile elements are installed after concreting.

The diameter of ducts for external unbonded post-tensioning should be generally of the same size as for internal bonded since the requirement for complete filling of the duct is the same.

Relevant properties of ducts may include:

- range of recommended friction and wobble coefficients;
- bond characteristics;
- minimum duct wall thickness;
- permissible minimum radius of tendon curvature (for wear resistance);
- recommended support spacing and details;
- ageing characteristics/durability of polymer materials (e.g. minimum oxidation induction time).

16.2.2.2.3 Technological aspects

The relevant properties of ducts should be declared in the system documentation and in technical approval documents where these exist.

16.2.2.3 Filling materials

16.2.2.3.1 General

Depending on the intended use, filling materials for prestressing tendons have to provide one or several of the following features:

- permanent corrosion protection to the tensile elements;
- bond to the tensile elements and transferring bond stresses to the duct and structure for bonded tendons;

– lubrication between the tensile elements and the duct to permit stressing of tendons;
– lubrication between the tensile elements and the duct and/or structure of permanently unbonded tendons.

Filling materials typically are either:

– cementitious grout;
– grease;
– wax;
– resin.

Resins and other filling materials may be considered if they provide the required features and if they offer equivalent performance to cementitious grout, grease or wax.

FRP tendons often use resins for filling and/or bonding to the structure.

More recently also gel has been used as filling material.

16.2.2.3.2 Performance requirements

Filling materials should be designed, mixed and installed such that they are fit for the intended purpose and that they have the expected durability.

Performance requirements for filling materials, testing procedures and corresponding acceptance criteria may be found for the different types of materials in:

– cementitious grout: *fib* Bulletin 20 [16-36], PTI Specification for grouting of post-tensioned structures [16-37], EN 445-447 [16-38][16-39][16-40], ISO 14824-1 to 3 [16-41][16-42][16-43].
– grease, wax: EAD 16-0004 [16-19], FIP Recommendations for corrosion protection of prestressing steel [16-44].

16.2.2.3.3 Technological aspects

Relevant technological aspects of the filling materials should be declared in the system documentation as required and in technical approval documents where these exist.

16.2.2.4 Quality control

The fabrication of post-tensioning system components and materials should be subjected to factory production control adapted to the importance and criticality of the particular component and material.

Guidance for a suitable quality control of post-tensioning system components and materials may be found in EAD 16-0004 [16-19].

The factory production control should be subjected to continuous internal control by the manufacturer, and continuous external control by an independent qualified body, including regular audits.

16.2.3 Protection of tendons

16.2.3.1 Temporary corrosion protection

Tensile elements and anchorages in all pretensioned and post-tensioned applications should be given suitable protection during fabrication, transport and storage on site. If the period between installation into the tendon and application of permanent corrosion protection exceeds values recommended for the expected environmental conditions and exposure, temporary corrosion protection should be provided.

Temporary corrosion protection may consist of suitable water-soluble oils. Attention should be paid that these water-soluble oils do not unacceptably reduce the bond characteristics of the tensile elements intended to be used in bonded tendons.

16.2.3.2 Permanent corrosion protection

Tensile elements and anchorages for all pretensioning and post-tensioning tendons should be given a suitable permanent corrosion protection adapted to the intended use, the expected environmental conditions and exposure, and the specified design life of the structure in which the tendons are placed.

The protection provided by the structure and/or other protective systems on the structure may be considered for the required level of protection on the tendon itself.

Surfaces of metal components exposed to the environment should be given a suitable corrosion protection. The required maintenance procedures and intervals should be specified in the project specification.

Guidance for the actual requirements, test procedures and acceptance criteria for different protection levels (PL1, PL2 and PL3) adapted to the protection provided by the structure may be found in *fib* Bulletin 33 [146].

[COMMENT 16-56]

For internal and external post-tensioning tendons the following solutions may, for example, be considered to provide the required protection:

– PL1: tendon with a metal duct and a cementitious filling material providing durable corrosion protection;
– PL2: tendon with PL1 plus a polymer envelope, enclosing the tensile element bundle over its full length (including the anchorages), and providing a permanent leak tight barrier;
– PL3: tendon with PL2 plus the integrity of tendon or encapsulation to be monitorable or inspectable at any time.

For pretensioning tendons the following solutions may, for example, be considered to provide the required protection:

– PL1: tendon with sufficient concrete cover adapted to the environmental conditions and exposure;
– PL2: tendon with PL1, and additionally protected with a suitable permanent coating and special protection of tendon ends;
– PL3: tendon with PL2, plus a suitable corrosion monitoring system.

FRP materials are not susceptible to classical corrosion and are generally resistant against a wide range of aggressive environments. In particular cases, certain environmental effects such as UV radiation and alkalinity may be of concern either to the resin or the fibres. Typical influences such as temperature, humidity and alkalinity should be included in the declaration of the supplier, who should make reference to independent test data.

16.2.3.3 Permanent corrosion protection of prestressing steel

Three protection levels for tendons are recommended, see Figure 16.2-1 [COMMENT 16-55]:

– PL1 for all tendons used in environments which have a relatively low aggressiveness and which are well protected by the structure;
– PL2 for all tendons used in all other combinations of environments and/or exposure, and protection provided by the structure not included in protection levels PL1 and PL3;
– PL3 for all tendons used in aggressive environments and/or severe exposure and with low protection provided by the structure.

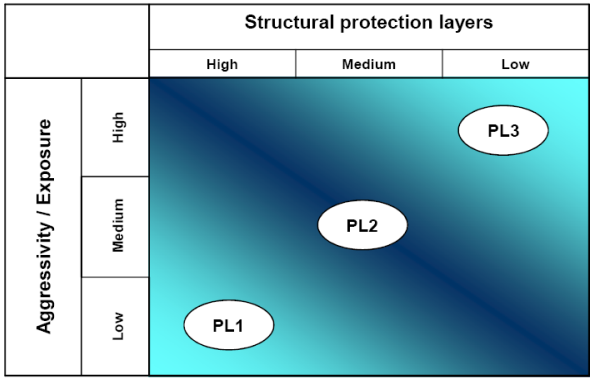


Figure 16.2-1: Tendon protection levels.

16.2.3.4 Permanent protection of FRP materials

FRP tendons should be given a suitable permanent protection adapted to the intended use, the expected environmental conditions and exposure, and the specified design life of the structure in which the tendons are placed.

Fire protection is typically provided by sufficiently thick cover with a suitable material or by supplementary protection to the member.

Resins in FRP tendons are particularly susceptible to the effect of fire.

16.2.3.5 Fire protection

Tensile elements and anchorages for all pretensioning and post-tensioning tendons should be given a suitable protection against the effect of fire, adapted to the intended use and exposure, and the specified fire rating of the structure or element in which the tendons are placed.

16.2.4 Stresses at tensioning, time of tensioning

16.2.4.1 Time of tensioning

Early application of prestress may prevent or reduce cracking of concrete due to shrinkage and temperature effects.

The minimum concrete strength required at the time when tensioning takes place depends mainly on the design of the anchorage, the provided local anchorage zone reinforcement, the edge distance of the anchorage and the spacing between adjacent anchorages.

The time when prestressing takes place should be fixed with due regard to:

- deformation conditions of the structure;
- safety with respect to local stresses and the compressive strength of the concrete;
- early application of a part of the prestress to reduce shrinkage effects.

The minimum concrete strength required at the time when tensioning takes place should be given in the approval documents and/or system documentation for the prestressing system concerned, and should be specified in the project specifications.

16.2.4.2 Tendons made from prestressing steel

The specified maximum force applies to the part of the tendon between end anchorages; that is, it does not apply to the part of the tendon in the stressing devices which do not form part of the permanent tendon.

In exceptional cases in post-tensioning where unforeseen deviation of frictional behaviour on the site occurs, it may be impossible to obtain the needed prestressing force under the limitations specified in this section. In such exceptional cases it is possible, if the actual tensile elements and prestressing technique allow it, to apply a higher stress at the end of the tendons. This stress should never exceed the value of $0.95\, f_{p0.1k}$ – see section 36.2.3.2.

The maximum tensile stress in the tendons made from prestressing steel, both pretensioning and post-tensioning, should generally not exceed the lower of the following values before transfer of prestressing to the concrete:

$$\sigma_{p0,\max} = 0.80\, f_{ptk} \qquad (16.2\text{-}1)$$

$$\sigma_{p0,\max} = 0.90\, f_{p0.1k} \qquad (16.2\text{-}2)$$

The maximum tensile force in the tendons, both pretensioning and post-tensioning, should generally not exceed the lower of the following values after transfer of the prestressing to the concrete:

$$\sigma_{p0,max} = 0.75\, f_{ptk} \quad (16.2\text{-}3)$$

$$\sigma_{p0,max} = 0.85\, f_{p0.1k} \quad (16.2\text{-}4)$$

16.2.4.3 Tendons made from FRP materials

The maximum tensile force in the tendons made from FRP materials should generally, after transfer of prestressing to the concrete, not exceed the permissible permanent prestress declared by the supplier, with consideration of stress rupture:

$$F_{p0,max} = \alpha\ F_{pkT} \quad (16.2\text{-}5)$$

F_{pkT} is the characteristic long term tensile strength of the tendon made of FRP for the declared design life of the tendon. It considers stress rupture for the applicable tendon/anchorage assembly and detrimental environment. It considers the strength retention criterion of section 16.2.2.1.2(b). Indicative values for the permissible long term stress of FRP are given in section 17.5.1.

The coefficient α should be declared by the supplier of the FRP tendons; α is expected to be on the order of 0.75.

16.2.5 Initial prestress

16.2.5.1 General

The value of the initial prestressing force (at time $t = 0$) at a given section of abscissa x along the tendon, is obtained by subtracting from the force at tensioning the different immediate losses described below.

The initial prestress (at time $t = 0$) is calculated taking into account the prestressing force and the permanent actions present at tensioning.

16.2.5.2 Losses occurring in pretensioning beds

The following losses should be considered in design:

- loss due to friction at the bends (in the case of curved tensile elements);
- losses due to seating of the anchoring devices (at the abutments) when anchoring on a prestressing bed;
- loss due to relaxation of the tensile elements during the period which elapses between the tensioning of the tendons and transfer of prestressing to the concrete.

Friction losses for deviated pretensioning tendons can be calculated similarly to post-tensioning tendons but neglecting the unintentional angular displacements, see section 16.2.5.3.

16.2.5.3 Immediate losses occurring during stressing

All values given in Table 16.2-1 below should be considered as indicative mean values.

The coefficient of friction μ is the product of the physical coefficient of friction μ_0 and the squeezing factor. This squeezing factor is dependent on the degree of filling of the duct. Where more exact investigations are not available, this squeezing factor can be assumed to be 1.3 to 1.35 for tendons filling the duct between 50% and 60%. The physical coefficient of friction μ_0 is influenced *inter alia* by the surfaces of tensile elements and ducts (micro- and macro-structures), rust, pressure, elongation of the tendon etc.

If more accurate values are not available and in the case of tensile elements and duct being both without rust, the values given below can be assumed, for μ with up to 50% filling of ducts.

Under site conditions, significant variations are possible. In the case of rust, variations of 50% or even higher may occur.

For the verification of the real values of prestressing losses at tensioning it is recommended to measure the transmission of prestressing force from one end of the tendon to the other (e.g. with lift-off testing at the dead-end anchorage) on some typical and/or critical tendons.

For external tendons, the effect of unintentional angular displacement may be neglected.

(a) Friction losses in case of bonded internal tendons made from prestressing steel

Table 16.2-1: *Typical friction coefficients μ for different types of prestressing steels and ducts.*

16.2.5.3.1 Losses due to the instantaneous deformation of concrete

Account should be taken of the loss in tendon force corresponding to the deformation of concrete:

- in the case of post-tensioning tendons, taking into account the order in which the tendons are stressed;
- in the case of pretensioning tendons, as a result of their action when they are released from the anchorages.

16.2.5.3.2 Losses due to friction (post-tensioning tendons)

In a cross-section which is at a distance x from a stressing anchorage, the stress $\sigma_{p0}(x)$ in the tendon being tensioned is lower than the stress at the anchorage $\sigma_{p0,max}$. The difference between these two stresses corresponds to the losses due to friction:

$$\sigma_{p0}(x) = \sigma_{p0,max} e^{-\mu(\alpha_\mu + k_\mu x)} \qquad (16.2\text{-}6)$$

where:

μ is the coefficient of friction between the prestressing steel and the duct;

α_μ is the sum of the angular displacements over a distance x, irrespective of direction or sign [radians];

k_μ is an unintentional angular displacement (per unit length) depending on the design layout (shape) of the tendon, stiffness of duct and spacing of duct supports [radians/m].

Values for μ and k_μ should be declared in the system documentation for the particular tendon details and deduced from previous experience or testing with the same type of materials and construction. These values μ and k_μ should be declared in technical approval documents where these exist.

With external prestressing, the friction is concentrated at deviation devices.

Type of prestressing steel	Corrugated metal duct	Corrugated plastic duct	Smooth steel pipe	Smooth plastic pipe
Cold-drawn wire	0.14–0.18	0.08–0.12	0.25–0.30	0.08–0.12
Strand	0.16–0.20	0.10–0.14	0.25–0.30	0.10–0.14
Ribbed bar	0.63–0.68			
Smooth and round bar	0.30–0.35			

The coefficient k_{μ} takes account of unintentional angular deviations. k_{μ} is also called the wobble coefficient. Its value depends on the quality of workmanship, stiffness of ducts and on the distance between supports of the tendon. Values for k_{μ} can be found in approval documents. The typical range may be assumed to $k_{\mu} = 0.005–0.01$ (m^{-1}). For internal tendons in precast segmental construction higher values for k_{μ} are recommended to be assumed in design to allow for unintended angular deviations (kinks) of ducts fixed to the formwork at the segment joints.

(b) Friction losses in the case of unbonded internal tendons made from prestressing steel

Tests and practical experience have shown that the friction coefficients μ and k_{μ} as listed below can be applied.

– For monostrands (individually greased and plastic sheathed strands; single or grouped): $\mu = 0.05–0.07$ $k_{\mu} = 0.006–0.01$ m^{-1}

– For multistrand or multiwire tendons inside plastic pipe and filled with grease: $\mu = 0.12–0.14$ $k_{\mu} = 0.004–0.008$ m^{-1}

– For dry multistrand or multiwire tendons (with dry air as subsequent corrosion protection) factors as for bonded internal tendons apply.

(c) Friction losses in the case of external tendons made from prestressing steel

– For bare dry strands or wires over steel saddle: $\mu = 0.25–0.30$ $k_{\mu} = 0$

– For lubricated strands or wires over steel saddle: $\mu = 0.20–0.25$ $k_{\mu} = 0$

– For dry strands or wires inside plastic pipe over saddle: $\mu = 0.12–0.15$ $k_{\mu} = 0$

– For bundle of monostrands (individually greased and plastic sheathed strands) over saddle: $\mu = 0.05–0.07$ $k_{\mu} = 0$

These values apply for saddle radii as given in section 16.2.9. For lower radii further test evidence may be needed.

(d) Friction losses in the case of tendons made from FRP materials

– Friction and wobble coefficients declared by the supplier of the FRP tendons should be considered for the calculation of immediate losses during stressing.

16.2.5.3.3 Losses caused by seating of the tensile elements

Seating of the prestressing steel in the anchorage causes a shortening (negative elongation) of the tendon with a corresponding loss of tendon force in the vicinity of the anchorage.

The values for seating to be taken into consideration are defined in the system documentation and/or approval documents for the prestressing system concerned.

Due to this seating, the highest stress along the tendon is no longer at the anchorage or tendon end.

Account should be taken of the loss which occurs during seating at the anchorages of post-tensioning tendons, that is during the operation of anchoring after tensioning.

Transfer of the prestressing force to the concrete in pretensioning tendons produces a loss in force in the tensile elements over the transfer length.

16.2.5.3.4 Effect of heat treatment curing

Heat treatment is mostly used in precasting, and therefore, applies particularly to precast-pretensioned applications. The typical procedure exposes the prestressing steel during a well-defined duration of some hours to elevated temperatures (typically less than 80–90°C).

For typical procedure the loss of prestress due to relaxation during the heat treatment may be equated to 75% of the total value of relaxation losses. The final loss of relaxation is slightly increased only. Figure 16.2-2 illustrates the effect of a typical heat treatment on very low relaxation and normal relaxation strand.

Two types of losses should be taken into account:

– reduction of stress in the tensile elements due to an acceleration of relaxation during heat treatment;
– direct thermal effect.

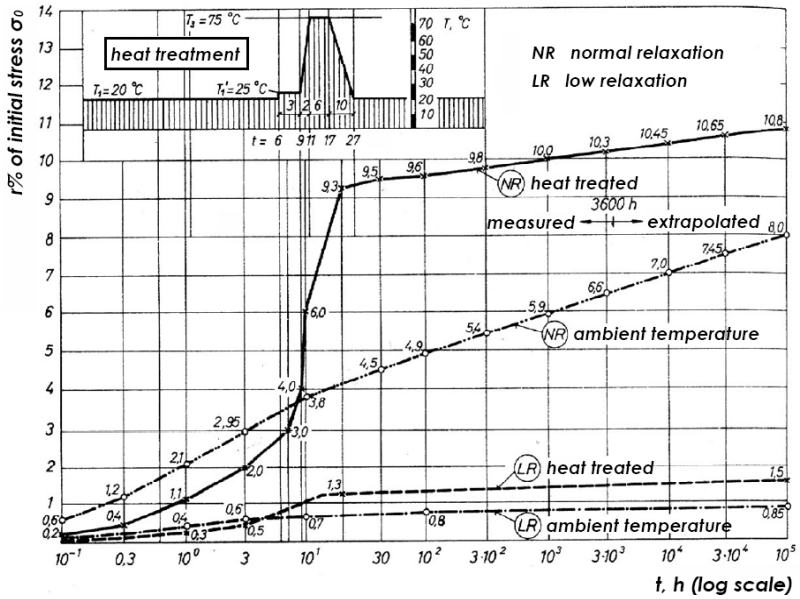


Figure 16.2-2: *Effect of typical heat treatment on relaxation losses of strands stressed initially to 80% of actual tensile strength (Ref:* Personal correspondence with A. Erdélyi*).*

(a) Relaxation losses

For non-typical heat treatment procedures, relaxation losses may be estimated by adding to the value of time a duration t_{eq} defined by:

$$t_{eq} = t_{p1}\ 1.14^{(T_{max}-20)} \qquad (16.2\text{-}7)$$

where:

T_{max} is the maximum temperature of the concrete during heat treatment in °C;

t_{p1} is the mean duration of the heating cycle, calculated by:

$$t_{p1} = \left(T_{max} - 20\right)^{-1} \int_0^{t1} \left[T(t) - 20\right] \mathrm{dt} \qquad (16.2\text{-}8)$$

where:

t_1 is the age of the concrete when its temperature returns to ambient temperature;

$T(t)$ is the temperature of concrete, in °C, at time t.

An overall increase of final relaxation may have to be considered depending on the duration of exposure to elevated temperature.

(b) Losses of direct thermal origin

Direct thermal effect is caused by:

– the dilation of concrete, when it is not bonded to the tensile elements;

– the restraint to the dilation of concrete presented by the tensile elements when they are bonded.

This type of loss does not exist with moulds supporting the tension of tendons and heated together with concrete.

The losses of direct thermal origin may be calculated by:

$$\Delta\sigma = \kappa E_p \alpha_p (T_{max} - T_0) \qquad (16.2\text{-}9)$$

where:

E_p is the elastic modulus of tensile elements;

α_p is the coefficient of thermal expansion of tensile elements;

T_0 is the temperature of tensile elements at tensioning;

T_{max} is the maximum temperature of tensile elements during heat curing;

κ is a coefficient which depends on effective bond of tendon to fresh concrete and the period of time before heat treatment starts. $\kappa = 0.9$ may be assumed conservatively. However, lower values have been measured down to $\kappa = 0.65$.

16.2.5.3.5 Other immediate losses

Account should be taken of all possible causes of immediate loss of tendon force related to the tensioning process or the equipment used for tensioning.

Data for calculation of the deformations of concrete under creep and shrinkage are given in section 14.6.1.6.4.

Ordinary reinforcement has an influence on the value of time-dependent shortening of concrete. The interaction may be estimated as described in CEB Bulletin 199 [16-45].

The reduction of strain in tensile elements due to time-dependent losses may be calculated by dividing the stress loss by the modulus of elasticity of tensile elements.

Basic data on the relaxation of tensile elements are given in section 16.1.7 for prestressing steel and section 17.6.3 for FRP materials.

16.2.6 Value of prestressing force during design life (time $t > 0$)

16.2.6.1 Introduction

The initial prestressing force in a tendon is the force existing in the tendon at the end of the stressing operation. The initial prestressing force on a prestressed element is obtained by considering all the forces existing in the tendons, at the end of the last stressing operation.

The prestressing force at a given time t is obtained by subtracting from the initial prestressing force the value of the time-dependent losses at this time t.

These losses are due to creep and shrinkage of concrete and relaxation of tensile elements.

16.2.6.2 Calculation of time-dependent losses made of prestressing steel

16.2.6.2.1 General

The time-dependent losses should be calculated by considering the following two reductions of stress:

- the reduction of stress, due to the reduction of strain, caused by the deformation of concrete due to creep and shrinkage, under quasi-permanent actions:
  - for bonded tendons, the local deformation at the level of the tendon has to be considered;
  - for unbonded tendons, the deformation of the whole structure between the anchorages of the tendons has to be taken into account;
- the reduction of stress within the tensile elements due to the relaxation of the prestressing steel under tension.

For strand, the difference between actual and specified tensile strength is in the order of a few percent, typically below 5%.

For initial stress values below 70% of ultimate tensile strength, a straight line through the 70% point may be assumed for interpolation with zero relaxation at an initial stress of 50% of specified ultimate tensile strength.

Test results over extended periods of time have confirmed that a linear relationship in a bi-logarithmic presentation between relaxation loss and time closely represent the actual behaviour. Figure 16.2-3 illustrates relaxation measurements made over a period of more than 40 years by: Müller and Zetterholm [16-46].

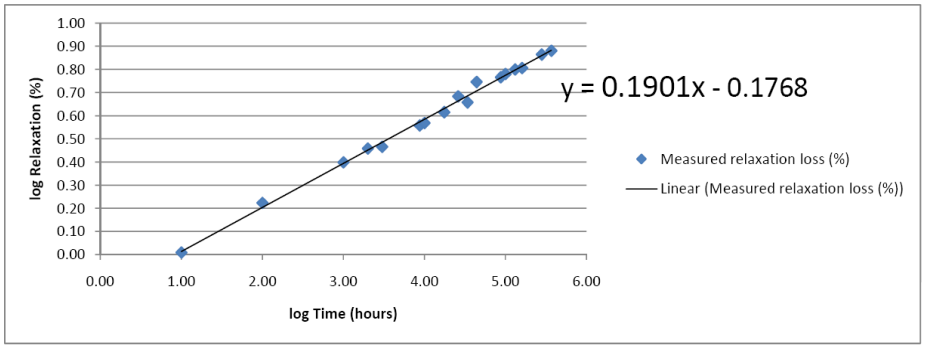
16.2.6.2.2 Effect of initial stress on relaxation loss

Maximum relaxation losses at 20°C and for a duration of 1000 hours are specified in relevant standards. Values for prestressing steel at initial stresses of 70% and 80% of actual ultimate tensile strength are given in section 16.1.7.1. For design purposes, these values may be assumed conservatively to apply at 20°C for initial stresses of 70% and 80% of the specified ultimate tensile strength of prestressing steel.

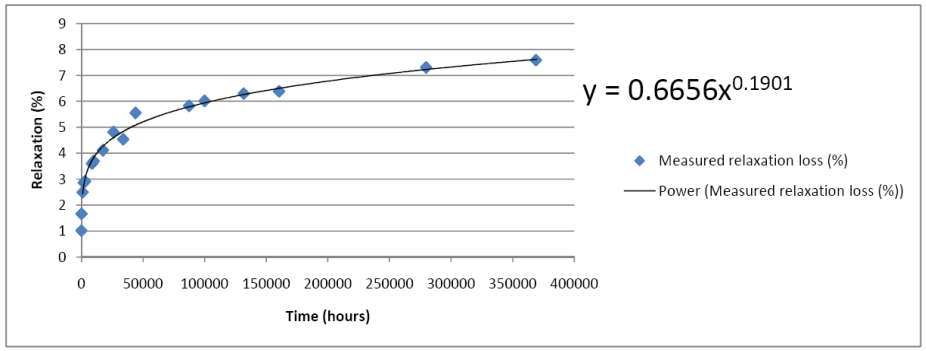
For initial stress values between 70% and 80% of tensile strength, the relaxation loss at 1000 hours may be estimated based on linear interpolation.

16.2.6.2.3 Effect of time on relaxation loss

For design purposes, a straight line relationship may be assumed in a bi-logarithmic presentation between relaxation loss and time.



(a) *Bi-logarithmic presentation of relaxation loss.*



(b) *Presentation of relaxation loss (power function).*

Figure 16.2-3: *Results of long duration relaxation measurements (Müller, Zetterholm [16-46]).*

Level I approximation typically applies at the time of design of structures when the producer of the prestressing steel, and hence the actual relaxation behaviour of the prestressing steel, is not known yet. For level I approximation the following approximation may be used for the relationship of relaxation and time:

– straight line or power line curve fitting through the specified relaxation loss at 1000 hours.

The final value of relaxation loss may be assumed to be reached at the end of the specified design life.

For design purposes the following level I and level II approximations for relaxation losses may be used:

– Level I approximation shall be applied when the assumed relaxation losses are not confirmed by test results for the actual prestressing steel;

For level I approximation of the loss due to relaxation, the following equation may be applied:

$$\rho_t = \rho_{1000} \, (t / 1000)^k \qquad (16.2\text{-}10)$$

where:

ρ_t is the relaxation after t hours;

ρ_{1000} is the specified relaxation after 1000 hours (Table 16.1-2);

k ≈ 0,16 (CEB Bulletin 213/214 [16-47]);

t time after tensioning in hours.

Level II approximation applies when the producer of the prestressing steel is known and relaxation test results of the actual batch of prestressing steel are available. For level II approximation the following approximation may be used for the relationship of relaxation and time:

(a) straight line best fit of actual test results up to at least 1000 hours in bi-logarithmic presentation, or

(b) power line best fit of actual test results up to at least 1000 hours in presentation of relaxation loss versus time:

(ii) $\log \rho_t = a \log t + \log b$

(ii) $\rho_t = b \, (t)^a \qquad (16.2\text{-}11)$

where a, b are coefficients from best fit curves.

– Level II approximation may be applied when the assumed relaxation losses are confirmed by test results for the actual batch of prestressing steel intended to be used. Level II approximation has to be determined as the best fit curve of actual test results at a given initial stress and temperature with a duration of at least 1000 hours.

16.2.6.2.4 Effect of temperature on relaxation loss

See Figure 16.2-4 for the typical effect of temperature on relaxation losses of prestressing steel. As shown, relaxation losses are accelerated, and overall magnitude of relaxation loss is increased when the temperature is higher than 20°C over an extended period of time.

See de Halleux 1970 [16-48] and de Halleux 1974 [16-49].

In a first approximation, the slope of the straight line best fit curves remains approximately constant for different temperatures. However, the curves for temperatures higher than 20°C shift upwards along the ordinate (Figure 16.2-4). This can be considered by an increase of the relaxation loss with an amplification factor, AF (Figure 16.2-5).

Relaxation losses for prestressing steel are specified in the relevant standards at a constant temperature of 20°C. For lower temperatures, these values may conservatively be assumed to apply.

In cases where tendons are exposed over a significant period of time to temperatures higher than 20°C, relaxation losses should be considered to develop more rapidly and increase in magnitude when compared to losses at 20°C.

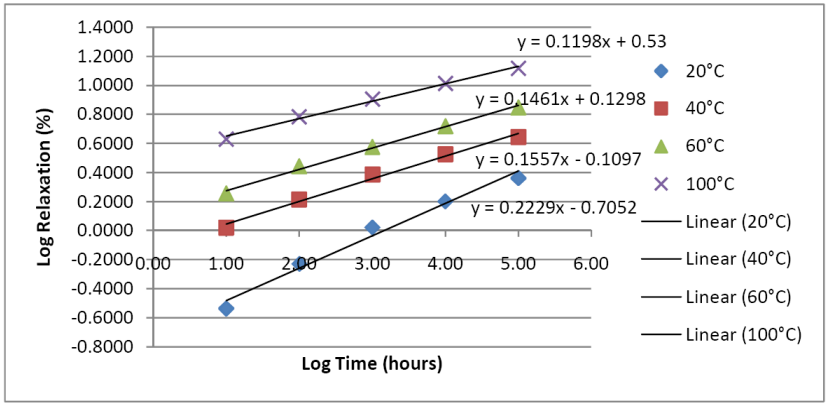


Figure 16.2-4: *Effect of temperature on very low relaxation prestressing wire and strand (data from Figure 16.1-4 shown in bi-logarithmic presentation).*

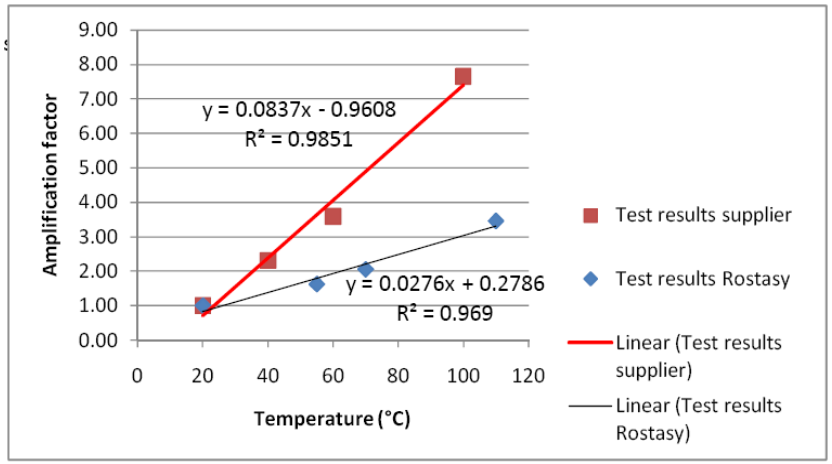


Figure 16.2-5: *Amplification factor for ρ_{1000} (T) for temperatures T > 20°C ("Test results supplier" = data from Figure 16.2-4; "Test results Rostásy" = data from Rostásy et al. [16-50]).*

For level I, the following approximation may be used for the relationship of relaxation and time:

$$\rho_t(T) = AF \times \rho_{1000} (t / 1000)^k \qquad (16.2\text{-}12)$$

where:

For design purposes the following level I and level II approximations for the effect of constant temperature on relaxation may be used:

– Level I approximation applies when no test results at the given constant temperature for the actual prestressing steel are available;

$\rho_t\,(T)$ is the relaxation after t hours at temperature $T > 20°C$;

ρ_{1000} is the specified relaxation after 1000 hours (Table 16.1-2);

AF is the amplification factor to account for the effect of $T > 20°C$ [COMMENT 16-67]:

$$AF = T / 20°C \qquad (16.2\text{-}13)$$

k $\approx 0{,}16$ as per section 16.2.6.1.2.

It should be noted that the above suggested amplification factor, AF, is not necessarily conservative. Hence, whenever temperature is significantly above 20°C over extended periods of time, and relaxation losses are important for the performance of the structure or member, it is strongly recommended to perform relaxation tests at the particular expected temperature.

For level II the following approximation may be used for the relationship of relaxation and time:

(a) straight line best fit of actual test results at the relevant constant temperature up to at least 1000 hours in bi-logarithmic presentation, or

(b) power line best fit of actual test results at the relevant constant temperature up to at least 1000 hours in presentation of relaxation loss versus time:

$$\log \rho_t\,(T) = a \log t + \log b$$

$$\rho_t\,(T) = b\,(t)^a \qquad (16.2\text{-}14)$$

where both a and b are coefficients from best fit curves as a function of the temperature T.

With at least one set of test results to at least 1000 hours at a constant temperature $T \geq 40°C$ a straight line approximation for the amplification factor, AF, may be created passing through AF = 1 at $T = 20°C$ and AF determined at $T \geq 40°C$.

The effect of elevated constant and variable temperatures has been described for example by Rostásy et al. [16-50].

– Level II approximation applies when test results either at the given constant temperature T or at one temperature level $T \geq 40°C$ for the actual prestressing steel are available.

Effects of variable temperature on relaxation losses should be suitably considered where relevant.

16.2.6.3 Calculation of time-dependent losses made of FRP

Relaxation losses of tendons made of FRP should be considered similarly to the methods described in section 16.2.6.2. A level II approximation should generally be used.

In exceptional cases, several sets of prestressing tendons (practically never more than two) should be considered separately. These cases should be identified by judgement. The criteria, to be simultaneously satisfied, are that:

- the effects of the two sets are of contrary senses;
- these effects have the same order of magnitude;
- the dispersions are relatively high and there are qualitative reasons why they should not be considered as correlated.

Some standards use factored/reduced effects of prestressing for fatigue design. However, whether or not to factor the effects of prestressing depends on the overall fatigue design concept.

In cases where the design value of the prestressing force influences the behaviour of the structure in a significant over-proportional way, the designer may consider providing in the design the possibility for one or several of the following options:

- provide access at both tendon ends to allow stressing of the tendons from both ends;
- provide spare anchorages and ducts for additional internal tendons as and when required (if these spare ducts are not used at the time of construction, they should be filled after successful completion of all stressing and grouting operations);
- provide spare anchorages and deviators for additional external tendons as and when required. These anchorages and deviators may allow prestressing force to be added, not only at the time of construction but also at a later stage during the service life of the structure.

16.2.7 Assumptions used for design

16.2.7.1 Design values of forces in prestressing for evaluation of structural performance

16.2.7.1.1 General

Prestressing is usually exerted by a set of tendons. The total permanent force exerted at a given section (abscissa x), and at a time t, by the whole set is considered as the prestressing force.

16.2.7.1.2 Design values for SLS and fatigue verifications

For all verifications relating to cracking (decompression included) and deformations and for the analysis of the fatigue effect, either the mean value of prestressing force or upper and lower characteristic values of the prestressing force may be taken as design value.

In cases where the design value of the prestressing force or any variation on site would influence the behaviour of the structure in a significant over-proportional way, the designer should take adequate precautions.

Elongation of multistrand tendons is measured on site before seating of the tendon in the stressing anchorage.

This procedure is generally applied for unbonded monostrand tendons and small bonded tendons used for post-tensioning of floors in buildings. Elongation of these tendons is measured on site only after anchorage seating.

The actual applied prestressing level does mainly effect the design value of the shear resistance, for bending the total amount of prestressing steel will be governing.

When the actual prestressing force is determined at a certain location the effect of losses as described in section 16.2.5 should be taken into account to estimate the prestressing force in other locations of the structure.

16.2.7.1.3 Design values for ULS verifications

For all verifications at ULS, the prestrain corresponding to the mean value of prestressing force at SLS should be taken and increased by the strain imposed onto the corresponding cross-section (bonded tendons) or on the overall tendon (unbonded tendons) at the relevant ULS load combination.

16.2.7.2 Design values of tendon elongations for evaluation of structural performance

For internal and external multistrand tendons, design values for tendon elongations should be calculated for the initial prestressing force – section 16.2.5 – before losses caused by seating of prestressing steel in the anchorage.

For internal unbonded monostrand tendons, design values for tendon elongations should be calculated by taking into account the losses caused by seating of the prestressing steel in the anchorage.

For pretensioning, the initial prestressing force – section 16.2.5.2 – before losses due to relaxation in the pretensioning bed and before heat treatment should be considered.

The assumed values for friction losses, μ and k, shall be declared.

16.2.8 Assumptions used for assessment

16.2.8.1 Prestressing systems without deterioration or defects

Assumptions according to section 16.2 may be used for assessment of structures with prestressing systems without deterioration or defects.

The maximum prestressing force assumed to be applied should be derived from the original design and construction records or, if not available from documented information for the applied prestressing system valid at the time of construction. If the prestressing level is not known, the effect of variation in the prestressing force should be subject to sensitivity analysis. The actual prestressing force may be measured by in-situ testing.

Detailing rules, in particular minimum edge distance and spacing of tendons, are given in section 30.13.7.2.

Based on experience, the following minimum radii of curvature of tendons have provided satisfactory behaviour in practice for tendons made from prestressing steel:

(a) internal bonded tendons:

$R_{min} = 2.8 \sqrt{(f_{ptk} A_p \text{ [MN]})} \geq 2.5 \text{ m}$ (16.2-15)

(b) internal unbonded monostrand tendons (greased and sheathed strands):

R_{min} = 2.5m for 0.6” and R_{min} = 2.0 m for 0.5” (16.2-16)

16.2.8.2 Prestressing systems with deterioration or defects

Effects of deterioration or defects should be carefully assessed and assumptions according to 16.2 adjusted where found to be relevant.

16.2.8.3 Prestressing systems with noncompliant properties

Effects of noncompliant properties should be carefully assessed and assumptions according to section 16.2 adjusted where found to be relevant.

16.2.9 Detailing rules for prestressing tendons

16.2.9.1 Pretensioning tendons

Minimum radii of tendon curvature at deviators in the pretensioning bed should be such that the tendon capacity at deviators complies with the tensile strength requirement for anchorage and coupling devices – see section 16.2.2.1.

Individual pretensioned tensile elements should be spaced such that they comply with requirements for bond transfer and placing and compacting of concrete.

Cover of tensile elements should be provided to ensure adequate durability but should not be less than required for bond / transfer of prestress.

16.2.9.2 Post-tensioning tendons

Minimum radii of tendon curvature for all types of tendons should be such that the tendon capacity in the curvature complies with the tensile strength requirement for anchorage and coupling devices – see section 16.2.2.1. The minimum radii of curvature given in the commentary may be deemed to comply with section 16.2.2.1.

(c) external tendons:

$$R_{min} = 1.4 \sqrt{(f_{ptk} A_p \text{ [MN]})} \geq 2.0 \text{ m} \quad (16.2\text{-}17)$$

(d) loop tendons:

$$R_{min} = 0.6 \sqrt{(f_{ptk} A_p \text{ [MN]})} \geq 0.6 \text{ m} \quad (16.2\text{-}18)$$

Loops are parts of tendons which are deviated by 180°, stressed simultaneously from both ends such that the prestressing steel inside the loop deviation does not move relative to the duct during stressing.

The minimum radius of curvature above recommended for internal bonded tendons limits the stresses on the concrete to values which do not require splitting or confinement reinforcement, in general. Lower values may be possible for specific systems (for instance proven according to [16-19] and stated in the ETA for a specific system).

The stresses on the concrete on the inside of the tendon curvature should be checked at the time of stressing and at ULS, and splitting or confinement reinforcement should be provided when required.

Minimum radii of tendon curvature should be declared by the tendon supplier in the system documentation and in technical approval documents where these exist.

Typically, the tendons are assumed to exit from the anchorage straight and perpendicular to the bearing plate. The minimum straight length depends on the specific type of anchorage and is specified in the post-tensioning system approval documents.

The tendon should exit in a geometry from the anchorage or coupling devices similar to the set-up tested for anchorage and coupling device performance – see section 16.2.1.2.

Typically, a minimum spacing of a duct diameter is provided. However, when groups of tendons are used, sufficient space for concrete compaction should be provided.

Ducts of groups of curved tendons shall be spaced such that the deviation forces from the curved tendons can be safely transferred around the adjacent ducts on the inside of the curve.

Typically, round ducts are provided with a minimum cover of 0,5 diameter when there is transverse reinforcement in the cover, and 1,0 diameter in other cases, but not more than 80 mm. For flat/rectangular ducts, either the small dimension, or 1/3 and ½ of the long dimension of the duct should be provided with and without transverse reinforcement, respectively.

Ducts of groups of tendons shall have minimum spacing which permits adequate placing and compacting of concrete.

Ducts of bonded tendons shall be provided with a minimum cover to permit transmission of bond forces.

References

[16-1] ISO 15630-3 ISO 15630-3:2019: Steel for the reinforcement and prestressing of concrete — Test methods — Part 3: Prestressing steel. ISO, 2019

[16-2] prEN10138 prEN10138, Prestressing steels, Part 1 General requirements, Part 2 wire, Part 3 Strand, Part 4 Bar.

[16-3] ASTM A416	ASTM A0416_A0416M-18, Standard Specification for Low-Relaxation, Seven-Wire Steel Strand for Prestressed Concrete, ASTM International .
[16-4] ASTM A421	ASTM A421/A421M-21, Standard Specification for Stress-Relieved Steel Wire for Prestressed Concrete. ASTM International
[16-5] ASTM A722	ASTM A722/A722M-18, Standard Specification for High-Strength Steel Bars for Prestressed Concrete. ASTM International
[16-6] JIS G3536	Steel wires and strands for prestressed concrete.
[16-7] JIS G3109	Steel bars for prestressed concrete.
[16-8] FIP State of the Art Report	Cryogenic behaviour of materials for prestressed concrete. FIP, 1982, available at *fib* FIP report at fib side.
[16-9] FIP Special Report SR 88/2	Special report SR 88/2: Assessment of mechanical properties of structural materials for cryogenic applications (99pp) - Draft Report by F. S. Rostasy, 1988. Link to report
[16-10] *fib* Bulletin 98	Acceptance of post-tensioning systems for cryogenic applications, *fib* Recommendation, 2021, ISBN: 978-2-88394-147-2
[16-11] Atienza and Elices	Atienza, J. M. and Elices, M,.Behaviour of prestressing steel after simulated fire: Fire-induced damage. *Construction and Building Materials*, Vol. 23, pp. 2923–2940, 2009
[16-12] Galvez et al	Galvez, F., Atienza, J. M., Elices, Behaviour of steel prestressing wires under extreme conditions of strain rate and temperature, *Structural Concrete*, No. 4, pp. 255–261., 2011
[16-13] CEB Bulletin 187	Concrete Structures under Impact and Impulsive Loading. 1988.
[16-14] ASTM A 1081	Standard Test Method for Evaluating Bond of Seven-Wire Steel Prestressing Strand. ASTM A1081/A1081M-15. West Conshohocken, PA: ASTM International, 2015
[16-15] Elices	Elices, M., Influence of residual stresses in the performance of cold-drawn pearlitic wires, *Journal of Materials Science*, Vol. 39, 2004, pp. 3889–3899, 2004
[16-16] ISO 14655	Epoxy-coated strand for the prestressing of concrete. ISO, 1999.
[16-17] XP A 35-037.	Steel products - Protected and sheathed high strength steel strands - Part 1 : general requirements - Produits en acier.
[16-18] NBNI10-008.	Voorspanwapeningen - Beschermde omhulde strengen [In Dutch].
[16-19] EAD-16-0004	European Assessment Document EAD-16-0004, Post-tensioning kits for prestressing of structures, EOTA, 2016
[16-20] EAD-16-0027	European Assessment Document EAD-16-0027, Special filling products for post-tensioning kits, Sept 2016, EOTA
[16-21] Architectural Institute of Japan.	Recommendation for design and construction of partially prestressed concrete (class III of prestressed concrete) structures.
[16-22] JIS K 2220.	Lubricating grease. 2018.
[16-23] JIS K 2246.	Rust preventive oils. 2018.
[16-24] JIS K 2265.	Determination of flash point.

[16-25] JIS K 6922-2. Plastics - Polyethylene (PE) moulding and extrusion materials - Part 2: Preparation of test specimens and determination of properties.

[16-26] *fib* Bulletin 89 Acceptance of stay cable systems using prestressing steels, *fib* Bulletin, 2019, ISBN: 978-2-88394-129-8

[16-27] SETRA. Recommendations recommendations of French interministerial commission on prestressing

[16-28] FIP 1982 Recommendations for the design and construction of prestressed concrete ground anchors, FIP Recommendation, 1982, ISBN 978-0-907862-19-2, FIP report on fib webside

[16-29] *fib* Bulletin 97 External tendons for bridges, *fib* State-of-the-art report, 2020, ISBN: 978-2-88394-145-8

[16-30] CWA 14646 Requirements for the installation of post-tensioning kits for prestressing of structures and qualification of the specialist company and its personnel, CEN, 2003

[16-31] FIP 1993 Recommendations for the acceptance of post-tensioning systems. FIP Recommendation, 1993, ISBN 978-1-874266-07-5, FIP report on fib Webside

[16-32] EN 523: 2003. Steel strip sheaths for prestressing tendons - Terminology, requirements, quality control.

[16-33] EN 524-1:1997. Steel strip sheaths for prestressing tendons - Test methods - Part 1: Determination of shape and dimensions.

[16-34] *fib* Bulletin 75. Polymer-duct systems for internal bonded post-tensioning. 2014.

[16-35] ISO 4200. Plain end steel tubes, welded and seamless; general tables of dimensions and masses per unit length. 1991.

[16-36] *fib* Bulletin 20 . Grouting of tendons in prestressed concrete. 2002.

[16-37] PTI M55.1-12. Specification for grouting of post-tensioned structures.2003.

[16-38] EN 445:2017. Grout for prestressing tendons - Test methods.

[16-39] EN 446:2017. Grout for prestressing tendons - Grouting procedures.

[16-40] EN 447:2017. Grout for prestressing tendons - Basic requirements.

[16-41] ISO 14824-1 Grout for prestressing tendons - Part 1: Basic requirements, ISO, 2012

[16-42] ISO 14824-2 Grout for prestressing tendons – Part 2: Grouting procedures, ISO, 2012

[16-43] ISO 14824-3 Grout for prestressing tendons – Part 3: Test Methods, ISO, 2012

[16-44] FIP 1996 Corrosion protection of prestressing steels, FIP Recommendation, 1996, ISBN 978-1-874266-26-6, Link to FIP Report on fib webside

[16-45] CEB Bulletin 199. Evaluation of the Time Dependent Behavior of Concrete, CEB, 1999.

[16-46] Müller and Zetterholm Müller, H.R. and Zetterholm, S, Results of extreme long duration of a relaxation test (42 years) on prestressing steel. *Proceedings of 1st fib Congress, Osaka*, pp. 385–390., 2002

[16-47] CEB Bulletin 213/214 CEB-FIP Model Code 90, CEB, 1993, ISBN 978-0-7277-1696-5

[16-48] de Halleux, B., 1970. Explication physique de l'influence de la temperature sur le fluage et la relaxation des aciers de precontrainte, *FIP Symposium, Prague*.

[16-49] de Halleux, B., 1974. Accelerated determination of the stress relaxation in prestressing reinforcement by an anisothermal relaxation test, *FIP VII Congress, New York*.

[16-50] Rostásy et al Rostásy, F.S., Thienel, K. Ch., et al, On prediction of relaxation of cold-drawn prestressing wire under constant and variable elevated temperature, *Nuclear Engineering and Design*, Vol. 130, pp. 221–227., 1991

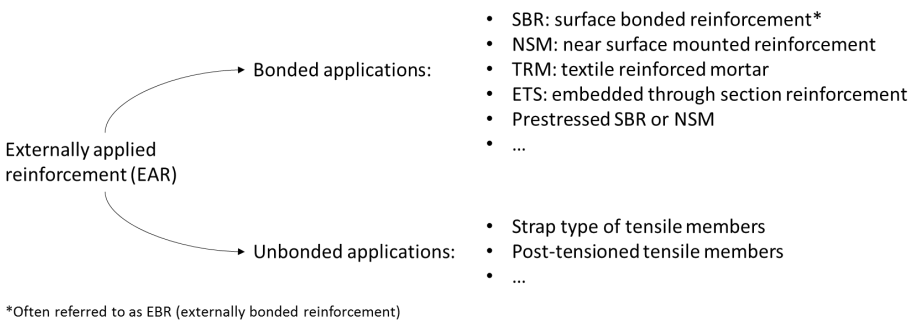
Fibres occupy the largest volume fraction (40–70%) and have an elastic modulus much higher than that of the matrix. As such, fibres are the principal stress-bearing constituent, while the resin transfers stresses among fibres and protects them.

In general, FRPs are characterized by high axial strength, high ratio of axial to transverse strength, limited ultimate strain, low weight, excellent chemical resistance and non-susceptibility to a wide range of aggressive media, electromagnetic neutrality, excellent fatigue characteristics (depending on fibre type), and limited ratio of long term to short term static strength for some fibre types.

Several types of fibres are used for FRP reinforcement, the most popular are: glass (GFRP), carbon (CFRP), aramid (AFRP), basalt (BFRP) and more rare as polyphenylene bezobisoxazole (PBO). Typical matrix materials are vinylester and epoxy, amongst other polymer matrices. Thermosetting matrices are often used, although FRP products based on thermoplastic matrices also exist. For rebar applications special concern is given on the surface treatment (Section 17.4.3) to achieve proper bond with the concrete.

For further information see *fib* Bulletin 40 [17-1].

For further information see *fib* Bulletin 90 [17-2]. Externally applied reinforcement techniques are listed in figure 17.1-1.



17 Non-metallic reinforcement

17.1 Scope definition and range of applicability

Non-metallic reinforcing elements consist of a large number of continuous, directionalized, organic or inorganic fibres, typically embedded in a polymeric matrix. Both the terms “non-metallic reinforcement” and “FRP (fibre reinforced polymer) reinforcement” are used for this reinforcement type.

The properties of FRP reinforcement basically depend on fibre and resin type and properties, constituent volume fractions, production parameters, shape (including bent FRP) and surface texture.

For rebar applications, amongst other, glass fibres (type electrical/chemical resistant) are used. For prestressing and externally applied reinforcement, carbon fibres are predominantly used.

For new structures pre-cured bars, tendons, grids or technical textiles are used to reinforce and prestress concrete elements.

To retrofit and strengthen existing structures, FRP elements - rather or not prestressed - are used as externally applied reinforcement (EAR). Mainstream EAR techniques are surface bonded reinforcement, often referred to as externally bonded reinforcement (EBR) and near-surface mounted (NSM) bars. The FRP may be in the form of:

- pre-cured strips or laminates for surface bonding
- sheets or fabrics applied by wet lay-up or pre-impregnated (prepreg) surface bonding
- pre-cured bars, small strips or profiles for NSM
- pre-cured tendons or strips for external post-tensioning

Figure 17.1-1: Externally applied reinforcement.

The methods of testing and certification of conformity are as defined in standards and recommendations at the national or international level. For the latter reference can be made to ISO 10406 [17-3] + [17-4], that covers in Part 1 methods of testing for FRP bars and grids and in Part 2 methods of testing for FRP sheets. At the European level, FRP performance characteristics and conformity can be declared via ETA (European Technical Assessment) on the basis of EADs (European Assessment Documents) issued for FRP. For FRP rebars reference can be made to EAD 260023-00-0301 [17-5] (for other FRP products such EADs are to be further developed). Protocols for testing and certification of conformity have also been issued in multiple national documents. For further information on performance characteristics of FRP see also the bulletins mentioned in section 17.1.

These quality control provisions are of importance to guarantee a consistent quality of the FRP, such as the coefficient of variation of the tensile strength of the FRP being less than 5%, as is normally the case for proper FRP products and systems. This is taken at the basis of the partial material safety factor for FRP (see section 17.8.1.1).

Site control provision have been made mainly available in national documents. For externally applied FRP reinforcement, this is also addressed in *fib* Bulletin 90 [17-2].

Typical fibre material notation is C (carbon), G (glass), A (aramid), B (basalt).

– grids, fabrics or technical textiles for TRM`

17.2 Provisions for quality control

Non-metallic reinforcement must comply with national or international product standards that specify their geometrical, mechanical and technological properties, such as the international product standards ISO 10406 [17-3] + [17-4].

The fabrication of non-metallic reinforcement must be subject to factory production control by the manufacturer, and continuous external control by an independent qualified body, including certification and regular audits.

Quality control provisions are also applicable to site control. They are of high importance to ensure proper in-situ application of FRP products and systems and so that the realisation can meet up to design expectations.

17.3 Designation

The designation of non-metallic reinforcing elements normally includes:

– fibre and matrix materials;
– nominal dimensions (e.g. diameter, width, thickness);
– characteristic tensile strength, modulus of elasticity and the ultimate strain in the direction of the fibres;
– relevant product standard.

Each product must be clearly identifiable with respect to this designation.

The definition of the nominal cross-section is for pre-cured forms typically based on the global nominal dimensions (diameter, thickness, width). Alternatively or for wet lay-up/prepreg forms, an equivalent dry fibre cross-section may be used (referring to the continuous fibres as principal stress bearing component). The latter is obtained as the ratio of the fibre mass per length and the fibre density. For FRP with multiple fibre directions, where a different amount of fibres per unit length is applied in different fibre directions, the definition of the nominal dry fibre cross-section always relates to the specified fibre direction, and more than one nominal cross-section (or nominal thickness) may be given depending on the fibre direction.

As design verifications are based on equilibrium of forces, strength values should always be used with their corresponding nominal cross-section, as declared on the product data sheets. If data sheets of FRP products are compared, the possible difference in definition of the nominal cross-section should be taken into account.

Surface texture of FRP rebars (for internal reinforcement or near surface mounted reinforcement) may be sanded or ribbed-like. FRP strips or sheets (for external bonding) often have no additional surface texture and are fairly smooth. Surface characteristics include the characteristics of the ribs or indentations or other surface deformations (e.g. sand-coating) by means of which bond with the concrete is achieved.

FRP to concrete bond quality is product specific, related to the surface characteristics. Further requirements are given in Part V.

17.4 Geometrical properties

17.4.1 Configuration

The configuration is characterized by the type of fibre(s) and matrix, fibre orientation(s), constituent fractions and surface texture. Constituent material fractions can either be given by mass (weight) or by volume.

17.4.2 Size

The size of non-metallic reinforcing elements is defined by a nominal diameter for circular bars or by the nominal cross-sectional dimensions for other products (e.g. thickness, width).

The difference between actual and nominal cross-sectional area must not exceed the limiting values specified in relevant product standards, or guidance documents specifying dimensional tolerances.

17.4.3 Surface characteristics

The surface of non-metallic reinforcement may be:

– plain;

– deformed.

17.5 Mechanical properties

The mechanical properties are defined on the basis of standard tests (see section 17.2).

17.5.1 Tensile strength, modulus of elasticity and ultimate strain

The characteristic values of

- tensile strength (f_f)
- percentage total elongation (ε_{fu}) at maximum stress (f_f) and
- creep rupture tensile strength ($f_{f,100a}$)

are respectively denoted f_{fk} , $\varepsilon_{fuk.}$, $f_{fk,100a}$

The mean value of secant modulus of elasticity is denoted (E_f).

The stress–strain response is quasilinear elastic for most FRP elements. The mechanical properties apply to the standard product, as tested (ISO 10406 [17-3] + [17-4]) on straight test coupons. The design considers effective characteristic strength values where relevant, for example at bent regions.

Due to the viscoelastic behaviour of FRP, its tensile strength is further defined by creep rupture under sustained tensile load, corresponding to the quasi-permanent load combination. The creep rupture strength is established (ISO 10406 [17-3] + [17-4]) at relevant exposure environment, and is considered for the applicable design service life. If not specified otherwise, for FRP reinforced structures, the creep rupture strength after 100 years is considered ($f_{f,100a}$). FRP creep rupture is considered in the design following the principles of Section 17.8.

As FRP reinforcement products are available in a range of strength and stiffness values, Table 17.5-1 gives exemplarily overall ranges.

Table 17.5-1: Indicative tensile properties of FRP reinforcement.

Property	CFRP	GFRP	AFRP	BFRP
Tensile strength f_f [MPa]	1000–3500	600–1600	900-1900	550-1800
Ratio $f_{f,100a}$ / f_f [%]	~90%	~40%	~50%	~60%
E-modulus E_f [GPa]	80–500	40–60	40-90	40-70
Ultimate strain ε_{fu} [%]	0.5–2.2	1.4–3.0	1.4-3.5	1.2-3.4

17.5.2 Compressive and shear strength

Given the generally limited compressive modulus of elasticity and the risk of microbuckling or kinking of the fibre within the restraint of the matrix material, The compressive or transverse shear properties for a particular reinforcing element, if needed for a particular application, should be given by the manufacturer,

non-metallic reinforcement is generally not used to resist high compressive stresses.

The interlaminar transverse shear strength of non-metallic reinforcement is basically dominated by the matrix and the fibres in off-axis directions.

FRP materials generally have very good fatigue resistance, though fatigue in FRP is regarded complex due to damage which can interdependently occur in the matrix, fibres, interfacial zones, etc. of an FRP. A further discussion on fatigue consideration in FRP RC concrete is provided in *fib* Bulletin 40 [17-1].

High modulus fibre composites have superior fatigue resistance. Cyclic tension fatigue strength of unidirectional CFRP exceeds that of prestressing steel, while that of GFRP is lower. Indicative values are given in Table 17.5-2.

Table 17.5-2: Indicative fatigue strength of reinforcement for $2 \cdot 10^6$ cycles.

Type of reinforcement	σ_{max}/f_{tk} (a) [–]	$\Delta\sigma$ (b) [MPa]
Prestressing steel	~ 0.60	~ 200
Carbon/vinylester (rod)	~ 0.60	> 350
Carbon/epoxy (strand)	~ 0.60	~ 310
Aramid/vinylester (rod)	~ 0.60	~ 235
E-glass/epoxy (rod)	~ 0.50	~ 75

(a) Applied maximum stress (σ_{max}) as a function of the characteristic tensile strength (f_{tk}) of the reinforcement.

(b) Stress range ($\Delta\sigma$) yielding fatigue failure at $2 \cdot 10^6$ cycles.

In general, FRP materials exhibit good to excellent resistance against chemical attack and FRP reinforcement for concrete must undergo environmental testing to satisfy certification requirements, such as proven GFRP rebar alkaline resistance.

who should also provide a description of the test method used to determine the properties.

In the absence of specific data on these properties, the possible contribution of the FRP reinforcement in this respect, shall be ignored.

17.5.3 Fatigue behaviour

If a non-metallic reinforcing element is subjected to a large number of load cycles, growth of internal or surface flaws may occur, resulting in a reduced mechanical strength compared to the short term static strength.

17.6 Properties describing / defining durability

17.6.1 Resistance against aggressive substances

Non-metallic reinforcement has excellent chemical resistance and non-susceptibility to a wide range of aggressive media. Unlike steel rebars, FRP cannot corrode.

The performance of FRP reinforcement may also be influenced by moisture, thermal stresses and UV radiation. For a detailed discussion on durability, see the bulletins mentioned in section 17.1. Field applications (of which several already ~20 years or more in service) have confirmed the excellent performance of GFRP rebars and CFRP EBR.

17.6.2 Creep behaviour

FRP reinforcement combines elastic fibres, which have excellent resistance to creep, with a viscoelastic polymer matrix, which may show significant creep deformations. As FRP tensile members normally have a high degree of fibre orientation, large fibre volume fractions and a high ratio of fibre over matrix stiffness, the tensile force shared by the matrix is extremely low, so that FRP creep deformations are negligible.

The permissible stress level against stress rupture depends on the fibre/resin system, alignment of the fibres, fibre volume fraction, as well as environmental exposure; so that they are product specific. Indicative values are given in Table 17.1-1.

Provided that stresses are maintained below the recommended permissible levels and the maximum service temperature is 20°C lower than the glass transition temperature, FRP creep deformations are generally negligible.

In reference to creep rupture (17.8.1.2), the long term maximum stress of non-metallic reinforcement is limited, as also specified in section 17.5.1.

17.6.3 Relaxation

Relaxation of GFRP and CFRP prestressing elements after 50 years of loading can be estimated as 4–14%, 2–10%, respectively. These values depend on the stress level and environmental influence.

Prestressing loss due to relaxation of FRP is compensated by a lower prestressing loss due to concrete shrinkage and creep (given the ratio of the modulus of elasticity E_f/E_c).

Relaxation of non-metallic reinforcement is to be considered for prestressing loss calculations.

17.6.4 Behaviour under elevated temperature and under extreme thermal conditions

The glass transition temperature T_g is of particular importance, as it reflects the change of molecular mobility of polymer materials. For factory processed FRP elements, the matrix generally has a T_g higher than 100°C. The T_g of cold-cured

Although fibres exhibit relatively high thermal stability, polymer resins are strongly affected by temperature. As a result, the material and bond properties of FRP are influenced by temperature and decrease drastically when reaching the glass

(ambient-cured) adhesives/saturating resins may be lower (typically in the range of about 50–80°C for epoxy).

FRP to concrete bond interaction is different from that of deformed steel rebars.

Thermoset resin based FRP elements are not bendable in situ. Bent FRP elements are factory made and pre-cured. Thermoplastic resin based FRP elements are bendable given proper application procedures.

The reduction of strength in the bent part can be order of magnitude of 30 to 50%, amongst other depending on the radius at bent corners and the type of fibres.

Given the anisotropy of FRP elements, different CTE values are obtained in the longitudinal and the transverse direction. Indicative values are given in Table 17.7-1. Note however that the CTE can significantly vary (amongst other depending on directions and volume of fibres, matrix type, considered temperature range, testing protocol, etc.), so that the values in Table 17.7.1 should be considered with care.

Table 17.7-1: Coefficient of thermal expansion (CTE).

transition temperature T_g. When designing against fire, sufficient concrete cover should be provided so that the glass transition temperature is only reached after the required fire resistance time. For external reinforcement systems, fire protection systems may be required, depending on the level of strengthening and the fire resistance of the unstrengthened structure.

17.7 Technological properties

17.7.1 Bond characteristics

Bond characteristics of non-metallic reinforcement relate to the surface characteristics, as specified in chapters 20 (for embedded rebars) and 22 (for externally applied reinforcement).

17.7.2 Bendability

The requirements concerning the bendability are specified in relevant product specifications.

The reduced FRP tensile strength of bend portions should be considered by a reduction factor η (in reference to equation 17.8-1), based on manufacturer specifications or preferably product declarations following European Assessment Documents.

17.7.3 Coefficient of thermal expansion

Values of the CTE for non-metallic reinforcement depend on the constituents and are product specific.

FRP type	Longitudinal CTE [10^{-6} °C^{-1}]	Transverse CTE [10^{-6} °C^{-1}]
CFRP	-2 to 0	≈ -25
GFRP	5 to 10	≈ -25
AFRP	-3 to -1	-40
BFRP	2 to 5	-25

For serviceability limit state (SLS) verifications, a linear stress-strain response $\sigma_f = E_f \varepsilon_f$ is considered, referring to the mean value of the secant modulus of elasticity E_f.

For ultimate limit state (ULS) verification, the design stress-strain curve is idealised by means of a linear response up to f_{fk} and ε_{fuk}. The slope of this design stress-strain curve refers to a modulus, defined as the ratio of f_{fk}/ε_{fuk}.

For ultimate limit state (ULS) verification of creep rupture, the sustained stress level is considered at quasi-permanent load combination.

The ratio $\eta = \varepsilon_{fue}/\varepsilon_{fum}$ accounts for non-environmental effects as follows. In many circumstances and in due consideration of the 2 points below, η can be considered equal to 1 as the effective ultimate FRP strain ε_{fue} expected in-situ will not significantly differ from the mean strain ε_{fum} obtained through uniaxial tensile

17.8 Assumptions used for design

17.8.1 Assumptions for evaluation of structural performance

17.8.1.1 Stress-strain diagram

For design purposes, an idealized stress-strain diagram according to Figure 17.8-1 is used.

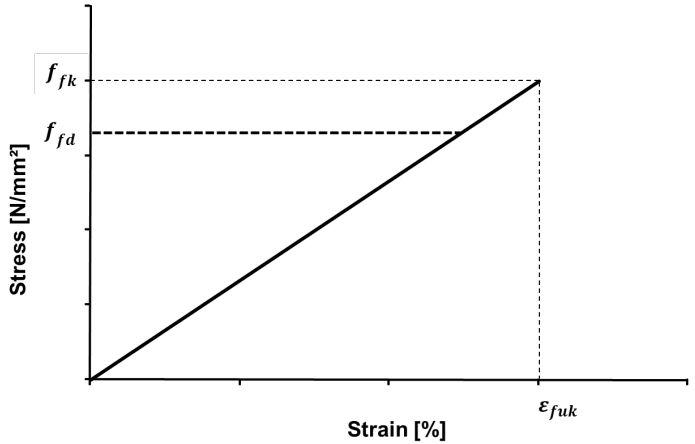


Figure 17.8-1: Constitutive model for the FRP tensile response.

The design value of the tensile strength is equal to:

$$f_{fd} = \eta \frac{f_{fk}}{\gamma_f} \qquad (17.8\text{-}1)$$

testing, as small variations are accounted for in the FRP material safety factor γ_f. This provided that the following conditions are met:

- The maximum service temperature at the location of the FRP is higher than *-40°C* and below T_g *-20°C*, and whereby T_g is considered as a lower bound value of the glass transition temperature range.
- For internal FRP rebars and for FRP EBR directly exposed to an outdoor environment an environmental conversion factor η_e is applied as defined in section 17.8.1.2.

In certain circumstances, the effective failure strain ε_{fue} may be significantly lower than ε_{fum}, for example when using bent reinforcing bars, wrapping plies around sharp corners, applying a high number of confining layers, or when the FRP is subjected to a multi-axial state of stress. A limiting value of the FRP failure strain may also be considered as a simplified design alternative. In this case, the ULS verification restricts excessive FRP deformations.

The partial safety factor has been derived on the assumptions of normal distribution of the FRP tensile strength with a maximum coefficient of variation of 5%, a target reliability index of 4,3 (ULS, 50 years, high consequences) or 2,05 (accidental situation, 50 years), an ISO 2394 [17-6] sensitivity factor of 0,8 (for dominating resistance parameter) and a model uncertainty factor of 1,1 (given the use of validated strength models for FRP design with inherent model uncertainty factors considered in the model calibration).

whereby, the material factor γ_f for FRP reinforcement is taken from section 6.4 and η is a non-environmental reduction factor considering the effective tensile stress or strain capacity and f_{fk} is the characteristic tensile strength of the FRP. The effect of the environment is considered in Eq. 17.8-2 and section 17.8.1.2.

For those design verifications in which creep rupture, in due consideration of the exposure conditions, can govern (e.g. flexural load bearing capacity), the provisions of section 17.8.1.2 are equally applicable. In this case the design can be based on the design value of the tensile strength equal to:

$$f_{fd} = min\left\{\eta\frac{f_{fk}}{\gamma_f}; \frac{f_{fk,100a}}{\gamma_f}\frac{\gamma_G G+\gamma_Q Q}{G+\psi_2 Q}\right\} \qquad (17.8\text{-}2)$$

whereby, $\eta_e f_{fk,100a}$ is the characteristic value of the creep rupture strength in the relevant exposure environment (section 17.8.1), $\gamma_G G+\gamma_Q Q$ is the ultimate limit state load combination and $G+\psi_2 Q$ is the quasi-permanent load combination.

γ_f for FRP reinforcement (FRP rebars and factory made FRP strips) is 1,3 in persistent/transient situation and is 1,1 in accidental situation. In case of in-situ wet lay-up made FRP sheets, a factor 1,4 should be considered instead of 1,3. For the case of seismic retrofitting and for bond verifications, specific values apply (chapters 19, 20, 27).

The FRP tensile creep rupture strength (considered at 100 years $f_{f,100a}$), follows from creep rupture curves established at exposure conditions relevant for the application. A schematic representation (in terms of characteristic values) is given in Figure 17.8-2.

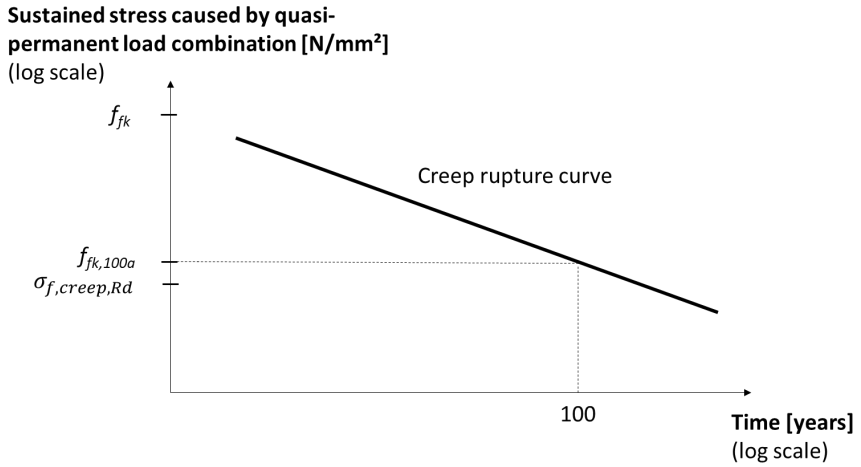


Figure 17.8-2: FRP tensile creep rupture design approach.

Given the high strength of FRP, relatively small cross-sectional areas are needed for ultimate limit state. Especially in flexure, ULS is rarely governing the design, e.g. as sufficient flexural stiffness should be provided to fulfil serviceability criteria. This may result in a considerably higher bending capacity.

The reason for strengthening might be prescribed from the requirement to enhance structural performance to persistent/transient loads, accidental loads (e.g. blast) or robustness demands. The degree of strengthening of properly designed retrofitted members is limited by the SLS and ULS provisions of this Model Code, further to economic feasibility. If this degree of strengthening also involves member collapse in case of accidental loss of the FRP, the designer should consider:

17.8.1.2 Creep rupture

Creep rupture of FRP reinforcement shall be prevented by limiting sustained stresses. All relevant quasi-permanent combinations of actions shall be used to verify creep rupture of composite members:

$$\sigma_{f,creep,E} \leq \sigma_{f,creep,Rd} \tag{17.8-3}$$

$$\sigma_{f,creep,Rd} = \frac{\eta_e \cdot f_{fk,100a}}{\gamma_f} \tag{17.8-4}$$

whereby, $\sigma_{f,creep,E}$ is the sustained stress caused by the quasi-permanent load combination, and $\sigma_{f,creep,Rd}$ is the design value of the stress limit for FRP creep rupture.

The relevant environmental exposure of the FRP is considered in establishing the creep rupture curve. If this is not the case an environmental conversion factor η_e is to be applied for FRP rebars and for FRP EBR directly exposed to an outdoor environment. Such environmental conversion factors are defined in *fib* Bulletin 40 or in national guidance documents. A typical value is 0,85 (prCEN/TS 19101: 2022 [17-7]).

17.8.1.3 Resistance hierarchy

An amount of FRP governed by SLS considerations, might produce a bending capacity higher than the shear capacity of the member, thus activating a brittle failure mode. This situation should be avoided by considering the resistance hierarchy as specified by the capacity design criteria.

Specific to strengthening existing structures with FRP, the design engineer shall duly verify:

– the minimum amount of steel reinforcement in the unstrengthened member in accordance to this Model Code, given that strengthening of plain concrete is out of scope of this document;

– the member against accidental loss of FRP, as to be aware of possible consequences of a given level of strength increase;

– proper protection against vandalism or accidental mechanical damage;
– local member collapse is not causing progressive collapse;
– end debonding is avoided through mechanical anchorage;
– fire protection design of the entire strengthened member.

FRP materials have no yield plateau like steel reinforcement. Warning by excessive deflections and cracking, are rather controlled by the curvature at the ultimate limit state and the related maximum depth of the compression zone.

In rare cases, typically in a design governed by SLS, it may be difficult to fulfil the condition of Eq. (17.8-4) while yet high safety factors between the acting (E_d) and the resisting (R_d) design load in ULS are obtained (hence also a high safety factor between the service load and the resisting design load). If $R_d > 1.2E_d$ the condition of Eq. (17.8-4) should no longer be fulfilled.

It should be noted that rotation capacity of the cross-section Eq. (17.8-4), does not relate to ductility capacity in terms of energy dissipation. Increased ductility can be obtained by means of confinement.

Based on the material properties of FRP, among which being an almost linear elastic material, some behavioural aspects can be expected of concrete structures reinforced or strengthened with FRP, which differ from structures traditional steel. A further discussion is provided in the bulletins mentioned in section 17.1.

– the impact of the local strengthening on global behaviour of the structure, including its behaviour at the state of progressive collapse.

17.8.1.4 Deformation capacity at the ultimate limit state of flexure

To secure adequate deformation capacity of a flexural member, with internal FRP reinforcement or with externally applied FRP reinforcement, the depth of the compression zone at ultimate limit state is limited:

$\xi \leq 0.45$ for concrete type C50/60 or lower

$\xi \leq 0.35$ for concrete type C55/67 or greater (17.8-4)

where:

$\xi = x/d$;

x the depth of the compression zone;

d the effective beam depth.

17.8.2 Other assumptions

The analysis and design of structures reinforced or strengthened with FRP is carried out by qualified engineers, familiar with FRP reinforcement.

The partial safety design format implies that the quality control provisions on the FRP materials and products, as well as their installation, are according to the specifications of this Model Code.

17.9 Assumptions used for assessment

Assessment of existing structures with embedded non-metallic reinforcement or with externally applied reinforcement shall be done in accordance to the design provisions of this Model Code.

FRP sample assessment testing will include at least measurement of the geometrical properties, description of the surface texture and tensile testing. This testing will be further complemented with relevant properties to fully allow verifications on the assumptions needed for the design.

In case of externally applied reinforcement, such as surface bonded reinforcement, bond verification testing will be performed (following procedures described in *fib* Bulletin 90 [17-2]).

Visual inspection of the FRP reinforcement will generally allow to identify the fibre type, yet will not allow to identify the FRP material properties. For the latter, reference should be made to 'as built' documentations rather or not supported by testing of FRP samples taken from the existing structure.

References

[17-1] fib Bulletin 40 FRP reinforcement in RC structures. 2007.

[17-2] fib Bulletin 90. Design of Concrete Members Strengthened with Externally Applied Reinforcement. 2019.

[17-3] ISO 10406:2015-P1 Fibre-reinforced polymer (FRP) reinforcement of concrete — Test methods — Part 1: FRP bars and grids, ISO, 2015

[17-4] ISO 10406:2015-P2 Fibre-reinforced polymer (FRP) reinforcement of concrete — Test methods —Part 2: FRP sheets, ISO, 2015

[17-5] EAD 260023-00-0301 Carbon, glass, basalt and aramid Fibre Reinforced Polymer bars as reinforcement of structural elements [Pending for citation in OJEU]

[17-6] ISO 2394: 2015 General principles on reliability of structures, ISO, 2015.

[17-7] prCEN/TS 19101: 2022 CEN/TS 19101:2022 Design of fibre-polymer composite structures, CEN, 2022

18 Fibre reinforced concrete

18.1 Scope definition and range of applicability

Fibre reinforced concrete (FRC) is a composite material characterized by a cement matrix and discrete fibres (discontinuous). Fibres can be made of steel, polymers, carbon, glass or natural materials.

Fibre materials with a Young's modulus that is significantly affected by ageing and/or thermo-hygrometrical phenomena in the range of environmental conditions in service are not covered by this Model Code since the cement composites made exclusively with such fibres should not be recommended for structural use.

Mixtures with fibres of different types and/or dimensions can also be used (called hybrid fibre reinforced concrete).

Structural design of FRC elements is based on the post-cracking residual strength provided by fibre reinforcement. Other cases, such as early age crack-control or spalling resistance in case of fire, are considered non-structural use of FRC.

For structural use, a minimum mechanical performance of FRC must be guaranteed.

Fibres can be used to improve the behaviour at SLS since they can reduce crack width, thereby improving durability.

Fibres can be used to improve the behaviour at ULS where they can partially or, if certain conditions are fulfilled, totally substitute conventional reinforcement.

The mechanical properties of a cementitious matrix are modified when fibres are added. However, elastic properties and compressive strength are not significantly affected by fibres, unless a high percentage of fibres is used. Design values for compression are described in 18.6.1.

Depending on their composition, FRC can show hardening or softening behaviour under uniaxial tension (Figure 18.1-1).

Unlike rebars or welded mesh, fibres are generally designed to slip without failing in tension. Fibres are active as soon as micro-cracks are formed in the concrete. The main advantage of adding fibres to concrete or mortar is that they generate a post-cracking residual tensile strength in combination with a relatively large tensile strain. As such, the material fibre reinforced concrete (FRC) is characterized by substantial ductility and toughness with respect to plain concrete.

The properties of the composite depend on the characteristics of the constituting materials, as well as on their dosage. Other factors such as the geometry, the volume fraction and the mechanical properties of the fibres, the bond between fibre and concrete matrix, and the mechanical properties of the matrix, significantly affect the FRC properties.

Due to differences in casting and vibration procedures, FRC flowability and geometry of the moulds, anisotropic fibre distributions may occur, the effect of which should be taken into account since fibre orientation affects FRC mechanical properties after cracking.

The rules in this chapter are based most of all on experience with steel fibre reinforced concrete (SFRC). Proper precautions must be taken when using these rules for concrete reinforced with those non-steel fibers covered by thus Model Code 2020.

In case of softening behaviour (a) the deformations localize in one crack. In case of hardening behaviour (b) multiple cracking occurs before reaching the peak value.

The relation between strain softening and strain hardening behaviour is shown in Figure 18.1-1. A FRC with softening behaviour in tension can develop hardening behaviour in bending. A FRC with softening behaviour in bending can exhibit a monotonically increasing load in the structure (Figure 18.1-2).

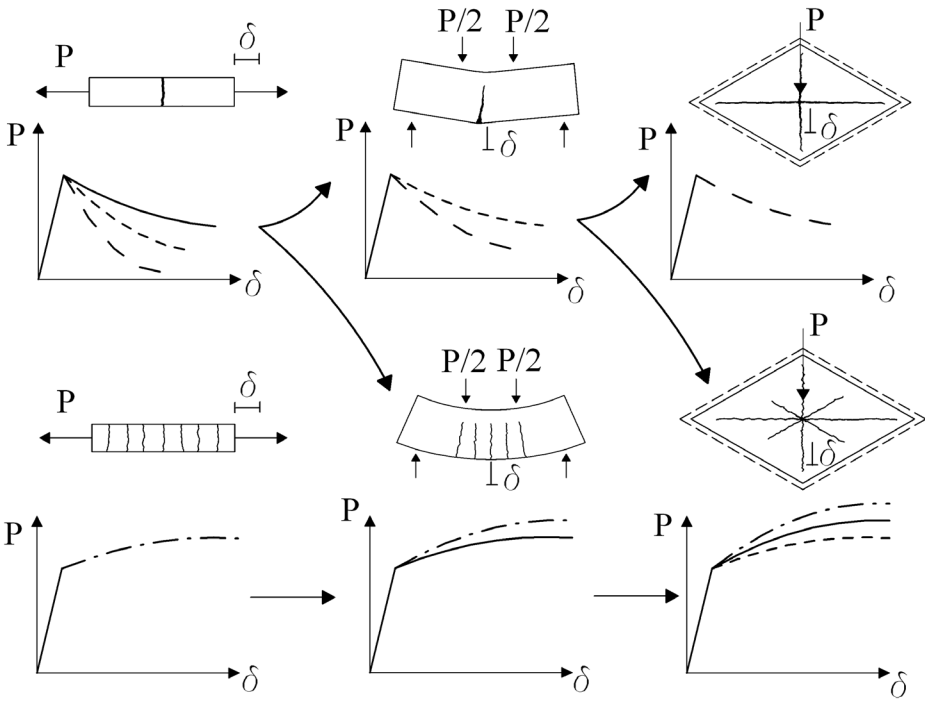


Figure 18.1-2: Different response of structures made of FRC having a softening or hardening behaviour under uniaxial tension or bending loads.

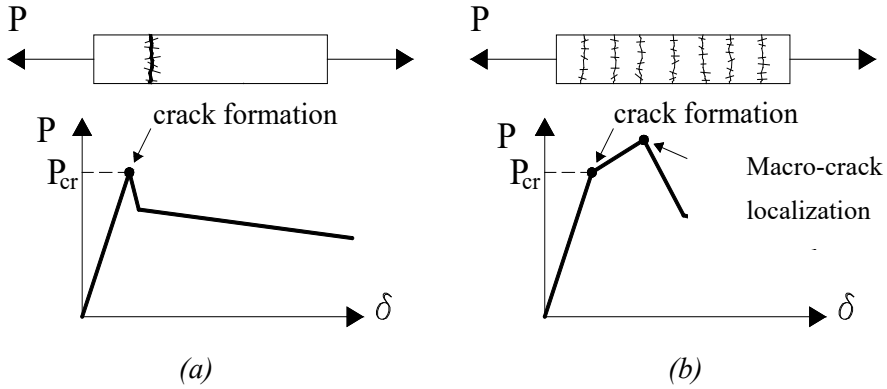


Figure 18.1-1: Softening (a) and hardening (b) behaviour in axial tension.

18.2 Material properties

18.2.1 Behaviour in compression

Fibres do not modify the compressive strength of the matrix, but reduce the brittleness in compression, especially in high or ultra-high strength concrete and in the cases where fibres are predominantly oriented orthogonally to the compressive force (Figure 18.2-11).

FRC is quite more ductile than plain concrete in compression: the average increase of energy absorption during a stress-strain test on a standard cylinder is significantly higher than that of the companion plain concrete (same strength class).

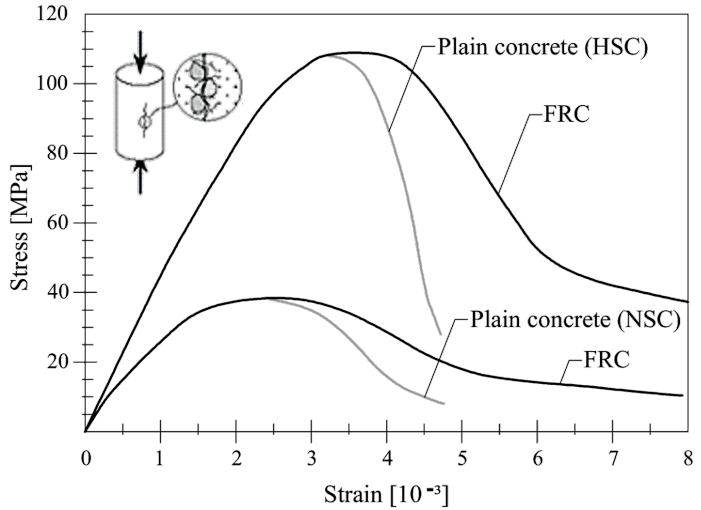


Figure 18.2-1: Main differences between plain and fibre reinforced concrete having both normal and high strength under uniaxial compression.

18.2.2 Behaviour in tension

Since the specimens are normally small, the number of fibres in the governing cross section will be small and their eventual different orientation is the main reason for the relatively high dispersion of the obtained results.

Bending tests can be carried out aiming at determining the load-deflection relation of the FRC. The results derived from these tests can be used to obtain the stress-crack width relationship of the material for design purposed by means of inverse analysis. This analysis is performed by imposing equilibrium, compatibility and kinematic conditions for numerous crack openings. A simplified method can be used as shown in Figure 18.2-2, where a linear strain profile is assumed, smearing the crack opening on a certain characteristic structural length l_{cs}, evaluated proportional to the net depth h_{sp}. Simplified relations of the type derived from inverse analysis are found in section 18.4.2

With regard to the characterization of the tensile behaviour, which is the most important aspect of FRC, various test methods are possible.

In general, uniaxial tensile testing is not advised for standard testing of new mixtures since these tests are difficult to carry out and interpret. Uniaxial tensile tests are especially suitable for determining the strain hardening behaviour.

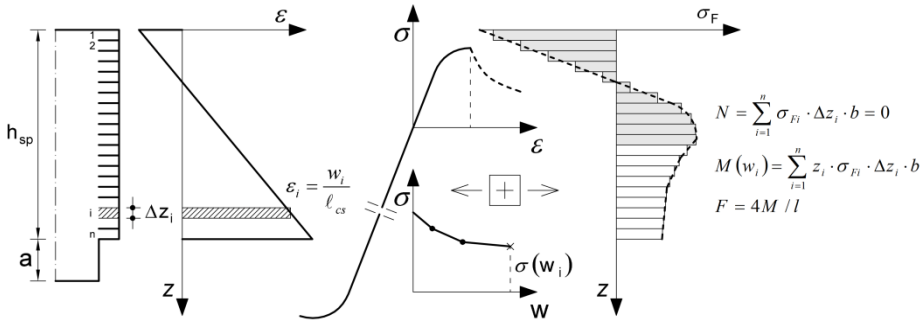


Figure 18.2-2: Inverse analysis of beam in bending performed to obtain stress-crack opening relation.

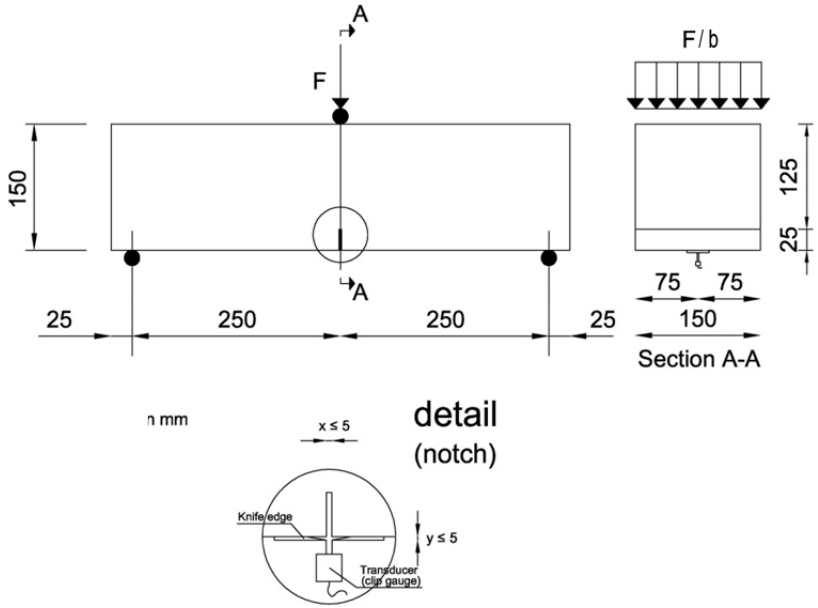


Figure 18.2-3: Test set-up required by EN 14651 [18-1] (dimensions in [mm]).

The definition of the nominal residual flexural strengths proposed in Eq. (18.2.1) is based on the adoption of an elastic stress redistribution on the notched

Nominal values of the material properties can be determined by performing a three-point bending test on a notched beam according to EN 14651 [18-1] (Figure 18.2-3). The diagram of the applied force (F) versus the deformation must be produced (Figure 16.2-4). The deformation is generally expressed in terms of crack mouth opening displacement (CMOD), which is the opening of the notch at the bottom face of the beam (Figure 18.2-4).

Parameters, f_{Rj}, representing the residual flexural tensile strength, are derived from the F–CMOD relationship, as follows:

$$f_{R,j} = \frac{3F_j \cdot l}{2b \cdot h_{sp}^2} \qquad (18.2\text{-}1)$$

where:

f_{Rj} [MPa] is the residual flexural tensile strength corresponding to CMOD = CMODj;

F_j [N] is the load corresponding to CMOD = CMODj;

l [mm] is the span length;

b [mm] is the specimen width;

h_{sp} [mm] is the distance between the notch tip and the top of the specimen (125 mm).

Other tests can be accepted if correlation factors with the parameters of EN 14651 [18-1] are proven to have statistical significance.

Strain hardening materials have to be classified with EN14651 [18-1], but different geometries closer to specific applications can be selected, (standard

cross section.

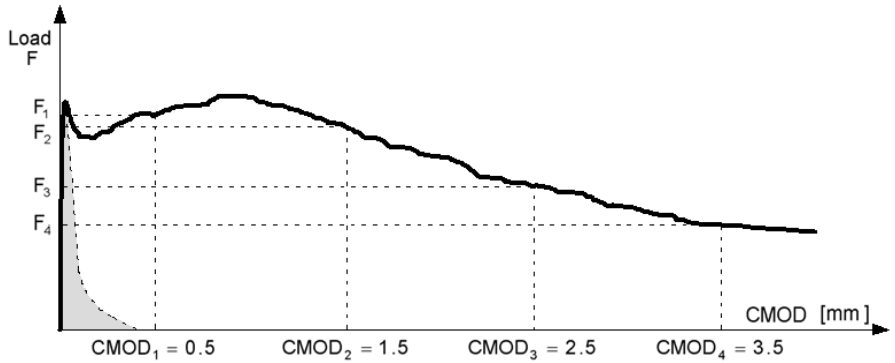


Figure 18.2-4: Typical load F–CMOD curve for plain concrete and FRC.

The characterization of hardening materials is also possible by using a three-point unnotched bending tests, following the procedure suggested by Lopez Martinez [18-4].

In case of synthetic and natural fibres, long term post-cracking behaviour can be affected by an additional creep of the fibres themselves. Creep tests can be carried out following the procedure suggested by Rilem TC 261-CCF [18-5], Llano-Torre & Serna [18-6].

As for HPC, for UHPC the creep coefficients are relatively small, with values usually lower than 2.

For structural applications with normal and high-strength concrete, the material classification is based on the characteristic post-cracking residual strengths.

For instance, a material denoted as "3b" has a strength f_{R1k} of 3MPa≤ f_{R1k} <4 MPa and the f_{R3k}/f_{R1k} ratio is in the range 0.7≤ f_{R3k}/f_{R1k} <0.9.

This recommendation is developed as a general basis for all types of fibre concrete. However, in case of high-performance fibre concrete, specific design rules could be adopted (JSCE Recommendations [18-7],NF P18-710 [18-11]).

bending on unnotched beams or tensile tests) to better identify the constitutive laws by inverse analysis NF P 18-470 [18-2], Baby et. al. [18-3], but they have to be correlated with the EN 14651 [18-1] test (Figure 18.2-3).

For both suitable type and volume of fibres (around or more than 1%) with a composition that leads to particle packing, strain hardening materials can be obtained. To properly identify and quantify the pre-peak hardening in tension, a uniaxial tension test carried out on unnotched specimens is required.

18.3 Classification

The post-cracking strength of FRC should be classified by an linear elastic behaviour considering the characteristic flexural residual strength values that are significant for serviceability (f_{R1k}) and ultimate (f_{R3k}) conditions, and, in particular, two parameters: f_{R1k} (representing the strength interval) and a letter a, b, c, d or e (representing the f_{R3k}/f_{R1k} ratio).

The limit of proportionality f_L, as defined in EN 14651 [18-1], can be determined by applying the following equation:

$$f_L = \frac{3 F_L l}{2 b \cdot h_{sp}^2} \qquad (18.3\text{-}1)$$

In order to define the characteristic value from the test results, the procedure suggested in EN 1990 [18-8] can be used. At first the standard deviation, s_x, is calculated from the mean value of the test results, x_m, the individual results, x_i, and the number of tests, n, from the relation:

$$s_x=\sqrt{\sum}(x_i-x_m)2)/(n-1) \qquad (18.3\text{-}2)$$

Then the coefficient of variation (COV), v_x, follows from:

$$v_x=s_x/x_m \qquad (18.3\text{-}3)$$

Now the characteristic value follows from:

$$x_k=x_m - k_n \cdot s_x \qquad (18.3\text{-}4)$$

where k_n is a parameter depending on the number of tests: k_n decreases with an increasing number of tests according to Table. 18.3-1.

Table 18.3-1: Parameters for determining the characteristic strength.

N	3	4	5	6	8	10	12	20	30	40
k_n, known	1.89	1.83	1.80	1.77	1.74	1.72	1,71	1.68	1.67	1.64
k_n, unknown	3.37	2.63	2.33	2.18	2.00	1.92	1.87	1.76	1.73	1.64

If data are available from a large production of the same material (with the same fibre content, fibre geometry, concrete rheology, production and casting process), the coefficient of variation can be considered as known. Also a log-normal distribution can be used. k_n unknown means that the coefficient of variation of the population is unknown. In that case the standard deviation of the test series itself is governing.

The strength interval is defined by two subsequent numbers in the series:

1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 10.0, 12.0, 14.0 [MPa]

while the letters a, b, c, d, e, correspond to the residual strength ratios:

a if $0.5 < f_{R3k}/f_{R1k} < 0.7$;

b if $0.7 \leq f_{R3k}/f_{R1k} < 0.9$;

c if $0.9 \leq f_{R3k}/f_{R1k} < 1.1$;

d if $1.1 \leq f_{R3k}/f_{R1k} < 1.3$;

e if $1.3 \leq f_{R3k}/f_{R1k}$.

The designer should specify the residual strength class and the f_{R3k}/f_{R1k} ratio as well as the material of the fibre.

COV is a meaningful statistical descriptor of the sensitivity of mixtures to heterogeneity of fibre distribution and orientation. If a FRC shows a low variability of the f_{Rj} values in a series of notched beam tests (very sensitive to fibre distribution and orientation), it is also expected the FRC is insensitive to local differences in fibre orientation and distribution within the structure. So, a low COV of the results of the notched beam tests can be seen as an advertisement for the reliability of a mixture, if the production, pouring and casting procedure used to prepare the specimen is similar to that used to fill the structure formwork.

The selection of 14 MPa as upper limit for f_{R1} takes into account the available materials in the market, when a 3D random distribution is considered.

The hardening FRCs are very often self-compacting materials and the material behaviour in the hardening stage is often orthotropic. This means that if checked in an orthogonal direction the material could present softening, because fibres are aligned in one main direction. Moreover, the extension of the hardening branch, that is responsible for the ductility, can be affected by size effect, which means that increasing the size of the specimen used to identify the material behaviour, the peak strength as well as the peak strain can reduce. For this reason, the use in the SLS design of the hardening branch has to be validated not only for small specimens, but also for real size elements. Uniaxial tension test can be carried out according to JSCE Recommendations [18-7].

The changes in the parameters of the stress-strain relation in Fig. (18.4-1) reflect the actual ductility of FRC depending on the value of $f_{R1,k}$, as shown in the database and subsequent analysis in the following references: Ruiz et. al. [18-9], De La Rosa et. al. [18-10].

The equations suggested are valid up to the classes declared (C120).

For UHPC the maximum aggregate size is mostly smaller than 5 mm and micro-fibres are characterized by small diameters (≤ 0.3 mm) and short lengths (≤ 20 mm).

For design according to this Code the following equations should be satisfied:

$$f_{R1k}/f_{Lk} > 0.4 \qquad (18.3\text{-}5)$$

$$f_{R3k}/f_{R1k} > 0.5 \qquad (18.3\text{-}6)$$

To guarantee a hardening behaviour in direct tension, a uniaxial tension test to check the ductility in the hardening branch should be performed: if localization of the deformations occurs for strain values lower than 1% (Fig. 18.4.7), the ultimate limit state is very often controlled by the softening behaviour occurring after the hardening pre-peak branch. In this case the material can be defined as a hardening material characterized by a limited ductility. The significant pre-peak hardening behaviour significantly improves the SLS performance of the structure, but the material response at the ULS is affected by the post-peak softening behaviour.

18.4 Constitutive laws

18.4.1 Uniaxial compression

The relation between stress σ_c and strain ε_c in compression in Fig. (18.4-1) may be used to model the response of FRC to short-term uniaxial compression provided the following modifications in the parameters are made:

For these reasons the specimen sizes could be significantly reduced, respecting the rules that guarantee the homogeneity of the material.

Four levels of approximation are defined. Level of approximation (LoA) I and II are suited for cross sectional analysis and mainly to evaluate the FRC contribution at ULS, while LoA III and IV are mainly suited for SLS.

LoA III and IV are also oriented to Finite Element Analysis in order to prevent spurious energy, allowing crack localization.

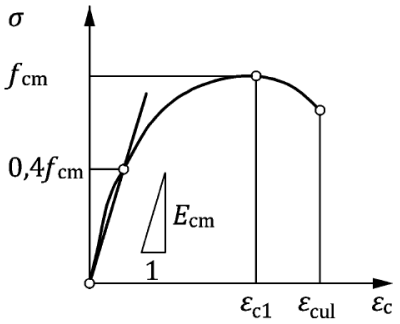


Figure 18.4-1: Stress-strain relation for fibre reinforced concrete in uniaxial compression

$$\varepsilon_{c1}(‰) = 0.7\, f_{cm}^{1/3}\,(1+0.03\, f_{R1k}) \qquad (18.4\text{-}1)$$

and, for $\varepsilon_{c1} \le \varepsilon \le \varepsilon_{cu1}$

$$\varepsilon_{cu1} = k\, \varepsilon_{c1} \qquad (18.4\text{-}2)$$

with

$k = 1+20/\sqrt{82 - 2.2\, f_{R1k}}$ [where f_{cm} and f_{R1k} must be inserted in MPa].

For hardening materials characterized by a very high compressive strength (>120 MPa) an elastic-brittle behaviour characterized by a peak compressive strength f_{ck} may be assumed for short-term compression design. Advanced rules may be used, e.g. to check elastic stability NF P18-710 [18-11].

18.4.2 Uniaxial tension

Stress-crack opening laws in uniaxial tension are defined for the post-cracking behaviour of FRC, Figure 18.4-2. The post-cracking tensile behaviour can be derived from the results of the notched beam tests, assuming the principle given in Figure 18.4-3.

Two simplified stress-crack opening constitutive laws may be deduced from the bending test results: a rigid plastic behaviour, or a linear post-cracking behaviour

When used in Finite Element Analyses approach, the pre-peak hardening branch (AB in Fig. 18.4-7) can be neglected, because in many cases it can also introduce a spurious energy dissipation, and a linear elastic behaviour up to the uniaxial tensile strength can be adopted.

The two simplified models corresponding to LoA I and II, cannot be used with Finite Element analyses, because they cannot reproduce crack localization because they fully neglect the matrix behaviour and reproduce only the pull-out resistant mechanism exhibited in the post-cracking regime and, therefore, they risk to introduce a significant spurious dissipated energy.

The rigid-plastic model (see Fig. 18.4-2) follows from the assumption of static equivalence as shown in Figure 18.4-3, that implies that f_{Ftu} results from the assumption that the whole compressive force is concentrated in the top fibre of the section.

$$M_u = \frac{f_{R3k} b h_{sp}^2}{6} = \frac{f_{Ftu} b h_{sp}^2}{2}$$

(hardening or softening) as schematically shown in Figure 18.4-2, where f_{Fts} represents the serviceability residual strength, defined as the post-cracking strength for serviceability crack openings (w_{SLS} = 0.5 mm), and f_{Ftu} represents the ultimate residual strength.

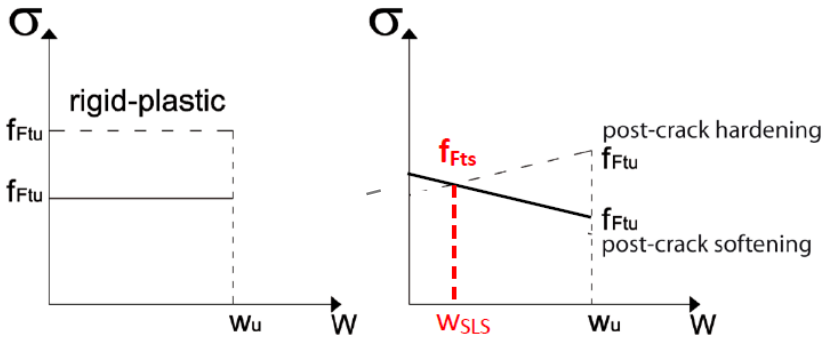


Figure 18.4-2: Simplified post-cracking constitutive laws: stress-crack opening (continuous and dashed lines refer to softening and hardening post-cracking behaviour, respectively).

Rigid-plastic model

According to the simplest approximation (LoA I), a rigid-plastic model (Fig. 18.4-2 Left) identifies a unique reference value, f_{Ftu}, to characterize the behaviour in the ULS. Such a value is determined as:

$$f_{Ftu} = \frac{f_{R3k}}{3} \qquad (18.4\text{-}3)$$

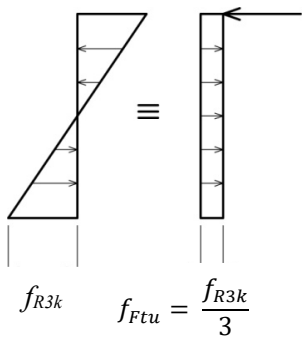


Figure 18.4-3: *Simplified model adopted to compute the ultimate residual tensile strength in uniaxial tension f_{Ftu} by means of the residual nominal bending strength f_{R3k}.*

The equation to compute f_{Ftu} at $w_u = CMOD_3$ is based on the assumption of rotational equilibrium at ULS, when a stress block in tension along the section is taken into account, as shown in Figure 18.4-3.

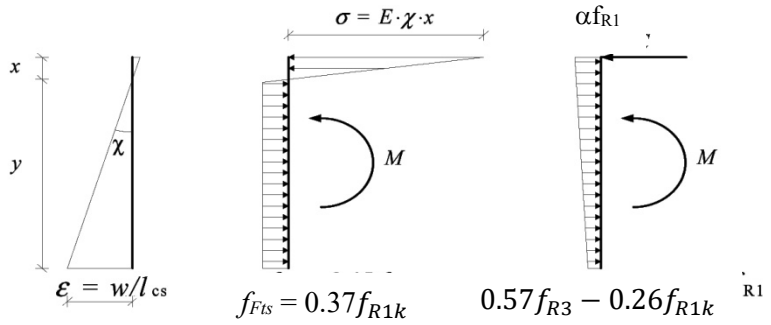


Figure 18.4-4: *Linear model: stress diagrams corresponding to a linear strain distribution (a), respectively for the determination of the residual tensile strength f_{Fts}, (b) at w = 0.5 mm (= $CMOD_1$ and (c) f_{Ftu} at w = 2.5 mm(= $CMOD_3$).*

The limit value w_u applies particularly to ULS verifications.

Linear model

A more advanced method of analysis (LoA II) is the linear post cracking model , which identifies two reference values, namely f_{Fts} and f_{Ftu}.

They have to be defined through residual values of flexural strength using the following equations:

$$f_{Fts} = 0.37 f_{R1k} \qquad (18.4\text{-}4)$$

$$f_{Ftu} = f_{Fts} - \frac{w_u}{CMOD_3}\left(f_{Fts} - 0.57 f_{R3k} + 0.26 f_{R1k}\right) \geq 0 \qquad (18.4\text{-}5)$$

where w_u is the maximum crack opening accepted in structural design; its value depends on the ductility required.

The equation for f_{Ftu} and $w_u \neq CMOD_3$ is obtained by considering a linear relation between points with abscissa $CMOD_1$ and $CMOD_3$, up to the point with abscissa w_u (Figure 18.4-5).

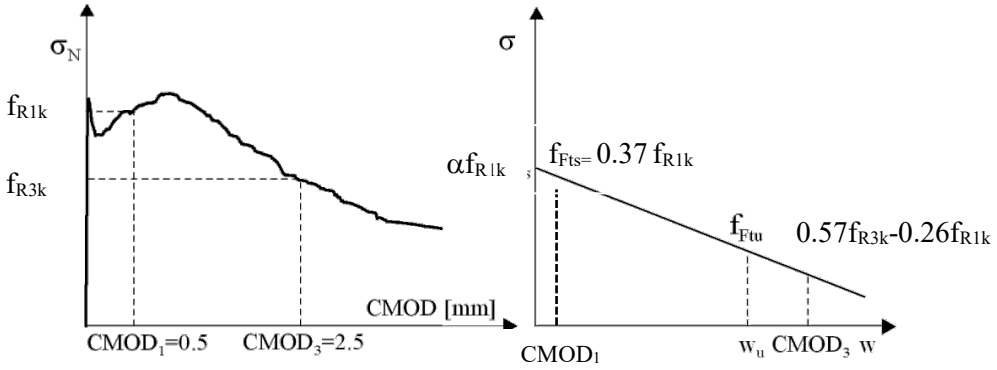
The stress value f_{Fts} corresponding to the crack opening $CMOD_1$ is determined from equilibrium, with the assumption that the compressive stress distribution is linear (Figure 18.4-4b) and that the tensile behaviour is elasto-plastic until a crack opening displacement corresponding to the serviceability limit state ($CMOD_1$):

$$M(CMOD_1) = \frac{f_{R1k} b h_{sp}^2}{6}$$

The coefficient introduced in Eq. (18.4-3) is affected by the Young's modulus of the material, but the variability range is very limited and therefore is here neglected and a unique value is assumed (see, Di Prisco et/ al/ 2011 [18-12]).

The stress value corresponding to the crack opening $CMOD_3$ is determined from equilibrium, with the assumption that the compressive stress resultant is applied on the extrados chord (Figure 18.4-4c) and that the tensile behaviour is rigid-linear:

$$M(CMOD_3) = \frac{f_{R3k} b h_{sp}^2}{6}$$



(a) (b)

Figure 18.4-5: Typical results from a bending test on a softening material (a); linear post-cracking constitutive law (b).

For numerical analyses, more advanced constitutive laws are recommended, including first crack tensile strength, with a specific care to the calibration of ultimate strain value.

α is the coefficient corresponding to $w = 0$ and can be computed according to the following equation:

$$a = 0.52 - 0.15\frac{f_{R3}}{f_{R1}}$$

When considering softening materials, the definition of the stress–strain law should be based on the identification of the crack width and on the corresponding structural characteristic length, l_{cs}, of the structural element. Thus, the strain can be assumed equal to:

$$\varepsilon = w / l_{cs} \qquad (18.4\text{-}6)$$

In elements with conventional reinforcement (rebars), when the usual plane section kinematic model is adopted l_{cs}, should be evaluated as:

$$l_{cs} = \min\{s_{rm}, y\} \qquad (18.4\text{-}7)$$

where:

s_{rm} is the mean distance between cracks, computed as suggested in Eq. 18.6-6:

y is the distance between the neutral axis and the tensile side of the cross-section (Figure 18.4-4a), evaluated in the elastic cracked phase by neglecting the residual tensile strength of FRC, and for a load corresponding to the serviceability state of crack opening and crack spacing.

If a different kinematic model is adopted as for Finite Element method, the structural characteristic length is correlated to the size of the finite element and requires a suitable calibration (*fib* Bulletin ZZ [18-13]). The ultimate tensile strength f_{Ftu} in the linear model depends on the required ductility, that is related to the allowed crack width. The ultimate crack width can be calculated as $w_u = l_{cs}\ \varepsilon_{Fu}$, by assuming ε_{Fu} is equal to 2% for variable strain distribution along the cross-section and 1% for constant tensile strain distribution along the cross-section. In any case, the maximum crack width may not exceed 2.5 mm.

In sections without traditional reinforcement under bending or under combined tensile-flexural and compressive-flexural forces with resulting force external to the section, y = h is assumed. The same assumption can be taken for slabs.

Strain hardening materials with $\varepsilon_P \geq 2\%$ (Figure 18.4.6) subjected to a variable strain distribution along the cross-section are considered ductile enough to be used neglecting localization even at ULS and setting $\varepsilon_{Fu} = \varepsilon_P$. For a constant tensile strain distribution along the cross-section the limit strain value becomes $\varepsilon_P \geq 1\%$.

The first and the second branch suggested in the pre-peak constitutive relationship and the post-peak crack propagation branch correspond to the behaviour of plain concrete until the intersection with the residual post-cracking behaviour which resumes fibre contribution due to pull-out.

An experimental approach to quantify the pre-peak behaviour of UHPC materials can also use a controlled tension stiffening test Khorami et. al. [18-14].

When f_{Fts} becomes closer to f_{ctm} (case II), a first localization corresponding to the matrix cracking occurs after the peak revealed by a jump, followed by a hardening branch correlated by fibre pull-out mechanism, as shown in figures 18.4-6b. The selection of the parameters α_s and β would require an optimization process aimed at minimizing the error with the experimental curve, by executing inverse analysis procedure. The equation proposed is aimed to simplify this procedure by imposing only one point of equilibrium at SLS for CMOD = 0.5 mm: the proposed values of α_s and β have been calibrated for this purpose.

For higher levels of approximation at ULS and for SLS, more refined stress-strain uniaxial relationships in tension could be applied.

LoA III

In the third Level of Approximation, 4 cases can be distinguished

Case (I) - For softening materials at SLS ($f_{Fts} \leq 0.8 f_{ctm}$), the same constitutive relationship adopted for plain concrete in uniaxial tension is used up to the peak strength f_{ctm}. In the post-cracking stage, a bilinear relation applies (Figure 18.4-6a).

The post-peak propagation branch (see BC in Figure 18.4-6a) is analytically described as:

$$\frac{\sigma - f_{ctm}}{0{,}2 f_{ct} - f_{ctm}} = \frac{\varepsilon - \varepsilon_P}{\varepsilon_Q - \varepsilon_P}, \text{ for } \varepsilon_P \leq \varepsilon \leq \varepsilon_C \qquad (18.4\text{-}8)$$

$$\text{with } \varepsilon_Q = \frac{G_F}{f_{ctm} \cdot l_{cs}} + \left(\varepsilon_P - \frac{0{,}8 f_{ctm}}{E_c}\right) \qquad (18.4\text{-}9)$$

where G_F represents the fracture energy of plain concrete, see Eq. (14.6-9).

Point A in the curves of Figure 18.4-6 (a), (b) is defined in Figure 18.4-6.

For softening materials, the residual strength (fourth branch) is defined by two points corresponding to ($\varepsilon_{SLS}, f_{Fts}$) and ($\varepsilon_{ULS}, f_{Ftu}$) where:

$$\varepsilon_{SLS} = CMOD_1 / l_{cs} \qquad (18.4\text{-}10)$$

$$\varepsilon_{ULS} = w_u / l_{cs} = \min(\varepsilon_{Fu}; 2.5/l_{cs}) \qquad (18.4\text{-}11)$$

with ε_{Fu} = 2% for variable strain distribution along the cross-section and 1% for only tensile strain distribution along the cross-section – see section 18.4.2.

Case (II): materials characterized by a residual strength f_{Fts} larger than $0.8 f_{ctm}$. In this case the cracking process shows localization, but the residual strength on uniaxial tension at SLS strain remains close to the peak strength: a three-branch post-peak relation defines the constitutive relationship. The first branch in the post-peak remains that corresponding to plain concrete, while the second branch (C'V) is analytically described as:

$$\frac{\sigma - \mathrm{b} f_{ct}}{\mathrm{s}_V - \mathrm{b} f_{ct}} = \frac{\varepsilon - \varepsilon_{C'}}{\mathrm{a}_s \varepsilon_{SLS} - \varepsilon_{C'}}, \text{ for } \varepsilon_{C'} \leq \varepsilon \leq \mathrm{a}_s \varepsilon_{SLS} \qquad (18.4\text{-}12)$$

Note that the upper limit introduced in the classification in section 18.3 for f_{R1k} is 14 MPa (section 18.3): for this value f_{Fts} is equal to 5.18 MPa. The compressive strength of this type of FRC in the range C90÷C120 MPa with f_{ctm} in the range of 5÷5.6 MPa.

The point C' in case II is introduced to model the crack localization that always takes place. The crack opening corresponding to point V depends on f_{Ftsk} and this dependence is controlled by the parameter $\alpha_s = \alpha_s(f_{R1k})$ with $\alpha_s \leq 1$.

In both cases (I) and (II) f_{ct} is always assumed equal to f_{ctm}.

It is worth to note that sometimes, in small specimens, the pre-peak branch can also extend up to 2-5‰, but this extension characterized by fine cracks and very small crack spacings, dominated by the progressive transition from the tensile matrix strength to the pull-out mechanism strength, can be significantly affected by size effects due to a not-homogeneous fibre distribution.

For these materials the effective tensile strength exceeds the matrix tensile strength and significantly depends on fibre amounts and their bond performance to surrounding medium.

The maximum tensile strength of the pre-peak behaviour f_{Ft} is usually significantly affected by fibre volume

In case of unavailable data for the softening branch of uniaxial tension test, the last branch can be identified also by means of EN 14651 [18-1] test by assuming:

$$e_D = \varepsilon_{FU} = 2.5\text{mm}/l_{cs}; \qquad (18.4\text{-}12)$$

$$w_D \cong 2.5\text{mm} - l_{cs}\varepsilon_R \qquad (18.4\text{-}13)$$

with $l_{cs} = 125$ mm, ε_R can be roughly assumed as $\varepsilon_R = \varepsilon_P - f_{Ft}/E_c$ that corresponds to a negligible damage, while the expression of the softening branch can be described by:

$$s(w) = aw^2 + bw + c \qquad (18.4\text{-}14)$$

where the constants a, b, c can be determined imposing the passage for B and D

$$w = 0 \quad \sigma = f_{Ft}; \qquad w = w_D \quad \sigma = f_{Ftu} \qquad (18.4\text{-}15)$$

and the condition that:

After heaving reached point V the material follows the pull-out branch characterized by softening with the slope of the branch (DE).

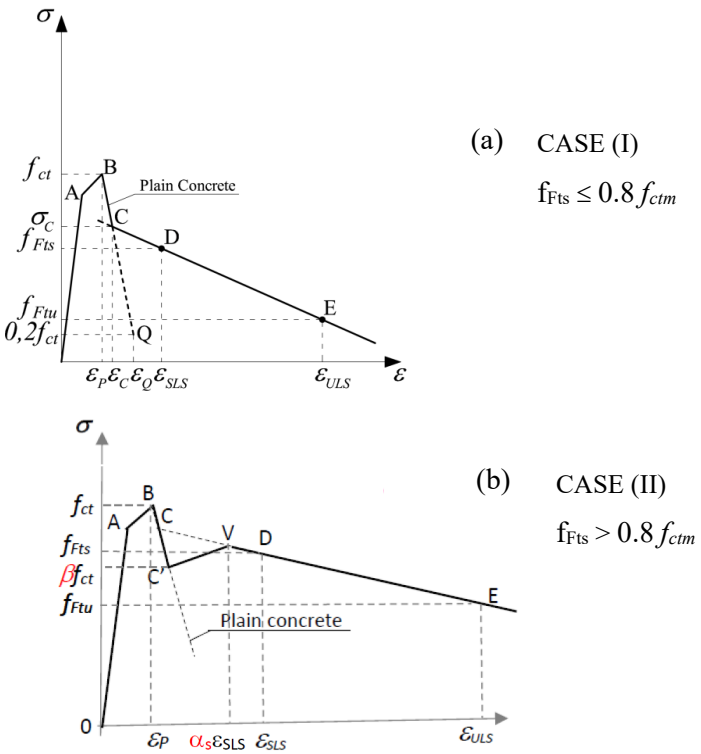


Figure 18.4-6: Stress–strain relations at SLS for softening (a) and quasi-plastic (b) behaviour of FRC.

The values α_s and β can be identified by means of equilibrium equations: may be assumed conservatively as $\alpha_s = 1$ and $\beta = 0.75$.

Case (III): For materials showing a hardening pre-peak behaviour exceeding f_{ctm}, the ε_{Pk} and the associated strength f_{Ft} have to be identified by means of the uniaxial tension test. The modulus of elasticity of the composite also grows, but its increase can be neglected for sake of simplicity.

The f_{Fl} tensile strength, where the material changes its stiffness and starts to accumulate damage often exceeds the matrix tensile strength and can be identified also by means of EN 14651 [18-1] test looking to the change of the initial slope. The

$$\frac{ds}{dw} = \frac{-f_{Ftu}}{(\frac{l_f}{2} - w_D)} \text{ for } w = w_D \qquad (18.4\text{-}16)$$

The expression of the constants a, b, c according to the assumptions described are:

$$a = \frac{f_{Ft}}{{w_D}^2} - \frac{\frac{l_f}{2} f_{FTu}}{{w_D}^2 (\frac{l_f}{2} - w_D)}$$

$$b = -2aw_D - \frac{f_{Ftu}}{\left(\frac{l_f}{2} - w_D\right)}$$

$$c = f_{Ft}$$

and can be determined by imposing iteratively the longitudinal and the rotational equilibrium for $w = w_D$, assuming a linear distribution of strains along the notched cross section, and as unknows the variables x and $\sigma(w_D) = f_{Ftu}$.

pre-peak behaviour usually is confined to SLS., because, in general, the maximum strain ε_P does not exceed 5‰; a bilinear law could also be associated to the softening regime experimentally identified up to a maximum crack opening variable, because of the contribution of the irreversible strains accumulated in the pre-peak behaviour (equal to ε_R; Figure 18.4-7). The characteristic structural length associated to the softening due to localization can be assumed equal to that of common FRCs. The last branch corresponding to that joining C and D points remains that associated to the pull-out mechanism on the localized crack. The equations (18.4-2, 18.4-3) are not valid to correlate the residual strengths in uniaxial tension due to pull-out to f_{R1k} and f_{R3k} residual strengths, because the pre-peak hardening has to be taken into account into the equilibrium.

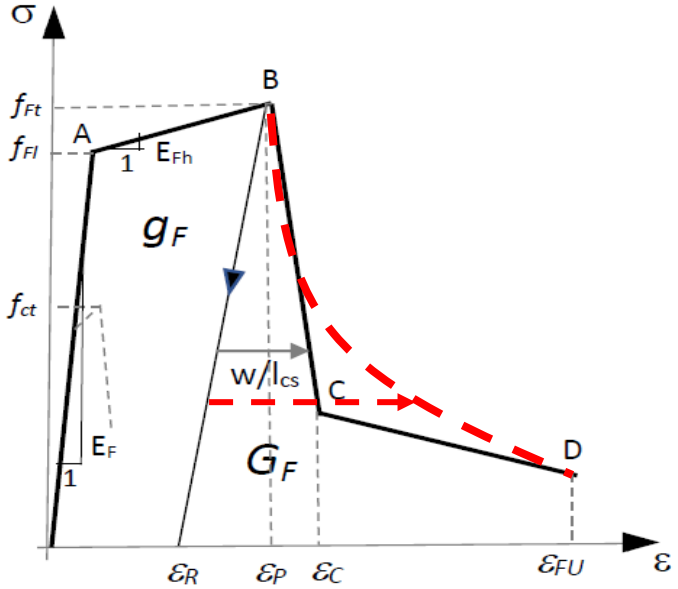


Figure 18.4-7: Stress–strain relations for UHPC.

Case (IV) When ε_{Pk} exceeds 1% in constant strain condition along the cross-section profile, no softening law is defined and the constitutive law reduces only to the pre-peak elasto-hardening branch.

LoA IV

The behaviour observed in the standard tests can deviate substantially (beneficial and non-beneficial) from the behaviour of the corresponding FRC in the structural element or structure. The main reason is related to fibre orientation that is affected by boundary conditions represented by the formworks and casting procedure. Fibre orientation can play a significant role especially in self-compacting concrete because a selected casting flow can easily induce a fibre orientation. Thus, the manufacturing method and the concrete consistency should be taken into account by the designer: following the same procedure adopted for the final production suitable preliminary specimens can be experimented, by applying the design by testing procedure Martinelli et. al. [18-15] , Ferrara et al [18-16], Zirgulis et. al. [18-17], Wille et. al. [18-18], Di Prisco et. al. 2016 [18-19], Di Prisco et. Al. 2013 [18-20], Schönlin [18-21].

Fibre orientation α_0 can be determined by means of special magnetic inductive devices on cored specimens or thin plates.

The residual tensile strength of steel fibre reinforced concrete is directly proportional to the number of fibres in the crack plane. The relation between the number of fibres in a given cross section, the geometrical characteristics and volume concentration of the fibres and the fibre orientation is given by:

$$N_f = a_0 \frac{V_f}{p\frac{d_f^2}{4}}$$

N_f number of fibres per unit cross-sectional area in a given cross section;

V_f fibre concentration by volume;

d_f fibre diameter;

The level of approximation IV is recommended for finite element analysis, where the stress-crack width relationship is determined from inverse analysis by fitting with the minimum error the force-CMOD registered in the EN 14651 [18-1] tests or in a specimen representing, as much as possible, the fibre distribution and orientation in the FRC to be designed. In smeared crack approaches the crack width is transformed in a tensile strain using a characteristic length that aims to avoid dependence of the results on the finite element mesh.

18.5 Orientation factor

A factor k_0 corresponding to the ratio between the fibre orientation α_0 in the structure and the fibre orientation $\alpha_0 = 0.58$ in the EN14651 [18-1] standard specimen is introduced:

$$k_0 = \frac{a_0}{0.58} \qquad (18.5\text{-}1)$$

In general, at least as a first stage for non-self-compacting FRC, a quasi-isotropic fibre distribution is assumed: in this situation the factor k_0 is assumed equal to 1. For precast elements by using "structural" tests able to reproduce carefully the casting procedure and the boundary conditions an experimental value of α_0 can be identified in bending tests or uniaxial tension tests.

The values $f_{Fts,ef}$ and $f_{Ftu,ef}$ represent the residual tensile strength values identified by bending tests according to EN14651 [18-1] and corrected taking into account the manufacturing method and the concrete workability:

$$f_{Fts,ef} = k_0 f_{Fts} \qquad (18.5\text{-}2)$$

$$f_{Ftu,ef} = k_0 f_{Ftu} \qquad (18.5\text{-}3)$$

For favourable effects, an orientation factor $k_0 > 1.0$ may be applied if experimentally verified. A maximum value of 1.5 may be assumed. When $k_0 > 1.0$ is applied in one direction, the k_0 in the orthogonal direction $k_{0\perp}$ has to be considered lower than 1.

For unfavourable effects, an orientation factor $k_0 < 1.0$ must be experimentally determined and applied.

α_0 factor representing the fibre orientation, with the following values;

for "standard" fibre orientations:

α_0 = 0.5 in the case of 3D random fibre orientation Dupont & Vandewalle[18-22]

α_0 = 0.64 in the case of 2D random fibre orientation (for any sectional

plane perpendicular to the fiber plane)

Experience has been gained in experimental quantitative assessment of the orientation factor using sawn prisms from a representative mockup, as compared to moulded specimens Simon et. al. [18-23] Schönlin [18-21].

The computation of N_f can be required if a check on the cracked plane has to be performed.

If no experimental measure is available possible values for slabs and walls are suggested in the bulletin ZZ [18-13].

The average strain under the maximum load in compressive tests is around 30% bigger than in that of the corresponding base concrete, which explains the increase of ε_{c2}. Similarly, the ultimate strain is calculated by making that the energy consumed in a rectangular post-peak branch of the stress-strain curve corresponds to the average increase of such energy in the database in the above references.

In a local check without any experimental investigation a value k_0 = 0.5 should be considered.

18.6 Assumptions used for design

18.6.1 Design values for compression

The strains ε_{c2} and ε_{cu2} for FRC in Eq. 30.1-1a + b can be taken equal to:

$$e_{c2} = 0.002*(1+0.03f_{R1k}) \quad (18.6\text{-}1)$$

$$\varepsilon_{cu2} = \varepsilon_{c2}*k \quad (18.6\text{-}2)$$

For hardening materials characterized by a very high compressive strength (>120 MPa) the compressive behaviour can be assumed for the sake of safety to be perfectly linear up to the peak value.

By reducing the safety coefficient γ_F on f_{cd}, the concrete, even if it exceeds the values of class C120, could be regarded as elastoplastic with the value of f_{cd} up to the peak strain ε_{cu2} corresponding to the f_{ck} / E_{cd}, computed as $E_{cd} = E_c/\gamma_F$.

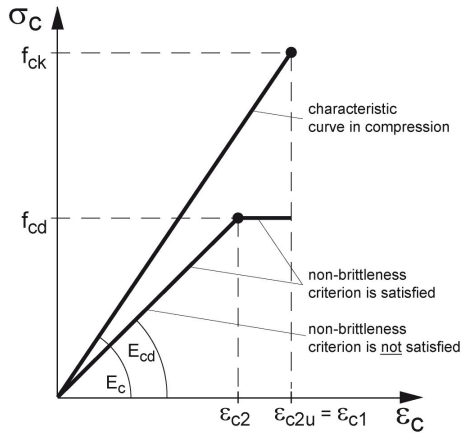


Figure 18.6-1: Design constitutive law for uniaxial compression of hardening materials with f_{ck} > 120 MPa.

18.6.2 Design values for uniaxial tension

The residual design values in uniaxial tension are affected by the member size and the number of fibres activated in the cracking process. Because of this the most relevant source of the large standard deviation which characterizes the composite is related to fibre number. For this reason, if a large volume of fibres/large concrete area is involved in the cracking propagation, the standard deviation is strongly reduced if the pre-peak behaviour is characterized by a stable crack propagation.

The design values of the residual strengths are therefore affected by a further coefficient denominated k_G. The value of this coefficient is affected by the cracking process volume and by its stability.

The factor k_G according to LoA I and II can be computed as:

$k_G = 1 + 0.5\ A_{ct}$ (18.6-3)

where:

A_{ct}, expressed in m^2 is the area of cracked concrete involved in the failure mechanism

The residual design strengths are therefore defined as follows:

$f_{Ftsd} = k_G\, f_{Fts,ef} / \gamma_F$ (18.6-4)

$f_{Ftud} = k_G\, f_{Ftu,ef} / \gamma_F$ (18.6-5)

The factor k_G can be evaluated according to LoA III and IV, remembering that it is affected not only by the volume of the cracking process, but also by the stability of the structure by computing it according to a limit analysis Colombo et. al. [18-24] or an evolutive non-linear analysis approach Di Prisco et. al. 2016 [18-25] as:

$$\kappa_G = \frac{\left(P_u(f_{Fts/u,random})\right)_k}{P_u(f_{Fts/uk})} \leq 0.9 \cdot \frac{f_{Fts/um}}{f_{Fts/uk}}$$

$\left(P_u(f_{Fts/u,random})\right)_k$ is the characteristic value of the ultimate loads computed regarding the material *heterogeneous*, specified by assigning several random variation of the mechanical properties f_{Fts} and f_{Ftu} in the discretized model of the structure.

$P_u(f_{Fts/u,k})$ is the characteristic value of the ultimate loads computed regarding the material *homogeneous,* by introducing the characteristic properties f_{Ftuk} and f_{Ftsk} in the constitutive relationships attributed to the whole discretized model of the structure.

Note: Both f_{Ftum} and f_{Ftuk} need to be known for the fibre concrete considered.

where γ_F is the safety coefficient for FRC.

18.6.3 Crack spacing

The average crack spacing for members reinforced with steel fibres and longitudinal bars should be calculated according to the Eq. (30.5-5) as:

$$s_{rm} = \left(k_c c + k_{\phi/\rho} k_{fl} k_b \frac{(f_{ctm} - f_{\mathrm{Fts,ef}})\phi}{\tau_{bms}\rho_{s,ef}}\right) \qquad (18.6\text{-}6)$$

where $f_{Fts,ef}$ is the effective residual tensile strength for the serviceability limit states, and the remaining parameters are as defined in section 30.5.2.4.4.2.

For hardening materials with limited ductility, $f_{Fts,ef}$ will be substituted by σ_C or $\sigma = \sigma(w=0.5mm)$. In case of hardening material s_{rm} is not defined.

The stress in the tensile reinforcement, σ_s, have to be calculated using the fibre contribution.

The stress and crack width limits in Table 30.5-1 may be applied to members reinforced with steel fibres.

In determining stresses and crack widths a portion of the cover $c_{f,dur}$ shall be disregarded from the concrete section. For design of SFRC in exposure classes XC2-XC4, XD1-XD3, and XS1-XS3, designed to be uncracked, the tensile strength in the outer most tension fibres shall be disregarded within a layer of $c_{f,dur}$ = 10 mm from the exposed surface.

For design of SFRC in exposure classes XC2-XC4, XD1-XD3, and XS1-XS3 designed to be cracked, the residual tensile strength in the outer most tension fibres shall be disregarded with a distance of $c_{(\mathrm{min,dur})}$ from the exposed surface, unless provisions (1) or (2) are satisfied:

(1) If protective measures are applied to the steel fibres to protect them for the design life of the structure, the residual tensile strength of the outermost tension fibres within depth $c_{f,dur}$ may be used.

(2) The residual tensile strength of fibres in the layer $c_{f,dur}$ may be used in temporary situations such as during the construction phase.

If the characteristic crack width calculated in accordance with chapter 30 is less than the crack width limit (w_{lim}) stated in Table 30.5-1, the values of $c_{f,dur}$ may be reduced as follows:

$$c_{f,dur} = c_{min,dur}\ w_{k,cal}/w_{lim} \geq 10 \text{ mm} \quad (18.6\text{-}7)$$

18.7 Provisions for quality control

The risk evaluation regarding stability in the concrete delivery and the casting work shall be carried out by the contractor. It is for safety reasons extremely important that fibre-continuity between casting-layers is assured The work should be planned and carried out so that interruptions, that can give layering (interfaces), do not occur.

The casting work shall be carried out so that possible obstacles (electrical tubes etc) do not create weakened zones with low amounts of fibres. The minimum distance between reinforcing bars should be 1.5 the bar diameter d_b.

At unloading and casting it shall be checked whether fibre balling does occur. If fibre balling occurs, this concrete shall be disregarded, and the production modified. Control and identity testing at the construction site may be performed according two possible alternative methods:

a) Process oriented control: in addition to the conformity control carried out by the concrete producer, the contractor is responsible for additional control of fibre content at the fresh state and air at the construction site, if available, and concrete compressive strength. If deviations occur, the Fibre Reinforced Concrete should be reclassified according to further specific tests on extracted samples. The control is assumed positive if:

$$V_{f,,min} \geq 0.8\ V_{fN} \quad (18.7\text{-}1)$$

$$V_{f,,av} \geq 0.9\ V_{fN} \quad (18.7\text{-}2)$$

where V_{fN} is the nominal percentage by volume declared for the mix, while $V_{f,,av}$ and $V_{f,,min}$ are respectively the average and the minimum percentage by volume measured on site. A minimum of three controls over the first 50 m^3 and then 1 control per 200 m^3 of FRC are suggetsed. For concrete strengths and air contents, the same rules adopted for plain concrete may be used.

b) Material oriented control: includes in addition casting of standard beams according to EN 14651 [18-1] at the construction site and residual strength identification, one per each 100 m^3, with a minimum of three tests. The minimum and average values should be considered. The control is assumed positive if:

The Double Punch Test (DPT) Molins et. al. [18-26] is a quite simple test, because it is a compression test that does not require any notch. The test control requires a ring strain extensometer, but it can be controlled also only by vertical displacement exerted by the piston. The advantages are related to the occurrence of at least three radial cracks randomly oriented, and therefore does not give any information on fibre distribution, but an average toughness correlated to at least three critical surfaces characterized by a normal vector at right angle with respect to the loading direction axis. A good correlation can be found between uniaxial tension test, three point bending tests according to EN 14651 [18-1] and DPT if the average values are taken into account and the residual strengths in the range 0.5-2.5 mm are considered Galeote et. al. [18-27].

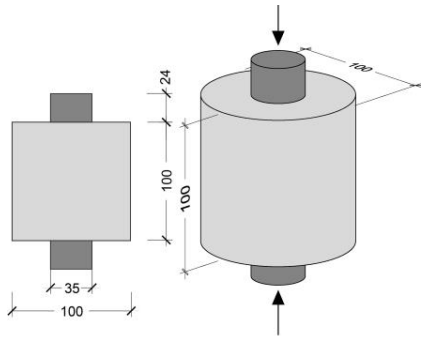


Figure 18.7-1: Double Punch Test for production control.

$$f_{R1,min} \geq 0.9 f_{R1,k}; f_{R3,min} \geq 0.9 f_{R3,k} \quad (18.7\text{-}3)$$

$$f_{R1,av} \geq 1.2\, f_{R1k}; f_{R3,av} \geq 1.2\, f_{R3k} \quad (18.7\text{-}4)$$

where $f_{Ri,min}$ and $f_{Ri,av}$ are the minimal and the average residual strengths measured on control tests, while f_{Rik} are the characteristic values of the declared class of the material.

The production control can be easily performed also by means of Double Punch Tests (Molins et. al. [18-26]), if a suitable correlation with EN14651 [18-1] standard test is preliminary calibrated based only on average values.

All concrete shall at arrival be visually inspected to confirm that it contains the right fibre type, and to get an opinion of the consistency, the fibre distribution (homogeneity) and to avoid fibre balling.

18.8 Assumptions used for assessment

18.8.1 Fibre reinforced concrete without deterioration or defects: fibre contribution

The assessment of material performance can be carried out by using the same tests proposed for plain concrete, with the exception of toughness in post-cracking regime. This performance is strictly correlated to fibre type, distribution and orientation. The extraction of prismatic samples can allow the identification of fibre distribution and orientation by planning bending tests oriented in different directions. Often, it could be easier to core cylindrical tests and by means of an inductive equipment detect fibre orientation and distribution Martinelli et. al. [18-15] and finally perform DPTs. Alternatively, it is possible to cut the cylinders, trace two notches and test the cylindrical samples by means of Double Wedge Splitting technique Di Prisco et. al. 2013 [18-28]. In case of thin-walled profiles a Non-Destructive technique allows the identification of fibre distribution and orientation by means of a magnetic device Torrents et. al. [18-29]Ferrara et. al. [18-30]. In case of slabs, a possible identification of the design performance, included the effect of k_0 and k_G, can be pursued by means of plate tests according to ASTM C 1550 [18-31] or similar tests, selecting smaller sizes Minelli & Plizzari [18-32] or plates

The Double Wedge Splitting Test consists in a classical Brasilian test on a preliminarly notched sample, to select the critical plane of interest:

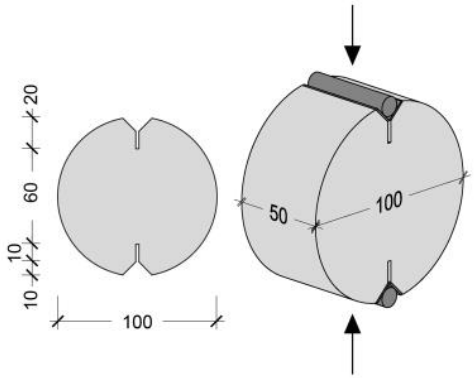


Figure 18.8-1: Double wedge splitting test specimen.

characterized by the same thickness of the on-site slab, by reproducing the same casting procedures followed on-site.

References

[18-1] EN 14651:2005 — Test method for metallic fibered concrete - Measuring the flexural tensile strength (limit of proportionality (LOP), residual)

[18-2] NF P 18-470. 2016. — Concrete – Ultra-high performance fibre-reinforced concretes – Specifications, performance, production and conformity.

[18-3] Baby, F., Gaybeal, B., et al., 2013. — UHPFRC tensile behavior characterization: inverse analysis of four-point bending test results', *Materials and Structures*, vol. 46 (8) , pp. 1337-1354.

[18-4] López Martínez, J. A. — Characterization of the tensile behaviour of UHPFRC by means of four-point bending tests, PhD thesis., DOI: 10.13140/RG.2.2.11442.40640

[18-5] Rilem TC 261-CCF. — Creep behavior in Cracked sections of Fiber Reinforced Concrete.

[18-6] Llano-Torre, A. and Serna, P., 2020. — Recommendation of RILEM Technical Committee 261-CCF: Test method to determine the flexural creep of Fibre Reinforced Concrete in the cracked state.

[18-7] JSCE Recommendations, 2006. — Recommendations for Design and Construction of High Performance Fiber Reinforced Cement Composites with Multiple Fine Cracks (HPFRCC).

[18-8] EN 1990: 2002. — Eurocode – Basis of structural design.

[18-9] G. Ruiz, Á. De La Rosa, E. Poveda., 2019. — Relationship between residual flexural strength and compression strength in steel-fiber reinforced concrete within the new Eurocode 2 regulatory framework. *Theoretical and Applied Fracture Mechanics* ,103.

[18-10] Á. De La Rosa, G. Ruiz, et al., 2019. — Study of the Compression Behavior of Steel-Fiber Reinforced Concrete by Means of the Response Surface Methodology. App. Sci. 9:24, 5330.

[18-11] NF P18-710: 2016. — National addition to Eurocode 2 - Design of concrete structures : specifique rules for ultra-high performance fibre-reinforced concrete (UHPFRC).

[18-12] di Prisco M. Colombo, M. et al., 2011 — Fibre-reinforced concrete in fib Model Code 2010: Principles, models and test validation (2013) Structural Concrete, 14 (4), pp. 342-361.

[18-13] *fib* Bulletin ZZ — Background document under preparation

[18-14] Khorami, M., Navarro-Gregori, J. et al., 2020. — Experimental methodology on the serviceability behaviour of reinforced ultra-high performance fibre reinforced concrete tensile elements, (2020) *Strain*, 56 (5), art. no. e12361.

[18-15] Martinelli P, Colombo M, P., et al., 2021. — Characterization tests for predicting the mechanical performance of SFRC floors: identification of fibre distribution and orientation effects., *Materials and Structures*, 2021, 54 (1), art. no. 3.

[18-16] Ferrara, L., Ozyurt, N. et al., 2011 — High mechanical performance of fibre reinforced cementitious composites: The role of "casting-flow induced" fibre orientation Materials and Structures, 44 (1), pp. 109-128.

[18-17] Zirgulis G., Svec, O. et al., 2016 — Importance of quantification of steel fibre orientation for residual flexural tensile strength in FRC, Materials and Structures (2016) 49:3861–3877

[18-18] Wille K., Viet Tue, N. et al., 2014 — Fiber distribution and orientation in UHP-FRC beams and their effect on backward analysis, Materials and Structures (2014) 47:1825–1838.

[18-19] di Prisco, M., Martinelli P. et al. 2016 — On the reliability of the design approach for FRC structures according to fib Model Code 2010: the case of elevated slabs, Structural Concrete, 17 (4), pp. 588-602.

[18-20] di Prisco M., Dozio D. et al., 2013. — On the fracture behaviour of thin-walled SFRC roof elements Materials and Structures, 46 (5), pp. 803-829.

[18-21] K Schönlin, 1988 — Determination of orientation, amount and distribution of fibers in fiber reinforced concrete, Beton-und Stahlbetonbau, 83(6) (in German)

[18-22] Dupont, D. and Vandewalle, L., 2005. — Distribution of Steel Fibers in Rectangular Sections. *Cement and Concrete Composites*, 27, 391-398.

[18-23] Simon A., Corvez D., Marchand P., 2013. — Feedback of a ten years assessment of fibre distribution using K factor concept, UHPFRC 2013, RILEM PRO 87, pp. 669-678.

[18-24] Colombo, M., Martinelli, P., et al., 2017. — On the evaluation of the structural redistribution factor in FRC design: a yield line approach, (2017) Materials and Structures, 50 (1), art. no. 100.

[18-25] di Prisco, M., Martinelli, P., et al., 2016. — The structural redistribution coefficient KRd: a numerical approach to its evaluation, (2016) Structural Concrete, 17 (3), pp. 390-407.

[18-26] Molins, C., Aguado. A., et al., 2009. — Double Punch Test to control the energy dissipation in tension of FRC (Barcelona test). *Materials and Structures,* 42(4).

[18-27] Galeote, E., Blanco, A., et al., 2017. — Correlation between the Barcelona test and the bending test in fibre reinforced concrete. *Construction and Building Materials*.

[18-28] di Prisco, M., L. Ferrara et al., 2013. — Double edge wedge splitting (DEWS): an indirect tension test to identify post-cracking behaviour of fibre reinforced cementitious composites." *Materials and Structures*, 46(11): 1893-1918.

[18-29] Torrents JM, Blanco A, et al 2012. — Inductive method for assessing the amount and orientation of steel fibers in concrete. *Mater Struct Constr*, 45:1577–1592.

[18-30] Ferrara, L., Faifer, M. et al.., 2012. — A magnetic method for non destructive monitoring of fiber dispersion and orientation in steel fiber reinforced cementitious composites-part 1: Method calibration, *Materials and Structures/Materiaux et Constructions*, 45 (4), pp. 575-589.

[18-31] ASTM C 1550 Standard Test Method for Flexural Toughness of Fiber Reinforced Concrete (Using Centrally Loaded Round Panel).

[18-32] Minelli, F. and Plizzari, G., 2011. A New Round Panel Test for the Characterization of Fiber Reinforced Concrete: A Broad Experimental Study." ASTM *Journal of Testing and Evaluation*, 39(5).

[192] SIA TL 2052. 2016 UHPFRC— Materials, design and construction.

19 Materials & systems for protection, repair and upgrading

19.1 Scope, definition and range of applicability

19.1.1 General

New concrete structures are designed, executed and maintained to meet a number of desired performances during their service life, such as being durable and having adequate strength. This is typically achieved by providing the concrete and reinforcement with suitable durability and mechanical properties.

However, under harsh environmental and/or mechanical actions, it may be more desirable technically and/or from economic and sustainability perspectives, to provide supplementary protection materials and/or systems as additional resistance measures. In these circumstances, protection materials are usually applied to the surfaces of concrete structures.

During their service life, existing concrete structures may be found to have inadequate durability and/or to have experienced deterioration such that they require structural repair to restore or enhance their strength and/or achieve their design service life. Also, there may be a need to make repairs for defects arising from inadequate design, specification or construction or the use of unsuitable construction materials. When the performance requirements of existing structures do not meet the present ones, upgrading, which includes measures for both enhancement of durability and mechanical performance, is necessary. The measures to enhance mechanical properties is called as strengthening.

Environmental actions, such as thermal effects, moisture, chloride in air/water and oxygen/carbon dioxide in air, can cause various forms of hazard and material deterioration, involving processes such as chloride ingress, carbonation, corrosion and frost damage. Some mechanical actions, such as abrasion by flowing water, can cause surface erosion and damage. For further details refer to sections 14.7 and 14.8.

Protection systems typically comprise a number of component products or measures which, when applied in accordance with the manufacturer's

Concrete is a durable and strong material when made with appropriate constituent materials used in suitable mix proportions. However, even if concrete is appropriately designed, produced, placed and maintained, it may need additional resistance under harsh environmental and/or mechanical actions, which can be provided by suitable supplementary protection materials and/or systems. Such protection materials and/or systems can be provided to both new and existing structures and, as environmental and mechanical actions are generally applied externally, these materials / systems are typically installed on the surface of the concrete.

Also due to unforeseen environmental and/or mechanical actions, including seismic effects, material degradation may take place in existing structures. In addition, the performance requirements for existing structures may become outdated, thereby requiring an enhancement of durability and/or mechanical performances to meet the revised/upgraded performance requirements. Thus, due to factors such as material degradation and outdated performance requirements, existing structures may require repair and/or upgrading interventions to be undertaken.

Appropriately selected, specified and installed protection materials and/or systems make an effective contribution in preventing degradation of structural performance and/or diminished durability in respect of defined hazards, and as component part of a repair and/or upgrading intervention.

As materials and systems for protection, repair and upgrading are provided externally to a structure, the effective bonding or connection of these materials and systems to the substrate concrete to achieve integral behaviour is an important consideration.

recommendations, form an overall system (i.e. with the component elements acting together) whose combined behaviours provide an overall system performance which meets the specified / defined requirements. A list of protection systems is provided in section 19.2.4.

However, where there has been degradation of structural performance during the service life of a structure by environmental and/or mechanical actions, such concrete structures could need repair or upgrading interventions to restore or upgrade the durability and mechanical performance of the structure in order to meet the required service life.

In order to select an appropriate protection material and system, the external deterioration factors or hazards need to be identified. The selected material and system for protection should indicate suitable properties (performances) for reducing the deterioration which might be caused by the external factor(s) being considered.

Instead of applying protection materials/systems as an additional measure, the application of highly durable concrete and reinforcement materials, such as UHPC and FRP reinforcement, may also be an option. The selected durable concrete and reinforcement should have a satisfactorily high degree of resistance against the deterioration by the considered external factor(s) / hazard(s).

A repair is an intervention taken to reinstate to an acceptable level the current and future performance of a structure or its component parts which are either defective, deteriorated, degraded or damaged in some way so that their performance level is below that anticipated by the designer. A defect is an unacceptable condition that may be in-built or the result of deterioration or damage.

Also, the durability, strength and/or other performances of an existing structure may not be satisfactory due to an uprating of the performance requirements above those which applied at the time of construction. In those situations, materials and/or systems for protection, repair and upgrading may form a valuable part of the overall solution. These measures can be used in both a reactive manner (eg. repair and upgrading) and/or a proactive manner (eg. prevention and upgrading).

19.1.2 Purpose of protection

Protection materials and systems are employed to provide shielding against external deterioration factors or hazards including:

- Thermal effects (high & low temperatures including fire, freezing, etc)
- Water effects
- Chloride ingress
- Gas (CO_2, O_2, etc) penetration
- Chemical ingress
- Physical effects (abrasion and impact)

After the external deterioration factor(s) / hazard(s) are identified, appropriate protection materials and/or systems shall be selected which can provide an appropriate degree of shielding.

19.1.3 Purpose of repair

Repair is undertaken to restore a degraded or deteriorating performance to the level intended by the designer and / or achieve satisfactory performance for the duration of the design service life. The repair intervention may be made in respect of an aspect of structural performance or some other form of performance, such as to address the implications of a reduced (degraded) performance of existing protection measures / shielding.

The restoration of the performance of degraded shielding measures, to enhance the durability of a structure or its component parts, can typically be undertaken using the same procedures as those originally used to introduce the shielding measures. The restoration of the degraded mechanical performance of a previous repair can typically be undertaken using the methods used for strengthening.

When undertaking strengthening, consideration has to be given not only mechanical properties of the strengthening materials but also to achieving an effective bond or connection between these materials and the substrate concrete, in order to achieve satisfactory integral behaviour.

Cementitious materials, including ordinary cementitious materials, can be used as materials for protection, repair and upgrading. Examples of this include their use to increase concrete cover thickness or concrete cross-section depth and cross-section area. There are various inorganic materials, whose protection capability is better than that of ordinary cementitious materials. These include fibre-reinforced cementitious (FRC) materials, high performance cementitious (HPC) materials and polymer-modified cementitious materials (PCM).

Silicate is used as a surface penetrant to enhance the protection properties of the concrete surface layer, so it acts more effectively as a protection layer.

19.1.4 Purpose of upgrading

Upgrading is undertaken to upgrade (enhance) the durability and / or mechanical performance of a structure or its component parts to satisfy revised performance requirements, such as its shielding and / or strength/stiffness, at a higher performance level than that originally intended by the designer and / or to achieve satisfactory performance for an extended service life period. The enhancement of mechanical performance is called as strengthening.

19.2 Types of materials & systems

19.2.1 Inorganic materials

The inorganic cementitious materials for protection, repair and upgrading include the following:

- Ordinary cementitious materials (concrete, mortar);
- Fibre-reinforced cementitious materials (concrete, mortar);
- High-performance cementitious materials (concrete, mortar);
- Polymer-modified cementitious materials (mortar).

Information relating to ordinary cementitious materials (concretes and mortars) is provided in chapter 14, while that for fibre-reinforced cementitious materials and high-performance cementitious materials is presented in chapter 18.

There are also inorganic materials, such as silicate, which may be used in addition to the cementitious materials listed above.

Polymers are used for surface coating materials, surface penetrants and crack injection materials. Polymers used as a surface coating are applied to the concrete surface. While surface penetrants are also applied onto the concrete surface, they penetrate into the surface layer of concrete to change (enhance) the properties of the concrete in that zone. Materials for crack injection are used to fill up the gap (void) created by the formation of cracks. Polymers are also used to enhance adhesion between materials applied for protection, repair and upgrading purposes and the substrate concrete.

Surface coating materials and surface penetrants may be provided as a protection to shield the concrete structure from harmful external substances. At the same time such surface coatings and penetrants should be permeable to water vapour to allow excess moisture to escape (evaporate) from the concrete.

Crack injection is conducted to prevent harmful substances from penetrating via cracks and to recover the mechanical properties of the concrete (e.g. tensile or shear stress transfer across / along cracks). The adhesive materials used need to possess a suitable force transfer capability for bonding externally applied repair and strengthening materials. The adhesive materials used can also be considered to perform as surface coating materials.

While steel can corrode, it can also be used to shield concrete from substances which cause deterioration. For example, concrete which is steel-encased will be protected from an aggressive environment (either wet or dry) which would otherwise attack and damage the concrete, provided the steel is suitably protected from the effects of corrosion. Non-metallic FRP, which does not experience corrosion, can also be used in a similar manner to shield concrete from attack by aggressive substances / aggressive environmental conditions.

External and near surface bonding is a method for strengthening and repair. This can be achieved by means of systems based on FRP or steel. Whereas FRP is typically non-metallic, alternative to steel plate, steel cord reinforced polymer can be considered as a special case of FRP being metallic instead of non-metallic.

19.2.2 Organic materials

The organic materials used for protection, repair and upgrading applications include polymers, as well as other materials which are not polymers.

The polymers used include various resins, such as those classified as thermosetting resins, such as epoxy, and thermoplastic resins, such as polyethylene. The other non-polymer materials used include monomers, organic solvents and surfactants.

19.2.3 Metallic and non-metallic materials

The metallic and non-metallic materials used for protection, repair and upgrading applications include the following:

- Steel plate plus, potentially, reinforcing and prestressing steels; and
- Fibre-reinforced polymers (FRPs).

To provide effective surface protection, the metallic (steel) and FRP materials would typically need to be in a sheet, plate or hollow section form.

Information relating to reinforcing and prestressing steels is provided in chapters 15 and 16, while that for FRPs is presented in chapter 17.

Externally bonded steel and FRP plate-like materials can be protective materials providing shielding against environments aggressive to concrete.

There are various electrochemical methods which are used as protection and repair systems including those listed below:

- Cathodic prevention against future reinforcing steel corrosion;
- Cathodic protection against current / future reinforcing steel corrosion;
- Re-alkalisation of carbonated concrete;
- Chloride extraction for the removal of ingressed chlorides;
- Electrochemical dehumidification for removal of ingressed moisture.

Corrosion inhibitors can be introduced into the concrete mix at the time of construction or as a supplementary application undertaken as a remedial intervention undertaken to an existing concrete structure.

Strengthening systems can be used also for the restoration of degraded mechanical properties (e.g. strengthening, stiffness enhancement). There are various methods/procedures which may be used as strengthening systems.

19.2.4 Protection and repair systems

Protection and repair systems include the following:

- Cathodic prevention;
- Cathodic protection;
- Re-alkalisation;
- Chloride extraction;
- Electrochemical dehumidification;
- Corrosion inhibitors.

The above systems can be applied as the upgrading measures to enhance durability.

19.2.5 Strengthening systems

The strengthening systems which might be considered include the following forms:

- External bonding;
- Top surface overlays / bottom surface (soffit) underlays;
- Jacketing;
- Bracing and/or moment-frame systems.

Information relating to external bonding and jacketing with steel and FRP plates / rods is provided in chapter 22, while that concerning overlaying / underlaying / jacketing with cementitious materials is presented in chapter 23.

19.3 Properties of materials and systems

The properties of the materials used to provide surface protection materials and systems need to be quantified to evaluate the protection they can provide to the structure / the concrete substrate after treatment. This would typically be achieved by laboratory testing or other suitable means.

In regard to the use of surface penetrants, the material performance properties should be established by testing of the concrete after its modification by the application of the surface penetrant.

The properties required for tensile strengthening materials are essentially the same as the tensile reinforcement materials installed in new structures, namely the reinforcing and prestressing steels and FRPs. Similar considerations apply to the materials acting in compression.

It is important to consider the interfacial behaviour between the added materials and the substrate concrete. In the case of externally bonded reinforcement, detailed consideration may need to be given to the transfer forces into the substrate concrete to avoid extreme interfacial deformations.

19.3.1 Durability properties

The material properties required for durable protection, repair and upgrading interventions depend, amongst other factors, upon the external deterioration factors / hazards that the structure is exposed to. Some examples of these are listed below, along with indications of the associated material properties / testing requirements:

- For thermal effects: heat transfer coefficient, heat conductivity;
- For water effects: water absorption rate, water permeability coefficient;
- For chloride ingress: chloride ion diffusivity;
- For gas penetration: gas permeability;
- For chemical ingress / attack: resistance to ingress of the relevant aggressive chemical species, the degree of degradation of the treated material / the protection material / system during an exposure test;
- For physical attack effects: resistance to relevant degradation process, the degree of degradation of the treated material / the protection material / system during an exposure test.
-

19.3.2 Mechanical properties

Repair and strengthening interventions undertaken to improve the mechanical performance of a structure would typically consider the performances listed below, but other factors might need to be taken into account depending upon the nature of the improvements in mechanical performance sought.

- Strength
- Stiffness
- Bonding achieved at interface with the substrate concrete

Materials for protection, repair and upgrading interventions, which are externally bonded or injected/inserted into the substrate concrete, should behave monolithically with the concrete substrate. In addition to verifying that the bond strength with the concrete substrate is satisfactory, it is necessary to maintain strain compatibility under the anticipated temperature- and/or moisture-induced strain variations in the concrete substrate. Strain and deformations associated with related movements, such as autogenous shrinkage, should be small.

The local strain induced in the surface protection material (coating) at locations where cracks may/have formed in the concrete substrate, should be controlled (strain limited) so as to avoid cracking of the coating, etc.

The properties of the materials used to provide surface protection materials and systems need to be quantified to evaluate the protection they can provide to the structure / the concrete substrate after treatment. This would typically be achieved by laboratory testing or other suitable means. Of particular importance are the long-term properties and performances influencing the durability of the surface protection materials and systems.

There are various materials and systems available for protection, repair and upgrading. Some of them are innovative and recently developed which may mean that they have limited information available concerning their long-term properties or effectiveness. In these circumstances, before they are selected and applied as intervention treatments, careful consideration should be given to the potential benefits and advantages that they may offer relative to any disadvantages and/or the uncertainties / risks that might potentially arise in respect of their future performance.

Information relating to reinforcing and prestressing steels is provided in chapters 15 and 16, while that for FRPs is presented in chapter 17.

19.3.3 Properties necessary for compatibility to substrate concrete

Bond strength and strain compatibility shall be checked to ensure that a satisfactory bond can be achieved and maintained between the substrate concrete and materials employed for protection, repair and upgrading interventions, especially those which are externally bonded or injected/inserted into the substrate concrete.

Information relating to the external bonding of steel and FRPs plates / rods is provided in chapter 22, while that for external bonding of ordinary cementitious materials is given in chapter 23 and that for anchorages in concrete is presented in chapter 25.

19.3.4 Durability

Materials and systems used for protection, repair and upgrading interventions are required to be effective for the design service life of the structure, which means that the protection materials and systems themselves are required to be durable for this period or be capable of being replaced / repaired / upgraded again should this become necessary.

19.4 Assumptions used for design

The assumptions for concrete, steel reinforcement, prestressing reinforcement, non-metallic reinforcement, and fibre reinforced concrete are provided in chapters 14, 15, 16, 17, and 18, respectively, which can be used for protection, repair, and upgrading.

The information needed for design, such as diffusivity, permeability, strength, stiffness, stress – strain relationship, creep, and thermal coefficient, is quantitative description of properties of materials and systems for protection, repair and upgrading, which are described in section 19.3. The properties assumed for design shall be obtained through reliable data tested by standardized method.

Geometrical information, such as thickness of material layer coating or bonded to substrate surface, cross-sectional area, and location of reinforcing material in matrix of externally bonded layer, is also provided for design.

19.5 Assumptions used for assessment

It is preferable to obtain the information needed for assessment regularly through inspection, especially for materials and systems for protection, repair and upgrading without enough data of long-term performance.

The information needed for assessment is the same as that for design. The information can be obtained directly from the existing structures with protection, repair and upgrading intervention.

When the obtained information indicates deteriorations and / or defects of materials and systems for protection, repair and upgrading, assessment should be conducted with consideration of deteriorations and / or defects.

In the case of materials and systems for protection, repair and upgrading, which do not comply with the current requirements, the information for assessment shall be obtained by the currently approved method.

References

[19-1] ISO 16311-1:2014. Maintenance and repair of concrete structures – Part 1: General principles.

[19-2] ISO 16311-2:2014. Maintenance and repair of concrete structures – Part 2: Assessment of existing structures.

[19-3] ISO 16311-3:2014. Maintenance and repair of concrete structures – Part 4: Execution of repairs and prevention.

20 Bond of embedded steel reinforcement: anchorages and laps

20.1 Introduction

Bond is the term used to denote the interaction and transfer of force between reinforcement and concrete. Bond influences performance of concrete structures in several ways. At the serviceability limit state, bond influences width and spacing of transverse cracks, tension stiffening and curvature. At the ultimate limit state, bond is responsible for strength of end anchorages and lapped joints of reinforcement, and influences rotation capacity of plastic hinge regions.

20.2 Design of anchorage and lapped joints in new structures

20.2.1 Scope

Within section 20.2 the term ‘embedded” denotes reinforcement with a minimum cover equal to or greater than the nominal bar diameter.

Design expressions presented are derived for conventional concretes, conventional ribbed bars and indented bars with a diameter (or equivalent diameter) up to 50 mm for ribbed and 20 mm for indented bars, and a relative rib area f_R or f_P of at least 0.05 (ISO 15630-1 [20-2]).

Where other concretes or reinforcement are to be used bond performance of these materials should be verified using the test specification for anchorage described in the present chapter (see section 20.6). Additional rules for fusion bonded epoxy coated bars are provided in 20.2-?.

Design expressions for anchorages and lapped joints are derived from data and analyses presented in *fib* Bulletin 72 [20-1].

Design expressions presented are derived for conventional ribbed bars and indented bars with the ratio between the rib top width in the direction of bar axis and rib spacing (b_r/c) smaller than 0.5.

Relative rib area f_r or f_P and or f_P may be determined with Standard procedure as proposed, for instance, in ISO 15630-1 [20-2] and EN15630 [20-3].

According to *fib* Bulletin 72 [20-1] strength of anchorages and lapped joints are governed by the same equations with possible different confining actions in specific case. However, to be better detail design requirements, anchorages and splices are described in different paragraphs.

Note that the resistance of two components acting in combination will generally be less than the sum of their resistances determined individually. The rules provided here make allowance for interaction effects.

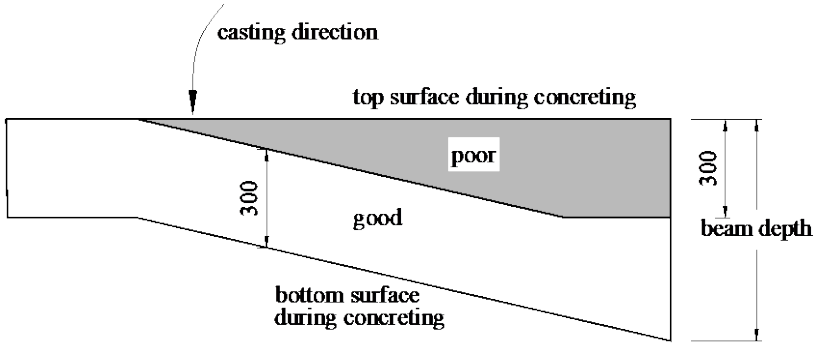


Figure 20.2-1: Definition of good and poor bond conditions

Anchorages at support reactions and under concentrated loads may be considered to be subject to transverse compression.

In the anchorage of longitudinal bars links are more effective than straight transverse bars (Fig. 20.2-2). Therefore, either wider longitudinal bar spacing or

20.2.2 Design of anchorages and lapped joints of rebars

20.2.2.1 Introduction

Minimum anchorage lengths are specified in section 20.2.2.3. Minimum lap lengths are specified in section 20.2.2.6.

Reinforcement may be anchored by bond alone or by a combination of bond along the straight portion of a bar together with a resistance provided by one of:

– a hook or bend (but only for bars in tension);

– welded transverse bar(s);

– a head welded or threaded to the end of the bar;

– bearing of the end of the bar on the concrete (only for bars in compression).

Only one additional resisting mechanisms may be considered to contribute to anchorage at a particular location.

Mechanical couplers may be used as an alternative to lapped joints.

'Good' bond conditions are defined as:

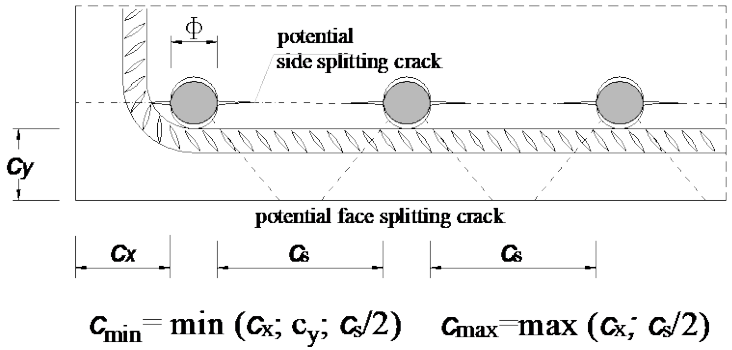
- bars with an inclination less than 45° to the horizontal which are up to 300 mm from the bottom of the formwork or at least 300 mm below the free surface during concreting.

Otherwise 'poor' bond conditions should be assumed

20.2.2.2 Minimum detailing requirements for anchorages and laps

Minimum cover $c_{min,b}$ related to bond should be equal to one bar diameter. For anchorage of a bundle, minimum cover is the equivalent diameter of the bundle (see Eq. (20.2-11)). Cover dimensions c_x and c_y are minimum covers to adjacent faces of the element. They may be determined on the basis that nominal cover c_{nom} for durability and fire resistance are provided (see section ??).

denser transverse reinforcement is required when straight transverse bars are adopted.



$c_{min} = \min(c_x; c_y; c_s/2)$ $c_{max} = \max(c_x; c_s/2)$

Figure 20.2-2: Definition of concrete cover and clear bar spacing.

Beams and columns are designated 'linear elements members' and slabs, walls and shells are designated 'planar' elements.

Minimum transverse reinforcement at laps and anchorages should be provided to keep the failure mode under control and to avoid brittle failures Reinforcement provided for other purposes (e.g. for shear resistance) may be included in computation of ΣA_{st}. However, bond failure is generally non-ductile even with relatively high proportions of confining reinforcement unless cover exceeds 5ϕ or transverse compression is present.

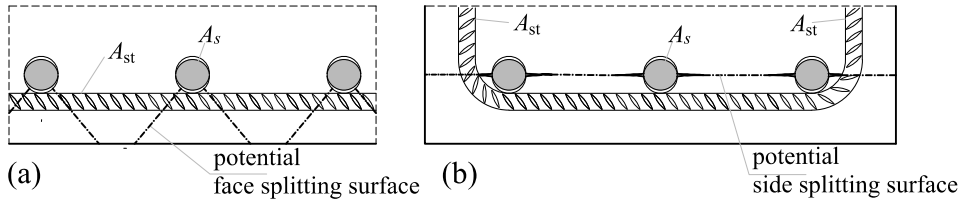


Figure 20.2-3: Calculation of minimum reinforcement: examples

When confinement is provided by straight bars (for face splitting crack as shown in Fig. 20.2-3a), ΣA_s is the area of a single anchored bar A_s (n_b=1) and ΣA_{st} is the total amount of transverse reinforcement crossing potential splitting surfaces.

Laps in linear elements shall be confined by links. Clear spacing c_s between anchored bars or bars belonging to different lapped joints confined by links shall be at least two times bar diameter except for rebars in zones subject to transverse compression, where the minimum clear spacing $c_{s,min}$ may be reduced to one bar diameter. In planar elements, where confining reinforcement comprises straight bars, clear spacing $c_{s,min}$ shall be at least three times bar diameter, except in zones subject to transverse compression, where the minimum clear spacing may be reduced to one bar diameter.

Minimum cover and clear spacing shall be compatible with the provisions related to maximum aggregate size (see chapter 14), durability (see section 30.6), and fire resistance (see section 30.4).

In linear elements, unless the bond zone is subjected to transverse compression, a minimum quantity of transverse reinforcement is to be provided within the anchorage and lap length.

Where the diameter Ø of the anchored bars is less than 20 mm and concrete is of grade C60 or below, transverse reinforcement or links provided for other reasons may be assumed sufficient to satisfy minimum requirements for confining reinforcement without further justification.

In other circumstances, Eq. (20.2-1) should be satisfied:

$$\sum A_{st} \geq \alpha_t \alpha_1 \sum A_s \qquad (20.2\text{-}1)$$

where

$$\sum A_{st} = n_g \cdot n_t \cdot A_{st} \qquad (20.2\text{-}2)$$

n_g is the number of groups links or items of confining reinforcement within the anchorage/lap length;

n_t is the total number of legs of links crossing a potential splitting failure surface at a section (Fig. 20.2-3b); for straight transverse reinforcement (Fig. 20.2-3a) $n_t = 1$

A_{st} is the cross-sectional area of one leg of a confining bar;

When the confinement is provided by links (for side splitting crack as shown in Fig. 20.2-3b), ΣA_s is the total cross-sectional area of all bars lapped or anchored at the section. Where bars of unequal size are lapped, it is the total area of the smaller lapped bars within the lapped pair and ΣA_{st} is the total amount of transverse reinforcement crossing the potential splitting surface.

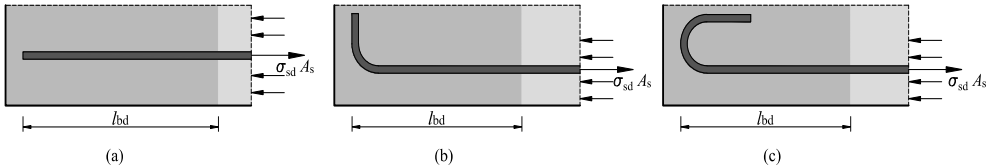


Figure 20.2-4: Anchorage length of straight bar (a) or with bar terminating in bend (b) or hook (c).

The provisions in 20.2.2.3 for anchorages and laps of deformed bars are valid for short term, sustained and fatigue loading. They are based on the results of a probabilistic study (*fib* Bulletin 72 [20-1] supplemented by Vollum and Goodchild, 2019 [20-4]) using the results of a large number of short-term tests with overlap splices failing by splitting and a reliability index $\beta = 3.8$ which is valid for 50 years. Therefore, it is assumed in 20.2.2.3 that the strength of splices is not influenced by long-term effects such as high sustained load, high cycle fatigue loading or tensile stresses induced in the concrete due to the restraint of deformations caused by temperature variations or concrete shrinkage.

According to section 14.6 the concrete tensile and compressive strengths might be reduced due to sustained or fatigue loading. Therefore, a reduction of the strength of anchorages and laps may be expected due to sustained or fatigue loading. It is expected that the reduction is smaller than for the concrete tensile strength since, due

ΣA_s is the total cross-sectional area of the bars to be anchored or lapped, according to Fig. 20.2-3; where lapped bars are of unequal size the total area of the smaller bars within the lapped pair may be used.

$\alpha_t = 0.5$ for bars up to and including size 25;

$\alpha_t = 0.5 + (\phi - 25)/50$ for $25 < \phi \leq 50$ mm;

$\alpha_t = 0$ for distribution reinforcement in the outer layer of planar members (walls and slabs);

$\alpha_1 = A_{s,cal}/A_{s,ef}$ for anchorage or lap-splice zones subject to transverse compression, where $A_{s,cal}$ is the calculated area of reinforcement required by the design and $A_{s,ef}$ is the area of reinforcement provided;

$\alpha_1 = 1.0$ in other circumstances.

20.2.2.3 Design anchorage length for bars in tension

A design anchorage length l_{bd} shall at least be provided to safely transfer the forces from a bar to the surrounding concrete without spalling or bond failures (Fig. 20.2-4).

Design anchorage length l_{bd} is measured to the end of a straight bar or to the outside of a hook or bend (Fig. 20.2-4).

For ribbed bars 8 mm $\leq \phi \leq$ 50 mm and indented bars $\phi \leq$ 14 mm, with 20 MPa $\leq f_{ck} \leq$ 75 MPa, $c_{min,b} \geq 1.0\phi$ (Fig. 20.2-2) and $\gamma_C = 1.5$, the design anchorage length in tension l_{bd} may be calculated from:

$$l_{bd} = \max\left[(k_{lbs} - k_{beb})\,\phi;\, k_{min}\right] \qquad (20.2\text{-}3)$$

where:

k_{lbs} = anchorage coefficient as provided in Table 20.2-1. Otherwise k_{lbs} shall be calculated using Eq. (20.2-4).

k_{beb} = coefficient for standard bends or end bearing, which may be determined according to section 20.2.2.4.

to the increased slip, a redistribution of steel tensile stresses along the bond length occurs during sustained or cyclic loading.

When designing a new structure, the concrete strength reduction due to sustained or cyclic loading will be partially offset by an increase in concrete strength with time.

In summary, a designer may consider increasing the calculated design bond length in critical applications by about 30% (new construction) or reduce the calculated anchorage or splice strength by about 20% (assessment of an existing structure). These recommendations are based on engineering judgement and are considered as conservative.

Confinement for cover, transverse reinforcement and transverse pressure in excess of that specified for Eq. (20.2-3) has a beneficial influence on bond strength.

Anchorages at support reactions may be considered to be subject to transverse compression. Confinement from transverse pressure initially has a strong beneficial influence on bond strength where it restrains a splitting failure mode. Once confinement is sufficient to restrain splitting, the rate of increase reduces. It will usually be beneficial to include the influence of transverse pressure when considering anchorage of reinforcement in deep beams and corbels (Fig. 20.2-5).

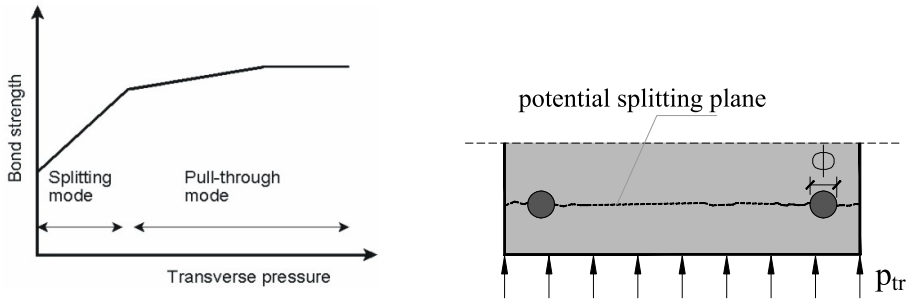


Figure 20.2-5: Influence of transverse pressure.

For ribbed bars with $\phi \leq 32$ mm, concrete strength $f_{ck} \geq 20$ MPa, anchored bars spacing $c_s \geq 3\ \phi$, concrete cover $c_x = c_y \geq \max(25,\ 1.5\ \phi\ +10$ mm), k_{lbs} may be calculated as:

$$k_{lbs} = k_{lbs,0} \bullet k_{fck} \bullet k_{\sigma sd} \qquad (20.2\text{-}5)$$

where,

$k_{lbs,0}$, k_{fck} and $k_{\sigma sd}$ may be taken from Table 20.2-1, or linearly interpolated between values therein.

k_{min} = coefficient for minimum lengths, which for anchorages shall be taken as
= max [10ϕ, 100 mm, $\sigma_{sd}/(8\ k_{tcPO} f_{ck}^{0.4})\ \phi$];

k_{tcPO} =0.5 for poor bond conditions and 1.0 for good bond condition.

Anchorage coefficient k_{lbs} shall be calculated according to Eq. (20.2-4). This Eq. (20.2-4) may also be used for a more detailed verification for situations within the scope of Table 20.2-1.

$$k_{lbs} = 67\ m\ ((\gamma_C/1.5)^{0.65}(25/f_{ck})^{0.45}(\phi/25)^{0.35}/(k_{conf}\ k_{tcSP}) \qquad (20.2\text{-}4)$$

where:

m = max [$\sigma_{sd}/435$; $(\sigma_{sd}/435)^{1.8}$]

k_{conf} = min ($\alpha_2+\alpha_3+1.5\ p_{tr}$, $3.2+0.04\ p_{tr}$)

where p_{tr} [MPa] is the value of the mean transverse compression (compressive stress is taken as positive)

k_{tcSP} = 1.0 for good bond conditions or k_{tcSP} = $0.4+0.6\ \sigma_{sd}/435 \leq 1.0$ for poor bond conditions, excluding the conditions for 'Special circumstances' as specified in Section 2.2.2.12;

σ_{sd} = design stress in the bar;

α_2 and α_3 represent the influence of passive confinement from cover (α_2) and from transverse reinforcement (α_3). Provided minimum detailing provisions in section 20.2.2.2 are satisfied, α_2 may conservatively be taken as 1.0 and α_3 may be conservatively taken as 0 when the minimum transverse reinforcement is provided

Eq. (20.2-4) is valid for 200 MPa $\leq \sigma_{sd} \leq$ 600 MPa; 20 MPa $\leq f_{ck} \leq$ 75 MPa, and $\phi \leq 50$ mm. Otherwise, anchorage (and laps) should be verified by other means. The value of f_{ck} to be used in Eq. (20.2-4) and Table 20.2-1 is limited to a maximum of 75 MPa. The value of ϕ is not to be taken less than 12 mm.

Table 20.2-1: *Coefficients* $k_{lbs,0}$, k_{fck}, $k_{\sigma sd}$

ϕ [mm]	8	10	12	14	16	20	25	28	32
$k_{lbs,0}$	42	42	42	45	47	51	55	57	60
f_{ck} [MPa]	20	25	30	35	40	45	50	60	≥75
k_{fck}	1.11	1.00	0.92	0.86	0.81	0.77	0.73	0.67	0.61
σ_{sd} [MPa]	≤ 200	300	400	435	500	600			
$k_{\sigma sd}$	0.46	0.69	0.92	1.00	1.28	1.78	Good bond conditions		
$k_{\sigma sd}$	0.68	0.85	0.97	1.00	1.28	1.78	Poor bond conditions		

Notes: values in Table valid for $f_{ck} \geq 20$ MPa, $c_s \geq 3\,\Phi$, $c_x = c_y \geq \max(25,\ 1.5\phi + 10$ mm), no account taken of confinement by links or transverse bars, or transverse pressure; $\gamma_c = 1.5$.

Cover parameters c_{max} and c_{min} are shown in Fig. 20.2-2.

Reinforcement provided for other purposes (e.g. for shear resistance) may be included in computation of K_{tr}.

Passive confinement from cover: straight reinforcing bars

Coefficient α_2, representing the confinement provided by concrete cover, is given by:

$$\alpha_2 = (c_{min}/\Ø)^{0.5} \cdot (c_{max}/c_{min})^{0.15} \qquad \text{for ribbed and indented bars} \quad (20.2\text{-}6)$$

where:

c_{min} = minimum of c_x, c_y and $c_s/2$ (see Fig. 20.2-2);
c_{max} = maximum of $c_s/2$ and c_x (see Fig. 20.2-2);
Definition of c_x and c_y can be found in Fig. 20.2-2.

Eq. (20.2-6) is valid for $\phi \leq c_{min} \leq 3.5\ \phi$, $1 \leq c_{max}/c_{min} \leq 5$.

Passive confinement from transverse reinforcement

When confining reinforcement (e.g. links, or transverse reinforcement) is arranged between the bar to be anchored and the free surface, confinement factor, α_3 shall be determined as follows:

$$\alpha_3 = k_d\,(K_{tr} - \alpha_t/50) \qquad (20.2\text{-}7)$$

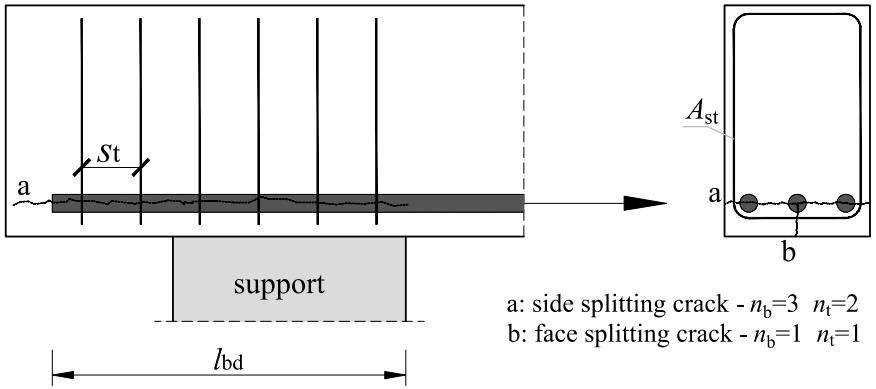


Figure 20.2-6: Location of links along an anchorage

The k_d factors in Fig. 20.2-7 are derived from k_m factors in Fig. 20.5-3 (see also Bulletin 72 [20-1]), taking account of the non-linear relationship between anchorage or lap length and the stress developed in the bar.

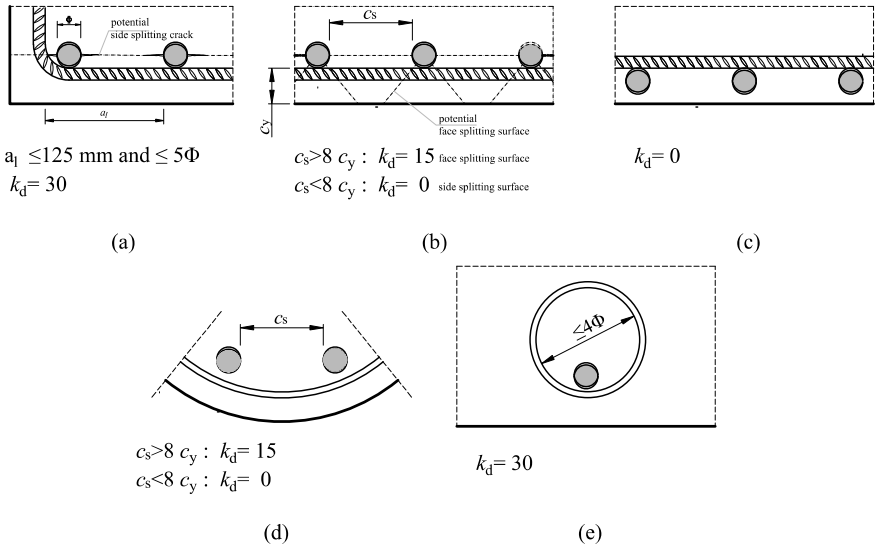


Figure 20.2-7: Transverse reinforcement factor k_d.

where (see Fig. 20.2-6):

$K_{tr} = n_t A_{st}/(n_b Ø s_t) \leq 0.05$ is the density of transverse reinforcement, relative to that of anchored or lapped bars;

n_b is the number of anchored bars (or pairs of lapped bars) in the potential splitting failure surface;

Ø is the diameter of the anchored bar or of the smaller of a pair of lapped bars (mm);

s_t is the spacing of links along the anchorage (Fig 20.2-6);

n_t is the number of legs of a link crossing a potential splitting failure surface at a section (Fig 20.2-6);

α_t is defined in section 20.2.2.2.

For considering the transverse reinforcement effect, no lapped or anchored bar shall be further than the lesser of 5Ø or 125 mm from a leg of a link which lies perpendicular to the plane passing through the bar axes, Fig 20.2-7.

k_d is an effectiveness factor dependent on the reinforcement detail, Fig. 20.2-7.

k_d = 30 where the legs of a link are perpendicular to the splitting plane, provided no anchored bar or pair of lapped bars are further than either 5Ø or 125 mm from where the leg crosses the splitting plane, Fig. 20.2-7a;

k_d = 30 where an individual bar or pair of lapped bars are wholly confined within a helix of internal diameter not exceeding 4 times the diameter of the lapped or anchored bar, Fig. 20.2-7e;

k_d = 15 where bars are confined by straight bars or helix within the cover thickness, provided that the clear spacing between the main bars is at least 8 times the cover – Figs. 20.2-7b and 20.2-7d;

k_d = 0 in other circumstances, Figs. 20.2-7b and 20.2-7c.

Laps in vertical legs of links should be avoided in areas of high shear in beams where inclined shear cracking may closely align to the direction of a lap.

The value of k_{beb} of 15 is based on the standard bend radius. For larger bend radii, see section 30.13.3.

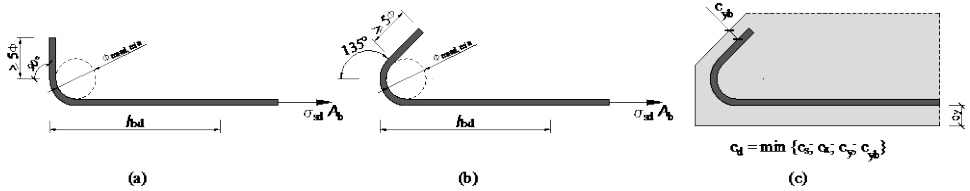


Figure 20.2-8: Standard 90° bent (a), hook anchorage in tension (b) and definition of nominal cover c_d (for definition of c_s, c_y and c_x see Fig. 20.2-5).

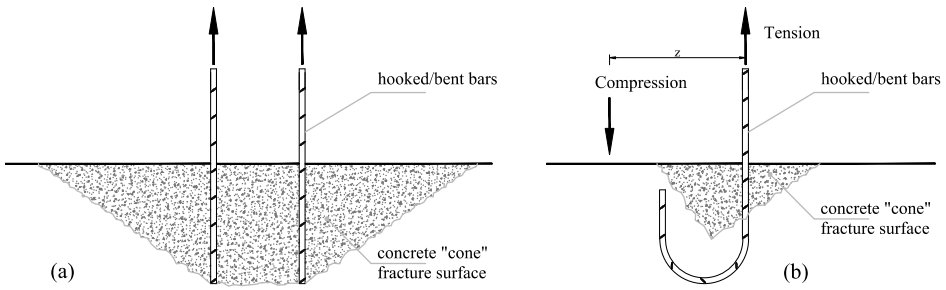


Figure 20.2-9: Cone failure hooked bars

Transverse tension

Transverse tensile stress (σ_{ctd}) reduces bond strength. However, good detailing practice will provide transverse reinforcement to resist tension perpendicular to potential splitting failure planes. When additional transverse reinforcement is applied to resist the transverse force, no reduction of anchorage or lap strength needs to be considered. Otherwise, the reduction may conservatively be taken as linearly proportional to σ_{ctd}/f_{ct}.

20.2.2.4 Contribution of hooks and bends to tension anchorages

End bends or hooks contribute to transfer of force for bars in tension. This contribution is taken into account by the coefficient k_{beb} which can be taken as 0 unless:

- for a standard hook or bend complying with Fig. 20.2-8 in tension anchorage. In this case, k_{beb} may be taken as = 15.

Where hooked bars are closely spaced and anchor a predominantly tensile force, the possibility of a concrete cone type failure mode should be considered (Fig. 20.2-9a).

The concrete cone failure mode may be modified when the tension force in the hooked or bent bar forms part of a moment couple (Fig. 20.2-9b).

In compression anchorages at least one item of transverse reinforcement should be positioned within the anchorage length no further than 50 mm or 2ϕ from the end of the bar. This requirement should be satisfied even if α_3 is conservatively taken as 0. Other items of transverse reinforcement should be spaced evenly throughout the anchorage length.

With the exception of compression bars of columns or walls in footings or equivalent locations where cover perpendicular to the bar axis is very high, bends and hooks are not permitted for compression reinforcement. The exceptions are footings and in exterior beam/column joints where the bend is toward the outside of the joint, the end cover parallel to the bar axis is at least 5.0 Ø and dense transverse reinforcement is provided in that cover over the entire length of the bend or hook.

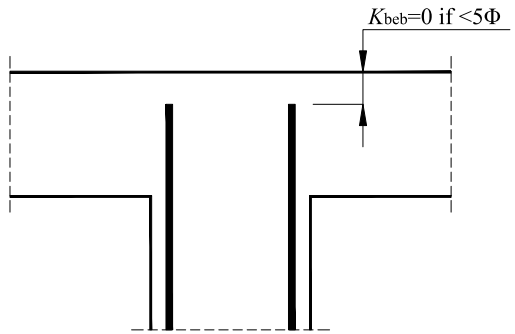


Figure 20.2-10: Minimum end cover for considering the contribution of end-bearing (k_{beb}) in compression bars

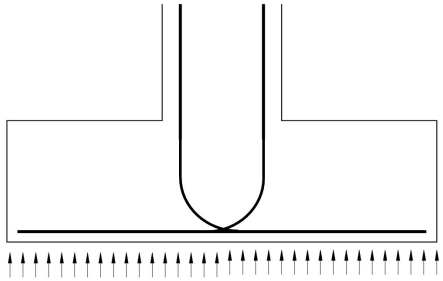


Figure 20.2-11: Column bars anchored in footing

20.2.2.5 Design anchorage length for bars in compression

In compression anchorages at least one item of transverse reinforcement is to be positioned within the anchorage length no further than 50 mm or 2ϕ from the end of the bar.

The anchorage length of bars in compression is given by Eq. 20.2-3 or Eq. 20.2-5. End bearing enhances strength of compression anchorages and but should only be considered to contribute where the end of the bar is no closer than 5ϕ (measured parallel to the bar axis) from an unsupported face (Fig. 20.2-10); in this case, may be taken as 15. Otherwise k_{beb}=0

In anchorages with bends or hooks coefficient k_{bed} should be taken as 0 unless:

for a standard hook or bend complying with Fig. 20.2-11 (with force reversed) in compression anchorage with a distance ≥ 5.0 ϕ to a free surface measured parallel with the line of the compressive force from any part of the bar to the free surface. In this case k_{beb} may be taken as = 15.

Where bars are lapped in regions of low stress – for example in the vicinity of points of contraflexure in continuous beams – the lap will not be required to develop yield under normal loading conditions, although it might be required to do so under accidental loadings for which a lower partial safety factor might be appropriate.

Lap failure may be extremely brittle, particularly where only minimum confining reinforcement is present.

A wider crack width may occur at the lap ends.

Tension laps may be detailed with up to 100 % of bars lapped at any section.

In cases of lapping of bars with different diameters l_{bd} may generally be based on the smaller diameter (although anchorage of the larger bar should be verified).

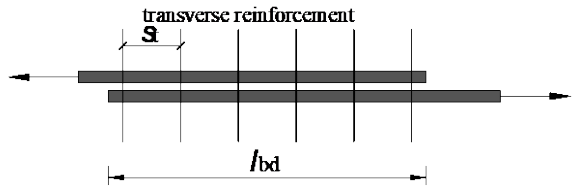


Figure 20.2-12: *Location of links along a lap (top view)*

When the bar stress at ends of a lap does not exceed 70% of the design strength of the bar then the lap region will not be required to behave in a ductile manner under the design loading and therefore no special provisions for deformation capacity are required.

Fig. 20.2-13 shows a continuous beam with laps adjacent to a support. As bond failure may be extremely brittle, the ductility of the section required for redistribution

20.2.2.6 Design of lap length in tension

Forces may be transmitted from one bar to another by:

- lapping of bars;
- mechanical couplers ensuring load transfer in tension and compression or compression only;
- full penetration butt welds or fillet weld.

The detailing of laps between bars shall be such that:

- the transfer of forces from one bar to the next is ensured;
- spalling of the concrete in the vicinity of the lap does not occur;
- cracks parallel to the lapped bars which affect the performance of the structure do not develop.

Lapped joints should be located away from regions of high stress whenever possible.

Where design or assessment is carried out using approaches based on linear elasticity with limited redistribution, theory of plasticity or on non-linear analysis, it shall be verified that adequate deformation capacity is available at laps where necessary.

A design lap length l_{bd} shall at least be provided to safely transfer the forces from a bar to the other bar without spalling or bond failures (Fig. 20.2-12).

Design lap length l_{bd} is measured from the end of one straight bar to the end of the other lapped bar, or from the outside of the hook or bend of one bar to the outside of the opposite hook, bend, or straight end of the other (Fig. 20.2-12).

The design lap length l_{bd} is calculated in accordance with Eq. (20.2-3), Eq. (20.2-4), or Eq. (20.2-5).

The coefficient k_{min} for minimum lap lengths shall be taken as:

$$k_{min} = \max [15\phi \text{ or } 150 \text{ mm};\ \sigma_{sd}/(8\, k_{tcPO} f_{ck}^{0.4})\ \phi]. \qquad (20.2\text{-}8)$$

To verify that adequate deformation capacity is available it is sufficient that one of the following three conditions is satisfied:

i) the lap (or anchorage) is located where bar stress is limited to no more than *70%* of the design yield strength of the bar;

ii) each lap (or anchorage) can develop a stress at least *30%* in excess of the design yield strength of the bar;

of support moments is to be achieved through yielding of reinforcement. Laps should therefore be designed to ensure bars develop strains in excess of yield.

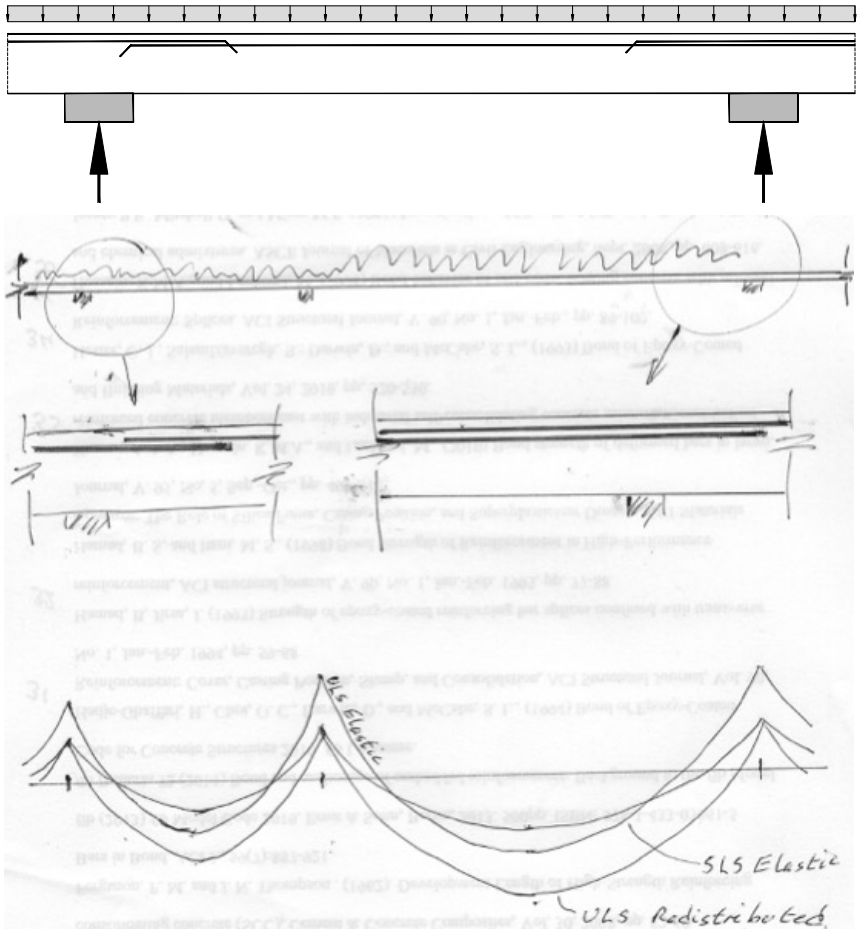


Fig. 20.2-13: Positioning of laps ~~in continuous beams~~ to allow the formation of plastic hinges away from supports.

Research[REF] has demonstrated brittle behaviour is significantly reduced if at least 2/3 of the area of reinforcement is continuous through the lap region.

Note that reinforcement itself must also have adequate deformation capacity.

iii) no more than 1/3 of the total area of reinforcement is lapped at the cross section. In case of multiple layers of tension reinforcement the 1/3 shall be evaluated per individual layer.

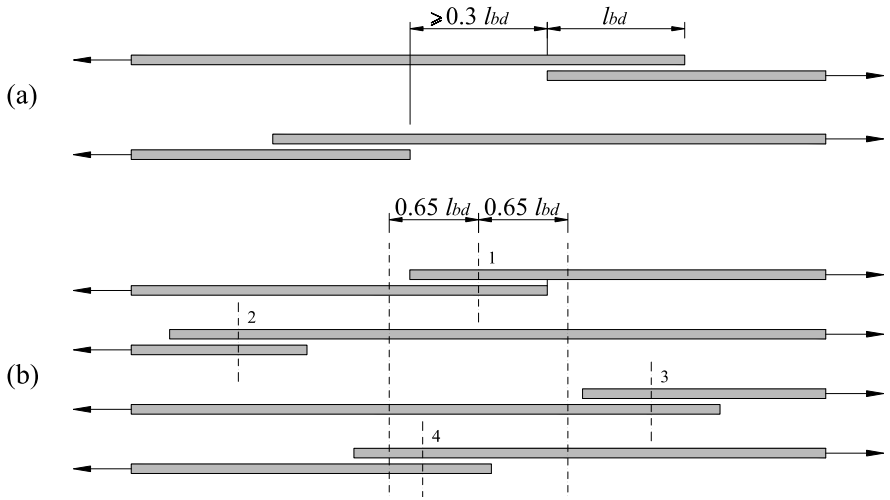


Figure 20.2-14: *Minimum distance between staggered laps (a) and example of percentage of staggered lap (b).*

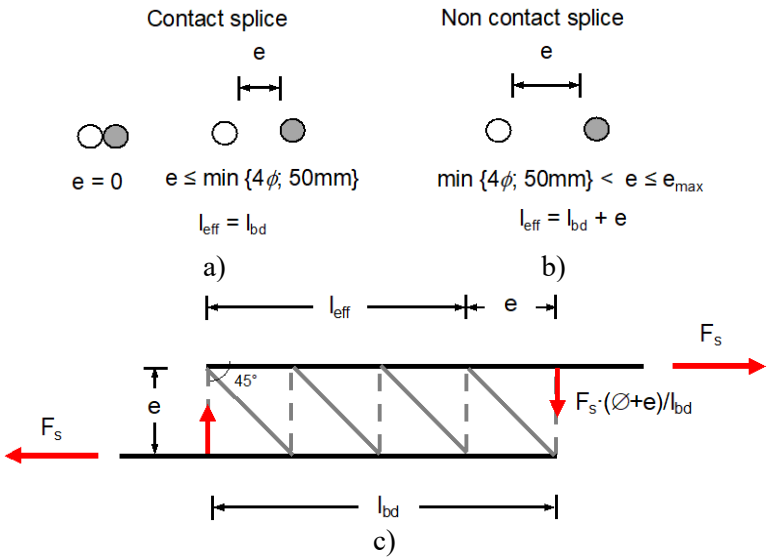


Figure 20.2-15: a) Contact splice; b) Non-contact splices; c) Strut-and-tie model to account for eccentricity in non-contact splices.

Where tension laps have to be positioned in the vicinity of plastic hinge locations, tension laps may be designed for f_{yd} if:

a) confining reinforcement is arranged according to Section 20.2.2.3 or
b) they are staggered so that the area of lapped bars ≤1/3rd of the total cross-section area of the reinforcing steel bars in linear members (beams and columns) or ≤ 50 % in planar members (slabs, walls and shells) or
c) justified by testing. In case of multiple layers of tension reinforcement the 1/3rd shall be evaluated per individual layer.

Otherwise, tension laps shall be designed for $1.3 f_{yd}$

If staggering of laps is chosen as an option (see (b), above), the distance between adjacent laps shall be at least 0.3 l_{bd} (see Fig. 20.2-14a).

The percentage of lapped bars should be calculated within 0.65 l_{bd} from the centre of any lap (see Fig. 20.2-14b).

For plastic hinge regions (except where lap-splicing is not allowed) proper detailing of lap-splice regions is required. This includes, (a) avoiding welding of reinforcement; (b) avoiding use of mechanical couplers that have not been tested with bars developing large inelastic strains: (c) provision of adequate confinement in the form of transverse reinforcement (either stirrups, hoops or externally bonded jackets.

Pairs of lapped bars should usually be in contact; where this is impractical the distance between a pair of lapped bars should be as small as possible (Fig. 20.2-15a). In case the clear distance e between lapping bars exceeds 50 mm or 4ϕ, the design lap length shall be increased by e. The definition on non-contact splices is given in Fig. 20.2-15b and additional forces due to the eccentricity between lapped bars shall be resisted by adequate reinforcement as determined by the truss-tie model shown in Fig. 20.2-15c. In cases where the clear distance between two bars exceeds the value e_{max}, no lapping is allowed. The value of e_{max} is defined as:

$$e_{max} = \min \{8\varnothing;\ 200 \text{ mm};\ 0{,}2 \cdot l_{bd}\} \quad . \tag{20.2-9}$$

For contact splices (see Fig. 20.2-15a) l_{bd} is calculated according to Eqs. (20.2-3), (20.2-4) or Eq. (20.2-5).

For non-contact splices see Fig. 20.2-15b and c):

$$l_{bd} = l_{eff} + e. \tag{20.2-10}$$

Usually, the horizontal legs of the stirrups used as minimum shear reinforcement are sufficient to resist the transversal force, F_s, due to eccentricity.

It is noted that the upper limit of the clear spacing in non-contact splices,e_{max}, is conservatively defined on the basis of the limited experimental evidence available (Sagan et al [20-5], Hamad et al [20-6]).

The minimum end cover is derived from source expressions and not from the design expressions given in this document. See *fib* Bulletin 72[20-1].

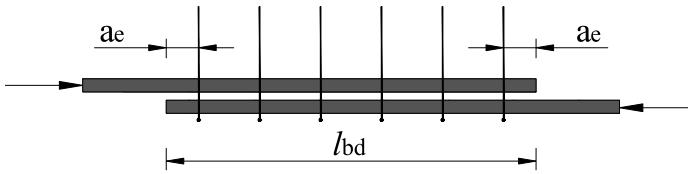


Figure 20.2-16: *Location of links near ends of lap splices in compression*

Addition of transverse reinforcement out of laps may increase the compressive lap strength. The laps in compression have been calibrated without reinforcement out of the laps.

with $l_{eff} = l_{bd}$ calculated according to equations Eq. (20.2-3), Eq. (20.2-4), or Eq. (20.2-5).

20.2.2.7 Design of lap length in compression

A design lap length l_{sd} between bars in compression shall be provided, at least equal to the anchorage lengths l_{bd} given in section 20.2.2.5.

All bars in compression may be lapped in one section and are to be designed for the maximum value of the reinforcement design stress σ_{sd}. In cases of lapping of bars with different diameters l_{sd} may generally be based on the smaller diameter (although lap of the larger bar should be verified).

End hooks and bends should not be used with compression laps.

For compression laps, at least one transverse bar or confinement link shall be placed at each end of the lap within a maximum distance a_e of 50 mm or 2ϕ of the lap ends, where ϕ is the diameter of the smaller lapped bar (Fig. 20.2-16). This requirement should be observed even if α_3 is conservatively taken as 1.0. Lap length l_{bd} has to be increased by 15% if such reinforcement is not provided. Other items of transverse reinforcement should be spaced evenly throughout the lap length.

20.2.2.8 Additional rules for bundled bars

Unless otherwise stated, the rules for individual bars also apply for bundles of bars. In a bundle, all the bars should be of the same characteristics (type and grade). Bars of different sizes may be bundled, provided that the ratio of larger to smaller diameter does not exceed 1.7.

The maximum number of bars in a bundle is 4.

Research [REF] on laps in bundles indicates that rules for lapping of individual bars may safely be used where one bar in a bundle is lapped at a section. The observation is consistent with theoretical approaches for splitting modes of bond failure. A simpler option for staggered anchorages of bars in a bundle consistent with findings of research would therefore be for individual bars to be terminated with their own anchorage length, determined according to their own diameter and covers (Fig. 20.2-17).

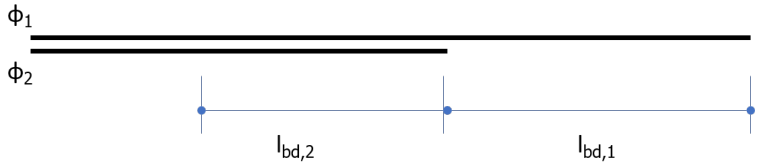


Figure 20.2-17 - Staggered anchorage of bars in a bundle, example of bundles with two bars in a bundle.

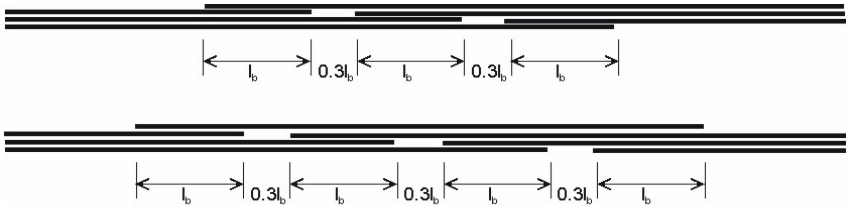


Figure *20.2.-18:* *Laps of individual bars in a bundle (schematic).*

There are several ways to connect a head end to the bar in a manner sufficient to develop the full yield strength of the bar. For example, welding, swaging, extruding, threading, and epoxy grouting methods can be effective. The supplier should provide data that demonstrate the method used for production provides satisfactory strength.

The bearing area of the head is the total area of the head minus the cross-sectional area of the bar and obstructions, if any. The anchorage length should be measured to the bearing face of the head and should thus not include the thickness of the head.

All provisions for anchorage of straight bars may be used also for bundles of bars according to Eq. (20.2-3), Eq. (20.2-4), or Eq. (20.2-5) with parameter ϕ replaced by an equivalent diameter of the bundle defined as:

$$\emptyset_b = \sqrt{\frac{4}{\pi} A_s} \qquad (20.2\text{-}11)$$

where A_s is the total area of all bars contained in the bundle.

When anchoring (or lapping) one bar in a bundle, the design anchorage (or lap length) should be based on its own diameter and covers according to Eqs. (20.2-3) and (20.2-5) (see Fig. 20.2-17). When anchoring (or lapping) more than one bar of a bundle, the design anchorage (or lap) length of those bars should be based on their equivalent diameter.

Where the bundle comprises two bars with an equivalent diameter $\emptyset_b$ not exceeding 32 mm, the bundle may be lapped at the same section with the equivalent diameter according to 20.2.11 used throughout all bond length calculations.

For bundles which consist of two bars with an equivalent diameter > 32 mm or of three bars, laps of individual bars within a bundle should be staggered in the longitudinal direction by at least $1.3l_b$ as shown in Fig. 20.2-18, where l_{bd} is based on a single individual bar. No more than four bars in any individual lap cross-section should be present; therefore, it is not possible to lap bars within a bundle of four bars.

Minimum cover and clear spacing dimensions for the bundle are to be used to determine α_2 (see equation (20.2-6)).

20.2.2.9 Headed bars

Headed bars utilize the principle of mechanical interlock (end bearing) to anchor and develop the reinforcement.

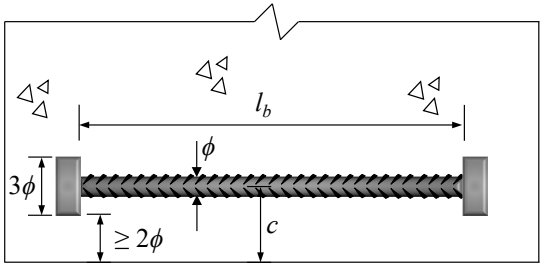


Figure 20.2-19: Condition (a), where the full yield capacity of the bar should be developed at the head

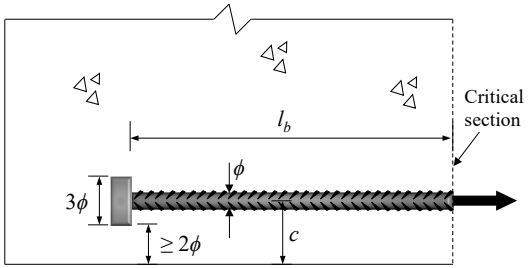


Figure 20.2-20: Condition (b), where the full yield capacity of the bar may be developed by a combination of anchorage by the head and bond along the length of the bar

While it is evident that the strength of an anchorage comprising a straight portion of bar plus a head is less than the sum of the strengths of the two components acting independently, as yet no consensus model for condition (b) is available.

When headed bars are adopted, two conditions may be considered:

a) the full yield capacity of the bar should be developed at the head, for example in double headed studs for shear or as wall ties.

Condition (a) requires that the other possible failure modes with double headed bars including bearing failure, bond splitting failure and side face blowout are not decisive. The bearing failure and the bond splitting failure may be precluded by a stud with head diameter 3 times that of the bar, where the minimum cover to the side of the head is not less than 2 bar diameters, spacing between bar centres is not less than 6 bar diameters, $f_{cd} > f_{yd}/40$, and transverse reinforcement is provided to prevent splitting failure (see Fig. 20.1-19).

The verification of side face concrete blowout maybe carried out in accordance with revised version of *fib* Bulletin 58 [20-7].

b) other circumstances in which the reinforcement is anchored by a combination of anchorage by the head and bond along the length of the bar (see Fig. 20.1-20).

Such circumstances may occur in the case of headed beam bars terminating into columns (beam-column joints) or headed column bars terminating into a foundation (column-foundation connection).

For design according to condition (b), all possible failure modes including steel yielding, bearing failure, bond splitting failure, side face blowout failure and concrete cone breakout failure shall be verified.

The verification of side face concrete blowout and concrete cone breakout may be carried out in accordance with *fib* Bulletin 58 [20-7].

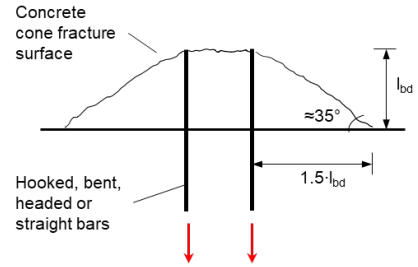
For condition (b), anchorage capacity may be determined in one of two ways:

i) the headed bar may be treated as if it was a bar terminated by a hook or bend (see 20.2.2.3 and Fig. 20.1-9) provided the net bearing area of the head is not less than 4 times the area of the bar and the head is affixed to the bar in a manner sufficient to develop the full yield strength of the bar;

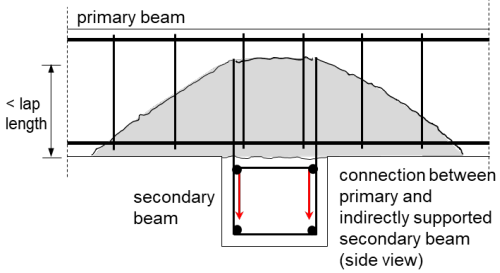
ii) as the capacity of the head alone with no contribution from bond along the straight portion of the bar according to subsection 20.2.2.3;

In all cases the head should have sufficient embedment beyond the most highly stressed point of the bar to preclude a premature concrete cone type failure.

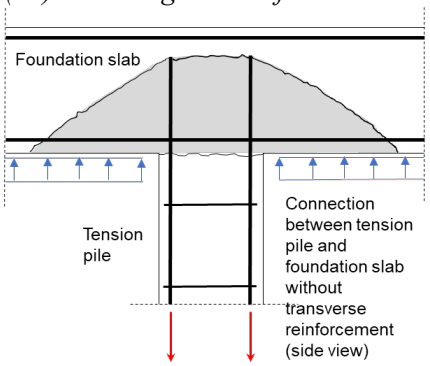
The anchorage lengths given in this document and resulting bond strength values are evaluated from the results of tests using test conditions where failure occurs by splitting or pullout; other failure modes (e.g., concrete cone or combined bond and concrete failure) are suppressed by a confined support or by adding a debonded length. These conditions may not always be present in reality (e.g., indirect support of a beam, exterior joint between beam and column or slab and wall, connection between column or wall and foundation slab). Therefore, under these conditions the possibility of concrete cone breakout, combined bond and concrete failure (straight bars only) and blowout failure (headed bars only) should be considered. This may be done by designing such connections according to the provisions given in the *fib* Bulletin 58 [20-7].



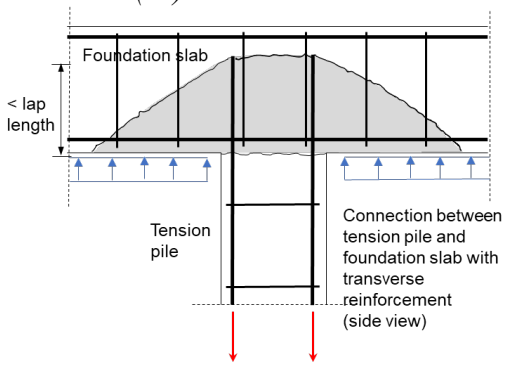
(a1) Anchorage or reinforcement



(a2) Beam-beam connection



(a3) Tensioned pile



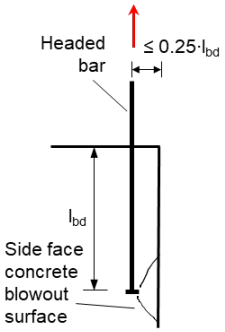
(a4) Tensioned pile

(a) Examples of tension loaded group of bars

20.2.2.10 Additional verifications

In addition to the verifications for anchorages according to sections 20.2.2.3, 20.2.2.4 and 20.2.2.9; verification of single or groups of straight, headed or hooked bars against possible concrete cone breakout failure and combined bond and concrete failure (straight bars only) or headed bars against side face blowout failure may be carried out in accordance with the rules given in *fib* Bulletin 58 [20-7].

For direct tension loads, a concrete cone forming an angle of approximately 35° with the surface may form (Fig. 20.2-21a). Such a situation might occur in applications such as an unconfined anchorage of reinforcement (a1), indirectly loaded beam through a secondary beam (a2) or tensioned pile anchored in a slab without transverse reinforcement (a3) or with transverse reinforcement (a4). By adding transverse reinforcement in the load taking member, the capacity of the anchorage can be increased.



(b) Headed bar close to an edge

Figure 20.2-21: Concrete breakout failure of end anchorages

Beam-column joints are critical components to maintain the integrity of the reinforced concrete frame structures. Under seismic action, their behaviour and performance depend strongly on the bond performance of longitudinal beam reinforcement. Sufficient anchorage of longitudinal beam reinforcement is essential to avoid premature bond failure of the joints.

Under seismic loads a plastic hinge may form within a distance of h_{cr} from the face of a beam (or slab) - column connection or a connection to the footing of any R.C. member, where inelastic bar strains are expected to develop in the longitudinal reinforcement, beyond the yielding limit. The length h_{cr} is determined depending on the ductility class (for example EN 1998-1 [20-8]). For highly dissipative regions, the rotation capacity of plastic hinges containing well confined lap-spliced longitudinal reinforcement may be governed by localized strain demands in the reinforcement extending from the adjacent connection. Owing to these high strain demands, local fracture of reinforcement may occur at the face of the support or inside the footing, at drift ratios that are less than those intended for this ductility class.

Under seismic loads, the longitudinal beam reinforcement in case of interior joints that extends on both sides of the column is subjected to compression from one side and tension on the other side (assuming compatibility of deformations between

Additionally, for headed bars, the possible side face blowout failure should be verified (Fig. 20.2-21b).

20.2.2.11 Anchorage in beam-column joints under seismic action

20.2.2.11.1 Anchorage in the joint

In high dissipative seismic structures, tension laps are not allowed in the plastic hinges required to form at the base of the first storey columns and lateral-load resisting walls, as well as in the beams in regions adjacent to beam-column joints and in the ends of coupling beams.

The longitudinal beam reinforcement shall be sufficiently anchored in the joints to ensure yielding under seismic actions.

a) Interior joints

For joints through which the longitudinal beam reinforcement extends on both sides, the column dimension in the direction of the beam reinforcement shall be at least equal to:

steel and concrete). This results in high bond requirements on the beam longitudinal reinforcement. The provision laid in Eq. (20.2-12) is expected to develop the beam longitudinal reinforcement passing through the joint considering the mean bond stress, mean yield stress of the rebar and taking into account an over-strength of 1.30. The additional 5ϕ takes into account the yield penetration in the rebar.

The established seismic design philosophy depends on reinforcement yielding as the dominant failure mode to ensure ductile behaviour in the system. Therefore, if the longitudinal beam reinforcement area provided exceeds what is required by the design calculation, the total reinforcement area should be considered to develop the design yield stress. Therefore, $l_{bd,y}$ is calculated for all the bars passing through the joint or anchored into the joint such that they can develop the design yield stress.

Although Eq. (20.2-3) considers the beneficial influence of transverse pressure on anchorage length, for anchorages through beam-column joints, the presence of axial load on the column is neglected, by setting $p_{tr} = 0$. This is owing to the absence of experimental evidence on the influence of varying transverse pressure (axial load) on the anchorage resistance of embedded bars in beam-column joints.

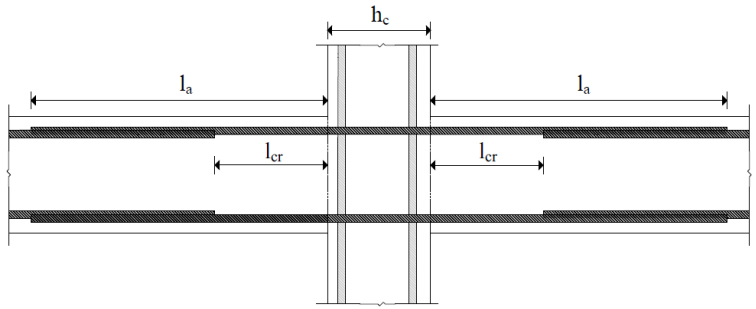


Figure 20.2-20: Anchorage in an interior joint

The requirement of projected length of the bend of the bar into an exterior joint ($l_{b,straight}$) is targeted to prevent:

(i) splitting failure in the plane of the hook;
(ii) concrete cone breakout from the hook;
(iii) bond failure under compression.

Note: the expression for $l_{bd,y}$ in Eq. (20.2-12) implicitly considers the influence of the hook through the value of $k_{beb} > 0$ in Eq. (20.2-3).
The longitudinal beam reinforcement should be anchored by bending into the joint to enable the formation of a stable strut mechanism.

$$h_c \geq max(0.6 l_{bd,y} + 5\phi; 20\phi; 300\ \text{mm}) \qquad (20.2\text{-}12)$$

where,

h_c is the dimension of the column in the direction of beam longitudinal reinforcement (see Fig. 20.2-20);
ϕ is the bar diameter;
$l_{bd,y}$ is the design anchorage length in tension required to develop the design yield stress in the reinforcement according to Eq. (20.2-3), taking $k_{beb} = 0$ and $p_{tr} = 0$.

a) Exterior joints

The longitudinal beam reinforcement shall terminate in the column or an extended stub beyond the column by bending the bar into the joint with a standard bend that has a straight length of not less than 10 times the diameter of the beam longitudinal reinforcement (Fig. 20.2-21). The straight part of the beam bar (projected length) terminating into the column or stub shall be as given by Eq. (20.2-13):

$$l_{b,straight} \geq max(0.6 l_{bd,y} + 5\phi; (20 - k_{beb})\phi; 300\ \text{mm}) \qquad (20.2\text{-}13)$$

where:

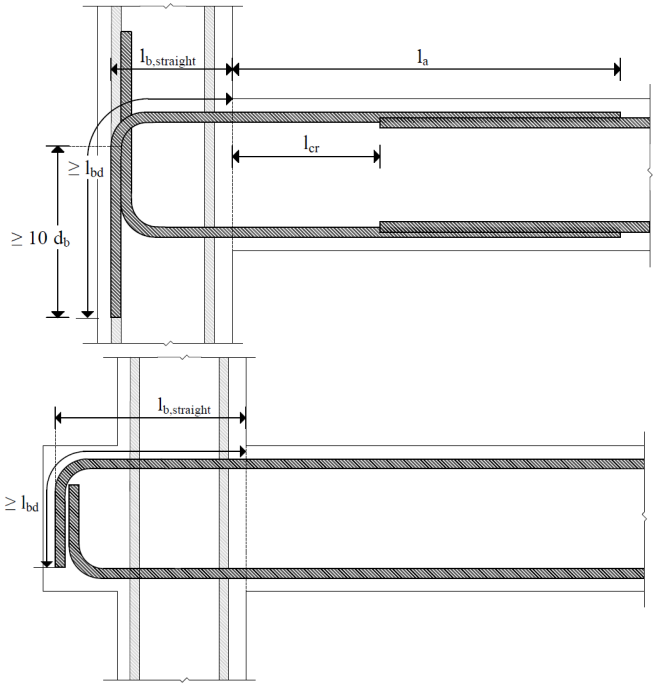


Figure 20.2-21: Anchorage in an exterior joint

From the requirement of assuring reinforcement yielding as the dominant failure mode, it follows that the plastic hinge in the beam forms within a certain distance from the face of the column, l_{cr}. This length has to be considered as critical due to the relatively high demand on the rotational capacity/ductility. Therefore, for high ductility class members, the splice of the longitudinal reinforcing bars in the beam shall start at a distance from the column face not less than l_{cr} (Figs. 20.2.20 and 20.2.21).

It is likely that the requirements laid by Eq. (20.2-11) or Eq. (20.2-12) are not satisfied by the anchorage in an existing reinforced concrete structure. The actual behaviour of the anchorage should be modelled according to the appropriate provisions of this chapter (section 20.3 and 20.5), within the framework of performance-based assessment approach. Where bond behaviour might be critical to the performance of the structure, an appropriate retrofit solution should be applied to reduce the demand on anchorage in beam-column joints. The addition of post-

$l_{bd,y}$ is the design anchorage length in tension required to develop design yield stress in the reinforcement according to Eq. (20.2-3) taking $p_{tr} = 0$;
ϕ is the bar diameter;
k_{beb} is the coefficient for standards bends, which may be determined from section 20.2.2.4.

20.2.2.11.2 Extension of bars beyond the face of the column

The top or bottom bars of a beam framing into the joint shall extend to a distance not less than $l_a \geq l_{bd,y}$ from the face of the column (Fig. 20.2-21).

For high ductility design, the top or bottom bars of a beam framing into the joint shall extend to a distance not less than $l_a \geq (l_{bd,y} + h_b)$ from the face of the column.

The splice of the longitudinal reinforcement in the beam shall start at a distance not less than $l_{cr} \geq h_b$ from the face of the column, where h_b is the beam depth (Fig. 20.2-21).

installed haunches is a solution to improve the anchorage of reinforcement in the corner. However, global retrofit solution (shear wall, and steel bracing) may be a general solution for seismic upgrade to reduce the anchorage capacity.

Table 20.2-2. *The values of the coefficient λ_{fi} (valid for $\gamma_{m,fi}$ = 1.0 and γ_m =1.5)*

Exposure duration	λ_{fi}			
[min]	$\eta_{fi} \leq 0.5$	$\eta_{fi} = 0.6$	$\eta_{fi} = 0.7$	$\eta_{fi} = 0.75$
0	1.00	1.00	1.00	1.00
15	1.00	1.00	1.00	1.00
30	1.00	1.00	1.00	1.00
60	1.00	1.00	1.10	1.20
90	1.00	1.20	1.40	1.50

Notes:

20.2.2.12 Bond of embedded steel reinforcement under special circumstances

In special circumstances, the casting position factor k_{tcSP} in Eq. (20.2-5) shall be equal to:

k_{tcSP} = 0.8 for all bars used in slipform construction unless it is shown that the vertical bars cannot move during casting;

= 0.7 for members executed under poor bond conditions of bentonite or similar slurries

For fusion bonded epoxy coated ribbed or indented bars, in Eq. (20.2-5) a lead coefficient of 84 shall be used in place of the coefficient of 67 and coefficient α_2 for confinement from concrete cover shall be taken as:

$$\alpha_2 = (c_{min} / Ø)^{0.8} \cdot (c_{max}/c_{min})^{0.15} \qquad (20.2\text{-}14)$$

20.2.2.13 Laps and anchorages in extreme thermal condition: fire (or elevated temperature)

The influence of fire on bond is generally accounted for by providing larger bond length than required under ambient conditions, based on the expected duration of fire exposure (required fire rating).

The anchorage or lap length, $l_{b,fi}$ deemed to withstand a defined fire exposure without failure can be calculated as:

$$l_{b,fi} = \lambda_{fi}\, l_{b,20} \qquad (20.2\text{-}15)$$

where, $l_{b,20}$ is the design anchorage or lap length of the reinforcing bar under ambient conditions (20 celsius degrees) and corresponds to l_{bd} given by Eqs. (20.2-3) and (20.2-4) or (20.2-5).

The values of the coefficient λ_{fi} can be read from Table 20.2-2.

i) the values of λ_{fi} recommended in Table 20.2-2 are derived assuming that the ratio of mean to characteristic strength is the same under ambient conditions and upon heating (i.e. that the scatter of bond strength is the same for ambient and hot state).

ii) The values of λ_{fi} take into account the reduced safety factors for design in case of accidental situation of fire ($\gamma_{m,fi}$) and reduction factor for the effect of actions under accidental loading η_{fi}.

iii) The values of λ_{fi} are derived based on results on beams tested in residual state (after cooling); these values are considered to be conservative for the hot state (during fire).

iv) The tests used to derive the values of λ_{fi} were performed without mechanical load during heating. It is, however, shown on pull-out specimens that a load up to 60% of the mean bond capacity does not adversely affect the bond performance (see section 20.5.4). In case of high design load (higher than 50% of the mean bond capacity), additional experimental tests and/or numerical analyses are required. In such cases, as a general guidance it may be assumed that when the value of variable actions exceed one third of the value of permanent actions, the factor λ_{fi} obtained using Eq. (20.2-16) should be multiplied by 1.25.

(v) In Table 20.2-2 the values of η_{fi} equal 1 indicate that in these cases the anchorage length is controlled by factors other than fire, or – in other terms – concrete grade and bar cover offer a sufficient protection to the reinforcement, for the specified fire duration.

Table 20.2-3. The values of the coefficient β_{fi} ; exposure to standard fire

Exposure duration	β_{fi}
[min]	*[-]*
0	1.00
15	0.80
30	0.68
60	0.45
90	0.35

In case other values for $\gamma_{m,fi}$ *and* γ_m are used, the coefficient λ_{fi} can be calculated as:

$$\lambda_{fi} = max\ \{(\eta_{fi} / \beta_{fi})\ (\gamma_{m,fi} / \gamma_m);\ 1\} \qquad (20.2\text{-}16)$$

where:

η_{fi} is the reduction factor for the design loads (at ULS), to be introduced in fire situation according to Section 30.4.2:

γ_m is the partial safety factor for the material under ambient conditions;

$\gamma_{m,fi}$ is the partial safety factor for the material under accidental loading;

β_{fi} is the reduction factor for bond capacity with respect to fire duration.

The values of β_{fi} shall be read from Table 20.2-3.

Alternatively, advanced numerical analyses should be carried out.

Advanced numerical analysis

In case of detailed numerical analysis, the latter effect (damage to concrete due to thermal gradients) might be automatically considered through the analysis. However, in that case a degradation of the material properties of concrete and steel as well as the degradation of bond strength due to elevated temperature needs to be

The thermal gradients and temperature at various locations in a section are calculated first by thermal analysis. The analysis should be able to consider the influence of thermal gradients on damage to concrete. The material model used for concrete and reinforcing steel should be such that it considers the influence of

considered in the material model. The degradation of bond stress-slip law due to elevated temperature can be taken into account using equations given in section 20.5.4.

The stress-dependence of fatigue effects on bond leads to a process of redistribution of stresses along the embedded length with the number of load cycles. Such a capacity of redistribution of stresses is generally responsible for different fatigue strength of laps and anchorages than the one corresponding to a uniform bond stress distribution. The shorter the embedded length, the smaller the capacity for redistribution of stresses.

Each welded joint should be capable of withstanding the shearing force given in section 15.7.3.

elevated temperature on their mechanical properties as laid out in the relevant sections of this code. The influence of elevated temperature on bond strength between ribbed reinforcing bars and concrete should be considered according to section 20.5.4.

20.2.2.14 Laps and anchorages under fatigue loading

Bond fatigue failure of laps and anchorages can take place if a high number of repeated load cycles is applied. In order to prevent such a failure mode, laps and anchorages should be placed at sections where the maximum load level is low enough in comparison with the load causing yielding of reinforcement.

The exact determination of bond fatigue strength of laps and anchorages is a complex task since fatigue depends on the stress level (see 18.4.1.3.5) and the bond stress distribution over the embedded length is nonlinear. In general, the fatigue degradation of bond leads to a process of redistribution of stresses along the embedded length with the number of load cycles. The estimation of fatigue strength of laps and anchorages is therefore complex but an approximate calculation can be evaluated with the S-N model of Eq. (18.1.15) by assuming a uniform bond stress distribution, thereby neglecting the capacity for redistribution of stresses.

Besides fatigue influence on bond, it must be considered that welding of steel reinforcement causes a significant reduction of fatigue strength of steel. Welding details should only be used in laps and anchorages subjected to fatigue loading if the welding process is approved for fatigue loading and the fatigue strength of the welding detail is taken into account in design.

20.2.3 Anchorage and lapped joints of welded fabric

20.2.3.1 Design anchorage length of welded fabric

The design anchorage length l_{bd} is calculated in accordance with Eq. (20.2-3) with $k_{beb} = 15$.

20.2.3.2 Design lap length of welded fabric

20.2.3.2.1 Lap length for main reinforcement

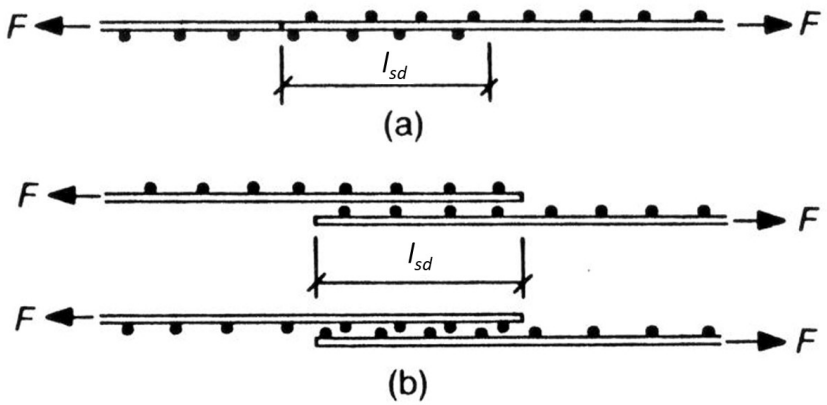


Figure 20.2-22: Lapping of welded fabric: (a) intermeshed fabric (longitudinal section); (b) layered fabric (longitudinal section).

The minimum number of welded cross wires over the lap length is:

$n = 1$ for fabric made of ribbed or indented wires

$n = 4$ ($A_{s,calc}/A_{s,ef}$) for fabric made of plain wires (n to be rounded up to the next whole number).

The design lap length is given by:

– with intermeshed fabrics (Fig. 20.2-22a)

$l_{sd} = l_{bd} \geq 250$ mm

where l_{bd} is calculated according to 20.2-3.

– with layered fabric (Fig. 20.2-22b):

$l_{sd} = l_{bd} + 2\phi \geq 250$ mm

where l_{bd} is calculated according to 20.2-3.

Splicing of welded fabric in structures assessed for fatigue loads should be done with intermeshed fabrics.

20.2.3.2.2 Laps in the transverse direction: secondary reinforcement

For intermeshed fabrics, provisions for the main welded fabric reinforcement apply (see section 20.2.3.2.1).

For layered fabrics, the length of lap is chosen from Table 20.2-4:

Table 20.2-4: Required lap lengths for splices of the secondary reinforcement (layered fabrics).

Diameter of wires	Lap lengths	Other requirements
Ø ≤ 6 mm	≥ 150 mm	at least 1 wire pitch within the lap
6 mm< Ø ≤ 8.5 mm	≥ 250 mm	at least 2 wire pitch
8.5 mm<Ø≤ 12 mm	≥ 350 mm	at least 2 wire pitch

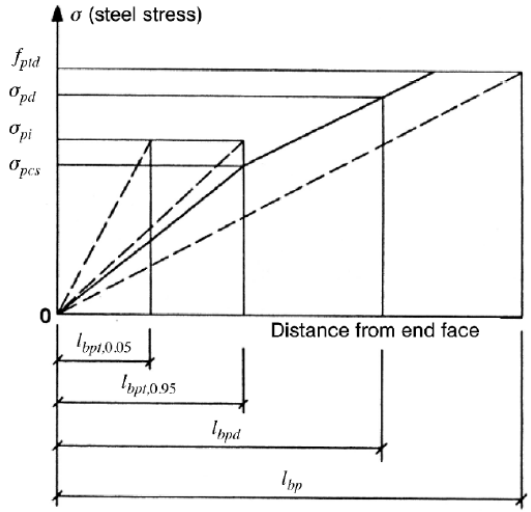


Figure 20.2-23*: Variation in steel stress along the anchorage zone of a pretensioned member: length parameters.*

Two different bond situations should be considered due to the transverse deformations of the tendon. “Push-in” along the transmission length, where the tendons become thicker at release and “pull-out”, which refers to the anchorage length where the opposite occurs when the steel stress is increased due to loading.

The clear spacing and concrete cover of pre-tensioned tendons shall ensure a proper placing and compacting of fresh concrete such that a sufficient bond can be attained between the concrete and the tendons.

20.2.4 Anchorage of pretensioned prestressing tendons

20.2.4.1 General

In anchorage zones of tendons, the following length parameters should be considered (Fig. 20.2-23):

- Basic anchorage length, l_{bp}, that is required to develop the full strength in an untensioned tendon (see section 20.2.4.3);
- Transmission length, l_{bpt}, over which the prestressing force (P_0) is fully transmitted to the concrete (see Section 20.2.4.4). There are two values of transmission length: $l_{bpt.0.05}$ for transverse stress verification, and $l_{bpt.0.95}$ for ULS verification (flexure, shear, anchorage);
- Anchorage length, l_{bpd}, over which the tendon force P_{pd}, in the ultimate limit state is fully anchored in the concrete (see section 20.2.4.5) l_{bpd} is the sum of two contributions: the transmission length $l_{bpt.0.95}$ and the length required to anchor the additional tendon stress due to external loading (flexural bond length). These two length values are computed at different ages of concrete;

The highest value of the bond strength of pretensioned prestressing tendons applies to the transmission length (i.e. the length required to introduce the prestressing force). Beyond that length, a lower bond strength has to be taken into account. As a result, a bilinear diagram has to be considered in the embedment length that is required to develop the design steel stress (Fig. 20.2-23).

20.2.4.2 Arrangement of tendons

The minimum spacing and cover for bond stress and to avoid splitting should be in accordance with Fig. 20.2-24 and Table 20.2-5. Other values of spacing and cover may be used, provided that a satisfactory behaviour in service and at ultimate limit state is demonstrated by testing or permitted by the specific European product standard with factory production control.

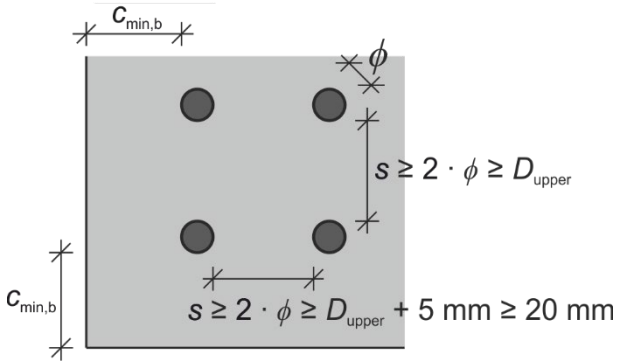


Figure 20.2-24: *Minimum clear spacing and cover for bond for pre-tensioning tendons (D_{upper} is the maximum size of aggregate).*

Table 20.2-5: *Minimum concrete cover $c_{min,b}$ for pretensioned tendons.*

Minimum spacing s	Minimum cover: $c_{min,b}$ / ϕ	
	Strand	Indented wire
$s = 2\ \phi$	3.0	4.5
$s \geq 2.5\ \phi$	2.5	4.0

Bundling of tendons should not occur in the anchorage zones without taking special provisions to cover splitting forces and taking into account the bond behaviour of bundles. Outside the anchorage zone up to three tendons may be bundled.

The factor A_{sp} / ($Ø\pi$) depends on the type of tendon:

$\frac{A_{sp}}{Ø\pi} = \frac{Ø}{4}$ for tendons with a circular cross-section;

$\frac{A_{sp}}{Ø\pi} = \frac{7}{36}Ø$ for 7-wire strands

20.2.4.3 Basic anchorage length

The basic anchorage length of an individual pretensioned tendon is:

$$l_{bp} = \gamma_c \frac{A_{sp}}{Ø\pi} \frac{f_{ptd}}{\eta_{p1}\eta_{p2} f_{ck}(t)^{1/2}} \qquad (20.2\text{-}17)$$

where:

A_{sp} is the cross-sectional area of the tendon;

f_{ptd} $= f_{ptk}/\ \gamma_S$ is the design tendon strength;

f_{ptk} is defined in section 16.1.5.2;

$Ø$ is the nominal diameter of the tendon;

η_{p1} takes into account the type of prestressing tendon;

$\eta_{p1} = 0.42$ for indented and crimped wires;

$\eta_{p1} = 0.36$ for 7-wire strands;

η_{p2} takes into account the position of the tendon, as follows:

$\eta_{p2} = 1.0$ for all tendons with an inclination of 45–90° with respect to the horizontal during concreting;

$\eta_{p2} = 1.0$ for all horizontal tendons which are up to 250 mm from the bottom or at least 300 mm below the top of the concrete section during concreting;

$\eta_{p2} = 0.7$ for all other cases;

$f_{ck}(t) = \beta_{cc}(t)^{2/3} f_{ck}$ is the concrete compressive strength at time t.

20.2.4.4 Transmission length

The transmission length of a pretensioned tendon is:

$$l_{bpt} = \alpha_{p1}\alpha_{p2}\alpha_{p3}\frac{\sigma_{pi}}{f_{ptd}} l_{bp} \qquad (20.2\text{-}18)$$

where:

σ_{pi} is the steel stress just after release;

α_{p1} considers the type of release:

$\alpha_{p1} = 1.0$ for gradual release;

$\alpha_{p1} = 1.25$ for sudden release;

α_{p2} considers the action effect to be verified:

$\alpha_{p2} = 1.0$ for calculation at ULS of anchorage length, shear capacity and moment capacity;

$\alpha_{p2} = 0.5$ for verification at SLS of transverse stress due to development and distribution of prestress in the transmission zone;

α_{p3} considers the influence of bond situation:

$\alpha_{p3} = 0.5$ for strands;

$\alpha_{p3} = 0.7$ for indented or crimped wires;

The use of narrow spaced stirrups or helices around the tendons and transverse prestressing may result in shorter transmission length. This is not considered due to lack of experimental data.

α_{p1} may be increased up to 1.35 for sudden release at the cut end of the prestressed element.

Tendon release that is obtained by sawing through the concrete and the steel should be considered as gradual release.

The transmission length can be estimated from the draw-in value (δ_e) of the tendons at the end face of the concrete member. However, when the concrete member is sawn from a longer production unit, the draw-in cannot be estimated properly. Assuming a linear steel stress along the transmission length, this draw-in may be:

$$\delta_e < 0.5\frac{\sigma_{pi}}{E_p} l_{bpt}$$

with $\alpha_{p2} = 1.0$ in Eq. (20.2-18) for l_{bpt}; E_p is the modulus of elasticity of the prestressing steel.

Note that $\alpha_{p2} = 0.5$ is associated with $l_{bpt,0.05}$ (highest value of bond stress) and $\alpha_{p2} = 1.0$ with $l_{bpt,0.95}$ (lowest value of bond stress) in Fig. 20.2-23.

The basic anchorage length is related to “pull-out” failure. The transmission length is connected to “push-in”.

If necessary, the required anchorage capacity may be obtained by additional end anchorages or non-prestressed reinforcement.

Increased values of the design anchorage length may be assumed for fatigue verification. This is not considered in the present document due to lack of experimental data.

The suitability and allowable use conditions for the anticipated field of application of a system for post-installation of deformed reinforcing steel bars should be verified by means of a prequalification procedure specified by an independent agency. The result of the evaluation corresponding to the prequalification procedure shall be documented in an approval or assessment document. The prequalification process consists of appropriate prequalification tests (see 20.2.5 (4)) at independent tests laboratories considering the type of loading and an assessment of test results by an independent organization. It is assumed that the technical product specification (e.g., approval or assessment document) contains all information for the installation and design of reinforcing steel bars required by this Model Code.

The provisions of this section apply only to uncoated ribbed straight reinforcing steel bars because no results of tests with coated post-installed reinforcing steel bars are available.

The rules apply to uncoated ribbed bars because limited research is available on indented bars (to be completed).

l_{bp} is the basic anchorage length in Eq. (20.2-17), computed at the time of release

20.2.4.5 Design anchorage length

The design anchorage length of a pretensioned prestressing tendon is:

$$l_{bpd} = l_{bpt} + l_{bp}\frac{\sigma_{pd} - \sigma_{pcs}}{f_{ptd}} \qquad (20.2\text{-}19)$$

where:

l_{bpt} is the transmission length adopting $\alpha_{p2} = 1.0$ ($l_{bpt0.95}$);

σ_{pd} is the tendon stress under design load ($\sigma_{pd} \le f_{ptd}$);

σ_{pcs} is the tendon stress due to prestress including all losses;

l_{bp} is the basic anchorage length in Eq. (20.2-17), computed at 28 days.

20.2.5 Anchorage and lapped joints of post-installed reinforcing bars

(1) This section applies to the anchorage and lapping of post-installed uncoated reinforcing bars, installed in normal weight reinforced concrete of compressive strength in a general ranging from f_{ck} = 12 MPa to f_{ck} = 50 MPa under static, quasi-static, sustained and seismic loading.
It is assumed that the post-installed rebar system has passed a prequalification procedure demonstrating the suitability to its intended use.
Post-installed reinforcing bars shall be anchored such that the bond forces are safely transmitted to the concrete avoiding longitudinal cracking or splitting.
Forces are transmitted from a post-installed reinforcing bar to a cast-in reinforcing bar by lapping. The detailing is done such that spalling of the joints and large cracks do not occur.
(2) The provisions in this section apply to systems for the post-installation of uncoated reinforcing steel bars comprising of:

- An uncoated ribbed straight reinforcing steel bar with properties according to section 15.4 and section 15.5;

The performance of post-installed reinforcing steel bars depends significantly on the quality of the installation and the level of inspection. The qualification should be verified by an independent organization (for details see e.g., guidelines of the Deutsche Institut für Bautechnik (DIBt 2020) [20-9] and the American Concrete Institute (ACI 2015) [20-10], covering the installation of bonded anchors).

The requirements on the bond strength may be considered as fulfilled if the mean bond strength of post-installed reinforcing bars under static, quasi-static and sustained load is $f_{bm,pi} \geq f_{bm,pi,rqd}$ with $f_{bm,pi,rqd}$ according to Table 20.2-6 when tested in appropriate prequalification tests and assessed according to EAD 330087-00-0601 [20-11] or ICC-ES AC308 [20-12].

Table 20.2-6: *Required minimum mean bond strength ($f_{bm,pi,rqd}$) of post-installed reinforcing steel bars*

Concrete strength, f_{ck} [MPa]	12	16	20	30	40	50
$f_{bm,pi,rqd}$ [1)][MPa]	7.7	8.9	10.0	12.2	14.1	15.8
[1)]Values for intermediate concrete strength classes may be linearly interpolated.						

For seismic loading in addition the bond strength after cyclic loading should fulfil the requirements given in EAD 330087-00-0601 [20-11] or ICC-ES AC308 [20-12].

A generally accepted qualification process of a system in case of fatigue loading is currently not available. Therefore, this type of loading is not covered.

– Tools required by the manufacturer's installation instruction for drilling and cleaning of the drill hole as well as for filling the borehole with mortar, especially for large bond lengths;
– An anchoring mortar with established suitability and properties according to 20.2.5 (4) and 20.2.5 (5) for the intended use, exposure condition and temperature.

(3) The design of post-installed reinforcing steel bars according to these provisions assume that the installation is performed according to the manufacturer's written installation instructions by qualified personnel and the inspection of the installation is carried out by qualified personnel.

(4) Post-installed reinforcing steel bars should develop at least the same mean bond strength $f_{bm,pi}$ as cast-in reinforcing steel bars under equivalent test conditions when subjected to static, sustained or seismic loading in appropriate pre-qualification tests at independent test laboratories. In the tests and the assessment of the test results the following influencing factors should be taken into account:

i. concrete alkalinity, environmental conditions (e.g. sulphurous atmosphere) as well as freeze/thaw cycles;
ii. intended types of loading (tension or tension and compression without consideration of end bearing, static, quasi static and sustained or static, quasi static, sustained and seismic);
iii. intended drilling methods (hammer drilling, diamond coring, compressed air drilling);
iv. tools necessary for drilling (e.g. drilling aids) and proper installation (e.g. cleaning and injection devices) for each drilling method, taking into account robustness of installation;
v. intended range of concrete strength (usually 12 MPa $\leq f_{ck} \leq$50 MPa);
vi. range of intended reinforcing steel bar diameters for each drilling method;
vii. maximum bond length for each injection tool as a function of the bar diameter;
viii. stiffness of anchoring mortar (bond splitting behavior in long anchorages);
ix. intended installation conditions:
    a) installation in dry concrete or dry and wet concrete,
    b) installation direction:
        a. vertical down only, or

b. vertical down and horizontal or
c. vertical down, horizontal and overhead;

x. minimum and maximum temperature in the concrete member during installation;
xi. minimum and maximum temperature in the concrete member in-service (maximum long-term temperature which occurs over a longer time and maximum short-term temperature which occurs during a short time during a day);
xii. crack width;
xiii. sustained load;
xiv. cyclic loading representing seismic actions (only for use under seismic loadings);
xv. bond resistance under fire exposure (if applicable).

Post-installed reinforcing steel bars are surrounded by anchoring mortar. Therefore, corrosion protection of the reinforcing steel bar is not ensured by the alkalinity of the concrete, but by the properties and density of the mortar layer. The ability of the anchoring mortar to protect the post-installed reinforcing steel against corrosion in service should be demonstrated by suitable tests (see EAD 330087-00-0601 [20-11] or ICC-ES AC308 (ICC-ES 2022) [20-12]).

(5) Post-installed reinforcing steel bars should demonstrate a suitable protection against corrosion taking into account relevant environmental factors and allowable chloride content of concrete.

Restrictions may be given in the technical product specification (e.g., approval or assessment document) in respect to the aspects listed in 20.2.5 (4):

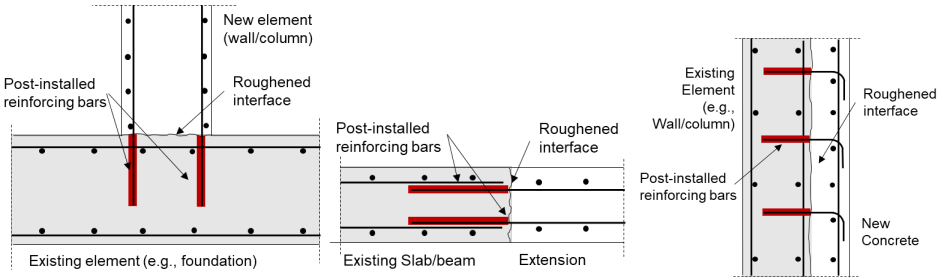


Figure 20.2-25: Examples of typical applications with post-installed reinforcing steel bars: a) starter bars loaded in compression; b) lap splices; c) dowels in shear friction interfaces.

(6) Connections between reinforced concrete elements using deformed post-installed reinforcing steel bars according to 20.2.5 (1) to 20.2.5 (4) are permissible for all applications where straight cast-in-place uncoated deformed reinforcing bars are allowed (see examples in Fig. 20.2-25). The design under static, quasi static, sustained and seismic loading can be conducted using the provisions for cast-in-place reinforcing steel bars in this code. However, restrictions that may be given in the relevant technical product specification (e.g., approval or assessment document) shall be taken into account in the detailing of the connection, its design and execution.

(7) If no larger values are specified in a technical product specification (e.g., an approval or assessment document), the minimum concrete cover of post-installed

The factors 0.06 $l_{bd,pi}$ and 0.08 $l_{bd,pi}$ in Table 20.2-7 take into account possible deviations from the alignment i.e., the inclination of the drilling tool versus the concrete surface during the drilling operation. These factors can be reduced to 0,02 $l_{bd,pi}$ if drilling aid devices ensuring a constant alignment during drilling operation are used. These values are based on experimental investigations carried out by Spieth [20-14] to make sure that the concrete cover can be kept.

The value 0.02 $l_{bd,pi}$ may be reduced if more effective drilling aid devices are employed than the one used by Spieth [20-14] and the effectiveness of these devices has been proven by suitable tests.

The given values for the clear distance between cast-in and post-installed reinforcement (in general equal to clear distance between spliced bars) as well as the values for the clear distance between post installed bars (in general equal to clear distance between neighbouring splices or started bars) should ensure that micro cracks in the concrete due to drilling (which cannot be avoided) do not reduce the anchorage or splice strength. They are valid for a ratio drilling diameter, d_0, to bar diameter $d_0/ø \leq 1.4$ ($ø \leq 10$mm) or $d_0/ø \leq 1.3$ ($ø \geq 12$mm) and a maximum aggregate size d_g= 20mm. For larger ratios $d_0/ø$, the term ø in Eq. (20.2-21) should be replaced by d_0 and for larger aggregates the clear distance $c_{s,min}$ according to Eq. (20.2-20a) or (20.2-20b) should be $\geq d_g$. If the size of the aggregates is not known d_g = 32 mm should be assumed.

The provisions are based on Spieth [20-14] who tested about 31 specimens with splices between cast-in and post-installed bars and 17 comparison splices between cast-in bars. The influence of drilling on microcracks between aggregates and cement paste was investigated by Fuchs & Sawade [20-15].

No research results are available with anchorages or lapped splices of post-installed reinforcing steel bars under transverse pressure. Therefore, conservatively, k_p = 1.0 should be used when calculating l_{bd} according to Eq. (20.2-21).

Eq. (20.2-21) assumes that under the same conditions the bond behaviour (bond stiffness and bond strength) of post-installed reinforcing bars is at least equal with the bond behaviour of cast-in reinforcing bars. When using post-installed systems with bond stiffness and/or bond strength lower than that of cast-in reinforcing bars the displacements (crack widths) at the loaded end of an anchorage or at the ends of a lapped joint of tensioned bars will increase compared to cast-in bars, even if the bond length is increased. This may not be acceptable at serviceability limit state.

reinforcing steel bars in all directions c_{min} should be detailed in accordance with Table 20.2-7, depending on the intended drilling method, the diameter of the reinforcing steel bar under consideration and the use of a drilling aid.

Table 20.2-7: Minimum concrete cover c_{min} for post-installed reinforcing steel bars.

Drilling method	Bar diameter ø (mm)	c_{min} without drilling aid (mm)	c_{min} with drilling aid (mm)
Hammer or diamond drilling	< 25 mm	30 mm+0.06 $l_{bd,pi} \geq 2$ ø	30 mm +0.02 $l_{bd,pi} \geq 2ø$
	≥ 25 mm	40 mm +0.06 $l_{bd,pi} \geq 2$ ø	40 mm +0.02 $l_{bd,pi} \geq 2ø$
Compressed air drilling	< 25 mm	50 mm +0.08 $l_{bd,pi} \geq 2$ ø	50 mm +0.02 $l_{bd,pi} \geq 2ø$
	≥ 25 mm	60 mm +0.08 $l_{bd,pi} \geq 2$ ø	60 mm +0.02 $l_{bd,pi} \geq 2ø$

(8) If no larger values are specified in a technical product specification (e.g., approval or assessment document) the following values should be used:
Clear spacing between parallel cast-in and post-installed reinforcing bars:
i) Hammer- or diamond drilling

$$c_{s,min} = \max (2\ ø\ ;\ 20\ \text{mm}) \qquad (20.2\text{-}20a)$$

ii) Compressed air drilling

$$c_{s,min} = \max (2\ ø\ ;\ 30\ \text{mm}) \qquad (20.2\text{-}20b)$$

The minimum clear spacing between parallel post-installed reinforcing bars is double the value according to the eqs. (20-20a) or (20-20b) for the applicable drilling method.

(9) The design anchorage length $l_{bd,pi}$ of post-installed reinforcing steel bars in tension shall be calculated as:

$$l_{bd,pi} = l_{bd} > l_{bd,min} \cdot \alpha_{lb} \qquad (20.2\text{-}21)$$

where:
l_{bd} = design anchorage length of cast-in deformed reinforcing steel bars evaluated according to Eqs. (20.2-3) and (20.2-5) but taking k_{tp} =1.0;
$l_{bd,min}$ = minimum anchorage length = 10ø ≥ 100 mm.

Furthermore, at ultimate limit state the strength of connections between concrete elements (e.g., connection between beam and column, slab and wall, column or wall and foundation) may be reduced even when increasing the bond length of tensioned bars because of the reduced shear strength in the interface.

Full size tests of overlap splices between cast-in and post-installed reinforcing bars in a tension specimen under static loading are available (Spieth [20-14]; Simons [20-16]). In these tests the bond strength of the post-installed reinforcing bars was lower than that of cast-in bars. The splice length was increased compared to cast-in bars. Based on these results a designer may consider to use a post-installed reinforcing bar system with a bond behavior inferior to that of cast-in reinforcing bars for splices with cast-in bars in regions with an almost constant moment, if the design anchorage length $l_{bd,pi}$ calculated according to Eq. (20.2-21) is increased (see Eq. (20.2-22)).

$$l_{bd,pi} = l_{bd} / k_b > l_{bd,min}\, \alpha_{lb} \qquad (20.2\text{-}22)$$

with:

k_b - ratio of mean bond strength of post-installed reinforcing bars evaluated from the results of suitable pre-qualification tests taking into account the parameters given in 20.2.5 (4) to required mean value $f_{bm,pi,rqd}$ according to Table 20.2-6, to be taken from the relevant technical product specification.

Based on the current knowledge k_b should be not smaller than 0.7 and the bond strength $f_{bm,pi}$ in C20/25 should be not smaller than 10 MPa. This corresponds to a factor k_b = 1.0 for concrete of strength class ≤ C20/25.

In general, the use of anchoring mortars with k_b = 1.0 is recommended for all concrete strength classes.

The factor α_{lb} accounts for accounts for a different load-bearing behaviour of cast-in place and post-installed rebars in cracked concrete. It should be evaluated from pre-qualification tests taking into account the parameters given in 20.2.5 (4) and/or taken from the relevant approval or assessment document (e.g., technical product specification), which is based on these criteria. In general, α_{lb} = 1.5 may be used.

Under static or quasi-static loading a designer may consider to use post-installed reinforcing bars with a bond behavior inferior to that of cast-in reinforcing bars for

α_{lb} = cracked concrete factor ≥ 1.0.

(10) The design anchorage length $l_{bd,pi}$ of post-installed reinforcing steel bars in compression should be determined as for cast-in reinforcing steel bars according to 20.2.5 (9), neglecting the effect of end bearing.

(11) Post-installed reinforcing steel bars may be lapped with straight cast-in deformed steel bars according to section 20.2.2.6 where the design anchorage

lapping with cast-in reinforcing bars if the design anchorage length $l_{bd,pi}$ is calculated according to Eq. (20.2-21).

The consequences of deterioration on structural capacity depends on both the intensity and location of the attack. For example, longitudinal cracking due to corrosion of reinforcement may impair transfer of force at laps and anchorages and thus degrade structural capacity. Loss of bond away from such locations will generally have little influence on ultimate capacity. Similar considerations apply in the transfer and anchorage length of prestressing strand and wires.

Loss of concrete cover from erosion, frost attack or fire damage will impair residual capacity of laps, anchorages and may extend transfer length of prestressing strand and wires.

Particular attention should be paid to areas where reinforcement is lapped or anchored when undertaking visual surveys. Any evidence of longitudinal cracking/spalling/delamination along laps or anchorages will require closer inspection and more detailed evaluation of residual capacities.

The formulae given in this section to evaluate the assessment value of steel stress f_{std} which can be transferred in a lap or anchorage depends on f_{ck}. To this aim, f_{ck} may be taken from the original design or construction records if available and if there is no doubt about the validity of this information and if no significant evidence of deterioration is present. Alternatively, f_{ck} should be estimated by in situ tests and information from test results can be combined with prior information if available. The reduction in bond strength due to corrosion, where relevant, which highly

length l_{bd} is replaced by $l_{bd,pi}$, calculated according to Eq. (20.2-22) but with $l_{bd,min}$ = 15ø. The lap length should be designed for the minimum concrete cover of either cast-in or post-installed reinforcing steel bars and the bond conditions of the cast-in reinforcing steel bars. If the clear distance between spliced bars exceeds the smaller of 50 mm or 4ø section 20.2.2.6 applies.
(12) For shear forces across interfaces as shown in Fig. 20.2-25c with post-installed reinforcing steel bars see chapter 23.
(13) In applications, where concrete breakout failure as well as combined bond and concrete failure cannot be ruled out, the design of the connection may be performed according to the provisions given in section 2.2.10.

20.3 Assessment of laps and anchorages of steel reinforcement in existing construction

20.3.1 Introduction

After several years of service life, structural elements may not satisfy the current safety requirements for several reasons, including material degradation (with particular reference to steel corrosion) or increased loads, by also considering the seismic actions that were non considered by building codes at the time of the original design. The assessment of existing structures requires a proper conceptual design and new approaches for evaluating their reliability by also considering the remaining expected service life. In addition, specific rules for older materials, as plain smooth bars, should be revised for a better assessment of old structures. Last, but not least, interventions in existing structures may require new technologies now available.

depends on the confinement to the bar, concrete quality and environment, should be assessed and f_{std} in the formulae given in this section should be decreased.

The deformations associated with the forces in reinforcement at B, C & D in the half joint detail shown in Fig. 20.3-1 differ and consequently the rate of development of stress in each set of reinforcement will also differ. As bond failure may be non-ductile, resistance of reinforcement at 'B' in Fig. 20.3-1 may start to reduce before resistance at 'C' and 'D' develops fully. It is not permissible to assume that all sets of reinforcement act at their individual maxima in a collapse analysis.

Note that reinforcement itself should also have adequate deformation capacity.

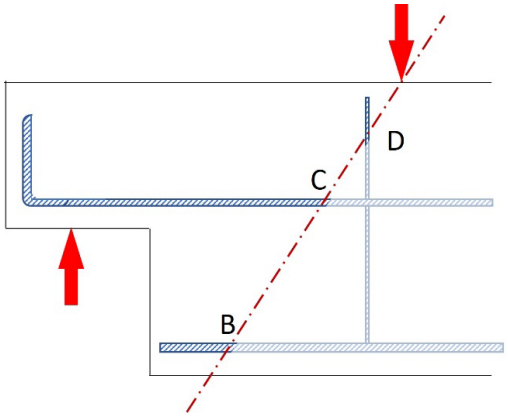


Fig 20.3-1: Illustration of deformability requirements.

k_{da} factors in Fig. 20.3-2 are derived from k_m factors in Fig. 20.5-3 (see also *fib* Bulletin 72 [20-1]), taking account of the non-linear relationship between anchorage or lap length and the stress developed in the bar.

k_{da}=12 when predominantly tensile force is generated in transverse reinforcement as a consequence of bond splitting; k_{da}=6 when predominantly dowel force is generated in transverse reinforcement as a consequence of bond splitting; k_{da}=0 when potential splitting cracks do not intersect transverse reinforcement.

Where two or more sets of reinforcement are to be taken to act in combination it should be verified that adequate deformation capacity is available. Specific consideration of deformation capacity of each component may be necessary unless it can be verified that all except the lap or anchorage undergoing the smallest deformation can develop a stress at least 20% in excess of the characteristic strength of the bar.

20.3.2 Anchorages and lapped joints of ribbed bars

The assessment value of steel stress which can be transferred in a lap or anchorage of ribbed bars complying with requirements of section 20.2.2 may be determined by Eq. (20.3.1).

Eq. (20.3-1) is valid for ribbed and indented bars having f_R and f_P, of at least 0.05. For bars not complying with this condition, the assessment value of steel stress should be reduced or these bars should be considered as plain bars.

Eq. (20.3-1) applies if:

- $l_b \geq 10\phi$ for anchorage (this could be reduced to 5ϕ at the supports or if visual inspections demonstrate that the area where the anchorage develops is uncracked in the present conditions) and 15ϕ for laps;

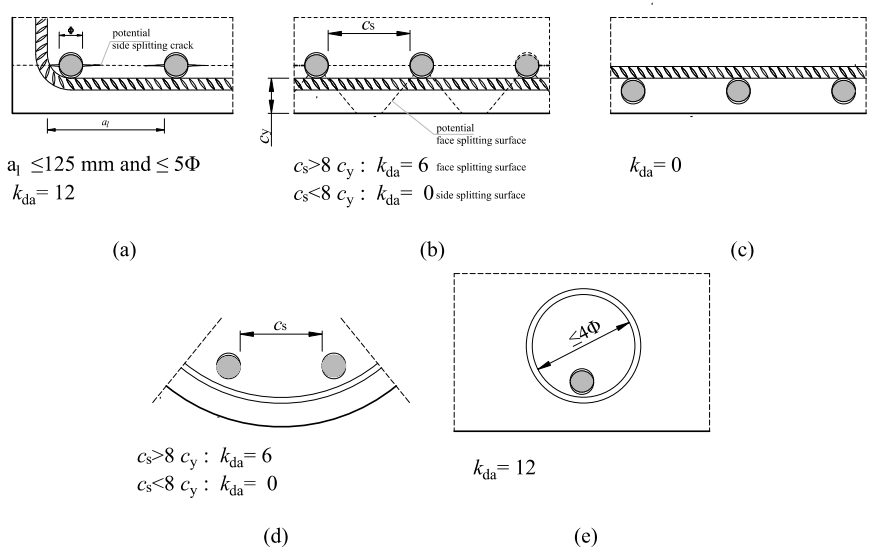


Figure 20.3-2: Transverse reinforcement factor k_{da}

Eq. 20.3-1 does not allow the beneficial contribution of transverse pressure on anchorage capacity to be considered. Eq. 20.2-5 may be applied also for assessment.

- $c_{min}/\phi \le 3.5$ and $c_{max}/c_{min} \le 5$.

$$f_{std} = \left\{ 6.5 \left(\frac{l_b}{\phi} + k_{beb} \right) \left(\frac{f_{ck}}{25} \right)^{0.45} \left(\frac{25}{\phi} \right)^{0.35} (\alpha_2 + \alpha_3) + \sigma_{eb,1} \right\} k_{tc}^{-1} \left(\frac{1.5}{\gamma_C} \right)^{0.65} \le f_{yd} \qquad (20.3\text{-}1)$$

but:

$$f_{std} \le \left\{ 43 \left(\frac{l_b}{\phi} + k_{beb} \right)^{0.55} \left(\frac{f_{ck}}{25} \right)^{0.25} \left(\frac{25}{\phi} \right)^{0.2} (\alpha_{2a} + \alpha_{3a}) + \sigma_{eb,2} \right\} k_{tc}^{-1} \left(\frac{1.5}{\gamma_C} \right)^{0.35} \le f_{yd} \qquad (20.3\text{-}2)$$

where:

k_{beb} = 0 for straight bars and 15 for bars which terminate in a standard hook or bend;

k_{tc} = 1.00 for 'good' bond conditions;

= 0.5+0.01 $l_b/\phi \le 1.0$ for 'poor' bond conditions;

α_2 is given by Eq. 20.2-6;

α_3 is given by Eq. 20.2-7;

$\sigma_{eb,1}$ and $\sigma_{eb,2}$ represent the contribution of a hook or bend termination or of end bearing in compression reinforcement;

f_{yd} is the assessment value of yield strength (Section 12).

$$\sigma_{eb,1} = 100 \left(\frac{f_{ck}}{25} \right)^{0.5} \left(\frac{25}{\phi} \right)^{0.2} (\alpha_{2a} + \alpha_{3a}) \qquad (20.3\text{-}3)$$

$$\sigma_{eb,2} = 67 \left(\frac{f_{ck}}{25} \right)^{0.5} \left(\frac{25}{\phi} \right)^{0.2} (\alpha_{2a} + \alpha_{3a}) \qquad (20.3\text{-}4)$$

where α_{2a} represents the contribution of confinement from concrete cover:

$\alpha_{2a} = (c_{min}/\text{Ø})^{0.25} (c_{max}/c_{min})^{0.1}$

α_{3a} represents the contribution of confinement from transverse reinforcement:

$\alpha_{3a} = k_{da}\ K_{tr}$;
where k_{da} is an effectiveness factor dependent on the reinforcement detail (see Fig. 20.3.2) and varying between 12 and 0.

Estimates of concrete strength f_{ck} and partial safety factor γ_C to be used in Eq. (20.3-1) and Eq. (20.3-2) should be derived using procedures given in Section xxx on assessment for materials properties.

If transverse reinforcement is not positioned within a distance equal to the lesser of 2ϕ or 50 mm from both ends of a compression lapped joint, then f_{std} calculated by Eq. (20.3-1) and Eq. (20.3-2) shall be reduced by 15%.

Except as stated in section 20.3.3, Eq. (20.3-1) and Eq. (20.3-2) is subjected to the same limits as Eq. (20.2-5) in the value of parameters that may be used.

If cover concrete is affected by deterioration due to fire, frost, ASR or any other strength deterioration mechanism, the actual concrete strength should be used for f_{ck}.

20.3.3 Types of old rebars

20.3.3.1 Introduction

Older reinforcement types include:

1) Plain surface bars:
   a. Hot rolled plain round, square, square chamfered, flat plate, 'T';
   b. Cold drawn round (small diameter only, typically ϕ<12 mm in older construction;
2) Type 1 deformed bars (Square twisted),
3) Type 2 deformed bars:
   a. Hot rolled with ribs of variety of forms;
   b. Hot rolled cold-twisted with ribs of variety of forms (Torbar);
   c. Indented;
4) Mesh;
5) Strands and wires.

1, 2 & 3 represent the main generic types of straight/bent bar for reinforced concrete.

Reinforcement may be produced by carbon steel or stainless steel and they may be zinc-coated or epoxy coated.

Older rebars commonly had no ribs or have rib patterns and surface geometries which differ from current norms and which consequently have different bond characteristics. This section provides strength for assessment of the capacity of laps and anchorages of straight plain round bars and plain square twisted bars in concrete in sound condition.

The capacity of a lap or anchorage of plain round bars, plain square section and of plain square section chamfered bars may be determined by section 20.3.3.2.

The capacity of a lap or anchorage of square twisted bars may be determined by Eq. (20.3-5).

Superficial corrosion need not be detrimental to bond of plain bars provided longitudinal cracking has not developed and there is no evidence of rust staining on the surface of the member

20.3.3.2 Anchorage and laps of plain surface bars

Where concrete cover is in sound condition and there is no evidence of longitudinal cracking from corrosion the assessment value of steel stress which can be transferred in a lap or anchorage of straight plain round or square bars (without coatings) may be determined by Eq. (20.3-5).

Eq. (20.3-5) applies if:

- $l_b \geq 10\phi$ for anchorage and 15ϕ for laps; $f_{std} \leq 300$ MPa and $c_{min}/\phi \leq 3$.

$$f_{std} = 9 \cdot k_{tc} \left(\frac{f_{ck}}{25}\right)^{0.5} \left(\frac{c_{min}}{\phi}\right)^{\eta_1} \left(\frac{l_b}{\phi} + k_{beb}\right)^{0.8} \left(\frac{\gamma_C}{1.5}\right)^{-\eta_2} \quad (20.3\text{-}5)$$

and $f_{std} \leq f_{yd}$

where:

f_{ck} in [MPa];

ϕ is the bar diameter and the bar size (equal to side dimension) for round and square bars, respectively;

k_{beb} = 0 for straight bars and 12 for bars which terminate in a standard hook or bend;

for plain and square bars in 'good' bond conditions, k_{tc} =1.0, η_1 =0.5 and η_2=1.0;
for plain round bars in 'other' bond conditions, k_{tc} = 0.4, η_1 = 0.15 and η_2 =1.7;
for plain square bars in 'other' bond conditions, k_{tc} = 0.5, η_1 = 0.15 and η_2 =1.7.

20.3.3.3 Anchorage and laps of square twisted bars

The assessment value of steel stress which can be transferred in a lap or anchorage straight square twisted (Type 1 deformed, without coatings) bars where the pitch does not exceed 15 times the bar size may be determined by Eq. (20.3-6).
Eq. (20.3-6) applies if:

- $l_b \geq 10\phi$ for anchorage and 15ϕ for laps; $f_{std} \leq 300$ MPa and $c_{min}/\phi \leq 3$.

$$f_{std} = 12.5 k_{tc} \left(\frac{f_{ck}}{25}\right)^{0.5} \left(\frac{c_{min}}{\phi}\right)^{\eta_1} \left(\frac{l_b}{\phi} + k_{beb}\right)^{0.8} \left(\frac{\gamma_C}{1.5}\right)^{-\eta_2} \text{ and } f_{std} \leq f_{yd} \quad (20.3\text{-}6)$$

where:

f_{ck} in [MPa];

k_{beb} = 0 for straight bars and 12 for bars which terminate in a standard hook or bend;

The values given in Table 20.3-1, have been obtained by using the reference model given in *fib* Bulletin 72 [20-1] for the evaluation of anchorage strength and by adapting it to the loss of concrete cover. A linear reduction in the bond strength with the reduction of concrete cover has been assumed.
The value of concrete cover, below which Table 20.3-1 applies, has been chosen equal to ϕ for the following reasons:

- to be consistent with the minimum value of concrete cover in the reference test database for the anchorage strength of plain bars;
- even if the minimum value of concrete cover in the reference test database (*fib* Bulletin No. 72 [20-1]) for anchorage strength of ribbed bars is 0.5ϕ the number of tests performed with $c = 0.5\phi$, is too low to have a statistical validity.

ϕ is the bar size (equal to side dimension);
k_{tc} = 1.0, $\eta_1 = 0.5$ and $\eta_2 = 0.8$ for 'good' bond conditions;
k_{tc} = 0.6, $\eta_1 = 0.5$ and $\eta_2 = 0.9$ for 'other' bond conditions.

20.3.4 Anchorage and lapped joints of rebars in case of non-compliant detailing

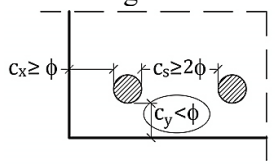
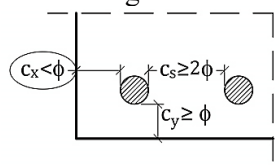
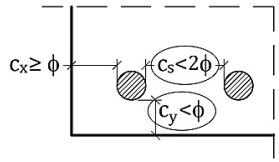
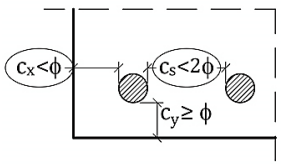
20.3.4.1 Anchorage and lap joints for $c_{min} < \phi$

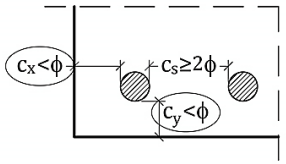
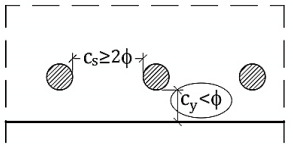
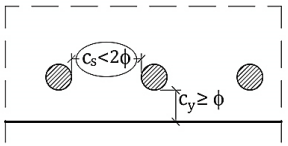
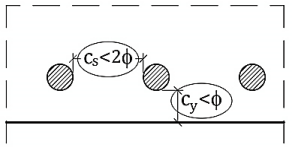
If $c_{min} < \phi$ the equations for anchorage of ribbed and indented (Eq. (20.3-1) and Eq. (20.3-2), plain bars (Eq. (20.3-5)) and square twisted (Eq. (20.3-6), in existing structures may be used with the following modifications:

- c_{min} should be replaced by ϕ ;
- f_{std} should be decreased by a factor k_c.

k_c should be taken from Table 20.3-1.
Edge and interior bars are defined in Table 20.3-1.

Table 20.3-1: Parameter k_c when $c_{min} < \phi$.

Case 1 Edge bar  $k_c = \left(0.5 + 0.5\frac{c_y}{\phi}\right)$	Case 2 Edge bar  $k_c = \left(0.5 + 0.5\frac{c_x}{\phi}\right)$
Case 3* Edge bar  $k_c = \left(0.5 + 0.5\frac{c_y}{\phi}\right)$	Case 4* Edge bar  $k_c = \left(0.5 + 0.5\frac{c_x}{\phi}\right)$

Case 5** Edge bar  $k_c = \frac{c_{min,xy}}{\phi}$ if $\frac{1}{3}\phi \le c_{min,xy} < \phi$ $k_c = \left(0.1 + 0.7\frac{c_{min,xy}}{\phi}\right)$ if $c_{min,xy} < \frac{1}{3}\phi$	Case 6 Interior bar  $k_c = \left(0.5 + 0.5\frac{c_y}{\phi}\right)$
Case 7 Interior bar  $k_c = \left(0.5 + 0.25\frac{c_s}{\phi}\right)$	Case 8* Interior bar  $k_c = \left(0.5 + 0.5\frac{c_y}{\phi}\right)$

*In cases no. 3, 4 and 8, bars having $c_s/2 < \phi$ shall be considered as bundled.
**In case no. 5, $c_{min,xy} = min(c_x; c_y)$.
Note: see Fig. 20.2-15 for the definition of c_s in laps.

20.3.5 Anchorage and lapped joints of rebars in case of degradation

20.3.5.1 Corrosion

20.3.5.1.1 General

Corrosion supposes the generation of iron oxides (rust) at the concrete/bar interface which directly influences the bond of the composite material. The oxides

The degree of the overall impact of the corrosion products depends in a first level, on the existence or not of transversal reinforcement. Further, corrosion effects are

generated by the corrosion produces radial stresses at the steel/concrete interface that influences the frictional component of bond.

Further corrosion usually leads to development of parallel to the bar cracking that generally reduces the resistance to the bursting forces generated by bond action.

In ribbed bars corrosion products may reduce the rib depth and the relative rib area.

Progressive corrosion may lead into spalling and delamination of concrete cover reducing its confining effect on bond. At high levels of deterioration, only a residual anchorage capacity may be present.

Loss of bond due to corrosion depends on cover to bar diameter ratio, concrete quality while structural safety after loss of bond also depends on the bar detailing, including the position and amount of transverse reinforcement and of the corrosion degree.

The values in Table 20.3-2 have been determined for a representative set of parameters, detailed in the table and with embedment length equal to 20 bar diameters, normal strength concrete and small, medium and large bar diameters. The use of the models is advised for other situations and in depth assessment.

The estimation of the effect of corrosion in bond strength of ribbed bars due to corrosion described in the Table 20.3-2 can be obtained with the models given below based in the corrosion attack penetration (loss in diameter due to corrosion) which can be determined as given in Section xxx and the different parameters affecting bond behaviour. A brief description of the models is given below:

Model A) Corrosion induced bond loss (Prieto et al., 2016 [20-17], Prieto et al., 2022 [20-18])

In this model, the capacity of a lap or anchorage of a corroded bar is provided. In this model the corrosion effect is considered by using the cross-section loss (% Cor), assumed as uniform along the bond length. In addition, corrosion effects are additionally considered with the variable *m* (values given in Table 20.3-3) that implicitly includes the effect of cracking depending on the range of cross section loss, bond condition and transverse reinforcement. In the case of transverse especially relevant when the anchorage length of the bars is placed outside the support zones since the external pressures in these support zones can counteract the effect of bond degradation due to corrosion.

The transverse reinforcement usually corrodes first due to the smaller concrete cover. This corrosion has to be properly taken into account as the beneficial effect of transverse reinforcement on bond of longitudinal bar will be compromised

Corrosion could affect bond strength in several ways.

20.3.5.1.2 Modelling of capacity of corroded anchorages and lapped joints

To provide general indications to designers, in addition to the two models included in the left column, a table containing set of residual anchorage capacity values has been prepared. The values are obtained from the lower bounds of the models results, for deformed bars.

The table provides the ratio of the maximum steel stress divided by the maximum steel stress in the absence of corrosion, for the anchorage of corroded reinforcement.

The following table can be used as a simplified estimation of the characteristic capacity of a lap or anchorage of a ribbed corroded bar.

reinforcement corrosion, this effect is considered through the loss of confinement through the variable K_{tr} using the remaining cross section.

The characteristic capacity of a lap or anchorage of a straight corroded bar, $f_{stk,corr,0}$, of the model can be obtained as:

$$f_{stk,corr} = 1.9 f_{ck}^{\frac{2}{3}} \cdot \frac{l_b}{\phi} \frac{\left[m\left(\frac{1}{\phi^2}+1\right)^9 \left(\left(\frac{\phi}{l_b}\right)^2+1\right)^{8.1} e^{-0.13\frac{f_{ck}}{40}} \left(\left(\frac{a}{\phi}\right)^4+1\right)^{0.06} \left(K_{tr}^2+K_{tr}+1\right)^{\frac{1}{2}} \cdot \left(\%Cor^2+1\right)^{-0.02} - 1 \right]}{\left(1-\frac{\%Cor}{100}\right)} \quad (20.3\text{-}7)$$

and $f_{stk,corr} \leq f_{yk}$ (20.3-8)

where a is the distance from the center of the anchored bar to the concrete surface ($a = c + \phi/2$);

l_b is the embedment length of the anchored bar considered [mm];

In the formulation, the nominal bar diameter (ϕ) of the anchored bar should be considered.

Table 20.3-3. *Variable m depending on bond conditions, confinement and corrosion.*

	Good bond conditions		All other bond conditions	
	Unconfined	Stirrups	Unconfined	Stirrups
No corrosion	1.27	1.32	1.24	1.28
0 < % Cor ≤ 5%	1.24	1.29	1.21	1.25
5 % < % Cor ≤ 20%	1.22	1.27	1.19	1.23

The design capacity of a lap or anchorage of a straight corroded bar of the model can be obtained as:

$$f_{std,corr} = \frac{f_{stk,corr,0}}{\gamma_M} \quad (20.3\text{-}11)$$

being γ_M equal to 2.15 for CC3 (derived for $\alpha_R = 0.8$ and an annual β of 3.7 for large cost of safety measures). For CC2 γ_M is equal to 1.8 (derived for $\alpha_R = 0.8$ and an annual β of 3.3 for large cost of safety measures). For CC1 γ_M is equal to 1.65 (derived for $\alpha_R = 0.8$ and an annual β of 3.1 for large cost of safety measures). The recommended value for existing structures is 3.7 (see section 11.3.2).

Table 20.3-2: *Ratio of the maximum steel stress divided by the maximum steel stress in the absence of corrosion (% of $f_{stk,sound}$)*

		No stirrups (% $f_{stk,sound}$)		Stirrups (% $f_{stk,sound}$) K_{tr} = 0.02		Stirrups (% $f_{stk,sound}$) K_{tr} = 0.04	
		c/Ø = 1	c/Ø = 2	c/Ø = 1	c/Ø = 2	c/Ø = 1	c/Ø = 2
Corrosion [% Cor]	0 %	100					
	2 %	80	75	80	75	85	80
	5 %	70	70	75		75	
	10 %	40	65	55	65	65	70

The maximum anchorage steel stress $f_{stk,corr}$ is obtained by dividing the maximum force developed along the anchorage R by the corroded reinforcement cross-section $A_{s,cor}$:

$$f_{stk,corr} = R / A_{s,cor} \quad (20.3\text{-}9)$$

with

$A_{s,cor} = A_s (1\text{-}\%Cor/100)$

% Cor is the percentual gravimetric loss of the corroded reinforcement along the embedment length (%Cor is equivalent to the cross-sectional loss) and can be obtained as follows:

$$\%Cor = \frac{W_s - W_c}{W_s} \cdot 100 \quad (20.3\text{-}10)$$

where

W_s is the weight of sound bar;

W_c is the weight of corroded bar;

The designer shall apply the appropriate partial safety factor for anchorages of corroded bars.

Model B) ARC (Anchorage in Reinforced Concrete, Blomfors et al, 2018 [20-19]) is a model for corroded reinforcement anchorage and lap capacity R_{ARC}, based on the MC2020 bond-slip model (see section 20.5.1), modified for the corrosion effects:

1. A reduced splitting strength is used with the cover cracked by corrosion:

$$\tau_{bu,split,red} = \eta_2 \cdot 6.5 \cdot \left(\frac{f_{ck}}{25}\right)^{0.25} \cdot \left(\frac{25}{\phi_m}\right)^{0.2} (1 + k_m \cdot K_{tr}) \qquad (20.3\text{-}12)$$

2. The local bond stress-slip curve of corroded reinforcement is approximated by shifting the uncorroded curve in the slip direction (Fig. 20.3-3):

$$s_{eff} = s + s_{eq} \qquad (20.3\text{-}13)$$

where s_{eff} is the effective slip, s is the mechanical slip and s_{eq} is the equivalent slip to account for the effect of corrosion, which is estimated as:

$s_{eq,nostir} = 2.9\ (\%\mathrm{Cor}/100)$ (mm) without stirrups;
$s_{eq,stir} = 13.6\ (\%\mathrm{Cor}/100)$ (mm) with stirrups.

3. The residual bond capacity for large slip values is given by:

$$\tau_{res,mod} = \begin{cases} (0.16 + 12K_{tr}) \cdot \tau_{bu,split,red} & for\ 0 \le K_{tr} \le 0.02 \\ 0.4 \cdot \tau_{bu,split,red} & for\ 0.02 < K_{tr} \end{cases} \qquad (20.3\text{-}14)$$

where all parameters are defined in defined in section 20.5.1.1.

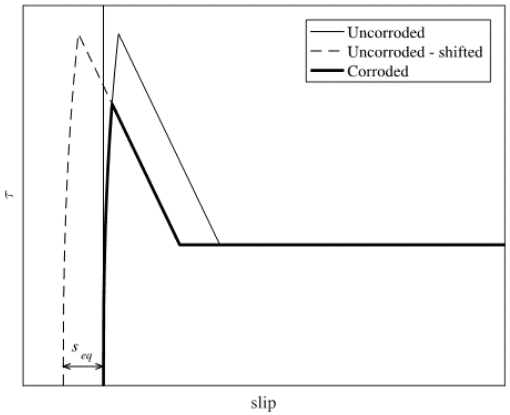


Figure 20.3-3: *Equivalent slip, s_{eq}, to account for the effect of corrosion in the bond stress-slip curve.*

The equations describing the bond transfer between reinforcement and concrete are numerically integrated along the bar, calculating $R_{ARC,k.}$ The design steel stress force $f_{std,corr}$ for the anchored reinforcement is obtained dividing $R_{ARC,k}$ by the corroded reinforcement cross section $A_{s,corr}$ and a partial safety factor without stirrups γ_M =3.4 and with stirrups γ_M =4.7, 4.9 and 5.2 for %Cor=5%, 10% and up to 20% for an annual β = 3.7 (Blomfors et al., 2019 [20-20]).

20.3.5.2 ASR

When laps and anchorages are restrained by links, expansion within the limits 0.45‰ (restrained) and 4.50‰ (unrestrained) induced by ASR has been shown to have no significant effect on bond strength.

Where transverse reinforcement is not present then maximum bars stress f_{std} may be reduced up to 50%.

Care is necessary in the assessment of residual resistance of anchorage zones of prestressed elements, particularly of pre-tensioned members in which the tendons are not contained by links.

20.3.5.3 Frost

Two types of damage to concrete may occur as a result of freezing and thawing:

- surface scaling;
- internal damage.

Surface scaling leads to a reduction in concrete cover to reinforcement. The effect on bond strength may be accounted for by using of a reduced cover in Eq. (20.2-6).

Residual bond capacity of ribbed bars not confined by links in freeze-thaw damaged concrete may be assessed using splitting tensile strength measurements on cores taken from the affected structure. Concrete compressive strength f_{ck} used in Eq. (20.2-5) may be substituted by residual concrete compressive strength after freeze-thaw attack.

-Notes:

i) The values of λ_{fi} recommended in Table 20.2-2 are derived assuming that the ratio of mean to characteristic strength is same under ambient conditions and upon heating (i.e. that the scatter of bond strength is same for ambient and hot state).
ii) The values of λ_{fi} take into account the reduced safety factors for design in case of accidental situation of fire ($\gamma_{m,fi}$) and reduction factor for the effect of actions under accidental loading η_{fi}. The values in Table 20.2-2 are derived for $\gamma_{m,fi} / \gamma_m = 1.0 / 1.5$.
iii) The values of λ_{fi} are derived based on results on beams tested in residual state (after cooling). These values are considered to be conservative for the hot state (during fire).
iv) The tests used to derive the values of λ_{fi} were performed without mechanical load during heating. It is, however, shown on pullout specimens that the mechanical load up to 60% of the mean bond capacity does not adversely affect the bond performance, see section 20.5.4.

In case of detailed finite element analysis, the effect of temperature on material properties and thermal gradients on damage to concrete is automatically considered through the analysis. However, a degradation of the material properties of concrete and steel as well as the degradation of bond strength due to elevated temperature needs to be considered in the material model. The degradation of bond stress-slip law due to elevated temperature can be taken into account using equations given in section 20.5.4.

Bond strength of ribbed bars is not degraded as severely where bars are confined by secondary reinforcement detailed in compliance with the requirements of this Code. Residual bond strength is expected to be at least 70% of that 'as constructed'.

20.3.5.4 Fire

If fire safety is relevant for an existing building, in case of existing anchorages and lap splices of deformed bars, the axial stress in the reinforcement shall be limited to values lower than the allowable axial stress under ambient conditions according to Eq. (20-3-15) below:

$$f_{fire} = \frac{f_{sta}}{\lambda_{fi}} \qquad (20.3\text{-}15)$$

where f_{sta} is the allowable design stress in the deformed reinforcing steel bars at ambient conditions according to Eq. (20.3-1) or Eq. (20.3-2).
The values of the coefficient λ_{fi} may be read from Table 20.2-2. In case other values for $\gamma_{m,fi}$ and γ_m are used, the coefficient λ_{fi} can be given by Eq. 20.2-16 where the values of reduction factor β_{fi} can be read from Table 20.2-3 and they are valid for standard fire.

Alternatively, detailed finite element analysis following the well-established and validated procedures can be used to verify the safety of existing lap splices in case of fire.
In case of finite element analysis, the thermal gradients and temperature at various locations in a section are calculated first by thermal analysis. The analysis should be able to consider the influence of thermal gradients on damage to concrete. The material model used for concrete and reinforcing steel should be such that it considers the influence of elevated temperature on their mechanical properties as laid out in the relevant sections of this code. The influence of elevated temperature on bond strength between ribbed reinforcing bars and concrete should be considered according to section 20.5.4.

20.4 Bond of reinforcement in case of repair and strengthening

When repairing existing structures special cementitious materials with fibre reinforcement may be used for existing and new rebars.

20.4.1 General guidance

20.4.2 Structural consequences of patch repairs

Patch repairs may be required to concrete structures when reinforcement corrodes. Typically, concrete around bars is broken out to at least 25 mm past the bars, leaving all or part of the reinforcement disbonded from concrete over the breakout length. Where bond is eliminated stress in reinforcement can no longer vary in response to variations in applied bending moment along a member. Without the lateral support from surrounding concrete cover the slenderness of bars in compression limits their ability to carry stress.

Temporary propping may be a significant addition to the cost of structural repairs and may often be avoided even when a member carries a substantial proportion of its design load. When repairs are required to a substantial proportion of the length of a member, the economics of repair will be improved if repair is carried out using longer breakout segments. Bars in tension may still contribute to member capacity despite being disbonded over a substantial portion of their length provided ends remain fully anchored. The pattern of stresses in the concrete will change, howeverFor members in flexure the maximum compressive strain in concrete will be increased and the possibility of an over-reinforced section failure is increased. ULS capacity of heavily reinforced sections (x/d~0.5) is more vulnerable to loss of local bond than that of lightly reinforced sections.

Loss of tension stiffening during breakout will result in increased overall deflection both during and following completion of repairs. Member stiffness tends to be enhanced on subsequent reloading after placement and hardening of repair concrete.

Where repair is carried out with a member under load (including self-weight) the width of some flexural cracks will undergo substantial increases while width of others reduces. Normal guidance on crack widths no longer applies in these circumstances. A few cracks may widen by an appreciable amount and require filling with an appropriate low viscosity resin.

Replacement reinforcement installed when a member is repaired under load will be at a lower strain than original reinforcement. This is of no consequence for member

Unless a member is fully propped during patch repairs due account shall be taken of changes in the distribution of stress in reinforcement that results when concrete is broken out and bond between bar and concrete is destroyed.

While concrete is broken out within an anchorage length from the end of a bar that part of the bar shall be considered to make no contribution to member capacity. The anchorage length to be considered shall be that required to develop the full design strength of the bar.

Reinforcement in compression shall be assumed inactive over the length of breakout plus an anchorage length to either side.

When concrete is broken out more than an anchorage length from the end of a bar acting in tension the bar may be considered to provide some contribution to member capacity, but account should be taken of changes in the distribution of stress in reinforcement and in the pattern of strains in the concrete that result from the loss of bond.

If a member is repaired while under load, it should be verified that the increase in deflections is acceptable. It will also be necessary to verify if crack widths have increased beyond acceptable limits.

Breakout of concrete near ends of prestressed pre-tensioned members causes the transfer length to extend into the member and it should be verified that stresses remain within acceptable limits.

Where replacement reinforcement is to be provided to compensate for loss of bar section as a consequence of loss of original bar section, the new reinforcement shall extend an anchorage length beyond the length over which it is required to compensate loss of original bar section.

behaviour provided a) variable loading is not greater than 50% of its design value and b) the section is not over-reinforced, i.e. provided $x/d < 0.5$.

It is generally understood that longitudinal reinforcement reduces the AAR expansion, whereas adequate transversal rebar may exert a beneficial confinement action on steel-concrete bond.
Still, the experimental evidence at present available does not allow to draw general conclusions and to propose specific design rules.

The reduction does not exceed 50%, and strength is restored within a few days after the treatment has been finished.

The bond perimeter of a post-tensioning strand bundle can be used if the layout of strands is known. Otherwise, the lower bound value of the perimeter of bundle can be approximated with equivalent circle with diameter $Ø_{eq}$ [Ref].

The factors $\eta_{p.1}$ and $\eta_{p.2}$ are calibrated according to minimum required strength of the grout. Minimum required strength is assumed to be 30 MPa obtained from 40 mm x 40 mm x 160 mm prism compression test in which the halves of prism are compressed after bending tests.

20.4.3 Electrochemical extraction of chlorides (ECE)

The correct application of electrochemical techniques for chloride extraction from or re-alkalization of concrete does not cause any significant long-term reduction in bond strength, except where:

- alkali aggregate reactions (AAR) are induced in concrete containing susceptible aggregates by the electrochemical process;
- an appreciable amount of corrosion has occurred prior to treatment. Plain round bars are likely to be particularly susceptible to loss of bond in these circumstances – see section 20.3.5.1.

There may be a temporary reduction in bond strength while treatment is underway. Considerations must be given to the need for temporary support or load restrictions during treatment.

20.4.4 Re-anchoring of strand bundles in case of tendon failure

The basic re-anchoring length of tendon consisting of a bundle of strands should be determined according to Eq. (20.2-15) by using the factor A_{sp} / ($Ø\pi$) for strand bundles:

$$\frac{A_{sp}}{Ø_{eq}*\pi} = 0.13 * Ø_{eq} \qquad (20.4\text{-}1)$$

in which $Ø_{eq} = \sqrt{\frac{4*A_{sp.tot}}{\pi}}$;
In case of tendons in cement grouted ducts in Eq. 20.2-15 $f_{ck}(t)$ is the grout compressive strength at time of tendon failure.
The factors in Eq. (20.2-15) are:
$\eta_{p.1} = 0.7$ for bundles of 7-wire strands
$\eta_{p.2} = 0.7$ for post grouted strand bundles in ducts

The factors in Eq. 20.2-16 is
$\alpha_{p3} = 0.7$ for strand bundles grouted in ducts

20.5 Local bond behaviour of steel reinforcement

20.5.1 Local bond-slip relationship for ribbed bars for predominantly static loading in case of no degradation

20.5.1.1 Local bond stress–slip model

Under well-defined conditions, it is possible to formulate an average "local bond" versus "local slip" relationship, for short anchorage lengths, for instance up 5 times bar diameter.
This section covers local bond behaviour of ribbed reinforcing bars which satisfy the requirements for classification as "high bond" in accordance with requirements of chapter 15.

For monotonic loading the reference value τ_b of the bond stresses between concrete and reinforcing bar for pull-out and splitting failure can be calculated as a function of the relative displacement parallel to the bar axis "s" as follows (Fig. 20.5-1):

$$\tau_b = \tau_{bmax} (s/s_1)^{\alpha} \quad \text{for} \quad 0 \le s \le s_1 \qquad (20.5\text{-}1)$$

$$\tau_b = \tau_{bmax} \quad \text{for} \quad s_1 \le s \le s_2 \qquad (20.5\text{-}2)$$

$$\tau_b = \tau_{bmax} - (\tau_{bmax} - \tau_{bf}) (s\text{-}s_2)/(s_3\text{-}s_2) \quad \text{for} \quad s_2 \le s \le s_3 \qquad (20.5\text{-}3)$$

$$\tau_b = \tau_{bf} \quad \text{for} \quad s_3 < s \qquad (20.5\text{-}4)$$

where the parameters are given in Table 20.5-1.

The bond stress–slip relationship depends on a considerable number of influencing factors including rib geometry (relative rib area), concrete strength, position and orientation of the bar during casting, state of stress, boundary conditions and concrete cover. The parameters given in Table 20.5-1 are valid for ribbed reinforcing steel with a relative rib area $f_r \ge 0.05\, f_{r,min}$ according to relevant international standards.
The bond stress–slip curves for confined and unconfined concrete presented in Fig. 20.5-1 can be considered applicable as an average formulation for a broad range of cases. Further reliability handling would be needed to derive design bond stress–slip curves.
The ascending part refers to the stage in which the ribs penetrate into the mortar matrix, characterized by local crushing and micro-cracking. A sustained plateau occurs only for confined concrete, during which advanced crushing and shearing off of the concrete between the ribs takes place. This level represents a residual bond capacity which is maintained only where a large concrete cover, dense transverse reinforcement or transverse compression is present to keep a certain degree of integrity intact. The descending branch refers to the reduction of bond resistance as concrete corbels between the ribs are sheared off. In the case of unconfined concrete, splitting failure occurs which is reflected by a sudden drop in bond stress before a constant residual level is reached. The peak value of bond strength in a splitting failure mode is denoted $\tau_{bu,split}$ – see Fig. 20.5-1 and Eq. (20.5-5).
With regard to the development of bond stresses, the following considerations apply:
reinforcement and concrete have the same strain ($\varepsilon_s = \varepsilon_c$) in those areas of the structure where the steel is in compression (outside eventual load introduction

areas) and in those areas where the steel is in tension in uncracked parts of the structure.

In cracked cross-sections, tension forces are transferred across the crack by the reinforcing steel. In general, the absolute displacements of the steel u_s and of the concrete u_c adjacent to a crack are different. Differences in displacements are similarly found along the transmission length l_{bpt} of pretensioned prestressed concrete members.

Due to the relative displacement $s = u_s - u_c$ bond stresses are generated between concrete and reinforcing steel or prestressing tendons. For $s < s_1$, the magnitude of these bond stresses depends predominantly on the slip s, but is influenced by the surface of the steel, concrete strength f_{cm}, the position of the reinforcing steel during concreting and by confinement from concrete cover, secondary reinforcement and the state of stress in the concrete around the bar. Between cracks or along the transmission length l_{bpt} a part of the tension force of the reinforcing steel at the cracked section is transferred into the concrete by bond (tension stiffening effect). The local change in relative displacement is characterized by the strain difference $ds/dx = \varepsilon_s - \varepsilon_c$.Depending on the selection of the coefficient α ($0 \le \alpha \le 1$) in Eq. (20.5-1) all usual forms of a bond stress–slip relationship can be modelled, starting from a bond characteristic with a constant stress ($\alpha = 0$) up to a bond stress–slip relationship with linear increasing bond stress ($\alpha = 1$).

The parameters from Table 20.5-1 to be applied are dependent primarily on the failure mode (pull-out or splitting). Further distinctions are made to account for differences in bond conditions within each failure mode.

The rationale underpinning the following rules is given in *fib* Bulletin 72 [20-1].

The values in Table 20.5-1, columns 3 to 6 (splitting failure) are derived from Eq. (20.5-5), which is derived from the semi-empirical formulation of bond presented in the *fib* Bulletin 72 [20-1] by setting $l_b/Ø = 5$, assuming a uniform bond stress over this length and evaluating $\tau_{bu,split}$ for $Ø = 25$ mm, $c_{max}/c_{min} = 2.0$, $c_{min} = Ø$ and $K_{tr} = 0.02$ in the case where bars are confined by stirrups or $K_{tr} = 0$ where no stirrups are provided.

$$\tau_{bu,split} = \eta_2 6.5 \left(\frac{f_{cm}}{25}\right)^{0.25} \left(\frac{25}{Ø}\right)^{0.2} \left[\left(\frac{c_{min}}{Ø}\right)^{0.25} \left(\frac{c_{max}}{c_{min}}\right)^{0.1} + k_m K_{tr}\right] \qquad (20.5\text{-}5)$$

where:

η_2 = 1.0 for good bond conditions;

= 0.7 for all other bond conditions (see section 20.2.2.1 for definition of bond conditions);

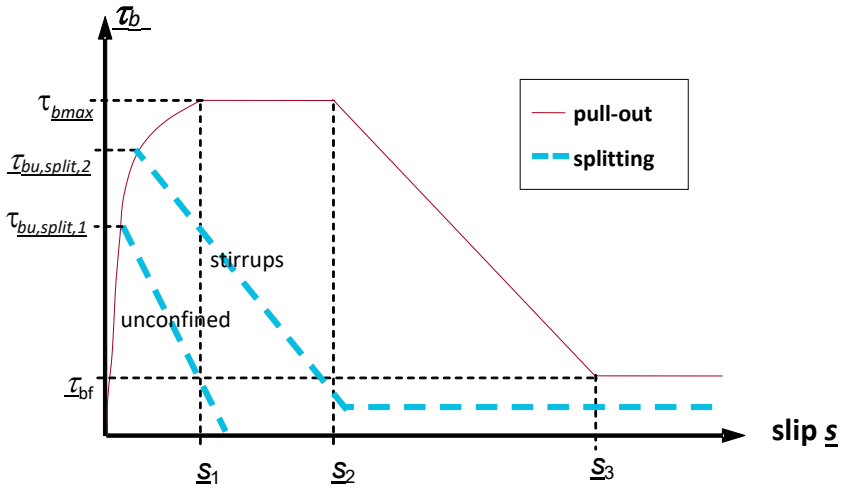


Figure 20.5-1: Analytical bond stress–slip relationship (monotonic loading). $\tau_{bu,split,1}$ and $\tau_{bu,split,2}$ denote peak local bond resistance in the absence ($K_{tr} = 0$) and presence, respectively, of confining stirrups. TO BE RE-DRAFT

f_{cm} is the mean cylinder concrete compressive strength [N/mm^2];

Ø is the diameter of the anchored bar considered [mm];

c_{min} = min{c_s/2, c_x, c_y};

c_{max} = max{c_s/2, c_x}.

k_m represents the efficiency of confinement from transverse reinforcement, and has a value of 12 where bars are confined inside a bend of links passing round the bar of at least 90° within a distance of 5ϕ (see Fig. 20.5-3). Where no confining reinforcement is provided between bars and the nearest face, k_m = 0, as shown in Fig. 20.5-3. Other values of k_m are described in Fig. 20.5-3. K_{tr} is defined in section 20.2.2.3.

Table 20.5-1: Parameters defining the mean bond stress–slip relationship of ribbed bars (according to Eqs. (20.5-1)–(20.5-4)).

	1	2	3	4	5	6
	Pull-out (PO)		Splitting (SP)			
	$\varepsilon_s < \varepsilon_{s,y}$		$\varepsilon_s < \varepsilon_{s,y}$			
	Good bond cond.	All other bond cond.	Good bond cond.		All other bond cond.	
			Unconfined	Stirrups	Unconfined	Stirrups
τ_{bmax}	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$
$\tau_{bu,split}$	—	—	$7.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$8.0\cdot\left(\frac{f_{cm}}{25}\right)^{0}$	$5.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$5.5\cdot\left(\frac{f_{cm}}{25}\right)^{0.2}$
s_1	1.0 mm	1.8 mm	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$
s_2	2.0 mm	3.6 mm	s_1	s_1	s_1	s_1
s_3	c_{clear} [1)]	c_{clear} [1)]	$1.2s_1$	$0.5c_{clear}$ [1)]	$1.2s_1$	$0.5c_{clear}$ [1)]
α	0.4	0.4	0.4	0.4	0.4	0.4
τ_{bf}	$0.40\,\tau_{max}$	$0.40\,\tau_{max}$	0	$0.4\,\tau_{bu,split}$	0	$0.4\,\tau_{bu,split}$

[1)] c_{clear} is the clear distance between ribs

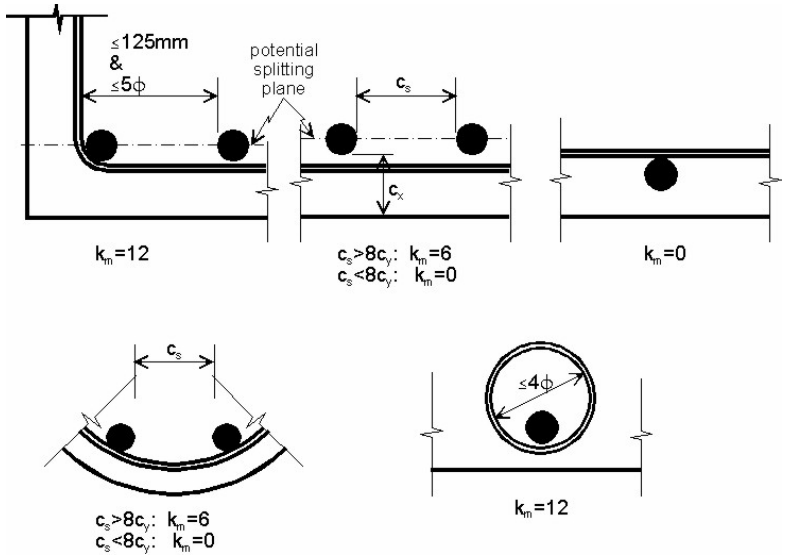


Figure 20.5-3: Confinement coefficients for transverse reinforcement

The values in Table 20.5-1, columns 1 and 2 (pull-out failure) are valid for well-confined concrete (concrete cover ≥ 5Ø, clear spacing between bars ≥ 10 Ø), or suitable confining reinforcement.

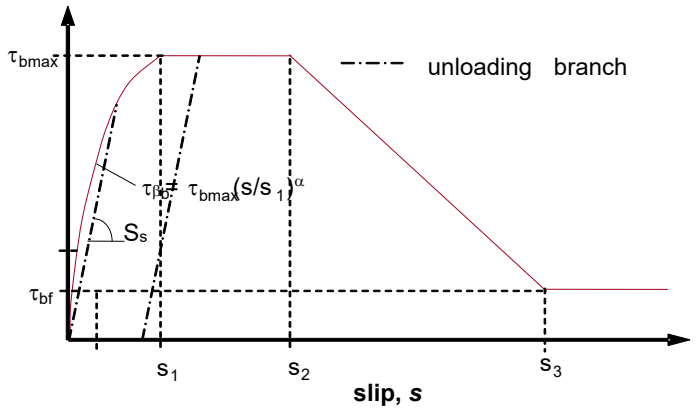


Figure 20.5-2: Unloading branch of the τ_b–s relationship.

The slope S_s of the unloading branch is taken as the secant modulus at a slip of 0.01 mm.
Unloading from a maximum slip of less than 0.01 mm is taken back to the origin, that is to zero slip.

The unloading branch of the bond stress–slip relationship is linear and valid for the increasing and horizontal part of the diagram. The slope S_s (see Fig. 20.5-2) is independent of the slip value s, and is given by:

$$S_s = 6.0\ \tau_{bmax}\ [\text{N/mm}^3] \qquad (20.5\text{-}6)$$

20.5.1.2 Influence of transverse cracking

For those parts of the reinforcing bar which are at a distance x ≤ 2 Ø from a transverse crack, the bond stress τ_B has to be reduced by the factor λ where:

$$\lambda = 0.5\, x/Ø \leq 1 \qquad (20.5\text{-}7)$$

20.5.1.3 Influence of yielding, transverse stress, longitudinal cracking and cyclic loading

The bond stress τ_b according to Eqs. (20.5-1)–(20.5-4) should be modified by the factors Ω_y, $\Omega_{p,tr}$ and Ω_{cr} and Ω_{cyc} in case of bar yielding, transverse pressure and cracking parallel to the bar axis and cyclic loading respectively:

$$\tau_{b,md} = \tau_{\mathrm{bm}}\, \Omega_{\mathrm{y}}\, \Omega_{\mathrm{p,tr}}\, \Omega_{\mathrm{cr}}\, \Omega_{\mathrm{cyc}} \qquad (20.5\text{-}8)$$

where:

$\tau_{\mathrm{b,md}}$ is bond stress according to the modified bond stress–slip curve;

τ_{b} is bond stress according to the bond stress–slip curve defined by Eqs. (20.5-1)–(20.5-4).

Ω_y is a factor which takes into account the influence of yielding of reinforcement

$\Omega_{p,tr}$ is a factor which takes into account the influence of transverse pressure

Ω_{cr} is a factor which takes into account the influence of longitudinal cracking

Ω_{cyc} is a factor which takes into account the influence of reversed cyclic loading

The modified bond stress $\tau_{\mathrm{b,md}}$ then replaces τ_{b} determined according to Eqs. (20.5-1)–(20.5-4).

20.5.1.3.1 Influence of yielding of reinforcement

If yielding of reinforcement occurs along the embedment length, the corresponding reduction of the bond stress is given by the factor Ω_y as:

$$\Omega_y = 1.0 \qquad \text{for } \varepsilon_s \leq \varepsilon_{sy} \qquad (20.5\text{-}9a)$$

$$\Omega_y = 1.0 - 0.85 \cdot (1 - e^{-5a^b}) \qquad \text{for } \varepsilon_{sy} < \varepsilon_s \leq \varepsilon_{su} \qquad (20.5\text{-}9b)$$

The bond stress–slip curve is considerably influenced by reinforcement yielding, transverse pressure and cracking along the bar, and by cyclic, repeated or sustained loading.

Bond resistance is decreased over the portion of an embedment length once reinforcement yield is reached, Fig. 20.5-3.

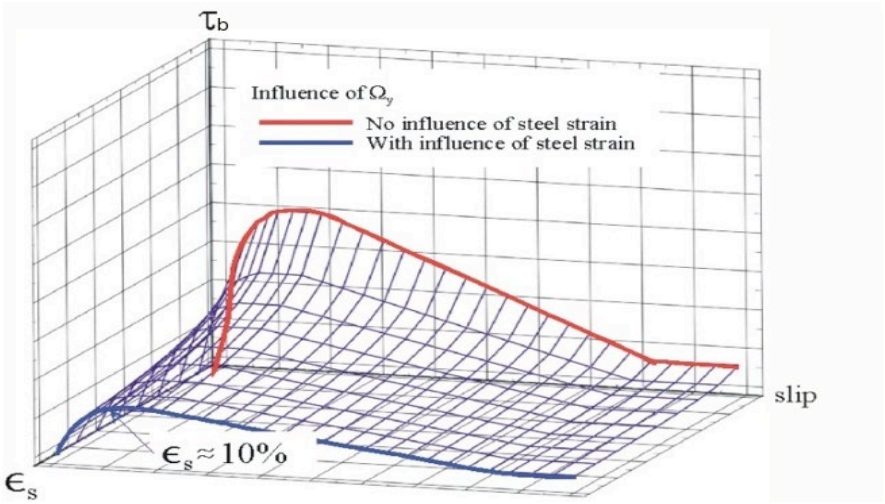


Figure 20.5-4-*: Influence of steel strains on local bond stress-slip relationship on case of pull-out failure. Bond resistance is decreased over the portion of an embedded length once reinforcement yield is reached.*

Bond resistance is decreased in the presence of transverse tension and increased in the presence of compressive stresses perpendicular to the bar axis – Fig. 20.5-4 and Eq. (20.5-11a) and Eq. (20.5-11b).

A negative quantity for p_{tr} in Eq. (20.5-11a) or (20.5-11b) denotes a compressive stress.

with

$$a = \frac{\varepsilon_s - \varepsilon_{sy}}{\varepsilon_{su} - \varepsilon_{sy}} \; ; b = \left[2 - \frac{f_{tm}}{f_{ym}}\right]^2 \qquad (20.5\text{-}10)$$

f_{tm} and f_{ym} are mean values of tensile and yield strength respectively.

20.5.1.3.2 Influence of transverse pressure

If transverse pressure perpendicular to the bar axis is present, the bond stress slip curve for pull-out failure should be modified by the factor $\Omega_{p,tr}$ as:

$$\Omega_{p,tr} = 1.0 - \frac{0.3 p_{tr}}{f_{ctm}} \qquad \text{for } f_{ctm} \geq p_{tr} \geq 0 \text{ MPa (tension)} \qquad (20.5\text{-}11a)$$

$$\Omega_{p,tr} = 1.0 - tanh\left[0.2 \frac{p_{tr}}{0.1 f_{cm}}\right] \qquad \text{for } p_{tr} \leq 0 \text{ MPa (compr.)} \qquad (20.5\text{-}11b)$$

where:

p_{tr} is the mean stress in the concrete (orthogonal to the bar axis) averaged over a volume around the bar with a diameter of 3 Ø.

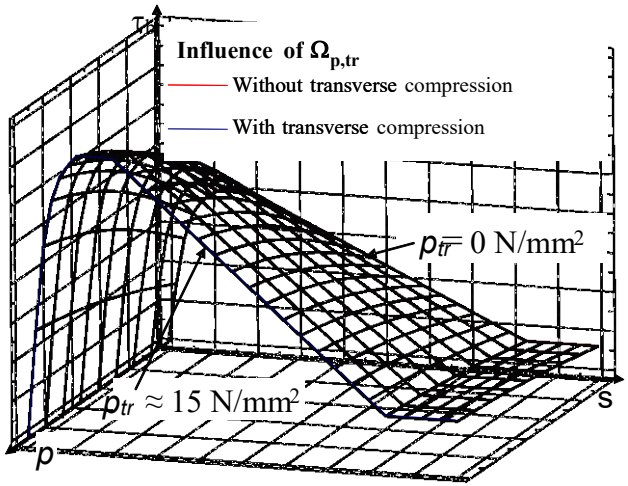


Figure 20.5-5: Influence of transverse pressure on local bond-stress slip relationship in case of pull-out failure.

A transverse tensile stress equal to the tensile strength of the concrete could split the concrete cover and leave zero bond resistance where there is no reinforcement crossing the splitting crack. Eq. (20.5-12) represents behaviour where reinforcement crosses potential splitting cracks and is not stressed above its yield strength.

Eq. (20.5-12) may be considered valid for cracks up to 0.5 mm wide.

Fig. 20.5-6 explains also the calculation of the dissipated energy during monotonic and cyclic loading (area under the bond stress-slip curve). In Fig. 20.5-7 the factor Ω_{cyc} is plotted as a function of the ratio Λ_{cyc}/Λ_0.

20.5.1.3.3 Influence of longitudinal cracking

If cracks parallel to the bar axis and transverse reinforcement are present, the bond stress slip curve for pull-out failure should be modified by the factor Ω_{cr}:

$\Omega_{cr} = 1.0$ where concrete is uncracked parallel to the bar axis

$\Omega_{cr} = 1 - 1.2\, w_{cr}$ where the concrete is cracked parallel to the bar axis and w_{cr} is the crack width [mm]. (20.5-12)

20.5.1.3.4 Influence of reversed cyclic loading

Reversed cyclic loading reduces the bond strength compared to monotonic loading as shown in Fig. 20.5-6.

After the bond stress has reached τ_{bmax} the ordinates (i.e. the stress values) of the bond stress slip curve for pull-out failure should be modified by the factor Ω_{cyc} on subsequent load cycles as:

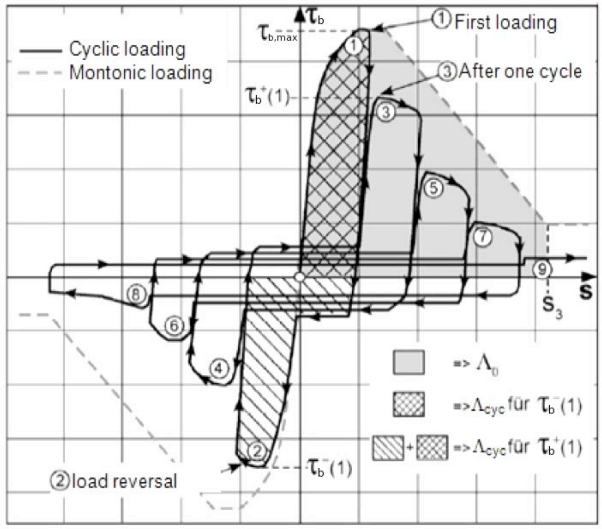


Figure 20.5-6: *Bond-stress slip relationship and definition of the dissipated energy under monotonic and cyclic loading.*

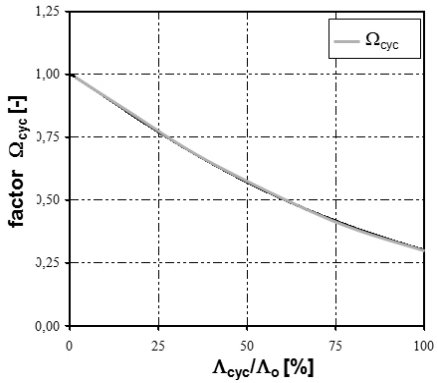


Figure 20.5-7: *Factor Ω_{cyc} as a function of the ratio Λ_{cyc}/Λ_0.*

$$\Omega_{cyc} = e^{\left(-1.2\cdot\left[\frac{\Lambda_{cyc}}{\Lambda_0}\right]^{1.1}\right)} \quad (20.5\text{-}13)$$

where:

Λ_{cyc} dissipated energy during cyclic loading (Fig. 20.5-7);

Λ_0 dissipated energy during monotonic loading (Fig. 20.5-7).

In the absence of test data, the β coefficient for epoxy coated ribbed bars should be taken as 50% of that for a "normal" ribbed bar.

The influence of bond on crack control and tension stiffening may be estimated from the secant modulus of local bond–slip behaviour measured in accordance with RILEM pull-out test procedures at a free end slip of 0.01 mm.

The terms "non-reversed repeated loading" or "fatigue loading" are used here to imply loading in which the bar stress does not change its sign and the bond stress does not exceed τ_{bmax} under short-term monotonic loading. This type of repeated loading produces a progressive increase of slip between steel reinforcement and concrete which may lead to failure at cyclic bond stress levels lower than the ultimate bond strength corresponding to monotonic loading. The rate of increase is influenced principally by the number of load cycles and the load level.

Provided that bond failure does not occur under repeated loading, then the preceding repeated loads do not adversely influence either bond-slip behaviour on subsequent monotonic loading to failure or bond strength at ultimate load.

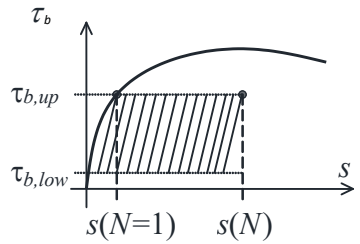


Figure 20.5-8: *Effect of repeated loading on the bond-slip curve*

20.5.1.4 Influence on serviceability

Rules for serviceability behaviour have been derived for ribbed bars with a relative rib area f_r within the range 0.05–0.07. More highly ribbed bars may be capable of developing higher bond stiffness, while coatings may cause a reduction. Coefficients β_{TS} representing the influence of bond in Eqs. (30.5-2), (30.5-11) and (30.5-12) and β in Eq. (30.5-25) in Section 30.5.2 may be modified in proportion to measured bond stiffness.

20.5.1.5 Influence of fatigue loading

Non-reversed repeated loading increases the relative slip between concrete and ribbed reinforcement with load cycles, Fig. 20.5-8. The increase of slip $s(N)$ after N load cycles can be calculated according to the following equation:

$$s(N) = s(N=1)\,N^b \qquad (20.5\text{-}14)$$

where $N = 1$ means quasi-static monotonic loading and factor b takes account of the load level as follows:

$$b = \begin{cases} 0.107 & , \tau_{b,up} / \tau_{bu} \le 0.51 \\ 0.45\left(\tau_{b,up} / \tau_{bu}\right)^{2.13} & , \tau_{b,up} / \tau_{bu} > 0.51 \end{cases} \qquad (20.5\text{-}15)$$

In the above equation, $\tau_{b,up}$ is the bond stress under maximum load level and τ_{bu} = min (τ_{bmax}, $\tau_{bu,split}$) is the monotonic bond strength under quasi-static loading

Eq. (20.5-16) has been derived for short pull-out test conditions, where the bond stress distribution along the embedded length can be considered uniform.
Eq. (20.5-16) is obtained by considering that fatigue failure initiates when the relative slip $s(N)$ reaches the peak s_1 referred to in Table 20.5-1.

For further information on bond under cyclic loading, see chapter 3, "Bond under repeated loading", of *fib* Bulletin 10 [20-21].

During the last few years, new types of concrete came into the market, including Fibre Reinforced Concrete (FRC), concrete with conventional aggregates, polymer concrete, etc.

Therefore, the rules proposed in section 20.2 and 20.3 cannot be applied to new types of concrete or different types of rebars.

In order to verify the applicability of the rules or the need of correction factors, a standard Bond Test is proposed in section 20.6.

Based on available literature the bond performance between steel reinforcement and LW (Low weight) concrete or RSFRC (Recycled Steel Fiber Reinforced Concrete) are linked to several parameters that need to be investigate with specific experimental tests (Section 20.6).

Only the pullout mode needs to be considered for plain surface bars. Many of the general points of principle for local bond modelling detailed in 20.5.1.1 are also applicable to plain surface bars and are not repeated here.

The parameters given in Table 20.5-2 are mean values. The scatter in measured slip is considerable, especially for small values of slip. For a given value of the slip the coefficient of variation of the bond stresses may amount to □pprox.. 30%. The

according to Table 20.5-1. Fatigue failure can occur if the number of applied load cycles (N) is sufficiently high to reach the fatigue resistance (N_f). For constant amplitude load cycles, bond fatigue strength can be evaluated with the following equation:

$$N_f = \left(\frac{\tau_{b,up}}{\tau_{bu}} \right)^{\frac{-1}{\alpha \cdot b}} \qquad (20.5\text{-}16)$$

where b is defined in Eq. (20.5-15) and α is the exponential factor defining the first branch of the quasi-static bond-slip curve (α = 0.4 can be used according to Table 20.5-1).

20.5.2 Local bond-slip relationship of steel reinforcement in non-conventional concretes

Anchorage and lap behaviour equations described in Section 20.2 are based on a wide database including worldwide experiments carried out on normal concrete on different types of rebars. Equations proposed in Section 20.2 are valid within a specific range of material properties.

20.5.3 Local bond-slip for plain (non-ribbed) surface bars

For monotonic loading the value τ_b of the bond stresses between concrete and reinforcing bar for pull-out failure can be calculated as a function of the relative displacement s parallel to the bar axis according to Eq. 20.5-17.

$$\tau_b = \tau_{bmax}\, Min\left[\left(s/s_1 \right)^{\alpha_r}, \left(s/s_1 \right)^{\alpha_f} \right] \qquad (20.5\text{-}17)$$

scatter is due to the use of different test specimens, the differences in the state of stress in the concrete surrounding the reinforcing bar, the different measuring techniques as well as to the different loading and deformation velocities. The heterogeneity of the concrete and the surface characteristics of the reinforcing bars (initial surface condition, whether millscale, surface corrosion etc., diameter) also have a significant influence on the τ - s relationship.

The parameters given in Table 20.5-2 are valid for plain (i.e. non-ribbed and non-twisted) reinforcing steel, depending on the main influencing factors: roughness of the bar surface, bond conditions and concrete strength. They are valid for confined concrete and are applicable only in loading states with concrete not subjected to lateral tension, in the elastic range of the reinforcement and for those parts of the reinforcing bar with the distance $x > 2Ø$ from a transverse crack. For those parts of the reinforcing bars with a distance $x \leq 2Ø$ from a transverse crack, the bond stress τ_b and the slip s are to be reduced by the factor λ according to Eq. (20.5-7).

Table 20.5-2 *Parameters defining the bond stress–slip relationship of plain surface bars in well confined conditions.*

	Hot rolled bars		Cold drawn bars and wires	
	Good bond conditions	All other bond conditions	Good bond conditions	All other bond conditions
s_1	0.25 mm	0.25mm	0.02 mm	0.02mm
α_r	0.2	0.2	0.2	0.2
α_f	-0.2	-0.2	-0.2	-0.2
η_4	$5.0(25/Ø)^{0.2}$	$2.5(25/Ø)^{0.2}$	$2.5(25/Ø)^{0.2}$	$1.25(25/Ø)^{0.2}$
τ_{bmax}	$\eta_4\sqrt{(f_{cm}/25)}$			

20.5.4 Local bond-slip under elevated temperature

Exposure to elevated temperatures leads to a degradation in the material properties of concrete such as the tensile strength, compressive strength and E-modulus. Consequently, the bond strength between reinforcement and concrete also degrades. The expression given in Eq. (20.5-18) provides a conservative estimate of the degradation in bond strength with elevated temperature (see Fig. 20.5-7). The expression is valid for both pullout failure and splitting failure and reaches a value of zero at 740°C.

For ribbed reinforcing bars, the degradation of bond strength between ribbed reinforcing bars and concrete due to the exposure to elevated temperature should be considered as:

$$\tau_{bmax,T} = \tau_{bmax,20} \cdot \left(1 - \frac{T-20}{720}\right) \text{ for } 20°C < T \leq 740°C \qquad (20.5\text{-}18)$$

where,

$\tau_{bmax,T}$ is the bond strength corresponding to an elevated temperature T (°C)

$\tau_{bmax,20}$ is the bond splitting strength according to Eq. (20.5-5) or the bond pullout strength given by Eqs. (20.5-1)–(20.5-4), whichever is smaller under ambient conditions.

T is the temperature, measured at the level of the reinforcing bar.

The bond stress-slip relationship between deformed reinforcement bar and concrete subjected to an elevated temperature T is given as:

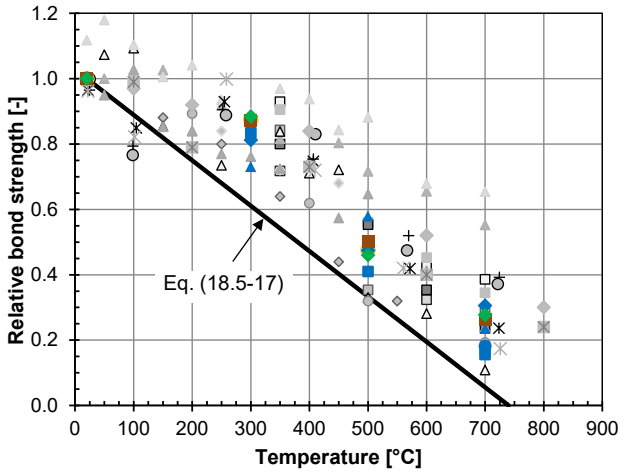


Figure 20.5-8: *Experimental evidence of bond strength degradation at elevated temperature in quasi-static thermal conditions and comparison with Eq. (20.5-18).*

Tests have shown that if the applied bond stress on the reinforcement under ambient conditions is less than 60% of its ambient bond strength, the influence of load on bond capacity at elevated temperatures is negligible. Since the design bond strength is typically approx. 50% of its mean bond strength, this influence is ignored in Eq. (20.5-18).

Exposure to elevated temperatures lead to a reduction in the bond stiffness. Evaluation of the test data shows that the degradation of secant stiffness corresponding to the first attainment of peak in case of can be considered as

(a) Bond pull-out failure

$$k_T = k_0\left(1-\frac{T-20}{5000}\right) \text{ for } T \le 300°C \quad (20.5\text{-}26)$$

$$k_T = k_0\left(0.94-\frac{T-300}{600}\right) \text{ for } T > 300°C \quad (20.5\text{-}27)$$

(b) Bond splitting failure

$$k_T = k_0\left(1-\frac{T-20}{800}\right) \quad (20.5\text{-}28)$$

where:

k_T is the bond stiffness corresponding to an elevated temperature T (°C)

k_0 is the secant stiffness corresponding to the first attainment of peak bond stress–slip defined by Eq. (20.5-1)–(20.5-4).

Temperature, T is measured at the level of the reinforcing bar.

$$\tau_{bT} = \tau_{bmax,T}\,(s/s_{1,T})^{\alpha} \quad \text{for } 0 \le s \le s_{1,T} \quad (20.5\text{-}19)$$

$$\tau_b = \tau_{bmax,T} \quad \text{for } s_{1,T} \le s \le s_{2,T} \quad (20.5\text{-}20)$$

$$\tau_b = \tau_{bmax,T} - (\tau_{bmax,T} - \tau_{bf,T})\,(s\text{-}s_{2,T})/(s_{3,T}\text{-}s_{2,T}) \quad \text{for } s_{2,T} \le s \le s_{3,T} \quad (20.5\text{-}21)$$

$$\tau_b = \tau_{bf,T} \quad \text{for } s_{3,T} < s \quad (20.5\text{-}22)$$

where, $\tau_{bmax,T}$ is given by Eq. (20.5-18). The values of $s_{1,T}$, $s_{2,T}$ and $s_{3,T}$ can be obtained as:

(a) For bond pullout failure

$$s_{1T} = 6.94 s_{1,20}\left(\frac{740-T}{5020-T}\right) \text{ for } 20°C < T \le 300°C \quad (20.5\text{-}23)$$

$$s_{1T} = 0.83 s_{1,20}\left(\frac{740-T}{864-T}\right) \text{ for } 300°C < T \le 740°C \quad (20.5\text{-}24)$$

where:

$s_{1,20}$ is the displacement s_1 under ambient conditions according to Fig. 20.5-1 and Table 20.5-1.

$s_{2,T} = 2s_{1,T}$

$s_{3,T} = c_{clear}$ = the clear distance between ribs

(b) For bond splitting failure

$$s_{1T} = 1.11 s_{1,20}\left(\frac{740-T}{820-T}\right) \quad (20.5\text{-}25)$$

where:

$s_{1,20}$ is the displacement s_1 under ambient conditions for splitting failure mode according to Table 20.5-1.

$s_{2,T} = s_{1,T}$

$s_{3,T} = 1.2\ s_{1,T}$ for no stirrups and $0.5c_{clear}$ for with stirrups case

20.5.5 Local bond-slip under fire

After a defined fire exposure duration (in minutes), the bond capacity, $\tau_{bu,fi}$ of a deformed reinforcing bar can be calculated as:

$$\tau_{bu,fi} = k_{fi}\ \tau_{bu,0} \qquad (20.5\text{-}29)$$

The values of the coefficient k_{fi} depend on the time of fire exposure and shall be read from Table 20.5-3.

The basic value of bond strength under ambient conditions, $\tau_{bu,0}$ is given by Eq. (20.5.5).

The values of slip of deformed reinforcing bar, $s_{i,fi}$ after an exposure to fire for a defined exposure duration can be obtained by using Eq. (20.5.30).

$$s_{i,fi} = k_{s,fi}\ s_{i,20} \qquad (20.5\text{-}30)$$

where:

$s_{i,20}$ is the value of slip (s_1, s_2 or s_3) under ambient conditions, to be read from Table 20.5-1. The recommended values for the slip modification factor $k_{s,fi}$ shown in Fig. 20.5-9.

The values of the coefficient k_{fi} shall be read from the Table 20.5-3.

Table 20.5-3. *The values of the coefficient* k_{fi}

Exposure duration [min]	**Value of k_{fi}**
0 min	1.0
15 min	0.5
30 min	0.4
60 min	0.33
90 min	0.30

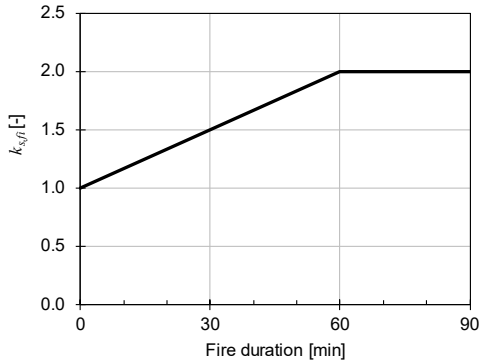


Figure 20.5-9: *Slip modification factor* $k_{s,fi}$ *for bond slip values*

20.6 Standard test method for anchorages

The test is intended to verify that existing provisions for anchorage and laps in the *fib* Model Code when non-conventional concretes and rebars are adopted (i.e. fibre reinforced concrete, recycled aggregate concrete, etc.). However, it could also be useful for testing bond characteristics of novel reinforcement.

This test method describes procedures to assess the bond strength (anchorage capacity) of steel reinforcing bars in concrete.

This test method is intended to determine the effects of concrete cover, anchorage length, transverse reinforcement, concrete strength, casting position on the bond strength of deformed steel reinforcing bars having a diameter of 16 mm. The test method allows a pressure transverse to the potential splitting plane to be applied.

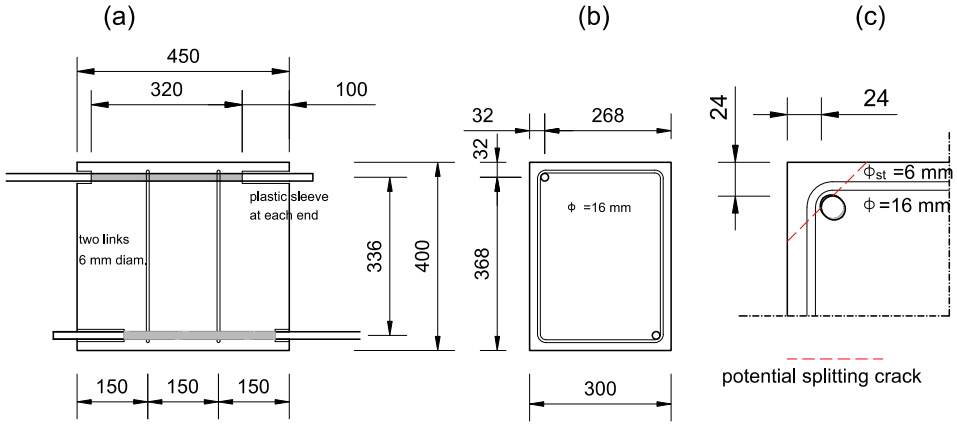


Figure 20.6-1: *Beam end test specimen.*
Adaptable to different cocnrete grade and bar diameter d16 and d32
dimension of the specimen in term of bar diameter
small corner bars in other corners – testin bars.

20.6.1 Test specimen

A typical test specimen is illustrated in Fig. 20.6-1. The specimen is 450 mm long, 300 mm wide and 400 mm deep.

Two bars are cast in each specimen (in the two opposite corners), one in a 'good' casting position and the other in 'poor' casting position. The ribs of the bar are oriented towards the specimen corner (see Fig. 20.6-1c).

The design aims to develop a mean stress equal to 435 MPa, and with a cover/diameter ratio of 1.50 a splitting failure mode is expected in normal strength concrete.

For a bar diameter of 16 mm, an anchorage length of 320 mm should be provided. Plastic sleeve (bond breakers) are placed at both bar ends. Two transverse stirrups having a diameter of 6 mm are placed along the anchorage length with a spacing of 150 mm.

Material characterization is based in standard methods.

Specimens should be cured in standard conditions (T=20°C and RH>95%) until the time of test.

20.6.2 Test set up and test procedure

A sketch of a suitable test set up is shown in Fig. 20.6-2. The steel bench should be designed to prevent any yielding of its components and its should be sufficient rigid to ensure that the applied load remains parallel to the bar axis during testing.

The loading system should be capable of measuring the applied tensile load to an accuracy of 2% of the applied force.

Compressive bottom portion of the specimen should be not greater than 100 mm (1/3 of the specimen depth).

Displacements of the loaded end of the longitudinal bar should be measured by two dial gauges od linear variable differential transformers (LVDTs) having an accuracy of 0.001mm.

The tensile load shall be applied to the anchored bar, as illustrated in Fig. 20.6-2. The loading rate shall be such that failure will not occur prior to 5 min after starting to load the test specimens.

At least six specimens (six bars in a 'good' casting position and six bars in 'poor' casting position) shall be tested.

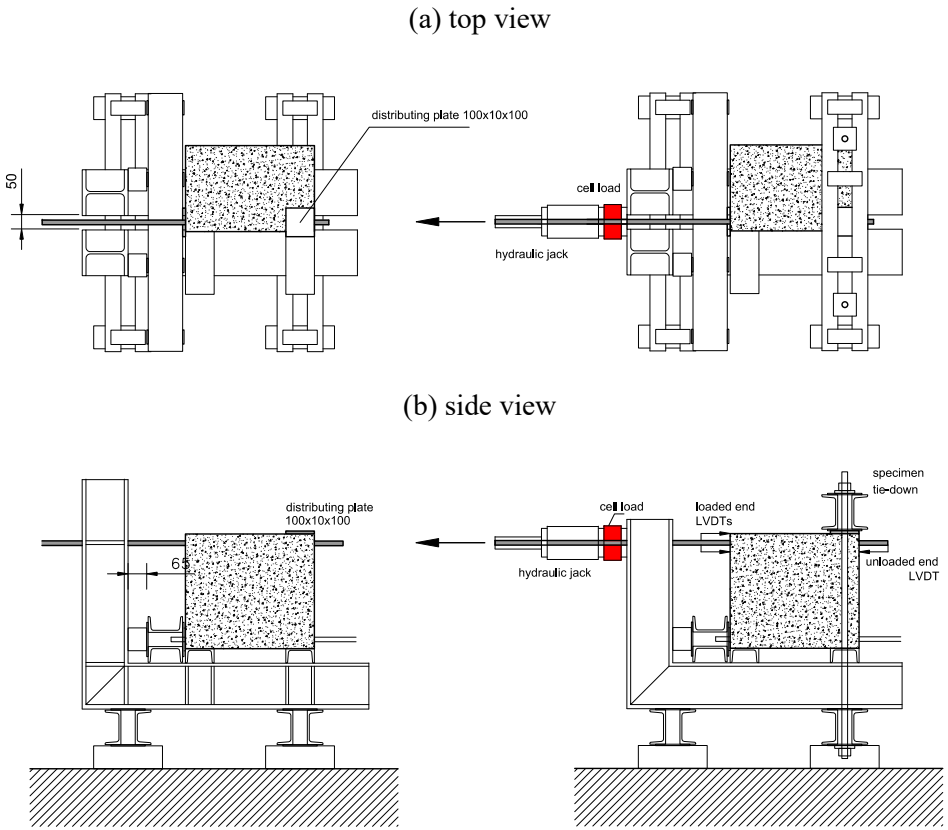


Figure 20.6-2. *Skecth of the test set up.*

20.6.3 Assessment of anchorage performance

Performance of the bar would be assessed against 4 measures, namely peak bar stress, bar stress at loaded end slips of 0.05 mm and at free end slip of 0.02 mm (crack control), free end slip at peak load, and the individual curves of force-slip relations. The splitting crack width shall be also checked both on the side surface and top surface of the specimen (see Fig. 20.6-1c).

The existing provisions for anchorages and laps in *fib* MC2020 may be applied to the tested non conventional concrete or new type of reinforcement if the averege

Do not oveweigth the limits for bars and concrete in Eq. 20.2-5.

Limitations of parameters…..

Only to NCC – f_r>0.05 and <0.09…

of maximum bar stress ($f_{sm,ex}$) is not lower than 0.85 the mean bar stress estimated by the Eq. (20.6-2) of *fib* Bulletin 72 [20-1]:

$$f_{sm,ex} \geq 0.85\ f_{stm} \tag{20.6-1}$$

$$f_{stm} = 54\left(\frac{f_{cm}}{25}\right)^{0.25}\left(\frac{l_b}{\phi}\right)^{0.55}\left(\frac{25}{\phi}\right)^{0.2}\left[\left(\frac{c_{min}}{\phi}\right)^{0.25}\left(\frac{c_{max}}{c_{min}}\right)^{0.1} + k_m K_{tr}\right] \tag{20.6-2}$$

Standard deviastion of test results….

Furthermore, the top cast ratio (the ratio between the mean anchorage strength of the bars near the top of the specimen and that of the bars close to the bottom surface) shall not be lower 0.7. to be consisten with roles of Section …

$$TC = f_{sm,ex}{}^{top} / f_{sm,ex}{}^{bot} \geq 0.7 \tag{20.6-3}$$

References

[20-1] *fib* Bulletin 72, 2010. Bond and anchorage of embedded reinforcement: Background to the fib Model Code for Concrete Structures 2010

[20-2] ISO 15630-1:2019 Steel for the reinforcement and prestressing of concrete - Test methods - Part 1: Reinforcing bars, rods and wire

[20-3] EN 15630

[20-4] Vollum & Goodchild Vollum, R., Goodchild, C., 2019, Proposed EN 1992 tension lap strength equation for good bond. *Structures*, 19, 5-18.

[20-5] Sagan et al V. E. Sagan, P. Gergeley, R. N. White: Behavior and Design of Noncontact Lap Splices Subjected to Repeated Inelastic Tensile Loading. ACI Structural Journal, V. 88, No. 4, July-August 1991, pp.420-431.

[20-6] Hamad et al B. S. Hamad, M. Mansour: Bond Strength of Noncontact Tension Lap splices. ACI Structural Journal, V. 93, No. 3 May-June 1996, pp. 2-11.

[20-7] *fib* Bulletin 58

[20-8] EN 1998-1: 2004. Eurocode 8: Design of structures for earthquake resistance – Part 1: General rules, seismic actions and rules for buildings.

[20-9] Deutsche Institut für Bautechnik 2020 Deutsche Institut für Bautechnik (DIBt 2020): Muster-Verwaltungsvorschrift Technische Baubestimmungen (Model administrative regulations for technical building regulations), Ausgabe 2020/1 (in German)

[20-10] ACI/CRSI 2015. American Concrete Institute (ACI 2015): ACI/CRSI, "American Concrete Institute and Concrete Reinforcing Steel Institute Certification Policies for Adhesive Anchor Installer,"American Concrete Institute, Farmington Hills, MI, May 22, 2015 revision, 12 pp.

[20-11] EAD 33087-00-0601, 2020. Systems for post-installed rebar connections with mortar.

[20-12] ICC-ES AC308. Test Program for Evaluating Adhesive Anchor Systems for Use in Cracked and Uncracked Concrete.

[20-14] Spieth, 2002. Spieth, H. (2002): „Tragverhalten und Bemessung von eingemörtelten Bewehrungsstäben" (load-carrying behaviour and design of post-installed reinforcement), PhD Dissertation, Universität Stuttgart (in German)

[20-15] Fuchs/Sawade, 1985. Fuchs, W.; Sawade, G. (1985): „Einfluß des Bohrverfahrens auf das Tragverhalten von Befestigungselementen" (Influence of drilling method on the load-carry capacity of fastenings), Report 8/5 – 85/12, Institute of Construction Materials, University of Stuttgart (in German)

[20-16] Simons, 2004. Simons (2004): Eingemörtelte Bewehrungsstäbe (WIT C100) Dehnkörperversuche mit gestoßenen Bewehrungsstäben Beton C50/60. University of Stuttgart, Institute of Construction Materials, Report No. E03/02 – E02502/1, March 9, 2004

[20-17] Prieto et al., 2016 Prieto M., Tanner P., Andrade C. (2016). Multiple linear regression model for the assessment of bond strength in corroded and non-corroded steel bars in structural concrete. Materials and Structures/Materiaux et Constructions, 49 (11), pp. 4749 - 4763

[20-18] Prieto et al., 2022 Prieto M., Tanner P., Andrade C. (2022). Model for the assessment of bond in corroded steel ribbed bars. 5th International Conference Bond in Concrete 2022: Bond Anchorage Detailing. Stuttgart, Germany, 25th—27th July 2022

[20-19] Blomfors et al., 2018. Blomfors, M., Zandi, K., Lundgren, K., & Coronelli, D. (2018). Engineering bond model for corroded reinforcement. Engineering Structures, 156, 394-410. doi:10.1016/j.engstruct.2017.11.030

[20-20] Blomfors et al., 2019. Blomfors, M., Larsson Ivanov, O., Honfi, D., & Engen, M. (2019). Partial safety factors for the anchorage capacity of corroded reinforcement bars in concrete. Engineering Structures, 181, 579-588.doi:10.1016/j.engstruct.2018.12.011

[20-21] *fib* Bulletin 10 Bond of reinforcement in concrete, *fib* State of the Art report, 2000, *fib*

21 Bond of embedded non-metallic reinforcement

21.1 General

Bond of non-metallic reinforcement is the term used to denote the interaction and transfer of force between fibre reinforced polymer (FRP) reinforcement and concrete. At the serviceability limit state, bond influences width and spacing of transverse cracks, tension stiffening and curvature. At the ultimate limit state, bond is responsible for strength at end anchorages.

The bond behaviour of FRP reinforcement to concrete depends mainly on the reinforcement geometry and surface characteristics. It varies from that of conventional steel reinforcement, mainly as a result of the following aspects:

- the modulus of elasticity of FRP is generally lower than that of steel, especially in the transverse direction.
- the shear stiffness of FRP is significantly lower than that of steel.
- the resistance offered by the surface deformations is a function of the mechanical properties of the resin matrix.

It is generally possible to obtain bond strengths for non-metallic reinforcement of similar or greater magnitude than for steel reinforcement. The bond characteristics of FRP rebars are determined in accordance to national or international documents, such as ISO 10406-1 [21-1], and in compliance with the quality control provisions of section 10.2.

21.2 Bond for predominantly static loading

21.2.1 Local bond stress-slip model in case of no degradation

The bond stress-slip relationship (monotonic loading) for deformed steel rebars is applicable for FRP reinforcement, provided that calibrated model parameters are used. Generally, the modified bond stress-slip relationship illustrated in Figure 21.2-1 is assumed. The model parameters are calibrated on the basis of experimental results for each FRP reinforcement bar.

The bond stress τ can be calculated as a function of slip s according to the following equations (see also Figure 21.2-1):

$$\tau = \tau_m \left(s / s_m\right)^{\alpha} \qquad \text{for } 0 \le s \le s_m \qquad (21.2\text{-}1a)$$

$$\tau = \tau_m - \tau_m \cdot p\left(s - s_m\right) / s_m \qquad \text{for } s_m \le s \le s_u \qquad (21.2\text{-}1b)$$

Though durability requirements can be satisfied with minimal concrete cover, the concrete cover should be designed to ensure appropriate bond stress transfer (e.g. to avoid splitting failures Figure 21.2-2). For detailed provisions reference is made to *fib* Bulletin 40 [21-2].

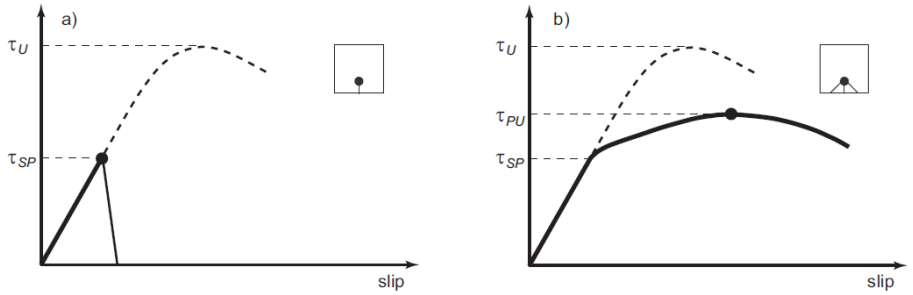


Figure 21.2-2: *Bond splitting: (a) cover crack induced failure (b) splitting off surrounding concrete.*

Bond of rebars with surface enhancements (e.g. deformed, sand coated, grooved) may be governed by the shear strength of the surface texture. For low strength concrete, concrete shear failures similar to those observed for deformed steel bars are more predominant.

Contrary to deformed steel reinforcement, long term effects can weaken the surface texture and impact the bond capacity. This is described through the long term bond strength f_{bdmax}, which is specific to the applied FRP rebar. A typical lower bond value of f_{bdmax} is 1,5 MPa.

η_2 is specific to the applied FRP rebar. A value of 0,65 is recommended if no specific information can be obtained from the manufacturer or through testing.

Generally, splitting failure is a common failure mode for densely FRP reinforced members. Considering the similarities in bond stress-slip relationship,

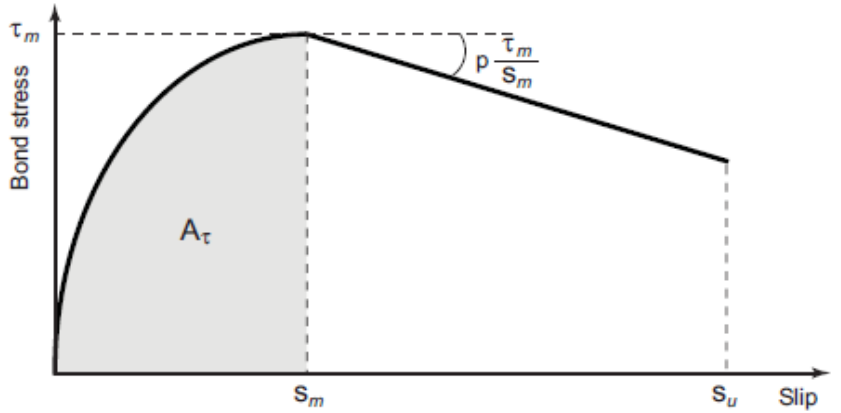


Figure 21.2-1: *Analytical bond stress-slip relationship.*

21.2.2 Ultimate bond stress in case of no degradation

The design value of the ultimate bond stress of surface treated FRP rebars reinforcing is given as follows, provided that no splitting occurs:

$$f_{bd} = 2{,}25 \cdot \eta_1 \cdot \eta_2 \cdot f_{ctd} < f_{bd,max} \qquad (21.2\text{-}1)$$

where:

f_{ctd} is the design value of concrete tensile strength. Due to the increasing brittleness of higher strength concrete, $f_{ck,0.05}$ should be limited here to the value for C60/75, unless it can be verified that the average bond strength increases above this limit;

$f_{bd,\max}$ is the limiting bond stress for FRP reinforcement surface failure under long term sustained loading, according to technical product specification;

η_1 is a coefficient related to the quality of the bond condition and the position of the bar during concreting, as already specified for steel reinforcement.

η_2 is related to the bar diameter and to the texture of the surface of the reinforcing bar. The value for η_2 shall be according to technical product specification.

The design value of the ultimate bond strength in the case of splitting will follow the provisions as for steel reinforcement.

provided the provisions in this chapter (including consideration of the different modulus of elasticity of FRP in the required anchorage length), splitting failure of FRP and steel rebars has been observed similar.

Long term effect on the bond behaviour of FRP reinforcement is still not well understood and predictive design models are not yet available. The long term bond strength can be determined on the basis of accelerated testing, and is to be declared per type of FRP rebar according to technical product specifications.

Sufficient concrete cover and/or fire insulation should be provided to limit the temperature at the level of the FRP rebars. Appropriate anchorage of the FRP rebars in cold regions of the structure should be considered at the design stage. A temperature dependant bond stress-slip relationship based on bond tests at different temperatures can be used where a more advanced analysis is required.

Exposure to freeze-thaw cycles generally does not affect the bond performance of FRP rebars, provided that the concrete is not prone to freeze-thaw deterioration.

21.2.3 Bond parameters in case of degradation

Bond degradation relates to the interfacial zone between the FRP rebar and the concrete, and is generally characterized by surface texture degradation of the FRP rebar. This results in a bond stress limitation (f_{bdmax}) which can be transferred over the entire service life, as specified in section 21.2.2.

21.3 Bond of embedded non-metallic reinforcement for non-static loading

Research estimating bond degradation under fatigue loading is not yet fully assessed.

21.4 Bond of embedded non-metallic reinforcement for extreme thermal conditions

Elevated temperatures impact the bond, whereby the temperature of the FRP rebar should be limited with respect to the glass transition temperature of the FRP, considering a lower bound value of the glass transition zone.

FRP rebars have superior inter-laminar shear strength, so that this is not governing the bond strength. For high concrete strength applications or special circumstances, bond verification will include inter-laminar shear as a possible bond failure aspect.

For FRP rebars the bond strength is not only dependant on the concrete strength, but also on the limiting bond stress for FRP reinforcement surface failure under long term sustained loading is considered.

21.5 Bond of embedded non-metallic reinforcement under special circumstances

Application in special circumstances will require bond verification through design by testing.

21.6 Anchorage and lapped joints behaviour of embedded non-metallic reinforcement

21.6.1 Design anchorage and lapped joint length

The calculation of the design anchorage length and lapped joints length of steel reinforcement is applicable for FRP reinforcement, provided that the bond properties as given in sections 21.2 to 21.5 are taken into account. This results in the following modification for the anchorage length:

$$l_{bd} = k_{lb} \cdot k_{cp} \cdot \varnothing \left(\frac{\sigma_{ftd}}{217}\right)^{\eta_\sigma} \left(\frac{25}{f_{ck}^*}\right)^{\frac{1}{2}} \left(\frac{\varnothing}{20}\right)^{\frac{1}{3}} \left(\frac{1{,}5\varnothing}{c_d}\right)^{\frac{1}{2}} \geq \left\{10\varnothing ; \frac{\varnothing}{4} \frac{\sigma_{ftd}}{f_{bd,max}}\right\} \qquad (21.6\text{-}1)$$

where:

$$f_{ck}^* = \left(\frac{f_{bd,max}}{0{,}315}\right)^{1,5} \leq 1{,}1 f_{ck} \qquad (21.6\text{-}1)$$

and where:

η_σ equals 1 for stresses limited to 217 MPa and 1,5 for stresses exceeding 217 MPa;

σ_{ftd} is the design value of the tensile stress in the FRP reinforcement to be anchored, and other parameters are as defined for steel reinforcement.

Due to the limited bending stiffness for FRP rebars bond influence factors related to bends shall be taken as 1,0, unless FRP rebar specific values are provided by experimental evidence.

FRP rebars cannot be welded and related bond influencing factors shall be taken as 1,0.

Laps of the FRP reinforcement to FRP reinforcement or other reinforcement shall be situated in zones where the calculated stress in the reinforcement at ultimate limit state is not more than 80% of the design strength.

21.6.2 Mechanical couplers

Mechanical devices assuring load transfer between FRP rebars in tension-compression, in compression or in tension may be used, subject to product specific FRP rebar – coupler systems.

References

[21-1] ISO 10406-1:2015. Fibre-reinforced polymer (FRP) reinforcement of concrete – Test methods – Part 1: FRP bars and grids, ISO, 2015

[21-2] *fib* Bulletin 40, 2007. FRP reinforcement in RC structures, *fib,* 2007

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The bond behaviour will depend on the type of external application, since the bond stress transfer process from the reinforcements to the concrete is dependent on the strengthening techniques, namely, externally bonded reinforcement (EBR, surface bonded) and near surface mounted reinforcement (NSM, bonded into grooves). In case of EBR made of steel plates, the approach described in this chapter can be followed (though the applicability of the proposed formulations relates to more limited experimental results available in literature for steel plates compared to FRP strips). The provisions for NSM reinforcement refer to FRP rebars (whereas applicability to steel rebars is not yet established).

Amongst other, the bond influences ultimate limit state resistance, by the potential occurrence of bond failure modes at anchorage or intermediate regions. At serviceability limit states the bond conditions influence deflections and cracking and deflections, via the tension stiffening effect.

The constitutive bond model for EBR, which is surface bonded, is assumed bilinear. The model parameters are basically related to the fracture energy G_f of the glued joint connection between the external reinforcement and the concrete, and have to be determined from experimental results.

22 Bond of externally applied reinforcement

Bond of externally applied reinforcement denotes the interaction and transfer of force between concrete and bonded FRP or steel reinforcement.

22.1 Local bond stress-slip model

22.1.1 Externally bonded reinforcement

The bond stress can be calculated according to the following equations (Figure 22.1-1):

$$\tau = \tau_m(s/s_m) \quad \text{for } 0 \le s \le s_m \quad (22.1\text{-}1a)$$

$$\tau = \tau_m\text{-}\tau_m(s\text{-}s_m)/(s_u\text{-}s_m) \quad \text{for } s_m \le s \le s_u \quad (22.1\text{-}1b)$$

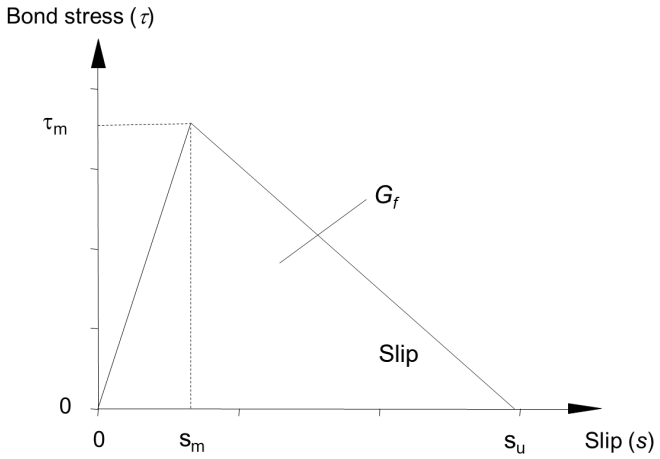


Figure 22.1-1: Bond stress-slip relationship for surfaced bonded reinforcement.

22.1.2 Near surface mounted reinforcement

The constitutive bond model for NSM reinforcement is affected by several parameters, among which the mechanical properties of the materials (reinforcement, groove filler and concrete substrate), the surface properties of the reinforcement and of the grooves, the shape of the strengthening system (bars or strips), the dimensions of the groove, the distance of the groove from the edge of the member, the distance between grooves and the depth of the reinforcement into the groove. The model parameters have to be calibrated on the basis of experimental results.

Depending on the splitting tendency of the surface mounted reinforcement, interfacial debonding or epoxy cover splitting is governing the failure of the joint. The latter is usually generated by bars of surface with high protrusions.

The bond stress-slip relationship is defined according to the following equations, in case of interfacial debonding (Figure 22.1-2):

$$\tau = \tau_m (s/s_m)^{\alpha} \quad \text{for } 0 \le s \le s_m \qquad (22.1\text{-}2a)$$

$$\tau = \tau_m (s/s_m)^{-\alpha'} \quad \text{for } s_m \le s \le s_u \qquad (22.1\text{-}2b)$$

In case of epoxy cover splitting failure mode, Eqs. (22.1-2) become as follows:

$$\tau = \tau_m (s/s_{m,sp})^{\alpha} \quad \text{for } 0 \le s \le s_{m,sp} \qquad (22.1\text{-}3a)$$

$$\tau = \tau_{f,sp} \quad \text{for } s_{m,sp} < s \le s_u \qquad (22.1\text{-}3b)$$

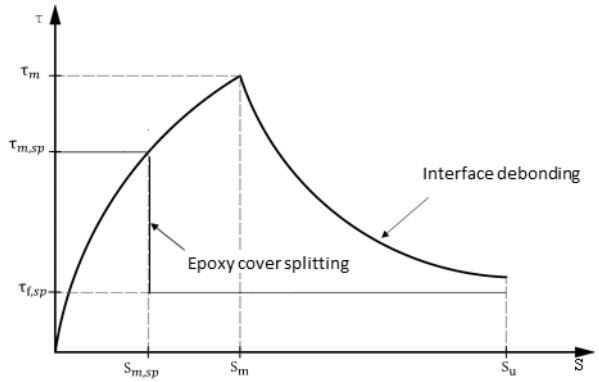


Figure 22.1-2: *Bond stress-slip relationship for the near surface mounted reinforcements.*

22.2 Bond and anchorage of externally applied reinforcement

Debonding of EAR can be predicted by considering the different bond failure modes that can occur, as outlined in the following clauses. This deals with bond failure at the extremities (anchorage), as well as debonding initiated at crack bridging.

Given the high shear strength of the adhesive and the good reinforcement-adhesive and adhesive-concrete adhesion, a cohesive failure through the concrete layer near the adhesive is the most common due to the relatively low tensile strength of the cement based material of this zone.

22.2.1 Surface bonded reinforcement

22.2.1.1 Bond failure modes

Depending on the starting point of the debonding process, the bond-critical failure modes in flexural and shear strengthening with EBR can be classified into the following main categories:

Bond failure can occur at the reinforcement-adhesive interface, through the adhesive, at the adhesive-concrete interface, through the concrete or being a mixture of them. As part of any flexural or shear strengthening design at ultimate limit state, the maximum force that may be transferred from the concrete to the external reinforcement must be evaluated. It can be assumed that bond interface failure will not occur under service load, as long as proper detailing and limitation of deflections and crack openings are considered in the design.

In cases where the FRP will reach its design tensile strain before the concrete crushes, failure normally occurs due to debonding (Figure 22.2-1) rather than

rupture. Debonding occurs through the concrete, which is the generally the "weak link" in terms of bond capacity.

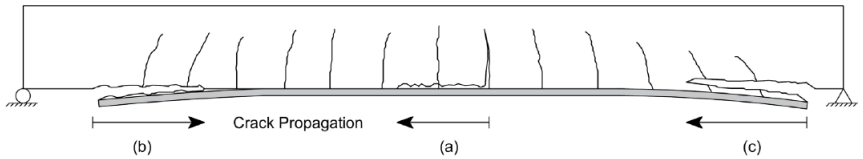


Figure 22.2-1: Debonding failure modes of a concrete member with externally bonded FRP: (a) Intermediate crack debonding; (b) end debonding; (c) concrete cover separation.

The assessment of k was done considering 280 bond tests on concrete elements externally bonded with different types of FRP materials. The range of variation of the main parameters influencing Eqs. (22.2-1) and (22.2-2) are: mean concrete strength f_{cm} = 15-62 MPa, elastic modulus of FRP E_f = 82-400 GPa, thickness of FRP t_f = 0.083-1.6 mm, 1-3 layers of FRP sheets, A_f=25-160 mm^2 of pre-cured strips, FRP to concrete width ratio k_b = 0.15-1.

In Eq. (22.2-1), f_{cm} is adopted since strengthening interventions have to be executed on existing structures and, usually, by means of in situ tests, the average values of strength can be more reliably assessed.

The values of k given here are valid for the mostly used epoxy resins and may deviate for other adhesives. Softer adhesives may, indeed, lead to a larger effective

– end debonding due to interfacial failure at the EBR extremities (anchorage zone);
– end debonding at the EBR extremities due to concrete cover separation (rip-off);
– intermediate crack debonding.

22.2.1.2 Fracture energy

Debonding failure is related to the fracture energy G_f of the interface, usually defined as the area under the bond-slip relationship, $\tau-s$. For whatever $\tau-s$ relationship, G_f depends on the strength properties of concrete and adhesive and on the surface's characteristics of the concrete and reinforcement. When debonding occurs in the concrete, the fracture energy can be expressed as a function of the bond strength, τ_m. Since τ_m depends on the concrete tensile strength, G_f can be directly expressed as a function of the compressive strength of concrete.

The following expression can be adopted for the evaluation of the fracture energy:

$$G_f = k^2 \cdot k_b^2 \cdot f_{cm}^{2/3} \qquad (22.2\text{-}1)$$

where the shape factor k_b (≥ 1) is defined as follows:

$$k_b = \sqrt{\frac{2-b_f/b}{1+b_f/b}} \geq 1 \qquad (22.2\text{-}2)$$

where:

b_f the width of the external FRP reinforcement
b the width of the strengthened element, or centre-to-centre spacing between successive FRP strips;
f_{cm} is the average value of cylindrical compressive strength of concrete, expressed in MPa;
k is a parameter calibrated by a statistical procedure on several bond tests. In order to calculate the 5% characteristic value of the fracture energy, G_{fk}, k can be assumed equal to 0.17 in Eq. (22.2-1).

bond length and an increased end debonding capacity as a result of the increased fracture energy.

Under the assumption of bilinear $\tau-s$ bond law, the effective bond length may be estimated according to the following expression:

$$l_e = \frac{\pi \cdot s_u}{k \cdot k_b}\sqrt{\frac{E_f \cdot t_f}{8 f_{cm}^{2/3}}} \qquad (22.2\text{-}3b)$$

where:

s_u is the ultimate value of slip of the bilinear $\tau-s$ relationship.

Assuming k = 0.17 and su = 0.25 mm, the 5% characteristic value of the effective bond length is obtained:

$$l_{ek} = \frac{0.25\pi}{0.17 \cdot k_b}\sqrt{\frac{E_f \cdot t_f}{8 f_{cm}^{2/3}}} \approx 1.5\,\frac{\pi}{k_b}\sqrt{\frac{E_f \cdot t_f}{8 f_{cm}^{2/3}}} \qquad (22.2\text{-}3c)$$

The end debonding analysis for interfacial debonding is conducted on the basis of the FRP anchorage capacity, which is determined according to the design by testing approach.

For FRP curtailment, a similar methodology as for curtailment of internal steel reinforcement can be followed. Hereby, a fully cracked state is assumed and the position of flexural cracks is not explicitly taken into consideration. Considering different levels of approximations, the position of flexural cracks is considered, so that the end debonding analysis is conducted at the flexural crack closest to the point of zero moment or at an arbitrary concrete element between cracks. These methods are described in section 6.2.1.4 of *fib* Bulletin 90 [22-1].

22.2.1.3 Effective bond length

The 5% characteristic value of the effective bond length l_{ek} may be estimated according to the following expression:

$$l_{ek} = 1.5\,\frac{\pi}{k_b}\sqrt{\frac{E_f \cdot t_f}{8 f_{cm}^{2/3}}} \qquad (22.2\text{-}3a)$$

where:

t_f is the thickness of the FRP reinforcement, being E_f and f_{cm} expressed in MPa, and t_f in mm.

22.2.1.4 Ultimate strength for end debonding – anchorage capacity

The characteristic value of the maximum tensile stress that a FRP reinforcement can support, f_{fbk}, can be obtained from the following equation:

$$f_{fbk}(l_b) = \beta_l(l_b)\sqrt{\frac{2E_f \cdot G_{fk}}{t_f}} \qquad (22.2\text{-}4)$$

where:

β_l is a factor that depends on the bond length, l_b:

$$\beta_l(l_b) = \begin{cases} \frac{l_b}{l_e}\left(2 - \frac{l_b}{l_e}\right) < 1 & if\ \ l_b < l_e \\ 1 & if\ \ l_b \geq l_e \end{cases} \qquad (22.2\text{-}5)$$

Bond lengths, l_b, shorter than the effective one, l_e, so that $\beta_l < 1$, may occur, for instance, in shear strengthening.

The design value of the tensile stress in the FRP reinforcement at anchorage debonding is:

$$f_{fbd}(l_b) = \frac{f_{fbk}(l_b)}{\gamma_{fb}} \qquad (22.2\text{-}6)$$

where the partial factor γ_{fb} is 1.5.

The design value of the tensile force in the FRP at debonding, F_{fbd}, is:

$$F_{fbd}(l_b) = \frac{\beta_l(l_b)\cdot b_f\sqrt{2E_f\cdot t_f\cdot G_{fk}}}{\gamma_{fb}} \quad (22.2\text{-}7)$$

In all equations, E_f is expressed in MPa and t_f in mm.

22.2.1.5 Ultimate strength for end debonding – concrete rip-off

When a shear crack occurs at the end of the EBR, it can propagate into concrete cover separation, related to the missing tension link between the internal and external reinforcement when considering the classical truss analogy for shear capacity of beams (Figure 22.2-2).

For avoiding concrete rip-off failure mode, it should be verified (figure 22.2-2):

$$V_{Ed} \le V_{Rd,rip-off} = \tau_{Rd} bd \quad (22.2\text{-}8)$$

where:

$$V_{Rd,rip-off} = 0.75\left[1+19.6\frac{(100\rho_s)^{0.15}}{a_f}\right]V_{Rd,c} \quad (22.2\text{-}9)$$

In Eq. (22.2-9) ρ_s is the reinforcement ratio of the internal tensile steel reinforcement, a_f is the distance between the end of the surface bonded reinforcement and the edge of the support and $V_{Rd,c}$ is the design shear resistance of concrete members without stirrups.

This debonding mode can also be avoided by providing a suitable shear strengthening at the end of the surface bonded reinforcement. The design of the shear strengthening aims at extending the existing shear links up to the level of the surface bonded reinforcement. The shear link can be designed to take the following design shear load, considering f_{fbd} according to Eq. (22.2-6):

$$V_{fwd} = f_{fbd}\, b_f\, t_f \tan\theta \quad (22.2\text{-}10)$$

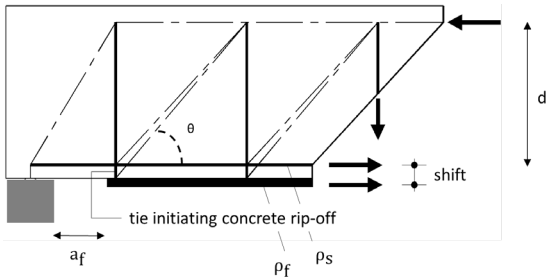


Figure 22.2-2: Geometric parameters for the formulation predicting the concrete rip-off failure mode.

22.2.1.6 Ultimate strength for intermediate debonding

According to the simplified FRP stress limit approach, at the ultimate limit state the maximum bending moment in a flexurally-strengthened member can be calculated assuming that in the EBR the tensile stress does not exceed the design value of the bond strength, $f_{fbd,IC}$, obtained by the corresponding 5% characteristic value, $f_{fbk,IC}$, by applying the following equations:

$$f_{fbk,IC} = k_{cr} \cdot f_{fbk} \quad (22.2\text{-}11)$$

$$f_{fbd,IC} = \frac{f_{fbk,IC}}{\gamma_{fb}} \quad (22.2\text{-}12)$$

where :

the characteristic value of the bond strength, f_{fbk}, is defined by Eq. (22.2-4);

considering β_l = 1, the coefficient k_{cr} has been assessed as equal to 1.8, and γ_{fb} is the safety factor for bond.

The corresponding value of the design strain in the FRP reinforcement, $\varepsilon_{fbd,IC}$, is:

$$\varepsilon_{fbd,IC} = \frac{f_{fbd,IC}}{E_f} \quad (22.2\text{-}13)$$

Eq. (22.2-11) is based on the maximum stress in the FRP in case of end debonding, corrected by a coefficient $k_{cr} > 1$. This is because the interfacial shear stresses relate to the FRP force gradient along the length of the beam, rather than the anchorage capacity at the FRP end. This implies that the value of the maximum FRP strain related to the intermediate crack debonding can be assumed higher than that pertaining to end debonding.

Alternative and more detailed approaches to prevent the debonding failure at intermediate cracks can be adopted based on the envelope line of tensile stress and on the force transfer between the concrete and surface bonded reinforcement. More information is provided in section 6.2.1.3.2 of the *fib* Bulletin 90 [22-1].

The value of k_{cr} has been assessed on the basis of a database of 214 results of beams externally reinforced in bending with FRP materials (164 cured in-situ systems and 50 pre-cured strips). For cured in-situ FRP systems, the relevant parameters vary in the following ranges: concrete width b = 75-960 mm, FRP width b_f = 30-480 mm, b_f/b = 0.17-1.0, FRP thickness t_f = 0.11 -2.55 mm, elastic modulus of FRP Ef = 21-390 GPa, mean cylinder compressive strength of concrete f_{cm} = 21-61 MPa. For specimens strengthened with pre-cured FRP strips, the same parameters vary in the following ranges: b = 180-800 mm, b_f = 25-280 mm, b_f/b = 0.13-1.0, t_f = 1.0-6.0 mm, Ef = 190-220 GPa, f_{cm} = 12.6-53.4 MPa. Many tests were performed by adopting a four-point bending scheme, though results of three-point bending tests are also present in the database. Since no relevant differences were observed in the values of k_{cr} when the two databases are considered separately, the

value 1.8 was assumed for beams strengthened with both cured in situ and pre-cured FRP materials.

The mostly used epoxy based resins are characterized by limited values of s_m (stiff bond behaviour). Thus, the linear elastic bond behaviour is limited. Nevertheless, for typical service load ranges the linear bond part is not exceeded. For higher value service loads, local loss of FRP-concrete bond initiated at positions of cracks in peak moment zones may lead to higher stresses in the steel reinforcement as the number of cycles increases, impacting the overall fatigue resistance (characterised by fracture of the main steel reinforcement).

22.2.1.7 Non-static bond behaviour

In general, the bond strength of EBR is not affected as long as stresses and slip are in the linear elastic part of the stress-slip curve. As such, fatigue damage of strengthened structures has been shown not to be significant if the FRP-strengthened RC member is exposed to typical service load ranges of 30% to 60% of the load at first yield.

Care should be taken in designing FRP strengthening when the new service loads are significantly higher than they were for the original structure (possibly beyond 60% of the load at first yield). Explicit fatigue verifications are described in detail in chapter 6 of *fib* Bulletin 90 [22-1].

22.2.2 Near surface mounted reinforcement

22.2.2.1 Bond failure modes

Given their relatively higher bond contact area per cross-sectional area, NSM systems are less sensitive to debonding phenomena compared to EBR systems.

The type and susceptibility of failure of a NSM reinforcement depend on its geometry, stiffness, surface characteristics and bond length, on the geometry and localization of the groove, and properties of the concrete substrate and of the adhesive. The influence of the geometry of the groove and reinforcement is provided by the k parameter represented in Figure 22.2-3. The bond is as effective as deeper the FRP is installed into the groove, and as higher is the ratio between the contact perimeter of the FRP to the adhesive (p_f) and the FRP's cross sectional area (A_f), expressed in ratio (p_f/A_f). Bond behavior also depends on the FRP axial stiffness (E_f A_f).

FRP-epoxy interface failure mode occurs when bond length is less than the effective bond length, which increases with the modulus of elasticity of the FRP, but is limited by the tensile strength of the concrete cover.

The epoxy-concrete interface failure mode is mainly observed for $k > 1.5$-2.0.

Tendency for splitting of the epoxy is favored by relatively small epoxy cover thickness ($k \cong 1$) and when using brittle adhesives, which are characterized by relatively low tensile strength (typical for cement-based adhesives). The propagation of the splitting to the surrounding concrete is promoted for limited concrete tensile strength.

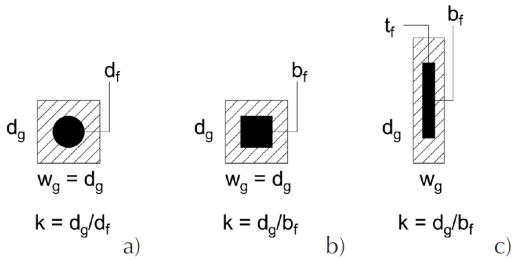


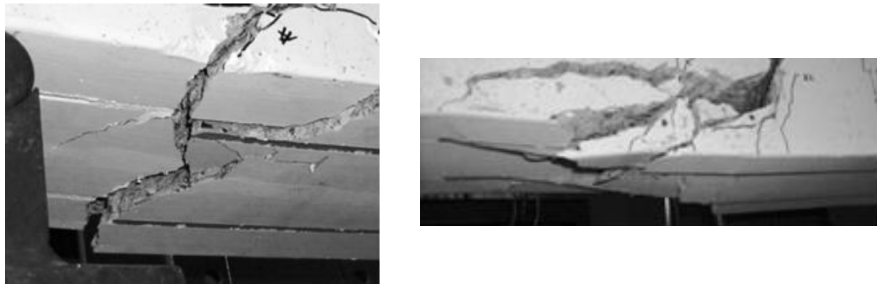
Figure 22.2-3: Definition of shape factor for a) round NSM bar, b) square NSM bars, c) rectangular NSM strips.

NSM reinforcements can fail by FRP-epoxy interface, epoxy-concrete interface, epoxy cover splitting, or concrete cover splitting:

- FRP-epoxy interface failure mode, which occurs when the bond length is smaller than the effective one, includes pure interfacial failure at the FRP surface (typical for FRP with smooth surface) and cohesive shear failure through the adhesive (typical for FRP with rough surface);
- The epoxy-concrete interface failure mode includes pure interfacial failure (grooves of smooth surface) and cohesive shear failure in the concrete (grooves of rough surface and with a large enough FRP-adhesive bond contacting area);
- The epoxy cover splitting failure mode is characterized by the formation of longitudinal cracks through the adhesive, which can also propagate through the concrete cover (mainly when grooves are relatively deep and adhesive cover thickness is large). For avoiding this type of failure mode, the shape factor, k, should be higher than 1.5;
- Concrete cover splitting can occur when a groove is too close to the edge of the concrete member and the concrete tensile strength is relatively low. Its occurrence can be minimized by disposing the groove at a distance from the edge higher than 2 or 4 times the width of the groove and the maximum aggregate size (depending on the arrangement of the FRP, exclusively in the tension soffit surface or also in the lateral faces).

22.2.2.2 Debonding mechanism in beams strengthened in flexure

The NSM systems applied for the flexural strengthening of RC beams can present the following failure modes: interfacial debonding and concrete cover separation.



a) b)

Figure 22.2-4: Concrete cover separation failure modes: a) strip end cover separation; b) cover separation induced by flexural or flexural-shear.

Further information can be found in section 6.3.5 'Shear strengthening in relation to flexural debonding' in *fib* Bulletin 90 [22-1].

Interfacial debonding includes all the bond failure aspects described in section 22.2.2.1. Depending on the loading configuration, interfacial debonding can occur at the anchorage zone of the NSM system or at an intermediate crack. The maximum NSM FRP strain capacity at interfacial debonding, established through design by testing, is provided in A5.1.2.2 of *fib* Bulletin 90 [22-1].

For close spacing of the NSM FRP reinforcement and limited tensile strength of the concrete cover, due to the detrimental group effect, concrete cover separation is governing as debonding failure mode (Figure 22.2-4).

Strip end cover separation (Figure 22.2-4a), also designated as concrete rip-off starts from the cut-off section and can be avoided by wrapping this end zone extremities of the NSM FRP systems by U shape end FRP straps, whose design is described in section 22.2.1.5.

Localized, flexural crack and flexural-shear crack induced cover separation are formed due to the high tensile strain field in the concrete cover surrounding the FRP induced by the propagation of a flexural or a flexural-shear crack in the high bending moment or high shear regions (Figure 22.2-4b). This intermediate cover separation can be prevented by applying U shear straps calculated for a shear force:

$$V_{fED} = max\left\{\frac{E_f A_f}{E_f A_f + E_s A_s} V_{Ed}\ ;\ V_{Ed} - V_{Rds}\right\} \qquad (22.2\text{-}14)$$

where:

V_{Ed} is the acting design shear load;

V_{Rds} is the design value of the shear resistance provided by the internal stirrups;

$E_f A_f$ is the elastic stiffness of the adhesively bonded longitudinal reinforcement, and

$E_s A_s$ is the elastic stiffness of the internal longitudinal steel reinforcement.

Detachment of the concrete cover along the side edges of the beam can occur when the NSM FRP reinforcement is located near these edges. This can be avoided by adopting the detailing recommendation provided in section 9.3 of *fib* Bulletin 90 [22-1].

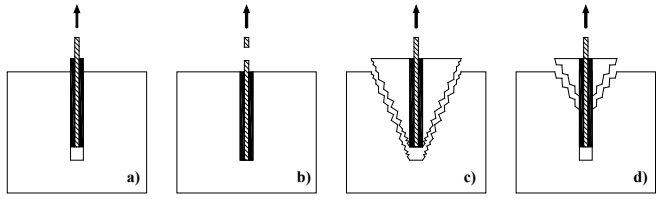


Figure 22.2-5: Possible failure modes of an NSM FRP reinforcement: a) debonding, b) tensile rupture, c) concrete semi-conical fracture, d) mixed shallow semi-cone plus debonding.

(a) (b)

Figure 22.2-6: Expulsion of the concrete cover.

Eqs. (22.2-15) and (22.2-16) are applicable to debonding (Figure 22.2-5a) or the possible prevention thereof in case of high bond capacity (Figure 22.2-6b). For attending the other failure modes, indicated in Figure 20.2-5c and 4d, in the evaluation of the l_e and f_{fb}, a detailed approach is described in Bianco et al. [22-2].

22.2.2.3 Debonding mechanism in beams strengthened in shear

When an NSM FRP shear strengthening reinforcement is crossed by a shear crack, depending of the effective bond length, geometry and properties of the FRP, and properties of the adhesive and surrounding concrete, the following type of failure modes can occur (Figure 22.2-5):

a) debonding;

b) tensile rupture of the FRP;

c) concrete semi-conical fracture;

d) mixed shallow semi-cone plus debonding, being this last one the most current.

By installing the NSM FRP reinforcement deep inside the grooves, there is a higher probability that tensile rupture is obtained, as long as the bond length is higher than the effective one. When the spacing of NSM FRP shear reinforcement is less than a certain limit, an outward expulsion of the concrete cover from the underlying beam core can occur (Figure 22.2-6). This failure mode can be avoided by adopting the detailing recommendations in Section 9.3 of *fib* Bulletin 90 [22-1].

22.2.2.4 Effective bond length and maximum tensile stress

The effective bond length of an NSM FRP reinforcement depends on its geometric and material properties, the local bond stress-slip law, $\tau - s$, (that can be determined experimentally or assessed from the recommendation in section A5.1.2.1 of *fib* Bulletin 90 [22-1]), and mechanical properties of the surrounding concrete. As indicated in section 22.1.2, the $\tau - s$ also depends on the type of bond failure mode. As for the case of EBR system, if a simple bilinear elastic-softening diagram $\tau - s$ is adopted (Figure 22.1-1), the effective bond length, l_e, is determined from the following equation:

$$l_e = \pi s_u \sqrt{\frac{E_f A_f}{8 G_f p_f}} \qquad (22.2\text{-}15)$$

where:

G_f is the fracture energy associated to the interfacial bond law, and

p_f and A_f are the perimeter and the cross-sectional area of the NSM FRP reinforcement, respectively.

The maximum tensile stress an NSM-FRP reinforcement can support at failure by debonding can be determined from:

$$f_{fb} = \sqrt{\frac{2E_f G_f p_f}{A_f}} \quad (22.2\text{-}16)$$

22.3 Mechanical anchorages for externally applied reinforcement

Debonding at the ends of the EAR can be avoided, or an enhancement of the debonding load can be achieved, using anchorage systems. Various solutions are available and can be designed for the specific case (e.g. transverse wrapping, spike anchors, steel or FRP bolts, etc.).

The ultimate value of the force transferred to the EAR system prior to debonding depends on the efficiency of the anchorage system. In the presence of mechanical anchorage, the design bond strength f_{fad} may be taken equal to:

$$f_{fad} = k_a f_{fd} \quad (22.3\text{-}1)$$

where:

f_{fd} is the design tensile strength of the EAR (either FRP or steel);

k_a is the effectiveness coefficient of the specific anchorage system ($k_a \leq 1.0$), provided on the basis of experimental results or specific analytical procedures.

Next to conventional mechanical anchorage systems for prestressed EAR, also innovative anchorage systems or procedures have been developed. Further information can be found in chapter 7 ‘Prestressed FRP Systems’ of RILEM TC 234 DUC [22-3]

In case of prestressed EAR, anchorage solutions are applicable specific to the commercially available prestressing systems and their anchorage procedures. In most cases, the external reinforcement is prestressed against the existing concrete structure. This involves the use of a mobile clamping system and auxiliary elements for the hydraulic jack. After the prestressing operation, mechanical anchorage at the ends of the FRP remain.

References

[22-1] *fib* Bulletin 90, 2019. Externally applied FRP reinforcement for concrete structures, *fib*, 2019.

[22-2] Bianco, V., Monti, G. et al., 2014. Design formula to evaluate the NSM FRP strips shear strength contribution to a RC beam. *Composites Part B: Engineering*, 56, 960-971.

[22-3] RILEM TC 234-DUC, 2016. Design Procedures for the Use of Composites in Strengthening of Reinforced Concrete Structures: State-of-the-Art-Report. https://link.springer.com/chapter/10.1007/978-94-017-7336-2_7

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In practice, the following aspects have to be considered:

- contamination of the concrete surface just before casting the overlay may reduce bonding significantly; likewise unfavourable climatic conditions (such as strong dry winds or solar radiation) can dry up the concrete surface and reduce bond strength;
- an appropriate roughening method shall be applied (e.g. high pressure water jetting), not harming the top surface of the concrete by inducing micro-cracking and allowing for laitance removal;
- inappropriate moisture content of the top surface of the concrete (above or below Saturated Surface Dry (SSD) condition) may locally affect the water-cement ratio of the overlay material and reduce bond strength;
- insufficient quality of the overlay concrete may cause larger constraint forces (e.g. due to shrinkage) and reduce bonding;
- insufficient quality of the overlay concrete placement may affect compaction of the overlay material near the interface and reduce bond strength;
- edge zones (where significant internal tensile and shear stresses may occur due to forces of constraint) have to be sufficiently secured.

23 Concrete to concrete

23.1 Definitions and scope

Concrete-to-concrete load transfer across interfaces has to be considered when two concretes are cast against each other at different times, usually after the hardening process of the older concrete is completed. While tensile loads have to be transferred via reinforcement when designing for the ULS, shear forces with their load direction parallel to the interface, normal forces perpendicular to the interface or a combination of both are to some extent transferred directly from concrete to concrete.

The topic is relevant in practice in the following situations:

- repair and strengthening of existing RC-members by means of new concrete layers;
- supplement of precast elements with additional concrete cast at the site;
- all situations at the site where, due to interruptions in the erection process, new concrete is cast against already completely hardened concrete;
- post-installed concrete elements (e.g. corbels) attached to existing members for the introduction of loads.

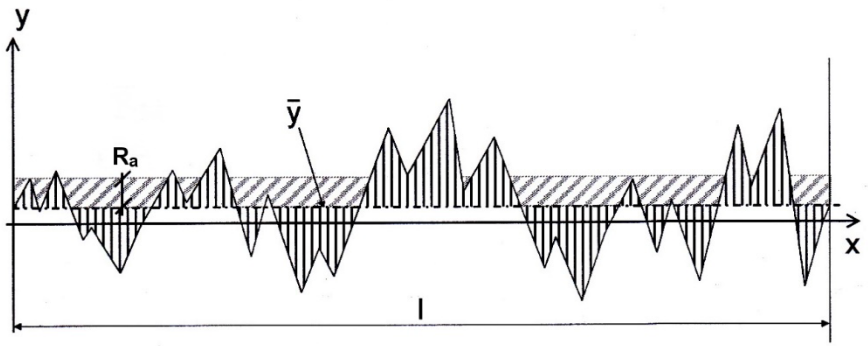

Figure 23.2-1: Average roughness R_a.

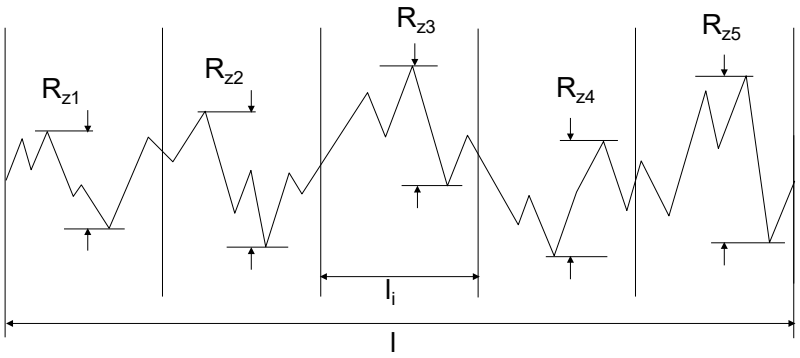

Figure 23.2-2: Mean peak-to-valley height R_z (l_i = l/5).

There exist several methods to measure and specify the roughness of a concrete surface. The sand patch method is a simple method widely used on the site. A defined volume of fine sand is spread on the surface; depending on the diameter of the circle the average roughness may be estimated as the mean height of the sand cylinder, that is, the sand volume divided by the circle's area The advantage of this

23.2 Interface roughness characteristics

There are several indicators to describe and quantify the roughness of a concrete surface. The most commonly used parameter is the mean roughness R_a (Figure 23.2-1) which represents the average deviation of the profile from a mean line ($\bar{y}$). It is determined as the mean value of profile heights along an assessment length l:

$$R_a = \frac{1}{l}\cdot\int_0^l \left|y(x)-\bar{y}\right|\cdot dx \approx \frac{1}{n}\sum_{i=1}^{n}\left|y_i-\bar{y}\right| \qquad (23.2\text{-}1)$$

$$\bar{y} = \frac{1}{l}\int_0^l y(x)\cdot \mathrm{d}x \approx \frac{1}{n}\sum_{i=1}^{n} y_i \qquad (23.2\text{-}2)$$

where:

l is the assessment length;

$y(x)$ is the profile height at position x.

Another frequently used parameter is the mean peak-to-valley height R_z (Figure 23.2-2), representing the average of maximum valley-to-peak-deviations R_{zi} within a certain number of assessment lengths, for example for 5:

$$R_z = \frac{1}{5}\cdot\sum_{i=1}^{5} R_{zi} \qquad (23.2\text{-}3)$$

Measuring the surface texture depth by the sand patch method works as follows: A volume of sand V is spread on the rough surface in a circular area with diameter D. The roughness parameter R_t follows from:

method is its simplicity and quickness. The disadvantages are that it is not very exact, only possible on top of horizontal or slightly inclined surfaces and that only the "peak-to-mean" roughness R_t ($\approx R_z/2$) can be determined.

$$R_t = \frac{4V}{\pi D^2} \qquad (23.2\text{-}4)$$

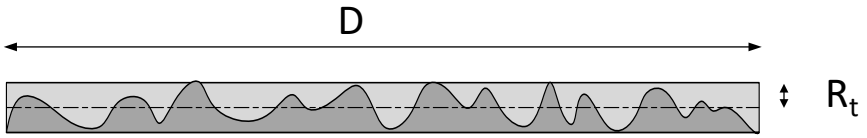

Figure 23.2-3*: Principle of sand patch method for the qualification of the roughness of an interface.*

Other advanced methods are due to their increased complexity preferably used for laboratory investigations. The advantage of these methods is that they allow for exact measurements of roughness profiles and therefore give a more detailed specification of the surface topography.

Examples are as follows:

- contact methods: mechanical-contact profilometer;
- non-contact methods: laser-triangulation;
- photogrammetry: digital imaging.

The simplified classification into the four categories very smooth, smooth, rough and very rough on the basis of the roughness R_t is not an exact method of categorization, but useful for practical design. However, the designer should be aware that identical values of the roughness R_t might still lead to different shear resistances due to differences in the actual surface topography and therefore engineering judgement is always required when determining the appropriate roughness category.

For design purposes, the surface roughness has to be classified into different categories. Depending on the roughness R_t derived from the sand patch method and the applied roughening method the categories in Table 23.2-1 may be defined as:

Table 23.2-1: Classification of surface roughness.

Category	$\boldsymbol{R_t}$ [mm]
Very smooth (e.g. cast against steel formwork)	Not measurable
Smooth (e.g. untreated, slightly roughened)	< 1.5 mm
Rough (e.g. sand blasted, high pressure water blasted etc.)	≥ 1.5 mm
Very rough (e.g. high pressure water blasted, indented)	≥ 3 mm

23.3 Mechanisms of shear transfer

Substantial experimental investigations have been performed since about 1960. The results show a very large scatter for the following reasons:

- The test setup affects the flow of forces within the specimen and a clear determination of the shear force distribution along the interface may be difficult even with small scale specimens;

The shear strength of interfaces between concretes cast at different times can be investigated experimentally on large scale members such as beams or slabs or on small scale specimens. There is a variety of small scale test setups such as tensile pull-off, slant shear, push-out or push-off shear tests. The main parameters decisive for the actual loadbearing capacity observed in tests are:

- interface roughness and surface texture;

– Depending on whether or not reinforcement or connectors cross the interface, the loadbearing behaviour may be completely different;
– Bond–slip and ultimate load are strongly influenced by the bond strength which, under laboratory conditions, can on the one hand be very good or, on the other hand, be intentionally eliminated by pre-cracking the specimen or reduced by the use of debonding agents in order to account for example for possible contamination at the site;
– In the case of an uncracked joint without reinforcement the shear stresses are transferred primarily into the edge zones so that the size and shape of the shear interface affect the ultimate shear strength τ_u.

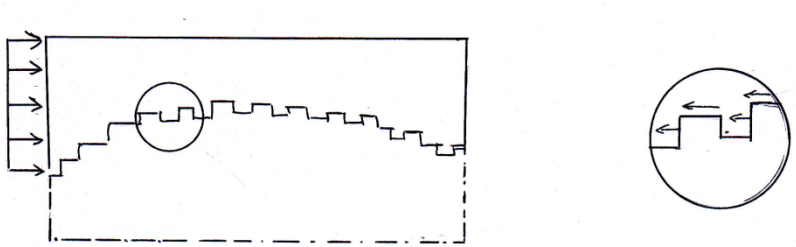

Figure 23.3-1: Simplified representation of the effects of adhesive bonding and mechanical interlocking.

Parameters influencing adhesive bonding and mechanical interlocking are, among others:

– preparation (roughening etc.) and cleanliness of interface;
– strength class of old and new concrete;
– quality of top layer of old concrete;
– porosity and moisture content of old concrete;
– quality, composition and properties of fresh new concrete;
– degree of shrinkage of new concrete;
– age of concretes.

The shear–slip characteristics of adhesive bonding and mechanical interlocking are characterized by a very stiff behaviour, only effective for shear slip values at about 0.05 mm or less when adhesive bonding predominates or somewhat more when mechanical interlocking contributes.

– cleanliness of surface;
– moisture content of the surface prior to casting the overlay;
– concrete strength and concrete quality;
– eccentricity/inclination of shear force;
– strong bond/pre-cracking/debonding before testing;
– ratio of reinforcement crossing the interface.

The main contributions to the overall shear resistance result from the following mechanisms:

– mechanical interlocking and adhesive bonding;
– friction due to:
  - external compression forces perpendicular to the interface;
  - clamping forces due to reinforcement and/or connectors;
– dowel action of reinforcement and/or connectors crossing the interface.

These mechanisms are activated at different interface shear slips.

Adhesive bonding and mechanical interlocking

Adhesive bonding and mechanical interlocking (Figure 23.3-1) may significantly contribute to the overall shear resistance provided the adhesion and interlocking effects are not reduced by contaminants along the interface. While adhesive bonding – that is adhesive forces due to chemical and physical bonding – can develop along smooth interfaces as well, mechanical interlocking requires an appropriate surface roughness. Taking into account the effect of adhesive bonding and mechanical interlocking for the ultimate bearing capacity implies that the related slip at failure is very small; for larger shear displacements the effects of adhesion and mechanical interlocking are considerably reduced due to loss of bond and fracture of protruding parts of the interface.

Compared to the other mechanisms contributing to interface shear transfer, adhesive bonding and to some extent also mechanical interlocking are sensitive to any contamination of the surface.

With increasing concrete strength, a moderate increase of the adhesive bond strength is observed. As an approximation, a linear relationship between bond strength and material's tensile strength may be established.

The effect of the frictional resistance in the case of rigid bond addresses actually the sticking contact situation, i.e. static friction, whereas non-rigid bond slip leads rather to sliding friction. The coefficient of the sliding friction situation is smaller than the effect of static friction, however, in the context of concrete to concrete bond design lower bound values are anyhow applied and no distinction is made for the sake of simplification.

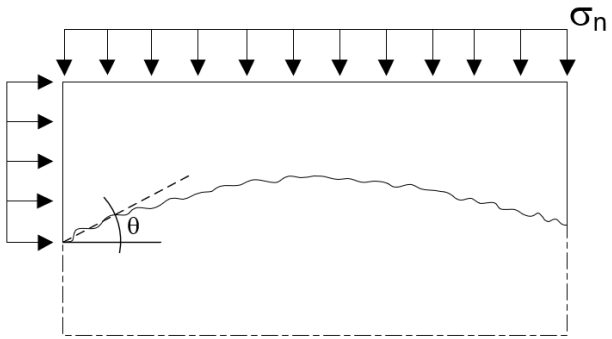

Figure 23.3-2: *Simplified representation of shear friction principle.*

According to a simplified shear friction theory (Figure 23.3-2), for a very smooth contact area (with only the general undulation) the shear resistance would be:

$$\tau = \sigma_n \tan\theta \qquad (23.3\text{-}1a)$$

Where:

Provided good bonding and mechanical interlocking are achieved by appropriate measures (clean surface, appropriate roughening, good concrete quality etc.) representative values for the mean shear resistance are for concrete grades ≤ C50:

Smooth interface:	0.5 – 2.0 N/mm^2
Rough interface (e.g. sand blasted):	1.5 – 3.0 N/mm^2
Very rough interface (e.g. high pressure water jetted):	2.5 – 4.0 N/mm^2

23.3.1 Shear friction

In the case of compression forces perpendicular to the interface, a so-called shear friction mechanism can develop depending on the roughness of the interface. Reinforcement and connectors can generate those compressive forces indirectly, since shear sliding normally goes along with joint opening, which leads to stretching of the reinforcement or connectors.

For a constant confining stress σc representative mean values for the coefficient of friction μ are for concrete grades ≤ C50 in the following ranges:

Very smooth interface:	**0.5** – 0.7
Smooth interface:	**0.6** – 1.0
Rough interface:	**0.7** – 1.2
Very rough interface:	**0.9** – 1.5

For design of interface shear transfer it is recommended to use for the beneficial effect of friction only the lower bound values (in bold).

$tan\theta$ is often replaced by the friction coefficient μ. Including the micro-roughness at the contact area (and mechanical interlocking effects respectively) results in the general basic expression:

$$\tau = \tau_a + \mu\sigma_n \qquad (23.3\text{-}1b)$$

The confining stress σ_c (compressive stresses are given positive sign) can as well be generated by reinforcing steel or connectors crossing the interface.

Shear–slip characteristics for friction at constant σc show a slight decrease of the shear resistance with increasing slip due to interface deterioration. If the confining action is obtained by reinforcement or connectors crossing the interface, the shear–slip relation may increase as long as the axial force in the reinforcement or connectors increases due to joint opening.

23.3.2 Dowel action

Shear–slip characteristics of dowel action show an increase of the shear resistance with increasing slip.

Dowel action of the connectors crossing the interface means first of all the bending resistance (Figure 23.3-3) which develops to its maximum with shear slips of up to approximately 0.1–0.2 times the bar diameter. With large slips the so-called kinking effect can be observed (Figure 23.3-3): in the interface intersection zone the inclined bar with large slips increasingly provides a contribution to the shear resistance due to the horizontal component of the tensile force in the bar.

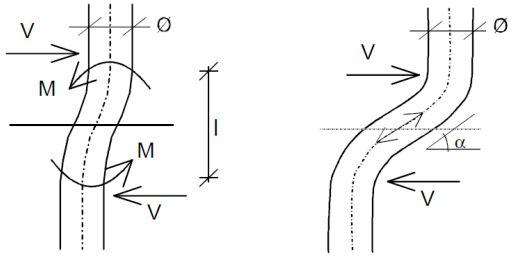

Figure 23.3-3: Dowel action - bending and kinking effect with large slips.

Dowel action refers to the bending resistance of connectors (reinforcing bars or dowels) crossing the interface: a shear slip along an interface leads to a lateral displacement between upper and lower connector ends, thereby inducing bending stresses in these bars which are superimposed by axial tensile forces due to the opening of the joint.

These tensile forces, on the other hand, do not allow for the full bending resistance resulting from the plastic moment of the bar to develop.

If there is no interface reinforcement present, the behaviour of an unreinforced joint is quite brittle; typically failure occurs with the loss of adhesion and mechanical interlocking with slips at around 0.05 mm or even less.

Reinforced joints with $\rho \geq 0.05\%$ (see definition of ρ in Eq. (23.4-4)) show a more ductile behaviour, depending on the reinforcement ratio and the interface roughness failure occurs typically with much larger slips (~0.5–1.5 mm).

A "rigid" bond–slip behaviour is to be expected when, in the case of smooth interfaces, no reinforcement or, in the case of rough interfaces, no or only small amounts of reinforcement, cross the interface. A noticeable amount of reinforcement ($\geq 0.05\%$) on the other hand allows for larger deformations until the ultimate failure load is reached, that is a "non-rigid" bond–slip behaviour where friction forces and dowel action are activated due to the slip.

23.3.3 Effects of interaction between the shear resisting components

In a real structure subject to shear loading the various mechanisms (i.e. adhesive bonding, mechanical interlocking, shear-friction and dowel action) interact, thereby affecting each other as a function of the shear slip. After failure of adhesive bond, with increasing shear slip also the mechanical interlock effect decreases quickly (reduction of micro-roughness and fracture of protruding parts of the interface, Figure 23.3-2). While also the shear-frictional resistance declines somewhat, the bending resistance of the connectors increases with larger slips.

23.4 Modelling and design

The ultimate resistance of an interface subject to shear forces may be approached by superposition of the single mechanisms of adhesion and mechanical interlocking, shear-friction and dowel action. As these mechanisms interact with each other and reach their maxima at different shear slips it is not possible to add them all together with their maximum values.

Depending on bond strength, roughness of the interface and degree of reinforcement either adhesive bonding (or mechanical interlocking respectively) or the other mechanisms of shear-friction and dowel action will dominate the overall loadbearing behaviour. Therefore two situations can be distinguished in general:

- strong adhesive bonding → "rigid" bond-slip behaviour:
  adhesive bonding is the main contributing mechanism to shear resistance;
- weak adhesive bonding → "non-rigid" bond–slip behaviour:
  shear-friction and dowel action are the main contributing mechanisms to shear resistance.

The single mechanisms may be summarized and approached as follows and given by the equations below.

23.4.1 Rigid bond-slip behaviour

Adhesion/interlocking + friction:

Besides other influencing parameters (see section 23.3) adhesive bond between concretes cast at different times is linked to the hydrophobicity of the existing concrete surface. Experimental investigations indicate that the adhesive bond strength can therefore be related among others to the roughness factor of Wenzel (ratio of the "true" contact area (following the profile) to the apparent (projected) contact area), whose identification in turn requires advanced measurement devices and depends on the density of the measuring grid.

Strong adhesive bond ("rigid" bond–slip) means that τa becomes predominant and any effect of the reinforcement becomes rather small.

In the case of horizontal shear transfer in composite concrete slab members (including top slabs of flanged beams such as T-beams) with cast-in-place toppings and intentionally roughened interface ($R_t \geq 1.00$ mm), the value of τ_a can be increased by 20%. The concrete contact surface shall be clean and free of laitance, dust or other adhesion-reducing particles and the surface preparation has to be subject to quality inspection.

Detailing of shear keys and definition of h_{key}: see section 23.6.

Background information on "Shear Friction" is provided in Baek et.al. [23-1] and on the "Extended Shear Friction Theory" in Randl 2013 [23-2]. Details on the dowel action effect are provided in Randl 2007 [23-3].

$$\tau_u = \tau_a + \mu \cdot \sigma_n \leq \beta_c \cdot f_{cc} \qquad (23.4\text{-}1)$$

where:

τ_a is the shear resistance due to adhesive bond/interlocking (see design values in section 30.1.3.7.2)

σ_n is (lowest) compressive stress resulting from a normal force acting on the interface (compressive stress given positive sign);

μ is the coefficient of friction (lower bound values are given in section 23.3);

β_c is a coefficient for the strength of the compression strut, recommended value on the safe side: $\beta_c = 0.3$;

Tensile stresses perpendicular to the interface reduce the adhesive bond, wherefore it is not recommended to consider rigid bond resistance in that case.

Shear keys: Design of interface shear transfer may be performed according to the "smeared approach" as per Eq. (23.4-1) considering the total area of the joint. However, large shear keys protruding in height h_{key} more than the maximum aggregate size may better be approached in design by addressing the contribution of the single keys. The joint's shear resistance may then be expressed via Eq. (23.4-1) by replacing the total joint area with the total sum of the relevant minimum base area of all keys, using a c_a-coefficient equal to 0.9 (which corresponds roughly to the pure shear strength of the concrete) and in case of external compression a coefficient of friction equal to 0.9.

23.4.2 Non-rigid bond-slip behaviour

If the adhesive forces cannot resist the shear loading, a transition to non-rigid bond-slip behaviour takes place and increasing displacements lead to an activation of the embedded reinforcement intersecting the joint. Thereby the bending resistance (i.e. dowel action) as well as a clamping effect resulting in frictional resistance are mobilized. In the case of rough interfaces, some interlocking effects can still contribute.

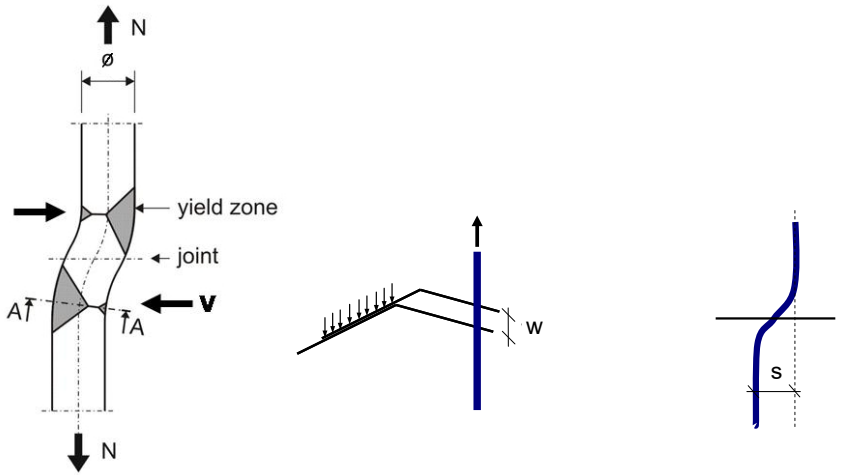

Figure 23.4-1: *Clamping force effect and dowel action.*

The upper limit of the bending resistance in Eq. (23.4-2) is derived on the basis of the "von Mises"-criterion (steel shear failure), usually becoming relevant only in high strength concrete.

Dowel action can only fully develop if not other premature failure modes such as concrete splitting govern. In order to prevent concrete splitting, a minimum side distance to the member surface of 5 $Ø$ (if no transverse reinforcement is provided along the edge) and a distance to the concrete edge in the direction of the acting interface shear force of at least 10 $Ø$ as well as a minimum spacing of the reinforcement or connectors of at least 15 Ø in the direction of the acting interface shear force is required. Upper limits for the spacing (maximum distance of interface reinforcement) are provided in section 23.6.2.

In case that sufficient anchorage is provided (see design section on anchorage of reinforcement) and full yielding under axial tension to be expected, the interaction

23.4.2.1 Dowel action

The resistance V_F of the reinforcement or connectors to an acting shear force may be approximated as follows:

$$V_F(s) \approx V_{F,\max}\sqrt{\frac{s}{s_{\max}}} = \kappa_{2,\max} \cdot A_s\sqrt{f_{cc} \cdot f_y}\sqrt{\frac{s}{s_{\max}}} \le \frac{A_s \cdot f_y}{\sqrt{3}} \qquad (23.4\text{-}2)$$

where:

$V_{F,\max}$ is the maximum value of dowel action as defined in Eq. (23.4-2): $V_{F,max} = \kappa_{2,max} \cdot A_s\sqrt{f_{cc} \cdot f_y}$;

$s_{\max}$ is the slip when $V_{F,\max}$ is reached: $s \le s_{\max} \approx 0.10\,Ø - 0.20\,Ø$;

$\kappa_{2,\max}$ is the interaction coefficient for flexural resistance at slip $s_{\max}$, where $\kappa_{2,\max} \le 1.6$ for circular cross-sections and C20–C50.

Interaction of tensile forces and bending leads to a reduction of the maximum possible dowel action, especially when the surface is rough so that substantial tensile forces are generated in the reinforcement/connectors. The reduction of the shear strength may be approached by multiplying $V_{F,\max}$ with the interaction factor κ_1 taking into account the degree of utilization:

$$V_F(s) = V_{F,max} \cdot \sqrt{\frac{s}{s_{max}}} \cdot \sqrt{1-\kappa_1^2} \qquad (23.4\text{-}3)$$

where:

$$\kappa_1 = \frac{\sigma_s}{f_y}$$

The reduction factor κ_1 accounts for the fact that the axial tensile force in the reinforcement/connectors may be reduced due to simultaneous bending (Figure 23.4-1) and/or reduced anchorage of the bars and, moreover, due to the fact

factor κ_1 may be set equal to 1.0. In this case, the dowel action effect disappears completely (Eq. (23.4-3)).

In the case of concrete overlay applications or other situations where the assumption of activation of the full tensile yield strength is not justified, it is generally recommended to apply reduced values for κ_1: For post-installed reinforcement or connectors with circular cross-sections, the interaction factor κ_1 may be assumed equal to 0.5 for rough and very rough interfaces (see background information on recommended coefficients in Randl 2013 [23-2].

Eq. (23.4-4) applies for deformed bars with a tensile yield strength in the range of 400 to 800 MPa.

In the case of plain bars, the dowel action effect can be generated, while the clamping force effect and the resulting friction resistance depend on the anchorage and bond properties of such bars.

Interlocking effects are acting together with adhesive bond, but can be observed in the case of deliberately roughened interfaces even in situations when adhesive bond is broken. This contribution to the overall interface shear resistance shall only be applied in the case of roughness categories “rough” or “very rough”.

While for “very smooth” interfaces dowel action of intersecting reinforcement or connectors represents the main resistance mechanism, with $\kappa_1 = 0$ and $\kappa_2 = 1.5$ as recommended coefficients, roughening of the interface leads to axial stresses in connectors with $\kappa_1 > 0$ and reduced κ_2 – values. For overlay situations and interface that interface shear failure may occur already at low slip values smaller than s_{max}: $\kappa_1 = \sigma_s/f_y \leq 1.0$.

23.4.2.2 Superposition

When superposing the different mechanisms, the following aspects have to be taken into account:

- the different mechanisms interrelate and thereby influence each other;
- interaction of tension and bending in the connectors leads to an interrelationship between clamping force and dowel action;
- maximum contributions of different mechanisms occur at different slips.

The ultimate shear stress at the interface resulting from the different single mechanisms can be described in an overall simplified approach as follows:

$$\tau_u = \underbrace{\tau_r}_{\text{Interlocking}} + \underbrace{\mu \cdot \left(\rho \cdot \kappa_1 \cdot f_y + \sigma_n\right)}_{\text{Shear friction}} + \underbrace{\kappa_2 \cdot \rho \cdot \sqrt{f_y \cdot f_{cc}}}_{\text{Dowel action}} \leq \beta_c \cdot f_{cc} \qquad (23.4\text{-}4)$$

where:

τ_r represents interlocking effects for rough interfaces;

ρ is ratio of reinforcement crossing the interface ($\rho = A_s/A_c$);

β_c is a coefficient for the strength of the compression strut, recommended value on the safe side: $\beta_c = 0.3$;

μ is the coefficient of friction (lower bound values applicable for design are given in section 23.3).

The interaction factors κ_1 and κ_2 in Eq. (23.4-4) take into account that:

- the reinforcement or connectors are subject to bending and axial forces simultaneously;

roughness categorized as “smooth” or rougher, it is recommended to limit κ1 to 0.5 and use the following corresponding κ_2 – values: $\kappa_2 = 1.1$ for “smooth” and 0.9 for “rough” and “very rough” surfaces. Higher κ_1 – values may be applied in situations where larger strains (or even yielding) may be argued, with the coefficient κ_2 reduced accordingly (Eq. (23.4-3)).

In the case of horizontal shear transfer in composite concrete slab members (including top slabs of flanged beams such as T-beams) with cast-in-place toppings and intentionally roughened interface ($R_t \geq 1{,}00$ mm), the value of c_r may be increased by 20%.

Tests performed with large scale specimens (e.g. beam tests) indicate that with rough interfaces (good bonding provided) under dynamic loading usually no significant strength reduction occurs compared to monolithic beams subject to fatigue loading.

However, if the adhesive bond resistance is exceeded under characteristic load (or likewise ~50% under fatigue loading), cracks are likely to occur. In that case, due to the relative displacements in the crack, more significant deteriorations have to be considered under fatigue loading:

- reduction of mechanical interlocking;
- reduction of shear-friction due to an increasing amount of fine grains;
- local plasticizing and deterioration of concrete around reinforcing bars/connectors crossing the interface.

Main factors affecting adhesive bond are the surface preparation of the substrate concrete and the fresh concrete properties of the new layer. A good bond is generally achieved when the concrete used for the overlay exhibits low kinematic viscosity in the fresh state. To estimate such characteristics slump-flow tests may be carried out.

- the maximum values of the different contributions occur with different slips.

The interlocking effect may be approached as follows:

$$\tau_r = c_r \cdot f_{ck}^{1/3} \qquad (23.4\text{-}5)$$

with:

c_r equal to 0.15 for rough and 0.3 for very rough interfaces on characteristic level ($c_r = 0$ for lower roughness categories). According design values are given in section 30.1.3.7, considering an overall safety factor of 1.5.

23.4.2.3 Additional design recommendations for interfaces subject to fatigue loading:

The main question concerning interface shear behaviour under fatigue loading is whether or not cracks along the interface are to be expected. In the case of monolithic behaviour (good bonding provided) the overall behaviour of the member can be judged according to section 30.2.2. However, as a simple approximation, a reduction of τ_a to 50% under fatigue loading may be assumed.

As soon as cracks appear, they tend to cause more significant deteriorations along the interface under dynamic loading than observed with monolithic concrete members. When cyclic shear loading is expected, an overall reduction of all contributing mechanisms to about 40% of the static resistance according to Eq. (23.4-4) is recommended, if no further evidence is available.

23.5 Verification of bond strength and consideration of advanced materials

There is evidence that advanced concrete materials such as high strength or ultra-high performance concrete usually lead to an improvement of the bond properties. This can be taken into account in design by applying the average concrete strength of substrate and overlay instead of the lower value of the two in Eq. (23.4-1). This

An optimal concrete viscosity in this context is achieved when the concrete can be classified as S4 or above according to the slump classes defined in EN 206-1 [23-4].

Compared to normal strength concrete, high strength concrete has a densified microstructure which leads to a densification of the material near the interface. In fibre reinforced concrete, reduced shrinkage or thermal effects on the interface can improve the bond strength.

The pull-off test is widely used and specified in different standards such as ASTM C1583 [23-5]. Observed ratios between shear bond and tensile pull-off bond strength are in the range from 0.6 to about 2 Zanotti and Randl [23-6], depending mainly on test setup and interface roughness. It is recognized that the test has some limitations, being sensitive to various factors such as system misalignment, improper gluing of the loading disc, drilling-induced damage, core geometry, and related stress disturbances, hence frequently leading to premature failure in the substrate. Nevertheless, the test can provide a conservative assessment of the bond quality.

In fact splitting tensile bond tests show usually less scatter and provide more consistent results, but it is rather difficult to gain appropriate specimens from core drills.

strength increase can only be applied in the context of rigid bond-slip behaviour and should be verified by testing on site (see below). The maximum interface shear resistance is limited by the pure shear strength of the weaker concrete which can be approximated roughly by its tensile strength.

Specialized concretes require higher quality concrete practice. In general, the quality of the interface and the bond may be more different from the quality of the bulk material overall and strict quality control should be put in place to ensure the quality of the interface. In particular, proper compaction of the concrete near the interface needs to be achieved to avoid a localized increase of porosity, which would jeopardize the improvement.

Testing: Due to its simplicity, the pull-off test is suitable for performance/quality assessment at the construction site.

A mean tensile bond strength of at least $1.2 \cdot c_a \cdot f_{ctm}$ should be achieved with this test, with a coefficient of variation $v_x \leq 0.20$. The values c_a and f_{ctm} are linked to the configuration used for calculation of the interface shear resistance, with f_{ctm} representing the mean tensile strength.

It can be expected that the test will not always lead to a pure adhesive failure. Typical failure modes include mixed adhesive-material failures. Material failures also occur; in that case, the actual bond strength exceeds the failure stress. However, as long as the tensile stress at failure is higher than the above given value, that is an indication that the desired bond quality and the specified bond strength are achieved.

23.6 Detailing

23.6.1 Arrangement of connectors

Reinforcement or dowels intersecting the interface are required if the shear load in the interface under consideration cannot be resisted by the adhesive/interlocking effect of the roughened joint surface alone. A stepped distribution of the connectors with respect to the interface shear load distribution may be used. In the case of smooth surfaces, the flexibility of the connectors even allows for a redistribution of forces and thus a uniform distribution of the connectors.

The recommended reinforcement ratio ρ_{min} has been derived based on the model that the shear force at loss of adhesion is taken up by shear-friction, dowel action and (only rough interfaces) partly by mechanical interlocking effects.

In the case of slabs, a total overall collapse of adhesion and interlocking effects does not have to be taken into account so that the minimum reinforcement may be reduced to ~50%.

In general, the spacing between such bars crossing the interface shall not exceed the following:

- shear transfer direction: 2,5 h ≤ 300 mm, where h is the overall depth of the slab;
- perpendicular to shear transfer direction: 5 h ≤ 750 mm (≤ 375 mm to the edge).

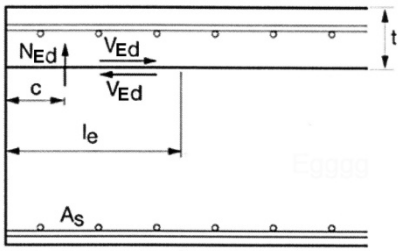

Figure 23.6-1: *Edge reinforcement.*

At the perimeter of a new concrete layer, the concrete dries out and tends to contract, thereby producing tensile forces and delamination perpendicular to the interface, leading eventually to cracking of the concrete. On the load side usually in practical design restraint forces along the perimeter are not taken into account because the realistic determination of the effects of differential shrinkage and

23.6.2 Minimum interface reinforcement

If interface reinforcement or dowels are required because the shear load in the interface under consideration cannot be resisted by adhesive bonding and interlocking effects alone, a minimum amount of reinforcement $\rho_{min} = A_{s,min}/A_c$ where $A_{s,min}$ is the intersecting reinforcement cross section and A_c the related interface area, should be foreseen in order to prevent brittle failure at loss of adhesion:

beam members: $\rho_{min} = 0.20\, f_{ctm} / f_{yk} \geq 0.001$;

slabs: $\rho_{min} = 0.12\, f_{ctm} / f_{yk} \geq 0.0005$.

23.6.3 Interface edge reinforcement

In order to counteract possible delaminating of the concrete overlay, reinforcement or dowels should be installed all along the perimeter if other adequate structural provisions are not taken. In the absence of more exact calculations, as an upper limit, the tensile cracking force of the new concrete overlay may be introduced as the maximum interface shear force to be expected along the edges (Fig. 23.6-1), and the reinforcement/connectors may be designed accordingly:

$$a_s = t \times f_{ctm} / f_{yk} \qquad (23.6\text{-}1)$$

where:

a_s is the cross section of the intersecting edge reinforcement per unit length;

t is thickness of the new concrete layer;

f_{ctm} is the tensile strength of the new concrete layer (reduction due to early age effects might be taken into account).

temperature gradients is difficult without knowing all the influencing factors. Therefore appropriate edge reinforcement is highly recommended.

The thickness of the additional concrete overlay is usually limited and lower than the height of the existing concrete member. Moreover, especially in bridge design, the requirements concerning the rather large concrete covers for bridges have to be fulfilled. Therefore the connectors usually need a special kind of end anchorage at the upper end. This end anchorage has to be designed in such a way that premature pull-out or concrete cone breakout can be excluded.

The minimum anchorage length of a connector (deformed reinforcing bar) with a diameter $\varnothing$ in the existing concrete should not be less than 6 $\varnothing$ in the case of smooth and 10 $\varnothing$ in the case of rough interfaces.

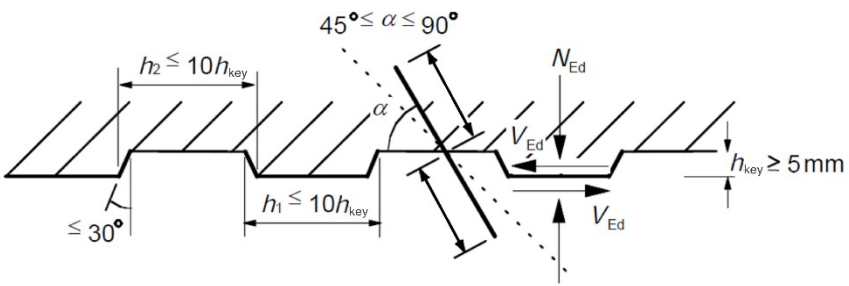

Figure 23.6-2: *Shear key geometry*.

23.6.4 Anchorage of interface connectors

The connectors have to be anchored appropriately in the old as well as in the new concrete in order to avoid premature pull-out failure.

If post-installed reinforcement or dowels are used in combination with the design approach based on the provisions given in section 23.4 and section 30.1.3.7, the tensile force to be anchored may be assumed as:

$$N_{Ed} = \kappa_1 \cdot A_s \cdot f_{yd} \qquad (23.6\text{-}2)$$

For detailed design of the anchorage loaded by N_{Ed} the possible failure modes known for bonded anchors have to be checked (see also Chapter 25).

23.6.5 Shear keys

A shear key should typically have proportions as recommended in Figure 23.6-2 in order to use the design values for very rough interfaces (Table 30.1-5). In addition, the base length h_1 of a key should be at least three times its height h_{key} (thus 3 $h_{key} \leq h_1 \leq 10\ h_{key}$).

References

[23-1] Baek et al. Baek, J.-W., Kim, S.-H. et al., 2020, Shear-Friction Strength of Low-Rise Walls with 600 MPa (87 ksi) Reinforcing Bars, *Structural Concrete*, 14:3, 230-241.

[23-2] Randl, N., 2013. Design recommendations for interface shear transfer in *fib* Model Code 2010, *Structural Concrete*, 14:3, 230-241.

[23-3] Randl, N., 2007. Load Bearing Behaviour of Cast-in Shear Dowels, *Beton- und Stahlbetonbau 102 (2007), Special Edition*, 31-37.

[23-4] EN 206:2016. Concrete – Specification, performance, production and conformity, CEN, 2016.

[23-5] ASTM C1583, 2020. Standard Test Method for Tensile Strength of Concrete Surfaces and the Bond Strength or Tensile Strength of Concrete Repair and Overlay Materials by Direct Tension (Pull-off Method).

[23-6] Zanotti and Randl Zanotti, C., Randl, N. , Are concrete-concrete bond tests comparable? Cement and Concrete Composites Volume 99, May 2019, Pages 80-88

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That mutual restraint between steel and RC members is not discussed in this chapter, however some concepts of anchor devices can be applied in these cases. Conversely the interconnection between steel and concrete/RC components that gives a composite behaviour is dealt with.

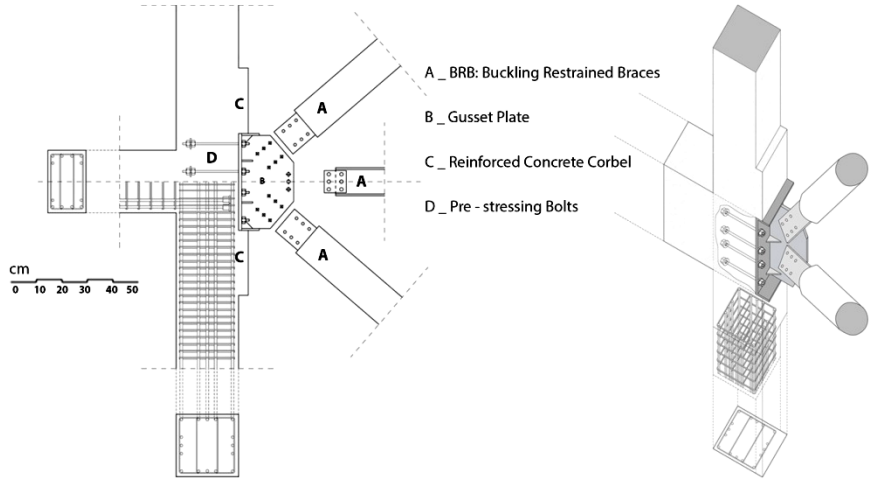

Figure 24.1-1: Connection between steel and RC elements.

Interaction between concrete and steel components can be classified as follows:

- adhesion (pure bond) between two materials;
- frictional interlock provided by peculiar shapes of the interface profile;
- mechanical interlock provided by specific treatments and deformations of the steel interface (i.e. indentations and embossments in sheeting);
- dowel action provided by anchor devices and systems.

In all cases the concrete-to-steel interfaces play a governing role with regard to the design of many composite steel-concrete structures and define the type and grade of interconnection between components. This section is intended to give

24 Concrete to steel by mechanical interlock

24.1 Classification of interaction mechanisms

Design of members with components made of concrete and steel, both hot-rolled and cold-formed, needs proper consideration of interaction mechanisms.

A variety of details exist, which depend on several aspects, such as the type of members to be connected, the actions to be transmitted and the design performance requirements.

A fundamental classification can be made according to the nature of interaction that is needed:

- mutual restraint between steel and concrete/RC members and/or substructures (Fig. 24.1-1);
- interconnection between components which allows steel element and concrete/RC elements to behave as a single composite structural member (Fig. 24.2-1).

In the first case, steel members are connected to concrete through mechanical devices and generate localized actions in the concrete/RC element.

In the second case, stiffness and strength of the interface or connection devices lead to the activation of the composite behaviour of the member trough a local or distributed interaction. The solutions to connect steel and concrete/RC elements can also consist of either post-installed fastenings, according to section. 30.14, or adhesive layers between the two parts.

general guidance, because the choice of the interaction action depends on the type of members and the type of loading.

Interaction between the interfaces aims at limiting relative displacements (slips), so that elastic and/or plastic analysis of cross-sections under both flexural and axial forces is allowed to apply to the composite element. Thus, this interaction strongly influences the overall structural response of members, that become composite elements, but also in the load introduction zones or in the critical regions, where high internal forces develop.

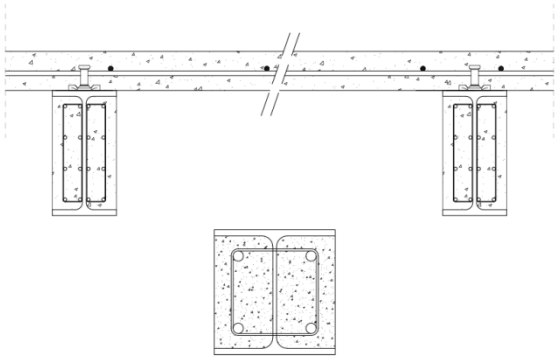

Figure 24.2-1: Examples of encased section

Relevant standards apply to production and design of steel sheeting as formwork; relevant rules are also available for the determination of the minimum nominal thickness of the steel sheeting.

An alternative solution that meets design requirements for composite members is represented by a sheeting with a re-entrant cross-section, so that frictional interface stresses can take advantage of lateral confinement actions due to shrinkage and

24.2 Bond of metal sheeting and profiles

24.2.1 Introduction

Combination of concrete and steel members through bond requires the development of composite action, which involves extended interface surfaces.

The shear transfer at the interface is due to three actions: chemical adhesion, interface interlock or active friction. Chemical adhesion acts separately from the other two mechanisms and is lost when interface slip starts. The friction action is active when the surface detachment is avoided and can be improved by load conditions and indentations/protrusions of the steel surface, adding interlocking to the interface. This last contribution that can occur in composite encased members, slabs made of metal sheeting and reinforced concrete slabs or profiles with a continuous connection device, is presented in the following.

24.2.2 Metal profiled sheeting

The use of metal sheeting is a common solution to realize composite slabs, as it takes advantage of the load bearing capacity of the cold-formed steel as temporary formwork supporting the hardening concrete in the construction stage.

The steel sheeting is usually profiled with ribs to improve its flexural stiffness and strength, but also mechanical interlock at the interface. However, a smooth sheeting is basically inappropriate for composite applications because its pure bond

flexural deformation.

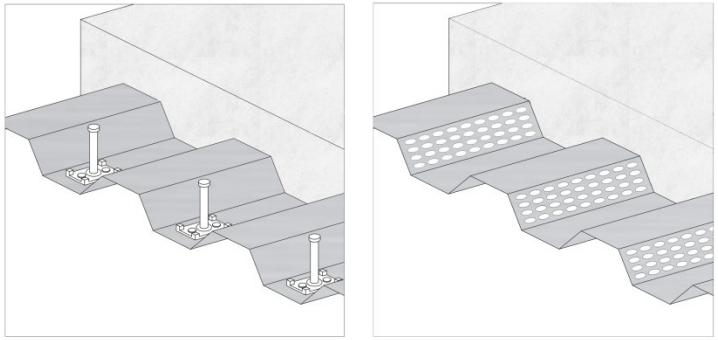

Figure 24.2-2:Metal sheeting with end connectors or embossments.

Relevant standards can be used to make sure that the requirements concerning the aspect ratio of steel components and/or flanges depending on the steel grade are satisfied.

The requirements for the structural behaviour of interfaces between steel and concrete depend on the type of cross-section, the design level of composite interaction, and on the load introduction mechanisms and details adopted.

is not enough to transfer shear forces throughout the contact interface in order to develop the composite action.

Conversely, mechanical interlock is improved by friction action provided by local deformations (indentations and protrusions) in the steel profile. Many solutions for the deformations of the profiles have been proposed, recently also small crown-shaped cuttings in the profile webs, as a replacement for the common embossments.

Mechanical connectors can be used in order to prevent end relative slip between the two components, as described in section 24.3.1. Deformation of end steel webs can be used as an alternative to additional steel devices when non-convex profiles are used. Example of solutions for improving the stress transfer at the interface are shown in Fig. 24.2-2.

24.2.3 Steel profiles

Several solutions can be used to increase the performance of steel profiles under combined loads (axial force, bending moment and shear force).

The structural performance of members can be increased by either fully encasing the steel profile or filling hollow steel sections.

In both cases, the effectiveness of the steel–concrete interface is critical for the development of composite action. Adequate structural performance is needed in order to ensure enough composite behaviour of the members and prevent local buckling of steel components.

Pure bond can give a good interaction for encased members but it is not suitable in the case of partially encased members subjected to flexure. Conversely, members in compression (e.g. composite columns) can take advantage of such an interaction mechanism. However, the load transfer regions can be critical.

Relevant codes report detailed data concerning characteristic strength values to be used in specific design contexts.

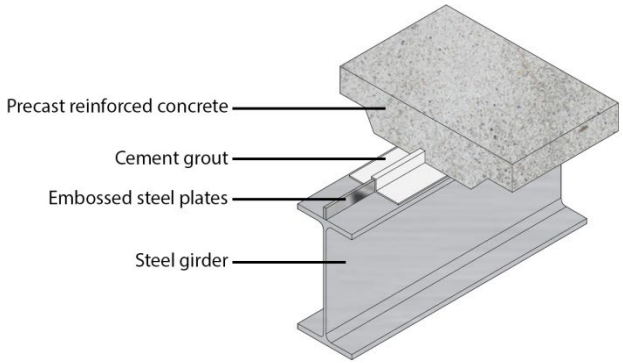

Figure 24.2-3: Steel profile with embossed steel plates.

Assuming some similar characteristics or defaults, it can be possible to compare the shear stress-slip relationship in different types of steel sheeting profiles. The ratio between the width of bottom opening, b_r, to the width of the top one, b_f, affects the shear-slip behaviour of the composite slab. Hence, the greater the b_r/b_f ratio the

24.2.4 Interface strength

As stated above, the interface strength depends on the type of shear transfer mechanism, such as chemical adhesion, interface interlock and active friction.

Chemical adhesive bond between concrete and steel has a strength of approximately 0.1 MPa and occurs over the total area between the steel profile and the adjacent concrete. Adhesion significantly drops as slippage starts and cannot be recovered. Thus full interaction exists up to reaching the adhesion capacity, so that the theory of elasticity can be used to determine the interface shear stresses. In metal sheeting the contact area is higher in the case of trapezoidal ribs and low deformed ribs. In a composite beam a possibility for activating the composite action is to improve the adhesion between the steel profile and the concrete through longitudinal embossed steel plates welded longitudinally to the upper flange of the steel girder (Fig. 24.2-3). In this case, the surface of the rib is roughened or holed.

Interface interlock is related to local passive restraint given by embossments and indentations of the steel profile. In metal deck the bond strength due to interface interlock is strongly influenced by the shape of the ribs, the thickness of the sheeting, the size and frequency of the embossments. The strength generated by this mechanism can range from zero for smooth trapezoidal ribs without embossment to about 0.8 MPa for deformed ribs with embossments. The strength levels are related to the area of the steel rib that is in contact with the concrete.

Active friction provides a contribution to the shear strength depending on the normal forces applied across the interface and by the surface treatment that govern the value of the friction coefficient. Common values of active frictional strength are about 0.003 MPa, much lower than values provided by adhesion and interface interlock. However, its contribution generally is not negligible, since it develops all over the interface surface.

24.2.5 Shear stress–slip relationships

The shear stress–slip relationship of the steel–concrete interface depends on the surface properties of the steel component.

In composite slabs, the longitudinal slip between steel sheeting and concrete is restrained by the shaping of the steel sheet and the embossments or indentations on

lower the bond strength.

Fig. 24.2-4 shows a typical result of a flexural test on trapezoidal sheeting, with a maximum strength, a plastic branch and a softening behaviour.

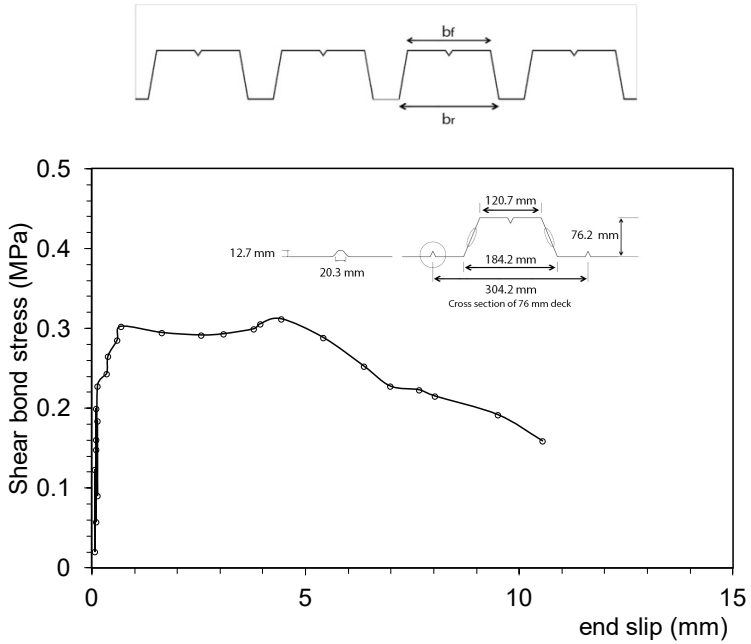

Figure 24.2.4: Typical experimental result of a flexural test on a trapezoidal steek sheeting. [24-1]

Fig. 24.2-5 shows typical shear bond-slip relations. The curves exhibit a strength increase until the ultimate (peak) shear stress value and a consequent descending curve until a stabilized shear stress value is reached. The peak is due to friction interaction (curve B for L-shaped deck) while the residual value is the actual interlock strength. During this last phase the steel deck stiffness, which depends on its thickness and the rib height, plays a key role. For design purposes, in analogy to similar interaction phenomena (i.e. bond of smooth reinforcing bars), in the theoretical stress–slip relations the peaking of the stress can be neglected and a linear elastic–plastic or rigid-plastic response with a plastic plateau at the residual stress level, can be assumed. The curve A in Fig. 22.2-5 can be obtained by tests on greased L-shaped steel sheeting, i.e. eliminating the friction interaction shown by the sheet surface. Their function is the same as corrugations on reinforcing bars for concrete, but the resistant mechanism is quite different. The bond strength between smooth steel sheets and concrete is low and allows slip at the interface. This deficiency is overcome by introduction of a shear transferring mechanism through mechanical interlock obtained with deformations of the surface.

The strength of the horizontal shear bond depends on many factors, among which the shape of the steel profile (ribs), type and spacing of embossments/protrusions/indentations, thickness of steel sheeting, arrangement of load, height/length ratio of the shear span (slenderness), natural clamping due to curvature under bending (in case of flexural behaviour), thickness of concrete, strain in the steel sheeting (i.e. loading level), and type of end anchorage. Because of these many influencing factors, it is not possible to provide representative design values that can be applied to all cases.

More in general three categories of steel sheeting can be defined: inverted L-shaped ribs, trapezoidal ribs, and re-entrant ribs. The steel sheeting with L-shaped ribs have excellent bond properties (i.e. until 0.8 MPa), but are structurally weak in terms of composite slab behavior; conversely the trapezoidal ribs give the best structural performance and the lower bond strength due to the tendency of detaching from concrete. Steel decking with re-entrant ribs is the intermediate solution for both the performance of structural behavior and composite action (bond strength).

The ribs shape influence the interlock strength. The re-entrant profiles provide greater bond strength (0.6-0.7 MPa) respect to trapezoidal shapes (0.1-0.3MPa).

The ranges of the interlock strength is due to the ribs shape but can be further improved by embossments, protrusions and indentations, which are still assessed in new specific products.

the curve B.

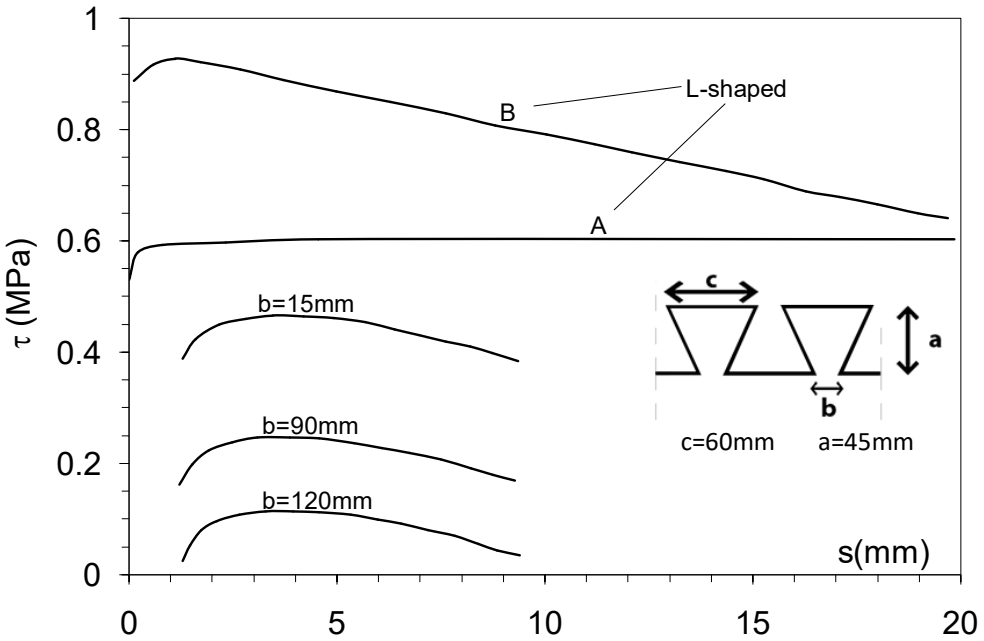

Figure 24.2-5: Shear stress–slip relationship for different types of sheeting in push test. the figures have to be reviewed.[24-2]

24.2.6 Influence of the type of loading

The structural behaviour of the interface is influenced by the type of loading. A satisfactory assessment of the influence of repeated loading and sustained loading has not been fully established yet.

Long term phenomena that develop in the concrete components should be taken into consideration, since they can influence the performance of the interface, especially when concrete filled members are used.

Repeated loading – that gives fatigue phenomena – needs stress and slip limitations at the interface level and should be taken into careful consideration.

Requirements for flexural tests on full scale members (type 1) – for example assessing the interface behaviour between steel and concrete components – can be found in relevant codes.

The experimental setup of shear bond tests on small scale elements (type 2) is similar to those used for tests on connection devices. Average forces and slip measured at the steel–concrete interface enable the determination of local constitutive relationships. The latter is influenced by the interface normal stresses depending on the type of the composite member: thus, specific force-controlled devices have to be used in order to reproduce realistic interface stress conditions.

Further, the head shear stud, as welded type, the **channel connector and T-connector** have been investigated such a long time (Fig. 24.3-1).

Damage tolerance at the interface level can become critical for specific applications.

24.2.7 Determination of properties by testing

As already introduced above the shear stress–slip constitutive relationships can be obtained from two main types of experiments:

– flexural tests on full scale members (type 1);
– shear bond tests (push test) on small scale elements (type 2).

24.3 Mechanical interlock

24.3.1 Introduction

Mechanical interlock is generally used for load transfer into concrete components and structures or to connect elements.

Over the years, many devices have been designed, however only the stud connectors have been widely investigated, thus the amount of information generated, including all the experimental values obtained from standard test of push-out, have led to fairly reliable formulations. For each typology of connectors, there are advantages and disadvantages; hence, their choice is driven by the specific purpose and according justified compromises.

24.3.2 Types of connectors

It's possible to classify the connectors into **four main groups**:

1 No-welded connection;
2 Welded connection;

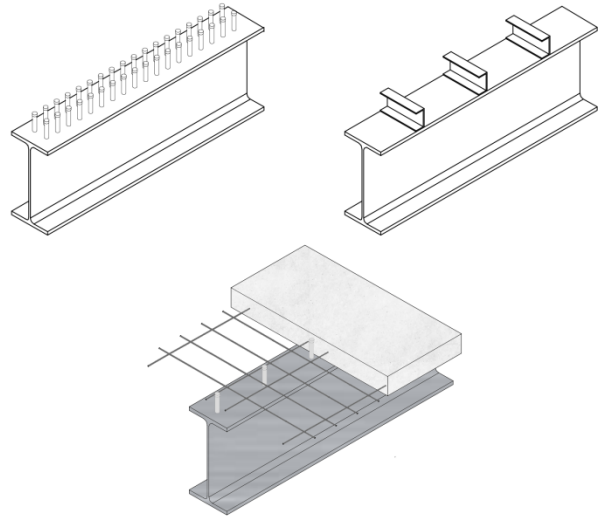

Figure 24.3-1: Studs and channel shear connectors.

Many other solutions have been developed and design provisions are still in progress to realize connectors that can be welded or, more conveniently, arise from the steel profile (Fig. 24.3-2). **Perfobond ribs** are innovative system developed to overcome the fatigue problem in bridges. Various parameters were investigated to predict shear strength: concrete strength, diameter and number of holes, height length and thickness of the rib, presence of added reinforcements. Also, the **Y-type perfobond** rib is a new type of connector device to complement the structural characteristics and workability of conventional perfobond rib shear connectors. The shear strength depends on concrete, thickness and bending of the rib, and reinforcement. **Pin-connector** is being investigated by experimental test; the height of a pin is only of about 3- 5 mm and the failure mode depends on various parameters such as pin distance and number, pin geometry, embedment depth and concrete and steel strength.

The most famous type of shear connector that can also arise from steel profile is the so-called **composite dowel**. Depending on the cutting line, there are different types of composite dowels (SA, PZ, CL, MCL and additional different shapes). A lack of standards for composite dowels with the economic clothoid and puzzle

3 Connectors arising from steel profile;
4 Connection by adhesion, interlocking and friction.

Among the **no-welded connection**, there are some devices used nowadays, as the bolted ones, but it still in need for further investigations about experimental test and formulations available to predict the shear strength.

The **head shear stud** is the most widely used type of welded connector, due to a simple and rapid application. Maybe it is one of the most investigated connectors over the years with shear strength formulations available in literature and codes, therefore a formulation is proposed in section 24.3.3.

Many other types of connectors can be realized as welded ones using various shapes of devices (as the channel ones) but some of the them can be obtained directly arising from the steel profile (as the composite dowel).

shapes has led to hesitate regarding applications. Different studies propose some formulations about capacity against concrete or steel failure, but the topic is still in progress.

Among new connectors, many are not much investigated, such as L-shape perfobond rib shear connector, oscillating perfobond strips, waveform strips, pyramidal shear connectors, hat shear connectors, cut in web.

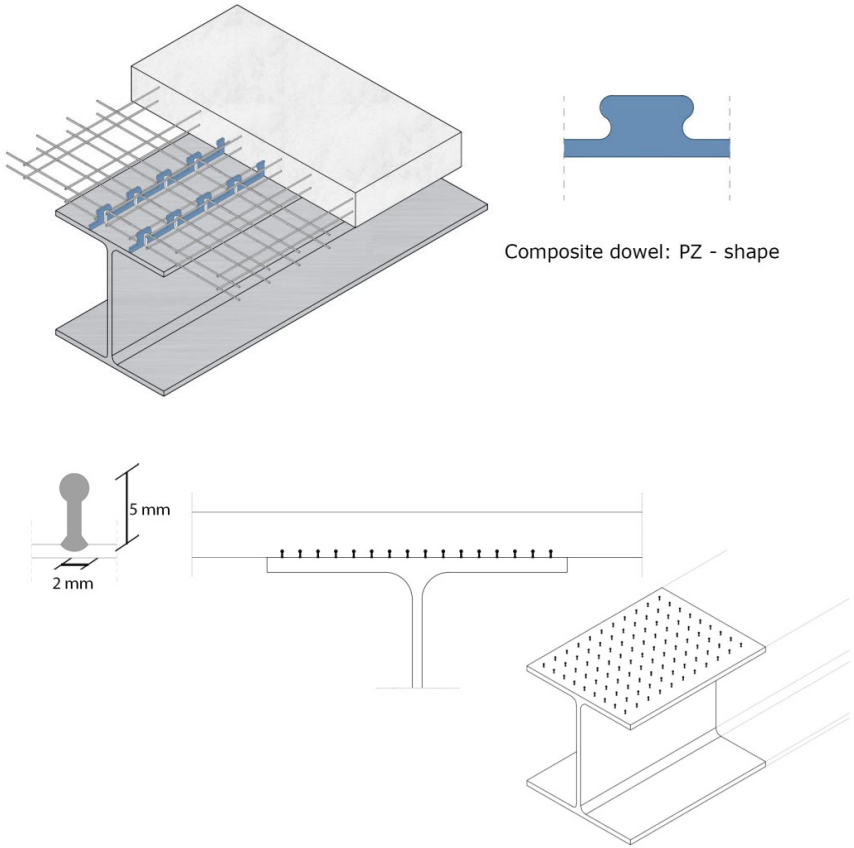

Figure 24.3-2: Examples of innovative connectors.

The connection between the concrete and the plates, which is realized by gluing the plates with a thin layer of epoxy resin, is characterized in general by good performance, but premature failures due to application detailing and debonding

For existing RC structures, externally bonded steel plates represent a traditional strengthening intervention aimed to improve both the flexural and the shear strength of the elements. The bond behaviour of such a strengthening system can be

phenomena due to an excess of tensile stress in the substrate can occur depending on various parameters.

It is well known that debonding phenomena generally occur within the concrete substrate and not in adhesive layer because of their higher mechanical properties in comparison with those of concrete. However, premature debonding may occur also if an appropriate surface preparation is lacking. In case of debonding in the concrete, the failure load is related to both the mechanical properties of concrete (tensile strength and fracture energy) and the axial stiffness of the external reinforcement (i.e. thickness of the plate and Young's modulus of the fibers).

The elastic properties of the adhesive can influence the bond stress distribution along the interface and, thus, the debonding, while its thickness appears to be less influent. Also for this application mechanical devices can be post-installed (Fig. 24.3-3).

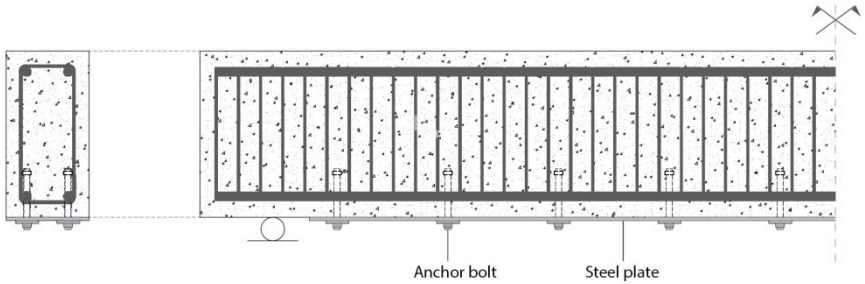

Figure 24.3-3*: Strengthening of RC elements with steel plates.*

considered analogous to that of Externally Applied Reinforcement (EAR) made of Fiber Reinforced Plastic (FRP) materials presented in section 30.8. The bond stress distribution is strongly influenced by the axial stiffness of the external reinforcement. In general, higher axial stiffness of the EAR corresponds to greater effective lengths necessary for fully developing the bond law, higher debonding loads, but lower strains in the reinforcement at debonding. However, when the bonded lengths of the EAR are lower than the effective ones, the debonding load seems to be roughly insensitive to the plate thickness.

The tensile strength of concrete plays a great role on the debonding load. Greater is the concrete strength, greater is the debonding load of glued plates and lower is the bonded length need to completely develop the bond law and sustain the debonding load. The effect of the concrete strength is less evident for bonded length greater than effective ones.

The effect of the plate-to-concrete width ratio is similar for FRP and steel EAR, nevertheless the isotropic behavior of the steel. The main difference between FRP and steel EAR is just related to their stiffness, since steel plates have usually thickness sensibly greater than those of the FRP sheets or plates available in the market, while the Young' moduli of FRP systems cover a wide range of values (the most common values range in 70-350 GPa) lower or higher than that of steel (210 GPa).

Nevertheless, the bond performance of steel EAR can be improved by means of anchoring devices, such as fastening that can easy applied on steel plates. Details on fastenings are reported in section 30.14.

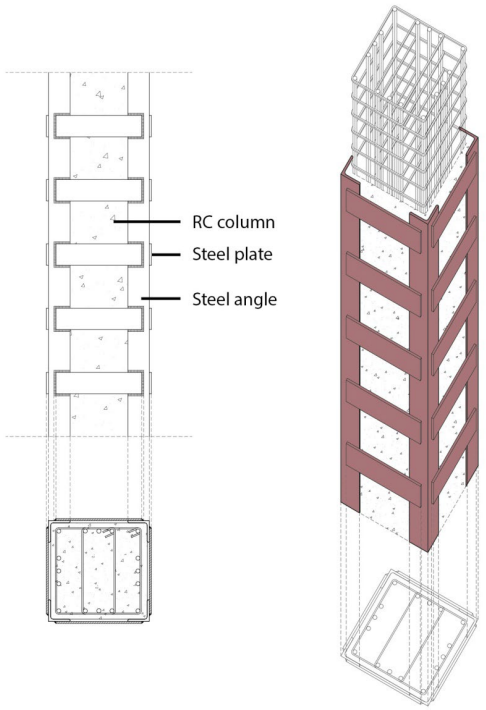

Figure 24.3-4: Strengthening of RC columns with steel cages

Failure can occur in the steel cage with the yielding and buckling of angles between two strips or yielding of horizontal strips, followed by the crushing of the RC column.

Failure modes and maximum axial load depend on strips spacing and width, thickness of the angles and strips, concrete strength, adherence between the angles and concrete.

The force components perpendicular to the interface arise during the loading process of composite member, but can play a secondary role if specific detailing of the devices and the connection system is provided. Therefore the main role of connectors is the transfer of shear between the concrete and steel parts.

Another typical application of external steel plates is the jacketing of columns to introduce a confinement effect on concrete (Fig. 24.3-4). This application involves the use of longitudinal plates (in rectangular sections four steel angles fixed at the corners and intermediate plates are used) along the elements and orthogonal steel strips welded to the longitudinal plates to realize a steel cage. The small gap between the steel cage and the surface of the concrete column is generally filled with cement mortar or epoxy grout to ensure full contact between the two material to realize a complete collaboration.

Depending on the structural details at the beam-column joints, the longitudinal plates can be considered as contributing to the axial and flexural strength or providing only to confinement effect.

24.3.3 Shear strength of stud connectors

Connection devices are widely used in composite structures, where structural combination of concrete and steel is achieved basically through a shear type (parallel to the interface) force transmission. As a result, common fastenings in composite structures are defined as “shear connectors”.

The parameters that influence the shear strength depends on the complex failure modes that can occur in connections realized with headed studs welded with a base collar.

Figure 24.3-5:Nelson headed stud.

The first experimental studies in '50 years already evidenced three types of failure: steel failures, where the stud diameter reached its yield point and failed; concrete failures, where the concrete surrounding the headed stud crushed; and mixed failures that included failure of both materials. Really each one type of failure depends on the geometric and mechanical characteristics of the system, but in general the connection failure can be due to concrete breakout around the studs (pray out) without yielding of the stud when squat devices are used. Therefore the entire strength of the connection can be exploited only when the height of the connector, respect its diameter, allows the development of a dowel effect with the formation of a plastic hinge in the steel stud; the ratio $h_c/\Phi \geq 4.5$ is a reliable limit to assure the exploiting of the connection strength. In this last case the experimental tests highlighted the possible failure of the stud shank or of the concrete around it.

The reference to the dowel action of rebars in concrete can be useful to understand the parameters involved in the phenomenon of plastic hinges formation in the steel stud and damage of concrete around it. However the numerical results and formulations are different because the studs are restrained on one side by a stiff plate (for example the flange of the steel beam) and also the effect of the welded collar can play a role.

Several formulas were obtained from many experimental and numerical studies (see [24-3]-[24-12]) and further studies are still in progress. The formulation proposed herein is based on experimental data regression of studs with $h_c/\Phi \geq 4.5$, Φ in the range 16-25mm, and materials in the following range 430 MPa $\leq f_t \leq$ 600 MPa and 20 MPa $\leq f_c \leq$ 55 MPa, in case of normal weight concrete and without

In general the shear resistance V_{con} of the headed stud welded to the steel plate depends on the following parameters:

$$V_{con} = V_{con}(A, f_{tk}, f_{ck}, h_c, \Phi) \qquad (24.3\text{-}1)$$

where:

A is the connector cross-section;

f_{ck} is the characteristic compressive strength of concrete;

f_{tk} is the characteristic ultimate tensile stress in steel;

h_c is the stud nominal height;

Φ is the stud nominal diameter.

Also the spacing of studs influence the response of the connection. Reducing the spacing between the studs the strength of the group could be lower than the sum of the strength of the singular studs, therefore a minimum step has to be established to apply the strength formulation neglecting the group effect.

The maximum strength of the concrete mechanism can be attained with a ratio $h_c/\Phi \geq 4.5$, but much higher values cannot give a further increase. A steel wire has to be inserted in the concrete slab not so far from the top of the stud.

The design shear strength of stud with efficient ratio $h_c/\Phi \geq 4.5$may be evaluated by the following formulation:

$$V_{con,d} = \frac{2.2\Phi^2\sqrt{f_{ck}\cdot f_{tk}}}{\gamma_v} < \frac{\pi\cdot\Phi^2\cdot f_{tk}}{4\gamma_v} \qquad (24.3\text{-}2)$$

where γ_v is assumed 1.3.

group effect. The formulation approach is the same of the first mechanical ones and it is based on a separate regression of experimental tests of steel shank failure (135specimens, COV 11%), that is more common, and concrete failure (41 specimens, COV 9%); the pray out failure is excluded. The partial safety factor of 1.3 was evaluated for both cases considering β=3.8.

The effects of the main factors can be accounted via a suitable reduction factor depending on the geometry of sheeting and stud dimensions and location.

Relevant codes provide limits for the use of the relationship in terms of the ratio b_r/h_r and the height of the headed stud compared to the height of the profiled sheeting (Figure 24.3-6).

The available formulations used to account for the influence of the sheeting on the shear strength of the studs were formulated mainly in the early 1990s. These formulations are not well suited for the currently used models of metal decking that are usually narrower and deeper than in the past.

Recent tests (DISCCO European project) showed that the shear stud resistance in composite slabs equipped with deep decking can be overestimated up to 30% by available formulations.

According to latest studies in currently used deck sheeting the classical shear resistance (due to concrete or steel shear failure) can be prevented by a bending failure of the rib. The bending resistance of the rib is given by a combination of the bending capacity of the concrete (Figure 24.3-7) and of the steel stud (Figure 24.3-8).

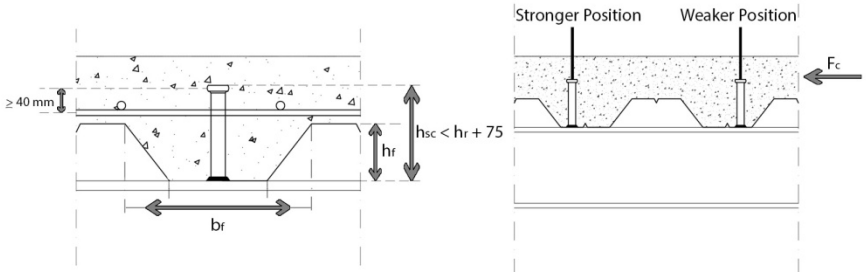

Figure 24.3-6: Sheeting geometry vs. headed stud performance.

If composite decks are used, the studs are placed within a rib, and their performance is fairly different from that described for the previous case of a solid concrete slab. Completely different aspects concern the concrete stiffness, degree of confinement and the resistance mechanism of the studs, which are loaded eccentrically.

The prime parameters affecting the stud behaviour are:

- the orientation of the ribs relative to the beam span;
- the rib geometry as characterized by the b_r/h_r ratio;
- the stud height h_c relative to the rib height h_r.

Note that design provisions for evaluating the shear strength of stud connectors in RC slabs or composite slabs are based on experimental results of tests performed on specimens with specific details that could be not present in existing structures. Some of the existing composite structures have been built in the past without suitable provisions, due to lacking of experimental evidence. In particular, studs without head were used and the role of the reinforcement in the slab for avoiding the failure of the concrete was not clear. Indeed the formulations of shear resistance of headed studs take into account the most effective parameters (e.g. concrete and steel strength, diameter and height of the stud). Nevertheless experimental tests showed the importance of the position of the steel mesh along the thickness of the slab and the key role of the head of the stud. The good performance of the connection can be ensured when double-layer reinforcement mesh is used and placed at a specific position respect to studs. A single layer mesh not intercepting the studs can give a relevant reduction of the connection resistance due to premature failure of concrete. Hence, it is clear that reinforcement mesh position respect the shank height is essential in achieving the design strength. Furthermore, experimental tests showed that depending on the elevation of reinforcement mesh, three failure modes can be observed: concrete failure, combined concrete failure and bent studs, stud shear-off from the steel flange. As the mesh position increases from intercepting the stud to being in flush with the top of the stud to above the stud, the ultimate load decreases. Furthermore, in specimens with un-headed shear studs the concrete failure is more likely to happen rather than stud bent failure, giving lower ultimate load.

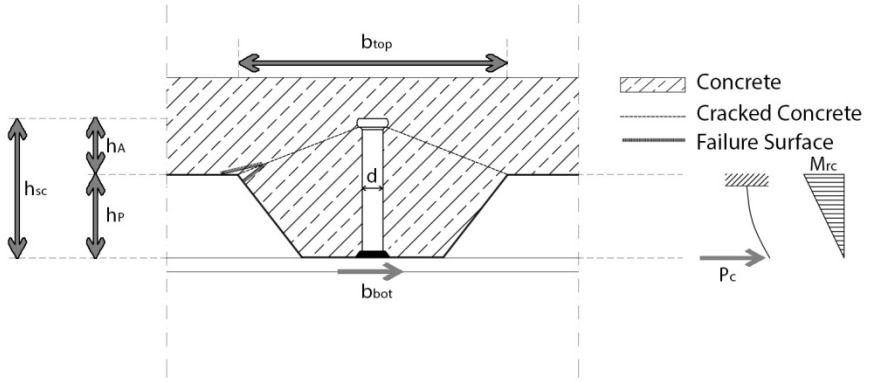

Figure 24.3-7: Resistance mechanism of concrete cone in bending.

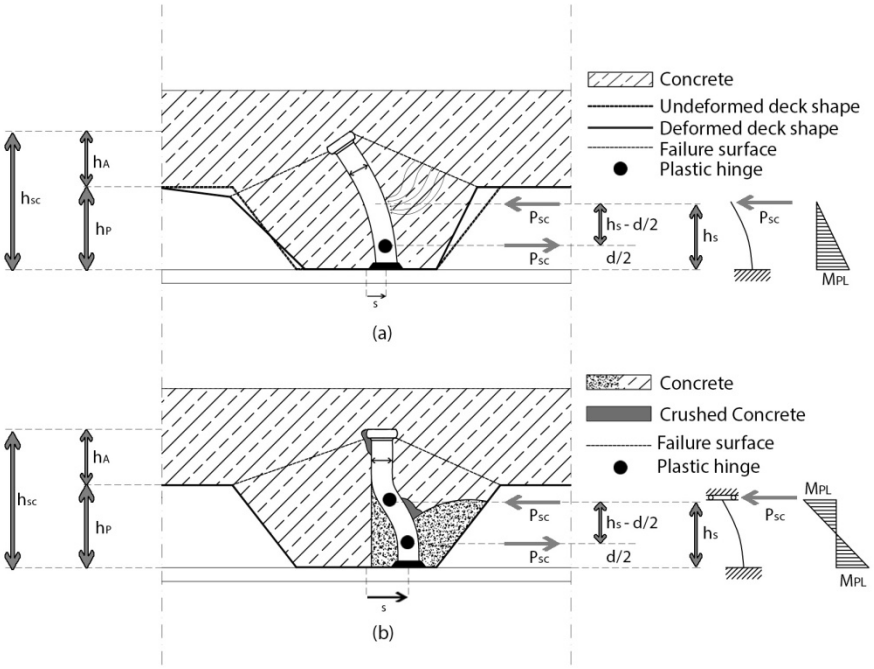

Figure 24.3-8: Resistance mechanisms of shear studs in bending.

Therefore, for existing composite structures it is not reasonable to refer to formulations calibrated on experimental tests with different details.

The so-called full connection condition is obtained if the design ensures that the ultimate flexural capacity of the composite member (beam or slab) is reached before interface failure.

A partially composite structural element is characterized by the property that the resistance of the shear connection is reached before flexural failure of the composite member. If a ductile connection is designed, the ratio between the shear capacity of the entire connection and the horizontal force to be transmitted between the steel and concrete components required to attain the complete collaboration, defines the degree of shear connection.

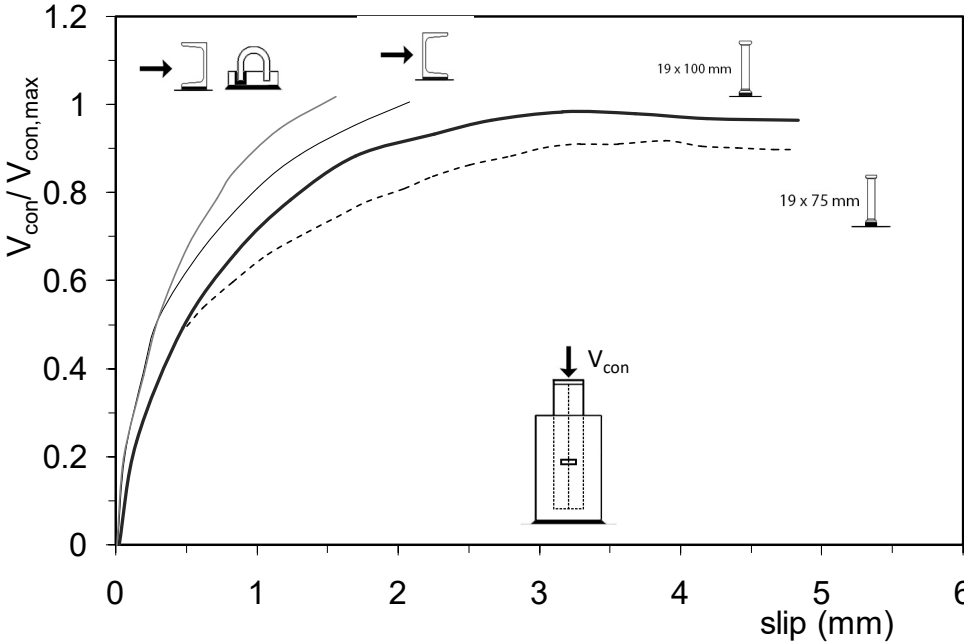

Figure 24.3-9:Non-dimensional force slip relationships for shear connectors.

24.3.4 Force-shear slip constitutive relationships

The strength of the connection between concrete and steel members may be influenced by the capacity of the anchors to redistribute loads among the devices connected by fixtures.

This emphasizes the relevance of the force–slip relationships on the overall performances of the composite substructure.

Structurally relevant aspects related to the force–slip relationships are:

- stiffness: a shear connection realizes either full interaction (the connection is “rigid” and no slip occurs under stress at the steel-concrete interface) or partial interaction (the connection is flexible and interface slip occurs);
- resistance: strength is required to the single connector or to the group of connectors in order to enable in a plastic approach the full transfer of forces related to the assumed structural mechanism;
- ductility: a connection is ductile if its deformation (shear and/or pull-out) capacity is adequate for a complete redistribution of the forces acting on the individual connector. In composite structures, the ductility demand depends on the member length and the degree of shear connection.

Modelling of the force–slip relationship (Fig. 24.3-9) can be carried out by evaluating specific tests on connectors that can be influenced by interface greasing of the steel profiles. In fact, due to chemical adhesion, force–slip relationships can show an initial rigid (zero slip) branch and then a very steep curve. Conversely, the presence of grease at the interface enables the generation of a response completely depending on the dowel action.

If the deformation of headed connectors in solid slabs is concerned, four relevant parameters can be mentioned:

- strength of the stud, $V_{con,max}$, as discussed in the previous section;
- ultimate slip s_m, corresponding to the maximum load $V_{con,max}$ of the stud;
- service slip s_e, corresponding to a half of the maximum load;

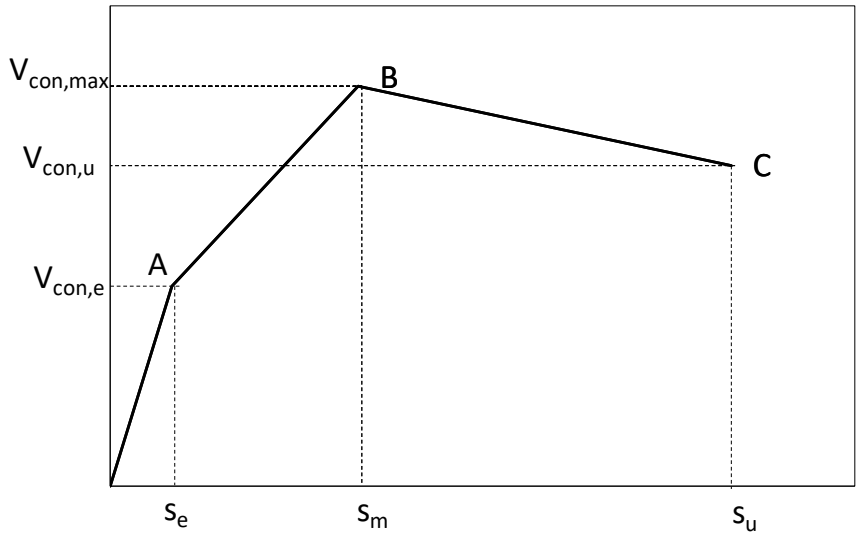

Figure 24.3-10: Force-shear slip constitutive relationships.

– slip capacity s_u evaluated on the post-peak descending branch of the relationship at 95% of the peak shear load.

Statistical analyses gave the following relationship for the ultimate slip and the slip capacity respectively:

$$s_m = (0.39 - 0.0023 f_c)\Phi \quad (24.3\text{-}3)$$

$$s_u = (0.48 - 0.0042 f_c)\Phi \quad (24.3\text{-}4)$$

The results are relevant for a range of concrete strengths between 20 MPa and 70 MPa, Φ being the stud nominal diameter.

Furthermore the service limit slip is more in general identified with the linear behaviour of the connection and the initial shear stiffness of the stud shear connection, K_{si} is defined as the mean stiffness at the applied load of $0.5V_{con,max}$:

$$K_{si} = \frac{V_{con,max}}{(0.16 - 0.0017 f_c)\Phi} \quad (24.3\text{-}5)$$

$$s_e = \left(80 \cdot 10^{-3} - 86 \cdot 10^{-5} f_c\right)\Phi \quad (24.3\text{-}6)$$

For refined modelling of composite construction advantage can be taken of theoretical curves that represent the force–slip relationships of studs up to the peak strength (Fig. 24.3-11):

$$V_{con}(s) = V_{con,max}\left(1 - e^{-\alpha_2 s}\right)^{\alpha_1} \quad (24.3\text{-}7)$$

where:

– s is the generic slip (mm);

– α_1 and α_2 are constants that enable a characterization of the curve.

The deformability of the connection can influence the deflection of the beams in serviceability condition especially in long spans as happens for bridge decks; therefore, in the design the maximum slip demand has to be limited.

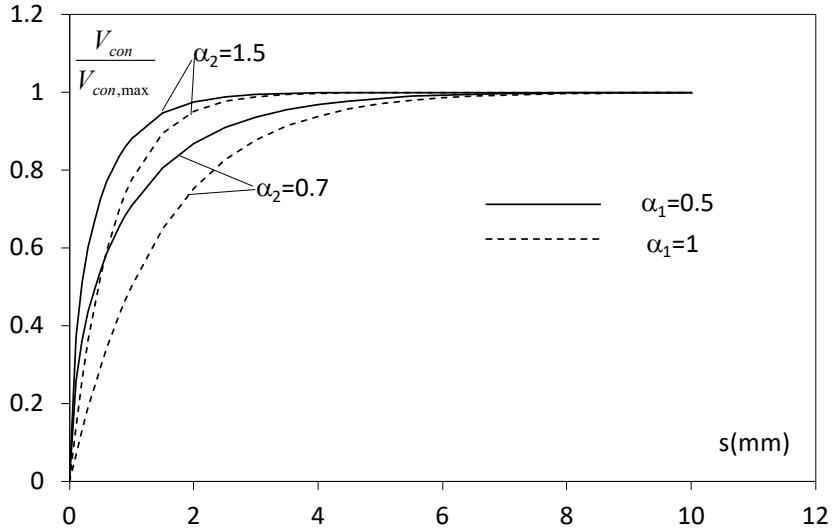

Figure 24.3-11: Shear–slip relationship depending on parameters α_1 and α_2.

Based on experimental results, α_1 ranges between 0.5 and 1, while α_2 ranges between 0.7 and 1.5. The effective value of the used studs has to be determined by testing or certificated by the manufacturer.

With regard to serviceability and deflection analysis of composite members, a linear approximation of the force–shear-slip relationship can be used. A possible definition of the linear stiffness can be based on the secant stiffness evaluated in correspondence with the slip and the shear force under serviceability conditions, that is at about 50–60% of the resistance.

24.3.5 Determination of properties by testing

Relevant standards can cover the need for design equations that relate material and geometrical parameters to the strength and behaviour of mechanical devices.

Since the equations do not always cover the selected connectors, experiments can supply design data.

The variables to be investigated include the geometry and the mechanical

properties of the concrete slab, the connectors and the reinforcement.

The resistance to types of loading, other than fatigue, may be determined by specific tests in accordance with relevant standards.

In the case of shear connectors for composite structures, specific push-off tests should be carried out such that the slabs and the reinforcement are suitably dimensioned in comparison with the beams for which the test is developed.

Thus, the length l of each slab should be related to the longitudinal spacing of the connectors in the composite structure; the width of each slab should not exceed the effective width of the beam's slab; the thickness of each slab should not exceed the minimum thickness of the slab in the member.

The applied load and the relative displacements between the single connector and the base concrete component should be measured. The slip should be measured at least until the load has dropped down to 20% of the maximum load. Average displacements can be used as reference values for the force–slip relationships. The characteristic resistance is evaluated reducing the maximum load of 10%.

The slip capacity of a specimen should be taken as the maximum slip measured at the characteristic resistance on the descending branch. The characteristic slip capacity should be taken as the minimum test value of the slip capacity reduced by 10% or determined by a statistical evaluation of all the test results.

References

[24-1] Abdullah and Easterling — Abdullah R., Easterling W. S. (2009), New evaluation and modeling procedure for horizontal shear bond in composite slabs, Journal of Constructional Steel Research 65 , 891-899.

[24-2] Oehlers and Bradford — Oehlers, D., Bradford, M.: Composite steel and concrete structural members fundamental behaviour, Ed. PERGAMON, (1995).

[24-3] Viest — Viest, I. M., (1956), Investigation of Stud Shear Connectors for Composite Concrete and Steel T-Beams, ACI Journal, Vol. 27, April, 1956.

[24-4] Ollgaard et al — Ollgaard, J.G., Slutter, R.G. and Fisher, J.W., (1971), "Shear strength of stud connectors in lightweight and normal-weight concrete", Eng. J., AISC, 8(2), 55-64.

[24-5] Hawkins — Hawkins, N. M., (1973), "The strength of stud shear connectors." *Australian Civ. Engrg. Trans., Vol. 15,* The Inst. of Engrs., Canberra, Australia.

[24-6] Oehlers and Johnson — Oehlers, D.J. and Johnson, R.P.: The strength of stud shear connections in composite beams, The Structural Engineer, (1987).

[24-7] Shim et al — Shim C., Lee P., Yoon T. (2004), Static behavior of large stud shear connectors. Engineering Structures Volume 26, Issue 12, October 2004, Pages 1853-1860

[24-8] Dig-Lee et al — Ding- Lee P., Shim C., Chang S. (2005). Static and fatigue behavior of large stud shear connectors for steel–concrete composite bridges. Journal of Constructional Steel Research 61; 1270–1285.

[24-9] Xue et al — Xue, W., Ding, M,. Wang, H., Luo, Z. (2008), Static Behavior and Theoretical Model of Stud Shear Connectors, Journal of bridge engineering, 10.1061/ (ASCE)1084-0702(2008)13:6(623).

[24-10] Pallares and Hajjar — Pallarés, L., F. Hajjar, J. (2010), Headed steel stud anchors in composite structures, Part I: Shear, Journal of Constructional Steel Research 66 (2010) 198-212.

[24-11] Han et al — Han Q., Wang Y., Xu J., Xing Y. (2015), Static behavior of stud shear connectors in elastic concrete–steel composite beams. Journal of Constructional Steel Research 113 (2015). 115–126.

[24-12] Jebara et al — Jebara K., Ožbolt J., Sharma A. (2020), Pryout capacity of headed stud anchor groups with stiff base plate: 3D finite element analysis. Structural concrete. 21 (2020). 905-916.

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25 Anchorages in concrete

25.1 Scope definition and range of applicability

The local and structural effects of anchorages should be considered in design. The design of anchorages in concrete should be performed according to the *fib* Bulletin YY [25-2]. Only anchors that have been prequalified for the intended use should be used.

The *fib* Bulletin 58 [25-1] covers the design of anchorages using the following types of anchors:

- cast-in components, such as headed anchors, anchor channels and shear lugs;
- post-installed anchors, such as torque-controlled and deformation controlled expansion anchors, undercut anchors, screw anchors, bonded anchors and torque-controlled bonded expansion anchors;
- post-installed and cast-in deformed reinforcing bars.

The suitability of an anchor for the intended use should be demonstrated by a standard or a technical approval or by the results of suitable prequalification tests.

The following design situations are covered:

- verification of ultimate limit state for predominantly static loading;
- verification of ultimate limit state for non-static loading (fatigue, seismic);
- verification of ultimate limit state for extreme thermal conditions (fire);
- verification of serviceability;
- verification of limit states associated with durability.

The *fib* Bulletin 58 [25-1] includes the local transmission of loads into the structure by anchorages without or with special anchor reinforcement. In the design of the structure the anchor loads and additional design requirements given in section 8 of the *fib* Bulletin 58 [25-1] should be taken into account.

Relevant codes of practice have to be used for design and structural detailing of fixtures.

Reliable fastening applications require qualified and experienced designers. Installations should be carried out by experienced and skilled personnel, and proper maintenance of the structure of the fixtures and of anchorages should be provided. Furthermore, the specified use of anchorages should not be changed during their intended service life without recalculation.

Anchors can act either as single components or as groups when common fixtures are used. Load transfer mechanisms can be either statically determinate or statically indeterminate.

Construction drawings should clearly give at least the following information:

- location of the anchorage in the structure including tolerances;
- number and type of anchors, including embedment length;
- spacing and edge distance of the anchors, including (positive) tolerances;

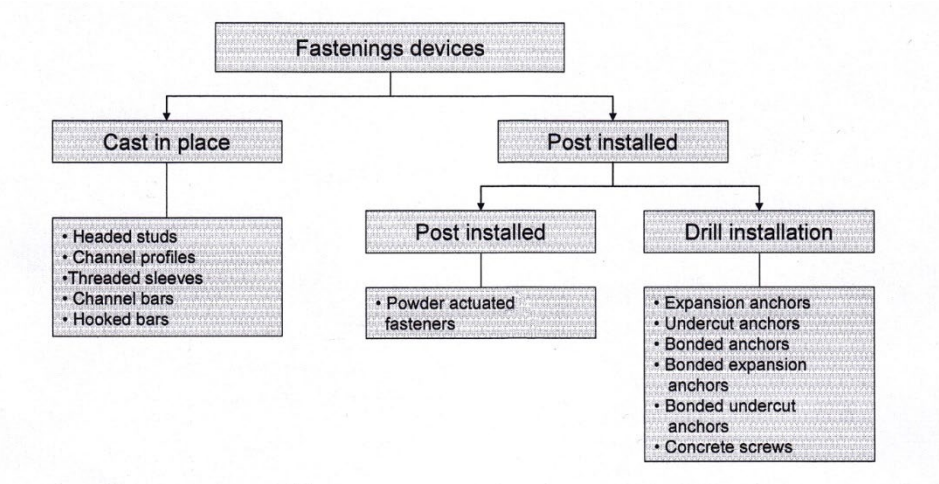

*Figure 25.2-1: Classification of anchorages.*FIGURE WORDING TO BE UPDATED (anchorages instead of Fastenings)

– position of the attachments on the fixture including tolerances;
– maximum thickness of any non-bearing layers below the fixture;
– any special installation instruction.

25.2 Classification of anchorages

Anchorages of steel to concrete may be classified as follows (Figure 25.2-1):

– cast-in-place devices;
– post-installed devices.

Cast-in-place devices are positioned in the formwork before the concrete is cast and thus can be used in members with dense reinforcement.

Post-installed systems may either be installed into drilled holes (drill installation) or be driven into the base material with impact energy (direct installation).

Different mechanisms can be activated to transfer tensile loads from the steel anchor to the surrounding concrete matrix:

– friction;
– mechanical interlock;
– dowel action;
– bond.

Cast-in-place systems such as headed studs, hooked bars and channel bars transfer loads mainly by mechanical interlock.

Similar mechanisms are activated in the case of undercut anchors, while in the case of expansion anchors, tensile forces are transferred to concrete basically by friction.

25.3 Provisions for quality control

[Not used at this time, to be updated in future edition]

25.4 Assumptions used for design

Provisions for strength evaluation of anchorages are given in chapter 30.14.

25.5 Assumptions used for assessment

Provisions for strength evaluation of anchorages are given in chapter 30.14.

References

[25-1] *fib* Bulletin 58 Design of anchorages in concrete, *fib* Guide to good practice, ISBN 978-2-88394-098-7, July 2011

[25-2] *fib* Bulletin YY Design of anchorages in concrete. Guide to good practice, Under preparation

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26 Conceptual design

26.1 General

26.1.1 Introduction

The conceptual design stage is the period during which identified needs are examined (i.e. the design challenge), requirements for potential solutions are defined, plausible answers are evaluated and, finally, a suitable structural concept for further design is conceived. Conceptual design is the art of conceiving and developing solutions satisfying the design challenge, namely the performance-based requirements associated with the three pillars of sustainable development, the defined requirements and identified constraints.

Conceptual design is employed for new structures and for interventions (eg. alterations, adaptations, renovation, upgrading of performance, extension of service life, etc) upon existing structures after an adequate structural assessment has been performed. The conceptual design also needs to be developed in conjunction with the life-cycle management (LCM) philosophy and the envisaged LCM activities,

The design process has different stages, as is indicated in chapter 10.

Conceptual design is the first step of the design process and it is generally considered to be the most important one. It is the stage where designers can display their creativity, where it is essential to come up with all of the ideas that may help find the optimum solution to a given design and construction challenge. The definition of a good concept in this design stage allows its improvement in subsequent phases. An inadequate choice at this stage may mean that the project is unable to be improved in later phases.

Without a valid well thought-through structural concept – the *'idea'* - defining a competent and harmonious functional solution to the performance requirements, there can be no established safety concept, no adequately defined behaviour or understanding of the flow of forces and, essentially, no solution to the defined design problem; without which a successful construction project cannot be realized.

Good and creative ideas are necessary in all the stages of the service life of the structure, namely during design, assessment and interventions, including life cycle management activities. While this chapter focuses upon the conceptual design stage, because it represents a crucial phase for delivering a successfully project, the concepts and procedures described equally apply to the design challenges associated with making adaptations, upgrading performance and performing other interventions upon existing structures, especially those which have experienced deterioration or damage.

Many iterations of the conceptual design process are commonly required to refine the design concepts to meet all of the requirements. The analytic tools applied at this stage to the investigation of the problem and evaluation of potential options may be relatively crude. The designer must employ his creative skills to produce the best possible result for a given problem (i.e. the design challenge). In order to successfully take forward this process, the designer must have a profound knowledge of engineering and of the available technologies. However, they must

The basic approach to performing conceptual design relies on decomposition and integration. Since design problems are large and complex, they have to be decomposed into sub-problems which are small enough to be solved. There are numerous alternative ways to decompose design problems, such as decomposition by functions of the facility, by spatial locations of its parts or by links among various functions or parts. Solutions to sub-problems must be integrated into an overall

also have a broad culture and knowledge in many related and important contextual aspects that are essential to face the holistic process that is required by conceptual design.

Further background information and illustrations on this topic can be found in other documents such as [Refs xx-yy]

The general principles concerning sustainable performance which are adopted in Model Code 2020 are defined in chapter 3.

Social performance of concrete structure concerns many matters including:

- Structural performance: Structural safety, serviceability, durability, robustness, structural resilience.

Other aspects of social performance may include many things, such as: Buildability, maintainability, demountability of structures and recycling of members, aesthetics, safety of workers and other persons, annoyance and inconvenience of neighbouring stakeholders, traffic congestion, communication and other operations, acoustic comfort, thermal comfort, local employment, as well as (potentially) many others.

Environmental performance of concrete structures supposes at least the consideration of the following aspects: use of resources, use of energy, use of land, emissions to air, water and soil, noise and vibration, waste production, species and ecosystem disturbance and others. In addition to the consideration of adverse environmental impacts, it may also be appropriate to consider any environmental benefits arising from the operation of the concrete structure.

Life-cycle economic performance concerns the cost and benefits arising from the concrete structure and includes matters such as: construction cost, operation cost,

solution. The integration and rationalization processes often create conflicts which must be identified and resolved.

Many ideas to solve the problem, taking into account the owner's programme and the expectations of the stakeholders, have to be proposed and explored during the conceptual design process in order to find a solution which complies in an optimal manner with the specified requirements. These ideas, even though lacking in detail, must be feasible and describe the solution from the points of view of sustainability, structural adequacy, functionality, construction, maintenance, economy and so on.

This phase shall also identify the more critical aspects which need to be more thoroughly developed in the following stages of the design process.

26.1.2 Consideration of sustainability

Sustainable development is one of the most important objectives of the Model Code 2020.

Sustainability can be described by means of three interdependent and mutually reinforcing pillars: namely the Social, Environmental and Economic pillars.

Sustainability is a holistic concept that involves many aspects that must be satisfied simultaneously and in a balanced way.

At the stage of conceptual design of new structures and interventions in existing structures the evaluation of sustainable performance shall be considered from the perspective of all three pillars of sustainability.

It is essential to consider changes and development of sustainable performance within the entire life of a structure.

maintenance cost, refurbishment cost, demolition or reuse cost, recycling cost, cost of externalities, cost of capital and support of local economy.

Many of these concepts are very well known by structural engineers and, of course, are taken into account in design, in general, and during the conceptual design stage, in particular.

Others are more recent concepts, such as robustness or resilience or environmental performance aspects; with the latter primarily focussed upon minimizing any adverse influence that the structures may have on the natural environment.

In any case, deep consideration of sustainability during the conceptual design stage, shall allow optimization of the sustainable performance of the created concrete structure.

The Institution of Structural Engineers, London in its publication “Design for zero” [xx] provides guidance, case studies and positive steps which can be taken to move structural engineering projects towards achieving net zero or carbon neutral design outcomes. The goal is durable low carbon constructed assets and buildings. Amongst others, the actions proposed include a focus on:

- Making carbon as important as safety in structural calculations
- Extending the life of existing constructed assets / resisting demolition
- The reuse of existing constructed assets as the starting point for new design
- Choosing appropriate low carbon materials, technologies & products
- Interrogating / challenging project briefs to better align them towards net zero.

Life cycle management (LCM) is the overall philosophy / strategy to be used in planning and managing a concrete structure through its life-cycle, particularly during its operation and use phase. Accordingly, the LCM strategy adopted shall be developed from the performance requirements for the new or altered structure. The LCM strategy will influence the development of the associated candidate design solutions / selection of relevant interventions, as appropriate. For more information see chapter 5.

26.1.3 Consideration of life-cycle management

During the conceptual design stage, it is important to identify the different factors that may affect the service life of the structure, so that they may be included in the design basis; in addition, it is relevant to find solutions that may mitigate such factors in order to be integrated into the adopted solution.

The first draft of the Durability Plan, see section 27.6, shall be prepared during the conceptual design stage. This plan identifies the most important durability aspects that must be taken into account in the project. In the case of simple or basic small projects, a simplified approach may be adopted during the development of the Durability Plan.

The cost of maintenance throughout service life directly depends on the adopted decisions in the different design stages and the LCM philosophy adopted. As shown in Figure 26.1-1, the more strategic decisions are initially taken, greater the benefits that can be obtained.

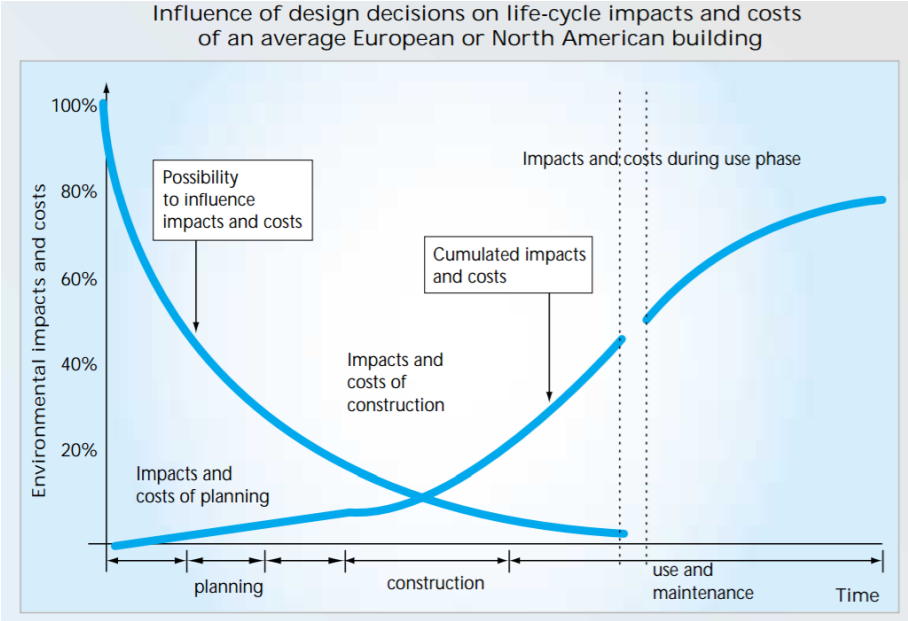

Figure 26.1-1: Impact of design decisions made at various stages of the life-cycle (Crawford 2011 [26-1] Rice [26-2] and/or Kohler [26-3] and/or Schlaich [26-4]).

During the conceptual design stage, it is necessary to propose details that will minimize the deterioration of the structure. For example, early consideration of the necessary drainage details and their operation will help guide their development in later design stages.

As further example, the provision of accessibility to the different structural members for their inspection will also be very useful for future developments in later design stages, such as the provision of purpose made concrete members to be collected during the life of the structure for destructive testing.

In this phase, it is essential to identify which are the structural members that will have a shorter service life and the auxiliary means that must be provided for their replacement.

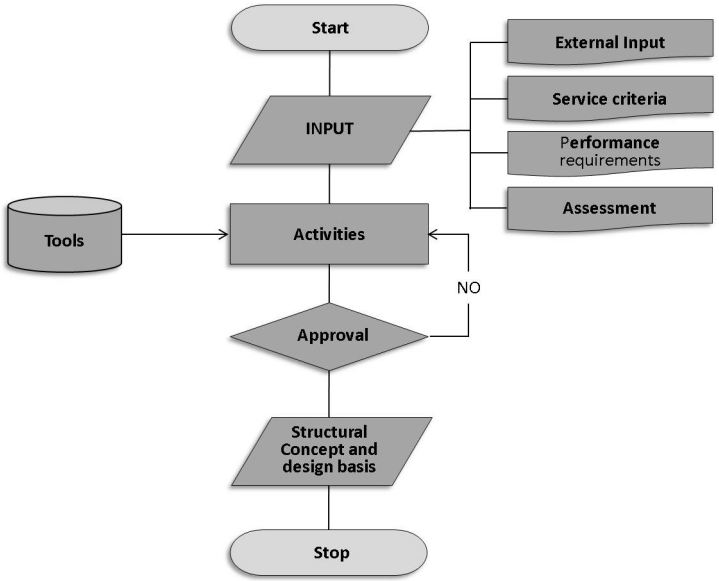

Figure 26.2-1: *Methodological flowchart for conceptual design of new structures and adaptations/interventions/restorations upon existing structures.*

The trial-and-error approach employed during the evolution of a conceptual design requires training and experience. The creative process involved can be learned [Norman et al [26-5]], Rice [26-2] and Schlaich [26-4].

The correct definition of the design challenge is essential. It is not possible to solve a problem which has not been adequately defined.

26.2 Methodology

Conceptual design is a creative process for which it is not easy to establish a methodology. Figure 26.2-1 illustrates a process which may provide some insight and be of assistance for carrying out the associated activities.

26.3 Input information

26.3.1 Introduction

Initial information must be established with regard to:

- general criteria;
- basic external input data and information upon significant constraints;

The definition of the problem is usually complex. It is an aspect that often takes shape as the conceptual design process progresses. In any case, for particularly complex projects, it is often necessary to be open to making updates and refining the definition of the basis of the design in later stages of the conceptual design process for the project.

It is not possible to establish a unique list of the necessary information for each project. Each project requires different types of data that must be identified and defined.

For more detailed information see section 10.4.5

To achieve the effective integration of the proposed structure into the existing societal context, it is essential to understand the associated physical, social and historical perspectives. A critical activity is to talk to the client and other stakeholders / professionals to obtain an appreciation of these perspectives, along with related insights into the task and the desired outcomes. A key part of this 'investigation' process is to visit the prospective site (or sites, if there are various locations involved) to gain an in-depth appreciation of the circumstances of the site(s). See *fib* Bulletin 51 [26-6].

The conceptual design proposal should be a rational response to the requirements and the above perspectives. It should:

Employ appropriate discretion, as well as displaying quality and elegance of design to facilitate acceptance of the work.

Satisfy the various functional uses of the work in an optimal and elegantly integrated manner in the final solution, to achieve balance and compatible with their respective requirements.

The integration of the structure into the landscape must be compatible with the general philosophy of the structure, and compatible and harmonious with the general landscape of the general circumstances.

- service criteria;
- performance requirements and constraints;
- assessment report, in case of interventions upon existing structures.

26.3.2 General criteria

- Owner and stakeholder requirements and expectations;
- sustainability;
- functional requirements;
- aesthetics;
- environmental integration (physical, social and historical);
- structural adequacy;
- redundancy, robustness and resilience;
- constructability;
- circular economy;
- feasibility;
- economy;
- minimum maintenance cost / environmental impact;
- etc.

If the basic external input data are not available to the designer, a process will need to be established so that it can be obtained either from the owner, the architect, the authorities or some other source or via an appropriate process instigated by the designer.

Basic data must be clearly specified in the Service Criteria Agreement, see Section 10.4.5.

There are new & emerging technologies which allow the procurement of very important information that must be integrated in this process (eg. Big Data, 3D photogrammetry, 3D scanners, etc.).

26.3.3 Basic external input data and information upon significant constraints

- Geotechnical data;
- topographical and bathymetrical data;
- climatological data;
- metocean data (eg. ocean wave characteristics and statistics);
- environmental data (eg. earthquake, hurricanes);
- constraints upon available material types and/or their properties;
- accessibility and transport facilities;
- local construction facilities and rules;
- in the case of buildings; architectural requirements, such as form, mass/volume, surface texture and colour.
- etc.

The service criteria have to be discussed and established with the owner or the architect; it must be approved by all and be clearly specified in the Service Criteria Agreement, see section 10.2.5.

26.3.4 Service criteria

- General aims for the use of the constructed works (efficiency, comfort, safety etc.);
- operational and maintenance requirements (efficiency, economy etc.);
- special requirements of the stakeholders (upgrading, replacement etc.);
- objectives of protection and special risks;
- loadings and loading combinations;
- environmental conditions;
- codes and regulatory requirements.
- etc.

26.3.5 Performance requirements

The performance requirements have to be established, proposed and explained by the designer to the owner and must be clearly specified in the Service Criteria Agreement.

- Performance requirements for sustainability
- performance criteria for safety and serviceability (including robustness, resilience and durability);
- service life constraints (temporary, replaceable, evolutive, long term);
- reliability constraints;
- etc..

26.3.6 Existing structures

The Assessment Report is the basic source of information which the designer has to study in detail when undertaking an adaptation / alteration / intervention upon an existing structure. This report shall provide the reasons why the works are needed and the justification for the adaptation / alteration / intervention procedures proposed. It is commonly better for the intervention methods to be selected after the results of conceptual design phase for the proposed works on an existing structure are known.

The design of new structures and works upon existing structures shall generally follow the same process of conceptual design activities described in this chapter, as they are typically valid for both circumstances.

Where adaptation / alteration / intervention works are to be performed upon an existing structure, the Assessment Report is expected to be a basic source of information for the designer of those works. See Assessment Report (see section 28.4)

The design of these works shall follow the different design phases normally adopted during the design of new structures.

26.4 Conception process

26.4.1 Activities

In general, activities carried out during the conceptual design stage of construction works are related to:

The conceptual design process can be characterized by a series of interactive activities:

- Constraint analysis and classification;
- environment analysis (including local politics and local traditions);
- general conception;
- choice of materials (considering economy and energy consumption for production and dismantlement);
- structural concept (structural logic, dimensions);
- integration and aesthetics (legibility, simplicity, proportions, equilibrium, shapes, detail philosophy);
- construction methods and sequences;
- preliminary construction and life-cycle cost estimates;
- comparison of alternative candidate solutions;
- series of presentation, explanation and discussions with the owner / architect;
- after acceptance by the owner, preparation of structural concept and the basis for design (drawings, notes, reports).

Attributes and tools such as the following may be employed during the conceptual design stage:

- Experience, plus insight from earlier work, feedback, database sources;
- intuition, feeling, understanding, sensitivity for the circumstances;
- creativity, imagination, capacity of simultaneous analysis and integration of different criteria and constraints taking into account their relative importance and significance;
- textbooks with canonical dimensions and proportion of acceptable structural solutions (Menn [26-7], Muttoni [26-8], Torroja [26-9]);
- databases of existing projects that help determine the main dimensions of the proposed solutions (Salonga & Gauvreau [26-10]);
- information from other similar examples previously designed;
- quick preliminary design/verification methods and simplified hand calculations estimates;

- Formulation, which refers to the definition or description of a design problem in broad terms, through the synthesis of ideas describing alternative concepts;
- Analysis, which refines the problem definition or description by separating important from peripheral information and by pulling together the essential detail; interpretation and prediction are usually required as part of the analysis;
- Search, which involves developing and revising a set of potential solutions for performing the specified functions, satisfying the user requirements and respecting the constrains;
- Decision, which means that each of the potential solutions is evaluated and compared to the alternatives until the best solution is obtained;
- Specification, which is to describe the chosen solution in a form is detailed enough for detailed development and implementation;
- Modification, which refers to the change in the solution or re-design if the solution is found to be insufficient or if new information becomes available.

26.4.2 The role of expertise, insight and tools

The conceptual design process aims to find acceptable design solutions for the project objectives and goals, its defined performance requirements, constraints and the associated opportunities provided by the circumstances.

The process is guided by the experience gathered in comparable design and construction works, along with the insight and intuition obtained in other relevant circumstances.

A variety of tools and aids may be used to assist the process, including those for visualization of candidate schemes and alternative options, basic dimensioning of members, preliminary evaluation of economic outcomes, etc.

Sketching is a powerful communication medium to exchange ideas which define the project or parts thereof. The development of ideas, concepts and design details by sketching may employ freehand sketches or could potentially involve the production of accurate drawings. Hand sketches allow a quick impression to be gained of the key aspects of the project in question.

- development of ideas, concepts and design details by sketching, with these ranging from freehand sketches to accurate drawings;
- In the later stages of the conceptual design process, for alternatives with more possibilities it is possible to produce 3D renders that enable the designer to see the whole project, or to study particular details of it;
- Physical and virtual models - Nowadays there are technologies, such as laser cutters and 3D printers, which can produce high-quality physical models at reasonable cost.

It is always interesting and generally beneficial to involve experts from other relevant fields. Depending on the type of structures involved, these might include architects, urban planners, landscape artists, archaeologists, historians, constructors, providers of technical equipment, etc.

The extent and content of the basis for design must be adapted to the importance of the construction works and the associated hazards and environmental risks, but it must always exist no matter how minor the project is considered to be.

Drawing to scale, to represent the true proportions of different members and members of the scheme, is the only way to have a clear understanding of the adopted dimensions and their relationship with other members in the proposed design. It also allows the rapid comparison of alternative options. Hand sketches are a very useful tool which fosters a hand-eye-mind cognitive relationship to create and express ideas, but designers may prefer computer aided / produced sketches to represent ideas;

In the later stages of the conceptual design process, for situations with a greater number of alternative possibilities, it is possible to produce 3D renders for visualisation purposes which enable the designer to see the whole project or to study particular details of it.

Physical and virtual models can also be used to visualise and communicate the vision and details of the proposed solution(s) to the design challenge.

26.5 Structural concept and basis for design

The structural concept derived from the conceptual design shall include:

- the chosen structural system and details of the associated flow of forces;
- information on the most important dimensions, construction material properties and construction details;
- comments on the envisaged methods of construction and phasing of the works;
- observations upon the provisions for circularity and the future dismantlement / recycling of the structure / its components / materials.

The structural concept derived from the conceptual design must be described in the basis for design, including the basis and requirements for the subsequent design phase, execution, use, life-cycle management and conservation activities.

The bases for design have to describe at least the following:

- The design service life;
- the service situations considered;
- the hazard scenarios considered;

The information contained in the report on the outcome of the conceptual design process for a building will might address the following matters:

- Summary of site related information (or sites, if there are various locations involved);
- Overview of any existing buildings/structures;
- Substructure design;
- Superstructure design;
- Drainage aspects;
- Design criteria for the structural and other aspects;
- How sustainability aspects are addressed;
- Summary and next steps information;
- Drawings;
- Indication of life-cycle costs and benefits evaluation;
- Indication of life-cycle environmental impacts and benefits; and
- Programme and 'information required / provision schedule'.

- the requirements of structural safety, serviceability and durability together with the measures needed to deliver them at an acceptable reliability for the design service life of the structure, including the division of responsibilities, processes, controls and corrective mechanisms;
- the assumed ground conditions;
- the important assumptions associated with the structural and analytical models;
- the accepted risks; and
- other conditions relevant to the design and life-cycle management of the concrete structure.

26.6 Outcomes from the conceptual design process

At the end of the conceptual design process, it will typically be necessary to produce a report, with some associated drawings and perhaps other supporting documents, to communicate the nature of the proposed design solution for the proposed new concrete structure or the alterations / interventions upon an existing concrete structure.

In simple terms, the report might be expected to cover matters such as:

- The brief - the client's requirements, how they evolved in the final brief;
- The design process & development – How it meets the requirements;
- The proposed solutions to the design challenge;
- Indication of life-cycle costs and benefits; and
- Indication of life-cycle environmental impacts and benefits

References

[26-1] Crawford, R. Life-cycle assessment in the built environment". Routledge. London. March 2011

[26-2] Rice, P. Rice, P, An engineer imagines, Artemis London, 1994.

[26-3] Kohler, Moffat Kohler, N., Moffat, S, Life-cycle analysis of the built environment, *UNEP Industry and Environment*, 17-21, 2003.

[26-4] Schlaich, J. Schlaich, J., Gabriel, K., On the conceptual design of structures – an introduction, *International Symposium on Conceptual design of structures,* Vol. 1: Methodology, p. 15 – 25, 1996.

[26-5] Norman et al Norman, J., Broadbent, O., Carr, J. F., De'Ath, R., Harpin, R., Knowles, G., Lloyd, I., Conceptual design of buildings, The Institution of Structural Engineers, pp. 254, 2020

[26-6] *fib* Bulletin 51 Structural Concrete Textbook on behaviour, design and performance, Second edition Volume 1: Design of concrete structures, conceptual design, Manual-textbook, ISBN 978-2-88394-091-8, November 2009

[26-7] Menn, C. Menn, C., Prestressed concrete bridges, Birkhauser Verlag, 1986.
Originally published in 1986 under the title "Stahlbetonbriicken" by Springer-Verlag, Wien.

[26-8] Muttoni, A. Muttoni, A., The art of structures, Introduction to the functioning of structures in architecture, EPFL Press/Routledge, 2011, EAN13 Book: 9782940222384.

[26-9] Torroja, E. Razón y Ser de Los Tipos Estructurales". Madrid: Instituto de la Construcción y del Cemento, 1957, (Available as : Consejo Superior de Investigaciones Científicas, 2010. ISBN: 978-84-00-08612-1)

[26-10] Salonga and Gauvreau Salonga, J., Gauvreau, P., Comparative study of the proportions, form, and efficiency of concrete arch bridges, *Journal of Bridge Engineering*, 19(3), 2014, 10.1061/(ASCE)BE.1943-5592.0000537

There are a number of phases in the development and use of a concrete structure. Commonly the sequence of phases and events through the life of such a concrete structure progresses largely as follows:

- Definition phase - where the owner's basic requirements and stakeholder needs are established with reference to the drivers arising from consideration of the overarching 'three-pillars' of sustainability.
- Conceptual Design phase – where the identified requirements are examined, potential solutions satisfying these are defined, evaluated and a suitable structural concept chosen for further detailed development. Many iterations of the process are commonly required to refine the solution concepts to meet the functional requirements and comply with associated sustainability/financial/other constraints.
- Design - usually involving preliminary and detailed design phases.
- Construction / Execution - the process whereby the structure is built.
- Operation and use - with life-cycle management activities / interventions being undertaken in a timely manner to maintain its functionality and ensure the ongoing adequacy of its performance; with these activities / interventions being recorded (e.g. Rebirth Certificate and associated documents) so the knowledge is utilised in subsequent LCM scenarios.
- Extended or alternative use – this is the process by which the useful life of the concrete structure can be extended or the concrete structure brought into a new use by being adapted, sold or via some other means.
- Disposal or re-use - the process by which a concrete structure is decommissioned and removed; being either dismantled (the components being re-used as far as possible) or demolished and the materials recycled. Issues of circularity / the circular economy shall be considered in the design and construction stages – refer chapter 39.

The observations made in this chapter are generally valid for both new concrete structures and for interventions being made upon existing concrete structures.

27 Approach to design

27.1 General

The approach to design shall be developed from the performance requirements for the new or altered concrete structure. These will influence the development of the candidate design solutions. To do this, the owner will typically need the support of a team of technical advisers, depending upon the expertise held by the owner.

Critical aspects of these considerations will be derived from the sustainability objectives defined for the concrete structure under the three pillars of sustainability (social responsibility, environmental quality and economic efficiency) discussed previously in chapter 3. The design strategy shall achieve an acceptable balance between these factors.

Thus, the design process shall respond to those factors in a coherent and integrated way providing input to the processes of design, construction, usage, and conservation of concrete structures, as well as, if appropriate, provision for their future adaptation to new uses / changed performance requirements.

The stages in the design and construction process are listed and commented upon in the informative left-hand column.

The *Conceptual Design* (refer chapter 26) shall consider these factors and the associated work flow in the development of appropriate design solutions.

Within the overall flow of the design and construction process chapter 27*: Approach to design* follows on from chapter 26: *Conceptual Design*. Accordingly, the content of chapter 27 assumes that a conceptual design has previously been developed.

Conceptual design is applicable for both new structures and interventions (eg. alterations, adaptations, renovation, upgrading of performance, extension of service life, etc) upon existing structures after an appropriate structural assessment has been performed.

27.2 Consideration of sustainability

Sustainability is a holistic concept involving many different aspects and associated performances which need to be satisfied in an appropriately balanced way. The general principles concerning sustainable development in Model Code 2020 are defined in chapter 3. The structural engineering perspectives are further developed in section 26.1 of chapter 26: *Conceptual Design*; along with the associated environmental and economic considerations.

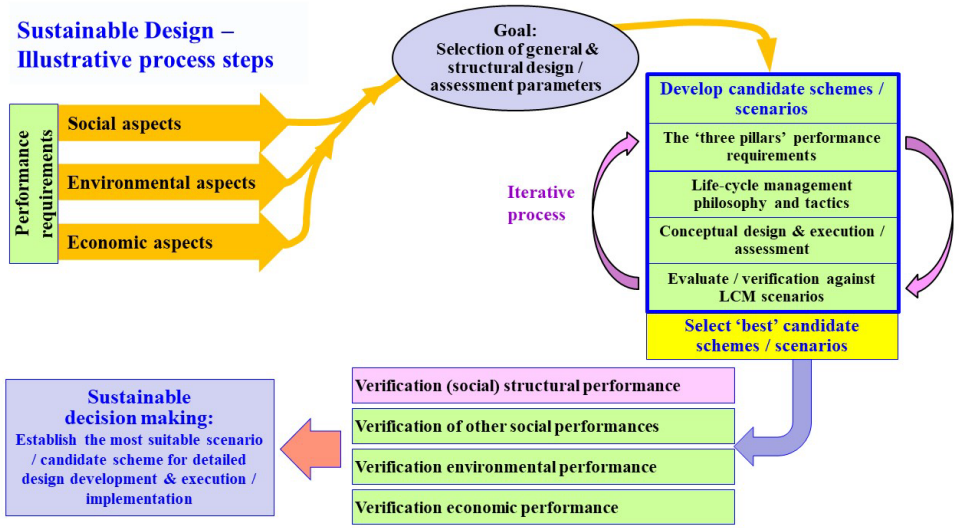

Figure 27.2-1: *Sustainable design – Illustrative process steps for general and structural design activities.*

In terms of the pressing goal to achieve net zero carbon emissions by 2050 (Net Zero 2050) needed to minimise global warming effects (UNSDG [27-1]), there is a need to greatly reduce the carbon footprint of construction generally. The Institution of Structural Engineers guide *Design for zero* (IStructE (Orr et al.) [27-2]) notes overdesign produces carbon-intensive structures.

Carbon considerations are now of considerable importance, potentially creating a tension with other aspects of design, which require thoughtful design and carefully balanced judgements about the conceptual design solutions chosen. BSI PAS 2080: 2016 *Carbon Management in Infrastructure* [27-3] contains a carbon reduction hierarchy and other guidance to help with decision-making. The associated "*Guidance Document for PAS 2080*" [27-4] provides additional insight.

Sustainable development is an overarching objective of Model Code 2020 which is defined through three interdependent and mutually reinforcing pillars: namely the objectives and performance requirements established under the pillars of social responsibility, environmental quality and economic efficiency.

The United Nations Sustainable Development Goals (UNSDG) provide a vision for a sustainable future and a plan of action for people, planet and prosperity UNSDG [27-1]. Account shall be taken of the focus this provides on development needs and to contributing positively to the achievement of Sustainable Life on our planet.

Figure 27.2-1 presents illustrative process steps for sustainable design, which are applicable to both general and structural design activities. Appropriately safe and reliable structural performance is a fundamental requirement for the satisfactory societal performance of a concrete structure.

The verification / evaluation of structural and other social performances are addressed in chapter 30: *Verification of structural performance* [e.g. structural safety, serviceability, durability, deterioration, monitoring, interventions etc.] and chapter 31: *Evaluation of other aspects of social performance* [e.g. health & quality of the built environment, aesthetics & cultural heritage aspects, safety & security, impact upon local community etc.]

The evaluation of environmental performance (impacts and benefits) by means of life-cycle analysis procedures is addressed in chapter 32.

The evaluation of economic efficiency utilising life-cycle cost and benefit analysis procedures is addressed in chapter 33.

Sustainable decision making is discussed in chapter 34: *Sustainability evaluation*.

The LCM strategy for the structure shall contribute to achieving an acceptable balance between the different factors considered under the three pillars of sustainability, as noted above – refer chapters 3 and 5.

ISO 22040 [27-5]: *Life cycle management of concrete structures* describes the general principles for implementing LCM of concrete structures throughout the stages of planning, design, execution, use and end-of-life, as well as the framework and procedures involved in delivering LCM.

The available conservation strategies and respective tactics are classified on the basis of their proactive versus reactive characteristics as follows:

- Category A: Proactive conservation activities;
- Category B: Reactive conservation activities;
- Category C: Situations where conservation activities are not feasible.

The end-of-life stage involves processes by which the concrete structure is decommissioned and removed; being either dismantled, with the components being re-used as far as possible, or demolished and the materials recycled. The objective shall be to reduce the amount of waste and promoting the reuse members / recycling of construction materials.

The principles of circularity / the circular economy require that planning be undertaken to facilitate 'higher-value' future uses of construction members and materials, particularly the reuse of structural members, where possible, in preference to simple recycling materials and 'down-cycling' processes.

In terms of integrated life-cycle design (sustainability design) the following outlines a procedural schema for the flow from performance / functional requirements, through technical specifications for the structure and on to the most suitable design alternative / candidate design solution / scenario.

27.3 Consideration of life-cycle management

The three pillars of sustainability provide a framework of requirements from which the Life-cycle management (LCM) philosophy and strategy for the concrete structure shall be derived.

LCM shall be used to make informed choices between various design, construction, use, conservation, adaptation and end of life options on the basis of formulating appropriate life cycle management scenarios (LCM scenarios) at the planning and design stages, as a means of ensuring the performance requirements are attained – refer chapter 5.

The risk of performance degradation of the structure during its use, as well as its impact on the environment, shall be considered during the formulation of the LCM scenarios.

LCM decisions shall be made so that the work at each stage of planning, design, execution, use, etc can be appropriately linked and managed on the basis of a consistent LCM philosophy and implementation activities.

In regard to LCM in the planning and design / assessment stages, the performance requirements of the structure shall be defined, along with the structural forms, materials and dimensional details so that the structure satisfies the performance requirements for the design working life at the chosen reliability level. The envisaged methods of maintenance / intervention shall be defined at this time. It is necessary to ensure that appropriate maintenance / intervention activities can be undertaken during the use stage.

Section 27.6 introduces the concept of 'Durability Planning' as an important component part of life-cycle management activities.

During the planning and design stages consideration shall also be given to the LCM activities required during the end-of-life stage of the proposed concrete structure.

27.4 Design process methodology

For integrated life-cycle design (sustainability design) of the overall structure / scheme, the performance / functional requirements shall be identified and, from those, appropriate technical performance properties shall

Step 1. Identification, definition and analysis of the performance / functional requirements of the owner and other stakeholders.

Step 2. Interpretation, analysis and optimisation of the required technical performance properties of the structure from the identified performance / functional requirements of the owner and other stakeholders (from Step 1). Identify appropriate design alternatives / candidate design solutions / scenarios.

Step 3. Specification of the technical performance properties of the structure based on the performance properties (from Step 2).

Step 4. Selection of most suitable design alternative / candidate design solution / scenario (from Steps 2 & 3).

Guidance on performing the tasks involved in Steps 1 to 4 is given in **[?????]**

Further details on performance-based approaches are given in chapter 4.

In regard to the evaluation of other aspects of social performance (chapter 31), some aspects to be taken into account include:

- Health & quality of the built environment – e.g. health issues and quality of performance within construction, service life, maintenance or dismantling.
- Safety and security – e.g. issues / accidents due to poor safety management.
- Aesthetics and cultural heritage – e.g. matters relating to the quality / acceptability of a structure's visual impact and the protection / preservation of cultural heritage (historical buildings and other forms of construction).
- Positive and negative impacts on local community / neighbouring entities, which might include beneficial effects (e.g. new employment, improved transportation networks, more attractive cityscape), but possibly also adverse impacts (e.g. noise, pollution, delays or disruptions associated with construction, maintenance, rehabilitation, repair activities, etc).

In the ***preliminary design phase***, checks upon serviceability limit states (SLS) and ultimate limit states (ULS) are needed. SLS checks may be crucial as serviceability aspects often govern the depth of slabs and beams. The load bearing capacity at ULS is typically verified by:

- loss of equilibrium of the (rigid body) structure or any part of it;
- failure by rupture, excessive deformation or loss of stability of the overall structure or any part of it.

be employed in the selection of suitable candidate design solutions / scenarios (chapter 34).

Life-cycle design / sustainability design, as described in chapter 3, forms the overarching performance-based framework for the design.

Critical consideration shall be given to the factors influencing the outcomes of the design development process, namely the functional requirements expressed in terms of:

- Societal and cultural aspects (refer chapters 3 and 31);
- Environmental considerations (refer chapter 32); and
- Economic and financial requirements (refer chapter 33);

along with the complexities associated with sustainable decision-making, as discussed in chapter 34: *Sustainability evaluation*.

A functional / performance- or outcomes-based approach (PBA) shall be used to define the requirements which have to be met, without defining the manner in which this shall be done. Thus, PBA designs and specifications shall focus upon defining appropriate performance or outcomes criteria and, accordingly, include the testing of relevant concrete properties and the evaluation of test results based on agreed conformity requirements.

Using the outputs from the above processes, the most appropriate design alternative shall be developed and refined to evolve the chosen solution. This will commonly employ four successive phases of iterative development activity, as listed below.

A simplified (illustrative) representation of the overall life-cycle phases and design process steps for a concrete structure are presented as Figure 27.4-1. The portrayed sequence represents the traditional approach to the development of a project, where a design is fully developed before being subsequently tendered for construction. Where other forms of procurement

In the preliminary design phase, linear elastic analysis, with or without moment redistribution, may be carried out. Simple checks made on the basis of the theory of plasticity might also be appropriate.

Before starting with the calculations, the engineer must carefully consider all relevant loading conditions. Referring to the effects on the structure, four kinds of actions may be distinguished: permanent, variable, seismic and accidental, plus imposed deformations.

Classifying actions by their duration leads to the following differentiation:

- Permanent (dead) loads, such as the self-weight of members, forces due to earth or water pressures, prestressing forces
- Variable (live) loads: imposed loads, traffic loads, wind loads, etc -which are of particular concern where they may result in fatigue effects.
- Accidental actions / hazards: seismic forces, impact, fire exposure, etc
- Imposed deformations & restraint: creep, shrinkage, settlement, etc.

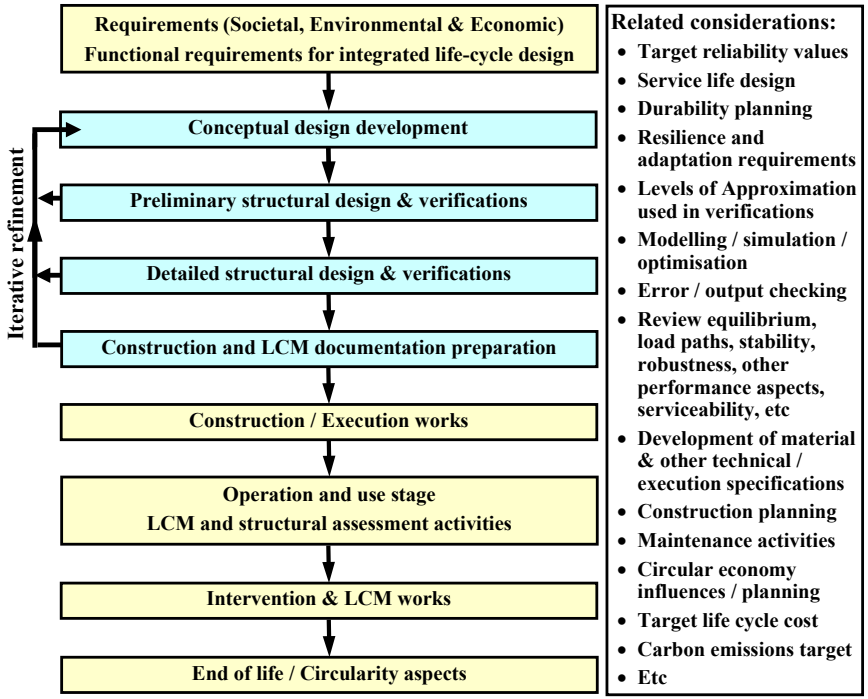

Figure 27.4-1: *Overall life-cycle phases and design process steps.*

are adopted, the sequence of activities and the associated documentation will need to be adjusted to account for the particular requirements of the procurement process. These may involve early contractor engagement via design and construct (ie. contractor lead with designer engaged or, say, the adoption of an alliance approach (e.g. owner with contractor and designer)

1. ***Conceptual design -*** The first stage develops potential candidate design solutions and establishes the type, scale, likely construction techniques and major features of the structure, such as the overall structural scheme, along with the layout, nature and types of members. The flow of forces shall be established and the geotechnical aspects investigated. This stage requires engineering flare, experience of similar projects, sufficient knowledge of the site(s) and local conditions, as well as a good understanding of relevant construction techniques and maintenance processes. This is the stage where the greatest positive impact on sustainability and carbon reduction can be achieved. It will also typically review the feasibility of the project under the client defined parameters. Chapter 26 provides further details on the activities involved.
2. ***Preliminary design*** – The second stage is where the chosen alternatives from the conceptual design are further developed and designed at a general level using preliminary calculations (typically employing lower Level of Approximation, LoA, methods) resulting in appropriate initial dimensioning of members based upon a satisfactory definition of the mechanical and geometrical properties for items such as the required cross-sectional areas of concrete members and reinforcement, concrete strength / mix requirements, appropriate foundation provision, etc. The structural analyses would typically be based on simplified models (lower LoA methods) of the overall behaviour. Consideration may need to be given to non-linear behaviour of a structure. For the combination of actions, sufficient load cases shall be examined for the critical design conditions to be established in the structural members. An outcome of this stage shall be clarification of the details of the preferred design solution.
3. ***Detailed design*** – In the third stage typically all structural details are designed and appropriate supporting calculations undertaken to verify the chosen design solution – refer chapter 30. Work starts from the member sizes and other details developed in the preliminary design stage. Normally, the third stage entails the greatest amount of design and calculation related work activities.

In regard to the ***detailed design phase***, particular factors to be considered may include issues such as:

- Ensuring adequate ductility, especially when plastic analysis adopted.
- If the material behaviour is idealised as being elastic, limited redistribution of moments may be applied.
- Non-linear analysis shall be performed by appropriately knowledgeable and experienced staff.
- If necessary, second order effects / associated deformations shall be taken into account where they may significantly affect the internal forces and moments or the overall stability of the structure.
- Consideration shall be given to the effects of possible imperfections in the geometry of the unloaded structure (geometric deviation).
- Cyclic loading may require dynamic / fatigue analyses to be made.
- Dimensioning of members and detailing reinforcement / prestressing shall take account of the flow of forces, the required structural behaviours, follow accepted best-practice arrangements and recognised site installation practicalities / limitations.
- Ensuring overall equilibrium is obtained, particularly during construction
- Ensuring all stages of the construction sequence are considered and account taken of all internal actions, so that the relevant limits states are not exceeded at intermediate stages.

The distribution of internal forces and moments (the stress resultants) shall be established for the structure, bearing in mind the flow of forces. Potential failure modes shall be identified and evaluated. This stage requires competence in calculation and modelling, appropriate knowledge of materials and the site(s) / local circumstances to perform the necessary design verifications.

As well as checks upon ultimate limit state (ULS) and serviceability limit state (SLS) situations, the checks shall include condition limit states (CLS) such as the applicable durability limit states (DLS).

Consideration shall be given to construction and LCM (maintenance and intervention) issues, as well as the practicalities implicit in on-site installation of materials (such as concrete) and of products (such as reinforcing bars), in recognition of the duty of care owed to 'constructors' to facilitate procedures and practices which allow safe construction and/or alterations to existing concrete structures, as well as how 'end of life' aspects are to be addressed.

4. ***Preparation of life-cycle management & construction documentation*** **-** This step uses the detailed design outputs to produce working drawings, technical specifications and other documentation required for construction (eg. construction method statements) and subsequent life-cycle management of a concrete structure.

   Depending upon the circumstances, this stage may also involve activities such as pilot works and/or full-scale trial constructions.

   Further information is available on the above topics including:

   - chapter 5 for life-cycle management aspects;
   - chapter 10 for quality & information management aspects;
   - chapter 30 for verification of structural performance;
   - chapter 35 for execution management;
   - chapter 36 for the execution of construction and adaptation works;
   - chapter 37 for the execution of intervention works;
   - chapter 38 for conservation; and.
   - chapter 39 for circularity – adaptation, dismantlement and re-use.

27.5 Basis of design

The Basis of Design is a very important document which is refined and further developed through the life of the project. During the conceptual design stage, it is necessary to formulate a Basis of Design document, and also to recognise that the document will change and evolve during the different stages involved in the creation of the project. Accordingly, it is not fully complete until the end of the project. The Basis of Design is a fundamental document which helps avoid errors and repetitions of work. It can also become a valuable reference after the completion of the project.

Assurance of structural performance is one of the most important design and engineering tasks, with key performance aspects including:

Structural safety – where the ULS are not exceeded and a structure (or structural member) achieves a (satisfactory) specified level of reliability over a stated period of time, including the effects of defined accidental phenomena. In some instances, this may be expressed in terms of the ratio (safety factor) between the actions which would cause collapse (or other similar form of structural failure) and the actions imposed upon the structure in service.

Structural equilibrium and stability aspects.

Robustness - where a structure subject to accidental or exceptional loadings or other actions (e.g. fire, explosion, impact) is able to sustain damage / local failure without experiencing damage disproportionate to the original cause.

Durability - which is the capability of a structure (or structural member) to satisfy, with planned maintenance, the design performance requirements over a specified period of time under the influence of the environmental actions.

Serviceability - which is the ability of a structure (or structural member) to perform adequately for normal use under the expected combination of actions.

Fire resistance - which is the ability of a structure (or structural member) to maintain its load-bearing capacity at or above a specified level of reliability under the heating effects of a defined fire scenario for a specified length of time (chosen standard duration).

The basic performance requirements for structural design are those relating to structural stability and safety (ULS), including fatigue and other dynamic effects, serviceability (SLS), durability and other condition limit states (CLS), robustness / response to accidental actions, fire resistance and, where appropriate, resilience and adaptability.

In addition, the designer shall address other aspects such as constructability, LCM activities (maintainability and anticipated interventions), circularity, as well as matters relating to damage limitation design etc, where these considerations are appropriate.

Performance requirements with respect to structural safety, serviceability and durability of a structure are usually expressed in terms of the structural reliability to be achieved.

Reliability is the ability of a structure, or structural member, to fulfil the specified requirements over the design service life. The treatment of reliability in the verification of structural performance is discussed in section 12.5.

Account shall also be taken of the influence of geotechnical / foundation aspects and the potential for differential movements occurring spatially across structure, and potentially also varying in magnitude with time.

A hazard identification / risk-informed approach shall be used where it can provide a useful framework for eliminating or reducing hazards to which the structure is exposed during construction or the in-use phase or at end of life.

In response to the required performances, there are many other factors which contribute to approaches adopted to create safe, serviceable and durable concrete structures. Taken together these factors form what is commonly referred to as ‘the basis of design’. These factors include the:

- Adoption of limit state design and the approaches employed
- Actions acting upon the structure, including accidental actions
- Choice of Consequence Class – see chapter 11
- Target reliability values to be adopted in various design situations
- Approaches to verification of structural and other performance
- Approach to service life design / durability aspects

- Approach to design for operation and maintenance
- Approach to modelling the construction sequence
- Approach to circularity – adaptation, dismantlement and re-use
- Approach to quality and information management

For the design of a new structure, it will typically be assumed that no structurally significant deterioration will occur due to environmental factors during the design service life. Appropriate measures are to be put in place to ensure this outcome.

In regard to existing structures, any deterioration shall be investigated and assessed, with estimates made of relevant safety and other performance criteria (ULS / CLS / SLS) on the basis of the assessed condition / site(s) investigation findings.

Limit state design - limit states and factors to consider could include:

- Structural design - ultimate limit states (ULS)
- Structural design – serviceability limit states (SLS) Stiffness
- Durability aspects - condition / durability limit states (CLS / DLS)
- Partial damage limit states (PDLS)
- Special considerations (Robustness, temporary conditions during construction, undertaking interventions and dismantlement activities)

Actions acting upon the structure – considerations may include:

- Permanent – predominantly static loads (self-weight of structure, imposed loads due to stored materials, water & 'fluids', snow, etc)
- Variable – non-static loads (variable imposed loads, traffic effects, wind loads, wave effects, etc)
- Geotechnical
- Extreme thermal (cold or cryogenic conditions)
- Combinations of actions (normal actions and/ or accidental actions)
- Accidental – extreme thermal (fire)
- Actions – Thermal & moisture induced strains, differential movements

Consequence Classes: The consequence class provides a classification of the consequences of failure of the structure or part of it, see Section 11.3.1.

Note: Previous design codes have adopted a design philosophy that there shall be **no** effective / substantive deterioration / reduction of performance over the design service life due to environmental factors. The same design philosophy / criterion is to be adopted for MC2020.

For further details upon the limit state approach and the various formats which may be adopted refer to section 11.1: *Principles of limit state approach*.

Ultimate limit states (ULS) – for further details refer Section 11.1.3.

Serviceability limit state (SLS) considerations could include stiffness related performances, such as dynamic behaviours.

To avoid damage or to ensure that damage is not disproportionate to the original cause, various measures can be enacted:

- Avoid, eliminate or reduce the hazards which the structure is exposed to
- Select a structural form with low sensitivity to the hazards considered
- Design to survive the accidental removal of a single structural member or damage to a limited part of the structure, with an acceptable degree of damage / deformation in a localised zone of the structure
- Avoid structural systems that could collapse without displaying warning indications
- Provide tying to connect structural members together to mobilise a combined response from a larger number of structural members

Robustness: The designer shall choose appropriate strategies for design against accidental action situations, as illustrated in Table 27.5-1:

Table 27.5-1: *Design strategies for accidental action situations* [After EN 1991-1-7 [27-6]].

Strategies based upon identified accidental actions	***Strategies based upon limiting the extent of localised structural failure***
AD1. Design structure to have sufficient minimum robustness	AD4. Enhanced redundancy (eg. mobilise alternative load paths)
AD2. Preventing or reducing the action (eg. apply protective measures)	AD5. Key element designed to sustain notional accidental action
AD3. Design structure to sustain action	AD6. Prescriptive rules (eg. integrity and ductility requirements)

Notes:

1. AD3 and AD5 represent resistance measures enhancing the strength / resistance of the structural members concerned
2. AD2 represent protection measure eg. avoiding the hazard such as the build-up of explosive substances, gases etc in the building / compartment.

Target reliability values – potential considerations include:

- Structural design – ultimate limit states (ULS)
- Structural design – ultimate limit state for fatigue ($ULS_{fatigue}$)
- Structural design – serviceability limit states as appropriate (SLS)
- Durability – condition / durability limit states (CLS / DLS)
- Partial damage limit state (PDLS)

Refer chapter 12 – *Principles of structural design and assessment.*

Approach to service life design / durability aspects - Factors to consider include:

- Length of the design service life
- Characteristics of the exposure environment – principal hazards / deterioration mechanisms / possibility of combined attack
- Hazard identification / hazard reduction / risk-informed approach
- Design strategies to be employed and for which structural members
  - Avoidance

- Resistance design - offered by chosen structural materials:
  - Deemed-to-satisfy
  - Partial-factor design
  - Full probabilistic design
- Design assisted by testing / monitoring

Verification of structural performance – possible approaches and circumstances include:

- Hazard identification / Risk-informed approach;
- Reliability based / Full probabilistic method;
- Partial-factor method;
- Global resistance method;
- Deemed-to-satisfy approach;
- Non-static loads – Fatigue;
- Non-static loads – Impact and explosion;
- Extreme thermal actions (hot and cold (cryogenic) situations);
- Serviceability;
- Design assisted by testing / monitoring;
- Numerical simulations;
- Robustness;
- Anchorages in concrete;

Verification of durability performance – possible approaches include:

- Risk informed;
- Reliability based / Full probabilistic method;
- Partial-factor method;
- Avoidance;
- Deemed-to-satisfy;
- Design assisted by testing / monitoring;

Verification of other performances include those for the following:

- Life-cycle cost;

For further details upon verification procedures and the various formats which may be adopted refer to chapter 11: *Structural performance evaluation framework*, section 11.4: *Special requirements,* and section 12.4: *Verification methods of structural performance* and section 12.5: *Verification formats*.

- Life cycle analysis.

27.6 Durability planning

27.6.1 Introduction

Under performance-based design the objective is to create a concrete structure which has the ability to fulfil the performance requirements for the specified design service life or extended service life at the required level of reliability.

To allow effective and efficient life-cycle management (LCM) the durability design and the associated conservation processes shall be adequately planned, documented and managed through the life of the structure – refer Chapter 5: *Life-cycle management.* The durability design and conservation planning shall be prepared taking account of the conservation categories which have been selected for the concrete structure and/or its component parts.

27.6.2 Formal durability planning

Durability planning is a formalised approach which applies durability knowledge during project design, execution / construction and operational maintenance activities to minimise the likelihood of premature deterioration.

Formalising the durability planning processes is intended to incorporate durability thinking into the wider aspects of design, execution / construction processes and, critically, in subsequent life-cycle management activities.

Durability planning work-flow employs a series of steps to achieve the design service life or extended service life. The following sequence would be for a project designed and with subsequent tendered construction, which was a common contract design and delivery process until recent times.

a) Concrete structure owner's brief (or of the responsible entity);

b) Life-cycle management / maintenance in operation and use phase;

c) Scope of works and technical criteria (SWTC) in tender documents;

Achieving the specified service life design objectives requires the durability aspects to be taken into account in the design, construction / execution and LCM processes, in order to avoid the premature deterioration of concrete structures. These issues and considerations are commonly referred to as *Durability planning,* which involves all construction materials and components used to create a concrete structure.

Conservation strategies and tactics are classified into three categories as described in Section 5.2; namely, 'A: *Proactive conservation*', 'B: *Reactive conservation*' and 'C: *No conservation*'.

An increase in the level of the performance requirements during the operational use stage may necessitate an intervention being made to enhance the performance or possibly the durability of the concrete structure.

Achieving durability is a world-wide concern and the need for improved durability has long been identified. One problem is that not all international and national technical standards incorporate appropriate provisions which account for the severity of all the macro / micro environmental exposure conditions or the deterioration mechanism(s) involved. The missing element in achieving satisfactory durability has been formal durability planning to coordinate and appropriately implement the available durability related technical knowledge.

An appropriate durability philosophy applied to the project design and delivery processes will contribute to achieving the sustainable design objectives, efficient capital investment, lowering the risk of unexpected or early deterioration / damage and, through appropriate design, execution and life-cycle management measures, help to achieve the concrete structure owner's required design life / extended design life and desired functionality.

The precise sequence of the durability planning processes and activities depends upon the form of contractual arrangement and documentation adopted for the works. Rather

than the traditional sequence of project design followed by tendered construction, many present-day projects may have early contractor engagement via design and construct (i.e. contractor lead with designer engaged), adopt an alliance approach (e.g. owner with contractor and designer) or employ some other form of contractual arrangement.

The durability planning inputs to achieve the design service life or extended service life are described in *fib* Bulletin XY [27-7]. Other contemporary guidance on durability planning also includes CIA Z7-01: *Durability Planning* [27-8].

Projects with early contractor engagement often adopt the processes and activities below.

1. Concrete structure owner's brief;
2. Life-cycle management activities;
3. Scope of works and technical criteria (SWTC) for the tender documents;
4. Contractor tender, which may include a conceptual design for durability;
5. Conceptual design studies for the durability approach;
6. Preliminary and detailed design studies for the durability approach;
7. Execution / construction durability approach;
8. Durability assessment report (DAR) documentation / deliverables for the above.

As the objective of durability planning is to avoid inadequate durability / unsatisfactory durability outcome, durability planning will typically involve processes such as:

- Reviewing deterioration / damage hazards and thresholds;
- Establishing the vulnerabilities of the structure to such durability hazards;
- Seeking to eliminate / reduce durability hazards and vulnerabilities, as far as possible;
- Proposing and/or reviewing LCM procedures for the potential hazards arising throughout the design service life, including the execution/construction phase.

d) Conceptual design;
e) Preliminary design;
f) Detailed design;
g) Project contractor tender;
h) Execution / construction;
i) Durability assessment report (DAR) deliverables for the above.

The durability planning requirements are to apply irrespective of the project contract delivery process.

The associated documentation and work activities will vary with the particular requirements of the chosen procurement process. The names given to the durability documentation may vary (i.e. durability assessment report, durability management plan) - refer to *Definitions* in chapter 2.

27.6.3 Durability planning – Requirement for expertise

Formalised durability planning requires expert durability input and technical analysis into the design and life-cycle management of concrete structures, including the preparation of the project related durability documentation.

The LCM risk assessment may be a relatively simple analysis. It shall include consideration of relevant hazards and their consequences. For example, such as the time and costs associated with construction defects, plus those of associated inspections and interventions, as well the associated implications of the resulting uncertainties arising during the design service life.

Structures requiring formal durability planning / expert input to the design and the LCM of concrete structures may include those where the concrete structure(s):

- has a long or very long specified design service life;
- provides socially or environmentally critical performances / functions.
- requires a high degree of reliability and / or low LCM / maintenance burden;
- is located in or exposed to aggressive environmental conditions;
- has members where the heat of hydration may cause problems;
- employs construction methods which could result in an unacceptable occurrence of defects and/or other quality problems potentially adversely impacting durability;
- utilises marginal or contaminated materials.

Examples of structures where the use of formal durability planning and expert durability input might not be required include: simple structures, temporary structures, those subject to a ‘mild’ exposure environment (i.e. those not exposed to aggressive exposure conditions) and structures employing historically proven durable design and construction approaches.

The formal report is termed the Durability Assessment Report: Conceptual Design (DAR: CD). The durability issues which might be addressed in the study include:

27.6.4 Structures requiring formal durability planning

The requirement for formal durability planning with expert durability input shall be established by means of a LCM / durability risk assessment.

The LCM / durability risk analysis and its outcomes for service life design shall consider design methodology, execution / construction processes and LCM requirements. These shall be in accordance with / inform the LCM planning and design / specification of the concrete structure(s).

27.6.5 Structures not requiring formal durability planning

Formal durability planning with expert durability input is not required for concrete structures where the LCM risk assessment outcome is judged to be ‘low’ (e.g. the deemed to satisfy provisions are expected to provide acceptable durability).

27.6.6 Conceptual durability design studies

In some complex projects an initial conceptual durability design study may be required to identify shortcomings in the available information which have

- Characterisation of the macro- and micro-environments.
- Determining the environment aggressivity (types and severities).
- The deterioration mechanisms affecting the construction materials.
- Deterioration models to determine future service life of materials.
- Appropriate materials for achieving the specified design service life at an acceptable life-cycle cost.
- Selection of structural and components suitable for the environment.
- Potential materials related construction difficulties.
- Drafting / reviewing preliminary specifications for durability matters.
- Develop cost effective inspection and testing programs.
- Define outline proactive operational and maintenance requirements.

to be resolved before a call for project tenders / bids can be made. This need may arise for:

- Contracts where project design is followed by tendered construction.
- During the contractor's tender for projects with early contractor engagement.

The outcome shall be presented in a formal report (see example left).

While a conceptual design study shall address all relevant durability matters, it may be appropriate for it to focus on some specific durability aspects. Such a study may also be used to establish whether a deemed-to-satisfy durability approach will be appropriate for the expected (design) circumstances.

Accordingly, the effort expended upon conceptual durability design, relative to that expended upon detailed durability design, may vary greatly from project to project. The flow and sequencing between these two processes will vary. Some projects may require very little effort to be expended in the conceptual design stage.

27.6.7 Detailed durability design studies

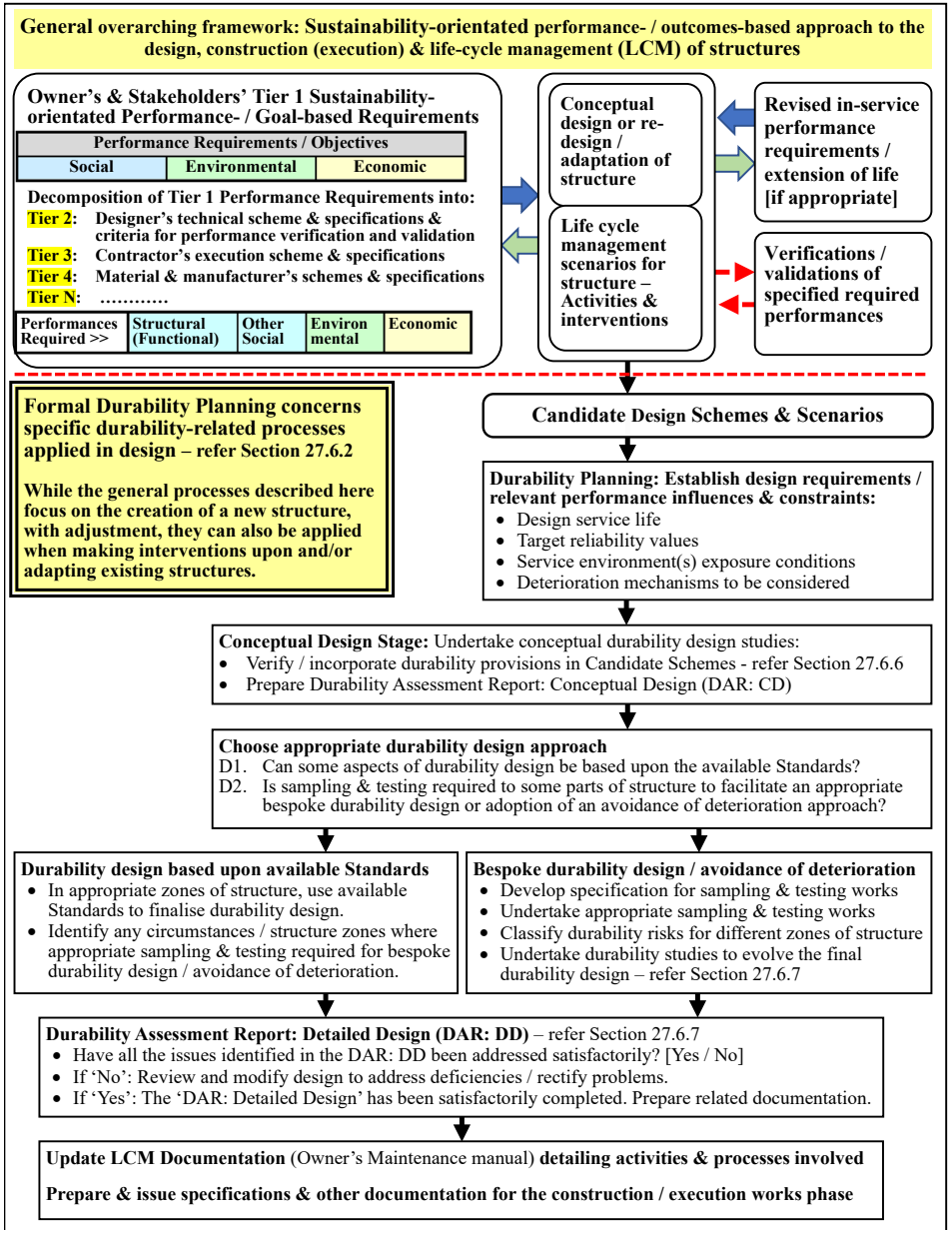

Figure 27.6-1: Durability process during design (after CIA Z7/01 [27-8], Figure 2.1).

27. Approach to design

Detailed design studies shall be undertaken to establish an appropriate durability approach. The outcome shall be presented in a formal report – the Durability Assessment Report: Detailed Design (DAR: DD), which shall address all relevant durability matters, such as the following:

- Verification that satisfactory durability will be achieved to satisfy the specified performance requirements, plus identify where a lack of durability technical knowledge and/or inadequate durability provision may prejudice the performance requirements being met.
- Modelling of deterioration processes as appropriate to support future service life design of structures/members for materials.
- Durability audit of the final design including drawings.
- Evaluation of the proposed materials to establish compliance with required performance criteria.
- Drafting and/or review of final specifications for durability matters.
- Durability review of design details to minimise and/or avoid adverse durability effects (e.g. rebar layout, fitting and fixtures, seals, flashings, waterproofing, cladding, etc.).
- Durability review of design details to minimise and/or avoid adverse durability effects of exposure (e.g. upon reinforcing bar layout, fitting and fixtures, seals, flashings, waterproofing, cladding, etc.).
- Durability review of materials specifications.
- Proposing and/or reviewing inspection, monitoring and testing programs to ensure prompt identification of premature deterioration, complemented by proactive LCM activities and interventions.
- Reviewing quality management requirements for durability sensitive critical elements of materials and workmanship.
- Proposing and/or reviewing LCM / maintenance procedures to ensure the structure will remain in the required condition at end of design service life for optimum life-cycle cost.

Figure 27.6-1 presents a flow chart of the processes delivering durability in the design phase with reference to deemed-to-satisfy durability provisions and exposure conditions to determine when additional durability assessment / provision is required. The durability process provides general guidance that can be applied to any design.

Expert durability input modelling includes consideration of concrete structure early-age thermal and later-age drying shrinkage crack control, deterioration mechanisms for concrete and reinforcements, waterproofing, coating performance, etc.

Materials evaluation includes detailed assessment of designer and contractor proposed materials, related current technical data and the historic performance of the materials.

The satisfactory achievement of durability shall be evaluated relative to the concrete structure Owners' specified durability / minimum performance requirements, the length of the design service life and the specified target reliability value(s).

27.6.8 Durability planning - execution aspects

The durability planning approach is applicable to all materials and components used in the creation and life-cycle management of concrete structures.

The execution/construction durability review requires continued and consistent expert durability input during construction, in accordance with the DAR: DD. Section 35.6 provides more details.

Additional exposure assessment may be required, for example involving further ground/groundwater sampling (eg. chemical composition), assessment of chloride ingress in adjacent structures, modelling of the influence of internal air CO_2 content / RH, etc.

Site trials may be required to prove concrete mix details, penetrability performance (e.g. chlorides, CO_2), development of heat of hydration and/or the ease of concrete placement / compaction

Perform reviews to ensure the durability design intent detailed in the Durability Assessment Report: Detailed Design (DAR: DD refer Section 27.6.7) will be achieved in the contractor's project execution / construction.

The contractor's compliance shall be evaluated with respect to the requirements defined in the project specifications and drawings, which shall be consistent with the DAR: DD durability design intent.

Prior to completion of execution / construction, durability verification (presented in a formal report, refer Section 35.6.4) is required to provide evidence that the durability design intent has been achieved by review of all durability non-compliances and repairs undertaken.

27.6.9 Durability planning - contribution to LCM

Life-cycle management / maintenance activities during use are an integral part of ensuring durability through the structure life-cycle and involve:

- A structure maintenance plan delivered by the contractor upon completion.
- A DAR: DD and the execution / construction durability verification formal report shall provide guidance on operational maintenance.
- Details of inspection audits and tests to be undertaken, plus assessment criteria.
- Details of maintenance materials and methods to be adopted.

Life-cycle management / maintenance activities during use are an integral part of ensuring durability through the structure life-cycle.

The operational maintenance provider shall be given the Durability Assessment Report: Detailed Design (DAR: DD, refer Section 27.6.7), and the execution / construction durability Verification Report (DAR: VR, refer Section 27.6.8), plus associated documentation such as the Birth / Re-Birth Certificate / maintenance plan / concrete structure management plan

Section 38.1.2 provides more details.

- LCM planning based on expected materials deterioration and costs to achieve acceptable performance predicted at appropriate intervals (not exceeding 10 years).
- Anticipated remedial works required during maintenance periods.
- Minor repairs to be completed within scheduled maintenance periods.
- Keeping records of defects and repairs throughout structure life-cycle.

Resilience is often colloquially described as '*the ability to bounce back*' after a disaster or mishap, such as a shock incident or extreme event.

Vulnerability is the possibility that a hazard or threat can cause damage, disruption or some form of harm to the entity or system concerned.

Ideally a structure shall also have an ability to adapt to longer-term changes in use / performance requirements or in the physical environment – see section 27.8.

The Intergovernmental Panel on Climate Change (IPCC) defines ***Vulnerability*** as "*the degree to which a system is susceptible to, and unable to cope with, adverse effects*".

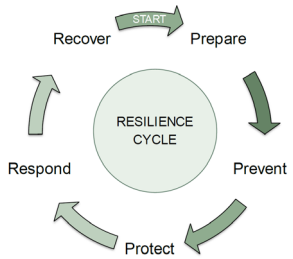

Figure 27.7-1: The resilience cycle.

27.7 Resilience

27.7.1 General

A resilient built environment is planned and appropriately designed to mitigate the effects of hazards. A key consideration is the time taken to recover functionality following in the aftermath of a short-term shock or extreme event.

When resilience (and adaptation) measures are deemed necessary, these requirements shall have an appropriate scope and resources provided at a suitable level, as established through hazard and vulnerability reviews and analyses.

Resilience is the ability of a system, community or society exposed to hazards to resist, absorb, accommodate and recover from the effects of a hazard in a timely and efficient manner, involving the preservation and restoration of its essential basic structures and functions, through the use of risk management techniques.

Societal resilience is achieved through the preservation and restoration of essential functions provided by the built environment and societal organisation.

For a society and its communities to be resilient in the face of the diverse range of hazards which they may potentially be exposed to, it is necessary for them to have a range of effective measures and procedures in place.

The resilience of society, its communities and of the elements of the built environment they rely upon shall be developed through the five sequential phases listed below.

- *Prepare* for disaster / shock event scenarios;
- *Prevent* by eliminating or mitigating against prospective hazards;
- *Protect* to resist the effects of disaster / shock events;
- *Respond* to these effects, and

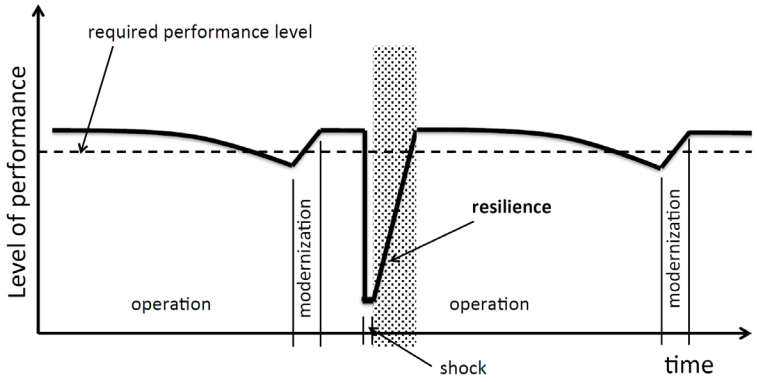

Figure 27.7-2: *Illustration of idealised resilient behaviour - the rapid 'bounce back' after an extreme event or shock incident.*

Table 27.7-1: Examples of potential hazards and threats.

Potential hazards and threats include:	
- Seismic events	- Landslides
- Flooding	- Volcanic events
- Tsunami	- Disease outbreaks
- Extreme weather (wind / snow / heat etc)	- Transport disasters
- Drought	- Malicious actions
- Explosions (deflagrations / detonations)	

Disaster risk is determined by the degree of overlap between hazard risk and the vulnerable system. Disaster risk reduction is the concept and practice of reducing disaster risks through systematic efforts to analyse and manage the causal factors of disasters, including reducing exposure to hazards, lessening vulnerability of people and property and improving preparedness for adverse events. Further information upon these issues is given in the "*Disaster Assistance Handbook*" [27-9] published by the American Institute of Architects.

The implications of a failure / lack of appropriate resilience have to be evaluated on the basis of the potential consequences associated with the particular threat / hazard scenario

- *Recover* from such effects in a timely and efficient manner.

Often these sequential phases are represented in the form of a resilience cycle – see Figure 27.7-1.

Effective preparatory risk management is essential for identifying potential hazards and threats. It can facilitate development of appropriate abilities before an extreme event, which aid an adequate response to events and help with successful recovery from them.

It is important to note that recovery may possibly involve some degree of short-term functional impairment – see Figure 27.7-2. This depends upon the severity of the shock or mishap experienced. It can also be influenced by any economic (or other) constraints which might apply to the availability of resources to respond to the extreme event or shock incident.

27.7.2 Resilient society and community requirements

The first step towards creating a resilient (and adaptable) society / community is to establish the scope of these requirements / the level at which they shall be provided, which requires hazard and vulnerability analyses and their review.

It is necessary for resilient societies and communities to have the ability to:

1. Effectively mitigate against a prospective disaster. Mitigation means taking measures which avoid / eliminate, reduce, resist and aid recovery from extreme short-term events or incidents, such as those listed in Table 27.7-1.
2. Successful mitigation requires that:
   - Potential hazards and threats are identified and evaluated. This shall be done by using effective preparatory risk management and vulnerability assessment procedures.
   - Appropriate preparatory activities are undertaken and that these are used to mitigate, prepare for, take preventative steps, provide protection against, develop plans and secure resources to be able to respond to and recover from an extreme events or incidents.
   - Appropriate societal, economic and environmental resources are available to implement the preparatory activities, along with timely and effective response actions.

(which might involve a combination or sequence of potential hazards) and the probability of its / their occurrence.

This approach is coherent with the adoption of performance-based or outcomes-based standards / codes of practice for design, specification and the assessment of existing concrete structures.

In order to rapidly resume the use of buildings and structures following a shock incident or event, it is essential to embrace all the associated aspects of:

- Avoidance, diminution or removal of identified threats or hazards
- Preparation for disaster event scenarios
- Resistance to the effects of disaster events, and
- Recovery following such an event

Consideration of the functionality of the structures / constructed system in conjunction with economic and social factors is central to making a holistic evaluation of its sustainability. Providing an appropriate level of robustness for exceptional events, and resilience for the subsequent recovery after such events, is clearly important. Central considerations in this will be human safety and wellbeing (and potentially environmental protection depending upon the nature of the building or structure concerned), social disruption, recovery and the life-cycle environmental and cost burdens that would arise from any subsequent repair / reconstruction activities.

More detailed consideration of these and other aspects relevant to highway structures is given in the report on the State of the art for the evaluation of resilience measures, BASt - FE 89.0320/2016 [27-10].

Other useful references include:

- Institution of Civil Engineers, Civil Engineering Special Issue on “Infrastructure resilience” [27-11];
- Tsunami loadings – see Macabuag et al. [27-12];
- Landslides – Hong Kong guidance on slope management (Geoguide 5[27-13]);
- Flood risk & mitigation design (FEMA 543 [27-14])
- High wind risk & mitigation design ([FEMA 543 [27-14])

27.7.3 Achieving structural resilience

As all concrete structures are essentially designed, constructed and managed through-life to meet one or more social functions, the resilience of a structural system shall be evaluated in terms of the disruption of those social functions and the subsequent recovery of that functionality.

Satisfactory structural resilience requires the ability to rapidly resume the functional use of buildings and structures following a shock incident or extreme event.

To achieve this requires the following matters to be adequately addressed:

- That all the aspects of preparation for and recovery following a shock incident or extreme event are in place.
- Appropriate design measures are in place for life safety and environmental protection considerations, e.g. by addressing disproportionate collapse requirements.
- Adoption of appropriate damage limitation design to reduce the need for repairs or reconstruction after a shock or extreme event. This effectively reduces the life-cycle environmental impact of the building or structure and improves life-cycle sustainability.
- Recognition of the spectrum of severity of potential events in the context of causing disruption to the intended social function of the structure. The time and effort involved in recovering from such events shall also be considered.
- Recognition of the range and nature of potential hazards / extreme events which may pose significant life-safety risks and result in associated damage. The extended time and effort involved in recovering from such events shall also be considered. This might include possible partial or total demolition and reconstruction.
- Mitigating / preventive actions to minimise potential malicious actions
- That design and construction account for and facilitate both envisaged LCM activities (e.g. maintenance) and recovery works following a shock incident or extreme event.

27.8 Adaptation

27.8.1 General

Where appropriate, effective preparatory risk management shall be used identifying potential hazards and threats in order to facilitate development of appropriate abilities to adapt to evolving requirements and the effects of a changing environment.

Where it is foreseen that societies and communities will need to adapt to longer-term changes in use or other evolving requirements, environment or circumstances, potential hazards and threats shall be identified and evaluated. This shall be done by using effective preparatory risk management and vulnerability assessment procedures.

Successful adaption requires that:

- Appropriate preparatory activities are undertaken and that these are used to develop plans to mitigate, prepare for, take preventative steps and implement appropriate counter-measures against the identified hazards / threats.
- Appropriate societal, economic and environmental resources are available to implement the preparatory activities.

27.8.2 Achieving structural adaptation

Structural adaptation is the ability of a concrete structure to meet gradually changing circumstances or the evolution of performance requirements. It needs to encompass the ability of the system or structure to sustainably accommodate such changing circumstances. Consideration shall be given to these altered performance requirements during the conceptual design stage.

In terms of longer-term structural adaption, consideration shall be given to the potential changes in the demands placed upon a structure. These might relate to factors such as ongoing environmental changes, evolving societal and functional needs, as well as associated changes in performance requirements.

A key consideration shall be the efficient and effective life-cycle use of materials and other resources.

Consideration shall be given to the timescales over which the various identified hazards and threats might develop, if not already an immediate concern, potentially allowing a phased approach to be employed in the development and implementation of the chosen adaptation strategy.

Examples of this need could be societal changes, evolution in performance requirements or the effects arising from climate change.

Some examples of possible adaptations include:

- Designing buildings to accommodate changes in their internal layout associated with potential future changes of use / functionality
- Designing buildings to facilitate the construction of additional floors / internal space
- Designing bridges with detailing to facilitate future widening;
- Designing foundations to withstand future water level rises / scour from flooding

As there are usually various ways of making structural alterations / adaptations the new performance requirements, whenever possible, these shall be considered during the conceptual design stage. In the case of a modular structure, it could be possible for relevant elements to be replaced / substituted to achieve the new performance requirements.

Key focus areas for structural adaptation are likely to include:

- Human safety and environmental protection
- Damage limitation design to facilitate the rapid resumption of use of structures after some form of extreme event or disaster
- Climate emergency effects
- Urbanisation, including the development of megacities

- Responding to humanitarian crises and natural hazards

References

[27-1] UNSDG — United Nations Sustainable Development Goals (UNSDG), United Nations – Resolution 70/1, 25 September 2015: *Transforming our world: The 2030 Agenda for Sustainable Development.* Available at https://www.un.org/ga/search/view_doc.asp?symbol=A/RES/70/1&Lang=E. [Accessed October 2020].

[27-2] Orr, J., Ibell, T., Smith, C., Cooke, M. — Orr, J., Ibell, T., Smith, C., Cooke, M., IatructE Guidance, Design for zero, 2021, Design for zero - The Institution of Structural Engineers (istructe.org)

[27-3] BSI PAS 2080:2016 — Publicly Available Specification PAS 2080: 2016: Carbon management in infrastructure, ISBN 9780580901553, BSI, London

New Version - Publicly Available Specification PAS 2080: 2022: Carbon management in buildings and infrastructure, BSI, London

Not sure if the scope of PAS 2080: 2022 has changed and whether the cited text only occurs in the PAS 2080: 2016 version

[27-4] Guidance Document for PAS 2080 — Guidance Document for Publicly Available Specification PAS 2080. The world's first specification for managing whole life carbon in infrastructure. Construction Leadership Council and The Green Construction Board, 2019.

Available from: https://www.constructionleadershipcouncil.co.uk/wp-content/uploads/2019/06/Guidance-Document-for-PAS2080_vFinal.pdf [Accessed October 2022]

[27-5] ISO 22040 — *ISO 22040:2021: Life cycle management of concrete structures*, ISO, 2021

[27-6] EN 1991-1-7:2006. — *Eurocode 1 - Actions on structures —Part 1-7: General actions — Accidental actions, CEN, Brussels.*

[27-7] *fib* Bulletin XY — Durability planning for concrete structures [IN PREPARATION, COM8]

[27-8] CIA Z7-01 — *Durability Planning,* which is a Recommended Practice Report forming part of the CIA Concrete Durability Series, Concrete Institute of Australia (CIA), Sydney, 2014.

[27-9] Disaster Assistance Handbook. — American Institute of Architects, Third Edition, 2017. http://content.aia.org/sites/default/files/2017-05/Disaster_Assistance_Handbook_050917.pdf

[27-10] BASt - FE 89.0320/2016. — State of the Art for the Evaluation of Resilience Measures, German Federal Highway Research Institute, BASt, 2017.

[27-11] ICE. — Institution of Civil Engineers, Special Issue – Infrastructure resilience, 165, CE6, 2012.

[27-12] Macabuag et al — Macabuag, J., Raby, A., Pomonis, A., Nistor, I., Wilkinson, S., Rossetto, T., Tsunami design procedures for engineered buildings: A critical review, *Proceedings of the Institution of Civil Engineers – Civil Engineering*, 171, 4, 166 – 178, 2018..

[27-13] Geoguide 5 — Guide to Slope Maintenance (Geoguide 5, 3rd Edition) (Continuously updated e-version released on 20 November 2018). Geotechnical Engineering Office, Civil Engineering and Development Department, HKSAR Government, 116 p.

[27-14] FEMA 543 — FEMA 543, Design guide for improving critical facility safety from flooding and high winds - Providing protection to people and buildings, Risk Management Series, January 2007.

Available at: https://www.fema.gov/sites/default/files/2020-08/fema543_design_guide_complete.pdf [Accessed October 2022]

28 Approach to Assessment

28.1 General

Assessment is a fundamental activity, coupled with Life-cycle management (LCM) procedures, which establishes the need for operational actions and intervention works (eg. alterations, adaptations, renovation, upgrading of performance, extension of service life, etc) upon existing structures

The adequacy of an existing concrete structure for continued service is verified by comparing its structural capabilities (eg load capacity, ability to resist / accommodate actions applied to the structure) with the current and/or future performance requirements, such as a defined set of loadings or actions (original or revised) and/or associated environmental actions.

Performance evaluation / verification may be undertaken to verify the adequacy of a concrete structure (or its component parts) – refer section 12.5:

a) To achieve its design service life, as envisaged at the time of design;

b) To achieve an extension of the planned length of service life or meet revised performance requirements (e.g. revised loadings and/or environmental actions or some aspects of the required functionality);

For structures constructed on the basis of previous (now outdated) design standards, the assessment shall use modern methods and be based upon up-to-date knowledge.

An increase in the level of the performance requirements during the use stage of a concrete structure may necessitate an intervention being made to increase the performance, including durability, of the concrete structure.

It may be necessary for a performance evaluation to:

a) consider the effects of any current deterioration / damage upon the adequacy of the concrete structure; and / or

b) make a prognosis of the future effects of any deterioration upon the adequacy of the concrete structure and/or the length of time during which its performance is anticipated to remain adequate.

The objectives of the performance evaluation (assessment) of an existing structure will typically be agreed in consultation with the client (owner) / stakeholders (eg. the owner, the relevant authorities, insurance companies, etc.) and be based upon defined performance levels. These might typically entail it's:

- safety performance level;
- continued functional performance for specific hazards, which may be defined in terms of specified loads and actions; and
- any special performance requirements defined by the client.

Account should be taken of permitted reductions in the target structural reliability level (β-value) appropriate for an existing concrete structure – refer section 11.3.2.3: *Target reliability levels for assessment of existing structures* and *fib* Bulletin 80 [28-1]

Further details on approaches to performance evaluation / structural assessment are given in ISO 13822 [28-2], ISO 16311 Part 2 [28-3], the JCI Guidelines on structural assessment [28-4] and in the CEN TS17440 report [28-5].

The process of performance evaluation would be expected to draw upon information obtained for and detailed in the Birth Certificate, and potentially also in the re-Birth Certificate, assuming that these are available. For further details on these topics see *fib* Bulletin 93 [28-6].

It is assumed that the associated survey and monitoring process will be able to furnish sufficient information to permit decisions to be made in respect of options for the management of the structure. If there is not sufficient information it may be necessary to initiate a detailed investigation.

An important aspect may be to verify compliance with the original performance requirements (eg. the loading/action levels) and usage conditions, including consideration of their potential influence on the structural behaviour.

For older structures / those constructed on the basis of previous (now outdated) design standards, the current adequacy (of the relevant parts) of the original design provisions shall be careful reviewed before they are accepted.

The evaluations performed shall take account of the conservation strategy and tactics defined at the time of design or re-design of the structure.

Consideration shall also be given to the full range of hazards the structure is exposed to (physical, environmental, etc) and the vulnerabilities it exhibits.

An overall evaluation of the extent of any material and/or structural performance reduction due to deterioration / damage shall be based on:

- records in the design file and the as-built-documentation, see chapter 10;
- results of survey and monitoring activities; and
- assessment of the nature, severity, significance and rate of deterioration.

28.2 Consideration of sustainability

The general principles concerning sustainable performance in MC2020 are given in chapter 3. Sustainability considerations would be of particular relevance if the client had requested an Assessment Report which required consideration of potential structure management activities / intervention works to address functional / performance deficiencies (Report 2, Fig 28.4-1).

The objectives and performance requirements established under the three pillars of sustainability; namely those of social responsibility, environmental quality and economic efficiency, define a set of interdependent performance requirements. This framework of requirements is the basis for the Life-cycle management (LCM) philosophy and strategy developed for the concrete structure.

28.3 Life-cycle management (LCM) considerations

It is important to identify the different factors which may affect the service life or performance of the structure, so that they may be adequately taken into account in the LCM strategy and during the assessment process.

During the in-use stage of the life of a concrete structure, periodic inspection / monitoring / diagnosis shall be conducted to confirm that the retained performance of the structure is adequate for the defined performance requirements.

Performance evaluation typically involves:

a. Analysing the available data to establish current performance of the concrete structure;

b. Making a prognosis of future performance, including identification of deterioration mechanisms and prediction of future damage;

c. Identifying the prospective need for, timing of and nature of any operational measures or physical interventions required.

The LCM strategy for the structure shall contribute to achieving an acceptable balance between the different factors considered under the three pillars of sustainability, as noted above – refer chapters 3 and 5.

As the LCM strategy is used in planning and managing a concrete structure throughout its life-cycle, particularly during its operation and use phase, account shall be taken of any previous or proposed changes in the performance requirements or adaptations made to the existing structure.

In regard to the activities which might form suitable elements of the LCM strategy defined for the concrete structure consideration shall be given to the adoption of appropriate operational measures, including use restrictions and the possibility of termination of use, as well as direct physical intervention measures involving repair / upgrading works, etc.

When there is an uprating of the defined performance requirements during the in-use stage, as an alternative to making a physical intervention to increase the performance / durability of the concrete structure, the assessment shall give

In response to the developing climate emergency, amongst other potential courses of action, upgrading an existing concrete structure for extended or alternative use may offer a more carbon efficient alternative to demolition and construction of a new replacement facility. This approach, which requires a structural assessment to be made of the concrete structures concerned, supports the aim that engineering design and life-cycle management processes move towards achieving a net zero carbon life-cycle. For further details on the various aspects of the LCM of concrete structures including structural assessment and the design and execution of interventions refer to ISO 13822 [28-2], ISO 16311 Parts 1 to 4 [28-2, 28-7 to -9], the JCI Guidelines on structural assessment [28-4] and the CEN TS17440 report [28-5].

consideration to load-carrying and other behaviours which might not have been considered in design (ie. those which tend to increase load-carrying capacity and other parameters such as flexural stiffness) as a basis for verifying that the concrete structure can satisfactorily meet the uprated performance requirements specified.

During the assessment the Durability Plan (refer section 27.7) shall be updated to take account of the current condition of the concrete structure and any changes in the performance requirements or adaptations made / proposed.

28.4 Performance / structural assessment procedure

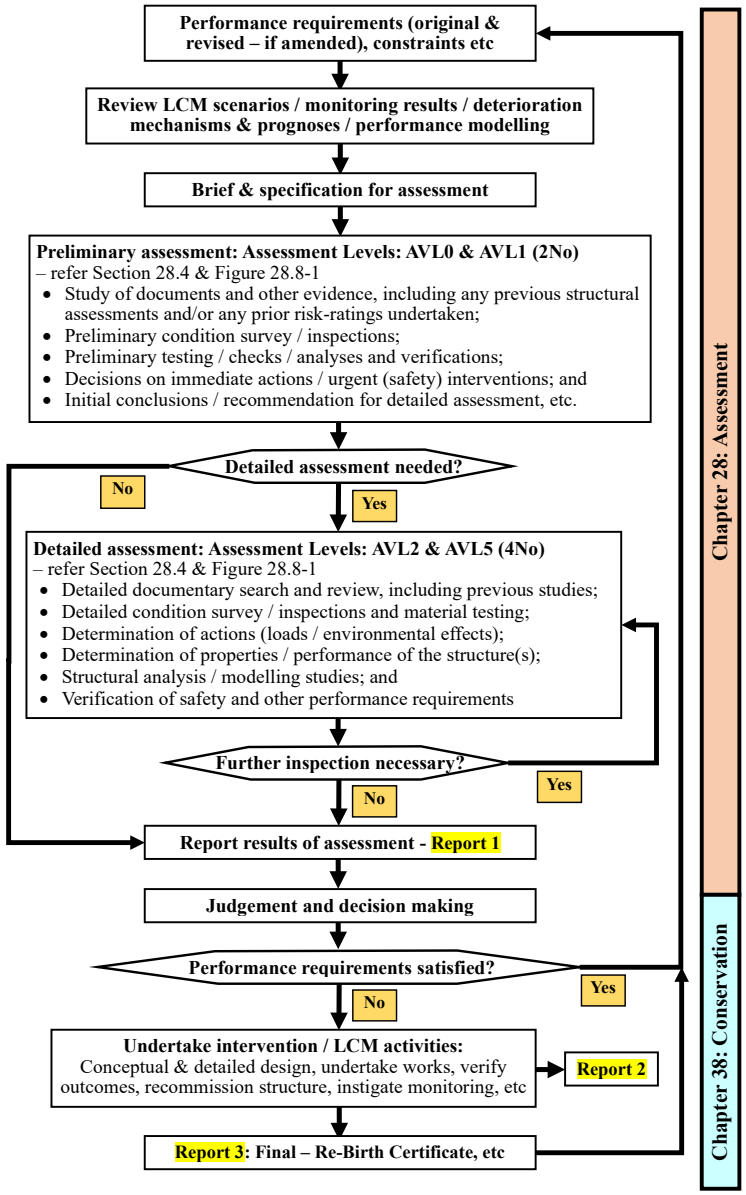

Figure 28.4-1. *General flowchart for the assessment of existing structures (adapted from ISO 13822, Annex B* [28-2]*).* **[To be Further update]**

The performance evaluation (structural assessment) is to be carried out taking into account the actual circumstances and condition of the structure (refer the flowchart presented in Figure 28.4-1) and generally would comprise the following steps a) to f), as outlined in ISO 13822 [28-2]:

a. Definition of the assessment objectives;

b. Assessment scenarios – relating to anticipated changes in structural conditions and/or action effects;

c. A preliminary structural assessment, which may involve:
   i. Study of documents and other evidence, including any previous structural assessments and/or any prior risk-ratings undertaken;
   ii. Preliminary condition survey / inspections;
   iii. Preliminary testing / checks / analyses and verifications;
   iv. Decisions on immediate actions / urgent (safety) interventions; and
   v. Initial conclusions / recommendation for detailed assessment, etc.

d. A detailed structural assessment, if required, which may involve:
   i. Detailed documentary search and review, including previous studies;
   ii. Detailed condition survey / inspections and material testing;
   iii. Determination of actions (loads / environmental effects);
   iv. Determination of properties / performance of the structure(s);
   v. Structural analysis / modelling studies; and
   vi. Verification of safety and other performance requirements.

e. Repetition of the above sequence, as appropriate / necessary.

f. The findings of the structural assessment shall be described in a structural assessment report [Report 1], but the following may also be necessary.

g. If required by client / owner when significant performance deficiencies are identified, the conceptual design of intervention works and the control of hazards and risks by use of operational actions [Report 2].

h. A final report after completion of the intervention works [Report 3]

Note to Figure 28.4-1: Report 1 is the end of the structural assessment process. However, Report 2 will be needed where the client requires detailed proposals for intervention works etc. For Report 3 refer to Bulletin 93 [28-6] for further details.

Whilst the incremental approach is aimed at minimising the investigation effort expended, the approach may not always represent good value when associated factors, such as the cost of gaining appropriate access to the relevant zones of the concrete structure, are taken into account.

Considering the potentially high cost of gaining access, it may be more appropriate to take the opportunity to perform additional inspections / testing to gather a variety of complementary data and samples. The additional cost may be relatively small.

These procedures may be applied to the structural assessment of both a specific structure and/or the assessment of a portfolio of concrete structures.

Preliminary assessment: The scope of the preliminary assessment may need to be adapted to suit the specific / various circumstances encountered as the activities progress. The goals of the preliminary assessment will typically include developing relevant damage / deterioration hypotheses in support of the initial conclusions and any recommendations for future actions / detailed assessment.

Detailed assessment: Depending upon the specific circumstances, special consideration may need to be given to matters such as:

- Updating the basis for the assessment, including refinement of the damage / deterioration hypotheses and prognoses of their future development, plus the potential evolution of damage and deterioration mechanisms, material properties, soil conditions, etc.;
- Evaluation of structural / other performances using results obtained by testing / other sources, including data from monitoring etc, as available;
- Decisions on any immediately required interventions / LCM activities;
- Conclusions and recommendations for future actions.

A core objective of structural assessment is to make estimates of relevant structural performance parameters, such as the current load-carrying capacity of a structure, to an appropriate degree of accuracy, whilst seeking to limit the effort (and cost) expended in achieving the required outcomes.

Accordingly, the process of undertaking a structural assessment, particularly of a deteriorated structure, is commonly taken forward using an incremental approach, which may involve a number of cycles of gathering and evaluating information. This step-by-step incremental approach is reflected in the adoption of the two assessment stages in Figure 28.4-1, namely the preliminary and the detailed assessment stages.

This approach recognises that the assessment of existing structures can be carried out using methods involving different degrees of sophistication and required effort. In most cases it is appropriate to start with simple conservative procedures and to only use more sophisticated procedures when the evaluated structural performance parameters (eg. load-carrying capacity) is insufficient.

The following provides supplementary observations to those given above for the main steps in the assessment process outlined in ISO 13822.

Preliminary assessment: The preliminary assessment shall normally consider the entire structure.

Detailed assessment: The scope of the detailed assessment shall typically be based on the findings of the preliminary assessment. However, if there has been an extended delay, the scope and activities originally planned for the detailed assessment shall be reviewed to take account of changes in circumstances.

Results of assessment: The outcomes from the assessment shall always be subject to appropriate plausibility checks, especially when there are notable discrepancies between the apparent performance of the structure and the assessment results (ie. an unexpectedly good or bad assessment outcome).

Assessment Report: A report documenting the results of the structural assessment shall be prepared, including appropriate supporting information. The contents might be expected to include the items listed in the (informative) left-hand column.

Results of the structural assessment: The report documenting the results of an assessment might be expected to include the following information:

- Scope of the assessment (and any limitations);
- Description of the structure and its anticipated usage (Utilisation plan);
- Investigations undertaken and results obtained;
- Analysis, verifications and interpretation of the findings;
- Review of intervention / LCM or operational change options;
- Conclusions and recommendations for future actions.

Report 1: Structural assessment report, plus brief preliminary indications of prospective interventions and operational actions to control / limit hazards.

Report 2: Structural assessment report, plus (where required by the client / owner) the development of detailed proposals for intervention works and operational actions to control / limit hazards – refer chapters 5, 8, 26, 27 & 38.

If the structural safety or serviceability are shown to be inadequate (currently or prospectively within the chosen study period), the results of the assessment shall be used to provide recommendations upon the need for LCM / operational actions (such as monitoring, temporary restrictions in use, long-term restrictions or change of use) and/or physical intervention works construction interventions (repairs, upgrading, reconstruction etc), or a combination thereof.

28.5 Condition survey and monitoring

Data may be gathered from the structure for condition assessment purposes by a variety of techniques which are used to undertake inspections, measurements, testing, sampling and monitoring activities. As such, these could be expected to utilise a variety of procedures including laboratory and in-situ procedures, structural health monitoring and associated techniques – refer sections 30.11 and 38.3.

A visual inspection, especially if undertaken at close-range, can provide a lot of valuable information about the nature and condition of the structure and its component parts to an experienced and knowledgeable observer. Such data, and insights gained from it, can be valuable in guiding the focus of the assessment, condition survey and monitoring activities, as well as whether more in-depth studies, specific tests, etc might be appropriate. The results of visual inspection may also establish that the condition of the structure is such that further studies are not merited.

The evaluation of the actual condition of a structure, including active deterioration mechanisms and existing damage / defects, shall take into account all the existing design and survey information. Where this documentation is inadequate, this shall be supplemented by additional information gathered from the structure for condition assessment purposes – for further details refer section 38.2

A visual inspection of an existing structure shall be conducted as it is an essential component of the structural assessment. A preliminary (reconnaissance) visual inspection is generally very beneficial in terms of the information gathered about the nature and current condition of the structure, especially about zones of damage and / or deterioration and whether the structure or any of its component parts might be in a potentially dangerous condition requiring an urgent intervention.

If the current and original design load effect can be assumed to satisfactorily represent the actual loading and if the current design load effect does not exceed the original one, it may be reasonable to assume that the actual load effect will have a suitably small probability of exceeding the load / action effects considered in design.

If so, the load and other action effects can be determined and the reliability can be verified in a simplified manner. As the material and action uncertainties are modelled on the basis of data from the literature and experience, this will only provide a preliminary indication of the safety and serviceability of the structure for the defined performance requirements during its intended remaining / extended service life.

Assuming that the design of the structure was performed correctly and the structure has been properly executed and maintained, the actual resistance value is considered not to be less than the required (design) structural resistance. Guidance for reliability assessment based on satisfactory past-performance is provided in section 28.9. Guidance for reliability assessment based on the partial factor method, a full-probabilistic method, and a risk-based method is provided in section 12.5.

The design information (after appropriate checking) will typically provide a detailed and valuable resource, providing insight into the prospective behaviour of the undamaged concrete structure.

All assumptions and data used in the structural analysis and verification shall be adequately documented, including updated information regarding:

- the structural form and the structural system details;
- the member geometry and dimension, as well as their consistency and agreement with the available design / construction documents;
- the construction materials and geotechnical properties – previous / contemporary values and those expected in the future (if different);
- contemporary action effects and influences, as well as those expected in the future (if different – consider potential climate change effects);
- the condition of the structure, in particular of those parts affected by deterioration mechanisms and damage / defects, and its behaviour;

28.6 Safety and serviceability - Performance assessment of an undamaged structure without deterioration

Safety and serviceability evaluations (assessments) shall consider all relevant loading and actions cases, together with the possible failure modes.

Assuming that the structural system under consideration can be satisfactorily defined and the structure is in an undamaged state, the structural resistances for the limit states used in design may be expected to provide an useful indication of the minimum magnitude of the member internal forces adopted in design; namely the values obtained from the load models multiplied by either the global safety factor (for older codes) or the partial safety factors (for recent codes) employed for design.

If the load and other actions models used for the original structural design or recent assessment are known, it is reasonable (after appropriate review and checking has been performed) to assume that the structure was designed / assessed in accordance with these models. If this is the case, it may be possible to adopt a simplified approach to evaluating (assessing) the current structural safety and serviceability performance of an undamaged concrete structure.

In this approach, the current load model, i.e. the one that represents the current loading and actions, is applied to the structural system and the load effects are compared with the values derived during the original structural design.

The verification process adopted shall take account of the objectives of the assessment and related matters, such as possible interventions, including:

- intended uses;
- remaining / required working / service life;
- performance requirements (including structural safety, robustness, serviceability, durability, etc);
- operational conditions (e.g. maximum in-service wind speed, protection against environmental influences, etc.);
- preferences concerning possible measures (e.g. change in use instead of constructional interventions, etc.);

- the relevant assessment situations, selected by considering the circumstances under which the structure is expected to fulfil the established and/or modified performance requirements.

The circumstances where detailed assessment may be requested might include:

- significant damage, displacement or deterioration is observed or severe degradation effects are predicted;
- when the structural system is poorly understood,
- a significant increase of the load effects is expected;
- the risk associated with local failures is estimated as high e.g. due to the structural significance, the danger of progressive collapse, etc.;
- repair would be very costly due to interruption of traffic or industrial operations, relocation of activities and/or users etc.

The methodology relies heavily on information from the design phase and/or gathered during detailed investigations.

While experienced inspectors are likely to know intuitively where the vulnerable zones will be, it is always helpful if they can be supported by the provision of suitable analytical studies / information.

Damage outside vulnerable zones can also trigger failures, but for these to occur the extent of damage needs to be significantly larger than that required in the vulnerable zones. If this seems likely, it is necessary to define any additional failure modes which could potentially be triggered by the observed damage.

- inspection and maintenance requirements (e.g. monitoring, application of protective coatings, etc.).

Detailed assessment is generally requested when a high risk (high failure probability and/or high consequences) is expected or costs of safety measures are excessive.

Chapter 30: *Evaluation of structural performance* provides details of verification methods for the different facets of load bearing capacity and other performances.

The partial-factor format provided in section 12.5.4 provides a practical partial-factor methodology to make assessments of existing structures which are undamaged / not suffering significantly from the effects of deterioration such as reinforcement corrosion. The method allows appropriate partial-factor values to be derived taking account of considerations including Consequence Class, target β-value, the length of the reference period (perhaps the remaining working life of the structure or other selected duration), the influence of permanent and variable loads / actions, model uncertainties, material parameters, as well as the updating of these via testing and sampling procedures.

Additionally, *fib* Bulletin 80 [28-1] presents examples of the application of the partial-factor methods, including the determination of load capacity, together with a comparative example of an assessment made utilising the full-probabilistic method.

28.7 Impact of structure specific observations – change from undamaged structure condition

The following need to be considered if the effects of deterioration and damage are to be appropriately considered in the assessment of a structure:

- Relevant failure modes, defined on the basis of the design documentation. These failure modes correspond to the critical load and other action effects / situations used in design; and
- The vulnerable zones for each failure mode.

Vulnerable zones are those segments and/or members of a structure in which deterioration / damage has the greatest adverse influence upon its safety and

The combination of the serviceability modes with those related to structural safety, needs an adequate consequence analysis, ie. a risk-based approach.

The levels-of-verification (Assessment Verification Level -AVL) approach is particularly valuable for the verification of the performance of existing structures as it allows the accuracy of the estimate of a structural members response (strength or another parameter) to be progressively refined, as more or better information becomes available.

The most appropriate Assessment Verification Level (AVL) for the case being considered will depend not only on the stage of verification process, e.g. preliminary or detailed assessment, but also on the complexity and importance of the member and the structure / under consideration. The AVL approach can be employed for the assessment of structural (load-carrying behaviour) and other forms of performance.

Details on the principles of the related Levels of Approximation (LoA) approach are given in Section 12.3.

Summary details of the attributes of the quantitative structural assessment verification levels - AVL1 to AVL5 – are presented in Table 28.8-1. See also Figure 28.8-1 which provides an overview of the full set of the structural assessment levels (AVL0 to AVL5).

serviceability. One vulnerable zone may be linked to several failure modes (such as flexural and shear modes).

28.8 Structural assessment verification levels

Analytical models having different degrees of sophistication can be employed to evaluate the structural and other performances of an existing concrete structure. These provide different levels of accuracy (or refinement) for the different magnitudes of analysis and verification effort expended.

A potential hierarchy of Assessment Verification Levels (AVL0 to AVL5) for ordering the assessment process is outlined below and illustrated diagrammatically in Figure 28.8-1. The illustrated arrangement is not intended to be definitive and the boundaries between the levels shall have a degree of flexibility. It will be noted that the assessment procedures vary in sophistication. It is recommended to start the assessment with simple but conservative low-level methods (AVL0 to AVL2, used during the Preliminary Assessment) and, where the assessment has not been satisfied, then progress to the use of the more refined upper assessment levels (AVL3 to AVL5, used during the Detailed Assessment).

Table 28.8-1. *Informative Summary of Attributes of Quantitative Structural Assessment Verification Levels - AVL1 to AVL5* [28-10].

Level AVL1: Measurement-based determination of load effects	**Level AVL2: Partial-factor method, based on document review and visual inspection**	**Level AVL3: Partial-factor method, based on supplementary investigation**	**Level AVL4: Assessment with structure-specific partial-factors**	**Level AVL5: Full probabilistic assessment**
Objectives	***Objectives***	***Objectives***	***Objectives***	***Objectives***
- Control of structural performance over a reference time period - Ensure selected performances remain within defined bounds – eg. SLS, fatigue limit values, deformation, stress, stress ranges - Control of variable loads / actions	- Preliminary assessment of safety and serviceability after extreme load damage or after deterioration damage - Determination of safety and serviceability for utility changes	- Determination of load carrying capacity and length of residual service life of damaged / deteriorated structures	- Adapting the partial-factor values on the basis of additional structure-derived information	- Determination of load carrying capacity and remaining service life of damaged structures, taking account of existing uncertainties. Typically, these are cases with uncommon levels of uncertainty or situations involving high risk & cost of safety measures, etc
Data acquisition	***Data acquisition***	***Data acquisition***	***Data acquisition***	***Data acquisition***
- Codes, project specifications, etc - Determination of load effects by measurement of structure performance values (such as deformations, stresses and dynamic parameters) under service or proof loading conditions	- Load & resistance data from design documents & codes, etc - Surveys, inspections - **Note:** Design documentation & models likely inadequate for deteriorating or damaged structures and/or extreme actions such as flooding, impact etc.	- In-depth review of action effects and/or material properties - Survey / investigation of material properties & dimensions (NDT informed by sampling & physical probing, laboratory tests etc) - Monitoring / test-based system identification - Load monitoring - Proof load tests	- In-depth review of action effects and/or material properties - Survey / investigation of material properties and dimensions, etc - Monitoring / test-based system identification (Eigenfrequencies, mode shapes, etc) - Load monitoring - Proof load tests	- In-depth review of action effects and/or material properties - Survey / investigation of material properties and dimensions - Monitoring / test-based system identification - Load monitoring - Proof load tests - Statistical characteristics of data
Structural analysis	***Structural analysis***	***Structural analysis***	***Structural analysis***	***Structural analysis***
- None undertaken	- Methods and models as used in the design process - Refined models (eg deterioration)	- Refined methods and models (FEM, Nonlinear analysis) - Adaptive FEM models	- Refined methods and models (FEM, Nonlinear analysis) - Adaptive FEM models	- Advanced models and methods – stochastic FEM models
Reliability verification	***Reliability verification***	***Reliability verification***	***Reliability verification***	***Reliability verification***
- Comparison of measured data / results with threshold values - Analysis of trends & correlations with external influences - **Note:** See Section 12.5 for details of MC2020 evaluation / verification methods	- Partial-factor methods	- Partial-factor methods - **Note:** For deteriorating / damaged structures and/or extreme actions, uncertainties are usually larger & probabilistic methods may need to be applied (or at least structure-specific / adjusted partial-factors)	- Partial-factors adjusted for the structure under investigation	- Probabilistic methods (FORM, SORM), simulation methods (Monte Carlo / Latin Hypercube Sampling) - Possibly risk optimisation (deriving structure-specific target reliabilities)

In these circumstances, the additional analytical effort may be able to demonstrate that a satisfactory level structural reliability or other form of performance is being achieved; thereby avoiding the often-considerable expense and societal disruption associated with the need to undertake a physical intervention to improve structural performance.

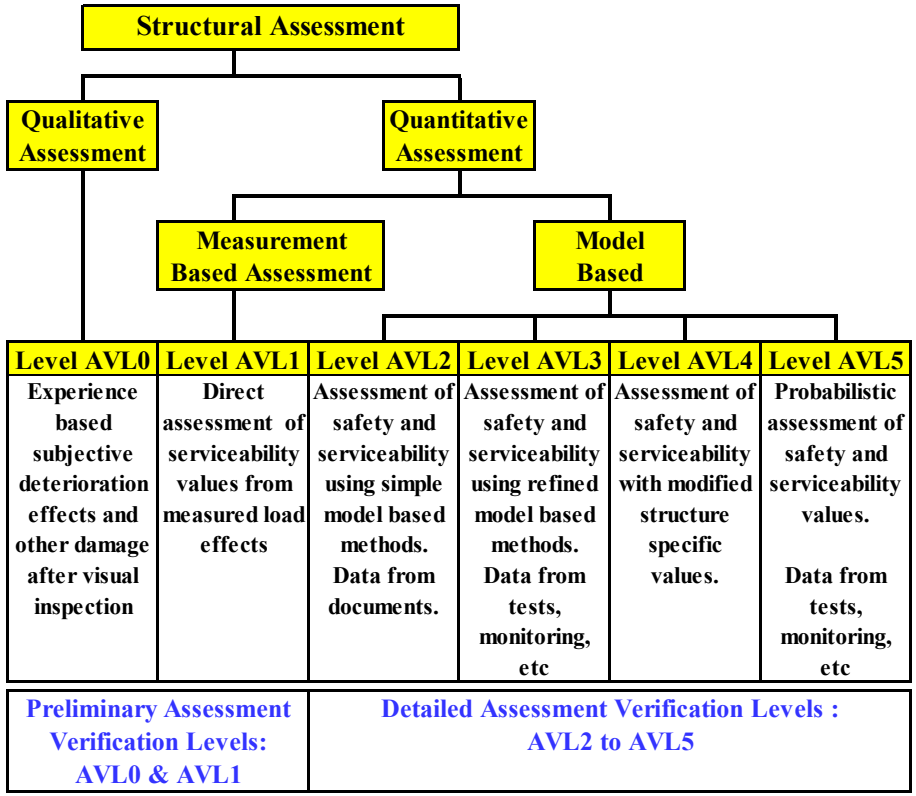

Figure 28.8-1. *Illustrative structural assessment levels* (after *the F08a Guideline of SAMCO Final Report 2006* [28-10]).

The lowest Assessment Verification Level (AVL0) can be used when high accuracy is not required, such as for preliminary design or structural assessment activities. This AVL may be able to demonstrate the satisfactory performance of the member / structure in question. The assessment activities for that member / structure would then be complete.

The highest Assessment Verification Level (AVL5) would be used when the greatest accuracy is required, such as when making an assessment of the global behaviour of a

Assessment Verification Level AVL0: ***Informal qualitative assessment:***

A preliminary assessment, essentially based on visual inspection and engineering experience, used to make a preliminary evaluation of the condition of a structure. The process is only able to recognise the possible effects of damage / deterioration, such as long-term deflections, the corrosion of reinforcing steel, degradation of concrete surface or fatigue effects once there are visible indications (eg. cracks, spalling).

Assessment Verification Level AVL1: ***Measurement-based determination of load effects:***

Assessment of serviceability by measurement of performance values and their comparison with threshold values. No structural analysis is carried out. The threshold values can be those given in codes or be individually specified. The assessment can be made under working loads or a defined proof load.

Assessment Verification Level AVL2: ***Partial-factor method, based on document review and visual inspection:***

Assessment of load-carrying capacity and serviceability is undertaken using information from codes, design, construction and visual inspection documents. Structural analysis is generally carried out using simple (design) methods, but more refined analysis methods may be used. Safety and serviceability verifications are based on partial-factors methods.

Assessment Verification Level AVL3: ***Partial-factor method, based on supplementary investigation:***

Assessment of load-carrying capacity and serviceability is undertaken using additional information gained from site-specific detailed surveys, investigations, non-destructive testing procedures and (potentially) load models, as appropriate. Structural analysis is carried out using refined methods and detailed models. Safety and serviceability verifications are based on partial-factors typically considering their values recommended for a class of existing structures such as the recommended values provided in section 12.5.4.2.3.3.

Assessment Verification Level AVL4: ***Structure-specific adjusted partial-factors:***

Verification of the load–carrying ***capacity*** is undertaken using site-specific adjusted partial factors, following the guidance in section 12.5.4.2.2. Structural properties, as well as external circumstances, can influence the safety evaluation.

critical structure. Undertaking a (simplified) AVL1 analysis would involve far less effort and cost than performing an intensive AVL4 analysis, which takes into account of a wider range of mechanisms influencing the behaviour / contributing to the transfer of the applied loads or other actions to the supports.

Practically, the partial factors may be adjusted for groups of structures with similar structural behaviour or load influences, where such a portfolio of structures exists (e.g. bridges owned by a particular authority, with similar design, execution, exposure, and maintenance).

Assessment Verification Level AVL5: Full probabilistic assessment:

Assessment made taking into account the statistical properties of all the basic variables. Probabilistic reliability analysis is used directly, instead of partial-factors. Uncertainties are modelled probabilistically. In exceptional cases, target reliability levels may need to be determined by probabilistic risk optimisation; see section 12.5.2.

28.9 Assessment based on satisfactory past-performance

If a structural assessment is to be based primarily upon satisfactory past performance, this typically implies that very few detailed structural calculations are likely to be performed. Accordingly, the assessment would primarily be a 'judgement' made on the basis of careful observations and survey results, supported by historical records, etc. This approach may be used by some practitioners adopting a 'filtering methodology', particularly for more minor lower risk structural elements, to reduce the effort (burden) and cost of undertaking an assessment.

However, if loads / actions are expected to increase or the exposure environment will become more severe and / or other relevant future circumstances are expected to change / become more onerous, such as an adverse change in the life cycle management / maintenance and care regime, then a structural assessment based on satisfactory past performance would not be appropriate. Thus, in those circumstances a more in-depth / more comprehensive, typically quantitative structural assessment process would be required.

While the informative comments seek to provide helpful guiding comments to assist practitioners to make judgements and deal with the many uncertainties and ambiguities which may be encountered in the process of undertaking structural assessments, caution must be exercised particularly where structures were designed and / or constructed to be particularly economical and may, as a result, not have the inherent safety margins in the limit states under consideration which might generally be expected by practitioners.

For further details refer to ISO 13822 [38-2].

Concrete structures designed and constructed based on earlier codes or in accordance with what was deemed to be good construction practice of the period when no structural codes existed may be assessed for safety and serviceability based on their satisfactory past performance considering all relevant acting actions, together with the possible failure modes.

When assessing concrete structures or parts of structures based on past performance, a detailed structural investigation shall be undertaken with regard to the limit states being considered/assessed, thereby reviewing:

- the structural form and the structural system details;
- the member geometry and dimension, as well as their consistency and agreement with the available design / construction documents;
- the construction materials and geotechnical properties; and
- the contemporary action effects and influences, as well as those expected in the future if different.

If an assessment based on past performance is carried out for the ultimate limit state, the following shall all be satisfied:

- careful inspection does not reveal any evidence of significant damage, displacement or deterioration;
- the structural system is understood, including the failure modes and the most critical parts of the structure;

- the structure has demonstrated satisfactory performance for a sufficiently long period of time during which extreme actions and unfavourable environmental effects corresponding to the assessment situations have most likely occurred;
- any predicted deterioration (taking into account the present conditions and planned maintenance) would not be expected to affect the reliability and/or the durability;
- there have been no changes for a period of time long enough to significantly increase the actions on the structure or affect its durability, and no such changes are anticipated;
- the risk (likelihood and consequence) associated with local failures can be classified as acceptable.

If an assessment based on past performance is carried out for the serviceability limit state, the following shall all be satisfied:

- careful inspection does not reveal any evidence of significant damage, deformation or other serviceability concerns;
- the structure has demonstrated satisfactory performance for a sufficiently long period of time for damage, distress, deterioration, displacement or vibration to occur;
- there will be no changes to the structure or its use that would significantly alter the actions and affect the serviceability;
- any predicted deterioration (taking into account the present conditions and planned maintenance) would not be expected to significantly affect the durability or serviceability performance.

28.10 Typical assessment cases and considerations

28.10.1 Introduction

The outcomes of structural assessment are used in the processes of:

- Reviewing the adequacy of the structure for continued service by comparing its current structural assessment to the specified performance

There are various types of load-carrying behaviours which may not have been considered in design, such as compressive membrane action (CMA) and the increase of the concrete compressive strength with time.

CMA is a phenomenon which occurs in slabs whose edges are effectively restrained against lateral displacement, thereby mobilising a form of arching action (compression membrane). If there is sufficient confining capacity, the failure load of a slab can be substantially higher than its flexural failure load (ie. many times). Also, if punching failure is the governing failure mode, the effect of CMA plays a role in substantially increasing the bearing capacity of the slab / laterally loaded wall element.

Note: Tensile membrane action (TMA) may also be a relevant structural behaviour phenomenon in some circumstances.

Moreover, there is the increase of the concrete strength in time exceeding the originally specified characteristic value, going along with a change in other concrete properties during the lifetime of a concrete structure.

In addition, advanced methods of structural analysis (eg. calibrated non-linear finite element analyses) are generally able to provide more accurate estimates of the load-bearing capacity of individual structural members, as well as of assemblages of members mobilising either system / whole-building behaviour.

The use of design models for assessment would lead to excessively conservative results because design models are generally simplified versions of more advanced theoretical models. This simplicity is an advantage for design, but the approach includes a higher degree of conservatism. In order to seek to avoid unnecessary financial costs and additional environmental impacts (embodied carbon) associated with strengthening, it is preferable to employ more advanced models. Examples of potential additional residual structural capacity which may not have been exploited in the original design include compressive membrane action in punching and tensile membrane action in slabs. See additional comments in section 28.10.1.

requirements, such as a defined set of loadings or actions (original or revised) and service-environment circumstances.

- Undertaking the above evaluation of adequacy for future service on the basis of a prognosis about the future deteriorated condition of the structure and/or changes in performance requirements, if this is an appropriate consideration.

LCM activities include a requirement to make estimates of current and future safety / structural capacity of existing concrete structures. Amongst other matters, appropriate material and structural models shall be adopted which can take account of structural behaviours and effects, such as:

- Load-carrying and other behaviours which may not have been considered in design – Consideration of these behaviours tends to increase assessed load-carrying capacity and parameters such as flexural stiffness; and
- Load-carrying behaviour in deteriorated members and structures or those which utilise inappropriate / non-compliant structural details / detailing - Consideration of these behaviours tends to decrease load-carrying capacity.

As these two sets of factors typically have opposing / counter-acting influences, it is important to have the ability to employ analytical models which allow the accuracy of the estimate of a structural members response (strength or other parameters) to be progressively refined when this is appropriate.

28.10.2 Assessment of structural safety of existing structures without significant damage or deterioration

A frequent objective is to determine the residual bearing capacity of an existing structure which is subjected to larger loads than those for which it was designed. Examples include bridges which are subjected to heavier traffic loads than foreseen at the time of design.

The Levels of Approximation (LoAs) / Assessment Verification Levels (AVLs) approaches, as described in section 12.3 and chapter 30, and potentially employing more sophisticated and less conservative methods than those which may have been used for design, is especially useful for the determination of

structural safety and other performances of existing structures without damage or deterioration.

This applies to all aspects of bearing resistance and all types of structural behaviour models (shear, punching, etc) - refer chapter 30 and section 12.5.

28.10.3 Assessment of structural safety of existing structures with non-compliant detailing

Consideration shall also be given to the potential influence that non-compliant detailing and construction defects / errors may have upon the safety and serviceability of individual members and the overall structure. This is particularly important for vulnerable zones of a structure as noted in Section 28.7.

A common concern is the assessment of the load-bearing capacity of structures or components with non-compliant details.

For these situations reference shall be made to section 30.1.12.

Reinforcement and other details in older structures, designed on the basis of previous (now outdated) structural codes, may not meet the requirements of contemporary structural design codes and/or associated product standards. In these circumstances the structure will be described as having non-compliant structural detailing and the reinforcement / other detailing classified as being inappropriate, with there being adverse implications for load capacity and structural ductility.

Examples of non-compliant structural details or detailing include structures with:

- Less than the (contemporary) minimum shear reinforcement provision and/or excessively large stirrup spacings / ineffective / deteriorated stirrups
- Plain reinforcing bars (ie. without ribs or indentations);
- Details inconsistent with the principles of modern strut and tie models;
- Inadequate / excessively small depths of concrete cover; and
- Inadequate / excessively short lap and/or anchorage lengths.

For further details refer *fib* Bulletin XX [28-11], particularly section 7.5: *Assessment of structures with non-conforming details under predominantly static and cyclic loading*.

28.10.4 Assessment of structural safety of existing structures suffering deterioration

When structures have experienced deterioration of materials and/or damage to components, an assessment shall be made of the potential adverse effects on bearing capacity and the other required performances (such as durability and serviceability).

The typical forms of deterioration include:

- Corrosion of reinforcement / prestressing strands and/or tendons and the associated adverse effects upon the:

- Reinforcement – concrete interface / prestressing strand – grout bond;
- Ductility of the steel and the structure; and
- The properties / performance of cracked concrete members

– Concrete degradation by alkali-silica reaction (ASR), various types of sulfate attack, freeze-thaw cycles, acid attack, etc.

Corrosion will result in varying degrees of reduction in the cross-sectional areas of reinforcing bars / prestressing strand and their ductility. The corrosion products, which occupy a volume larger than the original steel, can cause longitudinal cracks along the reinforcement which reduce its anchorage capacity. Concrete cracking may adversely affect durability, especially in severe exposure environments.

For further details refer *fib* Bulletin XX [28-11].

Although the partial-factor methods presented in *fib* Bulletin 80 [28-1] for the structural assessment of existing concrete structures are generally not applicable to those experiencing deteriorating, Annex A of *fib* Bulletin 80 [28-1] does provide suggestions about ways in which the effects of corrosion-induced damage may be taken into account.

The provision of design-type equations / models for failure modes specific to structures after intervention have been developed for some situations, such as debonding at the interface between the strengthening component (eg. a structural concrete overlay) and the substrate member.

Evaluation of the durability, bonding capacity and stiffness ratio of the materials used to make interventions, requires long-duration testing of the materials in appropriate ‘service’ environments. Often only extremely limited information is available concerning these in-service behaviours.

Guidance for reliability assessment of existing parts of upgraded structure based on the partial factor method is provided in section 12.5.4.2.3.3.2.

Detailed investigation and special testing of materials and deterioration phenomena may be required for the assessment of safety, durability and prediction of the evolution of deterioration / corrosion with time.

Material deterioration and associated design models are considered in chapters 14, 15 and section 30.1.11. Further considerations relating to material and structural testing, together with various forms of monitoring, are addressed in section 30.11.

The structural implications of material deterioration are considered in various sections of chapter 30 (ASR in section 30.1.11.2, Corroded reinforcement in 30.1.11.3, freeze-thaw effects in 30.1.11.4 and sulfate attack in 30.1.11.5).

Load carrying behaviours in deteriorated members and structures, can be different to those of members which have not experienced deterioration or damage. The magnitude and nature of the differences typically increases for a greater degree of deterioration / damage.

28.10.5 Assessment of structural behaviour and performance following an intervention

The adequacy of an existing concrete structure for continued service following an intervention shall be assessed by comparing its structural capabilities (eg load capacity, ability to resist / accommodate actions applied to the structure) with the current and/or future performance requirements (eg. a defined set of loadings or other actions and/or associated environmental actions).

In principle, the assessment of structural safety after intervention will generally involve checking that neither the ultimate limit state condition nor the fatigue failure limit state, using member cross-sectional analysis or other suitable means, will be reached during the remaining or extended design service life.

Similarly, it is necessary to assess that, after intervention, the structure will achieve the required durability for that period. This process shall consider both the original structure and those elements where interventions have been performed. These matters are discussed further in section 38.6.7.

References

[28-1] *fib* Bulletin 80	*Partial factor methods for existing concrete structures*, *fib* Technical Recommendation, *fib*, Lausanne, Switzerland, December 2016
[28-2] ISO 13822	ISO 13822. *Bases for design of structures - Assessment of existing structures.* Geneva, International Organisation for Standardisation (ISO, 2010).
[28-3] ISO 16311-2	ISO 16311-2. Maintenance and repair of concrete structures – Part 2: Assessment of existing concrete structures. Geneva, International Organisation for Standardisation (ISO, 2014).
[28-4] JCI Guidelines	JCI, Guidelines for assessment of existing concrete structures - 2014, Japan Concrete Institute, Tokyo, Japan, English version (JCI, May 2015).
[28-5] CEN TS17440	CEN/TC250/N2176: Assessment and Retrofitting of Existing Structures, Final Draft - Technical Specification TS17440 (A Structural Eurocodes document), European Committee for Standardization (CEN), Brussels, January 2020
[28-6] *fib* Bulletin 93	*Birth Certificate and through-life management documentation*, *fib* Technical Report, (*fib,* June 2020).
[28-7] ISO 16311-1	ISO 16311-1. Maintenance and repair of concrete structures – Part 1: General principles. Geneva, International Organisation for Standardisation (ISO, 2014).
[28-8] ISO 16311-3	ISO 16311-3. Maintenance and repair of concrete structures – Part 3: Design of repairs and prevention. Geneva, International Organisation for Standardisation (ISO, 2014).
[28-9] ISO 16311-4	ISO 16311-4. Maintenance and repair of concrete structures – Part 4: Execution of repairs and prevention. Geneva, International Organisation for Standardisation (ISO, 2014).
[28-10] SAMCO	Rücker W, Hille F and Rohrmann R: SAMCO Final Report 2006, *F08a - Guideline for the Assessment of Existing Structures*. Available from www.samco.org.
[28-11] *fib* Bulletin XX	Modelling of structural performance of existing concrete structures, State-of-the-Art Report, *fib* Task Group 3.2, *fib*, Lausanne [Draft]

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In particular cases, structural models may be based on experiments carried out in relation to the particular design or on a combination of testing and analytical calculations.

Typical D-regions are regions with static and/or geometric discontinuities, where plane sections do not remain plane after deformation, such as deep beams, beams/columns/slabs with geometrical discontinuities, bridge diaphragms, foundations and pile caps. Other examples of typical D-regions are the areas where structural components are connected (e.g. beam column, load introduction areas and supports).

Imposed deformations can result from differential settlements, temperature gradients or differences in humidity or from seismic actions.

With regard to the theory of plasticity, both the upper bound and the lower bound theorems of plasticity can be applied.

The application of the lower bound theorem of plasticity implies that a safe bearing mode is found, if a statically admissible bearing system applies in which, under the actions defined, the admissible stresses are nowhere exceeded. Examples of such systems are strut and tie models and the strip method, used for the design of slabs. The solutions found can be more or less economic, but represent a lower bound for the bearing capacity.

29 Structural analysis

29.1 Introduction

29.1.1 General approach

Structural analysis starts with a general determination of the actions in the structure, tracing the flow of forces and moments through the structure as a whole.

Structural analysis, both on a general and a more detailed level, comprises the determination of action effects such as internal stress fields, forces and moments, support reactions and deformations carried out on the basis of a structural model. To that aim, the structure can be subdivided into components, such as beams, slabs, walls and shells and connecting areas, like B- and D- regions. In B-regions the forces and moments vary gradually. The assumption that the sections remain plane after deformation (hypothesis of Bernouilli) is valid. In D-regions the forces and moments vary discontinuously, so the hypothesis of Bernouilli is no longer valid.

Analyses have to be carried out using idealizations of both the geometry and the behaviour of the structure. Idealizations must be appropriate to the case considered.

The effect of geometry and the properties of the structure and its behaviour at each stage of construction and service has to be considered in the design. Second order effects have to be taken into account where they are likely to affect the overall stability of a structure significantly and for the attainment of the ultimate limit state at critical sections.

The internal forces and moments in a structure follow from a system of loads or from imposed deformations or from a combination of both.

Internal forces, moments and deformations in statically indeterminate structures may be determined based on:

- theory of linear elasticity;
- theory of linear elasticity with limited redistribution;
- theory of plasticity;
- non-linear methods.

The application of the upper bound theorem of plasticity requires the adoption of a pattern of yield lines, generating a kinematic mechanism. The pattern that fails at the lowest load represents the bearing capacity. This method is particularly valuable for finding the bearing capacity of existing structures.

An example of a case where a detailed investigation is necessary in the SLS is the crack width control for leakage in a statically indeterminate structure. Examples of cases for the ULS are the control of rotation capacity and the determination of the magnification factor for verifying the stability of a structure.

The effect of time dependent behaviour of concrete (creep and shrinkage) must be accounted for according to the guidelines of section 29.3.3. Proper consideration must be given to the effect of relaxation of steel.

Except for seismic action, the effects of imposed deformations may be neglected in verifying structural safety if an adequate deformation capacity is ensured for all parts of the structure.

If detailed investigations are necessary for the determination of forces and moments in the serviceability limit state or the ultimate limit state (ULS), an analysis can be carried out with an adequately reduced stiffness of structural areas due to cracking.

29.1.2 BIM

[Not used at this time, to be updated in future edition]

29.2 Structural modelling

29.2.1 General

The static and geometrical boundary conditions as well as the transmission of support reactions must be taken into account when idealizing and delimiting the system.

Soil structure interaction must be considered appropriately.

29.2.2 Structural geometry idealization

Eq. 29.2-2 is based on assumption of an uncracked isotropic elastic analysis, but may be used at ULS as well as SLS.

29.2.2.1 General

For structural analysis, the structure has to be idealized using suitable models; examples are plane or space frames and B- and D-regions of structural components.

In the case of T-beams, the effective slab width depends on the web and the flange dimensions, the type of action, the span and the support conditions. The effective slab width may be estimated (figure 29.2-1) from:

$$b_{eff} = \Sigma b_{eff,i} + b_w \leq b \qquad (29.2\text{-}1)$$

where:

$$b_{eff,i} = 0.2\mathrm{b}_i + 0.1\mathrm{l}_0 \leq 0.2\mathrm{l}_0 \qquad (29.2\text{-}2)$$

The distance l_0 between the points of zero moment may be determined for usual cases according to figure 29.2-2, based on the following assumptions:

– the cantilever length is less than half the adjacent span;

– the ratio between adjacent spans is between 2/3 and 1.5.

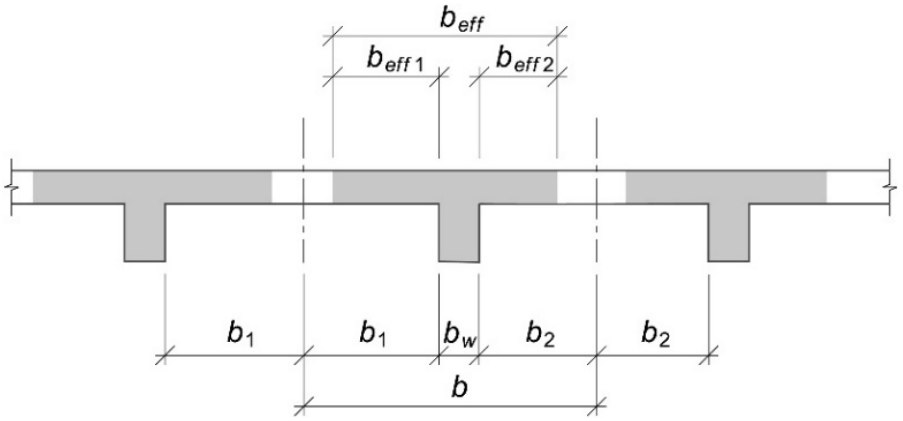


Figure 29.2-1: Effective slab width.

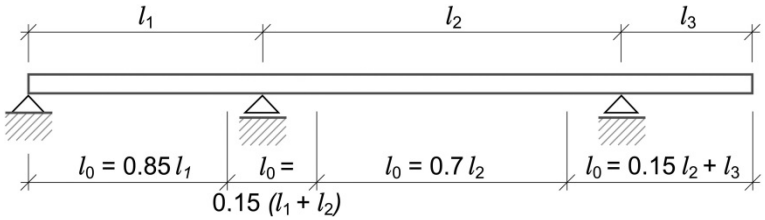


Figure 29.2-2: Relevant distances l_0 for the determination of the effective slab width

29.2.2.2 Geometric imperfections

Deviations in cross-sectional dimensions are normally taken into account in the material safety factors. Therefore these need to be included in the structural analysis.

The unfavourable effects of possible deviations in the geometry of the structure and the position of the loads have to be taken into account in the analysis of members and structures.

Imperfections must be taken into account for the verification of the ultimate limit state for persistent and accidental design situations. In the case of slender compression members, the second order effects and the influences of creep of concrete must be taken into account (section 30.1.7).

Imperfections need not be considered for the verification of the serviceability limit state.

In the case of bridge piers or highly stressed building columns, the inclination resulting from an unintended base inclination can be of importance for the dimensioning of the bracing structural members (e.g. floor slabs, bracings of buildings, bridge bearings). The effect of unintended inclination must be estimated and if necessary taken into consideration in the calculations.

Unless specified otherwise in the basis of design, the unintended inclination α_i of vertical compression members amounts to:

$$\frac{1}{200} \geq \alpha_i = \frac{0.01}{\sqrt{l}} \geq \frac{1}{300} \qquad (29.2\text{-}3)$$

where:

l denotes the height of the compression member or compression members standing on top of one another.

The definitions of l and m in Eqs.. (29.2-3) and (29.2-4) depend on the effect considered, for which three principal cases can be considered:

– effect on the single element: l = real length of structural member, $m = 1$;

In buildings, the average unintended inclination α_{im} of a group of vertical compression members can be estimated from:

$$\alpha_{im} = \alpha_i \sqrt{0.5(1.0 + \tfrac{1.0}{m})} \qquad (29.2\text{-}4)$$

– effect on stabilizing structure: l = height of building, m = number of vertical structural members that contribute to the horizontal force on the stabilizing structure;
– effect on floors or roofs which transmit horizontal loads: l = storey height, m = number of vertical elements in the storeys that contribute to the total horizontal force on the floor considered.

where:

m denotes the number of compression members which have to be included in determining the effect of unintended inclination – see Figure 29.2-3.

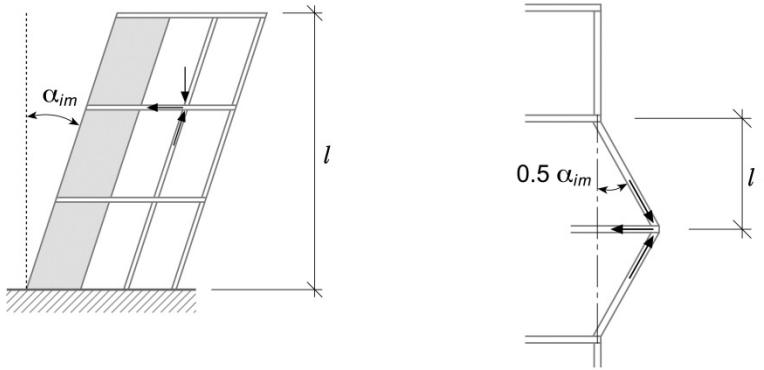


Figure 29.2-3: Geometrical imperfections.

29.2.3 Modelling of structural effects of time dependent behaviour of concrete structures

29.2.3.1 General

Shrinkage strains influence the state of deformation and induce stresses when they are restrained; they also cause stress redistributions in inhomogeneous and composite structures and sections and stress losses in prestressed structures.

The consequences of creep can be either beneficial or detrimental. On the one hand, creep exercises a beneficial effect by relieving undesirable stresses due to unintentional imposed strains like shrinkage, extreme initial temperatures, settlement of supports and yielding of restraints, etc. On the other hand, the long-term reliability of structures may be adversely affected as creep:

– increases by an important factor their initial deformations;
– reduces the beneficial effects of stresses artificially imposed to improve the performance of the structure with regard to serviceability, either causing

The inelastic strains due to creep and shrinkage of concrete may cause non-negligible variations of deformations and/or of internal actions and stresses in structures and structural elements.

The overall dimensions and the slenderness of structures and structural elements, in particular if combined with the adoption of thin sections, magnify their sensitivity to the time dependent behaviour of concrete. In fact, while the initial deformations are large, the creep amplification factor and the shrinkage strains are augmented, as a consequence of the drying in thin elements.

The time sequence of casting, loading and application of restraints, as well as the presence of important non-homogeneities exert a significant influence on the time dependent response.

prestress losses in structures prestressed with tendons or strands, or significantly jeopardizing stress corrections enforced by jacking;
- activates the delayed restraints in case of changes in the structural system after the application of sustained loads, inducing in some cases significant redistributions of internal actions and stresses that may lead to unfavourable increases of their values in some regions of the structures;
- reintroduces, for the same reason, a significant part of the internal actions and stresses due to self-weight that were provisionally eliminated in statically indeterminate structures by temporary reductions, during the construction stages, of their degree of restraint (e.g. bending moments in fixed end arches due to axial shortening, provisionally eliminated through the adoption of temporary hinges).

In non-homogeneous structures, creep induced stress redistributions transfer stresses from the parts of the structure creeping more to the parts creeping less, or from concrete to steel elements. Non-homogeneities may be due to differences in casting and loading ages, mixture proportions and components, size and shapes of structural elements and cross-section components, environmental conditions, etc. in the various concrete parts of the structure, and to the association of concrete and steel elements.

For example, in high-rise buildings, the combined effects of creep and shrinkage, of their non-uniform development and of sequential construction, besides generating several serviceability concerns regarding both structural and non-structural components, can cause also reductions of the safety margins with respect to ultimate limit state. This second concern can depend on the possible increase with time of action effects on some structural elements, and may become relevant especially when a limited ductility is available because of high axial loads, as in vertical elements, and of the use of high-strength concrete, or when buckling of slender steel elements must be feared. Influence of non-symmetrical time dependent vertical shortenings on gravity induced sidesway and its effects on structural reliability must also be investigated.

When checking the stability of compressed members the long-term deformations of concrete may approximately be taken into account as indicated in section 30.1.7 for Level II of approximation. When more refined analyses are required, the indications given in the same sub-clause for Level IV of approximation shall be respected.

For more information reference shall be made to CEB Design Manual "Structural Effects of Time-dependent Behaviour of Concrete" [29-1]; CEB Bulletin 215 [29-

Creep and shrinkage affect primarily the long-term serviceability and durability of structures. In particular, if the effects of time dependent strains are mainly in the sense of an increment of the deformations, the limit-state of deformations should be checked. If the state of stress is mainly influenced by time-dependent effects, the limit-states of stresses and of cracking should be checked.

Influence on the safety margins with respect to the ultimate limit state of strength depends on the ductile behaviour of the structure or structural element and can become a concern in presence of significant time dependent amplifications of action effects such as internal forces and moments in cases where these effects cannot be redistributed, especially when the ultimate limit state is governed by non-plastic failure of concrete.

In slender or thin structures or structural elements and whenever second order effects are of importance, the increase of deflections due to creep reduces the safety margins with respect to instability and may lead to creep buckling; the unfavourable influence of shrinkage should be considered as well.

The guidelines given in the following apply essentially to the verifications with respect to serviceability limit states.

2]; Section 4.1.6 of "Structural Concrete Textbook" fib Bulletin 52 [29-3]; "Analysis of Creep and Shrinkage Effects in Concrete Structures" to be published as ACI document 209.3R [29-4].

Structures may be conveniently classified according to their levels of sensitivity to time dependent effects. The lower levels are represented by small and simple structures. The higher levels refer to important, large and technically complex structures. Typical examples of highly sensitive structures are segmentally built large span cantilever and arch bridges joined at a later stage, cable-stayed bridges and structures, steel-tied arches, concrete or steel and concrete structures for high-rise and supertall buildings, steel-concrete composite beams or framed structures, structures with high ratios of prestressing and reinforcing steel, large span slender concrete arches and shells, structures stressed by jacking, off-shore, marine and nuclear structures, etc.

In general, a high sensitivity of the structure to the time dependent behaviour of concrete may be responsible of final uneconomies in its service life costs. A reduction of this sensitivity by proper design and construction provisions, which may require higher initial costs, besides contributing to the reduction of the margins of uncertainty in the assessment of long-term reliability, may result also in service life economies.

Excessive refinement is not warranted for low to moderate sensitivity structures, and in the preliminary and conceptual design stages of all types of structures Sophisticated and laborious analyses should be reserved to important, sensitive, and very sensitive structures in their final stages of design.

A refined creep and shrinkage prediction model ought always to be used for structures analyzed by sophisticated computational methods, while excessive refinement in the analysis should be avoided if the prediction of material properties is poor. The error caused by replacing an accurate prediction of creep and shrinkage values on the basis of the most realistic available models with a simple but crude estimation is often larger than the gain from replacing simplified analysis approaches with sophisticated approaches.

29.2.3.2 Levels of refinement of the analysis

When choosing the level of refinement for the analysis the following aspects should be considered:

– sensitivity of the structure to the time dependent behaviour of concrete,

– importance of the limit state under consideration,

– design stage,

– reliability of the information on material properties (prediction on the basis of the prediction models like e.g. the model given in sub-clause 14.6.1.6.4, or prediction accompanied by tests at early ages, or test extrapolation; mean cross-section behaviour or local rheological properties within the cross-section, etc.).

Important uncertainties affect both, and to large extent, the prediction of the material response and the evaluation of the consequent structural response.

Appropriate confidence limits should be better considered also in these cases on the basis of adequate estimations.

As evidenced in section 14.6.1.6.4, the deformation prediction suffers of important uncertainties, due to inherent scatter of creep and shrinkage strains, errors of the model and randomness of material properties and environment, and may result in a considerable prediction error.

In ACI 209.23-08 [29-5], besides the 1999 edition of the model of sub-clause 14.6.1.6.4, three other alternative mean-cross-section prediction models are presented, termed ACI 209R-92 [29-6], GL2000 [29-7] and B3 [29-8]. The last model predicts also basic creep (creep at constant temperature and moisture content), considered a material constitutive property independent of the specimen size and shape.

In cases of highly sensitive structures for which cracking is a dominant consideration for serviceability and safety (e.g. marine and nuclear structures), a realistic analysis of the time evolution of internal stresses and strains within cross-

A balanced attention should be given in any case to both the material properties problem and to the structural analysis problem.

29.2.3.3 Probabilistic and deterministic approach

The problem of evaluation of time dependent effects in concrete structures in the serviceability domain is statistical in nature, as most of the contributing factors are inherently random with significant coefficients of variation.

A deterministic approach based on mean prediction of time dependent effects may be adequate for low to moderate sensitivity structures, and whenever a refined analysis is not required.

A probabilistic approach is highly recommendable for sensitive structures, and becomes mandatory for very sensitive ones. The design of these structures should be based on predictions of extreme values of time dependent effects that are exceeded with a certain specified small probability, such as 5%. The probabilistic approach should take into account the various factors of uncertainty.

In what concerns the prediction of creep and shrinkage properties, reference should be made to the statistical indicators of the prediction model being considered. Updating the model parameters by testing is recommended in any case.

As for the evaluation of the consequent structural effects, the use of refined analyses contributes to the reduction of uncertainties.

29.2.3.4 Prediction models for concrete and significance of the analysis

When adopting a prediction model for creep and shrinkage of concrete of the type given in section 14.6.1.6.4, attention must be paid to its range of applicability as specified in section 14.6.1.6.4.2.

While this kind of models that predict the mean cross-sectional behaviour of a concrete member is generally suitable for the analysis of time dependent effects on the overall response of structures (with the exception of highly sensitive structures) in terms of internal actions, restraint reactions and deformations, it should be applied with some caution in the analysis of creep and shrinkage effects on local stress distributions within cross-sections, where larger errors may be introduced.

sections and throughout the structure by appropriate discretization techniques should be based on the description of local rheological properties (constitutive laws for a material point of the concrete), taking into account their intrinsic non-linearities, coupling with moisture and temperature distributions and non-linear effects of cracking. Reference shall be made to specialized literature [see e.g. Bažant et al. [29-9], and referenced literature. See also sub-clause 29.2.3.1 (b)].

For the range of validity of the linearity assumption, see section 14.6.1.6.4.3. Hygrotermal effects and cracking introduce major deviations from the principle of superposition. Non-linearities due to high stresses may be taken into account on the basis of section 14.6.1.6.4.3 (d).

Shrinkage strains may be considered as imposed stress independent strains.

In constitutive relation Eq. (14.6-57) or (29.2-5) strain or, respectively, stress may be considered as a functional of the previous stress (or strain) history alone.

If, on the contrary, the histories of the strain or, conversely, of the stress are prescribed in Eq. (14.6-57) or in equation (29.2-5), respectively, these equations represent Volterra hereditary integral equations for the determination of the corresponding stress or, respectively, strain histories.

Generalization to multiaxial stress of constitutive relation Eq. (14.6-57) requires the additional assumption of constancy of creep Poisson's ratio of concrete ν_c. For further information reference is made to Z. P., Bažant [29-10]; J. Salençon [29-11].

Rather, the results of such cross-sectional analyses should be considered to have essentially a nominal value, and not as realistic description of the actual stress values in the sections. However, for ordinary structures, in which fine cracking of concrete is not much of a concern, or ample prestress is provided, more refined analyses are normally not needed.

In the case of cross-sections composed of elements with different thickness or environmental exposure conditions (e.g. in box girders), the non-uniformity of drying can be captured, in a simple but rather approximate way, applying the mean cross-section prediction model for creep and shrinkage to each component separately.

29.2.3.5 Time dependent analysis based on aging linear viscoelasticity

Within the limits specified in sub-clauses 14.5.1.6.4. and 29.3.3.4, the time dependent analysis of concrete structures, for what concerns the evaluation of the effects of creep, may be based on the theory of aging linear viscoelasticity.

Aging linear viscoelasticity implies the validity of the principle of superposition of the responses in terms of strains or stresses to all imposed stress or strain increments applied in time.

29.2.3.6 Constitutive laws in aging linear viscoelasticity

The application of aging linear viscoelasticity gives the stress-strain constitutive relation in the form of the hereditary integral Eq. (14.6-57) representing the strain response to a sustained variable imposed stress history with initial value $\sigma_c(t_0)$. In this case, the time dependent behaviour of concrete is fully characterized by the compliance function $J(t,t')$.

Alternatively, with the same assumptions and range of validity, the stress response to a sustained variable imposed strain history with initial value $\varepsilon_{c\sigma}(t_0)$ may be written as:

$$\sigma(t) = \varepsilon_{c\sigma}(t_0)R(t,t_0) + \int_{t_0}^{t} R(t,t')d\varepsilon_{c\sigma}(t') \qquad (29.2\text{-}5)$$

where:

Different values of t_0 must be considered and repeated solutions of equation (29.2-6) must be performed to obtain $R(t,t')$ from $J(t,t')$. The procedure for numerical solution indicated in sub-clause 29.2.3.11 (a) shall be adopted.

The ranges of applicability of these simplifications are given in the corresponding sub-clauses.

The application of this assumption is based on the following considerations:

– the differences in the rheological properties of the concrete along the structure in terms of long-term values of creep and shrinkage strains, are usually small if compared to the large magnitude of these values obtained from prediction models;
– the influence of the non-homogeneities due to the presence of reinforcement is generally small in most prestressed structures, due to the small geometrical percentage of reinforcement and to the lack of cracking;

$R(t,t')$ is the relaxation function, representing the stress response at time t to a sustained constant unit imposed stress-dependent strain applied at time t'

$R(t,t')$ can be obtained from the compliance function $J(t,t')$ specified by the creep prediction model being considered as the stress response in Eq. (14.6-57) for a constant unit imposed stress dependent strain $\varepsilon_{c\sigma} = 1$. Therefore, the compliance and the relaxation functions are reciprocally related by the integral equation:

$$1 = R(t_0,t_0)\,J(t,t_0) + \int_{t_0}^{t} J(t,t')\,dR(t',t_0) \qquad (29.2\text{-}6)$$

29.2.3.7 Simplified approaches for time dependent analysis

When a highly refined analysis is not required, some convenient simplifications may be introduced:

– at the level of the structural model, by the introduction of the assumption of an effective rheological homogeneity for the concrete structure (section 29.2.3.8) or for the concrete part of a structure including steel structural elements (section 29.2.3.9) when performing the analysis of the overall behaviour of the structure;
– at the constitutive level, through the adoption of the approximate algebraic formulation of the age-adjusted effective modulus (AAEM) method for the constitutive Eq. (14.6-57) (section 29.2.3.10).

29.2.3.8 Effective homogeneous concrete structures with rigid or stress-independent yielding of restraints

A simplification of the structural model is to neglect the non-homogeneities in the rheological properties of the material and referring to an effective homogeneous concrete structure of averaged creep and shrinkage properties with rigid restraints.

Accordingly, a set of simple formulations for the determination of the overall time dependent response of the structure is obtained from the fundamental theorems of the theory of aging linear viscoelasticity. The system of the stresses $S(t)$ and of the deformations $D(t)$ of the structure, under sustained imposed loads or deformations or consequent to a modification of the restraint conditions, may then

– although this influence may be larger in ordinary reinforced concrete structures, as a consequence also of the interaction between creep and cracking, it may equally be disregarded in most cases, in particular if the analysis does not concern specifically local effects and the structure does not pertain to the high ranks of sensitivity;
– the stress-independent yielding of the restraints may be considered as a set of imposed deformations at the points of applications of the restraints of the structure;
– the case of restraints characterized by an elastic behaviour, or, equivalently, of the presence of steel structural elements considered as elastic elements, is discussed in sub-clause 29.2.3.9.

The formulations obtained from the assumption of effective rheological homogeneity are particularly apt for the serviceability assessment of low to moderate sensitivity structures, and, especially, in the preliminary and conceptual design stages for a wider class of structures. In fact, their sound theoretical fundaments guarantee their reliability from the conceptual point of view, while the reference to averaged rheological properties allows capturing the basic orders of magnitude. Therefore, if extreme and very special cases characterized by significant non-homogeneities in the concrete properties are excluded, the general trends of the time dependent phenomena under consideration are unquestionably seized.

Assuming the principle of superposition to be valid, the solutions presented here in sections *(a)* to *(d)* for each separate problem in terms of stress and strain histories may be superimposed.

be evaluated on the basis of the stresses $S^{el}(t)$ and deformations $D^{el}(t)$ for an elastic structure of constant reference elastic modulus E_c.

(a) Imposed loads

Under sustained variable imposed loads the elastic stresses are not modified by creep. The deformations at time t may be evaluated summing up the increments of the elastic deformations factored by the non-dimensional creep factor $E_c\, J(t,t')$ (1st theorem of aging linear viscoelasticity):

$$S(t) = S^{el}(t) \qquad (29.2\text{-}7)$$

$$D(t) = E_c \int_0^t J(t,t')dD^{el}(t') \qquad (29.2\text{-}8)$$

where:

$S(t)$ is system of the stresses (internal stresses, internal forces, restraint reactions) at time t,

$S^{el}(t)$ is elastic solution for the system of the stresses in the associated elastic problem at time t,

Under sustained constant loads creep does not alter the initial stresses in the structure. The initial deformations are followed by their creep induced gradual increase, the deformation history being related by an affinity to the compliance function (*creep problem* at the structural level).

For a normal weight ordinary structural concrete, the magnification factor $E_{ci}(t_0)J(t,t_0)$ at the end of a service life of 100 years can easily reach values in the range of 3 to 5 for typical ages t_0 at loading, depending on the creep prediction model, on the values of the various influencing parameters and on the reference value of the elastic modulus $E_{ci}(t_0)$ considered in the evaluation of the initial deformations (CEB Bulletin 215 [29-2] and *fib* Bulletin 52 [29-3]).

If reference is made to the prediction model of section 14.6.1.6.4.3, the magnification factor is $E_{ci}(t_0)J(t,t_0) = [1 + \phi(t,t_0)E_{ci}(t,t_0)/E_{ci}]$.

When applying Eq. (29.2-13) for the evaluation of the effects of cyclic time histories of the imposed deformations, e.g. when considering the effects of restrained annual thermal variations, reference to the comments of sub-clause 29.2.3.11 (a) is appropriate.

Under sustained constant imposed deformations creep does not alter the initial deformations. The initial stresses are followed by their creep induced gradual decrease, the stress history being related by an affinity to the relaxation function (*relaxation function* at the structural level).

For a normal weight ordinary structural concrete, the creep induced mitigation factor $R(t,t_0)/E_{ci}(t_0)J(t,t_0)$ lies in the range 0.1 to 0.3 at the end of a service life of 100

$D(t)$ is system of the deformations (internal strains, internal deformations, external displacements) at time t,

$D^{el}(t)$ is elastic solution for the system of the deformations in the associated elastic problem at time t,

E_c is reference elastic modulus for the associated elastic problem.

The response to a system of sustained constant loads imposed at $t = t_0$ is thus given by:

$$S(t) = S^{el,t_0} \quad (29.2\text{-}9)$$

$$D(t) = E_{ci}(t_0)J(t,t_0)D^{el,t_0} \quad (29.2\text{-}10)$$

where:

S^{el,t_0}, D^{el,t_0} are the elastic solutions for the system of the stresses and deformations calculated with the initial value $E_{ci}(t_0)$ of the elastic modulus.

(b) Imposed deformations

Under sustained variable imposed deformations the elastic deformations are not modified by creep. The stresses at time t may be evaluated summing up the increments of the elastic stresses factored by the non-dimensional relaxation factor $R(t,t')/E_c$ (2nd theorem of aging linear viscoelasticity):

$$D(t) = D^{el}(t) \quad (29.2\text{-}11)$$

$$S(t) = 1/E_c \int_0^t R(t,t')dS^{el}(t') \quad (29.2\text{-}12)$$

The response to a system of sustained constant imposed deformations imposed at $t = t_0$ is thus given by:

$$D(t) = D^{el,t_0} \quad (29.2\text{-}13)$$

$$S(t) = \frac{R(t,t_0)}{E_{ci}(t_0)} S^{el,t_0} \quad (29.2\text{-}14)$$

years for typical ages t_0 at loading, depending on the creep prediction model, on the values of the various influencing parameters and on the reference value of the elastic modulus $E_{ci}(t_0)$ considered in the evaluation of the initial stresses (CEB Bulletin 215 [29-2] and *fib* Bulletin 52 [29-3]). Therefore, as a result of relaxation, creep exerts a beneficial action in significantly relieving the stresses due unintentional sudden imposed deformations. On the other hand, it drastically reduces the beneficial effects of stress corrections artificially induced e.g. by jacking.

As a general indication, the formulations of Eqs. (29.2-15) to (29.2-17) show the importance of stress redistributions due to creep in case of changes in the structural system. In fact, the inspection of the diagrams of the redistribution function $\xi(t,t_0,t_1)$ shows that for average creep properties and for typical ages t_0 at loading and t_1 at change in structural system, characterizing usual construction sequences, the long-term values of ξ are contained in a range between 0.5 and 0.9, if reference is made to the prediction model of section 16.6.1.6.4.3. Higher values, in the range of 0.7÷0.9, are obtained for GL2000 [29-7] and B3 [29-8] models (CEB Bulletin 215 [29-2] and *fib* Bulletin 52 [29-3]).

As a result, the long-term distribution of stresses, internal actions and restraint reactions tends to approach those of the structure in its final structural system. Therefore, the creep induced stress redistribution must be accounted for in the serviceability analyses of the structure and may require additional reinforcement and/or the introduction of additional prestressing after the change in the structural system. When assessing the reliability with respect to the ultimate limit state, this redistribution must be properly considered in case of structures in which the internal actions cannot be redistributed due to limited available ductility, the collapse being governed by the brittle failure of concrete.

One possible strategy of reducing the creep induced stress redistribution consists in reducing the difference in the stress distributions corresponding to the two reference elastic configurations of the original and final structural systems, respectively. For instance, in prestressed structures under bending, in cases where the additional restraints prevent the delayed flexural deformations, this may be done by balancing the permanent loads in the original structural system by prestressing, thus reducing the corresponding flexural deformations. Therefore, the additional restraints have a limited effect in altering the original elastic stress distribution.

The delayed introduction of additional restraints is sometime conceived as an artifice to improve the behaviour of the structure under permanent loads, when the corresponding stress distribution in the original structural system is more favourable,

(c) Modification of restraint conditions after loading

In a structure subjected to sustained constant loads imposed at time t_0, whose initial structural system 1 is modified into a final structural system 2 by the introduction of additional restraints at time $t_1 \geq t_0^+$, the stress distribution evolves for $t > t_1$ according to Eq. (29.2-15) (3rd theorem of aging linear viscoelasticity):

$$S^2(t) = S^{el,1} + \Delta S^1(t) = S^{el,1} + \Delta S^{el,1}\xi(t,t_0,t_1) \qquad (29.2\text{-}15)$$

where:

$S_2(t)$ is system of the stresses at $t > t_1$ in the structure in the final structural system 2,

$S^{el,1}$ is elastic solution for the system of the stresses in the structure in the initial structural system 1,

$\Delta S^{el,1}$ is correction to be applied to the elastic solution $S^{el,1}$ in order to comply with the elastic solution $S^{el,2}$ corresponding to an application of the loads in the final structural system 2, i.e.:

$$S^{el,2} = S^{el,1} + \Delta S^{el,1}$$

$\xi(t,t_0,t_1)$ is redistribution function.

The redistribution function ξ is a non-dimensional factor whose values lie in the interval (0,1) (with $\xi = 0$ for $t = t_1$), and is related to the compliance function through the integral equation:

$$J(t,t_0) - J(t_1,t_0) = \int_{t_1}^{t} J(t,t')d\xi(t',t_0,t_1) \qquad (29.2\text{-}16)$$

(d) Multiple changes in the structural system

if compared to the one that would be obtained, for the same loads, in the final system. In such cases, the introduction of the additional restraints, intended to improve the response to live loads and the final robustness of the structure by an increase of its statical redundancy, substantially reduces the original benefits, because of the significant creep induced stress redistributions altering the original response to permanent loads (e.g. fixed concrete arches provisionally built as three-hinged arches with a centre line corresponding to the funicular curve for dead loads).

Different values of t_0 and t_1 must be considered and repeated solutions of Eq. (29.3-16) must be performed to obtain $\xi(t,t_0,t_1))$ from $J(t,t')$. The procedure for numerical solution indicated in sub-clause 29.2.3.11 (a) shall be adopted.

For further information reference is made to Chiorino [29-12].

Design aids. For a given creep prediction model, and for the corresponding compliance function $J(t,t')$, design aids can be provided for the evaluation of the related functions $R(t,t')$ and $\xi(t,t_0,t_1)$ in terms of set of graphs of these three functions and of the aging coefficient $\chi(t,t')$ introduced in sub-clause 7.2.4.10 (refer to CEB Bulletin 215 [29-2] for the creep prediction model of CEB Model Code 1990 [29-13]; a few examples for the models indicated in section 29.3.3.4 are given in *fib* Bulletin 52 [29-3]), or in terms of computational programs to be downloaded from web-sites (see e.g. www.polito.it/creepanalysis) or inserted in computational softwares. For further information reference is made to Sassone & Chiorino [29-14].

Reference shall be made to the specialized literature indicated in *fib* Bulletin 52, p. 53 [29-3], and to ACI 2090.3R Guide [29-4].

In cases where the transition from the initial to the final structural system is obtained by means of several different restraint modifications applied at times $t_i \geq t_0^+$ ($i = 1,\ldots,j$), the redistribution effects consequent to every single change in the structural system may be superimposed in time. Therefore, the system of the stresses evolves for $t>t_1$ according to Eq. (29.3-17) (4th theorem of aging linear viscoelasticity):

$$S^{j+1}(t) = S^{el,1} + \sum_{i=1}^{j} \Delta S^{el,i}\, \xi(t', t_0, t_i) \qquad (29.2\text{-}17)$$

where:

$\Delta S^{el,i}$ is correction to be applied, in the associated elastic problem, to the elastic solution $S^{el,i}$ in order to respect the geometrical conditions imposed by the additional restraints of structural system i+1, imagined as introduced before the loads.

29.2.3.9 Effective homogeneous concrete structures with additional steel structural elements

If the main cause of heterogeneity is represented by the presence of steel structural elements that may be considered equivalent to redundant elastic restraints, while the concrete part of the structure may still be approximately regarded as an effective homogeneous structure of averaged rheological properties, a set of simple formulations can be provided for the determination of the overall time dependent response of the structure for the cases of constant imposed loads or of constant deformations imposed at the points of applications of the restraints, or of a change in structural system (figure 29.3-4).

These formulations, which are expressed in matrix form, represent the extension to the case of elastic restraints of the aging linear viscoelastic formulations presented in sub-clause 29.2.3.8.

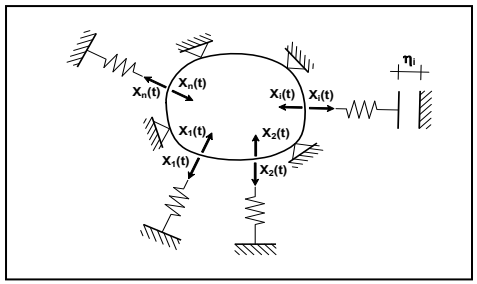


Figure 29.2-4: Schematic representation of a homogeneous structure with n redundant elastic restraints.

Although the theoretical condition of invariance can hardly be obtained in practical cases, due to construction sequences and differences in the creep properties along the structure, an appropriate stressing of the steel restraints can substantially reduce the variation in time of the stress state. This artifice, which may be responsible of higher initial costs as the steel restraints must be designed for higher forces, contributes to the reduction of uncertainties in the long-term reliability assessments and may result in service life economies.

In cable-stayed bridges, stressing of the stays is normally performed in more than one step to allow for adjustments, as required by usual cantilever sequential construction techniques. One other reason for stressing the stays is the need of eliminating the nonlinear mechanical behaviour of the stays due to cable sag. For more information reference is made to Casalegno et al. [29-15].

Typical structures that can be schematically modelled as effective homogeneous concrete structures with redundant elastic restraints are e.g. tied arches and frames, and cable-stayed bridges and structures. Although the final assessment of these types of structures - being inherently complex and usually characterized by intricate construction procedures and phases - must normally be performed by proper sequential approaches adopting the numerical techniques indicated in sub-section 29.2.3.11 and accounting for possible causes of non-linearity (e.g. due cable sag in stayed structures), the reference to this schematic model and the inherent formulations allows setting some general guidelines listed in the following, which are particularly useful in the preliminary and conceptual design stages.

(a) Constant imposed loads

Differently from the case of rigid restraints, for which the invariance of the state of stress is stated by Eq. (29.2-7), the initial elastic state of stress in the restraints and in the structure is significantly altered by creep. The higher the deformability of the restraints, the higher is the difference between the initial and long-term values. In the long term, the system of stresses tends to approach the solution corresponding to the case of rigid restraints.

A reduction of the time dependence of the state of stress under permanent loads when highly deformable restraints are adopted (e.g. high strength steel stays of small section as in cable-stayed bridges), can be obtained through a convenient stressing of these restraints. In fact, in the theoretical case of an effective homogeneous structure with elastic restraints introduced all at the same time, the invariance of the stress state is obtained forcing the restraints up to the values of the rigid restraints reactions.

(b) Constant imposed deformations

The stress losses are less pronounced with respect to the case of rigid restraints represented by Eq. (29.2-14). This is a consequence of the elastic energy stored in the system of the restraints. If the deformations are imposed at the points of applications of the restraints, the invariance of the stress state may be obtained as specified in (a).

(c) Modification of restraint conditions after loading

The theoretical solutions show that the system of elastic restraints contributes to a lower degree to the variation of the original system of the stresses in the structure,

Eq. (29.2-20), with the Eq. (29.2-21) for the aging coefficient χ, corresponds exactly to Eq. (14.6-57) for all one-step imposed action histories resulting from linear combinations of a creep and a relaxation problem, i.e. for all the strain histories of the type:

$$\varepsilon_{c\sigma}(t) = c_1 + c_2 J(t,t_0) = c_1 + c_2 \left[\frac{1}{E_c(t_0)} + \frac{\phi_{28}(t,t_0)}{E_{ci}}\right] = a + b\phi_{28}(t.t_0) \qquad (29.2\text{-}19)$$

where the time dependent part is related by an affinity to the compliance function $J(t,t_0)$ or, equivalently, to the creep coefficient $\varphi_{28}(t,t_0)$. This includes a broad range of strain (and corresponding stress) histories. With sufficient accuracy, its use may be extended to cover a large number of actual action histories in structures showing an initial finite or zero value at $t = t_0$ and a time dependent part varying at a gradually decreasing -rate over wide time intervals. In current use of AAEM method Eq. (29.2-18) is given a quasi-elastic incremental formulation relating the variations of the total strain $\Delta\varepsilon_c(t)$ and of the stress $\Delta\sigma(t)$ occurring over the interval (t_0^+,t) after the initial stressed state at time t_0^+. In both alternative procedures, the responses to multistep load histories can be obtained by superimposing the solutions for several one-step histories. For further information reference is made to Bazant [29-16]; Jirásek & Bažant [29-17].

Under these conditions, the AAEM method may be applied firstly to the analysis of the overall time dependent response of concrete structures that may be considered as homogeneous on the basis of the same assumption of effective rheological homogeneity discussed in section 29.2.3.8. The same ranges of applicability of the solution shall be considered.

The accuracy of the results remains normally satisfactory if the application is extended to cover the cases of heterogeneous structures consisting of concrete portions with different creep properties, and/or containing steel elements. However, in case of very complex structures and construction sequences, preference shall be attracting lower values of restraint reactions, with respect to the case of delayed additional rigid restraints discussed in sub-clause 29.2.3.8 (c).

29.2.3.10 Approximate algebraic formulation for the constitutive relation: age-adjusted effective modulus (AAEM) method

For a compliance function expressed in the form of Eq. (14.6-56) the hereditary integral constitutive relation of Eq. (14.6-57) may be written in the following equivalent algebraic form:

$$\varepsilon_c(t) = \sigma_c(t_0)J(t,t_0) + [\sigma_c(t) - \sigma_c(t_0)]\left[\frac{1}{E_{ci}(t_0)} + \chi(t,t_0)\frac{\phi_{28}(t,t_0)}{E_{ci}}\right] + \varepsilon_{cn}(t) = \frac{\sigma_c(t_0)}{E_{c,ef}(t,t_0)} + \frac{\sigma_c(t)-\sigma_c(t_0)}{E_{c,adj}(t,t_0)} + \varepsilon_{cn}(t) \qquad (29.2\text{-}18)$$

having introduced the aging coefficient:

$$\chi(t,t_0) = \frac{1}{1-R(t,t_0)/E_{ci}(t_0)} - \frac{1}{E_{ci}(t_0)J(t,t_0)-1} = \frac{E_c(t_0)}{E_{ci}(t_0)-R(t,t_0)} - \frac{E_{ci}}{E_{ci}(t_0)\phi_{28}(t,t_0)} \qquad (29.2\text{-}20)$$

the effective modulus:

$$E_{c,ef}(t,t_0) = \frac{1}{J(t,t_0)} = \frac{E_{ci}(t_0)}{1+[E_{ci}(t_0)/E_{ci}]\phi_{28}(t,t_0)} \qquad (29.2\text{-}21)$$

and the age-adjusted effective modulus:

$$E_{c,adj}(t,t_0) = \frac{E_{ci}(t_0)}{1+\chi(t,t_0)[E_c(t_0)/E_{ci}]\phi_{28}(t,t_0)} \qquad (29.2\text{-}22)$$

$\chi(t,t_0)$ varies relatively little with the age t_0 for sufficiently long elapsed times. For typical values of t_0 and other influencing parameters, its long-term values are contained in range between 0.6 and 0.9 for the prediction model of sub-clause 14.6.1.6.4 and in a narrower range for other models like GL2000 [29-7] and B3 [29-8].

The adoption of a fixed long-term value comprised in this narrower range, independently of the age at loading and of the creep properties of the structural element being considered, leads often to satisfactory accuracies in the evaluation of the long-term structural responses, particularly in the conceptual and preliminary

given to the numerical approaches of the General method illustrated in section 29.2.3.11.

The AAEM algebraic formulation of the constitutive relation in association with the assumption of plane sections shall be normally adopted for the estimation of stress redistributions due to creep and shrinkage in non-homogeneous and composite cross-sections of one dimensional elements, like prestressed concrete sections with prestressing and reinforcing steel in one or multiple layers, concrete-concrete and steel-concrete composite sections, etc. In fact, while on one hand the AAEM solutions are generally very accurate with respect to the use of the integral formulation (14.6-57) of the constitutive law, on the other hand reference to the comments of section 29.2.3.4 on the nominal character of these estimations is appropriate.

For any given compliance function $J(t,t')$, the aging coefficient $\chi(t,t')$ can be determined from Eq. (29.2-20) for different values t_0 of t' introducing the values of the relaxation function $R(t,t')$ calculated on the basis of the numerical procedure indicated in sub-clause 29.3.3.11 (a). For design aids refer to section 29.3.3.8.

Notation. In the referenced literature, and in ACI 209R-92 [29-6] and B3 [29-8] prediction models, the creep coefficient φ (noted also as ϕ) represents the ratio between the creep strain and the initial elastic strain $1/E_c(t_0)$ at the age t_0 at loading. The notation φ_{28} adopted here must be intended to correspond, on the contrary, to the definition of the creep coefficient in Eq. (14.6-55).

Reference is made to constitutive relation (14.6-57) as the majority of creep prediction model specify the compliance function $J(t,t')$. No real advantage would be obtained from the computational point of view if reference were made to the equivalent constitutive relation (29.3-5), even in the frame of the equilibrium method. In fact, the relaxation function $R(t,t')$ needs to be numerically calculated through Eq. (29.3-6), from the given compliance function.

The Volterra integral equations of structural creep problems can be solved analytically only for some simple forms of the compliance function $J(t,t')$. For the design stages and in the assessment of structures of low sensitivity to time dependent effects.

In such situations, it is often adequate to use the value χ=0.8.

29.2.3.11 General method

The most general and refined approach for the evaluation of creep and shrinkage structural effects in the frame of aging linear viscoelasticity consists in the incorporation of the constitutive relation (14.6-57) for concrete into the computational algorithms of continuum mechanics or of structures composed of beams.

The use of the aging linear viscoelastic model for the concrete portions of the structure and of the elastic model for steel leads to a system of Volterra integral compatibility or equilibrium equations, when the force or, respectively, the deformation method is adopted for the structural analysis.

A completely general, accurate and most effective approach is to obtain first the incremental form of Eq. (14.6-57) for a small time step by one of procedures

compliance function of the creep prediction model given at sub-clause 14.6.1.6.4, and for the other modern creep prediction models referenced in sub-clause 29.3.3.4, an incremental numerical solution of the type indicated here is required.

In consideration of the particular shape of the creep curves and of the typical imposed strain histories in most structural problems, both being characterized by increments at decreasing rate, if fluctuating strains, like e.g. cyclic thermal strains are excluded, it is possible to gradually increase the time steps Δt_k in order to reach the long-term response with an acceptable number of steps, considering that the first intervals should be of the order of fractions of the day (due to the high initial slope of the creep curves).

The same incremental numerical procedure shall be adopted for the solution of Eqs. (29.3-6) and (29.3-16) to obtain the relaxation function $R(t,t')$ and the redistribution function $\xi(t,t_0,t_1)$ from the given compliance function $J(t,t')$. The aging coefficient $\chi(t,t')$ can then be determined from Eq. (29.3-20). Further information and reference to computer programs in *fib* Bulletin 52 [29-3].

This storage requirement used to be a computational burden, but for modern computers it matters only for systems with a huge number of unknowns.

The rate-type form is also advantageous to deal with variable humidity and temperature, as it allows separating the effects of variable pore humidity or temperature on creep viscosity from those on the aging rate, and with material damage (fracture and distributed cracking) as they are rate-dependent processes. Only with the rate-type form it is possible to meet the thermodynamic restrictions and introduce physical concepts related to these processes. Reference should be made to the specialized literature (Bažant [29-9]; ACI document 209.3R [29-4]).

indicated in the following. The incremental form represents a linear elastic stress-strain relation with initial strains, in which the elastic moduli and initial strains vary through the body, and from step to step. Thus the problem of aging linear viscoelasticity gets converted, already at the constitutive level, to a sequence of elasticity problems. This eliminates any need to ever formulate the Volterra integral equations governing structural variables.

Any elastic finite element program, used repeatedly in a loop, can thus be generalized for creep, allowing the formulation of the solution in terms of a time-history.

(a) Incremental numerical solution based on the hereditary integral

An incremental form may be obtained by replacing the integral over the past stress or strain history with a sum.

The procedure is based on the approximation of the superposition integral of Eq. (14.6-57) with finite sums using the 2nd order trapezoidal integration rule. The time t is subdivided by discrete times $t_0, t_1, t_2,...t_i...t_k$ into sub intervals $\Delta t_i = t_i - t_{i-1}$ ($i = 1,2,...,k$) and at each step the average value of the compliance function $\frac{1}{2}[J(t_k,t_i) + J(t_k,t_{i-1})]$ is adopted in the calculation. In order to allow strain histories with an initial finite step, times t_0 and t_1 are assumed to be coincident, so that $\Delta t_1 = t_1 - t_0 = 0$, and consequently $\Delta\varepsilon_{c\sigma}(t_1) = \varepsilon_\sigma(t_0)$.

At each step the calculations of the previous steps must be stored, so that the entire history of stress and strain must be stored in computer memory.

(b) Incremental numerical solution based on rate-type creep laws

The computation can be made more efficient approximating the integral-type constitutive law with a rate-type relation based on Kelvin or Maxwell chains of spring-dashpot rheological models for aging materials. In that case, the history needs not to be stored because it is implied by the current values of a few hidden variables. The resulting differential equations are of second order. Rate-type laws are particularly helpful for the solution of structural problems by means of the finite elements method, because they are immediately compatible with this computational approach.

By appropriate confining of concrete the axial strength and the ductility are increased. If these properties are to be exploited in terms of calculations, it must be checked whether the ultimate resistance is still sufficient after spalling of the cover concrete, and it must be ensured that premature buckling of the longitudinal reinforcement is prevented by means of closely spaced stirrups.

The confining pressure σ_2 (with positive sign) can be calculated with the expressions:

$\sigma_2 = \omega_c f_{cd}(1 - \frac{s_c}{d_c})$ (circular cross-section confined by spiral reinforcement);

$\sigma_2 = \omega_c f_{cd}(1 - \frac{s_c}{d_c})^2$ (circular cross-section confined by circular hoops);

where $\omega_c = \frac{A_{sc} f_{yd}}{s_c d_c f_{cd}}$;

$\sigma_2 = \omega_c f_{cd}(1 - \frac{s_c}{a_c})(1 - \frac{s_c}{b_c})(1 - \frac{\Sigma b_i^2/6}{a_c b_c})$ (rectangular cross-sections)

where $\omega_c = min\left\{\omega_y = \frac{A_{sy} f_{yd}}{a_c s_c f_{cd}}, \omega_z = \frac{A_{sz} f_{yd}}{b_c s_c f_{cd}}\right\}$

and b_i is the centre line spacing along the section parameter of longitudinal bars (indexed by i) engaged by a stirrup corner or a cross-tie.

29.2.4 Modelling of confined concrete

Confinement of concrete results in a modification of the effective stress–strain relationship: higher strength and higher critical strains are achieved. The other basic material characteristics may be considered as unaffected by design.

In the absence of more precise data, the stress–strain relation given by Eqs (14.11-6)–(14.11-7) may be used, with increased characteristic strength and strains according to:

$$\frac{f_{ck,c}}{f_{ck}} = 1 + 3.5(\frac{\sigma_2}{f_{ck}})^{\frac{3}{4}} \quad (29.2\text{-}23)$$

$$\varepsilon_{c2,c} = \varepsilon_{c2}\left[1 + 5\left(\frac{f_{ck,c}}{f_{ck}} - 1\right)\right] \quad (29.2\text{-}24)$$

$$\varepsilon_{cu2,c} = \varepsilon_{cu2} + 0.2\sigma_2/f_{ck} \quad (29.2\text{-}25)$$

where σ_2 (= σ_3) is the effective lateral compressive stress at the ULS due to confinement, and ε_{c2} and ε_{cu2} follow from Table 14.11-1.

Confinement can be generated by adequately closed links or cross-ties, which reach the plastic condition due to lateral extension of the concrete – see Figure 29.2-5.

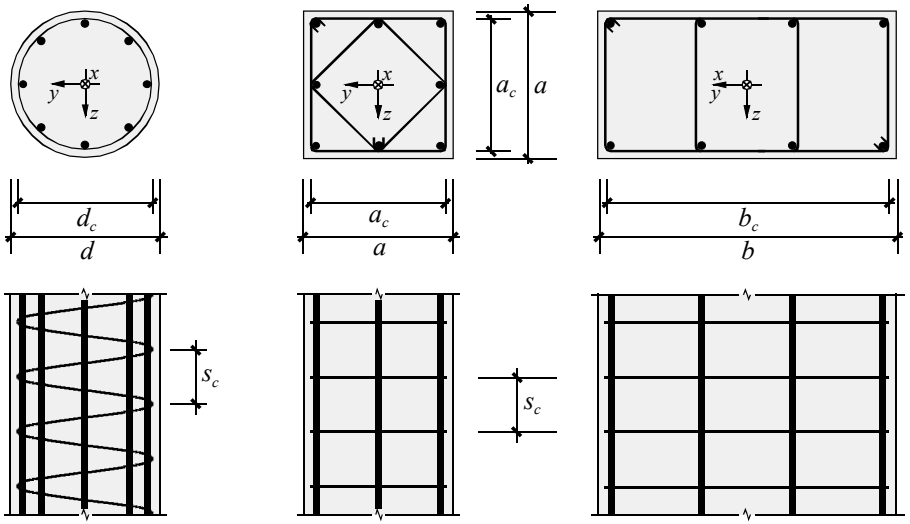


Figure 29.2-5:Compression members with confining reinforcement.

29.2.5 Modelling of partially loaded areas

For a uniform distribution of load on an area A_{c0} (Figure 29.2-6) the concentrated resistance force may be determined as follows:

$$F_{Rdu} = A_{c0} f_{cd} \sqrt{A_{c1}/A_{c0}} \leq 3.0 f_{cd} A_{c0} \qquad (29.2\text{-}26)$$

where:

A_{c0} is the loaded area;

A_{c1} is the maximum design distribution area with a similar shape to A_{c0}.

The design distribution area A_{c1} required for the resistance force F_{Rdu} should correspond to the following conditions:

– the height for the load distribution in the load direction should correspond to the conditions given in Figure 29.2-6;

– the centre of the design distribution area A_{c1} should be on the line of action passing through the centre of the load area A_{c0};

The dispersion of the concentrated forces causes biaxial or triaxial compression immediately under the load, whereas it produces transverse tension further away. Reinforcement should be provided for those tensile forces.

The increased stress according to Eq. (29.2-26) can only be combined with the increased stress due to confining action (29.2.4) if confirmed by appropriate test results.

For very large confining stresses or very small loading area, failure occurs due to wedging action under the loaded area, at a stress f_{cc}^{*} which can be significantly higher than $3f_{cd}$.

If no further data are available the average bearing capacity can be calculated with the expression:

$$\frac{f_{cc}^{*}}{f_{cc}} = 12.5\sqrt{(40/f_{cc})} \qquad (29.2\text{-}27)$$

(f_{cc} in MPa)

However, if limited penetration is considered, f_{cc}^{*} values not higher than $4f_{cc}$ should be taken.

In Figure 29.2-6 the following limits to h apply:

$h \geq a_2 - a_1$

$h \geq b_2 - b_1$

This approach implies that the "response relationship" is linear, and the assumption of reversible deformations is retained. The results are realistic only under the assumption that actions are low and members are uncracked.

For ULS verifications, existing practice allows the use of linear elastic analysis without direct verification of sufficient ductility. This is based on the assumption that there is enough ductility to balance the lack of compatibility. The method is normally used in combination with the uncracked cross-section of concrete

– if there is more than one compression force acting on the concrete cross-section, the designed distribution areas should not overlap.

The value F_{Rdu} should be reduced if the load is not uniformly distributed on the area A_{c0} or if high shear forces exist.

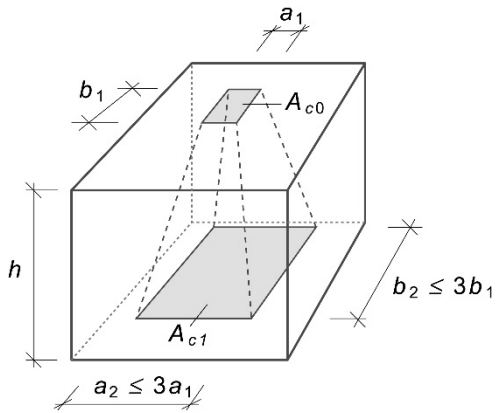


Figure 29.2-6: Load distribution for partially loaded areas.

29.2.6 Modelling of prestress applied to concrete

[Not used at this time, to be updated in future edition]

29.3 Analysis methods

29.3.1 Analysis based on linear elasticity

Analysis of elements based on the theory of linear elasticity may be used for both the serviceability and the ultimate limit states.

For the determination of the action effects, linear elastic analysis may be carried out assuming:

– uncracked cross-sections;

members; therefore it requires a definition of the geometry of the structure, but not necessarily of the reinforcement.

Cracked cross-sections may, however, be used if, in the limit state under consideration, a fully developed crack pattern can be expected. The results of a linear analysis are also used in the verification for the serviceability limit state.

If redistribution of moments is applied in determining the reinforcement, this may have an influence on the deflection and the crack width.

– linear stress–strain relationships;

– the mean value of the modulus of elasticity.

For determining the effect of imposed deformations at the ultimate limit state a reduced stiffness corresponding to cracked sections may be assumed. For the serviceability limit state, a gradual evolution of cracking should be considered.

29.3.2 Analysis according to linear elasticity with limited redistribution

Linear analysis with limited redistribution may be applied to the analysis of structural members for the verification at the ULS.

The influence of any redistribution of moments on other aspects of design has to be considered.

The moments at the ULS calculated using a linear elastic analysis may be redistributed, provided that the resulting distribution of moments remains in equilibrium with the applied loads.

The redistribution of bending moments without an explicit check on the rotation capacity may be applied for the members for the verification of ULS provided that the following conditions are fulfilled:

- all members are predominantly subjected to flexure (second order effects are negligible),
- in case of continuous beams or slabs, the ratio of lengths of adjacent spans is in the range of 0,5 to 2,0,
- the ratio δ_M of the moment after redistribution to the elastic bending moment complies with the value given by equation (29.3-1):

$$\delta_{\mathrm{M}} \geq \frac{1}{1 + 0{,}7\varepsilon_{\mathrm{cu}} \cdot E_{\mathrm{s}}/f_{\mathrm{yd}}} + \frac{x_{\mathrm{u}}}{d} \qquad (29.3\text{-}1)$$

$\geq 0{,}7$ where Class B or Class C reinforcement is used (see Table 5.5);

$\geq 0{,}8$ where Class A reinforcement is used (see Table 5.5).

In case of prestressed members, f_{yd} in Eq. (29.3-1) shall be replaced by:

$$f = \frac{\left(f_{\mathrm{pd}} - \sigma_{\mathrm{pm},\infty}\right)A_{\mathrm{p}} + f_{\mathrm{yd}}A_{\mathrm{s}}}{A_{\mathrm{p}} + A_{\mathrm{s}}} \qquad (29.3\text{-}2)$$

where:

$\sigma_{pm,\infty}$ is the long-term stress level in prestressing tendons at the state of zero (elastic) strain of the concrete at the same level.

All values refer to the section of the redistributed moment.

The arrangement of reinforcement in flat slabs should reflect the behaviour under service conditions, normally leading to a concentration of reinforcement over the columns. The requirements for maximum spacing of flexural reinforcement for solid slabs apply.

Linear analysis with redistribution with an explicit check on the rotation capacity may be applied for the verification of ULS provided that the rotation demand θ_{Ed} for the section with the plastic moment resistance is smaller than or equal to the rotation capacity θ_{Rd} of the section considered determined according to 29.3.3.3.

Redistribution should not be carried out in circumstances where the rotation capacity cannot be defined with confidence (e.g. in corners of frames with opening moments).

For the design of columns it should be checked whether the moment before redistribution is governing for the design.

29.3.3 Theory of plasticity

29.3.3.1 General

Plastic analysis should in general be based on the lower bound theorem of limit analysis. Plastic methods based on the upper bound theorem of limit analysis may be used, if it is known by experience that the type of assumed mechanisms can develop.

When applying methods based on the theory of plasticity it should be ensured that the deformation capacity of critical areas is sufficient for the envisaged mechanism to be developed.

The effects of previous applications of loading may generally be ignored and a monotonic increase of the intensity of the actions may be assumed.

For the application of the simplified procedure, it is supposed that the plastic hinge considered occurs as the first of those resulting in a kinematic system.

If the demand of rotation θ_s is calculated by integrating the curvatures between the plastic hinges, in general the application of a trilinear moment–curvature relation is appropriate. The moments caused by prestressing should be considered as a part of the effect of the load on the structure.

Plastic analysis shall only be used for reinforcement steel in Class B or C. Prestressing steel may be considered as Class B steel.

For plastic analysis the horizontal branch of the stress-strain diagram for the reinforcement (see Figures 15.10-3 and 16.1-6) shall be used when determining the sectional capacity of the cross sections.

29.3.3.2 Plastic analysis of beams, frames and slabs with the kinematic theorem

Plastic analysis without any check of the rotation capacity may be used for the ultimate limit state if all the following conditions are met:

– the area of tensile reinforcement is limited to such a value that, at any section where plastic hinges are expected to occur $x_u/d \leq 0.25$;

– the ratio of the moments at intermediate supports to the moments in the span is between 0.5 and 2.

Columns should be checked for the maximum plastic moments which can be transmitted by connecting members. For connections to flat slabs this moment should be included in the punching shear calculation.

When plastic analysis of slabs is carried out, account should be taken of any non-uniform reinforcement, corner tie down forces and torsion at free edges.

29.3.3.3 Rotation capacity

If continuous beams or continuous one-way slabs do not meet the conditions for which no check of rotation capacity is required, as formulated previously, a simplified procedure can be used. This procedure is based on a control of the rotation capacity. The rotation capacity is determined over a length of approximately 1.2 times the depth of the section. It is assumed that these zones undergo a plastic deformation (formation of yield hinges) under the relevant combination of actions. The verification of the plastic rotation in the ultimate limit state is considered to be fulfilled, if it is shown that under the relevant combination of actions the demand in rotation of the plastic hinge, calculated θ_s is smaller than or equal to the allowable plastic rotation $\theta_{pl,d}$ (Figure 29.3-1).

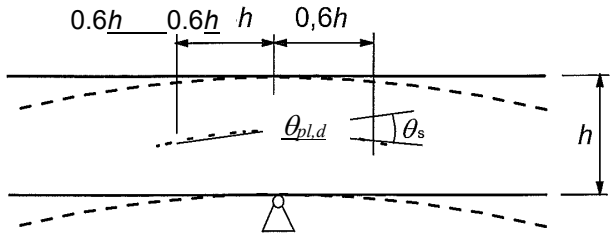


Figure 29.3-1:*Rotation capacity $\theta_{pl,d}$ of reinforced concrete sections for continuous beams and continuous one-way spanning slabs.*

The rotation demand follows from the integral of the curvatures where cracking of concrete should be considered and the tensile strength and tension stiffening may be considered.

The rotation capacity may be derived from Equation (29.3-3).

$$\theta_{\mathrm{Rd}} = \left(\left(k_{\mathrm{y}} \cdot k_{\mathrm{l,pl}} \cdot l_{\mathrm{cf}} + 2k_{\mathrm{d}} \cdot d \right) \cdot \left[\left(\frac{1}{r} \right)_{\mathrm{u,m}} - \frac{TS_{\mathrm{My}} \cdot \varepsilon_{\mathrm{yd}}}{(d - x)} \right] \right) \cdot \frac{1}{\gamma_{\theta}} \geq 0 \qquad (29.3\text{-}3)$$

where

$$k_{\mathrm{l,pl}} = \frac{1}{2} - \frac{\left(1 - \frac{1}{3} \cdot \frac{x}{d} \right)}{2 \cdot \left(1 - 0{,}4 \cdot \frac{x_{\mathrm{u}}}{d} \right)} \cdot \frac{f_{\mathrm{yd}}}{f_{\mathrm{s,ef}}} \qquad (29.3\text{-}4)$$

$$\left(\frac{1}{r} \right)_{\mathrm{u,m}} = TS_{\mathrm{Mu}} \cdot \min \left\{ \frac{\varepsilon_{\mathrm{ud}}}{(d - x_u)}; \frac{\alpha_{\varepsilon} \cdot \varepsilon_{\mathrm{cu}}}{x_u} \right\} \qquad (29.3\text{-}5)$$

$$TS_{\mathrm{Mu}} = 1 - \frac{1}{2\alpha} \cdot \left(1 - \frac{\varepsilon_{\mathrm{yd}}}{\varepsilon_{\mathrm{ud,ef}}} \right) \text{ when } \alpha \geq 1 \qquad (29.3\text{-}6)$$

$$TS_{\mathrm{Mu}} = \frac{\alpha}{2} \cdot \varepsilon_{\mathrm{ud,ef}} + \varepsilon_{yd}\left(1 - \frac{\alpha}{2} + \left(\frac{f_{\mathrm{s,ef}}}{f_{\mathrm{yd}}} - 1\right) \cdot \left(2 - \frac{1}{\alpha} - \alpha\right)\right) \text{ when } \alpha < 1 \quad (29.3\text{-}6)$$

$$TS_{\mathrm{My}} = 1 - 0{,}6\left[\frac{M_{\mathrm{cr}}}{M_{\mathrm{y}}}\right] \geq 0{,}4 \quad (29.3\text{-}6)$$

$$\alpha = \frac{\left(f_{\mathrm{s,ef}} - f_{yd}\right) \cdot A_s}{\pi \cdot \phi \cdot \frac{s_{\mathrm{m}}}{2} \cdot 0{,}3 \cdot f_{\mathrm{cm}}^{\frac{2}{3}}} \quad (29.3\text{-}6)$$

Where

l_{cf} is the distance between the two adjacent points of contraflexure;

$f_{\mathrm{s,ef}}$ Is the tensile stress in the reinforcement when M_{Rd} is found, assuming a stress-strain-relation for the rebar, see Figure 15.10-3, with a second ascending branch;

$\varepsilon_{\mathrm{ud,ef}}$ is the strain in the reinforcement with a stress equal to $f_{\mathrm{s,ef}}$;

M_{y} is the internal moment when the strain in the tension reinforcement equals $\varepsilon_{\mathrm{yd}}$

k_{y} =1

k_{d} =1

α_{ε} =1

γ_{θ} is a safety factor for model uncertainty, may be taken equal to 3,0.

29.3.3.4 Analysis with strut and tie models

Analysis with a strut and tie model is a method according to the lower bound theorem of plasticity (static method). According to this method a state of equilibrium

Orienting the struts to the compressive stress trajectories in the uncracked state, assuming linear elastic behaviour, aims at minimizing redistribution of forces after cracking, which could result in violation of service requirements (crack width and deformation) or even premature failure. If the redistribution of forces is minimized, for SLS and ULS the same strut and tie model is appropriate.

between external and internal forces has to be found, which fulfils the static boundary conditions and nowhere violates the yield conditions.

Strut and tie models consist of struts representing compressive stress fields, of ties representing the reinforcement and of connecting nodes. For SLS calculations, the struts should be oriented to the compressive stress trajectories in the uncracked stage based on linear elastic behaviour. The same approach should be applied for ULS calculations unless account is taken of the reduction in strength of the concrete strut with rotation away from the angle of the elastic compressive stress trajectory. The forces in the elements of a strut and tie model should be determined by maintaining the equilibrium with the applied loads in the ultimate limit state. The elements of strut and tie models should be dimensioned according to the rules given in section 30.1.6.

Strut and tie models may be used for the design of the reinforcement in continuity regions (B-regions) in the ULS (cracked state of beams and slabs) and for the design and detailing of discontinuity regions (D-regions) in the ULS. In general, D-regions extend up to a distance h from the discontinuity, where h is the largest cross-sectional dimension.

Verifications in the SLS may also be carried out using strut and tie models, for example, for the verification of steel stresses and crack width control, if approximate compatibility for strut and tie models is ensured; in particular the position and direction of important struts should be oriented according to the compression trajectories in the linear elastic stage.

The ties in a strut and tie model should coincide in position and direction with the corresponding reinforcement.

29.3.4 Non-linear analysis

Non-linear methods of analysis may be used for both ULS and SLS, provided that equilibrium and compatibility are satisfied and adequate non-linear behaviour for materials is assumed. The analysis may be first or second order.

Non-linear analysis should be carried out on the basis of the principles given in section 30.10. For predicting the mean behaviour mean values of the material characteristics should be used as defined in chapter 14.

References

[29-1] CEB Manual 142, 1984. Manual Structural Effects of Time-dependent Behaviour of concrete.

[29-2] CEB Bulletin 215, 1993. Structural Effects of Time-Dependent Behaviour of Concrete.

[29-3] *fib* Bulletin 52, 2010. Structural Concrete Textbook, second edition Volume 2.

[29-4] ACI 209.3R. Analysis of Creep and Shrinkage Effects in Concrete Structures.

[29-5] ACI PRC-209.2-08, 2008. Guide for Modeling and Calculating Shrinkage and Creep in Hardened Concrete.

[29-6] ACI 209R-92, 1997. Prediction of Creep, Shrinkage, and Temperature Effects in Concrete Structures.

[29-7] GL2000

[29-8] B3

[29-9] Bažant, Z. P. Li, G. H., Yu, Q. Prediction of creep and shrinkage and their effects in concrete structures: Critical appraisal", in: Tanabe et al. (eds), *Proc. Eigth Int. Conf. Creep, Shrinkage and Durability Mechanics of Concrete and Concrete Structures*.

[29-10] Bažant, Z. P., Theory of Creep and Shrinkage in Concrete Structures: a Précis of Recent Developments", *Mechanics Today*, Vol. 2, Pergamon Press, 1975, pp. 1-93.

[29-11] Salençon, J., Viscoélasticité pour le Calcul des Structures, *Presses des Ponts et Chaussées*, 2009.

[29-12] Chiorino, M. A. A Rational Approach to the Analysis of Creep Structural Effects", *ACI SP-227*, pp.107-141.

[29-13] CEB-FIP Bulletin 213/214, 1993. CEB-FIP Model Code 90.

[29-14] Sassone, M., Chiorino, M. A. Design Aids for the Evaluation of Creep Induced Structural Effects", *ACI SP-227*, 2005, pp. 239-259.

[29-15] C. Casalegno et al. Time dependent effects in cable-stayed bridges built by segmental construction, *Proceedings of 3rd fib International Congress*, Washington 2010.

[29-16] Z.P. Bažant. Numerical determination of long-range stress history from strain history in concrete", *Material and Structures*, Vol. 5, 1972, pp. 135-141.

[29-17] Jirásek, M., Bažant, Z.P. Inelastic Analysis of Structures, Wiley and Sons, 2002.

In contradiction to conventional steel reinforcement, FRP is not ductile. The designer should be familiar with the typical material behaviour of FRP and the extra assumptions used for design, as provide in section 17.8. Given the lack of a yield plateau in the FRP constitutive behaviour, analysis methods associated to plastic analysis and moment redistribution are omitted.

Figure 30.1-1 shows the possible range of strain distributions for concrete, reinforcing steel and prestressing steel. In the figure, the following limits are shown:

1 = reinforcing strain limit;

2 = concrete compression limit

30 Evaluation of structural performance

30.1 Evaluation of structural safety (ULS) for predominantly static loading

30.1.1 General

This subsection gives methods of verifying that, for a structure as a whole and for its component parts, the probability of an ultimate limit state exceeding the resistance of critical regions is acceptably small.

The determination of the partial safety coefficients and action effects is to be undertaken in accordance with the principles set out in chapter 11, 12 and 13.

Provisions given in this section apply to structures without deterioration or damage, for materials within the scope of part V and with reinforcement detailing complying with the rules given in section 30.13.

Provisions for structures with deterioration or damage are given in section 30.1.10. Provisions for structures with non compliant detailing are given in section 30.1.11.

In structural analysis for FRP reinforcement, both linear and non-linear methods of analysis can be applied, in reference to the FRP constitutive law described in Chapter 17. Plastic analysis and linear analysis with moment redistribution shall not be undertaken for members with FRP reinforcement.

30.1.2 Bending with and without axial force

30.1.2.1 Structures without deterioration or defects

30.1.2.1.1 Beams, columns and slabs

This subsection applies to undisturbed areas of beams, slabs and similar types of members for which sections remain approximately plane before and after loading. The discontinuity regions of beams and other members, where plane sections do not remain plane, may be designed and detailed according to subsection 30.1.6.

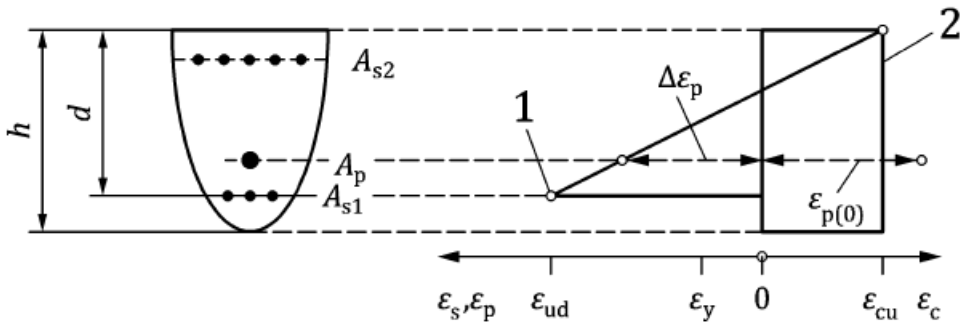


Figure 30.1-1: Possible strain distributions in the ultimate limit state.

Due to the lack of yielding of FRP bars, the overall deformation capacity of a FRP reinforced concrete structure can only be guaranteed by allowing for large ultimate strains of the reinforcement. This can be expressed by limiting the depth of the compression zone, or likewise by an equivalent minimum curvature.

Indicatively, both conditions can be combinedly considered in the design calculations through Eq. (17.8-2).

When determining the ultimate bending resistance of reinforced or prestressed concrete cross-sections, the following assumptions are made:

- plane sections remain plane;
- the strain in bonded reinforcement or bonded prestressing tendons, whether in tension or in compression, is the same as that in the surrounding concrete;
- the tensile strength of the plain concrete is ignored, in case of fibre reinforced concrete the uniaxial constitutive relationship defined in 18.4.3 (rigid plastic model) or in 18.4.4 and 18.4.5 (linear model) can be introduced, by using the design values defined in 18.6.4 and 18.6.5.;
- the stresses in the concrete are derived from stress–strain relations for the design of cross-sections as given in subsection 30.1.2.1.2 for plain concrete and 18.6.1 and 18.6.2 for fibre reinforced concrete;
- the stresses in the reinforcing and prestressing steel are derived from the design curves in subsections 15.10.2 and 16.1.9;
- the initial strain in the prestressing tendons is taken into account when assessing the stresses in the tendons.

For cross-sections with symmetrical reinforcement loaded by a compression force, the minimum eccentricity should be taken as $e_0 = h/30$ but not less than 20 mm, where h is the depth of the section.

To secure adequate deformation capacity of a flexural member with FRP rebars, the depth of the compression zone at ULS is limited as specified in Section 17.8.1.4.

The ULS design of members with FRP rebars shall ensure that:

- –the tensile strain in the FRP reinforcement is limited, in reference to the design tensile capacity (following equation 17.8-1);
- the sustained stresses under all relevant quasi-permanent combination of actions shall be limited so to prevent creep rupture of the FRP (following Eq. (17.8-3)).

Where the confining effect of FRP reinforcement is used in the design, the area of this reinforcement shall be reduced by the ratio E_f/E_s in any direction that confinement is considered.

The maximum value of the compressive strain accounts for non-linear influences due to time (see Malja et.al. [30.1-1]). The effect of the increased strain capacity for $f_{ck} > 50$ MPa compared to MC2010 is counterbalanced by the strength reduction factors α_{cc} and η_{fc} in the calculation of f_{cd} in Eq. (14.11-4).

The relationships given in Eq. (30.1-1a) and Eq. (30.1-1b) are not material stress-strain laws but are used to describe the stress distribution through the compression zones at ULS in combination with assumption that plane sections remain plane. These curves account for the influences of rate of loading and creep (linear and non-linear related to microcracking). For further references see: Rüsch [30.1-2] and Mattock et. al. [30.1-3].

30.1.2.1.2 Concrete stress distributions in compression zones

For the design of cross-sections, a choice can be made between two types of stress–strain relations. A parabola–rectangle relation (Figure 30.1-2a) is defined according to:

$$\sigma_c = f_{cd}\left[1-\left(1-\frac{\varepsilon_c}{\varepsilon_{c2}}\right)^2\right] \qquad \text{for } 0 \le \varepsilon_c \le \varepsilon_{c2} \qquad (30.1\text{-}1a)$$

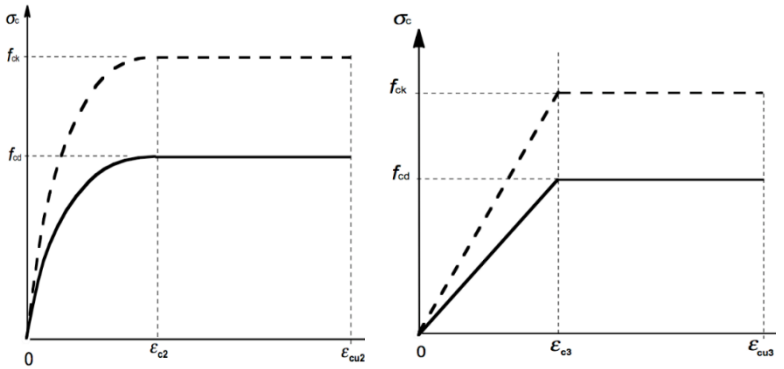
$$\sigma_c = f_{cd} \qquad \text{for } \varepsilon_{c2} \le \varepsilon_c \le \varepsilon_{cu2} \qquad (30.1\text{-}1b)$$

where:

$\varepsilon_{c2} = 0.002$ and

$\varepsilon_{cu2} = 0.0035$.

As alternatives, a bilinear stress distribution (based on the values ε_{c2} and ε_{cu2} , Figure 30.1-2b) or a rectangular stress distribution (Figure 30.1-2c) may also be used.



a) *b)*

Parameters η and λ referring to the strength reduction for f_{ck}> 50 MPa and to the dimension of the stress block according to MC2010 have been replaced by new formulation of f_{cd} (strength reduction applies not only for the compression zone, but also to columns and compression field in webs)

For the justification and background of this change, see: Moccia et al. [30.1-4].

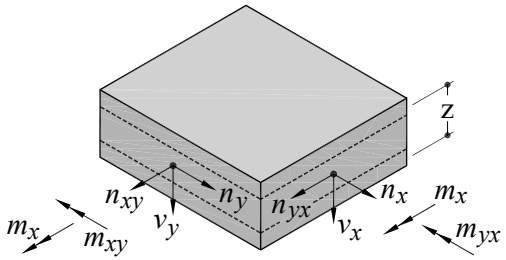
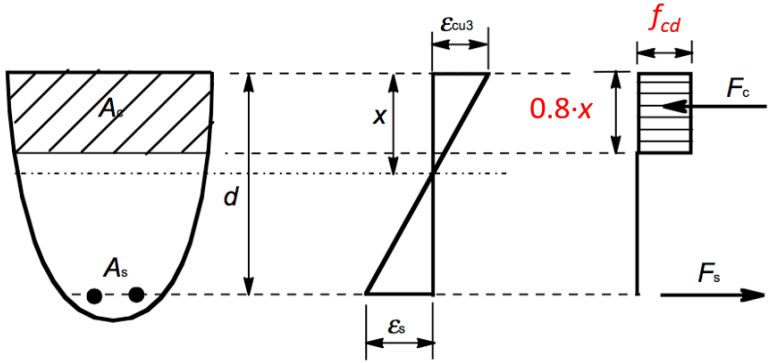


Figure 30.1-3: Three-layer plate model and stress resultants.

The subscript notations *inf* and *sup* refer to the inferior and superior faces of the element. The inferior face is the tensile face for an element in positive bending.



c)

Figure 30.1-2: Stress distribution in compression zones at ULS; (a) Parabola–rectangle distribution ; (b) bilinear distribution and (c) rectangular distribution.

30.1.2.1.3 Shells

Level I approximation

Shell elements may be modelled as comprising three layers (Figures 30.1-3 and 30.1-4). The outer layers provide resistance to the in-plane effects of both the bending and the in-plane axial loading, while the core layer provides a shear transfer between the outer layers.

The action effects of the applied loads are expressed as eight components, three moments per unit width, three axial forces per unit width and two shear forces per unit width in directions parallel to the orthogonal reinforcement. The stress resultants m_x, m_y, m_{xy}, n_x, n_y, n_{xy}, v_x, v_y produce the following forces per unit width on the element:

$$n_{xinf,sup} = \frac{n_x}{2} \pm \frac{m_x}{z} + \frac{v_x^2}{2v_o} cot\,\theta \qquad (30.1\text{-}2a)$$

$$n_{yinf,sup} = \frac{n_y}{2} \pm \frac{m_y}{z} + \frac{v_y^2}{2v_o} cot\,\theta \qquad (30.1\text{-}2b)$$

$$n_{xyinf,sup} = \frac{n_{xy}}{2} \pm \frac{m_{xy}}{z} + \frac{v_x v_y}{2v_o} cot\,\theta \qquad (30.1\text{-}2c)$$

The effective shear depth z may be taken as $0.9d$, where d denotes the distance between the extreme compression fibre and the mean planes of the reinforcement layers at the opposite face. The effective shear depth needs not be taken as less than $0.72h$.

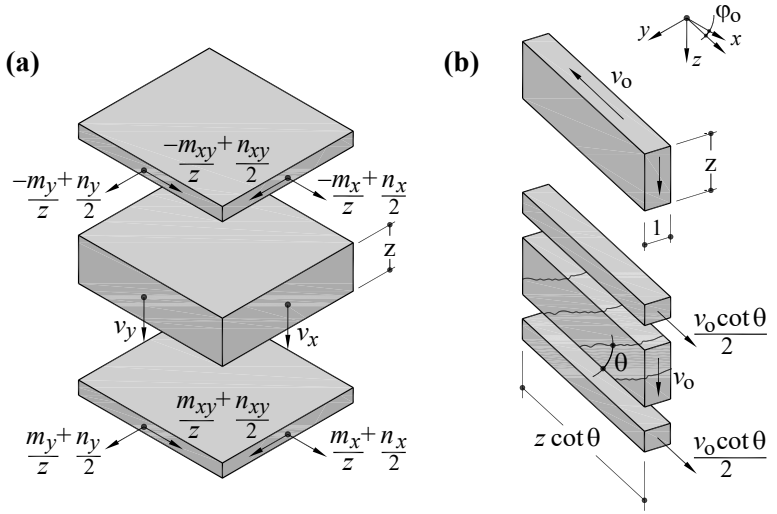


Figure 30.1-4: (a) Layer forces in sandwich model and (b) transfer of transverse shear force in uncracked and cracked core.

More information can be found in Marti [30.1-5].

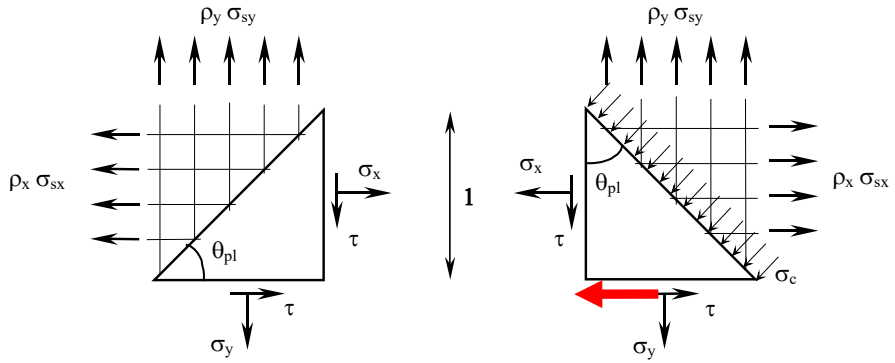


Figure 30.1-5: Stresses acting on and within a reinforced concrete element

where:

θ is the inclination of the compression stresses in the core layer;

z is the average lever arm between the forces in the x and y directions in the top and bottom layers and the effective shear depth, respectively;

v_o is the principal transverse shear force per unit length and follows from:

$$v_o = \sqrt{v_x^2 + v_y^2} \qquad (30.1\text{-}3)$$

For members with shear reinforcement, the angle θ is to be selected in accordance with section 30.1.3.3. For members without shear reinforcement, a value of $\cot\theta = 2$ may be used as is (implicitly) suggested in section 30.1.3.2.

If at least one principal stress is in tension, the outer layers may be designed as membrane elements according to plasticity theory such that (Figure 30.1-5):

$$\sigma_{sx} = \frac{1}{\rho_x}\left(\sigma_x + \tau\,cot\,\theta_{pl}\right) \le f_{yd} \qquad (30.1\text{-}4a)$$

$$\sigma_{sy} = \frac{1}{\rho_y}\left(\sigma_y + \tau\,cot\,\theta_{pl}\right) \le f_{yd} \qquad (30.1\text{-}4b)$$

$$\sigma_{cd} = \frac{\tau}{sin\,\theta_{pl}\,cos\,\theta_{pl}} \le k_\varepsilon \cdot f_{cd} \qquad (30.1\text{-}4c)$$

If no reinforcement has yielded and at least one principal stress is in tension:

$$k_\varepsilon = \frac{1.04}{1+0.00146\cdot\sigma_{si}} \le 1.0 \qquad (30.1\text{-}5a)$$

where σ_{si} is the maximum tensile stress (in MPa) in any layer of reinforcing steel ($i = x, y$)

To ensure that the ductility demand is met, the term $|\theta_{pl} - \theta_{el}|$ in Eq. (30.1-6) should not be greater than 15°, unless refined calculations are undertaken to justify a higher value.

In some cases, a verification with more than 3 layers can be more appropriate. A distribution of out of plane shear stress with respect to thickness will need to be assumed.

These provisions are based on the models presented in MC 2010 with appropriate updates reflecting current state of the art. [ONLY for DRAFT to be removed for final version]

Further background information on shear provisions treated in this section is given by Sigrist et al. [30.1-6].

If one or more layers of reinforcement yield:

$$k_\varepsilon = \left(1 - 0.032\left|\theta_{pl} - \theta_{el}\right|\right) \cdot \frac{1.04}{1+0.00146 \cdot \sigma_{si}} \leq 1.0 \qquad (30.1\text{-}5b)$$

where:

θ_{pl} is the compression field angle with respect to x-axis at the ULS;
θ_{el} is the first cracking angle with respect to the x-axis.

If both principal stresses are compressive:

$$|\sigma_{2d}| \leq k_\varepsilon \cdot f_{cd} \qquad (30.1\text{-}6)$$

where σ_2 is the minor principal (compressive) stress and ν may be taken as 1.0 or determined in accordance with subsection 5.1.6.

The shear core should be designed in accordance with section 30.1.3.

Level II approximation

A level II approximation may be made with a strain- based strength evaluation of the compressive concrete strength according to 30.1.3 and 30.1.6.

30.1.3 Shear

30.1.3.1 General

30.1.3.1.1 Design and assessment approaches

Design and assessment may be based on a stress field analysis or a strut-and-tie model, as outlined in section 30.1.6. Such models are especially suitable for the design of discontinuity regions (D-regions) at supports or transverse applied forces.

Alternatively, the following cross-sectional approach may be applied for the shear verification of the webs of beams and the core layers of slabs.

30.1.3.1.2 Design shear force and design shear resistance

The shear verification of a web or of a slab is based on following inequality:

The depth d denotes the effective depth in flexure which is defined as the distance from the extreme compressive fibre of the concrete to the resultant tensile force in the tensile reinforcing steel and tendons.

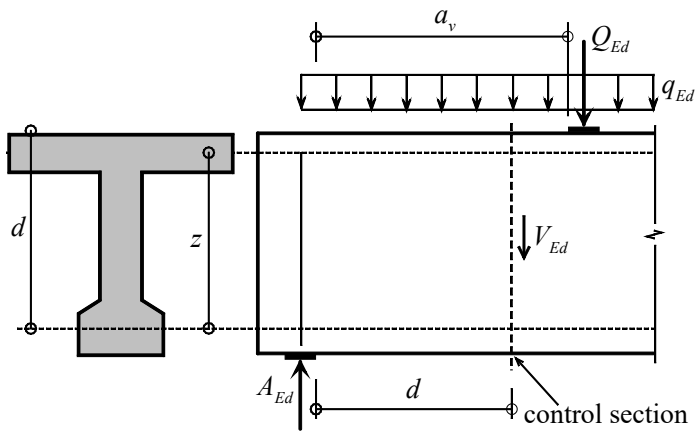


Figure 30.1-6: Definition of control section for sectional design.

For the verification of the region between point loads and linear supports of slabs, the width where the shear force due to point loads may be averaged according to this provision. The averaged shear force for this can also be determined approximately according to MC2010.

Where a concentrated load is applied to a slab near a support line, its capacity shall be checked for punching at the control perimeter around the loaded area, as described in

$$V_{Rd} \geq V_{Ed} \tag{30.1-7}$$

where:

V_{Rd} is the design shear resistance and

V_{Ed} is the design shear force.

The design shear force V_{Ed} shall in general be determined for control sections at a location d from the face of supports (see Figure 30.1-6) and from discontinuities of geometry or applied loads (concentrated loads).

Other control sections may be required, for example in case of varying web widths along a span, for non-uniform or significant concentrated loads, or at sections near points of curtailment of reinforcement.

Sections closer to supports than the distance d may be designed for the same shear force as at the control section provided that the member is directly supported.

For members without shear reinforcement, unless more refined modelling techniques are used to consider loads carried directly to a support through direct strut or arch action (see section 30.1.6), the following rules apply:

– the contribution of point loads applied within a distance of $d < a_v \leq 2d$ from the face of the support to the design shear force V_{Ed} may be reduced by the factor:

$$k_{dir} = \frac{a_v}{2d} \leq \frac{1}{2} \tag{30.1-8}$$

– in the case of point loads applied as close as $a_v < d$ from the face of the support, the design shear force V_{Ed} shall be calculated with $k_{dir} = 0.5$ as if the load was applied at $a_v = d$.

For members with shear reinforcement, loads carried directly to a support through strut or arch action may be considered according to 30.1.3.3.

When the shear force in slabs and shells is not constant along the control section, it may be averaged over the width not larger than $2d$ on each side from the peak of the shear force. If other internal forces are required for the calculation of the shear resistance, they may be also averaged over the same width.

section 30.1.5, and for shear at a control section taken parallel to the line of the support at the lesser of the distances equal to d and $a_v/2$ from the face of the support.

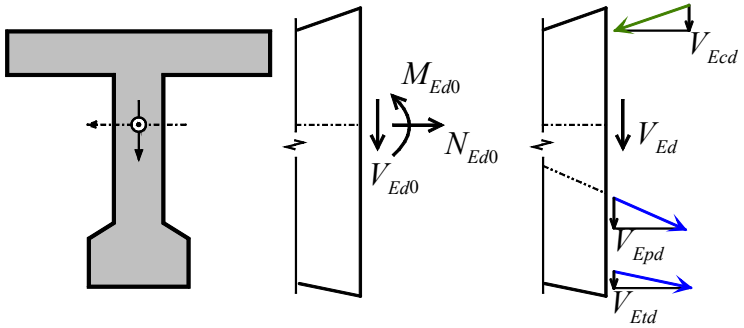


Figure 30.1-7: Contributions of inclined chord forces to design shear force (M_{Ed0}, V_{Ed0} and N_{Ed0} denote bending moment, shear and normal forces resulting from sectional analysis).

Membrane action due to internal or external restraints further increases the design shear resistance and, as a consequence, the shear reinforcement may be decreased; however, the compression stresses in the concrete are enhanced and should be checked. It is recommended to study such a situation with help of a strut-and-tie model (see subsection 30.1.6.).

For members containing ordinary steel reinforcement as well as prestressed tendons, the effective shear depth z_v may be taken as:

$$z_v = \frac{\left(z_s^2 A_s + z_p^2 A_p\right)}{\left(z_s A_s + z_p A_p\right)} \geq 0.72h \qquad (30.1\text{-}9)$$

where z_s and z_p are the distances between the centreline of the compressive chord and the reinforcement and tendon axes, respectively.

For determining V_{Ed} , the shear force from the sectional analysis V_{Ed0} may be reduced by favourable contributions resulting from any inclined tension chords (V_{Etd}), compression chords (V_{Ecd}) and prestressing tendons (V_{Epd}) – see Figure 30.1-7. In determining V_{Epd}, an eventual reduction in prestress due to the development length shall be considered.

Any unfavourable contributions from inclined chord and prestressing tendon forces shall be added to V_{Ed0}.

In the design for shear in beams and in slabs, the effects of axial tension due to creep, shrinkage and thermal effects in restrained members should be considered wherever appropriate.

For the effective shear depth z_v required for calculating the shear resistance V_{Rd} , a value of $0.9d$ may be assumed.

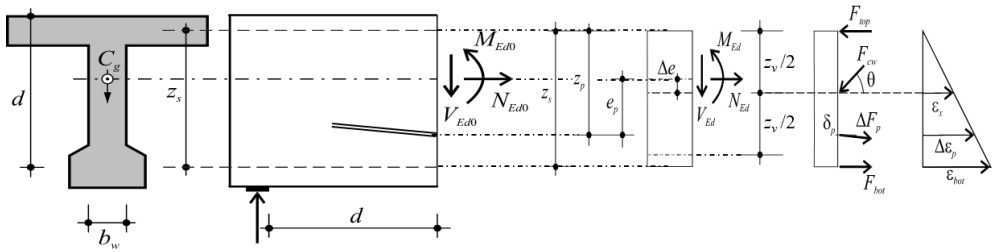


Figure 30.1-8: Definitions.

For members prestressed with bonded tendons, Eq. (30.1-10) is replaced by:

$$\varepsilon_x = \frac{\left(\frac{M_{Ed}}{z_v}+V_{Ed}+N_{Ed}\frac{(z_p-e_p)}{z_v}\right)}{2\left(\frac{z_s}{z_v}E_S A_S+\frac{z_p}{z_v}E_p A_p\right)} \geq 0 \qquad (30.1\text{-}11)$$

In members where the determined part of prestressing is considered as an internal action (i.e. when the determined part of prestressing is not included in the load combination used to determine V_{Ed}); the sectional forces are taken as:

$$\begin{aligned} M_{Ed} &= M_{Ed0} + M_{Pd} \\ N_{Ed} &= N_{Ed0} - F_p \cos\delta_p \\ V_{Ed} &= V_{Ed0} - F_p \sin\delta_p \end{aligned} \qquad (30.1\text{-}12)$$

where M_{Pd} denotes the design bending moment due to prestressing which includes a possible moment $M_{P,ind}$ resulting from static indeterminacy, that is $M_{Pd} = \pm F_p \cos\delta_p(e_p) + M_{P,ind}$. Analogously, shear and normal forces are affected.

The described model is based on the assumption that the shear force is carried by the web and by the inclination of the forces in the flanges. Nevertheless, there is evidence that near point loads, a limited fraction of the shear force can be carried by flanges in

The calculation of the shear resistance can require the state of strain to be taken into account. Within the framework of undertaking a cross-sectional analysis, the longitudinal strain (Figure 30.1-8) is calculated at the mid-depth of the effective shear depth or core layer being considered as follows:

$$\varepsilon_x = \frac{1}{2E_S A_S}\left(\frac{M_{Ed}}{z_v}+\frac{V_{Ed}}{s}\cot\theta + N_{Ed}\left(\frac{1}{2}-\frac{\Delta e}{z_v}\right)\right) \geq 0 \qquad (30.1\text{-}10)$$

where, as a first approximation, $\cot\theta$=2.0 may be assumed.

In the use of Eq. (30.1-10), the following conditions apply:

- M_{Ed} and V_{Ed} shall be taken as positive quantities and N_{Ed} as positive for tension and negative for compression.
- It is permissible to use a value of ε_x that is greater than half the yield strain of the longitudinal bars ($\varepsilon_{sy}/2$) but a more detailed cross-sectional analysis shall be undertaken. The strain ε_x shall not exceed 0.003.
- If the value of ε_x is negative, $E_s{\cdot}A_s$ in Eq. (30.1-10) shall be replaced by ($E_c{\cdot}A_{c,ten}+E_s{\cdot}A_s$) where $A_{c,ten}$ is the area of the tension chord due to bending
- For sections closer than d to the face of the support, the value of ε_x taken at d from the face of the support may be used.
- For sections within a distance $z_v/2$ of a significant bar curtailment, the calculated value ε_x shall be increased by a factor of 1.5.
- A_s comprises the main longitudinal reinforcing bars in the tensile chord; any distributed longitudinal reinforcement (longitudinal web reinforcement) is neglected.
- In calculating A_s (and A_p) the area of the bars that are terminated less than their development length from the section under consideration shall be reduced in proportion to their lack of full development.
- If the axial tension is large enough to crack the flexural compression face of the section, the calculated value of ε_x shall be multiplied by a factor of 2.0.

compression. This can be accounted for in design for instance by shifting the governing control section.

In case of a support that penetrates into the beam or slab, z is replaced with the effective depth, d_v in accordance with section 30.1.5.2.

In members without shear reinforcement, shear is resisted by a combination of aggregate interlock, residual tensile stress across inclined cracks, shear carried in the flexural compression zone and dowel action. Further information can be found in: Cavagnis et al. [30.1-7].

The level I equation is derived from the more general level II approximation with the assumption that the mid-depth strain at the control section may be taken as $\varepsilon_x \approx f_{yd}/(2E_s)$. For this reason, level I should not be used for a comparison to test results to check model accuracy, unless the experiments were designed to fail simultaneously in shear and flexure.

Additional background to Eqs. (30.1-15 – 17) can be found in Bentz and Collins [30.1-8] and Sigrist et al. [30.1.-6].

In higher strength concrete and lightweight aggregate concretes, the fracture surface may go through the aggregate particles, rather than around, reducing the crack roughness.

There is evidence that the shear resistance of members without shear reinforcement is influenced by the maximum size of the aggregate d_g. If the maximum aggregate particles, d_g, is not less than 16 mm, k_{dg} in Eq. (30.1-17) may be taken as $k_{dg} = 1.0$ as a

30.1.3.2 Members without shear reinforcement

General

The design shear resistance of a web or a slab without shear reinforcement is given by:

$$V_{Rd,c} = k_v \frac{\sqrt{f_{ck}}}{\gamma_C} b_w z_v \qquad (f_{ck} \text{ in MPa}) \qquad (30.1\text{-}13)$$

Where, for members with $z_v \geq 800$ mm, the value of $\sqrt{f_{ck}}$ may not be taken as greater than 8 MPa.

The longitudinal reinforcement in the flexural tensile chord at each section of interest shall be able to resist an additional force component due to the shear of:

$$\Delta F_{td} = V_{Ed} \qquad (30.1\text{-}14)$$

However, the total demand on longitudinal reinforcement doesn't need to exceed the demand due to maximum moment alone in the respective maximum moment region.

Level I approximation

For members with no significant axial load, with $f_{yk} \leq 700$ MPa and with a maximum aggregate size of not less than 10 mm:

$$k_v = \frac{0.4}{1+750 \cdot f_{yd}/E_s} \cdot \frac{1300}{1000+1.25 z_v} \qquad (z_v \text{ in mm}) \qquad (30.1\text{-}15)$$

where f_{yd} is the yield strength which has been used to design the flexural reinforcement.

Level II approximation

For the level II approximation, the design shear resistance is determined with:

$$k_v = \frac{0.4}{1+1500 \cdot \varepsilon_x} \cdot \frac{1300}{1000+k_{dg} \cdot z_v} \qquad (z_v \text{ in mm}) \qquad (30.1\text{-}16)$$

where the value k_{dg} accounts for roughness of the critical shear crack and should be calculated with:

safe estimate. In case an upper and a lower value of d_g are specified, the lower one should be assumed as d_g.

For members with $z_v \geq 800$ mm, constructed with $f_{ck} \geq 70$ MPa, k_{dg} according to Eq. (30.1-16) should be corrected as described on the right to account for the loss of aggregate interlock due to fracture of aggregate particles (d_g in Eq. (30.1-16) taken as zero), see for instance (Sherwood et al. [30.1-9].)

As given in Eq. (30.1-10), the longitudinal strain ε_x can be calculated as a function of the internal forces M_{Ed}, V_{Ed} and N_{Ed}. For designing new structures, this strain may be calculated directly as a function of the internal forces. For calculating the shear resistance of an existing structure, an iteration until the design value of the internal force corresponds to the resistance is needed to find ε_x.

Alternatively, closed-form expressions may be derived analytically from these equations allowing for a direct calculation of the shear resistance. For the case without axial force, following equation can be derived:

$$V_{Rd,c} = \left[\sqrt{1 + 7.8\frac{a}{d}\frac{\sqrt{f_{ck}}}{\rho_L(1000 + k_{dg}z)}} - 1\right]\frac{120\rho_L}{a/d}\frac{1}{\gamma_c}b_w z \qquad (30.1\text{-}18)$$

where a is the shear span which can be assumed as (M_{Ed} / V_{Ed} + 1.0)

Starting from the Critical Shear Crack Theory rather than the Modified Compression Field Theory, and implementing appropriate simplifications, the following equation included in FprEN 1992-1-1:2023 [30.1-10] (in a slightly modified form) can be derived:

$$V_{R,dc} = \frac{0.083}{\gamma_V}\left(\rho_l \cdot E_s \cdot f_{ck}\frac{d_{dg}}{\sqrt{a_{cs}\cdot d}}\right)^{\frac{1}{3}} \cdot b_w \cdot z \qquad (30.1\text{-}19)$$

where γ_V is the partial safety factor for shear, a_{cs} is the shear span and d_{dg} a parameter related to the aggregate size d_g (d_{dg}=32mm/k_{dg}). The derivation of this equation from the mechanical model can be found in Cavagnis et al. [30.1-11].

The shear force in planar members (slabs and shells) is a vector with a principal direction which can be calculated directly on the basis of the two components (see Eq. (30.1-3)) as:

$$tan\,\theta_v = \frac{v_y}{v_x} \qquad (30.1\text{-}20)$$

$$k_{dg} = \frac{32}{16+d_g} \geq 0.75 \qquad (30.1\text{-}17)$$

where d_g is the maximum size of the aggregate in mm.

If a member has $z_v \geq 800$ mm and $f_{ck} \geq 70$ MPa or uses lightweight concrete of any strength, $k_{dg} = 2.0$.

Level III approximation

In alternative to Eq. (30.1-10), the strain ε_x to be used in Eq. (30.1-14) may also be determined based on a more detailed cross-sectional analysis accounting for the principles of 30.1.3.1, or directly on the basis of a non-linear analysis of the member. For slabs and shells, the strain ε_x refers to the principal direction of the shear force.

The web reinforcement ratio ρ_w given by Eq. (30.1-21) corresponds to the minimum reinforcement ratio as defined in section 30.13.6. Members containing a lower reinforcement ratio are to be treated according to section 30.1.3.2. For members not complying with this condition (designed according to previous standards or cases where f_{ck} has significantly increased), there is evidence that there is a transition between the resistance without stirrups and the resistance of members with minimum shear reinforcement ratio. In absence of a mechanical model, a linear interpolation between the two resistances is appropriate (Angelakos [30.1-12]). Research has shown that the linear interpolation proposed above can underestimate the resistance of prestressed members (see Choi et al. [30.1-13]).

Size effects are limited in members with web reinforcement greater than that required by Eq. (30.1-21).

For the reduction of the minimum shear reinforcement for ductility classes higher than A, see Monney et al. [30.1-14].

The limitation on $\sqrt{f_{ck}}$ is provided due to the larger observed variability in shear strength of members with higher strength concrete.

This verification of the resistance of the inclined compression field is typically required near to supports, reduction of web width or other discontinuities.

The strength reduction factor k_ε accounts for the state of strain in the webs of beams or the core layers of slabs. An additional strength reduction factor accounting

30.1.3.3 Members with shear reinforcement

30.1.3.3.1 General

This subsection applies to members that meet the demand for minimum shear reinforcement according to:

$$\rho_w \geq 0.08 \frac{\sqrt{f_{ck}}}{f_{yk}} \qquad (f_{ck} \text{ and } f_{yk} \text{ in MPa}) \qquad (30.1\text{-}21)$$

The minimum shear reinforcement may be reduced by:

- 10% for ductility class B reinforcement;
- 20% for ductility class C or D reinforcement

according to ductility classes defined in 15.5.1.

The design shear resistance is then determined from:

$$V_{Rd} = V_{Rd,c} + V_{Rd,s} \leq V_{Rd,max} \qquad (30.1\text{-}22)$$

where:

$$V_{Rd,c} = k_v \frac{\sqrt{f_{ck}}}{\gamma_C} b_w z_v \qquad (30.1\text{-}23)$$

is the design shear resistance attributed to the concrete. In using this equation with members having z_v greater than 800 mm, the value of $\sqrt{f_{ck}}$ shall not be taken as greater than 8 MPa.

$$V_{Rd,s} = \frac{A_{sw}}{s_w} z_v f_{ywd} \cot\theta \qquad (30.1\text{-}24)$$

is the design shear resistance provided by shear reinforcement;

f_{ywd} is the design yield strength of the shear reinforcement;

θ is the inclination of the compressive stress field which depends on the LoA.

$$V_{Rd,\max} = \frac{k_\varepsilon f_{cd} b_w z_v}{\cot\theta + \tan\theta} \qquad (30.1\text{-}25)$$

is the maximum shear resistance related to crushing of concrete carrying the compression field;

for the brittleness of concrete (η_{fc} in MC2010 [30.1-16]) is already included in f_{cd} (see Eq. (14.11-4)) since it applies also to other cases.

For the case of shear reinforcement with $f_{ywk} > 600$ MPa, the verification for cracking at SLS can become governing.

For the assessment of an existing structure without change of use, the verification at SLS may be replaced by inspection.

In the case of shear reinforcement that is inclined relative to the beam axis, the Eqs. (30.1-24) and (30.1-25) should be replaced by:

$$V_{Rd,s} = \frac{A_{sw}}{s_w} z f_{ywd} (cot\,\theta + cot\,\alpha)\, sin\,\alpha \qquad (30.1\text{-}26)$$

And

$$V_{Rd,\max} = k_\varepsilon f_{cd} b_w z_v \frac{\cot\theta + \cot\alpha}{1 + \cot^2\theta} \qquad (30.1\text{-}27)$$

where α is the inclination of the stirrups with respect to the longitudinal axis.

Angles $\alpha > 90°$ (shear reinforcement inclined in the same direction as the compression field) should be avoided since the activation of the shear reinforcement requires significant strains in the web with potentially unacceptable cracking at SLS and stronger concrete strength softening (lower values of coefficient k_ε) at ULS.

Values of k_D depend on the material of the duct and whether it is grouted or not. Suggested values for design are:

- grouted steel duct: $k_D = 0.5$;
- grouted plastic duct: $k_D = 0.8$;
- ungrouted duct: $k_D = 1.2$.

Factor k_D may be reduced in presence of reinforcement transverse to the plane of the web.

In this simplified approach, it is assumed that the compression field crosses the tendons in the region of its maximum stress. A more refined analysis can be conducted by investigating the stress and the strength in the region where the compression field crosses the tendons.

In the case of stirrups that are inclined relative to the beam axis the Eq. (30.1-29) is be replaced by:

k_ε is the strength reduction factor of concrete carrying the compression field which depends on the LoA.

In Eq. (30.1-25), the effective web width b_w shall be taken as the minimum concrete web width within the effective shear depth z.

In the case of prestressing tendons with duct diameters $\varnothing_D \geq b_w/8$, the ultimate resistance of the compression struts shall be calculated on the basis of the nominal value of the web width:

$$b_{w,nom} = b_w - k_D \sum \varnothing_D \qquad (30.1\text{-}28)$$

where $\Sigma\varnothing_D$ is to be determined for the most unfavourable prestressing tendon configuration.

The longitudinal reinforcement in the flexural tensile chord at every section of interest is to be designed to resist the additional force due to shear of:

$$\Delta F_{td} = \frac{V_{Ed}}{2}(cot\ \theta - cot\ \alpha) \quad (30.1\text{-}30)$$

In the level III approach, where a concrete contribution $V_{Rd,c} \neq 0$ is considered, in the Eqs. (30.1-29) and (30.1-30) the design shear force V_{Ed} is replaced by V_{Ed}^* where:

$$V_{Ed}^* = V_{Ed} + V_{Rd,c} \quad (30.1\text{-}31)$$

The value of $\theta_{\min}$ is determined by the level of approximation.

The lower limit of θ (θ_{min}) can be justified by:

- the use of a fixed value of strength reduction factor k_ε (governing at LoAI) and
- the condition of preventing brittle rupture of shear reinforcement (governing at LoAII) or
- the avoidance of an explicit verification of the SLS in the web.

The upper limit of θ (cot θ = 1) is justified by:

- the condition of preventing concrete crushing before yielding of the shear reinforcement and
  the requirement of preventing anchorage failures at end supports.

Following Table 30.1-1 provides an overview of the concrete contribution $V_{Rd,c}$, of the minimum compressive stress field inclination θ_{min} and of the strength reduction factor k_ε to be assumed for the levels of approximation I-III.

Table 30.1-1: Overview of the design equations for different levels of approximation

Level of approximation	*Concrete contribution*	*Minimum compressive stress field inclination*	*Strength reduction factor for $V_{RD,max}$*
I	$V_{Rd,c} = 0$	$\cot\theta_{\min} = \text{fixed}$ Eqs (30.1-34-36)	$k_\varepsilon = \text{fixed}$ Eq. (30.1-37)
IIa	$V_{Rd,c} = 0$	$\cot\theta(\varepsilon_x)_{min}$ Eqs. (30.1-40-42)	$k_\varepsilon = f(\varepsilon_x, \theta)$ Eq. (30.1-43)
IIb	$V_{Rd,c} = f(\varepsilon_x)$ Eq. (30.1-45)	$\cot\theta = f(\varepsilon_x)$ $\theta = f(\varepsilon_x)$ Eq. (30.1-40)	$k_\varepsilon = f(\varepsilon_x, \theta)$ Eq. (30.1-43)

The level I approximation represents a variable angle truss model approach. For simplicity, the influence of shear transferred across diagonal cracks by aggregate interlock (i.e. $V_{Rd,c}$) is neglected in the definition of the inclination of compression field θ.

$$\Delta F_{td} = \frac{V_{Ed}}{2} cot\ \theta \quad (30.1\text{-}29)$$

However, the total demand on longitudinal reinforcement need not exceed the demand at the maximum moment location due to moment alone.

The limits of the compressive stress field inclination θ, relative to the longitudinal axis of the member (Figure 30.1-10), to be selected for Levels of approximation I and IIa, are:

$$1 \leq cot\ \theta \leq cot\ \theta_{min} \quad (30.1\text{-}32)$$

where θ may be chosen freely between these limits for design and assessment.

Values of cotθ < 1 are possible, provided that the reinforcement is designed according to 30.1.6 (design with stress fields and strut-and-tie models).

In case of level of approximation IIb, the inclination θ refers to assumed crack plane where the free body equilibrium is verified and is calculated as a function of the longitudinal strain.

Level I approximation

In the level I approach, the design shear resistance is given by:

$$V_{Rd} = V_{Rd,s} \leq V_{Rd,max} \quad (30.1\text{-}33)$$

For the case with simultaneous yielding of the shear reinforcement and failure of the inclined compression field, the shear resistance V_{Rd} results from the solution of Eq. (30.1-24) and Eq. (30.1-25) which yields following compression field inclination:

$$1 \le cot\,\theta = \sqrt{\frac{k_\varepsilon \cdot f_{cd}}{\rho_w \cdot f_{ywd}} - 1} \le cot\,\theta_{min} \qquad (30.1\text{-}38)$$

Eqs. (30.1-34 - 37) apply to cross-sections where the longitudinal strain ε_x remains below a value of 0.001.

In level II, the influence of strains on the shear resistance is accounted for explicitly. Two different approaches are provided:

- The level IIa approximation is based on a stress field approach consistent with 30.1.6 (Design with stress fields and strut-and-tie models) where the compression field inclination θ may be selected according to Eq. (30.1-32).
- The level IIb approximation is consistent with the Simplified Modified Compression Field Theory where the inclination θ refers to the assumed crack plane angle which depends directly on the longitudinal strain.

For a preliminary design or analysis, ε_x may be taken as 0.001.

but should not to be taken less than the resistance of the same member without shear reinforcement.

The minimum inclination of the compressive stress field is:

$\cot\theta_{\min} = 2.2$ for members with significant axial compression or prestress; (30.1-34)

$\cot\theta_{\min} = 1.7$ for reinforced concrete members; (30.1-35)

$\cot\theta_{\min} = 1.2$ for members with significant axial tension (30.1-36)

The width of the beam or web shall be checked for the selected inclinations of the compression stresses where k_ε is taken as:

$$k_\varepsilon = 0.55 \qquad (30.1\text{-}37)$$

Level II approximation

A level II approximation may be conducted according to two different approaches (IIa and IIb), both of which provide a safe estimate of the shear resistance.

Level IIa approximation

In the level IIa approach, the design shear resistance is given by:

$$V_{Rd} = V_{Rd,s} \le V_{Rd,max} \qquad (30.1\text{-}39)$$

but shall not to be taken less than the resistance of the same member without shear reinforcement.

Within the limits of Eq. (30.1-32) the inclination of the compressive stress field may be freely selected for design, or analytically determined for assessment.

The minimum inclination of the compressive stress field is defined by:

$$\cot\theta_{\min} = \cot\left(20° + 4000\varepsilon_x\right) \qquad (30.1\text{-}40)$$

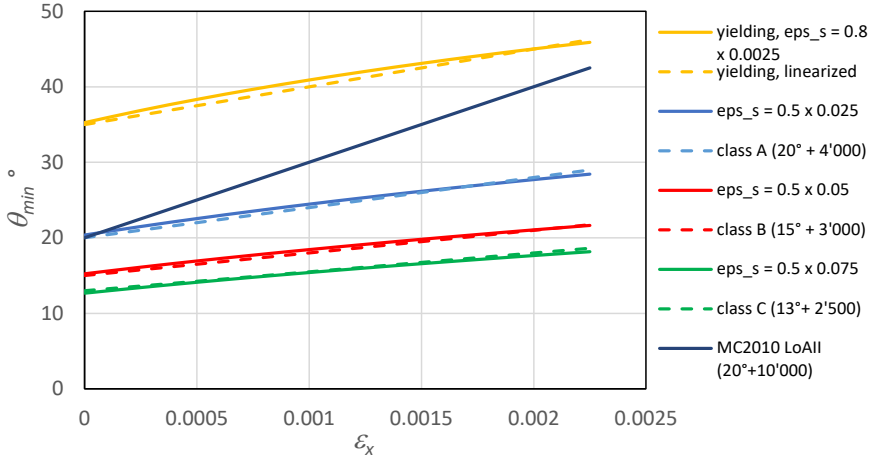


Figure 30.1-9: **Comparison to databases to be added, this Figure to be moved to the background document**

The strength reduction factor k_ε considers the influence of the state of strain in the web. This influence is important for prestressed members or members in compression, but less significant for reinforced elements and/or sections with higher θ and ε_x values.

For the calibration of the parameters in Eqs. 30.1-43 and 44, see Pejatovic et al. [30.1-15].

Level IIb approximation represents a general form of sectional shear equations and is based on the simplified modified compression field theory.

A comparison of the relative predictions for modelling levels I to III is shown in Figure 30.1-10.

where ε_x represents the longitudinal strain at the mid-depth of the effective shear depth as shown in Figure 30.1-9.

For shear reinforcement more ductile than class A according to 15.5.1, provided that cracking at serviceability limit state is verified explicitly, the minimum inclination of the compressive stress field may be reduced as follows:

$$\cot\theta_{\min} = \cot\left(15° + 3000\varepsilon_x\right) \quad \text{for ductility class B} \quad (30.1\text{-}41)$$

$$\cot\theta_{\min} = \cot\left(13° + 2'500\varepsilon_x\right) \quad \text{for ductility classes C and D} \quad (30.1\text{-}42)$$

The design shear resistance attributed to the concrete is neglected, that is $k_v = 0$. The width of the beam or web should be checked for the respective inclination of the compression stresses where k_ε is taken as:

$$k_\varepsilon = \frac{1}{1+110\cdot\varepsilon_1} \leq 1.0 \quad (30.1\text{-}43)$$

where:

$$\varepsilon_1 = \varepsilon_x + (\varepsilon_x + 0.001)\, cot^2\,\theta \quad (30.1\text{-}44)$$

The longitudinal strain ε_x at the mid-depth of the effective shear depth is calculated based on Eq. (30.1-11) or (30.1-10).

Level IIb approximation

In the level IIb approach, the design shear resistance is given by:

$$V_{Rd} = V_{Rd,s} + V_{Rd,c} \leq V_{Rd,max} \quad (30.1\text{-}45)$$

For strengths governed by $V_{Rd,c}$+$V_{Rd,s}$, $\cot\theta$ shall be taken as:

$$cot\,\theta \;\; = cot(29° + 7000\varepsilon_x) \quad (30.1\text{-}46)$$

while for cases governed by $V_{Rd,max}$, $\cot\theta$ may be selected within the ranges used in LoA IIa (Eq. (30.1-40-42)).

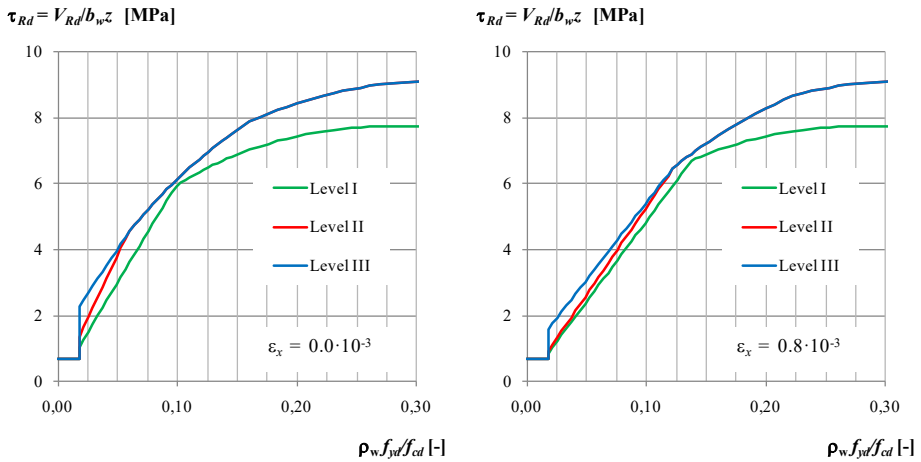


Figure 30.1-10: Comparison of levels I, II and III results for members with f_{ck} = 50 MPa (Note: for the curves shown in the figure the value f_{cd} is defined as $\eta_{fc} f_{ck}/\gamma_c$).

For members which meet the relative shear reinforcement spacing requirements of clause 30.13.6.2 but exceed the absolute spacing limit of 500 mm, a concrete contribution may be included if a partial size effect is accounted for. In such cases k_v can be taken from LoA II for members without shear reinforcement with the term $k_{dg} z_v$ in Eq. (30/1-16) replaced with (s - 300) where s is the spacing of the transverse reinforcement.

The use of tools based on advanced methods of analysis often requires extensive experience to ensure that safe and consistent results are obtained.

For determining the design shear resistance $V_{Rd,c}$ attributed to the concrete the following expression should be used:

$$k_v = \frac{0.4}{1+1500\varepsilon_x} \qquad (30.1\text{-}47)$$

The strain ε_x at the mid-depth of the effective shear depth is calculated with Eq. (30.1-10) or (30.1-11).

Level III approximation

The resistance of members in shear, or shear combined with torsion, may be determined by satisfying the applicable conditions of equilibrium and compatibility of strains and by using appropriate stress–strain models for the steel and for diagonally cracked concrete.

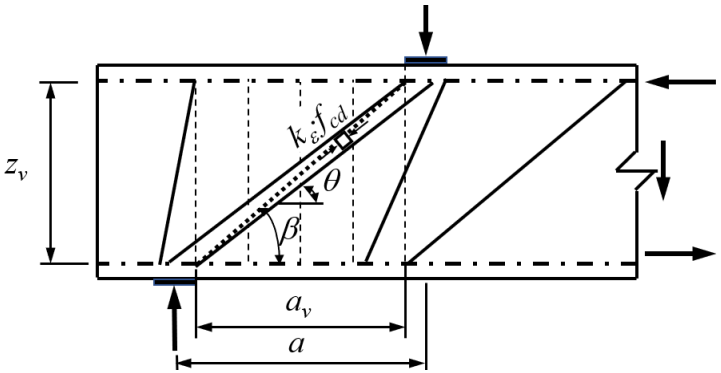


Figure 30.1-11: Loads carried directly to a support through strut or arch action according Levels of approximation I and II.

The maximum shear resistance according to Eq. (30.1-48) can be calculated by optimization varying $\cot\theta$. For Level I approximation, the optimum is obtained with:

$$\cot\theta = \cot\beta + \sqrt{1+\cot^2\beta} \le \cot\theta_{min} \qquad (30.1\text{-}50)$$

For Level II approximation (where k_ε depends on θ), a reasonable approximation is obtained assuming:

$$\cot\theta = 1.3 \cdot \frac{a}{z_v} \le \cot\theta_{y,c} \qquad (30.1\text{-}51)$$

where $\theta_{y,c}$ is the inclination according to Eq. (30.1-38).

This approach is consistent with section 30.1.6 (Design with stress fields and strut-and-tie models) and is based on the lower bound theorem of limit analysis. At the same time, it corresponds also to the upper bound theorem so that it can be considered as the exact solution according to limit analysis. The dotted line with the inclination β in Figure 30.1-11a represents the failure mechanism (yield line) according to limit analysis.

In case of multiple concentrated loads near to the support, all potential yield lines need to be verified (with the corresponding $\cot\beta$).

Further background information on the model treated in this subsection is given by Pejatovic et al. [30.1-15].

30.1.3.3.2 Loads carried directly to a support through strut or arch action

In cases where distributed loads are acting between the control section and the distance $z_v \cdot \cot\theta$ from the support, they may be deducted from the acting shear force V_{Ed} used to design the shear reinforcement according to Eq. (30.1-24).

In cases where concentrated loads are applied at a distance $a_v = z_v \cdot \cot\beta$ which is less than $z_v \cdot \cot\theta$ from a support as shown in Figure 30.1-11, the shear resistance according to Levels I and II approximation may be enhanced as follows:

$$V_{Rd} = k_\varepsilon f_{cd} \frac{\cot\theta - \cot\beta}{1+\cot^2\theta} \cdot b_w \cdot z_v + \frac{A_{sw}}{s_w} z_v \cdot f_{ywd} \cot\beta \le V_{Rd,max} \qquad (30.1\text{-}48)$$

In addition to the force increment in the tension chord due to shear according to Eq. (30.1-29), also the following moment ΔM_{Ed} should be added to M_{Ed} to determine the force in the tension chord:

$$\Delta M_{Ed} = \left(V_{Ed} - \frac{A_{sw}}{s_w} z_v \cdot f_{ywd} \cot\theta\right) \cdot \left(\frac{a}{2} - x\right) \qquad (30.1\text{-}49)$$

where:

- a is the distance between the axis of the support and the concentrated load (see Figure 30.1-11); and
- x is the distance between the axis of the support and the investigated cross section (see Figure 30.1-11).

Compression field inclinations with $\cot\theta < 1$ are allowed if the yield strength f_{ywd} in Eq. (30.1-48) and in Eq. (30.1-49) is replaced by the stress σ_{swd} which can be activated in the shear reinforcement:

$$\sigma_{swd} = E_s[\cot^2\theta \cdot (\varepsilon_x + 0.002) - 0.002] \qquad (30.1\text{-}52)$$

where ε_x is the longitudinal strain according to Eq. (30.1-10) with the resulting value of $\cot\theta$, at a cross section located midway between the support and the concentrated load (with $x = a/2$, ΔM_{Ed} according to Eq. (30.1-49) becomes zero).

Due to the interaction between the contribution of concrete $V_{R,c}$ and the contribution of residual tensile strength related to fibres $V_{Rd,FRC}$, $V_{R,c}$ and $V_{Rd,FRC}$ need to be determined based on the deformation of the member. This is considered in the following with 3 levels of approximation.

The equations are based on mechanical parameters and therefore are applicable for different types of fibres. However, all equations have been validated based on tests with steel fibres, therefore a preliminary check in case of different type of fibres is strongly suggested.

The parameter k_s consists of the multiplication of two components, the first accounts for behaviour of cracks in finding the pathways of least resistance and around fibre ends (this is k_f in the MCFT version of the 2010 Model Code equation for shear). Where the pathway is not fixed by boundary conditions or notches, the value of k_f is taken as 0.8. The second component accounts for fibre orientation bias differences between prism specimens tested according to EN14651 [30.1-17] relative to that of the shear failure and is denoted as k_0 (see chapter 18). The proposed value for coefficient k_s can be confirmed based on a reliability analysis.

The orientation bias factor is taken as k_0 =0.8. Higher values of k_0 may be justified through testing. Similarly, lower values of k_0 may be appropriate where strong fibre bias occurs and orientation that is less favourable to shear strength, such as may occur for self-compacting concrete, for example.

30.1.3.4 Shear in members in FRC

30.1.3.4.1 General

The shear verification of a web or of a slab is based on following inequality:

$V_{Rd} \geq V_{Ed}$

where:

V_{Rd} is the design shear resistance and

V_{Ed} is the design shear force.

30.1.3.4.2 Members without shear reinforcement

30.1.3.4.2.1 General

The design shear resistance of a member without shear reinforcement, is given by:

$$V_{Rd} = V_{Rd,c} + V_{Rd,FRC} \quad (30.1\text{-}53)$$

where:

$$V_{Rd,c} = k_{v,FRC} \frac{\sqrt{f_{ck}}}{\gamma_C} \cdot z_v \cdot b_w \quad (30.1\text{-}54)$$

where, for members with $z_v \geq 800$ mm, the value of $\sqrt{f_{ck}}$ may not be taken as greater than 8 MPa.

$$V_{Rd,FRC} = \frac{k_s \cdot f_{FTU}}{\gamma_F} \cdot cot\theta_F \cdot z_v \cdot b_w \quad (30.1\text{-}55)$$

is the contribution of residual tensile strength related to fibres, where:

$k_s = 0.64$

θ_F is the inclination of the SFRC compressive stress field; and

f_{FTU} is the characteristic value of the ultimate residual tensile strength for FRC in MPa, as defined below for the Level-of-Approximation adopted.

The following table provides an overview of the concrete contribution $V_{Rd,c}$, the fibre contribution $V_{Rd,FRC}$, of the angle θ and of the strength reduction factor to be assumed for the levels of approximation I-III.

Table 30.1-2: Overview of the design equations for different levels of approximation

Level of approximation	*Concrete contribution*	*FRC Contribution*	*Compressive stress field inclination*
I ($z_v \leq$ 800 mm)	$V_{Rd,c} = f(f_{yd}; E_s)$ Eq. (30.1-57)	$V_{Rd,FRC} = f(f_{FTUk})$	$\theta_F = f(f_{yd}; E_s)$ Eq. (30.1-59)
II ($z_v \leq$ 800 mm)	$V_{Rd,c} = f(\varepsilon_x)$ Eq. (30.1-63)	$V_{Rd,FRC} = f(f_{FTUk})$	$\theta_F = f(\varepsilon_x)$ Eq. (30.1-65)
III	$V_{Rd,c} = f(\varepsilon_x)$ Eq. (30.1-66)	$V_{Rd,FRC} = f(f_{FTwk})$	$\theta_F = f(\varepsilon_x)$ Eq. (30.1-67)

The Level I approximation assumes $\varepsilon_x = f_{yd}/(2E_s)$ and d_g = 10 mm. For this reason, level I should not be used for a comparison to test results to check model accuracy, unless the experiments were designed to fail simultaneously in shear and flexure.

Eqs. (30.1-57) and (30.1-62) are adapted from Eqs (30.1-15 and 16) (shear in RC members without shear reinforcement), where coefficient ζ accounts for a reduction of the concrete component due to the increased crack width that occurs in the presence of fibres at ULS.

For tensile softening FRC, f_{FTU} should not be taken as greater than the lower characteristic strength of the strength grade of concrete used. That is f_{FTU} taken as not greater than $0.7 \cdot 0.6 \cdot f_{ck}^{2/3}$.

Other approaches are available in the literature; in particular the following that is aligned to the approach presented in ANNEX L of FprEN 1992 [30.1-10].

For members not requiring shear reinforcement, having ordinary longitudinal reinforcement, the design shear strength can be computed as follows:

$$\tau_{Rd,cF} = \eta \frac{0.6}{\gamma_c}\left(100\rho_l f_{ck}\frac{d_{dg}}{d}\right)^{\frac{1}{3}} + f_{Ftud} \geq \eta\tau_{Rdcmin} + f_{Ftud} \qquad (30.1\text{-}60)$$

30.1.3.4.2.2Level I Approximation

The Level I approximation may be used for members with no significant axial load, with $f_{ck} \leq 70$ MPa, $f_y \leq 500$ MPa, with a maximum aggregate size of not less than 10 mm and $z_v \leq 800$ mm:

$$f_{FTU} = \frac{f_{R3k}}{3} \qquad (30.1\text{-}56)$$

but not taken as greater than the lower characteristic tensile strength of the concrete without fibres.

$$k_{v,FRC} = \frac{0.4}{\left(1+750\cdot\frac{f_{yd}}{E_s}\right)+\zeta} \cdot \frac{1300}{1000+1.25\cdot z_v} \ (z_v \text{ in mm}) \qquad (30.1\text{-}57)$$

$$\zeta = \left(20 - 7000 \cdot \frac{f_{yd}}{E_s}\right) \cdot \frac{f_{FTU}}{f_{ck}} \geq 0 \qquad (30.1\text{-}58)$$

$$\theta_F = 29° + 3500 \cdot \frac{f_{yd}}{E_s} \qquad (30.1\text{-}59)$$

where f_{yd} is the yield strength of the flexural tensile reinforcement.

where:

$0.4 \leq \eta = 1.2 - 0.5 f_{Ftuk} \leq 1.0$ (f_{Ftuk} in MPa) (30.1-61)

Steel fibres cannot be credited in calculating the shear strength for members resisting shear in combination with axial tension

An alternative approach for Levels I and II is to adopt a fixed width of the crack measured in the transverse direction; a value of $\underline{w}_u$ = 2.0 mm is recommended for $h \leq$ 800 mm. When substituted into the Eq. (18.4-5), f_{FTU} is then obtained as:

$$f_{FTU} = 0.46 f_{R3k} - 0.13 f_{R1k} \geq 0 \quad (30.1\text{-}62)$$

This replaces Eq. (30.1-56). To ensure a minimum level of softening, in this equation, f_{R1k} should be taken as the greater of that measured and $1.1 f_{R3k}$.

The limit of z_v = 800 mm is justified by the fact that for deeper members, crack widths > 2.0 mm can develop.

30.1.3.4.2.3 Level II Approximation

The Level II approximation may be used for members with $h \leq$ 800 mm:

For the level II approximation, the design shear resistance is determined with:

$$k_{v,FRC} = \frac{0.4}{(1+1{,}500\varepsilon_x)+\varsigma} \cdot \frac{1300}{1000 + k_{dg} \cdot z_v} \quad (z_v \text{ in mm}) \quad (30.1\text{-}63)$$

$$\varsigma = (20 - 14{,}000\varepsilon_x) \cdot \frac{f_{FTU}}{f_{ck}} \geq 0 \quad (30.1\text{-}64)$$

$$\theta_F = 29° + 7000 \cdot \varepsilon_x \quad (30.1\text{-}65)$$

f_{FTU} is the characteristic value of the ultimate residual tensile strength for FRC in MPa, according to Eq. (30.1-56) but not taken as greater than the lower characteristic tensile strength of the concrete without fibres.

ε_x is the longitudinal strain at the mid-depth of the effective shear depth and k_{dg} accounts for roughness of the critical shear crack, as defined in Eq. (30.1-17).

For members with significant axial tension, $V_{Rd,FRC} = 0$ should be assumed.

The shear strength need not to be taken less than the resistance of the same member without shear reinforcement.

The crack width is measured in the direction of opening which can be approximated as the crack width measured in the transverse direction of the member (perpendicular to the longitudinal axis). The tensile strength of the fibres crossing the crack for a given crack width (f_{FTU}) is determine from the linear model in 18.4.2 (i.e. Eq. 18.4-5).

The constitutive law that defines the stress as a function of the crack opening should be tensile softening. Where this does not occur, the hardening effect should be neglected (a rigid-plastic law should be assumed at level f_{FTs} according to 18.4.2 assuming w_u = 1.25 mm).

The use of steel fibres may assist in controlling the size effect with a possibility of lowering the minimum stirrup contribution required to mitigate the size effect. This is a subject for future research.

This requirement (same minimum shear reinforcement as in members without fibres) needs to be verified.

Shear reinforcement is not required in regions where $\tau_{Ed} \leq \tau_{RD,cF}$. Moreover, steel fibres cannot be credited in calculating the strength of members resisting shear in combination with axial tension.

Other approaches are available in the literature; in particular, the following that is aligned to the approach presented in ANNEX L of the FprEN 1992 [30.1-10]. For members with shear reinforcement, having classical longitudinal reinforcement, the design shear strength ($\tau_{Rd,SF}$) can be computed as follows:

30.1.3.4.2.4 Level III Approximation

For the Level III approximation, the design shear resistance is determined with:

f_{FTU} is the characteristic value of the ultimate residual tensile strength for the width crack determined by Eq. (30.1-68) in MPa, according to Eq. 18.4-5 but not taken as greater than the lower characteristic tensile strength of the concrete without fibres.

$$k_{v,FRC} = \frac{0.4}{1+1{,}500\varepsilon_x} \cdot \frac{1300}{1000 + k_{dg} \cdot z_v} \quad (z_v \text{ in mm}) \tag{30.1-66}$$

$$\theta_F = 29° + 7000 \cdot \varepsilon_x \tag{30.1-67}$$

$$w_u = (0.2 + 1000\varepsilon_x) \left(\frac{1000+k_{dg}d_v}{1300}\right) / cos\ \theta_F \geq 0.5 \text{ mm} \tag{30.1-68}$$

where:

ε_x represents the longitudinal strain at the mid-depth of the effective shear depth;

k_{dg} accounts for roughness of the critical shear crack, as defined in Eq. (30.1-17).

30.1.3.4.3 Members with shear reinforcement

30.1.3.4.3.1 General

This section applies to members that meet the demand for minimum shear reinforcement according to Eq. 30.1-21.

$$\tau_{Rd,sF} = \left(\eta_s \frac{A_{sv}}{s}\frac{z}{bd} f_{yd} + \eta_F f_{Ftud}\right) \quad (30.1\text{-}69)$$

where:

$\eta_s = 0.75$ and $\eta_F = 1.0$.

Due to the interaction between the contributions of concrete $V_{R,c}$, of residual tensile strength related to fibres $V_{Rd,FRC}$ and of shear reinforcement $V_{Rd,s}$, $V_{R,c}$ and $V_{Rd,FRC}$ need to be determined based on the deformation of the member. This is considered in the following with 3 levels of approximation.

The following table provides an overview of the concrete contribution $V_{Rd,c}$, the fibre contribution $V_{Rd,FRC}$, of the angle θ_F and of the strength reduction factor to be assumed for the levels of approximation I-III.

Table 30.1-3: Overview of the design equations for different levels of approximation

Level of approximation	*Concrete contribution*	*FRC Contribution*	*Compressive stress field inclination*	*Strength reduction factor for $V_{RD,max}$*
I ($z_v \leq 800$ mm)	$V_{Rd,c} = 0$	$V_{Rd,FRC} = f(f_{FTU})$	$\theta_F = fixed$ (same as LoA IIa in RC)	$k_\varepsilon = fixed$ Eq. (30.1-40)
II ($z_v \leq 800$ mm)	$V_{Rd,c} \neq 0$ (30.1-63)	$V_{Rd,FRC} = f(f_{FTU})$	$\theta_F = f(\varepsilon_x)$ (same as LoA IIb in RC)	$k_\varepsilon = f(\varepsilon_x, \theta)$ Eq. (30.1-43)
III	$V_{Rd,c} \neq 0$ (30.1-66)	$V_{Rd,FRC} = f(f_{FTwk})$	$\theta_F = f(\varepsilon_x)$ (same as LoA IIb in RC)	$k_\varepsilon = f(\varepsilon_x, \theta)$ Eq. (30.1-43)

The design shear resistance is determined from:

$$V_{Rd} = V_{Rd,c} + V_{Rd,FRC} + V_{Rd,s} \leq V_{Rd,max} \quad (30.1\text{-}70)$$

where:

$V_{Rd,c}$ is the design shear resistance attributed to the concrete as defined by Eq. (30.1-54), and depends on the Level of Approximation;

$V_{Rd,s}$ is the design shear resistance provided by shear reinforcement, as defined in Eq. (30.1-24) with $\theta = \theta_F$;

$V_{Rd,FRC}$ is the design shear resistance provided by (steel) fibres, as defined in Eq. (30.1-55) with $\theta = \theta_F$; and

$V_{Rd,max}$ is the maximum shear resistance related to crushing of concrete carrying the compression field, as defined in Eq. (30.1-25) with $\theta = \theta_F$;

θ_F is the inclination of the FRC compressive stress field.

The width of the beam or web should be checked for the respective inclination of the compression stresses where k_ε and $V_{Rd,max}$ are determined according to assumptions of the Level of Approximation adopted, which are the same as for members without fibres (section 30.1.3.3).

30.1.3.4.3.2Level I Approximation

The Level I approximation applies for members with $z_v \leq 800$ mm and without significant applied axial force.

For this level,

$V_{Rd,c} = 0$

f_{FTU} is given by Eq. (30.1-56).

Alternatively, in the Level II approach, Eq. (30.1-56) may be replaced with:

$$f_{FTU} = 0.46 f_{R3k} - 0.13 f_{R1k} \geq 0 \quad (30.1\text{-}71)$$

The crack width equation is for a level of shear reinforcement, sufficient to avoid the depth effect. This is taken as that for members without fibres. Further research is needed to determine if lower values may be adopted for members with fibres.

The minimum inclination of the compressive stress field may be assumed as for members without fibres (Eq. (30.1-34)).

30.1.3.4.3.3Level II Approximation

The Level II approximation applies for members with $z_v \leq 800$ mm and without significant applied axial force. For the concrete contribution to the shear capacity, use Eq. (30.1-60).

For this level, f_{FTU} is given by Eq. (30.1-56), but not taken as greater than the lower characteristic tensile strength of the concrete without fibres.

The inclination of the compressive stress field is:

$$\theta_F = 29° + 7000 \cdot \varepsilon_x \quad (30.1\text{-}72)$$

where ε_x represents the longitudinal strain at the mid-depth of the effective shear depth as shown in Figure 30.1-9.

30.1.3.4.3.4Level III Approximation

In the Level III approach, the design shear resistance is given by:

f_{FTU} is the characteristic value of the ultimate residual tensile strength determined according to Eq. (18.4-5) with the width crack determined by Eq. (30.1-23). In Eq. (18.4-5), f_{R3k} should not be taken as greater than f_{R1k}.

f_{FTU} should not be taken as greater than the lower characteristic tensile strength of the concrete without fibres.

$$k_{v,FRC} = \frac{0.4}{1+1{,}500\varepsilon_x} \quad (30.1\text{-}73)$$

$$\theta_F = 29° + 7000 \cdot \varepsilon_x \quad (30.1\text{-}74)$$

and the crack width is calculated from:

$$w_u = (0.2 + 1000\varepsilon_x)/cos\ \theta_F \geq 0.5 \text{ mm} \quad (30.1\text{-}75)$$

where ε_x is the longitudinal strain at the mid-depth of the effective shear depth as shown in Figure 30.1-9.

Given that FRP stirrups are not yielding, the special case of a compression field with $cot\theta = 1$ is considered and allowing for superposition of the shear contribution of the concrete and FRP stirrups (provided that the maximum strain in the FRP stirrups is restricted, to allow for sufficient aggregate interlocking).

Eq. (30.1-77) has been established for FRP bar diameters smaller than 25 mm and a bend radius within the range 3 to 4 times the bar diameter.

30.1.3.5 Shear in members with FRP reinforcement

For members requiring no shear reinforcement, the longitudinal FRP reinforcement can be considered similar to steel reinforcement, provided that the area of longitudinal FRP reinforcement is reduced by the ratio E_f/E_s in all equations.

For members requiring shear reinforcement, and further to considering an equivalent longitudinal reinforcement ratio $\rho_s E_f/E_s$ in the shear design equations associated to the concrete shear resistance, the shear resistance of the member shall be obtained as:

$$\tau_{Rd} = \tau_{Rdc} + \rho_w f_{fwRd} \leq \frac{\nu f_{cd}}{2} \quad (30.1\text{-}76)$$

where:

τ_{Rdc} is the design value of the shear resistance of the concrete;

ρ_w is the reinforcement ratio provided by the stirrups;

$\nu f_{cd}/2$ relates to the capacity of the compression field (with ν the strength reduction factor for the concrete strength capacity in the compression field) and;

f_{fwRd} is the design value of the effective capacity in the FRP stirrups, as follows:

$$f_{fwRd} = min\{f_{fd}; 0{,}004E_f\} \quad (30.1\text{-}77)$$

where f_{fd} is the design tensile capacity of the FRP (according to equation 18.8-1) taking into account the effective tensile capacity by the bended part of the FRP stirrup:

$$\eta_{bend} = (0{,}03 + 0{,}05\frac{r_f}{d_f}) \leq 1 \quad (30.1\text{-}78)$$

with r_f the radius of the bend, and d_f the diameter of the FRP bar.

For hollow core slabs and similar structural members the design shear resistance may be calculated on the basis of this subsection or alternatively of section 30.1.3.3; the higher of the results may be adopted as the capacity.

These Equations (for LoA I and II), which don't account for size effect, are potentially unsafe for members with $d > 500$ mm. They are proposed here since they should be used for hollow core slabs only (these precast solutions typically fulfil this limit).

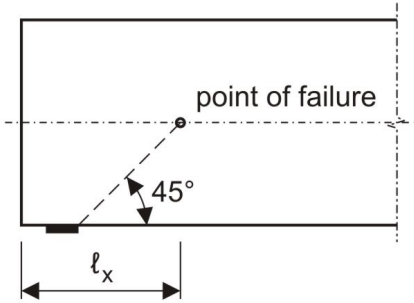


Figure 30.1-12: Basis for derivation Eq. (30.1-79).

30.1.3.6 Hollow core slabs

30.1.3.6.1 Introduction

In single span prestressed hollow core slabs without shear reinforcement, shear failure occurs when the principal tensile stress in the web exceeds the tensile strength of the concrete.

30.1.3.6.2 Level I approximation

The design shear resistance may be determined by:

$$V_{Rd,ct} = 0.8 \frac{I_c \cdot b_w}{S_c} \sqrt{f_{ctd}^2 + \alpha_l \cdot \sigma_{cp} \cdot f_{ctd}} \qquad (30.1\text{-}79)$$

where:

I_c is second moment of area;

S_c is first moment of area above and about the centroidal axis;

b_w is width of the cross-section at the centroidal axis;

σ_{cp} is concrete compressive stress at the centroidal axis due to prestressing, in the area where the prestressing force is fully introduced;

$\alpha_l = l_x / l_{bpt,95\%}$;

l_x follows from Figure 30.1-12;

$l_{bpt,95\%}$ follows from Eq. (7.13-5a).

30.1.3.6.3 Level II approximation

In a level II approximation, the design shear resistance is determined by:

$$V_{Rd,ct} = \frac{I_c \cdot b_w(y)}{S_c(y)} \left[\sqrt{f_{ctd}^2 + \alpha_l \cdot \sigma_{cp}(y) \cdot f_{ctd}} - \tau_{cp}(y)\right] \qquad (30.1\text{-}80)$$

where:

I_c is second moment of area;

$S_c(y)$ is first moment of area above height y and about the centroidal axis;

$b_w(y)$ is width of the cross-section at the height y;

y is the height of the critical point at the line of failure;

$\sigma_{cp}(y)$ is concrete compressive stress at height y and distance l_x;

$\tau_{cp}(y)$ is the shear stress in the concrete due to transmission of prestress at height y and distance l_x.

The concrete compressive stress at height y and distance l_x is determined from:

$$\sigma_{cp}(y) = \left[\frac{1}{A_c} + \frac{y_c - y}{I}\right] \cdot F_p(l_x) \qquad (30.1\text{-}81)$$

and the shear stress in the concrete due to transmission of prestress:

$$\tau_{cp}(y) = \frac{1}{b_w(y)} \cdot \left[\frac{A_c(y)}{A_c} - \frac{S_c(y)\cdot(y_c - y_{pt})}{I}\right] \cdot \frac{dF_p(l_x)}{dx} \qquad (30.1\text{-}82)$$

where:

y_c is height of concrete centroidal axis;

A_c is area of concrete cross-section;

$A_c(y)$ is concrete area above height y;

y_{pt} is height of centroidal axis of prestressing steel;

$F_p\ (l_x)$ is the prestressing force at distance l_x.

By varying y in the calculation, the lowest value of $V_{Rd,ct}$ in Eq. (30.1-80) is found.

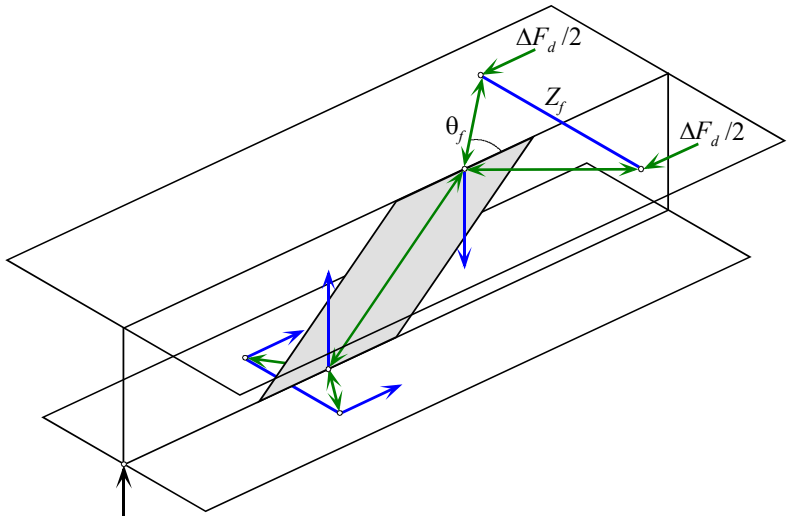


Figure 30.1-13: Strut-and-tie model for force introduction into the flanges.

Background information on this subject is given by Randl [30.1-18].

30.1.3.7 Shear between web and flanges of T-sections

The introduction of tensile or compressive forces into the flanges (Figure 30.1-13) creates shear forces at the transition of web and flanges, inducing corresponding transverse tensile and compressive forces in the flanges.

The spread of the compressive forces in the flanges shall be examined with the aid of stress fields. Recommended values for the angle of spread are:

$25° \leq \theta_f \leq 45°$ for compressive flanges; and

$35° \leq \theta_f \leq 50°$ for tensile flanges.

Unless a more detailed analysis is undertaken, reinforcement for the introduction of forces into the flanges is to be superimposed on that required for transverse bending.

In addition, the minimum transverse reinforcement shall not be less than that required by section 30.13.6.

The longitudinal flange reinforcement shall be anchored in accordance with the assumed stress field requirements.

30.1.3.8 Shearat the interface between concrete cast at different times

30.1.3.8.1 Introduction

In addition to the requirements formulated in sections 30.1.3.1–30.1.3.6 the shear stress at the interface between concrete cast at different times should also satisfy the following condition:

$$\tau_{Edi} \leq \tau_{Rdi} \qquad (30.1\text{-}83)$$

where τ_{Edi} is the design value of the shear stress in the interface, given by:

$$\tau_{Edi} = \beta \cdot V_{Ed}/(zb_i) \qquad (30.1\text{-}84)$$

where:

β is the ratio of the longitudinal force in the new concrete and the total longitudinal force either in the compression or tension zone, both calculated for the section considered;

z is the inner lever arm of the composed section;

b_i is the width of the interface and V_{Ed} is the shear force on the composed section.

If interacting bending moments are transferred via the interface, e.g. at vertical joints in flexural members, and parts of the cross section are subjected to tension, the interface shall be very rough or keyed. In this case, a non-rigid bond behavior (section 30.1.3.8.3) shall be considered for the design of interface shear transfer across the joint. Provided large crack widths are prevented by appropriate reinforcement design of the flexural beam, aggregate interlocking and friction can be regarded as contributing to the shear transfer via the joint, while the dowel action effect of the longitudinal (bending) reinforcement should not be used due to its usually significant utilization in tension and the close vicinity to the edge of the cross section.

30.1.3.8.2 Interface without reinforcement (rigid bond-slip behaviour)

If a "rigid" bond-slip behaviour is expected and very good adhesive bonding guaranteed on the site, the adhesive bond effect should be taken into account without superimposing effects of interface reinforcement. The most important precondition for the assumption of good adhesive bond is a well prepared and very clean concrete surface at the time of casting.

Adhesive bond resistance (c_a) as well as aggregate interlock effect (c_r) should only be applied for design if no tensile loading perpendicular to the interface is expected.

The design limit value τ_{Rdi} for the interface shear in Eq. (30.1-83) follows from:

$$\tau_{Rdi} = c_a \cdot f_{ctd} + \mu \cdot \sigma_n \leq 0.5 \cdot \nu \cdot f_{cd} \qquad (30.1\text{-}85)$$

where:

c_a is the coefficient for the adhesive bond;

μ is the friction coefficient from Table 30.1-5;

σ_n is the (lowest expected) compressive stress resulting from an eventual normal force acting on the interface.

The adhesion coefficients c_a actually depend on a variety of influencing parameters (see section 23.3); nevertheless the c_a factors given in Table 30.1-4 represent reasonable values on the safe side for the given roughness categories.

Special attention should be given to edge zones – see detailing rules in section 23.7.

For the definition of roughness of the classes distinguished in Table 30.1-4, see section 23.2.

The adhesion factor c_a depends on the roughness of the interface (see table 30.1-4; R_t is derived from the sand patch method).

Table 30.1-4: Coefficients for the adhesive bond resistance.

Surface characteristics of interface	c_a
Very rough (including shear keys) $R_t \geq 3.0$ mm	0.5
Rough (strongly roughened surface) $R_t \geq 1.5$ mm	0.40
Smooth (concrete surface without treatment after vibration or slightly roughened when cast against formwork)	0.20
Very smooth (steel, plastic, timber formwork)	0.025

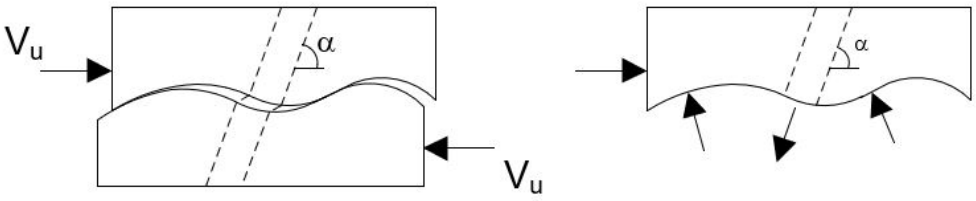


Figure 30.1-14: Transmission of shear forces across an interface intersected by reinforcing bars.

Eq. (30.1-86) relates to interfaces intersected by dowels or reinforcement and characterized by a rather non-rigid bond-slip behaviour.

Connectors may be omitted in interface regions where the design shear stress does not exceed the resistance given in Eq. (30.1-85).

Note that for a bar in tension, as explained in section 23.3.2, the tensile strength of the bar is reduced when the bar is also subjected to dowel action, as shown in Figure 30.1-15.

The detailing rules given in section 23.7 concerning embedment depth of connectors and minimum amount of steel cross-section should be obeyed.

Under fatigue or dynamic loads the values for c_a as found in table 30.1-4 have to be reduced to 50%.

30.1.3.8.3 Interface intersected by dowels or reinforcement

If strong adhesive bond cannot be guaranteed on the site or the design shear resistance provided by adhesive bond from Eq. (30.1-85) is lower than the design shear stress, interface connectors are required and the design limit value τ_{Rdi} follows from:

$$\tau_{Rdi} = c_r \cdot f_{cd}^{1/3} + \mu \cdot \sigma_n + \kappa_1 \cdot \rho \cdot f_{yd} \cdot (\mu \cdot \sin\alpha + \cos\alpha) + \kappa_2 \cdot \rho \cdot \sqrt{f_{yd} \cdot f_{cd}} \qquad (30.1\text{-}86)$$

where strength values are in N/mm^2 and:

c_r is the coefficient for aggregate interlock effects at rough interfaces;

κ_1 is the interaction coefficient for tensile force activated in the reinforcement or the dowels;

κ_2 is the interaction coefficient for flexural resistance;

μ is the friction coefficient;

ρ is the reinforcement ratio of the reinforcing steel crossing the interface;

σ_n is the (lowest expected) compressive stress resulting from an eventual normal force acting on the interface;

α is the inclination of the reinforcement crossing the interface (see Figure 30.1-14).

The coefficients for different surface roughness in interfaces reinforced with dowels or rebars are given in table 30.1-5.

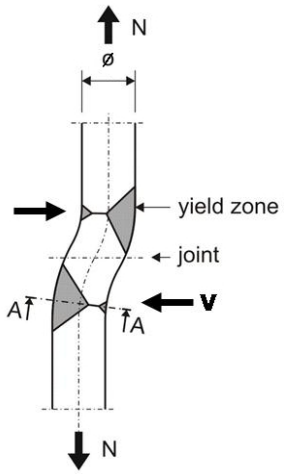


Figure 30.1-15: Dowel action under simultaneous tension and shear.

For the background of the values in table 30.1-5, see Randl [30.1-18].

The roughness of a concrete surface can be measured in various ways (see section 23.2). An appropriate way is the *sand patch method*, as depicted in Figure 30.1-16 (Kaufmann [30.1-19]). A volume of sand V is spread on the rough surface in a circular area with diameter D. The roughness parameter R_t follows from:

$$R_t[mm] = \frac{40 \cdot V}{\pi D^2} \qquad (30.1\text{-}87)$$

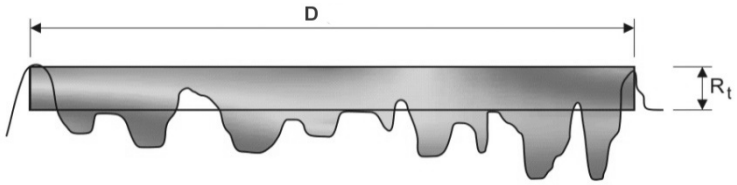


Figure 30.1-16: Principle of sand patch method for the qualification of the roughness of an interface.

Table 30.1-5: Coefficients for different surface roughness.

Surface Roughness	c_r	κ_1	κ_2	β_c	μ	
					$f_{ck} \geq 20$	$f_{ck} \geq 35$
Very rough* $R_t \geq 3.0$ mm	0.22	0.5	0.9	0.5	0.8	1.0
Rough $R_t \geq 1.5$ mm	0.11	0.5	0.9	0.5	0.7	
Smooth	0	0.5	1.1	0.4	0.6	
Very smooth	0	0	1.5	0.3	0.5	

* valid also for shear keys

Under fatigue or dynamic loads, the values for τ_{Rdi} according to Eq. (30.1-86) have to be reduced to 40%.

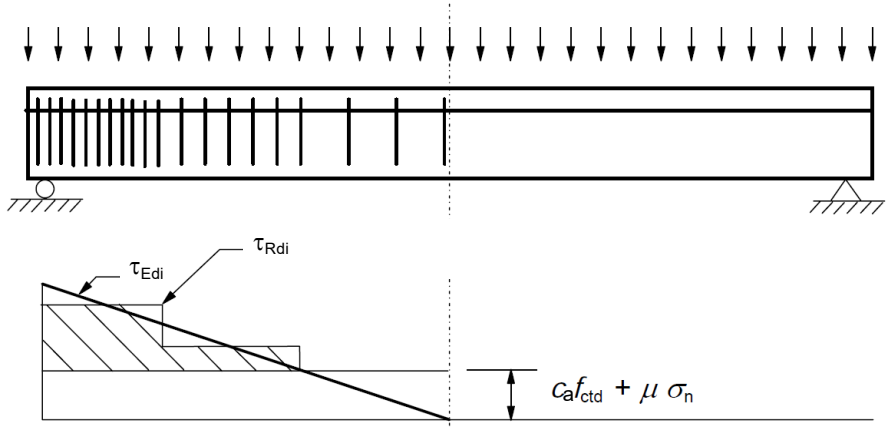


Figure 30.1-17: Shear diagram representing the required interface reinforcement.

A stepped distribution of the transverse reinforcement may be used, as indicated in Figure 30.1-17.

30.1.4 Torsion with and without axial force and / or bending

30.1.4.1 General

Where static equilibrium depends on the torsional resistance of elements of the structure, a full torsional design for both the ultimate and serviceability limit states shall be undertaken.

Where in structures torsion arises from consideration of compatibility only, and the structure is not dependent on torsional resistance for its stability, it will normally not be necessary to consider torsion at the ultimate limit state. In such cases minimum torsional reinforcement shall be provided in the form of stirrups, and longitudinal bars should be provided to prevent excessive cracking, as per the requirements of section 30.13.6.2.

The determination of the torsional resistance of box-girders and beams of solid cross-section is based on an ideal hollow cross-section as shown in Figure 30.1-18.

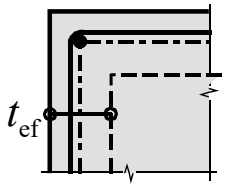


Figure 30.1-19: Minimum effective panel thickness.

In the case of combined action of torsion, bending and shear force in a solid section, the core within the idealized hollow cross-section may be used for the transmission of the shear forces.

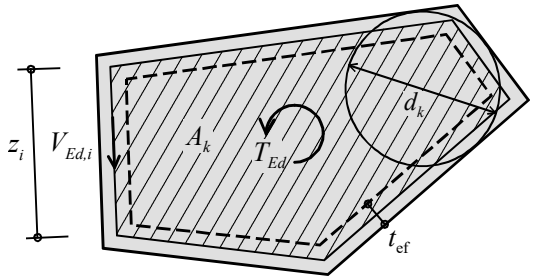


Figure 30.1-18: Definition of the ideal hollow cross-section.

If the internal forces and moments, the cross-sectional dimensions and the reinforcement do not change abruptly in the longitudinal direction, it may be assumed that the shear flow due to torsion is constant over the circumference of the effective cross-sectional area. The torsional moment T_{Ed} may then be resolved into equivalent panel forces such that:

$$V_{Ed,Ti} = \frac{T_{Ed}}{2A_k} z_i \qquad (30.1\text{-}87)$$

where A_k is the area within the centre line of the thin-walled effective cross-section, including inner hollow areas.

The provisions of section 30.1.3 apply analogously for the dimensioning of the reinforcement and checking of the panel dimensions. The effective panel thickness of solid cross-sections (Figure 30.1-19) may thereby be taken into account as:

$$t_{ef} \leq \frac{d_k}{8} \qquad (30.1\text{-}88)$$

where d_k denotes the diameter of the circle that might be inscribed at the most narrow part of the cross-section.

A minimum effective panel thickness $t_{ef,\min}$ of twice the distance between the concrete surface and the centre of the longitudinal reinforcement may be considered.

In the case of box-girders, the effective panel thickness corresponds to the wall thickness, if the wall is reinforced on all sides.

The longitudinal reinforcement due to torsion shall either be distributed evenly over the length of the panels or concentrated at the corners.

In the case of combined action of torsion, bending and shear force, the internal forces and moments are replaced by a statically equivalent set of normal and shear forces. The reinforcement is then determined according to the provisions of sections 30.1.2 and 30.1.3.

For box-girder sections the maximum resistance of a panel is given by $V_{Rd,c}$ for members without and $V_{Rd,\max}$ for members with shear reinforcement, respectively.

For other sections (such as rectangular cross-sections) the maximum resistance has to be checked by:

$$\left(\frac{T_{Ed}}{T_{Rd,\max}}\right)^2+\left(\frac{V_{Ed}}{V_{Rd,\max}}\right)^2\le 1 \qquad (30.1\text{-}89)$$

where $T_{Rd,\max}$ is calculated as:

$$T_{Rd,max}=k_\varepsilon\frac{f_{ck}}{\gamma_C}t_{ef}\,2A_k\,\sin\theta\cos\theta \qquad (30.1\text{-}90)$$

and the definitions of the parameters involved are given in section 30.1.3.3. Moreover $V_{Rd,\max}$ follows from Eq. (30.1-25).

For members with FRP reinforcement, the torsion provision of steel reinforcement apply, subject to the following alterations:

– $-f_{ywd}$ is replaced by f_{fd} (and considering η_{bend} following Eq. (30.1-78))

– cotθ is taken as 1.0

30.1.4.2 Torsion in members with FRC

The torsional resistance in the transverse and longitudinal directions of SFRC members should be taken as:

$$T_{Rd,swF}=2A_k\left(\eta_{sw}\cot\theta\frac{A_{sw}}{t_{eff}\cdot s}f_{ywd}+\eta_F f_{Ftud}\right)\ge 2A_k\left(\eta_{sw}\cot\theta\frac{A_{sw}}{t_{eff}\cdot s}f_{ywd}\right) \qquad (30.1\text{-}91)$$

$$T_{Rd,swF}=2A_k\left(\eta_{sw}\frac{\Sigma A_{sl}}{t_{eff}\cdot u_k\cdot\cot\theta}f_{yd}+\eta_F f_{Ftud}\right)\ge 2A_k\left(\eta_{sw}\frac{\Sigma A_{sl}}{t_{eff}\cdot u_k\cdot\cot\theta}f_{yd}\right) \qquad (30.1\text{-}92)$$

where:

η_{sw} = 0,75

η_F = 1,0.

Where a member is subjected to torsion in combination with shear and bending, fibre contribution to design resistances should only be as one of the two approaches:

- the fibre contribution is used to resist torsional action effects only;

Punching failures may develop with limited deformations (brittle behaviour). Therefore, the effects of imposed deformations (temperature, creep and shrinkage, settlements etc.) should be taken into account in design. The influence of imposed deformations may, however, be neglected if sufficient deformation capacity is provided. Strategies for increasing the deformation capacity are:

- choice of a sufficiently large supporting area and depth of slab in combination with low bending reinforcement ratios (rules are given in section 30.1.5.3);
- use of punching shear reinforcement (rules are given in section 30.1.5.3).

With flat slabs, safety against punching is particularly significant as the failure of one column can propagate to adjacent columns leading to a complete collapse of a structure. To avoid such progressive collapses, one (or both) of the following strategies should be adopted:

- increase of the deformation capacity of the potential failure zones (see above) to allow internal forces to redistribute;
- arrange integrity reinforcement for slabs with limited deformation capacity (rules are given in section 30.1.5.6).

Further background information on the punching shear models treated in this section is given by Muttoni et al [30.1-20].

- the fibre contribution is used to resist tensile forces arising from shear and bending action effects with fibre contribution to sectional resistance disregarded in determining axial tension and torsional resistance;

30.1.5 Punching

30.1.5.1 General

Punching can result from a concentrated load applied on a relatively small area of the structure. In flat slabs, punching shear failures normally develop around supporting areas (columns, capitals, walls). In other cases (e.g. foundation slabs, transfer slabs, deck slabs of bridges) punching failures can also develop around loaded areas. The rules presented hereafter for flat slabs apply by analogy to loaded areas.

30.1.5.2 Design shear force, shear-resisting effective depth and control perimeter

30.1.5.2.1 Design shear force

The design shear force with respect to punching (V_{Ed}) is calculated as the sum of design shear forces acting on a basic control perimeter (b_1).

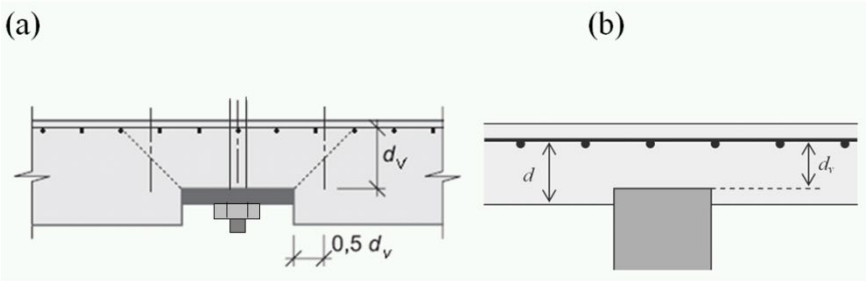


Figure 30.1-20: Effective depth of the slab considering support penetration (d_v) and effective depth for bending calculations (d).

For slabs without planned penetration of the columns into the slab, the reduction of the effective depth (Figure 30.1-20b) shall only be accounted for in cases where the penetration exceeds the tolerance defined in execution standards.

For flat slabs and footings, the design shear force is equal to the value of the support reaction reduced by the actions applied inside the basic control perimeter (such as gravity loads, soil pressure at footings and deviation forces of prestressing tendons).

30.1.5.2.2 Shear-resisting effective depth

The shear-resisting effective depth of the slab (d_v) is the distance from the centroid of the tensile reinforcement layers to the supporting area (Figure 30.1-20).

30.1.5.2.3 Basic control perimeter (b_1)

The basic control perimeter b_1 may normally be taken at a distance $0.5d_v$ from the supporting area (Figure 30.1-21 and Figure 30.1-22) and should be determined in order to minimize its length (Figure 30.1-21c). The length of the control perimeter is limited by slab edges (Figure 30.1-21d).

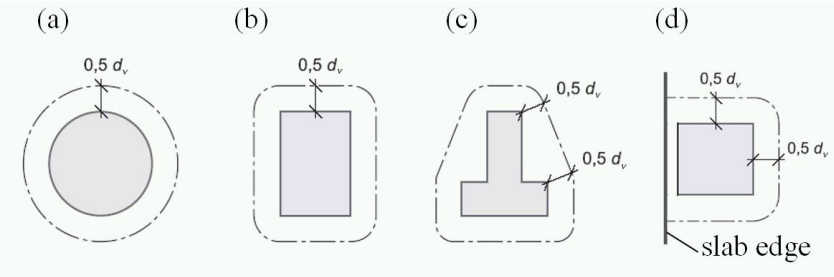


Figure 30.1-21: Basic control perimeters around supporting areas.

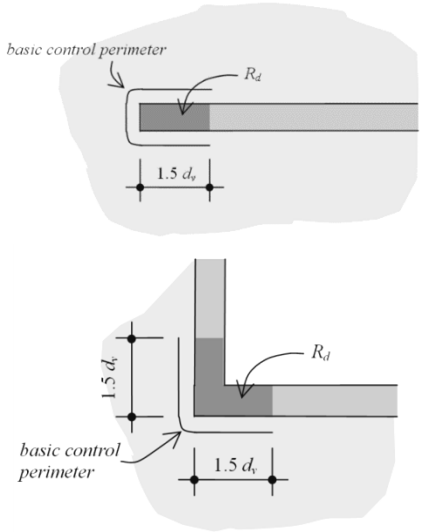


Figure 30.1-22: Basic control perimeter around wall end and wall corners.

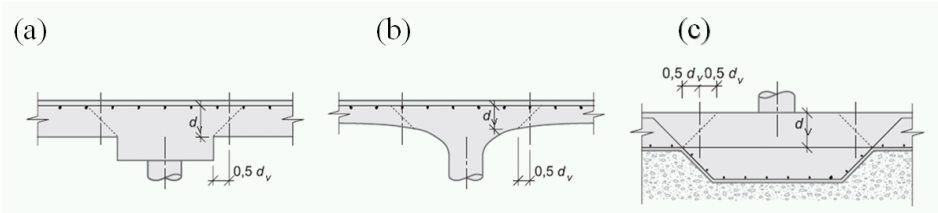


Figure 30.1-23: Choice of the potentially governing control perimeter.

In the case of slabs with variable depth, control sections at a greater distance from the supporting area can be governing (refer to Figure 30.1-23).

The shear-resisting control perimeter b_0 may be obtained on the basis of a detailed shear field analysis as:

$$b_0 = \frac{V_{Ed}}{v_{Ed,ave}} \qquad (30.1\text{-}93)$$

where $v_{Ed,ave}$ is the shear force per unit length averaged over a length of $2d_v$ on both sides of the peak of the shear force in the governing zone.

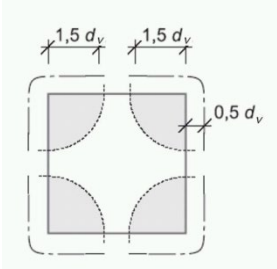
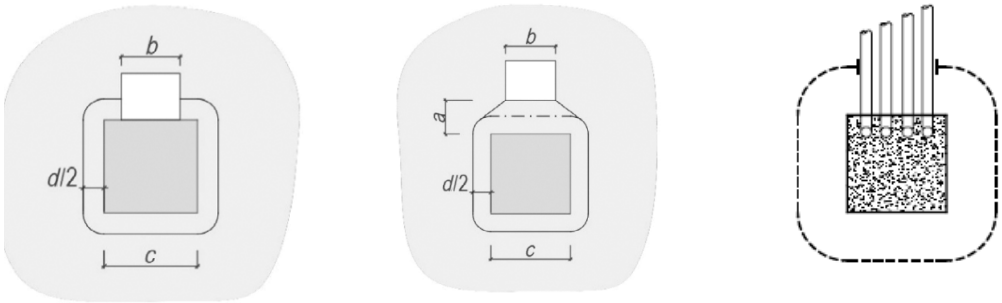


Figure 30.1-25: Reduction of basic control perimeter for large supporting areas ($b_{1,red}$).



(a1) (a2) (b)

Figure 30.1-26: Reduction of basic control perimeter ($b_{1,red}$) in presence of: (a) openings; and (b) pipes or inserts.

For the case of openings, the control perimeter should be chosen to minimize its length (see Figures 30.1-26(a1) and (a2)). For larger distances between the opening and the column, the control perimeter without accounting for the presence of the opening can become governing. For more detailed information, see dos Santon et al. [30.1-21].

30.1.5.2.4 Shear-resisting control perimeter (b_0)

For calculating the punching shear resistance, a shear-resisting control perimeter (b_0) is used. The shear-resisting control perimeter accounts for the non-uniform distribution of shear forces along the basic control perimeter.

A non-uniform distribution of the shear forces can result due to:

1. *Concentrations of the shear forces at the corners of large supporting areas.* This effect may approximately be taken into account by reducing the basic control perimeter ($b_{1,red}$) assuming that the length of its straight segments does not exceed $3d_v$ for each edge (Figure 30.1-25).
2. *Geometrical discontinuities of the slab.* In the presence of openings and inserts, the basic control perimeter ($b_{1,red}$) is to be reduced according to the rules of Figure 30.1-26.
3. *Concentrations of the shear forces due to moment transfer between the slab and the supporting area*. This effect may approximately be taken into account by multiplying the length of the reduced basic control perimeter ($b_{1,red}$) by the coefficient of eccentricity (k_e):

$$b_0 = k_e \cdot b_{1,red} \qquad (30.1\text{-}94)$$

4. *Presence of significant loads near the supporting area*. In cases where significant concentrated loads ($\geq 0.2V_{Ed}$) are applied near the supporting area (closer than $3d_v$ from the edge of the supporting area) the general procedure for calculating b_0 should be used, refer to Eq. (30.1-93).

Cast-in pipes, pipe bundles or slab inserts, where the distance from the supporting area is less than $5d_v$ should be arranged perpendicular to the control perimeter. In these cases, the control perimeter should be reduced in accordance to Figure 30.1-26.

The coefficient of eccentricity may be determined as a function of the moment transferred from the column to the slab as:

$$k_e = \frac{1}{1+\frac{1.1 \cdot e_u}{b_u}} \qquad (30.1\text{-}95)$$

where e_u is the eccentricity of the resultant of shear forces with respect to the centroid of the basic control perimeter – see Figure 30.1-27b – and b_u is the diameter of a circle with the same surface as the region inside the basic control perimeter. For design purposes, the location of the centroid of the basic control perimeter may be calculated by approximating its shape with straight lines, see Figure 30.1-27b.

For slabs with unsymmetric openings near to the column, the resulting eccentricity of the centroid of the control perimeter with respect to the resultant of the shear forces should be accounted for.

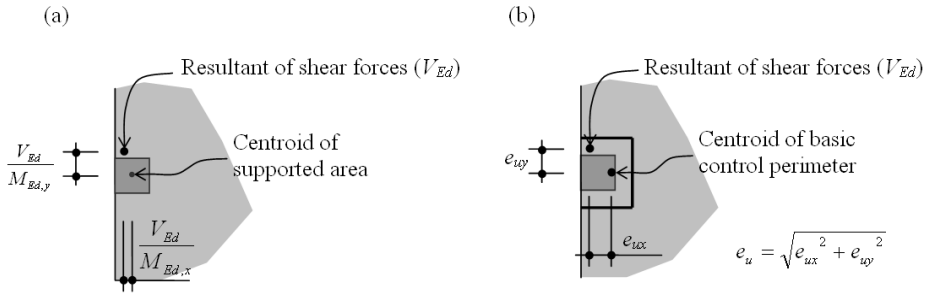


Figure 30.1-27: Resultant of shear forces: (a) location with respect to the centroid of the supporting area; and (b) approximated basic control perimeter for calculation of the position of its centroid and eccentricity between the resultant of shear forces and the centroid of the basic control perimeter.

In cases where the lateral stability does not depend on frame action of slabs and columns and where the adjacent spans do not differ in length by more than 25%, the following approximated values may be adopted for the coefficient k_e:

- 0.90 for internal columns;
- 0.70 for edge columns;
- 0.65 for corner columns;
- 0.75 for corners of walls (horizontal shear resisting members where the rules of Figure 30.1-22 apply).

The calculation of the punching shear strength is based on the critical shear crack theory.

More details on the failure criterion of Eq. (30.1-98) can be found in Muttoni [30.1-22]. It has to be noted that Eq. (30.1-98) refers to the design value (model uncertainties partially accounted for). For the mean value of Eq. (30.1-98), coefficient 1.5 should e replaced by 4/3 and coefficient 0.9 should be replaced by 15/32 (an explanation can be found in Yu et al. [30.1-23]).

The parameter ψ used for calculating the punching shear resistance refers to the rotation of the slab around the supporting area (rotation of the slab at a distance $2d$ form the face of the column with respect to the axis of the column due to the flexural deformation of the supporting strip, Figure 30.1-28). It can be calculated as a function of the average bending moment m_{Ed} over the support strip according to 30.1.5.4. For designing new structures, this can be done directly (as m_{Ed} refers to the moment for design of the flexural reinforcement). For calculating the punching shear resistance of an existing structure, since m_{Ed} at failure is unknown, an iteration is needed (by calculating $m_{Ed} = f(V_{Rd})$ according to 30.1.5.4). For non-axisymmetric cases, considering the maximum value of the rotation in both directions leads to a safe estimate of the punching shear resistance. The favourable effect of the redistribution of internal forces may be accounted for considering the geometric average of both rotations: $\psi = \sqrt{\psi_x \cdot \psi_y}$.

Alternatively, simplified closed-form expressions may be derived analytically from this theory as shown in Muttoni et al. [30.1.-24] allowing for a direct calculation of the punching shear resistance. For the case of symmetric punching, of internal columns with moderate dimensions, the following closed form expression included in FprEN 1992-1-1 [30.1-10]] (in a slightly different form) can be derived:

30.1.5.3 Punching shear strength

The punching shear resistance is calculated as:

$$V_{Rd} = V_{Rd,c} + V_{Rd,s} \geq V_{Ed} \qquad (30.1\text{-}96)$$

The design shear resistance attributed to the concrete may be taken as:

$$V_{Rd,c} = k_\psi \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \qquad (30.1\text{-}97)$$

with f_{ck} in [MPa].

The parameter k_ψ depends on the deformations (rotations ψ) of the slab and follows for the design value, from:

$$k_\varphi = \frac{1}{1.5 + 0.9 k_{dg} \cdot \psi_d} \leq 0.6 \quad \text{(d in [mm])} \qquad (30.1\text{-}98)$$

where d is the mean value [in mm] of the (flexural) effective depth for the x and y directions.

Provided that the size of the maximum aggregate particles, d_g, is not less than 16 mm, k_{dg} in Eq. (30.1-89) may be taken as $k_{dg} = 1.0$.

$$V_{Rd,c} = \frac{0.43}{\gamma_V}\sqrt{\frac{d_v}{b_0}}\left(\rho \cdot E_s \cdot f_{ck} \cdot \frac{d_{dg}}{\sqrt{d_v \cdot r_s}}\right)^{\frac{1}{3}} b_0 \cdot d_v \le 0.50\sqrt{f_{ck}} \cdot b_0 \cdot d_v \qquad (30.1\text{-}99)$$

where γ_V is the partial safety factor for shear and d_{dg} a parameter related to the aggregate size d_g (d_{dg}=32mm/k_{dg}).

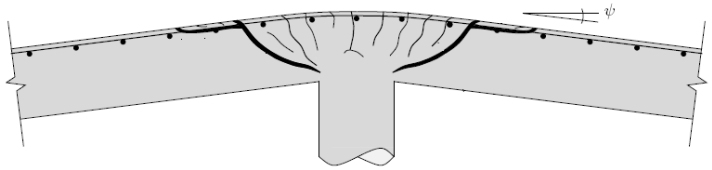


Figure 30.1-28: Rotation (ψ) of a slab.

There is evidence that the punching shear resistance is influenced by the maximum size of the aggregate (d_g). The value of k_{dg} in Eq. (30.1-100) is assessed as:

$$k_{dg} = \frac{32}{16+d_g} \ge 0.75 \qquad (30.1\text{-}100)$$

where d_g is in mm. If the maximum aggregate size, d_g, is not less than 16 mm, k_{dg} in Eq. (30.1-98) may be taken as k_{dg} = 1.0 as a safe estimate. For high strength and lightweight concrete, the aggregate particles may break, resulting in a reduced aggregate interlock contribution. In that case, the value d_g should be assumed to be 0.

The derivation of Eq. (30.1-104) can be found in Fernández et al. [30.1-25]. For inclined shear reinforcement or bent-up bars (Figure 30.1-29), Eq. (30.1101) is replaced by:

$$V_{Rd,s} = \sum A_{sw} k_e \sigma_{swd} \sin\alpha \qquad (30.1\text{-}102)$$

and Eq. (30.1-104) is replaced by:

$$\sigma_{swd} = \frac{E_s\psi}{6}(\sin\alpha + \cos\alpha) \cdot \left(\sin\alpha + \frac{f_{bd}}{f_{ywd}}\frac{d}{\varphi_w}\right) \le f_{ywd} \qquad (30.1\text{-}103)$$

The design shear resistance provided by the stirrups shall be calculated as

$$V_{Rd,s} = \sum A_{sw} k_e \sigma_{swd} \qquad (30.1\text{-}101)$$

Where ΣA_{sw} is the sum of the cross-sectional area of all shear reinforcement suitably anchored, or developed, and intersected by the potential failure surface (conical surface with angle 45°) within the zone bounded by $0.35d_v$ and d_v from the edge of the supporting area (Figure 30.1-29).

The term σ_{swd} refers to the stress that is activated in the shear reinforcement and may be calculated as:

$$\sigma_{swd} = \frac{E_s\psi}{6}\left(1 + \frac{f_{bd}}{f_{ywd}} \cdot \frac{d}{\varphi_w}\right) \le f_{ywd} \qquad (30.1\text{-}104)$$

where ϕ_w denotes the diameter of the shear reinforcement and f_{ywd} is its yield strength. The bond strength (f_{bd}) may be assumed as:

- $f_{bd} = \frac{1.4}{\gamma_C} f_{ck}^{2/3} \le \frac{5.5}{\gamma_C}$ MPa for ribbed and indented bars and

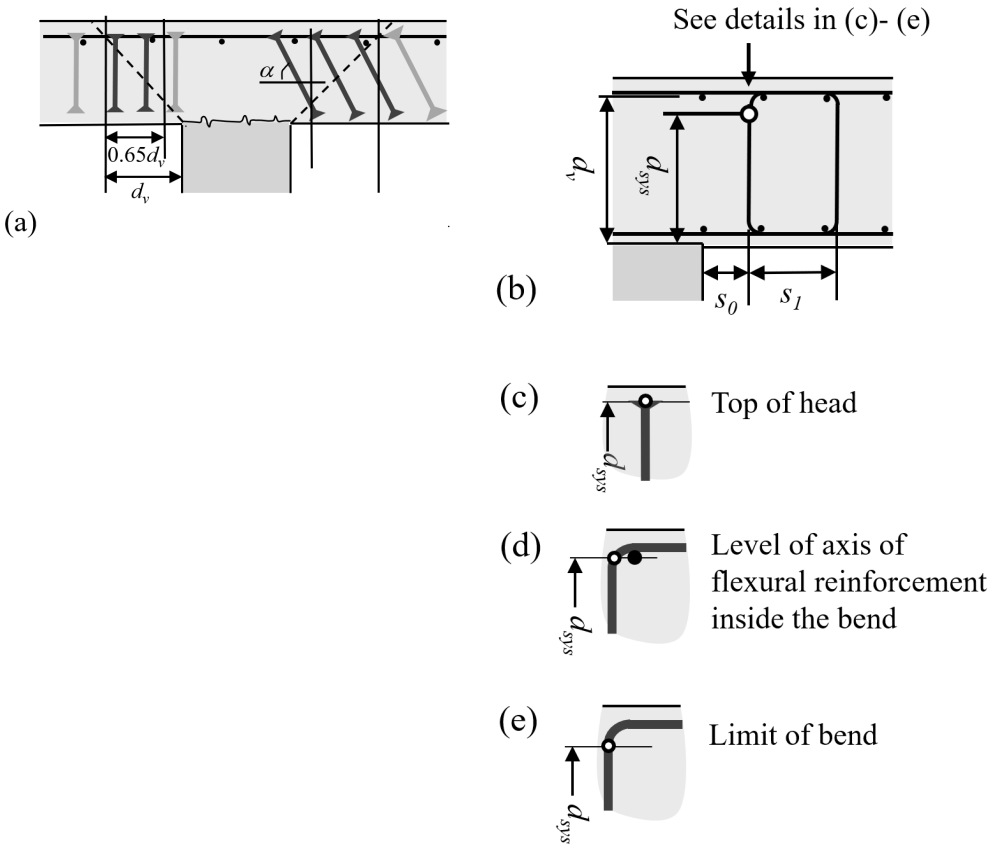


Figure 30.1-29: (a) Shear reinforcement activated at failure; and (b) – (e) definition of parameter d_{sys}..

Parameter k_{sys} depends on:

- The performance of anchorage of the shear reinforcement;
- $f_{bd} = 0$ for plain bars.

In cases where $V_{Rd,c}$ is neglected, σ_{swd} shall be assumed as f_{ywd}.

In order to ensure sufficient deformation capacity, in slabs with punching shear reinforcement a minimum amount of punching shear reinforcement is required such that:

$$\sum A_{sw} k_e f_{ywd} \geq 0.5 V_{Ed} \qquad (30.1\text{-}105)$$

The maximum punching shear resistance is limited by crushing of the concrete struts in the supporting area:

- The horizontal distance between the support edge and the upper anchorage of the shear reinforcement compared to the effective depth (distance s_0 in Figure 30.13-29b divided by d_v);
- The length of the control perimeter compared to the effective depth (ratio b_o/d_v).

Alternative values for the coefficient k_{sys} may be used provided that they are experimentally verified. For shear reinforcements not complying with the rules of section 13.13.6.3, lower values of k_{sys} shall be considered. For these cases, or for cases with more stringent detailing rules, k_{sys} may be determined on the basis of Eq. (30.1-107):

$$1.0 \le k_{swd} = 3 \cdot \left(\frac{d_{sys}}{d_v}\right)^{3/2} \cdot \left(\frac{b_{0,0}}{d_v}\right)^{1/2} \frac{1}{1+5\frac{s_{0,t}}{d_v}} \le 4.0 \qquad (30.1107)$$

where d_{sys} is defined in Figure 30.1-29(b)-(e), $b_{0,0}$ is the control perimeter located at the column edge and following the rules of b_0, and

$$s_{0,t} = 0.8 \cdot s_0 + 0.2 \cdot s_t \ge s_0 \qquad (30.1\text{-}108)$$

where s_t is the spacing of the shear reinforcement units in tangential direction (for further information, see Hernández et al. [30.1-26]).

Level I approximation is intended to be a safe estimate for a practical verification of punching shear resistance assuming that yielding of flexural reinforcement develops over the full width of the supporting strip (local flexural mechanism) and is consistent with Level II assuming m_{Ed}=m_{Rd}. Level I approximation should not be used for a comparison to test results to check model accuracy, unless the experiments were designed to fail simultaneously in punching and flexure.

Slabs calculated under this assumption comply with deformation capacity requirements stated in section 30.1.5.1.

The value of r_s may be approximated as 0.22 l_x or 0.22 l_y for the x and y directions, respectively, for regular flat slabs where the ratio of the spans (l_x/l_y) is between 0.5 and

$$V_{Rd,\max} = k_{sys} k_\psi \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \le \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \qquad (30.1\text{-}106)$$

The coefficient k_{sys} accounts for the performance of punching shear reinforcing systems to control shear cracking and to suitably confine compression struts at the soffit of the slab. It may be assumed as:

– k_{sys} = 2.4 for stirrups and bent-up bars with sufficient development length at the compression face of the slab and bent (no anchorages or development length) at the tension face;

– k_{sys} = 2.8 for studs (diameter of heads larger or equal than three times the bar diameter).

provided that reinforcement is detailed as per the provisions of section 30.13.6.3.

30.1.5.4 Calculation of rotations around the supporting area

30.1.5.4.1 Level I approximation

For a regular flat slab designed according to an elastic analysis without significant redistribution of internal forces, a safe estimate of the rotation at failure is:

$$\psi = 1.5 \cdot \frac{r_s}{d} \frac{f_{yd}}{E_s} \qquad (30.1\text{-}109)$$

where r_s denotes the position where the radial bending moment is zero with respect to the support axis.

2.0. In level I approximation, the maximum value of r_s in x – and y direction should be considered in Eq. (30.1109).

The derivation of Eqs. 30.1-109 and 110 can be found in Muttoni et al [30.1-20].

The average bending moment acting in the support strip (m_{Ed}) may be approximated for each reinforcement direction and support type as:

– for inner columns (top reinforcement in each direction):

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{2 \cdot b_s}\right) \qquad (30.1\text{-}111)$$

– for edge columns:

when calculations are made considering the tension reinforcement parallel to the edge:

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{2 \cdot b_s}\right) \geq \frac{V_{Ed}}{4} \qquad (30.1\text{-}112)$$

when calculations are made considering the tension reinforcement perpendicular to the edge:

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{b_s}\right) \qquad (30.1\text{-}113)$$

– for corner columns (tension reinforcement in each direction):

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{b_s}\right) \geq \frac{V_{Ed}}{2} \qquad (30.1\text{-}114)$$

In these equations, the term e_{ui} refers to the eccentricity of the resultant of shear forces with respect to the centroid of the basic control perimeter in the direction investigated ($i = x$ and y for x and y directions respectively, see Figure 30.1-27).

30.1.5.4.2 Level II approximation

In cases where significant bending moment redistribution is considered in the design, the slab rotation may be calculated as:

$$\psi = 1.5 \cdot \frac{r_s}{d} \frac{f_{yd}}{E_s} \cdot \left(\frac{m_{Ed}}{m_{Rd}}\right)^{1.5} \qquad (30.1\text{-}110)$$

where:

m_{Ed} is the average moment per unit length for calculation of the flexural reinforcement in the support strip (for the considered direction);

m_{Rd} is the design average flexural strength per unit length in the support strip (for the considered direction).

The rotation has to be calculated along the two main directions of the reinforcement.

The width of the support strip for calculating m_{Ed} is:

$$b_s = 1.5 \cdot \sqrt{r_{s,x} \cdot r_{s,y}} \leq l_{min} \qquad (30.1\text{-}115)$$

where close to slab edges, the width of the strip is limited to b_{sr} according to Figure 30.1-30. The same value for r_s as that for Level I approximation may be adopted.

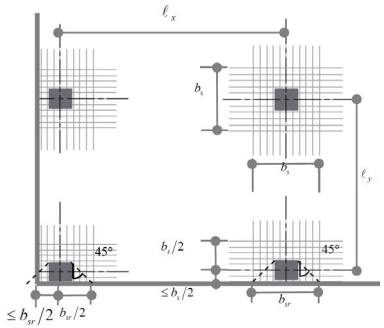


Figure 30.1-30: Support strip dimensions.

Slabs calculated under this assumption do not comply with deformation capacity requirements stated in section 30.1.5.1. Therefore, they need to be provided with integrity reinforcement.

The design average flexural strength per unit length in the support strip is to be calculated accounting for both ordinary and prestressing steel at yielding.

The beneficial effects of compressive membrane action and redistribution of internal forces in continuous slabs may be considered for internal columns by reducing the rotation ψ by coefficient

$$k_{\psi,cma} = 1 - \frac{2}{\gamma_C}\frac{m_{cr}}{m_{Rd}} \geq 1.0 \qquad (30.1\text{-}117)$$

where m_{cr} is the cracking moment of the slab calculated with concrete tensile strength f_{ctm} This beneficial effect may be considered provided that there are no openings in the slab in a distance $\leq 5d_v$ from the control perimeter.

Future details van be found in Einpaul et al. [30.1-27]

Level III approximation is recommended for irregular slabs or for flat slabs where the ratio of the span lengths (l_x/l_y) is not between 0.5 and 2.0.

Parameter m_{Ed} has to be calculated consistently with the method used for determining the flexural reinforcement and is to be determined at the edge of the supporting area maximizing m_{Ed}, see Figure 30.1-31.

Eq. (30.1-110) also applies to slabs with a flexural reinforcement that is increased over the supporting areas in order to increase their punching shear strength.

For prestressed slabs, Eq. (30.1-110) may be replaced by:

$$\psi = 1.5 \cdot \frac{r_s}{d}\frac{f_{yd}}{E_s} \cdot \left(\frac{m_{Ed}-m_{Pd}}{m_{Rd}-m_{Pd}}\right)^{1.5} \qquad (30.1\text{-}116)$$

where m_{Pd} denotes the average decompression moment over the width of the support strip (b_s) due to prestressing. Constrained forces and moments and losses due to shrinkage, creep and relaxation shall be taken into account.

30.1.5.4.3 Level III approximation

The coefficient 1.5 in Eqs. (30.1-113) and (30.1-114) may be replaced by 1.2 if:

- r_s is calculated using a linear elastic (uncracked) model;
- m_{Ed} is calculated from a linear elastic (uncracked) model as the average value of the moment for design of the flexural reinforcement over the width of the support strip (b_s).

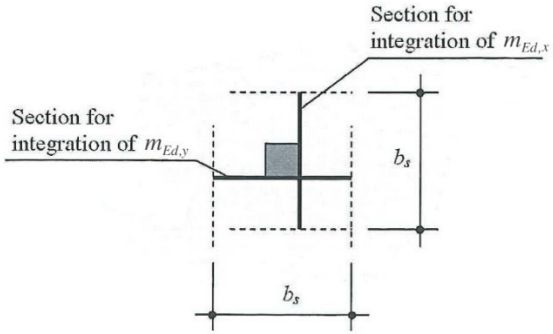


Figure 30.1-31: Example of sections for integration of support strip moments.

Analytical or numerical techniques (e.g. finite elements, finite differences etc.) may be used for level IV approximation. This is particularly relevant in cases where significant compressive membrane action can develop around columns and significant moment redistribution due to reinforcement yielding develop. Examples of this approach can be found in: Belletti et al. [30.1-28], [30.1-29], [30.1-30], Shu et al. [30.1-31], Abu-Salma et al. [30.1-32], [30.1-33], Setiawan et al. [30.1-34], [30.1-35].

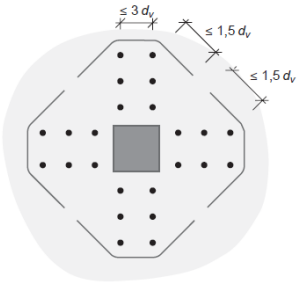


Figure 30.1-32: Reduced control perimeter and shear-resisting effective depth.

The width of the support strip may be calculated as in level II approximation taking $r_{s,x}$ and $r_{s,y}$ as the maximum value in the direction investigated. For edge or corner columns, the following minimum value of r_s has to be considered:

$$r_s \geq 0.67 b_{sr} \qquad (30.1\text{-}118)$$

30.1.5.4.4 Level IV approximation

The rotation ψ may be calculated on the basis of a non-linear analysis of the structure and accounting for cracking, tension-stiffening effects, yielding of the reinforcement and any other non-linear effects relevant for providing an accurate assessment of the structure. The governing value of ψ is the maximum relative rotation between the centre of the supporting area and a distance $2d_v$ from the control perimeter.

30.1.5.5 Punching shear resistance outside the zones with shear reinforcement or shearheads

The extent of the slab with shear-reinforcement may be determined by checking the resistance of the slab outside this region. Section 30.1.5.3 applies by accounting for a control perimeter with a maximum effective distance between two shear reinforcing elements of $3d_v$ (Figure 30.1-32).

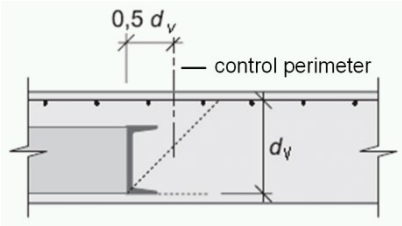
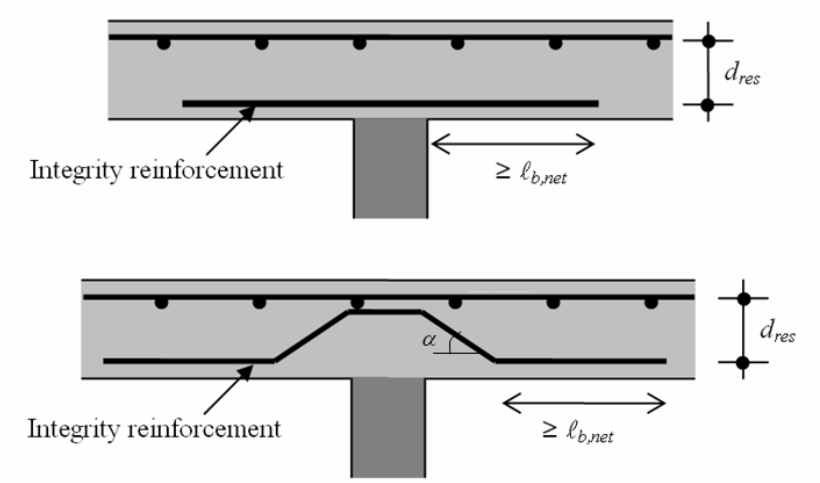


Figure 30.1-33: Shear-resisting effective depth and control perimeter accounting for shearhead penetration.

The design shear for calculation of the integrity reinforcement may be calculated on the basis of an accidental situation where progressive collapse has to be avoided.

(a)

(b)



The punching shear resistance of a slab outside of the shearhead is calculated on the basis of section 30.1.5.3 considering the shearhead as a rigidly supporting area. The shear-resisting effective depth shall account for the position of the shearhead in the slab as shown in Figure 30.1-33.

30.1.5.6 Integrity reinforcement

Slabs without shear reinforcement, or with insufficient deformation capacity, shall be provided with integrity reinforcement (Figure 30.1-34) to avoid progressive collapse.

The resistance provided after punching by the integrity reinforcement may be calculated as:

$$V_{Rd,int} = \sum A_s f_{yd} \left(f_t/f_y\right)_k \sin \alpha_{ult} \leq \frac{0.5\sqrt{f_{ck}}}{\gamma_c} d_{res} b_{int} \qquad (30.1\text{-}119)$$

where:

A_s refers to the sum of the cross-sections of all reinforcement suitably developed beyond the supporting area on the compression side of the slab or to well-anchored bent-up bars;

f_{yd} is the design yield strength of the integrity bars;

the ratio $(f_t/f_y)_k$ and parameter ε_{uk} are defined in subsection 15.5.1 and depend on the ductility class of the reinforcement;

α_{ult} is the angle of the integrity bar with respect to the slab plane at failure (after development of plastic deformations in the post-punching regime):

α_{ult}	*Type of integrity reinforcement*
0°	Straight bars, class of ductility: A
20°	Straight bars, class of ductility: B
25°	Straight bars, class of ductility: C or D
$\alpha \leq 40°$	Inclined or bent-up bars, class of ductility: B, C or D

(c)

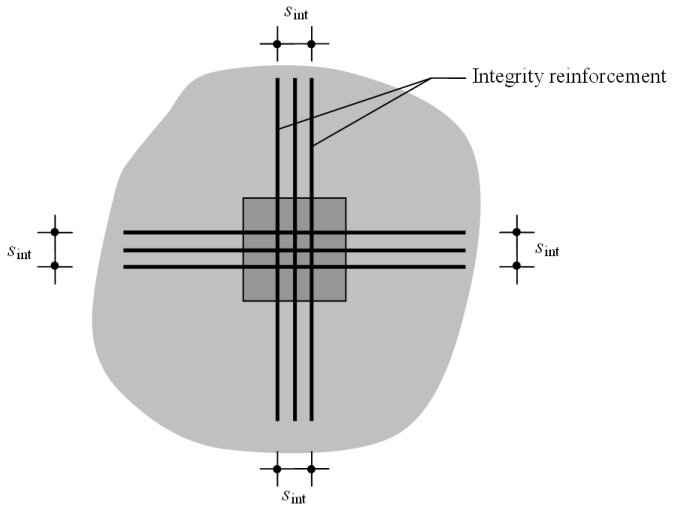


Figure 30.1-34: Integrity reinforcement: (a) straight bars; (b) bent-up.

The integrity reinforcement should at least be composed of four bars placed over the supporting area and correctly developed on the compression side of the slab. Post-tensioning tendons may also be considered as integrity reinforcement.

In order to allow full activation of the integrity bars, the diameter of the integrity bars (ϕ_{int}) has to be chosen such that: $\phi_{int} \leq 0.12\, d_{res}$.

Background information to this subsection can be found in Fernández et al. [30.1-36].

A more refined calculation of term $V_{Rd,F}$ may be conducted on the basis of a suitable evaluation of the average residual stresses of the fibres bridging the critical shear crack.

where α is the angle of the integrity bars with respect to the slab plane (before punching occurs), Figure 27.1-34;

d_{res} is the distance between the centroid of the flexural reinforcement ratio and the centroid of the integrity reinforcement, see Figure 30.1-34(a) and (b);

b_{int} is the control perimeter activated by the integrity reinforcement after punching. It may be calculated as:

$$b_{int} = \sum(s_{int} + \frac{\pi}{2} d_{res}) \qquad (30.1\text{-}120)$$

where the summation refers to the groups of bars activated at the edge of the supporting area and s_{int} is equal to the width of the group of bars (refer to Figure 30.1-34).

30.1.5.7 Punching of slabs in FRC

For FRC members, Eq. (30.1-96) can be replaced by:

$$V_{Rd} = V_{Rd,F} + V_{Rd,s} \qquad (30.1\text{-}121)$$

where $V_{Rd,F}$ represents the fibre reinforced concrete contribution to shear.

The design shear resistance attributed to the fibres may be taken as:

$$V_{Rd,F} = V_{Rd,c} + V_{Rd,f} \qquad (30.1\text{-}122)$$

where:

For that purpose, for slabs with longitudinal reinforcement, the ultimate crack opening $w_u = \psi_d/6$ is suggested, where ψ_d is calculated from section 30.1.5.4.

A detailed description of this approach can be found in Maya et al. [30.1-37] and Minelli et al. [30.1-38].

The background to this subsection can be found in *fib* Bulletin 100 [30.1-39].

$$V_{Rd,f} = \frac{f_{Ftuk}}{\gamma_F} b_0 d_v \quad (30.1\text{-}123)$$

f_{Ftuk} is the characteristic value of the ultimate residual tensile strength for FRC, calculated assuming w_u = 1.5 mm [MPa];

When reinforcement is needed for punching resistance, a minimum amount of fibres (and if applicable transverse reinforcement) is required in order to ensure sufficient deformation capacity at failure as follows:

$$V_{Rd,s} + V_{Rd,f} \geq 0.5 \cdot V_{Ed} \quad (30.1\text{-}124)$$

30.1.5.8 Punching of slabs with FRP reinforcement

For members requiring no punching shear reinforcement, the longitudinal FRP reinforcement can be considered similar to steel reinforcement, provided that the area of longitudinal FRP reinforcement is reduced by the ratio E_f/E_s in all equations.

For FRP punching shear systems, the relevant information shall be given through design by testing.

30.1.6 Design with stress fields and strut-and-tie models

30.1.6.1 General

Stress fields and strut-and-tie models represent the transmission of forces within a structural member. They are composed of the following elements (Figure 30.1-35):

- nodal regions or nodes: bi- or tri-axially stressed zones;
- compression fields or struts: in the form of constant-angle compression fields (parallel uniaxial compression stresses with uniform stress intensity) or fans (fan-shaped stress field of uniaxial compression with variable stress intensity);
- tension fields or ties: composed of distributed or concentrated reinforcement;
- chords: tension or compression linear elements with potentially variable level of force along their axis.

Stress fields (Figure 30.1-35 top) aim at representing a stress state in equilibrium with the external actions ensuring the fulfilment of the strength conditions of the

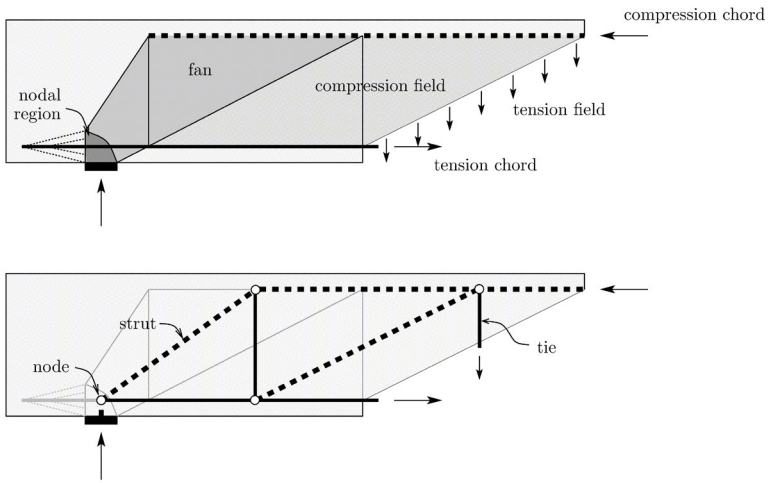


Figure 30.1-35: Basic elements for stress fields (top) and strut-and-tie modelling (bottom) for the end region of a girder.

Structures can be subdivided into B-regions, where the assumption of plane sections remain plane may be used (B for Beam/Bernoulli) and D-regions, where a non-linear strain distribution exists (D for Discontinuity/Detail, typically located at geometric discontinuities or near concentrated loads). B-regions can be designed according to sections 30.1.2-3 or with the provisions of this section. For D-regions, unless other suitable approaches are considered, they should be designed according to the provisions of this section.

Strut-and-tie models and stress fields may be used in a combined manner. Also, specific approaches based on them may be used, as the Stringer Panel Method (based, in its simplest form, on the arrangement of concentrated chords at the boundaries of a panel with constant compression and tension fields).

Stress fields and strut-and-tie models are based on the lower bound theorem of limit analysis, therefore sufficient deformation capacity is assumed. The applicability of these modelling approaches in case of a potentially brittle response shall be verified by means of a detailed analysis considering the degree of plastic redistribution required at ultimate limit state. Alternatively, comparison to suitable experimental evidence can be used to ensure the applicability of stress fields and strut-and-tie models to specific cases.

materials. The strut-and-tie models (Figure 30.1-35 bottom) depict the resultant of these forces.

The forces within a stress field or strut-and-tie model can be calculated with help of equilibrium conditions and the elements shall be verified by using sections 30.1.6.2-4 to ensure the fulfilment of strength conditions.

The clauses in this section are applicable provided that a minimum reinforcement amount according to Section 30.13.6.2 is arranged or that control of cracking is ensured by other means.

Strut-and-tie models and stress fields cannot be used for low-ductility class reinforcement (class A according to Section15.5.1 or, when phenomena like corrosion limit the ductility of the reinforcement to comparable values) unless a detailed verification of the strains in the reinforcement is performed.

30.1.6.2 Struts and compression fields

30.1.6.2.1 Introduction

The stress (σ_{cd}) in a compression field or strut developing within the concrete shall respect the following condition:

$$\sigma_{cd} \leq k_{\varepsilon} \cdot f_{cd} \qquad (30.1\text{-}125)$$

where k_{ε} is a strength factor accounting for the influence of the transverse strain state. This factor can be evaluated following a Levels-of-Approximation approach:

Due to generally lower stiffness and higher strengths of FRP reinforcement, the effectiveness factors for the concrete struts may need modification for structures with FRP to ensure a reasonably accurate evaluation of strength is made with some ability for a ductile response of a structure. The relevant information is provided in specialized literature.

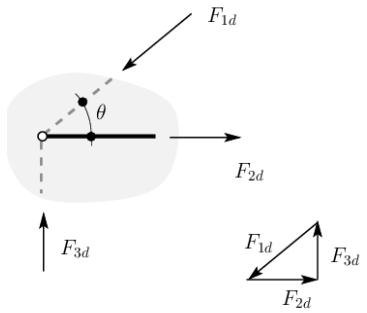


Figure 30.1-36: Angle θ between a strut and a tie.

The strength reduction factor k_ε accounts for the state of strain in the webs of beams or the core layers of slabs. An additional strength reduction factor accounting for the brittleness of concrete is already included in f_{cd} since it applies also to other cases.

The values proposed for Level II Approximation can be observed to be consistent with the shear design method for beams in section 30.1.3. In that case, the k_ε factor can be calculated consistently with the provisions of section 30.1.3 by assuming the longitudinal reinforcement at the onset of yielding ($\varepsilon_s \approx 0.002$):

$$k_\varepsilon = \frac{1}{1.0+110\cdot(\varepsilon_x+(\varepsilon_x+0.001)\,cot^2\,\theta)} \qquad (30.1\text{-}126)$$

The capacity of a compression field developing through cold joints shall comply with the provisions of section 30.1.3.8.

30.1.6.2.2 Level I approximation

A safe estimate $k_\varepsilon = 0.5$ can be assumed provided that the angle θ between the strut (resultant of compression field) and the tie is higher or equal than 25° (see Figure 30.1-36).

30.1.6.2.3 Level II approximation

A more refined estimate of the value of k_ε can be adopted depending on the value of the angle between the strut and tie as provided below:

- for compression fields and struts crossed or deviated by a tension tie with an angle:
  - $20° \le \theta < 25°$ $k_\varepsilon = 0.40$
  - $25° \le \theta < 35°$ $k_\varepsilon = 0.50$
  - $35° \le \theta < 45°$ $k_\varepsilon = 0.65$
  - $45° \le \theta$ $k_\varepsilon = 0.75$
- for compression fields and struts in a region with no imposed transverse tensile strain $k_\varepsilon = 1.0$

For verification of regions where significant plastic deformations or crack widths are expected (e.g. plastic hinges), a value $k_\varepsilon \leq 0.4$ shall be adopted. In these cases, a detailed evaluation by means of Level III approximation is advised.

For calculation of the principal strain value (ε_1), the concrete can be assumed to have a cracked response in tension and a bilinear (or more refined) response in compression.

30.1.6.2.4 Level III approximation

When the compression field develops in presence of principal transverse tensile strains ($\varepsilon_1 > 0$), the value of k_ε can be calculated at a given location in a detailed manner according to the following expression:

$$k_\varepsilon = \frac{1}{0.8+170\cdot\varepsilon_1} \leq 1 \qquad (30.1\text{-}127)$$

This expression applies provided that the state of strains and associated stresses can be assumed to be almost uniform (local analysis).

For confined concrete, values of k_ε consistent with Section [SECT_CONF.RESPONSE] may be adopted.

30.1.6.3 Ties and tension fields

The stress in the tension elements shall be lower or equal than its design strength.

For tension fields or ties composed of ordinary reinforcing steel, the design strength may be calculated as:

$$f_{yd} = \frac{f_{yk}}{\gamma_s} \qquad (30.1\text{-}128)$$

For prestressing steel, when considered as a state of initial strains in the tendon, its design strength may be calculated as:

$$f_{pd} = \frac{f_{p0.1k}}{\gamma_s} \qquad (30.1\text{-}129)$$

Otherwise (when prestressing is considered as a set of equivalent forces), the stress in the prestressing steel shall be limited to:

$$\Delta\sigma_P = f_{pd} - \sigma_P \qquad (30.1\text{-}130)$$

Where σ_P is the prestressing stress at the time of verification (accounting for any potential loss).

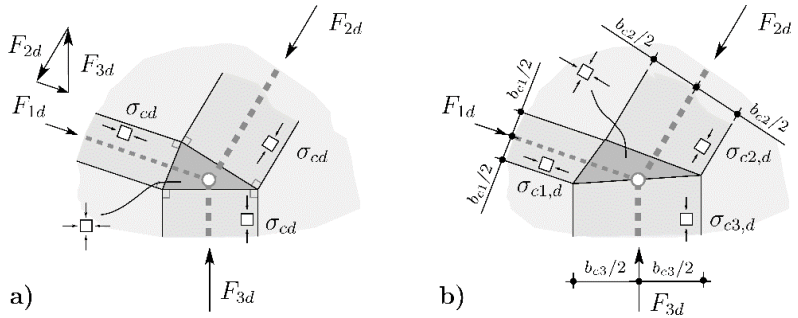


Figure 30.1-37: CCC node.

In case the converging struts have different stress levels (for instance due to different values of coefficient k_ε in the struts), spreading of the struts is allowed in order to ensure a hydrostatic in-plane stress state. Sufficient reinforcement has in this case to be provided.

30.1.6.4 Nodes

30.1.6.4.1 Introduction

Nodes ensure the transfer of forces amongst different concurring struts and ties. For nodes with three concurrent struts and ties, four cases can be distinguished:

- CCC nodes, where only struts reach the node
- CCT nodes, where only one tie is present
- CTT nodes, where only one strut is present
- TTT nodes, where only ties reach the node.

TTT nodes shall in principle be avoided unless specific considerations are performed on the value of coefficient k_ε and anchorage conditions

30.1.6.4.2 CCC Nodes

No check of the stress state within the nodal region is required as long as the struts comply with the stress limits in 30.1.6.2. The size of the nodal region can be determined assuming a maximum value $k_\varepsilon = 1$.

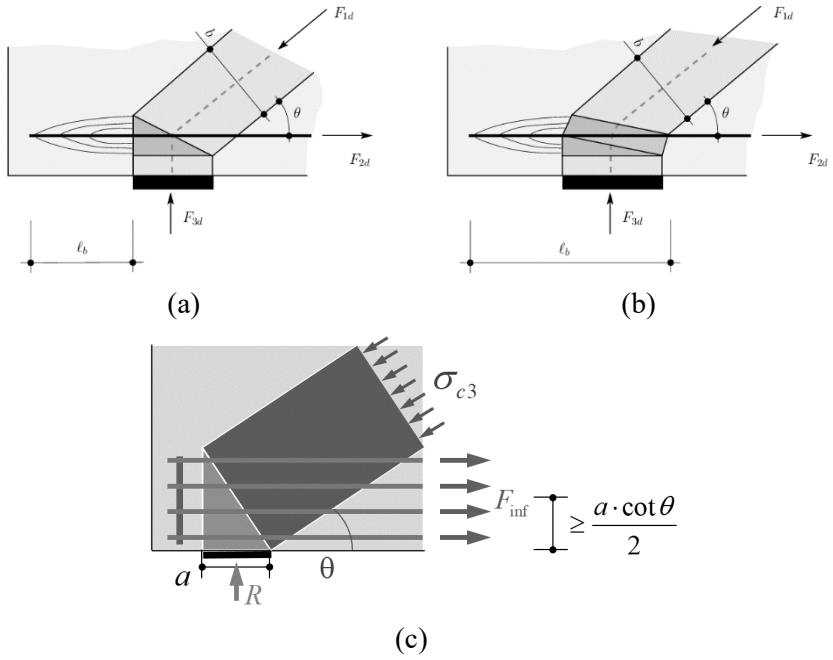


Figure 30.1-38: CCT node: (a) anchorage outside of the nodal region; (b) partial anchorage within the nodal region; and (c) anchorage for several layers of reinforcement.

Similar considerations on spreading of struts and use of fan-shaped compression fields as for nodes CCC can be performed. Arrangement of stirrups or transverse reinforcement in the anchorage length of the tension tie outside of the nodal region is required.

30.1.6.4.3 CCT Nodes

No checks are required if all concurring struts have the same stress level. This can be assumed when anchorage lengths are provided outside of the nodal regions (Figure 30.1-38a) and the tie is arranged with a sufficient cover ($c > a \cdot \cot\theta/2$, see Figure 30.1-38c). In this case, the size of the nodal region can be determined assuming a maximum value $k_\varepsilon = 1$. A smaller height of the nodal zone is also allowed provided that the stress state in the nodal region is verified

Full or partial anchorage of the reinforcement within the nodal region is also allowed (Figure 30.1-38b). In all cases, the strength reduction factors k_ε of the concurring struts outside of the nodal region are to be adopted according to 30.1.6.2 and the stress state of the nodal region may require to be checked.

In both cases, the anchorage length l_b defined in Figure 30.1-38 a-b should be verified according to section 20.

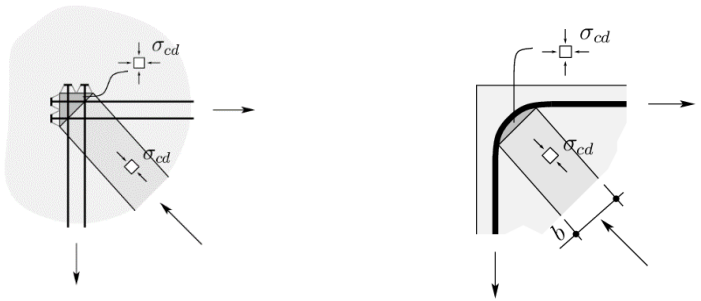


Figure 30.1-39: CTT node: (a) anchorage outside of the nodal region; and (b) bent bar.

Spreading of concentrated loads shall in principle be performed in accordance to a stress field consistent with the transfer of loads within the member. Guidance for the spreading of loads following the St-Venant principle (Figure 30.1-40a-b) may be a suitable approach.

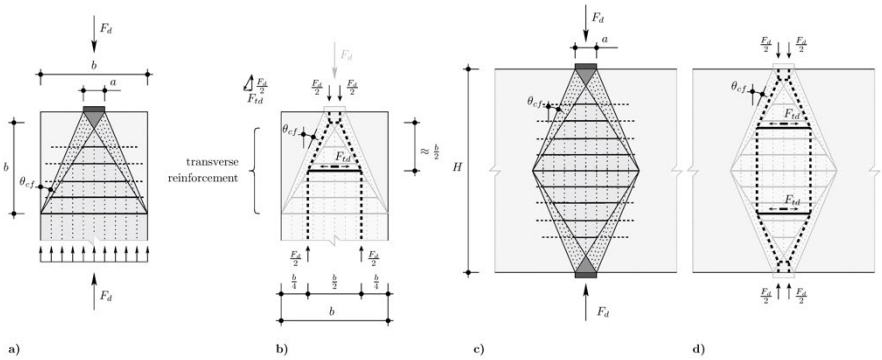


Figure 30.1-40: Spreading of concentrated forces, (a,b) stress field and force resultants for a narrow element; and (c,d) for a wide element.

30.1.6.4.4 CTT Nodes

No check on the stress state of the node is required if the anchorage lengths are provided outside of the nodal regions (Figure 30.1-39a). In this case, the size of the nodal region can be determined assuming a value of k_ε in accordance to 30.1.6.2.

If bent bars are used (Figure 30.1-39b) or if the strut is deviated along the anchorage length of the bar, the bent radius of the reinforcement is governing for the size of the nodal region (see section 30.13.3.4.1).

Minimum reinforcement shall be arranged unless control of cracking in the strut can be ensured by other means.

30.1.6.5 Introduction of concentrated loads

30.1.6.5.1 Spreading of concentrated loads in planar members

Spreading a compression force within a wide element requires arranging a reinforcement transversally to the direction of the force (see Figure 30.1-40):

$$F_{td} = \frac{F_d}{2} \tan \alpha \qquad (30.1\text{-}131)$$

where the spreading angle may be assumed as:

$$\tan \alpha = \frac{1 - a/b}{2} \qquad (30.1\text{-}132)$$

Other models for design of the reinforcement are also possible provided that the spreading angles and corresponding reinforcement layout are derived from an alternative valid stress field.

For the case shown in Figure 30.1-40c-d ($b > a + H/2$), the ratio a/b in Eq. (30.1-132) may be assumed equal to 0.

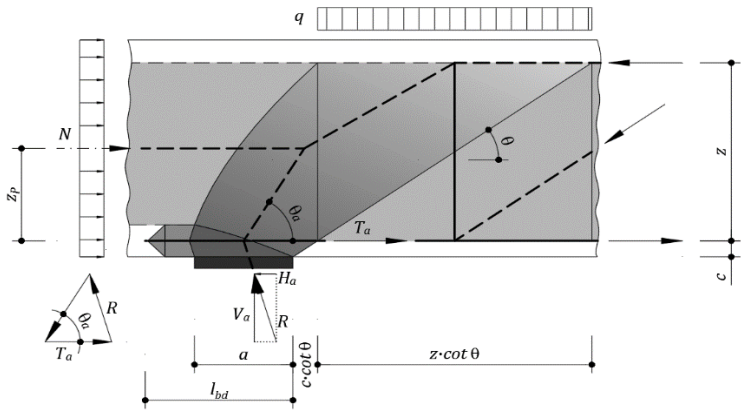


Figure 30.1-41: Possible stress field model for the free-body of a support region accounting for the different actions and anchorage conditions.

In tension, spreading of a concentrated tie may be performed to lead to suitable serviceability response.

30.1.6.5.2 Support regions

In this section, detailed provisions are given for design of support regions for beams subjected predominantly to distributed loads. Specific design clauses (see section 30.1.6.3) or tailored strut-and-tie models may be developed when concentrated loads are applied near a support.

Design of support regions may be performed on the basis of a free-body where a suitable stress field or strut-and-tie model is developed accounting for the support reaction and acting loads in the free-body as well as for the internal forces of the member at its boundaries.

For design of support regions, the resistance is usually governed by the check of the CCT node at the support plate. In the case of end supports of reinforced concrete beams, the tension force that needs to be anchored at the support region (Figure 30.1-41) may be calculated as:

$$T_{a,d} = V_{a,d} \cdot \frac{a+(z+2c)\cdot\cot\theta}{2\cdot z} + N_d \cdot \left(1 - \frac{z_P}{z}\right) + H_{a,d} \qquad (30.1\text{-}133)$$

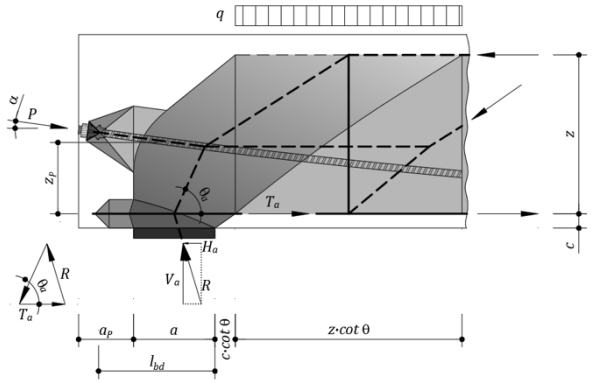
With N_d(+) positive for tension and the axial force can be assumed to be distributed over the section.

A support region with post-tensioning where the anchorage plate is located at sufficient distance to provide linear stress distribution across the section may be considered as a conventional support region (Figure 30.1-41). In this case, prestressing can be considered as an external compression force (accounting for prestressing losses N_d=P_∞ in Eq. (30.1-133).

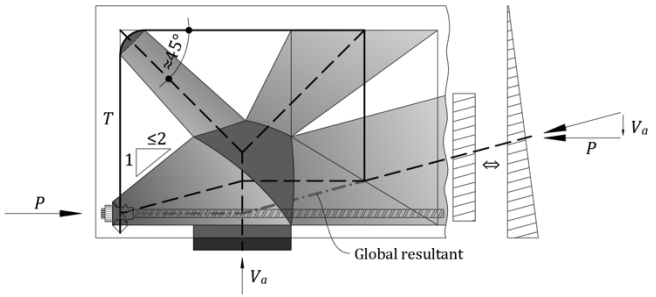
If the location of the anchorage plate is too close to the support, only a fraction of the prestressing force may be considered and the corresponding stress field or strut-and-tie model shall account for the location of the prestressing plate. This applies particularly to cases with post-tensioned tendons without eccentricity (Figure 30.1-42a), with eccentricity (Figure 30.1-42b) or for pre-tensioning (Figure 30.1-42c). In these cases, specific strut-and-tie models shall be developed by considering the level of prestressing and loading, the prestressing eccentricity, the length of overhang, type of prestressing (pre- or post-tensioning), the full or partial deviation of the compression strut carrying shear over the support and others.

Additional reinforcement or stirrups are required for spreading of the concentrated forces (Figure 30.1-42a) or due to the eccentricities of the tendons (Figure 30.1-42b). For pre-tensioning tendons, the vertical reinforcement required shall be spread over the anchorage distance (Figure 30.1-42c).

(a)



(b)



(c)

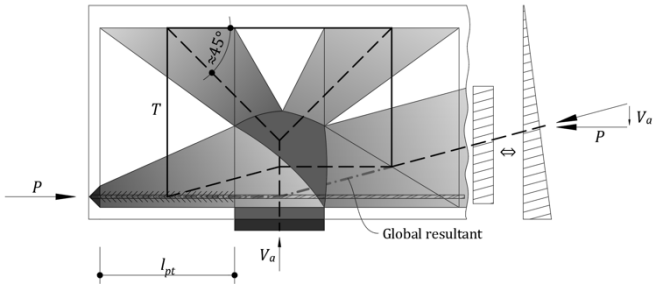


Figure 30.1-42: Example of detailing of prestressed support regions: (a) anchorage plate without eccentricity; (b) anchorage plate with eccentricity; and (c) eccentric bonded wires.

When the anchorage plate is located at a distance from the support lower than $2z_P$ (refer to a_P and z_P in Figure 30.1-42a), partial deviation of the compression strut is typically governing. In this case, in absence of a detailed analysis, the tension force in the tie according to Eq. (30.1-133) may be calculated in a simplified manner assuming $N_d=P_\infty\cdot\cos\alpha\cdot(a_P+a)/(2z_P+a)$ with $-a\leq a_P\leq 2z_P$ and by replacing $V_{a,d}$ by $V_{a,d}-P_\infty\sin\alpha$.

30.1.6.6 Design, assessment and strengthening

30.1.6.6.1 Design and assessment

Strut-and-tie and stress fields may be used both for design of new structures and for the assessment of existing structures provided that the conditions of applicability defined in 30.1.6.1 are fulfilled. Following a Levels-of-Approximation approach is highly recommended.

The use of strut-and-tie and stress field models are based on the lower-bound theorem of limit analysis and allow for safe design of new structures. When the stress field or strut-and-tie model is compatible with a kinematically-admissible mechanism (satisfying the upper-bound theorem of limit analysis), the calculated failure load corresponds to its theoretical maximum (exact solution according to limit analysis), which is relevant in many cases for assessing the strength of an existing structure.

For design, in many cases, good guidance (including serviceability response) for a suitable arrangement of strut-and-ties and stress fields can be obtained on the basis of the elastic-uncracked response of an element, or by analysis of the location of tension and compression zones by kinematical compatibility analysis.

The strut-and-tie method and stress field can be applied with different levels of refinement depending on the necessity to determine a safe and fast design or a refined and precise calculation of an existing structure. This can be performed for instance by following the Levels-of-Approximation approach, starting from simple solutions and only refining them (with higher-order LoA) upon necessity:

- Level I approximation (preliminary design): sketch of an equilibrium solution based on a simple strut-and-tie model allowing to determine the forces in the main ties and to verify the angles between struts and ties. Compatibility of deformations are not explicitly verified and can be considered by means of simplified considerations (refer to 30.1.6.2). Typically, the strut-and-tie model should follow the main transmission of forces, avoiding less important or secondary load paths that may increase the complexity of the model. Other required reinforcement may be calculated by local refinements of the model or specific considerations.
- Level II approximation (detailed design or brief assessment of an existing structure): refinement of the model selected for Level I considering relevant

secondary actions for transmission of forces and development of compression and tension fields in the critical regions. Compatibility of deformations shall not be explicitly verified and can be considered by means of simplified considerations (refer to 30.1.6.2)

- Level III approximation (refined design or assessment of complex structures): calculation of a suitable stress field accounting for the cracked response of concrete, compression softening due to concrete cracking, compatibility of deformations or other suitable criteria (as for example optimization approaches)
- Level IV approximation (refined assessment in case of elements with potentially insufficient deformation capacity or to have an accurate estimate of SLS response): calculation of a stress field accounting for equilibrium, compatibility of deformations and considering relevant tension-stiffening effects. Post-yielding behaviour of the reinforcement and compression softening should also be adequately considered to assess the deformation capacity of the structure.
-

Strut-and-tie and stress field models are useful tools to provide an insight of the flow of forces and to decide a strengthening strategy. Typical techniques for strengthening D-regions may consist of RC or UHPFRC jackets/layers, FRP strengthening, steel plates, post-installed bonded reinforecement, external post-tensinoning or others.

The constitutive stress vs crack opening relationship of the FRC material for design purposes should be determined according to section 18.4.

Modelling the load-deformation behaviour of fibre-reinforced structures requires predicting thoroughly the crack widths (potentially by means of a LoA IV modelling approach). Crack widths should be computed using crack spacings estimated on the basis of the reinforcement and FRC performance.

30.1.6.6.2 Considerations for strengthening

Strut-and-tie and stress field models may be used to design the strengthening of an existing structure provided that brittle structural responses are avoided.

In general, this requires assessing the level of deformation and associated structural damage prior to strengthening and to consider them during the activation of the strengthening. The analysis of the strengthening activation shall in principle consider compatibility of deformations unless sufficient ductility and capacity to redistribute stresses can be ensured.

30.1.6.7 Fibre-reinforced concrete

The use of strut-and-tie and stress fields is permitted for design and assessment of fibre-reinforced concrete structures provided that the conditions to develop smeared cracking can be ensured.

The fibres can be considered as an additional tension field, with tensile resistance only due to the fibre bridging contribution (tensile strength of the concrete matrix should be neglected as for stress fields and strut-and-tie models without fibres). Such fibre bridging contribution is in general dependant on the crack opening and therefore on crack spacing.

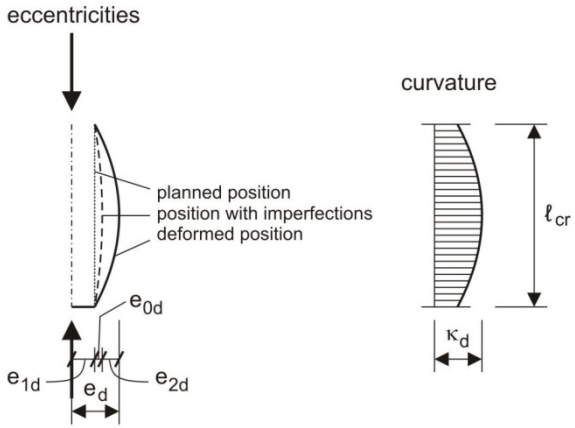


Figure 30.1-43: Compression resultant with eccentricities and curvature variation.

Fibres may be used in ties in place of the transverse reinforcement. Fibres may substitute up to 30 % of the total required main longitudinal reinforcement in ties.

A maximum stress of $f_{Fts,ef}$ and f_{Ftud} may be assumed for serviceability and ultimate limit state respectively in the concrete areas involved.

In absence of specific data, the value of coefficient k_ε can be conservatively estimated on the basis of section 30.1.6.2.

30.1.7 Compression members

30.1.7.1 Stability of compressed members in general

Where the behaviour of a member is significantly influenced by second order effects (Figure 30.1-43), the verification is carried out for the deformed system and the dimensioning values of the actions. The geometrical imperfections according to section 29.2.2 have to be taken into account. Furthermore, the influence of cracking, non-linear deformation of the structural materials and time-dependent deformations should be considered.

The dimensioning value of the bending moment is:

$$M_d = -N_d e_d \qquad (30.1\text{-}134)$$

The maximum eccentricity e_d, that is the maximum distance between the compression resultant and the deformed axis of the compression member, may be determined as:

$$e_d = e_{0d} + e_{1d} + e_{2d} \qquad (30.1\text{-}135)$$

where e_{0d} is eccentricity due to imperfections, being the greater value of:

$$e_{0d} = \alpha_i \, l_0/2 \text{ and } e_{0d} = d/30 \qquad (30.1\text{-}136)$$

where l_0, is the effective length of the compressed member, see for example Figure 30.1-44. The value of α_i may be estimated as:

$$\frac{1}{200} \geq \alpha_i = \frac{0.01}{\sqrt{l_0}} \geq \frac{1}{300} \; (l_0 \text{ in m}) \qquad (30.1\text{-}137)$$

The first order eccentricity e_{1d} may be estimated according to:

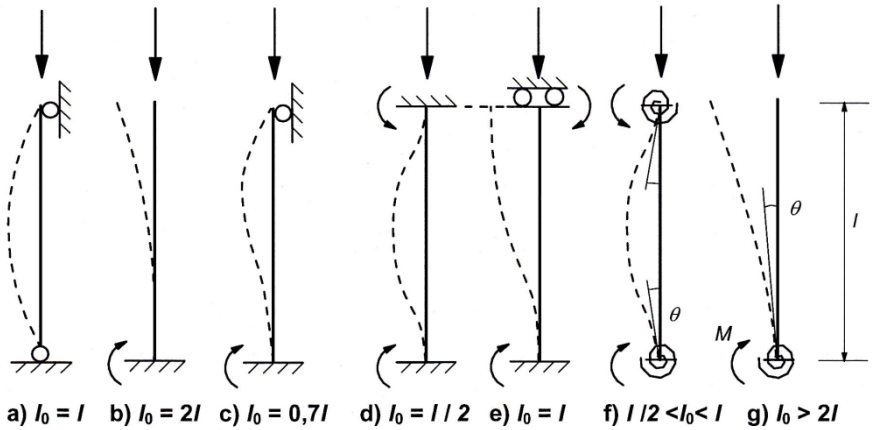


Figure 30.1-44: Examples of different buckling modes and corresponding effective lengths for isolated members.

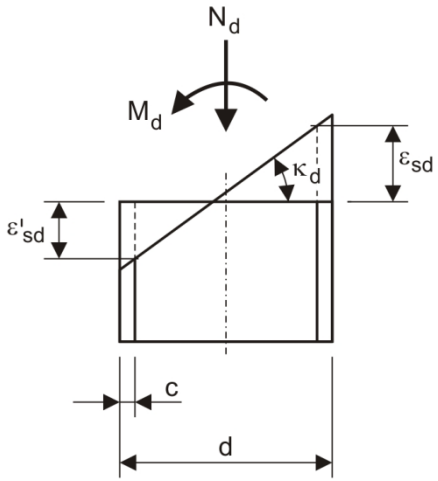


Figure 30.1-45: Strain plane corresponding to maximum curvature.

$$e_{1d} = \frac{M_{1d}}{-N_d} \quad (30.1\text{-}138)$$

The eccentricity due to the deformation of the compression member e_{2d} may be calculated by:

$$e_{2d} = \kappa_d \frac{{l_0}^2}{c_0} \quad (30.1\text{-}139)$$

where κ_d is the maximum design curvature (Figure 30.1-45):

$$\kappa_d = \frac{\varepsilon_{sd} - \varepsilon'_{sd}}{d - 2c} \quad (30.1\text{-}140)$$

and c_0 is the integration factor accounting for the curvature distribution along the member.

30.1.7.1.1 Level I approximation

The value of the integration factor may be assumed as:

Eq. (30.1-143) is based on interpolation using the interaction diagram, which is normally used as a design aid to determine the reinforcement in cross-sections subjected to a normal force and a bending moment. Figure 30.1-46 shows a simplified representation of such a diagram. At point B, the reinforcement yields at both sides of the column, so that the curvature is $\kappa = \varepsilon_{yd}/(0.45d)$. At point A, the curvature $\kappa = 0$. So, the curvature in point C can be obtained by interpolation from Eq. (30.1-143).

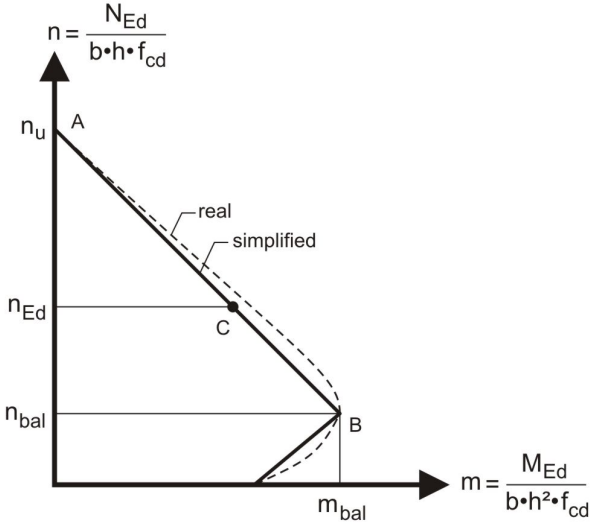


Figure 30.1-46: Simplified representation of interaction diagram.

$$c_0 = \pi^2 \qquad (30.1\text{-}141)$$

The maximum design curvature may be obtained from Eq. (30.1-128) with:

$$\varepsilon_{sd} = \frac{f_{yd}}{E_s} \text{ and } \varepsilon_{sd}' = -\frac{f_{yd}}{E_s} \qquad (30.1\text{-}142)$$

30.1.7.1.2 Level II approximation

A more accurate value of the maximum design curvature may be obtained from the equation:

$$\kappa_d = \left(\frac{n_u - n_d}{n_u - n_{bal}}\right) \cdot \frac{\varepsilon_{yd}}{0.45d} \qquad (30.1\text{-}143)$$

where:

$n_u = 1 + \omega$;

$n = N_{Ed}/(A_c f_{cd})$;

n_{bal} = value of n at maximum moment resistance ≈ 0.4 (point B in Figure 30.1-46);

$\omega = A_s f_{yd}/(A_c f_{cd})$.

The long term deformations due to creep and shrinkage of the concrete may approximately be taken into account as pre-curvature of the cross-section. The maximum curvature according to Eq. (30.1-143) shall be increased by adding the irreversible portion:

$$\kappa_{d,\infty} = \frac{|\varepsilon_{c\infty}|}{d} \qquad (30.1\text{-}144)$$

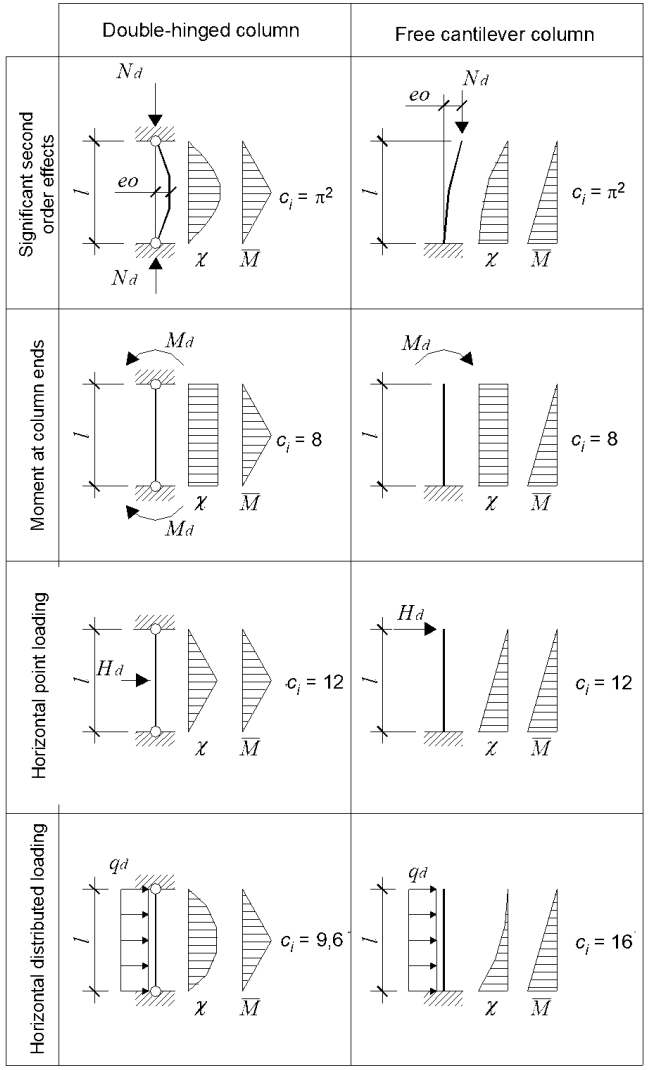


Figure 30.1-47: Values of integration factors c_i as a function of the load type and the boundary conditions.

For the background of Eq. (30.1-145), see Schertenleib et al. [30.1.-40].

30.1.7.1.3 Level III approximation

A more refined value of the integration factor c_0 may be calculated on the basis of the values of the various integration factors c_i for each action according to:

$$c_0 = \pi^2 \cdot \frac{N}{N_{cr}} + \frac{\sum_{i=1}^{n} M_i}{\sum_{i=1}^{n} \frac{M_i}{c_i}} \cdot \left(1 - \frac{N}{N_{cr}}\right) \qquad (30.1\text{-}145)$$

where the factors c_i are given in Figure 30.1-47.

30.1.7.1.4 Level IV approximation

A refined calculation of second order effects may be determined using an analysis that accounts for non-linear behaviour of concrete in compression, cracking, creep and shrinkage, reinforcement yielding and other non-linear effects important to the change in behaviour over time and loading state.

30.1.7.2 Biaxial eccentricities and out-of-plane buckling

For members with rectangular cross-sections, separate verifications in the two principal planes y and z are permissible, if the point of application of N_{Ed} is located close to one principal axis, for example within the hatched zones in Figure 30.1-48. The ratios of the corresponding eccentricities e_{y1}/b and e_{z1}/h have to satisfy one of the following conditions:

$$(e_{z1}/h)/(e_{y1}/b) \leq 1/4; \text{ or} \tag{30.1-146}$$

$$(e_{y1}/h)/(e_{z1}/b) \leq 1/4 \tag{30.1-147}$$

where the eccentricities e_{y1} and e_{z1} are those in the directions of the section dimensions b and h, respectively, and include an imperfection allowance e_{0d}, as defined in Eq. (30.1-91).

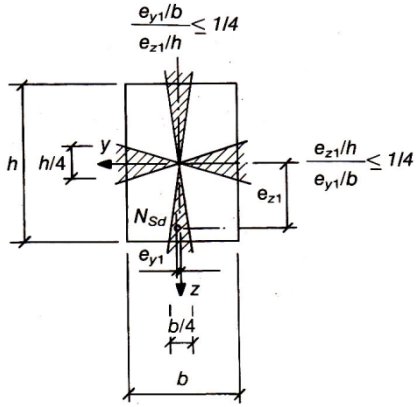


Figure 30.1-48: Condition for separate verification in the two principal planes.

If the criteria expressed by Eqs. (30.1-146) and (30.1-147) are not satisfied, the cross-section should be designed for biaxial bending, including the second order effects in each direction. In the absence of a refined cross-sectional analysis for biaxial bending, the following simplified criterion may be used:

$$\left(\frac{M_{Edx}}{M_{Rdx}}\right)^a + \left(\frac{M_{Edy}}{M_{Rdy}}\right)^a \leq 1.0 \qquad (30.1\text{-}148)$$

where:

$M_{Edx/y}$ is the design moment around the respective axis, including nominal second order moments;

Situations where a check is necessary are, for example, slender precast beams during transport and erection, and beams with insufficient lateral bracing in the finished structure. In such cases, geometric imperfections should be taken into account.

$M_{Rdx/y}$ is the moment resistance of the cross-section in the respective direction;

a is an exponent which is 2 for circular and elliptical cross-sections and for rectangular cross-sections follows from:

N_{Ed}/N_{Rd}	0.1	0.7	1.0
a	1.0	1.5	2.0

N_{Ed} is the design value of axial force;

N_{Rd} is the design axial resistance of section = $A_c f_{cd} + A_s f_{yd}$.

30.1.8 Design of 2D members

[Not used at this time, to be updated in future edition]

30.1.9 Lateral instability of beams

A check of the lateral instability of beams is relevant in long beams where lateral bracing is lacking.

A lateral deflection of $l/300$ should be assumed as a geometric imperfection in the verification of beams in unbraced conditions, where l is the total length of the beam. In finished structures, bracing from connected members may be taken into account.

Second order effects with regard to lateral instability may be ignored if the following condition is fulfilled:

$$\frac{l_{0f}}{b} \leq \frac{50}{(h/b)^{1/3}} \qquad (30.1\text{-}149)$$

where:

l_{0f} is the unbraced length of the compression flange;

h is the total depth of the cross-section of the beam in the central part;

b is the width of the compression flange.

For background see Foster et al. [30.1-41].

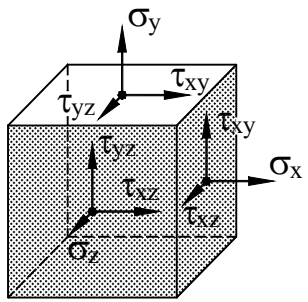


Figure 30.1-49: *3D stresses at a point defined in the orthogonal xyz axis system.*

In the application of this design method, the xyz axes are taken to correspond with reinforcing directions. The normal stresses applied at a point in a reinforced concrete solid element are carried by reinforcing steel and/or the concrete while shear stresses are carried by the concrete alone. Given that the applied stress tensor has been determined, for example, by 3D finite element solid modelling, the Mohr's circles of applied stress may be plotted, as shown in Figure 30.1-50. Within the circles the stress points (σ_i, S_i) are also plotted where $i = x, y, z$. As the reinforcing steel cannot carry shear stress it follows that the points denoting the concrete stresses (σ_{ci}, S_{ci}) should fall within the hatched region of the concrete stress circles where $\sigma_{ci} = \sigma_i - \rho_{s.j}\sigma_{s.j}$ and $S_{ci} = S_i$.

In the xyz space σ_x, σ_y and σ_z are, by definition, normal to the yz, xz and xy planes, respectively. The magnitudes of the shear stresses on these planes are given by

$$S_x = \sqrt{\tau_{xy}^2 + \tau_{xz}^2}; S_y = \sqrt{\tau_{xy}^2 + \tau_{yz}^2}; S_z = \sqrt{\tau_{xz}^2 + \tau_{yz}^2}$$

30.1.10 3D solids

30.1.10.1 Stress limit requirements

Considering a stress tensor for a reinforced concrete 3D element (Figure 30.1-49), the applied stresses on an element may be replaced by equivalent stresses in the concrete (subscript *c*) and in the reinforcement (subscript *s*), according to:

$$\begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} = \begin{bmatrix} (\sigma_{cx}+\sigma_{sx}) & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & (\sigma_{cy}+\sigma_{sy}) & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & (\sigma_{cz}+\sigma_{sz}) \end{bmatrix} \quad (30.1\text{-}150)$$

where *x*, *y* and *z* are the axis directions of the orthogonal reinforcing steel.

The equivalent reinforcement stresses are limited by:

$$|\sigma_{s.j}| \le \rho_{s.j} f_{yd.j} \quad (30.1\text{-}151)$$

where $\rho_{s.j}$ $(j = x, y, z)$ are the reinforcement ratios in the x, y and z directions, respectively.

The concrete stresses (ordered as $\sigma_{c3} \le \sigma_{c2} \le \sigma_{c1}$ as shown in Figure 30.1-50) are required to satisfy:

$$-\sigma_{c3} \le \nu f_{cd} \quad (30.1\text{-}152)$$

If no reinforcement has yielded and at least one principal stress is in tension then:

$$\nu = \frac{1.18}{1.14+0.00166\sigma_{si}} \le 1.0 \quad (30.1\text{-}153)$$

where σ_{si} is the maximum tensile stress (in MPa) in any layer of reinforcing steel $(i = x, y, z)$.

If one or more layers of reinforcement yield:

$$\nu = (1 - 0.032|\delta_i|) \cdot \frac{1.18}{1.14+0.00166 f_{yd}} \quad (30.1\text{-}154)$$

where δ_i is given by Eq. (30.1-161).

If all principal stresses are compressive, ν may be taken as 1.0 or determined in accordance with section 14.6.1.3.

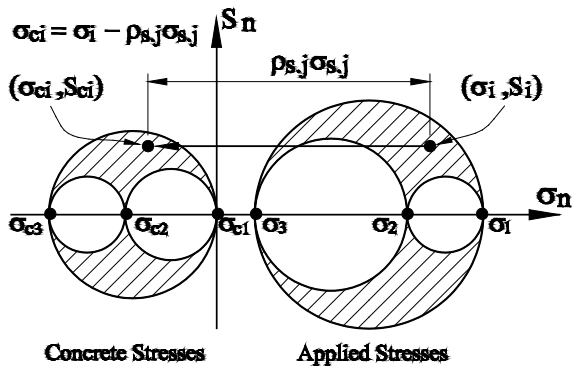


Figure 30.1-50: Compression field for 3D stress at a point.

In developing solutions, the designer should account for the limitations of the concrete material. In a solid subject to a constant ratio of normal and shear stresses (with at least one tensile principal stress), before cracking the stress field in the concrete remains relatively elastic and the stresses in the reinforcement are negligible. After cracking the tensile stresses in concrete on the principal plane are nil and correspondingly the stresses in reinforcement increase. After yielding of reinforcement in one direction, to ensure the equilibrium, the forces within reinforcing steel are redistributed by means of a variation of angles of cracking plane up to yielding along other directions of reinforcement. Tangential stresses appear along the cracking modified plane, that should be transferred by concrete through a combination of interlock and dowell mechanisms that reduce the concrete strength. Concrete elements have a limit on the amount of redistribution that can be achieved. As a rule, concrete elements should not be pushed far beyond that which is "natural". Designers should critically examine the load path being assumed, to ensure that a sufficient level of ductility is available to meet the demands of the imposed tractions. It is suggested to limit $|\delta i| \leq 15°$ (Figure 30.1-51) and correspondingly the concrete strength will be reduced up to 50%. For design purposes a deviation between direction of initial cracking plane and modified ones implies a saving of reinforcement and ductility availability; on the contrary a design procedure with $\delta=0$ produces no yielding in reinforcement and maximum bearing capacity on concrete side. In case of use of the model for verification, first the minimum ν value should be calculated from Eq. (30.1-

30.1.10.2 Ductility requirements

The ductility demand can be assessed by comparing the principal stress directions (Figure 30.1-51) of the resulting concrete stress tensor with that of the applied, factored loads.

The direction cosines of the principal stresses of the loading tensor $\boldsymbol{n_i} = \{n_{ix}, n_{iy}, n_{iz}\}$ (i = 1, 2, 3) are:

$$n_{ix} = \frac{-c_{iy}c_{iz}}{C} \quad (30.1\text{-}155)$$

$$n_{iy} = \frac{-c_{ix}c_{iz}}{C} \quad (30.1\text{-}156)$$

$$n_{iz} = \frac{-c_{ix}c_{iy}}{C} \quad (30.1\text{-}157)$$

where $C = \sqrt{c_{ix}^2 c_{iy}^2 + c_{ix}^2 c_{iz}^2 + c_{iy}^2 c_{iz}^2}$ and where:

$$c_{ix} = (\sigma_x - \sigma_i)\tau_{yz} - \tau_{xy}\tau_{xz} \quad (30.1\text{-}158)$$

$$c_{iy} = (\sigma_y - \sigma_i)\tau_{xz} - \tau_{xy}\tau_{yz} \quad (30.1\text{-}159)$$

$$c_{iz} = (\sigma_z - \sigma_i)\tau_{xy} - \tau_{xz}\tau_{yz} \quad (30.1\text{-}160)$$

152) then from Eq. (30.1-154) the corresponding δ value is derived and for this value the reinforcement stresses are verified, respecting the limitations of Eq. (30.1-151).

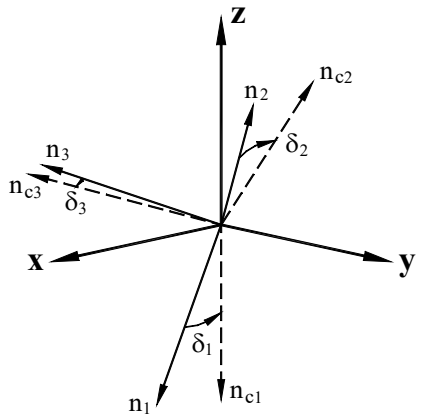


Figure 30.1-51: Comparison of concrete principal stress directions and the principal stress directions due to the applied tractions for the case of optimum reinforcement.

When assessing an existing concrete structure the properties of the structural materials are not defined as specifically as is the case for the design of a new structure. For the design of a new structure the 28-days concrete strength class is known, including all related properties of the concrete. Furthermore, for the design of a new structure the properties of the chosen type of reinforcement are specified in detail, whereas, in a structure to be assessed a good definition of material properties is a first step in the process of evaluating structural resistance. The concrete material properties usually exhibit a pronounced variability on account of simultaneous and potentially counteracting mechanisms such as the continuous hydration of the cement paste and deteriorating mechanisms. Additionally, the relationships between the various concrete

The direction cosines of the principal concrete stresses $\mathbf{n_{ci}}$ are calculated from Eqs. (30.1-155) to (30.1-157) and Eqs. (30.1-158) to (30.1-160) with σ_{cx}, σ_{cy} and σ_{cz} substituted for σ_x, σ_y and σ_z, respectively.

The enclosed angles between the concrete stresses and those from the applied loading δ_i ($i = 1, 2, 3$) are given by:

$$\delta_i = cos^{-1}|n_{ix}n_{cix} + n_{iy}n_{ciy} + n_{iz}n_{ciz}| \qquad (30.1\text{-}161)$$

The rotational demands as indicated by the differences in the stresses resulting from the applied loading and those in the concrete should be limited to ensure sufficient ductility capacity to meet the demands.

30.1.11 Evaluation of structural safety (ULS) for deteriorated structures

30.1.11.1 General

Deterioration of concrete structures can occur due to various reasons. The properties of the concrete may suffer from internal effects, like alkali silica reaction and internal sulfate attack. Moreover, external influences can play a role: the alkalinity of the concrete around the bars, which fulfils a protecting function for the reinforcement, can be lost due to carbonation of the concrete or due to chloride ingress to the reinforcement exceeding a critical level. Other external processes that may play a role are frost-thaw cycles, external sulfate attack or leaching and acid attack.

For the evaluation of structural resistance of existing concrete structures and its expected development, as much as possible use will be made of the same basic models as used for the design of new structures. If necessary those models will be

properties of a structure in the field may differ from the standard empirical relationships for new, sound concrete.

The deterioration processes may depend on the composition of the structural materials used, and on the environmental conditions, which are not always the same throughout the structure, leading to spatial variability of the mechanical properties. Corrosion of reinforcement may lead to a reduction of the cross-section of the reinforcing and prestressing steel, but as well to a reduction of the ultimate strain. This may limit the force redistribution capacity of the structure, originally assumed in the design. Moreover corrosion may lead to spalling of the concrete cover, and as such to a reduction of the effective cross-sectional area of the concrete member and at the same time the elimination of its protecting function. This means that not only the assessment of the actual structural safety should be the target of investigation, but as well its development in time.

extended to take the effect of corrosion of reinforcing and prestressing steel on the bearing capacity into account.

Alkali Silica Reaction (ASR) takes place when reactive aggregates and alkali pore solution in concrete undergo a chemical reaction and form a hygroscopic gel as a reaction product. When this reaction product absorbs water, it expands. Expansion causes concrete to crack as the tensile stresses imposed by the expanding ASR gel on the surrounding concrete overcome the tensile strength of concrete.

Structures affected by ASR display cracking due to ASR-induced expansions. ASR has varying effects on the concrete mechanical properties such as the compressive strength, tensile strength, and modulus of elasticity. Mechanical properties of ASR-affected concrete can be determined by extracting cores out of an ASR-affected structure. However, since the cores extracted out of a structure lose their structural context, i.e. confinement stressed, the mechanical test results cannot be directly used to inform structural evaluation.

Understanding the fundamentals of the expansion mechanism is key to implementing an effective monitoring strategy and a conservative structural evaluation methodology. To that extent, this section of MC 2020 covers practices that have been employed in the United States, Europe and various countries in Asia that are actively dealing with ASR-affected civil infrastructure.

ASR is a bulk expansion mechanism as discussed in section 30.1.11.2.1. Clear understanding of the root cause of the degradation experienced by a concrete structure is necessary for developing and implementing an effective monitoring strategy and for performing structural evaluation.

ASR-affected structures commonly display map cracking. The cracking pattern is influenced by the restraint conditions. In this context restraint is provided by the reinforcement provided in the concrete structure (i.e. internal restraint) and boundary conditions of the structural component under investigation. The directions in which there is minimal or no restraint are the directions where ASR-affected expansions reach their maximums. Consequently, the inner crack pattern depends heavily on the restraint condition. Nevertheless, map cracking is typically observed on the surface of ASR-affected structures. In addition to map cracking, in some cases, ASR-gel exudation,

30.1.11.2 Structures affected by alkali silica reaction

30.1.11.2.1 General

Alkali Silica Reaction (ASR) is a chemical degradation mechanism that impacts existing concrete structures. ASR-affected structures shall be diagnosed, monitored and assessed by using the code language and additional guidance provided in the commentary of this section.

ASR causes bulk expansion of concrete. The provisions provided in all sections of 30.1.1.2.3 are valid within the context of experimental data that informed these provisions. Provisions of 30.1.1.2.3 and its subsections are valid for volumetric expansions that are less than 1.5%.

30.1.11.2.2 Diagnosis for ASR

ASR-affected structures, shall be diagnosed by using petrographic analysis. A representative number of cores shall be extracted from affected portions of the structure under investigation and a petrographer shall examine the cores to positively identify the root cause of the degradation mechanism as being ASR

presence of dark reaction rims and other features of ASR may readily be observable on the surface. Observing such crack patterns, and ASR features is the starting point of diagnosis. Ultimately, petrographic analysis is necessary to properly identify the root cause. In some cases, restrained shrinkage cracking and Delayed Ettringite Formation (DEF) induced cracking may produce crack patterns that look quite similar to cracking due to ASR and visual observation by itself is not sufficient to conclude that distress is directly attributed to ASR.

Presence of ASR gel in hardened concrete, as observed in petrographic analysis, is necessary. However, this condition is not sufficient to attribute distress to ASR. Cracking observed in the microstructure of concrete should be tied to ASR gel pockets and expansions observed by those gel pockets. It is possible to identify some ASR gel in many concrete structures but presence and extent of ASR may or may not be sufficient to cause distress, (i.e. cracking) observed in the structure.

30.1.11.2.3 Extent of Condition Analysis : ASR

An extent of condition analysis is necessary prior to implementing a monitoring strategy and taking subsequent actions. The extent of condition analysis is aimed at:

(i) identifying all areas of the structure displaying degradation,
(ii) performing additional petrographic analyses to confirm that all visually observed damage can be attributed to ASR, and
(iii) classifying different areas affected by ASR into classes of different severity.

With respect to item (i) above, it is important to conclude that all damage observed by the structure relates to ASR. In some cases, the visual damage observed may relate to multiple degradation mechanisms. For example different parts of a structure may be affected by ASR and/or DEF depending on the initial curing conditions, maximum temperatures reached within the structural core of the components. Larger components, and hence larger heats of hydration during initial curing, may result in elevated temperatures and trigger delayed ettringite formation. DEF-affected structures have the potential of experiencing higher degrees of expansion and therefore may require different monitoring frequencies. Alternatively, structural components subjected to different environmental exposure and/or structural loading conditions may suffer from additional effects including but not limited to corrosion of reinforcing steel, carbonation of concrete, fatigue damage to reinforcement anchorage or mechanical interlock of aggregate and hence concrete contribution to shear strength, etc.

An extent of condition analysis shall be performed on ASR-affected structures. Once the observed distress is positively attributed to ASR in an area, the extent of this condition, i.e. the ASR-damage, shall be evaluated for the remaining portions of the structure.

With respect to item (ii) above, it is important to clearly understand the role of petrography. Petrographic analysis is necessary to positively identify and attribute all observed damage to ASR. In the event multiple degradation mechanisms are present, considerations beyond those, which are articulated in this section of the Model Code will be necessary.

With respect to item (iii) above, and as discussed in code and commentary portions of this section 30.1.11.2, it is necessary to classify ASR-damage into different categories to facilitate an effective structural evaluation. A structure may have areas that show (a) no degradation, (b) mild degradation, (c) moderate degradation and (d) severe degradation. Both monitoring frequencies and conclusions of the structural evaluation may be different for those areas. Ultimately, structural retrofit may be necessary in some cases where damage is severe and structural safety margins are low. In such cases, classification of damage and structural evaluation that is based on the levels of approximation (LoA) process described in section 30.1.11.2.12 are key to allocation of resources appropriately to areas that require different degrees of attention.

Monitoring ASR-affected structures is key to ensure safety of those structures while they performed their intended service function. ASR-induced cracking and structural expansions are the key reasons for any impact to structural capacity. Measurement of those expansions once a structure is diagnosed with ASR and ensuring the limits of applicability for the structural evaluations carry primary importance. Excessive ASR expansions leading to levels of expansion not bracketed by data reported in the literature may limit the applicability of the structural evaluation. Further, sudden changes in the expansions monitored can signal the fracture of the reinforcement that is restraining the ASR expansions. The presence and maintenance of the integrity of reinforcement is key to maintaining structural safety.

ASR is typically discovered once visual manifestations are present. It is also typical that most structures are not instrumented at the time of original construction. Therefore, estimating expansions-to-date, is the first step in expansion monitoring. To this end a structural engineer can utilize one or more of the following techniques individually or in combination:

i. A conservative estimate of the expansions can be obtained by using crack indexing, in most cases. Summation of the widths of the cracks on a straight line with a sufficient length (gage length) can be performed. Dividing that sum with the gage length will help identify a reasonable and conservative estimate of expansions that occurred in that direction to date. In cases where the

30.1.11.2.4 Monitoring ASR-Affected Structures

ASR-affected structures shall be monitored to identify and quantify progression of damage. Monitoring expansions and/or engineering strains in three principal directions is necessary to inform validity to the structural evaluations performed in compliance with this section.

Use of crack indexing, mechanical measurements, mechanical property degradation to estimate expansions in one or more directions under consideration shall be permitted.

chemical prestress is being calculated and used in structural analysis, an over-estimation of expansions to date may lead to an over-estimation of chemical prestress force. In such analyses, it would be more conservative to discount the beneficial effects of prestress or conservatively ignore such effects in their entirety.

ii. In directions where crack indexing is not feasible, for example through-thickness direction in structural walls and slabs, a reasonable and conservative estimate can be obtained by extracting cores and conducting mechanical tests on the unconfined concrete cores. Once a core is extracted out of a structure, it loses its structural context (i.e. confining stresses acting on it) and relaxes to its "unconfined state." As discussed in section 30.1.11.2.5, mechanical properties of plain (unconfined) concrete are greatly impacted by ASR. Modulus of elasticity of concrete and tensile strength of concrete are much more susceptible to ASR than the compressive strength of concrete. Therefore, reduction in modulus of elasticity and/or tensile strength of concrete can directly be used to estimate the expansions to date in the direction of cores that are taken. The trend lines presented in Figures 30.1-52, 30.1-53 and 30.1-54 may be used in lieu of data directly applicable for the structure under consideration. In cases where extensive sampling and data are present for the structure being evaluated, those data sets may be used to directly inform the decision-making process. Loss in modulus of elasticity or tensile strength can conservatively be based on original mechanical property values based on calculations performed on original design documents and compressive strength of concrete specified in those documents (LoA II). Alternatively, and consistent with a higher level of approximation (LoA III), areas of the same structure or structural component can be used to obtain reference cores, and after lack of ASR damage is confirmed for that area, the mechanical properties measured for that area can be used as baseline numbers.

Once expansions-to-date are established, future expansions can be measured by taking mechanical measurements on pins/targets installed on the structure and its components. Mechanical measurements are typically more accurate than crack indexing. With that stated, depending on the level of approximation utilized in the engineering analysis the use of crack indexing can be completely appropriate, if not preferred by the engineer.

Damage rating index (DRI), a technique based on petrographic examination of cores, can be used as a semi-quantitative to assess progression of ASR. However, it has been reported by several researchers and industry guidance documents (e.g. FHWA

Guidance Document) that DRI values calculated by different petrographers for the same structural component, in fact for the same core, can greatly vary. This level of variability calls the reliability of this technique into question. Structural consequences of ASR are related to expansions. Measuring/monitoring expansions provide the most direct input to appraisal of structural capacity.

The impact of ASR on plain concrete can be calculated on the basis of free expansions. Figures 30.1-52, 30.1-53 and 30.1-54 (adopted from Martinez [30.1-42]) serve to illustrate the impact of ASR on mechanical properties of plain concrete. As can be observed in these figures modulus of elasticity and tensile strength of concrete are impacted to a greater extent than the compressive strength. It is for these reasons that modulus and tensile strength reductions seen in cores extracted from structural components serve as better indicators of expansions experienced in the direction of core extraction. Conversely, for the fact that compressive strength is impacted the least, this property is not suitable to estimate the expansion-to-date.

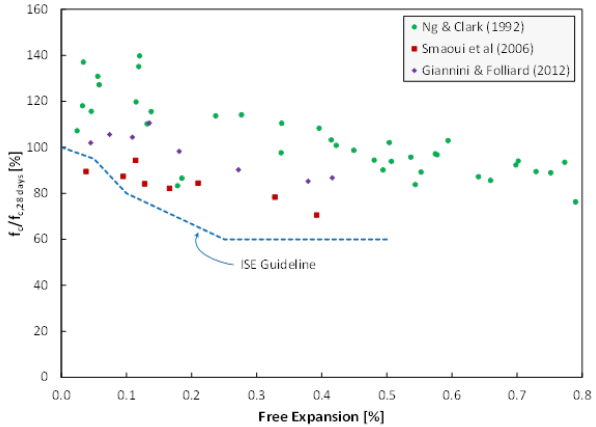


Figure 30.1-52. Impact of ASR on Compressive Strength of Concrete (Adopted from Martinez [30.1-42]).

30.1.11.2.5 Impact of ASR on Plain Concrete

The impact of ASR on plain concrete shall be calculated on the basis of charts provided in Figure 30.1-52 – 30.1-54.

The use of reduced mechanical properties of plain concrete, i.e. cores extracted from a structure, in structural evaluation shall be permitted if beneficial effects of confinement are explicitly taken into account by experimentally validated constitutive models. Use of reduced mechanical properties, while neglecting the structural context (i.e. confinement provided by chemical prestress due to action/reaction between expanding concrete and reinforcement), will result in gross underestimation of structural capacity, and hence shall be avoided in all cases except grossly over-designed structures.

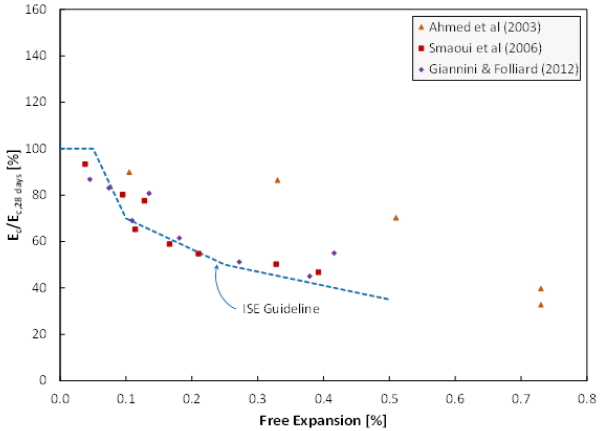


Figure 30.1-53. Impact of ASR on Modulus of Elasticity of Concrete (Adopted from Martinez [30.1-42]).

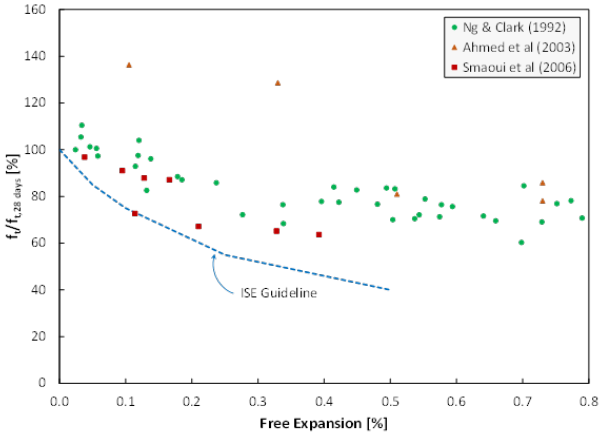


Figure 30.1-54: Impact of ASR on Tensile Strength of Concrete (Adopted from Martinez [30.1-42]).

Reduced compressive strength, tensile strength and modulus of elasticity, if used in structural analysis without modelling beneficial effects of confinement, will result in gross underestimates of structural capacity. While conservative, and therefore consistent with a LoA I, typically this approach will result in unacceptable results

unless the original structure is grossly over-designed. Such examples exist in cases where typical loads do not control the original designs, e.g. architectural features etc.

The impact of ASR on axial capacity of reinforced concrete in compression is insignificantly small. The largest adverse impact reported in the literature (Chana & Korobokis[30.1-43]) is 18%. The 18% reduction in axial compressive capacity of the columns tested by Chana & Korobokis [30.1-43] can be explained on the basis of chemical prestress. As ASR-prone concrete used in their experimental program expanded, it placed longitudinal column bars in tension. The longitudinal bars in tension applied compressive loads on the concrete section, thereby imposed an additional demand on the columns.

Industry-accepted variability, and ASTM-accepted tolerance, in compressive strength of concrete as measured by concrete cylinders without ASR damage, is about 8-10%. With that variability in compressive testing and the chemical prestress effect discussed above, it can be concluded that the impact of ASR on axial compressive capacity of concrete within the structural core is negligibly small.

With above considerations noted, the interaction between the structural core and reinforcing cage should be modelled. In other words, the negative impact of chemical prestress on axial compressive capacity has to be accounted for in structural evaluation.

In Columns ASR may lead to cracking in the unconfined concrete cover and sometimes spalling of the cover while the presence of ties or spiral reinforcement effective confines the structural core. Consequently, buckling of the axially loaded rebars should be considered in structural evaluation. If the spacing of transverse reinforcement is too high in relation to the longitudinal bar diameter, buckling of the bars in compression may reduce the axial capacity of the column under consideration (Bayrak, 1999 [30.1-44]).

ASR does not impact axial tensile capacity of reinforced concrete. The quantity and strength of reinforcement dictate the tensile capacity of reinforced concrete. The mechanical properties of concrete do not influence the axial tensile capacity.

30.1.11.2.6 Impact of ASR on Axial Capacity of Reinforced Concrete Elements

The impact of ASR on axial compressive capacity shall be considered to be negligibly small.

The adverse influence of chemical prestress to structural capacity in axial compression shall be accounted for in structural analysis.

The impact of ASR on axial tensile capacity of reinforced concrete shall be taken as zero.

Research on the flexural capacity of reinforced concrete (Giannini et al. [30.1-45], MPR 4273 [30.1-46]) indicated that the structural capacity of reinforced concrete sections subjected to flexure-critical loads was not adversely impacted. Further, slight increases observed in flexural capacity of reinforced concrete sections in members displaying extreme levels of ASR damage could be explained by the aforementioned researchers on the basis of ASR-induced chemical prestress. In other words, axial compression improves flexural capacity for compressive loads less than the balanced failure condition.

Research on shear strength of reinforced concrete (Deschenes et al [30.1-50]) indicated that the structural capacity of reinforced concrete sections subjected to shear-critical loads was not adversely impacted. Deschenes et al. noted shear strength increases both for short (Strut-and-Tie Shear) and long (Beam Shear) increased by about 16%. This increase can be explained on the basis of chemical prestress.

Research on the shear strength of reinforced concrete beams without transverse reinforcement (Martinez [30.1-42], and MPR 4273[30.1-46]) showed increased shear strength with increasing levels of ASR damage. This observation is counter to a 20-25% reduction in shear strength observed in small scale beams tested by Ahmed et al. [30.1-47] and bridge deck specimens harvested by Den Uijl et al. [30.1-48]. Specimens tested by Ahmed et al. were highly-scaled and suffered from “edge effects” where large cracks easily penetrated through the cross sections of beams that were 75-mm.-wide and 125-mm. deep. Den Uijl et al. [30.1-48] tested sections of highway slabs they harvested from in service ASR-affected bridges. The impact of specimen harvesting from bridges, loss of structural context as specimens were harvested and additional damage that may have occurred during transportation to a structural testing facility, all contributed to observed structural capacities. Further, other environmental effects and fatigue loading resulted from repetitive load application may have influenced the test results of Den Uijl et al. [30.1-48]. On the basis of these observations, while direct application of their test results to a number of bridges in Netherlands is justified, the extrapolation of their results to other application is overly conservative. Instead, the generic study performed by Martinez [30.1-42], and highly-repeatable tests conducted at different expansion levels, inform the MC 2020 provisions.

30.1.11.2.7 Impact of ASR on Flexural Capacity

The impact of ASR on flexural capacity shall be considered to be negligibly small.

Chemical prestress does not influence the flexural capacity of a reinforced concrete section. With that stated, and for the purposes of simplicity, chemical prestress shall not be taken into account in structural analysis.

30.1.11.2.8 Impact of ASR on Shear Strength

The impact of ASR on shear capacity of members with a sufficient quantity of shear reinforcement shall be considered to be negligibly small.

The impact of ASR on shear capacity of members with no shear reinforcement shall be considered to be negligibly small.

Research on punching shear capacity of slabs (Ng and Clark, 1992 [30.1-49]) indicated that the reduction in punching shear capacity of slabs experiencing advanced levels of ASR is negligibly small. In the worst case reported the reduction was measured to be 9% and this reduction is less than the variability observed in the testing program, indicating no substantial impact on punching shear strength. Further, putting the test results of Ng and Clark [30.1-49] in context with those obtained by Deschenes et al. [30.1-50] and Martinez [30.1-42] by using critical shear crack theory (CSCT), it is possible to explain the observed behaviour, i.e. no capacity reduction due to ASR, with the presence of chemical prestress.

Research on anchorage of reinforcing bars in plain concrete (Chana [30.1-51]) indicates that the pull-out capacity of deformed reinforcing bars cast in plain concrete is reduced by 40%. Such a reduction in capacity is significant. This is partly a function of the test method employed by Chana [30.1-51] in 1989 and partly due to the fact that the concrete tensile ring that surrounds a reinforcing bar as a result of the force transfer, and as explained by Tepfers, is compromised due to ASR cracks. Current state-of-the art involves beam tests rather than pull-out tests performed early on. Performing beam tests will likely reduce the impact of ASR seen in pull-out tests.

Research on anchorage of deformed bars in bi-axially reinforced concrete beams (MPR 4273 [30.1-46]) indicates that ASR does not adversely impact the anchorage capacity. These findings have been further validated by (Zhychkovska and Sheikh [30.1-52]. Presence of reinforcement strengthens the concrete tensile ring (Tepfer's model) that forms around the anchored bar and mitigates the adverse effects of ASR cracks. Likewise, deformed bars anchored in reinforced concrete elements that are reinforced in three directions is not compromised by ASR.

30.1.11.2.9 Impact of ASR on Punching Shear Strength

The impact of ASR on shear capacity of members with a sufficient quantity of shear reinforcement shall be considered to be negligibly small.

The impact of ASR on punching shear capacity of members with no shear reinforcement shall be considered to be negligibly small.

30.1.11.2.10 Impact of ASR on Anchorage of Deformed Reinforcing Bars

The impact of ASR on anchorage of deformed reinforcing bars in plain concrete shall be considered to be 40%. This reduction in capacity is associated with the ASR-induced volumetric expansion limit 1.5% given in section 301.11.2.1. Linear interpolation between 40% reduction and no reduction for 0% volumetric expansion shall be permitted.

The impact of ASR on anchorage of deformed reinforcing bars in reinforced concrete elements in compliance with the minimum reinforcement quantities of Model Code 2020 shall be considered to be negligibly small.

30.1.11.2.11 Potential of Reinforcing Bar Fracture in ASR-Affected Concrete

Over 30 structures in Japan have experienced the fracture of reinforcing bars at 90-degree bar bends as a result of prolonged, severe ASR expansions, coupled with other degradation mechanisms due to high concentration of chlorides in concrete Miyagawa et al.[30.1-53]. An examination of reinforcing bar details that experienced fracture at bar bends was conducted by Webb [30.1-54]. Webb confirmed the findings of Miyagawa et al.[30.1-53] and reported that initial bar bending process by using mandrel diameters that are smaller than those required by Model Code 2020 is the root cause of the reinforcing bar fracture. A great majority of the structures constructed in Asia, Europe and the Americas easily comply with the Model Code 2020 rules on bar bends. With that stated, some of the very old inventory constructed in the early parts of the 20th century may have bar bend radii that may render an ASR-affected structure prone to rebar fracture. In the context of ASR, this phenomenon must be investigated.

As Miyagawa et al. [30.1-53] indicated *"...As long as the reinforcement is not fractured due to ASR-related expansion, the safety of a structure is not considered to be seriously compromised. However, the safety of a structure becomes questionable when confinement of the concrete degrades due to fracture of reinforcement."*

LoA I analysis can be performed by making conservative assumptions. In this level of approximation the most damaged location in the structure can be assumed to represent all areas of the structure and the impact of ASR on structural performance can be performed on the basis of this assumption. The use of linear and nonlinear analysis techniques to estimate the additional demands imposed on structural components can be deemed appropriate by the engineer provided that the objectives of structural analysis are consistent with the simplifying, albeit conservative, assumptions made. The method of analysis chosen by the structural engineer has to be benchmarked against experimental data available in the literature. For the purposes of benchmarking large-scale tests should be preferred over small-scale tests, since large-scale specimens will be less impacted by alkali leaching and surface effects seen in all experimental programs that focused on ASR.

LoA I analysis typically provides overly conservative estimates of demands and capacities. In some cases where the structural design is not governed by load application, LoA I analysis may prove to be satisfactory to ensure structural adequacy. If such is not the case a more refined analysis (LoA II) can be performed. In this level

The potential of reinforcing bar fracture in ASR-affected concrete structures shall be examined. If reinforcing bar bend diameters of the ASR-affected structures are in compliance with those of Model Code 2020, the potential of reinforcing bar fracture shall be taken as negligibly small.

30.1.11.2.12 Analysis of ASR-Affected Reinforced Concrete Structures

ASR-affected concrete structures shall be analysed to determine the effects of additional loads or deformations imposed by ASR expansions. The analysis techniques employed shall be consistent with the objectives of the analysis, and determined by the structural engineer in charge.

Within the Levels of Approximation (LoA) framework of Model Code 2020, the use of equivalent linear and nonlinear techniques shall be permitted.

of approximation, a thorough mapping of ASR cracks and identification/classification of areas with varying levels of ASR damage will be necessary. Portions of the structure subjected to water ingress or exposure or high levels of relative humidity are likely to display greater levels of ASR damage, i.e. ASR-induced expansions. The severely-, moderately-, and slightly-damaged portions of the structure should be modelled in such a manner that different expansions/displacements can be imposed on different parts. This level of refinement is likely to reduce demands imposed on the structure by ASR, as it provides a more realistic approximation between a numerical model and actual structure experiencing ASR-damage. Once again, equivalent linear or nonlinear techniques can be employed depending on the primary objectives of analysis.

LoA III analysis necessitates modelling the interaction between the expanding concrete and reinforcement explicitly, in addition to making accommodation for variation of ASR damage throughout the structure. This level of refinement is only justified in checking portions of a structure to check reinforcement yielding or in areas of low structural margin to gain improved insights on the reinforcing cage, ASR-damaged concrete interaction.

LoA IV analysis is the highest level of analysis possible to get the best estimate, albeit with the least level of conservatism implicit in structural analysis. This level of refinement requires modelling nonlinear material response of concrete, reinforcing bars, and effects of confinement explicitly and through experimentally-verified models. Computational effort required by this level of refinement may or may not be justified depending on the cost of the required structural retrofits. In other words, it may be simpler and more cost-effective to deploy structural retrofit than try to conduct LoA IV analysis, given the need for experimental verification or benchmarking.

30.1.11.3 Structures with corroded reinforcement

30.1.11.3.1 General

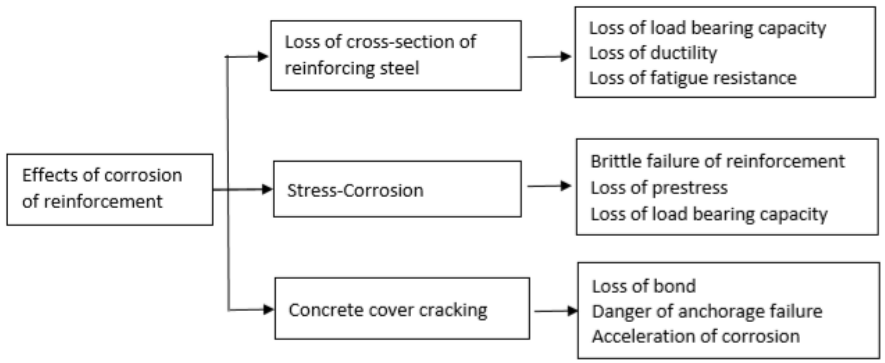


Figure 30.1-52: Effects of reinforcement corrosion on structural resistance (CEB Bulletin 189 [30.1-55]).

Research on the relation between corrosion and crack width, see (Blagojevic [30.1-56] showed that pitting corrosion can occur even in cracks with moderate widths in an aggressive environment, especially if the crack width varies under the effect of a non-constant load (like traffic), which counteracts the possible effect of self-healing.

"Moderate" and "low" values of corrosion have been defined to apply approximately for medium bar diameters with 5% weight loss and 0.25mm depth of corrosion [Blomfors et al. [30.1-57]). These figures are only indicative. For bars with small diameters the same degree of corrosion penetration will cause a relatively larger loss of cross-sectional area, but will produce a smaller volume of oxides (corrosion products) and therefore less spalling of the concrete. For bars with a larger bar diameter

In order to be able to assess the influence of the effect of corroded steel on the bearing capacity, it should be verified whether the corrosion is uniformly distributed along the reinforcing bars, which may occur in the case of carbonation, or is localized, which may happen in case of pitting corrosion due to chloride penetration. In the first case the effect of corrosion results predominantly in a reduction of the cross-sectional area of the steel, whereas the ultimate strain will not be significantly influenced. In case of pitting corrosion the geometry of the pits and their distribution along the reinforcing and prestressing steel plays a role, since here not only the reduction of the tensile capacity of the reinforcing steel is relevant but also the reduction of the ultimate strain (section 15.6). In the case of parallel bars it is important whether pits occur in the same cross-section (like caused by a wide crack in a region with chloride ingress). This information is relevant to be able to make a reliable analysis of the structural behaviour.

30.1.11.3.2 Bending resistance of structural members with corroded reinforcement subject to bending

30.1.11.3.2.1 Members subjected to bending

When the bending resistance of a structure with corroded reinforcement is determined, the following aspects should be considered:

- The effect of reduction of concrete strength due to the splitting effect caused by the volume increase of the steel due to corrosion;
- The reduction of the cross-sectional area of the reinforcing steel and its ductility;
- The reduction of bond strength due to corrosion.

For low to moderate levels of corrosion the effect of the expansion of the reinforcement in the compression zone can be taken into account by reducing the concrete compressive strength with a factor $k_c = 0.75 \cdot \eta_{fc}$ where $\eta_{fc} = (30/f_c)^{1/3}$ is a factor taking into account the more brittle behaviour of such a concrete for $f_c > 30$ MPa. As a conservative approach, it can be assumed that in the

the opposite occurs, with a reduced loss of cross-sectional area, but potentially a larger extent of cracking.

Guidance on the determination of the residual steel cross-section is given in section 15.11.2.1.

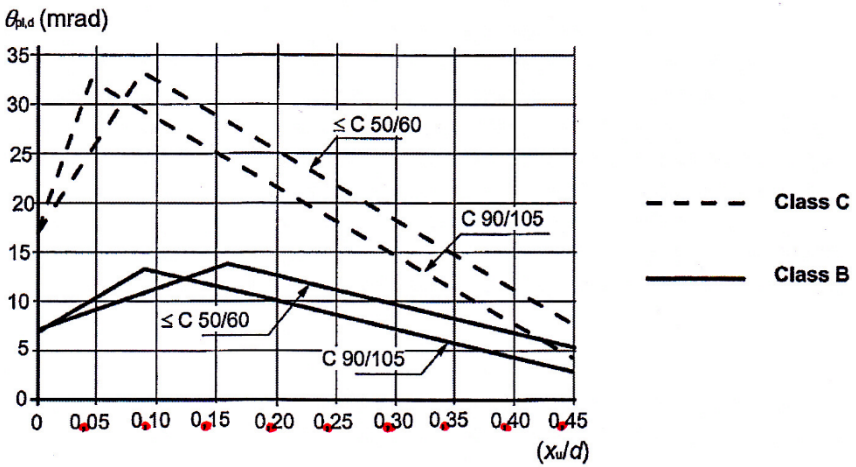


Figure 30.1-56: Allowable rotation $\theta_{pl,d}$ for reinforced concrete beams or slab with uncorroded reinforcing steel bars class B and C

compressive zone the concrete cover should be ignored in the calculation of the bending resistance.

For the tensile resistance of the reinforcement, the residual cross-section should be determined using the minimum bar cross-sections measured along the whole length of the tensile chord. In case of broken stirrups the compressive reinforcement may not be fully effective and should not be considered in the calculations. For the determination of the yielding tensile force in the tensile zone only the reduction of the cross-sectional area of the bars has to be taken into account. A reduction of the yield stress is not required.

Highly corroded beams with significant bond deterioration, cover spalling or delamination should be analysed without the usual assumption of perfect bond.

30.1.11.3.2.2 Rotation capacity

In the design of statically indeterminate beams and slabs often use is made of the possibility of redistribution of bending moments. To this aim the rotation capacity of the plastic hinges should be checked. The allowable rotation is calculated as the product $1.2h \cdot \theta_{pl,d}$ where h is the effective depth of the section and $\theta_{pl,d}$ is the allowable rotation, which can be read, for reinforced beams or slabs, from a diagram, see Fig. 30.1-56. However, for the steel classes B and C the following limit values apply:

Class B: $(f_t/f_y)_k \geq 1.08$ and $\varepsilon_{uk} \geq 5$ ‰;

Class C: $(f_t/f_y)_k \geq 1.15$ and ≤ 1.35 and $\varepsilon_{uk} \geq 7.5$ ‰;

It should be verified whether the corroded steel in the plastic hinge region (with a defined length of 1.2 h) satisfies those demands, taking into account the shape of the pits, the distance between the pits and the hardening capacity of the reinforcing steel used (see also section 15.11.2).

30.1.11.3.2.3 Bearing capacity of members with corroded reinforcement subjected to bending and normal force

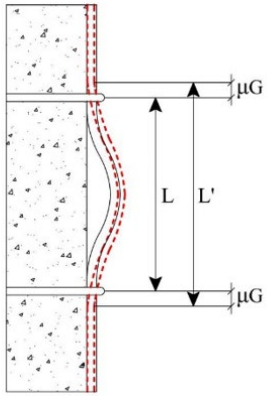


Figure 30.1-57: Rebar buckling mechanism (uncorroded (solid black) and corroded (dashed red) line.

Eq. (30.1-164) is an extension of Eq. (30.1-16), in which the bond coefficient k_b has been introduced, and were A_s has been replaced by $A_{s,det}$. Both the values k_b and $A_{s,det}$ are relevant for the flexural shear resistance because they determine the width of the governing bending shear crack, and as such the contribution of aggregate interlock. Pitting corrosion can occur in bars crossing flexural shear cracks, so that especially the local effect of cross-sectional bar reduction should be considered, Figure 30.1-58.

For reinforced concrete members subjected to bending and normal force conservative assumptions for the residual cross-sections and the material properties should be made. Strength calculations should include cross-sectional reductions of steel and concrete. When stirrups are broken the compressive reinforcement may not be considered in the calculations. If stirrups are not broken it should be checked whether their confining action (bar diameter and stirrup distance) is sufficient to prevent buckling of the bars in compression, Figure 30.1-57. Here it should be noted that due to corrosion the diameter of longitudinal bars and stirrups is reduced, so that a gap between those bars arises, which has to be overcome in order to create a resisting mechanism (L' versus L). In case of pitting corrosion an equivalent strength and stiffness should be derived on the basis of pit geometry, distance and hardening behaviour of the steel after yielding.

30.1.11.3.3 Shear resistance of members with corroded reinforcement

30.1.11.3.3.1 Shear resistance of members without shear reinforcement with corroded longitudinal reinforcement

In section 30.1.3 the design shear resistance of the web of a beam, or slab without shear reinforcement is given by:

$$V_{Rd,c} = k_v \frac{\sqrt{f_{ck}}}{g_c} z \cdot b_w \qquad (30.1\text{-}162)$$

There are two levels of approximation. According to the second level the parameter k_v follows from:

$$k_v = \frac{0.4}{1+1500e_x} \times \frac{1300}{1000+k_{dg}\cdot z} \quad (z \text{ in mm}) \qquad (30.1\text{-}163)$$

In this equation ε_x is the longitudinal strain calculated at mid-height of the effective shear depth. For corroded longitudinal reinforcement the expression:

$$e_{x,det} = \frac{1}{2E_s \cdot A_{s,det}} \cdot \frac{1}{k_{bond}} \left\{ \frac{M_{Ed}}{z} + V_{Ed} + N_{Ed}\left(\frac{1}{2} \pm \frac{D_e}{z}\right) \right\} \qquad (30.1\text{-}164)$$

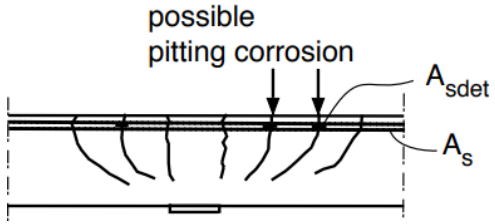


Fig. 30.1-58: Relevance of $A_{s,det}$ on shear capacity in case of pitting corrosion.

The value k_{bond}= 0.75 is an approximate value, derived from data for bond of reinforcing steel from of *fib* Bulletin YY "Modelling the structural performance of existing structures concrete" chapter 6.1.2.2.3, [30.1-58]), see also Section 20.3.5.1.2..

Test on the shear capacity of beams with smooth plain bars showed that there is hardly any reduction of the shear capacity compared to beams with ribbed reinforcing steel. The reason of this is that in beams with smooth plain bars the reinforcement does not transmit bond forces into the concrete, so that no inclined shear cracks can develop, and as a result no shear failure will occur. However, plane bars in structures normally still show substantial bond as a result of adhesion, so that this phenomenon does no occur. For the shear resistance of structures with plain bars reference is made to the *fib* Bulletin YY [30.1-58].

applies. Here the coefficient k_{bond} is a bond factor, with which the effect of reduced bond due to corrosion can be accounted for. For non-corroded bars k_{bond} = 1.0. For bars with a moderate level of corrosion (lesser of equal to 5% weight loss) k_{bond} = 0,75 may be assumed.

$A_{s,det}$ is the cross-sectional area of the bar reduced by deterioration (in this case corrosion). In case of pitting corrosion $A_{s,det}$ is the minimum cross-sectional area of the tensile reinforcement in the area where flexural shear cracks occur.

30.1.11.3.3.2 Shear resistance of structural concrete members with shear reinforcement subject to corrosion

According to section 30.1.3.3 for the determination of the shear resistance of members with shear reinforcement three levels of approximation are considered. For the assessment of the shear resistance of members with corroded steel the level of Approximation IIa appeared to be most appropriate.

This implies that only the contribution of the shear reinforcement is considered, with:

$$V_{Rds,det} = z \cdot \cot\theta \frac{A_{sw,det} \cdot f_{ywd}}{s} \qquad (30.1\text{-}165)$$

where the angle θ should not be smaller than determined according to Eq. (30.1-36a).

The reduced cross section $A_{sw,det}$ for pitting corrosion should consider the maximum pit depth for all stirrups (Coronelli [30.1-59]).

The reduction of the cross-sectional area of the shear reinforcement A_{sw} to an effective value of $A_{sw,det}$ can be caused by corrosion of the vertical legs of stirrups, but as well by corrosion of other parts of the stirrup e.g. the upper part as a result of chloride deterioration from the top of the member. This does not necessarily imply a total loss of the effect of the damaged stirrups, because adapted truss mechanisms are possible, where bond between the vertical stirrup legs and the concrete may restore a part of its effect (Figure 30.1-59).

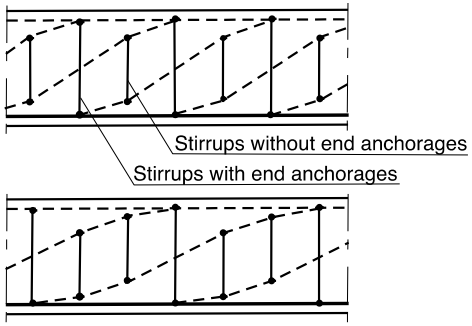


Figure 30.1-59: Truss models incorporating defective stirrups (Regan et al [30.1-60].

In Eq. (30.1-165) $A_{sw,det}$ is the reduced effective cross-section of the shear reinforcement in the area considered, due to corrosion. The value ε_x in Eq. (30.1-40) follows from Eq. (30.1-164).

The upper limit of the shear resistance $V_{Rd,det,max}$ follows from:

$$V_{Rd\,max,det} = k_{c,det}\frac{f_{ck}}{\gamma_c} b_{w,det} \cdot z_{det} \cdot \sin\theta \cdot \cos\theta \qquad (30.1\text{-}166)$$

where:

$f_{ck,det}$ characteristic concrete compressive strength reduced by deterioration;

$b_{w,det}$ is the width of the web, reduced by deterioration;

z_{det} is the inner lever arm of the cross-section considered, taking into account damage of the cover concrete, reducing the contribution of the concrete compression chord to the shear resistance.

$k_{c,det}$ strength reduction factor defined as $k_{c,det} = k_{\varepsilon,det} \cdot \eta_{fc}$.

where:

$$h_{fc,det} = \left(\frac{30}{f_{ck,det}}\right)^{\frac{1}{3}} \leq 1.0 \qquad (30.1\text{-}167)$$

and:

$$k_{\varepsilon,det} = \frac{1}{1.2 + 55\varepsilon_{1,det}} \leq 0.65 \qquad (30.1\text{-}168)$$

where:

$$\varepsilon_{1,det} = \varepsilon_{x,det} + \left(\varepsilon_{x,det} + 0.002\right)\cot^2\theta \qquad (30.1\text{-}169)$$

30.1.11.3.4 Punching shear capacity

The punching shear resistance of non-deteriorated slabs may be determined according to section 30.1.5.3.

The level II approximation in section 30.1.5.4.2 gives a more advanced expression for the calculation of ψ with Eq. (30.1-104).

Here it should be noted that a significant bending moment distribution is considered in design. This, however, requires sufficient ductility of the reinforcement. As shown in section 15.11.2 the ductility can significantly be reduced by pitting corrosion. Therefore the application of this equation should not be carried out without a carefull consideration of the degree of corrosion in the flexural reinforcement.

In chloride-attacked concrete there is a possibility that the upper reinforcement is subject of uniform or pitting corrosion. This may have an effect both on the longitudinal reinforcement and the shear reinforcement.

In case the longitudinal reinforcement is corroded in such a way that due to its volume increase a substantial part of the concrete cover has been lost or longitudinal cracks occur, the bending capacity is influenced by the loss of bond, anchorage and reduction of the rebar cross-section. In case of pitting corrosion this holds true as well for the ductility of the steel and the redistribution capacity of the structure (see section 15.11.2).

With regard to punching shear the anchorage of the stirrups may have been reduced by corrosion of the upper legs of the stirrups (like in the case of shear, described in 30.1.11.3.3 for stirrups in beams). In such a case the contribution of the stirrups to the punching shear resistance may not have been fully lost, because the stirrups are still partially bonded to the concrete. If the vertical legs are not corroded, for a bond stress f_{bd} the stress σ_{wd} in an individual stirrup leg can still reach values between 0 and

$$\sigma_{swd,max} = \frac{f_{bd} \cdot d_s}{0.25 f_w} \qquad (30.1\text{-}170)$$

where d_s is the height of the stirrup and ϕ_w its diameter. $\sigma_{swd,max}$ is the maximum stress that can be activated in the stirrups. The averaged contribution of the stirrups to the punching shear resistance is then:

$$\sigma_{swd,av} = \frac{2 f_{bd} \cdot h_s}{f_w} \qquad (30.1\text{-}171)$$

This limitation should be added to the condition $\sigma_{swd,max} \leq f_{ywd}$ in Eq. (30.196) in case of corroded stirrup upper legs.

With regard to the contribution of the concrete term $V_{Rd,c}$ it should be noted that this term depends on the rotation ψ, through the factor k_ψ in Eq. (30.1-95), which on its turn depends on the longitudinal reinforcement, and as such is sensitive to a loss of cross-section, but also to a possible loss of bond (e.g. by a loss of concrete cover, or the formation of longitudinal cracks).

According to the Level 1 approximation in section 30.1.5.4.1 for a regular flat slab designed according to an elastic analysis without significant

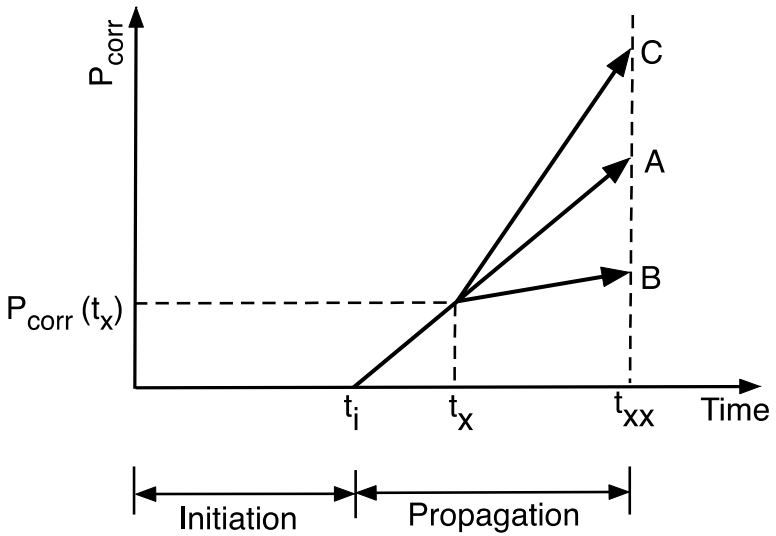


Figure 30.1-60: Velocity of corrosion P_{corr} (reduction of bar diameter per year) and extrapolation to future (Andrade [30.1-61]).

The velocity of corrosion evolves in relation with the climatic/environment conditions but the average annual corrosion is rather constant due to the repetition of the climatic seasons. This trend can change due to cracking of the concrete cover due to the expansive character of the iron oxides. This cracking may induce an acceleration of the corrosion, if the structure is in regular contact with water (line C) or can slow

redistribution of internal forces, a safe estimate of the rotation at failure for a non-deteriorated slab can be determined with Eq. (30.1-104).

In Eq. (30.1-104) the rotation ψ is calculated assuming no bond between longitudinal reinforcement and concrete, so it is basically suitable to estimate the value of ψ in case of loss of bond due to deterioration effects. The loss of cross-section of the reinforcement due to deterioration is introduced through the value of r_s.

30.1.11.3.5 Estimation of the remaining service life of structures with corroded reinforcement.

In the previous sections the actual bearing capacity of structures with corroded reinforcement was determined. If the demands for structural safety at the time of evaluation are still satisfied, it may be wondered how long this situation will last.

In order to be able to make a prediction of the remaining service life it is necessary to know the rate of corrosion in the coming period. Figure 30.1-60 shows the reduction of the rebar diameter $\Delta\phi$ due to corrosion in time. The reduction of the rebar diameter due to corrosion starts at the end of the initiation period t_i. If the reduction of the rebar diameter, measured at the time t_x is equal to $\Delta\phi(t_x)$, the velocity of corrosion P_{corr} can be expressed as:

$$P_{corr} = \Delta\phi(t_x)/(t_x - t_i) \qquad (30.1\text{-}172)$$

The time t_i used in this expression can be determined if the actual position of the carbonation or the chloride threshold front is known, by calculating the number of years necessary to arrive at the external rebar surface.

Knowing the velocity of corrosion, the loss of rebar diameter at a future time t_{xx} follows from:

$$\Delta\phi(t_{xx}) = \Delta\phi(t_x) + P_{corr}/(t_{xx} - t_x) \qquad (30.1\text{-}173)$$

Here it is assumed that the velocity of corrosion is constant, which leads to the linear relation A in Figure 30.1-60. Depending on the variability of the circumstances, like change of environmental conditions, or the effect of cracking, however, an increased (line C) or a decreased (line B) corrosion process is possible.

down the corrosion because in the cracked area drying occurs faster than in the non-cracked zones (line B).

Table 30.1-6a,b give values for the representative velocity of corrosion $P_{corr,REP}$ for various exposure classes [Manual Contecvet [30.1-62])

Table 30.1-6a: Values for $P_{corr,REP}$ [μm/year] for carbonation-induced corrosion [Manual Contecvet [30.1-62]].

Exposure class	Mean value	St.Dev.
Sheltered	2	3
Unsheltered	5	t

Table 30.1-6b: Values for $P_{corr,REP}$ [μm/year] for chloride induced corrosion [Manual Contecvet [30.1-62]].

Exposure class	Mean value	St. Dev.
Wet	4	6
Cyclic dry-wet	30	40
Airborn sea water	30	40
Submerged	4	7
Tidal zone	50	100

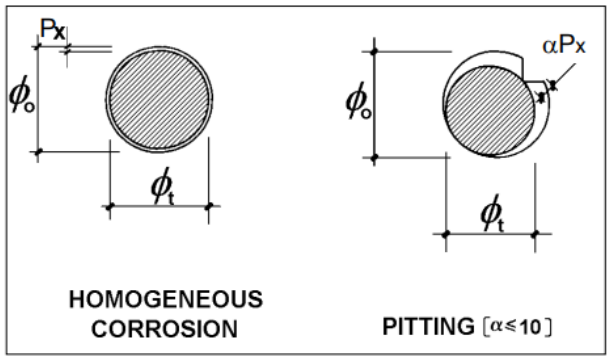


Figure 30.1-61: Cross-sections for homogeneous and pitting corrosion.

The fact that corrosion of reinforcing bars can appear as uniform corrosion or as local corrosion requires different types of characterization. For uniform corrosion the loss of weight of a rebar in time may be considered as a representative parameter to determine the loss of cross-sectional area of the rebar and as such of the tensile resistance. To this aim the reduction factor M_{corr} is used, defined as:

$$M_{corr} = (m_0 - m_1)/m_0 \qquad (30.1\text{-}174)$$

Where m_0 is the mass of the steel bar over the length considered and m_1 is the mass of the steel bar with the same length after corrosion.

For local corrosion the pitting factor α has been defined as the ratio between the maximum pit depth to the average corrosion depth of the steel bar:

$$\alpha = P_{l,max}/P_{l,av} \qquad (30.1\text{-}175)$$

where $P_{l,max}$ is the maximum corrosion pit depth in the whole steel bar and $P_{l,av}$ is the average corrosion depth in the whole steel bar, see Figure 30.1-60.

The pitting factor α can be obtained by geometrical measurement. If not, an expected value of about 10 may be used [Manual Contecvet [30.1-62]].

With the factor α the residual cross section of the steel after pitting corrosion can be determined, Figure 30.1-61.

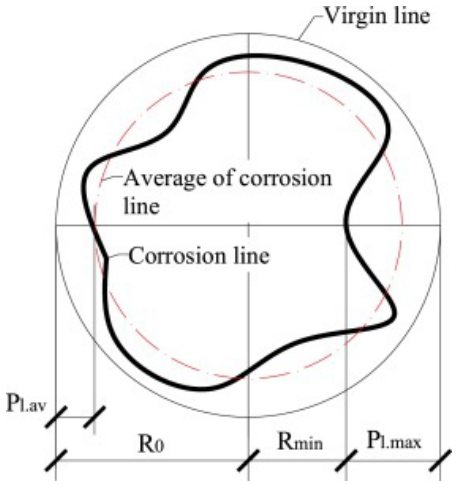


Figure 30.1-62: Cross-sectional schematic representation of corrroded bar.

The type of damage which can occur due to the stress development, sketched before, depends essentially on the boundary conditions. E.g. ice is often formed at a certain depth, where the influence of de-icing salts is exhausted. At a higher depth the temperature has not dropped far enough to cause freezing. Once freezing starts at a certain depth, water is attracted from the vicinity to the ice layer which thus will grow (according to the ice crystallization pressure) and will cause spalling already during the freezing period, but more likely at the start of the thawing period. In that case surface spalling can occur whereas the remaining concrete stays unaffected.

It is as well possible that the volume expansion of the freezing water, restrained by the surrounding concrete, cannot be accommodated in the pore system. Thereby, tensile stresses are initiated and micro and macro cracks are introduced into the concrete body, which leads to a type of severe damage known as internal frost damage. This mechanism affects the compressive strength, tensile strength, modulus of elasticity, fracture energy, and the bond between the reinforcing bars and surrounding concrete in damaged regions.

30.1.11.4 Structures damaged by freeze-thaw effects

30.1.11.4.1 General

There are three principal types of degradation for the freeze-thaw effect, which can appear as:

a) Scaling, which is generally limited to the outer skin, directly in contact with frost and deicing salts, resulting in loss of the outer concrete layer (scaling) which causes a reduction of the cover;

b) Spalling at a certain depth in the cover or at the level of the reinforcement, by which also the cover decreases or even becomes zero.

c) Internal cracking, resulting in a decrease of the concrete strength and an increase in permeability. This degradation form leads to a crack pattern which is sometimes hard to distinguish from the crack pattern caused by ASR.

The major stress development due to the freeze-thaw effect can be summarized as follows:

1. The first increase of (internal) stress occurs during the freezing period, as a result of the expansion of water when it turns into ice: especially the degree of saturation and the permeability of the concrete determine the effect of this mechanism on the formation of damage (surface scaling);
2. The second development of (internal stresses) occurs during thawing: ice shrinks at a decrease of the temperature and when the temperature increases after the freezing period, ice expansion can lead to damage, especially when the freezing period was long, and the ice in the pores had the opportunity to grow (surface spalling);
3. The third generation of (internal) stresses could be called “ice crystallization stresses”: it is the pressure that can build-up in a pore due to ice-growth and the fact that the interface stress between ice and (salt) water or vapour is different.

It is believed that there is a strong correlation between the level of frost damage and the dynamic modulus of elasticity (Petersen et al. [30.1-63], Zandi et al. 2009-1 [30.1-64], Zandi et al. 2009-2 [30.1-65]). Such a correlation can be suggested using the available test data in the literature; see Figure 30.1-63. (This should be regarded as a trend-line, because E_{dyn} is very sensitive to changes in the moisture content, temperature and the existence of microcracks.

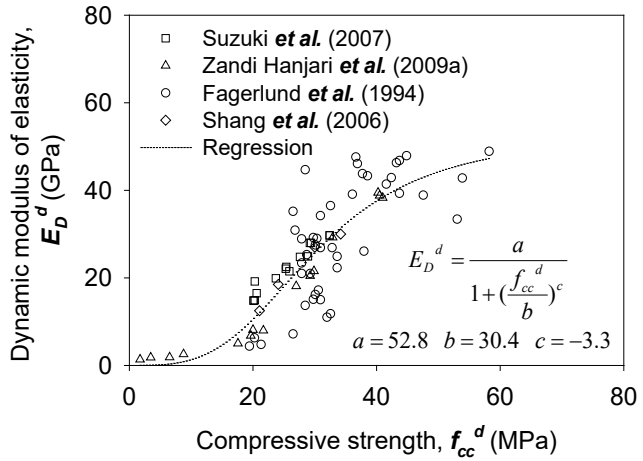


Figure 30.1-63: Correlation between compressive strength and dynamic modulus of elasticity for frost-damaged concrete (for internal frost damage).

30.1.11.4.2 Analyzing the mechanical behaviour and determining the load bearing capacity

A methodology to analyze the mechanical behavior and to determine the remaining load-carrying capacity of reinforced concrete structures affected by frost damage is presented. This methodology is based on the premise that the effect of internal frost damage can be modeled as changes in material and bond properties. Moreover, the effect of surface scaling can be taken into account as a change in geometry, such as a reduction in concrete cross-section when surface scaling leads to cover spalling while the remaining concrete is assumed to be unaffected. The methodology is restricted to the prediction of the mechanical behavior of a structure with an observed amount of damage at a given time; hence, the formation of the damage over time is not included.

In the assessment of the damaged structures, the compressive strength and the dynamic modulus of elasticity, as indicators of damage, must be measured in each individual case. The compressive strength can be evaluated by compression tests on a few drilled cores, supplemented with non-destructive tests to determine the extent of the damaged region. The dynamic modulus of elasticity can be evaluated using either an ultrasonic transit time measurement or a fundamental transverse frequency test.

Frost damage changes the internal structure of concrete, by causing micro and macro cracks, which lengthen the travel time of an ultrasonic wave through the damaged concrete. Thereafter, the effect of frost damage on material properties, such as the stress-strain response in compression, the stress-crack opening relation in tension and the bond-slip behavior, is estimated using the damage indicators.

Finally, the behavior of the damaged structure can be studied with a finite element (FE) analysis. Within this context, suggestions are given on how the effect of frost can be considered as changes in the effective material and bond properties. Reference is made to several tests performed and models developed by several authors, in order to clarify different aspects of the methodology.

30.1.11.4.3 Mechanical properties of concrete damaged by freeze-thaw effect

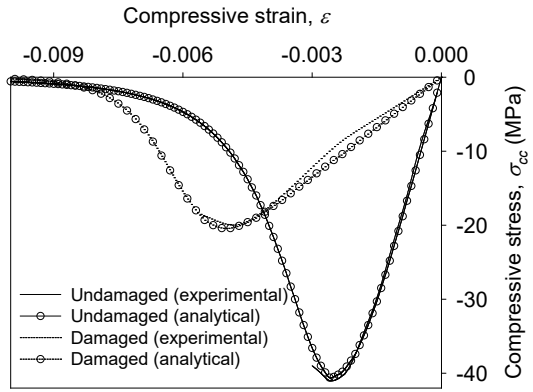


Figure 30.1-64: Stress-strain relation for frost-damaged concrete in compression: the level of frost-damage corresponds to 50% reduction in compressive strength (Zandi et al. 2009-1 [30.1-64]).

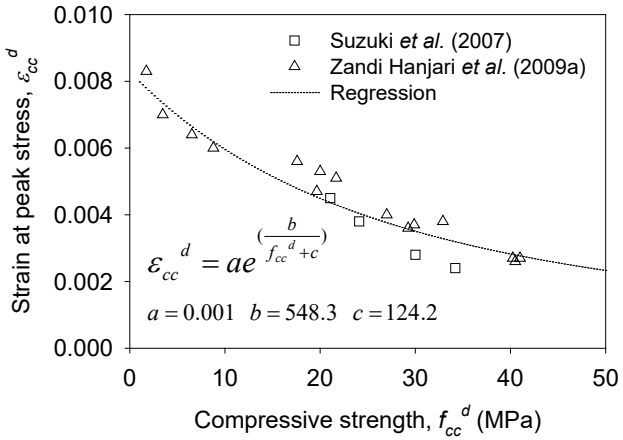


Figure 30.1-65: Estimation of the strain at peak stress, using the damage indicator $f_{cc}^{\ d}$.

30.1.11.4.3.1 Behaviour in compression

The cracks caused by internal frost damage influence the response of the concrete compression. The damaged concrete exhibits a considerably lower initial E_c modulus, a relatively larger strain at the peak stress and a more ductile response in the post-peak behaviour compared with the undamaged concrete, Figure 30.1-64.

A stress-strain relation, based on undamaged concrete with an extension to damaged concrete, using damage parameters is:

$$\sigma_{cc} = f_{cc}^{\ d}\left(\frac{\varepsilon}{\varepsilon_{cc}^{\ d}}\right)\cdot\left(\frac{n}{n-1+\left(\frac{\varepsilon}{\varepsilon_{cc}^{\ d}}\right)^{n\cdot k}}\right) \qquad (30.1\text{-}178)$$

where σ_{cc} and ε_{cc} are the stress and the strain in concrete, respectively, and $\varepsilon_{cc}^{\ d}$ is the strain at the maximum stress. The correction factors, n and k, are defined as follows:

$$n = \frac{E_c^{\ d}}{E_c^{\ d} - E_0^{\ d}} \qquad (30.1\text{-}179)$$

$$k = \begin{cases} 1 & \text{for} \quad 0 < \varepsilon < \varepsilon_{cc}^{\ d} \\ 0.67 + \dfrac{f_{cc}^{\ d}}{62} & \text{for} \quad \varepsilon > \varepsilon_{cc}^{\ d} \end{cases} \qquad (30.1\text{-}180)$$

where $E_c^{\ d}$ and $E_0^{\ d}$ are the secant and tangential modulus of elasticity of the damaged concrete, respectively. The strain at the maximum stress $\varepsilon_{cc}^{\ d}$ can be estimated by means of experimental data, Figure 30.1-64, and the secant modulus is calculated as $f_{cc}^{\ d}/\varepsilon_{cc}^{\ d}$. An approximation of the tangential modulus of elasticity of the damaged concrete based on the dynamic modulus of elasticity is given in Figure 30.1-65, where $E_c^{\ d}$ and $E_0^{\ d}$ are the secant and tangential modulus of elasticity of the damaged concrete, respectively.

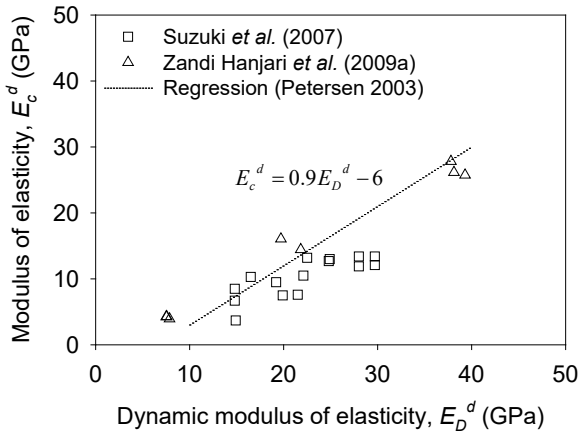


Figure 30.1-66: Estimation of the tangential elastic modulus for the damaged concrete, using the damage indicator E_D^d (Petersen [30.1-66]).

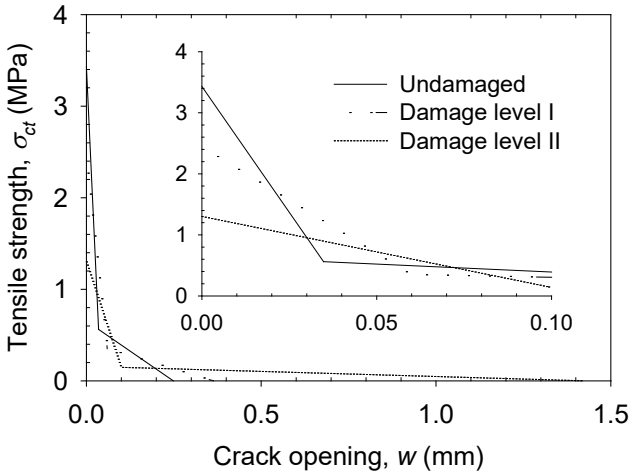


Figure 30.1-67: Bi-linear $\sigma_{ct} - w$ relation: the damage levels I and II correspond to 25 and 50% reduction in compressive strength (Zandi et al. 2009-2 [30.1-65]).

30.1.11.4.3.2 Behaviour in tension

The effect of frost damage on the concrete tensile strength is slightly lower than on the compressive strength.

For an analysis with nonlinear FEM models also the stress-crack opening relation and the fracture energy are important properties, see Figure 30.1-67 and 30.1-68.

The bilinear relation between the tensile stress σ_{ct} and the crack opening w for frost-damaged concrete was estimated by using inverse analysis carried out on wedge splitting test results. The tests were performed on cylinder specimens of 100 x 100 mm, damaged to two levels equivalent to 25% and 50% reduction in compressive strength, reflected by damage levels I and II indicated in Figure 30.1-67.

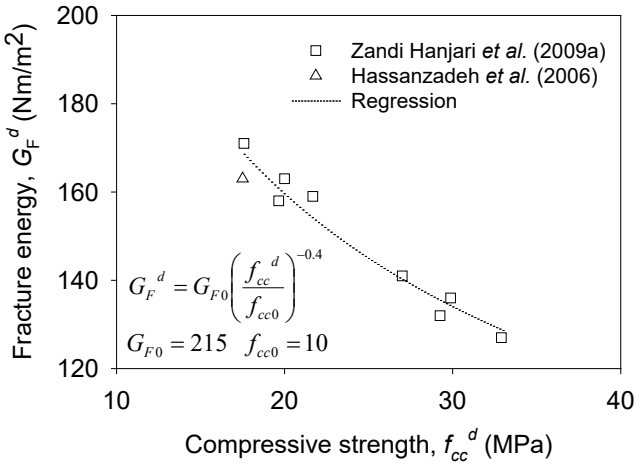
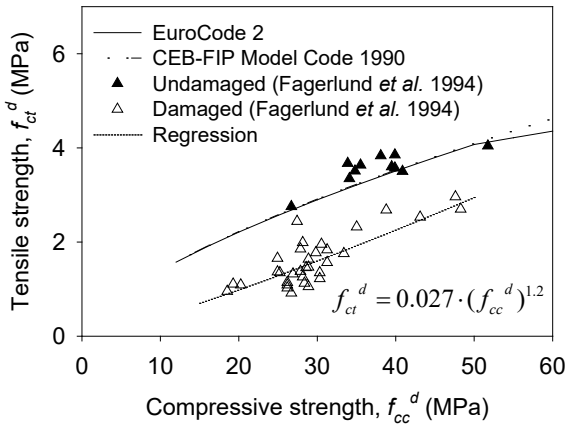


Fig. 30.1-68: Effect of frost damage on fracture energy (relations valid for internal damage) (Zandi et al. 2009-2 [30.1-65]).

(a)



30.1.11.4.3.3 Relation between compressive and tensile strength

In Figure 30.1-69(a) and (b) the relation between the concrete tensile strength and the cube compressive strength for undamaged and damaged specimens are compared. The specimens were subjected to 7 freeze-thaw cycles, each with a duration of two days and a minimum temperature of -18 ^{0}C. Different degrees of saturation were applied.

For undamaged concrete the following relation between concrete cube compressive strength (determined on cubes 150 mm^3) and concrete tensile strength (determined from splitting tests on cubes multiplied with a factor 0.9) are often used.

$$f_{ct} = 0.27 \cdot f_{cc}^{2/3} \qquad (30.1\text{-}181)$$

For concrete damaged by freeze thaw cycles the relation

$$f_{ct} = 0.027 \cdot f_{cc}^{1.2} \qquad (30.1\text{-}182)$$

was found by curve-fitting (see Figure 30.1-69a and b)

(b)

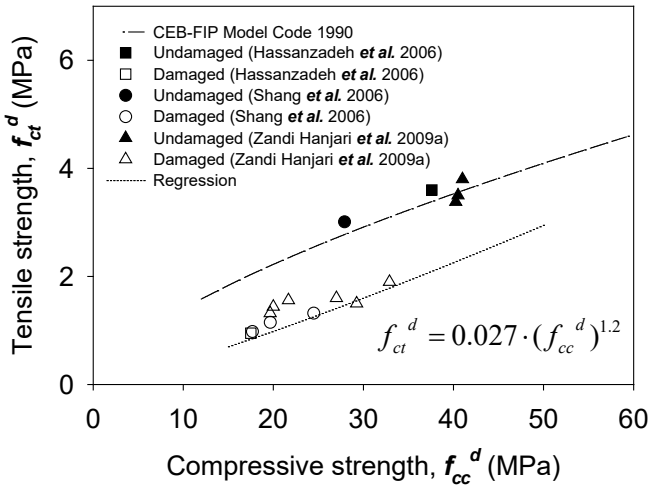


Figure 30.1-69: Compressive strength versus tensile strength.of frost-damaged concrete from (a) Fagerlund [30.1-67] and (b) other experimental investigations.

It is clear that the tensile strength of the damaged concrete is markedly lower than the values that would result from the equation for undamaged concrete.

The relation for the damaged concrete due to freeze-thaw effects turned out to be independent of environmental conditions and only depend on the amount of damage, i.e. the compressive strength of the damaged concrete at a given time.

30.1.11.5 Structures subjected to sulfate attack

Sulfate attack of concrete can be subdivided into internal and external attack.

Internal attack can occur when a sulfate is already present in the concrete in the production (mixing) stage. A special type of internal sulfate attack is delayed ettringite formation. This occurs in concrete subjected to elevated temperatures during production. Delayed ettringite formation causes expansion of the concrete and can lead to serious damage. The word “delayed” indicates that ettringite could not form (as in the usual process) during the cement hydration, because of an overly elevated temperature (ettringite is destroyed over 70^0 C). This can show up several weeks, months or even years after the casting. Damage to the concrete occurs when the ettringite crystals exert an expansive force within the concrete as they grow. The material expands, similarly to what happens in case of an alkali aggregate reaction, creating a network of cracks at the concrete surface. This leads to a loss of strength all over the structure. The strength loss can be estimated on the basis of drilled cores.

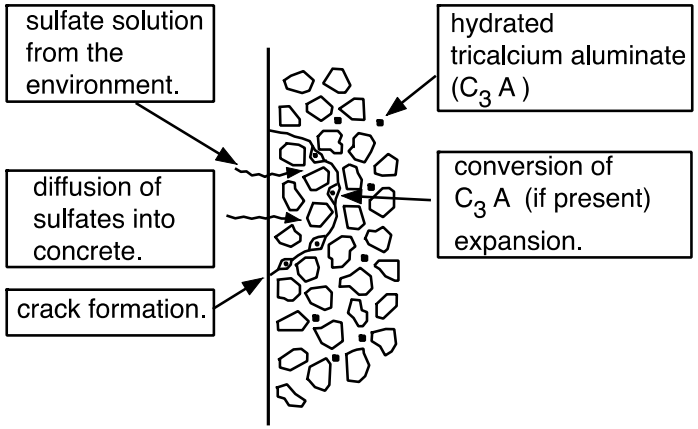


Figure 30.1-70: Schematic representation of external sulfate attack (CEB Bulletin 182 [30.1-68]).

External sulfate attack can only occur if the following conditions apply:

High permeability of the concrete. Not only the water-cement ratio but also the type of cement plays a role. Blast-furnace cement generally leads to a concrete with a lower permeability;

An environment with a high level of sulfates, where also the type of sulfate plays a role: magnesium sulfate is generally more aggressive than other forms of sulfate. Calcium sulfate is usually the least reactive;

The presence of water.

External attack

The most common type of sulfate attack is the external sulfate attack. This reaction typically occurs where water containing dissolved sulfate penetrates the concrete. External sulfate attack can occur when soil and groundwater contain a sufficient concentration of sulfates, which react with the cementitious paste of the concrete. Other sources of sulfate which can cause sulfate attack include seawater and bacterial action in sewers.

Generally, deterioration starts at the contact zone between the concrete and the sulfate containing environment, forming a propagating front. (Fig. 30.1-70). While concrete ahead of this front is essentially normal, concrete behind this front is completely changed in composition and texture, with its integrity significantly lost. The chemical reactions behind the front are highly expansive and will generate stresses exceeding the tensile strength of the concrete, resulting in cracking and disintegration of the concrete. Sulfate attack can be progressive: as long as sulfate ions can penetrate the concrete surface, further reaction may occur throughout the concrete interior.

The reactions may vary in type or severity but commonly include:

Extensive cracking;

Expansion;

Loss of bond between the cement paste and the aggregate.

Those changes result in an overall loss of concrete strength in the area behind the front, Figure 30.1-71.

Sulfate attack can be progressive: as long as sulfate ions can penetrate the concrete surface, a further reaction may occur throughout the concrete.

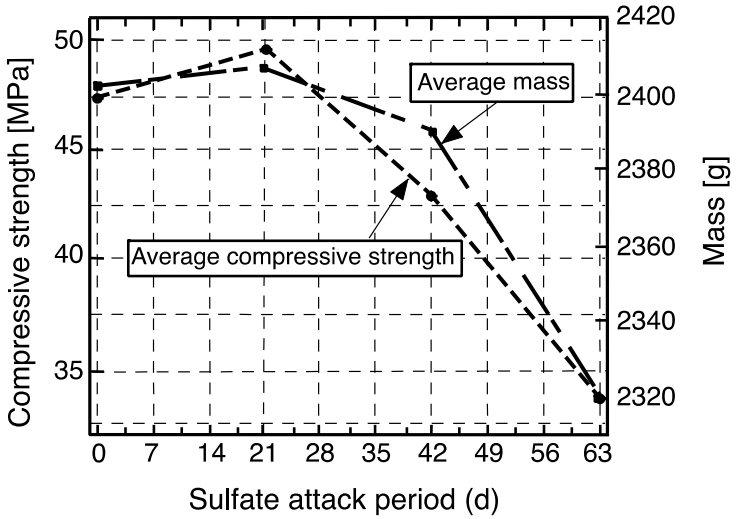


Fig. 30.1-71: Variation of mass and compressive strength in concrete exposed to wetting-drying cycles with sulfate attack.

Figure 30.1-71 shows the variations in mass and compressive strength in concrete exposed to drying-wetting cycles with sulfate attack [Tian & Han [30.1-69]] It was shown that the variation in mass experienced two periods. In the early period of sulfate attack, the mass of concrete specimens increased. This was because the solution reacted with the cement hydration products, filled the material's pores, and increased its mass. In the late period of sulfate attack, the mass continued to decrease and hydration products gradually dissolved from the surface and specimens became loose and sandy. The uniaxial compressive strength of the specimens presented an "up-down" trend as the sulfate attack period progressed. Initally, the hydration expansive products ettringite, gypsum, and sulfate crystals filled the pores to improve the specimens'density and strength. Then, the expansion force of ettringite and gypsum and the crystallization pressure produced sulfate crystallization starting to act on the pore walls. When the pressure of the pore wall exceeded the concrete's tensile strength, many internal microcracks appeared and decreased the uniaxial compressive strength.

The first building codes date from more than 100 years ago. Since the first design recommendations were made, significant technical development has occurred. Improvements in the design recommendations were found to be necessary on the basis of the findings of subsequent / new research, development of more advanced materials and practical experience gained with many types of structures. This has resulted in:

- Better understanding of the behaviour of concrete structures under different types of loading and environmental conditions;
- Adjusted regulations to take into account differences in applied materials (e.g. higher steel strength for reinforcement, ribbed instead of plain bars);
- The introduction of new design criteria, such as those related to durability;
- The introduction of improved structural materials allowing more advanced applications, requiring more refined and comprehensive design rules;
- The need for more appropriate structural safety provisions.

30.1.12 Evaluation of structures with non-compliant detailing

30.1.12.1 General

If the current governing code for structural design is used for the assessment of an existing structure it may turn out that the existing structure satisfies the detailing demands of the original (old) code but not those of the now current governing one. The detailing of such structures is defined as “non-compliant”.

Where non-compliant detailing is identified, the behaviour of the respective region should be investigated, including its bearing resistance and its ductility.

If the detailing of the structure is locally non-compliant, its effect on the overall structural behaviour should also be investigated.

This should result in a decision as to whether the structural safety requirements are met or not.

Various possibilities can occur:

a) Structures designed using former codes may have a satisfactory residual bearing capacity, even in spite of the effect of non-compliant detailing. So it might be that the structure still satisfies the demands expressed in the current governing design recommendation.
b) The structure does not meet the requirements of the current code governing design, but the difference is limited and the structure has demonstrated satisfactory performance over many years and that it functions appropriately. In such a case a slight decrease of the structural safety factors for an existing structure can be accepted, whereas the prospective satisfactory performance over the remaining service life is a point of further consideration. In these circumstances reference should be made to chapter XXX.
c) The structural safety is below an acceptable level. In this case the structure should be strengthened to meet the requirements for future use under the governing loading conditions or closed to further use.

Remarkably experiments on the bearing capacity of structural elements reinforced with plain longitudinal reinforcing steel, subjected to shear or punching, do not show a lower resistance compared to ribbed bars. If the steel is really smooth and is not affected by corrosion, the resistance may be even better than obtained with ribbed or indented reinforcing bars. The explanation for this is that formation of inclined shear cracks is hindered by the absence of bond in the longitudinal reinforcement, so that predominantly bending cracks occur.

Whether a critical situation with regard to fatigue can occur depends not only on the bending radius of the steel bars, but as well on the yield strength of the reinforcing steel. In the past often steel with lower yield strengths was used (e.g. 200 MPa), which had its influence on the maximum value of the loads causing the variation in the steel stress.

For the design of detailing regions, the principle of strut & tie design was introduced in the *fib* Model Code 1990. Since then, this method of designing "discontinuity regions" has been adopted in most national codes. When designers became familiar with this way of designing, the designs became more rational than before. However, many discontinuity regions designed in the period before the introduction of the strut and tie method show suboptimal or even unsuitable solutions.

30.1.12.2 Non-compliant details influencing structural safety

Poorly conceived structural details may have a direct influence on structural behaviour and, consequently, on the applicability of calculation methods. Inappropriate structural details primarily have an effect on bond and fatigue strength of reinforcing bars, ductility, fire resistance and/or durability.

Non-compliant details to be considered include:

- Smooth or weakly ribbed reinforcement associated with lower bond strength, are inferior to indented steel with regard to crack width control. Furthermore smooth an weakly ribbed reinforcement can result in larger deflections of structures than obtained with indented steel under the same loading conditions. The lower bond can lead as well to a reduction of the flexural bearing capacity, However, lack of bond of the longitudinal steel generally does not lead to a decrease of shear or punching shear capacity in members without shear reinforcement;
- Reinforcing bars with low ductility properties, which can lead to reduced ductility of the structure;
- A small concrete cover which may result in reduced bond strength in the longitudinal reinforcing bars (see section 20.3.4), but also an increased ductility at the structural level;
- Small bar spacings or a high number of bar layers, which can result in reduced bond strength and a higher risk of concrete cover spalling over large surface areas (see section 20.3.4);
- An excessively small bending radius of reinforcing bars in the anchorage region, which may result in reduced fatigue strength and increased transverse tensile forces;
- An excessively pronounced staggering of longitudinal reinforcement, which can lead to loss of bending resistance of structural members (see section 2.2.6);
- Insufficient anchorage of transverse reinforcement in flexural tensile zones, which can lead to loss of anchorage capacity;
- Shear reinforcement not enclosing the longitudinal reinforcement reducing the shear capacity;
- Shear reinforcement following the shape of the outer surface of structural members with variable width;
- Indirect supports with no, or insufficient, suspension reinforcement;
- Dapped-end beams ("half-joints") with insufficient anchorage of reinforcement and/or insufficient suspension reinforcement.

An example of the use of an expression based on realistic behaviour is the determination of interface shear resistance (Figure. 30.1-72) by the basic equation:

$$\tau_u = \tau_a + \mu \cdot (\rho \cdot k_1 \cdot f_y + \sigma_n) + k_2 \cdot \rho \cdot \sqrt{f_y \cdot f_{cc}} \qquad (30.1\text{-}183)$$

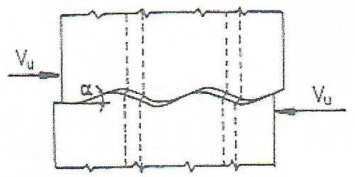


Figure 30.1-72: Transmission of interface shear across a rough concrete interface.

Eq. (30.1-183) is a simplified version of Eq. (30.1-83) in section 30.1.3.8.3.

The term τ_a in Eq. (30.1-183) represents the contribution of adhesion to the interface shear capacity.

- Supports with insufficient sliding capacity due to unsuitable design.
- Insufficient reinforcement which does not accord with minimum reinforcement requirements in bending, shear or punching.

30.1.12.3 Failure mode assessment

The structural safety assessment should establish the governing failure mode. It should be evaluated, in particular, if local failure will be announced by excessive deformation.

Potential brittle failure modes are, in general, related to shear failure, e.g. punching shear, one-way shear in slabs without transverse reinforcement and failure of compression fields in girders. It may be taken into account that shear forces leading to shear failure may not be able to develop because of the limitations provided by the flexural load capacity of the structure or parts of it.

Determination of shear forces associated with the flexural failure load should consider the overstrength of materials in the determination of flexural resistances, and should verify that there is sufficient plastic deformation capacity for attaining the associated flexural failure loads.

For the assessment of the structural behaviour of areas with non-compliant details, appropriate models should be used which transparently reflect the role of all parameters involved. Such models can also reveal the influence of the non-compliant aspects.

Empirical models, derived from test results, should be handled with care, because the role of single parameters influencing the behaviour is mostly not clearly recognizable.

In order to be able to determine the residual capacity of structural members according to this Model Code behavioural models with different Levels of Approximation are described. The higher levels of approximation can indeed reveal a “hidden safety margin” but should also be handled with care if they are used for structures with inherent weaknesses, which may limit the residual capacity. Also models with a high level of approximation have mostly been calibrated against test on elements without non-compliant results, in which the governing design criteria are fulfilled (like the presence of minimum reinforcement).

The second term represents the shear stress that can be transmitted across the concrete interfaces due to the clamping action of the reinforcing steel to the interface (reinforcement counteracts opening and sliding of the rough interfaces).

The third term is the contribution of dowel action of the steel bars. The expression was derived for ribbed steel bars.

For a detailed description of the symbols in Eq. (30.1-183) reference should be made to section 30.1.3.8.

If the ribbed steel were to be replaced by plain steel of the same diameter, the first and the third term remain the same. Only the clamping effect of the steel is reduced, which affects the second term.

Assuming that the bond of the plain steel bars is 50% of that of ribbed steel bars, the interface shear capacity of a similar detail with plain bars can be determined by reducing the term $\rho \cdot k_1 \cdot f_y$ in Eq. (30.1-183) to 50% of its value for deformed bars.

For further information on the use of strut and tie models in assessment, reference is made to section 30.1.6.6.

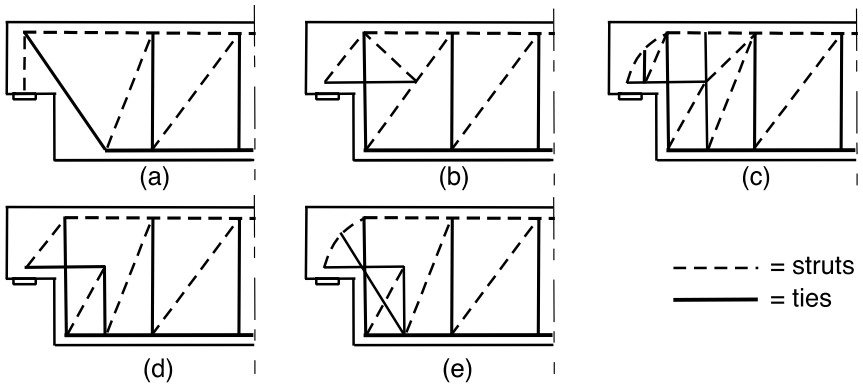


Figure 30.1-73: Strut and tie models for reinforced half-joints: (a) diagonal model, (b), (c) and (d) orthogonal models and (e) combination model. (Desnerck et al, [30.1-70]).

A relatively simple method of assessment with a wide applicability is the strut and tie method. For the design of new structures, this method is very suitable for optimum detailing of load-introduction regions (i.e. regions where loads are applied).

Existing structures to be assessed often show sub-optimum reinforcement detailing in those regions. In that case, the strut and tie method can be used as well, but now looking for alternative ways for the transmission of forces tailored to the given reinforcement configuration. An example is given in Fig. 30.1-73.

For more complicated cases a NLFEM analysis may be appropriate. This method is especially powerful when combined with experiments. For the application of NLFEM analysis reference is made to section 30.11.

Finally, also proof loading a structure with non-compliant details is an option: it can be considered in the case that available behavioural models, in combination with the relevant material properties may provide a reliable estimation of the structural bearing capacity.

The occurrence of corrosion induced damage to reinforcing bars has led to a continuous increase of the required concrete cover in codes over many years.

Not fulfilling requirement with respect to cover thickness or reinforcement bar diameter and/or spacing will influence the results of the SLS assessment as given in section 30.5.4, however it will not always lead to actual serviceability related problems. If a structure has demonstrated satisfactory performance over many years with respect to serviceability (no signs of corrosion etc) the need for further investigation of the actual structure depends on the foreseen remaining life time and mainly the intended use of the structure. If for instance loading conditions do not change, further detailed investigation might not be required.

30.1.12.4 Non-compliant details influencing service life

The most frequently observed non-compliant detail in concrete structures is an insufficient thickness of concrete cover. In the past the protecting effect of the concrete cover has been greatly overestimated during many years.

To assess the state of the structure the actual concrete cover should be measured. In case of doubt about durability, drilled cores may give further information.

If site-based observations show indications of possible serviceability problems drilled concrete cores can also be used for the determination of chloride profiles (i.e. as associated with the ingress of chlorides into the concrete members). Also it is important to consider the effects of carbonation (eventually in combination with chlorides) at the depth of the reinforcement, which is where the corrosion in the reinforcing bars /prestressing strands takes place.

Additional information can be obtained from water and air penetration tests and electric resistivity tests. Based on such data a prediction can be made of the remaining service life. For further background information on this subject reference is made to Chapters XX.

30.1.13 ULS verification of interventions

30.1.13.1 Interventions to increase the flexural resistance

[Not used at this time, to be updated in future edition]

30.1.13.2 Interventions to increase the load carrying capacity of columns

[Not used at this time, to be updated in future edition]

30.1.13.3 Interventions to increase the shear resistance

Post-installed shear reinforcement consists to add bars, straps, plates of different materials. This method included external and internal reinforcement, bonded and unbonded to the concrete. This type of reinforcement plays a similar role than stirrups to carry shear. They have to be properly anchored to the structure with, for example, mechanical anchorages, bonding adhesive or both. This method can increase both member shear capacity and ductility.

Flexural reinforcement consists to add reinforcement such as steel or FRP bars or plates mechanically anchored or bonded to the structure. The added reinforcement can increase flexural stiffness as well as member flexural lever arm, which can increase flexural and shear capacity.

Enlarging member’s cross-section consists to cast additional concrete to increase member dimensions. Added concrete may have improved mechanical properties. Stirrups and flexural reinforcement can be added into the enlarged cross-section to improve shear and flexural capacity.

Other methods can be used to reduce shear load, such as reducing the member span.

30.1.13.3.1 General

Different methods can be used to increase the capacity of existing structures for one-way shear. These methods typically include the following:

a) Post-installed shear reinforcement;
b) Flexural reinforcement;
c) Enlarging member's cross-section;
d) Methods can take advantage of prestressing.

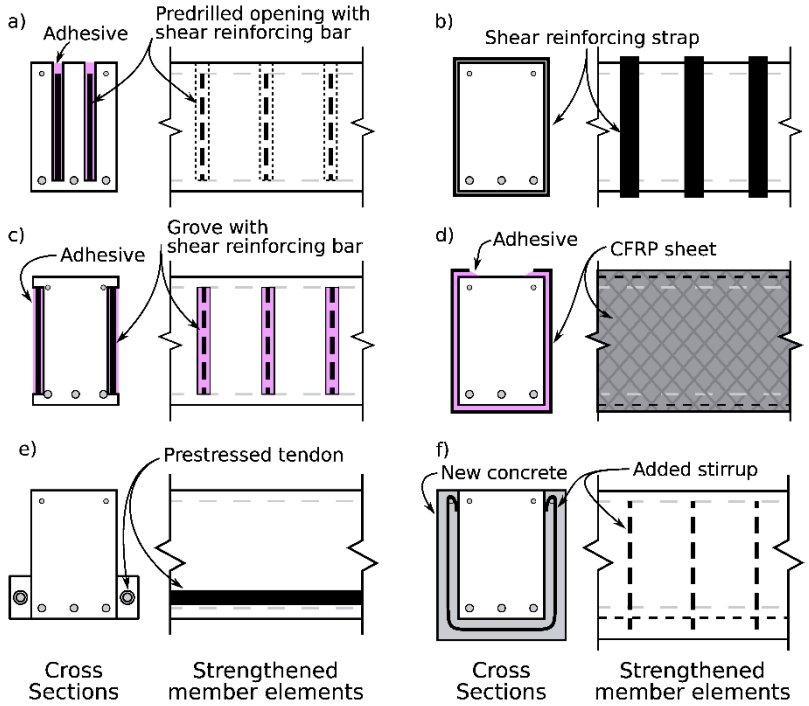


Figure 30.1-74: Examples of strengthening methods of an existing member for one way shear: (a) post-installed bonded shear reinforcing bars; (b) post-installed unbonded shear reinforcing straps; (c) post-installed near surface mounted bars; (d) post-Installed bonded CFRP sheets; (e) flexural reinforcement with prestressing; (f) member widening with the stirrups.

Reference: (Adhikary & Mutsuyoshi [30.1-71]; Bastien-Masse & Bruhweiler [30.1-72]; Chaallal et al. [30.1-73]; Colalillo & Sheikh [30.1-74]; Dias & Barros [30.1-75]; El-Shafiey & Atta [30.1-76]; Fiset et al. 2016 [30.1-77] Fiset et al. 2019 [30.1-78]; Mofidi et al. [30.1-79]).

Designers must consider the effects of the adopted method on the deficient member as well as the whole structure. Designers must also pay attention to the strengthened member ductility and potential failure mode which can switch from ductile to brittle.

30.1.13.3.2 Design of shear strengthening

Generally, provisions given in 7.3.3 applied for members strengthened for one-way shear, but each case shall be investigated in a detailed manner. Some

Attention must also be paid to the compatibility between materials and the effects of shrinkage, creep and thermal strain.

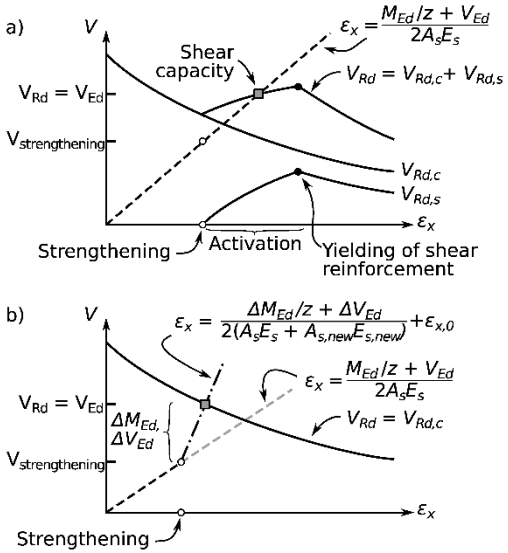
Additional comments:

To a):

- According to the adopted method, interaction between stirrups and post-installed shear reinforcement may reduce their contributions to shear capacity $V_{Rd,s}$;
- In case of bonded reinforcement without mechanical anchorage, bond stress shall be verified;
- In case of external bonded reinforcement or near-surface mounted reinforcement, concrete spalling or cover splitting shall be considered;
- Size effect in shear may need to be considered for deep members with low amount of bonded shear reinforcement installed at a large spacing or members with unbonded shear reinforcement.

To b):

- Special attention should be addressed to the potential reduction of the deformation capacity of the member and the load distribution in statically indeterminate structures;



general principles listed below may need to be considered regarding the employed shear strengthening method:

a) Post-Installed shear reinforcement:
   - Shear resistance provided by shear reinforcement in Eq. (7.3-11) may consider both the stirrups and post-installed shear reinforcement;
   - The shear resistance attributed to post-installed shear reinforcement shall consider their activation and the member deformation at the time of strengthening as illustrated in Figure 30.1-75;
   - Bond between concrete and post-installed reinforcement, anchorage stiffness and anchorage capacity shall be considered in the design;
   - Maximum longitudinal spacing allowed for post-installed shear reinforcement shall consider the expected failure mode;
   - Minimum amount of shear reinforcement given in Eq. (7.3-22) may be adapted to consider both the stirrups and post-installed shear reinforcement contributions;

b) Flexural strengthening:
   - Calculation of the longitudinal strain has to consider the sequence of loading and strengthening as illustrated in Figure 30.1-75;
   - Shear resistance calculation may be adapted to consider the increase of shear depth, z, caused by flexural strengthening;

c) Member widening:
   - Materials compatibility, creep and shrinkage shall be taken into account in the design;
   - Shear resistance determined from Eq. (7.3-11) may be adapted to consider the mechanical properties of each material;
   - The calculation of the longitudinal strain shall consider the loading sequence for each material (existing and new part of the section);

d) Prestressing:
   - Activation of post-installed shear reinforcement may consider the effect of prestressing when applied;
   - Calculation of the strengthened members properties, cracking and deformation may be adapted to consider prestressing;

Figure 30.1-75: The effect of the member deformation at the time of strengthening and the activation of shear reinforcement for a member with (a) post-installed shear reinforcement and (b) flexural strengthening.

For strengthening of existing slabs against punching, some methods allow increasing the strength but reduce the deformation capacity (more sensitive to imposed deformations) while others enhance both the strength and deformation capacity.

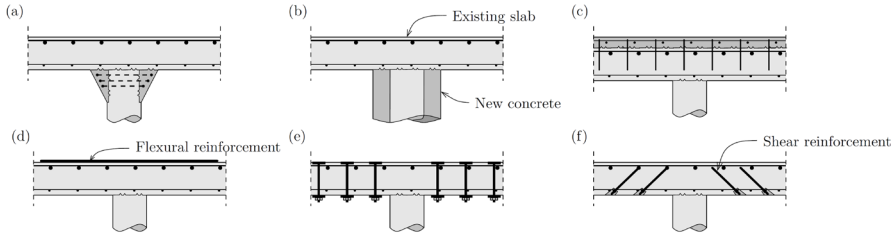


Figure 30.1-76: Strengthening of existing slabs against punching shear: (a) concreting or installing of steel-precast capital;(b) widening of column; (c) addition of upper concrete layer; (d) addition of glued flexural reinforcement; (e) post-installed shear reinforcement with mechanical anchorage; and (f) bonded post-installed shear reinforcement.

A description of possible strengthening techniques can be found in:Fernandez Ruis et al [30.1-80], Lapi et al [30.1-81].

The reduction of the acting shear force corresponding to the vertical component of longitudinal prestressing force may be accounted for;

30.1.13.4 Interventions to increase the punching shear resistance of slab-column connections

30.1.13.4.1 General

For existing structures showing insufficient punching shear strength, different approaches can be used to enhance the capacity of slab-column connections. These methods normally comprise:

a) Enlargement of the supporting area (as column capitals or widening, Figure 30.1-76a-b);
b) Post-installing shear reinforcement (bars with mechanical anchorage or bonded to the slab, Figure 30.1-76e-f);
c) Increasing the flexural capacity (increase of effective depth and/or of flexural reinforcement ratio, Figure 30.1-76c-d);
d) Post-tensioning techniques (flat slab or local in the connection area).

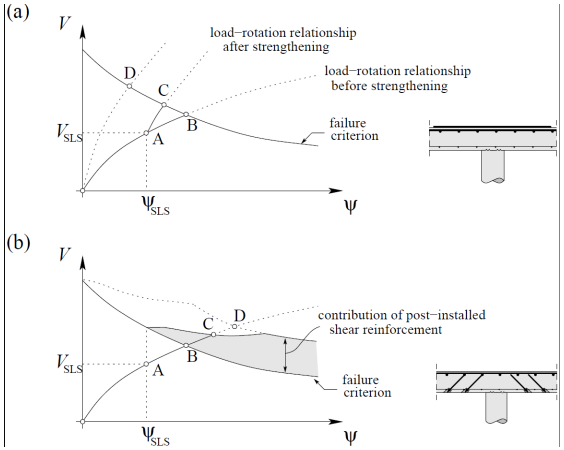


Figure 30.1-77: Modified load-rotation relationships and failure criteria for a) case of increased flexural capacity and b) post-installed shear reinforcement.

Additional comments:

to a) Anchorage of flexural reinforcement (distance to new control perimeter, 30.13.5.3) can become governing and shall be verified.

to b) In case of bonded post-installed reinforcement without mechanical anchorage, the bond stress shall be verified;

to c) - special attention should be addressed to the potential reduction of the deformation capacity of the slab-column connection after strengthening.
- the increase of strength observed in tests on isolated specimens might not be obtained in a continuous flat slab. A consistent design accounting for the global bending response of the system is required. A detailed description of this limitation can be found in: Faria et al. 2014 [30.1-81].

Particular applications can be found in Faria et al. 2011 [30.1-82].

30.1.13.4.2 Design of strengthening against punching

Each case shall be investigated in a detailed and tailored manner. Some general principles for design of the strengthening systems listed in 30.1.10.4.1 are given below:

a) Enlargement of the supporting area:
   - the load-rotation relationship according to 30.1.5.4 may be assumed as unchanged;
   - the resistance according to Eq. (30.1-98) may be calculated on the basis of the augmented control perimeter increased according to geometry of the enlarged supporting area

b) Post-installed shear reinforcement:
   - the load-rotation relationship according to 30.1.5.4 may be assumed as unchanged;
   - the resistance according to Eq. (30.1-98) may be calculated accounting for the contribution of the post-installed reinforcement (activation of reinforcement, punching outside of the shear-reinforced area and maximum punching capacity);
   - For the activation of the post-installed reinforcement, Eq. (30.1-104) shall be adapted to account for the level of rotation ψ_{SLS} at the moment of strengthening (Figure 30.1-55b);

c) Increase of flexural capacity
   - the load-rotation relationship according to 30.1.5.4 may be adapted accounting for the increased flexural capacity. In this case, the increased stiffness shall be accounted for starting from the level of rotation ψ_{SLS} at the moment of strengthening as shown in Figure 30.1-55a.
   - in case of increase of the effective depth (based on the centroid of of existing and additional reinforcement), the resistance according to Eq. (30.1-98) may be adapted;

d) Post-tensioning techniques
   - The reduction of the acting shear force corresponding to the vertical component of the prestressing force at control perimeter may be accounted for;
   - The influence of additional prestressing on the flexural behaviour of the slab-column connection may be considered according to Eq. (30.1-117).

For all strengthening solutions, the maximum punching shear resistance according to 30.3.5.3 shall be calculated with a parameter k_{sys} verified with tests or based on technical approvals.

References

[30.1-1] Malja et. Al. Malja X, Ebrahimi, Motlagh HR, Fernández Ruiz M, Muttoni A. Influence of sustained loading on resistance and deformation capacity of reinforced concrete members in compression. Structural Concrete. 2022. https://doi.org/10.1002/suco.202200571

[30.1-2] Rüsch H. Rüsch H., Researches toward a general flexural theory for structural concrete, Journal of the ACI, July 1960, 57-1, pp. 1-28

[30.1-3] Mattock et.al. Mattock A. H., Kriz L., B., Hognestad,, Rectangular stress distribution in ultimate strength design, Journal of the ACI, February 1961, 57-43, pp. 875-928

[30.1-4] Moccia et al Moccia, F., Yu, Q., Fernandez Ruiz, M., Muttoni, A., Concrete compressive strength: From material characterization to a structural value. *Structural Concrete*, 2020, *22(1):*

[30.1-5] Marti Marti, P. , Design of Concrete Slabs for Transverse Shear. *ACI Structural Journal*, 1990.

[30.1-6] Sigrist et al Sigrist, V., Bentz, E., Fernandez Ruiz, M., Foster, S., Muttoni, A., Background to the *fib* Model Code 2010 Shear Provisions – Part I: Beams and Slabs. *Structural Concrete*, 2010, 14.

[30.1-7] Cavagnis et al. Cavagnis, F., Fernandez Ruiz, M., Muttoni, A , An analysis of the shear transfer actions in reinforced concrete members without transverse reinforcement based on refined experimental measurements. *Structural Concrete*, 2018, 19:49 – 64.

[30.1-8] Bentz and Collins Bentz, E. C., Collins, M. P.: Development of the 2004 CSA A23.3 Shear Provisions for Reinforced Concrete. Canadian Journal of Civil Engineering, vol. 33, No. 5, 2006, pp. 521–534.

[30.1-9] Sherwood et al Sherwood, E.G., Bentz, E., Collins, M. P., Evaluation of Shear Design Methods for Large, Lightly-reinforced Concrete beams. *Solid Mechanics and its Applications*, 2006, 140: 153 – 164.

[30.1-10] FprEN 1992-1-1:2023 CEN, “Eurocode 2 - Design of Concrete Structures - Part 1-1: General Rules - Rules for Buildings, bridges and civil engineering structures (Final Draft FprEN 1992-1-1:2023)”. CEN European Committee for Standardization Brussels, Belgium, 2023, 405 p.

[30.1-11] Cavagnis et al. F. Cavagnis, J. T. Simões, M. Fernández Ruiz, A. Muttoni, “Shear strength of members without transverse reinforcement based on development of critical shear crack”, ACI Structural Journal, vol. 117, no. 1, 2020, p. 103-118.

[30.1-12] Angelakos Further Details Required

[30.1-13] Choi et al Choi et al., 2021, Further Details Required

[30.1-14] Monney et al. Monney F. et al.: Influence of amount of shear reinforcement and its post-yield response on the shear resistance of reinforced concrete members. Structural Concrete. (2022). 1–33.

[30.1-15] Pejatovic et al. Pejatovic M, Ruiz MF, Muttoni A. Design of slender and squat reinforced concrete members with shear reinforcement. Structural Concrete. 2022. 1–17.

[30.1-16] *fib* Bulletin 65 & 66. Model Code 2010 – Final draft, Volume 1 & 2.

[30.1-17] EN 14651:2005. Test method for metallic fibrered concrete – Measuring the flexural tensile strength (limit of proportionality (LOP), residual).

[30.1-18] Randl Randl, N., Design recommendations for interface shear transfer in *fib* Model Code 2010. *Structural Concrete*, 2013, 14 (3).

[30.1-19] Kaufmann Kaufmann, N., Das Sandflächenverfahren. *Strassenbautechnik*, 1971, 3.

[30.1-20] Muttoni et al. Muttoni, A., Fernandez Ruiz, M., Bentz, E., Foster, S., Sigrist, V. , Background to *fib* Model Code 2010 shear provisions – part II: punching shear. 2013, *Structural Concrete*, 14(3): 204 – 214.

[30.1-21] dos Santos et al. dos Santos JB, Muttoni A, de Melo GS. Enhancement of the punching shear verification of slabs with openings. Structural Concrete. 2022. 1–18.

[30.1-22] Muttoni Muttoni A., Punching shear strength of reinforced concrete slabs without transverse reinforcement, ACI Structural Journal, V. 105, N° 4, 2008, p. 440-450

[30.1-23] Yu et al. Yu Q., Simões J.T., Muttoni A., Model uncertainties and partial safety factors of strain-based approaches for structural concrete: example of punching shear, *Engineering structures*, 2023.

[30.1-24] Muttoni et al. Muttoni A, Fernández Ruiz M, Simões JT. The theoretical principles of the critical shear crack theory for punching shear failures and derivation of consistent closed-form design expressions. *Structural Concrete*. 2017;1–17

[30.1-25] Fernández et al. Fernández Ruiz M., Muttoni A., Applications of the critical shear crack theory to punching of R/C slabs with transverse reinforcement, *ACI Structural Journal*, Vol. 106 N° 4, pp. 485-494, USA, 2009

[30.1-26] Hernández et al. Hernández Fraile D., Simões J. T., Fernández Ruiz M., Muttoni A., A mechanical approach for the maximum punching resistance of shear-reinforced slab-column connections, *Proceedings of the fib Symposium*, Lisbon, 2021, pp. 1628-1639.

[30.1-27] Einpaul et al Einpaul, J., Ospina, C., Fernandez Ruiz, M., Muttoni, A., Punching Shear Capacity of Continuous Slabs, 2016. *ACI Structural Journal*, 113(4).

[30.1-28] Belletti et al B. Belletti, M. Pimentel, M. Scolari, J.t C. Walraven, “Safety assessment of punching shear failure according to the level of approximation approach”, *Structural Concrete*, vol. 46, 2015, pp. 366-380

[30.1-29] Belletti et al B. Belletti, A. Muttoni, S. Ravasini, F .Vecchi, “Parametric analysis on punching shear resistance of reinforced-concrete continuous slabs”, *Magazine of Concrete Research*, 2019, p. 1-14

[30.1-30] Belletti et al B. Belletti, J. C. Walraven, F. Trapani, “Evaluation of compressive membrane action effects on punching shear resistance of reinforced concrete slabs,” *Engineering Structures*, vol. 95, 2015, pp. 25-39.

[30.1-31] Shu et al J. Shu, B. Belletti, A. Muttoni, M. Scolari, M. Plos, “Internal force distribution in RC slabs subjected to punching shear”, *Engineering Structures* 153, 2017, p. 766–781

[30.1-32]	Abu-Salma et al	D. Abu-Salma, R. L. Vollum, L. Macorini, "Design of biaxially loaded external slab column connections", Engineering Structures, vol. 249, 2021, 113326, https://doi.org/10.1016/j.engstruct.2021.113326
[30.1-33]	Abu-Salma et al	D. Abu-Salma, R. L. Vollum, L. Macorini, "Modelling punching shear failure at edge slab-column connections by means of nonlinear joint elements," Structures, vol. 34, 2021, pp. 630-652
[30.1-34]	Setiawan et al	A. Setiawan, R. L. Vollum, L. Macorini, B. A. Izzuddin, "Efficient 3D modelling of punching shear failure at slab-column connections by means of nonlinear joint elements," *Engineering Structures*, vol. 197, 2019, 109372, https://doi.org/10.1016/j.engstruct.2019.109372
[30.1-35]	Setiawan et al	A. Setiawan, R. L. Vollum, L. Macorini, B. A. Izzuddin, "Numerical modelling of punching shear failure of reinforced concrete flat slabs with shear reinforcement," *Magazine of Concrete Research*, vol. 73, no. 23, 2021, pp. 1205-1224
[30.1-36]	Fernández et al	Fernández Ruiz M., Mirzaei Y., Muttoni A., Post-Punching Behavior of Flat Slabs, *ACI Structural Journal*, V. 110, No. 5, September-October 2013, pp. 801-811.
[30.1-37]	Maya et al.	Maya L.F. , Fernández Ruiz M., Muttoni A., Foster S.J., Punching shear strength of steel fibre reinforced concrete slabs, *Engineering structures*, Vol. 40 n° 3, 2012, pp. 83-94,
[30.1-38]	Minelli et al.	To be completed
[30.1-39]	*fib* Bulletin 100.	*fib* Bulletin 100, Design and assessment with strut-and-tie models and stress fields: from simple calculations to detailed numerical analysis, fib, 2021, ISBN 978-2-88394-151-9, p. 235
[30.1-40]	Schertenleib et al.	Schertenleib P., Muttoni A., Schwartz J., "Members with compressive axal forces" (in French and German), chapter of "Introduction to the Swiss Code SIA 262", Swiss Engineers and Architects Documentation, D0182, Zurich, 2003, p. 67-77
[30.1-41]	Foster et al	Foster, S., Marti, P. Mojsilovic, N., Design of Reinforced Concrete Solids Using Stress Analysis. *ACI Structural Journal*, 2003, 100(6): 758 – 764.
[30.1-42]	Martinez	MArtinex, G.A., Effects of alkali-silica reaction on the shear behaviour of reinforced concrete beams without transverse reinforcement, *PhD thesis,* University of Texas at Austin, 2016.
[30.1-43]	Chana & Korobokois	Chana, P. S., Korobokois, G. A., Bond strength of reinforcement in concrete affected by alkali silica reaction, Phase ii, Crowthorne, Transport and Road Research Laboratory (TRRL), Contractor Report 233, 1991.
[30.1-44]	Bayrak	Bayrak 1999, Further Details Required
[30.1-45]	Gianini et al	Gianini et al, 2011., Further Details Required
[30.1-46]	MPR-2473.	Seabrook Station – Implications of Large-Scale Test Program Results on Reinforced Concrete Affected by Alkali-Silica Reaction, *Quality Assurance Document*, 2016.
[30.1-47]	Ahmed et al.	Ahmed et al., 1998., The static and fatigue strength of reinforced concrete beams affected by alkali-silica reaction, ACI Materials Journal 95(4), 376-387, 1998
[30.1-48]	Den Uijl et al	Den Uijl, J.A., Kaptijn, N., Walraven, J.C., "Shear capacity of flat slab bridges affected by ASR", Proceedings of the 11th International Conference on ASR in Concrete, Quebec City, Canada, 2000, pp. 1129-1138.

[30.1-49] Ng & Clark — Ng and Clark 1992, Further Details Required

[30.1-50] Deschenes et al. 2009 — Deschenes et al. 2009, Further Details Required

[30.1-51] Chana — Chana, P. S., Bond strength of reinforcement in concrete affected by alkali silica reaction, *TRRL Contractor Report,* CR141, 1989.

[30.1-52] Zhychkovska & Sheikh — Zhychkovska O. and Sheikh S., Bond and Anchorage Behavior of Alkali-Silica Reactive Concrete, 2021, ACI Structural Journal, 118 (3), 279-293

[30.1-53] Miyagawa et al — Miyagawa, T., Seto, K, Sasaki, K, Mikata, Y., Fracture of Reinforcing Steels in Concrete Structures Damaged by Alkali-Silica Reaction-Field Survey, Mechanism and Maintenance, *Journal of Advanced Concrete Technology*, 4(3): 339 – 355, October 2006.

[30.1-54] Webb — Webb, 2011, Further Details Required

[30.1-55] CEB Bulletin 189. — CEB-FIP Model Code 1990 – Supplementary Documents for the 1st Pre-draft.

[30.1-56] Blagojevic — Blagojevic, A., The influence of cracks on the durability and service life of reinforced concrete structures in relation to chloride-induced corrosion, PhD Thesis, TU Delft, 2016

[30.1-57] Blomfors et al — Blomfors, M., Zandi, K., Lundgren, K., Coronelli, Engineering bond model for corroded reinforcement, *Engineering Structures*, 156: 394 – 410, 2018.

[30.1-58] *fib* Bulletin YY — *fib* Bulletin: Modelling the structural performance of existing structures concrete, TG3.2, Bulletin under preparation

[30.1-59] Coronelli — Coronelli D, Resistance of corroded RC beams: Extending fib Model Code 2010 models, STRUCTURAL CONCRETE (ISSN: 1464-4177) V.21,5, 1747-1762, 2020, https://doi.org/10.1002/suco.201900311

[30.1-60] Regan et al — Regan, Kennedy Reidt (2004).

[30.1-61] Andrade — Further Details Required

[30.1-62] Manual Contecvet, 2001. — Further Details Required

[30.1-63] Petersen et al — Petersen, L., Lohaus, L., Polak, M. A., Influence of Freezing-and-Thawing Damage on Behaviour of Reinforced Concrete Elements, *ACI Materials Journal*, 104 (4): 369 – 378, 2007.

[30.1-64] Zandi et al 2009-1 — Zandi, K., Utgennant, P., Lundgren, K., Frost-damaged concrete: Part 1. Material Properties, *4th International Conference on Construction Materials: Performance, Innovation and Structural Implications*, 2009.

[30.1-65] Zandi et al 2009-2 — Zandi, K., Utgennant, P., Lundgren, K., Frost-damaged concrete: Part 2. Bond properties, *4th International Conference on Construction Materials: Performance, Innovations and Structural Implications*, 2009.

[30.1-66] Petersen — Petersen 2003., Further Details Required

[30.1-67] Fagerlund — Fagerlund, 1994. Further Details Required

[30.1-68] CEB Bulletin 182. — Durable Concrete Structures – CEB Design Guide, Second Edition, 1989.

[30.1-69]	Tian & Han	Tian, W., Han, N., Experiment Analysis of Concrete's Mechanical Property Deterioration Suffered Sulfate Attack and Drying-Wetting Cycles, *Advances in Materials Science and Engineering*, 2017.
[30.1-70]	Desnerck et al	Desnerck, P., Lees, J.M., Morley, C.T. , Strut and tie models for deteriorated reinforced half-joints, *Engineering Structures*, 161: 41-54, 2018.
[30.1-71]	Adhikary & Mutsuyoshi	Adhikary, B. B., Mutsuyoshi, H. , Shear Strengthening of Reinforced Concrete Beams Using Various Techniques. Construction and Building Materials, 2006, 20(6): 366-373.
[30.1-72]	Bastien-Masse & Brühwiler.	Bastien-Masse, M., Brühwiler, E., Contribution of R-UHPFRC Strengthening Layers to the Shear Resistance of RC Elements, 2016, *Structural Engineering International*, 26(4): 365 – 374.
[30.1-73]	Chaallal et al	Chaallal, O., Mofidi, A., Benmokrane, B., Neale, K., Embedded Through-Section FRP Rod Method for Shear Strengthening of RC Beams: Performance and Comparison with Existing Techniques. 2011, *Journal of Composites for Construction*, 15(3): 374 – 383.
[30.1-74]	Colalillo & Sheikh	Colalillo, M. A., Sheikh, S. A., Behavior of Shear Critical Reinforced Concrete Beams Strengthened with Fiber-Reinforced Polymer-Experimentation. 2014, *ACI Structural Journal*, 111(6): 1373 – 1384.
[30.1-75]	Dias & Barros	Dias, S. J. E., Barros, J. A. O., Shear Strengthening of T Cross Section Reinforced Concrete Beams by Near-Surface Mounted Technique, 2008. *Journal of Composites for Construction*, 12(3): 300 – 311.
[30.1-76]	El-Shafiey & Atta	El-Shafiey, T., Atta, A., Retroffiting of Reinforced Concrete Beams in Shear Using External Prestressing Technique, 2012. *Magazine of Concrete Research*, 64(3): 201 -2011.
[30.1-77]	Fiset et al 2016	Fiset, M., Bastien, J., Mitchell, D., Methods for Shear Strengthening of Thick Concrete Slabs, 2016. *Journal of Performance of Constructed Facilities*, 31(3), 10.
[30.1-78]	Fiset et al 2019	Fiset, M., Bastien, J., Mitchell, D. , Shear strengthening of concrete members with unbounded transverse reinforcement, 2019. *Engineering Structures*, 180, 40 – 49.
[30.1-79]	Mofidi et al	Mofidi, A., Chaallal, O., Cheng, L., Shao, Y., Investigation of Near Surface Mounted Method for Shear Rehabilitation of Reinforced Concrete Beams Using Fiber-Reinforced Polymer Composites, 2016. *Journal of Composites for Construction*. 20(2), 14.
[30.1-80]	Fernandez Ruiz et al	Fernandez Ruiz, M., Muttoni, A., Kunz, J., Strengthening of Flat Slabs Against Punching Shear Using Post-Installed Shear Reinforcement, 2011. *ACI Structural Journal*, 434 – 442.
[30.1-81]	Lapi et al	Lapi, M., Ramos Pinho, A., Orlando, M., Flat slab strengthening techniques against punching-shear, 2019. *Engineering Structures*, 180: 160 – 180.
[30.1-82]	Faria et al 2014	Faria, D. M., Einpaul, J., Ramos, A. P., Fernandez Ruiz, M., Muttoni, A.On the efficiency of flat slabs strengthening against punching using externally bonded fibre reinforced polymers, 2014. *Construction and Building Materials*, 73: 366 – 377.
[30.1-83]	Faria et al 2011	Faria, D. M., Lucio, V., Ramos Pingo, A., Strengthening of flat slabs with post-tensioning using anchorages by bonding, 2011. *Engineering Structures*, 33(6): 2025 – 2043.

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The S-N relations given in section 30.2.2.1 are verified for concrete stored in a constant environment of approximately 20°C, 65% relative humidity. For pure compressive fatigue of concrete, relations for concrete with permanent high moisture content and/or submerged in water are additionally given.

For further information see section 14.6.2.1.1 and *fib* Bulletin XX.

The fatigue strength of steel is given both for a normal environment and for a marine environment.

30.2. Evaluation of structural safety (ULS) for non-static loading

30.2.1. Scope

As a typical “non-static” loading, fatigue design with/without deterioration or defect is determined in section 30.2.2. The structural safety under impact and explosion is presented in section 30.2.3. Seismic design is provided in later section 30.3.

30.2.2. Fatigue design and assessment

30.2.2.1. Structures without deterioration or defect

30.2.2.1.1. Scope

The following design rules apply for the entire service life of concrete structures without previous deterioration or defects. The rules for reinforcing and prestressing steel should be applied if more than 10^4 load repetitions are expected; neither low-cycle fatigue nor seismic design are covered. The following design rules should not be applied for lightweight concrete and concrete with recycled aggregate as long as evidence has not been given.

The verification of the design principle (see section 12.5.4.3) can be performed according to four methods with an increasing refinement. The methods according to level II, level III and level IV approximation are given in sections 30.2.2.1.3, 30.2.2.1.4 and 30.2.2.1.5. The models for the analysis of stresses in reinforced and prestressed concrete members under fatigue loading are treated in subsection 30.2.2.1.2 as well as concrete stress gradients.

Section 30.2.2.1.6 deals with shear design and in 30.2.2.1.7 a method for calculating the increased deflections under fatigue loading is given. The relevant combination of loads is treated in section 12.5.4.3.

If FRP - reinforcement is considered for fatigue verification, the relevant information shall be given through design by testing or provided by local specifications like for instance an European Technical Product Specification. A member with FRP reinforcement is defined as subjected to predominantly static loading if the maximum stress range is within 10% of f_{fk}, with a maximum stress of $0{,}5 \cdot f_{fk}$ for a maximum of 2×10^6 cycles.

Indicative values for the fatigue resistance of FRP are provided in section 17.5.3.

In this section no design regulations for FRP reinforcement are given.

30.2.2.1.2. Analysis of stresses in reinforced and prestressed members under fatigue loading

The results of experiments consistently show that as the number of load cycles increase, the concrete strain associated with the maximum stress level increases with the number of load cycles, as does the strain associated with the lower stress level but at a somewhat lower rate. This has the effect of significantly reducing the stiffness of the concrete, and a redistribution of the stresses in the concrete. Stresses that are in the most highly-stressed regions are redistributed to reinforcing steel that is present in this region and to adjacent concrete regions that were less stressed.

Linear elastic models may generally be used, and reinforced concrete in tension is considered to be cracked. The ratio of moduli of elasticity for steel and concrete may be taken as $\alpha = 10$.

In the case of prestressed members it should be verified if the relevant section is sensitive to cracking. This holds true if any combination of loads (see subsection 12.5.4.5) causes tensile stresses at the concrete surface. In that case the stress ranges for reinforcing steel and prestressing steel should be calculated assuming the cracked state.

The effect of differences in bond behaviour of prestressing and reinforcing steel has to be taken into account. Unless a more refined method is used, this can be done using a linear elastic model for stress calculation which fulfils the compatibility in strains and multiplying the stress in the reinforcing steel by the following factor:

$$\eta_s = \frac{1 + \left(A_p / A_s\right)}{1 + \left(A_p / A_s\right)\sqrt{\xi\left(\varphi_s / \varphi_p\right)}} \qquad (30.2\text{-}1)$$

where:

η_s is the factor which increases the stress in the reinforcing steel due to differences in bond behaviour between prestressing and reinforcing steel;

A_s is the area of reinforcing steel;

A_p is the area of prestressing steel;

ϕ_s is the smallest diameter of the reinforcing steel in the relevant cross-section;

For post-tensioned members the following values may be used:

$\xi = 0.2$ for smooth prestressing steel;

$\xi = 0.4$ for strands;

$\xi = 0.6$ for ribbed prestressing wires;

$\xi = 1.0$ for ribbed prestressing bars.

For pretensioned members the following values may be used:

$\xi = 0.6$ for strands;

$\xi = 0.8$ for ribbed prestressing steels.

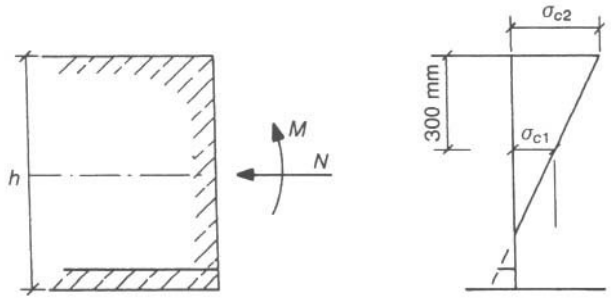

Figure 30.2-1: Definition of stress σ_{c1}, σ_{c2}.

For level I approximation, see section 12.5.4.3.2. This is just a qualitative verification that no variable action is able to produce fatigue.

For γ_{Sd}, $\gamma_{s,fat}$ *and* $\gamma_{c,fat}$, see section 12.5.4.3.3.

ϕ_p is the diameter of the prestressing steel (for bundles an equivalent diameter $1.6\sqrt{A_p}$ is chosen, where A_p is the cross-section area of the bundle);

ξ is the ratio of bond strength of prestressing steel and high-bond reinforcing steel.

The stress gradient for concrete in the compression zone of a cracked section may be taken into account by multiplying the maximum stress in the compression zone by a factor η_c, equal to:

$$\eta_c = \frac{1}{1.5 - 0.5|\sigma_{c1}| / |\sigma_{c2}|} \qquad (30.2\text{-}2)$$

where:

η_c is the averaging factor of concrete stresses in the compression zone considering the stress gradient;

$|\sigma_{c1}|$ is the minimum absolute value of the compressive stress within a distance no more than 300 mm from the surface under the relevant load combination of actions (Figure 30.2-1);

$|\sigma_{c2}|$ is the maximum absolute value of the compressive stress within a distance no more than 300 mm from the surface under the same load combination as that for which $|\sigma_{c1}|$ is determined (Figure 30.2-1).

For concretes with $f_{ck} \geq 55$ N/mm² the factor η_c is limited to ≥ 0.85

30.2.2.1.3. Level II approximation: the simplified procedure

30.2.2.1.3.1. General

This procedure is only applicable to structures subjected to a limited number of acting stress cycles ($n \leq 10^8$).

30.2.2.1.3.2. Steel

The fatigue requirements will be met if the maximum calculated stress range under the frequent combination of loads, max $\Delta\sigma_s$, satisfies the condition:

$$\gamma_{Sd} \cdot max\,\Delta\sigma_s \le \Delta\sigma_{Rsk} / \gamma_{s,fat} \quad (30.2\text{-}3)$$

where:

$\Delta\sigma_{Rsk}$ is the characteristic fatigue strength at 10^8 cycles.

Values for $\Delta\sigma_{Rsk}$ are given in Tables 30.2-1 and 30.2-2.

30.2.2.1.3.3. Concrete

Detailed fatigue design need not be carried out if the maximum calculated stresses under the frequent combination of loads, $\sigma_{c,\max}$ (compression), $\sigma_{ct,\max}$ (tension), respectively, satisfy the following criteria:

Compression

$$\gamma_{Sd} \cdot \sigma_{c,max} \cdot \eta_c \le \omega_w \cdot 0.45 f_{cd,fat} \quad (30.2\text{-}4)$$

where:

$\sigma_{c,\max}$ is the maximum compressive stress in the concrete; see section 12.5.4.3.3.

ω_w is a coefficient taking into account the moisture content and environment of concrete

η_c is an averaging factor considering the stress gradient Eq. 30.2-2;

$f_{cd,fat}$ is the design fatigue reference compressive strength for concrete.

The design fatigue reference compressive strength is defined as follows:

$$f_{cd,fat} = 0.85\beta_{cc}(t) \cdot f_{ck}\left(1 - \frac{f_{ck}}{400}\right) / \gamma_{c,fat} \quad (30.2\text{-}5)$$

where:

The design fatigue reference compressive strength $f_{cd,fat}$ has been introduced to take into account the increasing fatigue sensitivity of concrete with increasing compressive strength.

The coefficient 0.85 takes into account the effect of high mean stresses during loading and of actual frequencies of loading, which are in most practical cases significantly lower than those applied in experiments.

Eq. (30.2-6) is valid for dry condition.

When the unique value Q can be chosen satisfactorily (e.g. as fatigue equivalent), this method is a more precise assessment than the simplified procedure.

For γ_{Sd}, $\gamma_{s,fat}$ *and* $\gamma_{c,fat}$, see section 12.5.4.3.4.

$\beta_{cc}(t)$ is a coefficient which depends on the age of concrete t at the beginning of fatigue loading, to be taken from section 14.6.1.6.1, Eq. (14.6-46).

For structures submerged in water, offshore-structures in the splash zone, structures that are permanently connected with wet ground and structures with comparable moisture content coefficient $\omega_w = 0.75$ shall be used. In other cases $\omega_w = 1.0$ shall be used.

Tension

$$\gamma_{Sd} \cdot \sigma_{ct,max} \leq 0.33 f_{ctd,fat} \qquad (30.2\text{-}6)$$

where:

$\sigma_{ct,\max}$ is the maximum tensile stress in the concrete; see section 12.5.4.3.3;

$f_{ctd,fat}$ is the design fatigue reference tensile strength of the concrete.

The design fatigue reference tensile strength is defined as follows:

$$f_{ctd,fat} = f_{ctk,min} / \gamma_{c,fat} \qquad (30.2\text{-}7)$$

where:

$f_{ctk,min}$ is the lower bound value of the characteristic tensile strength, see section 14.6.1.2.1.

30.2.2.1.4. Level III approximation: verification by means of a single load level

30.2.2.1.4.1. General

This method takes into account the required service life with a foreseen number of acting cycles, n. This number intervenes in the verification with the maximum fatigue effects of the action, Q, as defined in section 12.5.4.3.4.

When it is considered necessary to carry out fatigue tests to determine the performance of reinforcing steel, the tests should be made according to standardized procedures, as described in ISO 15630-1 [30.2-1], ISO 15630-2 [30.2-2] and ISO 15630-3 [30.2-3].

The S–N relation for steel consists of segments of the form $(\Delta\sigma_{Rsk})^m \cdot N = \text{const.}$ (see Figure 30.2-2). Values for the S–N relations are given in Table 30.2-1 and Table 30.2-2.

The values given in Tables 30.2-1 and 30.2-2 are characteristic and do not incorporate partial safety factors. These values or higher values must be validated by appropriate approval documents.

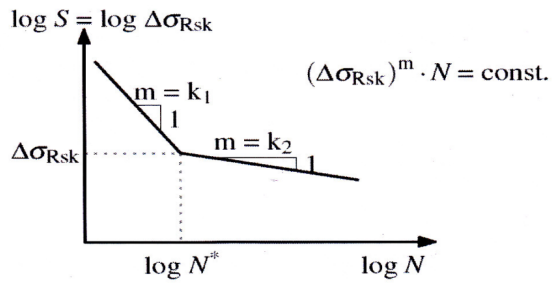

Figure 30.2-2: Shape of S–N relations for steel.

Where appropriate information is provided by specific approval documents for the steel to be used, higher fatigue strength values may be used accordingly.

Data on the fatigue behaviour of bars with diameters larger than 40 mm are scarce. Therefore no data for these bars are given here.

30.2.2.1.4.2. Steel

The fatigue requirement will be met if the calculated maximum acting stress range, max $\Delta\sigma_s$, satisfies the condition:

$$\gamma_{Sd} \cdot max\,\Delta\sigma_s \le \Delta\sigma_{Rsk}(n) / \gamma_{s,fat} \qquad (30.2\text{-}8)$$

where:

$\Delta\sigma_s$ is the steel stress range under the acting loads;

$\Delta\sigma_{Rsk}(n)$ is the stress range relevant to n cycles obtained from a S–N relation.

Table 30.2-1: Parameters of S–N relations for reinforcing steel (embedded in concrete).

	N^*	Stress exponent		$\Delta\sigma_{Rsk}$ (MPa)[g]	
		k_1	k_2	at N^* cycles	at 10^8 cycles
Straight and bent bars $D \ge 25\phi$					
$\phi \le 16$ mm	10^6	5	9	210	125
$\phi > 16$ mm[a]	10^6	5	9	160	95
Bent bars $D < 25\phi$[b]	10^6	5	9	—[c]	—[c]
Shipped as coiled and re-straightened bars					
$\phi \le 12$ mm	10^6	5	9	150	—[e]
$12 < \phi \le 16$ mm[d]	10^6	5	9	130	—[e]
Welded bars[b], including tack welding and butt joints, and mechanical connectors	10^7	3	5	50	30

Marine environment[b),f)]	10^7	3	5	65	40

[a] The values given in this line represent the S–N relation of a 40 mm bar; for diameters between 16 and 40 mm, interpolation between the values of this line and those of the line above is permitted.

[b] Most of these S–N relations intersect the curve of the corresponding straight bar. In such cases the fatigue strength of the straight bar is valid for cycle numbers lower than that of the intersection point.

[c] Values are those of the according straight bar multiplied by a reduction factor ξ depending on the ratio of the diameter of mandrel D and bar diameter ϕ: $\xi = 0.35 + 0.026D/\phi$.

[d] The value is valid for $\phi 16$ or less.

[e] The values should be validated by experiment.

[f] Valid for all ratios D/ϕ and all diameters ϕ.

[g] In cases where $\Delta\sigma_{Rsk}$ values calculated from the S–N relation exceed the stress range $f_{yd} - \sigma_{min}$, the value $f_{yd} - \sigma_{min}$ is valid.

The values given in Table 30.2-2 are on the safe side compared to the strength values for the basic material given in section 16.1.5.4.

The reduction of the $\Delta\sigma_{Rsk}$ values for curved tendons compared with the values of straight tendons is due to fretting corrosion, which results from the lateral pressure and slip between prestressing strands and/or ribs of the steel sheaths.

Table 30.2-2: Parameters of S–N relations for prestressing steel (embedded in concrete).

	N^*	Stress exponent		$\Delta\sigma_{Rsk}$ (MPa)[1]
		k_1	k_2	at N^* cycles
Pretensioned steel				
Straight strands and wires	10^6	5	9	185
Post-tensioned steel				
Single strands in plastic ducts	10^6	5	9	185
Straight tendons or curved tendons in plastic ducts	10^6	5	10	150
Curved tendons in steel ducts	10^6	5	7	120
Splicing devices	10^6	5	5	80

1 In cases where the S–N relation intersects that of the straight tendon, the fatigue strength of the straight tendon is valid.

Characteristic S–N relations for concrete can be used without any restriction for frequencies higher than 0.1 Hz. For lower frequencies, the fatigue life should be reduced – see chapter 3 in CEB Bulletin 188 [30.2-4] for guidance.

For the assessment of $\sigma_{c,\max}$, $\sigma_{c,\min}$, and $\sigma_{ct,\max}$, see section 12.5.4.3.4 using the fatigue equivalent or frequent value of the variable action Q.

Eqs. (30.2-9) - (30.2-12) are valid for pure compressive fatigue loading.

30.2.2.1.4.3. Concrete

The fatigue requirements under cyclic loading will be met if the required lifetime (number of acting cycles) is lower than or equal to the number of cycles to failure:

$$n \le N$$

where:

n is the foreseen number of acting cycles during the required design service life;

N is the number of resisting stress cycles, to be calculated from S–N relations given below.

(a) Pure compression

For $S_{cd,\max} < 0.90$ and $0 \le S_{cd,\min} \le 0.8$, the following equations apply:

$$\log N_1 = \left[\frac{8}{Y-1}\left(S_{cd,max}-1\right)\right] \qquad (30.2\text{-}9)$$

$$\log N_2 = \left[8+\frac{8\ln(10)}{Y-1}\left(Y-S_{cd,min}\right)\log\left(\frac{S_{cd,max}-S_{cd,min}}{Y-S_{cd,min}}\right)\right] \qquad (30.2\text{-}10)$$

With

$$Y = \frac{0.45+1.8S_{cd,min}}{1+1.8S_{cd,min}-0.3S_{cd,min}^2}$$

where:

(a) if $\log N_1 \le 8$ then $\log N = \omega_w \cdot \log N_1$; (30.2-11)

(b) if $\log N_1 > 8$ then $\log N = \omega_w \cdot \log N_2$ (30.2-12)

where:

$\sigma_{c,\min}$ is the minimum compressive stress;

$\sigma_{c,\max}$ is the maximum compressive stress;

$S_{cd,\min} = \gamma_{Ed}\sigma_{c,\min}\ \eta_c / f_{cd,fat}$ is the minimum compressive stress level;

$S_{cd,\max} = \gamma_{Ed}\sigma_{c,\max}\ \eta_c / f_{cd,fat}$ is the maximum compressive stress level;

η_c is an averaging factor considering the stress gradient Eq. (30.2-2);

$f_{cd,fat}$ is the design fatigue reference compressive strength, see Eq. (30.2-5);

ω_w is a coefficient taking into account the moisture content and environment of concrete.

For $S_{cd,\min} > 0.8$, the S–N relations for $S_{cd,\min} = 0.8$ are valid.

For structures submerged in water, offshore-structures in the splash zone, structures that are permanently connected with wet ground and structures with comparable moisture content coefficient $\omega_w = 0.75$ shall be used. In other cases $\omega_w = 1.0$ shall be used.

(a) Compression – tension with $S_{cd,max}$ and $S_{td,max} < 0.90$ *and* $\sigma_{ct,max} \leq 0.026|\sigma_{c,max}|$

$$log\, N = 9\left(1 - S_{cd,max}\right) \qquad (30.2\text{-}13)$$

Eqs. (30.2-13) and (30.2-14) are valid for dry condition. Special attention should be paid when members are submerged in water or in high moisture content condition where water gets into the member through cracks in tension areas.

If Eq. (30.2-13) is applied, it may be assumed that the concrete always fails in compression.

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(c) Pure tension and tension-compression with $S_{cd,max}$ *and* $S_{td,max} < 0.90$ *and* $\sigma_{ct,max} > 0.026|\sigma_{c,max}|$

$$log\, N = 12\left(1 - S_{td,max}\right) \qquad (30.2\text{-}14)$$

where:

$\sigma_{ct,\max}$ is the maximum tensile stress;

$S_{td,\max} = \gamma_{Sd} \cdot \sigma_{ct,\max} / f_{ctd,fat}$ is the maximum tensile stress level

$f_{ctd,fat}$ is the design fatigue reference tensile strength, see Eq. (30.2-7).

30.2.2.1.5. Level IV approximation: verification by means of a spectrum of load levels

This method takes account of the required service life, the load spectrum (which is divided into j blocks) and the S-N relations.

Fatigue damage D is calculated using the Palmgren–Miner summation:

The partial coefficients are applied in this procedure as follows:

For steel, values N_{Ri} are calculated from the S–N relations given in Tables 30.2-1 and 30.2-2 using an increased stress range $\gamma_{Sd} \cdot \gamma_{s,fat} \cdot \Delta\sigma_{Si}$.

For concrete, values N_{Ri} are calculated directly from the S–N relations given in section 30.2.2.1.4.

For steel D_{lim} = 1.0.

For concrete, the Palmgren-Miner rule is only a very rough approximation of the fatigue failure under changing maximum and minimum stress levels. Therefore, the value of the Palmgren-Miner sum indicating fatigue failure for concrete is varying in various codes from 0.2 to 1.0. Because of the absence of better approach, for concrete D_{lim} can be taken as $D_{lim} = 1.0$.

If $V_{max}/V_{ref} \leq 0.4$, no verification is needed.

Factor η is intended to consider the different loading rate of laboratory fatigue tests and real structures. Under laboratory loading conditions, average factor has been found η = 1.1. A value of η = 1.0 is recommended for design.

It is known that the travelling loads, such as traffic loads, and some environmental conditions, such as the wet condition and frost attack, would reduce the fatigue life of punching significantly. Eq. (30.2-16) is applicable only to the loading at a fixed point without such environmental conditions.

$$D = \sum_{i=1}^{j} \frac{n_{Ei}}{N_{Ri}} \qquad (30.2\text{-}15)$$

where:

D is fatigue damage;

n_{Si} denotes the number of acting stress cycles associated with the stress range for steel and the actual stress levels for concrete;

N_{Ri} denotes the number of resisting stress cycles at a given stress level.

The fatigue requirement will be satisfied if $D \leq D_{lim.}$

30.2.2.1.6. Shear design

30.2.2.1.6.1. Members without shear reinforcement

The fatigue requirements will be met if, under cyclic loading the number of cycles corresponding to the required service life (n) is smaller than or equal to the resisting numbers of cycles to failure (N):

$$n \leq N$$

N should be calculated from the fatigue strength function given below:

$$N = \left(\frac{1 - \frac{V_{min}}{V_{ref}}}{\frac{V_{max}}{V_{ref}} - \frac{V_{min}}{V_{ref}}} \right)^{m} \qquad (30.2\text{-}16)$$

where:

V_{min} is the minimum shear force under the basic combination of non-cyclic loads (permanent loads including prestress), which is equivalent to the quasi-permanent load combination for SLS;

V_{max} is the maximum shear force including the basic combination of non-cyclic loads combined with the frequent fatigue-relevant cyclic load;

$V_{ref} = \eta V_{Rd,c}$ is the reference shear strength, where $V_{Rd,c}$ is the quasi-static monotonic shear strength, calculated according to section 30.1.3, and η takes account of strain-rate effects in the reference shear strength;

If it can be demonstrated that in the structural member no shear cracks will occur under the relevant combination of loads, fatigue in shear need not be verified. This is because stresses in shear reinforcement is negligibly small if no shear crack occurs.

According to Eq. (30.2-17), the strain in shear reinforcement increases under fatigue loading due to the accumulated damage induced by shear crack propagation under fatigue loading. The value of 0.5 for $k_{c,f}$ can be assumed according to the relevant technical document, JSCE Standard Specifications for Concrete Structures [30.2-5].

Shear reinforcement shows lower fatigue strength due to the local deformation at intersection with shear crack and bent portion. The value of 0.5 can be assumed for $k_{s,f}$ based on the experimental evidence in accordance with the relevant technical document, JSCE Standard Specifications for Concrete Structures.

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The stress in the shear reinforcement should be calculated according to chapter 12, assuming the following inclination of the compression struts under fatigue loading:

$$\tan\theta_{fat} = \sqrt{\tan\theta} \qquad (30.2\text{-}18)$$

where:

θ_{fat} is the angle between the web compression and the chords valid for verification of the reinforcement.

For assessment of the θ value, see section 30.1.3.

According to Eq. (30.2-18), the strut inclination θ_{fat} is steeper than the value θ used for design under static loading. This takes account of the higher value of θ in the SLS.

The fatigue reference strength is to be reduced in the same way as the compressive strength of the concrete subjected to simultaneously acting compressive and transverse tensile forces.

m=17 for beam shear failure and for slab at punching shear failure.

30.2.2.1.6.2. Members with shear reinforcement

The stress in the shear reinforcement should be calculated according to section 30.1.3, assuming the following reduction in shear resistance attributed to concrete under fatigue loading:

$$V_{\max} = k_{c,f} \cdot V_{Rd,c} + V_{Rd,s} \qquad (30.2\text{-}17)$$

where:

$k_{c,f}$ is the reduction factor of shear resistance attributed to concrete.

The stress in shear reinforcement to calculate $V_{Rd,s}$ is $k_{s,f}\Delta\sigma_{Rsk}(n) + \sigma_{w,\min}$. $\sigma_{w,\min}$ is the stress in shear reinforcement under minimum shear force, $V_{\min}$ and $\Delta\sigma_{Rsk}(n)$ is the fatigue strength of steel reinforcement according to Eq. (30.2-8). $k_{s,f}$ is the reduction factor for fatigue strength of shear reinforcement.

The resistance of compressive struts can be verified using Eq. (30.2-4) or Eqs. (30.2-9) and (30.2-10) reducing the fatigue reference strength given in by Eq. (30.1-23) in section 30.1.3.3.1. The compression of web concrete subjected to fatigue loading should be calculated using the angle θ (see section 30.1.3).

30.2.2.1.7. Increased deflections under fatigue loading in the SLS

Under cyclic loading, progressive deflection can occur in reinforced concrete members in addition to the deflection produced by creep. The cyclic effect can be calculated from:

$$a_n = a_1\left[1.5 - 0.5exp\left(-0.03n^{0.25}\right)\right] \quad (30.2\text{-}19)$$

where:

a_n final deflection after n cycles;

a_1 deflection in the first cycle due to the maximum load including effects of shear strains;

n is the number of cycles.

30.2.2.2. Structures with deterioration or defects

30.2.2.2.1. Corrosion

The combined effect of a corrosive environment and repeated loads may be more severe than the isolated fatigue effect referred to in previous clauses. In marine environments, the fatigue strength of reinforcing steel can be calculated with the reduced parameters given in Table 30.2-1. Nevertheless, a detailed study of each particular case is recommended as the fatigue strength of reinforcing and prestressing steels can be more severely affected in case of localized pitting damage.

In addition, corrosion may significantly affect bond between concrete and steel, thereby reducing the available bond strength and eventually lead to cover cracking. A reduced bond strength τ_{bu} should be taken into account to evaluate the fatigue resistance of bond following the models of section 20.3.5.1.

30.2.2.2.2. Freeze/thaw damage

Freeze/thaw damage reduces fatigue strength of concrete in compression and tension. The degree of fatigue strength reduction is greater than that of concrete

The available studies on combined effects of freeze/thaw damage and fatigue show the accelerated fatigue strength reduction. In those studies, concrete was

exposed to freeze/thaw cycles first and then applied to loading cycles. The fatigue strength, which is normalized by the static strength of concrete with freeze/thaw damage, shows the greater reduction. However, the fatigue strength may be improved when loading cycles is applied under freezing temperature. This is because the concrete strength increases under freezing temperature, which makes water in concrete ice. It is noteworthy to state that the fatigue strength normalized by the increased static strength under freezing temperature is smaller than the fatigue strength of concrete without damage. This is because the ice in concrete which enhances the concrete strength can melt during fatigue loading.

There have been reports on the damaged reinforced concrete decks of highway bridges. They found punching shear failure although the traffic volume was less and serious freeze/thaw damage on top of the concrete decks. Experimental studies in the past disclosed that fatigue punching shear of concrete deck with freeze/thaw damage is much lower than that without freeze/thaw damage.

It has been reported that the fatigue lives of RC slabs or PC girder damaged by ASR are not always shorter than structures without ASR. Furthermore, there is a report describing enhancement of fatigue resistance or RC beams by ASR. This is probably due to the chemical prestress action. See Ahmed et.al 1998 [30.2-6] and Maeshima et al 2016 [30.2-7].

It has been reported that the decrease of compressive strength of concrete due to fire damage tends to recover to a certain level in time passing. However, the relation between degree of fire damage and intensity of repetitive load has not been clarified yet.

without freeze/thaw damage. For assessment of existing structures, it should be considered that effects of freeze/thaw damage is greater for the part closer to the surface of concrete structure.

30.2.2.2.3. Alkali-silica reaction

Severe damage due to ASR surely decrease fatigue strength as well as static capacity of concrete members, even though slight damage can have both positive and negative impact on fatigue. The relation between degree of ASR damage and intensity of repetitive load should be carefully examined for assessment.

30.2.2.2.4. Fire damage

Fire damage, especially caused by more than 300 degree Celsius, degrease fatigue strength as well as static capacity of concrete members due to degrease of concrete strength. It should be carefully examined for assessment.

30.2.2.2.5. Combined deterioration modes

Combined deterioration, such as corrosion of rebar coupled with freeze/thaw damage of cover concrete, accelerate each damage due to interactions in long duration.

Fatigue strength as well as static capacity of the members should be carefully examined for assessment.

30.2.2.2.6. Non-specified deterioration or defects

Any unexpected deformation, cracks, spalling and any defects can decrease fatigue strength due to damage acceleration depending on time, even if the influence on the static capacity is small. The cause of deterioration should be investigated, then refer 30.2.2.2.1 - 30.2.2.2.5.

30.2.2.3. Structures with non-compliant detailing

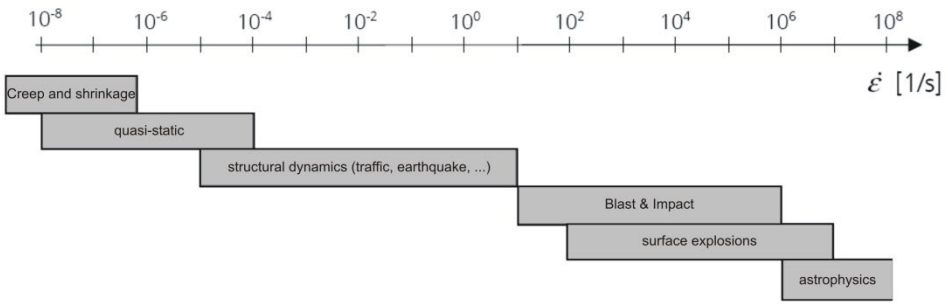
[Not used at this time, to be updated in future edition]

30.2.3. Impact and explosion

30.2.3.1. Structures without deterioration or defects

30.2.3.1.1. General remarks

The following chapter concerns situations where the accidental scenario is defined. Loads caused by impact and explosions are characterized as high dynamic loads inducing strain rates which can be significantly higher than those caused for example by traffic loads or even earthquake loads (Figure 30.2-3).



The increasing strain rates caused by impulsive loadings produce special effects in the structure which must be taken into account in the design process:

- inertia effects; resonance effects (namely, internal forces differ from those calculated in a quasi-static regime, neglecting the acceleration term in Newton's equation);
- spalling and scabbing effects; penetration and perforation; change of the failure mode with punching and shear instead of bending due the transient distribution of internal forces.

Critical structures that are designed to resist high dynamic loads are for example:

- infrastructures for mobility (underground, railway stations, airports, rocksheds)
- nuclear power plants;
- shelters for planes and ammunition;
- office buildings with a high risk of being attacked, such as embassies and military buildings;
- chemical factories;
- piers of bridges which can be hit by trains or vehicles.

An extreme load, which is generated by detonation of a cased charge, is characterized by a combination of blast (overpressure) and fragmentation. The former is caused by the explosive charge and the latter – by fracture of the (commonly metal) casing. The peak overpressure and impulse of the cased charge are lower than those, which are generated by a bare charge (without casing) of the same mass and type, but the effect of the fragments should be considered.

The effect of the fragments is twofold: their penetration causes damage to the structure and thus weakens its resistance, and their impact can add an actual physical load due to the momentum transferred to the structure by the fragments. Although a structure, which is exposed to the extreme load is not necessarily struck simultaneously by the fragments and the blast wave, for most stand-off distances (more than few meters) the fragments strike the structure first, thus causing the structure to be already damaged when responding to the blast load.

Common design procedures decouple the two effects and deal separately with the blast and fragmentation loadings (e.g., US Army 2008 [30.2-8], Leppänen 2005 [30.2-9]). However, experimental evidence indicates that there is a synergistic effect of the combined loading of blast and fragments on structures, i.e., the structural response to these combined loadings and the damage caused by them can be more severe than those caused by the sum of their separate actions.

Figure 30.2-3: Strain rates for different loading velocities.

High dynamic loads can be caused by:

- impact of vehicles, trains or airplanes;
- impact of projectiles and missiles;
- impact of debris;
- surface burst explosion;
- blast caused by nearby or distant explosions;
- explosions in rooms, including multiple reflections.

It is possible to recognise three different regimes associated to the structure response to a given specific impulse: quasi-static, dynamic and impulsive regimes. They are exemplified in Figure 30.2-2, looking to the structure as a one degree of freedom system.

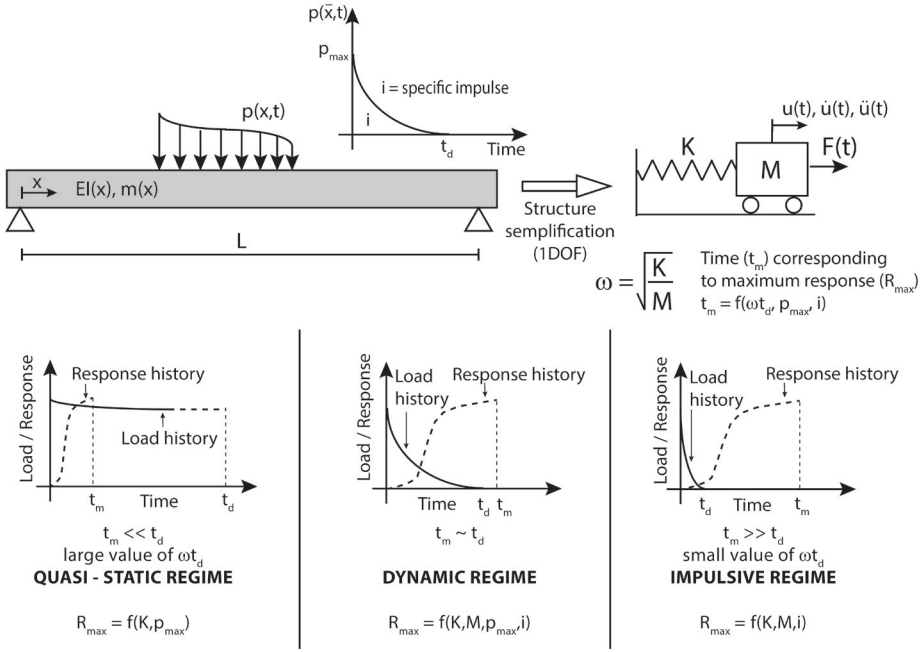


Figure 30.2-4: Loading regimes related to the structure response vs. load function.

For higher strain rates, a dynamic calculation is required. For known phenomena such as free-air burst explosions or plane impact, idealized load–time curves or pressure–time curves are available in codes like UFC 3-340-02 [30.2-10] or DIN 25449 [30.2-11] which can be adapted to the respective situation.

30.2.3.1.2. Effective loads

30.2.3.1.2.1. General

The characteristic loads strongly depend on the type and intensity of the explosion or impact so that a general definition of the load is not possible. The following approaches can be adopted:

- quasi-static equivalent loads (level I);
- standardized pressure–time curves (level II);
- computing loading histories (level III).

For some configurations, loads can be obtained from literature or special codes (level I or II). In other cases, numerical calculations with the use of hydrocodes, FE-codes or tests are necessary to get load data (level III).

Figures 30.2-5 and 30.2-6 give examples for pressure–time curves for loads caused by free-air burst explosions.

For the determination of internal forces and stresses in the structure, three levels of analysis are distinguished.

30.2.3.1.2.2. Level I

For quasi-static equivalent loads, a linear or non-linear static calculation can be done using the same static models which are used for dead and live loads.

30.2.3.1.2.3. Level II

For pressure-time curves, a linear or non-linear dynamic calculation using a time-step method is required.

30.2.3.1.2.4. Level III

For a detailed modelling of impact or explosion by hydrocode or FE-calculation, the internal stresses can be obtained directly from this calculation. As an alternative,

The effect of a free air burst explosion can be modelled using a standard curve as in Figure 30.2-5. The load value and loading duration depend on several factors such as the distance between the explosive and the concrete structure and the type and quantity of the explosive material.

Figure 30.2-6 gives a more simplified pressure-time-function without taking account of the suction phase which can be used for structures with a high weight and a high stiffness which are loaded by low pressures.

pressure–time curves can be derived from the results of such calculations and then used for a calculation according to level II.

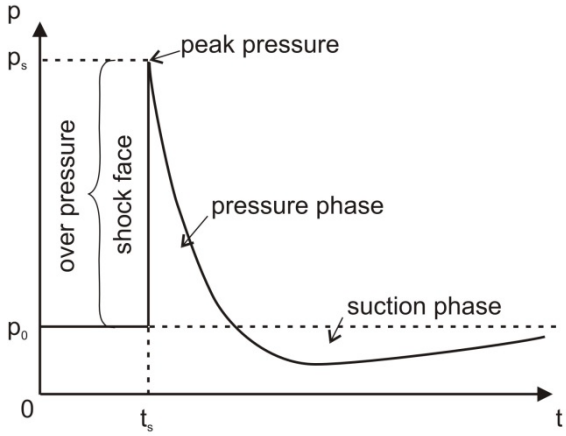


Figure 30.2-5: Principle pressure-time-curve caused by an air burst explosion.

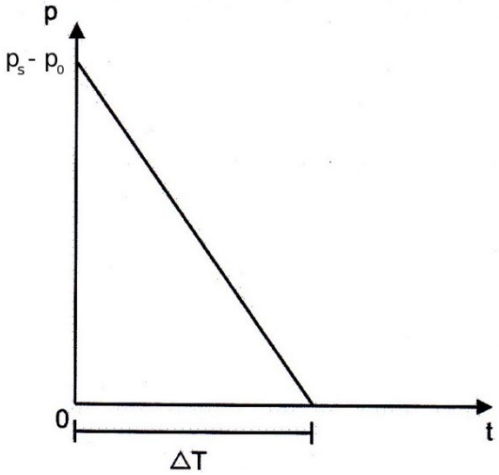


Figure 30.2-6: Free air burst explosion.

The calculation methods for level I and II analysis are similar to those for seismic design, see section 30.3.

Validity of this approach however requires:

- verification of local failure mode, such as penetration, spalling, perforation, punching, possibly activated due to transient high internal forces;
- fulfilment of special detailing rules (e.g. rebars lacing, etc.) aimed at avoiding local failure modes activation.

These provisions have not been sufficiently substantiated for lightweight and recycled concrete.

Loading at high strain rates usually results in very high stresses. However, when estimating the structural resistance, special, observable effects should be taken into account. The materials, concrete and steel, exhibit an increased resistance with increasing strain rates (section 14.6.2.2). When the computation adopts kinematic models, which are not able to take into account micro-inertia effects, it is possible to modify the constitutive relationships introduced in (section 14.6.2.2) by taking account of Dynamic Increase Factors as an artificial way to take into account these

30.2.3.1.3. Verification of structural safety

30.2.3.1.3.1. General

For the dimensioning of RC structures for overall stresses such as bending moments, normal and shear forces, principally the same procedures can be used as for seismic loads. Three different approaches are available:

- Level I: static analysis with ULS verification of internal forces (section 30.2.3.1.3.2);
- Level II: elastic dynamic analysis with verification of internal forces (section 30.2.3.1.3.3);
- Level III: mechanically non-linear dynamic analysis considering structural redistribution (section 30.2.3.1.3.4).

30.2.3.1.3.2. Static dimensioning (Level I)

For static dimensioning with regard to internal forces the following procedures can be used:

- linear or non-linear calculation of the static internal forces calculated with a load defined according to level I of effective loads;
- dimensioning of the structure with the usual methods adopted for static loads;
- dimensioning with normal values for the strength of the concrete and reinforcement without dynamic increase factors.

30.2.3.1.3.3. Dynamic dimensioning (Level II)

With regard to dimensioning concrete structures for internal forces caused by dynamic loads, the following considerations apply:

- the internal forces caused by dynamic loads (level II, III) have to be calculated using dynamic numerical calculations. The actions are determined by means of elastic analysis;
- the reinforcement and the concrete in any cross-section can be dimensioned for bending with the same methods as used for static loading, considering an

effects – see Figures 30.2-7 and 30.2-8. The related curves for the increase of the strength of concrete and of reinforcing steel are given in relation to the strain rates.

The behaviour of concrete in 213ompressionn and tension is based on the Eqs. (14.5.1.114 – 14.5.1.118) given in chapter 14.

The equations that describe the yield and ultimate stress of reinforcing steel (Figure 30.2-8) are listed below and can be found in [30.2-12].

$$DIF = (\dot{\varepsilon} / 10^{-4})^{\alpha} \qquad (30.2\text{-}20)$$

where:

$\alpha_{fy} = 0.074 - 0.040\ (f_y/414)$;

$\alpha_{fu} = 0.019 - 0.009\ (f_y/414)$;and

f_y is the static yield strength of the reinforcing bar [Mpa].

This formulation is valid for reinforcing steel having yield stresses between 290 and 710 Mpa and for strain rate between 10^{-4} and 10 s^{-1}. For higher strain rate experimental validation is needed.

All of these data are derived from tests and can be used also in calculations on level III, except if micro-inertia effects are taken into account explicitly.

increase of strength of concrete and reinforcement steel (DIF values) according to Figures 30.2-7 and 30.2-8, under high dynamic strain rates;

- the use of the dynamic increase factors for reinforcing bars is only possible if the steel has sufficient ductility, as for seismic requirements – see Table 30.2-3;
- the structure should be dimensioned for bending with normal forces, shear and torsion for the internal forces.

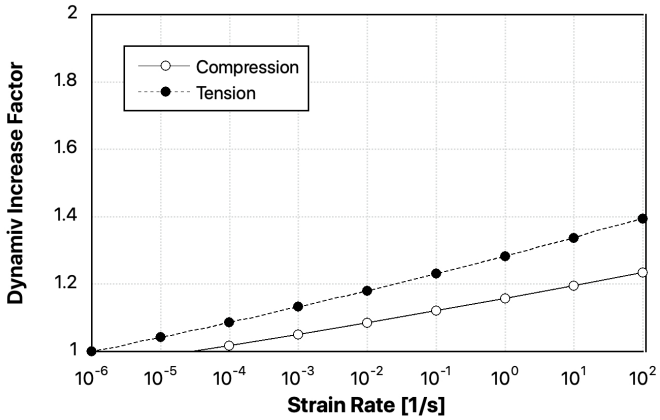


Figure 30.2-7: Increase of compressive and tensile strength as a function of strain rates.

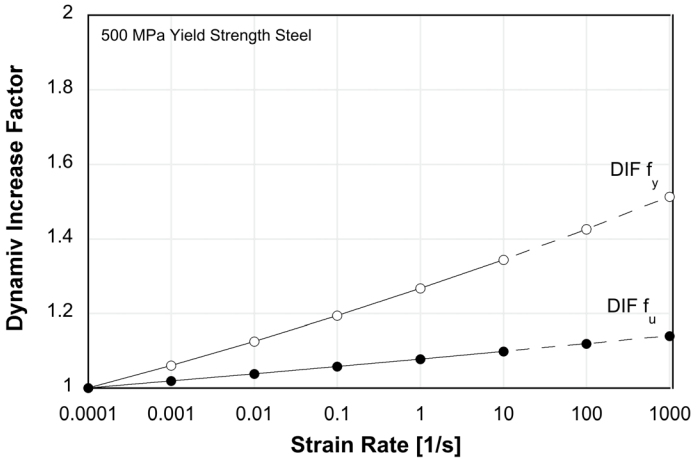


Figure 30.2-8: Increase of reinforcing steel strength as a function of strain rates.

A simplified approach for the increase of material strength is given in Tables 30.2-3 and 30.2-4 (UFC 3-340-02 [30.2-10]) as proposed by Mayrhofer [30.2-13].

$$Z = r/W^{1/3} \ [m/kg^{1/3}] \qquad (30.2\text{-}21)$$

where:

r is the distance between explosion epicentre and structure [m];

W is the weight of explosives as TNT equivalent [kg];

with

$z \leq 0.5$ for nearby explosion;

$z > 0.5$ for distant explosion.

The calculation of large structures such as nuclear power plants is usually carried out at level II. For level II calculations the DIFs in Tables 30.2-3 and 30.2-4 should be used, because local strain rates are not known.

Table 30.2-3: Dynamic increase factors (DIFs) f_{dy}/f_y and f_{dt}/f_t for the design of reinforced concrete elements.

Type of stress	**Distant design range**			**Nearby design range**		
	Reinforcing bars		Concrete	Reinforcing bars		Concrete
	f_{dy}/f_y	f_{dt}/f_t	f'_{dc}/f'_c	f_{dy}/f_y	f_{dt}/f_t	f'_{dc}/f'_c
Bending	1.17	1.05	1.19	1.23	1.05	1.25
Diagonal tension	1.00	—	1.00	1.10	1.00	1.00
Direct shear	1.10	1.00	1.10	1.10	1.00	1.10
Bond	1.17	1.05	1.00	1.23	1.05	1.00
Compression	1.10	—	1.12	1.13	—	1.16

For prestressing steel, strands or wires, a dynamic increase factor f_{dy}/f_y should always be taken as 1.

Table 30.2-4: Dynamic design stresses for the design of reinforced concrete elements.

Type of stress	Type of reinforce-ment	Maximum support rotation, Θ_m (degrees)	Dynamic design stress	
			Reinforce-ment f_{ds}	Concrete, f_{dc}
Bending	Tension and compression	$0 < \Theta_m \leq 2$ $2 < \Theta_m \leq 6$ $6 < \Theta_m \leq 12$	f_{dy} [1] $f_{dy} + (f_{du} - f_{dy})/4$ $(f_{dy} + f_{du})/2$	f'_{dc} —[2] —[2]
Diagonal tension	Stirrups	$0 < \Theta_m \leq 2$ $2 < \Theta_m \leq 6$ $6 < \Theta_m \leq 12$	f_{dy} f_{dy} f_{dy}	f'_{dc} f'_{dc} f'_{dc}
Diagonal tension	Lacing	$0 < \Theta_m \leq 2$ $2 < \Theta_m \leq 6$ $6 < \Theta_m \leq 12$	f_{dy} $f_{dy} + (f_{du} - f_{dy})/4$ $(f_{dy} + f_{du})/2$	f'_{dc} f'_{dc} f'_{dc}
Direct shear	Diagonal bars	$0 < \Theta_m \leq 2$ $2 < \Theta_m \leq 6$ $6 < \Theta_m \leq 12$	f_{dy} $f_{dy} + (f_{du} - f_{dy})/4$ $(f_{dy} + f_{du})/2$	f'_{dc} —[3] —[3]
Compression	Column	—[4]	f_{dy}	f'_{dc}

[1] Tension reinforcement only.

[2] Concrete crushed and not effective in resisting moment.

[3] Concrete is considered not effective; shear resisted by reinforcement only.

[4] Capacity is not a function of support rotation.

30.2.3.1.3.4. Non-linear dynamic analysis considering structural redistribution (Level III)

In the numerical calculation of the internal forces on level III the impact or the explosion or the impact process itself is part of the numerical calculation. So, special codes are necessary, by which the relevant phenomena such as wave propagation, large deformations and fracture can be modelled adequately. This can be done for

For dynamic dimensioning with regard to ductility the following considerations apply:

– The deflections caused by dynamic loads have to be calculated by use of a dynamic numerical calculation.

example by hydrocodes or special finite element codes. On level III the structure is modelled in a very detailed way, local strain rates are usually available.

In the design phase, the engineer has several options to reduce the high dynamic loads. If there is sufficient space, this can be done by increasing the distance between the structure and the threat, or by barriers. If this is not possible, the materials have to be selected and characterized and the structure has to be designed for the dynamic loads concerned.

There are two general options for design concerning loads with high strain rates when failure should be taken into account. One possibility is to define alternative load paths in the structure. This leads to a safe design only if the quantity of the loads can be calculated with sufficient accuracy. The alternative solution is the use of defined breaking points or areas in the structure to make sure that no progressive collapse can occur. In this case, local damage is accepted but the collapse of the total structure has to be avoided.

- As for earthquake design, the plastic rotations shall be limited (Table 30.2-4).
- The dynamic design will be performed considering non-linear constitutive relationship where stresses are limited to the values given in Table 30.2-4. With these parameters the generalized stress-deformation behaviour can be determined for each cross-section, for example to establish moment–curvature diagrams.

The use of DIF is correlated to the specific discretization of the structure. When the contribution of micro-inertia is already taken into account in the discretization, the constitutive relationship to be adopted is that proposed in section 12.6.

30.2.3.1.4. Structural detailing and other measures

30.2.3.1.4.1. Possibilities for a reduction of the loads in the design phase

In the design stage other measures could also be considered to reduce the danger of structural failure due to dynamic loads, such as:

- measures to secure sufficient distance between structure and threat;
- security walls and venting panels for a reduction of pressure;
- sandwich cross-sections with hard and soft layers, fuse supports or hinges to get a sufficient dissipation of energy, adapted to the special case of loading;
- symmetric design of the lateral load-resisting elements in plane, as for seismic design to avoid high local stresses.

30.2.3.1.4.2. General recommendations for detailing

With regard to detailing the following possibilities may be considered:

- use concrete with reduced Young's modulus;
- use concrete with high toughness, as fibre-reinforced concrete able to limit also spalling and scabbing effects and projectile generation;
- generate alternative load paths and redundant structural systems;
- ensure high ductility of cross-sections and connections;

Generally, it is important that the cross-sections, as well as the complete structure, have a high ductility. This involves the choice of the material, the layout of the connections and of the cross-sections as well.

Strengthening of existing structures can be done by the same methods as used in seismic design.

– use steel quality as required for seismic design;
– use adequate confinement to secure the cross-sections.

30.2.3.1.4.3. Strengthening of existing reinforced concrete structures

The following methods for strengthening existing structures may be considered:

– increase of strength by implicating external reinforcement, for example carbon retrofitting;
– application of sandwich elements for energy dissipation;
– limitation of debris by, for example, an additional layer of textile reinforced concrete.

References

[30.2-1] EN 15630-1:2019. Steel for the reinforcement and prestressing of concrete – Test methods – Part 1: Reinforcing bars, rods and wire.

[30.2-2] EN 15630-2:2019. Steel for the reinforcement and prestressing of concrete – Test methods – Part 2: Welded fabric and lattice girders.

[30.2-3] EN 15630-3 Steel for the reinforcement and prestressing of concrete – Test methods – Part 3: Prestressing steel.

[30.2-4] CEB Bulletin 188. Fatigue of Concrete Structures, 1988.

[30.2-5] JSCE **JSCE Standard Specifications for Concrete Structures [FURTHER DETAILS MISSING]**

[30.2-6] Ahmed e.a. . 1998 Tarig Ahmed, Eldon Burley, and Stephen Rigden, ”The State and Fatigue Strength of Reinforced Concrete Beams Affected by Alkali-Silica Reaction”, ACI material journal, Vol.95, issue 4, pp. 376-388, 1998.

[30.2-7] Maeshima e.a., 2016 Maeshima, T, Iwaki, I, Koda, Y：” Influence of alkali silica reaction on fatigue resistance of RC bridge deck”, Proceedings of the 11th *fib* International PhD Symposium in Civil Engineering, *fib* 2016

[30.2-8] US Army, 2008. UFC 3 - 340 - 02. Unified Facilities Criteria (UFC) - Structures to resist the effects of accidental explosions. U.S. Army Corps of Engineers (HQUSACE), Naval Facilities Engineering Command (NAVFAC), and Air Force Civil Engineer Support Agency, (AFCESA), December 2008.

[30.2-9] Leppänen, 2005. “Experiments and numerical analyses of blast and fragment impacts on concrete”, International Journal of Impact Engineering, Vol 31 (7), 843-860, 2005

[30.2-10] UFC 3-340-02. Unified Facilities Criteria (UFC) – Structures to Resist the Effects of Accidental Explosions, 2008.

[30.2-11] DIN 25449, 2016. Reinforced and prestressed concrete components in nuclear facilities – Safety concepts, actions, design and construction.

[30.2-12] Malvar, L. J. Malvar, L. J. - Review of Static and Dynamic Properties of Steel Reinforcing Bars, 1998,

[30.2-13] Mayrhofer, C, 2006. “Grundlagen zu den Methoden der dynamischen Grenztragfähigkeitsberechnung bei terroristischen Ereignissen”, 2006, *Workshop “BAU-PROTECT”*.

30.3 Seismic design, assessment or retrofitting

30.3.1 Introduction - format of the verifications

Seismic actions impose deformations on structures. The structure then produces the internal forces and moments which correspond to these deformations on the basis of its stiffness and resistance.

Under certain conditions linear elastic analysis may be used for the determination of the deformations due to the seismic action, but not of the internal forces (see section 30.3.2.2).

Action effects, S_{Ed}, in seismic situations and the corresponding resistances, R_d, for the verification of the ULS of resistance according to section 12.5.4.2.4 are deformations demands for ductile modes of behaviour and failure (e.g., for flexure with limited axial load), or internal forces for brittle ones (notably, for shear):

- deformation action effects, S_{Ed}, are determined through nonlinear analysis according to 30.3.2.4, or - under certain conditions - via linear elastic analysis according to 30.3.2.2.
- deformation resistances are normally plastic chord rotations at member ends, established according to 30.3.3.
- shear force action effects, S_{Ed}, are determined through nonlinear analysis according to 30.3.2.4, or by capacity design calculations according to 30.3.2.3.
- shear force resistances, R_d, are determined according to section 30.1.3, with the special rules of subclause 30.3.4.1 applying within flexural plastic hinges.

For the verification of the SLS of deformations in seismic situations according to section 12.5.4.5.2.5[**missing section**], a is normally the chord rotation at a member end and is verified according to section 30.3.4.1.

30.3.2 Determination of seismic action effects through analysis

Nonlinear methods of analysis should normally be used for seismic actions, taking into account the dynamic response of the structure and representing the action through the system of structural deformations or inertia loads it produces.

30.3.2.1 Modelling

The model of the structure should represent the distribution of its stiffness, mass and resistance, so that all significant deformation shapes and inertia forces are properly accounted for.

As the seismic action induces large inelastic rotation demands to beam ends, the effective slab width on either side of the web at the ends of T-beams in Figure 29.3-1, $b_{eff,i}$, is taken as the lesser of 25% of the beam span and of the mid-distance to the adjacent parallel beam, b_i. All slab bars which are parallel to such a beam and fall within this width are considered as fully effective as longitudinal reinforcement of the beam's end section.

30.3.2.1.1 Primary and secondary elements

Structural elements which contribute little to the lateral strength and stiffness of the structural system, or whose strength and stiffness against seismic actions is not considered as reliable, may be designated as secondary elements, which are not part of the lateral force resisting system. Their compliance criteria are less safe-sided than those of the rest, which are categorised as primary. In return, primary elements should be verified against the most unfavourable seismic action effects from two analysis models: one in which the strength and stiffness of secondary elements against seismic actions is neglected and another in which it is included.

If some of the elements are designated as secondary ones, two types of analyses should be carried out:

- primary elements should be verified against seismic action effects computed from a model which neglects the strength and stiffness of secondary elements against seismic actions;
- secondary elements should be verified against seismic action effects computed from a model which realistically accounts for their strength and stiffness against seismic actions.

In new structures, elements which are outside the scope of the code may be treated as secondary for convenience. In retrofitting, existing elements which are not upgraded may also be classified as secondary.

It is up to the designer to decide which elements to classify as secondary ones. However, the requirement to verify primary elements on the basis of an analysis which neglects the contribution of secondary elements, sets an effective ceiling to the number, type and location of the elements classified as secondary.

30.3.2.1.2 Stiffness

For new structures, the characteristic value of the yield stress of steel may be used as mean yield strength of steel reinforcement.

Seismic deformation demands are underestimated, if sources of flexibility are neglected and stiffness is overestimated. To this end, tension stiffening is neglected, as it diminishes with load cycling. Slippage of the tension bars from their anchorage in joints or foundation elements increases the member's apparent flexibility, owing to the fixed-end-rotation of the end section it produces.

Central in seismic design for controlled inelastic response is the hypothesis of a bilinear monotonic force-deformation law of the lateral-load-resisting system as a whole and of its individual components in primary (monotonic) loading. The effective stiffness to the yield-point should therefore be used as element elastic stiffness.

The chord rotation at the end of a one-dimensional member, θ, is the angle between the normal to the end section and the chord connecting the member ends at the member's displaced position (Figure. 30.3-1). It includes flexural and shear deformations along the member, as well as the fixed-end-rotation of the end section due to slippage of longitudinal bars from their anchorage zone outside the member length. In the linear elastic regime the member deformations that determine – together with any loading between the ends – the bending moments and shears along the member are the chord rotations at the two member ends.

30.3.2.1.2.1 General rules

Mean values of material properties – including strengths – should be used in the calculation of stiffness.

For both the ULS and the SLS the elastic stiffness should be based on fully cracked sections of those members which are expected to yield in the limit state of interest, without tension stiffening. The fixed-end-rotation of the member's end section due to slippage of longitudinal bars from their anchorage zone outside the member length should also be taken into account.

The secant stiffness to the yield-point should be used as effective elastic stiffness of those members which yield at the limit state of interest.

For one-dimensional concrete members (including slender walls) the secant stiffness to the yield-point may be estimated as:

$$EI_{eff} = \frac{M_y L_s}{3\theta_y} \qquad (30.3\text{-}1)$$

where:

M_y is the yield moment

θ_y is the chord rotation at yielding of the member end

$L_s = M/V$ is the shear span (moment-to-shear-ratio) at that end section in the seismic situation.

The stiffness of an element may be taken to be equal to the mean value of $M_y L_s/(3\theta_y)$, at the two ends of the element, for positive and negative ((sagging or hogging) moments there.

The yield moment may be estimated by section analysis assuming linear-elastic behaviour of the materials and using as yield criterion yielding at the centroid of the tension chord.

If longitudinal bars, ribbed or smooth, are lap-spliced starting at the end section of a member, the yield moment and all other resistance, stiffness and deformation capacity properties of the member should be computed with both bars in a pair of lapped compression bars fully counting as compression reinforcement.

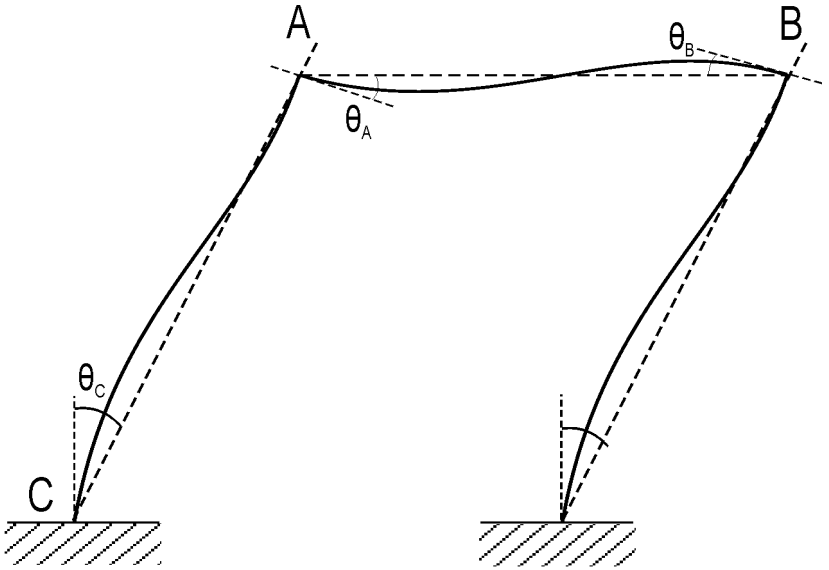


Figure 30.3-1: Illustration of the definition of chord rotation at member ends

Estimation of EI_{eff} before the analysis for the seismic action may be based on presumed values of L_s equal to 50% of the clear length of beams between columns and of columns between beams in the plane of bending, as well as of bridge piers fixed against rotation by the deck in the plane of bending. For the strong direction of walls in buildings, the value of L_s within a storey may be taken equal to 50% of the height from the wall's base section in that storey to the top of the wall in the building. In members cantilevering within the plane of bending L_s is the member clear length.

An average value of EI_{eff} at: (a) the two ends where it frames into other members in the plane of bending and (b) for positive and negative moments may be used for a member.

Application of equation (30.3-1) presumes that the longitudinal reinforcement of the end sections is known. In new structures this may entail dimensioning first this reinforcement from the non-seismic situations and minimum reinforcement and estimating then how much it may increase for the verifications in the seismic situation. As the value of EI_{eff} depends weakly on the amount of longitudinal reinforcement, use may also be made of empirical expressions giving the ratio of EI_{eff} to the uncracked gross-section stiffness for different types of members in terms of known parameters

If a member is wrapped near the yielding end in a FRP jacket, the yield moment and curvature, M_y and ϕ_y should be computed taking into account the enhancement of concrete strength due to confinement by the FRP. To this end, the elastic Modulus of concrete may be calculated from the value of the enhanced concrete strength, f_{cc}, eg, as $E_c=10000(f_{cc}(\text{MPa}))^{1/3}$

(e.g., the shear-span-to-depth-ratio at the end of the member, the mean axial stress, N/A_c, the ratio of mean bar diameter to the section depth, etc.). An example is the expression:

$$\frac{EI_{eff}}{E_c I_c} = a\left(0.8 + ln\left[max\left(\frac{L_s}{h}; 0.6\right)\right]\right)\left(1 + 0.048\, min\left(50MPa; \frac{N}{A_c}\right)\right) \qquad (30.3\text{-}2)$$

where the effect of slippage is included and coefficient a assumes the values:

- a = 0.09 for beams and members with T-, U-, H- or hollow rectangular section;
- a = 0.08 for rectangular columns;
- a = 0.115 for circular columns or piers and rectangular walls;

45°-cracking near the member end precedes flexural yielding if the value of $V_{Rd,c}$ is less than the shear force at flexural yielding, M_y/L_s.

The shear deformation of the member at the time it yields in flexure may be taken equal to:

$0.0019[1+h/(1.6L_s)]$ in beams or rectangular columns (30.3-3a)

$0.0011[1+h/(3L_s)]$ in walls or piers with a box section (30.3-3b)

$0.0025\ \max[0; 1-L_s/(8D)]$ in circular piers or columns (30.3-3c)

Eq. (30.3-1) applied with so-computed values of θ_y and with the prestress taken as part of the actions, underestimates, in general, the effective stiffness of prestressed components with bonded tendons. Hence, its use is safe-sided in the context of displacement-based seismic design or assessment

In a member with ribbed longitudinal bars the fixed-end rotation due to slippage of the tension bars from the region beyond the member length may be computed from:

$$\theta_{slip} = \frac{\phi \cdot d_b \cdot \sigma_s}{8\tau_b} \qquad (30.3\text{-}4)$$

where:

30.3.2.1.2.2 Rules for members with ribbed (deformed) bars, continuous in the potential plastic hinge end region

At the end of a member with ribbed bars which are continuous throughout the potential plastic hinge region adjoining the end section, θ_y is the sum of:

1. a flexural component, equal to $\phi_y(L_s+z)/3$ if 45°-cracking of the member precedes flexural yielding of its end section, or to $\phi_y L_s/3$ if it does not;
2. a shear deformation, which normally increases with decreasing member shear-span-to-depth ratio, L_s/h; and
3. the fixed-end-rotation due to slippage of longitudinal bars from the zone outside the member length, at yielding of the end section, $\theta_{slip,y}$..

ϕ is the curvature at the end section;

σ_s is the stress of the tension bars at the end section;

d_b is the mean diameter of the tension bars;

τ_b is the mean bond stress along the straight anchorage length of the tension bars outside the member length.

At yielding of the end section ϕ and σ_s may be taken equal to their yield values, ϕ_y and f_y, while for ribbed bars τ_b (in MPa) may be taken equal to $\sqrt{f_c}$(MPa), giving a value of θ_{slip} denoted as $\theta_{slip,y}$.

The increase in the member's apparent flexibility due to this fixed-end-rotation is equivalent to adding a rotational spring at the end section, with stiffness $M_y/\theta_{slip,y}$, where M_y is the yield moment there (proportional to ϕ_y). If such a spring is added, then the term $\theta_{slip,y}$ is not included in θ_y.

30.3.2.1.2.3 Rules for members with ribbed longitudinal bars, lap-spliced in a potential plastic hinge end region, starting at the end section

To account for the effect of a short lapping, l_o, of the straight ends of ribbed bars on the yield moment and curvature of a member with or without a FRP jacket around the region of the lap splice, the tension bars may have their yield stress reduced to:

$$f_{sm} = min\left(\sqrt{\frac{l_o}{l_{oy,min}}};1\right) f_{ym} \tag{30.3-5}$$

with:

$$l_{oy,min} = \frac{0,25 d_b \cdot f_{ym}}{max\left(\frac{c_{min}}{d_{bL}};0,7\right)\left(1+4\frac{t_f}{R_c}\frac{E_f}{E_c}\right)^2 f_{ctm}} \tag{30.3-6}$$

where:

- f_{ctm} may be taken as f_{ctm}(MPa)=0.3(f_{ck}(MPa))$^{2/3}$;
- c_{min} is the minimum concrete cover of lapped bars, or half the clear distance to the closest lap-spliced bar, whichever is smaller;

- t_f is the thickness of a FRP jacket, if any;
- E_f the elastic modulus of FRP;
- R the radius of FRP at chamfered corners of a rectangular section or around a circular one;
- E_c the elastic modulus of concrete.

The effect of lap splices on the chord rotation at yielding of the end section of a member with ribbed longitudinal bars having straight ends lap-spliced starting at the end section, may be accounted for as follows:

- The yield curvature should be computed according to the rules for members with lap-spliced ribbed bars;
- the term which accounts for shear deformations should be multiplied by the ratio of the yield moment as computed taking into account the effects of the lap splice, to its value neglecting them;
- to include or not the tension-shift, z, in the flexural component of θ_y, the end moment at diagonal cracking, $L_s \cdot V_{Rc}$, should be compared to the value of M_y that accounts for the splicing according to the general rules for members with lap-spliced ribbed bars.
-

30.3.2.1.2.4 Rules for members with smooth (plain) longitudinal bars, lap-spliced in a potential plastic hinge end region, starting at the end section

If the ends of smooth bars with standard 180° hook are lap-spliced over a length l_o starting at the end section, both bars in a pair of lap-spliced compression bars count as compression reinforcement.

If the ends of smooth bars at a lap-splice are straight without hooks, the yield moment may still be estimated from section analysis with elastic material laws till yielding, with both bars in a pair of spliced compression bars counting fully as compression reinforcement and the yield stress of the lap-spliced tension bars taken as:

$$f_{eff} = min\left(1; l_o / l_{oy,min}\right) f_{ym} \qquad (30.3\text{-}7)$$

where:

$$l_{oy,min} = 0.5 d_b \cdot f_{ym} / \sqrt{f_{cm}} \qquad (30.3\text{-}8)$$

where:

l_o is the length of the lap splice in [mm]

d_b is the mean diameter of the tension bars [mm];

f_{ym} and f_{cm} in MPa.

The maximum stress that a bar can develop ahead of a standard hook or bend may be taken as (cf. Eq. (3-17) in *fib* Bulletin 72 [30.3-1]):

$$f_o(MPa) = 60\sqrt{\frac{f_{cm}(MPa)}{25}}\left(\frac{20}{d_b(mm)}\right)^{0.2} \cdot \left(min\left(2; \left(min\left(3.5; \frac{c_{min}}{d_b}\right)^{0.25} \cdot min\left(5; \frac{c_{max}}{c_{min}}\right)^{0.1} + k_{conf}\, min\left(0.05; \frac{\frac{A_{sw}}{s} + \frac{E_f t_f}{E_s}}{d_b n_b}\right)\right)\right)\right) \quad (30.3\text{-}9)$$

where, with reference to Figure 30.3-2a:

- c_{min} $=\min(c_x; c_y, c_s/2)$, $c_{max}= \max(c_x; c_y, c_s/2)$
- n_b number of pairs of lapped bars in the potential splitting failure surface;
- k_{conf} effectiveness factor depending on the type of confinement of a potential splitting crack along the lapped bars:
  - k_{conf} $=12$ for confining reinforcement crossing the potential splitting surface and/or FRP jacket as in Figure 30.3-2b (net distance $\leq 5d_b$);
  - k_{conf} $= 3$ for reinforcement within the cover c_y and/or FRP jacket as in Figure 30.3-2c ($c_s \geq 8c_y$);
  - k_{conf} $= 0$ in other circumstances.
- A_{sw}/s total confining reinforcement area crossing potential splitting crack per unit length along the lapped bars;
- t_f total thickness of confining FRP jacket (if any) crossing potential splitting crack;

- E_s, E_f elastic modulus of steel or FRP (if any).

In the general case where a smooth bar is lap-spliced over a straight lapping l_o on one side of a member's end section followed by a standard hook or bend and is anchored on the other side with another standard hook or bend preceded by straight embedment depth l_b, and the maximum tensile stress of the bar which can be transferred by the hook or bend according to Eq. (3.30-9) is denoted by f_o on the lap-splice side or by f_b on the anchorage side, the maximum stress that can develop in the bar at the end section in question is given by:

$$f_{eff} = min\left(1; f_b / f_{ym} + l_b / l_{oy,min}; f_o / f_{ym} + l_o / l_{oy,min}\right) f_{ym} \qquad (30.3\text{-}10)$$

For straight end(s) of the bar(s) on the lap-splice or the anchorage side of the section, f_o or, respectively, f_b should be taken equal to zero in Eq. (30.3-10).

In a member with smooth longitudinal bars, the shear component of θ_y, under rule 2 in section 30.3.2.1.2.2 may be calculated as in a member with continuous ribbed bars.

Rules 1 and 3 in section 30.3.2.1.2.2 are replaced as follows:

The part of the chord rotation at yielding which is not due to shear deformations may be taken equal to the sum of:

1. the flexural deformation of the uncracked shear span, $M_y L_s/(3EI_g)$, with EI_g denoting the stiffness of the uncracked gross concrete section;
2. a fixed-end-rotation due to slippage of the tension longitudinal bars along their length within the member;
3. a fixed-end-rotation due to slippage of the tension longitudinal bars from the zone outside the member length;

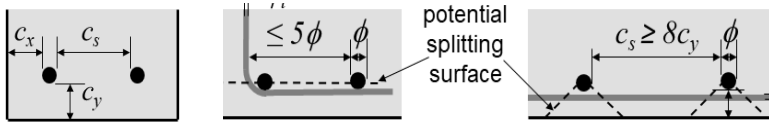
The fixed-end-rotations under 2 and 3 above may be estimated assuming a linear reduction of the tension stress in the bar from its effective yield stress at the yielding end section, at a rate of $f_y/l_{oy,min}$ per unit distance from the end section along the bar, with $l_{oy,min}$ given by equation (30.3-8).

If the bar ends with a standard hook within that distance, its stress ahead of the hook should not be taken greater than the maximum bar stress which can be anchored by that hook.

In members with smooth longitudinal bars, flexural cracking of the sections where the moment attains its maximum value (including the end sections at the face of joints with other members) controls the apparent stiffness till one of these sections yields. Till then the rest of the member remains essentially uncracked, in flexure and in shear.

The sum of the deformation components under 1 and 2 is the counterpart of the deformation component given by rule 1 in section 30.3.2.1.2.2.

The deformation component under 3 is the counterpart of that given by rule 3 in section 30.3.2.1.2.2, but is computed differently.



(a) (b) (c)

Figure 30.3-2: Anchored or spliced bars: (a) notation for geometric variables; (b) case for k_{conf}= 12 in Eq.(30.3-9); (c) case for k_{conf}= 3 in Eq. (30.3-9).

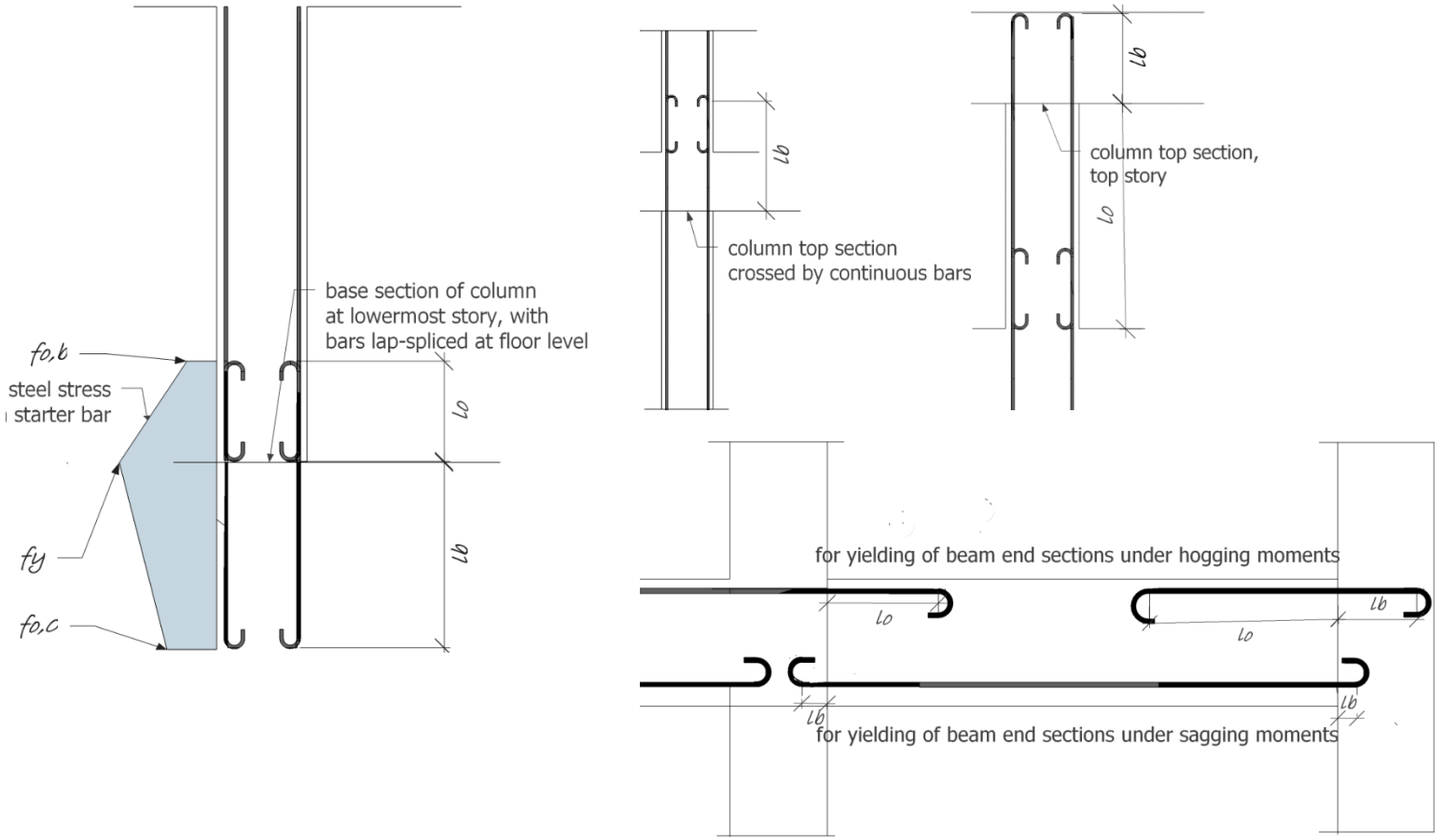


Figure 30.3-3: *Examples of embedment depth of smooth (plain) bars with hooked ends beyond the member's length and of anchorage or lap-splice length within the member.*

So, the total fixed-end-rotation at a yielding end section of a member (sum of deformation components under 2 and 3 above), denoted by $\theta_{slip,y}$, may be estimated as follows:

$$\theta_{slip,y} = \frac{\phi_y}{2} \left(\begin{array}{l} min\left[l_{oy,min}; l_b \right] \cdot \left(\frac{f_{eff}}{f_y} + min\left[\frac{f_b}{f_y}; max\left(0; \frac{f_{eff}}{f_y} - \frac{l_b}{l_{oy,min}} \right) \right] \right) + \\ min\left[l_{oy,min}; l_o \right] \cdot \left(\frac{f_{eff}}{f_y} + min\left[\frac{f_o}{f_y}; max\left(0; \frac{f_{eff}}{f_y} - \frac{l_o}{l_{oy,min}} \right) \right] \right) \end{array} \right) \quad (30.3\text{-}11)$$

where ϕ_y is the yield curvature at the end section considered, $l_{oy,min}$ is given by Eq. (30.3-8) and f_{eff} by Eq. (30.3-10), l_b is the embedment length of the bars in the joint or the foundation *outside* the member's length and $f_{o,b}$ the bar stress ahead of the hook on that side of the end section, see Eq. (30.3-9); l_o is the distance to the nearest hook along the tension bar *within the member* and $f_{o,0}$ the maximum stress the bar can develop ahead of that hook, given by Eq. (30.3-9) (see Figure 30.3-3 for examples).

30.3.2.1.2.5 Rules for members with a concrete jacket

For members strengthened with a concrete jacket, the following simplifying assumptions may be made:

- the jacketed member behaves monolithically, with full composite action between old and new concrete;
- the full axial load is assumed to act on the jacketed member, no matter whether it was originally applied to the old member alone;
- the properties of the jacket's concrete are assumed to apply over the full section of the jacketed member;
- the longitudinal bars of the old member may be taken into account for the purposes of modelling and verification of the jacketed member at their location in the section of the composite member and with their strength properties, as intermediate reinforcement between the tension and the compression chords.

Then, the yield moment, the chord rotation at yielding, the effective stiffness and the hysteretic energy dissipation during the cyclic response of the jacketed member may be taken equal to the values calculated under the assumptions above, regardless of the measures to connect the jacket to the old concrete.

Being flexible, flat-slab-frames without a stiff bracing system are prone to second-order (P-delta) effects and do not protect partitions from drift-induced damage. They should be combined with walls and/or beam-column frames and may not be appropriate as primary elements in high seismicity areas (see also Fardis 2022 [30.3-2])

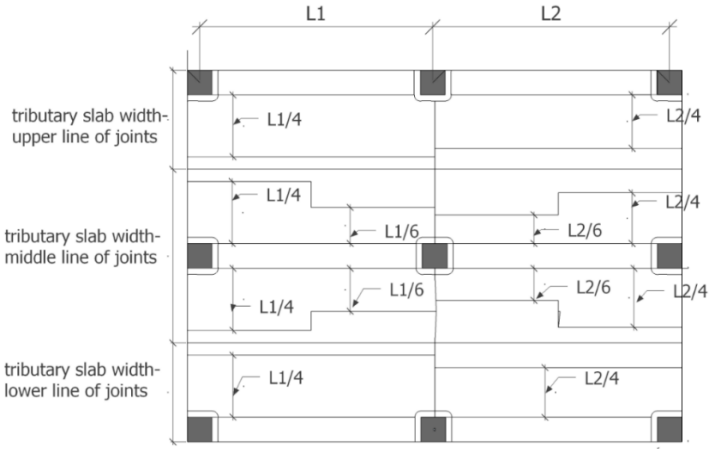


Figure 30.3-4*: Geometry of flat slabs: control perimeter around columns, tributary slab width and width of equivalent beams.*

The width of equivalent beams is chosen so that Eq. (30.3-1) gives for them an effective stiffness on average equal to the apparent secant stiffness of the slab to yielding. It is larger on one side of exterior joints than at interior ones because of a smaller increase of an edge joint's apparent flexibility by torsional deformation.

The longitudinal reinforcement ratio is taken equal to that of the support strip across the whole effective width of the horizontal element, no matter whether this width is wider than the support strip.

With the effective width and the reinforcement ratios specified in this sub-clause for the horizontal elements of the equivalent frame, the lateral stiffness of the flat-slab frame and the capability of the flat slab to transfer moments to the vertical elements by a combination of flexure and torsion are on average emulated.

30.3.2.1.2.6 Special rules for flat-slab frames in buildings

A flat-slab frame may be considered as a primary element which provides earthquake resistance as an equivalent moment-resisting frame comprising:

1. the vertical elements (columns or structural walls) which are monolithically connected to the flat slabs; these are considered as column elements monolithically connected at floor levels to the horizontal elements according to 2 below;
2. prismatic horizontal elements with:
   - depth equal to that of the slab in the vicinity of the connected column;
   - longitudinal axis connecting the centroids of the cross-section of adjacent column elements,
   - longitudinal reinforcement ratios at top and bottom equal to those of the support strip between the connected columns.
   - effective width on each side of this axis which extends beyond the edge of the supporting columns by the smallest of (a) to (c) at the side considered:
     a) one-quarter of the clear span between columns at interior slab-column joints, or one-third of it at joints which are at, or next to, the slab's edge (Option 1), or, for simplicity, one-quarter of the clear span irrespective of location (Option 2).
     b) the mid-distance to the nearest other columns of the floor;
     c) the distance to the edge of the slab.

The horizontal elements of the flat-slab frame should be considered in the structural model for the analysis in the seismic design situation, in the deformation-based verifications of the flat-slab (see 30.3.3.2) and in its force-and-deformation-based verification against punching shear (see 30.3.3.10), with the properties computed from the effective cross-sectional dimensions and the longitudinal reinforcement ratios specified in 2 of the present sub-clause.

A simple and cost-effective way of adding walls for seismic retrofitting is to infill with concrete selected frame bays, especially on the perimeter. For convenience in modelling and in design of the retrofitting, the resulting composite wall should act as fully monolithic. For monolithic behaviour, the new wall is often constructed thicker than the existing beams and columns, encapsulating them; the new reinforcement is placed outside the old members and detailed as in a new wall, and the slabs are pierced for continuity of the new concrete and the vertical bars across floors. This choice is convenient for design and detailing but may give a wall which is much stronger than necessary for the retrofitting and difficult to provide a foundation to.

A more rational and cost-effective alternative is to provide to the new wall a web with thickness not larger than the width of the old beams and columns around it. For the connection, the dowels crossing the interface should extend into the new web long enough to double as starter bars of the web's vertical and horizontal bars. A dowel of diameter d_d which doubles as starter bar to n web bars of diameter d_{bw} develops a tensile stress of $\sigma_s = n \cdot f_{yd} \left(d_{bw} / d_d\right)^2$ and its maximum design shear resistance, $F_{0,max} = \left(\pi \cdot d_d^2/4\right)\sqrt{f_{cd} \cdot f_{yd}}$ is reduced to:

$$F_{d,\max} = F_{0,\max}\sqrt{1-\left(\frac{\sigma_s}{f_{yd}}\right)^2} = F_{0,\max}\sqrt{1-n^2\left(\frac{d_{bw}}{d_d}\right)^4} \qquad (30.3\text{-}12)$$

Dowels crossing Section 0 normally have higher tensile strength per linear meter of horizontal interface than the vertical reinforcement of the new web. So, Section d has

30.3.2.1.2.7 Rules for walls added by infilling a bay of an existing frame

If the new web has a thickness not larger than the width of the old beams and columns around it, its connection to the surrounding frame members should be rigid, in order for the composite wall to behave like a monolithic wall. The rules for the yield properties and the effective stiffness of the composite wall are variations of those for monolithic walls.

Two sections are considered in each storey of the composite wall (Figure 30.3-5):

– The base section (index 0). If the new web is connected to the frame beam or footing below through dowels crossing section 0 and fully anchored below, these dowels serve as web vertical reinforcement of the section;
– The section where dowels connecting the RC infill to the beam or footing below stop (index: d); its distance from the base section is the embedment length of dowels in the new web, $l_{0,d}$.

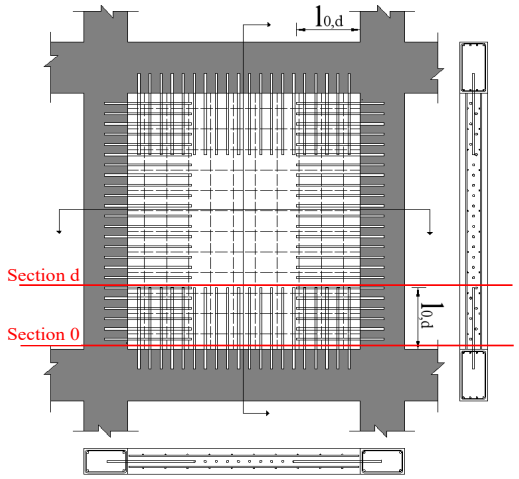


Figure 30.3-5: Connection of new wall web to surrounding frame members, with dowels all-along the interface, doubling as starter bars lap-spliced to smaller diameter web bars over the lap length of the latter. Sections 0 and d are candidates for being critical in flexure.

If the vertical bars of the columns are lap-spliced, over a length l_o less than the minimum lapping from Eq. (30.3-6), the effective steel stress of the tension chord

lower yield strength than Section 0 and, despite its lower acting moment, may yield first, controlling the wall's yield moment.

Material properties to be used in calculations of stiffness and flexural deformation capacity and moment resistance are those of the old members, except for the vertical web bars which are new. Properties to be used in the verification of the web and it's interfaces in shear are those of the new materials.

The ratio of M to θ is considered constant until the storey yields in flexure.

When bending is mainly in a single plane, it may be sufficient to use for each member end a moment-(chord) rotation model within that plane. The effect of significant variation of the axial load during the response (e.g., in the exterior columns of tall frames) on the moment-rotation behaviour should be taken into account. Simplified treatment of the coupling between the two directions of bending in vertical elements of three-dimensional models is normally acceptable.

is the minimum of the effective stresses obtained from Eq. (30.3-5) for the column bars and the vertical bars of the new web.

Section analysis gives the yield moment and the yield curvature of Sections 0 and d, $M_{y,0}$, and $\phi_{y,0}$ or $M_{y,d}$, and $\phi_{y,d}$, respectively. If $L_{s,0}$ is the shear span (M/V ratio) at Section 0, the effective yield moment at the base of the storey is equal to $M_y = \min[M_{y,0}; M_{y,d}(1+l_{0,d}/L_{s,0})]$.

Section 0 cracks first and controls the cracked stiffness. So, for the wall storey, the secant-to-yield-point stiffness is $EI_{eff} = M_{y,0}L_{s,0}/3\theta_{y,0}$, where $\theta_{y,0}$ is the chord rotation at the base of the wall for yielding of Section 0, computed according to clause 30.3.2.1.2.2.

The wall storey yields when the moment at Section 0 reaches the value $M_y = \min[M_{y,0}; M_{y,d}(1+l_{0,d}/L_{s,0})]$, which is possibly less than $M_{y,0}$. The chord rotation at Section 0 at apparent yielding of the wall, θ_y, is the value attained when the moment at the base becomes equal to M_y and is derived from the effective stiffness of the storey, EI_{eff}, as $\theta_y = M_y L_{s,0}/3EI_{eff}$.

Verification in shear may be carried out as though the composite wall were monolithic. For the verification of the interfaces in shear according to 30.1.3.7.2, the last term in Eq. (30.1-77) should be multiplied by $\sqrt{1-n^2\left(\frac{d_{bw}}{d_d}\right)^4}$, where n is the number of web bars of diameter d_{bw} to which a dowel of diameter d_d doubles as starter bar.

30.3.2.1.3 Models of member hysteretic behaviour and energy dissipation

30.3.2.1.3.1 Introduction

One-dimensional members (including slender walls) designed or retrofitted for earthquake resistance normally have all their inelastic action concentrated in flexural plastic hinges at their ends, where they are connected with other components. As a minimum, the nonlinear model of such a member should employ a force-deformation relation for these end regions – notably, a relation between the end moment and the corresponding (chord) rotation.

In the end, all components (including those of an existing structure being retrofitted for earthquake resistance) are normally verified to remain elastic if they are brittle, or to have some margin against their ultimate deformation - after which the drop in resistance is significant – if ductile. So, the reduction in resistance after ultimate strength may be neglected and the force-deformation relation in primary loading taken bilinear, especially if the initial post-yield strain-hardening is also neglected, with the full post-yield behaviour taken perfectly plastic. It is on such a straight post-yield branch that the limit deformation is verified, with corrective measures taken if this limit is exceeded.

The envelope is often idealized as a multi-linear curve with corner points specified in terms of a member's geometry and material properties and serves as a skeleton of the peaks of full force-deformation loops in cyclic loading, possibly with corner points evolving with cycling of the deformation.

Hysteresis rules play also an important role for the estimation of residual deformations of members (for local damage) or of the structure as a whole (e.g., the permanent tilt) after the earthquake.

A strong earthquake will find non-prestressed concrete members already cracked. owing to gravity loads, restraint of thermal or drying shrinkage and/or past earthquakes. So, a model which is linear until yielding, as in Figure 30.3-6, is appropriate for them. However, the hysteretic energy dissipation in small cycles before yielding should be accounted for in the analysis model.

As a minimum, nonlinear models for components should use a bilinear force-deformation law in primary (virgin or monotonic) loading. Positive post-yield stiffness (due to strain-hardening) may be neglected and elastic-perfectly plastic behaviour assumed instead. Strong post-yield softening or cyclic decay of the resistance may be taken into account through negative post-yield stiffness.

Force-deformation models of members should realistically represent the envelope of experimental force-deformation loops and the associated hysteretic energy dissipation in large response cycles as a function of the deformation amplitude.

The rules for unloading, reloading or small-cycle behaviour should reflect the hysteretic energy dissipation in cyclic tests of members as a function of the deformation history.

The energy dissipated in a force-displacement hysteresis loop, E_h, normalized by the elastic deformation energy $E_{el}=F_{max}\delta_{max}/2$, at the cycle's peak force and displacement, gives an "equivalent" viscous damping ratio, $\zeta = E_h/(4\pi E_{el})$.

30.3.2.1.3.2 Hysteretic energy dissipation before yielding

Concrete members dissipate energy even in small amplitude pre-yield cycles. In pre-yield load cycles the "equivalent" viscous damping ratio, ζ, may attain quite high values during the cycles which induce cracking, but decreases thereafter.

Mean values of the "equivalent" viscous damping ratio, ζ, are about 8% in walls or 5.5% in all other types of members, well-detailed for ductility and earthquake resistance or not, independently of the continuity and surface characteristics of longitudinal bars, of possible retrofitting with a jacket of concrete or FRP, etc., and of deformation amplitude, at least in the range of stabilized cracking before yielding.

Bilinear unloading-reloading parallel to the elastic and post-yield branches in monotonic loading is characteristic of steel but not of structural concrete, producing unrealistically large hysteretic damping.

Unloading from the envelope after yielding is nearly linear, almost until the force reverses sign; its average stiffness from the envelope to zero force is gradually reduced relative to the "elastic" stiffness from the origin to the yield point as the ductility ratio (: ratio of peak displacement to yield displacement) at the point on the envelope where unloading starts increases (see Figure 30.3-6).

In members designed and detailed for earthquake resistance, having continuous or properly lap-spliced ribbed bars, (re)loading in the opposite direction may be taken to follow a nearly straight force-deformation branch from the residual deformation to the peak force and deformation reached in the past in that direction of loading (Figure 30.3-6).

In members with smooth bars or short lap-splices or in joints where most of the deformation is due to bond-slip , as well as in members controlled by shear (e.g., those with low shear-span-to-depth ratio), reloading from the residual deformation takes place initially at a lower stiffness, increasing afterwards and leading again to the peak force and deformation reached before in that direction of loading. The resulting inverted-S shaped, "pinched" loops are narrower and dissipate less hysteretic energy.

The "equivalent" viscous damping ratio, ζ, may be expressed as a fraction of the "equivalent" viscous damping ratio of a "reference hysteresis model" (shown in Figure 30.3-6 with a dashed line) which has:

– a bilinear primary (or virgin, or monotonic) loading curve, consisting of an elastic branch to the yield point and a post-elastic branch with constant hardening ratio p, serving as skeleton of the peaks of full unload-reload loops;
– linear unloading from the peak point on the post-elastic branch of the primary curve with the elastic stiffness (i.e., parallel to the elastic primary branch) until the horizontal axis is reached at a residual deformation d_r;
– linear, peak-oriented reloading, i.e., from the residual deformation after unloading, straight to the peak point ever reached before on the post-elastic branch in the current direction of reloading.

30.3.2.1.3.3 Hysteretic energy dissipation after yielding

The force-deformation law in primary loading should be supplemented with unloading-reloading (“hysteresis”) rules, realistically reflecting the amount of post-yield hysteretic energy dissipation and the reduction of unloading and reloading stiffness with increasing peak deformation of a cycle (“stiffness degradation”) which characterises concrete components. The decay of resistance with load cycling should be included if it is significant (e.g., in brittle or poorly detailed components).

After yielding, hysteretic energy dissipation, as expressed by the "equivalent" viscous damping ratio, ζ, increases with increasing ductility factor ratio μ (: ratio of peak chord rotation in a cycle to the chord rotation at yielding).

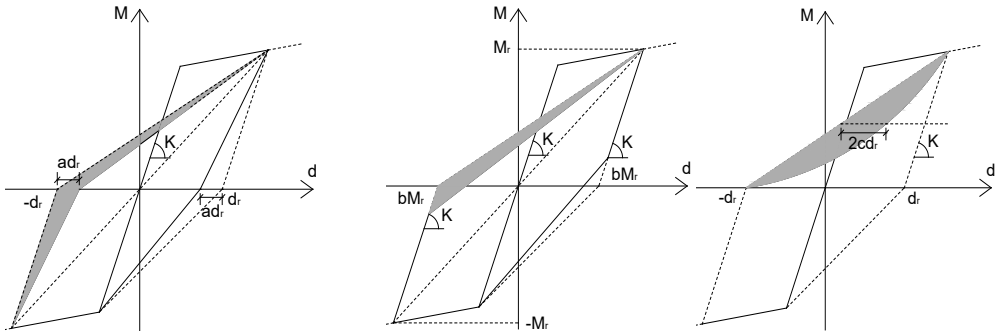


Figure 30.3-6: Simple hysteretic force-deformation models and loss in energy dissipation with respect to the "reference hysteresis model" due to: (a) degradation of unloading stiffness (modified Takeda model); (b) "early reload"; (c) "pinching" with parabolic reloading.

The "reference hysteresis model" gives an "equivalent" viscous damping ratio, ζ_o, equal to:

$$\zeta_o = \frac{1-p}{\pi}\left(1-\frac{1}{\mu}\right) \qquad (30.3\text{-}13a)$$

The same dependence on p and μ is retained for the "equivalent" viscous damping ratio, ζ, which is taken to be a fraction (1-λ) of ζ_o, with λ called "loss in energy dissipation":

$$\zeta = (1-\lambda)\zeta_o = (1-\lambda)\frac{1-p}{\pi}\left(1-\frac{1}{\mu}\right) \qquad (30.3\text{-}13b)$$

Loss in energy dissipation with respect to the "reference hysteresis model" may take place in various ways. Three typical ones are shown in figure 30.3-6:

a) degrading unloading stiffness (as in the “modified Takeda model”), controlled by unloading parameter *a* (Figure 30.3-6(a));
b) "early reload" (or "aborted unloading") in Figure 30.3-6(b), controlled by unloading parameter *b*;
c) "pinching" of the loop, shown in Figure 30.3-6(c) to be in the form of parabolic reloading, controlled by pinching parameter *c*.

In Figure 30.3-6 the ratio of the shaded to the triangular area enclosed by the chord connecting the positive to the negative peak on the primary branches and by the single

unload-reload path between these peaks in the "reference hysteresis model" is the loss in energy dissipation. So, for given "loss in energy dissipation" parameter λ, we have:

$$a + b + 4c/3 = \lambda \quad (30.3\text{-}14)$$

The "loss in energy dissipation" parameter λ may be taken as:

$$\lambda_0 = \lambda_m - 0.12\frac{L_s}{h} - 0.15(100\rho_{tot}) + \frac{2}{3}(100\rho_f \frac{E_f}{E_c}) + 0.36\frac{l_{oy,min}}{l_o} - 0.03\frac{N}{A} - 0.2(100\rho_w) \quad (30.3\text{-}15)$$

where:

- λ_m = 1.1 in rectangular beams or columns with ribbed bars;
- λ_m = 1.35 in circular columns with ribbed bars;
- λ_m = 0.8 in walls, rectangular or not, or members of box-section, with ribbed bars;
- λ_m = 1.25 in rectangular beams or columns with smooth (plain) bars;
- ρ_{tot}: total longitudinal reinforcement ratio;
- L_s: shear span (M/V ratio) at the end section of the member where a plastic hinge may form;
- ρ_f: geometric ratio of any FRP jacket wrapped around end region of the member, in the cross-sectional direction of the member considered;
- E_f: elastic modulus of any FRP wrapped around end region of the member;
- E_c: elastic modulus of concrete;
- $l_{oy,min}/l_o$ = 0 if all longitudinal bars are continuous in the end region considered; if they are not, then:
  - l_o lapping within the end region where a plastic hinge may form;
  - $l_{oy,min}$ minimum lap-length which is necessary to avoid adverse effect of lap-splicing on the yield moment and effective stiffness of the member:
    - from Eq. (30.3-6) for ribbed bars with straight lapped ends;
    - from Eq. (30.3-8) for plain ones with straight lapped ends;
  - ρ_w **[to be declared]**
  - N/A in [MPa]

  one-sixth of the value from Eq. (30.3-8) for plain bars with 180°-hooks at their lapped ends.

30.3.2.1.3.4 Combination of hysteretic energy dissipation and viscous damping in the model

Viscous damping is commonly used not only to represent hysteretic energy dissipation in the members themselves, but also to model the energy dissipated in components which are not included in the structural model (infill walls, other non-structural elements, or structural ones that are not part of the lateral-load-resisting system) or are taken as rigid (eg, floor diaphragms).

Index 0 in λ_0 means that this value of λ can only be used with 0% viscous damping for pre-yield hysteretic energy dissipation. In other words, when pre-yield hysteretic energy dissipation in the member itself is accounted for through a dedicated hysteretic model which runs and dissipates energy throughout the pre-yield stage of the member and is switched-off at a member end once the member yields at that end. If global viscous damping is employed throughout the analysis, the hysteretic model used for members should reflect the balance between the energy dissipation implied by Eq. (30.3-14) and global viscous damping. If an "equivalent" viscous damping ratio denoted by ζ (%) is included in the global viscous to account for the hysteretic energy dissipation in the members themselves before they yield, the loss in energy dissipation, should be increased to λ_ζ, as follows:

$$\lambda_\zeta = \lambda_0 + a_m \cdot \zeta(\%) \qquad (30.3\text{-}16)$$

where:

- $a_m = 0.06$ in rectangular beams or columns with ribbed bars;
- $a_m = 0.05$ in circular columns or walls - rectangular or not - or members of box-section with ribbed bars;

$a_m = 0.03$ in rectangular beams or columns with smooth (plain) bars.

This is the "equal displacement rule" at the level of member deformations. For a single-degree-of-freedom oscillator with elasto-plastic force-deformation law in monotonic loading the "rule" states that the maximum displacement in the inelastic seismic response is about the same as that of an the elastic oscillator with the same mass, damping and elastic stiffness.

Concrete structures have fundamental periods in the range where the "equal displacement" rule gives fairly accurate results for an "equivalent" single-degree-of-

If the force-deformation model used for the member is linear from the origin to the yield point, account should be taken of the hysteretic energy dissipated by the member before one or both of its ends yield in flexure. This can be conveniently accomplished by introducing viscous damping in the model for the analysis

Viscous damping is typically introduced at the level of the entire structure, with the global damping matrix taken as a linear combination of the global mass and (initial or tangent) stiffness matrices (Rayleigh damping). Global viscous damping is retained throughout the nonlinear analysis of the response, i.e., after members enter the post-yield range. If part of the global viscous damping is meant to reflect energy dissipated in the member itself, measures should be taken to avoid double counting by the hysteretic model.

30.3.2.2 Linear elastic analysis for the calculation of seismic deformation demands (action effects)

Member inelastic flexural deformations due to the seismic action may be determined through linear elastic analysis with damping 5% of critical, under the condition that they are not concentrated at certain parts of the structure (notably at one side of a building in plan, or in one or few building storeys or bridge piers) but are distributed fairly uniformly throughout the structure.

freedom oscillator and applies well on average for member deformations (albeit with considerable deviations for individual members).

The end section of a strong column framing into weaker beams, or of a strong beam framing into weaker columns, does not yield in the seismic situation, even when their M_E/M_R-ratio exceeds 1.0. Except at such sections, the ratio M_E/M_R is about equal to the corresponding ductility ratio demand for the chord rotation at the end of the member (: peak inelastic chord rotation demand divided by the corresponding value at yielding of the end section).

In the following M_R denotes a moment resistance component with vector normal to the shear force of interest, index b is used for beams, c for columns, w for walls and l_{cl} is the clear length of a beam, column or bridge pier:

Applicability of linear elastic analysis for the estimation of inelastic flexural deformations may be checked by inspection of the distribution over the structure of the ratio of the moment from the linear analysis at member end sections, M_E, to the corresponding moment resistance, M_R.

The pertinent moment-deformation relations should be used to calculate moments at member ends from inelastic flexural deformations estimated via linear elastic analysis.

30.3.2.3 Calculation of force demands through Capacity Design

Force demands on brittle components or failure modes, such as those due to shear, can be calculated through capacity design:

– when linear elastic analysis is used for the estimation of deformation seismic action effects,

– when the type and level of modelling used for nonlinear seismic analysis does not lend itself for estimation of force demands in brittle components or failure modes with sufficient accuracy.

In capacity design the maximum force demands on brittle components or modes of behaviour or failure are estimated from equilibrium and from the moments delivered to the component in question at its connections to the others. Each of these moments is taken equal to the minimum of:

(a) the moment component with vector normal to the shear force of interest from the analysis, M_E,

(b) the moment resistances at plastic hinge formation - multiplied by $\gamma_{Rd} \geq 1.0$ for overstrengths not explicitly modelled (e.g., due to steel strain hardening) - with plastic hinges taken to form at those two opposite faces of a joint where the weaker elements frame into.

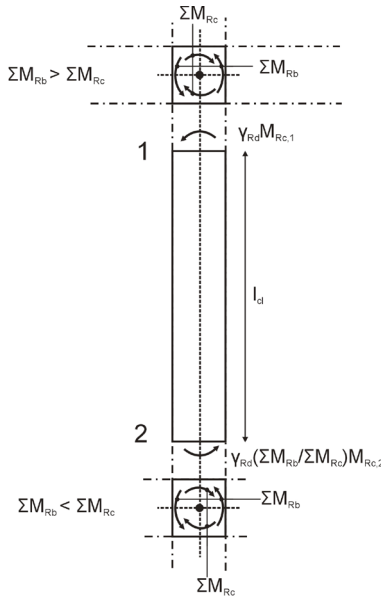


Figure 30.3-7: End moments for capacity design shears in frame column.

(1) The capacity-design shear between ends 1 and 2 of a frame column, bridge pier or similar member is:

$$V_{CD} = \frac{min\left[M_{E,c1};\ \gamma_{Rd}\cdot M_{R,c1}\cdot min\left(1;\ \frac{\sum M_{R,b}}{\sum M_{R,c}}\right)_1\right] + min\left[M_{E,c2};\ \gamma_{Rd}\cdot M_{R,c2}\cdot min\left(1;\ \frac{\sum M_{R,b}}{\sum M_{R,c}}\right)_2\right]}{l_{cl}} \quad (30.3\text{-}17)$$

where:

M_E and M_R enter with positive values;

ΣM_{Rc} or ΣM_{Rb} are the sums of moment resistances of the columns or the beams framing into the joint, respectively;

At the connections with the foundation or of a pier with a deck integral with it, it is always: $\Sigma M_{Rb}/\Sigma M_{Rc} > 1$;

For convenience, the moment resistances of columns or piers, M_{Rc}, are computed for the axial force due to gravity loads alone.

(2) The capacity-design shear at distance x from end section i of a frame beam (the other end denoted by j) is:

$$V_{CD}(x) = V_{g+\psi 2q,o}(x) + \frac{min\left[M_{E,bi}^{-};\ \gamma_{Rd}\cdot M_{R,bi}^{-}\cdot min\left(1;\frac{\sum M_{R,c}}{\sum M_{R,b}}\right)_i\right] + min\left[M_{E,bj}^{+};\ \gamma_{Rd}\cdot M_{R,bj}^{+}\cdot min\left(1;\frac{\sum M_{R,c}}{\sum M_{R,b}}\right)_j\right]}{l_{cl}}$$

(30.3-18)

where:

all moments enter as positive, but index (-) is used for moments inducing tension at the top flange of a beam (hogging ones) and (+) for those inducing tension at the bottom (sagging);

$V_{g+\psi 2q,o}(x)$ is the shear force at section x due to the quasi-permanent gravity loads $g+\psi_2 q$, with the beam considered as simply supported (index: o).

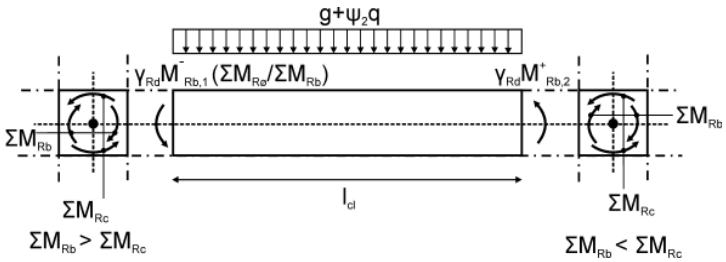


Figure 30.3-8: End moments for capacity design shear in a frame beam.

(3) In a joint where a horizontal element (a beam – including a foundation beam – a bridge deck, etc.) and a vertical one (frame column, structural wall, bridge pier, etc) frame into each other, the nominal shear stress, v_j can be derived through capacity-design:

either from the horizontal shear force in the joint, V_{jh}, as: $v_j = \frac{V_{jh}}{b_j h_{jc}}$ (30.3-19a)

or

from the vertical shear force in the joint, V_{jv}, as: $v_j = \frac{V_{jv}}{b_j h_{bj}}$ (30.3-19b)

where:

b_j is effective joint width in the orthogonal horizontal direction;

h_{jc} is the section depth of the vertical element in the joint in the the section depth of the vertical element in direction of V_{jh}, and

h_{bj} is the section depth of the horizontal element in the joint.

If $\sum M_{Rb} < \sum M_{Rc}$, $v_{j,}$ is controlled by the horizontal shear force in the joint:

$$V_{jh} = \sum M_{Rb} \left(\frac{1}{h_{bj}} - \frac{1}{h_{st}} \cdot \frac{L_b}{L_{bn}} \right) \qquad (30.3\text{-}20a)$$

Otherwise, i.e., if $\sum M_{Rb} > \sum M_{Rc}$, it is governed by the vertical shear force:

$$V_{jv} \approx \sum M_{Rc} \left(\frac{1}{h_{cj}} - \frac{1}{L_b} \cdot \frac{h_{st}}{h_{st,n}} \right) \qquad (30.3\text{-}20b)$$

where:

h_{st}, $h_{st,n}$ are theoretical and clear height of the vertical element, respectively,

L_b, L_{bn} are theoretical and clear span, respectively, of the horizontal element (beam or deck), and

If the moments from the analysis, M_{Eb}, M_{Ec}, are such that $\sum M_{Eb} < \sum M_{Rb}$ and $\sum M_{Ec} < \sum M_{Rc}$, v_j may be estimated either from:

- the joint's horizontal shear, V_{jh}, from Eq. (30.3-20a) using in it $\sum M_{Eb}$ instead of $\sum M_{Rb}$,

or

- the vertical shear, V_{jv}, from Eq. (30.3-20b) using there $\sum M_{Ec}$ instead of $\sum M_{Rc}$

(4) In flat-slab-frames considered as primary elements for the seismic action, slab-column joints should be designed for the vertical shear force from the analysis alongside "unbalanced" column moment in each one of the two orthogonal directions, taken equal to the smaller of:

1. The sum of column moment resistances at the interfaces with the slab above/below the joint;
2. The slab's flexural resistance at a support section normal to the plane of bending, computed:

a) either from the full width of the slab tributary to the joint (Figure 30.3-9):
   - at interior joints, as 80% of the sum of moment resistances in sagging and hogging bending;
   - at edge joints, as 75% of the hogging moment resistance;

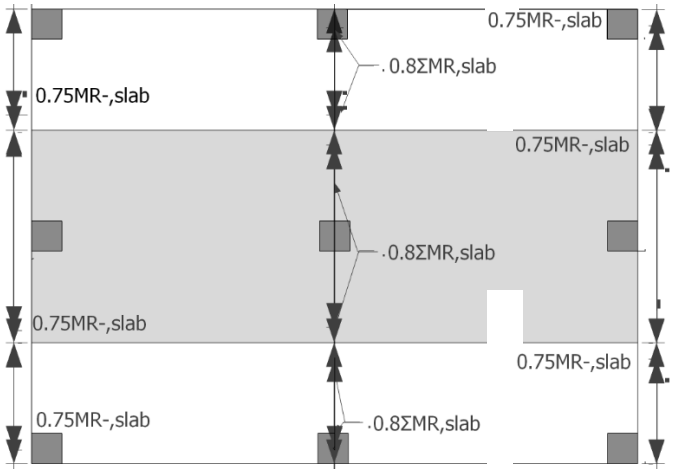


Figure 30.3-9: Moments transferred from column to flat slab, as determined by capacity design from the flexural resistance of the full tributary slab width.

b) or from the support section of "equivalent beams" framing in the joint in the plane of bending and having width according to section 30.3.2.1.2.6 (Figure 30.3-4):
   - at interior joints, as 1.25-times the sum of sagging and hogging moment resistances of the "equivalent beam";
   - at corner edge joints, as 80% of the hogging moment resistance of the "equivalent beams" according to Option 1 in 2(a) of section 30.3.32.1.2.6, or 100% for Option 2.
   - at edge joints, as at corner ones, but using the sum of hogging and sagging moment resistances, instead of just the hogging ones, for moments with vector at right angles to the edge.

(a) above applies in case the flat slab is modelled with Finite Elements across its full width, while (b) addresses the case where "equivalent beams" are used instead.

(5) Capacity-design shears along the height of a multi-storey wall cannot be established only from equilibrium moment capacities of the wall at the base and of

Whenever capacity-design effects cannot be determined solely from the moment capacities at plastic hinge formation on the basis of equilibrium, they may

beams framing into it at floor levels. They may be conveniently estimated, instead, at elevation z above the base (top of foundation or of a rigid basement) as follows:

For "squat" walls (with height-to-length ratio $h_w/l_w \leq 2$):

$$V_{CD,w}(z) = min\left(1; \frac{M_R(z=0)}{M_E(z=0)}\right) \cdot V_E(z) \qquad (30.3\text{-}21a)$$

For "slender" walls ($h_w/l_w > 2$), taking also into account higher mode effects on shears after yielding at the base:

$$V_{CD,w}(z) = \sqrt{min\left(1; \frac{M_R(z=0)}{M_E(z=0)}\right)^2 + \sum_{i>1}\left(\frac{V_{E,i}(z=0)}{V_{E,1}(z=0)}\right)^2} \cdot V_{E,1}(z) \qquad (30.3\text{-}21b)$$

where:

$V_E(z)$ shear force from modal response spectrum analysis for the seismic action of interest,

$V_{E,i}(z)$ shear force from modal response spectrum analysis for the seismic action of interest, due to normal mode i,

$V_{E,1}(z)$ shear force from modal response spectrum analysis for the seismic action of interest, due to the normal mode with the largest modal participating mass in the direction of the shear force considered; that mode has index i=1,

$M_R(z=0)$: moment resistance at the base section (z=0) with vector at right angles to the shear force considered, for the value of axial load due to the quasi-permanent gravity loads alone,

$M_E(z=0)$: moment at the base section (z=0) with vector at right angles to the shear force considered, from modal response spectrum analysis for the seismic action of interest.

Unlike linear elastic analysis, which may be relied upon – under certain conditions – to estimate seismic deformations but not internal forces, nonlinear analysis may be used to estimate all types of seismic action effects.

be found assuming that the seismic action effects at the instant the moment capacities at plastic hinge are reached are proportional to the corresponding outcomes of the linear elastic analysis for the seismic situation.

30.3.2.4 Nonlinear analysis for the seismic action

Nonlinear dynamic analysis, with solution of the equations of motion in the time-domain, is the reference analysis method for seismic actions. The action should be specified as a suite of independent seismic events in terms of time-histories of the ground motion components (see section 12.5.4.1.5).

Under certain conditions, nonlinear static ("pushover") analysis may be applied, under gradually increasing unidirectional lateral forces.

Nonlinear models should be employed, at least for those components expected to enter the inelastic regime in the seismic situation of interest.

The elastic stiffness of components and the viscous damping associated with the elastic regime should be the same as in linear elastic analysis.

30.3.3 Cyclic plastic rotation capacity of prismatic members

30.3.3.1 Definitions

Under seismic loading material failure at the local level (even rupture of a tension bar) does not constitute by itself member failure. The plastic hinge fails owing to local material failures gradually accumulating during cycling of the imposed deformations.

In seismic situations the verification at the ULS of one-dimensional members (including slender walls) takes place in terms of inelastic flexural deformations in plastic hinges forming at member ends.

Deformation measures used in the verifications should represent the behaviour of the plastic hinge region as a whole.

The global safety format in section 12.5.5 is appropriate for the ULS verification of a plastic hinge region, considered as an element.

The dependence of the plastic rotation capacity on each basic variable (material property or geometric dimensions) is not always monotonic and does not lend itself to application of partial factors on these variables.

The plastic part of the chord rotation at a member end is essentially the same as the plastic hinge rotation there, plus the post-yield part of the fixed-end-rotation, θ_{slip}, due to slippage of longitudinal bars from their anchorage zone outside the member length.

An appropriate deformation measure for the plastic hinge region is the plastic part of the chord rotation at the member end, θ_{pl}.

Under cyclic loading ${\theta^{pl}}_u$ is the value of θ_{pl} beyond which an increase in imposed deformation cannot increase the moment resistance above 80% of its maximum ever value.

The ultimate value of θ_{pl}, denoted as ${\theta^{pl}}_u$, is conventionally identified with a 20% drop in moment resistance.

30.3.3.2 Members with continuous ribbed bars - including members with FRP jacket around the plastic hinge region

The expected (mean) value of the plastic part of the ultimate chord rotation at a member end may be estimated as:

a_{sl} is equal to 0 if the tension bars are fixed, e.g., by welding, at the section of maximum moment, or if that section is on an axis of symmetry of the geometry and the loading, or under similar conditions.

As corrosion of reinforcement reduces its strength and ductility, reduces the diameter and cross-sectional area of bars and can split open their cover, it reduces the member's rotation capacity. Whenever relevant and feasible, this effect should be taken into account (see Vecchi and Belleti [30.3-3]).

For more information on ultimate strain criteria reference is made to Grammatikou et al 2016 [30.3-4].

$$\theta_{u,m}^{pl} = \left(\phi_u - \phi_y\right) L_{pl} \left(1 - \frac{L_{pl}}{2L_s}\right) + a_{sl} \cdot \Delta\theta_{slip,u-y} \qquad (30.3\text{-}22)$$

where:

ϕ_u and ϕ_y are the ultimate and the yield curvature, respectively, of the end section,

L_s is the shear span (M/V ratio) at the member end,

L_{pl} is the "plastic hinge length".

$a_{sl}\Delta\theta_{slip,u-,y}$ is the post-yield part of the fixed-end-rotation due to slippage of the tension bars from their anchorage zone outside the member length, from Eq. (30.3-30) with a_{sl} a coefficient, equal to 0 if such slippage is not physically possible, or to 1 if it is.

ϕ_u and ϕ_y are determined from section analysis.

– for ϕ_y, linear-elastic stress-strain relations may be assumed until yielding of the tension or the compression chord.

– for ϕ_u, the parabola-rectangle diagram of section 14.11.1.5 and Figure 14.11-5 should be used for concrete in compression and the idealised one of chapter 15 and Figure 15.10-3 (with linear strain-hardening) for the reinforcing steel. Calculation of ϕ_u should take into account all possible failure modes:

(a) rupture of tension reinforcement in the full, unspalled section;

(b) exceedance of the concrete ultimate strain ε_{cu2} at the extreme compression fibres of the unspalled section;

(c) rupture of tension reinforcement in the confined core after spalling of the cover;

(d) exceedance of the ultimate strain $\varepsilon_{cu2,c}$ of the confined core after spalling.

Failure mode (b) governs over (c) or (d) if the moment resistance of the confined core exceeds 80% of that of the full unspalled and unconfined section.

The following failure criteria may be used for the purposes of the determination of ϕ_u:

(i) Before spalling of the concrete cover:

Bonded tendons of prestressed components do not buckle under cyclic loading. It is safe-sided to take their strain at rupture as equal to that applying in monotonic loading, as function of the number of prestressing tendons in the tension zone.

Owing to the large local inelastic strains and surface cracking that develop in a bar when it buckles in one half-cycle of loading, the bar may rupture during the subsequent tensile half-cycle under a strain much lower than its nominal elongation at maximum force (which in practical applications may be taken equal to its characteristic value, $\varepsilon_{u,k}$). Moreover, being erratic and unpredictable under cyclic loading, bar buckling of one out of several bars in a section may take place early, leading then to bar rupture.

FRP wrapping prevents bar buckling, which typically precedes bar rupture in members without FRPs.

The reduction of the failure strain of the bars in the tension zone of a section with increasing number of bars in Eqs. (30.3-22), (30.3-24b) is of probabilistic nature and follows the functional form of the dependence of the mode of the distribution of the minimum value with the size of the sample.

There is a tendency for the ultimate strain of the extreme fibres of the confined core after spalling of the cover under cyclic loading to increase with decreasing section depth, h, in the unspalled section, or depth, h_o, of the confined core of the spalled section.

the failure strain of $N_{b,tens}$ bars in the tension zone of a section, having uniform elongation $\varepsilon_{su,nom}$ at tensile strength in a standard coupon test:

- in monotonic loading:

$$\varepsilon_{su,mon} = \left(1 - 0.3\sqrt{ln\, N_{b,tens}}\right) \cdot \varepsilon_{su,mon} \qquad (30.3\text{-}23)$$

- in cyclic loading:
  - in a section without wrapping in FRP:

$$\varepsilon_{su,cy} = 0.4\varepsilon_{su,nom} \qquad (30.3\text{-}24a)$$

  - in a section wrapped in FRP:

$$\varepsilon_{su,cy,FRP} = 0.6\varepsilon_{su,nom}\sqrt{1 - 0.15\, ln(N_{b,tens})} \qquad (30.3\text{-}24b)$$

- for unconfined concrete:

$$0.0035 \le \varepsilon_{cu} = \left(\frac{18.5}{h}\right)^2 \le 0.01 \qquad (30.3\text{-}25a)$$

where h in [mm]

- for FRP-confined concrete:

$$\varepsilon_{cu,FRP} = \varepsilon_{cu} + a_f \beta_f \, min\left(0.5; \frac{0.6\varepsilon_{u,f} \cdot E_f}{f_{cm}}\rho_f\right)\left(1 - min\left(0.5; \frac{0.6\varepsilon_{u,f} \cdot E_f}{f_{cm}}\rho_f\right)\right) \qquad (30.3\text{-}25b)$$

where:

$$a_f = 1 - \frac{\left(b_y - 2R\right)^2 + \left(b_x - 2R\right)^2}{3b_x \cdot b_y} \qquad (30.3\text{-}26)$$

with b_x, b_y denoting the sides of rectangle circumscribed around the section and R the radius of the corners of the section (b_y=b_x=2R=D in circular sections).

- β_f=0.115 for CFRP, GFRP, β_f=0.1 for AFRP.

The failure strain of FRP is about equal to 1.5% for CFRP or ARFP and 2% for GFRP.

The ratio of bar diameter to stirrup spacing in Eq. (30.3-27) reflects the importance of buckling of bars during a compressive half-cycle of their stress-strain response in precipitating bar rupture in a subsequent tensile half-cycle.

ε_{su} drops with increasing s_h/d_b, because in cyclic loading bars break in tension after they buckle in a previous half-cycle;

ε_{su} increases with increasing number of bars at the extreme compression fibers, because these bars share with the member its curvature in the longitudinal direction (ie, their convex side faces inwards); to buckle outwards, their curvature must be reversed, which is unlikely for the intermediate bars before the confined core disintegrates, but is possible for the corner ones, which can buckle sideways. The larger $N_{b,compr}$ is, the more the intermediate bars between the two corner ones are.

Eq. (30.3-28a) was fitted to monotonic and cyclic test results together, as they don't have systematic differences but cannot support separate expressions; however, having a common expression increases the scatter of the data with respect to it. In contrast, Eqs. (30.3-28b) and (30.3-28c) were fitted to cyclic tests alone; strictly speaking they apply only to cyclic loading.

The general strength model for confined concrete under concentric compression in section 29.2.4 is assumed to apply to the extreme compression fibres of the confined core under bending with axial force. However, experimental results show that, if applied to those fibres, the associated model for ultimate strain of confined concrete in concentric compression leads to low values of the ultimate curvature under cyclic loading. Eqs. (30.3-28) are not meant to give the experimental values of this strain; they refer exclusively to the extreme compression fibers at the instant of ultimate curvature of a section under flexure with or without axial load; they have just been fitted to the results of such tests, so that, if used in section analysis along the lines set out herein, the computed ultimate curvatures under cyclic flexure fit the experimental data in an optimal way.

- $\varepsilon_{u,f}$: failure strain of FRP
- E_f, ρ_f: elastic modulus and geometric ratio of FRP in the loading direction,

(ii) After spalling of the concrete cover (including FRP-wrapped sections after the ultimate strain of Eq. (30.3.-25b) has been reached):

– for the tension bars:

$$\varepsilon_{su} = \frac{4}{15}\left(1+3\frac{d_b}{s_h}\right)\left(1-0.75e^{-0.4N_{b.compr}}\right)\varepsilon_{su,nom} \qquad (30.3\text{-}27)$$

where:

s_h/d_b is the stirrup-spacing-to-bar-diameter ratio,

$N_{b,compr}$ is the number of bars at the extreme compression fibres, taken equal to 2 in a circular section.

– for the confined concrete core inside closed steel ties with at least 135°-hooks:

- in a rectangular compression zone (uniaxial bending):

$$\varepsilon_{cu,c} = \varepsilon_{cu} + 0.04\sqrt{\frac{a\cdot\rho_w\cdot f_{ywm}}{f_{cm}}} \qquad (30.3\text{-}28a)$$

- in a circular section:

$$\varepsilon_{cu,c} = \varepsilon_{cu} + 0.07\sqrt{\frac{a\cdot\rho_w\cdot f_{ywm}}{f_{cm}}} \qquad (30.3\text{-}28b)$$

- in a triangular compression zone (biaxial bending):

$$\varepsilon_{cu,c} = \varepsilon_{cu} + 0.12\sqrt{\frac{a\cdot\rho_w\cdot f_{ywm}}{f_{cm}}} \qquad (30.3\text{-}28c)$$

where:

ρ_w is the ratio of transverse reinforcement in the direction of bending (or the minimum in the two transverse directions for biaxial bending)

f_{ywm} is the average yield stress of transverse reinforcement

f_{cm} is the average compressive strength

α is the confinement effectiveness factor:

α may be estimated as:

for rectangular sections:

$$\alpha = \left(1 - \frac{s_h}{2b_o}\right)\left(1 - \frac{s_h}{2h_o}\right)\left(1 - \frac{\sum b_i^2 / 6}{b_o \cdot h_o}\right) \quad (30.3\text{-}29a)$$

for circular sections with circular hoops:

$$\alpha = \left(1 - \frac{s_h}{2D_o}\right)^2 \quad (30.3\text{-}29b)$$

for circular sections with spiral reinforcement:

$$\alpha = \left(1 - \frac{s_h}{2D_o}\right) \quad (30.3\text{-}29c)$$

where:

s_h denotes the centreline spacing of stirrups,

D_o, b_o, h_o are the confined core dimensions to the centreline of the hoop;

b_i is the centreline spacing along the section perimeter of longitudinal bars (indexed by i) engaged by a stirrup corner or a cross-tie.

Between yielding of the end section and the ultimate curvature there under cyclic loading, the yielding of the tension bars penetrates into their anchorage zone, increasing the fixed-end-rotation of the end due to slippage of longitudinal bars from their anchorage by:

- in monotonic loading:

$$\Delta\theta_{slip,u-y} = 9 d_b \left(\phi_u + \phi_y\right) \quad (30.3\text{-}30a)$$

- In cyclic loading:

$$\Delta\theta_{slip,u-y} = 4.25 d_b \left(\phi_u + \phi_y\right) \quad (30.3\text{-}30b)$$

Only for specimens with rectangular section are the tests on members which do not conform to seismic codes (eg, having widely spaced ties without at least 135°-hooks) sufficient for statistical analysis. These tests show that poor details do not have a clear effect on the monotonic flexure-controlled ultimate deformation, but reduce the cyclic one. The adverse effect of poor detailing is much more pronounced on the ultimate curvature than on the ultimate chord rotation. Measurements of fixed-end-rotations in non-conforming columns are not sufficient for fitting own expressions of the type of Eq. (30.3-31a); so, Eqs. (30-.3-31b) are taken to apply. In that case, the plastic hinge length that needs to be used in Eq. (30.3-22), alongside the smaller ultimate curvature, turns out to be 30% higher in non-conforming members − a paradox at first sight.

With ϕ_u, ϕ_y and $\Delta\theta_{slip,u-,y}$ determined as above, the "plastic hinge length" L_{pl} of members detailed and designed according to codes for seismic design for ductility may be estimated as:

– for members with section consisting of rectangular parts, with web width b_w and depth h (beams, rectangular columns or walls and members of T-, H-, U- or hollow rectangular section):

  - in monotonic loading:

$$\frac{L_{pl}}{h} = 0.34\left(1+1.1\,min\left(9;\frac{L_s}{h}\right)\right)\left(1-0.5\sqrt{min\left(2.5;max\left(0.05;\frac{b_w}{h}\right)\right)}\right)\left(1-0.5\,min(0.7;\nu)\right) \quad (30.3\text{-}31a)$$

  - In cyclic loading:

$$\frac{L_{pl}}{h} = 0.3\left(1+0.4\,min\left(9;\frac{L_s}{h}\right)\right)\left(1-\frac{1}{3}\sqrt{min\left(2.5;max\left(0.05;\frac{b_w}{h}\right)\right)}\right)\left(1-0.45\,min(0.7;\nu)\right) \quad (30.3\text{-}31b)$$

– for circular columns or piers with diameter D in cyclic loading:

$$\frac{L_{pl}}{D} = 0.7\left(1-min(0.7;\nu)\right)\left(1+\frac{1}{7}min\left(9;\frac{L_s}{h}\right)\right) \quad (30.3\text{-}32)$$

where:

$$\nu = \frac{N}{A_c \cdot f_{cm}}$$

The plastic hinge length from Eq. (30.3-31a) applies also to members which do not conform to codes for seismic design for ductility, subjected to monotonic loading. However, for such members under cyclic loading, the plastic hinge length that needs to be used in Eq. (30.3-22) alongside the smaller ultimate curvature, should be 30% higher than in code-conforming members, , Eqs. (30.3-31) and (30.3-32):

$$L_{pl,non-conform,cy} = 1.3L_{pl,Eq} \quad (30.3\text{-}33)$$

where:

$L_{pl,Eq}$ to be determined according to Eq. 30.3-31a or 30.3-31b

FRP jackets heal any adverse effects of deficient detailing on the cyclic flexural rotation capacity.

Two practically equivalent empirical expressions for beams, rectangular columns or walls and members of T-, H-, U- or hollow rectangular section, are:

$$\theta_{u,m}^{pl} = 0.0242\left(1+0.7a_{sl}\right)\left(1-0.45a_{st}\right)\left(1-0.5a_{cy}\right)\left(1-0.22a_{nc}\right)*$$
$$\left(1-0.41a_{w,r}\right)\left(1-0.31a_{nr}\right)\left(0.2^{min(0.7;\nu)}\right)\left(min\left(50;f_{cm}\right)\right)^{0.1}* \qquad (30.3\text{-}34a)$$
$$\left[min\left(9;\frac{L_s}{h}\right)\right]^{0.35}\left[\frac{max\left(0.01;\omega'\right)}{max\left(0.01;\omega_{tot}-\omega'\right)}\right]^{0.25}\cdot 24^{max\left(\frac{a\cdot\rho_s\cdot f_{ywm}}{f_{cm}};\left(\frac{\alpha\cdot\rho\cdot f_u}{f_{cm}}\right)_f\right)}\cdot 1.225^{100\rho_d}$$

and

$$\theta_{u,m}^{pl} = a_{sl}\cdot\Delta\theta_{u,slip} + 0.0183\left(1-0.45a_{st}\right)\left(1-0.42a_{cy}\right)\left(1-0.08a_{nc}\right)*$$
$$\left(1-0.048\,max\left(4;min\left(8,\frac{h}{b_w}\right)\right)\right)\left(0.225^{min(0.7;\nu)}\right)\left[min\left(50;f_{cm}\right)\right]^{0.2}* \qquad (30.3\text{-}34b)$$
$$\left[min\left(9;\frac{L_s}{h}\right)\right]^{0.4}\left[\frac{max\left(0.01;\omega'\right)}{max\left(0.01;\omega_{tot}-\omega'\right)}\right]^{0.175}\cdot 8.5^{max\left(\frac{a\rho_s f_{ywm}}{f_{cm}};\left(\frac{\alpha\rho f_u}{f_{cm}}\right)_f\right)}\cdot 1.565^{100\rho_d}$$

where:

$\theta^{pl}_{u,m}$ is in rads, f_c in MPa;

a_{sl} =0 if slippage of the tension bars from their anchorage zone outside the member length is not physically possible, a_{sl}=1 if it is (cf. Eq. (30.3-22));

a_{st} =1 for cold-worked longitudinal bars (Class A steel), a_{st}=0 for hot-rolled or tempcore bars (Class B, C or D steel);

a_{nc} =0 if the member conforms to recent generation seismic codes or has its ends wrapped in FRP jackets, a_{nc}=1 if it doesn't;

$a_{w,r}$ =1 for rectangular walls, $a_{w,,r}$ = 0 in all other cases;

$a_{w,nr}$ =1 for T-, H-, U- or hollow rectangular section, $a_{w,nr}$=0 for rectangular one;

b_w is width of one web, even in cross-sections with two or more parallel webs;

ν $=N/bhf_{cm}$
with b the width of the compression zone and N positive for compression;

ω_{tot} $=\Sigma\rho f_{ym}/f_{cm}$,

If the ends of a non-conforming members are wrapped in FRP jackets, there is no need to increase the plastic hinge length according to equation (30.3-33).

$\theta^{pl}_{u,m}$ may also be estimated through purely empirical expressions, which are associated with less model uncertainty.

ω' $=\rho' f_{ym}'/f_{cm}$: mechanical ratio of all longitudinal bars and of the compression reinforcement, respectively;

L_s/h $=M/Vh$: shear-span-to-depth ratio at section of maximum moment;

α is confinement effectiveness factor from Eq. (30.3-29a);

ρ_s $=A_{sh}/b_w s_h$, f_{ywm} ratio of transverse steel parallel to the plane of bending and its yield stress, respectively;

$(\alpha\rho f_u/f_{cm})_f$: confinement term due to FRP wrapping (if there is one):

$$\left(\frac{\alpha\cdot\rho\cdot f_u}{f_{cm}}\right)_f = a_f c_f \, min\left[0.4;\frac{0.6\varepsilon_{u,f}\cdot E_f\cdot\rho_f}{f_{cm}}\right]\left(1-0.5\,min\left[0.4;\frac{0.6\varepsilon_{u,f}\cdot E_f\cdot\rho_f}{f_{cm}}\right]\right) \qquad (30.3\text{-}35)$$

with:

$\varepsilon_{u,f}$, E_f, ρ_f: failure strain, elastic modulus and geometric ratio of FRP in the loading direction;

a_f from Eq. (30.3-26),

c_f =1.15 for GFRP;

c_f =1.9 for CFRP, c_f=0.9 for AFRP in Eq. (30-.3-34a);

c_f =2.8 for CFRP, c_f=0.95 for AFRP in Eq. (27.3-34b);

ρ_d steel ratio of diagonal bars (if any) along each diagonal of the member in side view.

Further background information can be found in Grammatikou et al 2018 [30.3-5].

For prestressed components with bonded tendons Eqs. (30.3-34) may be applied (they are safe-sided), if the prestressing tendons are included in the calculation of ω_{tot} and ω' and the prestressing force in N.

30.3.3.3 Members with lap-spliced ribbed bars - including those with FRP jacket around the plastic hinge region

The lap length, $l_{ou,min}$, beyond which the ultimate chord rotation is not reduced due to the lap splicing may be taken equal to:

$$l_{ou,min} = \frac{d_b f_{ym}}{\left(1+a_c a_n a_s\sqrt{\frac{d_b}{2R_c}}\,min\left(3;\frac{p_c}{f_{ctm}}\right)\left(1-\frac{1}{6}min\left(3;\frac{p_c}{f_{ctm}}\right)\right)\right)f_{ctm}} \qquad (30.3\text{-}36)$$

where:

- a_n factor for confinement effectiveness within a section, equal to 1 in a circular section, or to n_{restr}/n_{tot} in a rectangular section with n_{restr} lapped bar pairs restrained at corners or hooks of ties or inside a chamfered corner of a FRP jacket, out of a total of n_{tot} bar pairs;
- a_s factor for confinement effectiveness along the length of the member, equal to 1 for confinement by a FRP jacket; for confinement by ties, it is equal to the product of the first two terms in Eq. (30.3-29a), with the centreline dimensions of the rectangular perimeter tie, b_o, h_o, replaced by the diameter of the circular hoop, D_o, in circular columns;
- a_c confining medium factor, equal to a_c=7.5 for confinement by steel ties, a_c=9.5 for confinement by a CFRP jacket, a_c=10.5 for a GFRP jacket and a_c=12 for confinement by an AFRP one;
- R_c confining medium radius, equal to the bending radius of the steel tie or of the FRP jacket which confines the lap splice;
- p_c confining pressure on the lap splice, equal to $A_{sh}f_{ywm}/(s_hR_c)$ for confinement by steel ties, or to $t_f f_{u,f}/R_c$ for confinement by a FRP jacket of thickness t_f.

If $l_o < l_{ou,min}$, the ultimate plastic rotation of a member with any cross-sectional shape may be estimated from the general expression based on curvatures and a plastic-hinge length, L_{pl}, Eq. (30.3-22). The effect of the lap splice may be considered to be limited to the ultimate curvature, ϕ_u, which should be computed applying the general rule of including both bars in a pair of lapped compression bars in the compression reinforcement and reducing the maximum elongation of the extreme tension bars at ultimate conditions due to steel failure as follows:

$$\varepsilon_{su,ribbed,laps} = \min\left(1; l_o / l_{ou,\min}\right)\varepsilon_{su} \geq \left(l_o / l_{oy,\min}\right) f_y / E_s \qquad (30.3\text{-}38)$$

with ε_{su} from Eqs. (30.3-23), (30.3-24a), (30.3-24b) -whichever applies.

30.3.3.4 Members with smooth bars, continuous or lap-spliced - including those with FRP jacket around the plastic hinge region

The flexure-controlled ultimate chord rotation of members with smooth bars can be estimated by adding to the chord rotation at yielding of the end section (see

If the compression zone is rectangular and at right angles to the web of the member, lap splicing may be taken to reduce the plastic ultimate rotation in Eq. (30.3-34a) as follows:

$$\theta^{pl}_{u,ribbed,laps} = min\left(1.4\sqrt{\frac{l_o}{l_{ou,min}}} - 0.4; 1\right)\theta^{pl}_{u,m} \qquad (30.3\text{-}37)$$

with $\theta^{pl}_{u,m}$ according to Eq. 30.3-34a.

If Eq. (30.3-34b) is used, the term reflecting the fixed-end rotation should be computed with the values of the ultimate curvature, ϕ_u, and the yield curvature, ϕ_y, which account for the lap-splice; the rest should be reduced in the manner of Eq. (30.3-37).

If the lap-splice region is wrapped in FRP, the term in the exponent of the confinement factor in Eqs. (30.3-34) which is due to the FRP and given by Eq. (30.3-35) should be ignored if the total number of bars around the perimeter is 10 or greater.

If the section consists of rectangular parts, the empirical expression for continuous ribbed bars, Eq. (30.3-34a) may be modified as follows (with factor a_{nc} taken equal to 0,

as the effect of non-conformity is incorporated in the modification factors) for members with smooth bars:

– for members with continuous smooth bars, with or without FRP wrapping:

$$\theta_u^{pl} = 0.7\theta_{u,m}{}^{pl}{}_{,ribbed\ bars} \quad (30.3\text{-}39a)$$

– for members with lap-spliced smooth bars with hooked ends, with or without FRP wrapping:

$$\theta_u^{pl} = 0.6a^{pl}{}_{lap\ hooked} \cdot \theta_{u,m}{}^{pl}{}_{,ribbed\ bars} \quad (30.3\text{-}39b)$$

– for those with lap-spliced smooth bars with straight ends with or without FRP wraps:

$$\theta_u^{pl} = 0.6max\left[0;\ min\left(1; l_o/\left(20d_b\right) - 2/3\right)\right]\theta_{u,m}{}^{pl}{}_{,ribbed\ bars} \quad (30.3\text{-}39c)$$

where coefficient in Eq. (3.30-35) is reduced to 75%, 55% or just 10% of its values for ribbed bars, if the smooth bars are continuous (see Eq. (30.3-39a)), lap-spliced with hooked ends (Eq. (30.3-39b)) or lap-spliced with straight ends (Eq. (30.3-39c)), respectively.

In fact, on average, the chord rotation capacity in the few dozen cyclic tests of jacketed columns available is about 20% higher than the value estimated using the physical approach of Eqs. (30.3-22) to (30.3-33) and about 15% higher than the value from the empirical expressions, Eqs. (30.3-34).

section 30.3.3.2), a plastic part, $\theta_{u,}{}^{pl}$.

30.3.3.5 Members strengthened with a concrete jacket

With the assumptions listed in section 30.3.2.1.2.5, the plastic chord rotation capacity of the jacketed member may be taken equal to the value calculated under these assumptions, no matter the measures to connect the jacket to the old concrete.

30.3.3.6 Walls added by infilling a bay of an existing frame

If the new web has thickness not larger than the width of the old beams and columns around it and its connection to the surrounding frame members aims at a rigid interface according to section 30.3.2.1.2.7, the plastic chord rotation capacity of the composite wall may be taken equal to the value calculated considering it as a monolithic wall.

If a member reaches its monotonic shear resistance before it yields in flexure, it fails in shear in a brittle manner.

For further background reference is made to Biskinis and Fardis 2020-1 [30.3-6] and Biskinis and Fardis 2020-2 [30.3-7].

The maximum aggregate size is taken equal to zero, owing to the rapid deterioration of aggregate interlock with load cycling.

It is reminded for convenience that:

$$\eta_{fc} = \left(\frac{30}{f_{cm}}\right)^{\frac{1}{3}} \leq 1 \qquad (30.3\text{-}40)$$

With f_{cm} in [MPa]

30.3.4 Cyclic shear resistance of members and of their joints

30.3.4.1 Cyclic shear resistance of prismatic members with shear reinforcement

The shear resistance of a prismatic member in the seismic situation can be estimated according to the present clause both inside and outside flexural plastic hinges.

In the seismic situation, the shear resistance may be computed only using Levels II or III of approximation in section 30.1.3, modified for seismic loading according to the present section 30.3.4.1.

The maximum aggregate size is taken as zero.

Equation (30.1-25) is modified as follows:

– after flexural yielding:

$$V_{Rd,max} = k_{\varepsilon}\eta_{fc}\, min\left(f_{cd}; 100\right)\left(b_w z\right)\frac{cot\,\theta}{1+cot^2\,\theta} \qquad (30.3\text{-}41a)$$

– before flexural yielding:

$$V_{Rd,max} = k_{\varepsilon}\eta_{fc}\, min\left(f_{cd}; 50\right)\left(b_w z\right)\frac{cot\,\theta}{1+cot^2\,\theta} \qquad (30.3\text{-}41b)$$

Where:

f_{cd} the design concrete compressive strength in[MPa]

An upper limit is set to the yield stress of transverse reinforcement, as a function of it's ratio, ρ_w:

$$f_{ywd} = min\left(f_{ywk}; 6/\rho_w\right) \qquad (30.3\text{-}42)$$

In Eq. (30.3-41b) k_{ε} is replaced by:

$$k_{\varepsilon} = \frac{1}{1.6\left(1+110\dot{\varepsilon}_1\right)} \qquad (30.3\text{-}43)$$

where:

$V_{Rd,N}$ is the contribution to shear resistance of the transverse component of the force in the diagonal strut which transfers the axial force from the compression zone of the top section of the vertical element to that of the bottom section, the element being in skew-symmetric bending in the seismic situation (V_N in Figure 30.3-10, taken from CEB/FIP Model Code 90 [33.3-8]).

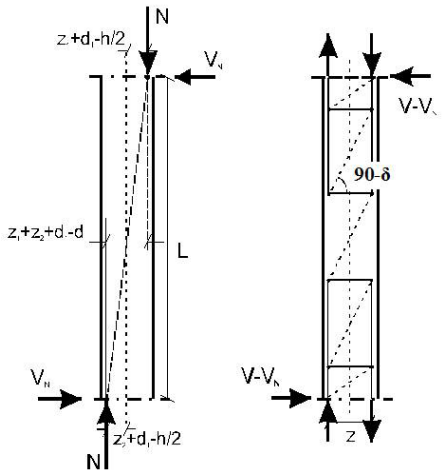


Figure 30.3-10: Contribution to shear resistance of transverse component of diagonal strut force which transfers axial force between the compression zones of the end sections of a vertical element in skew-symmetric bending in the seismic situation (from CEB/FIP Model Code 90 [33.3-8]).

$$\varepsilon_1 = \varepsilon_x + (\varepsilon_x + 0.001)\cot^2\theta \qquad (30.3\text{-}44)$$

and ε_x is the longitudinal strain at section mid-depth.

In the Level II approximation, the angle of the compression field is limited as follows:

$$20^o = \theta_{min} \le \theta \le 45^o \qquad (30.3\text{-}45)$$

while an explicit contribution of the axial force is added:

$$V_{Rd} = V_{Rd,s} + V_{Rd.N} \qquad (30.3\text{-}46)$$

where:

- in vertical members, with axial force N_{Ed} in the seismic design situation (positive for compression, zero for tension), clear length L, depth h and neutral axis depth x:

$$V_{Rd,N} = \frac{h-x}{L} N_{Ed} \qquad (30.3\text{-}47a)$$

- in horizontal members:

$$V_{Rd,N} = 0 \qquad (30.3\text{-}47b)$$

In the Level III approximation Eq.(30.4-45) applies for the angle of the compression field.

While Eq. (30.3-46) is extended as follows:

$$V_{Rd} = V_{Rd,s} + V_{Rd,c} + V_{Rd.N} \qquad (30.3\text{-}48)$$

Zones expected to enter the inelastic range are:

– the end zones near connections to stronger members – as in columns or walls connected to stronger beams or to the foundation, and in beams connected to stronger columns or walls - and

– the plastic hinge zones according to a nonlinear analysis.

The chord rotation ductility factor is the maximum value of the chord rotation at a member end during the response, divided by the chord rotation of the same end at yielding.

Zones expected to remain elastic during the response are:

– those where the value of M_{Ed} is less than the yield moment, and

the member end zones near connections to weaker members – as in columns or walls connected to weaker beams or in beams connected to weaker columns.

Eqs. (30.1-23) and (30.1-41) apply.

If the angle of the compression stress field θ is less than the angle β, defined as:

$$\cot\beta = \cot(L_s/h) \qquad (30.3\text{-}49)$$

then, the shear resistance may be calculated from the following:

$$V_{R,sd} = V_{Rc,d} + V_{RN.d} + k_\varepsilon \eta_{fc} f_{cd} b_w z \frac{cot\,\theta\vartheta - cot\,\beta}{1 + cot^2\,\theta\vartheta} + \frac{A_{sw}}{s} z \cdot \sigma_{swd}\, cot\,\beta \qquad (30.3\text{-}50)$$

where:

$$\sigma_{swd} = E_s\left[\left(\varepsilon_x + 0.002\right)\,\cot^2\theta - 0{,}002\right] \le f_{ywd} \qquad (30.3\text{-}51)$$

The longitudinal strain at section mid-depth may be obtained for the purposes of Eqs. (30.3-45) and (30.3-51) from Eq. (30.1-10), using as M_{Ed} and V_{Ed} their values in the seismic situation, except in member end zones which are expected to enter the inelastic range.

In those zones the values of M_{Ed} and V_{Ed} from a nonlinear analysis according to section 30.3.3.4, should be multiplied by the chord rotation ductility factor at the nearest member end, as obtained from the same analysis.

If linear elastic analysis according to section 30.3.2.2 is used for estimation of the deformation demands, the longitudinal strain at section mid-depth of members or parts thereof which are expected to remain elastic may be obtained from Eq. (30.1-10), using the values of M_{Ed} and V_{Ed} from the analysis divided by the chord rotation ductility factor at the nearest member, as obtained from the same analysis. Moreover, the value of N_{Ed} should be that due to the gravity actions in the seismic design situation. That value of the axial force should be used also in the contribution of the inclined compression force in columns or walls according to Eq. (30.3-47).

The following limit applies to the value of longitudinal strain:

$$\varepsilon_x \le 0.02 \qquad (30.3\text{-}52)$$

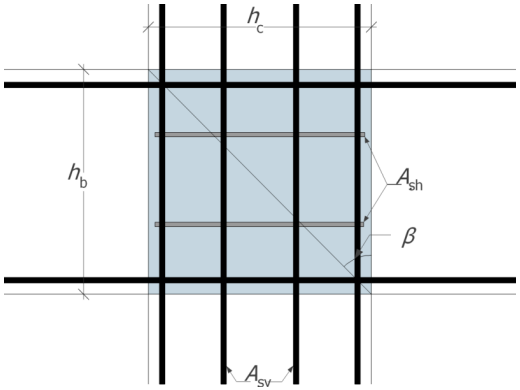


Figure 30.3-11: Notation for the geometry and the reinforcement of beam-column joints

The parameters in Figure 30.3-11 are defined as:

β angle between the diagonal of the joint and the vertical direction,

h_b, h_c beam and column section depth in the joint,

A_{sh}, A_{sv} cross-sectional area of horizontal and vertical reinforcement in the joint, placed between the extreme longitudinal bars of the beam and the column, respectively

Further background information can be found in Fardis [30.3-9].

Joints are essentially immune to oblique cracking due to restrained imposed deformations. Moreover, non-seismic actions are very unlikely to produce high shears in the joint core. So, a strong earthquake is expected to find the joint uncracked.

30.3.4.2 Shear resistance of joints

30.3.4.2.1 Introduction

The present clause applies for the calculation of the shear resistance of joints between horizontal and vertical elements (beams and columns in frames, piers and the deck in a bridge) at the ULS in the seismic situation.

The shear strength of a joint increases if the joint is connected to another horizontal member, such as a slab or a transverse beam or both, at right angles to the axis of the horizontal member being considered.

Vertical reinforcement between the extreme bars of the vertical element in the plane of loading counts as joint vertical reinforcement. Hoops should be placed as joint horizontal reinforcement.

The effective width of the joint, b_j, should be taken as the average of the widths of the beam and the column at right angles to the direction in which the shear resistance of the joint is calculated:

$$b_j = \frac{b_b + b_c}{2} \tag{30.3-53}$$

where b_b, b_c are the beam and column section widths in the joint.

30.3.4.2.2 Joint shear at diagonal cracking

The design value of the horizontal shear force in a joint at first cracking, $V_{jh,cr,}$ should be taken as a lower limit to its design shear strength.

$V_{jh,cr,}$ may be estimated assuming that:

– at mid-depth of the joint the vertical normal stress due to the axial force in the column, N_v,- taken positive for compression – is uniform over the full horizontal cross-sectional area of the column, A_c,

– any normal force, N_h, in the beam (due to prestressing, as in pier-to-deck joints in bridges, taken positive for compression), produces uniform horizontal normal stresses over a vertical section defined by the column width and the beam depth; and

– the horizontal shear force is uniformly distributed over a horizontal section having depth in the direction of the shear force equal to the column depth in the same direction, hc, and width in the orthogonal direction equal to the effective width of the joint, bj, defined above.

– the joint, considered as unreinforced and uncracked so far, cracks diagonally when the average shear stresses at its centre reaches a biaxial failure envelope in principal stress space, comprising:

  – a straight line between the design values of the concrete strength in tension and in compression, f_{cd} and f_{ctd}, respectively, in the tension-compression regime;

  – a tension cut-off at f_{ctd} in biaxial tension.

If: $1+\frac{N_h}{b_c h_b f_{ctd}} \geq \frac{N_v}{A_c f_{cd}} \geq -\left(1-\frac{N_h}{b_c h_b f_{cd}}\right)\frac{f_{ctd}}{f_{cd}}$ then:

$$V_{jh,cr} = \frac{f_{ctd} f_{cd}}{f_{ctd}+f_{cd}} \sqrt{\left(1+\frac{N_h}{b_c h_b f_{ctd}}-\frac{N_v}{A_c f_{cd}}\right)\left(1-\frac{N_h}{b_c h_b f_{cd}}+\frac{N_v}{A_c f_{ctd}}\right)} b_j h_c \qquad (30.3\text{-}54)$$

If $-\left(1-\frac{N_h}{b_c h_b f_{cd}}\right)\frac{f_{ctd}}{f_{cd}} > \frac{N_v}{A_c f_{cd}} \geq -\frac{f_{ctd}}{f_{cd}}$ and: $\frac{N_h}{b_c h_b f_{ctd}} \geq -\frac{f_{ctd}}{f_{cd}}$ then:

$$V_{jh,cr} = \sqrt{\left(\frac{f_{ctd}}{f_{cd}}+\frac{N_h}{b_c h_b f_{cd}}\right)\left(\frac{f_{ctd}}{f_{cd}}+\frac{N_v}{A_c f_{ctd}}\right)} b_j h_c \qquad (30.3\text{-}55)$$

If $\frac{N_h}{b_c h_b f_{cd}} + \frac{f_{ctd}}{f_{cd}} < 0$, or $\frac{f_{ctd}}{f_{cd}} + \frac{N_v}{A_c f_{ctd}} < 0$ then:

$$V_{jh,cr} = 0 \qquad (30.3\text{-}56)$$

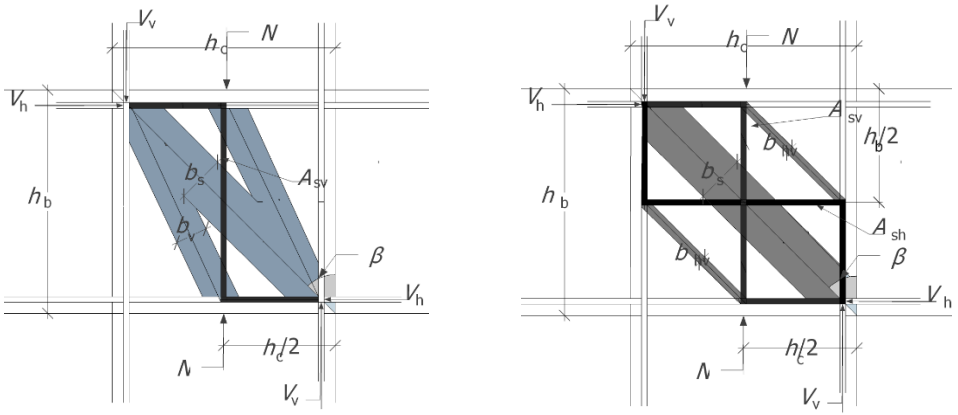


Figure 30.3-12: Simple Strut-and-Tie Models for (a) joint with only vertical reinforcement and/or column axial load; and (b) with horizontal and vertical reinforcement (or just an axial load in the column, if there is no vertical reinforcement).

If there is vertical reinforcement in the joint and/or axial compression in the column, but no horizontal reinforcement or axial compression in the beam, the simple STM in Figure 30.3-12(a) lumps the full cross-sectional area A_{sv} of vertical reinforcement in a Tie at column mid-depth and connects its two ends to the far corner of the joint through two steep side-struts oblique to the central diagonal strut.

The simple STM in Figure 30.3-12(b) lumps the two types of joint reinforcement in two ties along the axes of symmetry of the joint and connects the ends of the ties on adjacent sides through two side-struts parallel to the central diagonal strut.

With the notation:

b_s width of the central diagonal compression strut within a vertical plane,

b_v width of side-strut in a joint with no horizontal reinforcement and only vertical reinforcement or axial force in the column (Figure 30.3-12(a)),

b_{hv} width of side-strut in a joint with horizontal reinforcement and either vertical one or axial force in the column or both (Figure 30.3-12(b))

f_{yhd} yield stress of horizontal joint reinforcement,

30.3.4.2.3 Level-of-Approximation (L-o-A) I

Simple Strut-and-Tie Models (STMs) can provide an estimate of the shear resistance of the joint, at first approximation. In them, the horizontal and the vertical shear force components in the joint V_h and V_v are considered as point forces applied at diagonally opposite corners of the joint.

f_{yvd} yield stress of vertical joint reinforcement,
N_v axial force in the column, positive for compression,
N_h normal force in the beam (due to prestress, as in pier-to-deck joints in bridges), positive for compression,
η_{fc} ratio of effective concrete strength in the struts to the uniaxial concrete strength.

The horizontal component of the design shear resistance, due to the central strut and the two side struts, is:

- In a joint with only vertical reinforcement and/or column axial load (Figure 30.3-12(a)):

$$V_{Rhd} = max\left(\begin{matrix} V_{jh,cr}; \\ b_s b_j \eta_{fc} f_{cd} \sin\beta + min\left(max\left(A_{sh} f_{yhd} + N_h; 0\right); \dfrac{max\left(A_{sv} f_{yvd} + N_v; 0\right)}{\cot\beta}; \dfrac{b_v b_j \eta_{fc} f_{cd}}{\sqrt{1+\cot^2\beta}} \right) \end{matrix}\right) \quad (30.3\text{-}57)$$

  The width of the central diagonal strut, b_s, may be taken equal to the following fraction of the length of the diagonal of the joint in elevation:

  - In interior joints: 1/6,
  - In exterior joints: 1/9.

  The width of each side-strut, b_v, may be taken equal to the following fraction of the length of the diagonal of the joint in elevation:

  - In interior joints: 1/30,
  - In exterior joints: 1/50.

- In a joint with horizontal and vertical reinforcement (or just an axial load in the column, if there is no vertical reinforcement)

$$V_{Rhd} = max\left(\begin{matrix} V_{jh,cr}; \\ b_s b_j \eta_{fc} f_{cd} \sin\beta + min\left(max\left(A_{sh} f_{yhd} + N_h; 0\right); \dfrac{max\left(A_{sv} f_{yvd} + N_v; 0\right)}{\cot\beta}; \dfrac{b_{hv} b_j \eta_{fc} f_{cd}}{\sqrt{1+\cot^2\beta}} \right) \end{matrix}\right) \quad (30.3\text{-}58)$$

  The widths of the central diagonal strut, b_s, and of the side-struts, b_{hv}, may be taken the same, equal to the following fractions of the length of the diagonal of the joint in elevation:

  - In interior joints: 1/6,
  - In exterior joints: 9%.

By contrast, the simple STMs described above lump the tension and compression stresses in the joint in discrete elements – the ties and the struts.

Then, the surface of cracks parallel to the compression field is stress-free.

σ_{sh} is the stress in the horizontal reinforcement in the joint, obtained from the horizontal strain at the centre of the joint, which is in turn estimated from the principal strain along the compression field (assumed to be equal to -0.002) and from the vertical strain at the centre of the joint, given by Eqs. (30.3-67) and (30.3-68) for σ_{sv}, the stress in the vertical reinforcement in the joint. As σ_{sv} depends on $V_{Rhd,CF}$,iterations are needed.

ε_1 is the principal tensile strain considered to take place at right angles to the compression stress field in the concrete and estimated from the principal strain along the

30.3.4.2.4 Level-of-Approximation (LoA) II

The stress field in the joint may be considered as continuous and the joint reinforcement as smeared within the joint volume.

When the shear strength of the joint is attained, the direction of the compression field, at an inclination θ to the vertical, can be taken as a principal stress direction for the concrete and a principal strain direction for the joint volume.

The horizontal component, $V_{Rhd,}$ of the design shear resistance after cracking may be taken as the maximum resistance over all values of the compression field inclination, θ, calculated as follows:

$$V_{Rhd} = max\left(V_{jh,cr}; V_{Rhd,min} + V_{Rhd,CF}\right) \quad (30.3\text{-}59)$$

where:

– In interior joints:

$$V_{Rhd,min} = 1.15 f_{ctd} b_j \sqrt{h_c h_b} \quad (30.3\text{-}60)$$

– In exterior joints:

$$V_{Rhd,min} = 0.45 f_{ctd} b_j \sqrt{h_c h_b} \quad (30.3\text{-}61)$$

$V_{Rhd,CF}$ may be calculated as follows :

– For values of $\theta < \beta$:

$$V_{Rhd,CF} = min\left(\frac{max\left(A_{sv} f_{yvd} + N_v; 0\right)}{cot\,\theta}; \; max\left(A_{sh}\sigma_{sh} + N_h; 0\right) + \left(cot\,\theta - cot\,\beta\right) sin^2\,\theta \cdot \eta_{fc} k_{\varepsilon,j} f_{cd} h_c b_j\right) \quad (30.3\text{-}62)$$

with:

$$\sigma_{sh} = E_s\left[\left(\varepsilon_v + 0.002\right)\,\cot^2\theta - 0.002\right] \le f_{yhd} \quad (30.3\text{-}63)$$

$$\varepsilon_v = \sigma_{sv} / E_s \quad (30.3\text{-}64)$$

$$\sigma_{sv} = \left[\; V_{Rhd,CF} \cot\,\theta - N_v\right] / A_{sv} \le f_{yvd} \quad (30.3\text{-}65)$$

compression field (taken equal to -0,002) and from the vertical strain ε_v at the centre of the joint.

$$k_{\varepsilon,j} = \frac{1}{1.25\left(1.2 + 55\varepsilon_1\right)} \quad (30.3\text{-}66)$$

with ε_1 from:

$$\varepsilon_1 = \varepsilon_v + \left(\varepsilon_v + 0.002\right)\cot^2\theta \quad (30.3\text{-}67)$$

– For values of $\theta \geq \beta$:

$$V_{Rhd,CF} = min\left(\frac{max\left(A_{sh}f_{yhd} + N_h; 0\right)cot\,\theta}{cot\,\beta}; \frac{max\left(A_{sv}f_{yvd} + N_v; 0\right)}{cot\,\beta} + \left(tan\,\theta - tan\,\beta\right)cos^2\,\theta \cdot \eta_{fc}k_{\varepsilon,j}f_{cd}h_c b_j\right) \quad (30.3\text{-}68)$$

with:

$$\sigma_{sv} = E_s\left[\left(\varepsilon_h + 0.002\right)\tan^2\theta - 0.002\right] \leq f_{yvd} \quad (30.3\text{-}69)$$

$$\varepsilon_h = \sigma_{sh} / E_s \quad (30.3\text{-}70)$$

$$\sigma_{sh} = \left[V_{Rhd,CF}\,\tan\,\theta \cdot \cot\,\beta - N_h\right] / A_{sh} \leq f_{yhd} \quad (30.3\text{-}71)$$

Eq. (30.3-66) applies with:

$$\varepsilon_1 = \varepsilon_h + \left(\varepsilon_h + 0.002\right)\tan^2\theta \quad (30.3\text{-}72)$$

The upper limit of Eq. (30.3-43) applies to the yield stress of the horizontal and vertical joint reinforcement.

If the joint is connected to a slab and/or a transverse beam, the value of $k_{\varepsilon,j}$ from Eq. (30.3-66) is multiplied by the ratio of the total surface area of the two lateral faces of the joint to the surface area of the exposed part of those face (i.e., the part which is not connected to the slab and/or the transverse beam). No upper limit is set to the value of this ratio.

30.3.4.3 Cyclic punching shear resistance of slabs

Coefficient k_{sys} accounts for the ability of punching shear reinforcing systems to control shear cracking and confine compression struts at the soffit of the slab.

30.3.4.3.1 Cyclic punching shear resistance without shear reinforcement

Under cyclic loading, the design shear resistance attributed to the concrete may be taken according to Eq. (30.3-73):

$$V_{Rd,c} = \frac{0.4\left(100\rho_l\right)^{1/4}\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \qquad (30.3\text{-}73)$$

Where

d_v is the mean value [in mm] of the slab's effective depth in the *x*- and *y*-directions,

ρ_l is the tension reinforcement ratio in the support strip (geometric mean in the *x*- and *y*-directions).

30.3.4.3.2 Cyclic punching shear resistance of slabs with shear reinforcement

The punching shear resistance under cyclic loading is calculated as:

$$V_{Rd} = V_{Rd,c} + 0.6V_{Rd,s} \geq V_{Rd,s} \qquad (30.3\text{-}74)$$

where $V_{Rd,s}$ is the design shear resistance provided by the shear reinforcement.

The maximum punching shear resistance for cyclic loading, as limited by crushing of the concrete struts in the supported area, is given by:

$$V_{Rd,max} = k_{sys} V_{Rd,c} \leq \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \qquad (30.3\text{-}75)$$

with:

$$k_{sys} = 1.15\frac{d_{sys}}{d} + 0.63\left(\frac{b_{1,red}}{d}\right)^{1/4} - 0.85\frac{s_o}{d_{sys}} \qquad (30.3\text{-}76)$$

where d_{sys} is the distance of the soffit of the slab to the top head of a shear stud or bolt, or to the start of the bend at the top of a link, or to the centre of a top bar inside such a bend, and s_0 is the distance of the perimeter of the column to the nearest link, shear stud, bolt, or other piece of shear reinforcement.

30.3.5 Verifications

30.3.5.1 SLS verifications of flexural deformations

Deformation measures used in the verifications should represent the behaviour of the plastic hinge region as a whole.

For the verification of the SLS of deformations in seismic situations the demand is the chord rotation at a member end. Its value is determined through nonlinear analysis according to section 30.3.2.4 or via linear elastic analysis according to section 30.3.2.2, if applicable.

– if the operativeness limit state is being verified, it is appropriate to use as limit valua for the demand the value of the chord rotation at yielding, θ_y, of the member end of interest.

If the immediate use limit state is being verified, an appropriate limit is equal to $2\theta_y$.

30.3.5.2 ULS verifications of inelastic flexural deformations

The global safety format is appropriate for the ULS verification of a plastic hinge region, considered as an element.

The design value of $\theta^{pl}{}_{u}$ for the verifications is obtained from its mean value, $\theta^{pl}{}_{u,m}$, as:

$$\theta^{pl}{}_{u,d} = \theta^{pl}{}_{u,m} / \gamma_{Rd} \qquad (30.3\text{-}77)$$

For the near collapse (NC) limit state, the values of γ_{Rd} are as follows:

γ_{Rd} = 1.0, for secondary elements;

γ_{Rd} = 1.5, for primary elements:

- with ribbed bars, continuous or lap-spliced in the plastic hinge region, or
- or with smooth bars, continuous or with hooked lapped ends in the plastic hinge region,

γ_{Rd} = 2, for primary elements with smooth bars and straight ends lapped in the plastic hinge region,.

The control or predominant period of the elastic response spectrum is the period at which both the spectral velocity and the spectral acceleration assume their maximum values.

In jointed structural systems the inelastic demand is concentrated within the critical connections between the precast concrete elements, as a result of the opening and closing of a gap at the interface between elements.

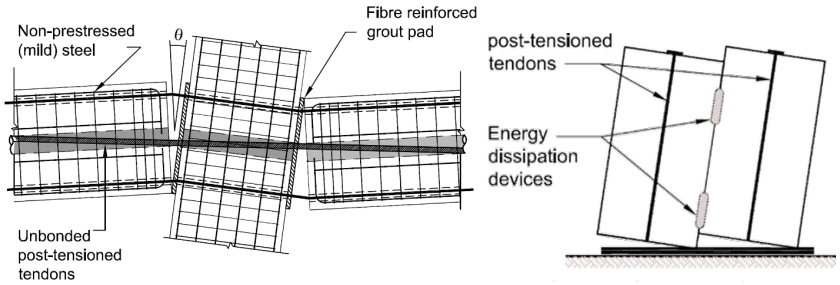


Figure 30.3-13: Jointed hybrid post-tensioned frame and coupled wall system.

Further background on the concept is given in:

- [30.3-10] Bianchi et al
- [30.3-11] fib Bulletin 27
- [30.3-12] NZCS
- [30.3-13] NZS3101

The same values of γ_{Rd} apply to the Life Safety limit state, but after multiplying by the ratio of spectral values at the control period of the elastic response spectrum, for the Near Collapse and the Life Safety LSs, not be taken for the present purposes less than 2.0.

30.3.5.3 ULS verifications of shear forces

Shear is verified at the Near Collapse (NC) ULS, using a design value of shear resistance obtained from design values of material properties, $V_{R,d}$.

30.3.6 Special provisions for low-damage design with unbonded post-tensioning

30.3.6.1 The design concept

Precast concrete elements with dry joints and unbonded post-tensioning may form a ductile structural system which can sustain strong earthquakes with little or no damage, if any.

The unbonded post-tensioned tendons in the precast elements which cross their interfaces with other elements should be designed to remain elastic during the design seismic action.

The structural system, capable of providing both lateral resistance and gravity load bearing capacity, may be a moment resisting frame, a wall- or a dual (wall-frame) system.

- [30.3-14] Priestley et al
- [30.3-15] Pampanin et al
- [30.3-16] Pampanin

Cost-effective external dissipators, with easy access, inspection and if needed, replacement after an earthquake, have been developed. They consist for example of axial, tension-compression yielding mild steel short-bar-elements, machined down to the desired "fuse" dimension (acting as the "weakest link of the chain") and inserted and grouted (or epoxied) in a steel tube acting as anti-buckling restrainer.

Either metallic and/or other advanced materials (e.g. shape memory alloys, visco-elastic systems) can provide alternative types of dissipation mechanisms (elasto-plastic due to axial or flexural yielding, friction, visco-elastic). Self-centring/dissipative high-performance systems, referred to as advanced flag-shape systems (AFS), combine alternative forms of displacement-proportional and velocity-proportional energy dissipation (i.e. yielding, friction or viscous damping) in series and/or in parallel with the main source of re-centring capacity (unbonded post-tensioned tendons, mechanical springs or Shape Memory Alloys (SMA) with super-elastic behaviour).

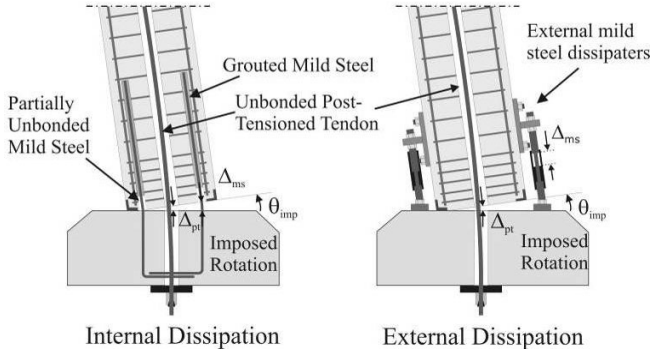


Figure 30.3-14: *Energy dissipation systems.*

Hybrid systems are jointed structural systems combining self-centring capabilities due to unbonded post-tensioning and the axial load, with energy dissipation capabilities thanks to ordinary (non-prestressed) reinforcement or external replaceable dissipators.

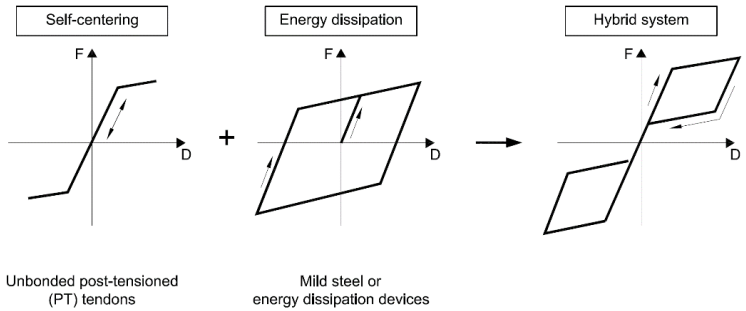


Figure 30.3-15: Idealized flag-shape hysteretic rule for a hybrid system.

The critical sections occur at the interfaces of two connected precast concrete members (beam-to-column, wall-to-wall) or between precast concrete members and the foundation. The opening and closing of the gap causes much reduced or negligible damage in the adjacent structural elements. Since the unbonded tendons are designed to remain elastic, there will also be negligible or no residual deformation of the structure. The unbonded tendons will close the joints and provide a self-centring capacity.

The rocking mechanism of a hybrid beam-column is shown in figure 30.3-16.

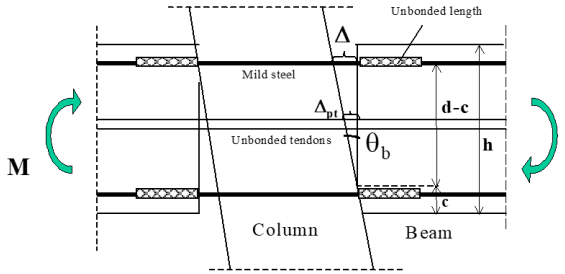


Figure 30.3-16: Rocking mechanism of a beam-column hybrid connection.

The governing behaviour is a controlled rocking and dissipative mechanism, with a characteristic flag-shape hysteresis rule *(*figure 30.3-15*)*.

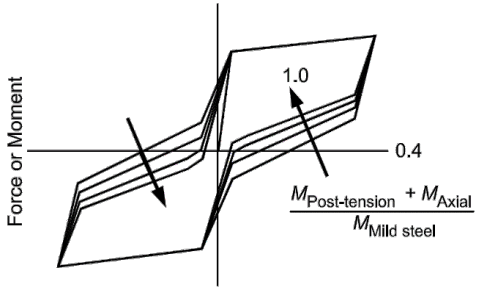


Figure 30.3-17: Effect of moment contribution ratio on shape of hysteresis loops.

30.3.6.2 General design approach

Capacity design can be used to achieve a weak-beam strong column inelastic mechanism or prevent brittle (e.g. shear) shear mechanisms in elements and connections (see clause 30.3.3.3(1) to (3)).

A combination of self-centring and energy dissipation contributions of a hybrid structural system should be designed to meet deformation limits and reduce or even prevent residual deformations.

Full self-centring of a general jointed connection can be achieved by selecting in design an appropriate value of the moment contribution ratio λ, defined as:

$$\lambda = \frac{M_{pt} + M_N}{M_s} \geq \alpha_o \qquad (30.3\text{-}78)$$

where M_{pt}, M_N, and M_s are the contributions to flexural strength of the post-tensioned tendons, the axial load and the non-prestressed steel reinforcement or energy dissipating devices with respect to the centroid of the concrete compression resultant in the section, respectively, $\alpha_o \geq 1.25$ is an overstrength factor for the non-prestressed steel reinforcement or the energy dissipating devices.

The energy dissipation capacity provided by the flag-shape hysteresis rule typical of hybrid systems, should be evaluated in terms of the equivalent viscous damping ratio, ζ, by interpolation between a pure dissipative system (equivalent monolithic system with elasto-plastic or near elasto-plastic behaviour) and a pure self-centring system (an unbonded post-tensioned system with non-linear elastic behaviour).

$$\zeta_{Hybrid} = \frac{M_{pt} + M_N}{M_{total}} \cdot \zeta_{PT} \cdot + \frac{M_s}{M_{total}} \cdot \zeta_{monolithic} \qquad (30.3\text{-}79)$$

Action effects, both in terms of forces, moments and deformation shall be determined based on non-linear analysis.

Specific modifications to the section/connection analysis are required when compared to a monolithic cast-in-situ system with ductile plastic hinges, in order to capture the moment-rotation behaviour of the controlled rocking and dissipative connection as described in the following sections

30.3.6.3 Shear transfer at critical connections

Shear forces shall be transferred at the critical dry connections by shear keys, concrete corbels, metallic corbels or other means for which there is experimental or analytical evidence of satisfactory performance.

Friction induced by post-tensioned tendons at critical connections shall not be taken into account in design to transfer shear due to gravity loads. Transfer of shear due to the seismic action may rely on friction at interfaces induced by tendons, if the tendons remain elastic at that level of seismic action.

Shear capacity shall be in any case at least equal to the design shear force due to the factored gravity loads.

Torsion moments in the beam, i.e. due to the weight of a floor system orthogonal to the primary frame, shall be transferred at the critical connections by means of shear keys, concrete or metallic corbels/brackets, dowel action of the non-prestressed reinforcement or other means for which there is experimental or analytical evidence of satisfactory performance.

30.3.6.4 Provisions for the post-tensioning tendons

Either post-tensioned strands or threaded bars may be used for post-tensioned tendons. Anchorages shall withstand a minimum of 50 cycles of loading for which the load in each cycle is varied between 40 % and 80 % of the minimum specified strength of the prestressing tendons.

The profile of post-tensioned tendons may be straight or draped to follow the bending moment diagram.

Design does not put limits on the length of the unbonded post-tensioned tendons. The unbonded length shall be clearly defined and controlled as per design requirements.

The location of the tendons in the critical beam section where the gap may open and close can vary according to the design requirements. Concentric prestressing in the beam is preferred in high seismicity regions, for easier control of the additional elongation under seismic loading.

The initial prestress shall be limited by a lower bound, in order to guarantee sufficient moment contribution M_{pt} for full self-centring capacity, and by an upper

bound in order for the tendons to remain elastic at a target storey drift level, while still providing self-centring.

The upper bound limit for the initial prestress may be obtained from:

$$f_{pt,initial} \leq 0.9 f_{pty} - E_{pt}\varepsilon_{pt} \quad (30.3\text{-}80)$$

where

$f_{pt,\,initial}$ is the initial prestress, after losses

f_{pty} is the yield strength of prestressed tendons

E_{pt} is the modulus of elasticity of tendons, and

ε_{pt} is the additional strain in the tendons due to lateral drifts/displacements.

30.3.6.5 Calculation of flexural strength at given storey drift levels

The flexural strength at different storey drift levels can be derived following a procedure to determine the complete moment-rotation section behaviour for connections/systems with local strain-incompatibilities such as:

(a) Partially bonded or unbonded tendons;
(b) Unbonded length of non-prestressed reinforcement or energy dissipation devices;
(c) Hybrid connection with combinations of the above.

For unbonded tendons compatibility between strain in the tendons and strain in the concrete does not apply at any given section.

Member compatibility and section equilibrium conditions shall be met.

Moment-Rotation Procedure (Monolithic beam analogy):

The flowchart of the simplified moment-rotation procedure in Figure 30.3-18, is intended to reflect section analysis in monolithic members. For given rotation θ, the depth of the neutral axis, x, corresponds to a unique solution respecting both the equilibrium equations at section level and compatibility conditions at a member level.

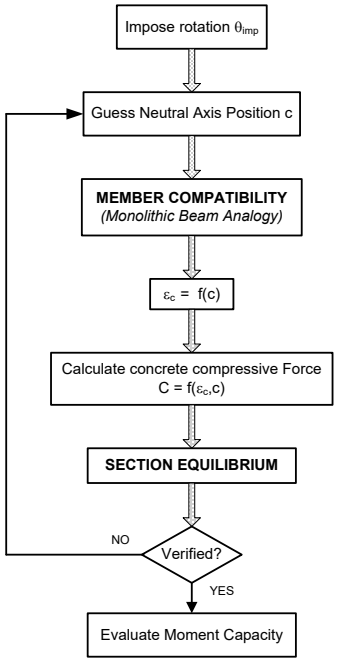


Figure 30.3-18: Flow chart of moment-rotation procedure in presence of strain incompatibility.

Calculation of the strain in the non-prestressed reinforcement or in any additional energy dissipation devices shall rely on section compatibility only if fully bonded conditions are present. For partial debonding the section/connection analysis in the monolithic beam analogy or a similar approach validated experimentally shall be followed.

The gap opening (rotation θ) shall be used to estimate the additional elongation of tendons, non-prestressed reinforcement or energy dissipation devices, assumed to be proportional to distance from the neutral axis.

The strain level in the unbonded post-tensioned tendons due to the gap opening shall be calculated as:

$$\varepsilon_{pt}(\theta) = \frac{n \cdot \Delta_{pt}}{\ell_{ub}} \qquad (30.3\text{-}81)$$

where n is the total number of joint openings at beam-column interfaces along the beam involving the tendons, and ℓ_{ub} the unbonded length in the tendons.

The total strain in the unbonded post-tensioned tendons at a given gap rotation shall be taken equal to the sum of the initial strain due to the prestress and the additional strain due to the gap opening:

$$\varepsilon_{pt,tot} = \varepsilon_{p,i} + \varepsilon_{pt}(\theta) \qquad (30.3\text{-}82)$$

When energy dissipation is assigned to longitudinal non-prestressed reinforcement, a debonded length, ℓ'_{ub}, can be adopted in the beam adjacent to the interface in order to avoid premature fracture of the reinforcement. In these conditions, the strain in the steel due to the gap opening may be calculated as:

$$\varepsilon_s(\theta) = \frac{\left(\Delta_s + 2\Delta_{sp}\right)}{\ell'_{ub}} \qquad (30.3\text{-}83)$$

where Δ_{sp} is the part of the gap opening due to the strain penetration of the non-prestressed steel reinforcement assumed to occur at both ends of the unbonded region, ℓ'_{ub}.

A simplified formula that may be used is:

$$\varepsilon_s = \frac{\left(\Delta_s + \frac{2}{3}\ell_{sp}\varepsilon_y\right)}{\left(\ell'_{ub} + 2\ell_{sp}\right)} \qquad (30.3\text{-}84)$$

where

ℓ_{sp} is the strain penetration length, which may be assumed equal to 10 times the diameter of the reinforcing bar.

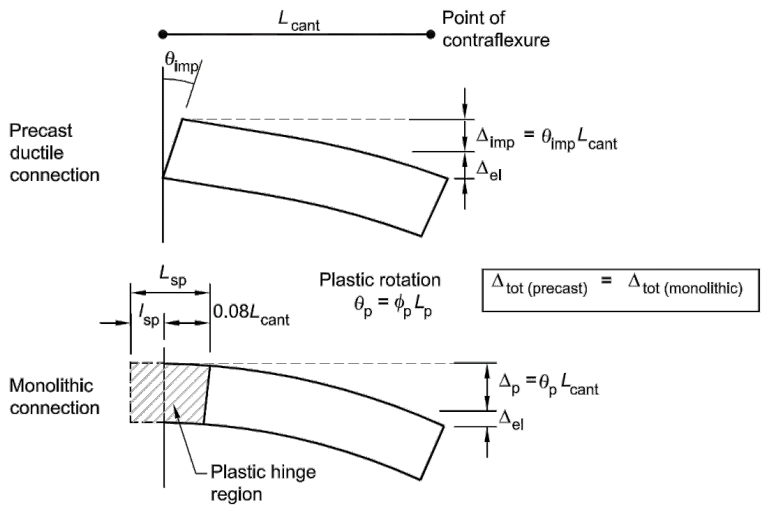


Figure 30.3-19:Monolithic beam analogy for member compatibility condition.

Diaphragm action in the floor system can be significantly impaired by displacement incompatibility between floor and lateral-load resisting systems, leading to extensive damage. Loss of diaphragm action or collapse of the floor due to loss of seating can occur if inappropriate design measures are taken.

At the design level storey drift, the strain in the non-prestressed reinforcement or in the dissipation devices should not exceed 90 % of the ultimate deformation capacity; that is $\varepsilon_s(\theta) \leq 0.9\ \varepsilon_u$.

The neutral axis depth x and the concrete compression strain ε_c corresponding to a given level of storey drift or gap opening rotation may be obtained by an iterative procedure assuming member compatibility conditions. The compressive strain in the concrete at the extreme fibre, ε_c, may be estimated from the following expression, which satisfies member compatibility conditions:

$$\varepsilon_c = \left[\frac{(\theta L_{cant})}{\left(L_{cant} - \frac{L_p}{2}\right) L_p} + \phi_y \right] c \qquad (30.3\text{-}85)$$

where

L_p is the plastic hinge length of an equivalent monolithic connection (including strain penetration component)

L_{cant} is the distance between the column interface and the point of contraflexure (length of the beam cantilever), and

ϕ_y is the yield curvature of the section in an equivalent monolithic connection.

30.3.6.6 Floor-to-lateral-load resisting system compatibility

The expected relative vertical displacements and connection forces between lateral resisting systems and diaphragms should be computed. Special connection details able to accommodate the relative displacements and sustain the increased forces shall be adopted.

In order to minimize additional relative displacements and forces, the connections shall be placed in regions of relatively limited vertical displacement incompatibility and dimensioned to accommodate the expected deformations without affecting their capacity to transfer the inertia forces to the lateral load resisting systems.

The design of the floor-to-lateral resisting systems collectors shall account for the expected overstrength coming from a possible diaphragm inelastic response, as well as for the increased floor acceleration values due to higher mode effects.

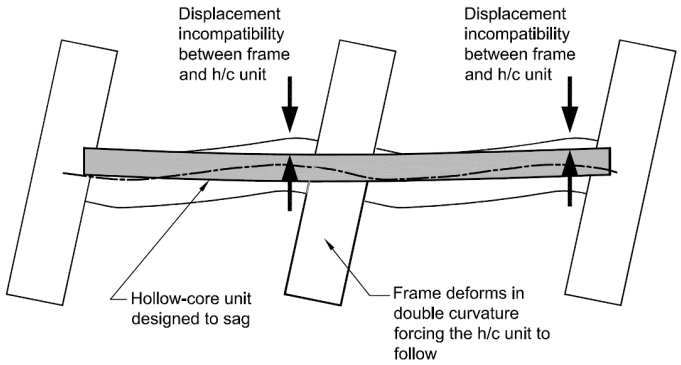


Figure 30.3-20: Vertical displacement incompatibility between floor and frame systems.

Vertical displacement incompatibility between the lateral-load resisting systems and the floor (see Figure 30.3-20 for frame-floor action and 30.3-21 for coupled-wall floor action) could be accommodated by locating the collectors (either in the form of ordinary reinforcement, or mechanical couplers) in regions of minimum displacement incompatibility and/or by assigning them adequate flexibility in the vertical plane. The two "displacement incompatible" systems can be disconnected in regions of high incompatibility, maintaining the coupling where more appropriate and needed to transfer the inertia forces.

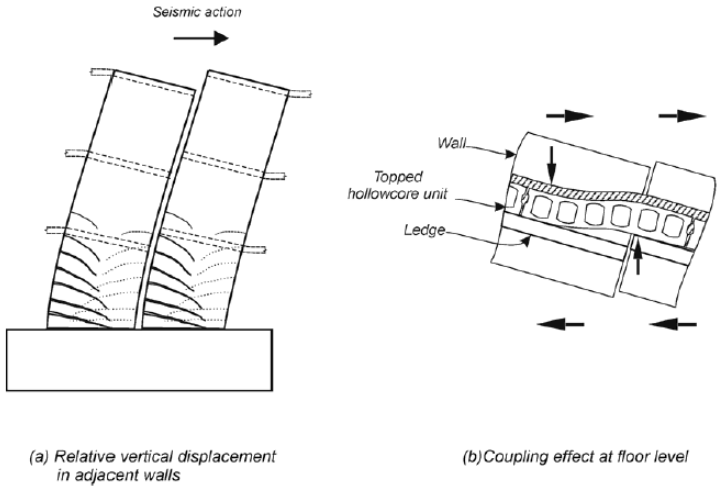


Figure 30.3-21: Vertical displacement incompatibility between floor and coupled-wall system.

The floor-to-lateral resisting systems connectors may be assigned inelastic behaviour with energy dissipation, if adequate evidence of satisfactory global performance from analytical studies on the overall system (floor-lateral resisting system interaction) are provided.

References

[30.3-1] *fib* Bulletin 72. Bond and anchorage of embedded reinforcement: Background to the *fib* Model Code for Concrete Structures 2010.

[30.3-2] Fardis 2022 M.N. Fardis Flat slabs as primary seismic elements in second-generation Eurocode 8. Bulletin of Earthquake Engineering. Vol. 20, pp. 4025–4063, 2022.

[30.3-3] Vecchi and Belleti Vecchi F., Belletti B. 2021 Capacity assessment of existing RC columns. Buildings 11(4), 161, https://doi.org/10.3390/buildings11040161

[30.3-4] Grammatikou et al 2016 S. Grammatikou, D.E. Biskinis and M.N. Fardis, Ultimate strain criteria for RC members in monotonic or cyclic flexure. Journal of Structural Engineering, ASCE, Vol. 142(9), 04016046, Sept. 2016. https://doi.org/10.1061/(ASCE)ST.1943-541X.0001501

[30.3-5] Grammatikou et al 2018 S. Grammatikou, D.E. Biskinis and M.N. Fardis. Flexural rotation capacity models fitted to test results using different statistical approaches. Structural Concrete, Vol. 19(2), April 2018, 608-624. https://doi.org/10.1002/suco.201600238

[30.3-6] Biskinis and Fardis 2020-1 D.E. Biskinis and M.N. Fardis. Cyclic shear resistance for seismic design, based on monotonic shear models in fib Model Code 2010 and in the 2018 draft of Eurocode 2. Structural Concrete, Vol. 21(1), Feb. 2020, 129-150. https://doi.org/10.1002/suco.201900037

[30.3-7] Biskinis and Fardis 2020-2 D.E. Biskinis and M.N. Fardis. Cyclic shear resistance model for Eurocode 8 consistent with the second-generation Eurocode 2. Bulletin of Earthquake Engineering, Vol. 18(6), Apr. 2020, p. 2891-2915. https://doi.org/10.1007/s10518-020-00807-1

[30.3-8] *fib* MC90

[30.3-9] Fardis M.N. Fardis A Levels-of-Approximation approach to seismic design or assessment of beam-column joints in shear. Structural Concrete, Vol. 22(3), June 2021, p. 1259-1284. https://doi.org/10.1002/suco.202000336

[30.3-10] Bianchi et al Bianchi, S.,Ciurlanti, J.,Pampanin, S., 2021, Comparison of traditional vs low-damage structural and non-structural building systems through a cost/performance-based evaluation, Earthquake Spectra, 37(1): 366-385, https://doi.org/10.1177/8755293020952445

[30.3-11] *fib* Bulletin 27 Seismic Design of Precast Concrete Building Structures. State-of-the-Art report, Bulletin No. 27, Lausanne, 2004

[30.3-12] NZCS NZCS, 2010 "PRESSS Design Handbook" (Editor: S. Pampanin), NZ Concrete Society, Wellington

[30.3-13] NZS3101 NZS3101:2006, Concrete Design Standard. Standards New Zealand, Appendix B (Normative) "Special provisions for the seismic design of jointed precast concrete systems", Standards New Zealand

[30.3-14] Priestley et al Priestley, M. J. N., Sritharan, S., Conley, J. R. and Pampanin, S., 1999. Preliminary results and conclusions from the PRESSS five-story precast concrete test building. PCI Journal, 44(6), 42-67. https://doi.org/10.15554/pcij.11011999.42.67

[30.3-15] Pampanin et al Pampanin, S., Priestley, M.J.N, Sritharan, S., 2001 "Analytical Modeling of the Seismic Behaviour of Precast Concrete Frames Designed with Ductile Connections", Journal of Earthquake Engineering, Vol. 5, No.3, pp.329-367 https://doi.org/10.1080/13632460109350397

[30.3-16] Pampanin Pampanin, S., 2012. Reality-check and Renewed challenges in Earthquake Engineering: Implementing low-damage structural Systems – from theory to practice, Bulletin of the New Zealand Society for Earthquake Engineering, 45(4), pp. 137-160, December, https://doi.org/10.5459/bnzsee.45.4.137-160

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The codes propose several nominal fire curves in the design process to represent the action of fire. Most often used are the ISO 834 [30.4-1] fire curve, the ASTM E119 [30.4-2] curve, the hydrocarbon fire and the external fire curve. All curves consist of a simple relationship giving on temperature (the temperature of the gases in the compartment) as a function of time. They are thus representing a fully developed (or post-flashover) fire. For a large compartment, such a situation is not encountered before a significant amount of time has elapsed since the very beginning of the fire. This initial period of time is thus not taken into account in the calculated fire resistance whereas, as far as safety of people is concerned, this is the most important period. In fact, it is the only period during which evacuation from the fire compartment is possible.

The cooling down phase of the fire is generally not taken into account in the nominal fire curves. In fact, when a particular fire resistance time is required, no consideration is given to the period beyond this duration.

These relationships hardly depend on the particular characteristics of the situation for which the design is performed. The quantity of combustible material, the dimensions of the compartment and the conditions of ventilation, for example, are not taken into account.

If these characteristics have to be taken into account, it is necessary to apply fire safety engineering methodology, where the fire risk is considered and the exposure is determined based on the actual conditions and utilization. A fire curve, indicated as parametric (or natural) fire, which includes the cooling phase, is estimated for the actual room/structure. It should be noted that the fire safety in this case is dependent on the assumptions for the determination of this fire exposure.

30.4 Verification of structural safety (ULS) for extreme thermal conditions

30.4.1 General

[Not used at this time, to be updated in future edition]

30.4.2 Fire design

30.4.2.1 Introduction

Fire design includes consideration of a transiently changing material in interaction with its exposure: fire, temperatures, thermal characteristics and the impact on the structural behaviour.

Fire design is necessary in order to achieve a suitable performance in the case of fire and to minimize the damage from a fire event.

The present chapter on fire contains fire design principles, and calculation methods and it outlines the design of selected important structural elements. Finally compartmentation is briefly discussed.

In the present section the fire risk and the fire exposures have not been specified. Even though these topics are very important, it is out of the scope of this Model Code to specify loads and exposures.

Normal structural design is based on design of components. The safety format takes the component considerations as a basis.

For fire design of concrete structures, it is strongly recommended to consider the behaviour of the entire structure including interaction between deformations and materials decay in parts of the structure exposed to fire. This may in some case lead to critical load situations in other parts of the structure not directly exposed to fire.

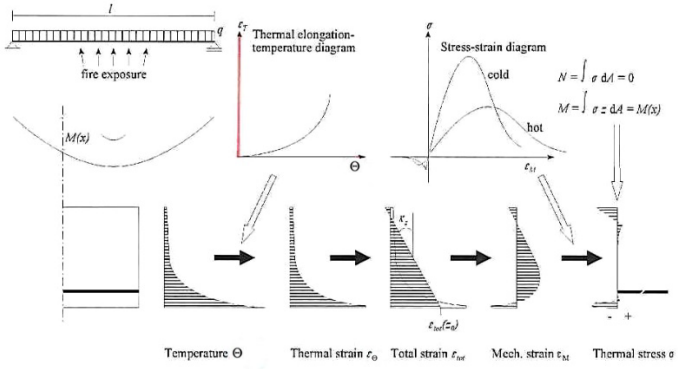


Figure 30.4-1: Graphical representation of strains and stresses in a simply supported beam exposed to fire along the bottom face.

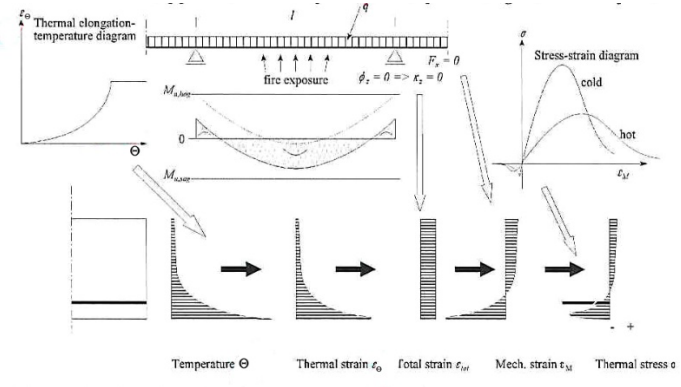


Figure 30.4-2: Graphical representation of strains and stresses in a continuous beam exposed to fire along the bottom face.

The symbols in Figure 30.4-2 are defined as:

$M_{u,hog}$ is the moment at support;

$M_{u,sag}$ is the mid-span moment;

F_x is the axial force in x direction;

κ_z is the curvature in z direction;

φ_z is the angular rotation in z direction.

30.4.2.2 Direct and indirect effects of fire

The most direct effect of a fire on a structure is that the temperature in the structure will increase, in the first phase, and then decrease progressively as the fire decreases until extinction. During the heating phase, heat is introduced into the structure by a combination of convection from the surrounding gas and radiation.

During a fire the temperature profile over the depth of a concrete member, is far from uniform. The main reason for this is the gradual heat conduction from the exposed surfaces to the internal part of the cross-sections of structural members. The main influencing factors are the thermal properties of the concrete i.e. thermal conductivity, specific heat and coefficient of thermal expansion. Because the material characteristics change as a function of temperature, the material properties at various locations in the cross-section will transiently change. This influences the cross-sectional behaviour and results in internal stresses, deformations and possibly in restraint forces which depend on the boundary conditions

In a single member that is completely free to expand indirect actions appear at the local level, in the form of self-equilibrated stresses. These stresses are caused by the nonlinear distribution of the thermal strains on the section (Figure 30.4-1).

Indirect actions are those effects of actions that arise from restrained thermal expansion. For example, a beam that cannot freely expand longitudinally will be subjected to an increase in the axial force especially during the first minutes of the fire (Figure 30.4-2).

Thermal gradients in concrete slabs generate large deflections that are responsible for significant membrane tension effects to develop. In this case, thermal expansion and the ensuing curvatures and deflections have a positive effect on the stability.

Restraint to axial expansion induces axial compression forces in beams which, depending on the position of the restraint, can have a favourable effect (restraints close to the bottom of the end sections), or an unfavourable effect (restraints close to the top of the end sections), in both simply supported and continuous beams (favourable effect = reduction of the tensile stresses and of the bending moments).

Restraint to thermal expansion increases the compressive force in columns but the effect may not be as detrimental as expected if the behaviour of the building as a whole is taken into account.

Restraint to thermal bowing dramatically modifies the bending moment diagram in continuous beams or slabs with a clear tendency to have more negative bending than under ambient conditions.

30.4.2.3 Fire design principles

30.4.2.3.1 Ultimate limit state

The fire situation is an accidental situation that requires only verifications against the ultimate limit state (as opposed to the serviceability limit state). Ultimate limit states are those states associated with structural collapse or other similar forms of structural failure such as loss of equilibrium, failure by excessive deformation, formation of a kinematic mechanism, rupture or loss of stability.

In the semi-probabilistic approach, the design against the ultimate limit state is based on the comparison between the resistance of the structure calculated with the design values of the material properties, on the one hand, and the effects of the design values of the actions on the other hand, as in Eq. (30.4-1).

$$R_{d,fi}(X_{d,fi}) \geq E_{d,fi}(F_{d,fi}) \qquad (30.4\text{-}1)$$

where:

$R_{d,fi}$ is the design value of the resistance in the case of fire;

$X_{d,fi}$ is the design value of the material properties in the case of fire;

$E_{d,fi}$ is the design value of effects of actions in the case of fire;

$F_{d,fi}$ is the design value of the actions in the case of fire.

The resistance and the effects of actions are both based on characteristic values of geometrical data, usually the dimensions specified in the design, for example cross-sections. Geometrical imperfections such as a bar out of straightness or frame initial inclinations are represented by design values.

Different actions generally occur simultaneously on the structure. In an accidental situation, they have to be combined as follows:

- design values of permanent actions;
- design value of the accidental action;
- frequent or quasi-permanent value of the dominant variable action;

The design value of the accidental action that has been mentioned previously does not appear in Eq. 30.4-1 because, in the case of fire, the fire action is not of the same form as the other actions. It does not consist of some static load that could be added to the dead weight or to the wind load. The fire action consists of indirect effects of actions induced in the structure by differential and/or restrained thermal expansion.

Table 30.4-1 given here (Table A1.1 of Eurocode 0 [30.4-3], Table A1.1 in Annex A1) gives the relevant ψ factors for the fire situation in buildings.

Table 27.4-1: Coefficients for combination ψ for buildings

Action	ψ_1	ψ_2
Imposed load in buildings		
category A: domestic, residential	0.5	0.3
category B: offices	0.5	0.3
category C: congregation areas	0.7	0.6
category D: shopping	0.7	0.6
category E: storage	0.9	0.8
Traffic loads in buildings		
category F: vehicle weight ≤ 30 kN	0.7	0.6
category G: 30 kN < vehicle weight < 160 kN	0.5	0.3
category H: roofs	0.0	0.0
Snow loads for H < 1000 m amsl	0.2	0.0
Wind loads	0.5	0.0

The rationale for using 1.0 as a partial safety factor for material properties and for the actions is based on the theory of conditional probabilities. Assuming that the probability of failure at ambient condition meets a particular target value, the following possibilities exist:

P(failure at ambient conditions) ≤ Target value

The probability that the structure ever fails in a fire is the product of two probabilities: the probability that a severe fire occurs and the probability that this fire causes failure.

– quasi-permanent values of other variable actions.

When it is not obvious how to determine which one among the variable actions is the dominant one, each variable action should be considered in turn as the dominant action, which leads to as many different combinations to be considered.

In the case of fire, and if the variability of the permanent action is small (i.e. in most cases), the following symbolic equations hold:

$$\sum_{j\geq 1} G_{k,j} + P + \psi_{1,1}\, Q_{k,1} + \sum_{i>1} \psi_{2,i} Q_{k,i} \qquad (30.4\text{-}2a)$$

$$\sum_{j\geq 1} G_{k,j} + P + \psi_{2,1}\, Q_{k,1} + \sum_{i>1} \psi_{2,i} Q_{k,i} \qquad (30.4\text{-}2b)$$

where:

G_k, Q_k are the characteristic values of the permanent and variable actions, respectively;

P refers to prestressing ;

ψ_1 is the coefficient for the frequent value of a variable action;

ψ_2 is the coefficient for the quasi-permanent value of a variable action.

Eq. 30.4-2a is based on the frequent value and Eq. 30.4-2b on the quasi-permanent value for the dominant variable action.

The design values of the material properties, $X_{d,fi}$, are described by the general equation:

$$X_{d,fi} = \frac{X_k(\Theta)}{\gamma_{M,fi}} \qquad (30.4\text{-}3)$$

where:

$\gamma_{M,fi}$ is the partial safety factor for material property in fire design, normally taken as 1.0;

Θ is the temperature.

P(failure in fire condition) = P(there is a fire) × P(failure caused by this fire)

In this equation it is assumed that the coupled probability of having a fire and and the conditional probability of having a failure under that fire is the same as the probability of having a failure at ambient temperature. This is why more favourable values of the partial safety factors are used in the fire situation, as well as in any accidental situation.

P(failure in fire condition) ≤ Target value

From the above equations it follows:

P(failure caused by the fire) ≤ Target value / P(there is a fire)

An explanation of why the material factors are taken equal to 1.0 in fire, instead of 1.2 in accidental conditions, is that in many accidental conditions (e.g. earthquakes) the uncertainty is taken care of partly in the material (γ_m = 1.2) and partly in the actions. In fire, however, the uncertainty is totally taken care of in the actions, by means of very severe temperature–time curves (standard fire), or rather refined and realistic temperature–time curves (parametric/natural fires).

30.4.2.3.2 Heating of materials

After having defined or determined the fire exposure, and in order to be able to evaluate the modifications of the material and the structural effects, it is necessary to investigate the thermal field, which depends on the fire exposure and on the thermal properties of the material, as well as on the structural context.

The thermal properties of the material will not be discussed here.

The direct consequence of the heating of the concrete is a modification of the material properties such as a decrease of the strength and of the stiffness, as well as the generation of additional deformations associated with the stress level during first heating. These deformations are usually called load-induced thermal strain (LITS) or transient creep in the literature. Furthermore, thermal elongation is a direct result of the temperature increase, which is not uniformly distributed on the section. These strains and their non-uniform distribution have several effects on the behaviour of concrete members and the effect may be different depending on section type. The most important effects have to do with: spalling, elongation and deflections.

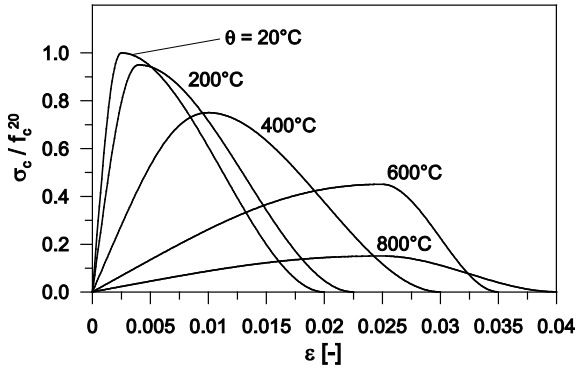


Figure 30.4-3: Example of stress–strain relationships of concrete under compression at elevated temperatures.

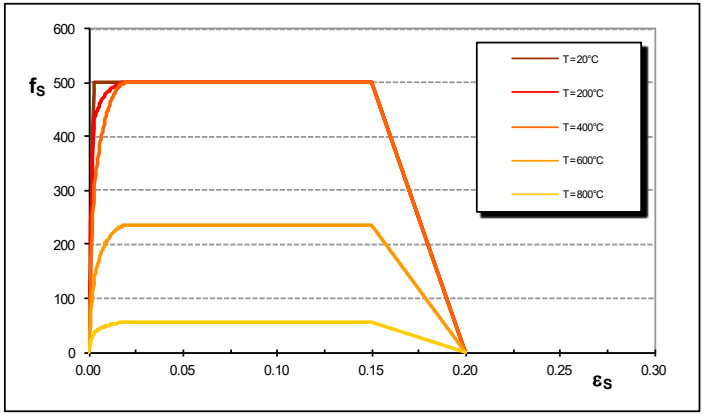


Figure 30.4-4: Stress–strain relationships of reinforcing steel at elevated temperatures.

30.4.2.3.3 Concrete and steel

The present section concerns design of concrete structures, and the detailed description of concrete as a material is beyond the scope of this section. Therefore only some principal features are mentioned with respect to concrete material and fire design.

Concrete is in itself a versatile material, which, based on the mix-design including the admixtures, can exhibit many different characteristics, depending on the actual requirements. In the case of fire the material properties change because of the progressive thermal damage, so the structure or the cross-section actually consists of a multitude of different concrete layers with different material properties. These properties constantly change during and after the fire.

The strength as well as the stiffness of steel and concrete are reduced by a temperature increase. For ease of use, and for numerical purposes mainly, the stress-strain laws are conventionally expressed by making reference to the stress-related strain, that lumps together the different strain components that take place upon heating (Figures 30.4-3 and 30.4-4).

The modifications of the material properties comprise the following:

Compressive strength is the most extensively analysed property of concrete. The strength at room temperature, the water/cement ratio, the type of cement, the maximum aggregate size and the rate of heating appear to have little influence on the relative reduction, expressed in terms of percentage of the original strength. The type of aggregate has an influence, the decrease being less important with calcareous or lightweight aggregates compared to siliceous aggregates. The aggregate/cement ratio also has an effect, with the reduction being proportionally smaller for lean mixes than for rich mixes. Finally, the reduction highly depends on the test procedure, with more favourable results obtained when a certain compression stress level is maintained during heating (pre-loaded tests), as is generally the case in reinforced concrete columns.

The modulus of elasticity of concrete is influenced in the same way by the factors previously mentioned for the compressive strength. The reduction as a function of the temperature is larger than for the compressive strength because of the simultaneous decrease of the compressive strength and increase of the strain at the peak stress.

Steady state creep is of importance essentially for service conditions, that is at temperatures below 150°C applied for very long periods, in concrete reactor vessels

for example. In a fire situation, the creep rates observed under steady state conditions are very much less important than the creep values observed under transient temperature conditions.

Transient thermal strain is the particular deformation that occurs in concrete during first heating under load. It is influenced mainly by the aggregate type, the aggregate/cement ratio and the curing conditions; air-cured and oven-dried specimens exhibit a significantly lower transient strain than water-cured specimens.

The tensile strength of concrete has a tendency to decrease faster with the temperature than the compressive strength, but less than the elastic modulus.

The fracture energy of concrete has a tendency to keep constant or to slightly increase with temperature, but how to define and measure the fracture energy is still a highly debated issue, in spite of the many test results.

The stress–strain relation of concrete in compression reflects the modifications due to elevated temperatures of the compressive strength, of the modulus of elasticity and of the strain at the peak stress. Moreover, stress-strain relations for concrete may include the contribution of creep and transient strain through increased deformability (implicit models). As for the ultimate strain at 20°C, its value is defined in such a way that the M–N envelopes at the ultimate limit state obtained by means of an incremental-iterative procedure be in agreement with those obtained with the traditional approach based on strain limitations, and on the usual parabola–rectangle (or bilinear) stress–strain curve.

The thermal properties of concrete are also modified by a temperature increase. The thermal conductivity is normally reduced, whereas the specific heat is increased by an elevation of the temperature. As a result, the thermal diffusivity decreases with increasing temperatures.

The yield strength of steel is reduced by a temperature increase. The relative reduction does not depend on the value of the yield strength at room temperature but varies with the type of steel, hot-rolled (less temperature sensitive) or cold-worked (more temperature sensitive) reinforcing steel, quenched and tempered or cold-worked prestressing steel (very temperature sensitive).

The elastic modulus of steel is also reduced at elevated temperatures, somewhat faster than the yield strength.

The thermal properties of steel are also modified by a temperature elevation, with a decrease of the thermal conductivity and a slight increase of the specific heat.

This is not particularly relevant in concrete structures because the amount of steel is generally so low that it hardly influences the temperature distribution.

The geometrical size of concrete and steel is modified by a temperature increase. This is the well-known *thermal expansion*. The expansion is not a linearly increasing function of the temperature. The order of magnitude of the thermal expansion may reach 1% at very high temperatures (up to 800°C). This phenomenon plays an important role in the behaviour of structures, because it induces either large displacements that may generate geometrical second order effects or indirect effects of actions if the expansion is restrained.

The bond strength between concrete and steel has been shown to decrease with temperature similar to the tensile strength rather than to the compressive strength, although the experimental results are not all in agreement with each other. Experience has yet very rarely produced evidence of failures by debonding in reinforced structures. The problem is more critical in prestressed pretensioned members.

Spalling in concrete structures is a very important phenomenon linked to high temperatures. Different types of behaviour are often called spalling, from the very progressive sloughing off at the surface that progressively exposes the inner part of the section and the reinforcing bars to elevated temperatures, to the explosive spalling that suddenly completely destroys the material. Extensive research activity is still going on in order to understand and mitigate this phenomenon, the problem becoming more crucial with the introduction of high strength concretes because of their more closed and dispersed porosity, which favours the increase of vapour pressure in the pores. The factors most often mentioned as playing a negative role in spalling are: fast temperature increase, high moisture content, high compression stress level, young age, low porosity, thin members and geometrical effects (corner spalling). Some of these are related to the material itself, but it seems that structural effects also play a role in this phenomenon.

30.4.2.4 **Calculation method**

30.4.2.4.1 Sectional analysis

For the approach based on the use of tabulated data, see EN 1992-1-2 [30.4-4], section 5.

For the approach based on the use of the 500°C isotherm method, see EN 1992-1-2 [30.4-4], Annex B1. For the approach based on the zone method, see EN 1992-1-2 [30.4-4], Annex B2.

For the approach based on the use of stress–strain, temperature-dependent laws, see EN 1992-1-2 [30.4-4], section 3.2.2.1.

See EN 1992-1-1 [30.4-5].

See EN 1992-1-2 [30.4-4].

30.4.2.4.1.1 *Introduction*

Sectional analysis is a common approach to verify the fire resistance. The bearing capacity of reinforced concrete sections subjected to fire is usually evaluated by means of different approaches:

- by using tabulated data (first-level method);
- by using the reference-isotherm method or the zone method (second-level methods);
- by using temperature-dependent stress–strain relationships within the framework of an incremental-iterative procedure (third-level method).

In the present section focus is given to non-linear analysis. Different issues are addressed:

- the use of incremental-iterative procedures (advanced method) and non-linear analysis in fire conditions;
- the validity of the well-known 500°C isotherm method in fire conditions, under an eccentric axial force;
- the relevance of thermal deformations and the ensuing eigenstresses in section-based approaches.

30.4.2.4.1.2 *Tabulated data*

Tabulated data allow designers to give a quick response in many practical cases with well-defined boundary conditions. On the other hand, this approach does not allow the designer to refer to materials' properties and fire scenarios other than ordinary concrete and the standard ISO 834 [30.4-1] Fire Curve. Neither the mechanical nor the thermal aspects of the problem are explicitly addressed by this approach.

30.4.2.4.1.3 *Reference isotherm method (500°C isotherm)*

The reference isotherm method is based on reasonable assumptions, and was originally devised for reinforced concrete sections subjected to pure bending, where the failure is generally controlled by the yielding of the tensile reinforcement.

Investigations show that the larger the fire duration, the worse is the agreement between the 500°C isotherm method and the advanced method, particularly under large axial forces. Of course the two methods give almost the same results in the case of less temperature-sensitive large sections. The bearing capacity in pure compression at low values of the fire duration is slightly overestimated by the isotherm method, which on the whole is rather conservative.

For the temperature-dependent stress–strain curves, see EN 1992-1-2 [30.4-4], section 3.2.2.1.

The reference isotherm method (or effective section method) is based on the assumption that concrete is fully damaged above the temperature of 500°C, while it is fully effective (fully undamaged) for temperatures below 500°C. On the contrary, the mechanical decay of the reinforcing steel is explicitly introduced.

This method may be applied within the context of limit analysis, by assuming the parabola–rectangle stress–strain curve at ambient temperature for the concrete, with the usual strain limitations, and by considering only the undamaged part of the concrete section.

This method yields results that are very close to the advanced method for sections subjected to bending; in addition, it shows that (a) in sections exposed to positive bending, the decay of the ultimate moment is mostly due to the decay of the steel close to the exposed (bottom) face, and (b) in sections exposed to negative bending, the decay of the ultimate moment is due to the reduction of the effective depth.

30.4.2.4.1.4 *Zone method*

The zone method retains the philosophy of the 500°C isotherm method, but considers a more complex and realistic effective section, whose size depends on the temperature distribution. Also the characteristics of the concrete in the effective section (compressive strength and Young's modulus) depend on the temperature distribution. In order to perform the calculation, the section is divided into a finite number of zones. The temperature is determined in the centroid of each zone, on the basis of the thermal analysis. The method is more complex than the 500°C isotherm method, but yields better results, especially in the case of pure compression and thin webs exposed to the fire on both sides. Moreover, the method allows to consider second-order effects, by introducing proper correction coefficients to the stiffness of the element..

30.4.2.4.1.5 *Advanced method – incremental-iterative procedure*

The incremental-iterative procedure is based on the temperature-dependent stress–strain curves. At first, a thermal analysis is performed in order to determine the temperature distribution in the section, and thus the thermal damage at each point, for any given fire duration. The mechanical properties of concrete and steel at each point may then be related to the maximum temperature reached locally by

The application of the incremental-iterative procedure is illustrated in Figure 27.4-5.

This procedure is rather time consuming when compared with the previous methods (500°C isotherm and zone methods). Nevertheless, non-linear analysis based on strain limitations cannot be used with the stress–strain curve proposed in EN 1992-1-2 [30.4-4], section 3.2.2.1, because the attainment of the ultimate strains in one of the two materials does not correspond – in general – to the attainment of the ultimate bearing capacity.

means of the temperature-dependent stress–strain curves. In this way, the section is considered as a composite section, consisting of many different materials, whose properties and spatial distribution are related to the thermal field.

The next step is to determine the ultimate value of the bending moment M_u for a suitable set of values N_u of the axial force. This is done by working out, for each value N_u, the corresponding moment–curvature diagram of the section, by means of an incremental iterative procedure (see also 30.4.2.4.2). Once the moment–curvature diagram is known, the maximum value of the bending moment is the ultimate bending moment M_u, corresponding to the assigned value of the axial force N_u. The couples of values (N_u,M_u) identify as many points in an M–N domain, and the interaction envelope is obtained by connecting these points.

The sectional capacity is underestimated if the analysis is not carried out properly, i.e. by adopting the aforementioned incremental iterative procedure. It seems that in order to achieve a reliable result with non-linear analyses, the incremental iterative procedure will have to be followed.

30.4.2.4.2 The role of thermal strain

Including the effect of the thermal strains requires a clear understanding of the various assumptions concerning the different strain components acting on the section, that is the total strain ε_{tot}, the thermal strain ε_Θ and the stress-induced strain ε_σ:

- the hypothesis that plane sections remain plane is still valid, but with reference to the total strain ε_{tot};
- the usual non-linear analysis techniques may still be applied, provided that the limitations are referred to the stress-induced strain ε_σ.

To work out the moment–curvature diagrams for any given value N of the axial force, the following incremental iterative procedure is applied, for a number of values of the curvature χ:

For the constitutive laws, see EN 1992-1-2 [30.4-4].

- for each value of χ, a tentative deformation ε_0 is assumed in the centroid of the section, and the corresponding total deformation ε_{tot} is determined at each point of the section, assuming that plane sections remain plane;
- the corresponding stress-related strain ε_σ is determined at each point of the section:

  $\varepsilon_\sigma = \varepsilon_{tot} - \varepsilon_\Theta$
- the value of the stress σ is determined at each point, by using the constitutive laws;
- the axial force N' is evaluated: if $N' = N$ (within a prefixed tolerance), the bending moment M is calculated, and a next value of χ is considered;
- if $N' \neq N$, the procedure is repeated until $N' = N$, by varying ε_0; then, the bending moment M is calculated, and a next value of χ is considered.

30.4.2.4.3 Plastic analyses

It is stated in Eurocode 2 (EN 1992-1-2 [30.4-4]) that an important question is whether load redistributions between different sections of a member in bending may be accepted in the case of fire, these redistributions being allowed by the plastic behaviour of both the reinforcement and the concrete not only at room temperature but (even more) at high temperature. One of the key conditions for this plastic behaviour is the ductility of the section, which is the capacity of the section to keep on developing the plastic bending moment, when the curvature increases to very high values. This seems to be the case according to some numerical examples that show how the ductility of a section tends to increase during a fire, even if some evidences reported in the literature testify to a reduction of the ultimate strain of steel for medium temperature values (200-500°C). The effects on the rotation capacity are not fully clear, although it seems that the effect may be of minor importance, given that the behaviour of structural members is governed by the most heated sections (where the temperatures in the steel are higher than the abovementioned values).

The main difference between the hot and cold situations is the ratio between the ultimate plastic moment and the first-yielding moment. This ratio is much higher in fire, which means that much higher rotations have to take place before the full plastic moment is reached. This is in no way a contradiction to what is generally observed during laboratory tests, where the failure of RC structures is often accompanied by very large displacements.

The theory of plasticity gives a theoretical validation to the fact that several effects leading to self-equilibrated stress distributions may be neglected in non-linear numerical analysis. Among these effects, it is worth mentioning:

(a) those occurring either in the construction phases or during the service life at room temperature, before the fire starts (due for instance to shrinkage, creep and thermal strains), and

(b) those occurring during the fire (due to creep and thermal expansion). A consequence of neglecting these effects is that the strains, stresses and tangent moduli that are computed in any given point of a structure are only approximate – or "mean" – values compared to the "true" values that would be computed if all these effects were taken into account. The computed values are indeed based on the hypothesis of a "virgin" initial stress distribution, which is far from reality.

Neglecting these self-equilibrated stress distributions has some limitations, since it is justified only as long as the ensuing displacements are small. Hence, the effects of thermal expansion during the fire should be taken into account.

Another strain component that may affect deformations – second order effects included – in concrete structures submitted to fire is the transient-creep strain.

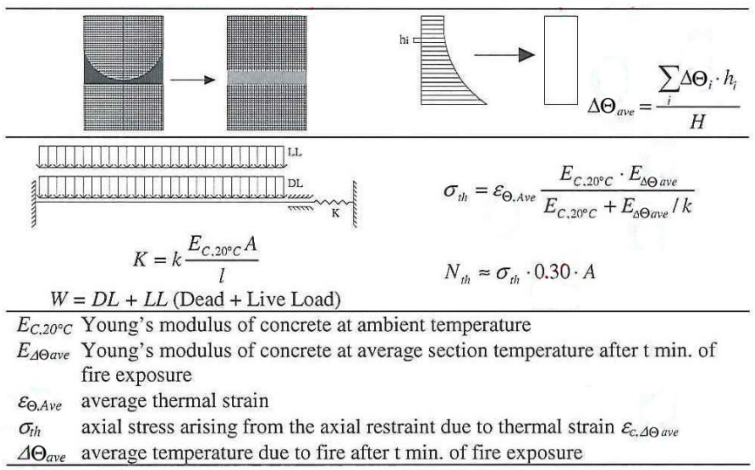


Figure 30.4-7: Evaluation of the axial force in axially restrained beams.

Comments to verification of the method:

In the case of no axial restraint (K = 0), the ultimate bending capacity of the critical sections at any given fire duration is generally underestimated. As a result, plastic analysis underestimates the ultimate load-carrying capacity of a beam, thus leading to conservative results.

The assumed axial force in axially restrained beams results in a lower-bound estimate of the actual force (as given by non-linear analysis). Hence, the proposed method underestimates the effects of the axial restraint.

In most cases, the ultimate bending moment of fully axially restrained beams is overestimated, particularly close to the end sections. As a result, plastic analysis leads – in most cases – to a non-conservative estimate of the ultimate load-carrying capacity.

30.4.2.4.3.1 *Use of plastic analysis*

Within the scope of fire design applications, the objective of plastic analysis is generally to evaluate the load-carrying capacity of a beam during a fire.

According to plastic analysis the verification is positive if the ultimate load at the

The plastic (e.g. ultimate) moments at critical sections may be determined according to various sectional analysis methods, such as the 500°C isotherm method or the zone method.

In axially unrestrained beams, the application of plastic analysis is straightforward and generally leads to a conservative estimate of the ultimate load.

For axially restrained beams, the ultimate bending moment depends on the axial force developed during the fire (Figure 30.4-7). Its value may be determined by performing the analysis of the entire structure. Since this procedure involves sophisticated models and heavy computing, the axial force has to be evaluated in a simpler way, based on fire effects (i.e. thermal field of the section), and on the actual axial restraints of the member in question, before performing the mechanical analysis of the section. An approximate estimate of the axial force, to be adopted in plastic analysis, may be performed via the simplified procedure outlined in the following:

- based on the results of the thermal analysis after a time t of exposure to the fire, the average temperature distribution at each level along the section is determined and the average temperature in the section is found;
- the axial force ensuing from the restrained thermal elongation, corresponding to the mean temperature, is evaluated (a) by computing the longitudinal stress σ_{th} arising in a beam of stiffness $K_{beam} = (E_{\Theta ave}\ A)/L$, axially restrained by a spring of stiffness $K = k\ (E_{20}\ A)/L$, as a consequence of the average thermal elongation $\varepsilon_{\Theta ave}$; and (b) by multiplying such a stress by a suitably reduced cross-section A'. The following definitions/values should be adopted: $A' = 0.30A$, where A is the cross-section area of the beam; $E_{\Theta ave}$ (for the evaluation of σ_{th}) is the elastic modulus corresponding to the average temperature of the section; k has to do with the number of equal spans connected to the heated beam (e.g. k = 1.00, 0.50, 0.33 for a two-span, three-span and four-span continuous beam, with one end spam in fire); $N_{th} = \sigma_{th}\ A'$ applied in the geometric centroid of the section. At any temperature T, excluding creep and

For partially restrained beams, plastic analysis leads in most cases to acceptable results, that is conservative or slightly non-conservative results.

It is observed that in plastic analysis, by completely neglecting the effects of the axial restraint, the estimate of the ultimate loadbearing capacity of a beam is always on the safe side.

For further reference and explanation, see *fib* Bulletin 46 [30.4-6] M–N envelopes for different cross sections and fire exposures are exemplified in Figure 30.4-8.

For the integrity requirements, see EN 1991-1-2 [30.4-4].

transient strain, the elastic modulus may be calculated by multiplying the elastic modulus in virgin conditions (E_{20}) by the normalized decay inferred from the σ–ε curves given in the EN 1992-1-2 [30.4-4]: $E_\Theta = E_{20} \cdot (E_\Theta/E_{20})_{EC2\text{-}Fire}$;

– the plastic (i.e. ultimate) moments of the critical sections are determined by means of either the 500°C isotherm method or the zone method considering also the axial force.

Plastic analysis is a simple and straightforward method that is very sensitive to the evaluation

(a) of the plastic moments at the inner intermediate supports, and

(b) of the effects of the axial restraints.

Neglecting the effects of the axial restraints always leads to a conservative estimate of the ultimate load-carrying capacity in statically redundant beams.

30.4.2.5 **Structural elements**

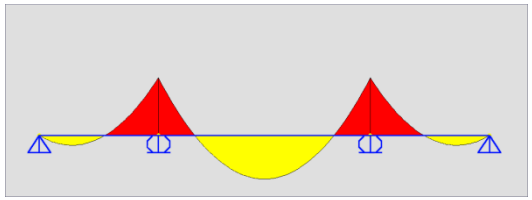
In the following sections some important conclusions concerning the fire design of structural elements are given.

Not only the structural resistance (R) will have to be estimated and documented according to the fire design principles (see section 30.4.2.3) but also the insulation requirements (I) and the integrity requirements will have to be documented. This may require a detailed investigation of the deformations.

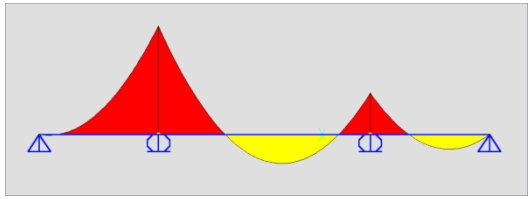
30.4.2.5.1 Beams and slabs

Hence, neglecting the axial force in the design of reinforced concrete beams under fire conditions leads to a conservative estimate of the fire resistance.

The collapse of continuous reinforced concrete beams is generally controlled by the intermediate support sections, where concrete damage may lead to anticipated section failures.



(a)



(b)

Figure 30.4-8: Redistribution of the bending moments from the ordinary condition (a) to the fire situation (b) in the case of fire exposure of the first span.

The following requirements and recommendations are the result of recent studies:

It is recommended to thoroughly consider all the sectional forces when designing beams and frames. It has been observed that axial forces in the design of reinforced concrete beams (predominantly exposed to bending moment) under fire conditions may contribute to an increased fire resistance.

During the fire, the bending moment can shift upwards due to the negative bending moment induced by thermal gradients (Figure 30.4.8). This effect will occur at an early stage in fire before the stiffness degradation takes place.

In such a situation, top reinforcement is required in the mid-span sections to avoid a premature collapse during a fire, especially in (one-way) slabs.

The shear capacity in continuous beams may become critical even though it would not be critical at ambient temperature. Especially for (one-way) slabs the shear forces may become critical as result of the fire.

For beams which are partly restrained in the axial direction, axial displacements increase rapidly in earliest phases of the fire, whereas they keep constant with rather low values after the fire-induced stiffness degradation has taken place. Also the axial force increases rapidly at the beginning of a fire and moderately later on.

Second order effects have a marginal influence on rather massive RC members subjected to bending; on the contrary, in rather thin members like one-way slabs, second-order effects may induce shear forces larger than those at ambient temperature.

30.4.2.5.2 Frames

The following requirements and recommendations are the result of recent studies.

All the sectional forces should be thoroughly considered in the design of frames.

Neglecting the effects of beam thrust in the design of RC frames may lead to highly non-conservative results, because of the increasing bending and shear in the columns during the first 30 minutes of fire exposure.

It should be observed that even if concrete above 500°C has little residual strength, it still contributes markedly to the inertia of the section, because of its peripheral location.

In some cases, considering the zone method instead of the 500°C isotherm method, may yield more reliable results as it app

For horizontal members, failure in a standard test is assumed to have occurred when the deflection reaches a value of $L/20$ (L = clear span of the specimen) or the rate of deflection (mm/min) exceeds a prefixed value (for instance $L^2/9000d$ where

It is required to thoroughly consider and model the support and foundation conditions of the frames in the estimation of the fire resistance of the structure. The thermal effects on bending and shear depend on the type of the foundation adopted in column design, since – for instance – isolated footings provide less rotational stiffness than continuous foundation beams and even less than 2D foundations mats. Consequently, bending and shear in fire increase less in the first case.

Second-order effects have a marginal influence on rather massive reinforced concrete frames subjected to bending.

Detailing rules for the columns similar to those generally adopted in seismic design seem to be suitable also in fire design. In fact, the adoption of densely spaced closed stirrups (hoops) is instrumental in improving section strength and ductility under combined bending and axial force as well as the shear ensuing from the expansion of beams (that are usually much stiffer than columns), and helps in controlling concrete spalling.

30.4.2.5.3 Columns

Reduced M–N envelopes taking into account second order effects are still hardly available in the literature, and only qualitative sketches can be found.

It may be possible – as a simplified and rough model – to evaluate the slenderness on the basis of the effective section using the reference isotherm method; in the case of normal strength concrete (NSC) the 500°C isotherm may be adopted, whereas a lower value should be chosen in the case of high performance concrete (HPC) – see also section 30.4.2.4.1.

For a more detailed consideration, the reduction of the modulus of elasticity should also be taken into account.

Except for the slenderness considerations, the columns may be considered in accordance with the M–N envelopes discussed above

30.4.2.6 **Compartmentation**

Compartmentation has traditionally been defined according to the concept of fire resistance, with reference to collapse (R criterion), fire penetration (E criterion), and excessive heat transfer (I criterion).

d is the effective depth in mm). The rate-of-deflection criterion only applies once the deflection has reached a value of $L/30$.

The origin of the deflection limits are unclear, although they seem to be related to the need of preventing excessive damage in laboratory tests; as a matter of fact, tests have demonstrated that the loadbearing capacity may be maintained for deflection values much larger than this limit. For concrete floor members, compartmentation failure is generally related to the insulation capacity rather than to the loadbearing capacity.

The origins of the definitions and of the values concerning the limits of deformability, integrity and insulation are unclear and may deserve to be worked over in order to reflect the situation in real fires.

However, the basic idea is that, for any given fire duration, structural members should behave properly according to the original design philosophy, which means, for instance, that slabs and walls should remain almost flat during a fire.

The purpose of subdividing spaces into separate fire compartments is twofold. Firstly, it prevents any rapid fire spread, which could trap occupants of the building and secondly it restricts the overall size of the fire.

Continuity at the junctions of the fire-resisting members enclosing a compartment shall be guaranteed. Typically, this would be the junction between a wall (either loadbearing or non-loadbearing) and a floor.

Loss of integrity is deemed to occur when gaps and fissures allow the ignition of a cotton pad on the unexposed side, or when gaps are wider than 25 mm.

Insulation failure is deemed to occur when the mean temperature exceeds by more than 140°C its initial value, or the local temperature on the unexposed face exceeds by 180°C its initial value.

30.4.2.7 FRP reinforcing bars

In FRP reinforced concrete elements the reinforcing bars are embedded in concrete and the reinforcement cannot burn due to the lack of oxygen, but the resin will soften at temperatures beyond the glass transition temperature T_g. At high temperatures the decay of bond for FRP bars is more significant than that experienced by steel bars.

The thermal field in FRP-RC members should be assessed by solving the heat propagation problem due to radiation and convection from the fire to the exterior surface of the member, in which the presence of protective materials is to be considered. The presence of FRP bars may be neglected in the thermal model due to their small size. From a design point of view, the task of the designer may be facilitated through the use of simplified methods for the assessment of temperatures without having to carry out detailed thermal numerical analysis.

The temperature dependent properties of the FRP in terms of tensile stress-strain behaviour and bond behaviour shall be established through design by testing and are required for a more comprehensive fire analysis.

Note, that for the specific case of FRP properly anchored in cold zones, the reduction in bond in the fire-exposed region has significant implications on the flexural deformability, since unbonded reinforcement can lead to the development of fewer, larger flexural cracks and higher overall deflections. However,

The use of FRP reinforcing bars should be considered with care for structures in which fire resistance is vital to maintain structural integrity. In such case an explicit fire resistance calculation should be performed:

$$E_{fi,d,t} \leq R_{fi,d,t} \qquad (30.4\text{-}4)$$

in which $E_{fi,d,t}$ and $R_{fi,d,t}$ are the design effect and capacity in fire situations at time t of fire exposure, respectively.

In a simplified conservative design it shall be verified, through thermal analysis, that the concrete cover is such that the temperature at the location of the FRP rebars remains limited to $T_g - 20$°C, in which T_g is considered as a lower bound value of the glass transition temperature range.

In a more comprehensive design, the analysis is generally performed by separating the thermal and mechanical response models. The thermal analysis will allow to establish the acting temperature in the load resisting materials (concrete and FRP). The mechanical analysis, at time t of fire exposure, shall apply temperature dependant constitutive models, and the associated reduced mechanical properties, in applying the structural capacity models.

As a specific case, the fire design may consider members with FRP properly anchored in cold zones, so to allow for an upper bound fire endurance of the FRP,

experimental tests showed that such cracks do not result in localized heating of the FRP bars. Bar temperatures recorded during fire tests largely depend on the concrete cover, whereas properly anchored FRP rebars' fire endurance depends on fibre strength at high temperatures.

In the design of structures for storing refrigerated liquefied gases (RLG) generally a distinction is made between "double containment" tank systems and "full containment tank systems". Double containment tank systems consist of a closed primary and an open top secondary container. The primary container is a single containment RLG structure. Under normal operating conditions it contains both the refrigerated liquid and the associated vapours. The secondary containment is often an open top concrete wall serving two basic functions:

- protection of the primary container from external loads under normal operating conditions;
- containing the leakage from the primary container, but not the vapour generated from such leakage under accidental spill conditions.

A full containment tank systems consists of an open top primary and a closed top secondary container. Under normal operating conditions the primary tank contains the refrigerated liquid and vapour, while the secondary tank contains vapour and provides protection to the primary container. Under accidental spill conditions the secondary container contains the leakage from the primary container and contains or controls the vapour generated from such leakage.

Storage structures for refrigerated liquefied gases require taking into account different scenarios for coping with the increased risk of such structures.

which mainly depends on the fibre type (instead of T_g). This case is conditional to the following:

- the anchorage of the FRP rebars is located in cold (or protected) zones, so that suitable anchorage of bars at the ends is guaranteed once T_g has been reached in the fire-exposed zone of the member and the resin softening reduces adhesion at the FRP-concrete interface;
- the use of overlapping bars in the zone exposed to fire is omitted.

30.4.3 Cryogenic design

30.4.3.1 General

Design of structures under cryogenic circumstances is most relevant for the design of containment structures for refrigerated liquefied natural gas (LNG). Concrete is particularly well suited for the storage of cryogenic liquids because most of its properties and its behaviour improve substantially as temperature is lowered into the cryogenic range. In this section the most important properties of concrete in cryogenic conditions are treated.

30.4.3.2 Design loads to be considered in the design of structures for refrigerated liquefied gases

Design loads to be considered for the design of structures containing refrigerated liquefied gases are:

A primary container should at least remain liquid tight under hydrotesting, operational loads and operational loads in combination with a specified basic seismic load. The primary container should also retain its containment capability under specified seismic loading (SSE = safe shutdown earthquake) and aftershock events (SSE_{aft}. – safe shutdown aftershock event). For the secondary container, under all conditions, including but not limited to spill, SSE and SSE_{aft} plus spill events, the structural integrity of the wall should be maintained. The secondary containment should be designed for SSE_{aft} while containing the total volume of spilled product with sufficient prestressing force and reinforcement provided to prevent through-thickness cracking of concrete during the SSE_{aft}.

Details about these scenarios can be found, for example, in ACI 376 [30.4-7].

Thermal shock loading will not cause any appreciable loss in strength if the relative humidity (RH) is less than a critical value of about 85%. For higher values of RH, a thermal shock can cause a significant loss of strength only if:

(a) the concrete is water-saturated and/or

(b) a high w/c ratio has been used.

(a) *regular loads:* dead load, product pressure and weight, prestressing forces, installation loads, construction specific loads, wind;

(b) *incidental loads:* testing, commissioning and decommissioning loads;

(c) *accidental loads*: seismic loads, explosion and impact, fire, environmental loads;

(d) *imposed deformation*: loads due to thermal and moisture gradients, loads induced by shrinkage and creep, differential settlement, thermal and moisture gradient loading under spill conditions.

30.4.3.3 Failure mechanisms to be regarded in the design of structures for storing refrigerated liquefied gases

The following ULS criteria apply:

– loss of overall equilibrium;

– failure of critical regions;

– instability resulting from large deformations;

– instability resulting from plastic or creep deformation.

The following SLS criteria apply:

– cracking and spalling;

– deformations;

– corrosion of reinforcement or deterioration of concrete;

– vibrations;

– tightness.

30.4.3.4 Concrete material properties under cryogenic conditions

30.4.3.4.1 Concrete compressive strength

The influence of the moisture content is demonstrated in Figure 30.4-9.

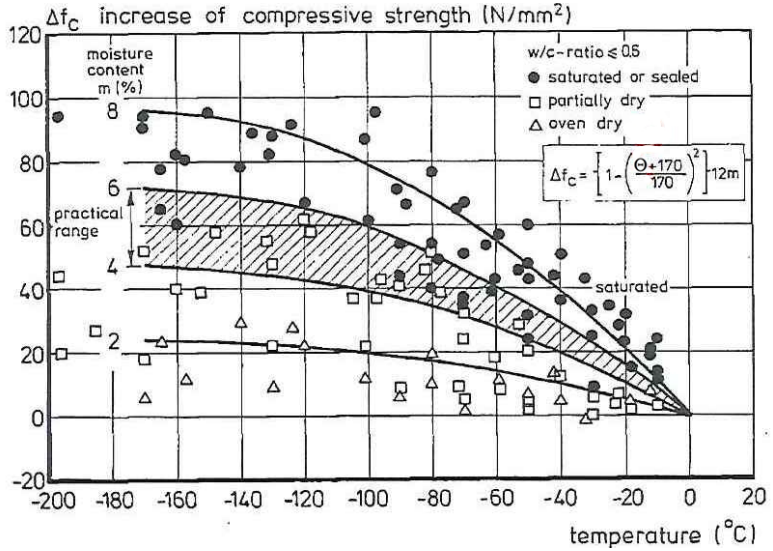


Figure 30.4-9: Increase of compressive strength as a function of temperature and moisture content (Van der Veen [30.4-8]).

For an overview of the mechanical properties of concrete under cryogenic conditions see Report H-216 [30.4-9].

The concrete cylinder compressive strength under a temperature Θ may be formulated as:

for Θ > −120°C: $f_c(\Theta, mc) = f_c + 12 \cdot mc \cdot [1 - (\frac{\Theta+170}{170})^2]$ (30.4-5a)

for Θ ≤ −120°C: $f_c(mc) = f_c + 10.7 \cdot mc$ (30.4-5b)

where:

Θ is the temperature in °C;

mc is the percentage of moisture content by weight.

30.4.3.4.2 Compressive stress–strain relation

The compressive stress–strain relation is a function of the temperature and the moisture content. The strain $\varepsilon_{fc}(\Theta)$ at the ultimate strength at a temperature Θ may be expressed by:

$$\varepsilon_{f_c}(\Theta) = \varepsilon_{f_c}(\Theta = 20°C) + \kappa \cdot \Delta\varepsilon_{f_c}^{max}(\Theta = -60°C) \quad (30.4\text{-}6)$$

where:

$\Delta\varepsilon_{f_c}^{max}(\Theta = -60°C)$ is the maximum increase in strain experienced at −60°C and the coefficient κ is defined as:

for Θ > –60°C $\kappa = 1 - (\frac{\Theta-60}{60})^2$ (30.4-7a)

for −170°C < Θ < −60°C $\kappa = \frac{\Theta-170}{110}$ (30.4-7b)

In Eq. (30.4-6) it may be assumed that $\varepsilon_{f_c}(\Theta = 20°C) = 0.2\%$ and $\Delta\varepsilon_{f_c}^{max}(\Theta = -60°C) = 0.1\%$ for Portland cement-based concretes and 0.15% for blast furnace cement-based concretes.

The following relationship may be used for predicting the effect of temperature on the stress–strain relation:

$$\sigma_c(\Theta) = f_c(\Theta, mc) \cdot [1 - (1 - \frac{\varepsilon_c(\Theta)}{\varepsilon_{fc}(\Theta)})^n] \quad (30.4\text{-}8)$$

where:

σ_c (Θ) and ε_c (Θ)	are stress and strain at a given point of the stress–strain curve;
$\varepsilon_{f_c}(\Theta)$	is the strain at ultimate strength defined in Eq. (30.4-6);
n	is an exponent defined in Eqs. (30.4-9a, b, c):

at Θ = 20°C	$n = 2$	(30.4-9a)
for -170°C < Θ < 20°C	$n = 1 + \frac{\Theta+170}{170}$	(30.4-8b)
at Θ = −170°C	$n = 1$	(30.4-9c)

30.4.3.4.3 Splitting tensile strength

The splitting tensile strength $f_{c,split}(\Theta)$ may be derived from the cylinder compressive strength $f_c(\Theta, mc)$ according to the relation:

$$f_{c,split}(\Theta, mc) = C \cdot f_c(\Theta, mc)^{0.67} \quad (30.4\text{-}10)$$

The coefficient C varies between 0.3 and 0.56 in case of air-dried and water-saturated concrete.

30.4.3.4.4 Modulus of elasticity

The modulus of elasticity increases with decreasing temperature, but not as significant as the concrete strength. However, there is significant scatter.

Neville [30.4-10] reported on the E modulus at a temperature of −190°C which was about 1.7 times as high as when measured at room temperature. Van der Veen [30.4-8] reported an increase of 1.15, 1.5 and 1.8 for air-dried, sealed and water-saturated samples, respectively, exposed to −165°C.

30.4.3.4.5 Coefficient of thermal expansion

The coefficient of thermal expansion decreases with decreasing temperature. For dry concrete this decrease is about 10%. The response of concrete stored at an RH above 86% will show a sudden decrease in the coefficient of thermal expansion. This

decrease is dominated by the content of free moisture. For moist concrete, at a temperature of −100°C even an expansion can occur.

30.4.3.4.6 Creep

The creep of concrete decreases with decreasing temperature. At a temperature of −30°C creep is about 50% of that measured at room temperature. However, because the elastic strains are also reduced with a reduction of temperature, the influence of temperature on the ratio between elastic and creep strains almost cancel each other out. Hence the general relationship between instantaneous (elastic) and long term strain values still remains the same as at room temperature.

30.4.3.4.7 Bond and crack width control

For background information about bond and crack width control, see Van der Veen [30.4-8].

The bond properties improve at decreasing temperature. Crack width control, carried out for room temperature conditions, gives a reasonable, slightly conservative, approach for the determination of crack widths in structures at cryogenic conditions.

References

[30.4-1] ISO 834-11:2014. Fire resistance tests – Elements of building construction – Part 11: Specific requirements for the assessment of fire protection to structural steel elements.

[30.4-2] ASTM E119-20. Standard Test Methods for Fire Tests of Building Construction.

[30.4-3] EN 1990:2003. Eurocode – Basis of structural design.

[30.4-4] EN 1992-1-2:2004. Eurocode 2: Design of concrete structures – Part 1-2: General rules – Structural fire design.

[30.4-5] EN 1992-1-1:2004. Eurocode 2: Design of concrete structures – Part 1-1: General rules and rules for buildings.

[30.4-6] *fib* Bulletin 46. Fire design of concrete structures – structural behaviour and assessment, 2008.

[30.4-7] ACI 376. Requirements for Design and Construction of Concrete Structures for the Containment of Refrigirated Liquefied Gases.

[30.4-8] Van der Veen, C. Cryogenic bond stress-slip relationship, 1990, TU Delft, *PhD thesis*.

[30.4-9] Report H-216. Assessment of mechanical properties of structural materials for cryogenic application, Technical University of Braunschweig.

[30.4-10] Neville **1995 [MORE REFERENCE INFORMATION REQUEIRED]**

As the stiffness of FRP rebars might be significantly lower than that for steel rebars, SLS is predominantly governing the design of members reinforced with FRP.

In the case of structures with intervention, the interface and connection between original part and intervening part may be assumed to be intact for serviceability evaluation if the evaluation of structural safety is conducted.

Extended background information on verification of serviceability can be found in *fib* Bulletin 92 [30.5-1].

The verification of SLS is performed under service load conditions and the operational failure probability to exceed the limit state is about a thousand times higher than that of ULS (see Table 11.3-7). However, if the SLS criteria are exceeded, this does not yet mean that the structure should be rejected.

30.5 Evaluation of serviceability (SLS) of RC and PC structures

30.5.1 General

Evaluation of the serviceability (SLS) is primarily presented for the structures which are made of materials without any defects or deterioration and their design completely corresponds to the requirements of this Model Code. The procedures of evaluation in these cases are described in the section 30.5.2.

If the embedded FRP reinforcement is used, the principles of verification of serviceability limit states apply, but the criteria and material laws and bond properties have to be modified.

If the materials exhibit some defects or deterioration or the structures do not comply with the requirements of this Model Code, the models described in the section 30.5.2. remain valid, but their parameters should be modified. The proposed modifications are mentioned in sections 30.5.3 and 30.5.4.

In the case of structures with intervention, such as strengthening and repair, the methods described in section 30.5.2 or 30.5.3 may be used by considering the effects of added materials as reinforcement and/or matrix. However, the difference in material properties from steel as reinforcement and concrete as matrix should be taken into account.

30.5.2 Structures without deteriorations of defects

30.5.2.1 Requirements

It should be demonstrated that the structure and the structural elements perform adequately in normal use. To meet this requirement the serviceability limit states (SLS) should be verified.

Depending on the type and function of a structure or a structural element the verification of different serviceability limit states may be relevant, such as the limitation of:

Exceeding the limit state of stresses or limit state of cracking may lead to limited local structural damage mainly affecting the durability of the structure, its tightness or its appearance.

Excessive deformations may produce damage in non-structural elements or load-bearing walls and may affect the efficient use or appearance of structural or non-structural elements.

Vibrations may cause discomfort, alarm or loss of functionality.

For special problems, other suitable combinations may be agreed by client and designer.

In general, for the limit state of deformations, the mean value of the prestressing force at the time considered is appropriate, while for the limit state of cracking the upper or lower value (P_{sup} or P_{inf}) according to section 12.5.4.1.5 is suitable.

If the value of the prestressing force is known (e.g. from site measurement) the mean value of the prestressing force may be also used for the crack width analysis.

Linear or non-linear methods may be used. For most SLS problems, linear analysis is sufficient. If, however, a non-linear analysis is carried out for ULS, the action effects under service loads may be calculated by the same model. Plastic analysis is unsuitable for SLS calculations.

– stresses (see subsection 30.5.2.3);
– crack widths (see subsection 30.5.2.4);
– deformations (see subsection 30.5.2.5);
– vibrations (see subsection 30.5.2.6).

30.5.2.2 Design criteria

For SLS verifications the partial safety factors are generally taken equal to 1.0.

The combination of loads to be considered depends on the type of SLS and on the specific problem. It is suitable to utilize one of the combinations given in section 12.5.4.2.3.4 using representative values of variable actions. These representative values may be:

– characteristic value;
– combination value;
– frequent value;
– quasi-permanent value.

The relevant values of the prestressing force depend on the type of SLS and the problem considered. Prestressing force values to be considered are suggested in section 12.5.4.1.5.2.

For structural analysis any appropriate method may be used, which takes account of the material behaviour under service loads.

In selecting appropriate stress limits, the effect of the absolute dimensions of the member should be considered. Lower limits will be appropriate for larger members due to size-effects.

Further information is given in section 30.5.2.4.

Stresses are calculated using section properties corresponding to either the uncracked or the fully cracked condition, whichever is appropriate.

Where the maximum tensile stress in the concrete calculated on the basis of an uncracked section under the characteristic combination of loads exceeds f_{ctm} (see Table 14.11-1), the cracked state should be assumed.

Where the section is assumed to be uncracked, the whole concrete section is considered to be active and both concrete and steel are assumed to be elastic, both in tension and compression.

In specific cases, for example in incremental launching with precast elements, a minimum compressive stress may be required.

30.5.2.3 Stress limitation

30.5.2.3.1 Introduction

Under service load conditions, limitation of stresses may be required for

- tensile stresses in the concrete;
- compressive stresses in the concrete;
- tensile stresses in the steel.

The limitation of tensile stresses in the concrete is an adequate measure to reduce the probability of cracking.

The limitation of compressive stresses in concrete aims at avoiding excessive compression, producing irreversible strains and longitudinal microcracks (parallel to the compressive strains), which increase the permeability of cover and reduce durability.

Tensile stresses in reinforcement should be limited with an appropriate safety margin below the yield strength, preventing uncontrolled cracking which could result in unacceptable appearance and durability issues.

In calculating the stress, account should be taken of whether the section is expected to crack under service loads. Moreover, the effects of creep, shrinkage, relaxation of prestressing steel and differential temperatures should be considered.

Stresses in the FRP reinforcement under quasi-permanent load combination shall be limited, to prevent creep rupture. This condition is verified according to section 17.8.1.2.

30.5.2.3.2 Tensile stresses in the concrete

Depending on the limit state considered, various stress limitations may apply. The limit state of decompression is the most relevant. Stresses may be calculated on the basis of a homogeneous uncracked concrete section. The contribution of reinforcement to the area and the section modulus of the cross-section may be considered.

As a rule, the limit state of decompression should be required, if cracking or reopening of cracks has to be avoided under a given load combination.

In a beam, the state of decompression is reached when the section under consideration is compressed and the extreme fibre concrete stress is equal to zero.

The occurrence of longitudinal cracks may lead to a reduction in durability. In the absence of other measures (such as an increase of concrete cover), it is recommended to limit the compressive stress for exposure classes XD, XS and XF (section 12.5.6.2). However, no limitation in serviceability conditions is necessary for stresses under bearings and anchorages through mechanical devices (e.g. anchor plate of prestressing tendons).

The limit of $0.6f_{ck}$ (t) is not a sharply defined value. Consequently, in the corresponding verification, the prestressing force may be represented by its mean value. On the other hand, the prestressing force and concrete strength should be considered in the verification with their values at the time at which the maximum stresses are reached.

If creep is likely to significantly affect the behaviour of the member considered (e.g. with regard to loss of prestress, deformation, validity of the structural analysis) an alternative measure would be to limit the stress to $0.4\,f_{cm}(t)$. However, the limitation may be taken as a value between $0.4\,f_{cm}(t)$ and $0.6\,f_{cm}(t)$ for verifications relating to a transient situation (e.g. during construction) depending on the duration of the loading.

Creep effects in a cracked cross-section may be considered by assuming a modular ratio of 15 for situations where more than 50% of the stress arises from quasi-permanent actions. Otherwise, creep effects may be ignored.

This requirement will be met provided that, under the characteristic combination of loads, the tensile stress in the ordinary reinforcement does not exceed $0.8f_{yk}$. Where the stress is due entirely to imposed deformations, a stress of $1.0f_{yk}$ is acceptable.

30.5.2.3.3 Limit state of decompression

The limit state of decompression is defined as the state where concrete stresses are below or equal to zero in all principal directions.

30.5.2.3.4 Compressive stresses in the concrete

Excessive compressive stress in the concrete under service load may lead to longitudinal cracks and high and hardly predictable creep, with serious consequences e.g. to prestress losses. When such effects are likely to occur, measures should be taken to limit the stresses to an appropriate level.

If the stress does not exceed $0.6f_{ck}(t)$, under the characteristic combination, longitudinal cracking is unlikely to occur.

If the stress exceeds $0.4\,f_{cm}(t)$, under the quasi-permanent combination of actions, non-linearity should be considered for the assessment of creep deformation (see section 14.6.1.6.4.3).

30.5.2.3.5 Steel stresses

Tensile stresses in the steel under serviceability conditions, which could lead to inelastic deformations of the steel should be avoided as this will lead to large, permanently open cracks. The allowed stress in prestressing steel is in accordance with section 16.2.4.2

Stress verifications should be carried out for fatigue effects in partially prestressed members.

Cracks may be due to other causes such as plastic shrinkage or chemical reactions accompanied by expansion of the hardened concrete. The avoidance and the control of the width of such cracks is not covered by this chapter.

It should be noted for the determination of the crack width in practice that the phenomenon of cracking is of a highly probabilistic nature. Calculated design crack width (w_{cal}) is a value for comparison with nominal limit values (see definition of w_{cal} and w_{lim} in section 30.5.2.4.3). The actual crack widths observed on the real structure may differ from these nominal calculated values and shall not be basically used to verify compliance.
However, the following methodology to record and assess cracks within a structure may be used:

1. Determination of actual mean crack width within a given structure. The actual mean crack value is a mean value calculated from measured values along the following items.
    - measurement at the surface to the closest 0.05mm;
    - at a minimum of 10 points having same restraint;
    - at randomly selected locations.
2. Determination of a comparable allowable crack width on the basis of the calculated design crack width (w_{cal}).
    - 2.1 The mean value of the calculated crack width is obtained from w_{cal} by dividing it by1.7 and by multiplying by a correcting factor considering higher concrete cover (c_{nom}/c_{min}) which should not exceed 1.5;
    - 2.2 The maximum value of the calculated crack width is given by multiplying the calculated design crack width (w_{cal}) by a factor

30.5.2.4 Limit state of cracking

30.5.2.4.1 Requirements

30.5.2.4.1.1 General

It should be ensured that, with adequate probability, cracks will not impair the serviceability and durability of the structure.

Cracks do not, per se, indicate a lack of serviceability or durability; in reinforced concrete structures, cracking due to tension, bending, shear, torsion (resulting from either direct loading or restraint of imposed deformations) is often inevitable and does not necessarily impair serviceability or durability.

Design crack width may be specified to satisfy requirements concerning functionality (section 30.5.2.4.1.2), durability (section 30.5.2.4.1.3) or appearance (section 30.5.2.4.1.4).

1.2 (effect of random distribution of the crack width) and by a correcting factor considering higher concrete cover (c_{nom}/c_{min}).

3. The actual mean crack width (determined in item 1) should not exceed the mean value of the calculated design crack width determined in item 2.1 and the maximum measured crack width should not exceed the maximum valued of the calculated design crack width determined in item 2.2.

Example: case of structural element with c_{nom} = 50 mm (assumed minimum cover for durability = 40 mm) and with w_{cal} = 0.3 mm.

- Expected mean actual surface crack (obtained by measurement according to item 1) should be less than (0.3/1.7)·(50/40) = 0.22 mm;
- Expected maximal actual surface crack (obtained by measurements according to item 1) should be less than 0.3·1.2·(50/40) = 0.45 mm.

Loss of functionality occurs, for instance, in liquid retaining or containment structures, where leakage can occur if the crack widths are too large.

For containment structures, a concrete compression zone with a minimum height may be specified, or a lining may be provided in more demanding cases.

The durability of a structure is predominantly governed by the thickness of the concrete cover and the quality of the concrete if the crack widths are limited to the values of w_{lim} given in section 30.5.2.4.3.

The design crack width limit (w_{lim}) may be increased by multiplying it by a coefficient equal to the ratio between the actual cover and the assumed minimum cover for durability purposes (c_{nom}/c_{min}) and limited to 1.5, to allow for the increase of superficial crack widths in cases with actual cover larger than the minimal cover required for durability. However, the designer's attention should be drawn to cases with large covers (e.g. c_{nom} exceeding 60mm) and high reinforcement ratios that could lead to excessive cracking.

30.5.2.4.1.2 Functional requirements

The functionality of the structure should not be impaired by the cracks formed.

In relevant cases, nominal crack widths (w_{lim}) may be agreed with the client.

30.5.2.4.1.3 Durability

The durability of the structure during its intended lifetime should not be impaired by the cracks formed.

In some particular cases, more severe demands should be formulated in agreement with the owner, such as if de-icing agents are frequently used on top of tension zones in reinforced or partially prestressed structures.

Usually, a crack width up to 0.4 mm is acceptable. Larger cracks may be allowed, but the limits should be based on the agreement among the client, designer and contractor before the construction starts.

The concrete may have a higher strength than that which was ordered. A higher tensile strength of the concrete may lead to larger crack widths for constrained elements subjected to imposed deformations. Another uncertainty is the development of the tensile strength in time due to the continued hydration of the cement, in combination with the loading scenario.

At all sections which are expected to be subjected to significant tension (due to imposed or restrained deformation, combined or not combined with direct loading), a minimum amount of reinforcement should be placed, to ensure that the reinforcement does not yield immediately after cracking in SLS.

This applies also to prestressed members in regions where tension is expected to develop in concrete in SLS.

FRP reinforcement being not susceptible to rebar corrosion, the crack width is generally not as important for members with FRP reinforcement. It is rather controlled by aspects related to aesthetics, or in particular cases also where fasteners are presents, where crack width impacts water tightness levels, etc.

30.5.2.4.1.4 Appearance of the structure

The appearance of the structure should not be unacceptable because of cracking.

30.5.2.4.1.5 Uncertainties

For structures which are highly sensitive to cracking (e.g. architectural concrete, concrete in highly aggressive environments, containment structures, etc.), conservative values of the design parameters should be assumed to account for uncertainties related to the actual local concrete tensile strength, and to the highly stochastic nature of the cracking phenomenon.

30.5.2.4.2 Design criteria against cracking

(a) The specific requirements of sections 30.5.2.4.1.2 to 30.5.2.4.1.3 may be met by an appropriate limitation of crack widths. This may be achieved either by means of more refined analytical procedures (sections 30.5.2.4.4 and 30.5.2.4.5) or by simplified conservative formulations (section 30.5.2.4.6).

(b) The rules given in this chapter only apply if stress limitation requirements according to section 30.5.2.3 are fulfilled.

The provisions for steel reinforcing may be applied for concrete with FRP rebars, where the value of E_s is replaced by E_f. in all relevant equations.

Different combinations of actions may be considered under particular conditions.

(a) For reinforced concrete members, for exposure classes XC, XD, , XS (as specified in section 12.5.6.2), a w_{lim} = 0.30 mm may be assumed satisfactory under the quasi-permanent combination of actions with respect to both appearance and durability, in the absence of specific requirements (e.g. water tightness).
For exposure class X0, this limit may be relaxed provided that it is not necessary for reasons other than durability.

When de-icing agents are expected to be used on top of tensioned zones of reinforced concrete elements, appropriate w_{lim} values should be specified in accordance with the client, depending on the thickness of concrete cover and quality of the concrete, and of additional protective layers.

(b) For prestressed members, if more detailed data are not available, the crack width limit values presented in Table 30.5-1 may be used.

Table 30.5-1: Crack width limits (w_{lim}) (mm) for reinforced members and for prestressed members with bonded prestressing steel

	RC, PL2, PL3	**PL1**
X0	0.4	0.4
XC	0.3	0.2
XD	0.3	$\sigma < 0^*$
XS	0.3	$\sigma < 0^*$

For PL1, PL2 and PL3 see section 16.2.3.3.

*Stress in concrete at the level of prestressed reinforcement

For structural members subjected to exposure classes XD or XS and bending, no tension is allowed in a distance of 100 mm from the duct or the perimeter of the prestressed reinforcement towards the tension face. This value may be reduced to 25 mm in cases where the overall height of the member is below 400 mm.

30.5.2.4.3 Limitation of crack width

In order to meet the demands with regard to functionality, durability and/or appearance, the crack width has to satisfy the following condition:

$$w_{cal} \leq w_{lim} \qquad (30.5\text{-}1)$$

where:

w_{cal} denotes the design crack width calculated as in section 30.5.2.4.4. under the appropriate combination of actions (section 11.1.2) considered at the concrete surface;

w_{lim} denotes the nominal limit value of crack width considered at the concrete surface, which is specified for cases of expected functional, appearance-related or in some particular cases durability-related consequences of cracking. The nominal limit value of crack width corresponds to the nominal concrete cover c_{nom} according to section 30.13.3.

For ordinary FRP reinforced members in absence of appearance conditions, fasteners, punctual wheel pressure, lap slice or freeze thaw, the crack limitation can be relaxed to values up to 0,7 mm.

With regard to crack width limitation for fluid-tightness, different requirements may apply for various cases (0.15–0.25 mm). Special recommendations should be used. The requirement generally depends on the consequences of leakage and the pressure of the fluid:

– if leakage should be limited to a small amount, and some surface staining is acceptable, w_{lim} = 0.20 mm may be used as a limit for self-healing cracks. Otherwise w_{lim} < 0.2 mm may be more appropriate;

– if leakage should be minimal, no continuous cracks are allowed and a compression zone greater than 30 mm or 1.5 times max. aggregate size should be available in the characteristic load combination.

Whether self-healing of cracks can occur depends on the chemical composition of the fluid, type of cement, water pressure, time after subjecting to water pressure etc.

Longitudinal cracks due to the corrosion of steel bars are not covered by these criteria; they should be avoided by provisions for ensuring durability.

For the control of longitudinal cracks (parallel to the main steel bars), the thickness of the concrete cover as well as, where necessary, the secondary reinforcement (skin reinforcement) transverse to the main steel bars should be appropriately selected (as a function of their diameter), in order to secure the full development of the bond resistance estimated without consideration of longitudinal cracking.

30.5.2.4.4 Calculation of crack width in reinforced concrete members

30.5.2.4.4.1 General

Figure 30.5-1 shows the basic behaviour of a reinforced prism subjected to increasing axial deformation. In the cracks, the steel has to carry the full tensile load. To both sides of the crack the load is partially transmitted to the concrete. At a distance $l_{s,max}$ at both sides of the crack, the undisturbed situation is reached again. Under increasing deformation, more cracks are formed gradually.

The crack width calculations are based on the basic case of a prismatic reinforced concrete bar, subjected to axial tension. Adjustment factors are used to estimate the width of flexural cracks. With regard to the behaviour under increasing tensile strain in SLS, three stages are distinguished:

– the uncracked stage;

– the crack formation stage;

– the stabilized cracking stage.

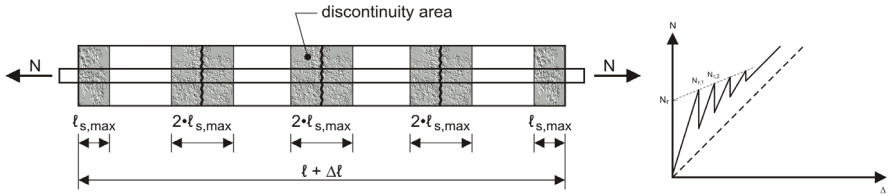


Figure 30.5-1: Behaviour of a reinforced prismatic bar subjected to imposed deformation

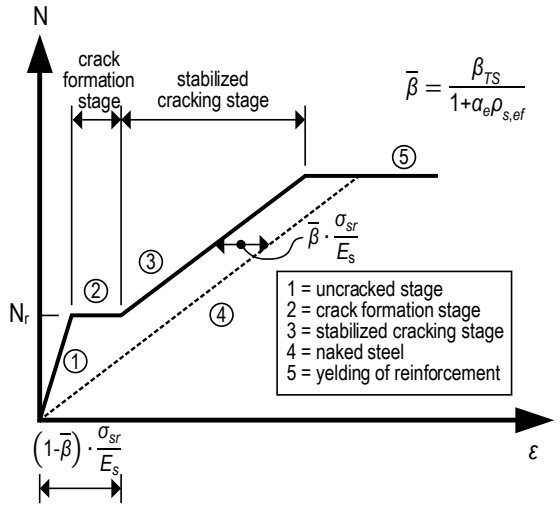


Figure 30.5-2: Simplified load – strain relation for a centrically reinforced member subjected to tension

Figure 30.5-2 shows a simplified representation of the load–deformation behaviour of a centrally reinforced member, restrained at the ends, subjected to tension or imposed deformation. According to the simplification, in the crack formation stage (2) the axial tensile force does not increase. When enough cracks have been formed to ensure that no undisturbed areas (white areas in Fig. 30.5-1) are left, the tensile strength of the concrete cannot be reached anymore between the cracks, so that no new cracks can appear. This is the start of the stabilized cracking stage (3). In this stage, no new cracks are formed but existing cracks widen. Finally, the steel will start yielding at stage (5).

For the model, as a simplification line (2) has been assumed to be horizontal, corresponding to a constant tensile cracking force:

For carrying out crack width calculations, it is necessary to know whether the crack formation stage or the stabilized cracking stage applies. Basically, the crack width results in both stages from the strain difference between concrete and reinforcement in the vicinity of the viewed crack. When going from the crack formation stage to the stabilized cracking stage the strain difference increases whereas the related length for integration of the strain differences decreases according to the real crack spacing in the stabilized crack stage.

The models presented below are simplified. The bond between reinforcement and concrete along the transfer length $l_{s,max}$ is assumed to be constant (Fig. 30.5-3). If a more detailed analysis is required, the bond stress may be assumed dependent on the slip of the bar [proposed Bulletin].With regard to ease of use as well as higher prediction quality, two approaches are given for the calculation of the crack width:

1) An approach on basis of the stabilized crack pattern (30.5.2.4.4.2), whereby the changed conditions of the crack formation stage are expressed by differences in the ratio of mean concrete tensile strength to mean bond strength and other modification factors (see Table 30.5-2).

2) A simplified approach (see section 30.5.2.4.6) derived from the general formulation with conservative assumptions from which the maximum bar diameter or bar spacing is derived for a given limiting crack width and a given stress value in the reinforcement.

The distinction between crack formation stage and stabilized cracking stage may be done with Figure 30.5-2, whereby the stabilized cracking stage applies when the load is larger than the cracking load ($N > 1.3\ N_r$). For imposed deformations, the crack formation stage applies when the mean strain satisfies the following condition:

$$\varepsilon = \frac{\Delta L}{L} \leq \frac{\sigma_{sr}(1-\beta_{TS})}{E_s} \cong (1-\beta_{TS}) \frac{f_{ctm}}{E_s \rho_s} \qquad (30.5\text{-}2)$$

where:

σ_{sr} is the stress in reinforcement in the crack formation stage;

$N_r = A_c\ f_{ctm}\ (1 + \alpha_e\ \rho_s)$

ρ_s is the reinforcement ratio (for tensile bar = A_s/A_c),

A_c is the area of the cross-section.

In case of the tie, $\sigma_{sr} = N_r/A_s$, in case of a cross-section subjected to bending, σ_{sr} is the stress in steel in the crack calculated from the equilibrium of internal forces in the section at the moment of the crack origin, i.e. at the moment when the concrete tensile stress σ_{ct} reaches the limit value $\sigma_{ct} = f_{ctm}$.

In reality this line will not be horizontal but inclined, ranging from the first crack for $f_{ctk,0.05}$ to an upper value of $f_{ctk,0.95}$ (sawtooth line in Fig. 30.5-1 right).

The approach with the horizontal line (2) is regarded as accurate enough, considering the influence of a number of uncertainties, such as the accuracy of placement of reinforcement, the real effective tensile strength and the influence of construction quality.

Under imposed deformation the stabilized cracking stage is usually not reached.

The transmission of forces in a disturbed area adjacent to a crack is shown in Fig. 30.5-3. Again, the relations are simplified by linearization.

β_{TS} is a coefficient to assess the mean steel strain over $l_{s,max}$, depending on the type of loading (instantaneous, long term, repeated, see Table 30.5-2);

$l_{s,max}$ is the transfer length defined in the Figs. 30.5-1 and 30.5-3.

If the mean strain is larger than this value, the stabilized cracking stage applies.

30.5.2.4.4.2 Calculation of crack width

The design (or calculated) crack width w_{cal} may be determined by:

$$w_{cal} = k_{1/r}\ s_{r,max}(\varepsilon_{sm} - \varepsilon_{cm}) \qquad (30.5\text{-}3)$$

where:

$k_{1/r}$ is a factor which accounts for the increase of crack width with cover due to curvature in bending. In case of pure tension, it simplifies to 1.0. For combined bending and axial force, it may be determined from Eq. (30.5-4):

$$k_{1/r} = \frac{h-x}{d-x} \qquad (30.5\text{-}4)$$

where:

h is the height of the section;

d is the effective depth of the section;

x is the depth of the neutral axis of the cracked section;

$s_{r,max}$ is the max. spacing between cracks in the stabilised cracking stage or a double of the transfer length in crack formation stage, calculated with Eq. (30.5-5);

ε_{sm} is the average steel strain over the length $l_{s,max}$;

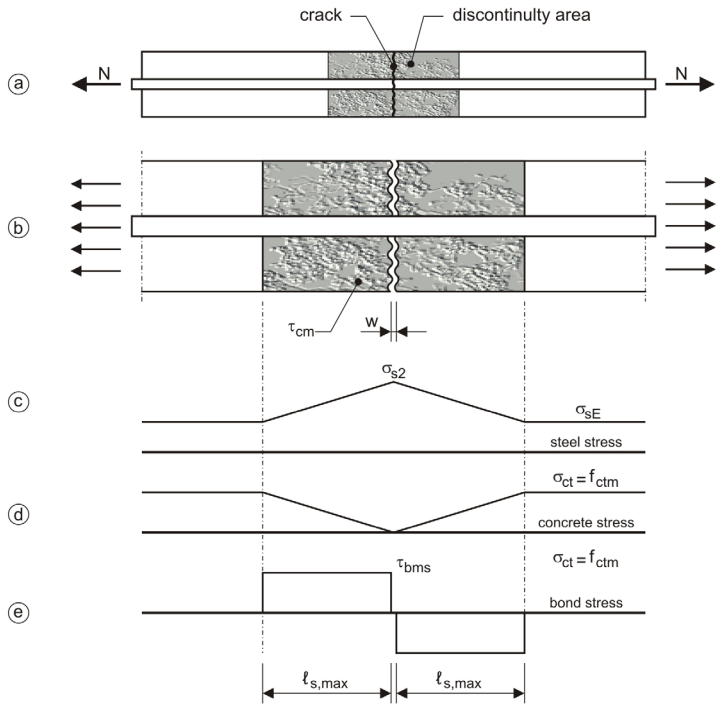


Figure 30.5-3: Steel, concrete and bond stresses in disturbed area in the crack formation stage (simplified representation): (a) centrically reinforced tensile member with crack; (b) discontinuity area; (c) steel stress development in discontinuity area; (d) concrete stress development in discontinuity area; (e) mean bond stress in discontinuity area.

The factor β_w=1.7 allows to transform a mean crack spacing at the stabilized cracking stage into a characteristic value, i.e. $s_{r,max} = \beta_w s_{r,m}$, where $s_{r,m}$ is a mean crack spacing.

Stress in reinforcement σ_s is calculated from the structural analysis of the structural element or it may be given by Eq. (30.5-13) in the crack formation stage.

The effect of cover on mean crack spacing (i.e. $k_c\, c$ term) is explained by the fact that a smaller number of internal cracks (also known as *Goto* cracks) propagate to the concrete surface as the concrete cover increases.

ε_{cm} is the average concrete strain over the length $l_{s,max}$;

ε_{cs} is the strain of the concrete due to unrestrained shrinkage.

The maximum crack spacing in the stabilised cracking stage (or double of transfer length in crack formation stage) may be evaluated using the Eq. (30.5-5). The accuracy of this model may be improved, e.g. if variable bond stress along the transfer length is introduced [proposed Bulletin]. The model is based on the calculation of the mean crack spacing in the stabilized cracking stage. It was calibrated using experimental results.

$$s_{r,max} = \beta_w \left(k_c c + k_{\phi/\rho} k_{fl} k_b \frac{f_{ctm}\phi}{\tau_{bms}\rho_{s,ef}} \right) \qquad (30.5\text{-}5)$$

where:

β_w is a factor which may be taken equal to 1.7 for the stabilized cracking stage and equal to 2.0 for the crack formation stage;

σ_s is the steel stress in the outer layer of the reinforcement determined assuming a fully cracked section;

k_c is an empirical parameter to quantify the influence of the concrete cover; as a simplification, k_c = 1.5 may be assumed;

c is the maximum of the vertical and horizontal clear concrete covers of the superficial bar;

$k_{\phi/\rho}$ is a parameter quantifying the influence of bond τ_{bms}; if a constant value of τ_{bms} over the transfer length is assumed, $k_{\phi/\rho}$ =0.25;

k_{fl} is a factor to account for stress distributions before cracking; its value may be determined according to Eq. (30.5-6);

$$k_{fl} = \frac{1}{2}\left(1 + \frac{h - x_g - h_{c,ef}}{h - x_g}\right) \qquad (30.5\text{-}6)$$

where:

x_g is the depth of the neutral axis before cracking;

$h_{c,ef}$ is the effective height according to 0 or 0;

k_b is a factor to account for the effect of casting position on crack spacing;

$$k_b = \begin{cases} 0.9 & \text{for reinforcement located in good casting conditions} \\ 1.2 & \text{for reinforcement located in poor casting conditions} \end{cases} \qquad (30.5\text{-}7)$$

For rectangular cross sections subjected to pure bending Eq. (30.5-6) simplifies to:

$$k_{fl} = \frac{h - h_{c,ef}}{h}$$

For sections subjected to pure tension, Eq. (30.5-6) simplifies to 1.00.

In the absence of a more refined model, as appropriate, may be used to assess the effective concrete area in tension. For single bars, the calculation should be done for the most unfavourable bar of the perimeter of the tensioned part of the section. When individual tension areas of different bars overlap the effective reinforcement ratio should be considered for the group of bars, as shown in Fig. 30.5-5.

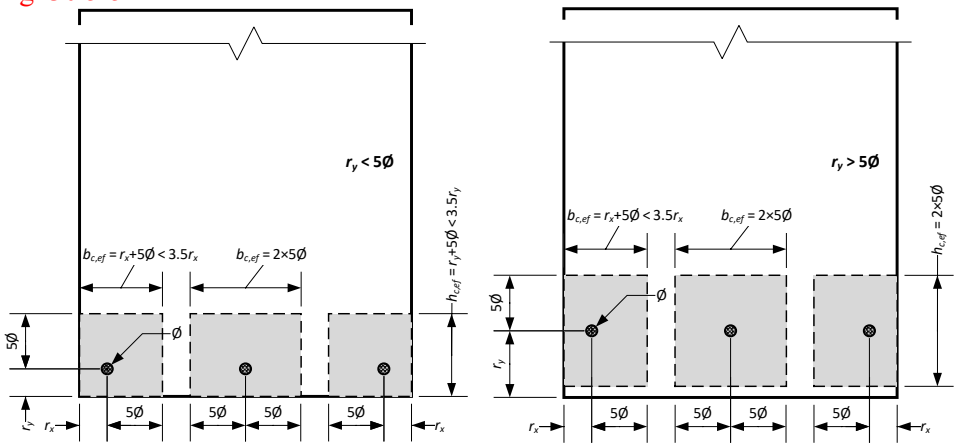


Figure 30.5-4: Effective tension area of concrete $A_{c,ef}$ around a single bar (local concept).

τ_{bms} is the mean bond strength between steel and concrete (Table 30.5-2);

$\rho_{s,ef}$ is the effective reinforcement ratio according to Eq. (30.5-8):

$$\rho_{s,ef} = \frac{\frac{\pi\phi^2}{4}}{A_{c,ef,bar}} \text{ or } \rho_{s,ef} = \frac{A_s}{A_{c,ef,group}} \qquad (30.5\text{-}8)$$

with $A_{c,ef,bar}$ = effective area of concrete in tension around a single bar (Fig. 30.5-4 and Eq. (30.5-9));

$$\begin{aligned} A_{c,ef,bar} &= b_{c,ef} \cdot h_{c,ef} \\ b_{c,ef} &= min(r_x + 5\phi; 10\phi; 3.5r_x) \\ h_{c,ef} &= min\left(r_y + 5\phi; 10\phi; 3.5r_y\right) \le h - x \\ b_{c,ef} &= min(r_x + 5\phi; 10\phi; 3.5r_x) \end{aligned} \qquad (30.5\text{-}9)$$

and $A_{c,ef,group}$ = effective area of concrete in tension around a group of bars (Fig. 30.5-5);

$$\begin{aligned} A_{c,ef,group} &= b_{c,ef} \cdot h_{c,ef} \\ h_{c,ef} &= min\left(r_y + 5\phi; 10\phi; 3.5r_y\right) + (n_l - 1)s_y \le h - x \\ b_{c,ef} &= b \end{aligned} \qquad (30.5\text{-}10)$$

where:

A_s is the total area of the bars within the group;

r_y is the cover to the centre of the bar in the y direction;

x is the depth of the neutral axis;

n_l is the number of reinforcement layers;

s_y is the spacing in the y direction between bars in the tensile zone;

r_x is the cover to the centre of the bar in the x direction;

s_x is the spacing in the x direction between bars in the tensile zone.

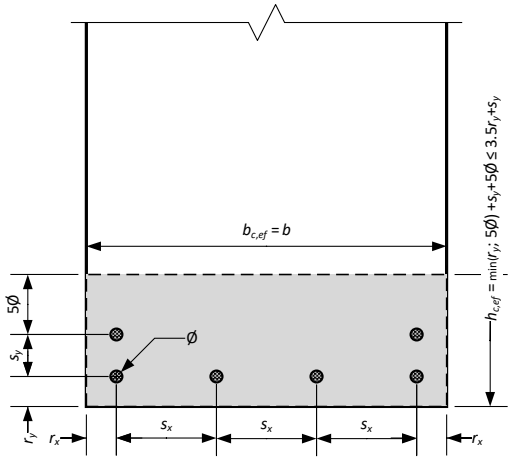


Figure 30.5-5: Effective tension area of concrete $A_{c,ef}$ around a group of bars (to be applied when effective tension areas of adjoining bars overlap)

In the elements where the imposed strain is restrained along the edges, Eq. (30.5-12) applies. It is applicable only in cases where the edge restraint has a dominating role, e.g. in low long walls cast on the stiff foundation. If a wall is higher, the mechanism is more complex and a combination of edge and end restraint should be considered. More detailed information can be found e.g. in [30.5-17]

In case of FRC the maximum crack spacing is computed as:

$$s_{r,max} = \beta_w s_{rm} \qquad (30.5\text{-}\#\#)$$

where s_{rm} is computed according to Eq. (18.6-6).

For an element subjected to direct loads (stabilized cracking) or subjected to imposed strains (crack formation phase) where end restraint dominates the relative mean strain in Eq. (30.5-11) follows from:

$$\varepsilon_{sm} - \varepsilon_{cm} = \frac{\sigma_s - \beta_{TS}\sigma_{sr,ef}}{E_s} \ge \frac{\sigma_s}{E_s}(1 - \beta_{TS}) \qquad (30.5\text{-}11)$$

For an element subjected to restrained imposed strains (crack formation phase) and restrained at the edges, the relative mean strain accounting for the effect of shrinkage in Eq. (30.5-12) follows from:

$$\varepsilon_{sm} - \varepsilon_{cm} = R_{ax}\varepsilon_{free} - \beta_{TS}\frac{f_{ctm}}{E_c} \qquad (30.5\text{-}12)$$

where:

σ_s is the steel stress in the crack;

$\sigma_{sr,ef}$ is the steel stress in a crack in the crack formation stage, which is given by (Eq. (30.5-13)):

$$\sigma_{sr,ef} = \frac{f_{ctm}}{\rho_{s,ef}}(1 + \alpha_e \rho_{s,ef}) \quad (30.5\text{-}13)$$

where:

α_e is the modular ratio $= \frac{E_s}{E_c}$;

β_{TS} is an empirical coefficient to assess the mean strain of steel with respect to concrete depending on the type of loading (Table 30.5-2);

R_{ax} is the restraint factor (Eq. (30.5-14)):

$$R_{ax} = 1 - \frac{\varepsilon_{restr}}{\varepsilon_{imp}} \quad (30.5\text{-}14)$$

ε_{restr} the strain which develops in the restrained element;

ε_{imp} the value of the imposed strain (i.e. unrestrained shrinkage, unrestrained temperature strain); The ratio $\varepsilon_{restr}/\varepsilon_{imp}$ may be estimated according to linear elastic analysis;

ε_{free} is the imposed strain which develops after the construction stage when restraint is applied (e.g. precast elements).

For long-term effects, shrinkage strains will tend to increase the relative strain between steel and concrete in a magnitude equal to $(1\text{-}\beta_{TS})\varepsilon_{cs}$. However, shrinkage is not considered in the crack formation stage because, shrinkage strains will generate tensile forces between cracks which will reduce the cracking force of the section (see Fig. 30.5-6). As the axial force in an element restrained at the ends in the crack formation stage cannot be higher than the cracking force, the reduction in the axial force as new crack forms will compensate for most of the increase in crack width due to the development of shrinkage. For the stabilized cracking phase, development of shrinkage will lead to the formation of a new crack between the initially stabilized cracks (reduction in the transfer length) and this will also offset the increase of crack width due to the development of shrinkage. These considerations justify the absence of shrinkage from the crack width formulation.

Table 30.5-2: Values for τ_{bms}, β_{TS} and η_r for deformed reinforcing bars.

Load type	***Crack formation stage***	***Stabilized cracking stage***
Short-term, instantaneous loading	$\tau_{bms} = 1.8 \cdot f_{ctm}(t)$	$\tau_{bms} = 1.8 \cdot f_{ctm}(t)$
	$\beta_{TS} = 0.6$	$\beta_{TS} = 0.6$
Long-term, repeated loading	$\tau_{bms} = 1.35 \cdot f_{ctm}(t)$	$\tau_{bms} = 1.8 \cdot f_{ctm}(t)$
	$\beta_{TS} = 0.6$	$\beta_{TS} = 0.4$

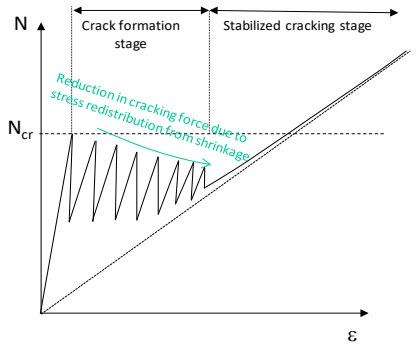


Figure 30.5-6: Reduction in the cracking force due to the development of shrinkage strains between cracks.

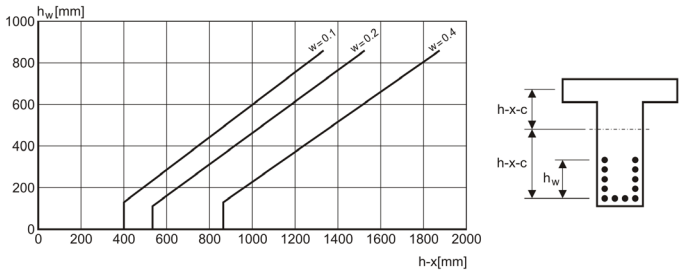


Figure 30.5-7: The height of the web that should be provided with crack distribution reinforcement to prevent wide cracks outside the effective tension area (illustrative diagram).

For a more refined calculation, an adequate bond stress-slip law may be introduced to determine the average bond stress.

If different bar diameters are used in the tensile area, the value ϕ in Eq. (30.5-5) may be replaced by ϕ_{eq} according to:

$$\phi_{eq} = \frac{\Sigma n_{s,i}\phi_i^2}{\Sigma n_{s,i}\phi_i} \qquad (30.5\text{-}15)$$

where:

$n_{s,i}$ is the number of bars with diameter ϕ_i.

The effective area of concrete in tension $A_{c,ef}$ accounts for the non-uniform normal stress distribution by bond forces in the concrete cross-section.

With the method given in this section, the design crack width within the effective area of concrete in tension is controlled. Outside this area, larger cracks may occur; this can be avoided by providing crack distribution reinforcement (Figure 30.5-7).

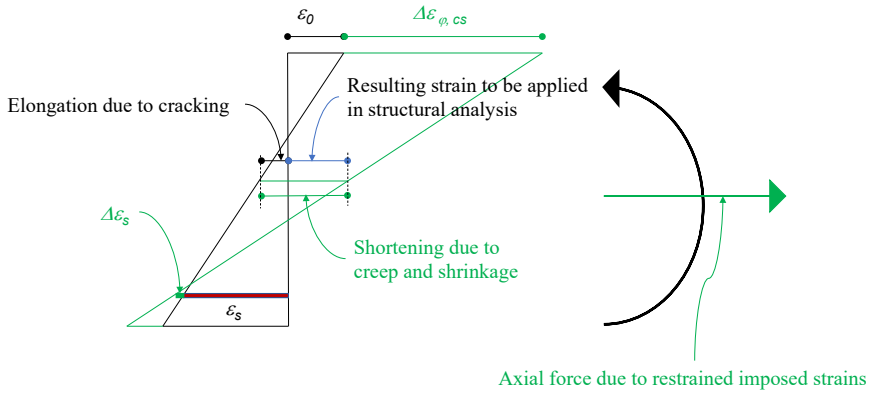


Figure 30.5-8: Change in the strain plane due to time-dependent effects.

30.5.2.4.4.3 Combined effects of load and imposed deformations

Figure 30.5-8 shows the strain plane due to a bending moment applied at $t=t_0$ and its variation due to the effects of creep and shrinkage. Due to cracking, at the centroid of the uncracked element (which is normally the reference point used for structural analysis) an elongation occurs. If the element under consideration is restrained longitudinally, this will produce a compressive axial force in the element. Due to creep and shrinkage the most compressed fibres will significantly increase their compressive strain ($\Delta\varepsilon_{\varphi,cs}$). At the level of the reinforcement there will also be a slight increase of the tensile strain ($\Delta\varepsilon_s$). This is due to creep relaxation of concrete stresses which will lower the stress neutral axis and slightly reduce the lever arm between the centroid of the compressive stresses and the centroid of the reinforcement, thereby requiring an increase in the forces to maintain equilibrium with the externally applied forces. The result of the development of creep and shrinkage will be a shortening of the centroidal fibre of element. If the element is restrained longitudinally this will result in a tensile axial force that will, in most cases, compensate for the compressive force developed due to cracking.

Creep and shrinkage will also generate a variation in curvature. The structural effects of the variation in axial strain and curvature (i.e. a variation in bending moments and a variation in the axial force) may be determined by applying the variations in strain and curvature obtained from sectional analysis at the different sections of the element, on the structural model, with properties accounting for creep of concrete (for example by using the effective modulus of elasticity for concrete) and cracking of reinforced concrete. In general, when the element is subjected to restrained deformations (crack formation stage), the maximum axial force will not exceed the cracking force which will be reduced by self-equilibrating tension stresses produced at the sectional level by shrinkage.

The effect of temperature is simpler. It may be assumed that the dilation coefficient of concrete and steel is the same. In this case all fibres of the section will expand or contract in the same amount. If the element is restrained longitudinally, a compressive or tensile force, respectively, will be generated. In this case, as the variation in temperature is considered as an instantaneous action no account for creep effects is needed.

Crack width calculations may then be carried out considering the additional effects of the variations in axial force and bending moment according to 30.5.2.4.2.

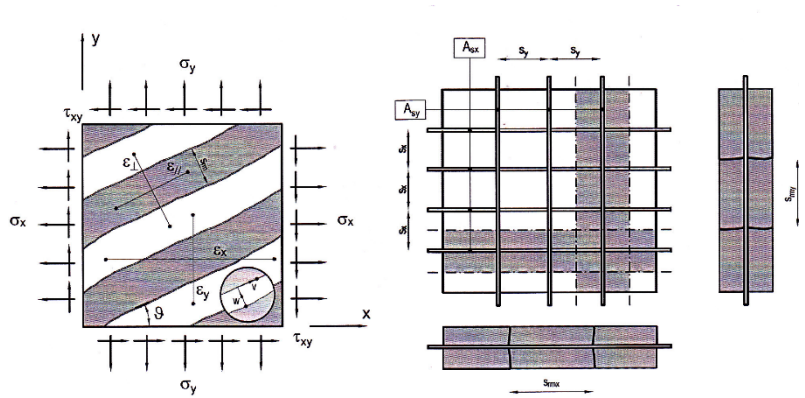


Figure 30.5-9: Basis for calculation of crack width for reinforcement deviating from the direction orthogonal to the crack.

Crack width calculations of orthogonally reinforced members are based on assuming that the orthogonal reinforcements act as tension chords in-between skew cracks formed as depicted in Figure 30.5-9. The parameters $(\sigma_{s,x}, \sigma_{s,y}, \varepsilon_2, \theta)$ necessary to determine the skew crack widths by Eqs. (30.5-16) and (30.5-18) are explicitly provided in this subchapter, in which the concrete in compression and the steel reinforcement in the cracked member are assumed to behave as linear elastic materials.

30.5.2.4.4.4 Calculation of crack width in orthogonally reinforced members

If the cracks in a member reinforced in two orthogonal directions are expected to form at an angle which differs substantially (>15°) from the direction of the reinforcement, the approximation by Eqs. (30.5-16) and (30.5-18) may be used to calculate $s_{r,max}$ and ε_{sm} -ε_{cm}.

$$s_{r,max,\theta} = \left(\frac{\cos\theta}{s_{r,max,x}} + \frac{\sin\theta}{s_{r,max,y}}\right)^{-1} \qquad (30.5\text{-}16)$$

where:

θ denotes the angle between the reinforcement in the x direction and the direction of the principal tensile stress; and may be determined from linear elastic analysis. A better approximation than using linear elastic analysis may be obtained by solving Eq. (30.5-17) for θ.

$$(\tan\theta)^4 + \frac{\sigma_x}{\tau_{xy}}(\tan\theta)^3 - \frac{\sigma_y}{\tau_{xy}}\frac{\rho_{s,ef,x}}{\rho_{s,ef,y}}\tan\theta - \frac{\rho_{s,ef,x}}{\rho_{s,ef,y}} = 0 \qquad (30.5\text{-}17)$$

where:

τ_{xy} is the shear stress;

σ_x is the normal stress in x direction;

σ_y is the normal stress in y direction;

$\rho_{s,ef,x}$, $\rho_{s,ef,y}$ are effective reinforcement ratios in the reinforced orthogonal directions according to Eq. (30.5-8);

$s_{r,max,x}$, $s_{m,max,y}$ denote the characteristic crack spacing in the two orthogonal directions, calculated according to Eq. (30.5-5).

The relative mean strain in Eq. (30.5-11) follows from:

$$\varepsilon_{sm} - \varepsilon_{cm} = (\varepsilon_{sm,x} - \varepsilon_{cm,x}) - (\varepsilon_{sm,y} - \varepsilon_{cm,y}) - \varepsilon_2 \qquad (30.5\text{-}18)$$

where:

ε_2 is the principal compressive strain and may be determined according to:

$$\varepsilon_2 = -\frac{|\tau_{xy}|}{E_{cm}\sin\theta\cos\theta} \qquad (30.5\text{-}19)$$

$(\varepsilon_{sm,x}\text{-}\varepsilon_{cm,x})$,$(\varepsilon_{sm,y}\text{-}\varepsilon_{cm,y})$ denote the relative mean strain in the reinforced orthogonal directions and may be determined in

conjunction with Eq. (30.5-11), however, adjusting for the fact that the tension stiffening for biaxial stress conditions in general differs from the case of uniaxial stress conditions, and may be taken as:

$$\varepsilon_{sm,x} - \varepsilon_{cm,x} = \frac{\sigma_{s,x} - \beta_{TS,x}\sigma_{sr,x,ef}}{E_s} \geq \frac{\sigma_{s,x}}{E_s}(1 - \beta_{TS,x}) \qquad (30.5\text{-}20)$$

$$\varepsilon_{sm,y} - \varepsilon_{cm,y} = \frac{\sigma_{s,y} - \beta_{TS,y}\sigma_{sr,y,ef}}{E_s} \geq \frac{\sigma_{s,y}}{E_s}(1 - \beta_{TS,y}) \qquad (30.5\text{-}21)$$

where:

$\sigma_{s,x}$ is the reinforcing steel stress at the crack in x direction taken as $(\sigma_x + \tau_{xy}\tan\theta)/\rho_{s,ef,x}$;

$\sigma_{s,y}$ is the reinforcing steel stress at the crack in y direction taken as $(\sigma_y + \tau_{xy}\cot\theta)/\rho_{s,ef,y}$;

$\beta_{TS,x}$ is taken as $\beta_{TS}(s_{r,max,\theta}/\sin\theta)/s_{r,max,x}$;

$\beta_{TS,y}$ is taken as $\beta_{TS}(s_{r,max,\theta}/\cos\theta)/s_{r,max,y}$;

30.5.2.4.5 Calculation of crack width in prestressed concrete members

The calculation of crack width concerns structural members with bonded prestressing reinforcement.

In general, the calculation of crack widths for prestressed structural members follows the procedure and the formulas given for reinforced concrete members in section 30.5.2.4.4.

For the general case of combined reinforcement consisting of reinforcing steel and prestressing steel, the prestressing steel is replaced by an equivalent cross-sectional area of reinforcing steel, which takes the lower bond quality of the prestressing steel into account. So, Eq. (30.5-8) is extended to the more general case of reinforcing and prestressing steel by replacing $\rho_{s,ef}$ by $\rho_{s+p,ef}$, where:

When prestressed and non-prestressed types of steel are simultaneously used, since the bond behaviour of prestressing tendons is different from the bond behaviour of deformed reinforcing bars, different steel stresses will be developed in each type of steel. For the calculation of the equivalent effective reinforcing ratio $\rho_{s+p,ef}$ both equilibrium and compatibility are respected, Figure 30.5-10 and Figure 30.5-11.

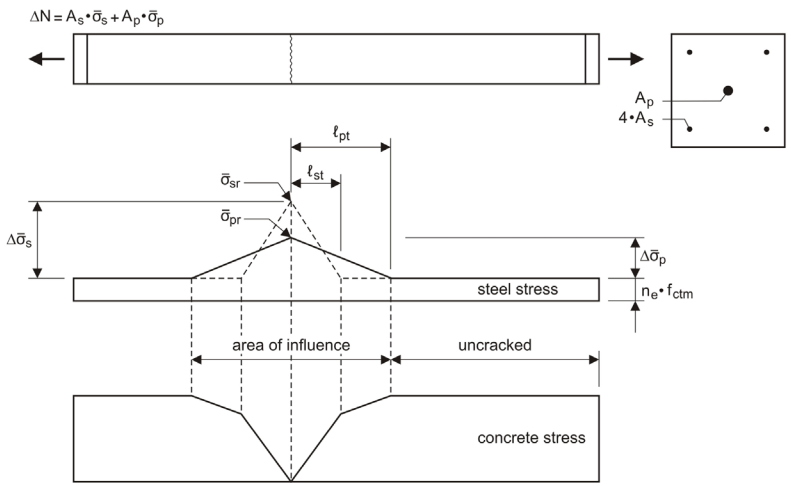


Figure 30.5-10: Crack formation stage: development of the steel and concrete stresses beyond decompression for a combination of reinforcing steel and prestressing steel.

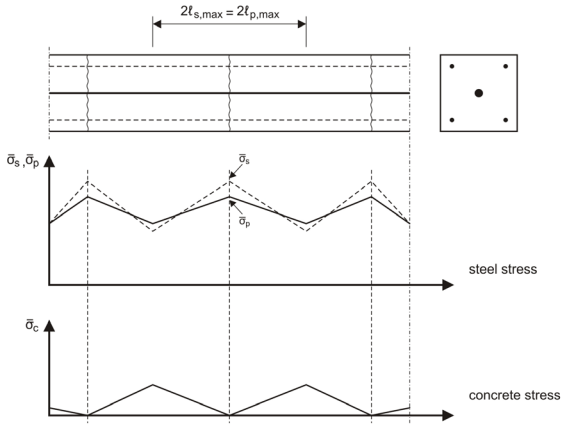


Figure 30.5-11: Stabilized cracking stage: development of the steel and concrete stresses beyond decompression for a combination of reinforcing steel and prestressing steel.

$$\rho_{s+p.ef} = \frac{A_s + \xi_1^2 A_p}{A_{c,ef}} = \rho_{s,ef} + \xi_1^2 \rho_{p,ef} \qquad (30.5\text{-}22)$$

where ξ_1 is a bond factor according to:

$$\xi_1 = \sqrt{\frac{\tau_{bmp}}{\tau_{bms}} \frac{\phi}{\phi_{p,eq}}} \qquad (30.5\text{-}23)$$

and the equivalent diameter $\phi_{p,eq}$ of the prestressing steel is given as:

$$\phi_{p,eq} = \frac{4\,A_p}{u_p}$$

where:

$A_p = \Sigma A_{p,i}$ is the total area of the prestressing steel;

$u_p = \Sigma u_{p,i}$ is the total equivalent perimeter of the prestressing steel;

where:

$u_{p,i} = 1.6 \cdot \pi \cdot \sqrt{A_{p,i}}$ for bundles;

$u_{p,i} = 1.75 \cdot \pi \cdot \varphi_{wire}$ for 7-wire strands;

$u_{p,i} = 1.20 \cdot \pi \cdot \varphi_{wire}$ for 3-wire strands;

where ϕ_{wire} is the single wire diameter in the strands.

In the absence of more detailed data, approximate values for τ_{bmp}/τ_{bms} may be taken from Table 30.5-3.

Table 30.5-3: *Bond factors τ_{bmp}/τ_{bms} for different types of prestressing steel.*

Surface condition	Pretensioned steel	Post-tensioned steel	No bond
Smooth wire	0.40	0.20	0
Strand	0.60	0.40	0
Indented wire	0.80	0.60	0
Ribbed bar	—	1.00	0

For reinforced or prestressed slabs subjected to bending without significant axial tension, no special measures to control cracking are needed, provided that the overall depth of the slab does not exceed 200 mm.

30.5.2.4.6 Minimum reinforcement for crack control in FRC

For controlling the crack width in the elements under bending, if needed, a minimum reinforcement should be applied, at least equal to:

$$A_{s,\min} = k_c \cdot k \cdot (f_{ctm} - f_{Ftsm}) \frac{A_{ct}}{\sigma_s} \qquad (30.5\text{-}\#\#)$$

where:

f_{ctm} is the average value of the tensile strength of the concrete matrix;

f_{Ftsm} is the average value of the residual strength of FRC;

A_{ct} is the tensile part of the concrete cross-section, evaluated by considering a stress field at elastic limit;

σ_s is the maximum tensile stress in the reinforcement in the cracked state, that can be considered equal to the yielding stress of the steel;

k_c is a coefficient taking account of the stress distribution in the cross-section just before cracking and the change of the inner lever arm; for rectangular cross-sections $k_c = 1$;

k is a coefficient taking into account of non-uniform self-equilibrating stresses, leading to a reduction of the cracking force:

$k = 1.0$ for webs with $h \leq 300$ mm or flanges with width ≤ 300 mm;

$k = 0.65$ for webs with $h \geq 800$ mm or flanges with width ≥ 800 mm; for intermediate values interpolation can be applied.

When $A_{s,min}$ is negative, the minimum reinforcement can be due only to the fibre reinforcement.

30.5.2.4.7 Simplified control of cracking

Under well specified conditions, the fulfilment of the requirements in sections 30.5.2.4.1, 30.5.2.4.2 and 30.5.2.4.3 may also be achieved by means of appropriate practical rules:

– when small depth elements subjected mainly to bending are considered, no special measures are needed for crack control;

The minimum reinforcement guarantees that the tensile force carried by concrete just before the origin of the crack is transferred by steel reinforcement in which the stress will be $\sigma_{s,lim} \leq f_{yk}$, immediately after cracking.

Eq. (30.5-24) is derived from the model of 30.5.2.4.4 by adopting the following simplifications:

- the height of the effective section is conservatively taken as $3.5r_y$;
- tension stiffening effects are assumed to reduce the mean relative concrete stress by 10% with respect to the value determined on the basis of a fully cracked section;
- the ratio τ_{bm}/f_{ctm} is assumed to be equal to 1.8;
- $k_{1/r}$ is determined, conservatively, assuming pure flexure.

Eq. (30.5-25) follows from and is equivalent to Eq. (30.5-24) since bar diameter and bar spacing are related by the definition of reinforcement ratio.

- under the condition that the minimum reinforcement is provided, the design crack width may be kept to acceptably low values, if appropriately chosen bar diameters or bar spacings are used. The rules given in 30.5.2.4.4 may be complied with by restricting either the bar diameter (ϕ) according to Eq. (30.5-24), or the bar spacing (s_l) according to Eq. (30.5-25).

$$\phi \leq \frac{2.1\,\rho_{s,ef}}{\frac{r_y}{d}\,k_{fl,simpl}\,k_b}\left(\frac{w_{\lim,cal}}{\beta_w\,k_{1/r,simpl}\,0.9\frac{\sigma_s}{E_s}} - 1.5\,c\right) \qquad (30.5\text{-}24)$$

where:

$$k_{fl,simpl} = \begin{cases}\left(1 - 3.5\frac{r_y}{h}\right); \; if\ one\ face\ is\ in\ compresion\\ 1.00\,; if\ both\ faces\ are\ in\ tension\end{cases}$$

$$k_{1/r,simpl} = \begin{cases}25\left(\frac{h}{d} - 1\right)\rho_{s+p,ef} + 1.15\,\frac{h}{d} - 0.15; \; for\ bending\\ 1.00; \; for\ tension\end{cases}$$

$$s_l \leq \frac{3.45\,\rho_{s,ef}}{\frac{r_y^2}{d}k_{fl,simpl}^2\,k_b^2}\left(\frac{w_{\lim,cal}}{k_w\,k_{1/r,simpl}\,0.9\frac{\sigma_s}{E_s}} - 1.5\,c\right)^2 \qquad (30.5\text{-}25)$$

30.5.2.5 Limit states of deformation

30.5.2.5.1 General

30.5.2.5.1.1 Requirements

The deformations should be accommodated by other connected elements such as partitions, glazing, claddings, services or finishes. In some cases, a limitation may be required to ensure the proper functioning of machinery or apparatus supported by the structure, or to avoid ponding on flat roofs.

To establish deformation limits is not within the scope of this Model Code.

Deflections that could damage adjacent parts of the structure should be limited. For the deflection increment after their installation, span/500 is normally an appropriate limit for quasi-permanent loads. In general, the appearance and general utility of the structure can be impaired, if the calculated sag of the element, relative to supports, subjected to quasi-permanent loads exceeds span/250 (cantilever span/125). Pre-camber may be used to compensate some or all of the deflection.

In-service deformations (deflections and rotations) may impair:

- the appearance of the structure;
- the integrity of non-structural parts;
- the proper functioning of the structure or its equipment.

To avoid excessive deformations, appropriate limiting values should be respected.

The provisions developed for steel reinforcement may be applied for concrete elements with embedded FRP rebars, where the value of E_s is replaced by E_f in all relevant equations.

The long-term deformations are calculated for the quasi-permanent load combinations. The instantaneous deformations should be calculated for the characteristic load combination. For the calculation of camber, the quasi-permanent load combinations are usually considered at buildings. At bridges or technological structures, a limit for the deflection induced by traffic (applied) load should be also considered, usually about span/600.

In order to calculate camber, the mean values of the material properties may be used.

It should be emphasized that the aim of the SLS check is to guarantee that the deflection is not larger than the allowable value rather that to calculate a precise value of the actual deflection.

The actual deformations may differ considerably from the calculated values, particularly if the values of the applied moments are close to the cracking moment. The difference will depend on the dispersion of the material properties, the ambient conditions, the loading conditions, the previous loading conditions, the restraints at the supports and so on.

Attention should be paid to cases where the basic assumptions of plane sections and uniformly distributed stresses across the section may not be adequate, such as

30.5.2.5.1.2 Combinations of actions

The combinations of actions to be considered depend on the criteria in question and are defined in section 30.5.2.2.

Possible pre-cracking due to relevant loading situations during construction or along service life should be taken into account.

30.5.2.5.1.3 Data for materials

The values of the material properties to be applied depend on the criteria in question. In order to prevent damage due to deformations, prudent values of the material properties should be used.

30.5.2.5.1.4 Modelling

Depending on the precision needed, appropriate deformation models should be used, as described in the following subsections.

30.5.2.5.2 Deformations due to bending with or without axial force

30.5.2.5.2.1 General methods (LoA III)

The most general method to assess deformations is to perform a numerical non-linear analysis capable of calculating the instantaneous and time-dependent deformations, taking into account the effects of applied loads and prestressing, the non-linear behaviour of concrete and steel and the time-dependent deformations due to creep and shrinkage of concrete and the relaxation of prestressing steel.

In structures where the construction process requires an accurate estimation of deflections, a step-by-step analysis should be performed, including the sequence of

in the case of shear lag effects in large prestressed structures. For prestressed concrete members, it may be necessary to control deflections assuming unfavourable deviations of the prestressing force and the dead load.

Contribution of tensioned concrete between cracks (tension-stiffening effect) should be modelled by appropriate constitutive laws.

For the usual level of concrete stresses at service, the principles of linear viscoelasticity are accepted (see subsection 29.2.3).

General methods are described in the CEB Bulletin 158 [30.5-2], CEB Bulletin 235 [30.5-3] and *fib* Bulletin 92 [30.5-1].

A different expression for the interpolation coefficient ζ may be adopted for prestressed members.

Use of the Effective Modulus Method is among the most widespread procedures for deflection calculation. The total deformation including creep is obtained by using an effective modulus of elasticity of concrete according to:

$$E_{c,ef} = \frac{E_c}{1+\varphi} \qquad (30.5\text{-}28)$$

where:

φ is the creep coefficient at the age of loading.

Shrinkage curvatures may be assessed by:

$$\frac{1}{r_{cs}} = \varepsilon_{cs} \cdot \alpha_e \cdot \frac{S}{I} \qquad (30.5\text{-}29)$$

where:

$1/r_{cs}$ is the curvature due to shrinkage;

S is the first moment of area of the reinforcement about the centroid of the section;

I second moment of area of the section;

α_e is the effective modular ratio = $E_s/E_{c,ef}$.

loading and possible changes in geometry, structural scheme, support conditions and the evolution of material properties with time.

Members loaded above the cracking load level are expected to have an average behaviour between the uncracked and the fully cracked stages.

A general approach to carry out a sectional analysis taking into account the average deformation of cracked members is based on the interpolation between the uncracked and fully cracked states, according to the expression:

$$a = \zeta a_{II} + (1-\zeta) a_I \qquad (30.5\text{-}26)$$

where:

a is the deformation parameter considered which may be, for example, a strain, a curvature or a rotation. As a simplification, it may also be taken as deflection;

a_I, a_{II} are the values of the deformation parameter calculated for the uncracked and fully cracked conditions respectively;

ζ is an interpolation coefficient (allowing for the effect of tension stiffening at a section) given by the expression:

$$\zeta = 1 - \beta \left(\frac{\sigma_{sr}}{\sigma_s}\right)^2 \qquad (30.5\text{-}27)$$

β is a coefficient accounting for the influence of the duration of loading or repeated loading on the average strain:

- $\beta = 1.0$ for a single short term loading;
- $\beta = 0.5$ for sustained loads or multiple cycles of repeated loading;

σ_s is the highest stress having occurred in the tension reinforcement calculated on the basis of a cracked section under the load considered;

σ_{sr} is the stress in the tension reinforcement calculated on the basis of a cracked section under the loading conditions that cause first cracking.

After sectional analysis, deflections and axial displacements may be obtained by appropriated integration of curvatures and axial strains along the member length.

Further information is also given in Model Code 1990 [30.5-4], *fib* Model code 2010 [30.5-5], Eurocode 2 (EN 1992-1-2, 2004) [30.5-6].

The method detailed in this section is consistent with those exposed in *LoA III*, representing a simplification in its application. It is appropriate for simply supported or continuous flexural members with or without side cantilevers, and for fixed-end cantilevers, with constant usual cross sections (rectangular, T, I) and variable reinforcement along the element length. The rotation necessary to calculate the end cantilever deflection may be interpolated using also Eq. (30.5-26). The approach in *LoA II* may be general enough to cover most current practical cases of buildings and other reinforced concrete structures, avoiding cumbersome calculations, while keeping precision compared to general models of *LoA III*. For more complex cases, as for example thick elements where the distribution of shrinkage strains is markedly nonlinear, numerical methods as those indicated in *LoA III* are recommended.

No explicit integration of curvatures is needed. Instantaneous and long-term deflections can be obtained under constant or variable load history. Members can be reinforced with steel or FRP reinforcement. Effects of cracking, tension stiffening, distribution of reinforcement along the element length, as well as creep and shrinkage of concrete are take into account.

Factor k_e takes into account the influence on the effective element inertia of the portions of the beam that are not cracked, and depends on the bending moment law and on M_{cr}/M_a. For elements with low reinforcement ratios and high strength concrete, not using factor k_e may provide excessively conservative values of deflections.

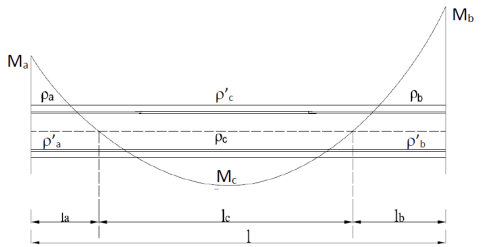


Figure 30.5-12*: Bending moment diagram defining l_a, l_b and l_c.*

30.5.2.5.2.2 Simplified method for RC structures LoA II

The instantaneous deflection may be estimated by interpolating the deflections computed in the uncracked and fully cracked states, a_I and a_{II}, respectively, through Eq. (30.5-26), adopting a secant modulus of elasticity of concrete E_c=0.9E_{c0}, and the following interpolation coefficient ζ:

$$\zeta = k_e\left[1-\beta\left(\frac{M_{cr}}{M_a}\right)^2\right] \qquad (30.5\text{-}30)$$

where, for simply supported and continuous beams, k_e is given by

$$k_e = 1 \qquad for\ \frac{M_{cr}}{M_a} \le 0.5$$

$$k_e = 2\left(1-\frac{M_{cr}}{M_a}\right) \qquad for\ 0.5 < \frac{M_{cr}}{M_a} \le 1 \qquad (30.5\text{-}31)$$

While for cantilevers:

$$k_e = 1-\left(\frac{M_{cr}}{M_a}\right)^2 \qquad for\ 0 < \frac{M_{cr}}{M_a} \le 1 \qquad (30.5\text{-}32)$$

where:

M_{cr} cracking moment obtained by using f_{ctm};

M_a maximum bending moment along the member length under the characteristic load combination, even in the case of estimation of deflections under the quasi permanent load.

To compute a_I, the stiffness of the gross section may be used, while to compute a_{II}, a representative fully cracked stiffness for the member should be used, obtained using average tensile and compressive reinforcement ratios, ρ_m, ρ'_m along the element length (see Fig. 30.5-12), as follows:

$$\rho_m = \frac{\rho_a l_a+\rho_c l_c+\rho_b l_b}{l} \qquad (30.5\text{-}33)$$

$$\rho'_m = \frac{\rho'_a l_a+\rho'_c l_c+\rho'_b l_b}{l} \qquad (30.5\text{-}34)$$

where:

ρ_a, ρ'_a reinforcement ratio of tensile and compressive reinforcement at support A;

The following simplified values may be adopted for l_a, l_b and l_c:

– for simply supported members: l_a=l_b=0; l_c=l;

– for end spans of continuous members:

$l_a = 0.2l(1 - 0.5\Psi)$; $l_b = 0$; $l_c = 0.2\ l\ (4 + 0.5\Psi)$

– for interior spans of continuous members:

$l_a = l_b = 0.25\ l(1 - 0.5\Psi)$; $l_c = 0.5\ l(1 + 0.5\Psi)$

where ψ is the ratio between the live load and the total load.

For beams with T-shaped cross section, ρ_m and ρ'_m should be determined multiplying the reinforcement ratios ρ_a , ρ'_a, ρ_b *and* ρ'_b, in the negative bending moment zones by the ratio $b_w/b_{f,eff}$, being "b_w" the web width and "b" the flanges effective width.

The use of a multiplicative factor for creep deflections avoids calculation of mechanical properties of the age transformed section.

Eqs. 30.5-37 and 30.5-38 for creep and shrinkage deflections, respectively, in cracked members are derived from a non-linear time dependent sectional analysis, assuming a constant strain of the tensile reinforcement along the time under permanent loads. This assumption is very close to the experimentally observed behaviour of cracked sections under permanent loads (Marí et al 2010 [30.5-7], Marí et al. 2019 [30.5-8], *fib* Bulletin 92 [30.5-1]

When a non-constant loading history takes place (i.e. when different permanent loads are applied at different times, t_i), the creep deflection at a final time, t, may be obtained by adding the deflections produced by each permanent load, using the corresponding creep coefficient $\varphi(t,t_i)$.

A more simplified approach consisting of using a weighted creep coefficient $\varphi^*=\Sigma(\varphi(t,t_i)\cdot g_i)/\ \Sigma g_i$ provides similar results.

A more precise value for k_s may be obtained from:

$k_s = 1$ for simply supported members;

ρ_b, ρ'_b reinforcement ratio of tensile and compressive reinforcement at support B;

ρ_c, ρ'_c reinforcement ratio of tensile and compressive reinforcement at mid-span;

l_a,lb lengths of those parts subjected to negative moments;

lc lengths of the part subjected to positive moments;

l element length.

The time dependent deflection is the sum of creep and shrinkage deflections:

$$a(t) = a_\varphi + a_{cs} \qquad (30.5\text{-}35)$$

The creep deflection is obtained by applying a multiplicative factor to the instantaneous deflection due to quasi-permanent loads.

For uncracked members:

$$a_\varphi = a_{i,qp}(t_0) \cdot \varphi \qquad (30.5\text{-}36)$$

For cracked members:

$$a_\varphi = a_{i,qp}(t_0)\frac{x_0}{d}\frac{0.8\ k_t\ \varphi}{1+12\alpha\rho'_m} \qquad (30.5\text{-}37)$$

where:

a_i, instantaneous deflection under the quasi-permanent load

x_0/d instantaneous neutral axis depth at time t_0, computed with ρ_m, ρ'_m

φ concrete creep coefficient at the age of loading ;

$\alpha=E_s/E_c$ modular ratio between steel (or FRP) and concrete;

ρ'_m average amount of compressive reinforcement;

k_t factor to account for the influence of the uncracked parts of the beam $k_t = 0.3 + 100\rho_m \le 1$ ·

$k_s = 0.7 - 4\left(\frac{l_c}{l} - 0.7\right)^2$ for end spans of continuous members;

$k_s = 2.5\frac{l_c}{l}\left(1 - \frac{l_c}{l}\right)$ for interior spans of continuous members.

where l_c is the length of the portion of beam subjected to positive (Fig. 30.5-12).

The method proposed in this paragraph is fully compatible with the general method (see 30.5.2.5.2.1), applied using the rheological models included in MC 2020. The method is conservative for elements with low reinforcement ratios when high strength concretes are used.

The shrinkage deflection may be assessed by:

$$a_{cs} = k_s k_t \psi_{cs} \frac{l^2}{8} \quad (30.5\text{-}38)$$

where:

k_s integration factor to obtain the deflection from the curvature due to shrinkage, which depends on the support conditions, and may be taken as 1 for simply supported members, 0,7 for end spans of continuous members, and 0,85 for interior spans of continuous members

ψ_{cs} reference element curvature due to shrinkage. For uncracked members Eqs. 30.5-29 may be used, while for cracked members is given by:

$$\psi_{cs} = \frac{\varepsilon_{cs}}{d}\frac{1}{1+12\alpha\rho'_m} \quad (30.5\text{-}39)$$

where

ε_{cs} is the shrinkage strain;

l span length.

30.5.2.5.2.3 Simplified method for RC structures LoA I

For rectangular sections, long-term deflections may be determined from linear elastic analysis using gross concrete section properties and assuming long term properties (i.e. $E_{c,ef}$) according to Eq. (30.5-40):

$$\delta = k_I\left[\delta_{LOADS} + k_s\delta_{\varepsilon_{cs}}\right] \quad (30.5\text{-}40)$$

where:

δ_{loads} is the linear elastic deflection, determined for uncracked conditions, due to the quasi-permanent combination of actions

$\delta_{\varepsilon cs}$ is the linear elastic deflection, determined for uncracked conditions, due to differential shrinkage computed by using the curvature given by:

$$\left(\frac{1}{r}\right)_{\varepsilon_{cs}} = \frac{E_s}{E_{c,ef}}\varepsilon_{cs}\frac{S_s}{I_g} \quad (30.5\text{-}41)$$

where:

S_s is the first order moment of area of the provided tension and compression reinforcements with respect to the centroid of the gross concrete cross section;

I_g is the second order moment of area of the gross concrete cross section;

k_I is an adjustment coefficient to account for the effect of cracking, for tension stiffening and for the fact that creep deformations are less than proportional to the creep coefficient in cracked sections. k_I may be estimated from:

$$k_I = \begin{cases} 1{,}00 \ if \ M_a < M_{cr} \\ 1 - \zeta + \zeta \frac{I_g}{I_{cr}} \ if \ M_a \geq M_{cr} \end{cases} \quad (30.5\text{-}42)$$

with:

$$\frac{I_{cr}}{I_g} = 2.7\left(\alpha_{e,ef}\rho\right)^{0.6}\left(\frac{d}{h}\right)^3 \quad (30.5\text{-}43)$$

where:

ζ is given by Eq. (30.5-30);

k_s is a coefficient to account for the effect of cracking on the shrinkage deflection. k_s may be estimated from:

$$if \ M_a < M_{cr} \rightarrow k_s = 1.00$$
$$if \ M_a \geq M_{cr} \rightarrow k_s = 455\rho^2 - 35\rho + 1{,}6$$

$\alpha_{e,ef}$ is the long-term modular ratio: $\alpha_{e,ef} = \frac{E_s}{E_{c,ef}}$;

ρ is the tensile reinforcement ratio $\rho = \frac{A_s}{bd}$, where, A_s, is the tensile reinforcement.

Two-way slabs may be very susceptible to construction history and redistribution of moments due to cracking and time effects. As a consequence, an accurate assessment of deflections is not always feasible. Theoretical consideration of the influencing aspects can be carried out through *LoA III* general methods based on non-linear time-dependent FEM analysis. Simplified procedures based on crossing beams or bands are described elsewhere (CEB Bulletin No. 158 [30.5-2]). Although less precise, these methods allow to analyse the slab using either the cross sectional or beam analyses in *LoA III* and *LoA II*, provided the distribution of moments is previously computed by linear-elastic FEM analysis or any other available procedure. Alternatively, displacements obtained from linear-elastic

FEM analysis of the slab can be used and transformed considering cracking and time-dependent effects in two-way slabs.

The values of Table 30.5-4 may be extrapolated to other support conditions by multiplying them by the cubic root of the ratio of the linear elastic deflection of a simply supported beam of the same span and the linear elastic deflection of the actual structure

For flanged sections where the ratio of the flange breadth to the rib breadth exceeds 3, the values of *l/d* given by Table 30.5-4 should be multiplied by 0.8.

For beams and slabs, other than flat slabs, with spans exceeding 7 m, which support partitions liable to be damaged by excessive deflections, the values of *l/d* given by Table 30.5-4 should be multiplied by 7/*l* (with *l* in metres).

For slabs where the greater span exceeds 8.5 m, and which support partitions liable to be damaged by excessive deflections, the values of *l/d* given by Table 30.5-4 should be multiplied by 8.5/*l* (with *l* in metres).

The curvature at time t is defined as:

$$\frac{1}{r}(t) = \frac{\varepsilon_{c,bottom}(t) - \varepsilon_{c,top}(t)}{h} \qquad (30.5\text{-}44)$$

where $\varepsilon_{c,bottom}(t)$ and $\varepsilon_{c,top}(t)$ are the strains at time *t* in the bottom and top fibres of the cross-section and h is the height of the section.

30.5.2.5.2.4 Verification of deflections without calculation

Provided that reinforced concrete beams or slabs in buildings, subjected to predominantly uniformly distributed loads, are dimensioned in compliance with the limits of span to effective depth ratio given in Table 30.5-4, their total deflections may be considered as not exceeding the limits set by l/250. In such cases it is not required to explicitly verify the deflections.

Table 30.5-4: *Basic ratios of span/effective depth for reinforced members without axial force.*

Structural system	Required mechanical reinforcement ratio								
	ω_r=0,3			ω_r=0,2			ω_r=0,1		
	LL/TL			LL/TL			LL/TL		
	60%	45%	30%	60%	45%	30%	60%	45%	30%
Simply supported beam, one-way spanning simply supported slab	15	14	12	17	15	13	22	19	17
End span of continuous beam or one-way slab	20	18	16	22	20	17	29	25	22
Interior span of beam or one-way spanning slab	23	21	18	26	23	20	33	29	26
Cantilever	7	7	6	8	7	6	10	9	8

For a maximum total deflection of l/a, the values of the table 30.5-4 should be multiplied by k_a=250/a. Interpolation between values of table 30.5-4 may be used.

30.5.2.5.2.5 Simplified methods for PC structures

Under the quasi-permanent load combination, the structure is considered to be uncracked.

The strains at time *t* are calculated in a way that takes into account the effect of self-weight, the initial prestress, other quasi-permanent actions and the stress variation due to creep, shrinkage and relaxation according to the following expression:

$$\varepsilon_c(t) = \varepsilon_{cs}(t,t_0) + \sigma_c(t_0)\cdot\frac{1+\phi(t,t_0)}{E_{cm}} + \frac{\sigma_c(t)-\sigma_c(t_0)}{E_{cm}}\cdot[1+\chi\cdot\phi(t,t_0)] \qquad (30.5\text{-}45)$$

where:

$\sigma_c(t_0)$ is the concrete stress at the initial time t_0 due to prestress and load effects;

$\sigma_c(t)$ is the concrete stress at time t.

For long-term loading the ageing coefficient χ may be assumed to be equal to 0.8.

30.5.2.5.3 Deformation due to shear forces

The effects of shear on the deformations of linear elements and slabs may be ignored where these are likely to be less than 10% of those due to bending.

The analysis of the deformations due to shear forces can be performed under three methods to which correspond increasingly high level of approximation.

30.5.2.5.3.1 General method (LoA III)

The most general method to assess deformations is to perform a numerical non-linear analysis capable of calculating both the contributions to the mean deformation of each block due to shear, such as the increment of mean curvature and the mean shear strain. The displacements of each point of the structure may be subsequently determined through the integration of the mean deformations of the blocks, for example by applying the principle of virtual works.

30.5.2.5.3.2 Simplified method - LoA II

From the experimental observations it results that shear does not produce only a transversal deformation, as assumed in Saint-Venant theory, and which can be taken into account passing through the shear strain, but in a cracked RC beam it produces also longitudinal deformations of the tension and compression chords. These longitudinal deformations come from the truss behaviour of cracked RC element. In effects, because of the truss behaviour, the web is subjected to an axial tensile force, that produces an increase of the deformation of the tension chord and a decrease of the deformation of the compression chord.

When a general method (*LoA III*) is applied, both states I and II are considered.

In state I, which corresponds to the behaviour of an uncracked zone subjected to a bending moment M, the mean curvature $1/r_1$ and the mean shear strain γ_1 may be calculated in accordance to the elastic theory.

For state II, which corresponds to the behaviour of a cracked zone ($M > M_r$), different models have been proposed in the literature based on "smeared models" to determine shear deformations (Vecchio & Collins [30.5-9]; Debernardi et al. [30.5-10], Ueda et al. [30.5-11]).

In the *fib* Bulletin 92 [30.5-1] a reference of the models present in the literature is presented.

Experimental evidence (Debernardi et al. [30.5-10]; Ueda et al., [30.5-11]) indicates that generally the contribution to the overall deformation of the element produced by the mean shear strain is significantly higher than that produced by the mean curvature after shear cracking.

Furthermore, both the concrete contribution given by the concrete section and the inclination of diagonal concrete strut, θ vary in a complex way, depending on the acting shear as well as on various parameters, such as the bending-to-shear ratio, longitudinal and transverse reinforcement ratios, class of concrete, geometry of the section. However, the values of concrete contribution and inclination of the diagonal concrete struts can be determined "a priori", in closed form, through simplified approaches (Debernardi et al. [30.5-12]; Ueda et al., [30.5-11]).

In the *fib* Bulletin 92 [30.5-1] a reference of the simplified approaches is presented.

A simplified method may also be applied, which is based on the same considerations of the general method, but simplified formulae for the assessment of the concrete contribution for shear and the inclination of diagonal concrete strut. Furthermore, the contribution of the shear to the mean curvature may be neglected, while only the contribution of the mean shear strain is taken into account.

In this way it is possible to avoid an onerous iterative calculation by adopting a simplified approach, though less refined, but in a closed form.

30.5.2.5.3.3 Influence of shear deformation for pre-dimensioning of RC beams LoA I.

When a greater precision is not required, in case of RC beams subjected to uniform load the shear contribution is expected lower than 10% and may therefore be ignored if:

$$\frac{10-B/B_w}{10-(B/B_w)_{min}}+\frac{L/d-6}{(L/d)_{max}-6}\geq 1 \qquad (30.5\text{-}47)$$

where:

L/d	is the span/effective depth ratio of the beam;
B/B_w	is the width-to-web thickness ratio of the section;
$(B/B_w)_{min}$ and $(L/d)_{max}$	are basic values given in Table 30.5-5.

This simplified rule applies when $L/d \geq 6$.

Table 30.5-5: Basic ratios of width-to-web thickness and span/effective depth for reinforced members without axial compression or tension.

For practical purposes, the immediate calculation of the deflection taking into account the shear contribution may be performed in a preliminary way referring to the equation:

$$f_{total} = f_{bending}\left(1+\frac{f_{shear}}{f_{bending}}\right) \qquad (30.5\text{-}46)$$

where $f_{bending}$ is calculated taking into account only the flexural effect according to rules given in 25.5.2.5.2 and the *"deflection due to shear / the deflection due to bending"* ratio is determined through a graphic method based on the results of an extensive parametric analysis carried out involving the main parameters, as described in *fib* Bulletin n. 92[30.5-1].

Structural system	$(B/B_w)_{min}$			$(L/d)_{max}$		
	ρ_s 0.5 %	ρ_s 1.0 %	ρ_s 1.5 %	ρ_s 0.5 %	ρ_s 1.0 %	ρ_s 1.5 %
Simply supported beam	2	0.5	0.25	16	21	23
End span of continuous beam	1	0.5	0.25	22	27	29
Interior span of the beam	1	0.5	0.25	28	30	31

30.5.2.6 Vibrations

30.5.2.6.1 General

Vibrations of structures may affect the serviceability of a structure as follows:

- functional effects (discomfort to occupants, affecting operation of machines, etc.);
- structural effects (mostly on non-structural elements, as cracks in partition walls, loss of cladding, etc.).

Vibrations can be caused by several variable actions, for example:

- rhythmic movements made by people, such as walking, running, jumping and dancing;
- machines;
- waves due to wind and water;
- rail and road traffic;
- construction work such as driving or placing by vibration of sheet piling;
- driving of precast piles as well as blasting work**.**

Vibrations that endanger the structure, such as very large deflections due to resonance or the loss of resistance due to fatigue, should be included in the verification for ULS of the structure.

30.5.2.6.2 Vibrational behaviour

To secure satisfactory behaviour of a structure subject to vibrations, the natural frequency of vibration of the relevant structure should be kept sufficiently apart from critical values which depend on the function of the corresponding building, see Table 30.5-6. The natural frequency shall be far enough from the critical frequency. If a critical frequency of the structure is lower than the imposed frequency, a more detailed analysis of the dynamic response including damping is required.

Table 30.5-6: Critical frequency in structures subjected to vibrations caused by movements of people.

Structures	**Frequency** (s^{-1}) f_{crit}
Gymnasia and sports halls	8.0
Dance rooms and concert halls without permanent seating	7.0
Concert halls with permanent seating	3.4
Critical work areas	1,0
Residence	1.4 – 4.0
Office	4.0
Workshop	8.0
Structures for pedestrians and cyclists	See below*

*Natural frequencies between 1.6 and 2.4 s^{-1} and between 3.5 and 4.5 s^{-1} are to be avoided in structures for pedestrians and cyclists. Joggers can also cause vibrations in structures with natural frequencies between 2.4 and 3.5 s^{-1}.

For the comfort of pedestrians, the accelerations of the structure or its parts should not exceed:

0.7 m/s^2 in vertical frequency, 0.2 m/s^2 in horizontal frequency, or 0.4 m/s^2 in horizontal frequency induced by exceptional loading by a crowd of people.

More detailed rules for design footbridges by dynamic load may be found in (*fib* Bulletin 32, page 21 [30.5-13]).

E.g., in EN 1991-2 [30.5-14].

If the structure is stiff (i.e., the deflections are small), the sensitivity to vibrations is reduced.

The vibrational behaviour of structures can be influenced by the following measures:

- changing the dynamic actions;
- changing the natural frequencies by changing the rigidity of the structure or the vibrating mass;
- increasing the damping features etc.

Special attention should be paid to dynamic behaviour of cable stayed and suspension bridges. Especially the stays of cable stayed bridges are sensitive to vibrations due to wind or rain action.

Rules for dynamic analysis of road and railway bridges may be found in appropriate design codes.

Indirectly, the vibrations may be limited by setting limits on deflections due to live loads.

30.5.2.7 Verification of serviceability limit states by numerical simulation

30.5.2.7.1 Scope

Nonlinear finite element analysis may be applied for verification of serviceability limit states of a wide range of concrete structures. Instantaneous effects of concrete cracking as well as time-dependent effects due to aging, creep, shrinkage and environmental actions may be included in the verification of serviceability states. Crack propagation is typically the most significant nonlinear effect and should be included in the serviceability verification.

30.5.2.7.2 Constituents of crack models

The crack propagation may be simulated by finite element analysis based on nonlinear fracture mechanics. The constituents listed below are typically used in the numerical models of cracking in form of model parameters or options.

Crack propagation in concrete can be simulated by the nonlinear fracture mechanics implemented within the finite element method. The constitutive model of crack behaviour is described by the crack opening law, which is derived from experiments. It reflects a quasi-brittle property of heterogeneous materials. An example of such a law is shown in Figure 30.5-13 which is typically used for normal strength concrete. For other materials, such as UHPC, FRC other appropriate formulations are used.

30.5.2.7.2.1 Crack opening law

Crack opening law describes the stress-crack width relationship. The main parameter is the fracture energy G_f required to create a unit area of the stress-free crack.

30.5.2.7.2.2 Bond-slip law

The bond-slip law describes the interface behaviour of reinforcement to concrete by a relationship between bond stress and bond-slip on the reinforcement surface.

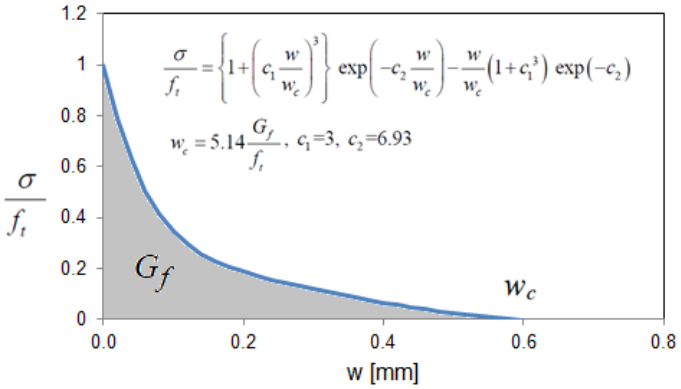


Figure 30.5-13: Example of exponential crack opening law. COD – crack opening displacement, σ – stress normal to the crack face, f_t – tensile strength, w_c – crack opening after complete stress release, G_f – fracture energy. (Hordijk [30.5-15]).

All parameters of the crack opening law, f_t, G_f, and function shape, affect the resulting crack pattern, spacing and widths.

30.5.2.7.2.3 Tension stiffening

Tension stiffening describes a contribution of cracked concrete to the stiffness of reinforcement. It describes an average response of cracked concrete and thus it is relevant to the coarse finite element meshes.

30.5.2.7.3 Mesh size effect and element types

The application of the modelling tools has limits for the numerical model discretization, namely the mesh size and element type.

The rough rule for the crack width simulation requires a mesh size approximately reflecting the concrete heterogeneity represented by the aggregate size, and a minimum of three elements between cracks. This leads to the mesh size in the range 15 – 50 mm. In such meshes, the fracture model based on the crack opening law combined with the bond-slip model can predict mean cracks well.

In the case of large element sizes, which are typically slab, shell, and beam elements, usage of fracture parameters only leads to an overestimation of the crack width. This is due to a lack of concrete contribution between cracks. In these cases, the crack spacing, and the crack width, may be estimated by applying the method according to section 30.5.2.4.4.

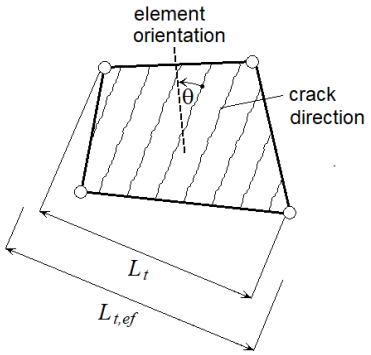


Figure 30.5-14: Crack band definition in 2D quadrilateral finite element with smeared cracks.

Two basic approaches are used in the finite element modelling of cracking:

Smeared crack model. The crack is modelled by orthotropic damage to concrete within the volume assigned to the integration point, or element. The damage region has a finite size, referred to as a crack band. An illustration of the smeared crack concept for a 2D stress state is shown in Figure 30.5-14. During the crack propagation, the tensile strain localizes within a crack band. The crack opening w is related to the fracture strain ε_f in the crack band by the transformation:

$$w = \varepsilon_f \cdot L_{t,ef} \quad with \quad L_{t,ef} = \alpha \cdot \gamma(\theta) \cdot L_t \qquad (30.5\text{-}48)$$

in which $L_{t,ef}$ is the effective crack band and L_t is the projection of element size in the crack direction. Parameter α reflects element type, parameter $\gamma(\theta)$ is a function of the crack orientation θ. The purpose of this provision is to reduce the effect of mesh orientation on the crack direction.

Two formulations are available for the smeared crack model: (a) *Fixed crack* model, in which the crack direction is determined at the crack onset as the normal to the principal tensile stress. In the course of a consequent loading, the principal stress can rotate, while the crack direction stays fixed. This can generate shear stress on the crack face and a formulation of the shear resistance of the cracked concrete becomes important. (b) *Rotated crack* model keeps the crack direction coincidental with the normal to the principal stress during the analysis. Consequently, the crack direction rotates during the analysis. Although the rotated crack model is more simple, in many cases it leads to unrealistic crack patterns. Therefore, the fixed crack model is generally preferred.

Discrete crack model, in which the crack is formed on the interface between two finite elements. The fracture properties are described by the interface crack opening law. In contrast to the smeared model, the crack and related damage are modelled as a discontinuity. This approach requires a re-meshing and a mesh refinement near the crack tip during the crack propagation, to reflect the crack direction not aligned with the initial element mesh. This method is typically used for unreinforced structures, such as dams, where only a limited number of major cracks occur.

The discrete crack approach can be also used within the extended finite element method (XFEM), in which a discontinuity is inserted as an additional layer into the existing finite element mesh. This improves the computational efficiency because it does not require a re-meshing, while it preserves the advantage of the discrete crack formulation.

Specification of defects is similar as that in the section 30.1.2.2. The consequences for SLS verification are highlighted in the following sections

Existing structures where no intervention is planned may require verification of load carrying capacity and SLS when their purpose is changing, the loading is changing, or the change of purpose requires using of more stringent code requirements.

In existing structures which exhibit some deterioration, the verification of ULS and SLS may be needed, so that any intervention is expected to be designed to meet the requirements for future use of the structure.

In existing structures where the repair or rehabilitation was executed the verification of ULS and SLS is required. The contribution of new materials used in the structure as well as their progressive activation during the loading process should be considered in the analysis.

30.5.2.7.4 Deflections

Analysis of deflections may be performed based on the models described in section 30.5.2.5. This allows a realistic simulation of the cracking effect on the stiffness reduction. Time effects, such as creep and shrinkage, aging, and corrosion may be included. This approach may be used for the assessment of structural life.

30.5.3 Structures with deterioration or defects

30.5.3.1 Structures where SLS are to be verified

Existing structures are exposed to mechanical loading, to physical effects (shrinkage, creep, swelling, etc.) chemical attack, high temperature impact and unfavourable environmental conditions, which may result in degradation of materials. The parameters related to the verification of the serviceability limit states (stresses, crack widths and deformations) may change with time.

Sometimes it is necessary to check if the serviceability limit state conditions are satisfied. These situations may be as follows:

- Existing structures where the conditions of use change and no intervention is planned
- Existing structures where repair or strengthening is expected
- Existing structures after repair or rehabilitation

The principles defined in following sections are applicable also for structures without deterioration, where some interventions took place.

The degradation of materials is defined on the basis of assessment of the existing structure, which is a subjective process. It is necessary to consider the accuracy of estimates of degradation processes and to specify an appropriate method for verification of SLS criteria.

Existing structures can suffer from deterioration of individual materials (concrete or steel) which can be visible or not. Some structures can also suffer from defects like excessive deflections, poor integrity of joints or support movements. A detailed assessment is necessary for specification of causes and possible future interventions.

During the assessment of the structure, it should be decided if deteriorated concrete should be removed or if only reduced parameters will be used in structural analysis.

The parameters of existing concrete (or of a new replacing material) should be defined in the assessment report.

Deterioration processes are dependent on time, and therefore usually it is necessary to specify the time development of the future performance or a single value of the SLS parameter at a specific time (e.g., at the end of the service life).

30.5.3.2 Types of deterioration or defects

30.5.3.2.1 Deterioration and defects of concrete

There is a variety of deterioration processes, which may be considered. If the structure is assessed, future actions should be defined, before SLS criteria are verified.

The most important deterioration processes of concrete are:

- Carbonation of concrete (causing corrosion of steel);
- Chloride ingress into concrete;
- Chemical attack (acid or pure water attack, biological attack);
- Freeze – thaw damage;
- Expansive reactions (Alkali-silica reaction, internal{external sulphatic reaction);
- High temperature effects;
- Cracking of concrete due to corrosion of steel;
- Accidental overloading and mechanical damage;
- Combined deterioration effects.

The deterioration of concrete most often results in a reduction of strength and stiffness of concrete especially in surface areas.

Corrosion of reinforcing steel is often a cause of spalling of concrete and longitudinal cracks.

Usually, the corrosion induced by water and oxygen is considered, which is often observed at reinforced concrete structures. The local corrosion due to hydrogen was identified at prestressing steel. It becomes a cause of a brittleness and a possible local failure. Local reduction of the cross-section of the steel bar/strand leads to reduction of its strength and also to reduction of the ultimate strain. The failure (at smaller ultimate load) can happen at small elongation of the reduced section of the bar/strand, which makes the failure brittle.

Reduction of the cross section of bars and of the bond has a direct effect on the reduction of the load carrying capacity of anchorages and joints using laps.

The local damage of prestressing steel is very dangerous, and it is difficult to discover. There are almost no non-destructive methods for identification of local corrosion of prestressing steel.

Drop of the prestressing force along the length of the element may result in increased deflections and cracking. However, the local corrosion of prestressing steel usually cannot be recognised by increasing deflections. Significant growth of cracks due to local corrosion appears usually when the structure or the prestressing unit fails. It cannot be expected that growing deflections or wider cracks will become a warning before failure.

The lack of concrete parts, caused for example by cover spalling, should be properly considered in the geometrical modelling of the element. Cover spalling has not only effects on the geometry of the element but also on the modelling of bond behaviour. A detailed modelling of geometry and bond is especially important in numerical methods.

30.5.3.2.2 Deterioration and defects of steel

Reinforcing steel is a subject of corrosion which is caused by carbonation of concrete or ingress of chlorides into concrete. Corrosion may be spread over a larger area or may be located at specific points. The corrosion of reinforcing steel results in:

- Reduction of the active cross-sections of the reinforcement (resulting in reduction of ultimate strain and/or ultimate strength of reinforcing bars and/or prestressing units);
- Reduction of the bond between steel and concrete.

If applicable, mechanical damage of steel (e.g., after a vehicle impact) should be considered.

Corrosion of prestressing steel leads to a reduction of the prestressing force. Local corrosion of prestressing steel can result in a significant loss of ductility and load carrying capacity of prestressing units.

30.5.3.3 Parameters for SLS verification

30.5.3.3.1 Geometrical parameters

Actual geometry of structural members should be used in the analysis. Actual position and cross-sectional area of reinforcing bars should be used, with consideration of the changed bond properties.

Special attention has to be paid to supporting conditions, quality of joints and position of loading.

Compression strength and elastic modulus may be taken according to tests on bored cores. Non-destructive testing is recommended to increase amount of data and to reduce the statistical scatter of its statistical features (expectation, variance).

Tensile strength may be measured by tear off tests or by transformation of the splitting strength measured on cores.

Creep and shrinkage are strongly dependent on the age of concrete and also on the temperature development (especially in early ages of concrete). If no more precise data are available, the models according to the section 14.6.1.6.4.4 may be used.

In the SLS analysis the bond strength between steel and concrete is assumed using a bond-slip diagram, or in a simplified way, by considering its dependence on the tensile strength of concrete. (Table 30.5-2). If no more precise data are available, the values from the Table 30.5-2 may be used, provided the reduction due to corrosion of steel is applied.

The local damage of prestressing steel may result is a loss of ductility and strength due to stress concentration at the location of the damage. Small elongation may lead to significant stress growth at that point finalizing in the failure. As a consequence of the local reduction of the cross-section, prestressing steel tends to lose its ductility, the strength is reduced and the steel becomes brittle.

30.5.3.3.2 Concrete parameters

Material parameters of concrete should be preferably derived from tests on concrete of the existing structure.

Creep and shrinkage of old concrete is smaller, appropriate values should be estimated.

If a part of the cross-section is removed and replaced by a new material the difference in material stiffness, rheological behaviour, bond properties and load histories should be considered

30.5.3.3.3 Steel parameters

Material parameters of steel itself do not change significantly with time. The corrosion results in geometrical changes (non-uniform reduction of the cross-section, local eccentricity, etc.) which appear as a significant change of the ultimate strength and ultimate strain and may lead to the brittleness of reinforcing and especially of prestressing steel (see section 15.11.2.1). The brittle failure of especially prestressing steel could lead to a sudden failure of the structure.

The elastic modulus and yield strength may be measured on specimens taken from structure or from the original design parameters of the steel used in the structure.

The corrosion influences preferably dimensions and ultimate strain, as mentioned in the section 30.5.3.3.1 and 15.11.2.1.

Bond strength may be significantly reduced by corrosion of steel (see section 20.3.5.1). The parameters applicable in the analysis should be specified by testing in actual conditions.

30.5.3.3.4 Prestressing steel parameters

Prestressing steel also does not change its mechanical material parameters with time, but the local corrosion may significantly influence the structural behaviour.

The stress drop due to stress relaxation in prestressing steel is dependent on the stress level and time elapsed from prestressing. The dependence of the relaxation coefficient on temperature should be considered.

High attention has to be paid to anchorage of prestressing units.

The prestressing force in the section is usually hard to estimate. Indirect estimates based on magnetoelastic behaviour or deformation measurements may be used.

The loading history should be considered. In usual cases, the principle of superposition of the effects of individual loading cases, which may be active on different structural systems, may be used. Linear analysis is applicable as a reasonable simplification, especially in verification of SLS.

The sections after repair are modified (they may be composed of different materials) and they should be considered as composite sections. The reinforcement may be strengthened and this should be considered in the prediction models.

Progressive loading of the structural elements should be considered.

Stresses in reinforced concrete sections are calculated under the assumption of cracked sections. Effective modulus method may be used for long-term effects.

In statically indeterminate structures, the loss of steel area reduces the stiffness and modifies its distribution along the elements length. Redistributions of internal forces, increments of stresses and deflections may take place.

If a numerical (non-linear) analysis is used, the limits of stresses do not apply for singularities and local stresses in anchorage areas and similar locations.

30.5.3.3.5 Loading of the structures

Existing structures or elements are subjected to loading. Only in exceptional cases, the individual elements may be totally unloaded and then loaded in new conditions. The subsequent development of loads acting on corresponding structural systems should be identified and then used as an input into the structural analysis.

The appropriate loading combinations are used as mentioned in section 30.5.2.2.

30.5.3.4 Models for SLS verification

In general, the principles of SLS verification are identical to those used for new structures (section 30.5.2). The input parameters may be modified according to the degree of degradation of individual materials.

During interventions, the structural systems are modified and thus shall be considered in the models.

Modifications of sections, changes in the supporting conditions, prestressing of members and sequence of loading and unloading should be included in the structural model for SLS verification.

30.5.3.4.1 Stress limitation

Stresses in concrete, steel and prestressing steel in existing structures are induced by actual loading and unloading history. The input parameters should be taken according to the actual state of individual parts of the structure as described in the section 30.5.3.3. The actual prestressing force should be estimated on the basis of measurements made within structural assessment and/or on the investigations of the history of the structure.

Stress analysis is usually made under the assumption of linear analysis (allowing application of principle of superposition). If necessary, a non-linear numerical

In existing structures, the cracks already exist. If no intervention is assumed, and the crack widths do not satisfy the criteria, it should be decided whether or not wide cracks should be repaired. This decision may be based on the observed durability performance or on appearance criteria, as applicable.

If an intervention is assumed, the expected new cracks should theoretically satisfy the crack width limits and the decision if already existing cracks which exceed these limits, should be repaired is necessary.

The decision on repairing of wide existing cracks should be made on the basis of future use of the structure and environmental conditions.

The input parameters applied in the crack width analysis are as follows:

concrete:

Tensile strength (f_{ctm}), modulus of elasticity (E_c), shrinkage strain (ε_{cs});

steel:

analysis may be used. In this case, the correct loading history should be applied and the principle of superposition is not applicable.

Limits for stresses in concrete and steel in SLS are identical to those, which are applicable at new structures (section 30.5.2.3)

30.5.3.4.2 Limit states of cracking

30.5.3.4.2.1 Requirements

Requirements specified for new structures are applicable for also existing structures (section 30.5.2.4.1).

30.5.3.4.2.2 Limitation of the crack width

The criteria for limits of the crack width in deteriorated structures are identical to those used for structures without deterioration or defects (table 30.5-1).

30.5.3.4.2.3 Calculation of crack width in reinforced and prestressed members

The principles and methods described in the section 30.5.2.4.4 and 30.5.2.4.5 are valid also for structures with deterioration. The input parameters should reflect the actual conditions of the structure.

In case of externally applied reinforcement, it should be considered.

Modulus of elasticity (E_s), Bond strength (τ_{bms}, τ_{bmp});

geometry:

Dimensions of the concrete cross-section, Cross-sectional area of steel (A_s), diameter of reinforcing bars/prestressing units.

These parameters should be modified according to the actual conditions.

The bond slip law of steel in the deteriorated structure should be defined on the basis of the survey.

Based on the bond slip law the mean bond strength may be determined.

The bond strength of corroded bars is strongly dependent on the confinement.

The actual bond parameters of steel, which is corroded or exhibits deviations from the usual ribbed steel, should be considered (see section 20.2).

The most important input parameters are:

- Geometrical parameters:

  dimensions of the section, area of steel reinforcement, position of the steel reinforcement;

- Concrete parameters:

  modulus of elasticity, tensile strength, time dependent effects (creep and shrinkage);

- Steel parameters:

  modulus of elasticity, yield strength, stress relaxation at prestressing steel.

Preferably these parameters should be modified in accordance with the results of structural assessment of the existing structure.

30.5.3.4.3 Limit states of deformation

Existing structures exhibits deformations. Deformations should be measured and compared with limiting criteria and with calculated values. The realistic loading should be used for the analysis used for the purpose of this comparison. Differences should be explained.

Deflection analysis is based on identical principles and methods, which are applicable for new structures (section 30.5.2.5).

The criteria for deformation limits should consider the initial deformation of the existing structure.

The criteria for the deformation should consider the behaviour of the non-bearing structures installed after intervention executed on the existing structure.

The input parameters should be modified according to the actual state of the structure (including deterioration).

The time dependent effects should be considered including the loading history.

If an intervention is designed, the new materials should be added at the appropriate time and loading steps in the structural analysis. (Earlier loading effects acting on the existing structure should be considered).

The input parameters based on measurements on site may be obtained by the statistical evaluation. If no more accurate procedure is used, methods following the EN 1990, Annex D [30.5-16] may be used.

For assessment of existing structures, partial safety factors may be modified according to the *fib* Bulletin 80 [30.5-17].

30.5.3.5 Uncertainties

Input parameters for the SLS analysis are derived from the survey and assessment of the existing structures. The accuracy of their specification is dependent on the number of tests and on the degree of deterioration of the structure. This uncertainty should be considered in their specification.

30.5.4 Structures with non-compliant detailing

30.5.4.1 General

Existing structures with non-compliant detailing involve the structures which do not exhibit deterioration or defects but their reinforcement or minimum dimensions do not correspond to the actual code requirements.

Examples of structures where detailing is not compliant with the MC2020 requirements

- Structures which are extremely slender (not satisfying the requirements on detailing rules);
- Structures with low or extremely high reinforcement ratio;
- Structures, where anchorage lengths are not satisfactory;
- Structures, where overlapping length lapps joints is not sufficient, or transversal reinforcement is not sufficient, etc.;
- Structures with low concrete cover.

30.5.4.2 Effect of non-compliant detailing on verification of SLS

It may be expected that the impact of non-compliant detailing to the verification of ULS criteria will be more critical, than that to the verification of SLS.

If deformations or crack widths exceed the limits, any possible intervention should be done in dependence of ULS verification and evaluation of conditions of function and durability.

An assessment of the existing structure is a basis for evaluation of effects of non-compliant detailing on verification of SLS. Crack widths and deformations of the existing structure should be evaluated with respect to the actual loading level and design loading level, in particular.

If loading of the structure is increased, the ULS criteria shall be satisfied first (considering the non-compliant detailing). After that the SLS criteria will be verified for the increased load.

If ULS criteria are satisfied for the structure with non-compliant detailing, it may be assumed that the methods for SLS verification are applicable also for these structures.

If a reasonable agreement is achieved, and ULS criteria are satisfied, the methods and procedures described in the section 30.5.2 are applicable also for increased loads or for structures after intervention.

If a reasonable agreement is not found, a more detailed assessment of the existing structure is necessary and the reasons for disagreement should be found. It can be expected that the intervention will be necessary.

If the crack widths or deformations exceed the limiting criteria after increasing load or after intervention, the consequences for function, appearance and durability should be specified, considering the actual conditions. If necessary, the intervention should be modified.

Interventions (if applicable) should be designed in such way so that the deficiencies due to non-compliant detailing would be eliminated.

30.5.4.3 Verification of SLS

The crack widths and deformations at existing structure should be calculated using the methods described in the section 30.5.2 and the results should be compared with those observed on the site.

After increasing of load or after intervention, the methods of SLS verification described in the section 30.5.2 may be used, provided the ULS criteria are satisfied.

If plain surface bars are used the reduced bond strength shall be considered. The bond slip relation according to the section 20.3.3 may be used.

References

[30.5-1] *fib* Bulletin 92. Serviceability Limit State of Concrete Structures, 2019.

[30.5-2] CEB Bulletin 158. Cracking and Deformations, 1985.

[30.5-3] CEB Bulletin 235. Serviceability Models, 1997.

[30.5-4] Model Code 1990 CEB-FIP Model Code 1990, CEB Bulletin 213/214, 1993

[30.5-5] *fib* Bulletin 65/66. Model Code 2010, 2012.

[30.5-6] EN 1992-1-2:2004. Eurocode 2: Design of concrete structures – Part 1-2: General rules – Structural fire design.

[30.5-7] Mari et al 2010 Mari A., Bairan J., Duarte N., Long-term deflections in cracked reinforced concrete flexural members, Engineering Structures, 2010, Vol. 32 (3), 829–842.

[30.5-8] Mari et al 2019 Marí, A., Torres, L., Oller, E., Barris, C. (2019), Performance-based slenderness limits for deformations and crack control of reinforced concrete flexural members, Engineering Structures, 2019, Vol. 187, 267-279.

[30.5-9] Vecchio & Collins Vecchio, F. J., Collins, M. P., The modified compression field theory for reinforced concrete elements subjected to shear. ACI Journal, 1986, 83, No.2, 219-231.

[30.5-11] Debernardi et al Debernardi, P. G., Guiglia, M., Taliano, M. , Shear strain in B-regions of beams in service. ENGINEERING STRUCTURES, vol. 33, p. 368-379, ISSN: 0141-0296, doi:10.1016/j.engstruct.2010.10.017.

[30.5-11] Ueda et al Ueda, T., Sat, Y., Ito, T., NishIizono, K, .Shear Deformation of Reinforced Concrete Beam, Journal of Materials, Concrete Structures and Pavements, JSCE, No.711/V-56, pp.205-215, August 2002.

[30.5-12] Debernardi et al Debernardi, P. G., Guiglia, M., Taliano, M., Parametric analysis of the influence of shear on the deflection of RC beams in service. MAGAZINE OF CONCRETE RESEARCH, vol. 64, p. 335-350, ISSN: 0024-9831, doi: 10.1680/macr.11.00013.

[30.5-13] *fib* Bulletin 32. Guidelines for the design of footbridges, 2005.

[30.5-14] EN 1991-2:2003. Eurocode 1: Actions on structures – Part 2: Traffic loads on bridges.

[30.5-15] Hordijk. Hordijk, D. A., Local Approach to Fatigue of Concrete, dissertation, Delft University of Technology, The Netherlands, 1991, ISBN 90/9004519-8.

[30.5-16] EN 1990:2002. Eurocode – Basis of structural design.

[30.5-17] *fib* Bulletin 80. Partial factor methods for existing concrete structures, 2016.

[30.5-18] Röhling, S., Meichsner, H. Rissbildungen im Stahlbetonbau, Ursachen, Auswirkungen, Maßnahmen. Fraunhofer IRB Verlag, Stuttgart, 2018, ISBN 978-3-8167-9645-9

Further information is given in *fib* Bulletin ZZ [30.6-1].

It is possible that several deterioration mechanisms may occur simultaneously. As a matter of fact, there may be a combination of exposure categories leading to the simultaneously different safety formats (Coupled mechanisms detailed in section 30.6.8).

The Table 30.6-1 summarizes the applicability at this time of the different levels of approximation described in section 14.3.2 for each deterioration mechanisms developed in the following sections

Table 30.6-1: Presently available approaches for SLD assessment.

	LoA1	LoA2	LoA3	LoA4	Avoidance
Carbonation induced corrosion	✓	★	★	✓	✓
Chloride-induced corrosion	✓	★	★	✓	✓
Freeze-Thaw	✓	✓	-	-	✓
Chemical attacks	✓	★*	-	-	✓
ASR	✓	✓	-	-	✓
DEF	✓	✓	-	-	✓
Abrasion	★	★	-	-	✓

* Depending on the nature of the chemical attack;

✓ Existing rules: rules available in most of the Codes and/or models assumed being mature enough for being implemented by skilled users;
★ Limited rules: only available in some national codes, or still under validation and calibration;
- Unavailable: ongoing research.

30.6 Evaluation of condition limit states (CLS) associated with durability

30.6.1 Introduction

30.6.1.1 General

Verification of limit states associated with durability may be carried out according to one of the following safety formats/levels of approximation:

- Concrete composition prescriptive requirements (LoA1);
- Rules by tests and measurements (Performance indicators) (LoA2);
- Partial factor method (Semi-probabilistic format) (LoA3);
- Full probabilistic (LoA4);
- Avoidance-of-damage approach.

A detailed description of these levels is given in section 14.3.

Concrete and reinforcement can be exposed to many aggressive environments and contaminants (section 14.8.1). Within section 30.6, the following deterioration mechanisms are addressed:

- Carbonation-induced corrosion;
- Chloride-induced corrosion;
- Freeze-thaw resistance;
- Chemical attacks (sulfate and magnesium ions, acids, microbiological, aggressive CO_2 and ammonium, (pure water), leaching by soft water, chlorides in groundwater and deicing salts, seawater, and food production and processing);
- Alkali-aggregates-reaction
- Delayed-ettringite formation;
- Abrasion, erosion, and cavitation.

In the following subsections, the aims of the Model Code related to the durability of concrete structures are presented.

Carbonation is considered to induce a general and homogeneous corrosion while the chlorides would start by producing localized attacks and pits.

Carbonation and corrosion rates optimum appear at very different, even contradictory, environmental conditions. In indoor conditions, around the optimum rate of carbonation, corrosion rate is expected to be very low nearly negligible. In outdoor sheltered conditions, the carbonation rate may be equal, higher, or lower than indoor rate, with a corrosion rate usually low unless the averaged RH outdoors is above 80%. While under outdoor unsheltered conditions, the carbonation rate is often lower compared to the carbonation rate under indoor condition, the corrosion rate after initiation may be low to very high depending on the degree of wetting. In arid conditions the corrosion rate will be very low such that corrosion damage is unlikely. In areas of intermittent rain, it may be very high because concrete absorbs water quickly but dries only slowly.

For deterioration mechanisms, other than reinforcement corrosion, concrete composition prescriptive rules, or rules by tests and measurements (durability indicators), are detailed in sections 30.6.4, 14.8.4, 30.6.5, 14.8.5, 30.6.6, 14.8.6, 30.6.7 and 14.8.7. In present *fib* MC2020 their modelling has been considered not always mature enough and neither fully calibrated to allow implementation either by full probability or partial safety format for Service Design Life assessment.

As far as steel reinforcement corrosion deterioration mechanism is considered, two approaches for durability design regarding carbonation and/or chlorides induced corrosion are proposed.

a) The design service life equals only the initiation of corrosion phase

$$t_{SL} = t_{ini}$$

t_{SL} is the period of service life in years

t_{ini} is the time of corrosion initiation in years

b) The design service life equals the initiation phase and a certain level of the corrosion phase (until a certain level of corrosion is reached)

$$t_{SL} = t_{ini} + t_{prop}$$

t_{prop} is the time of corrosion limited by a certain level of corrosion loss in years.

The design service life, the limit state and the accepted failure probability are decided by the owner and/or by local building legislation in some countries. Typical values are 25, 50, 75, 100 or 120 years.

Concrete structures service life can be verified by each of the different levels of approximation described in section 14.3.2 as experience of use of durability indicators and implementation of time-explicit models starts to be gathered worldwide. Models and relevant durability indicators are described in sections 30.6.2, 14.8.2, 30.6.3, 14.8.3.

30.6.1.2 FRP materials

A further discussion on the durability of FRP reinforcement and a design for durability format, is provided in *fib* Bulletin 40 [30.6-2]. The limit state associated with durability is integrated in the design by means of the following considerations:

– an environmental factor η_e is considered in the design value of the FRP tensile strength (section 17.8.1.1 + 17.8.1.2);

Direct contact of FRP reinforcement with steel reinforcement should be avoided, as this might enhance steel corrosion.

Though FRP materials have generally good durability performance and are not susceptible to rebar corrosion, their durability should still be considered with care in concrete (such as exposure to moisture and concrete alkalinity).

– the ultimate limit state of creep rupture considers the acting environmental conditions (section 17.8.1.2);

– bond degradation during service life is considered through a limiting bond stress, considering the acting environmental conditions (section 21.1.3).

Because corrosion induced by carbonation or chlorides does not occur for FRP reinforcement, a minimum concrete cover due to environmental conditions is not necessary.

In those situations, care should be taken to consider the exposure environment of the concrete element. The situation should be analysed case by case, although particular attention is recommended to how permeable to vapour gas or the moisture the external layers are, as the durability can be threatened if water condensates in the concrete surface or on its interior.

On the other hand, such multi-barrier systems could be an additional protection to prevent future concrete degradation, providing their durability is equal or longer than that of the concrete, or they are renovated periodically.

Carbonation implies the diffusive ingress of atmospheric gaseous carbon dioxide into the concrete air-filled pores, and its reaction with the reactive Calcium-bearing phases in the cement paste, which lowers the pH of the pore solution down to around 8 and makes the rebars vulnerable to corrosion. Further explanations are given in *fib* Bulletin ZZ [30.6-1].Using the limit state “carbonation front reaches the surface of the rebar” as proxy for limit state corrosion initiation might in some cases be very conservative (Angst et al. 2020 [30.6-3]).

30.6.1.3 Multibarrier systems

The provisions of the present chapter consider the concrete surfaces in direct contact with the exposure environment. However, there are many situations where the concrete is covered by different types of protective or finishing materials: paints, coatings, membranes, geotextiles, brick, plaster, etc. or the concrete is forming part of a multilayer or multi barrier system. That is, the element is a sandwich where concrete is only a layer covered, by one or several faces, by other materials.

There is no guidance (apart from the instructions of the suppliers) on how to proceed in these cases in relation to the durability of the concrete layer and no data have been found of general applicability.

30.6.2 Carbonation induced corrosion

30.6.2.1 General

The limit state for service life design of reinforced concrete structures related to carbonation-induced corrosion can be reached either when:

- The carbonation front reaches the surface of the rebar (proxy for limit state corrosion initiation, or
- Active corrosion has caused a certain corrosion attack / penetration level.

It is to be noted that the carbonation front is commonly ascribed to the depth of the colourless part given by a pH indicator, which is not always accurate as the carbonation front does not always progress in a sharp manner.

A corrosion penetration of 50 µm as average/indicative value of homogeneous corrosion has been considered as Condition limit state associated to durability (see section 11.3.2.3.7) with a low probability of cracking.

This CLS was adopted as basis for the durability provisions in EN 1992-1-1: 2023 [30.6-4].

The carbonation rate under constant conditions (see Eq (30.6-1)) can be written as a function of two concrete properties: CO_2 effective diffusion coefficient and the amount of carbonatable products in the binder. The carbonation rate k is also denoted "V_{CO2}" or "A".

According to Gehlen [30.6-5] and *fib* Bulletin 34 [30.6-6] the parameters k_e, k_c, and W(t) given in Eq. (30.6-5) can be expressed as:

$$k_e = \frac{\left(1 - RH_{real}^{fe}\right)^{ge}}{\left(1 - RH_{ref}^{fe}\right)^{ge}} \qquad (30.6\text{-}3)$$

$$k_c = \left(\frac{t_c}{7}\right)^{b_c} \qquad (30.6\text{-}4)$$

$$W(t) = \left(\frac{t_0}{t}\right)^{\frac{(p_{Sr}.\mathrm{ToW})^{bw}}{2}} \qquad (30.6\text{-}5)$$

The source of quantification and description of the above-mentioned equations and Eq. (30.6-5) are given in Table 30.6-2.

30.6.2.2 Physical models for carbonation-induced corrosion

30.6.2.2.1 Initiation period

The carbonation depth X_c (t) under constant conditions may be estimated by the following equation:

$$x_c(t) = k \cdot t^b \qquad (30.6\text{-}1)$$

where:

$x_c(t)$ is the carbonation depth at the time t in [mm];
k apparent natural carbonation rate [mm/year^b];
t time [year];
b the time exponent b ≤ 0.5 [-]; default is b = 0.5;

For b=0.5, equation 30.6.1 can be written also as follows:

$$x_c(t) = \sqrt{\frac{2D \cdot c}{a}} \cdot t^{0.5} \qquad (30.6\text{-}2)$$

c is the CO_2 concentration of the testing condition [$kgCO_2/m^3$];
D is the CO_2 effective diffusion coefficient [mm^2/year];
a is the amount of product available to react with CO_2 [$kgCO_2/m^3$].

An updated version of the basic square root model Eq. (30.6-1) is suggested in *fib* Bulletin 34 [30.6-6] to consider the wetting events through the introduction of the

Table 30.6-2: Description of model parameter.

	Description	Unit	Source for quantification
$C_{s,d}$	CO_2 concentration of the ambient air	[kg/m³]	Measurement Without pollution through industry or other sources:
$R_{ACC,0,}^{-1}$**	Inverse effective carbonation resistance of concrete	[mm²/year kg^{-1} m³]	Laboratory test and correction factor**
RH_{real}	relative humidity of the carbonated layer	[%]	Weather station
ToW	Time of Wetness	[d]	Weather station
p_{SR}	Probability of driving rain	[-]	Weather station
t_c	Period of curing	[d]	Standard or execution documents
RH_{ref}	reference relative humidity	[%]	Predefined*
t_0	time of reference	[years]	Predefined*
t	Time	[years]	Time variant up to the defined design service life
f_e	exponent (quantified through regression analysis)	[-]	Predefined*
g_e	regression exponent	[-]	Predefined*
b_c	regression exponent	[-]	Predefined*
k_t	regression parameter	[-]	Predefined*
ε_t	error term	[mm²/year kg^{-1} m³]	Predefined*
b_w	regression exponent	[-]	Predefined*

*See *fib* Bulletin ZZ [30.6-1] for guidance on predefined values.

**According to the recent work carried out in fib TG8.3, Chapter 5.1 [draft in progress] [30.6-7] and as a preliminary indication, it should be noted that for $R_{ACC,0}^{-1}$ and ε_t, a different time unit are used. To set the correct unit [(mm²/yr)/(kg/m³)] a conversion factor F must be introduced: F= 315.576 [(mm²/yr)/(10^{-11}m²/s)].

weather function.

$$x_c(t) = \sqrt{2 \cdot k_e \cdot k_c \cdot (k_t \cdot R_{ACC,O}^{-1} + \varepsilon_{t,d}) \cdot C_{s,d}} \cdot W(t) \cdot \sqrt{t} \qquad (30.6\text{-}6)$$

where:

k_e environmental function [-];
k_c execution transfer parameter [-];
k_t is a regression parameter [-];
$R^{-1}_{ACC,0}$ is the inverse carbonation resistance of concrete [mm².year^{-1}.kg^{-1}.m³];
$\varepsilon_{t,d}$ error term for inaccuracies which occur conditionally when using the accelerated test method in [mm².year^{-1}.kg^{-1}.m³];
$C_{s,d}$ is the CO_2 concentration of the ambient air [kg/m³];
$W(t)$ weather function [-].

In the structures, after depassivation and initiation of active corrosion, the propagation model can be divided into two parts (or more) with different corrosion rate (Figure 30.6-1 and *fib* Bulletin ZZ [30.6-1]).

This change in the slope of P_{corr} versus time can be produced due to cracking of the cover (in general parallel to the bar direction) due to the expansive character of the iron oxides or to the formation of semi-protective rust. This cracking may accelerate the corrosion progress if the structure is in continuous contact with water or can slow down the corrosion because in the cracked area drying occurs faster. A bi-linear or non-linear evolution of the P_{corr} with time can be modelled with a two-step propagation relation as shown in Figure 30.6-1:

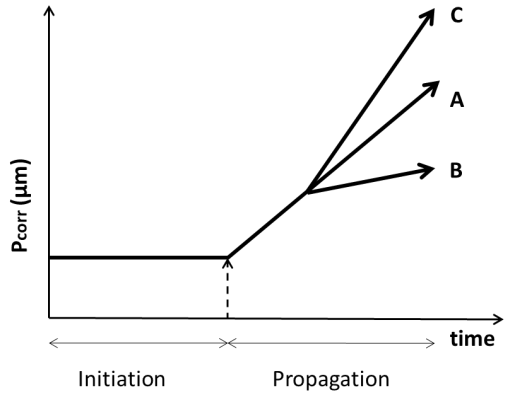


Figure 30.6-1. Linear (line A) and two-step (bi-linear) corrosion propagation models (line B with a lowering and line C with an increase with time).

30.6.2.2.2 Propagation period

The model of the propagation period is given in Eq. (30.6-7) taken from section 30.2.3.6:

$$t_{prop} = \frac{\phi_0 - \phi_t}{V_{corr}} = \frac{P_{corr}}{V_{corr}} \qquad (30.6\text{-}7)$$

where:

V_{corr} is the averaged value of the corrosion rate per year in the period considered in [µm/year];

P_{corr} is the penetration of corrosion attack (accumulated corrosion) during the period considered [µm];

ϕ_0 is the initial diameter [µm];

ϕ_t is the diameter at time t [µm].

The V_{corr} in [µm/year] can be expressed in electrical units as I_{corr} [µA/cm^2]. The equivalence between them is:

$$V_{corr} = 0.0116\, I_{corr} \qquad (30.6\text{-}8)$$

If the corrosion rate cannot be measured, the concrete electrical resistivity is an adequate alternative. The equivalence existing between the corrosion rate (in µm/year) and electrical resistivity is:

$$V_{corr} = \frac{K_{corr}}{\rho} = \frac{0.0116 \cdot 26}{\rho} = \frac{0.3016}{\rho} \qquad (30.6\text{-}9)$$

where:

ρ is the electrical resistivity in [Ω·m];

K_{corr} is the factor of equivalence between the corrosion rate and electrical resistivity.

As examples, the following indicators are specified as related to the carbonation initiation. Regarding limiting values given for the material properties in general we can state:

- Air or oxygen permeability;
- The diffusion coefficient when a limiting value is prescribed with no time-explicit-model. The test can be carried out under natural or accelerated conditions;
- The buffering capacity;
- Water sorptivity;
- Electrical Resistivity.

To date, there is no international agreement on threshold values for these durability indicators to deduce the performance of the material in relation to the cover depth or the target reliability.

30.6.2.3 Concrete composition prescriptive rules

The concrete prescriptions rules for carbonation induced corrosion are usually related to the concrete cover and the specific concrete requirements for the relevant exposure condition. These requirements are related to:

- Provisions on concrete material properties (see section 14.8.2.3)
- The minimum concrete cover to reinforcement and a permitted allowance for deviations,
- The calculated maximum crack width.
-

30.6.2.4 Rules by tests and measurements (Performance indicators)

30.6.2.4.1 Performance indicators considering the initiation period

Among the indicators of concrete performance that may be mentioned include:

- The carbonation rate.

- The total water porosity.

- pH level (using pH indicators such as phenolphthalein or thymolphthalein…).

With respect to the limiting values of these indicators:

- The carbonation rate threshold value depends on the cover depth and can be calculated from Eq. (30.6-1) if the carbonation depth is measured at some specified time, the carbonation depth is evaluated according to tests described in section 14.8.2.4.1;
- Regarding the water porosity, there are several classifications following national standards;
- The pH indicator limiting value is used in the determination of the carbonation rate.

All these indicators are included in Rilem Recommendations 2003 [30.6-8], Rilem Recommendations 2004 [30.6-9], and CPC11.2 [30.6-10]) as shown in Tables 30.6-3 and 30.6-4 and 30.6-5. Further details are provided in *fib* Bulletin ZZ [30.6-1].

Table 30.6-3: Typical ranges of potentials of normal steel in concrete (Volts CSE) Rilem recommendations (2003) [30.6-8].

water saturated concrete without oxygen	-0.9 to -1.0 V
wet, chloride contaminated concrete	-0.4 to -0.6 V
humid, chloride free concrete	+0.1 to -0.2 V
humid, carbonated concrete	+0.1 to -0.4 V
dry, carbonated, or non-carbonated concrete	+0.2 to 0 V

Note that potential ranges can be altered for some binder types such as slag.

Table 30.6-4: Typical ranges of current density (I_{corr}) and corrosion rate (V_{corr}). Rilem recommendations (2004) [30.6-9].

Corrosion level	I_{corr}(µA/cm²)	V_{corr} (µm/year)	Condition
Negligeable	< 0,1	< 1	Passive state
Low	0,1 – 0,5	1 – 5	Depassivated in low humidity concrete
Moderate	0,5 – 1	5 – 10	Active corrosion in medium level humidity
High	> 1	>10	Large areas corroding in wet concrete

Table 30.6-5: Typical ranges of electrical resistivity values given to describe the corrosion level. Rilem recommendations TC-116-PCD [30.6-8]+[30.6-9].

Resistivity	Corrosion Level
>1000 – 2000 Ω.m	The corrosion rates will be low. The concrete is dry
>500 – 1000 Ω.m	The concrete is moderately dry. Low corrosion rate
>100 – 500 Ω.m	Concrete is wet. Corrosion rate will be moderate
<100 Ω.m	Wet concrete of high porosity. High corrosion rate

30.6.2.4.2 Performance indicators considering the propagation period

An indicator of the corrosion risk is the electrical potential. Resistivity, and corrosion rate are indicators of the velocity of the corrosion propagation. It is recommended to measure all the beforementioned indicators.

Note that the electrical potential is a more qualitative test method compared with electrical resistivity and corrosion rate.

30.6.2.5 Semi-probabilistic format

30.6.2.5.1 Initiation period

Safety factors for the carbonation depth have been studied in *fib* Bulletin 34 [30.6-6], DAfStb 622 [30.6-11], DAfStb 623 [30.6-12], LNEC E 465 [30.6-13], Izquierdo [30.6-14]. Partial safety factors determined with the different approaches are summarized in the second column of Table 30.6-6.

Note that this table provides the partial safety factors obtained for the indicated reliability target value in column 6 including a propagation period (t_{ini} of Eq. 30.6-10 is then replaced by t_{SL}) as well in some cases the one resulting on a penetration corrosion attack of 50 µm.

Table 30.6-6: Partial safety factors.

Safety margin for concrete cover [mm]	γ_f [-]	γ_{RH} [-]	γ_{RACC} [-]	γ_{corr} [-]	β_{ini} and $\beta_{ini+corr}$ [-]	source
$\Delta a = 1.64 \cdot c_\sigma$ Usually, Δa = 10 mm	1.3	-	-	–[1)]	≈1.5	DAfStb 622/623 [30.6-11]+[30.6-12]
$\Delta a \geq 5$	-	1.3	1.5	-[1)]	≈1.7	*fib* Bul. 34 [30.6-6]
$\Delta a \geq 0$	2.3	-	-	1.0	≈.5	LNEC E 465 [30.6-13]
$\Delta a \geq 0$	2.0	-	-	1.0	≈1.2	LNEC E 465 [30.6-13]

c_σ = standard deviation of concrete cover

β_{ini} : reliability index for the initiation period

$\beta_{ini+corr}$: reliability index for the initiation and corrosion propagation period

[1)]see Eq. 30.6-16 or γ_{corr} = 2.0.

In the case of using the carbonation model as presented in Eq. (30.6-5), the following equations give the expressions of the design values of the parameters $k_{e,d}$ and $R_{ACC,d}^{-1}$:

$$k_{e,d} = \frac{\left(1 - \frac{RH_{real}^{fe}}{\gamma_{RH}}\right)^{ge}}{\left(1 - RH_{ref}^{fe}\right)^{ge}} \qquad (30.6\text{-}12)$$

$$R_{ACC,d}^{-1} = R_{ACC,O}^{-1} \cdot \gamma_{RACC} \qquad (30.6\text{-}13)$$

Where the corresponding partial safety factors are listed in Table 30.6-6, columns 3 and 4.

The following inequality is considered to satisfy the limit state of depassivation by carbonation:

$$c_d \geq \gamma_f \cdot x_{c,k}(t_{ini}) = \gamma_f \cdot k_k \cdot t_{ini}^{0.5} \qquad (30.6\text{-}10)$$

where:

c_d is the design value of the concrete cover [mm];
γ_f is the safety factor of the carbonation depth [-];
$x_{c,k}$ is the characteristic value of the carbonation depth [mm];
t_{ini} is the time to initiation [year], e.g. t_{SL} - $t_{prop;}$
k_k characteristic value of the apparent natural carbonation rate [mm/ year$^{0.5}$].

The design value of the concrete cover, representing the uncertainty in the geometry is:

$$c_d = c_{nom} - \Delta a \qquad (30.6\text{-}11)$$

where:

c_{nom} is the nominal concrete cover [mm];
Δa is a safety margin of the concrete cover [mm];
c_d is the design value of concrete cover.

Corrosion propagation follows carbonation under wet-dry events. Under this exposure condition, when the steel is depassivated, the corrosion rate may exhibit a CoV up to 200%, while it is typically around 100% (usually under a lognormal distribution).

For the design of new structures partial safety factors for the corrosion propagation are suggested in Table 30.6-6 and in Eq. (30.6-14) (a value of 1.5 is considered there).

To date, there are not still developed partial factors for the concrete resistivity.

On the other hand, in particular cases it is possible to follow prEN1990:2020 [30.6-15], for a log-normal distributed variable with large coefficient of variation CoV (>25%) and the FORM sensitivity factor α, the equation to calculate the partial factor $\gamma_{d,corr}$ for the propagation period is:

$$\gamma_{d,corr} = \frac{V_{corr,k}}{V_{corr,\mu}} e^{\left(\frac{1}{2}ln\left(1+CoV_\gamma^2\right)-\alpha_\gamma \cdot \beta_\gamma \cdot \sqrt{ln\left(1+CoV_\gamma^2\right)}\right)} \quad (30.6\text{-}15)$$

where:

$V_{corr,\mu}$	is the average value of the corrosion rate in the particular exposure class;
$V_{corr,k}$	is the characteristic value of the corrosion rate in the particular exposure class;
α	FORM sensitivity factor for the V_{vorr};
β	reliability index (the target value should be agreed on with the stakeholder);
CoV	coefficient of variation.

In the case of lacking data, the following values can be applied into the Eq. (30.6-16).

Table 30.6-7: Sensitivity factor and CoV applicable to the V_{corr}.

Sensitivity	Coefficient of Variation
α	CoV
0.7	100-150%

30.6.2.5.2 Propagation period

The corrosion propagation model is shown in Eqs. (30.6-7) and (30.6-8). The expression for the design value of the propagation period assuming a single constant corrosion rate from the initiation of the corrosion, is:

$$t_{prop} = \frac{1}{\gamma_{corr}} \frac{P_{corr}}{V_{corr,k}} \quad (30.6\text{-}14)$$

where:

P_{corr}	design value of the penetration of corrosion attack expected until a certain limit state is reached, for example cracking or spalling, reduction of bond strength [mm];
$V_{corr,k}$	characteristic value of corrosion rate in the particular exposure class [µm/year];
γ_{corr}	partial safety factor for the corrosion rate [-];
t_{prop}	the design value of the propagation time [year].

In the full probabilistic format, the probability p that the carbonation depth is larger than the cover depth is calculated. This probability should be smaller than the target probability p_0 or the corresponding reliability index higher than the target reliability β_0.

Regarding the limit state of solely depassivation of steel reinforcement, the literature provides a wide range of suggested target β_0-values. A value of $\beta_0 = 1.3$ is often used assuming that there will be no major repairs required during the design life, although this will also depend on the actual structure and reinforcement detailing.

In contrast to verification of ultimate and serviceability limit states in most literature only one reliability class is considered for durability. However, depending on the consequences (economic, societal and/or environmental), a differentiation in reliability classes may be considered.

The use of $\beta_0 = 1.5$ for the depassivation period in combination with a well-specified part of the subsequent propagation period is assumed to result into a broadly similar safety level than considering $\beta_0 = 1.3$ with depassivation only.

For carbonation induced corrosion, due to the homogenous corrosion and more excessive and voluminous amount of rust formed, usually cracking or spalling is the limit state considered for the service life/condition limit states associated to durability. Ranges of P_{corr} leading to cracking depend among others on the ratio between rebar diameter and concrete cover thickness and are $22 < P_{corr} < 200$ µm (see Bulletin ZZ [30.6-1]).

30.6.2.6 Full probabilistic format

30.6.2.6.1 Initiation period

In a full probabilistic approach, with respect to depassivation of reinforcing steel by carbonation, to avoid corrosion initiation with a defined probability, Eq. (30.6-16) shall be fulfilled:

$$\mathrm{P}\{\} = \mathrm{P}_{dep} = \mathrm{P}\left\{c - x_c\left(t_{ini}\right) < 0\right\} < \mathrm{P}_0 \qquad (30.6\text{-}16)$$

where:

$\mathrm{P}\{\}$	is the probability that depassivation occurs [-];
c	is the concrete cover [mm];
$x_c\,(t_{ini})$	is the carbonation depth at the time t_{ini} [mm];
t_{ini}	is the time of corrosion initiation [years];
P_0	is the target failure probability [-].

The models for the carbonation depth introduced in section 30.6.2.2 are used for the probabilistic design. The model parameters are replaced by random variables. Random variables are described by distribution functions and distribution parameters. They are given section 14.8.2

30.6.2.6.2 Propagation period

For the propagation period, to avoid the corrosion cracking with a defined probability, Eq. (30.6-16) related to initiation period is fulfilled, or the following Eq. (30.6-17) should be fulfilled:

$$\mathrm{p}\{\} = \mathrm{p}\left\{t_{SL} - t_{ini} - t_{prop} \leq 0\right\} < \mathrm{P}_0 \qquad (30.6\text{-}17)$$

where:

$\mathrm{p}\{\}$	is the probability that carbonation-induced cracking occurs;
t_{SL}	is the design service life [years];
t_{ini}	is the time of corrosion initiation[years];

Statistical distributions of average corrosion rates in dependency of exposure classes are presented in *fib* Bulletin ZZ [30.6-1].

In existing structures suffering corrosion, the value of P_{corr} would depend on the structural consequences related to the SLS and ULS.

When studying carbonation induced corrosion both the initiation and the propagation periods should be treated probabilistically.

For each method adopted to delay the corrosion induced carbonation, it is recommended to refer to the relevant standard.

These methods delay the corrosion initiation although in different degrees and depending on the CO_2 concentration and relative humidity. Not all the methods give the same extension in service life and would depend on the product. Designers are addressed for more specialized literature.

More details on the chloride ingress mechanisms are given in *fib* Bulletin ZZ [30.6-1].

t_{prop} is the propagation period [years] to reach a limit corrosion depth (P_{corr}), equal to 50µm;

P_0 is the target failure probability.

30.6.2.7 Avoidance of corrosion initiation by carbonation

Carbonation is a natural process that cannot be completely avoided. However, there are some procedures that can reduce the carbonation kinetics:

- Coatings with a high CO_2 diffusion resistance
- Increase the resistance of the steel against depassivation in neutral environment (stainless steel, galvanized steel)
- Concrete is so impermeable to gaseous CO_2 that carbonation don't start
- Corrosion inhibitors added in the mix or applied in existing structures
- Concrete coatings in new non-contaminated structures,
- Applying Cathodic Protection.

The extension of the service life by the chosen method has to be verified by the designer, theoretically or by experiment.

30.6.3 Chloride-induced corrosion

30.6.3.1 General

Chloride ions can penetrate through the concrete cover by different mechanisms among which diffusion is the most predominantly considered. The chlorides are depassivating ions (corrosion promoters) regarding the steel.

The limit state for service life design of reinforced concrete structures related to chloride induced corrosion can be indicated either when:

Eq. (30.6-18) is based of Fick's second law of diffusion.

Refer to *fib* Bulletin ZZ [30.6-1].

$C_{s,\Delta x}$ act as a function of the available chloride source which is taken to be a constant action. For the case with $\Delta x = 0$, $Cs_{,0}$ is the chloride concentration at the surface of the member at the time of observation as a function of the available chloride source which is taken to be a constant action (surface chloride concentration).

Detailed explanation of the model parameters and recommendations to their statistical distribution have been depicted in *fib* Bulletin 34 [30.6-5]and *fib* Bulletin 76 [30.6-16].

The apparent chloride diffusion coefficient $D_{app}(t)$ which represents the material resistance has been dealt with in section 14.8.3.6.1.1.

- A critical chloride content reaches the surface of the rebar (initiation period of corrosion), or
- The rebar has lost a certain cross section by corrosion (propagation period of corrosion).

Guidance for selecting limit state and target reliability level for service life design is given in chapter 11.

30.6.3.2 Physical models for chloride-induced corrosion

30.6.3.2.1 Initiation period

The time- and depth-dependent chloride concentration in concrete can be calculated by:

$$C(x,t) = C_0 + \left(C_{S,\Delta x} - C_0\right) \cdot erfc \frac{x - \Delta x}{2 \cdot \sqrt{D_{app}(t) \cdot t}} \qquad (30.6\text{-}18)$$

where:

$C(x, t)$ is the chloride concentration at depth x at time t [% weight of binder];

C_0 is the initial chloride content of the concrete [% weight of binder];

$C_{s,\Delta x}$ is the chloride concentration at depth Δx (equivalent surface chloride concentration) [% weight of binder].

Δx skin zone; is the depth range in which the chloride penetration behaviour may deviate from the behaviour according to Fick's law of diffusion owing to intermittent chloride attack [m];

x is the depth with a corresponding chloride content $C(x, t)$ [m];

t is the concrete exposure duration [s];

$erfc$ is the complementary Gauss error function (= 1-erf);

$D_{app}(t)$ is the apparent chloride diffusion coefficient of the concrete [m²/s]. For each t, the variable is assumed constant over entire exposure period.

30.6.3.2.2 Propagation period

The consideration of the propagation period in the design service life and the associated model can be found in section 30.6.2.2.2.

If the corrosion rate cannot be measured, the concrete electrical resistivity is an adequate alternative. The equivalence existing between the corrosion rate and electrical resistivity is provided in Eq. (30.6-8).

30.6.3.3 Prescriptive rules on concrete composition and construction aspects

In prescriptive specifications of codes, adequate durability with respect to chloride-induced corrosion is deemed satisfied by ensuring compliance with limiting values for all or part of parameters related to the concrete compositions, which are presented in section 14.8.3.3.1, and to the construction aspects as follows:

- The minimum concrete cover to reinforcement and a permitted allowance for deviations,
- Maximum calculated crack width.

These provisions are given depending on design service life and exposure conditions.

The prescriptive requirements are found in standards where the code committee is the responsible for the calibration of the method and establishing the limiting values for the parameters depending on the exposure situation.

For most standards it is not explicitly stated if the prescriptive limiting rules for concrete consider only the initiation period of reinforcement corrosion by controlling the chloride penetration resistance of the concrete or also include parts of the corrosion propagation. It is not clear what level of reliability is assumed for reaching the design service life.

The nominal value for the concrete cover is the dimension given to the constructor in the project specification (i.e. on drawings) and is assumed to represent the mean value of the cover depth. The safety margin Δc_{dev} (allowance for deviation) ensures that the great majority (in operational standards often assumed as the 95% fractile) of the cover thickness for the reinforcement bars are larger than the minimum cover used as a basis for the service life design. ISO 22966 [30.6-17] assumes Δc_{dev} = 10 mm if no other values are given in the execution specifications. Further details are provided in *fib* Bulletin ZZ [30.6-1].

Currently, there is a lack of precise definition and feedback for some durability indicators. At present, there is no well-known correlation between the durability indicators, the concrete cover, and the exposure conditions for chloride-induced corrosion in order to use the durability indicators to assess the durability of the structures reasonably. Designers should clarify the meaning of the prescribed indicator and whether specified limit values are considered as design or characteristic values.

For the compressive strength, its meaning is related to the structural calculations and the specified limits are typically characteristic values as partial safety factors for design are considered by the codes.

If the indicators selected are being used as controlling parameter of time-explicit model, then the LoA should be that of probabilistic or partial factor approaches.

fib Bulletin ZZ [30.6-1] provides examples for the models' inputs statistical distributions and partial factors calibrated based on a specific database, target reliability and exposure situation. These values cannot be used as universally valid.

30.6.3.4 Rules by tests and measurements (Performance indicators)

30.6.3.4.1 Performance indicators considering the initiation period

Several concrete characteristics or properties can be assessed as indicators of the concrete resistance to chloride penetration (see section 14.8.3.4.1). When specifying them, their definition and scope should clearly be stated.

The value of the indicators shall be linked to the exposure class (see section 14.5.6.2) related to exposure conditions of the structure, the cover depth, and the target reliability.

30.6.3.4.2 Performance indicators considering the propagation period

The indicators of the corrosion propagation risk and velocity are detailed in section 30.6.2.4.2.

30.6.3.5 Semi-probabilistic format

30.6.3.5.1 Initiation period

The following two limit state equations can be considered:

$$C_{Crit,d} - C_d\left(c_d, t_{SL}\right) \geq 0 \qquad (30.6\text{-}19)$$

$$c_d - x_{crit,d}(t_{SL}) \geq 0 \qquad (30.6\text{-}20)$$

where:

$C_{crit,d}$	is the design value of the critical chloride content [%] by weight of binder];
$C_d(c_d, t_{SL})$	is the design chloride content at the depth of the minimum concrete cover at time t_{SL} [% by weight of binder];
c_d	is the design concrete cover; corresponds to the minimum concrete cover c_{min} [m];
$x_{crit,d}(t_{SL})$	is the design depth of the critical chloride content at time t_{SL} [m];
t_{SL}	is the design service life [year].

The same transport model used in the fully probabilistic format, Eq. (30.6-18), is employed in the semi-probabilistic design format.

30.6.3.5.2 Propagation period

See section 30.6.2.6.2.

Corrosion propagation follows depassivation when the corrosion is active. Under wet-dry exposure conditions, when the steel is depassivated, the corrosion rate may exhibit a CoV up to 400%, while it is typically around 100%. In submerged and tidal exposures, the variation can be much smaller and 50% can be a reasonable CoV.

As corrosion propagation is very rapid in tidal and submerged conditions (when corrosion is active) no partial safety factors are suggested for those exposures. To date, there are not still developed partial factors for the concrete resistivity.

30.6.3.6 Full probabilistic format

30.6.3.6.1 Initiation period

In full-probabilistic design format, statistically distributed values of the model variables are used instead of design values like for the semi-probabilistic approach.

To avoid depassivation with some probability during the design service life, the following inequality should be:

$$P\{\} = P_{dep} = P\left\{t_{SL} - t_{ini}\left(c, t_{SL}\right) < 0\right\} < P_0 \qquad (30.6\text{-}22)$$

where :

t_{ini} is the initiation period [years].

This general equation describing the limit state of depassivation has been developed by RILEM, see RILEM Technical Committee 130-CSL [30.6-18].

This rather conservative limit state is therefore normally linked to a corresponding relaxed target reliability level for failing, often in the order of 10^{-1} to 10^{-2}. See also section 11.3.2.3.6.

The limit state for chloride-induced corrosion of the reinforcement is the point at which a critical chloride content is reached at the surface of the rebars.

The critical chloride content (see also section 14.8.3.6.1.2) can be expressed as:

- "acid soluble or total or overall" chloride content that are the chlorides that are the sum of the chlorides in the pore solution and those that are bound with the cement phases and
- "water soluble" or "free" chloride content are those of the pore solution that are the only ones being able to induce corrosion.

In the Codes the limiting values are expressed as total chloride content or as soluble chlorides. The relation between free and total chlorides is not constant as it depends on the concentration itself, the type of cement, potential phase changes and pore solution composition, and of the temperature and humidity as well as on the pH (concomitant carbonation). In section 14.8.3.6.1.2 the critical chloride content has been considered as overall chloride content.

The use of $\beta_0 = 1.5$ for the depassivation period in combination with a well-specified part of the subsequent propagation period is assumed to result into a broadly similar safety level than considering $\beta_0 = 1.3$ with depassivation only.

An exception can be made in the atmospheric zones (depending on the environmental conditions of the exposure site), where the concrete is not fully saturated

The following inequality shall be fulfilled:

$$P\{\} = P_{dep} = P\left\{C_{Crit} - C\left(c, t_{SL}\right) < 0\right\} < P_0 \qquad (30.6\text{-}23)$$

where:

$P\{\}$ is the probability that depassivation occurs;
C_{Crit} is the critical chloride content to achieve depassivation of the reinforcement [wt.-%/binder content];
$C(c, t_{SL})$ is the chloride content at depth c and time t_{SL} [wt.-%/binder content];
c is the concrete cover [mm];
t_{SL} is the design service life [years];
P_0 is the target failure probability.

The critical chloride content or chloride threshold is the chloride content which causes depassivation of the surface of the rebars.

The limit state equation can also be described as follows:

$$P\{\} = P_{dep} = P\left\{c - x_{crit}\left(t_{SL}\right) < 0\right\} < P_0 \qquad (30.6\text{-}24)$$

where:

$x_{crit}(t_{SL})$ is the depth of the critical chloride content at time t_{SL} [m].

The variables c, C_{crit}, $C(c, t_{SL})$ and $x_{crit}(t_{SL})$ shall be quantified in a probabilistic approach (cf. *fib* Bulletin 34 [30.6-6] and 76 [30.6-16]).

30.6.3.6.2 Propagation period

Due to the much higher corrosion rates `produced in the presence of chlorides, the consideration of a propagation period can be neglected.

and for which the duration of a propagation period until reaching a pit depth of P_{pit} = 500 µm, indicating the limit state of durability, is important (See section 30.6.2.6.2).

For each method adopted to delay the chlorides induced corrosion, it is recommended to refer to the relevant standard.

These methods delay the corrosion initiation although in different degrees and depending on the chloride concentration and relative humidity. Not all the methods give the same extension in service life and would depend on the product. Designers are addressed for more specialized literature.

30.6.3.7 Avoidance

Harmful chloride penetration and/or corrosion initiation can be avoided by a proper selection of measures as listed:

- Coatings with a high Chloride diffusion resistance
- Increase the resistance of the steel against depassivation, i.e. Increase of chloride threshold (stainless steel, galvanized steel)
- Concrete is so impermeable to chlorides that they do not penetrate (ultra-high-performance concrete)
- Corrosion inhibitors added in the mix or applied in existing structures
- Applying Cathodic Protection.

30.6.4 Freeze-thaw resistance

30.6.4.1 General

The exposure of concrete structural components to sub-zero temperatures in service could result in internal cracking and scaling and thus in a loss of strength due to moisture transport and the expansion of water on freezing. In addition, freeze/thaw action also increases the sorptivity of the concrete, causing increased moisture ingress. The deterioration of concrete caused by freeze-thaw attack with de-icing agents is related to complex processes associated with physical and chemical changes in the pore solution, binder paste matrix and aggregate. It results in scaling, i.e. external damage.

At present, no validated time-dependent model exists for the calculation of the resistance of a given concrete in a structural component to the action of freeze-thaw cycles or freeze-thaw cycles combined with de-icing agents.

Current design aims at avoiding damage by the specification of concrete composition for a particular service environment and standard testing methods characterizing freeze-thaw and freeze-thaw de-icing salt action.

The concrete composition prescriptive approach is based on calibration to long-term experience. It considers:

- Limitations to the porosity of the concrete (maximum water/binder-ratio, minimum cement content, and required curing conditions, where the later are important for especially the porosity of the exposed surface). Alternatively, it can be expressed as a minimum strength requirement.
- Limitations related to the binder composition, e.g. cement type and the use of additions like silica fume, fly ash, and slag.
- Limitations on types of aggregates (on their own should be frost resistant)
- In some national provisions, requirements related to the air void structure, i.e. total air content and spacing factor.
- In some national provisions, adequate performance can be demonstrated by accelerated freeze-thaw testing, and this documentation can partly or fully replace the above-mentioned limitations/requirements.

Note that the deemed-to-satisfy requirements can be categorized as being both deemed-to-satisfy and avoidance-of-deterioration when designing new structures.

30.6.4.2 Limit state of freeze-thaw resistance without de-icing agents/sea water

30.6.4.2.1 Introduction

This limit state covers freeze-thaw damage causing loss of mechanical properties, cracking, scaling, and loss of cross section without de-icing agents or sea water.

30.6.4.2.2 Concrete composition prescriptive rules

For structures exposed to freezing and with the potential for a higher level of moisture than the critical degree of saturation, a design according to the prescriptive rules method is the normal procedure. This is typically based on a set of requirements on the concrete composition (depending on relevant local standards):

- Cement type and maximum water/binder-ratio.
- Minimum strength and/or minimum cement content.
- Freeze-thaw resistant aggregates.
- Air entrainment.

Examples of performance indicators for freeze-thaw are:

- The relative dynamic modulus (RDM). A RDM = 1 (100%) indicates that no damage has occurred after being subject to *n* accelerated freeze-thaw cycles. A RDM < 1 indicates some degree of internal cracking. Acceptable values are typically provided in national guidance documents.
- The expansion of concrete specimens after a given number of freeze-thaw cycles, with a reference to a threshold provided in national guidance documents. (e.g. $\Delta\varepsilon$ = 400 µm/m is given in France).

The air void spacing factor, with a reference to a threshold provided in national guidance documents. (e.g. 250 µm is given in some European countries).

The limit state function presented remains conceptual, to illustrate how freeze-thaw could be addressed. It is based on the premise that concrete will not be damaged by frost action if the degree of saturation S_{ACT} remains below a critical degree of saturation S_{CR}.

Such an approach should also consider the time/weather dependant variation of the degree of saturation, temperature history of the concrete, as saturation without freezing will not result in damage.

This approach is not used in practice with the preferred method being deem-to-satisfy with or without additional performance indicators.

An example of a design procedure is given in Annex B of *fib* Bulletin 34 [30.6-6]. The approach described is mostly theoretical/academic, and not used in common practice.

More details on the mechanism of damage can be found from (Fagerlund [30.6-19]; Fagerlund 1997 [30.6-20]; Penttala [30.6-21], Pigeon et al. [30.6-22]; Scherer & Valenza [30.6-23]).

30.6.4.2.3 Rules by tests and measurements (performance indicators)

Concrete mix designs can also be accepted subject to positive result obtained in direct performance tests, in which concrete specimens are subject to accelerates freeze-thaw cycles (in a variety of different configurations). Examples of performance tests are given in section 14.8.4.

30.6.4.2.4 Probabilistic format

The following conceptual limit state function should be fulfilled:

$$P\{\} = P_{FT,I} = P\left\{T\left(t < t_{SL}\right), S_{CR} - S_{ACT}\left(t < t_{SL}\right) < 0\right\} < P_0 \qquad (30.6\text{-}25)$$

where:

$P\{\}$ is the probability that freeze-thaw induced internal structural damage occurs;

S_{CR} is critical degree of saturation [-];

$S_{ACT}(t)$ is actual degree of saturation at time t [-];

$P\{\}$ is the probability that scaling occurs;

$T(t)$ is concrete temperature in [K];

t_{SL} is the design service life [years];

$P0$ is target failure probability.

Other methods may be used, provided that the basic principles formulated in chapters 11 and 12 are fulfilled.

Concrete mix designs can also be accepted if positive results are obtained in direct performance tests, in which concrete specimens are subject to accelerates freeze-thaw cycles (in a variety of different configurations). Examples of performance tests are given in section 14.8.4.

30.6.4.2.5 Avoidance-of-deterioration approach

Generally, the avoidance-of-deterioration method implies that deterioration process will not take place due to the separation of the environmental actions from the structure or component, such as:

– Avoiding freezing temperatures,

– Avoiding ingress of external sources of moisture

30.6.4.3 Limit state of freeze-thaw resistance with de-icing agents/sea water

30.6.4.3.1 Introduction

This limit state cover freeze-thaw damage causing scaling and loss of cross section with de-icing agents or sea water.

30.6.4.3.2 Concrete composition prescriptive rules

Section 30.6.4.2.2 applies here.

30.6.4.3.3 Rules by tests and measurements (Performance indicators)

A common performance indicator for freeze-thaw surface scaling is the cumulative scaled material per square metre of concrete for a given number of accelerated freeze-thaw cycles.

Acceptable values are typically provided in national guidance documents.

The limit state function presented remains conceptual, to illustrate how freeze-thaw damage might be used to assess the service life.

It is assumed that the critical level of scaling resistance changes with age, depending on exposure and type of concrete.

An example of a design procedure is given in (Lay & Schießl 2003 [30.6-24]).

More details on the mechanism of damage can be found from (Bager & Jacobsen [30.6-25]; Kang et al. [30.6-26]; Linmark [30.6-27]; Setzer [30.6-28]; Valenza & Scherer [30.6-29]).

See comment for section 30.6.4.2.2.

At the time of publishing Model Code 2020, probability safety format models, and partial safety factors models are still under discussion. These approaches for service life design (SLD) are therefore not feasible. Although a number of models have attempted to describe the deterioration due to sulfate attack for example, the complex nature of

30.6.4.3.4 Probabilistic safety format

The following limit state function should be fulfilled:

$$P\{\} = P_{FT,S} = P\left\{T\left(t < t_{SL}\right), S_{C,CR} - S_{C,ACT}\left(t < t_{SL}\right) < 0\right\} < P_0 \qquad (30.6\text{-}26)$$

where:

$P\{\}$	is the probability that freeze-thaw induced surface scaling occurs;
$S_{C,CR}$	is the critical level of scaling[kg/m^2];
$S_{C,ACT}(t)$	is the actual degree of scaling at time t [kg/m2];
$P\{\}$	is the probability that scaling occurs;
$T(t)$	is concrete temperature in [K];
t_{SL}	is the design service life [years];
$P0$	is target failure probability.

Other methods may be used, provided that the basic principles formulated in chapters 11 and 12 are fulfilled.

30.6.4.3.5 Avoidance-of-deterioration approach

Section 30.6.4.2.5 applies here.

30.6.5 Chemical attack

30.6.5.1 General

The following classification of aggressive species is used to describe chemical attacks:

- Sulfate and magnesium ions;

the mechanism, which involves both precipitations, cracking and dissolution, makes it difficult to adopt any one approach to the modelling of deterioration of concrete in sulfate environments.

The attack in the presence of Mg ion is particularly severe, because of the resultant depletion of $Ca(OH)_2$ from the surface zone, which leads to C-S-H decalcification and eventually the formation of a hydrous magnesium silicate (M-S-H), which is non-cementitious. Further details are given in *fib* Bulletin ZZ [30.6-1].

Acid attack on concrete is brought about by hydrogen ions in water, which convert compounds of calcium in the cement to a salt of the type of acid present. The attacking medium may be classified as a function of the solubility of the acid's salts (calcium salt, mainly), and the type of acid (strong acids or weak acids, including organic types). Further details are given in *fib* Bulletin ZZ [30.6-1].

- Acids: (i) inorganic acids, (ii) organic acids. The latter may be naturally present in soil, in stored materials in contact with concrete (for example silage), or from food production, biogas production, and farming;
- Sewage and wastewater;
- Aggressive CO_2;
- Aggressive ammonium;
- Soft water;
- Chlorides in groundwater;
- Seawater;
- Food production and processing.

30.6.5.2 Description of mechanisms and available physical models

30.6.5.2.1 Sulfate and magnesium ion attack

The attack by sulfates leads to the formation of products such as gypsum and ettringite, which initially fill the porosity near the surface of the concrete, but after a point, start exerting expansive stresses – that lead to cracking of the surface zone.

30.6.5.2.2 Inorganic and organic acid attack

The degree of degradation of concrete caused by acid attack is defined by a deterioration depth with respect to the original surface. It comprises the depth of material removed by abrasion and/or crystallization pressure and the depth of corroded material remaining on the concrete surface.

In structures containing or transporting aged sewage, slimes may develop on surfaces below the liquid level, which decompose organic material in the sewage with the production of hydrogen sulphide gas. Further details are given in *fib* Bulletin ZZ [30.6-1].

Special attention may need to be paid to sites contaminated by industry or agriculture, in particular ammonium ions originating from fertiliser or refuse. Further details are given in *fib* Bulletin ZZ [30.6-1].

Leaching is a process that progresses as a front from the external concrete surface. It consists of a diffusive process leading to the release of hydroxides and calcium ions from the alkaline pore solution and consequent dissolution of hydrated cement phases. Further details are given in *fib* Bulletin ZZ [30.6-1].

Detailed information related to attacks by chlorides in groundwater and deicing salts are given in *fib* Bulletin ZZ [30.6-1].

30.6.5.2.3 Microbiological induced deterioration in sewage and wastewater

Microbiological induced deterioration can lead to very low pH-values. Very strong action can be expected where the H_2S can escape from the water, for example at the end of pressure pipes.

30.6.5.2.4 Aggressive CO2 and Ammonium attack

Ammonium salts can be highly aggressive to concrete, but damaging concentrations in soil are rarely found (BRE Special Digest 1 [30.6-30]). The effects of normal concentrations of ammonium in soil have not been known to cause significant damage to concrete.

30.6.5.2.5 Soft water attack and leaching

As a result of dissolving the cement paste component by soft water attack and/or leaching, the mechanical performances of the concrete are altered, and the risk of reinforcement corrosion increases.

30.6.5.2.6 Attack by chlorides in groundwater and deicing salts

There are no specific models for the alteration of the cementitious matrix by chlorides in groundwater. Chloride diffusion can be modelled using the solution to Fick's Second Law for a time-varying diffusion coefficient, as is applied for marine exposure (section 30.6.3).

Concrete exposed to seawater can be subject to various chemical and physical actions (Neville 2011): chemical attacks, salt weathering, and abrasion by leaching. Sulfate and magnesium ions are present in significant concentration in sea water and have a bearing on chemical processes which occur when the concrete is exposed. Further details are given in *fib* Bulletin ZZ [30.6-1].

Detailed information related to food production and processing are given *fib* Bulletin ZZ [30.6-1].

The minimum requirements for the parameters concerned depend on the environmental actions. The intensity of the actions is defined by classifying the environment according to the type and severity of exposure. The relevant exposure classes from ISO 22965-1 [30.6-31] are XA which are split into three sub-groups depending on concentrations of harmful species. Further details are given in *fib* Bulletin ZZ [30.6-1].

30.6.5.2.7 Seawater attack to plain concrete

There are no specific models to describe the alteration of concrete by seawater attack, as it is a complex term composed of a number of actions, both physical and chemical. In general, sea-water is not harmful for the concrete itself.

30.6.5.2.8 Food production and processing

Concrete in structures for agriculture or food production may be subject to attack by various chemical agents, such as dissolved CO_2, organic acids, and/or ammonium resulting from the production process that is largely based on the microbial activity, and by the microbial activity itself.

30.6.5.3 Concrete composition prescriptive rules

In prescriptive specifications of codes, adequate durability with respect to chemical attacks is deemed satisfied by ensuring compliance with limiting values for all or part of the following parameters:

- The permitted cement type (amount of C_3A, SO_4, …)
- The minimum percentage of supplemental cementitious materials
- The minimum binder content,
- The maximum water/cement ratio,
- The minimum compressive strength

Concrete composition prescriptive rules for leaching are described in section 14.8.5.3.

There are number of tests for sulfate attack, which give different limits of expansion for failure in the test – these are used primarily in material selection for the concrete. An example from the ASTM codes is provided in section 14.8.5.

In soft water attack, where the leaching progress is determined in solvents with pH close to neutral, the following indirect and direct performance indicators are summarized as follows:

Indirect performance indicators

- Hydraulic conductivity;
- Water permeability.

Direct performance indicators are dependent on the model applied such as:

- Leaching rate;
- Effective diffusion coefficient;
- Porosity (and tortuosity).
-

30.6.5.4 Rules by tests and measurements (Performance indicators)

Concrete mix design can be accepted subject to positive result obtained in accelerated testing of concrete samples (e.g. “torture performance tests” (refer section 14.8.5), with the condition that acceptance criteria would have been defined.

30.6.5.5 Semi-probabilistic format

The leaching depth in the semi-probabilistic approach can be assessed by Eq. (30.6-27):

$$x_d > \gamma_f \cdot x_L\left(t \le t_{SL}\right) \tag{30.6-27}$$

where:

x_d is the design value of the tolerable loss of concrete cover regarding strength and creep or the design value of the concrete cover regarding corrosion initiation due to calcium leaching in mm;

x_L is the leaching depth in mm;

γ_f is the safety factor.

t the exposure time to leaching

Table 30.6-8 summarizes the distributions for the parameters used in the models given for leaching in *fib* Bulletin ZZ [30.6-1].

Table 30.6-8: Distribution functions for leaching parameters.

Parameter	Distribution function	CoV [%]
Porosity ϕ [%]	Normal	8-20
Leaching rate L [mm/year$^{0.5}$]	Normal, Lognormal	20-60

30.6.5.6 Fully probabilistic format

The following inequality shall be fulfilled to guarantee a correct performance in leaching:

$$p(x - x_L(t \leq t_{SL}) < 0) < p_0 \qquad (30.6\text{-}28)$$

where:

p is the probability of failure;

p_0 is the target failure probability;

x is the tolerable loss of concrete cover regarding mechanical properties or the value of the concrete cover regarding corrosion initiation due to calcium leaching;

$x_L\,(t)$ is the leaching depth in [mm] at time t.

30.6.5.7 Avoidance of deterioration

In most cases, the use of dense and impermeable concrete can restrict the chances of ammonium salts attack originating from fertiliser or waste.

An appropriate “additional protective measure” (APM) will protect concrete against higher levels of sulfate than can be catered for just by specification of the concrete. Thus, a special protection cover (e.g. epoxy resin, steel-plates, special cementitious coatings) can prevent concrete direct exposure to chemical attacks.

For inorganic and organic attacks as well, for situations in which the rate of attack is too great for compensation by low w/c or sacrificial concrete, the structure can be further protected by an APM, among which paint coatings of appropriate formulation (Technical Note 68 [30.6-32]) can be highly effective.

30.6.6 Internal expansion mechanisms

30.6.6.1 Alkali–aggregate reactions

30.6.6.1.1 General

Alkali aggregate reaction refers to the deleterious expansion that might occur when reactive siliceous or carbonaceous aggregates combine with the available alkalis (Na_2O and K_2O) and Calcium in the concrete. Measures for minimizing the risk of AAR are developed in section 14.8.6.1.

30.6.6.1.2 Concrete composition prescriptive rules

Chemical deterioration in the form of alkali-aggregate reaction (AAR) is a slow-evolving but progressive process leading to expansion and cracking of concrete. Two main types of AAR are currently recognized based on the type of reactive minerals involved in the reaction: the alkali silica reaction (ASR) and the alkali-carbonate reaction (ACR).

The alkali-carbonate reaction is a chemical reaction between the alkali from the cement paste and certain carbonate rocks, particularly calcitic dolomite and dolimitic limestones, present in some aggregates. The reaction is usually accompanied by dedolomitization and expansion of the affected aggregate particles, leading to abnormal expansion and cracking of concrete in service.

The predominant type of AAR is the alkali-silica reaction, which is a chemical reaction between the alkali (Na2O and K2O) from the cement paste and certain siliceous minerals present in some aggregates. The product of the reaction is a gel which swells in the presence of moisture leading to pressure build-up that causes macroscopic expansion.

ASR impact on mechanical properties of concrete is developed within the section 30.1.11.2.

International examples of alkali limits for controlling expansion from reactive aggregate are given in *fib* Bulletin ZZ [30.6-1].

The total alkali content of cement is often expressed as Na_2O equivalent = Na_2O + 0.658·K_2O; it is generally agreed that the limitation of alkali content in the concrete is beneficial when potentially reactive aggregates are present.

Considering long-term performance also the potential alkali leaching from aggregates need be considered.

Due to possible differences in local climate and geology, care should be taken in transferring experience from one region to another.

In this situation, assessment of affected existing structures is generally based on conventional SLS / ULS assessment with partial factors, combined with a best estimate description of ASR-induced damage/ageing, expansions and cracking which modify the structural response (Godart & Wood [30.6-33]), Godart and De Rooij [30.6-34]). A sensitivity study with respect to the parameters describing the ASR effects, and check of predictions using monitoring data is recommended (Mahut, Fasseu et al.[30.6-35]), which can help in selecting appropriate margins.

Within this approach, one or a combination of the following remedies are normally required based on calibration to long term experience:

- minimise the amount of alkalis originating from cementitious materials and other sources;
- limitations to the alkali-reactivity of the aggregate or use of non-reactive aggregate;
- use of aggregates with limited alkali leaching;
- use of composite cements or of additions like silica fume, fly ash, metakaolin and slag;
- use of lithium nitrite/hydroxide and other chemical additions;
- keeping the humidity in the concrete below that needed to support the reaction (for instance, dry indoor environment or ensure proper drainage for outdoor structures).

30.6.6.1.3 Rules by tests and measurements (Performance indicators)

Aggregate or concrete mix design can be accepted subject to positive result obtained in accelerated testing of mortar or concrete samples (e.g. “expansion performance tests” (see *fib* Bulletin ZZ [30.6-1]) assessing either the aggregates or a given concrete mix potential expansion).

30.6.6.1.4 Partial safety factor format

At the time of publishing this Model Code, a partial safety factor approach for safety and serviceability assessment of ASR-affected structures is not feasible.

The reasons include:

- Insufficient statistical knowledge in the description of the scattered reaction-related structural effects
- Nonlinear couplings between a number of parameters (moisture, temperature, applied stresses) and the reaction consequences (induced expansive strains / pressure) which prevent extending the known safety formats.
-

To support an alkali–aggregate reaction, reactive aggregate, sufficient humidity and alkalis need to be present in the concrete member. If one or more of these conditions are missing, avoidance of reaction may be achieved.

The use of linings and protective coatings as well as moisture barriers (coatings and penetrants) can reduce or cut off the moisture ingress, reducing then the potential for expansion. The use of joints can also relieve the stresses due to expansion and lower the cracking.

Note: experience has shown that the use of coatings and penetrating sealers may retard the reaction but are unlikely to avoid it if all the essential ingredients are present.

Treatment of the reaction at the material scale is certainly difficult, there is however increasing experience in mitigating damaging effects at the structural scale:

- Effective sealants
- Stress release

30.6.6.1.5 Full probabilistic format

At the time of publishing this Model Code, a probabilistic approach for safety and serviceability assessment of ASR-affected structures is not feasible.

30.6.6.1.6 Avoidance-of-deterioration approach

The means of avoiding deterioration may include the use of coatings and penetrants that restrict the penetration of moisture into the concrete structure.

Additional possible mitigation measures at the structural scale, which require careful justification and adaptation to the case concerned, may include (Germain [30.6-36]):

- Jacketing of expansive zones by either steel plates, FRP, precast or cast-in-place (possibly prestressed or fibre-reinforced) concrete shells
- Releasing excessive induced compressive stresses by sawing notches, which is mainly applicable to non-reinforced concrete structures such as dams (Omikrine-Metalssi et al. [30.6-37]).

30.6.6.2 Delayed ettringite formation

30.6.6.2.1 General

Delayed Ettringite Formation (DEF) is a cause of disorders that can cause quite severe damage in concrete structures and can occur depending on achieved temperature during initial hardening period, humidity and concrete mix design (chemical composition of cement (including alkali content), use of additions, etc.). Measures for minimizing the risk of DEF are developed in section 14.8.6.2.

- Jacketing
- Local substitution of the concrete (in case DEF concerns a limited zone in the structure)

ACI 201.2R-16, amongst other national standards, provides some guidelines, , for minimizing the risk of DEF when control of the concrete below range [65°C - 70°C] is not possible.

Performance tests have been made available (Pavoine & Brunetaud [30.6-38]) and included in technical guidance documents in several countries (Godart,et al. [30.6-39]) so that acceptability can be assessed for a concrete mix-design associated to a given temperature evolution at early age (as a representative envelope of expected temperature histories that the concrete will be submitted to in the structure / precast elements to be produced).

If concrete is exposed to high temperature over a certain period in its early life (e.g. around 1 day for precast element and up to 1 week or more for massive cast in place elements), experience has shown that if the temperature exceeds about 70 ºC and if other key parameters are present, depending on humidity and concrete mix design (alkali content, chemical composition of cement). In general concrete with blended binders are more robust for high temperatures. ISO 22966 [30.6-17] has introduced an upper allowed peak temperature of 70ºC unless it is documented that with the combinations of materials used, higher temperatures have no significant adverse effect.

30.6.6.2.2 Concrete composition prescriptive rules

Unless specified otherwise, the peak temperature of the concrete within a component exposed to a wet or cyclic wetting environment should not exceed a range of [65°C- 70°C], unless data are provided to prove that, with the combination of cementitious materials used, higher temperatures will have no significant adverse effect on the service performance of the concrete. See section 14.8.6.2.

30.6.6.2.3 Rules by tests and measurements (Performance indicators)

Concrete mix design can be accepted subject to positive result obtained in accelerated testing of mortar or concrete samples (e.g. “expansion performance tests” (see *fib* Bulletin ZZ [30.6-1]) assessing a given concrete mix potential expansion.

30.6.6.2.4 Partial safety factor format

The reasons include:

- Insufficient statistical knowledge in the description of the scattered reaction-related structural effects;
- Nonlinear couplings between a number of parameters (moisture, temperature, applied stresses) and the reaction consequences (induced expansive strains / pressure) which prevent extending the known safety formats. A sensitivity study with respect to the parameters describing the DEF effects, and check of predictions using monitoring data is recommended (Mahut et al. [30.6-35]), which can help selecting appropriate margins Detailed information is provided in *fib* Bulletin ZZ [30.6-1].

Additional possible mitigation measures are described in section 30.6.6.1.6.

At the time of publishing this Model Code, a partial safety factor approach for safety and serviceability assessment of DEF-affected structures is not feasible. In this situation, assessment of affected existing structures is generally based on conventional SLS / ULS assessment with partial factors, combined with a best estimate description of DEF-induced damage/ageing, expansions and cracking which modify the structural response.

30.6.6.2.5 Full probabilistic format

At the time of publishing this Model Code, a probabilistic approach for safety and serviceability assessment of DEF-affected structures is not feasible.

30.6.6.2.6 Avoidance-of-deterioration approach

Delayed ettringite formation can be avoided if one or more of these conditions are missing:

- high temperatures at early age;
- critical contents of SO_3 and C_3A of cement;
- alkalis content and enough water and sufficient calcium hydroxide in the porous solution need to occur in the concrete member.

If one or more of these conditions are missing, avoidance of the reaction is achieved.

30.6.7 Abrasion, erosion, and cavitation

30.6.7.1 General

Abrasion, erosion, and cavitation result in a gradual reduction of the concrete surface (cover) and therefore to a reduced service life due to ingress of harmful substances like chlorides or carbon dioxide (section 14.8.7). Due to wear, the load application behaviour is also affected. In case of significant wear, the structural performance of the affected member may be compromised due to loss of section depth.

30.6.7.2 Concrete composition prescriptive rules

The classification of exposure classes does not differentiate between abrasion, erosion and cavitation.

According to the European Standard EN 206 [30.6-40] (which does not address the wear of concrete) in the German national application document DIN 1045-2 [30.6-41] three exposure classes XM1, XM2 and XM3 are defined. Concerning these three different exposure classes corresponding limitations of values of concrete composition (compressive strength, minimum cement content, water/cement ratio and type of aggregate) as well as the curing and the surface treatment (finishing or application of hard material) are prescribed. Further details are provided in *fib* Bulletin ZZ [30.6-1].

30.6.7.3 Rules by tests and measurements (Performance indicators)

Concrete mix design can be accepted subject to positive result obtained in accelerated testing of concrete samples (e.g. “torture performance tests” (refer section 14.8.8) subject that acceptance criteria would have been defined.

In some national codes classification of concrete traffic floors are given (e.g. CCI) considering their abrasive exposure. For each floor class maximum permissible abrasion is given for different test methods (e.g. ASTM C779/C779M [30.6-42]). Further details are provided in *fib* Bulletin ZZ [30.6-1].

30.6.7.4 Semi-probabilistic approach

At the time of publishing this Model Code, models for the description of abrasion, erosion and cavitation are still under discussion. A partial safety factor approach for SLD is therefore not feasible.

30.6.7.5 Full Probabilistic format

At the time of publishing this Model Code, design models for the description of abrasion, erosion and cavitation are still under discussion.

30.6.7.6 Avoidance-of-deterioration

A particular protection cover (e.g. epoxy resin, steel-plates, basalt slabs, and special cementitious coatings) can decelerate abrasion, erosion, and cavitation.

Furthermore, avoiding the access of most of the bed load of the river into the hydraulic structures by means of appropriate construction measures and a hydraulically optimized geometry with less intense exposure due to the flow conditions decreases erosion significantly. In other situations, such as coastal and offshore structures this is not possible, however good detailing practice including attention to the component´s geometry can mitigate against erosion damage.

30.6.8 Coupled deterioration mechanisms

30.6.8.1 General

Amongst the possible interactions, the main coupled effects, which are the most encountered on site, discussed in this chapter are:

- Erosion by mechanical abrasion, frost scaling and leaching effect on the reinforcement corrosion

When dealing with coupled effects, common practice, is to design with the aim to avoid the damage by the most detrimental attack (quicker and heavier). The less salient deterioration mechanism is assumed to have a lower impact or damage at longer times.

- Alkali Silica Reaction (ASR) impact on reinforcement corrosion
- Leaching/chemical attack effect on frost scaling and internal frost resistance
- ASR effect on internal frost resistance
- Carbonation and chloride penetration
- Sulfate and chloride ingress
- Sulphides and carbonation

It is highly important to consider not only the most deleterious mechanism, but also the possible counterbalance of them giving rise to a favorable situation. Further information is given in *fib* Bulletin ZZ [30.6-1].

A special case concerning the frost scaling pertains to the concrete elements such as bridge piers partially immersed in cold sea region. In this case, the concrete surface can also be subject to the action of frost scaling with the frost source coming directly from seawater. In fact, the extent of scaling would be less serious than deicing frost applied on road since the latter normally produces a solution with salt concentration much higher. The surface scaling by seawater frost will promote the chloride ingress and the subsequent corrosion of reinforced steel, so this effect is to be considered in durability design for long service lives. Further details are provided in *fib* Bulletin ZZ [30.6-1].

Further information on the effect of ASR on reinforcement corrosion is provided in *fib* Bulletin ZZ [30.6-1].

The principles can be applied as well to existing structures, after the identification of the predominant mechanism and the establishment of the combined rates of deterioration.

30.6.8.2 Effect of erosion by mechanical abrasion, frost scaling and leaching on the reinforcement corrosion

Mechanical abrasion, frost scaling, and leaching shorten the initiation period for active corrosion.

30.6.8.3 Effect of Alkali Silica Reaction on reinforcement corrosion

The cracking of reinforced concrete caused by alkali silica reaction (ASR) increases the incidence of reinforcement corrosion by accelerating the ingress of CO_2 (carbonation) and chloride ions, which may shorten the corrosion initiation period.

The corrosion rate is affected by the time of wetness that usually is longer in the presence of the gels produced by the ASR.

Further information on the effect of leaching on frost scaling and internal frost resistance is provided in *fib* Bulletin ZZ [30.6-1].

30.6.8.4 Effect of leaching on frost scaling and internal frost resistance

The simultaneous action of leaching and low temperatures shortens the period for the detrimental action of the two types of frost attack.

30.6.8.5 Effect of ASR on internal frost resistance

Low temperatures decelerate the expansion due to ASR. However, once the expansions are produced, the generated cracks will hold water which will accelerate the damage once frozen.

ASR increases the cracks on the concrete despite the denser microstructure developed due to the produced gel during the reaction. The latter retains water, which leads to concrete with higher degree of water saturation. Further details can be found in *fib* Bulletin ZZ [30.6-1].

ASR can reduce the resistance of concrete to cyclic freezing and thawing, even if the concrete is adequately air entrained. If cracks induced by ASR become saturated the freezing water will propagate and widen the cracks. Horizontal surfaces, such as pavements, are particularly vulnerable to this combination of processes.

30.6.8.6 Effect of joint carbonation and chloride penetration

In practical cases, the impact of carbonation on chloride ingress will exit in two typical cases: concretes exposed in marine atmosphere and concrete surfaces subject alternatively to deicing salts application and the subsequent dry periods. In both cases, the carbonation tends to promote the chloride ingress thus the corrosion initiation of embedded steel bars. The humidity condition determines the order of importance of carbonation impact. Further details can be found in *fib* Bulletin ZZ [30.6-1].

It is to be noted that when a structure is designed to resist chloride ingress (concrete quality and cover), it is overprotected against carbonation ingress to the reinforcement.

The sequences of carbonation and chlorides attacks depend on the curing period and their interaction period in the concrete. The longer the curing period is the slower the progress of carbonation. Then, if chlorides are in contact with the concrete, its further drying would be more difficult, and the carbonation rate would be smaller and the coupling effect much less important.

For shorter curing periods the concrete would start to carbonate before the chloride ions enter in contact with the concrete. These ions will penetrate further due to the coarsening of the pore structure after carbonation and the lower binding capacity.

In soils, it is also important to consider the possibility of damage by salt crystallization, either due to very high concentrations of salt above the saturation limit or due to wet-dry periods.

In addition to that, the pH of the chemicals in the soil should be considered because of the possible leaching /lowering of pH effect in addition to the penetration of salts.

It is to be noted that currently most used concrete mix designs for marine structures are developed with blended binders (ground granulated blast-furnace slag, pulverized fly ash, silica fume, etc.).

Table 30.6-9 gives an estimation of the value of sacrificial cover required for a range of carbonation covers combined with c_{loss} values. The calculated values have been rounded up to the nearest 5 mm and there is an additional level of conservatism in the design, as Haifeng (2013) showed that carbonation of concrete slowed the rate of H_2S attack. Since the concrete will be carbonated, the actual rate of H_2S erosion will be slower than assumed.

The estimated sacrificial cover will be added to c_{min} and Δc to give the nominal cover. For the example consider a case study of a structural element of a sewage network. From Table 30.6-9 the carbonation cover is 50mm and the sacrificial cover required is 70mm. Without allowing for any loss for abrasion, the nominal cover is 130mm (50+70+10), by applying Table 30.6-9, the 130mm cover is reduced to 100mm (50+40+10).

30.6.8.7 Effect of joint sulfate and chloride ingress

The mutual influence between sulfate and chloride ions attack is usually produced when the concrete is in contact with sea water or saline groundwater. It is important to emphasize that in field exposure, it has not been detected significant cases of sulfate attack to the concrete in sea water, while the corrosion of reinforcement is evident in numerous marine infrastructures. Then, the use of low C_3A cements is much less important than to prevent the chloride penetration. Cements with low C_3A content have in general higher chloride diffusion coefficients due to their reduced binding ability (Andrade et al 1999 [30.6-43]).

30.6.8.8 Sulphides and carbonation

Hydrogen sulphide attack is rarely the only mechanism occurring on a structural element. Very often, carbonation acts in conjunction with H_2S.

When calculating the carbonation cover, the assumption of carbonation starting from day one is made. A sacrificial cover will be added to the concrete cover of c_{loss} [mm], which in theory will be eroded away after 120 years. The sacrificial concrete is protecting the concrete cover provided for durability from being exposed to CO_2 and therefore no carbonation will occur until the sacrificial concrete is itself carbonated or eroded away (depending on whether H_2S or carbonation is the most aggressive mechanism). This adds a significant level of conservatism to the design.

Due to the low levels of chlorides, combined chloride-induced corrosion with H_2S is not considered as a potential risk and has not been considered.

Table 30.6-9: Estimated sacrificial cover for combined carbonation and H_2S for a case study of a structural element of a sewage network.

Carbonation cover (mm)	Sacrificial cover required for given c_{loss} (mm) for different depths of carbonation													
	5	**10**	**15**	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**	**60**	**65**	**70**
10	0	5	10	10	15	20	25	30	35	40	45	50	55	60
15	0	5	5	10	15	20	25	30	30	35	40	45	50	55
20	0	5	5	10	15	15	20	25	30	35	40	45	50	55
25	0	5	5	10	10	15	20	25	25	30	35	40	45	50
30	0	5	5	10	10	15	20	20	25	30	35	35	40	45
35	0	5	5	10	10	15	15	20	25	25	30	35	40	45
40	0	5	5	10	10	15	15	20	20	25	30	35	35	40
45	0	5	5	10	10	15	15	20	20	25	30	30	35	40
50	0	5	5	10	10	10	15	20	20	25	25	30	35	40

By assuming that the H_2S loss will be uniform over the life of the structure (conservative assumption as it is unlikely to occur in the early life of the structure) and the carbonation rate is a function of the square root of time, then if the rate of carbonation is reassessed on a yearly basis to allow for the loss of cover, it will take 68 years before the durability cover starts to carbonate. It is therefore reasonable to reduce the cover so that the combined attack results in a design life of 120 years rather than 188 years. However, it is important that bond is maintained therefore at the 120 year service life the cover should be at least $c_{min,b}$.

References

[30.6-1] *fib* Bulletin ZZ — Background document for Duarbility related issues in MC2020, *fib* Bulletin under preparation

[30.6-2] *fib* Bulletin 40 — FRP reinforcement in RC structures, *fib* Technical Report, 2007

[30.6-3] Angst et al — Angst et al 2020, further details for reference to be given by M. Geiker

[30.6-4] prEN1992-1-1:2023 — prEN 1992-1-1 Design of concrete structures – Part 1-1 General rules, rules for buildings, bridges and civil engineering structures, Sept 2022

[30.6-5] Gehlen, C. — Probabilistische Lebensdauerbemessung von Stahlbetonbauwerken-Zuverlässigkeitsbetrachtungen zur wirksamen Vermeidung von Bewehrungskorrosion, PhD thesis. RWTH Aachen, 2000.

[30.6-6] *fib* Bulletin 34 — Model code for service life design of concrete structures., 2006.

[30.6-7] *fib* Bulletin TG8.3 — fib TG8.3, Chapter 5.1 [draft report in progress]

[30.6-8] RILEM Recommendations 2003 — Electrochemical techniques for measuring metallic corrosion Half-Cell Potential Measurements – Potential Mapping on Reinforced Concrete Structures -Vol 36 (2003), B. Elsener, C. Andrade, J. Gulikers, R. Polder, M. Raupach.- p 461-471, Rilem 2003

[30.6-9] RILEM Recommendations 2004 — Electrochemical Techniques for Measuring Metallic Corrosion. Recommendations Test methods for on-site corrosion rate measurement of steel reinforcement in concrete by means of the polarization resistance method." Materials & Structures 37(273):Andrade C, Alonso C, Gulikers J, Polder R, Cigna R, Vennesland Ø, Salta M, Raharinaivo A, Elsener B., p 623-643, Rilem 2004.

[30.6-10] RILEM CPC11.2. — Absorption of water by capillarity, 2nd ed., IRAM 1871. (2004))

[30.6-11] DAfStb Issue 622. — Durability design rules for XC-exposed structural elements (in German) Greve-Dierfeld (2016).

[30.6-12] DAfStb Issue 623. — Durability design with partial safety factors including a benchmark of deemed-to-satisfy rules (chloride, carbonation), Greve-Dierfeld, Gehlen, 2010

[30.6-13] LNEC E 465:2007. — Concrete – methodology for estimating the concrete performance properties allowing to comply with the design working life of reinforced or pre-stressed concrete structures under environmental exposures XC and XS. MOPTC – Laboratório Nacional de Engenharia Civil, Portugal.

[30.6-14] Izquierdo — Izquierdo D., Basis of design for a probabilistic treatment of reinforcement corrosion processes- 2001 Polytechnic University of Madrid. Faculty of Civil Engineering.

[30.6-15] prEN1990:2020 — Eurocode - Basis of structural design, CEN,

[30.6-16] *fib* Bulletin 76. — Benchmarking of Deemed-to-Satisfy Provisions in Standards – Durability of Reinforced Concrete Structures Exposed to Chlorides, 2015.

[30.6-17]	ISO 22966:2009.	Execution of concrete structures.
[30.6-18]	RILEM Technical Committee 130-CSL.	Durability design of concrete strucutures. Edited by A. Saria and E. Vesikary.
[30.6-19]	Fagerlund	Fagerlund, G., Intenfal frost attack – state of the art. *RILEM Proceedings 34*, *Frost resistance of concrete* , p. 321 – 338.
[30.6-20]	Fagerlund 1997	Fagerlund, G., On the service life of concrete exposed to frost action. *Freeze-Thaw Durability of Concrete*, 1997, p. 21 – 41.
[30.6-21]	Penttala	Penttala, V. , Surface and internal deterioration of concrete due to saline and non-saline freeze–thaw loads. Cement and Concrete Research 36. 921 – 928.
[30.6-22]	Pigeon et al	Pigeon, M., Pleau, R., Durability of concrete in cold climates. London. 244 p, 1995.
[30.6-23]	Scherer & Valenza	Scherer, G. W., Valenza, J., Mechanisms of Frost Damage, *Materials Science of Concrete,* p. 209 – 246, 2005.
[30.6-24]	Lay & Schießl	Lay & Schießl 2003, further details to be given
[30.6-25]	Bager & Jacobsen	Bager, D. H., Jacobsen, S., A conceptual model for the freeze-thaw damage of concrete. *3rd Nordic Research Seminar in Lund, Frost resistance of building materials*, 1999.
[30.6-26]	Kang et al	Kang, Y, Hansen, W & Borgnakke, C. , Effect of air-void system on frost expansion of highway concrete exposed to deicer salt, 2012, International Journal of Pavement Engineering, 13:3, 259-266.
[30.6-27]	Linmark	Linmark, S , Mechanisms of Salt Frost Scaling of Portland Cement-bound Materials: Studies and Hypothesis", Ph.D. thesis, Report TVBN 1017, Lund Inst. Tech., Lund, Sweden, 1998, 286 p., 1998
[30.6-28]	Setzer	Setzer, M. J., Micro-Ice-Lens Formation in Porous Solid. *Journal of Colloid and Interface Science*, 243, 193 - 201.
[30.6-29]	Valenza & Scherer	Valenza, J. J., Scherer, G. W., A Review of Salt Scaling: II. Mechanisms*, Cem. Concr. Res.*, 37, p. 1022 – 1034, 2001.
[30.6-30]	BRE Special Digest 1	Special Digest 1:2005. Concrete in aggressive ground, BRE Construction Division (UK).
[30.6-31]	ISO 22965-1: 2007.	Concrete – Part 1: Methods of specifying and guidance for the specifier.
[30.6-32]	Technical Note 68.	Sulfate-resisting concrete, Cement, Concrete & Aggregates Australia, May 2011.
[30.6-33]	Godart & Wood	Godart, B., Wood, J.., Appraising structures affected by the alkali–aggregate reaction. Construction Materials, 169(3), June 2016, pp. 162–171, DOI: 10.1680/jcoma.15.00068.
[30.6-34]	Godart & De Rooij	Godart, B., De Rooij, M.. , Diagnosis, Appraisal, Repair and Management. – Chapter 5 of the book Alkali-Aggregate Reaction in Concrete: A World Review, 2017. Editors: Alan B. Poole and Ian Sims, CRC Press, 768 Pages, ISBN 9781138027565.
[30.6-35]	Mahut & Fasseu et al.	Mahut ,B., Fasseu, P. et al. , Aide à la gestion des ouvrages atteints de réactions de gonflement interne, guide technique, collection Techniques et méthodes des Laboratoires des Ponts et Chaussées, novembre 2003, 68 pages.
[30.6-36]	Germain	D. Germain (ed.), Protection et réparation des ouvrages atteints de réaction de gonflement interne du béton, 2010, Recommandations provisoires, LPC guidance document, LCPC, Paris.

[30.6-37]	Omikrine-Metalssi et al	Omikrine-Metalssi O., Seignol J.-F., Rigobert S., Toutlemonde F., Modelling the cracks opening-closing and possible remedial sawing operation of AAR-affected dams, Engineering Failure Analysis, 36 (2014), pp. 199-214.
[30.6-38]	Pavoine &Brunetaud	Pavoine, A., Brunetaud, X., Développement et validation d'une méthode d'essai vis-à-vis de la formation différée de l'ettringite : Application à l'étude d'impact de différents paramètres sur la cinétique et l'amplitude de gonflement, 2010. LPC technical report OA 65, LCPC, Paris, France.
[30.6-39]	Godart et al.	Godart B., Divet L. et al. , Recommandations pour la prévention des désordres dus à la réaction sulfatique interne, technical guidelines, IFSTTAR, Paris, France, 2017 (english version published in 2018 : https://www.ifsttar.fr/ressources-en-ligne/librairie/collections-ifsttar/ouvrages-edites-par-lifsttar/recommendations-for-preventing-disorders-due-to-delayed-ettringite-formation/).
[30.6-40]	EN 206:2013+A1:2016.	Concrete - Specification, performance, production and conformity.
[30.6-41]	DIN 1045-2:2008-08,.	Concrete, reinforced and prestressed concrete structures – Part 2: Concrete – Specification, properties, production and conformity – Application rules for DIN EN 206-1.
[30.6-42]	ASTM C779/C779M	Standard Test Method for Abrasion Resistance of Horizontal Concrete Surfaces, ASTM, 2019
[30.6-43]	Andrade et al 1999	Andrade, C., Castellote, M.; Alonso, C.; Gonzalez, C., Relation between colourimetric chloride penetration depth and charge passed in migration tests of the type of standard ASTM C1202-91. Cement and Concrete Research 29 (1999), pp 417-421.

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A distinction should be made between structures with linear elements (statically determinate and indeterminate beams or frames), where the stress redistribution is limited to few sections, where strain localization may occur and structures with a higher degree of redundancy, where stress redistribution occurs in multiple cracks (as in slabs).

In linear structures there may be critical zones where a model based on a plane section can be assumed, by spreading the localized relative rotation in defined regions characterized by a prescribed length in which the curvature is assumed constant. In these zones the curvature jumps to larger values (post-peak values) with respect to the values reached in the adjacent regions (where the curvature corresponds to the unloaded elastic values), although the bending moment is the same to respect the rotational equilibrium.

For thin walled structures, fibre orientation may be influenced by the reduced thickness (wall effect), which is significantly dependent on the casting direction. In these structures, material properties can be better determined by performing "structural tests".

In thin walled structures subjected to bending (shell structures), due to the reduced thickness, the ultimate strain should be significantly reduced in order to satisfy ductility requirements expressed in terms of ultimate strain.

The design rules are not intended to be used for slabs on grade, or for temporary sprayed concrete linings, or to design applications in which increased resistance to plastic shrinkage, increased resistance to abrasion or impact are aimed for. The design rules refer to applications, covered in the scope of EN 1992-1-1:2004 [30.7-1].

Due to the nature of FRC, design supported by testing leads to optimized solutions in terms of structural performance versus cost.

30.7 Evaluation of safety, serviceability and condition limit states associated to durability for FRC structures

30.7.1 General

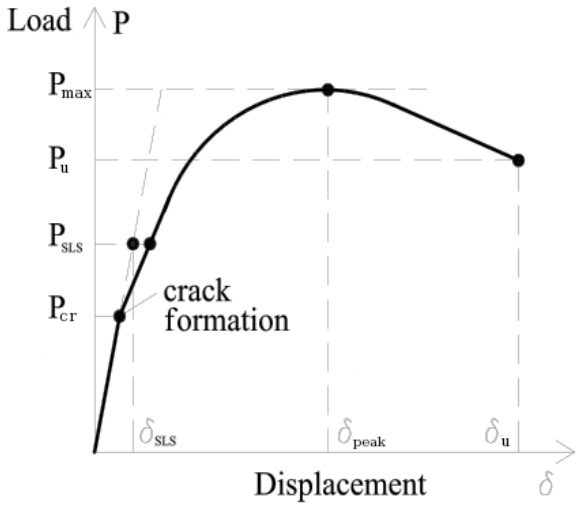

Figure 30.7-1: Typical load (P) – displacement (δ) curve for a FRC structure.

In the case of linear elements, without conventional reinforcement, subjected to axial tension with small eccentricity (i.e. without compressive stresses in the section), in addition to the limitations provided by Eqs. (30.7-1) and (30.7-2), FRC should have a hardening behaviour in tension (see section 18.1, Fig. 18.1-1).

Further information on the design equations, as treated in this section, is given by Di Prisco et al. [30.7-2].

30.7.2 Design principles

Structural design must satisfy requirements for resistance and serviceability for the expected service life of FRC elements. The ductility requirement in bending can be satisfied by minimum conventional reinforcement (see also section 30.13; Fig. 30.7.1).

In all FRC structures without the minimum conventional reinforcement, one of the following conditions must be satisfied:

$$\delta_u \geq 20\ \delta_{SLS} \qquad (30.7\text{-}1)$$

$$\delta_{peak} \geq 5\ \delta_{SLS} \qquad (30.7\text{-}2)$$

where δ_u is the ultimate displacement, δ_{peak} is the displacement at the maximum load and δ_{SLS} is the displacement at maximum service load computed by performing a linear elastic analysis with the assumptions of uncracked concrete and initial elastic Young's modulus.

Usually, δ_u is related to the maximum deformation requirement of the structure.

The ultimate load P_u should always be higher than the load at crack initiation P_{cr} and higher than the maximum service load P_{SLS}.

30.7.3 Verification of safety (ULS)

30.7.3.1 Bending and/or axial compression in linear members

The bending failure stage is supposed to be reached when one of the following conditions applies (Fig. 30.7-2); the meaning of the variables is found in sections 18.4 and 18.6):

– attainment of the ultimate compressive strain in the FRC, ε_{cu};
– attainment of the ultimate tensile strain in the steel (if present), ε_{su};
– attainment of the ultimate tensile strain in the FRC, ε_{Fu}.

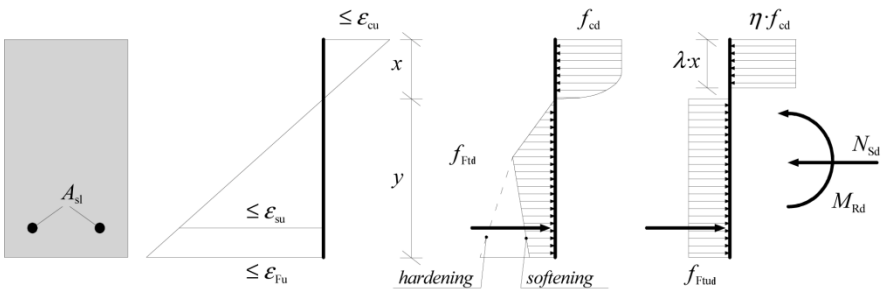

Figure 30.7-2: ULS for bending moment and axial force: use of the simplified stress/strain relationship (λ and η coefficient in accordance Eqs. (14.11-8) to (14.11-11) in section 14.11.1.5)

30.7.3.2 Shear in beams

30.7.3.2.1 Beams without longitudinal and shear reinforcement

When FRC with tensile-hardening behaviour is used and members without both conventional longitudinal and transverse reinforcement are considered, the principal tensile stress, σ_1, must not be higher than the design tensile strength:

$$\sigma_1 \leq \frac{f_{Ftsk}}{\gamma_F} \tag{30.7-3}$$

where:

f_{Ftsk} is the characteristic value of the residual tensile strength for crack widths at the serviceability limit state accounting for fibre orientation determined with Eq. (18.4-5) for w_u = 1.5 mm [MPa];

γ_F value found in section 12.5.4.2.3.2.

f_{Ftuk} is to be determined by an axial-tensile test.

For redundant structures, such as slabs, the ultimate deformation determines the ultimate limit state condition. From that the corresponding crack opening (w_u) can be computed, which can be in the order of magnitude of the SLS crack opening.

30.7.3.2.2 Beams with longitudinal and shear reinforcement

Regulations for beams with longitudinal and shear reinforcement is given in Section 30.1.3.4.

30.7.3.3 Walls

30.7.3.3.1 Walls without conventional reinforcement

For 2D elements loaded in their plane, it is possible to check the limit state criteria for SLS and ULS by means of the biaxial domain of failure, where the uniaxial tension strengths are reduced to f_{Fts} or $f_{Ftu,}$ according to the limit state considered.

30.7.3.3.2 Walls with conventional reinforcement

In FRC structures, the fibre contribution can be accounted for by non-linear finite element analyses or strut-and-tie models with the constitutive laws defined in section 18.4

In FRC structures satisfying minimum requirements (Eqs. (18.3-4) and (18.3-5)), secondary conventional reinforcement can be omitted.

30.7.3.4 Slabs

30.7.3.4.1 Members without reinforcement

For slab elements without conventional reinforcement (Fig. 30.7-3) with predominantly bending actions, the strength verification can be done with reference to the resisting moment, m_{Rd}, evaluated by considering a rigid plastic relationship (Figure 30.7-2c):

$$m_{\mathrm{Rd}} = \frac{f_{\mathrm{Ftud}} \cdot t^2}{2} \qquad (30.7\text{-}4)$$

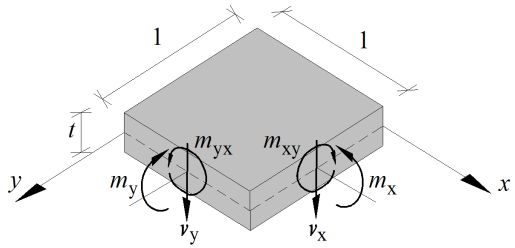

Figure 30.7-3: Actions in a slab element.

When a linear analysis is performed, the maximum principal moment should be lower than m_{Rd}.

When a limit analysis is performed, m_{Rd} can be regarded as the reference value.

Shear in FRC slabs without conventional reinforcement or prestressing is not regarded as dominant unless significant load concentrations occur close to the support.

30.7.3.4.2 Members with reinforcement

The verification of FRC elements with conventional reinforcement can be done with a non-linear analysis (e.g. limit analysis, non-linear finite element analysis).

30.7.4 Verification of serviceability (SLS)

30.7.4.1 Stress limitation

In structural FRC elements having a tension softening behaviour after cracking, the tensile stress verification is not necessary if the element is verified at the ULS.

In structural FRC elements having a tension hardening behaviour after cracking, the tensile stress verification must be done by imposing the limitation:

$$\sigma_t \leq 0.6 f_{Ftsk} \quad (30.7\text{-}5)$$

where σ_t is the principal tensile stress and f_{Ftsk} is the characteristic value of f_{Fts} given in Eq. (18.4-4).

-For members constructed with joints to avoid uncontrolled cracking, it should be ensured that brittle failure of these members does not lead to collapse of the structure.

Members using FRC and designed in accordance with this clause do not preclude the provision of steel reinforcement needed to satisfy serviceability, nor reinforcement in certain parts of the members.

Any reinforcement may be taken into account for the local verification of ultimate limit states as well as for the checks of the serviceability limit states.

Members subject to imposed deformations should comply with provisions of 30.13.6.2, 30.13.6.3, 30.13.6.4.

30.7.4.2 Crack width in members with conventional reinforcement

For combinations of fibres and reinforcing steel the limit values of the crack width as given in Table 30.5.1 .can be used. With regard to possible corrosion of the fibres near the surface of the structure, a maximum crack width of 0.2mm under the quasi-permanent load combination is generally considered as a suitable limit value for steel fibre concrete under severe circumstances, like chloride exposure(see Marcus-Meson et. Al. [30.7-3])

30.7.4.3 Minimum reinforcement for crack control

For controlling the crack width in the elements under bending, if needed, a minimum reinforcement should be applied, at least equal to:

$$A_{s,\min} = k_c \cdot k \cdot (f_{ctm} - f_{Ftsm}) \frac{A_{ct}}{\sigma_s} \qquad (30.7\text{-}6)$$

where:

f_{ctm} is the average value of the tensile strength of the concrete matrix;

f_{Ftsm} is the average value of the residual strength of FRC;

A_{ct} is the tensile part of the concrete cross-section, evaluated by considering a stress field at elastic limit;

σ_s is the maximum tensile stress in the reinforcement in the cracked state, that can be considered equal to the yielding stress of the steel;

k_c is a coefficient taking account of the stress distribution in the cross-section just before cracking and the change of the inner lever arm; for rectangular cross-sections $k_c = 1$;

k is a coefficient taking into account of non-uniform self-equilibrating stresses, leading to a reduction of the cracking force:
$k = 1.0$ for webs with $h \leq 300$ mm or flanges with width ≤ 300 mm;
$k = 0.65$ for webs with $h \geq 800$ mm or flanges with width ≥ 800 mm;
for intermediate values interpolation can be applied.

When $A_{s,min}$ is negative, the minimum reinforcement can be due only to the fibre reinforcement.

References

[30.7-1] EN1992-1-1:2004. Design of concrete structures - Part 1-1 : General rules and rules for buildings

[30.7-2] Di Prisco, M., Colombo, M., Dozio, D. Fibre-reinforced concrete in *fib* Model Code 2010: principles, models and test validation, 2013. *Structural Concrete*, 14:4, 342 – 361.

[30.7-3] Marcus-Meson, V., Michel, A., Solgaard, A., Fischer, G., Edvardson, C., Skovhus, T "Corrosion resistance of steel fibre reinforced concrete – A literature review', technical University of Denmark, published in the Proceedings of fib Symposium 2016 and in Cement and Concrete Research 103 (2018), pp. 1-20

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For a definition of externally applied reinforcement and a further discussion on FRP material properties, reference is made to chapter 17.

Note that the formulations in section 30.8 have not been prepared for hybrid reinforcing methods, combining EBR and NSM. It is however possible to use hybrid combinations, such as NSM for the flexural and EAR for shear in beams or NSM for the flexural strengthening of RC columns and EAR for the shear and confinement.

The designer should be familiar with the typical material behaviour of FRP (chapter 17) and the extra assumptions used for design, as provide in section 17.8.

Given the lack of a yield plateau in the FRP constitutive behaviour, analysis methods associated to plastic analysis and moment redistribution are omitted.

For further details on the provisions in section 30.8, reference is made to *fib* Bulletin 90 [30.8-1].

At each critical cross section of the strengthened member, it shall be ensured that its design resistance is higher than the design acting forces of all relevant load combinations.

30.8 Evaluation of safety, serviceability and condition limit states associated to durability for RC structures with externally applied reinforcement

30.8.1 General

This section applies to the design of existing buildings and civil engineering works in reinforced and prestressed concrete, repaired or strengthened by means of externally bonded reinforcement (EBR) or near surface mounted reinforcement (NSM).

Only strengthening systems will be applied which comply to product standards (section 17.2) and with quality control provisions in place on the practical execution (*fib* Bulletin 90, Chapter 10 [30.8-1]).

In structural analysis involving FRP reinforcement, both linear and non-linear methods of analysis can be applied, in reference to the FRP constitutive law described in chapter 17. Plastic analysis and linear analysis with moment redistribution shall not be undertaken for members with FRP reinforcement.

The structural analysis shall further consider the resistance hierarchy considerations provided in section 17.8.1.3, including the verification against accidental loss of the external reinforcement.

30.8.2 Evaluation of structural safety (ULS) for predominantly static loading

30.8.2.1 Bending with and without axial force

30.8.2.1.1 General

The capacity of the strengthened member is assessed by means of traditional cross-sectional analysis, imposing equilibrium of axial forces and bending moments, considering the constitutive laws of the different materials, and adopting strain

Given the high tensile capacity of FRP, a limited cross-sectional area is needed to fulfil ULS. As a result the increase in stiffness is relatively low, while higher FRP cross-sectional area (and associated flexural stiffness) might be required for SLS.

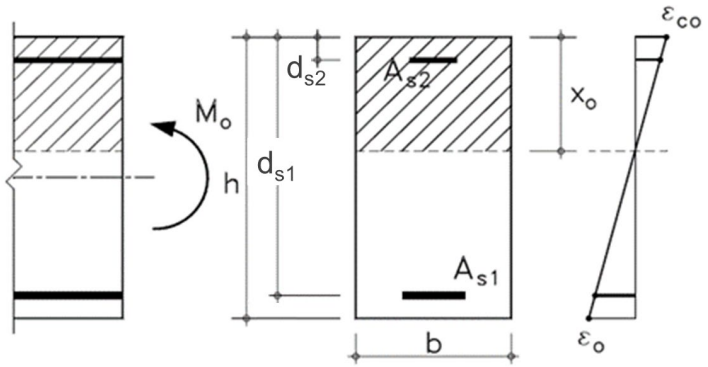

Figure 30.8-1: Initial strain in the concrete substrate where FRP will be applied.

In case of strengthening uncracked sections, the value for ε_o will be small and can be neglected.

compatibility on the assumption that full composite action between the FRP and the concrete substrate is preserved. Furthermore, the analysis will verify FRP debonding.

Note that the SLS of deflection control might be governing in the design, and can be considered as a starting point rather than ULS.

30.8.2.1.2 Members with EBR subjected primarily to bending

(1) Initial strain at the moment of applying the strengthening

The load level during strengthening is taken into account through the acting strains in the cross section at the moment of FRP application (Figure 30.8-1).

The strain in the concrete substrate where FRP will be applied, ε_o, should be estimated for the applied bending moment at the strengthening stage, M_o:

$$\varepsilon_o = \frac{M_o (h - x_o)}{E_c I_o} \tag{30.8-1}$$

where x_o is the depth of the neutral axis obtained by solving the following equation (Figure 30.8-1):

$$\frac{1}{2} b x_o^2 + (\alpha_s - 1) A_{s2} (x_o - d_{s2}) - \alpha_s A_{s1} (d_{s1} - x_o) = 0 \tag{30.8-2}$$

In this equation $\alpha_s = \frac{E_s}{E_c}$ is the modular ratio of the steel. In equation (30.8-1), I_o is the moment of inertia of the cracked section:

$$I_o = \frac{1}{3} b x_o^3 + (\alpha_s - 1) A_{s2} (x_o - d_{s2})^2 + \alpha_s A_{s1} (d_{s1} - x_o)^2 \tag{30.8-3}$$

If M_o is less than the cracking moment of the section, M_{cr}, the uncracked moment of inertia should be used in Eq. (30.8-1) unless by observation the member is already in a cracked state due to previous loading history.

(2) Resisting moment of the strengthened cross-section

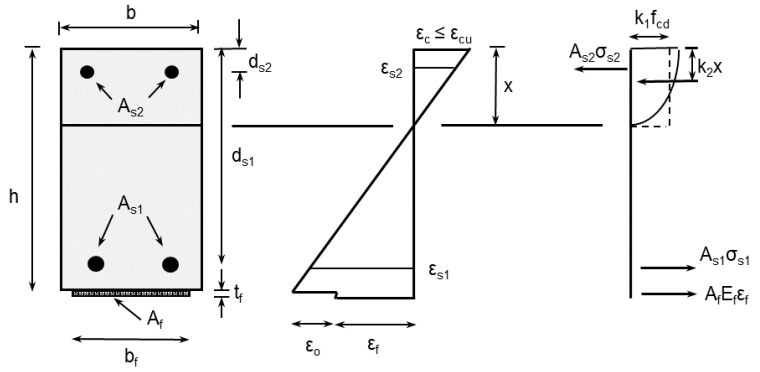

Figure 30.8-2: Cross section analysis: a) geometry; b) strain distribution; c) internal forces.

Figure 30.8-2 and Eqs. (30.8-5) to (30.8-9) assume an equivalent rectangular-parabolic concrete stress block (though similar equations can be applied e.g. assuming a rectangular concrete stress block).

The effective depth of the member with respect to the FRP reinforcement can be defined as $d_f = h + t_f/2$ d_f or simplified $d_f \approx h$.

A traditional cross-sectional analysis is conducted as also indicated in Figure 30.8-2. Hereby, the deformability criterium of section 17.8.1.4 shall be taken into account.

As a starting point to anticipate the verification of intermediate crack debonding, following section 30.8.2.1.2(3), the design shall be based on a maximum stress in the FRP equal to:

$$\sigma_f \le min\left\{f_{fbd,IC}; f_{fd}\right\} \tag{30.8-4}$$

where $f_{fbd,IC}$ is the design value of the FRP stress corresponding to intermediate crack debonding, as defined in section 22.2.6, and f_{fd} is the design value of the FRP tensile strength, as defined in section 17.8.1.1.

The depth of the neutral axis, x, is obtained by solving the following force equilibrium equation:

$$k_1 f_{cd} bx + A_{s2}\sigma_{s2} = A_{s1}\sigma_{s1} - A_f E_f \varepsilon_f \tag{30.8-5}$$

where the concrete stress block parameters are a function the concrete strain $0 < \varepsilon_c \le \varepsilon_{cu}$:

$$k_1 = \begin{cases} 1000\varepsilon_c\left(0.5 - \dfrac{1000}{12}\varepsilon_c\right) & for\ \varepsilon_c \le 0.002 \\ 1 - \dfrac{2}{3000\varepsilon_c} & for\ 0.002 \le \varepsilon_c \le 0.0035 \end{cases} \tag{30.8-6a}$$

$$k_2 = \begin{cases} \dfrac{8 - 1000\varepsilon_c}{4\left(6 - 1000\varepsilon_c\right)} & for\ \varepsilon_c \le 0.002 \\ \dfrac{1000\varepsilon_c\left(3000\varepsilon_c - 4\right) + 2}{2000\varepsilon_c\left(3000\varepsilon_c - 2\right)} & for\ 0.002 \le \varepsilon_c \le 0.0035 \end{cases} \tag{30.8-6b}$$

and whereby the strain in the concrete (ε_c) and the stresses in the reinforcements ($\sigma_r = E_r\varepsilon_r$, with subscript r representing the FRP (f), lower steel (s1) or upper steel (s2) reinforcement, respectively) follow from strain distribution.

Assuming that the failure is governed by crushing of the concrete, the strains can be expressed as:

$$\varepsilon_c = \varepsilon_{cu} \tag{30.8-7a}$$

$$\varepsilon_f = \varepsilon_{cu}\frac{h - x}{x} - \varepsilon_o \le min\left\{f_{fbd,IC}; f_{fd}\right\} / E_f \tag{30.8-7b}$$

$$\varepsilon_{s1} = \varepsilon_{cu} \frac{d_{s1} - x}{x} \leq f_{yd} / E_s \quad (30.8\text{-}7c)$$

$$\varepsilon_{s2} = \varepsilon_{cu} \frac{x - d_{s2}}{x} \leq f_{yd} / E_s \quad (30.8\text{-}7d)$$

If the condition expressed in Eq. (30.8-7b) is not fulfilled, it can be assumed that the failure is governed by the maximum FRP stress, so that the strains become:

$$\varepsilon_c = (\varepsilon_{fu} + \varepsilon_o) \frac{x}{h - x} \leq \varepsilon_{cu} \quad (30.8\text{-}8a)$$

$$\varepsilon_{fu} = min\left\{ f_{fbd,IC}; f_{fd} \right\} / E_f \quad (30.8\text{-}8b)$$

$$\varepsilon_{s1} = (\varepsilon_{fu} + \varepsilon_o) \frac{d_{s1} - x}{h - x} \leq f_{yd} / E_s \quad (30.8\text{-}8c)$$

$$\varepsilon_{s2} = (\varepsilon_{fu} + \varepsilon_o) \frac{x - d_{s2}}{h - x} \leq f_{yd} / E_s \quad (30.8\text{-}8d)$$

After evaluation of x, through Eqs. (30.8-5) to (30.8-8), the resisting bending moment is determined as:

$$M_{Rd} = \sigma_{s1} A_{s1} \left(d_{s1} - k_2 x \right) + \sigma_{s2} A_{s2} \left(k_2 x - d_{s2} \right) + \sigma_f A_f \left(h - k_2 x \right) \quad (30.8\text{-}9)$$

(3) Debonding verification: intermediate crack-induced debonding

In the simplified FRP stress limit method the intermediate crack debonding is considered as $\sigma_s \leq f_{fbd,IC}$, whereby the latter is the design value of the FRP stress corresponding to intermediate crack debonding, as defined in section 22.2.1.6.

Improved level of approximation methods for intermediate crack-induced debonding have been established, for which reference is made to section 6.2.1.3 of *fib* Bulletin 90 [30.8-1].

Intermediate crack debonding can be prevented further by the use of extra shear straps, for which reference is made to section 6.3.5.2 of *fib* Bulletin 90 [30.8-1].

(4) Debonding verification: end anchorage debonding

The anchorage capacity of the FRP at its ends shall be designed according to section_22.2.1.4, and an associated effective bond anchorage length according to section 22.2.1.3.

When calculating the resisting moment, under the assumption of full composite action between the FRP and the existing member, it is recommended to limit the FRP stress to anticipate possible intermediate crack-induced debonding. This is applied in section 30.8.2.1.2(2).

Higher accuracy methods for verification of crack-induced debonding evaluate the state of FRP stress and associated bond stresses along the length of the strengthened member, considering both uncracked and cracked sections and where relevant looking into the combined effect of high acting moments and shear forces..

The maximum anchorage capacity of the surface bonded FRP is limited to that corresponding with the effective bond length. For longer bond lengths the anchorage capacity does not further increase, as a results of which it is important to anchor the surface bonded reinforcement in regions with low moments and hence with low FRP force demand. Alternatively, extra mechanical anchorages can be applied.

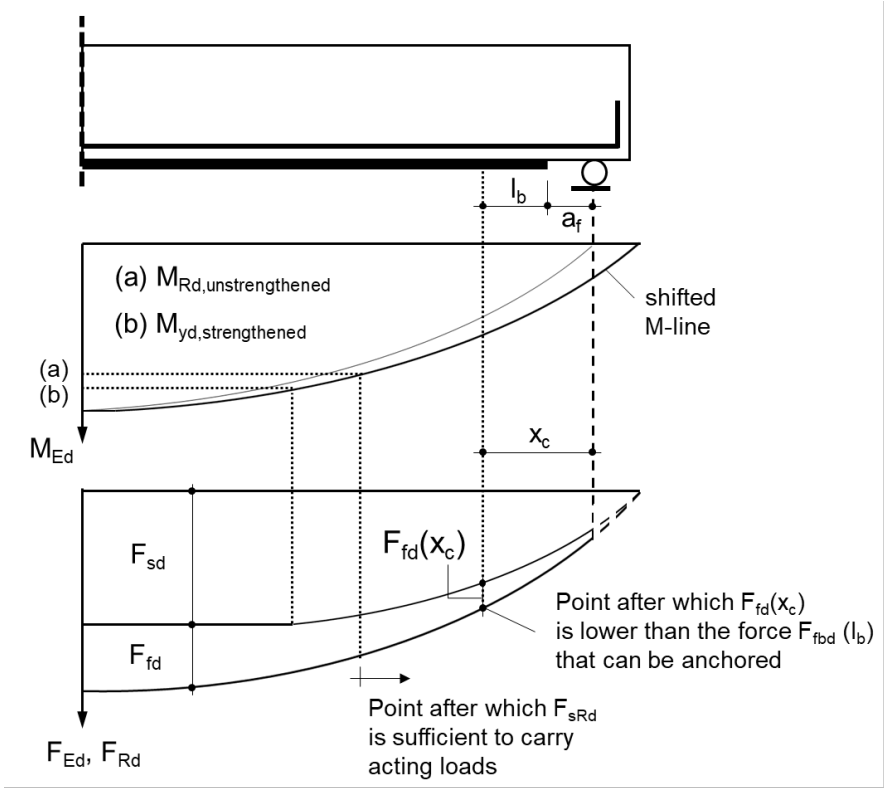

Figure 30.8-3: Schematic representation of the end debonding analysis, assuming a fully cracked state.

The surface bonded reinforcement shall be curtailed as follows:

- the point of curtailment shall be determined along a shifted moment line
- the point of curtailment shall consider the force demand in the FRP at that point and the ability to provide sufficient additional bond anchorage length to anchor the FRP at that point;
- in case the bonded reinforcement is also required for deflection control, it shall be curtailed outside the cracked region of the member.

Similar to curtailment for internal reinforcement, the curtailment of surface bonded reinforcement is evaluated assuming the fully cracked state (Figure 30.8-3). This verification is composed by the evaluation of the maximum distance $a_f + l_b$ from the support, at which the FRP may be applied without risk of debonding. The tensile stress in the FRP in this section, σ_f, should be less than the resisting stress $f_{fbd}(l_b)$ defined in section 22.2.1.4 (Eq. (22.2-6)). The stress in the FRP is evaluated for the acting moment M_{Ed} at the location $(a_f + l_b)$ and considering a shifted moment line. This is done by cross-sectional analysis similar as described in section 30.8.2.1.2(2).

As an improved level of approximation method, the point of curtailment can be calculated by means of a more refined analysis in reference to end anchorage at an arbitrary concrete element between 2 cracks (section 6.2.1.4.2 of *fib* Bulletin 90 [30.8-1]).

The curtailment point should further fulfil the following conditions:

- The point of curtailment should be in the zone where the resisting tensile force of the internal reinforcement (unstrengthened cross-section) is sufficient to carry the acting moment in the strengthened state (as indicated in Fig. 30.8-3).
- In case the externally applied reinforcement is also needed for deflection control, the FRP should span at least the zone where concrete can be cracked, meaning that l_b is in an uncracked zone.

(5) Debonding verification: end anchorage cover separation

The resistance of the strengthened member associated to concrete rip-off cover separation at the end of the surface bonded reinforcement, and its prevention thereof, is designed according to section 22.2.1.5.

Differently from EBR (see section 30.8.2.1.2(2)), and given the higher bond capacity of NSM, as a starting point the full contribution of the NSM reinforcement can be assumed, up to an FRP stress level $\eta \leq 0{,}8$.

The effective depth with respect to the FRP reinforcement is defined as $d_f = h - (t_s - b_f/2)$, with t_s the depth of the groove (Figure 30.8-4).

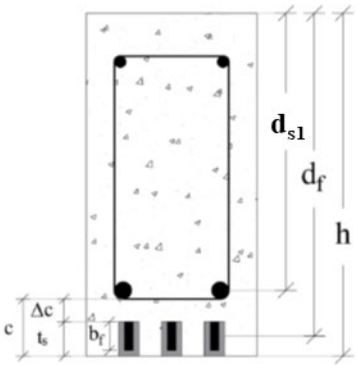

Figure 30.8-4: Variables at the level of the cross section.

30.8.2.1.3 Members with NSM subjected primarily to bending

In case of NSM FRP reinforcement, the same principles as for EBR (section 30.8.2.1.2) are adopted, in further reference to the NSM bond provisions given in section 22.2.2.

For the resisting moment of the strengthened cross-section the design shall be based on a maximum stress in the FRP equal to:

$$\sigma_f \leq f_{fd} \qquad (30.8\text{-}5)$$

where f_{fd} is the design value of the FRP tensile strength, as defined in section 17.8.1.1 and considering a value $\eta \leq 0{,}8$.

In terms of debonding verification the provisions of sections 22.2.2.2 and 22.2.2.3 apply.

The curtailment of the NSM reinforcement can be conducted according to section 30.8.2.1.2(2) and Figure 30.8-3, whereby the tensile stress in the FRP at the location $(a_f + l_b)$ should be less than the resisting stress $f_{fbd}(l_b)$ defined in section 22.2.1.4 (Eq. (22.2-6)).

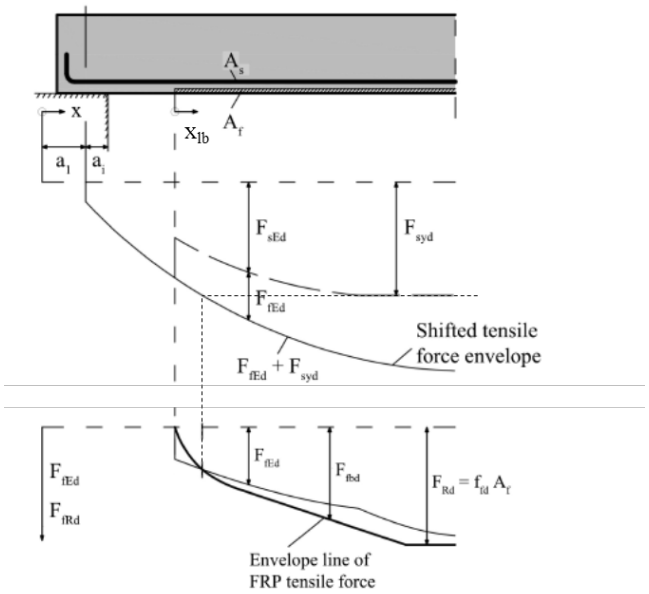

Figure 30.8-5: End debonding analysis for NSM strips.

Given the different debonding failure aspects which can occur along the length of the NSM, alternatively, an integrated approach can be followed which considers the envelope line of the FRP force, as indicated in Figure 30.8-5:

$$F_{fEd} \le F_{fRd} \tag{30.8-10}$$

where F_{fEd} is the acting force in the FRP and F_{fRd} is the resisting force along the length of the NSM:

$$F_{fRd} = n_f F_{fbd} \le f_{fd} A_f \tag{30.8-11}$$

with n_f the number of NSM bars, A_f the total cross-sectional area of NSM reinforcement and F_{fbd} the resisting tensile force in an NSM rebar considering debonding. For the latter, a capacity model has been tentative proposed, in reference to the notations in Figures 30.8-4 and 30.8-5:

$$F_{fbd} = \begin{cases} 0.95 b_f \tau_{b1d} \sqrt[4]{a_r x_{lb}} \left(0.4 - 0.0015 x_{lb}\right) & for \quad x_{lb} \le 115 \text{ mm} \\ 0.95 b_f \tau_{b1d} \sqrt[4]{a_r x_{lb}} \left(26.2 - 0.065 \tanh\left(\frac{a_r}{70}\right)\left(x_{lb} - 115\right)\right) & for \quad x_{lb} > 115 \text{ mm} \end{cases} \tag{30.8-12}$$

with the units in N and mm. In this equation a_r is the distance from the longitudinal axis of the reinforcement to the free edge of the beam's cross section in mm (≤150mm), while τ_{b1d} is the design bond strength:

$$\tau_{b1d} = \frac{1}{\gamma_{fb}} min\left\{a_{ba}\tau_{bak}; a_{bc}\tau_{bck}\right\} \tag{30.8-13a}$$

in which a_{ba} is a product specific factor for long-term behaviour of the adhesive in accordance with testing approvals (=0.5 for pre-dimensioning), and a_{bc} is a product specific factor for long-term behaviour of the concrete in accordance with testing approvals (=0.9 for pre-dimensioning). In Eq. (30.8-13a), τ_{bak} and τ_{bck} are the characteristic bond strength of the adhesive and FRP and of the concrete with the adhesive:

$$\tau_{bak} = k_{sys} \sqrt{\left(2 f_{atk} - 2\sqrt{f_{atk}^2 + f_{ack} f_{atk}} + f_{ack}\right) f_{atk}} \tag{30.8-13b}$$

$$\tau_{bck} = k_{bck} \sqrt{f_{cm}} \tag{30.8-13c}$$

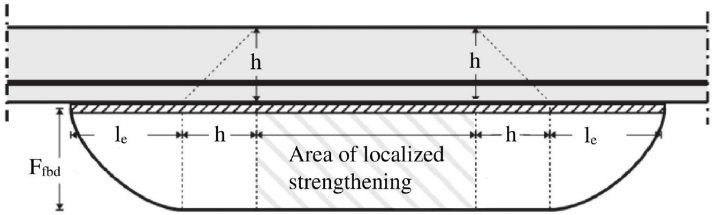

Figure 30.8-6: Externally bonded FRP for localised strengthening.

The confinement reduces the lateral expansion, resulting in a confining stress that enhances both the ultimate compressive strength and strain of concrete.

Differently from steel based confinement, FRP deforms elastically up to very high stresses exerting an increasing confining action on concrete until the fibres rupture.

Different constitutive laws have been proposed in the scientific literature as well as in the technical recommendations. Among them, the Lam and Teng model can be adopted. Figure 30.8-7 compares this FRP confined concrete model with the parabolic-rectangular concrete stress block which can be adopted for unconfined concrete.

where f_{atk} and f_{ack} are the characteristic value of the tensile and compressive strength of the adhesive, respectively, in MPa; k_{sys} is a product specific factor for the adhesive bond failure in accordance with testing approvals (=0.6 for pre-dimensioning), and k_{bck} is a product specific factor for the concrete bond failure in accordance with testing approvals (=4.5 for pre-dimensioning).

30.8.2.1.4 Localized flexural strengthening interventions

For localised flexural strengthening interventions (Figure 30.8-6), the bond length on both sides of the region requiring strengthening shall be equal to at least the member height plus the effective anchorage length (l_e). In the absence of a more refined calculation, the maximum tensile force carried out by the FRP in this localised strengthened region is:

$$F_{fbd} = b_f t_f f_{fbd} \qquad (30.8\text{-}14)$$

Where: the effective bond length (l_e) and the corresponding bond capacity (f_{fbd}) are considered as defined in chapter 22.

30.8.2.1.5 Strengthening columns through confinement

FRP wrapping may be used to confine columns, activating a multiaxial state of stress that can increase axial load capacity. It is worth noticing that it can increase the shear capacity too (see next section). Depending on the induced confinement level, the wrapped concrete can benefit of a considerable increase of the ultimate axial strain and its strength can increase as well.

(1) Constitutive law for FRP-confined concrete

The stress-strain relationship ($\sigma_c - \varepsilon_c$) consists of two branches:

1) nonlinear for axial strains lower than $\varepsilon_{c2,c}$, whereas the latter is slightly greater than the corresponding strain $\varepsilon_{c2} \approx 0{,}002$ for unconfined concrete;

2) linearly increasing up to the ultimate axial strain $\varepsilon_{cu,c}$.

$$\sigma_c = E_c\varepsilon_c - \frac{(E_c - E_2)^2}{4f_{cd}}\varepsilon_c^2 \qquad \text{for } 0 \le \varepsilon_c \le \varepsilon_{c2,c} \qquad (30.8\text{-}15a)$$

$$\sigma_c = f_{cd} + E_2\varepsilon_c \qquad \text{for } \varepsilon_{c2,c} \le \varepsilon_c \le \varepsilon_{cu,c} \qquad (30.8\text{-}15b)$$

where

f_{cd} the design value of the unconfined concrete strength

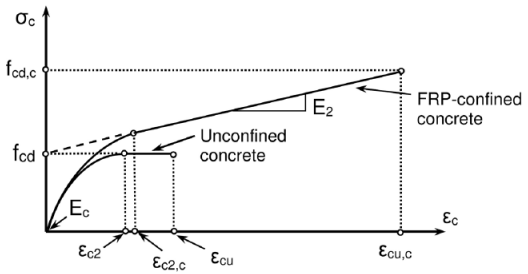

Figure 30.8-7: Stress-strain model for FRP-confined concrete in compression.

For square or rectangular sections, corners should be rounded so to introduce the confinement pressure in proper way (Figure 30.8-8) and to increase the strengthening lay-out efficiency factor α_f given in section 30.8.2.1.5(2). Preferably, the radius shall not be less than 20 mm.

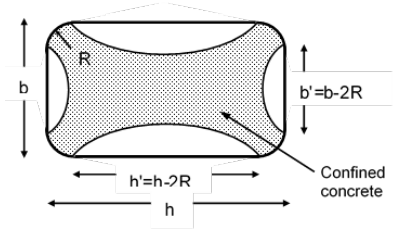

Figure 30.8-8: *Dimensional parameters for confinement of rectangular cross-section.*

$\varepsilon_{c2,c}$ defined by equalising Eqs. (30.8-15a-b) and solving it for $\varepsilon_c = \varepsilon_{c2,c}$

E_2 is the slope of the 2nd branch of the curve, defined as:

$$E_2 = \frac{f_{cd,c} - f_{cd}}{\varepsilon_{cu,c}} \qquad (30.8\text{-}15c)$$

The confined concrete behaviour and strength is mainly dependant on the lateral confining pressure:

$$f_{ld}^* = \frac{2t_f}{D^*} f_{fd} \qquad (30.8\text{-}16a)$$

where the fibre thickness (t_f) that can be evaluated as function of the thickness of one layer (t_o) and the number (n) of the layers, as follows: a) $t_f = nt_o$ for $n = 1, 2\ or\ 3$, and b) $t_f = n^{0.85} t_o$ for $n \geq 4$; and D^* is the diameter of the circular section or the equivalent diameter of rectangular sections with same cross-sectional area; and f_{fd} is the design value of the effective FRP ultimate strength, as defined in section 17.8.1.1 and where an effective strain factor $\eta = \eta_l^*$ should be considered to account for the duration of loading, environmental effects, possible creep effects in the concrete, effect of buckling of the longitudinal steel reinforcement, and effect of radius of rounded corners. If no detailed data are available and a more rigorous analysis is not possible, the reduction factor η_l^* may be taken equal to:

$$\eta_l^* = \begin{cases} 0.5\left[\frac{R}{50}\left(2 - \frac{R}{50}\right)\right] & R < 50\ mm \\ 0.5 & R \geq 50\ mm \end{cases} \qquad (30.8\text{-}16b)$$

with R the radius of rounded corners for rectangular sections.

As a function of lateral confining pressure (f_{ld}^*), the ultimate values of both axial strain ($\varepsilon_{cu,c}$) and stress ($f_{cd,c}$) are defined as follows:

$$\frac{\varepsilon_{cu,c}}{\varepsilon_{c2}} = 1.75 + 12\sqrt{\frac{h}{b}}\left(\alpha_f \frac{f_{ld}^*}{f_{cd}}\right)\left(\frac{\varepsilon_{fu,h}}{\varepsilon_{c2}}\right)^{0.45} \qquad (30.8\text{-}17a)$$

$$\begin{aligned} \frac{f_{cd,c}}{f_{cd}} &= 1 + 3.3\left(\alpha_f \frac{f_{ld}^*}{f_{cd}}\right)\left(\frac{b}{h}\right)^2 && if\ \ \alpha_f \frac{f_{ld}^*}{f_{cd}}\left(\frac{b}{h}\right)^2 \geq 0.07 \\ \frac{f_{cd,c}}{f_{cd}} &= 1 && if\ \ \alpha_f \frac{f_{ld}^*}{f_{cd}}\left(\frac{b}{h}\right)^2 < 0.07 \end{aligned} \qquad (30.8\text{-}17b)$$

where dimensional parameters are as defined in Figure 30.8-8, considering $b/h = 1$ for circular cross-sections, and α_f the strengthening lay-out efficiency factor given in section 30.8.2.1.5(2).

The lateral pressure by the FRP (f_{ld}^{*}) is most ideal in the case of a circular cross-section and being fully wrapped in an orthogonal configuration (tensile fibre direction perpendicular to the longitudinal axis of the column). For non-circular cross-sections as indicated in Figure 30.8-8, for partial wrapping (e.g. to still allow exchange of moisture over the concrete surface) as indicated in Figure 30.8-9a for helicoidal wrapping or Figure 30.8-9b for orthogonal wrapping; the lateral pressure is less effective. This is considered by an effectiveness factor applied to the lateral pressure ($\alpha_f f_{ld}^{*}$).

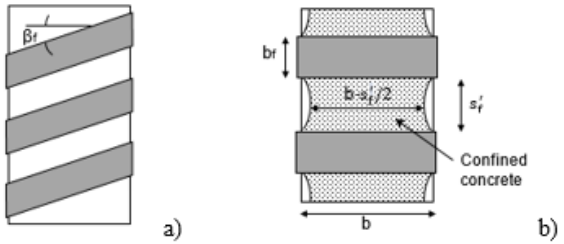

Figure 30.8-9: Confining discrete strips: (a) helically applied, (b) orthogonal applied.

In columns under predominant flexure (with very low axial loads) the effect of confinement, in terms of strength increases, is minimal, as illustrated qualitatively by the bending moment-axial load interaction diagram reported in Figure 30.8-10.

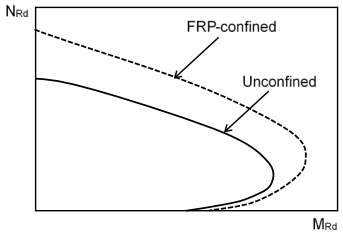

Figure 30.8-10: Qualitative bending moment – axial load interaction curves for FRP-confined columns.

(2) Strengthening lay-out efficiency factor

The effectiveness of the confinement depends on the shape (and slenderness ratio b/h), being fully or partially wrapped and the angle of wrapping (orthogonal strips versus helicoidal wrapping):

$$\alpha_f = \alpha_n \alpha_s \alpha_a \tag{30.8-18a}$$

with α_n the effect of a rectangular section, α_s the effect of partial wrapping, and α_a the effect of the angle of the main tensile direction of the FRP with respect to the longitudinal axis of the column:

$$\alpha_n = 1 - \frac{(b'-2R)^2 + (h-2R)^2}{3bh} \tag{30.8-18b}$$

$$\alpha_s = \left(1 - \frac{s_f'}{2b}\right)\left(1 - \frac{s_f'}{2h}\right) \tag{30.8-18c}$$

$$\alpha_a = \frac{1}{1 + (\tan \beta_f)^2} \tag{30.8-18d}$$

where all dimensional parameters are as defined in Figures 30.8-8 and 30.8-9.

(3) Section capacity calculation

Once the stress-strain model, for FRP-confined concrete, is defined, the verification of FRP-confined columns can be carried out based on traditional analytical or numerical methods (e.g. Section 30.1.2.1).

For slender columns, section capacity can be limited by second order effects that need to be take in account.

In the case of combinations of axial force and flexural moment, where the flexural moment prevails, it is recommended to apply rather flexural strengthening or a combination of both, as indicated in section 30.8.2.1.6.

In case of seismic actions, the most relevant contribution of the confinement concerns the increase of the strain capacity, which improves the ultimate curvature of the section. Design calculations shall be conducted according to section 30.3.

The main benefit of the longitudinal FRP is to increase the flexural resistance of the member, whereas the FRP wrapping confines the concrete, increases the bond strength between FRP and concrete and prevents buckling of the externally applied longitudinal FRP reinforcement. Transverse FRP is also essential for shear, which prerequisites for supporting the increased flexural strength.

Figure 30.8-11 emphasizes that if there is no confinement, flexural strengthening of RC columns with externally applied longitudinal FRP reinforcement is effective only at low levels of axial load. If, on the other hand, confinement is present, flexural strengthening is possible even at high levels of axial load.

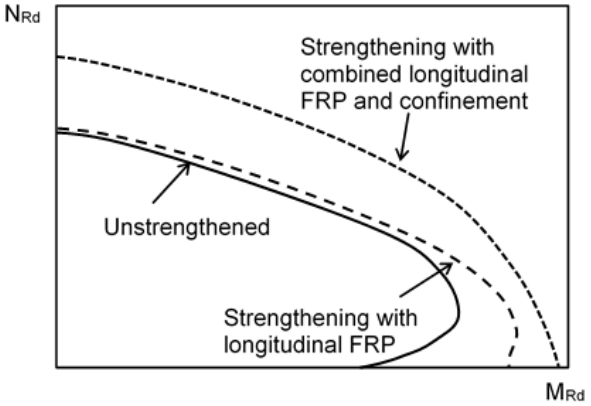

Figure 30.8-11: Qualitative bending moment – axial load interaction curves for columns strengthened in flexure.

The member shear capacity can be increased by applying external stirrups (shear links) by adhesive bonding. The additional reinforcement bridges the shear crack and enhances the shear capacity. Such enhancement is maximised if the reinforcement forms a closed force system. Open systems may also be used as long as the reduced effectiveness due to debonding is taken into account.

30.8.2.1.6 Strengthening columns in flexure

The design of columns strengthened in flexure with longitudinal FRP, with or without confinement, can be carried out from first principles, using the conventional cross section analysis. The analysis of the capacity can be facilitated through interaction diagrams, which have the general form given in Figure 30.8-11.

It should be emphasized that if flexural enhancement is required at the column ends and not within the span, the longitudinal FRP must be sufficiently anchored at the column ends, such as exemplary indicated in Figure 30.8-12.

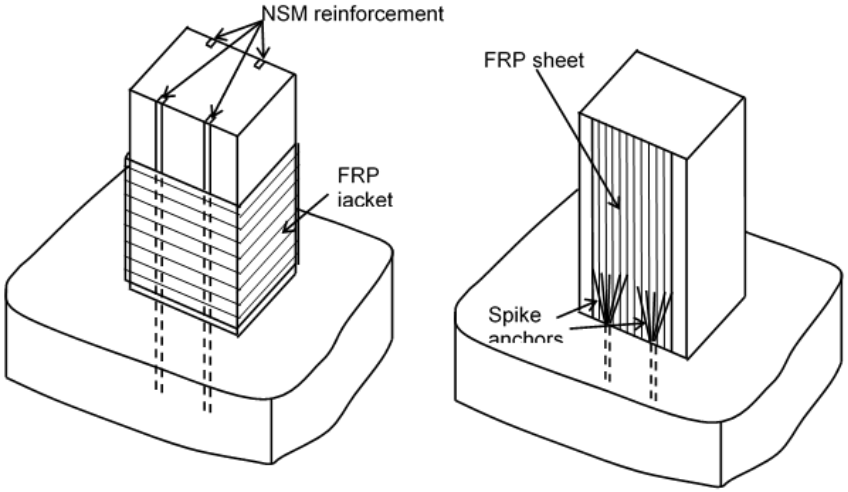

Figure 30.8-12: Flexural strengthening with NSM (left) or EBR (right) - wrapping may be applied in both cases.

30.8.2.2 Shear

The shear resistance of a member with shear strengthening may be obtained by superposition, whereby the design value of the shear force carried by the externally applied reinforcement ($V_{Rd,f}$) is added to the shear capacity of the member before strengthening. The latter is obtained by following the regulations from section 30.1.3.3 and whereby it is also verified that the shear strengthening is not exceeding the limit associated to crushing of the compressive field.

The effective thickness is calculated from the number of FRP layers placed in the jacket, n, and the thickness of a single layer, t_o, as $t_f = nt_o$ for $n = 1, 2\ or\ 3$, and as $t_f = n^{0.85}t_o$ for $n \geq 4$.

Maximum efficiency is obtained for external stirrups with closed configurations. For practical reasons, also 3-sided configurations are used, in which the external reinforcement encloses that part of the member with the main longitudinal tensile reinforcement. EBR shear strengthening configurations only applied to the side faces of the member should be avoided due to lack of effectiveness. Given its higher relative bond capacity, for NSM side face shear strengthening can be considered.

The additional contribution resulting from the shear strengthening is given as follow:

$$V_{Rd,f} = \frac{A_{fw}}{s_f} h_f f_{fwd} \left(cot\,\theta + cot\,\alpha\right) sin\,\alpha \qquad (30.8\text{-}19)$$

where

A_{fw} = area of externally bonded shear reinforcement measured perpendicular to the direction of the fibres (=$2b_f t_f$ in the case of symmetrically applied strips of width b_f and effective thickness t_f),

s_f = spacing of strips (measured in the direction of the member axis),

h_f = height of FRP crossed by the shear crack (taken equal to $h - 0.1d_{s1}$ in the case of full-depth FRP, and with d_{s1} the effective depth with respect to the longitudinal internal tensile reinforcement),

$cot\theta$ expressing the angle of the compression field (see section 30.1.3.3.1)

α = angle between main tensile direction of the external stirrups and the member axis perpendicular to the shear force,

f_{fwd}= design value of the tensile stress in the FRP intersected by the shear crack in the ultimate limit state (effective tensile strength).

The value of f_{fwd} depends on the configuration of the externally applied reinforcement, and is defined as follows.

For closed FRP or 3-sided FRP anchored in the compression zone:

$$f_{fwd} = k_a f_{fd} \qquad (30.8\text{-}20a)$$

with k_a a reduction factor associated to the mechanical anchorage, as defined in section 20.3 (for closed systems and sufficient overlap $k_a = 1$; for 3-sides configurations with mechanical anchorage $k_a \leq 0{,}9$), and f_{fd} the design value of the effective FRP ultimate strength, as defined in section 17.8.1.1 and where in the absence of more detailed information an effective strain factor $\eta \leq 0{,}5$ is considered according to Eq. (30.8-16b).

For 3-sided FRP (U-configuration), without extra anchorage, the effective FRP stress is also determined by possible debonding:

$$f_{fwd} = min\left\{f_{fbwd}, f_{fd}\right\} \qquad (30.8\text{-}20b)$$

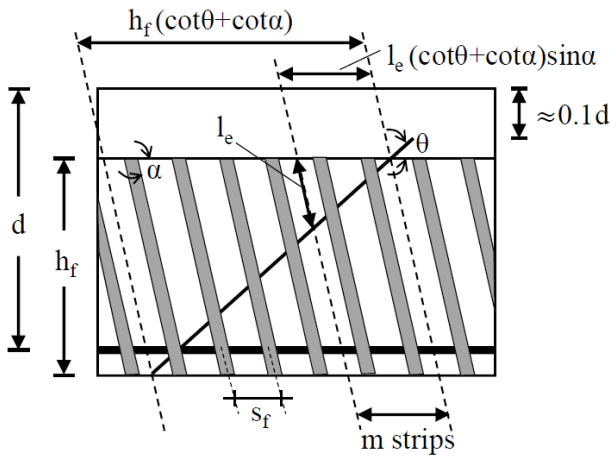

Figure 30.8-13: Strips crossed by the shear crack.

The case of full area bond (continuous FRP sheets) may be treated as a special case of FRP strips, with $s_f = b_f/sin\alpha$, so that Equations 30.8-21a till 21c become:

$$f_{fbwd} = \left[1 - \frac{1}{3}\frac{l_e}{(h_f/sin\alpha)}\right]\frac{f_{fbk}}{\gamma_{fb}} \qquad \text{for } h_f/sin\alpha \geq l_e \qquad (30.8\text{-}21d)$$

$$f_{fbwd} = \frac{2}{3}\frac{(h_f/sin\alpha)}{l_e}\frac{f_{fbk}}{\gamma_{fb}} \qquad \text{for } h_f/sin\alpha < l_e \qquad (30.8\text{-}21e)$$

The external torsional reinforcement may consist of continuous spirals (strips with a spacing s_f) or closed loop straps with an angle α = 90° with respect to the longitudinal axis of the member (and with $s_f = b_f$ in case of full area bond). By preference such closed configurations are used, in reference to f_{fwd} as defined by Eq. (30.8-21a). Other torsional strengthening configurations with EBR or NSM, adapted to the specific torsional crack pattern to be bridged, may also be realized when the dimensions of the cross section is sufficiently high for assuring enough resisting bond length; however such a system requires confirmation by testing.

with f_{fd} as defined for closed FRP systems, and with f_{fbwd} the design value of the FRP resistance corresponding to debonding (Figure 30.8-13):

– for $h_f/sin\alpha \geq l_e$ and $l_e \leq s_f/[(\cot\theta + \cot\alpha)\sin\alpha] \leq h_f/\sin\alpha$, i.e. if all strips intersected by the shear crack have bond length ≥ l_e:

$$f_{fbwd} = \frac{f_{fbk}}{\gamma_{fb}} \qquad (30.8\text{-}21a)$$

– for $h_f/sin\alpha \geq l_e$ and $s_f/[(cot\theta + cot\alpha)sin\alpha] < l_e$, i.e. if some of the strips intersected by the shear crack have bond length ≥ l_e and some others have bond length < l_e:

$$f_{fbwd} = \left[1 - \left(1 - \frac{2}{3}\frac{m \cdot s_f}{l_e(cot\theta + cot\alpha)sin\alpha}\right)\frac{m}{n}\right]\frac{f_{fbk}}{\gamma_{fb}} \qquad (30.8\text{-}21b)$$

– for $h_f/sin\alpha < l_e$ and $s_f/[(\cot\theta + \cot\alpha)\sin\alpha] \leq h_f/\sin\alpha$, i.e. if all strips intersected by the shear crack have bond length < l_e:

$$f_{fbwd} = \frac{2}{3}\frac{n \cdot s_f}{l_e\left[(cot\theta + cot\alpha)sin\alpha\right]}\frac{f_{fbk}}{\gamma_{fb}} \qquad (30.8\text{-}21c)$$

where, in reference to the notations explained in Figure 30.8-13, n = number of strips crossed by the shear crack, evaluated as the integer of $h_f(cot\theta + cot\alpha)/s_f$, m = number of strips for which the bond length is less than l_e, evaluated as the integer of $l_e(cot\theta + cot\alpha)sin\alpha/s_f$; and where l_e is the effective bond length and f_{fbk} is characteristic maximum bond strength, as defined in chapter 20.

30.8.2.3 Torsion

Similar as for shear strengthening (section 30.8.2.2), is the resistance of a member with torsional strengthening obtained by superposition, whereby the design value of the torsional resistance carried by the externally applied reinforcement ($T_{Rd,f}$) is added to the resistance of the member before strengthening (section 30.1.4).

For members of rectangular cross section with dimensions b and h, the contribute of the externally applied FRP reinforcement to the torsional resistance is:

$$T_{Rd,f} = \frac{2b_f t_f}{s_f} b \cdot h \cdot f_{fwd} cot\theta \qquad (30.8\text{-}22)$$

where f_{fwd} is the design value of the average stress in the FRP intersected by the diagonal crack in the ultimate limit state (effective strength), calculated from Eq. (30.8-21a) (for closed loop strengthening configurations).

In case of combined torsion, T_{Ed}, and shear, V_{Ed}, the following limitation shall be met:

$$\frac{T_{Ed}}{T_{Rd,max}} + \frac{V_{Ed}}{V_{Rd,max}} \leq 1.0 \qquad (30.8\text{-}23)$$

where $T_{Rd,max}$ and $V_{Rd,max}$ may be determined according to Eq. (30.1-84) and Eq. (30.1-25).

30.8.2.4 Punching

When determining the punching shear resistance of the concrete in accordance with section 30.15, the bonded reinforcement shall not be considered in determining the area of the longitudinal tensile reinforcement A_{sl}, unless experimental verification of the specific FRP system under consideration suggests differently.

In case punching strengthening is needed, specific FRP systems and punching shear resistance formulations should be applied, for which reference is made to the relevant literature and design by testing.

30.8.2.5 Design with strut and tie models

Only the force in anchored externally bonded reinforcement may be used as the tie force in such reinforcement.

30.8.3 Evaluation of structural safety (ULS) for non-static loading

FRP materials generally have good fatigue resistance (chapter 17), so that the fatigue resistance of the internal steel is governing. The fatigue analysis shall be conducted accordingly, whereby:

– the acting stresses in the internal reinforcement should be considered for the non-static loading acting on the member after strengthening;

Local loss of FRP-concrete bond initiated at positions of peak moment (e.g. at crack positions) may lead to higher stresses in the steel reinforcement as the number of cycles increases, and will impact the fatigue verification. For fatigue bond behaviour of externally applied reinforcement further refence is made to section 22.2.1.7.

Seismic retrofitting by means of externally bonded reinforcement is an effective measure to increase ductility and deformation capacity by suppressing premature failure modes that usually occur in members with absence of proper seismic detailing.

For the evaluation of structural safety for extreme thermal conditions (fire but also temperature higher than the glass transition temperature of the resin, T_g) it is necessary to consider (i) the possible presence of fire protection systems, which can improve the fire endurance of the strengthening system, and (ii) the strengthening degree, because the FRP strengthening system could be necessary to sustain load at ULS and not necessary to sustain load expected during the accidental action.

Strengthened elements without fire protection

In this case, the strengthening of the concrete element will be lost very quickly due to the high temperature. The fire resistance may be evaluated using tabulated data (minimum dimensions and minimum concrete cover for different types of elements), or using a simplified calculation method (thermal analysis and mechanical analysis with reduced material properties and/or modified section geometry) or using a general calculation method (thermal and mechanical analysis with temperature dependent material constitutive laws). Particular attention should be paid to the value of the acting load in the accidental condition.

Strengthened elements with fire protection

– it should be verified that the bond stresses are in the linear elastic part of the stress-slip curve (as is normally the case for load ranges below 60% of the load at first yield).

In case of concerns, a more explicit consideration of the (bond) fatigue of the external reinforcement can be conducted in accordance to *fib* Bulletin 90 (Section 6.8) [30.8-1].

30.8.4 Seismic retrofitting

Design provisions for seismic retrofitting with externally applied reinforcement are incorporated in Section 30.3.

30.8.5 Evaluation of structural safety (ULS) for extreme thermal conditions

Proper fire design will consider evaluation of the individual FRP strengthened member, as well as reflect possible effects on the whole structure.

If no special measures (like protective insulating linings) are taken, the FRP strengthening system will be lost during fire due to the weakening of the adhesive. In such cases, fire resistance verification of the unstrengthened element in the accidental design situation must be performed, considering the relevant acting loads during fire (and given that these loads reflect the fact that the member is strengthened at the start of the fire scenario). The rules that are applicable to reinforced and prestressed concrete elements, may also be applied here (for evaluation of structural safety for extreme thermal condition reference is made to Section 30.4).

When fire protection is provided for a strengthened element, the fire resistance can be calculated by considering the presence of the FRP strengthening system, but a refined calculation method should be used, to consider the temperature dependent material properties, and the effectiveness of the protecting system. This is requires a detailed performance-based fire design. Alternatively, dimensioning of the protection system that guarantee the limitation for the temperature rise, during a certain time could be sufficient, when traditional fire safety approaches are used (e.g. single members analyses).

For moderate strengthening degrees, the strengthening is not needed in an accidental design situation. Hence, after checking this, the fire protection system mainly focuses on the concrete member to limit temperatures in the concrete and steel.

For high strengthening degrees, the fire protection system mainly focuses to limit temperatures to ensure bond interaction between the concrete and the strengthening system in the event of fire. Fire insulation schemes should be properly designed, and their effectiveness should be proven.

For performance-based fire design of FRP-strengthened structures reference is made to the specialized literature (e.g. Bisby and Stratford [30.8-2], Palmieri et al. 2013 [30.8-3], Del Prete et al. [30.8-4]).

30.8.6 Evaluation of serviceability (SLS)

30.8.6.1 Stresses

Under service load conditions it is required to limit stresses in the concrete, steel and FRP to prevent damage or excessive creep of the concrete and adhesives, steel yielding and excessive creep or stress rupture of the FRP.

The tensile creep rupture strength, follows from creep rupture curves established at exposure conditions relevant for the application.

Calculations to verify the serviceability limit state in the case of flexural strengthening may be performed according to a linear elastic analysis of both the

(1) Stress limitations in the materials

The stress limitations for the concrete and internal steel, as provided in Section 30.5, shall be applied.

As a result of the internal steel yield stress limitations and considering the strain compatibility, the stress in the FRP (EBR or NSM) under the characteristic combination of loading is likewise limited to:

$$\sigma_f \leq 0.8 f_{yk} \frac{E_f}{E_s} \qquad (30.8\text{-}24a)$$

and, when the stress is caused by an imposed deformation, to:

$$\sigma_f \leq f_{yk} \frac{E_f}{E_s} \qquad (30.8\text{-}24b)$$

Most importantly, to avoid tensile creep rupture, the stress in the FRP under quasi-permanent load combination shall be limited. This condition is verified according to section 17.8.1.2.

(2) Calculation of stresses in case of flexural strengthening

The acting stresses can be determined with reference to the presence of a moment M_o acting on the section before strengthening, and an additional moment M applied

uncracked (state 1) and the cracked sections (state 2). Formulations are applicable for both EBR and NSM reinforcements.

The neutral axis x_{o2} and the moment of inertia I_{o2} of the unstrengthened section are taken as defined in section 30.8.2.1.2(1). Likewise, the neutral axis x_2 and the moment of inertia I_2 of the strengthened cracked section are given by:

$$\frac{1}{2}bx_2^2 + (\alpha_s - 1)A_{s2}(x_2 - d_{s2}) - \alpha_s A_{s1}(d_{s1} - x_2) - \alpha_f A_f(d_f - x_2) = 0 \quad (30.8\text{-}27a)$$

$$I_2 = \frac{1}{3}bx_2^3 + (\alpha_s - 1)A_{s2}(x_2 - d_{s2})^2 + \alpha_s A_{s1}(d_{s1} - x_2)^2 + \alpha_f A_f(d_f - x_2)^2 \quad (30.8\text{-}27b)$$

where, the modular ratios α_s and α_f take into account the evolution of creep phenomena in the concrete, distinguishing between verifications under short term and long loading period, for the relevant service load combinations, as also further elaborated in section 30.8.6.3.

To protect the internal steel reinforcement from corrosion and to guarantee the functionality of the member, the crack width limitation provisions associated to unstrengthened members remain valid.

after strengthening under the design service loading conditions. Thus, the total maximum moment applied on the strengthened section is:

$$M_k = M_o + M \quad (30.8\text{-}25)$$

If the moment M_o is higher than the cracking moment, M_{cr}, of the section, the analysis can be done referring to the cracked inertia (state 2) before and after the application of the FRP reinforcement. The stresses can be evaluated adding the contributions of the two moments M_o and M according to the following expressions:

$$\sigma_c = \sigma_{c,M_o} + \sigma_{c,M} = \frac{M_o}{I_{o2}}x_{o2} + \frac{M}{I_2}x_2 \quad (30.8\text{-}26a)$$

$$\sigma_s = \sigma_{s,M_o} + \sigma_{s,M} = \alpha_s\frac{M_o}{I_{o2}}(d_{s1} - x_{o2}) + \alpha_s\frac{M}{I_2}(d_{s1} - x_2) \quad (30.8\text{-}26b)$$

$$\sigma_f = \sigma_{f,M} = \alpha_f\frac{M}{I_2}(d_f - x_2) \quad (30.8\text{-}26c)$$

where, reference is made to section 30.8.2.1 for the applied notations, I_{o2} and I_2 are the moment of inertia of the unstrengthened and strengthened cracked section, respectively, x_{o2} and x_2 are the depth of the compression zone of the the unstrengthened and strengthened cracked section, respectively, and α_s and α_f are the modular ratio of the steel and the FRP reinforcements, respectively.

If the moment at the strengthening intervention, M_o, is lower than the cracking moment, M_{cr}, but the total moment M_k is higher than M_{cr}, the moment of inertia I_1 of the uncracked section has to be used instead of the inertia I_{o2} in Eqs. (30.8-26).

If both M_o and M_k are lower than M_{cr}, the loading condition is not critical in terms of SLS, though it should be noted that this case will normally not occur in typical applications.

30.8.6.2 Cracking

(1) Crack width limitation

The crack limitations for reinforced or prestressed concrete members as provided in section 30.5.2.4.3 remain applicable to the strengthened members.

For members strengthened in flexure with EBR or NSM, new cracks will appear in between existing cracks due to the additional tensile stress transfer to the concrete provided by the external reinforcement. Hence, more diffuse crack patterns with smaller crack widths are observed, which often makes the verification of crack width not necessary.

The assessment of the crack width considers that both the internal and external reinforcement are combinedly bridging the crack.

Depending on the desired level of approximation, different formulations for the assessment of the crack spacing and average strains can be applied. Further to the simplified approach presented in this section (based on statistical treatment of data obtained from tests for externally bonded reinforcement, and whereby it is assumed that the internal longitudinal reinforcement bars are reasonably closely spaced next to each other (spacing ≤ 5(c_{s1}+ϕ_s/2), reference is made to *fib* Bulletin 90 (Section 7.3) [30.8-1], for alternative calculation methods.

The coefficients k_i in Eq. (30.8-29b) were calibrated (Ceroni and Pecce, [30.8-5]) considering 40 experimental results, whereby the elastic modulus and the thickness of the FRP reinforcements used in the collected tests varied in the following ranges: E_f = 65-230 GPa, t_f = 0.100-0.167 mm, while the average compressive strength of concrete f_{cm} varied between 29-92 MPa.

(2) Calculation of the crack width

The characteristic value of the crack width of an EBR or NSM strengthened member can be evaluated following a similar approach as for unstrengthened members:

$$w_k = s_{r,max}\left(\varepsilon_{rm} - \varepsilon_{cm}\right) \qquad (30.8\text{-}28)$$

where $s_{r,max}$ is the crack spacing and $(\varepsilon_{rm} - \varepsilon_{cm})$ is the difference between the average values of tensile strain in the tensile reinforcements, ε_{rm}, and in the concrete, ε_{cm}.

The crack spacing $s_{r,max}$ considers the upper bound 95% characteristic value:

$$s_{r,max} = s_{rk,95\%} = k \cdot s_{rm} \qquad (30.8\text{-}29a)$$

where, k is a coefficient experimentally assessed (proposed value $k = 1{,}6$) and s_{rm} is the average value of crack spacing. The mean crack spacing s_{rm}, can be calculated as:

$$s_{rm} = k_1 + k_2 \frac{A_{c,ef}^{k3} \cdot \phi_s}{A_s^{k4} + \left(\frac{A_f E_f}{E_s}\right)^{k5}} \qquad (30.8\text{-}29b)$$

Where:

c_{s1} is the concrete cover on the internal longitudinal steel reinforcement

ϕ_s diameter of the internal steel reinforcement

E_s elastic modulus of internal steel reinforcement,

A_f the area of the external FRP reinforcement, expressed in mm^2

E_f the elastic modulus of the external FRP reinforcement, expressed in MPa

$A_{c,ef}$ is the effective area of concrete in tension surrounding the group of steel reinforcement bars (Eq. 30.5-10, Figure 30.5-5).

and whereby the coefficients k_i in Eq. (30.8-29b) can be evaluated by means of statistical analysis of experimental results. For FRP EBR reinforcements, these values can be taken as $k_1 = 20$ mm, $k_2 = 4$, $k_3 = 0.5$, $k_4 = 0.75$, $k_5 = 0.75$.

The strain difference $(\varepsilon_{rm} - \varepsilon_{cm})$ in Eq. (30.8-28) can be calculated similar as for internal steel reinforcement (section 30.5.2.4.4.2, Eq. 30.5-11):

$$\varepsilon_{rm} - \varepsilon_{cm} = \frac{\sigma_s - \beta_{TS}\sigma_{r,ef}}{E_s} \geq \frac{\sigma_s}{E_s}\left(1 - \beta_{TS}\right) \qquad (30.8\text{-}30a)$$

$$\sigma_{r,ef} = \frac{f_{ctm}}{\rho_{ef,eq}}\left(1 + \alpha_e \rho_{ef,eq}\right) \qquad (30.8\text{-}30b)$$

Where:

σ_s is the tensile stress in the steel reinforcement according to section 30.8.6.1,

β_{TS} is a factor dependent on the duration of the load (0.6 for short term and 0.4 for long term loading),

α_e is the modular ratio (E_s/E_c)

$\rho_{ef,eq}$ is the total equivalent reinforcement ratio with respect to the effective area of concrete in tension, calculated as:

$$\rho_{ef,eq} = \frac{A_s}{A_{c,ef}} + \frac{E_f}{E_s}\frac{A_f}{A_{c,ef}} \qquad (30.8\text{-}31)$$

30.8.6.3 Deformation control

(1) Deformation limitation

Deflections in strengthened flexural members should fulfil the deflection limitations as set forth in section 30.5.2.5.1.1 and whereby care should be taken to ensure that the limits are appropriate for the particular structure considered or that specific requirements, such as in ISO 4356 [30.8-6], are applied.

Small amounts of EBR or NSM FRP reinforcement significantly increase the ultimate load and, thus, small cross-sectional areas A_f are needed to satisfy the verifications at the ULS. This results in a relatively low axial stiffness $E_f A_f$ of FRP reinforcement, that is often insufficient to reduce curvatures and deflection in the strengthened member under the acting service load conditions. Hence, the required cross-sectional areas A_f might rather be governed by the SLS of deflection, over that of the ULS.

The tension stiffening effects in strengthened members due to both the internal steel and the FRP reinforcement can be evaluated with reference to the concrete element between two consecutive cracks, by assuming the presence of a local slip between both the internal steel and the external FRP and concrete A system of differential equations is obtained by means of equilibrium and compatibility conditions aimed to evaluate the strain and stress distribution at each cross section and, subsequently, the moment-curvature relation. Despite their accuracy, these procedures are complicated from a computational point of view and their use for design practice is difficult. At the other hand, it has been demonstrated that applying the approach provided in this section results in reasonably reliable predictions.

The flexural stiffness $E_c I$ and the modular ratios α_s and α_f take into account the evolution of creep phenomena in the concrete, distinguishing between verifications under short term and long loading period, for the relevant service load combinations. As such, the long term effects under quasi-permanent load combinations are considered in reference to $\alpha_{s\infty} = E_s/E_{c\infty}$, $\alpha_{f\infty} = E_f/E_{c\infty}$, $E_{c\infty} = E_c/(1+\varphi)$, and φ is the creep coefficient of the concrete (to be taken relevant for the age of the structure. This approach assumes that nonlinear creep phenomena are avoided, through the stress limitations of section 30.8.6.1 and given the fact that the long term behaviour is primarily governed by the creep of the concrete in compression (as bond stresses in the interface with the external reinforcement are normally low under service loads and in the linear elastic part of the bond stress-slip curve). In case of concerns, a more rigorous evaluation of long term deflections might be relevant.

Given that the strengthening is often applied to old concrete structures, creep is seldom a controlling factor in the dimensioning of the strengthened elements, unless these elements are relatively new, so that the concrete can still be expected to develop substantial creep deformations. In this respect, Eqs. (30.8-32) can be refined further, so to associate the most relevant flexural stiffness to the corresponding part of the action long term moment.

(2) Calculation of deflections

The approach to calculate the deflections as a tension stiffening based weighted average of deflections under the uncracked and cracked state, remain applicable for EBR or NSM strengthened members. In this respect the approaches given in section 30.5.2.5 can be applied. Hereby, the moment of inertia of the strengthened section is considered, as well as the existing deflection at the moment M_o when applying the FRP strengthening:

$$\delta_1 = k_M l^2 \frac{M_k}{E_c I_1} \qquad (30.8\text{-}32a)$$

$$\delta_2 = k_M l^2 \left(\frac{M_o}{E_c I_{02}} + \frac{M_k - M_o}{E_c I_2}\right) \quad \text{for } M_k > M_o \qquad (30.8\text{-}32b)$$

where, k_M is a coefficient depending on the load bearing system, and whereby the deflection of the uncracked section δ_1 can be considered for the total moment $M_k = M_o + M$ (given the limited influence of the external reinforcement on the moment of inertia of the uncracked section I_1), and the deflection of the cracked section δ_2 considers the moment of inertia of the section before strengthening I_{02} applied to M_o and the moment of inertia after strengthening I_2 applied to $M_k - M_o$. For the different parameters, reference is also made to sections 30.8.2.1.2(1) and 30.8.6.1(2).

The interface between the concrete and the extra reinforcement is the critical component to the effectiveness and durability of most structural strengthening applications, as this is the location where the transfer of stresses occurs. Next to considering environmental conditions, only compatible and correctly installed strengthening systems will be applied so not to hamper the durability performance.

A further discussion on durability considerations of EBR or NSM strengthened members is provided in *fib* Bulletin 90 (Chapter 4 and Chapter 10, Section 10.2) [30.8-1].

The externally applied reinforcement generally does not eliminate existing or latent problems such as steel corrosion, water leakage, high chloride values, etc, although under specific circumstances some of these problems are alleviated. These aspects should be verified. If needed the strengthening has to be preceded by concrete repair and internal steel protection techniques.

In the case of EBR a substantial part of the concrete surface might be covered by the bonded reinforcement, and influence the dampness of the member. Even if the adhesive ensures good bond conditions in humid environment, the dampness of the structure can still have other negative influences, such as risk of internal steel corrosion (involving spalling of the concrete cover), higher susceptibility to freeze/thaw damage of the concrete, etc., if no other special provisions are taken into account.

30.8.7 Evaluation of condition limit states (CLS) associated with durability

The limit state associated with durability is integrated in the design by means of the following considerations:

- an environmental factor η_e is considered in the design value of the FRP tensile strength (section 17.8.1.1);
- the ultimate limit state of creep rupture considers the acting environmental conditions (sections 17.8.1.2 and 27.8.6.1).

Furthermore, specific precautions are obliged to avoid durability issues, as follows:

- the glass transition temperature of the strengthening system (especially the bond interface) is such that the service temperature is limited to $T_g - 20°C$, in which T_g is considered as a lower bound value of the glass transition temperature range;
- the external FRP reinforcement is protected from UV radiation through protective paint (unless the FRP has proven UV resistance);
- direct contact between CFRP reinforcement and internal steel reinforcement is avoided, as this might enhance steel corrosion;
- where relevant, precautions are taken against possible occurrence of mechanical damage and vandalism.

Proper concrete repair is executed preceding the application of the strengthening by means of EBR or NSM.

In humid environment it is verified that the internal humidity of the structure (dampness) is not negatively affected by the bonded reinforcement and vice versa.

References

[30.8-1] *fib* Bulletin 90. Externally applied FRP reinforcement for concrete structures, 2019.

[30.8-2] Bisby and Stratford Bisby, A. L., Stratford, T., Design for fire of concrete elements strengthened or reinforced with fibre-reinforced polymer: State of the art and opporunities from performance-based approaches, 2013. *Canadian Journal of Civil Engineering*, 40:11.

[30.8-3] Palmieri et al Palmieri, A., Matthys, S. and Taerwe, L, Fire endurance and residual strength of insulated concrete beams strengthenened with near surface mounten reinforcement, Journal of Composites for Construction, 17(4), 454-462., 2013

[30.8-4] Del Prete et al Del Prete, I., Bilotta, A., Nigro, E., Performances at high temperature of RC bridge decks strengthened with EBR-FRP, 2015. *Composites Part B Engineering*, 68: 27 – 37.

[30.8-5] Ceroni and Pecce Ceroni, F., Pecce, M., Design provisions for crack spacing and width in RC elements externally bonded with FRP, 2009. *Composites Part B Engineering*, 40(1): 17 – 28.

[30.8-6] ISO 4356: 1977. Bases for the design of structures – Deformations of buildings at the serviceability limit states, ISO, 1977.

Examples of accidental and/or exceptional events to be considered (in a direct or indirect way) include:

1) sudden damage caused by explosions, collisions, impact, extremely high water pressure, fire or earthquakes, consequences of human errors, etc.;
2) continuous damage associated with deterioration or fatigue.

Design for resistance to disproportional collapse should be considered in the early stages of the design process.

On one hand, ductility requirements are expected to improve both seismic performance and structural robustness. By contrast, possible conflicts between seismic and robustness designs may originate from capacity design criteria, particularly the strong-column/weak-beam design according to the hierarchy resistance criterion in current seismic design provisions. This could lead to systems that cannot be able to provide the required performances for robustness.

In case the robustness is assessed through a risk analysis then all elements should be involved.

Consequences include human, economic and environmental consequences, which can be related to each other by monetarization.

Direct consequences are those resulting from damage states of individual component(s) and are generally confined to the direct effects of a hazard.

Indirect consequences are related to a loss of system functionality as a result of direct consequences.

30.9 Design for robustness

30.9.1 General

It should be adequately assured that no accidental and/or exceptional events or damage to structural components would result in disproportional consequences for the structural system or even collapse of the whole structure during its lifetime. Hence, the robustness of the system must be adequately considered.

Design for robustness should be an integral part of the conceptual design phase of the structure taking into account different strategies that can be applied (Andre e.a. [30.9-1]).

An acceptable design for robustness is not necessarily achieved by seismic design.

In general, the design of a structure for robustness can involve the identification of:

- hazardous events H which may affect the structure (see section 30.9.3);
- Local (direct) damage D;
- Systemic damage S (follow-up/indirect damage), encompassing progressive collapse;
- Direct C_{dir} and indirect C_{ind} consequences.

The total risk R_{tot} in relation to accidental and/or exceptional events can be based on the following equation, considering the addition of risk associated to local (direct) damage and systemic (follow-up) damage:

$$R_{tot} = \sum_i \sum_j C_{\text{dir},ij} P\left[D_j \middle| H_i\right] P[H_i] + \sum_i \sum_j \sum_k C_{\text{ind},ijk} P[S_k | D_j \cap H_i] P[D_j | H_i] P[H_i] \quad (30.9\text{-}1)$$

where:

$P[H_i]$	the probability of occurrence of hazardous event H_i;
$P[D_j \mid H_i]$	the probability of (direct) damage D_j conditional on hazardous event H_i;
$P[S_k \mid D_j \cap H_i]$	the probability of systemic damage S_k conditional on the damage D_j and hazardous event H_i;
C_{dir}	the direct consequences;
C_{ind}	the indirect consequences.

This equation enables the identification of different strategies.

Conditional risk assessment can also be executed in case of a specified hazard and/or damage.

Consequence Classes categorise the severity level of the (direct and indirect) consequences due to failure of a structural member or of a limited part of the entire structure. Consequence Classes are typically defined based on the type of structure, profile of occupancy and functionality. The same qualitative definition for

The following strategies for reducing the risk of disproportionate collapse can be distinguished:

- Reducing one or more of the probabilities $P[H_i]$ of the occurrence of hazards (see section 30.9.4.4);
- Reducing one or more of the probabilities $P[D_j|H_i]$ of (direct) damage (see section 30.9.4.5);
- Reducing one or more of the probabilities $P[S_k \mid D_j \cap H_i]$ of systemic damage (see section 30.9.4.2);
- Reducing direct C_{dir} and indirect C_{ind} consequences (see section 30.9.4.3).

30.9.2 Design considerations

The design considerations for robustness should be based on the Consequence Class of the structure. Particular design considerations may be required for existing structures (refer to section 30.9.7).

Consequence Class can be used for robustness as for reliability differentiation, see Consequence Classes CC1 (Low-risk), CC2 (Medium-risk) and CC3 (High-risk) and examples for buildings and bridges in 11.3.1.2. Structures complying with the criteria for more than one Consequence Class should be designed considering the most onerous Consequence Class. Depending on the type and properties of the structural system, different areas in an asset may be permitted to be classified independently in different Consequence Classes and hence different design approaches with regards to robustness (e.g. office and sensitive equipment areas in industrial facilities, offices and auditoria in educational buildings, auxiliary and seating areas in stadia).

Nevertheless Consequence Classes deal with the severity of the consequences given an adverse event occurs and they have been used traditionally in design codes to define the target reliability level implicit in the structural design. In cases where the consequences are not negligible and the triggering adverse event is uncertain or unknown it may be necessary in design codes to specify supplementary design provisions, namely design for robustness as they become more important.

In case of the medium-risk group of structures, a distinction can be made between medium-low and medium-high risk structures considering the expected consequences for uncertain and critical adverse events.

The recommended design considerations are

– For low-risk group structures, where negligible consequences are expected regarding all possible adverse events:

  Provided that the structure has been designed and constructed for satisfying stability under normal use, no additional measures are necessary to achieve robustness.

– For medium-low risk group structures, where minor consequences are expected regarding well defined identified adverse events and negligible consequences are expected regarding uncertain and critical adverse events:

  Use indirect design methods such as prescriptive rules (section 30.9.4.6) or the key element method, dependent on the type of construction.

– For medium-high risk group structures, where minor consequences are expected regarding well defined, identified and likely to occur adverse events and minor to moderate consequences are expected regarding uncertain and critical adverse events:

  In addition to the design requirements for medium-low risk group structures, use direct design methods (section 30.9.4).

A systematic risk assessment generally combines qualitative and quantitative analyses towards the risk evaluation before and after the risk mitigating measures are implemented. A cost-benefit assessment for each of the risk reduction measures is recommended as well as a review of the residual risk and a check of the sensitivity of the assumptions made in the risk assessment. The communication of the results and conclusions from the systematic risk analysis with all stakeholders is essential.

A traditionally used criterion to judge if a building structure is adequately robust is to limit the admissible failure area given a loss of a single column to 15% of the floor area or 100 m^2 whichever is smaller in each of two adjacent storeys. However, this recommendation originates in the aftermath of the Ronan Point collapse and relates to the typical area of one bay and its occupancy and should therefore be considered with care. A similar approach can be used for bridges, for which an acceptable collapsed area should be determined.

For example, in case of frame type structures, vertical members, like columns and walls, should run (without interruption or strong capacity reduction) from the foundations to the top of the building. In order to obtain the maximum capacity in terms of ductility of diaphragm elements, similar span values should be adopted along the longitudinal and transversal direction of the in-plan view of the building.

The adequateness relates to the goal of achieving the correct hierarchy of failure modes, i.e. prevent brittle failure (punching shear). Thicker or thinner thickness therefore is case specific and also relates to the transverse reinforcement.

Although generally advantageous for most design situations, structural continuity might increase the consequences following a local member failure in an unforeseen accidental situation, e.g. due to horizontal force transmission to adjacent bays in structural systems such as frames. In such cases, it might be considered to

– For high-risk group structures, where minor to moderate consequences are expected regarding well defined, identified and likely to occur adverse events and moderate to serious consequences are expected regarding uncertain and critical adverse events:

  Use direct design methods following the recommendations from a systematic risk assessment taking into account both identified and unidentified hazards. The risk assessment indicates the design strategies (section 30.9.4) to be followed depending on the hazards, likelihood and consequences as well as tolerable risks.

Design considerations should ensure that the structure is adequately robust.

To this aim masses, stiffness and member capacities should be, as much as possible, uniformly distributed both in plan and in elevation.

For each design requirement, recommended for each consequence class, detailing rules should be coherent with the basic assumption of the adopted design method.

Detailing rules should assure ductile failure modes of structural members, characterised by the formation of plastic hinges in beams or yield lines in continuous slabs. Therefore, ductile failure mode should precede brittle failure mode, like shear failure modes in beams and slabs or punching shear failure modes in slabs. Adequate transversal reinforcement in beam and slabs or adequate slab thickness have to be designed to obtain this hierarchy of resistance in structural members.

Prescriptive rules, like tie methods, are based on the assumption of continuity of longitudinal rebars. Therefore, adequate anchorages and lap length have to be designed to obtain an anchorage resistance higher than the axial capacity of rebars.

build predefined failure mechanisms (structural fuses) into the system in order to limit collapse propagation (section 30.9.4.3).

Continuity or a generally undesired actual static system can arise due to load-bearing and load-transmission of secondary or non-structural components (e.g. walls, façades and heavy-duty suspended ceilings and their substructures). Such elements should be accounted for in the conceptualisation of the structure's load-bearing system, if they are deemed to affect it.

Robustness should be adequately considered in the maintenance strategy of a structure and, when it is re-evaluated, be based on the contemporary state of the structure's condition, the anticipated loads and environmental effects.

The accidental actions and/or events to be assumed in the design can differ substantially from project to project. Therefore, it is difficult to specify the design scenarios in a standardised manner, in particular when hazard-specific design is used.

Possible hazards that cannot be specified include hazards that are not identified or known hazards that cannot be quantified.

In general cases, such input data is incomplete and/or imprecise because some hazards are unforeseen or their magnitude is difficult to predict. Therefore, the application of a direct design method against (identified) accidental actions may in

30.9.3 Design scenarios for robustness

30.9.3.1 Design scenarios

Design for robustness should involve the identification and assessment of design scenarios, i.e. each design scenario relating to a set of events/conditions occurring during the construction or lifetime of the structure leading to a state of the system for which the effect of disproportionate consequences should be assessed. Such design scenarios can be identified based on specified accidental actions (section 30.9.3.2), notional damage or loads (section 30.9.3.3) and should be discussed with the relevant stakeholders. Notional damage is typically specified as cases of initial local failure.

In case of design for identified hazards, the design scenarios consider prescribed actions (such as accidental actions, see section 30.9.3.2) and/or condition states and the resulting effect on the structure.

In case possible hazards cannot be specified, they are expressed by notional actions or notional damage.

30.9.3.2 Design for specified accidental actions

Hazard-specific design requires the identification and quantification of all abnormal events that could possibly affect the structure and the resulting actions on it.

some cases require to be complemented by elements of hazard-unspecific design (indirect design methods), in particular by the assumption of notional damage.

Attention should be paid to the definition of the hazard-unspecific design cases, which should not deviate from the likely damage scenarios the structure would be subjected to by the occurrence of accidental actions. For example, if the identified accidental action is the impact of a vehicle on two columns, it is not adequate in general to design the structure for the case where only one column is notionally removed.

In general, a structure should be designed considering the worst case scenario from both hazard-specific and hazard-unspecific accidental actions.

EN 1991-1-7 [30.9-2] provides guidelines for modelling of impact loading and explosions occurring inside buildings, which are accidental actions with high destructive potential. Several types of impact on buildings and bridges are considered, providing reference values and locations of impact forces that are modelled as either equivalent static loads or dynamic loads depending on the Consequence Class of the structure. In case of explosions, EN 1991-1-7 [30.9-2] takes into account both pressures directly acting on structural members and pressures transferred to structural members from non-structural members.

The load combination to be used when designing the structure for specific accidental actions is given in section 12.5.4.2.3.4.

Target reliabilities for design against specific accidental actions (gas explosions, impact) are suggested in section 11.3.2.2.

When structures are checked for specific accidental actions and/or to resist local damage, load combination rules should reflect the low probability of concurrence of the accidental action and the design live loads, i.e. partial factors which are lower than those applied for ULS verifications.

30.9.3.3 Design for actions from unspecified hazards

30.9.3.3.1 Notional damage scenarios

Notional damage scenarios can include:

– Notional removal scenarios;

– Notional deterioration scenarios.

The analysis of the structural system under a notional damage scenario should be performed considering the accidental load combination as defined in section 12.5.4.2.3.4.

These members are typically, but not exclusively, perimeter columns and/or load bearing walls between the ground and first levels. It may also include interior load bearing elements in vulnerable locations.

When the performance objective that no initial damage shall occur is specified (see section 30.9.4.3), notional damage cannot be specified as a hazard scenario and only the design methods event control, protection and local resistance are applicable.

30.9.3.3.2 Notional removal scenarios

In case of notional removal scenarios, structural elements are notionally removed and the structure is checked for disproportionate consequences, e.g. applying an alternative load path design or the identification of key elements. Preliminary structural analyses may be used to identify the critical elements to consider to be notionally removed.

Structural elements to be notionally removed can include:

- One or more columns;
- One or more panels or nominal wall length;
- Any other elements judged vital to the structural performance.

Notional removal scenarios are usually considered to be related to the failure of a connection or structural member due to an unidentified hazard.

The analysis of the structural system under a notional removal scenario is commonly executed through a static analysis. In cases where dynamic behaviour is dominating the structural response (e.g. referring to the structural response of the intact system), the analysis should consider dynamic effects in a simplified (e.g. energy-balance based) or extensive way.

30.9.3.3.3 Notional deterioration scenarios

In case of notional deterioration scenarios, the geometrical and/or material properties of one or more structural elements are notionally reduced and the structure is checked for disproportionate consequences.

A value often referred to in standards for a notional load for buildings is 34 kN/m^2. This nominal value was recommended after the Ronan Point (UK) accident, triggered by a gas explosion. The value should be treated with caution in case the design scenario considered is considerably different, e.g. in case of the design of a key element, especially in case where the failure of the key element relates to large consequences.

Design for notional loads may be completely inefficient if the effects of unidentified actions/influences are not covered by such notional loads.

The adoption of a particular strategy for designing a given structure for robustness may lead to a conceptual solution with structural features which may be beneficial for some hazard scenarios but detrimental for others, depending on the structural system, the abnormal triggering-event, the magnitude and location of the initial failure or the type of collapse. Therefore, in many cases, an appropriate way for economically meeting all structural safety and robustness requirements should be based on a combination of different design strategies as appropriate. For this purpose, a pragmatic approach can be adopted (Tanner & Hingorani [30.9-3]):

– Adoption of continuous structural systems with a ductile behavior.

30.9.3.3.4 Notional loads

Notional loads are generally specified as a uniformly distributed equivalent static load.

These loads should be applied in the most unfavourable direction for the element under consideration.

When applying notional actions to a column or a beam, the value of the actions should be determined based on the contributing wall and floor area.

30.9.4 Design strategies

30.9.4.1 General

Generally, design strategies for robustness can be divided into two types:

– Direct design methods;

– Indirect design methods.

The method to be applied depends on the design considerations described in section 30.9.2.

– Provision of either alternative load paths (section 30.9.4.2) or predefined collapse mechanisms (section 30.9.4.3) for the event of local failure of selected structural key elements (section 30.9.4.5) due to unidentified accidental actions.

– If alternative load paths are provided and verified for the case of structural key member failure, the associated collapsed area (A_{col}) can be deemed negligible and human safety criteria can be neglected. In this case, the choice of the target reliability for the structural member can be based on economic optimization principles only (Table 11.3-5).

– If predefined collapse mechanisms (fuses) are inbuilt and verified or in the case of non-redundant structural systems (e.g. statically determinate structures), the collapsed area (A_{col}) can be readily established and human safety criteria should be applied for design or assessment of key elements (see Eqs (11.3-3), (5.2-9) for normal load conditions and Eq. (11.3-5), (5.2-10) for risk scenarios involving gas explosions).

Guidance on the verification of alternative load paths and predefined collapse mechanisms (fuses) is provided in section 30.9.6.

The direct design methods explicitly aim to limit the effect of local failure. They require structural analyses in order to evaluate the performance of the structure for a certain damage scenario and can start from one or more of the following design strategies:

– Alternative load path strategy (section 30.9.4.2)

– Consequence reduction strategy (section 30.9.4.3)

In case the previous approaches do not lead to an adequate level of robustness considering reasonable investments, one can opt to focus the design in reducing the probability of failure of critical elements by means of event control strategy (see section 30.9.4.4) and/or the specific load resistance strategy, e.g. related to key element design (see section 30.9.4.5).

The indirect design methods do not explicitly consider the ability of a structure to sustain an accidental and/or exceptional load effect (section 30.9.4.6).

The alternative load path strategy considers the situation that one or more structural elements have been damaged and have no more load bearing capacity. For the remaining part of the structure is required that for a specified short period of time the structure withstands the associated actions with an acceptable reliability (see section 30.9.6).

Structural integrity is related to ensuring the element continuity and enhances the system behaviour. This is obtained through an appropriate level of tying between structural elements in order to avoid physical separation.

For buildings these can be as external walls and partitions.

Common calculation procedures for an alternative load path analysis are:

– a linear static procedure considering suitable factors for dynamic and non-linear effects;
– a nonlinear static procedure considering a suitable factor for dynamic effects;
– a linear dynamic procedure considering a suitable factor for non-linear effects;

30.9.4.2 Alternative load path strategy

The alternative load path strategy explicitly considers the resistance to progressive collapse (i.e. 'indirect' or 'follow-up' failure) when the level of damage is specified. The alternative load path strategy focuses on reducing the probability $P[S|D \cap H]$.

Measures associated with the alternative load path strategy include:

– Providing redundancy/integrity;

– Providing ductility.

In frame type structures, resisting mechanisms which can minimize the risk of progressive collapse can include:

a) Bending of the beam where the column has failed; this mechanism is generally ineffective and seldom adopted since the beams have to be over-dimensioned.
b) Vierendeel behaviour of the frame over the failed column.
c) Arch effect of the beams where the column has failed; this mechanism is effective in case the neighbouring structure is sufficiently stiff to limit horizontal displacements.
d) Catenary/tensile membrane behaviour of beams/slabs, bridging the damaged column by means of large rotations and displacements.
e) Contribution of non-structural elements.

The transition of bending action in beams and slabs to tensile membrane (or catenary) action is commonly considered to be a building's last line of defence against progressive collapse.

If the designer relies on one of these resisting mechanisms for robustness, these should be demonstrated through the analyses and design considerations in 30.9.2.

– a nonlinear dynamic procedure.

–

The consequence reducing measures should be selected on the basis of their risk reduction and the related costs necessary to achieve it.

Horizontal progression of collapse can be limited effectively by dividing the structure into independent structural systems by means of so-called ‘structural fuses’. The type and location of structural fuses should be appropriately chosen and suitably designed, detailed, executed and verified.

Structural fuses are particularly effective in large, low buildings but less effective for tall buildings. In case of the latter, compartmentalization generally involves the installation of strong floors intermittently over the height of the building.

Examples of changing the context of the building can be related to:

– Architectural and structural variables such as the shape and static scheme of the building;
– Organisational variables such as relocation of key business operations to more protected and/or robust parts of the building;
– Self-rescue and rescue by others;
– Backup facilities.

Examples of other non-structural mitigation measures include the organisation of an efficient emergency response including fire-fighting teams, police, rescue teams, nearby hospitals, exercises, feedback from technical experts.

30.9.4.3 Consequence reduction strategy

The consequence reduction strategy aims to limit unacceptable (disproportionate) consequences (C_{ind} and/or $P[S|D \cap H]$) associated with damages D.

Measures associated with the consequence reduction strategy can include:

– Structural segmentation/compartmentalization;
– Changing the context of the structure;
– Other non-structural mitigation measures.

30.9.4.4 Event control strategy

The event control strategy consists of preventing the occurrence of the hazard H, or, limiting the occurrence rate of the hazard to an acceptable value and requires that the hazard (or spectrum of hazards) is identified. This strategy focuses on reducing

the probability P[*H*] and does not increase the intrinsic resistance of a structure to unacceptable damage.

Measures associated with event control might include:

– Changes in the building site or access to it;

– Restricting the use of the structure;

For example avoiding high risk areas.

For example limiting the use of certain areas for specific utilizations or prohibiting the storage or transport of explosives or other hazardous material sources.

– Installation of early-warning systems for the hazard under consideration;

For example monitoring systems for wind, gas, fire development, etc.

– Installation of passive measures;

For example barriers to prevent vehicle collision, dykes, fire insulation, etc.

– Quality management to prevent human errors;

– Maintenance.

For example in case of deterioration.

30.9.4.5 Key element design strategy

The key element design strategy aims at preventing or limiting the local damage *D* (often referred to as 'direct damage') caused by a certain hazard *H*, such as blast pressures. Therefore, the key element design strategy is a hazard specific approach focussing on reducing the probability P[*D*|*H*].

Key elements are structural elements and/or connections which are essential to the resistance of the structure. Failure of a key element typically results in significant consequences since, in its absence, the structure is usually unable to develop adequate alternative load paths (see section 30.9.4.2), unless a combined design strategy is applied as suggested in section 30.9.4.1.

Examples of potential key elements include structural members on the lower floor levels that are closest to exterior vehicle threats, piers of continuous bridges, cables in cable supported structures, etc.

Reliability criteria for key components in buildings, depending on the expected failure consequences, can be found in section 11.3.2.2.4 for structural design (see Eqs. (11.3-4) for normal load conditions and Eqs. (11.3-5) for risk scenarios involving gas explosions) and 11.3.2.3.3 for assessment of existing structures.

Measures associated with specific load resistance might include:

– Increasing the element resistance;

– Increasing the element stiffness;

– Using active or passive isolation techniques such as base isolation of the structure;

Structural ductility can be achieved by providing ductility at the level of the system, the elements and the material.

The key element design strategy can be supplemented with providing sufficient structural ductility.

Material ductility can be achieved by material strain-hardening and/or by material deformation capacity.

Ductile detailing can be achieved by using continuous bottom reinforcement over supports, confinement at joints, adequate ties to allow for load transfer, etc.

Prescriptive rules are assumed to enhance the robustness implicitly. Formulas given by e.g. EN 1991-1-7 [30.9-2] (and UFC 4-023-03 [30.9-4]) are straightforward and derived from specific cases, which makes their generalization sometimes questionable.

Prescriptive rules with a more explicit nature could be used based on analytical, numerical, empirical models, as e.g. in Izzudin & Siu [30.9-5].

For framed structures, prescriptive rules consist in providing horizontal and/or vertical ties and for load bearing wall systems in providing effective anchorage of floors and roofs to walls.

An explicit quantification of robustness might be considered in case of:

– The comparison of design alternatives;
– The follow-up of robustness during the lifetime of the structure.

30.9.4.6 Prescriptive rules

The application of prescriptive design rules is an indirect design method, not performance based and can be considered as part of the design considerations in section 30.9.2.

The use of prescriptive design rules is limited to the cases mentioned in section 30.9.2.

The most common prescriptive rules relate to providing horizontal and/or vertical tension ties.

30.9.5 Robustness quantification

Robustness quantification should be based on the comparison of:

1) The amount and extent of initial damage;
2) The corresponding follow-up consequences.

Robustness metrics can be divided into the following levels with decreasing complexity:

When determining the design value of the resistance, R_d, reduced values for partial factors can be taken into account for material properties and resistance model uncertainties, associated with accidental design situations (see section 12.5.4.2.3). Nevertheless, structural design for robustness might still be uneconomic. This could be mitigated by using a risk-based approach to the design of the remaining structure, factoring into the reliability verifications the conditional probability of a system failure following a local failure. Tentative values for this probability are 0.01-0.1 (see Ellingwood [30.9-6], Ellingwood & Dusenberry [30.9-7]).

In case structural fuses have been inbuilt (section 30.9.4.3), the additional internal forces and moments, $\Delta E_{d,col}$, should conservatively be established based on the upper design value of the fuse strength, $R_{d,sup,fuse}$. In addition, fuse failure must be verified. For the relevant situations after the accidental event ($A = 0$), the design value of the action effects in the assumed fuses, $E_{d,fuse}$, should exceed the upper design value of the corresponding resistance, $R_{d,sup,fuse}$:

$$R_{d,sup,fuse} < E_{d,fuse}$$

In existing structures, measures associated with the conceptual layout (e.g. design of alternative load paths in case of local member failure) cannot normally be put into practice without costly structural interventions. Unequivocal identification

- Risk-based quantification based on a complete risk analysis in which the consequences are divided into direct and indirect consequences (see section 30.9.1);
- Reliability based quantification derived from e.g. the reliability of the structural system in the damaged and in the undamaged situation;
- Deterministic quantification based on structural parameters such as the load-bearing capacity, stiffness, damage energy, etc.

30.9.6 Robustness criteria

An operational criterion for verification of the robustness of a structure assuming a local damage due to failure of one or several of its structural members is given as:

$$E_d\left(A=0,\Delta E_{d,col},\ldots\right) \leq R_d \qquad (30.9\text{-}2)$$

where the combinations to be considered for the establishment of the design value of the effects of actions, E_d, refer to situations after an accidental event (no explicit accidental action: $A = 0$). However, dynamic effects due to sudden element collapse must be taken into account when redistributing internal forces and moments to the remaining part of the structure (e.g. by means of catenary action). Such redistribution results in additional internal forces and bending moments $\Delta E_{d,col}$.

30.9.7 Robustness considerations for existing structures

Particular design considerations for robustness are needed for existing structures where different types of constraints may affect the level of robustness that can be achieved.

of all relevant hazards and hazard scenarios is therefore particularly important for assessing the robustness of existing structures.

Typical changes of use that might result in a significant increase in risk include changes of a building from a private to a public use.

Typical alterations include excavations of basements, removal of internal supports, changes in the layout of secondary and non-structural building components (e.g. wall partitions), creation of openings or introduction of mezzanine floors.

For example, the application of prescriptive rules (section 30.9.4.6) might be feasible for small alterations (i.e. retrofitting tying systems) whereas for other cases direct design approaches are needed to demonstrate that the retrofitted structure can achieve satisfactory levels of robustness.

Two common design situations are identified:

– Change of use: a higher Consequence Class may be adopted in the design considerations (section 30.9.2) if the change of use results in a significantly higher risk.

– Alterations and extensions: these works should not reduce the level of robustness of the existing structure.

In situations where there are large unknowns on the structural properties with potential hidden defects in the existing structure, design considerations for robustness for a high risk group described in section 30.9.2 are recommended.

In cases of existing structures the strategies

– Consequence reduction (section 30.9.4.3)

– Event control (section 30.9.4.4)

– Key element upgrade (section 30.9.4.5)

should be mainly considered in order to improve the structural robustness since other design strategies described in section 30.9.4 might be unfeasible when dealing with existing structures..

References

[30.9-1] André et al — André, J., Anghileri, M., Belletti, B., Biondini, F., Caspeele, R., Demonceau, J.-F., Izzuddin, B., Martinelli, P., O'Connor, A., Parisi, F., Sio, J.A., Thienpont, T.Guidance on the design for structural robustness. Support to the implementation, harmonization and further development of the Eurocodes. Publications Office of the European Union, Luxembourg, 2022

[30.9-2] EN 1991-1-7:2006. — Eurocode 1: Actions on structures – Part 1-7: General actions – Accidental actions.

[30.9-3] Tanner & Hingorani — Tanner P, Hingorani R. , Approach to Robust and Elegant Bridge Design. Structural Engineering International. 2021;31:4, 638-648

[30.9-4] UFC 4-023-03: 2006. — Design of Buildings To Resists Progressive Collapse.

[30.9-5] Izzudin & Sio — Izzudin B.A., Sio J., Rational Horizontal Tying Force Method of Practical Robustness Design of Building Structures. Engineering Structures. 2022;252:113676

[30.9-6] Ellingwood. — Ellingwood B. , Mitigating Risk from Abnormal Loads and Progressive Collapse. Journal of Performance of Constructed Facilities. 2006;20:315-23.

[30.9-7] Ellingwood & Dusenberry — Ellingwood B., Dusenberry DO , Building Design for Abnormal Loads and Progressive Collapse. Computer-Aided Civil and Infrastructure Engineering. 2005;20:12; ASCE 7.10.

30.10 Evaluation assisted by non-linear finite element analyses

30.10.1 Scope

Non-linear Finite Element Analysis (NLFEA) can be applied for determining the structural capacity of concrete structures and concrete structural elements, during design of new structures and assessment or strengthening of existing structures. Various types and ages of concrete and reinforcements with or without deterioration effects and possible non-compliant detailing can be considered. The considered actions can be a combination of time-independent and time-dependent actions. The considered actions can include permanent, imposed, and environmental actions.

Typical application areas of NLFEA for concrete structures are those where simplified analytical models fail to accurately predict structural resistances, e.g. where non-linear behaviour related to material properties and/or geometry significantly affects structural reliability or where load effects are difficult to assess due to interaction of structural members or dynamic nature of the phenomenon. Specific examples include:

- Quantifying the inelastic deformation capacity of a concrete structure for quasi-static push-over loading;
- Determination of the distribution of action effects in a structure for further capacity control of cross-sections, when significant concrete cracking and/or reinforcement yielding influences the load paths;
- Assessing the seismic capacity of an existing structure under earthquake action, including, if needed, also non-structural elements;
- Assessing the shear capacity of an existing structure with outdated reinforcement detailing;
- Assessing the activation of a second bearing path in a design of a bridge;
- Assessing the risk of thermal cracks in massive concrete structures such as dams, foundations and tunnels, during the early hardening process in concrete, due to restraints and uneven temperatures;
- Simulating the failure mode and loading capacity of post-tensioned fibre reinforced concrete slab without conventional longitudinal reinforcement for distributed loading;
- Assessing the bearing capacity of a tunnel section subject to fire and temperature dependent material deterioration;
- Simulating the behaviour of a building subject to impact or explosion;

- Verifying serviceability limit state requirements of a concrete outer wall of a liquefied natural gas storage tank, at low temperatures under a liquid spill condition;
- Assessing the failure mode and capacity of disturbed regions with complex detailing and boundary conditions, such as anchorage zones, subjected to concentrated loads;
- Assessing the residual capacity of corroded structures with deteriorated steel and concrete properties due to carbonation and/or chloride penetration;
- Simulating the capacity of a building, including soil-structure interaction effects, for seismic actions.

The purpose of using NLFEA can be related to the various stages of the life cycle of concrete structures, whenever consideration of non-linear structural behaviour becomes essential. These stages include:

- research and development;
- development of design codes;
- design of new structures;
- assessment or strengthening of existing structures;
- forensic engineering.

and the purpose of performing NLFEA will vary accordingly.

Within this scope, section 30.10.2.2 focusses on determining model uncertainties of NLFEA results. Section 30.10.3.2 covers the treatment of the expected failure mode and the assignment of input values to the parameters describing the failure mode. The safety formats for NLFEA are treated in section 30.10.3.5.

The purpose of including NLFEA in a project or activity can be manifold. NLFEA should always be complemented by analytical or experimental validations and verifications.

The analyst should define a strategy for obtaining a solution from the NLFEA: a solution strategy, see section 30.10.2. This details the applied NLFEA method and should be fully documented.

The scope of section 30.10 is the design of new concrete structures and the assessment of existing concrete structures, within the application areas mentioned above, using a displacement-based or a force-based finite element method, adopting an incremental-iterative technique for quasi-static loading conditions and adopting explicit or implicit time integration scheme for dynamic loading.

30.10.2 Methods for NLFEA

30.10.2.1 Definition and verification of solution strategy

The analyst should define a solution strategy for NLFEA. The solution strategy should be suitable for the actual problem at hand. It should be recognized that different problems might require different solution strategies.

The selected solution strategy will always depend on what is available in the software that is used. This section does not give specific requirements for the solution strategy, but rather introduces certain minimum requirements that should be fulfilled. After selecting the solution strategy, the analyst is always responsible for *validating* the accuracy, *verifying* the objectivity and *demonstrating* the applicability with respect to the problem at hand.

Only at a high abstraction level there is only one NLFEA method. From the practical standpoint, however, a number of NLFEA methods can be distinguished. Usually, the choice of a certain finite element program implies which NLFEA method is used. After selecting the specific NLFEA method, a user usually still has substantial freedom on how to set up a model and an analysis, thus implementing a so-called solution strategy.

Incremental-iterative techniques are usually used to find equilibrium. Use of non-iterative techniques is rare. Either implicit or explicit time integration methods are frequently used to find dynamic equilibrium, both involving time increments.

The solution strategy also refers to the choice of displacement-based versus force-based finite elements. Displacement-based elements have the advantage of a smoother inclusion in the general solution strategy at the structural level, since it is always displacement-based, however, a refined mesh is generally required, especially in the critical zones, which increases the number of elements in the model. On the other hand, force-based elements have the advantage of providing an exact equilibrated solution by using a smaller number of elements, however, they require more internal iterations to achieve compatibility.

Suitable material models could be based on relevant parts of this Model Code or on available literature, and could usually cover the following or other effects relevant to the problem at hand:

- Strength degradation due to lateral cracking;
- Strength increase due to lateral confinement;
- Concrete-to-concrete interaction;
- Concrete-to-reinforcement interaction;
- Effects of cyclic loading;
- Effects of high strain rates;

The solution strategy comprises all the choices related to i) kinematic compatibility, e.g. finite element types and sizes, ii) equilibrium, e.g. iterative solution method, and iii) material models, e.g. constitutive relations for concrete and reinforcement steel. The solution strategy also covers other problem-specific choices related to, for example, finite element discretization, representation and modelling of boundary conditions and interfaces to other surrounding structures, idealization of the structural geometry, idealization of the applied actions, and identification of failure, see section 30.10.3.2.

The material model for concrete should treat effects of cracking, lateral tensile and compressive stresses, and load reversals to a level of refinement reflecting the nature of and the knowledge about the problem at hand. The selected model should be validated by comparing results from experiments to analysis results for simple models relevant to the problem at hand.

– Effects of deterioration mechanisms.

Although strongly connected to the selected material models, the objectivity of the results also depends on the selected element types, for example

– beam and shell elements;
– 2D and 3D continuum elements;

the interpolation order, for example:

– linear or quadratic interpolation functions;

and on the iterative solution method, for example:

– modified or full Newton-Raphson;
– load- or displacement-controlled analyses;
– arc-length method;
– line search;
– Quasi-Newton methods;
– Newmark methods for implicit time integration.
–

The capacity may be defined in different ways depending on the type of structure and the type of loading, e.g. ultimate load bearing capacity in case of static actions, and ultimate deformation capacity in case of seismic actions.

The modelling uncertainty should be treated as a log-normally distributed random variable defined as $\theta = R_{\text{exp}}/R_{\text{NLFEA}}$, where R_{exp} and R_{NLFEA} are the experimental and the predicted capacities. R_{NLFEA} is evaluated based on the material parameters given in the experimental report or paper, or derived from the former, and on the nominal values of geometrical parameters. All the physical factors that are not explicitly accounted for in R_{NLFEA} will contribute to the estimated modelling uncertainty. This implies that $\theta = R_{\text{exp}}/R_{\text{NLFEA}}$ as treated herein includes contributions from physical variability of the experiment, e.g. lab conditions, spatial variability of material properties within the specimen and the uncertain relation between material parameters.

The material model for reinforcement steel should allow to represent the cyclic elastic-plastic behaviour observed in experiments, to the desired level of accuracy for the problem at hand.

The analyst is responsible for verifying that the selected solution strategy is able to produce objective results upon element mesh size refinement and load increment size refinement.

30.10.2.2 Validation of solution strategy and assessment of modelling uncertainty

The modelling uncertainty, which represents the accuracy of the selected solution strategy in predicting the capacity, should be assessed by comparing NLFEA predictions to benchmark experimental results. The benchmark experiments should as a minimum cover structurally significant subsets of the problem at hand, showing a similar type of structural behaviour and failure process. The selected suite of benchmark experiments should contain an adequate number of experiments to limit the statistical uncertainty in the estimate. Note that the nature of the structural behaviour and failure process of the benchmark experiments narrows the scope of application of the solution strategy and the estimated modelling uncertainty.

The modelling uncertainty can be represented by one of the following:

- a mean value and a standard deviation or coefficient of variation;
- or a modelling uncertainty factor derived from these two quantities.

By performing n benchmark analyses, the sample mean and variance can be calculated using the regular estimators $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s_y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$, where $y_i = \ln\left(R_{\text{exp}}/R_{\text{NLFEA}}\right)_i$. The mean and the variance of $y = \ln\theta$ should be calculated using the following expressions, where $\nu_y = n - 1$ are the degrees of freedom:

$$\mu_y = \bar{y} \tag{30.10-1}$$

$$\sigma_y^2 = \frac{\nu_y(\nu_y+2)}{(\nu_y-2)(\nu_y+1)} s_y^2 \tag{30.10-2}$$

The expressions above are result of Bayesian updating of a non-informative prior with unknown mean and variance and include the statistical uncertainty related to a limited number of benchmark analyses.

Furthermore, the mean and coefficient of variation of θ should be calculated using the following expressions, where the errors of approximation are less than 2% for $V_\theta < 0.2$:

$$\mu_\theta \approx \exp(\mu_y) = \exp\bar{y} \tag{30.10-3}$$

$$V_\theta \approx \sigma_y \tag{30.10-4}$$

The partial factor for modelling uncertainty should be calculated using the following expression:

$$\gamma_{\text{Rd}} = \frac{1}{\mu_\theta}\exp(\alpha_R \beta_{\text{LS}} V_\theta) \tag{30.10-5}$$

Ideally, the NLFEA should yield unbiased predictions, i.e., the mean of θ should approach unity, while its coefficient of variation is generally not reducible.

Table 30.10-1 contains prior parameters for the distribution of $y = \ln\theta$. Bayesian updating should be used to update the prior parameters using the results from the benchmark analyses described above.

Table 30.10-1: Prior parameters for the distribution of $y = \ln\theta$.

s_y'	ν_y'	$\bar{y}'$	n'
0.10	6.2	0.02	1.4

The updated parameters for the modelling uncertainty should be calculated using the following expressions:

$$n'' = n' + n \tag{30.10-6}$$

30.10 Evaluation assisted by non-linear finite element analysis

The modelling uncertainty should be properly accounted for before conclusions are drawn based on the analysis results. The solution strategy should be applied consistently in all the benchmark analyses and furthermore during solution of the problem at hand. The modelling uncertainty should be accounted for by following one of the two approaches below, see also the separate sub-sections covering the different safety formats in section 30.10.3:

1. Estimate the parameters of the modelling uncertainty by updating the prior parameters in table 30.10-1, and
   a. either calculating γ_{Rd} assuming a log-normal distribution when using the Partial Factors Method, section 30.10.3.3,
   b. or include the parameters when calculating γ_R in the Global Factor Method, section 30.10.3.4,
   c. or use the parameters directly in a sampling-based method for the resistance side, section 30.10.3.5.
2. In exceptional cases when using the Partial Factors Method and where the solution strategy is not validated through benchmark analyses, a value $\gamma_{\text{Rd}} \geq 1.35$ should be used.

$$\nu_y'' = \nu_y' + \nu_y + 1 \quad (30.10\text{-}7)$$

$$\bar{y}'' = \frac{1}{n''}(n\bar{y} + n'\bar{y}') \quad (30.10\text{-}8)$$

$$\nu_y'' s_y''^2 = \nu_y s_y^2 + \nu_y' s_y'^2 + \frac{nn'}{n+n'}(\bar{y} - \bar{y}')^2 \quad (30.10\text{-}9)$$

$\bar{y}''$ and $s_y''^2$ should be input to the expressions for μ_θ and V_θ determined according to Eq. (30.10-3) and (30.10-4).

By using the above expressions, the estimate of the modelling uncertainty will be strengthened when the number of benchmark analyses is increased, thus promoting a thorough validation of the solution strategy. Figure 30.10-1 demonstrates this by plotting an estimated γ_{Rd} for unbiased predictions as a function of ν_y. Note that no strict lower limit is put on the number of benchmark analyses as long as the results from the benchmark analyses are used to update the prior parameters in table 30.10-1.

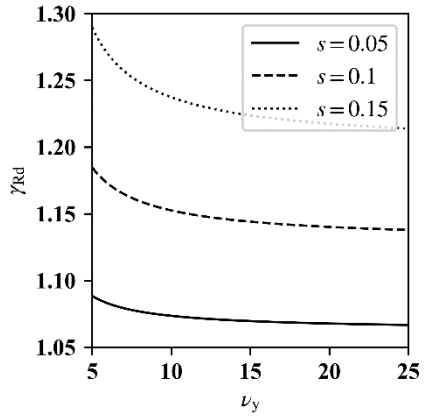

Figure 30.10-1: γ_{Rd} calculated for unbiased predictions ($\bar{y} = 0.0$) and different values of the sample standard deviation s as a function of the degrees of freedom ν_y.

In case the solution strategy is selected from literature where also results from benchmark analyses are presented, γ_{Rd} could be based on such results provided that the scope of application is unchanged.

NLFEA predictions could also be qualitatively compared to hand calculations. Since hand calculations are also model predictions based on abstractions of the reality, such comparisons cannot be used to quantify the modelling uncertainty of the NLFEA predictions.

For solution strategies that are not validated through benchmark analyses, the given value of γ_{Rd} can be interpreted as follows. The prior parameters in table 30.10-1 correspond to a modelling uncertainty factor $\gamma_{Rd,w} = 1.15$, for the within-model component of the uncertainty, assuming, $\alpha_R = 0.4 \cdot 0.8 = 0.32$, $\beta_{LS} = 3.8$, and a reference period of 50 years, representing the uncertainty of the model outcome with one selected solution strategy. Similarly, $\gamma_{Rd,b} = 1.18$ can be estimated for the between-model component of the uncertainty, representing the uncertainty of the prediction if the solution strategy was selected randomly. For solution strategies that are not validated through benchmark analyses, a value of $\gamma_{Rd} = 1.15 \cdot 1.18 \approx 1.35$ is therefore obtained. See also: Engen et al. [30.10-1].

NOTE: the given value of γ_{Rd} could slightly change when the target reliability index and corresponding reference period is agreed upon.

The objective of non-linear analysis is to simulate the real structural behaviour and to evaluate a representative value of the capacity. The uncertainty due to random variation of material properties, and other parameters affecting the capacity, including geometrical parameters and detailing, is described by suitable probabilistic models. In addition, model uncertainty must be considered separately.

The design condition is compatible with the one applied within the partial factors design method, see section 12.5.4.

The capacity variable R can represent a load resultant, displacement, acceleration or any other quantity describing the overall structural response.

The design value of the capacity, R_d, is estimated using one of the safety formats summarized in Table 30.10-2, that is:

I) the Partial Factors Method;

II) the Global Factor Method, or

III) the Sampling-based Method.

In the *Partial Factors Method*, the design capacity is determined using local partial factors applied to the random variables affecting the overall structural capacity. In the *Global Factor Method*, the design capacity is determined using a

30.10.3 Assessment of structural capacity by NLFEA

30.10.3.1 General

The design principles based on the global resistance method described in section 12.5.5 apply, whereas the character of numerical simulation by non-linear analysis must be considered appropriately, including a proper definition of model uncertainties, as described in sections 30.10.2.1 and 30.10.2.2.

The design condition is considered in the form of Eq. (12.5-30) with separated uncertainties for load and capacity:

$$F_d \leq R_d \qquad (30.10\text{-}10)$$

where:

F_d is the design value of the load variable,

R_d is the design value of the capacity variable, which is estimated by means of one of the methods described in the sections 30.10.3.3 to 30.10.3.5, and summarized in Table 30.10-2.

Table 30.10-2: Safety formats for estimating the capacity with NLFEA.

	R_d	Notes

global factor applied to the capacity. In the *Sampling-based Method*, the design capacity is determined as function of the required reliability index or failure probability. The choice of a relevant method for the calculation of the design capacity depends on engineering judgement while taking into consideration the capacity model and possible uncertainties.

I: Partial Factors Method	$R_{\mathrm{d}} = \frac{R_{\mathrm{NLFEA}}(\boldsymbol{x}_{\mathrm{d,NLFEA}})}{\gamma_{\mathrm{Rd}}}$	γ_{Rd} is introduced in section 30.10.2.2.
II: Global Factor Method	$R_{\mathrm{d}} = \frac{R_{\mathrm{NLFEA}}(\boldsymbol{x}_{\mathrm{m}})}{\gamma_{\mathrm{R}}}$	γ_{R} is introduced in section 30.10.3.4.
III: Sampling-based Method	$P(\theta R_{\mathrm{NLFEA}}(\boldsymbol{x}) \leq R_{\mathrm{d}}) = \Phi(-\alpha_{\mathrm{R}}\beta)$	Probability distribution for θ is introduced in section 30.10.2.2, and other basic variables $\boldsymbol{x}$ are discussed in section 30.10.3.5.

30.10.3.2 Identification of failure based on NLFEA results

In general terms, *failure* is the exceedance of a certain limit state. In structural systems, a clear distinction must be made between local failure(s) and global failure. Identification of the local failure(s) from the NLFEA results, and how they affect the global failure, is a prerequisite for a further application of the safety formats in the following sections. Local failure could be defined based on exceedance of local limits for materials, e.g. strengths and limit strains, or mechanisms, e.g. resistance and limit deformability. The global failure *mode* could be defined based on either *a*) a mechanical consequence of (concurrent) local failure *mechanism(s)*, or *b*) the exceedance of a purposely chosen phenomenological measure of the overall capacity of the structural system.

An acceptance criterion for the capacity should always be reliability-based, allowing for proper treatment of the uncertainties of the problem at hand. A reliability-based acceptance criterion forms the basis for establishing quantitative measures. Examples of acceptance criteria are:

- maximum tensile stress to avoid cracking;
- maximum load bearing capacity leading to failure of the first structural component;
- maximum load bearing capacity before global failure;
- maximum deflections;
- maximum earthquake intensities before attaining a given Limit State;
- maximum leakage rates.

Note that acceptance criteria as described here are problem- and project-dependent and are usually results of functional requirements set by the client.

From a NLFEA, the analyst should identify action levels where either there is lack of convergence or the analysis aborts due to divergence. During post-processing of the results, the analyst should interpret whether these issues were due to physical failure or numerical failure and identify at which load levels the relevant physical failure occurred.

This process would include for example a close and integral study of:

- the convergence behaviour in terms of required number of iterations and obtained convergence norms per step, in case an incremental-iterative technique is adopted;

A thorough investigation of the material, component or system behaviour at or around failure, shall provide a mechanical explanation of the relationship between the observed local failure *mechanism(s)* and the global failure *mode*, also with the aim of identifying the governing set of material parameter(s). The sequence of local failure mechanisms leading to the global failure mode should be investigated thoroughly.

- the total displaced shape of the structure at, or immediately after, peak response. Rather than showing the total displacements, the incremental displacements of only the last loading stage or load increment could be used to obtain more insights;
- the role of each local failure mechanism within the global system;
- the distribution of local failure mechanisms and their development sequence within the system under increasing intensity of the action;
- the position of local failure events along the global curve describing the overall system capacity;
- contour plots of principal tensile and compressive strains in concrete;
- vector plots of principal strains and stresses in concrete;
- crack orientation and distribution plots for the concrete;
- axial stresses and (plastic) strains in reinforcement;
- slipping of reinforcement bars and concrete-to-concrete interfaces;
- other measures of physical damage in the structure.

To ensure that all relevant local failure mechanisms have been identified, the sensitivity of the global failure mode to the values of the material properties can be investigated e.g. by performing additional NLFEA with alternating combinations of mean and design values for the sets of material properties related to the local failure mechanisms.

In the Partial Factors Method, the structural analysis is performed by using the design values of the material strengths. This corresponds to an event where the structure has materials with strengths according to a low fractile of the respective distributions. The intention is to estimate the resistance given this event. Although the estimated capacity in specific cases would deviate from the estimate using e.g. first order reliability methods, past experience has shown that the system of partial factors leads to satisfactory overall reliability levels, except in those cases where an increase of material strengths leads to a decrease of the global capacity.

The design values of the basic variables are found using a set of partial factors that correspond to a given limit state. Since the modelling uncertainty is globally

The analyst should make sure that all relevant local failure mechanisms are identified according to the previous paragraph and should recognize that system failure could be due to concurrent local failure mechanisms.

30.10.3.3 Partial Factors Method

In the Partial Factors Method, the design value of the capacity is given as:

$$R_d = \frac{R_{NLFEA}\left(\boldsymbol{x}_{d,NLFEA}\right)}{\gamma_{Rd}} \qquad (30.10\text{-}11)$$

where:

$\boldsymbol{x}_{d,NLFEA}$ are the design values of the basic variables, calculated based on partial factors applicable for NLFEA and the relevant limit state according to Table 30.10-3 and Eq. (12.5-8b).

Table 30.10-3: Partial factors for the basic variables applicable for design of new structures using the Partial Factors

accounted for on the capacity level via γ_{Rd}, the partial factors for the basic variables should not include the effect of the modelling uncertainty. The partial factors $\gamma_{\mathrm{M,NLFEA}}$ are calculated based on Eq. (12.5-14) in section 12.5.4.2.3, neglecting the terms μ_θ and V_θ.

In case of existing structures, the partial factors should be based on Eq. (12.5-15) where information from the structures is used as input, and still neglecting the terms μ_θ and V_θ.

If the capacity is sensitive to variations in the geometrical parameters, the design values of the geometrical parameters should be used. In other cases, the nominal values could be used.

The Partial Factors Method corresponds to the ordinary method for ultimate limit state design based on partial safety factors, except that the modelling uncertainty is accounted for on the global capacity level instead of in the expressions for the partial factors of the materials.

In the Global Factor Method, the design capacity is obtained as the global capacity, estimated by performing a NLFEA with the mean values of the material parameters affecting the local behaviour, divided by the global factor γ_{R}. The global capacity can be approximated by a lognormal distribution with two parameters: the mean R_{m} and the global coefficient of variation V_{R}.

Two or more concurrent local mechanisms reaching failure at approximately the same global action level contribute significantly to the system reliability and should therefore be considered in the global factor.

Method for NLFEA. The partial factors correspond to a target reliability $\beta_{LS} = 3.8$, and a reference period of 50 years.

Limit state	$\gamma_{\mathrm{c,NLFEA}}$	$\gamma_{\mathrm{s,NLFEA}}$
Ultimate limit state	1.46	1.20

All other material parameters that are necessary for the analysis should be derived from the design material strengths. See section 14.6.

In the Partial Factors Method, γ_{Rd} should be calculated according to section 30.10.2.2.

30.10.3.4 Global Factor Method

In the Global Factor Method, the reference value for the capacity variable is taken as the mean value, R_{m}, corresponding to the attainment of a global failure mode as defined in section 30.10.3.2 and the expression for the design capacity becomes:

$$R_{\mathrm{d}} = \frac{R_{\mathrm{m}}}{\gamma_{\mathrm{R}}} \approx \frac{R_{\mathrm{NLFEA}}\left(\boldsymbol{x}_{\mathrm{m}}\right)}{\gamma_{\mathrm{R}}} \qquad (30.10\text{-}12)$$

where it is assumed that the NLFEA run with mean values, $\boldsymbol{x}_{\mathrm{m}}$, of the basic variables gives the mean value of the capacity R_{m}. The mean values of the material parameters refer to the expected in-situ material properties, see section 12.5.4.2. For structures sensitive to geometrical variability, governing geometrical variables should be included in $\boldsymbol{x}$, and thereby influencing $V_{R_{\mathrm{x}}}$ calculated in Eq. (30.10-15); while for other structures the geometrical variables should be represented by their nominal values.

The global failure can be attained either through a *single* failure mechanism or through *two or more* concurrent local failure mechanisms. The failure mechanisms can have a variable degree of correlation, through the basic variables governing the respective failure mechanisms.

Example 1: Consider a beam with low or no web reinforcement, in which NLFEA identifies diagonal tension failure and concurrent anchorage failure. This is a series system. The global mean capacity R_m equals the simulated resistance with mean parameters, based either on diagonal tension failure or anchorage failure, whichever occurs first.

Example 2: Consider a structural frame in which NLFEA identifies failure mechanisms developing concurrently in two columns at a storey. This is a parallel system. The global mean capacity R_m equals the simulated resistance with mean parameters, based on failure either in the first or the second column, whichever occurs last.

The Global Factor Method requires that the solution strategy be properly validated beforehand, meaning that the modelling uncertainty factor should be based on the results from the benchmark analyses performed by the analyst. These results should be used to update the given prior parameters for the modelling uncertainty in Table 30.10-1. Note that the parameters of the modelling uncertainty should reflect the structural behaviour and failure process of the problem at hand.

For design of new structures and assessment of existing structures, the variabilities that should be included should follow section 12.5.4.2.

30.10 Evaluation assisted by non-linear finite element analysis

In case of a *single* failure mechanism, the global factor is:

$$\gamma_R = \frac{1}{\mu_\theta} exp\left(\alpha_R \beta_{LS} V_R\right) \qquad (30.10\text{-}13)$$

where:

μ_θ is the mean value of the modelling uncertainty;

α_R is the FORM sensitivity factor for the capacity;

β_{LS} is the target reliability index of the limit state of interest;

V_R is the global coefficient of variation.

Values for α_R and β_{LS} are found in section 11.3.2.

In case of *two* (nearly) correlated concurrent failure mechanisms, the global factor coincides with the global factor for a single failure mechanism.

In case of *two or more* (nearly) uncorrelated concurrent failure mechanisms, the global capacity depends on whether the mechanisms are arranged in series or in parallel, and a more thorough assessment is needed.

– In case of *two* (nearly) uncorrelated concurrent failure mechanisms, the global factor differs depending on whether they are arranged in series or in parallel.

– $$\gamma_R = \begin{cases} \frac{1}{\mu_\theta} exp\left(\ln\sqrt{2}\, V_R + \alpha_R \beta_{LS} V_R\right), \text{for series systems} \\ \frac{1}{\mu_\theta} exp\left(\frac{1}{2}\alpha_R \beta_{LS} V_R\right), \text{for parallel systems} \end{cases} \qquad (30.10\text{-}14)$$

For series systems, R_m in Eq. (30.10-12) is the lower global capacity of the two mechanisms and for parallel systems, R_m is the higher global capacity of the two mechanisms.

In case of a nearly linear relation between the basic variables $\boldsymbol{x}$ and the capacity R, V_R can be expressed as:

$$V_R = \sqrt{V_{R_x}^2 + V_\theta^2 + V_G^2} \qquad (30.10\text{-}14)$$

where V_G pertains to the geometrical variability, see section 12.5.4.2, V_θ pertains to the modelling uncertainty, see section 30.10.2.2, and V_{R_x} pertains to the basic variables in $\boldsymbol{x}$, and can be expressed as:

$$V_{R_x} = \frac{1}{1.645} ln \frac{R_m}{R_k} \approx \frac{1}{1.645} ln \frac{R_{NLFEA}(\boldsymbol{x}_m)}{R_{NLFEA}(\boldsymbol{x}_k)} \quad (30.10\text{-}15)$$

with $\boldsymbol{x}_k$ being the characteristic 5%-fractiles of the basic variables.

30.10.3.5 Sampling-based Method

In the Sampling-based Method, the design capacity is estimated using a probabilistic analysis on the capacity side of Eq. (30.10-10).

The design value of the capacity is found from the following relation:

$$P\left(\theta \cdot R_{NLFEA}(\boldsymbol{x}) \le R_d\right) = \Phi\left(-\alpha_R \beta_{LS}\right) \quad (30.10\text{-}16)$$

where:

In the Sampling-based Method, the design capacity is estimated from the probability density function of the capacity estimated using a fully probabilistic analysis. The probabilistic analysis based on numerical simulation with random sampling can be briefly described as follows:

1. **Definition of a numerical model** based on the non-linear finite element method according to section 30.10.2. Such a model describes the capacity function $R_{NLFEA}(\boldsymbol{x})$ and is used to perform deterministic analysis of the capacity R for a given set of basic input variables $\boldsymbol{x}$ (e.g. material properties, geometry and detailing).
2. **Randomization of the basic variables**. Note that the basic variables of the capacity function can influence the demand function, F (e.g. due to pre-stressing, dead load, period shift and energy dissipation in seismic conditions). Random properties are defined by a random distribution type and its parameters (mean and standard deviation). The randomization can be done by two methods: (a) *Random variables*, where the parameter is constant within a sample (structure), but changes between samples. (b) *Random field*s, where the parameter is randomly variable within a sample. Eventual correlations between random variables should be considered appropriately.
3. **Probabilistic analysis of the capacity**. This can be performed numerically, e.g. either by Monte Carlo sampling or by Latin Hypercube sampling (LHS). The outcome of this analysis are estimates of the capacity distribution function and its parameters. Note that the outcome of such analyses is highly dependent on the number of samples, such that the number of samples should be selected carefully.
4. **Evaluation of design capacity** based on the reliability index β or the probability of failure.

The sensitivity factor, α_R, and the target reliability index, β_{LS}, should be selected according to the appropriate limit state and reference period, see section 11.3.2.

The probabilistic method requires that the solution strategy be properly validated on beforehand, meaning that the modelling uncertainty factor should be based on the results from the benchmark analyses performed by the analyst. These results should be used to update the given prior parameters for the modelling uncertainty.

$R_{\mathrm{NLFEA}}(\boldsymbol{x})$ is the estimated capacity based on the basic variables $\boldsymbol{x}$;

θ is the modelling uncertainty;

$\Phi(\cdot)$ refers to the cumulative standard normal distribution;

α_{R} is the FORM sensitivity factor for the capacity;

β_{LS} is the target reliability index of the limit state of interest.

In the sampling-based method, the modelling uncertainty is represented by a suitable distribution for θ. The parameters of the distribution for θ should be based on updating of the prior parameters given in Table 30.10-1 with results from benchmark analyses.

30.10.4 Determination of load effects by NLFEA

NLFEA could be considered when determining load effects for further capacity control of critical sections in the following cases:

- Where the global load paths in the structure are affected by concrete cracking or reinforcement yielding.
- Where the magnitude of the actions is affected by concrete cracking or reinforcement yielding, e.g. in case of deformation actions.
- For slender members in compression, sensitive to second order effects.

The material parameters used in NLFEA for determining action effects should refer to the expected in-situ material properties, see section 12.5.4.2.

References

[30.10-1] Engen et al — Engen, M., Hendriks, M. A. N., Monti, G., Allaix, D. L. , Treatment of modelling uncertainty of NLFEA in fib Model Code 2020, *Structural Concrete*. 2021;22;3202-12.

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Testing and monitoring techniques may provide:

– direct information: information on the parameters of the assessment model obtained through direct measurements of model parameters (e.g. direct measurements of the properties used in the structural assessment collected during inspection, testing or monitoring)

– indirect information: information on the parameters of the assessment model obtained through measurement or observation of indicators (e.g. measurement or observation of element or system condition collected during inspection, testing or monitoring) related by models to the properties used in the structural assessment.

In a general sense, observations from testing and monitoring may also be used in order to improve and further develop specific provisions in the Model Code.

This chapter does not cover the routine control tests on reinforced concrete industrial products.

Verification assisted by testing of structural material(s) may refer to all kinds of tests including e.g. mechanical, physical and chemical tests, that are expected to give additional structure-specific information about the structural material(s). In this respect, two particular cases may be distinguished in practice:

– testing of *non-traditional structural materials used in design of new structures*, when e.g. for a particular material composition the basic variables related to material characteristics in the design model(s) need to be determined by testing or when an adequate material model(s) may not be initially available or considerable modification of material model(s) is sought.

30.11 Verification of structural performance assisted by testing and monitoring

30.11.1 Introduction

30.11.1.1 General objectives

Information from testing and monitoring can be used in determining the structural performance of new elements or structures and in updating the assessment of structural performance of the existing elements or structures, using evaluation models provided in Chapter 30.

This chapter contains provisions for information gathering and for the use of structure-specific information in verification of structural performance during the design of new structures and the assessment of existing structures.

The following methods of gathering information are considered:

– testing of structural material(s) (see section 30.11.2),

– testing of structural element(s) and/or system(s) (see section 30.11.3),

– monitoring the performance of structural material(s), structural element(s) and/or system(s) (see section 30.11.4)

– analysis of past performance of structural element(s) and/or system(s).

Verification of structural performance assisted by testing of structural material(s) is considered here as a procedure where testing on a limited series of representative material samples is used to obtain specific properties of structural materials using specified testing procedures. General provisions for testing of structural material(s) are given in section 30.11.2.1. Observations (direct information) from testing of structural material(s) can be used for:

– estimation of the basic variables related to material characteristics in the design model(s) for new structural element(s) and/or system(s) or updating the basic variables related to material characteristics in the assessment

– testing of *structural materials in/from existing structures*, when the characteristics of materials in existing structures are not known and/or their behaviour is not well understood, or when an improvement of the material model is sought for particular service condition or degradation.

Tests to obtain specific material properties using specified testing procedures may include e.g. in-situ Ultrasonic Pulse Velocity testing of the modulus of elasticity of the concrete in existing structures, laboratory testing of compressive strength on concrete cores taken from existing structures, or laboratory testing of samples of new materials.

Verification assisted by testing of structural element(s) and/or system(s) may cover all kinds of load tests on either scaled-down models, on special specimens or on the entire structure itself. In this respect, three particular cases may be distinguished in practice:

– testing of *(small) repeatedly produced structural elements for new structures* (e.g. prefabricated elements), when an improvement of the design model is sought: a number of full-scale specimens are tested, taking into account all actions and influences during the lifetime of the structure where these elements will be incorporated with an aim to prove a higher design value of the resistance then derived by using the general applicable design equations of the current design standards.

– testing of *(larger) unique structural elements for new structures,* when an adequate design model may not be initially available or considerable modification of design models is sought: a comparatively lower number of tests are carried out simulating the real life conditions with the aim to determine the performance level of the design.

– testing of *(larger) existing structural elements or systems,* when an adequate assessment model may not be initially available or considerable modification of assessment models is sought to improve its accuracy: in-situ load testing is carried out simulating the real life conditions with the aim to determining the performance level or to update the action effect model for the assessment.

model(s) for existing structural element(s) and/or system(s) (see section 30.11.2.2),

– determining the material behaviour model for design of new elements or structures or updating the material behaviour model for the assessment of the existing elements or structures on the resistance side, taking into account the intrinsic variability of material properties and the material behaviour model uncertainties (see section 30.11.2.2).

– determination of the design values of basic variables related to material characteristics in the design model(s) for new structural element(s) and/or system(s), or updating the assessment values of basic variables related to material characteristics in the assessment model of the existing structural element(s) and/or system(s), which is the basis for determination or updating of the partial factor γ_M for the basic variables related to material characteristics, used in the limit state verifications at the semi-probabilistic level (see section 30.11.2.3).

Verification of structural performance assisted by testing of structural element(s) and/or system(s) is considered here as a procedure where load testing on a limited series of representative specimens is used for the determination of the response or evaluation of the performance of structural element(s) and/or system(s). General provisions for load testing are given in section 30.11.3.1. Observations (direct and indirect information) of the structural response gathered from load testing of element(s) and/or system(s) can be used for:

– establishing the design value of the resistance and/or the design value of serviceability properties of new structural element(s) and/or system(s) for given loading conditions by load testing (till failure) (see section 30.11.3.5.5 and 30.11.3.5.6),

– assessing directly the compliance with reliability requirements with regard to the limit state under consideration by proof-load testing of the existing elements or structures (see section 30.11.3.6.5),

– assessing directly the compliance with performance requirements under service load by diagnostic testing of the existing elements or structures (see section 30.11.3.7.5),

– calibrating the action effect model for design of new elements or structures and updating the action effect model for the assessment of the existing elements or structures (see section 30.11.3.7.6),

– calibrating the resistance model for design of new elements or structures and updating the resistance model for the assessment of the existing elements, by

quantifying from load testing till failure the uncertainties in parameters used in resistance models (see section 30.11.3.7.7),

– structural identification based on measurement of structural response during diagnostic load testing, typically in the context of a physics-based finite element model (see section 30.11.3.7.8).

Verification of structural performance assisted by monitoring is considered here as a procedure where the data gathered from monitoring feed appropriate key-performance indicators and serve as input for structural diagnostics and structural assessment procedures. General provisions for monitoring of structural material, structural element(s) and/or system(s) are given in section 30.11.4.1. Observations (direct and indirect information) gathered from monitoring of element(s) and/or system(s) can be used for:

– diagnostics based on SHM data, which may be considered as a trigger for decisions regarding conservation or life cycle management of existing structural element(s) and/or system(s) (see section 30.11.4.5.3),

– updating of the assessment (FEM) model of the existing structural elements or systems, accounting for the uncertainty of the (FEM) model (see section 30.11.4.5.4).

Verification of structural performance may also be based on past performance, which is considered here as a procedure where the failure probability of existing structural element(s) and/or system(s) is determined by accounting for evidences of survival to measured loads or to loads under normal operational conditions of which statistical knowledge is available. Principles of the verification of structural performance of structural element(s) and/or system(s) based on their past performance are still under development.

The decision of implementing testing and/or monitoring and the choice of the testing and/or monitoring strategy should be based on the maximisation of the expected benefit from its use. In section 30.11.5 general principles are given for the decision-making regarding the implementation of testing and/or monitoring.

The accuracy of verifications assisted by testing and monitoring largely depends on data quality and management, which should be consistent with the principles of Quality and Information Management given in Chapter 6.

Observations gathered from monitoring of loads and exposures can be used to determine actions on structures, which can then be the basis for determination of the load models that can be used for the verification of structural performance. This aspect is however beyond the scope of the Model Code. For more information regarding the background of the calibration of traffic load models based on data from monitoring campaigns reference is made to e.g. G. Sedlacek et al. [30.11-1], X. Y. Zhou et al. [30.11-2] and other.

Analysis aimed at determining the failure probability by accounting for a surviving load situation require thorough understanding of the structural behaviour, actions and uncertainties, which should be carefully considered before any conclusions can be drawn from evidences of past performance. When deterioration is ascertained in a structure (e.g. in an inspection), evidences of satisfactory past performance might give false (excessive) confidence that the structure would exhibit an enhanced (higher load) capacity under the ultimate load / limit state conditions in the future. For more information refer to CEN/TS 17440 [30.11-3].

The expected benefits shall be appraised prior to implementing testing and/or monitoring and the formulation of the benefits might depend on the case at hand.

The entire process of verification of structural performance shall be conducted in connection with the data quality assurance, that may be evaluated against a set of relevant criteria, including: completeness, accuracy, precision, consistency of information, etc.

Examples of such structural members and structural systems or special design problems may include:

- the economic importance of a project or of an element to be mass produced justifies the effort of verification assisted by testing;
- the performance evaluation model as provided in the code, in some particular conditions, is judged to be clearly too inaccurate.
- special design problems, such as:
  - assemblages of reinforced concrete prefabricated elements,
  - units with special shape (polyhedric panels, bunkers).
- cases that deviate from the provisions of this Model Code with respect to e.g.:
  - geometric shapes;
  - (combinations of) new materials;
  - particular detailing and/or connections;
  - severe actions.

Examples of special design problems studied experimentally may be:

- prestress losses;
- special production processes (accelerated curing etc.).

Testing of structural material(s) may also part of the routine control tests on reinforced concrete industrial products, however this objective falls outside the scope of this chapter.

Testing of structural material(s) can also serve as a support of research and development for new materials and technologies used in structural applications.

30.11.1.2 Scope of application

Verification of structural performance assisted by testing or monitoring may be pursued instead of verification by calculation when:

- the calculation models given in this Model Code are thought to lead to unnecessary uneconomic results of design or assessment,
- in cases where greater precision in design or assessment is desirable,
- in case of special design or assessment problems within the Model Code's range of application which the calculation models given in this Model Code are thought to be insufficient to cover,
- in case of structural materials, structural members and/of structural systems that are not included in the Model Code's range of application.

In very particular cases assessment of the resistance of structural members or structural systems by load testing (till failure) may also be used for special design problems within the Model Code's range of application.

30.11.2 Verification of structural performance assisted by testing of structural material(s)

30.11.2.1 General provisions

Testing of structural materials involves the essential examination of the properties of materials used in the construction of a new structure or in assessment of the existing structure. The specific objectives of testing of structural materials may include:

- determination of material properties and evaluation of the material's response to different environmental conditions and loading scenarios;
- verification of the material's compliance with the specified requirements and assessment of the material's suitability for a particular structural application or design;

- detection of any defects or flaws in the material that could affect its performance or durability;
- identification of any changes in the material properties over time due to aging or exposure to different conditions;
- collecting data for calibration and validation of numerical models used for simulation and design.

30.11.2.2 Scope of application

Section 30.11.2 covers only those cases involving testing of structural materials in which a statistical interpretation of test results is possible. In general, provisions in section 30.11.2 apply to all structural material that are within the scope of this Model Code.

30.11.2.2.1 General requirements

The material behaviour model and material properties shall be determined by testing of material specimens of defined size and conditioning, which shall be sampled according to prescribed rules and subjected to an agreed-upon testing procedure, with the results evaluated according to specified procedures. For determination by testing of the properties of materials within the scope of this Model Code, reference is made to the provisions in Chapters 14, 15, 16, 17 ,18 and 19, and other relevant references indicated to in these chapters. For determination by testing of the properties of interfaces within the scope of this Model Code, reference is made to the provisions in Chapters 20, 21, 22, 23, 24 and 25, and other relevant references indicated to in these chapters.

30.11.2.3 Reporting

The reporting of results of testing of structural materials shall be provided accurately, clearly, unambiguously and objectively, and shall cover all information relevant to the interpretation of the results of the tests.

In cases where the structural material falls outside the scope of this Model Code, the verification of structural performance with the aid of testing of structural materials should be thoroughly evaluated before drawing any conclusions about how the properties of the materials affect the structural performance of new or existing structural elements or systems

As an aid in the calculation and uniform recording of test results a standard report form or work sheets are often used, and a facsimile of the form may be included in the relevant standards for specific methods for testing of material properties.

Reporting sampling shall include, where relevant:

- the date and time of sampling and identification of the personnel performing sampling;
- data to identify and describe the sample (e.g. number, amount, name);
- a reference to the sampling plan and sampling method incl. identification of the equipment used, diagrams or other equivalent means to identify the sampling location when appropriate etc.;
- details of any environmental conditions during sampling or transport that affect the interpretation of the test results;
- information required to evaluate measurement uncertainty for subsequent testing or calibration;
- deviations, additions to or exclusions from the sampling method and sampling plan, if relevant.

The technical records for each laboratory activity shall contain sufficient information to facilitate if possible, identification of factors affecting the measurement result and its associated measurement uncertainty and enable the repetition of the laboratory activity under conditions as close as possible to the original.

When evaluating measurement uncertainty, all contributions which are of significance, including those arising from sampling, shall be taken into account using appropriate methods of analysis. An uncertainty statement relates to a particular result obtained in a laboratory carrying out the test, as opposed to precision and bias statements which are mandatory parts of the method itself and are normally derived from an interlaboratory study conducted during development of the test method.

Reporting shall include the following general information: title, the laboratory's name and address, the location where the laboratory activities were performed, and the customer's name and contact information, the date(s) of the laboratory activity and the date of the report's issuance.

Reporting shall include identification of the method used, indicating which procedure was used and, if the test method permits variation in operating or other conditions, a statement regarding the particular conditions used in the.

Reporting shall include reference to the sampling plan and sampling method used, where these are relevant to the validity or application of the results. Where relevant, the sampling method shall be described, including information about selection of samples or sites, the sampling plan, and about preparation and treatment of sample(s) to yield the required item(s) for subsequent testing or calibration.

Reporting shall include the test results. Where necessary for the interpretation of the test results, the report shall also include:

- information on specific test conditions, such as environmental conditions;
- a statement of conformity with requirements or specifications;
- the measurement uncertainty i.e. an estimate of the magnitude of systematic and random measurement errors, presented in the same unit as that of the measured or in a term relative to the measured (e.g. percent);
- opinions, interpretations or additional information which may be required by specific methods, authorities, customers etc.

Reporting shall be reviewed and authorized prior to release.

Due attention should be given to selection of the information sources used in the preliminary assessment and in the detailed assessment, recognizing that limited level of knowledge about the material properties and material behaviour can limit applicability of high level of approximation approaches, often used in the detailed assessment. When the original design documents are available and no serious deterioration, design errors or construction errors are observed or suspected, the characteristic value in accordance with the original design may be used. If appropriate, destructive or non-destructive inspections should be performed and evaluated using statistical methods.

For further information see also *fib* Bulletin 80 [30.11-4].

30.11.2.4 Determination of material properties and material behaviour model(s)

Strengths and other material properties to be considered as basic variables are represented by their characteristic values X_k or by their mean values.

The material behaviour model for design of new elements or structures and the basic variables related to material characteristics, may be estimated from testing data, taking into account the intrinsic variability of material properties, the uncertainty of the test method and the uncertainties of the material behaviour model. The approach to determination of material behaviour models is further elaborated in section 30.11.3.5.7.

Test results are used to determine the design values of the basic variables related to material properties to be used in the verification models of structural performance.

For more information, see ISO 2394 [12-8] and ISO 13822 [12-9].

When assessing existing structures, the material properties and the material behaviour model must account for the actual state of the material. If tests are conducted on the existing structure to assess the mechanical characteristics of materials (or update geometric or durability parameters), the representative values of basic variables may be determined consistently with the provisions given in section 12.5.4.1.5.3.

30.11.2.5 Derivation of design values of basic variables related to material characteristics

Following determination of representative values of basic variables related to material characteristics, the design (or assessment) values X_d of the material resistance variable X shall be determined for the design and assessment situation consistently with the provisions in Chapter 12. The design values of basic variables related to material characteristics can be determined:

– by assessing the characteristic value from test results and by applying the partial factor γ_M and conversion factors, where appropriate, or

– by assessing directly the design value from test results

If tests are conducted on an existing structure to assess the mechanical characteristics of materials (or update geometric or durability parameters), the partial factors γ_M may be updated based on the actual mean values and coefficients of variation consistently with provisions given in section 12.5.4.2.2.1. Updating partial factors should be done with due consideration of the magnitude of the coefficients of variation of the measured parameters and if they are significantly larger than the values normally assumed as reference value for new structures, than the partial factors should be updated.

30.11.3 Verification of structural performance assisted by testing of structural element(s) and/or system(s)

30.11.3.1 General provisions

Verification assisted by testing involves subjecting structural element(s) and/or system(s) to a controlled application of known actions, where the outcomes of a limited series of tests are used to determine the response or evaluate the performance of structural members or structural systems.

Different types of testing can be employed, depending on the objectives of the testing:

– load testing to support design (design assisted by testing), which is a controlled load testing that allows for the verification of the design assumptions related to resistance of the structural element or system by comparing the actual performance during testing with the designed performance. In case of non-compliance of the design performance with the actual performance, the design assumptions need to be updated. For provisions on the objectives, preparation, execution and interpretation of load testing (till failure) to support design, reference is made to section 30.11.3.5.

– proof load testing, which is a controlled load testing with a target load that the structural element(s) and/or system(s) must sustain in order to give evidence of the satisfactory performance. Proof load testing is used to directly demonstrate that the structural element(s) and/or system(s) fulfils the

The descriptions in this section address load tests in reinforced and prestressed concrete bridges and buildings. Since the loads for which bridges and buildings need to be assessed are different, the practice of load testing is different in bridges and buildings. The underlying philosophy, however, is the same for bridges and buildings.

Careful consideration should be given to the selection of the appropriate type of testing of structural element(s) and/or system in order to achieve the desired results effectively.

Proof load tests on reinforced concrete bridges can be carried out to demonstrate compliance with the requirements of the ultimate limit state or the serviceability limit state. For prestressed concrete bridges, proof load tests can be carried out to demonstrate compliance with the serviceability limit state requirements. However,

proof load testing for verifications of the ultimate limit state is not recommended, since the generation of cracks is irreversible.

For buildings in reinforce concrete, proof load tests can be carried out on structural element(s) and/or system(s) to demonstrate compliance with the serviceability limit state and the ultimate limit state requirements. In case of prestressed concrete, a careful preparation and identification of stop criteria is necessary to avoid unwanted cracking of the structural elements.

If the target load was not achieved during proof load testing, the results of such partial proof load testing may be used to verify structural performance. However, it's important to note that provisions for verification assisted by partial proof load testing are still under development. Therefore, the procedures for carrying out this type of verification should be carefully considered before any conclusions are drawn.

For bridges, diagnostic load tests are more common than proof load tests and assessments are often carried out by the combination of a diagnostic load tests in-situ and verification by models.

For buildings, proof load tests are more common than diagnostic load tests.

Examples of tests which generally cannot be statistically interpreted e.g. due to a limited number of test trials:

- tests on frictional losses of prestress made by measurements of forces transmitted by tendons on the structure during construction;
- tests for the determination of uncertainties related to static equilibrium made by measurements of the smaller reaction of supports;

requirements of the limit state under consideration which is done by applying the factored loads in a controlled manner. The target load in a proof load test is the magnitude of load that is required to demonstrate that the structure can carry the code-prescribed factored loads without signs of distress. For provisions on objectives, preparation, execution and interpretation of proof load testing, reference is made to section 30.11.3.6.

- diagnostic load testing, which is a controlled load testing to measure structural responses for a set of loading conditions that correspond to service load levels in order to provide a basis for verifying or determining specific response characteristics of the structural element(s) and/or system(s). Diagnostic load testing is used to verify if the overall structural performance or the response of a certain structural element under the applied (service) load is according to the design (for example prior to opening of the structure), to assess certain specific aspects of structural behaviour (such as e.g. action effects under (service) load), and/or to gather information to validate or calibrate models employed for verification of structural behaviour. For provisions on objectives, preparation, execution and interpretation of diagnostic load testing, reference is made to section 30.11.3.7.

30.11.3.2 Scope of application

With respect to load testing to support design (design assisted by testing), Section 30.11.3 covers only those cases involving testing of (new or existing) structural element(s) and/or system(s) in which a statistical interpretation of test results is possible. In cases in which a statistical interpretation of results is not possible, a procedure based on a combination of testing and a priori knowledge may be applied.

In cases where a statistical interpretation of results is not feasible, combined procedures may be employed:

– the Bayesian approach, which provides a formal framework for combining experimental tests with a priori knowledge, which is especially important in case of a small number of tests;

– in case of testing of structural element(s) and/or system(s), prior knowledge may be based on various sources of information, including numerical models of the structural response of interest.

In this context, 'response' may mean stiffness or strength characteristics.

With respect to proof load testing and diagnostic load testing, a priori knowledge of the response to be updated by the results of the load tests is required.

30.11.3.3 General requirements

Special care is necessary for several reasons:

– computational design versus ultimate limit states often covers indirectly serviceability as well and intermediate limit states (which for simplicity are not codified): a modification of the codified provisions for ULS might result in deficiencies in this respect;

– verification assisted by testing is not able to account for some uncertainties (e.g. on structural analysis or secondary effects due to the environment) which are covered by code provisions, nor to justify a reliability format more precise than the codified one.

Verification of structural performance assisted by testing of structural element(s) and/or system(s) must be done with special care and reasonable checks should be exercised to verify the results and conclusions.

All possible critical conditions, actions and possible influences expected during the lifetime of the structure must be appropriately reproduced during testing of structural element(s) and/or system(s).

The testing procedure must be developed by the structural engineer (e.g. expert in designer or assessment) and be approved by the relevant counterpart (e.g. examiner or inspector nominated by the client).

A plan of testing should be developed by the designer, and should contain the objective of testing and all indications necessary for the sampling, manufacturing or extraction of the specimens, the execution of the tests and their evaluation.

Testing must be performed by qualified institutions with staff experienced in planning, executing and evaluating tests.

For guidance on how to report diagnostic and proof load testing of bridges, Alampalli et al. [30.11-5] can be consulted.

30.11.3.4 Reporting

If testing is used to assist in verification, the corresponding report should include, at a minimum, the following types of information, if relevant to the particular case:

- general information and responsibilities:
  - the name of the part asking for the verification assisted by testing,
  - the scope of the verification assisted by testing,
  - date and time of the testing,
  - names and affiliations of personnel involved in the verification assisted by testing including names and affiliations of personnel deployed during the test and responsible measurement engineer,
- description of structural element or structural system tested:
  - in case of laboratory testing: technical description of the laboratory specimens (dimensions, materials, fabrication technique, pre-conditioning, number of specimens etc.),
  - in case of in-situ testing: technical description of structure (name of structure, location, year of construction, representative photograph, type of structure and material etc.),
  - overview of available information (drawings, material investigation reports, original design calculations, incl., if applicable, information on repairs, inspection reports, current condition assessment of structure (for existing structures), preliminary assessment calculations (for existing structures) or description of analytical calculations (for new structures), etc.),
- information from preparation of testing including description of testing procedure:
  - primary and secondary objectives of the testing,
  - motivation for type of testing,
  - description of models used to prepare the test, determination of load layout, magnitude and load position with target load, expected response to load during test,

- testing protocol, instrumentation plan (including stop criteria if relevant), and safety plan with risk analysis for the execution,

– information from execution of testing:
  - observations during testing including all measurements of basic and nominal variables, the actions process, the deformation of the specimens, crack pattern, the failure mode and the critical material, photos and/or video recordings, etc.,
  - observations of discrepancies between available information and actual state of the structure,
  - ambient conditions (humidity and temperature),
  - deviations from planned load magnitudes and positions,
– analysis of test results:
  - review of measurements, post-processing of measurement data, derivation of aspects from measurements (i.e. position of neutral axis, stresses, moments, forces), and if relevant, e.g. for proof-load testing, evaluation of stop and acceptance criteria, comparison between field measurements and model, development of field-verified model, evaluation of stability of load-carrying mechanisms, development of assessment model,
  - outcomes of the analysis in terms of the objectives of verification assisted by testing
– final recommendations.

30.11.3.5 Load testing to support design (design by testing)

30.11.3.5.1 Objectives

The aims of the load testing to support design (design by testing) are:

– to establish directly the resistance or serviceability properties of structural elements or structural systems under specified load conditions
– to obtain values, required for the design, of parameters governing the response of structural members and structural systems under specified load conditions with respect to a certain limit state.

A design assisted by a testing procedure should ensure that the design leads to the same level of reliability as this Model Code. To this end, all possible critical

If some of these variations and conditions are expected to have a systematic effect on the structural response under investigation, their complete reproduction during

testing of structural element(s) and/or system(s) must be secured. In this respect, see for example:

- the in-time variation of the basic variables (see section 30.11.3.5.2.1),
- the condition affecting the conversion factor η (see section 30.11.3.5.5).

The dimensions of the specimens should, if possible, cover the entire range of the probable variation of the dimensions of the structural element.

When scaled-down specimens are used, special attention should be paid to the influence of aggregate size, size and number of rebars, concrete workability and so on, on the response.

Specimens taken from production may be used either for preliminary tests or in the case of small structural elements repeatedly produced under factory conditions defined in section 7.11.3.

Temperature, humidity among others are further influences that can affect the behaviour of the specimens and their response.

conditions, actions and possible influences expected during the lifetime of the structure must be appropriately reproduced in the laboratory.

30.11.3.5.2 Preparation and execution of testing

30.11.3.5.2.1 General provisions

The experimental procedure may assist in verification of either an ultimate limit state or a serviceability limit state. In either case, a sufficient level of knowledge of the relevant response mechanism and parameters is sought.

30.11.3.5.2.2 Test specimens

The specimens should preferably be dimensioned as close to full scale as possible, so that size effects do not affect the validity of the test results.

Otherwise, similitude laws, such as on the basis of fracture mechanics, must be appropriately taken into account.

The specimens should be specifically manufactured for the testing to simulate as good as possible the mechanical and physical material properties which can be expected during the service life of the structure.

30.11.3.5.2.3 Testing protocol

Load testing to support design involves controlled application of known actions. The actions may be direct forces (loads in general), imposed deformations or other influences varying in space and/or in time.

The actions process (“load path”) should, if relevant, also include data regarding rate effects (e.g. load application velocity, the order of application of each influence).

The necessary number of tests is higher where not only mean values but also standard deviations are to be statistically assessed. For more information, refer to ISO 12491:1997 [30.11-6], Kersken-Bradley et al. [30.11-7].

Quantitative and qualitative observations are highly recommended in order to check these assumptions, made implicitly or explicitly, or to allow alternative evaluations afterwards. If, during testing, a given condition is found to influence considerably the results, this condition must be more systematically studied.

When a direct measurement is not possible, an indirect measurement is allowed. Then the respective conversions should be used, by introducing the necessary new variables of the conversion factors or by increasing the variance of the relevant variable. In the case of indirect measurement of a variable, the difference of the statistical properties between the populations in the laboratory and on site must be taken into account.

Since the response of the structural member is in general identified in terms of load intensity at which the limit state is reached, the actual value of the limit load should be measured with additional care.

It is recommended that measurements are redundant in order to make a mutual checking of the results possible.

The scheduled actions process should be fully determined by a set of load (or occasionally of other influencing) parameters. The actions process must be selected so that they are representative for the anticipated scope of application of the structural member.

30.11.3.5.3 Number of tests

The number of tests carried out should be sufficiently large in order to lead to results with satisfactory small statistical uncertainty.

30.11.3.5.4 Analysis of test results

30.11.3.5.4.1 Information from execution of testing

Special care should be taken in order to check that all assumptions made in the preparation of testing (see section 30.11.2.6.) are satisfied.

The actual values of all basic and nominal variables included in the calculation model should be, as far as practicable, determined by direct measurements for each specimen and each experiment. If not, the resulting uncertainty should be accounted for.

The actual values of the imposed actions have to be recorded during the test and especially at the “critical point” of the limit state considered.

During the tests, systematic measurements must be carried out concerning the deformations (elongation, deflection, rotation) and other relevant aspects of structural response.

If relevant, the characteristic values $\mathbf{X_k}$ are substituted in R_k by other representative values depending on the combination and/or the limit state.

The material partial factor γ_M is to be adopted in accordance with this Model Code as:

$$\gamma_M = \gamma_S \qquad (30.11\text{-}7)$$

or

$$\gamma_M = \gamma_C \qquad (30.11\text{-}8)$$

For numerical values of the γ-factors according to the failure mode and to the material decisive for the failure see chapters 11 and 12.

The differences covered by the factor η may, for example, be:

- loading time (when concrete strength is critical, a factor equal to 0.85 should be applied at least to account for sustained loads);
- support conditions;
- humidity conditions and their alternations;
- differences in geometry;
- differences in workmanship and/or curing conditions.

Influences of systematic character should be covered by the model and not by η, for example in case of a brittleness factor such as $(1 - f_{ck}/250)$.

Those conversion conditions included in γ_M should not be duplicated when assessing η.

Only influences of secondary importance may be accounted for by means of conversion factors estimated in a more or less empirical way.

Experts' opinions may be used in evaluating η provided that all systematic differences are taken into account in the tests.

Generally, the characteristic value R_k (usually 5% fractile) is defined as the value for which:

30.11.3.5.5 Determination of design value of the response for ULS by load testing

The design value of the response for the ultimate limit states (ULS) verification is given by the expression:

$$R_d = \eta\, R_k / \gamma_M \qquad (30.11\text{-}6)$$

where:

R_k is the characteristic response defined statistically on the basis of the test results;

γ_M is the material partial safety factor adopted according to the failure mode of the material decisive for the bearing capacity;

η is a conversion factor, taking into account the differences between testing conditions and conditions in the actual structure. If a more detailed model is known, accounting separately for the role of both materials, separate conversion and safety factors may be used accordingly;

The characteristic value R_k may be calculated from test results taking into consideration the statistical uncertainty as a result of a limited number of tests.

$$P\,(R < R_k) = 0.05 \qquad (30.11\text{-}4)$$

For details, see ISO 12491:1997 [30.11-6].

γ_{Rd} values for SLS are generally different than those for ULS. Some minor aspects of a conversion factor may as well be accounted for by this γ_{Rd} factor.

γ_{Rd} should be increased according to experience in cases where the scatter of d in Eq. 30.11.3 is high (e.g. its coefficient of variation is greater than 0.15) or the expected variability of the basic variables is significantly higher on site than in specimens, or the value of the conversion factor on site conditions seems to be less than the already adopted η value.

If such a model is not available prior to testing, trial preliminary tests should be carried out in order to facilitate the identification of parameters with significant influence on the response.

The response depends in general on a set of measurable quantities. The quantities which are random are referred to as basic variables **X**. The quantities which are considered as deterministic are referred to as nominal variables **W**. Nominal variables may be constant or vary in a predetermined manner.

The calculation can either be based on a predictive Bayesian estimate or on assuming a specific confidence level (normally the confidence level is taken equal to an approximate value of 0.75).

A priori knowledge, number of tests and scatter of the test results should be taken into account in the partial factor γ_M.

It is also possible to make a direct estimation of the design value on the basis of the statistical results.

30.11.3.5.6 Determination of design values of serviceability properties for SLS by load testing

The design values of serviceability properties for the serviceability limit states (SLS) verification are given by the expression:

$$R_d = R_k / \gamma_{Rd} \qquad (30.11\text{-}9)$$

where:

R_k is the characteristic response defined statistically on the basis of the test results

γ_{Rd} is a complementary model uncertainty factor intended to cover differences between the testing conditions and the actual ones in the structure, which cannot be accounted by d_k or by the conversion factor η.

30.11.3.5.7 Determination of models of the structural response by load testing

An empirical or physics-based calculation model should be available prior to testing, in order to evaluate the response. This model is referred to as “prior calculation model” g_R.

$$R_t = g_R\,(\mathbf{X}, \mathbf{W}, \mathbf{D}) \qquad (30.11\text{-}1)$$

where:

R_t is the response of the available theoretical model;

X is the vector of basic variables with known statistical distribution;

NOTE: The terminology used in this section and the following ones, with regard to the variables, is not exactly the same as elsewhere in this Model Code because the character of some variables (fundamental or not) is not yet established at this stage. Some variables indeed, although random, may have very little influence on the response.

In case of unidentified significant parameters, a larger scattering of results is expected.

Special care is necessary because statistical parameters of basic variables do not have an intrinsic character. They generally depend on the considered statistical population, which may be very different at the level of the Code, at the level of the specimens and at the level of the expected application of the test result.

Special mention should be made of the quasi-deterministic concrete strength of the specimens which exhibit a very low variability. Thus, in the relevant model, the characteristic strength of in-structure concrete should be made equal to the characteristic strength of the specimen concrete.

Aspects to be considered are for example:

– the in-time development of the concrete strength in a positive or negative way (e.g. by further hardening after 28 days, negative e.g. by effects of deterioration), depending on the concrete composition, ambient conditions and/or exposure;

– the possible decrease of the tensile strength of concrete due to hydrothermal cyclic conditions;

– the possible decrease of steel strength, ductility or fatigue performance due to corrosion conditions.

The coefficients **D** may also be considered as random variables, but in order to check the validity of the equation (30.11-1) in a first step, the coefficient **D** should be given the best value $\mathbf{D_m}$.

For simple models of the type

$$R_t = d\, g_R(\mathbf{X}, \mathbf{W}) \qquad (30.11\text{-}3)$$

W is the vector of known nominal variables;

D is the vector of unknown coefficients to be determined by the testing.

All quantities affecting the response should be present in the model either as basic variables or as nominal variables.

For the basic variables included in the calculation model, statistical parameters have to be known. If these values are not known, it is recommended to estimate them by preliminary tests. Where information is available only from a limited sample, the statistical uncertainty associated with the number of tests should be taken into account.

The in-time variation of the basic variables should be taken into account in the model.

30.11.3.5.7.1 Estimation of the unknown coefficients **D**

The estimation of the coefficients **D** to be determined by testing (see equation (30.11-1) may be based on a statistical analysis such as:

– least square methods;
– maximum likelihood methods;
– Bayesian regression analysis.

where:

d is a single parameter (i.e. model uncertainty) and the distribution of d is a central t-distribution. When elaborating the test a distinction should be made between variable x_i that can be measured in each test and variable x_i that cannot be measured.

Often it is known that coefficients can only take values larger or smaller than unity.

Generally, the characteristic value R_k (usually 5% fractile) is defined as the value for which:

$$P\,(R_t < R_k) = 0.05 \qquad (30.11\text{-}4)$$

Accordingly, appropriate statistical analysis must be conducted or the following simplified approximation must be applied:

$$R_k \approx d_k\, g\,(\mathbf{X_k}, \mathbf{W}) \qquad (30.11\text{-}5)$$

Proof load testing is a method of assessment that can be particularly interesting for structures with large uncertainties, that can be related to the (lack of) information

For linear models or models which can be transformed to linear, coefficients should generally be assumed as normally distributed.

Where no linear models are obtained, a Taylor expansion in the vicinity of the expected design point may be used to avoid non-linear regression analysis.

When coefficients of applied models are to be estimated, prior information on those coefficients should be derived from physical considerations.

30.11.3.5.7.2 Estimation of characteristic value of response R_k

The characteristic value R_k may be calculated from test results taking into consideration the statistical uncertainty as a result of a limited number of tests.

The calculation can either be based on a predictive Bayesian estimate or on assuming a specific confidence level (normally the confidence level is taken equal to an approximate value of 0.75).

30.11.3.5.7.3 Estimation of the design value of response R_d

The design value of the response is determined according to the approach reported in section 30.11.3.5.5 (ULS conditions) and section 30.11.3.5.6 (SLS conditions).

30.11.3.6 Proof load testing

30.11.3.6.1 Objectives

The objective of proof load testing is to verify the compliance with the limit state requirements of the codes by demonstrating directly that the response measured

available about the structure, to the effect of deterioration on the structural capacity and to the overall structural behaviour at load levels beyond the serviceability state.

Several countries have national guidelines or standards for proof load testing. In some cases, only bridges or only buildings are addressed, and in other cases, only diagnostic or only proof load testing is addressed. Examples of relevant codes include ACI 437 [30.11-8], DAfStb [30.11-9], AASHTO [30.11-10], and guidance is provided in Alampalli et al. [30.11-5] and *fib* Bulletin X TC3.2 [30.11-11].

In the case of proof loading at the ultimate limit state, the fact that the structure (or elements of the structure) does not fail during the test verifies that the limit state was not exceeded, provided that it can be shown that all load-carrying mechanisms that occur during the test can also be counted on in the future (e.g., frozen bearings / effect goes away after replacing bearings etc).

Structural modelling can aid in interpreting partial proof load testing results, but careful considerations are necessary before drawing conclusions.

Since load testing requires detailed preparation to make sure the objectives of the test can be met, the reader may want to refer to documented case studies of proof load tests by Lantsoght et al. [30.11-12], [30.11-13], [30.11-14], [30.11-15]. For step-by-step flowcharts describing proof load tests on bridges, the reader is referred to Alampalli et al. [30.11-5]. Guidance on how to carry out a proof load test on a prestressed concrete bridge for serviceability limit state see Aguilar et al. [30.11-16].

For buildings the number of (adjacent) spans to load in order to create the relevant load effects needs to be in line with the test objectives. For bridges, the number of positions should be in line with the test objectives.

during proof load testing does not exceed the specified limit state magnitude. In a proof load test, the limit state of interest can be an ultimate limit state (for reinforced and prestressed structures), or a serviceability limit state (for prestressed structures).

In proof load testing, a load representative of the factored live load, the so-called target proof load, is applied to support verification of structural performance. The target load is the load required to demonstrate that the structure can carry the code-prescribed factored loads without showing any distress.

In proof load testing, the limit load must be determined which is the load at which specified distress is observed.

When the target load is reached in proof load testing without the limit load being reached, the verification can be considered successful and the structure has been shown to be able to carry the code-prescribed factored loads without signs of distress.

When the limit load is reached before the target load is reached, further careful analysis is necessary to determine the maximum service load on the structure, but this process should be approached with caution as provisions for verification assisted by partial proof load testing are still under development.

30.11.3.6.2 Preparation and execution of testing

30.11.3.6.2.1 Calculation model

Prior to conduct a proof load testing, the target load and the limit load must be determined.

The position of the load and the target proof load are related to the limit state of interest and should be in line with the test objectives. Different types of load arrangements may be employed depending on the limit state which is being investigated.

The stop criteria should be related to the type of structure (girder, slab,…), the material (reinforced or prestressed concrete), the failure mode of interest, and the uncertainties on the structure, its state, and the failure mode of interest,. The stop criteria should be expressed in terms of measurable responses, should be developed together with the instrumentation plan, and should consists of a set of criteria to allow for redundancy. Stop criteria should be monitored closely during the test and inform the decision if further loading is allowable.

The relevant stop criteria depend on the type of structure and the expected governing failure mode. Typically, a set of criteria is prescribed. These could include Lantsoght et al. [30.11-17], Lantsoght et al. [30.11-18] limiting strain in the concrete, limiting strain in the reinforcement steel, limiting crack width, changes in stiffness, qualitative evaluation of deflected shapes as well as load-deflection responses, or wave-based parameters. Stop criteria for flexure are well-defined and available in existing codes and guidelines (in Germany DAfStb Richtlinie [30.11-9] and Spain Ministerio de Fomento [30.11-20]).

Stop criteria for shear are the topic of current research.

Several methods of load application are possible for proof load tests. The use of dead loads is not recommended, since it may be unsafe as loads cannot be removed quickly when a stop criterion is reached. If dead loads are used during a proof load test in a building, scaffolding may need to be considered. A second possibility for load application in bridges is the use of specialized vehicles, very heavy trucks, or the combination of army tanks on trucks. A third option is the use of hydraulic jacks. This method is most suitable in combination with the application of a loading protocol. An example of distributed dead load is the use of water tanks or controllable water level in a tub. The application of dead load however is only recommended when it can be done in a safe manner and when unloading can occur quickly when there are signs that further and/or sustained loading may lead to permanent damage or collapse. Using hydraulic jacks is usually a safer method.

Several load levels may be applied, and several load cycles at each load level are necessary to check linearity and reproducibility of response measurements. The highest load that is to be applied is the target proof load. This load should be sufficiently large, so that it results in the same sectional moment or force as the factored load combination for which the structure is assessed.

For proof load testing, stop criteria need to be identified prior to the test. A stop criterion is defined as the threshold after which further loading will result in irreversible damage to the structure. When such a threshold is reached during testing, further loading may lead to permanent damage or collapse of the structure.

At this moment, proof load testing of shear-critical structures is not recommended yet.

30.11.3.6.2.2 Testing protocol

To prepare for a proof load test, the loading protocol should be established, taking into consideration the load application methods and the position of the load, the target proof load and the limit load, and the stop criteria determined with the calculation model.

The loading protocol should describe the different load levels and cycles that will be applied. For proof load testing, at least four load levels should be used and all structural responses should be checked at each load level before a higher load level can be attempted. A loading protocol can be monotonously increasing or cyclic. A cyclic load protocol requires for a better quantification of structural responses under loading and unloading.

Most codes and national guidelines prescribe that first, one has to calculate the target proof load, and then apply the load in increments to reach the target. Proof loading beyond the target is only recommended when it really is the objective of the test to find the onset of nonlinear behaviour. The described approach certainly would not be recommended for prestressed girders.

Proof loading beyond the target can be managed by the use of acoustic noise emission sensors, which identify the development and increase of micro-cracking in the concrete component of a reinforced concrete structure as the applied load is increased towards failure. Of course, to employ this form of testing there needs to be certainty that the failure mode will be ductile, and definitely not brittle.

Additional practical steps during the preparation of a load test include developing a planning for the day(s) of all on-site activities (form the first on-site preparatory activities to the final dismantling of sensors), setting up the safety plan, restricting access to the structure during testing, and carrying out a risk analysis of the on-site activities.

At the moment, the required number of proof load tests is still a topic of research. For buildings, research results on this topic are given by Lantsoght et al. [30.11-17].

In DAfStb Richtlinie [30.11-9] there is now a guidance table for buildings.

A second important step in the preparation of a proof load test is the determination of stop criteria. The development of the stop criteria goes hand in hand with the development of the instrumentation plan, which should serve the purpose of monitoring the structural responses under increased loading, evaluating the stop criteria at every load level and during load increases, and addressing any additional load test objectives.

Increasing the load after reaching a stop criterion is generally not permitted, and is never permitted without the explicit approval of the structure's owner.

During the execution of a proof load testing, an estimate of the mean capacities should be used to evaluate the risks during the test.

30.11.3.6.2.3 Special considerations

Important aspects during proof load testing is to include the communication with all parties (e.g. owner, structural engineer, etc.) involved during the load test and all safety protocols, which should be communicated to all parties involved in the test.

During proof load testing of bridges, temporary closure of the bridge is recommended. Traffic management and/or rerouting should be arranged prior to the test. For load testing of buildings, closure of the building and temporary shoring should be used to allow for safe testing procedures.

30.11.3.6.3 Number of tests

The number of proof load tests to be carried out on the same structure depends on the type of structure, geometry of the structure, expected variability of material properties and the considered ultimate limit state. The required number of proof load tests should be evaluated prior to the test, when the loading protocol is developed.

Composite action between structural and non-structural components may mean that the structural components appear to perform differently than if they were tested in isolation. Load sharing occurring between individual structural components forming a larger element can produce different results, floors formed from individual precast concrete beam of plank components are a particular case. Such influences may need to be taken into account when load testing regimes are designed and when interpreting the results of load tests.

Non-structural elements often have a significant influence upon the observed stiffness and strength of both individual structural members and the overall structure under normal and accidental load situations, especially where damage has been caused and alternative load paths are mobilised.

Accordingly, it is unwise to extrapolate load test results beyond the load level applied to a structure during a proof load test.

30.11.3.6.4 Analysis of test results

The load effects in the governing cross-section caused by the externally applied load i.e. the quantities such as the sectional moment or force induced by the applied load, are the evaluated outcome of the test. This resulting sectional moment or force can then be considered as the lower bound of the capacity of the structure or structural element.

30.11.3.6.5 Assessing performance level of the existing elements or structures by proof-load testing

In the evaluation stage, the post-processing of all field measurements takes place together with a more detailed analysis of the test observations and measurements. For a proof load testing, this stage also includes the final evaluation of the stop criteria. In addition, acceptance criteria related to the long-term behaviour of the structure and its service life need to be addressed.

A proof load test generally gives a direct outcome: if the test is successful (i.e. the target proof load can be carried by the structure without signs of distress), it is directly demonstrated that the structure can carry the code-prescribed factored loads that correspond to the target proof load for the failure mechanisms targeted by the proof-load test. However, before a final conclusion can be made, it must be evaluated if all load-carrying mechanisms that contributed during the proof load test can be counted on in the long-term. If there are mechanisms that may cease to contribute over time, additional analysis may be necessary.

The final step of the evaluation includes reflecting on the original verification objectives and formulating recommendations to the owner of the structure.

30.11.3.7 Diagnostic load testing

The diagnostic load test does not give information with regard to the capacity, but it provides information to evaluate particular aspects of performance, such as height of the compression zone, load distribution among structural elements, unintended composite action.

Diagnostic load testing can be static or dynamic. A static load test addresses the static behaviour and static properties of the structure, while a dynamic load test addresses vibrational properties, impact factor for bridge live loads, and other dynamic properties of the structure.

In some countries, a diagnostic load test is required on bridges prior to opening to demonstrate that the bridge behaves as designed under live loading. Elsewhere, the use of a diagnostic load test can be required to verify the performance of a structure that was designed using new materials, construction methods, or using a new type of structural elements.

For step-by-step flowcharts describing diagnostic and proof load tests on bridges, the reader is referred to Alampalli et al. [30.11-5].

Typically, for diagnostic load testing of bridges, trucks are used. As such, the loading on the trucks should be determined, the number of trucks to be deployed, and the load paths of the trucks. The load should be large enough that measurable responses can be obtained during the load test.

Typically, for diagnostic load testing of bridges, trucks are used. As such, the loading on the trucks should be determined, the number of trucks to be deployed, and the load paths of the trucks. The load should be large enough that measurable responses can be obtained during the load test.

The following structural responses can be typically considered to be measured during the load test:

- deflection;
- rotation;

30.11.3.7.1 Objectives

Verification assisted by diagnostic load testing involves applying a controlled load that corresponds to service load levels, recording structural responses, and can serve to verify the overall structural performance or the response of a specific structural element under the applied load.

In particular, it may allow for verification if the structure responds as intended under service load (see section 30.11.3.7.5), enable determination of the actual action effect model (see section 30.11.3.7.6) under the applied load, enable determination of the structural model for assessment of structural capacity by NLFEA (see section 30.11.3.7.7), or allow structural identification (see section 30.11.3.7.8).

30.11.3.7.2 Preparation of testing

In a diagnostic load test, the limit state of interest is evaluated by using the field-verified model and resulting assessment model.

To prepare for a diagnostic load test, the magnitude of the load that will be used during the test should be determined.

Additionally, to prepare for a diagnostic load test, the instrumentation plan should be developed. This plan should cover the necessary instrumentation to measure the structural responses that should be evaluated to address the test objectives.

– strains;
– crack opening/closing (e.g. crack width, crack length, crack position, etc.);
– support settlements;
– temperature and relative humidity.

This difference may be given in a national code or guideline. For example in Spain, see Ministerio de Fomento [30.11-19] and [30.11-20], the maximum measured deflection in a load test on a prestressed concrete bridge cannot exceed the analytically determined deflection by more than 10%, and for reinforced concrete bridges this value is 15%. If the measured deflection is less than 60% of the

30.11.3.7.3 Number of tests

The number of diagnostic load tests to be carried out on the same structure depends on the type of structure, geometry of the structure, expected variability of material properties and the considered ultimate limit state. The required number of diagnostic load tests should be evaluated prior to the test, when the loading protocol is developed.

30.11.3.7.4 Execution of testing

The loading protocol shall described the different load levels and cycles that will be applied. For diagnostic load testing, it is recommended to carry out trial runs before the truck paths in order to verify proper functioning of all sensors. A loading protocol can be monotonously increasing or cyclic. A cyclic load protocol requires for a better quantification of structural responses under loading and unloading.

Important aspects during diagnostic load testing also include the communication with all parties (e.g. owner, structural engineer, etc.) involved during the load test and all safety protocols, which should be communicated to all parties involved in the test.

During diagnostic load testing of bridges, temporary closure of the bridge is recommended. Traffic management and/or rerouting should be arranged prior to the test. For load testing of buildings, closure of the building and temporary shoring should be used to allow for safe testing procedures.

30.11.3.7.5 Verification of response at service load levels

In order to verify the overall structural performance or the response of a specific structural element under the applied load by diagnostic testing, the difference between the field test and the model should be small, and the responsible engineer should be able to explain the source of the differences.

analytically determined deflection, the discrepancy should be further investigated. What constitutes of an acceptable prediction depends on the material used, the structural system, and even the sensing technique deployed. It may be difficult to quantify an acceptable % of error here.

It should be recognised that environmental conditions, especially temperature, can have a large influence on the static and dynamic measurands. These influences must be considered when evaluating the structural properties.

30.11.3.7.6 Determination of the action effect model

Information on structural responses under controlled loading obtained during diagnostic load testing may be used calibrating the action effect model based on quantifying from diagnostic load testing the uncertainty of the load effect model.

30.11.3.7.7 Determination of the structural model for assessment of structural capacity by NLFEA

Information on structural responses under controlled loading obtained during diagnostic load testing may be used to develop a field-verified NLFE model which matches the field test results as close as possible.

This field-verified model is a model improved with the measured structural responses in the field. When formulating this model, for all mechanisms that contribute to carrying load during the test it should be evaluated if these contributions can be accounted for long-term and if these contributions exist under higher loads. In addition, results of inspections and material testing should be used as basis for the setup of this model.

An assessment model can be created from this field-verified model by removing load-carrying mechanism that are not reliable in long-term or at higher load levels, and by introducing the relevant factors and using the approach to assessment of the structural capacity by NLFEA (section 30.10.3.1). For general recommendations regarding the approach to assessment the reader is referred to Chapter 28.

30.11.3.7.8 Structural identification

The best practices for structural identification were summarised as a six-step analysis-experiment-decision integration cycle by the ASCE Technical Committee on Structural identification of constructed systems, see Çatbaş et al. [30.11-21], as follows:

- Observation and conceptualisation;
- A-priori modelling;
- Controlled experimentation;
- Processing and interpretation of data;
- Model calibration and parameter identification; and
- Utilisation of the calibrated model (i.e. Digital-twin for simulations and decision-making).

When system identification is applied periodically or even permanently, the time variance of structural properties by the deterioration processes or other damage causing events, can be identified and monitored. The structural model can then be updated using the new measurement data taken.

Structural identification and performance evaluation procedures have been employed to establish the causes of performance deficiencies of highway bridges, see for example Ye et al. [30.11-22].

Access to both an a-priori model (used for designing the field experiments) and a model (i.e. a digital twin) calibrated by parameter identification and used to perform simulations, allowed options for corrective measures to be explored and appropriate interventions to be (confidently) proposed.

Structural identification procedures enable to use response from diagnostic load to refine the structural model so that it reflects the behaviour observed for the actual structure. Typically, structural identification seeks a mechanistic and quantitative description of a structural system in the context of a physics-based finite element model.

Structural identification procedures are also an efficient tool for damage detection, localization and quantification.

The static response like displacements (e.g. deflections, inclinations) measured under defined loads, as well as dynamic response like acceleration, can be used to identify structural - such as stiffness - or modal, such as natural frequencies and mode shapes – parameters of the structural model.

If dimensions and material properties of the real structure cannot be estimated by direct measurement or testing (e.g. due to inaccessibility, hidden damage, etc), structural properties such as stiffness of structural members and joints, flexibility of hinges or bearing conditions can be estimated by system identification and model updating procedures.

30.11.4 Verification of structural performance assisted by monitoring

30.11.4.1 General provisions

Verification assisted by monitoring is considered here as a procedure where tests and sensor systems under controlled or not controlled loads (LIT See Part II. Figure 1) on limited series of representative specimens, structural components or systems are used for the measure of the actions, of the response of the material, the structural members or the structural systems.

Verification assisted by monitoring refers to specific measures to be undertaken to assess the condition of structures There are various cases in which this may be required, such as e.g.:

- large structures for which monitoring with extensive use of sensors is as preventive measure to control their performance in order to avoid disasters (e.g. dams);
- situations where there is a change of intensity of actions on the structure (change of function, increase in traffic or live loads, etc.);
- situations where there is a sudden change in structural performance due to damage (earthquakes, fire, explosions, etc.);
- situations where there is a gradual change in structural performance due to deterioration (steel corrosion, ASR, fatigue, etc.)
- structures that fall outside the Model Code provisions and cannot be assessed using conventional methods (e.g. analytical or numerical methods), thus requiring special procedures to verify their performance.

Redundancy can be provided in various ways, depending upon the importance of the measured parameter and its location. One way is to provide additional instrumentation of the same type located in parallel or, alternatively, by additional

Monitoring refers to frequent or continuous, normally long-term, observation or measurement at predetermined intervals of structural condition, actions or structural response.. Monitoring is considered to enable the observation of the material and structural response evolution during the lifetime under environmental and general conditions (including critical conditions) and actions.

Usually, verification assisted by monitoring will be combined with testing of representative specimens, structural components or systems. Testing refers to a discontinuous experimental observation of the structural response or the use of samples to test e.g. material properties. as discussed in section 30.11.2 and 30.11.3.

30.11.4.2 Scope of application

Monitoring can be applied for determining the structural performance of concrete structures and concrete structural elements, during and for the assessment or strengthening of existing structures. Another objective of monitoring and/ or testing is the estimation of model input parameters under real on-site conditions to predict structural performance.

Monitoring can be applied to any existing structure or component. However, it is in particular of importance for those existing structures that are not included in the Model Code's range of application.

In cases in which a statistical interpretation of results is not possible, a procedure based on a combination of testing and monitoring, as discussed in section 30.11, and numerical verifications, as discussed in section 30.10, can be applied.

Monitoring, frequently in combination with testing, may also support the life cycle management, see chapter 34.

30.11.4.3 General requirements

Redundancy of instrumentation in critical areas and/or in respect of important monitored performance characteristics, can reduce vulnerability to sensor or system malfunctions.

sensors positioned at complementary parts of the member or structural system in question. In particularly important circumstances, it might be wise to provide additional measurement systems that operate on different principles (e.g. strain gauging and deflection measurement, or two different methods of measuring deflection or strain, etc.). It can also be useful, where practicable, to perform occasional calibration checks against some form of independent external measurement system (e.g. verification of the monitored deflection results against those obtained via an independent laser system).

With respect the need to identify different forms of structural and material behaviour, it might be appropriate for sensor distribution to be considered at the macro-, meso- and micro-scales (as defined previously in section 34.2.3 in relation to the diversity of the service environmental conditions experienced). Considerations on selecting appropriate sensor locations might also potentially be linked with the risk-based approach to identifying hazards, evaluating potential failure modes and recognising vulnerable points.

Care needs to be taken with the selection, positioning and orientation of the sensors forming the monitoring system, as this will influence the sensitivity of that monitoring system to the behaviours and effects being experienced by the structure. Care is also required to ensure the correct handling and processing of the measured data.

Figure 1.1 of *fib* Bulletin 33 [30.11-23] presents illustrative hazard scenarios for prestressing steel in a 'typical' concrete box girder bridge, providing indications of potentially 'weak points' where water (possibly contaminated with chlorides and/or other substances) could gain access to the tendons, cause corrosion and thereby adversely affect the structural performance / safety of the bridge. Figure 1.1 of *fib* Bulletin 33 [30.11-23] illustrates how a risk-based approach to identifying hazards and potential 'weak' or vulnerable points, combined with an evaluation of the relevant potential failure modes and the resulting consequences, would assist in identifying the most appropriate locations for inspections, testing and monitoring activities.

Ref to 34.2.3.

Ref to TB 3.3 (*fib* Bulletin 22 [30.11-24]).

It is recommended that monitoring systems incorporate a degree of redundancy in the monitoring instrumentation in order to allow cross-checking of the outputs of sensors to allow verification of critical in-service sensor readings and the detection of malfunctions of the monitoring system, so avoiding possible misinterpretation of structural behaviour.

Consideration should be given to the benefits of additional sensor locations and measurements. These can help improve the degree of resolution of the behaviour being experienced by a structure, while enhancing the redundancy within the monitoring system / the measured data set.

Monitoring may be performed through various types of techniques and can be classified in action and reaction monitoring depending on the parameters to measure. (LIT see Part II table 3 to modify for other types of structures including buildings, bridges, and tunnels)

Caution should be exercised when interpreting the collected data, recognizing that measured data obtained by a structural monitoring system does not necessarily display all aspects of the actual behaviour being experienced by a structure (i.e. reality).

Consideration should be given to employing a risk-based approach to identify the most appropriate locations for monitoring activities. Locations from which monitoring data is collected must be carefully selected so that the desired information about the condition or performance can be obtained, keeping in mind factors such as:

- the environmental conditions, usage conditions and the nature of the threats / hazards posed;
- the likely mechanism(s) and rate of deterioration;
- locations of greatest hazard (e.g. the locations subject to the most severe exposure conditions, potential deterioration / damage 'hot spots');
- locations where the structure is most vulnerable to deterioration or damage (e.g. half-joints in bridges); and
- the conservation strategy and tactics, and the inspection testing and monitoring regimes defined at the time of design or re-design.

Results of a monitoring campaign should be reported in a standard fashion to reduce time and cost associated with review and archiving of a campaign.

30.11.4.4 Preparation, execution and reporting

Preparation and execution of the monitoring campaign should adequately address and the reporting should contain at least:

- specification and documentation of the responsibilities:
  - who designed the monitoring campaign ?
  - who installed monitoring system and executed monitoring ?
  - who evaluates the data ?
  - who developed the evaluation algorithm?
- specification and documentation of general information:
  - the objective of monitoring campaign;
  - the type of monitoring campaign;
  - specification of monitoring equipment and associated tools like software; version and date of the release.
- specification and reporting of testing or/and monitoring preparation and checking:
  - consistent use of units;
  - material or/and structural parameters;
  - type, number, and if appropriate, the sensor and testing scheme, a plot of the monitoring or/and testing layout;
  - description and plot of the testing procedure and its constraints, loading, environmental conditions and its boundaries/limitations;
  - miscellaneous data necessary to re-process the tests or monitoring if necessary;
  - in case the behaviour of the material or/and structural parameters recorded by the used testing or/and monitoring are not obvious, like is e.g. the case for materials with advanced lateral effects, a report with the testing or/and

monitoring results of single tests or sensors with e.g. well defined strain paths is strongly recommended.

– specification and reporting of monitoring set up:
  - information about the test and monitoring set up as well as the material or concrete structure (e.g. sensor types, alignment of sensors, measuring amplifier, product data sheet etc., cabling, data acquisition temperature compensation, magnetic shields, sampling rate, etc.);
  - the loading scheme and schedule;
  - time used for the testing or/and monitoring (only if significant);
  - the used analyses or assessment techniques and its associated performance indicators/ thresholds;
  - discuss warnings issued during testing or monitoring and motivate why these can be ignored;
  - convergence behaviour and stability and accuracy of the monitoring system itself;
  - the number of significant points and its causes in the testing or monitoring history;
– specification and reporting of monitoring results:
  - a plot of the test or sensor readings of the observed indicators for all load cases;
  - monitoring fields, and history data of significant points in the structure;
  - a comparison of the results of the testing/monitoring with the expected outcome; for instance based on an analysis of simplified model or section analysis;
  - discussion of the validity of the results both in qualitative and quantitative sense.
– specification and reporting of post-testing/monitoring checks to asses possibility of a different, and sometimes more dangerous, failure mode such as shear failure. The testing/monitoring results should be checked for:
  - regions with critical strain situations;
  - regions with open cracks;
  - determining crack width from crack strain and equivalent length;

- possible shear failure, especially when beam or shell elements are tested or monitored.

30.11.4.5 Structural Health Monitoring

30.11.4.5.1 Objectives

Structural Health Monitoring (SHM) refers to use of monitoring technique for the continuous observation and evaluation of structural integrity. SHM may involve short-term, periodic or continuous measurement of time-variant parameters such as displacements, strains and stresses, crack width (as a proxy for damage) and vibration characteristics etc., with the aim of detecting changes in structural properties. In some installations, alarms can be triggered when limit condition states are reached or exceeded.

The main objective of SHM is to support the decisions of asset owners and operators in the through-life management of structures, with regard to:

– the verification of the compliance of the structure against normative requirements,

– the support to planning of maintenance activities and rehabilitation interventions,

– the support to decisions related to lifetime extension,

– verification of repair or rehabilitation project efficacy.

In case of short-term monitoring, the monitoring system is temporary installed and the parameters of interest are measured for a limited period of time. This type of monitoring is recommended for the assessment of the structural performance after an accidental event or after retrofitting interventions or for the duration of the nearby construction works (or other hindering activities) or during a load test of a structure or for the monitoring of actions.

In case of periodic monitoring, the monitoring system is installed at regular intervals over time for a limited time period. This type of monitoring is recommended for tracking the evolution over time of degradation and damage phenomena or for the monitoring of the evolution of actions.

In case of continuous (or permanent) monitoring, continuous (or permanent) monitoring: the monitoring system is operational for long periods of time. This type of monitoring is recommended for the control of the structural integrity in operational and accidental situations.

SHM can be applied to newly constructed structures as a tool for life-time monitoring. In these circumstances the technique can estimate the structural properties of the original structure in an undamaged state. This allows future data to be compared with this benchmark, facilitating the direct identification of changes in the structural properties.

SHM can also be applied to existing structures when their performance is assessed as being "just sufficient" or "just insufficient" (i.e. potentially being marginally adequate or inadequate for the defined performance requirements). SHM allows such structures to be carefully observed in a continuous / semi-continuous manner. This allows to implement remedial actions to enable the structure to safely remain in-service, thereby avoiding the need for them to be demolished, with all the

social disruption, cost and adverse environmental impacts that could be involved with such a course of action.

Monitoring campaigns may also be used to evaluate the development of actions over time.

For more background regarding definition of monitoring objectives refer to Rytter [30.11-25], Allaix et al. [30.11-26], Allaix [30.11-27]. For further details upon SHM refer to Sparkes & Webb [30.11-28], Reynders et al. [30.11-29] or Sousa et al. [30.11-30].

Information acquired by inspection, testing and monitoring the condition of the structure can greatly improve the accuracy of performance prediction by more accurately assessing the variability of the input parameters, which are typically assumed to be random variables. This can help reduce the uncertainty associated with the input parameters and aid characterization of their variability. Ref to 34.2.5.1

The specific monitoring objectives, related to this main objective, commonly include:

- the evaluation of the development of the condition of structure during construction and operation incl.:
  - detecting the existence of damage to the structure (detection);
  - locating where the damage has occurred (localization);
  - identifying the types of damage experienced; and
  - quantifying the severity of the damage (quantification).
- identification of the cause of structural performance deficiencies (e.g. presence of defects, damage or deterioration,
- the evaluation of the development of the structural behaviour during operation,
- the evaluation of the performance of non-inspectable details or elements,
- the early warning for critical conditions, critical structural response or critical actions.

It is recommended that monitoring systems incorporate a degree of redundancy in the monitoring instrumentation in order to allow cross-checking of the outputs of sensors to allow verification of critical in-service sensor readings and the detection of system malfunctions, so avoiding possible misinterpretation of structural behaviour.

Caution should be exercised when interpreting the collected SHM data. Structural modelling can be a powerful tool to assist with the interpretation of the measured data and help understand the behaviour being experienced by a structure.

30.11.4.5.2 Use of SHM in combination with survey

30.11.4.5.2.1 General

Tools and techniques for monitoring are often combined with tools for inspection, and testing, which may also be employed as a part of a monitoring procedure when applied as components of a periodic / repetitive survey regime.

To reduce uncertainties about load and resistance parameters, site specific data should be used within the assessment process whenever practicable. This data is

Surveys may include e.g.:

- Verification whether the asset / structure being inspected is consistent with the available documentation for the design and any structural alterations / interventions made whilst in-service. This is achieved by checking site findings with available specifications, drawings and other documents;
- Structural integrity (e.g. for hidden damage or inhomogeneity) using, for instance, impact echo testing or dynamic monitoring.
- Material strength using tension and compression tests on material samples taken from the structure, plus on-site testing using accelerometer hardness test, rebound test hammer, pull-out tests, pull-off tests, etc.;
- Cross-sectional and longitudinal geometry changes (i.e. effects of damage) due to overloading (e.g. visible cracks, ruptures), other damage processes (e.g. fatigue cracks) and from deterioration processes (e.g. corrosion, spalling, delamination) using lasers, ultrasonic devices, slide gauges, electronic gauges, etc.;
- Parameters influencing the dead load and the superimposed dead load (e.g. material densities, permanent equipment, unplanned variations / errors in structure geometry);
- Parameters influencing durability / length of service life of the structure (e.g. environmental conditions, carbonation depth, chloride content of concrete, achieved concrete cover depth, location of zones of low concrete cover depth, corrosion of reinforcement) using pH-indicator- test, quantitative chloride analysis of samples, electrochemical tests, etc.; and
- Serviceability (e.g. surface crack widths, excessive deflection and/or deformed profile, excessive vibration).

obtained from site inspections and material testing. There are a large variety of methods which may be employed, starting with simple visual or photographic inspection procedures and extending to various high-end non-destructive/non-invasive imaging techniques.

Inspections are especially useful for the detection and investigation of deterioration/damage processes, such as corrosion and fatigue (see below), and for detection of changes in the structural systems. Therefore, it is necessary to conduct inspections regularly/periodically.

Material tests may be undertaken to determine the strength parameters (e.g. compressive, tensile, shear strength) of the construction materials used. The tests which can be conducted on site can be semi-destructive or non-destructive in nature. Alternatively, material samples can be recovered from the structure and tested at an approved / certified laboratory.

30.11.4.5.2.2 Tools and techniques for surveys

A survey is a process, often involving visual examination but which may utilize various forms of sampling and localized condition testing (e.g. measurement of depth of cover to reinforcement), whereby information is gathered about the form and current condition of a structure or its components. Tools and techniques for surveys can involve a wide range of procedures. Typically, they are likely to include a combination of visual observations, material sampling and possibly selected non-destructive and non-invasive testing methods, as well as procedures including structural health monitoring and associated techniques.

Similarly to monitoring, survey regimes can be classified on the methods to be used, their frequency, the planning and timing of inspections/surveys, as shown in Figure 30.11-2.

An inspection is a primarily visual examination, often at close range, of a structure or its components with the objective of gathering information about their form, current condition, service environment and general circumstances.

The testing undertaken is usually non-invasive or non-destructive in nature.

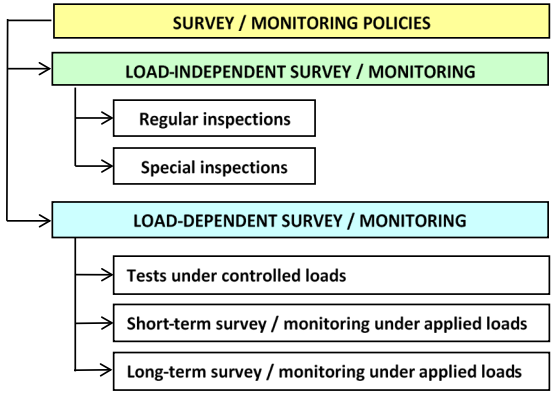

Figure 30.11-2: Survey / monitoring strategies differentiated on the basis of the measurement of load related effects.

30.11.4.5.2.3 Study of documents

Some degree of investigation / due diligence review should be undertaken of the available documents to verify that they provide a satisfactory / acceptable (valid) representation of the structure under consideration.

In addition to information about the specific structure in question, there is generally a large volume of supplementary explanatory information which relates to systems of construction, codes of practice, contemporary advice and guidance notes, textbooks and papers and associated matters such as patents. This type of information is most likely to be available from professional institutions and specialised sources such as research establishments, trade associations, libraries and manufacturers. Ref to 34.2.5.3

Much useful information about a structure is contained in the documents prepared during its design and construction (e.g. design calculations, material specification, material delivery documentation, etc.). Reports of earlier surveys and maintenance works, plus documents concerning subsequent modifications and repairs will provide an insight into the history, use, maintenance and problems encountered with the structure whilst in service.

Past and present codes can provide guidance on the nature and magnitude of loads which a structure might have been designed for. Information on environmental conditions / loadings may be available from design and / or re-design studies (possibly for adjacent structures), inspection and monitoring reports.

30.11.4.5.3 Diagnostics based on SHM data

Diagnostics of structures is intended as the process of reviewing data gathered from monitoring, eventually in combination with other sources of data such as inspections and testing, that enables the evaluation of the current condition of the structure, as well as the prediction of damage and prognosis of future performance.

The condition survey results are usually compared to specified requirements and standards for determining whether the behaviour of the item or activity is meeting those performance targets.

Identification and evaluation of active damage or deterioration processes is essential in diagnostics based on SHM. Observable / measurable indications of such processes can provide a basis for anticipating the future development of damage / deterioration / change in behaviour of a structure / constructed asset, along with the associated effects upon its in-service performance.

As activities required for conservation of a concrete structure to meet its design service life must be planned when designing the structure, this process would be expected to lead to the identification of deterioration mechanisms, which might potentially adversely affect the structure or its components.

Data-driven methods extract damage indicators (DIs) relying solely on the recorded data and on picking up anomalies in damage sensitive indicators, with respect to those corresponding to regular operating conditions (e.g. the shifts or the differences in data sets). Where possible, data-driven methods for damage detection should be used in combination with physics-based models. Data-driven methods are attractive for adoption within real-time damage identification, preferably used in combination with physics-based models.

Model-based methods most typically rely on the use of data to update parameters of a numerical, often finite element (FE), model. They are usually less attractive for real-time damage identification, due to the computational effort of the updating process, but the availability of a FE model allows for higher-end damage characterization and, if a degradation model is available or can be defined based on monitored data, they can further provide salient information regarding the remaining service life of the structure. The FE models used for damage detection must

In this Model Code, diagnostics of structure based on SHM is understood to include all activities performed to gather information regarding the form and nature of the structure, current or potential deterioration mechanisms and/or change in performance of the structure or service conditions, which are needed to evaluate conformity with the design (data) for actions and/or material and/or product properties. Diagnostics of structure comprises damage detection, actions evaluation and identification of system limitations. Diagnostics of a structure would also seek to gain an understanding of the (previous) circumstances which have led to the development of that state, together with the associated mechanisms causing damage or system limitations Provisions in this section refer to provisions in section 34.2.1 and section 34.3.2.

For evaluation of the development of condition of structure is necessary to identify the deterioration and/or damage mechanisms which are affecting, or might potentially affect, the structure or its components. To identify the existence, cause, localization, and magnitude of damage, factors such as external actions and influences (e.g. imposed loadings), the effects of induced and/or restrained deformations, and the environment to which the structure is exposed (incl. variations in exposure conditions) must be considered and the (current, and previous, if available) condition and performance data should be analysed for any signs of anomalies that may be observed or detected.

Damage detection can be performed by data-driven methods or by model-based methods. The choice of the method for damage detection based on SHM data should be motivated, taking into consideration the measurement uncertainties of the sensors and the uncertainties (e.g. originating from approximations and numerical errors) introduced during data acquisition, analysis and pre-processing.

adequately and with the appropriate level of detail represent the effect of anticipated damage on structural behaviour.

DIs and PIs can be of qualitative (e.g. presence of cracks) or quantitative type (e.g. crack width), and they can be obtained from observations during inspections (e.g. qualitative indicators obtained from visual inspection), testing or structural monitoring.

For references regarding the use of DIs and PIs during the assessment of the condition and the performance of structures, and during the decision-making with regard to maintenance see COST TU1406 [30.11-31].

Data collected from monitoring used for model updating can be subdivided into two categories:

- direct information: direct measurements of the properties used in the structural assessment,
- indirect information: measurements of indicators which are related by models to the properties used in the structural assessment.

During identification of damage, damage indicators (DIs) and performance indicators (PIs) should be determined. DIs are observations or are derived from observations, and can serve for damage detection, damage localisation and/or damage characterisation, usually represented by a scalar value (e.g. the crack width, carbonation depth, dynamic properties of the structure extracted from measured response histories by modal analysis). PIs are observations or are derived from observations, and can describe the property of the structure and/or of the aspect of its performance, used to qualify fitness of the structure for its purpose during service life. A sub-set of PIs that carry most significant of information about present performance of a structure and are used to qualify present fitness of the structure for its purpose are referred to as key performance indicators (KPIs).

The choice of the DIs and PIs for diagnostics based on SHM should be such that their combined evaluation represents an unambiguous and exhaustive characterisation of the actual structural condition and performance.

Diagnostics based on SHM data may be considered as a trigger for decisions regarding conservation or life cycle management of existing structural element(s) and/or system(s), see chapter 34. Principles of the decisions regarding conservation or life cycle management based on DIs PIs are still under development.

30.11.4.5.4 Updating of the assessment (FEM) model of the existing structural element(s) or system(s)

Model updating generally aims to calibrate unknown system properties which appear as parameters in numerical models, based on the observed behaviour of the system of interest. Models may describe the structural response to actions or the evolution of damage and deterioration over time.

Observations from monitoring can be used in the assessment by combining it with prior probabilistic models of the parameters governing the limit state conditions under investigation, leading to the posterior probabilistic models of those parameters. These prior probabilistic models represent the knowledge about the structure before performing any investigation and they might be based on the as-built documentation of the structure.

The updating of structural models based on indirect information is an inverse problem, as it reverses the standard relationship between parameters and output of a model; the objective is to obtain the parameters that produce a certain given output. As such, these techniques are often prone to be ill-posed and ill-conditioned, as the existence, uniqueness and stability of a solution of inverse problems cannot be guaranteed.

For examples of reliability assessment of bridge based on monitoring see Frangopol et al. [30.11-32].

To explore the feasibility of the testing and/or monitoring, the responsible engineer and structure's owner should also consider the following aspects:

- necessity and possibility to access and/or close the structure during testing and/or monitoring (including, if necessary, during application of instrumentation and during removal of all equipment)
- expected ability of testing and/or monitoring to give insight in the main open questions regarding the structure
- availability of required personnel, testing equipment, instrumentation etc.

The decision-making process regarding the implementation of testing and monitoring should include:

- preliminary analysis
- definition of testing and monitoring objectives
- definition of the information required to reach the monitoring objectives
- evaluation of fitness-for-purpose of monitoring
- selection of the data analysis methods and interpretative models
- selection of parameters to be measured
- evaluation of expected magnitude of measurement variations

Procedure for updating of the assessment (FEM) model of the existing structural element(s) or system(s) based on monitoring data is similar to the procedure of structural identification, described in section 30.11.3.7.8.

The methodologies used for updating models based on information from condition survey should also cover the uncertainty of the model parameters. In the context model updating of finite element (FE) models, the model parameters describe material parameters, geometry dimensions and boundary conditions.

After calibration, the model has to be checked with measured data which are not applied for calibration (e.g. additional data sets from inspections or testing).

30.11.5 Decision-making regarding the implementation of testing and/or monitoring

The decision of implementing testing and/or monitoring and the choice of the testing and/or monitoring strategy should be based on the optimisation of the expected costs and benefits from its use with respect to the testing and/or monitoring objectives. The extent of the analysis, the formulation of the cost and benefit, and the optimization objective and complexity of the optimisation procedure depend on the case at hand. The expected costs and benefits shall be quantified conditional on the choice of the testing and/or monitoring strategy.

The testing and/or monitoring strategy should be carefully designed to determine the scope and frequency of testing and monitoring, as well as the methods and tools needed to ensure the accuracy and reliability of the results. The testing and/or monitoring strategy should answer the questions what, where, when, how long and how to test and/or monitor, and it should account for:

- the uncertainty of the material of structural performance
- the costs of structural failure
- the cost of testing and/or monitoring
- the possible outcomes of testing and/or monitoring

– definition of required measurement accuracy and frequency
– design of the monitoring system
– choice of the monitoring strategy
– definition of the installation, data acquisition, data pre-processing and monitoring systems maintenance plans

For more information related to decision-making process regarding the implementation of structural monitoring, the choice of the monitoring strategy and design of the monitoring system refer to [Allaix et al. [272]].

For more information related to cost-benefit analysis for SHM systems refer to [Sousa et al.[265]].

– the possible risk mitigation actions taken based on the outcomes of testing or monitoring and the corresponding costs.

Determining the aims of testing and monitoring is a crucial step for choosing the testing and/or monitoring strategy. First, the testing and monitoring objectives need to be identified, then the information required from testing and/or monitoring should be specified, and then the type of testing and/or monitoring to be carried out and data analysis approach to be followed should be selected and their feasibility of such should be explored. Once it has been shown that it is feasible to meet the objectives of testing and/or monitoring, more specific preparations can be developed.

References

[30.11-1] Sedlacek G., Merzenich G., Paschen M., Bruls A., Sanpaolesi L., Croce P., Pratt M., Leendertz M. — Background document to EN 1991: Part 2 - Traffic loads for road bridges and consequences for the design. *JRC Scientific Technical Report*. JRC, 2008

[30.11-2] Zhou X. Y., Schmidt F. & Jacob B. — Extrapolation of traffic data for development of traffic load models: assessment of methods used during background works of the Eurocode. *IABMAS 2012. 6th International Conference on Bridge Maintenance, Safety and Management*. Lac de Come. Italy. July 2012. available at : https://hal.science/hal-01213632

[30.11-3] CEN/TS 17440:2020 — Assessment and retrofitting of existing structures, CEN, 2020

[30.11-4] *fib* Bulletin 80 — Partial factor methods for existing concrete structures - Recommendation. *International Federation for Structural Concrete (fib)*, Lausanne, ISBN 978-2-88394-120-5. December 2016

[30.11-5] Alampalli S., Frangopol D.M., Grimson J., Kosnik D., Halling M., Lantsoght E., Weidner J.S., Yang D.Y., Zhou Y.E. — Primer on Bridge Load Testing. Transportation Research Circular TRC E-C257. *Transportation Research Board*. Washington DC. November 2019. available at: https://www.trb.org/Main/Blurbs/179887.aspx

[30.11-6] ISO 12491:1997. — Statistical methods for quality control of building materials and components. *International Organization for Standardization (ISO)*. 1997

[30.11-7] Kersken-Bradley M., Maier W., Rackwitz R., Vrouwenvelder, A. — Estimation of structural properties by testing for use in limit-state design, general principles for planning, sampling and evaluation for the elementary case. JCSS Working document. *Joint Committee on Structural Safety (JCSS)*. 1990

[30.11-8] ACI 437.2M-13. — Code Requirements for Load Testing of Existing Concrete Structures (ACI 437.2M-13) and Commentary. *American Concrete Institute (ACI)*. 2013.

[30.11-9] DAfStb Richtlinie. — Belastungsversuche an Betonbauwerken, *Deutscher Ausschuss for Stahlbeton.* 2000.

[30.11-10] AASHTO. — The manual for bridge evaluation with 2016 interim-revisions 2nd edition. *American Association of State Highway and Transportation Officials (AASHTO).* 2016.

[30.11-11] *fib* Bulletin TG3.2 — TITLE ?, *International Federation for Structural Concrete (fib)*, Lausanne, 2023

[30.11-12] Lantsoght, E., Okomus, P. — Evaluation of Concrete Bridge Behaviour through Load testing – International Perspectives. SP-323. *ACI Symposium*, 2018.

[30.11-13] Lantsoght, E. — Diagnostic and Proof Load Tests on Bridges - Editorial. Frontiers in Built Environment. 6(173). 2020. https://doi.org/10.3389/fbuil.2020.586704,

[30.11-14] Lantsoght, E. (ed.) — Load testing of bridges: Current Practice and Diagnostic Load Testing. *Structures and Infrastructure*, Vol. 12, 2019.

[30.11-15] Lantsoght, E. (ed.) — Load testing of bridges: Proof Load Testing and the Future of Load Testing. *Structures and Infrastructure,* Vol. 13, 2019.

[30.11-16] Aguilar , C. V., Jauregui, D. V., Newtson, C. M., Weldon B. D., Cortez, T. M. — Load Rating a Prestressed Concrete Double-Tee Beam Bridge without Plans by Proof Testing. *Transportation Research Record*. 2522(1): 90 – 99, 2015.

[30.11-17] Lantsoght, E., Van der Veen, C., Hordijk, D. A. — Proposed stop criteria for proof load testing of concrete bridges and verification. *IALCCE 2018*, 2018.

[30.11-18] Lantsoght, E., Yang, Y., Van der Veen, C., Hordijk, D. A., De Boer, A. — Stop Criteria for Flexure for Proof Load Testing of Reinforced Concrete Structures. *Frontiers in Built Environment.* 5: 47,2019.

[30.11-19] Ministerio de Fomento. Instrucciones para la Puesta en carga de estructuras (pruebas de carga provisionales), 2009.

[30.11-20] Ministerio de Fomento. Direccion General de Carreteras, Recomendaciones para la realizacion de pruebas de carga de reception en puentas de carretera, 1999.

[30.11-21] Çatbaş F.N., Kijewski-Correa T. & Aktan (ed.) Structural Identification of Constructed Systems: Approaches, Methods, and Technologies for Effective Practice of St-Id. *American Society of Civil Engineers (ASCE)*. Ed. Çatbaş F.N., Kijewski-Correa T., Aktan, A.E.. ISBN (print): 978-0-7844-1197-1ISBN (PDF): 978-0-7844-7647-5. 2013.

[30.11-22] Ye, S., Lai, X., Bartoli, I., Aktan, A.E. Technology for condition and performance evaluation of highway bridges. *Journal of Civil Structural Health Monitoring*. 10. 10.1007/s13349-020-00403-6. May 2020

[30.11-23] *fib* Bulletin 33. Durability of post-tensioning tendons. *International Federation for Structural Concrete (fib).* Lausanne. 2005.

[30.11-24] *fib* Bulletin 22 Monitoring and safety evaluation of existing concrete structures. *International Federation for Structural Concrete (fib).* Lausanne. 2003

[30.11-25] Rytter, A. Vibrational Based Inspection of Civil Engineering Structures, *Dept. of Building Technology and Structural Engineering*, Aalborg University. *Fracture and Dynamics*, Vol. RP314, No. 44.

[30.11-26] Allaix, D.L., Bigaj-van Vliet, A.J., Daró, P., Mancini, G., Strauss A., Bergmeister, K., Köhler, J. Formulation of the EC mandate to CEN and plan of approach for the execution of the mandate. *H2020 CSA IM-SAFE.* report D5.1. available at: https://im-safe-project.eu/publications/formulation-of-the-ec-mandate-to-cen-and-plan-of-approach-for-the-execution-of-the-mandate/ . 2023

[30.11-27] Allaix, D.L Technical background materials complementing the EC mandate for CEN. *H2020 CSA IM-SAFE*. report D5.2. available at: https://im-safe-project.eu/publications/technical-background-materials-complementing-the-ec-mandate-for-cen/ . 2023

[30.11-28] Sparkes, P., Webb, J. Structural health monitoring in civil engineering. *CIRIA*. Report C788. ISBN: 978-0-86017-893-4. June 2020.

[30.11-29] Reynders, E., Teughels, A., De Roeck, G. Finite element model updating and structural damage identification using OMAX data. *Mechanical Systems and Signal Processing,* 2010. 24(5): 1306 – 1323.

[30.11-30] Sousa, H., Wenzel, H., Thons, S. Quantifying the Value of Structural Health Information for Decision Support, *COST action TU1402.*

[30.11-31] COST TU1406 Quality Specifications for Roadway Bridges, Standardization at a European level BridgeSpec). COST ACTION TU1406 (2015): available at: https://www.cost.eu/actions/TU1406/

[30.11-32] Frangopol, D. M., Strauss, A., Kim, S. Bridge Reliability Assessment Based on Monitoring, 2008. *Journal of Bridge Engineering*, 13(3).

Performance specifications are often high level type specifications. For analysis and judgement, these high level specifications need to be processed and detailed to such an extent that smart specifications and/ or criteria allow engineers to apply available tools for analysis. Compliance with the detailed specifications and/ or criteria enable the fulfilment of the high level performance specification.

Guidance for structuring the process of design and construction/ mitigation can be found in ISO 15288 [30.12-1]. This approach, demonstrated to be feasible and applicable in design, construction and maintenance/ interventions of civil engineering projects, contains a V-shaped figure/V-model portraying:

1) decomposition of the performance requirements on the left-hand leg of the V;
2) integration of the delivery of the required performances at different scales / elements / subsystems requirements on the right-hand leg of the V;

30.12 Evaluation of effectiveness of new structures, measures and interventions

30.12.1 General

Upon construction or completion of new structures and interventions, structures must meet specific requirements to assure that the performance at any time during the service life complies with governing performance requirements. Governing performance requirements must be based on, and should be in compliance with, the demands of the stakeholders and statutory responsibilities and requirements.

In addition to the compliance with the performance requirements/ specifications, also the measure of effectiveness should be considered. The measure of effectiveness can be derived from the hierarchy of alternative solutions (design, construction, maintenance, intervention). The hierarchy is set by judgement how alternative solutions rank in terms of cost, value creation and risk exposure.

30.12.2 Scope

This section deals with the evaluation of compliance and effectiveness of new structures, measures and interventions

30.12.3 Performance Specification Compliance

To assure compliance of the structure (upon completion or after intervention) with the performance specifications, a structured process of design and construction/ intervention is required. This process should consist of a breakdown analysis of high level performance requirements/ specifications (on a Life Cycle Bases, so inclusive of the operational and maintenance phase), a sequential order of design development from initial to detailed design, and a structured integration of components and elements during construction/ intervention, all supported by a consistent application of verifications and validations.

3) verification/validation of performances /maintainence activities.

Figure 30.12-1 illustrates a practical application of the Systems Engineering philosophy for new structures with boxes as presented in this Model Code. For interventions, the same philosophy can be applied.

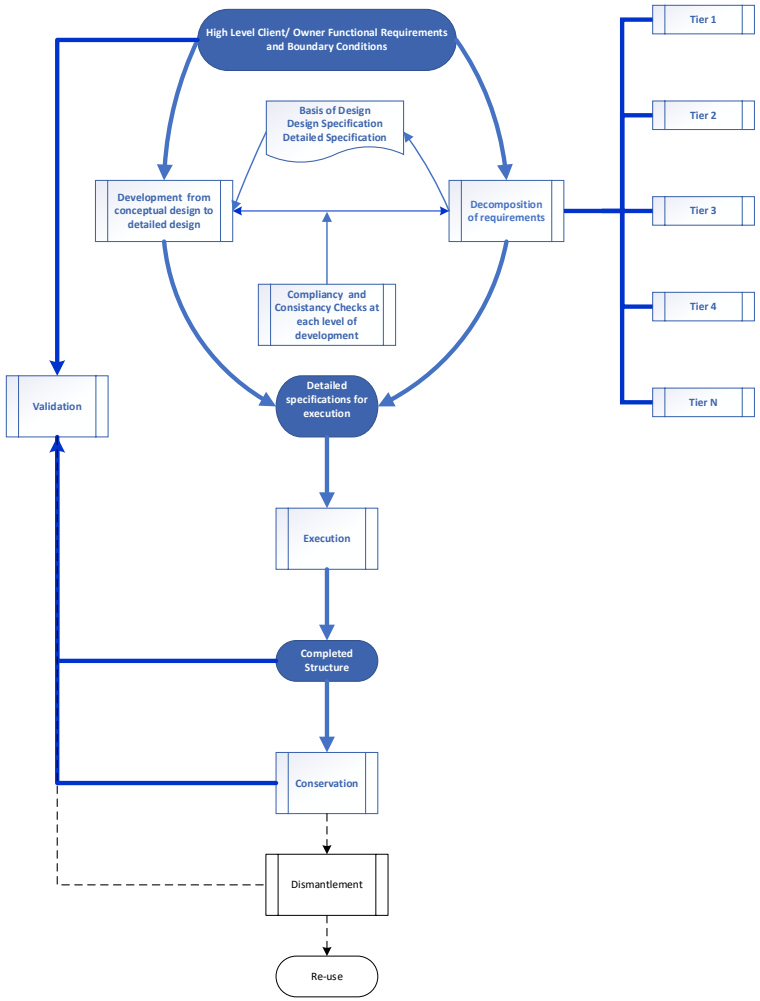

Figure 30.12-1:

At each step from conceptual design to detailed design, verifications must be performed to demonstrate compliance with the specifications. To allow this verification process the specifications should be decomposed through a break down analysis appropriate to the level of the design development.

Upon completion of construction or intervention, compliance with the performance specification must be demonstrated. Validation is an appropriate approach. Validation can be performed by testing of the completed structure/ intervention. If testing is disproportional, compliance with the functional requirements can be demonstrated by verification and appropriate quality management during construction/ intervention. In such case appropriate aspects and criteria must be available as well as reliable analysis tools and/ or records of quality management during construction/ interventions. In such case validation by a chain of verifications is achieved. This valiadation process could be the Birth Certificate / Re-Birth Certificate concept mentioned in various *fib* Bulletins, but especially Bulletin 93 [30.12-2]

For a set of (functional) requirements, including all applicable aspects of sustainability, and boundary conditions, several solutions are generally possible. From each viewpoint to be considered/ angle of approach, different schemes may result. From the tension or strengthening potential arising from the different angles of approach, a selection of solutions, for further development or choice, can be concluded. A schematic flowchart of such process is given in Figure 30.12-2.

The ranking methodology depends upon the level of concept development and as such on the level of reliability of available information. At a conceptual level of detail a trade-off matrix (ToM) might the appropriate tool to judge each available alternative against the applicable requirements/ specifications. The ToM generally has the alternatives along one axis and all the relevant aspects which make the difference between the alternatives along the other axis. In the boxes an appraisal (a mark) will be given. Various aspects may have a different weighted factor; the boxed value will then be the mark multiplied by the weight-factor. By adding up all the boxed values, a score per alternative is obtained which indicates the ranking between the alternatives.

30.12.4 Effectiveness Analysis

Concepts, the realization thereof and subsequent interventions should comply with all requirements in an effective manner. Effectivity in this respect is considered as the balance between cost, value creation and risk but under the prerequisite that all requirements are met. To perform an effectivity analysis , alternative solutions should be developed and ranked. The ranking provides a tool to select the preferred solution (with respect to design/ construction and maintenance/ intervention).

At a more developed stage of design, more reliable information will be available to justify a more numerical weighting of aspects.

To avoid suboptimizations and to allow integral optimization, it is important to consider all aspects of the service life chain being treated holistically

At all levels of ranking, risk must be considered. At early stages of concept development this could be done by awareness of the level of complexity of both design, construction or intervention. At a more detailed level of available information, a more numeric approach is justified. With the risk analysis, remedial actions should be considered to create an integral understanding of the impact of risk in terms of variation of cost, compliance with requirements, feasibility of anticipated value creation and time management both of design and construction/ intervention.

In case of a numerical analysis, different approaches are available, ranging from all aspects expressed in a common unit to different units for the major components of the ranking (cost&risk, value creation, and sustainability aspects) in combination with a weighting factor between the units applied.

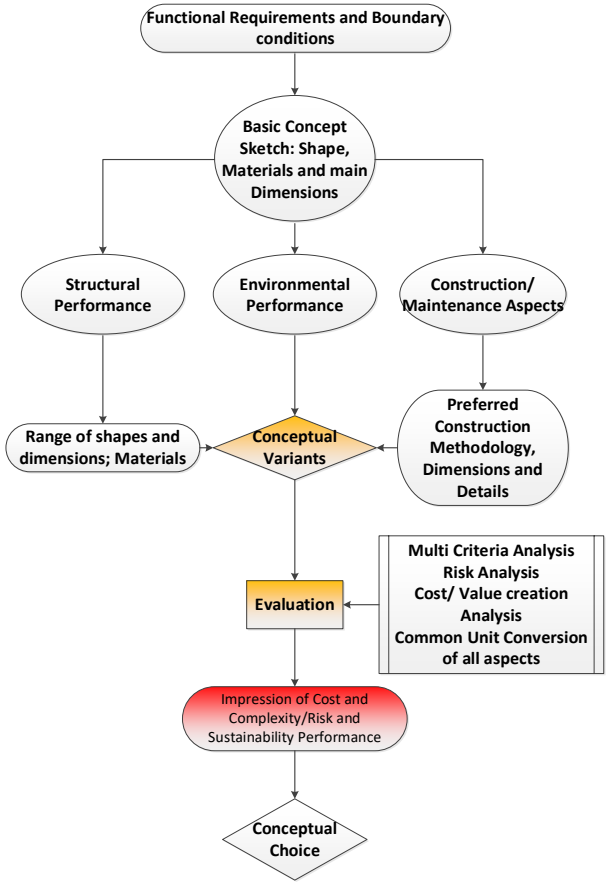

*Figure 30.12-2:***[Add Title]**

References

[30.12-1] ISO/IEC/IEEE 15288:2015. Systems and software engineering – System life cycle processes.

[30.12-2] *fib* Bulletin 93. Birth Certificate and Through-Life Management documentation.

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In this chapter no detailing rules explicitly for FRP reinforcement (embedded or externally applied) are given, though the main principles of detailing remain applicable.

30.13 Detailing

30.13.1 General

30.13.2 Basic principles of detailing

The dimensions of structural members should be chosen in relation to the amount and position of the reinforcement and to the properties of the structural materials. The position and composition of the reinforcement should allow a qualitatively perfect casting and compaction of the concrete. Special attention should be paid to the placing of the reinforcement.

The detailing shall correspond to the structural model underlying the structural analysis. The influences of any deviations between the structural model and the actual properties of the structure should be taken into account.

If the reinforcement is used for the earthing of construction works, the measures have to be discussed with the responsible electrical engineers, whereby particular attention should be paid to corrosion protection.

The relevant information for FRP reinforcement detailing shall be given through national specifications, further to the detailing provisions provided in *fib* bulletin 40 [30.13-1] (embedded reinforcement) and Bank [30.13-2], and *fib* bulletin 90 [30.13-3] (externally applied reinforcement). Considerations on bond and anchorage length are given in Chapters 21 (embedded) and 22 (externally applied).

30.13.3 Positioning of reinforcement

30.13.3.1 General

The provisions of this subsection apply to reinforcing bars with diameters ≤ 40 mm. Bar diameters above 32 mm should only be used in exceptional cases; the principles of detailing should be adapted if necessary.

Generally Δc_{dev} can be taken equal to 10 mm. Where a quality assurance system is used, including measurement of the concrete cover, the allowance in design for deviation Δc_{dev} may be reduced to a value 10 mm $\geq \Delta c_{dev} \geq$ 5 mm. Where a very accurate measuring device is used for monitoring and non-conforming members are rejected (e.g. precast elements) the value Δc_{dev} can be further reduced.

Table 30.13-1 gives indicative values for minimum cover with respect to durability demands, assuming moderate concrete strength (from C20/25). The values apply for a service life of 50 years. For a service life of 100 years, the values of c_{min} should be increased. The necessary increase of the cover depends on the time

30.13.3.2 Cover of reinforcement

The cover of reinforcement c_{nom} refers to minimum values which have to be complied with if no special protective measures are taken or no special circumstances exist. Permissible dimensional tolerances are given in section 14.5.4.1.5.4 (see Tables 14.5-4 to 14.5-6).

The concrete cover shall be specified on the drawings. It is defined as an absolute minimum cover, c_{min} plus an allowance in design for deviation Δc_{dev}:

$$c_{\text{nom}} = c_{min} + \Delta c_{dev} \qquad (30.13\text{-}1)$$

The cover is the largest of four values, depending on:

– technological requirements;
– bond and anchorage requirements;
– fire requirements;
– durability requirements.

The highest value applies. The cover chosen should be shown on the drawings.

In order to ensure good casting, the cover should be at least equal to the maximum aggregate particle diameter plus 5 mm.

In order to ensure the transfer of the bond forces between concrete and reinforcing bars, c_{min} shall be larger than:

– in case of single bars: the diameter of the reinforcing bar;
– for bundles: the equivalent diameter $Ø_n$ – see Eq. (30.5-15).

In case of post-tensioned bars c_{min} shall be larger than:

– for circular ducts: the duct diameter;
– for rectangular ducts: the smaller dimension or half the largest dimension.

For pretensioned strands or wires c_{min} shall be larger than:

– 1.5 the diameter of the strand or plain wire;
– 2.5 the diameter of an indented wire

In order to fulfil the fire resistance requirements, reference is made to section 30.4.2.

For durability, the minimum cover depends on the concrete resistance against the ingress of chemical substances in combination with the aggressivity of the

dependent nature of the ingress of chemicals with a detrimental effect. In general, the cover thickness should be increased with about 30%. The cover can be reduced if a concrete with smaller permeability is used. This reduction can be more substantial if the water cement ratio decreases and the degree of hydration increases. Also quality control plays an important part. Another possibility is to design the concrete mixture for low permeability.

Table 30.13-1: Minimum cover for reasons of durability (indicative values).

Environmental class	X0	XC1	XC2 XC3	XC4	XD1 XS1	XD2 XS2	XD3 XS3
Reinforced concrete	10	15	25	30	35	40	45
Prestressed concrete	10	25	35	40	45	50	55

environment and the intended service life of the structure. For verification of durability requirements, see section 30.6.

In the case of structural members in which concrete is cast directly against soil, the minimum value of the cover on the reinforcement is $c_{nom} \geq 75$ mm; in the case of soil preparation $c_{nom} \geq 50$ mm.

If the cover of reinforcement exceeds 50 mm, it may be appropriate (e.g. regarding fire action) to provide a skin reinforcement.

In the case of architectural concrete surfaces, in particular those with a treated surface, particular attention has to be paid to the cover of the reinforcement.

30.13.3.3 Minimum bar spacing

The spacing of reinforcing bars and prestressing tendons should be sufficient to allow perfect casting and compaction of the concrete. Splices of reinforcement should be staggered if necessary.

The clear distance between parallel reinforcing bars and prestressing tendons should be at least equal to the maximum aggregate particle diameter +5 mm and the diameter of adjacent bars. It should be at least 20 mm.

In the case of dense reinforcement, the arrangement of bar bundles with a maximum of three contacting bars running in the same direction is permitted.

The minimum mandrel diameters d_2 given in this clause should be fulfilled to avoid cracks or failures of the reinforcement. These values are for steel in compliance with EN 10080 where neither rupture nor visible cracks should occur for 180° bend tests according to ISO 15630-1 with following mandrel diameters: $d_{mand} = 3Ø$ for $Ø \leq 16$ mm and $d_{mand} = 6Ø$ for $Ø > 16$ mm. For steels compliant with other standards, the minimum mandrel diameters may have to be adapted accordingly.

For re-bends, the following minimum mandrel diameters have to be complied with: $d_2 = 6Ø$ for $Ø \leq 16$ mm; $d_2 = 9Ø$ for 16 mm $< Ø \leq 25$ mm and $d_2 = 11Ø$ for $Ø > 25$ mm. These requirements are for steels in compliance with EN 10080 and bent with a temperature ≥ 10 °C (testing limit according to ISO 15630-1).

30.13.3.4 Forms and bends

30.13.3.4.1 Regulations for design

Compliance with the minimum mandrel diameters (Figure 30.13-1) prevents bending cracks in the reinforcing steel and failure of the concrete enclosed by the bend. The mandrel diameters have to be specified by the design engineers.

To avoid damaging the reinforcement, the following minimum mandrel diameters of bars without welds or of bars and welded fabrics with welds at least 3Ø away from the bend have to comply with:

– for bends $d_1 = 15Ø$;

– for end hooks, angle hooks, loops and stirrups:

$d_2 = 4Ø$ for bars ≤ 16 mm;

$d_2 = 7Ø$ for bars > 16 mm

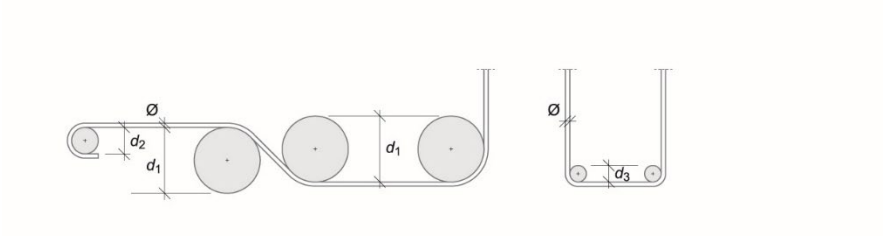


Figure 30.13-1: Mandrel diameters.

No welds are permitted in the immediate vicinity of bends.

In the case of welded reinforcement, the weld should fulfil the requirements as given in Section 36.1.4.

In the case of bend mesh reinforcement, the cross bars should be located outside the bending zone.

Verification of stresses inside the bent to avoid concrete damage may be omitted for:

- stirrups and links in compliance with 30.13.6.2;
- standard hook and bend anchorages complying with Figure 20.2-9 where:

- the anchorage of the bar does not require a length more than 5ϕ past the end of the bend;
- there is a clear distance $c_x \geq 1{,}5\phi$ from an edge parallel to the bend (see Figure 20.2-5);
- there is a clear distance between bars $c_s \geq 3\phi$ according to Figure 20.2-5; and
- provided that $f_{yk} \leq 500$ MPa and $f_{ck} \geq 25$ MPa, all bends with an angle $\alpha_{bend} \leq 45°$ at a clear distance c_x
- $\geq 2{,}5\phi$ from an edge parallel to the bend, a clear distance between bars $c_s \geq 5\phi$ and the length of the straight segments between multiple bends is not shorter than 4ϕ.

In all other cases the steel stress should be verified according to Eq. (30.13-2).

Specifying the minimum radii of curvature of prestressing tendons the technical documentation of the prestressing system should be considered. In addition, it may be necessary to check the local concrete stresses. See section 16.2.9.2 which gives minimum radii of curvature for post-tensioning tendons.

30.13.3.4.2 Assessment of non-compliant detailing

In case of non-compliant detailing with respect to mandrel diameters the assessment value of the steel stress σ_{sa} should be verified to avoid concrete failures inside the bend according to following equation:

$$\sigma_{sa} \leq 0.65 f_{cd} \frac{d_{mand}}{\emptyset} + \frac{\sqrt{f_{ck}}}{\gamma_C} \left(\frac{d_{dg}}{\emptyset}\right)^{1/3} \left(\frac{c_d}{\emptyset} + \frac{1}{2}\right) \left(32 \frac{45^\circ}{\alpha_{bend}} + 0.7 \frac{d_{mand}}{\emptyset}\right) \quad (30.13\text{-}2)$$

where:

σ_{sa}	is the maximum steel stress in the assessment
d_{mand}	is the mandrel diameter applied;
$\emptyset$	is the bar diameter;
d_{dg}	$= 16 + d_g \leq 40$ mm
d_g	is the maximum size of the aggregate

In the past minimum mandrel diameters applied according to former codes may have been smaller than values given in section 30.13.3.4.1. Due to lower allowable steel stresses in older types of reinforcement this may not imply that concrete failure inside the bend is likely to occur.

For the background of Eq. (30.13-2), see Monney et al. [30.13-4].

c_d is the nominal cover or half the clear space between adjacent bars (see Figure 20.2-9)

30.13.3.5 Anchorage

This subsection covers anchorage of ribbed reinforcing bars and meshes of reinforcing steel which satisfy the requirements for classification as “high bond reinforcement” in accordance with section 15.4.2.

This subsection does not cover bars forming part of a bundle.

The rules here are limited to the following:

- characteristic yield strength of reinforcement ≤ 500 MPa;
- minimum requirements for confining reinforcement and cover in section 20.2.2.2 are satisfied; where the diameter Ø of the anchored bars is smaller than 20 mm, transverse reinforcement or links provided for other reasons may be assumed sufficient to satisfy the minimum requirements for confining reinforcement without further justification;
- on flexural members, loading is predominantly uniformly distributed along the span;
- no applied tensile stress acts transverse to the axis of the anchored bar.

If any of these conditions is not satisfied the detailed rules of section 20.2.2 shall be used.

The dimensioning value of the bond length may be taken as:

$$l_b = \alpha_1 \cdot \beta_b \cdot \left(\frac{25}{f_{ck}}\right)^{0.45} (\varnothing/25)^{0.35} \geq k_{min} \qquad (30.13\text{-}3)$$

where:

l_b has been derived as a simplification of section 20.2 and Eq. (20.2-3);

k_{min} minimum bond length, see 20.2.2.3;

$\alpha_1 = A_{s,cal} / A_{s,ef}$ for anchorages in zones AS, AB and AC;

where:

$A_{s,cal}$ is the calculated area of reinforcement required by the design;

$A_{s,ef}$ is the area of reinforcement provided;

This subsection gives rules which are simplified with regard to the relations given in section 20.2.2.

The more refined rules of section 20.2.2 may permit shorter anchorage lengths.

α_1 = 1.0 in other circumstances;

β_b is a factor given in Table 30.13-2 for anchorage length appropriate to bond zones classified as follows:

Type AS : straight bars in tension in zones near the ends of members and in which the support reaction or load from upper storeys is transferred through the anchorage zone in a direction perpendicular to the plane passing through the axes of the anchored bars.

Type AB : tension bars which terminate in a hook or bend in zones near the ends of members and in which the support reaction or load from upper storeys is transferred through the anchorage zone in a direction perpendicular to the plane(s) passing through the axes of the anchored bars.

Type AC : column and wall bars anchored near the middle of footings or pilecaps and acting in compression under all design loadings.

Type RS : straight bars in all other situations.

Type RB : anchorages of bars terminating in a hook or bend in all other situations.

Figure 30.13-2 provides illustrations of representative situations for each type of bond zone.

Table 30.13-2: Bond length factors for anchorages [to be updated based on chapter 20]

Bond zone	**Bond length factor β_b**			
	Casting position			
	"Good"		"Poor"	
	c_{min} / Ø ≥ 2.5,	c_{min} / Ø ≥ 1.0,	c_{min} / Ø ≥ 2.5,	c_{min} / Ø ≥ 1.0,
AS				
AB				
AC				
RS				
RB				

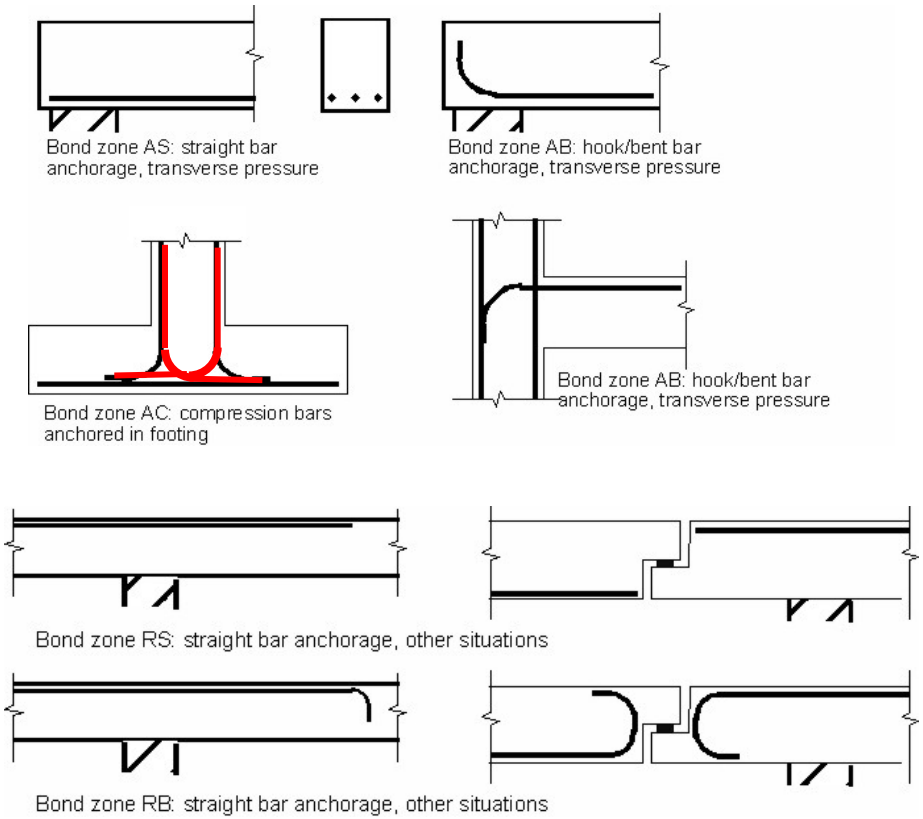


Figure 30.13-2: Classification of anchorage bond zones.

This subsection gives rules which are simplified with regard to the relations given in section 20.2.

Casting position is defined in section 6.1.3.2. Minimum cover c_{min} is defined in Figure 6.1-2 and Figure 6.1-15 for straight and bent/hooked bars respectively.

$\beta_{b,min}$ shall be taken as ## for type RS and RB bond zones, or ## for types AS, AB and AC.

30.13.3.6 Lapped joints

This subsection covers lapped joints or splices of ribbed reinforcing bars and meshes of reinforcing steel which satisfy the requirements for classification as "high bond" in accordance with requirements of section 15.4.2.

The more refined rules of section 20.2_ may permit shorter lap lengths.

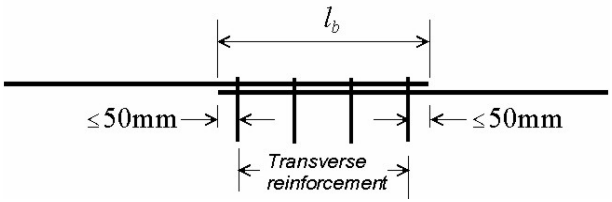


Figure 30.13-3:Location of links near ends of lap.

The contribution of end bearing in compression laps and anchorages is included in the average bond strength given by Eq. (30.13-3).

Values for β_b in Table 30.13-3 include a partial safety factor of 1.5 on bond strength.

Wherever possible, splices should be arranged in zones of low stress.

For bar diameters > 12 mm, lap splices should, if possible, be staggered so that, in a cross-section, not more than one-third of force in the reinforcement needs to be transferred by a lap.

The rules in this subsection are limited to the following:

- characteristic yield strength of reinforcement ≤ 500 MPa;
- minimum requirements for confining reinforcement and cover in section 20.2.2.2 are satisfied; where the diameter Ø of the anchored bars is smaller than 20 mm, transverse reinforcement or links provided for other reasons may be assumed sufficient to satisfy minimum requirements for confining reinforcement without further justification;
- at least one item of transverse reinforcement should be positioned within a lap length no further than 50 mm from the end of both the lapped bars; other items of transverse reinforcement should be spaced evenly throughout the lap length (Figure 30.13-3);
- no applied stress transverse to the axis of the lapped bars.

If any of these conditions are not satisfied, the detailed rules of section 20.2.2.2 shall be used.

The dimensioning value for the lap length is given by Eq. (30.13-3), where the value of β_b is given in Table 30.13-3.

Bond zones for lapped splices are classified as follows:

Type LS : straight bars in tension where the splice will be subject to a stress not exceeding 50% of the characteristic strength of the reinforcement at the ultimate limit state;

Type LL: straight bars in tension not classified as type LS;

Type LC: straight bars in compression under all load combinations.

Figure 30.13-4 provides illustrations of representative situations for each type of bond zone.

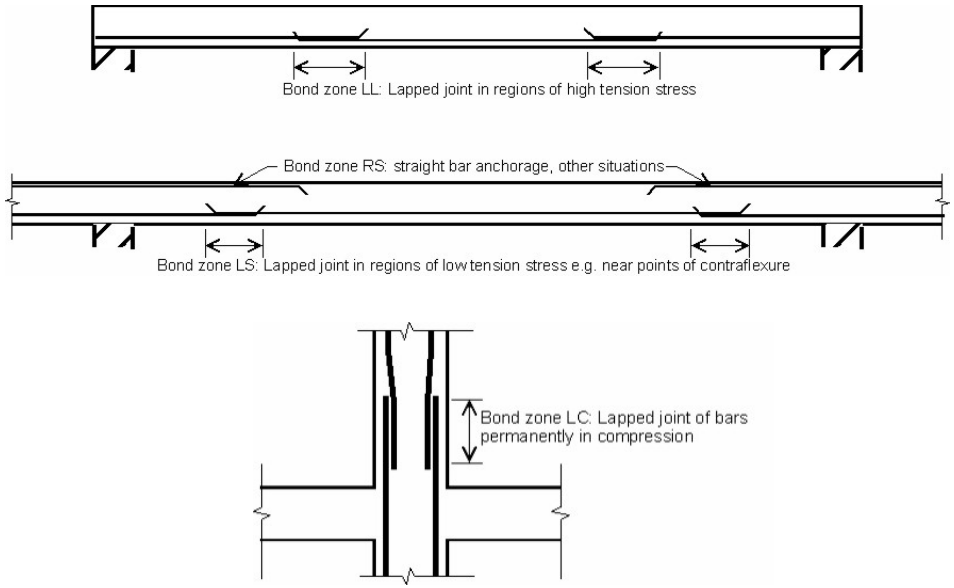


Figure 30.13-4: Classification of lapped splice zones.

Table 30.13-3: Bond length factors for lapped joints [to be updated based on chapter 20]

Bond zone	Bond length factor β_b			
	Casting position			
	"Good"		"Poor"	
	Confinement			
	c_{min} / Ø ≥ 2.5	c_{min} / Ø ≥ 1.0	c_{min} / Ø ≥ 2.5	c_{min} / Ø ≥ 1.0
LS				
LL				
LC				

Mesh reinforcement is not covered by this subsection. See sections 6.1.4.2 and 6.1.4.3.

30.13.3.7 Deviations and curvatures

In the case of curved or kinked tension or compression chords, the effects of the deviation forces shall be considered. The same provision apply for curved post-tensioning tendons near the concrete surface.

Deviation forces acting toward the surface of the concrete, as shown in Figure 30.13-5 should in general be resisted by means of additional transverse reinforcement. If this is not provided it shall be verified that the deviation forces at yielding of the steel can be resisted by the cover concrete. In this specific case the

concrete tensile strength may be taken into account with a maximum value according to:

$$f_{ctd} = \frac{1}{3}\frac{f_{ctk0.05}}{\gamma_c} \qquad (30.13\text{-}4)$$

The strength according to Eq. (30.13-4) has to be reduced by 50% in case of plastic deformations in the reinforcement.

If the deviation forces have to be resisted by the concrete, the inaccuracies of execution (thinner cover concrete, smaller spacing between reinforcements, non-uniform curvature) shall be taken into account for the verification.

For the calculation of the concrete tensile strength the effective width b_u may be assumed to be equal to:

$$b_u = s - Ø \leq 2\sqrt{3(c_{nom} + \frac{Ø}{2})} \qquad (30.13\text{-}5)$$

where s = spacing between reinforcing bars and Ø = bar diameter.

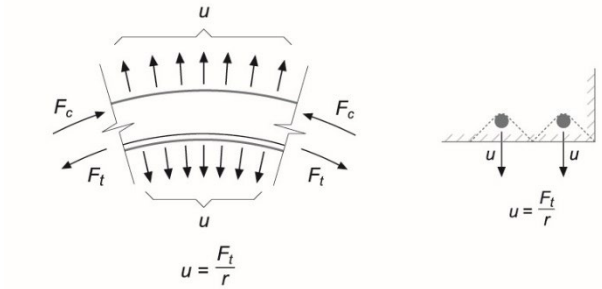


Figure 30.13-5: Equilibrium in curved tension and compression chords (r = radius of bend).

In the case of reinforcing bars which are curved or bent parallel to outer surfaces, the effects of the transverse tensile forces need to be examined.

Transverse tensile forces according to Figure 30.13-6 should be resisted by means of transverse reinforcement (e.g. stirrup or U-shaped reinforcement). If no transverse reinforcement is provided, it shall be verified that the transverse tensile forces due to the yield tensile force can be resisted. The concrete tensile strength may be taken into account according to Eq. (30.13-4) at most. No verification is necessary if the lateral cover of reinforcement corresponds to at least three times the bar diameter.

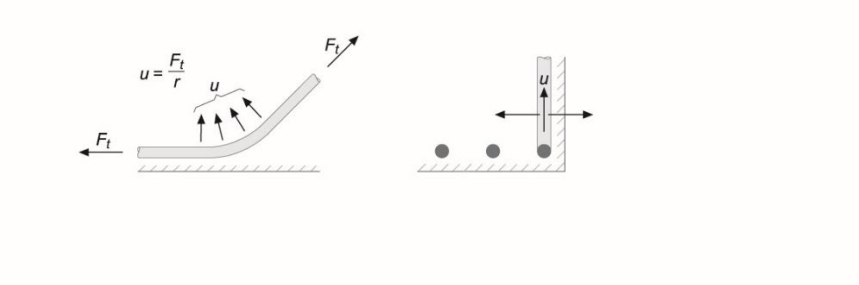


Figure 30.13-6: Transverse tensile forces in the case of bent-up reinforcing bars.

30.13.4 Prestressed structures

30.13.4.1 General

With regard to the transmission of the prestressing force into the concrete for post-tensioned structures, two zones are distinguished:

1) The area immediately around and behind the contact area, where the prestressing force is transmitted to the concrete: dimensions and reinforcement detailing are specified in Technical Approvals (e.g. ETA) of the prestressing system. The reliability for the corresponding detailing is the responsibility of the system holder.
2) The zone where the prestressing force is spread over the full cross-section. Design and detailing of the reinforcement in this zone is the responsibility of the designer.

Force spreading zones shall be detailed in such a way that the dimensioning value of the prestressing force according to chapter 16 can be introduced into the structure.

The flow of forces should be examined using suitable models, neglecting the tensile strength of the concrete. Splitting forces have to be resisted by means of reinforcement.

A minimum reinforcement should in any case be placed in anchorage and coupling zones to avoid the occurrence of harmful cracks.

The Eqs. (30.13-6a) and (30.13-6b) follow from Eq. (20.2-16), section 20.2.3.4, for a sudden release of the prestressing force, and a position of the tendon in the good bond area of the member.

Anchorages and couplings should if possible be positioned in areas in which only small stress variations occur.

30.13.4.2 Anchorage of prestressing wires and strands

A practical expression for the upper bound value of the transmission length l_{bpt} is:

$$l_{bpt,95\%} = \frac{0.10\phi_s\sigma_{pi}}{f_{ctd}} \quad \text{for strands} \qquad (30.13\text{-}6a)$$

$$l_{bpt,95\%} = \frac{0.15\phi_s\sigma_{pi}}{f_{ctd}} \quad \text{for indented wires} \qquad (30.13\text{-}6b)$$

The design value of the anchorage length l_{bpd} is:

$$l_{bpd} = l_{bpt,95\%} + \frac{A_p(\sigma_{pd}-\sigma_{pcs})}{\pi\cdot\varnothing_s\cdot f_{bpd}} \qquad (30.13\text{-}7)$$

where:

σ_{pd} is the tendon stress under the design load;

σ_{pcs} is the tendon stress after all losses;

σ_{pi} is the initial prestress;

f_{bpd} $= \eta_{p1}\ \eta_{p2}\ f_{ck(t)}^{1/2}$ (30.13-8)

where:

η_{p1} takes into account the type of prestressing tendon;

$\eta_{p1} = 0.42$ for indented and crimped wires;

$\eta_{p1} = 0.36$ for 7-wire strands;

η_{p1} = 0.21 for bundles of 7-wire strands;

η_{p2} takes into account the position of the tendon, as follows:

$\eta_{p2} = 1.0$ for all tendons with an inclination of 45–90° with respect to the horizontal during concreting;

$\eta_{p2} = 1.0$ for all horizontal tendons which are up to 250 mm from the bottom or at least 300 mm below the top of the concrete section during concreting;

$\eta_{p2} = 0.7$ for all other cases;

$\eta_{p2} = 0.7$ for post grouted strand bundles in ducts;

$f_{ck}(t) = \beta_{cc}(t)^{2/3} f_{ck}$ is the (grout) compressive strength at time of the release.

30.13.5 Bearings and joints

In the case of unreinforced concrete structures, the arrangement of expansion joints may be required in order to compensate for the effects of shrinkage and temperature changes.

In the case of reinforced concrete structures, expansion joints may have to be arranged if the effects of shrinkage, creep, temperature changes and non-uniform settlements cannot adequately be rendered harmless by the provision of crack distributing reinforcement.

In order to reduce the effects of shrinkage, it needs to be examined whether shrinkage joints for the temporary partition of larger structural members or other measures have to be specified.

Bearings of important structural elements should be permanently protected against moisture. They should be easily accessible and easy to replace.

Expansion joints should, if possible, be designed in such a way that moisture cannot reach the joint.

30.13.6 Structural members

30.13.6.1 Unreinforced structural members

Unreinforced concrete is used for solid structural members that are primarily subjected to compression and are not subjected to any significant seismic or dynamic loads. Examples include:

- foundations;
- arches and vaults;
- retaining walls;

– walls;

– compression members.

The dimensioning value of the compressive strength f_{cd} shall be reduced by 20% for verifying the structural safety of unreinforced structural members.

When verifying structural safety, a state of equilibrium between the internal and external forces should be assumed, neglecting the concrete tensile strength. Kinematic compatibility has to be ensured. Any constrained deformations and displacements have to be taken into account. External forces (reactions) should be taken into account if these act even in case of small deformations (e.g. frictional forces, abutment forces).

For structural members of secondary importance, the structural safety has to be verified taking into account the concrete tensile strength according to

$$f_{ctd} = 0.5 \frac{f_{ctk,0.05}}{\gamma_c} \qquad (30.13\text{-}9)$$

In addition, it should be ensured that the formation of cracks does not result in failure of the structural member.

The structural safety of unreinforced compression members may be verified in analogy to section 30.1.7. The dimensioning value of the compressive strength shall be reduced as indicated above.

30.13.6.2 Beams and T-beams

A beam is defined as a bearing element, spanning in one direction, with a width smaller than five times the cross-sectional depth.

Sections containing less reinforcement than $A_{s,min}$ should be considered as unreinforced.

The area of longitudinal reinforcement should not be taken as less than:

$$A_{s,min} = 0.26 \frac{f_{ctm}}{f_{yk}} b_t d \qquad (30.13\text{-}10)$$

where b_t is the width of the tension zone.

Adequate reinforcement shall be provided to cope with restraining effects which have been neglected in the structural analysis.

– In the application to new structures, the rules of this subsection should be complied with in full.

Where shear reinforcement is required (see section 30.1.3), the minimum area of shear reinforcement shall be:

$$A_{sw,min} = 0.08 \sqrt{f_{ck}} \frac{b_w s_w}{f_{ywk}} \qquad (30.13\text{-}11)$$

– In the application for the assessment of existing structures, the detailing rules presented in this subsection may be relaxed provided that the assessment on strength in cases of non-compliance is assessed fully.

The minimum amount of shear reinforcement (Eq. (30.13-11)) is required to ensure that failure does not occur immediately upon shear cracking and that truss action can develop.

In case of FRC the minimum shear reinforcement should be provided by either stirrups (7.3-22), or fibres.

Eq. (30.13-12) is based on steel fibre concrete research, and should be checked for other types of material.

In beams, at least 50% of the shear reinforcement should consist of closed stirrups, because the inclined struts tend to exert a lateral force on the outer longitudinal reinforcing bars (Figure 30.13-7). In slabs those lateral forces can be accounted for by transverse bars.

In the case of wide webs, the web width may be taken into account up to a maximum of 400 mm. In the case of web widths > 500 mm, stirrups with more than two legs shall be used.

In beams, stirrups shall generally be provided. Their spacing s_w shall not exceed $0.75d$ or 500 mm.

For FRC the minimum amount of conventional shear reinforcement (stirrups) is not required if the following condition is fulfilled:

$$f_{Ftu,ef} \geq 0.08 \sqrt{f_{ck}} \qquad (30.13\text{-}12)$$

where:

$f_{Ftu,ef}$ is the characteristic value of the ultimate residual tensile strength for FRC, by considering w_u = 1.5 mm according to Eq. (18.4.-5) [MPa].

This allows limiting the development and the diffusion of the inclined cracking and, as a consequence, can ensure sufficient member ductility.

When the above-mentioned limitation is not applied, conventional shear reinforcement (stirrups) have to be introduced.

Transverse reinforcement provided for shear and torsion may consist of a combination of:

– stirrups or ties perpendicular to the axis of the member, enclosing the longitudinal tension reinforcement and the compression zone;

– cages, ladders, welded wire fabric and so on which are cast-in without enclosing the longitudinal reinforcement but are properly anchored in the compression and tension zones. Such reinforcement may only be applied in combination with stirrups enclosing the longitudinal reinforcement, carrying at least 50% of the total force in the shear reinforcement;

– longitudinal bars bent to provide an inclined portion having an angle of 30° or more with the longitudinal bars and crossing potential diagonal cracks. However, only the centre three-quarters of the inclined portion of these bars may be considered effective;

– headed shear reinforcement demonstrated to be able to achieve the yield strength of the bar;

– spirals.

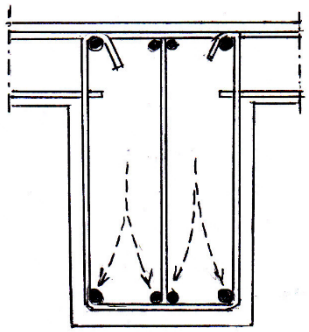


Figure 30.13-7:Lateral forces on outer bars due to strut action.

The stirrups shall enclose the longitudinal tensile reinforcement. They shall be anchored in such a way that the necessary stirrup forces can act over the height of the lever arm of the internal forces.

In the support areas, at least 25% of the total chord reinforcement required in the span should be fully anchored.

In the area of negative moments of T-beams, a significant portion of the tensile reinforcement shall be concentrated over the web. In order to limit crack widths, a certain proportion should also be distributed over the adjacent slab. The effectiveness of the longitudinal reinforcement distributed outside of the web shall be ensured by means of transverse reinforcement.

Adequate transverse reinforcement, the cross-section of which should amount to at least 0.2% of the slab cross-section, shall be provided in the compression slab to ensure the shear connection.

Minimum reinforcement for ductility behaviour of cross sections in FRC structures

In members designed as parts of reinforced concrete structures, minimum reinforcement $A_{s,min}$ shall be provided to:

a) ensure distributed cracking and to handle forces from restrained deformations where not considered explicitly in the design;
b) ensure sufficient deformation capacity to contribute to structural robustness by allowing alternative load paths;

A slab is a bearing element, spanning in one or two directions. A one-way bearing element is considered to be a slab if the width of the cross-section is equal or larger than five times the overall cross-sectional depth.

– In the application to new structures, the rules of this subsection should be complied with in full.
– In the application for the assessment of existing structures, the detailing rules presented in this subsection may be relaxed provided that the bearing resistance in cases of non-compliance is fully assessed.

c) avoid failures due to unpredicted cracking;
d) ensure applicability of limit design models;
e) ensure constructability

$M_{R,min} \cdot (N_{Ed,min}) \geq M_{cr} \cdot (N_{Ed,min})$ shall always be provided for beams and slabs.

To guarantee this statement in case of pure bending in beams:

- $A_{s,min} \geq \max [0.26 f_{ctm}/f_{yk}; 0.2(f_{R1k} - f_{R3k})/f_{yk}]$ (30.13-13)

Minimum torsion reinforcement for FRC structures

The minimum torsion reinforcement ratio $\rho_{Fw,min}$ for members reinforced with steel fibres, requiring longitudinal and transverse reinforcement may be taken as follows for both types of reinforcement:

$$\rho_{Fw,min} = \rho_{w,min} - \frac{f_{Ftu,ef}}{f_{yk}} \geq 0.3 \frac{f_{ctm}}{f_{yk}} \quad (30.13\text{-}14)$$

A clear bar distance $\geq k_F$ times the fibre length should be used, where the suggested value of k_F is 2,0 unless a different value is experimentally checked.

30.13.6.3 Slabs

In the zones of largest moments, the bar spacing of the main reinforcement shall not exceed the lesser of 1.2 times the slab thickness and 300 mm.

This requirement is not applied to FRC plates if the material is characterized by $f_{Ftu,ef} > 0{,}3 f_{ctm}$.

The minimum reinforcement of slabs shall be specified in accordance with the serviceability requirements. Transverse reinforcement shall not be less than 20% of the longitudinal reinforcement.

The reinforcement details should aim to avoid localization of cracks (anchorage, splices etc.) as well as ensure optimum efficiency of transverse reinforcement.

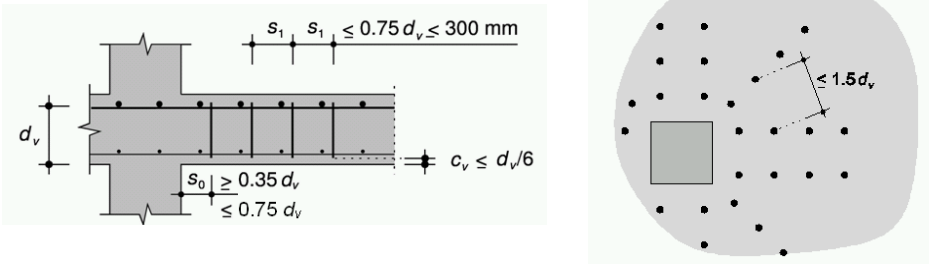


Figure 30.13-9: Detailing rules for punching shear reinforcement.

For the case of slabs without shear reinforcement, at least one-half of the bending reinforcement required at the points of maximum moment shall be fully anchored beyond the extent of the supports.

The shear reinforcement in slabs may be fully replaced by steel fibres if:

$$\frac{f_{Ftu,ef}}{f_{yk}} \geq \rho_{wmin} \qquad (30.13\text{-}15)$$

Free slab edges shall be reinforced with bent-up longitudinal reinforcement or with stirrup reinforcement of at least Ø10 mm, in accordance with Figure 30.13-8.

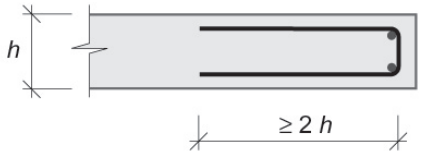


Figure 30.13-8: Free slab edges.

For punching shear, the development length and splices in flexural reinforcement shall not be located inside the perimeter defined by the minimum of:

- a line at 2.5d_v from the control perimeter of the supported region;
- the position of the line of contraflexure of radial bending moments.

For members with punching shear reinforcement, the design equations of section 30.1.5.3 are applicable provided that:

- a minimum of two shear reinforcing elements are provided in the radial direction;
- the geometry and type of shear reinforcement guarantees anchorage at both ends;
- the distance between the first shear reinforcing element and the face of the support is larger than or equal to 0.35d_v and smaller than or equal to 0.75d_v (Figure 30.13-9); any shear reinforcement closer to the support than 0.35d_v shall not be considered for resistance;
- the maximum spacing between shear reinforcing elements in the radial direction is not larger than the smaller value of 0.75d_v and 300 mm;

- the maximum distance between concentric shear reinforcing elements at the level of the second radial reinforcement does not exceed $1.5d_v$;
- the maximum concrete cover at the compression side of the slab does not exceed $d_v/6$;
- the maximum diameter of the shear reinforcement $Ø_{max}$ does not exceed the values given in Table 30.13-4.

Table 30.13-4: Maximum diameter of punching shear reinforcement as a function of slab effective depth d_v.

d_v [mm]	$Ø_{max}$ [mm]
< 160	—
160–180	14
181–220	16
221–260	18
261–340	20
341–600	25
> 600	30

Minimum reinforcement for ductility behaviour of cross sections in FRC structures

In members designed as parts of reinforced concrete structures, minimum reinforcement $A_{s,min}$ shall be provided to:

a) ensure distributed cracking and to handle forces from restrained deformations where not considered explicitly in the design;
b) ensure sufficient deformation capacity to contribute to structural robustness by allowing alternative load paths;
c) avoid failures due to unpredicted cracking;
d) ensure applicability of limit design models;
e) ensure constructability

$M_{R,min} \cdot (N_{Ed,min}) \geq M_{cr} \cdot (N_{Ed,min})$ shall always be provided for slabs.

To guarantee this statement in case of pure bending in elevated slabs:

- $A_{s,min} \geq \max [0.13\, f_{ctm}/f_{yk};\ 0.1(f_{R1k} - f_{R3k})/f_{yk}]$ (30.13-16)

30.13.6.4 Compression members

In the case of storey-high compression members, the minimum dimensions according to Table 30.13-5 should generally be complied with.

Table 30.13-5: Minimum dimensions of compression members.

	Cast in place [mm]	Precast element [mm]
Compression members (side ratio up to 4:1)	200	150
Reinforced walls	150	100
Unreinforced walls	120	-

In compression members, the minimum longitudinal reinforcement ratio is 0.2%. In the case of large column cross-sections, a wall thickness of at least 200 mm according to Figure 30.13-10 may be taken into account in determining the minimum reinforcement ratio.

The minimum reinforcement should be distributed proportionally around the circumference of the column cross-section and shall have a diameter of at least 8 mm.

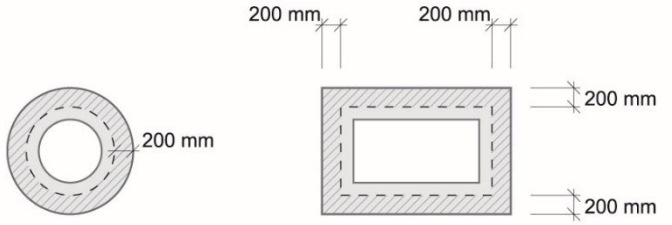


Figure 30.13-10: Minimum wall thickness for determining the minimum reinforcement of compression members.

In walls, the minimum required cross-section of the vertical reinforcement is 0.2% of the concrete cross-section required for structural safety. The vertical bar spacing shall not be greater than either twice the wall thickness or 300 mm.

Compression members with a reinforcement ratio below that stipulated above should be dimensioned like unreinforced structural members according to section 30.13.5.1.

The maximum longitudinal reinforcement ratio in compression members should in general not exceed 0.04 outside lap regions, unless it is shown that the integrity of the concrete is not affected, and the full strength is achieved at the ULS. This limit should be increased to a maximum of 0.08 at laps. In the case of large reinforcement ratios, special detailing and execution measures have to be taken and the stirrup reinforcement should be strengthened.

The longitudinal reinforcing bars of compression members shall be prevented from local buckling by means of stirrups. If the reinforcement of the compression zone reaches the yield strength in the ultimate limit state, the corner bars as well as every second longitudinal bar shall be enclosed with hooks or additional stirrups.

The spacing s of the stirrups and hooks shall fulfil the following requirements:

$$s \leq 15\phi_{s_{l,min}}$$

$$s \leq a_{min} \qquad (30.13\text{-}17)$$

$$s \leq 300\text{mm}$$

where a_{min} is the minimum cross-sectional dimension and $\phi_{sl,min}$ is the minimum diameter of the longitudinal reinforcing bars.

In compression members with a polygonal cross-section, a longitudinal bar shall be provided at least at every corner.

In the zones where forces are applied, in the area of splices and where the cross-section of compression members changes, additional stirrups shall be provided in order to resist transverse tensile forces. The bar diameter of the stirrups shall be at least 1/3 of the diameter of the thickest longitudinal bar.

The provisions for compression members apply analogously to the stirrup reinforcement in walls.

The horizontal reinforcement of walls shall be dimensioned according to the requirements of section 30.5 but shall not be less than 25% of the vertical reinforcement.

In the hinge regions of columns constructed with concrete of strength classes of C60 or higher, and without steel fibres, stirrups shall be detailed such that an equivalent minimum core confining pressure of $0.01 f_{ck}$ is achieved where the confining pressure is determined in accordance with section 7.2.3.1.6. Confining reinforcement shall be continued beyond $1.2h$ to each side of sections with maximum moments. Alternatively, 2% by weight of end-hooked fibres may be used to provide sufficient sectional ductility to compression members of high strength concrete.

30.13.6.5 Lightly reinforced FRC structures

This clause provides additional rules for fibre reinforced structural members where the minimum reinforcement provided is less than the requirements in 30.13.6.2, 30.13.6.3, 30.13.6.4.

Fibres shall not be used to replace reinforcement across a construction joint.

The residual tensile strength of FRC shall be disregarded at construction joints.

To avoid brittle failure, it may only be applied for statically indeterminate structures such as: elastically supported structures, anchored underwater concrete slabs, piled slabs, shell-type components and monolithic, prefabricated containers, segmental lining.

30.13.7 Special aspects of precast concrete elements and composite structural members

30.13.7.1 General

In designing precast concrete elements and structures assembled out of these, the final state as well as the construction states during transport and assembly should be

taken into consideration. The development of the concrete strength over time should also be taken into account.

In the case of precast concrete elements and their connections, the technical requirements relating to production and assembly as well as the static requirements should be taken into account when specifying the dimensional tolerances. In particular, the effects of deformations due to creep and shrinkage of the concrete as well as temperature changes should be adequately taken into account.

Panels of facade plates with a multilayered structure should, as far as possible, be able to move freely within their plane. The panels should be connected by means of fatigue-proof and corrosion-resistant fastenings. The connecting elements should, as far as possible, be:

– chemically and physically compatible;
– protected against chemical and physical influences;
– fireproof in accordance with the structure.

30.13.7.2 Arrangement of tendons

(1) The spacing and concrete cover of pre-tensioned tendons shall be such as to ensure that placing and compacting concrete can be carried out satisfactorily and that sufficient bond can be attained between the concrete and the tendons.

(2) The minimum spacing and cover for bond stress and to avoid splitting should be in accordance with Figure 30.13-11 and Table 30.13-6.

Other values of spacing and cover may be used, provided that satisfactory behavior in service and at ultimate limit state is demonstrated by testing or permitted by the specific product standards with factory production control.

Note by TS: The values presented here differ compared to old MC2010 Table 7.13-10 (section 7.13.6.5.4)

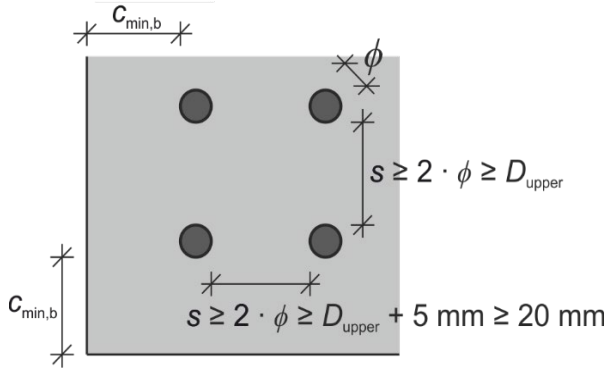


Figure 30.13.-11: Minimum clear spacing and cover for bond for pre-tensioning tendons.
Note: D_{upper} is the maximum size of aggregate.

Table 30.13-6: Minimum concrete cover $c_{min,b}$ for pretensioned tendons.

Minimum spacing s	Minimum cover: $c_{min,b} / \phi$	
	Strand	Indented wire
$s = 2\ \phi$	3.0	4.5
$s \geq 2.5\ \phi$	2.5	4.0

(3) Bundling of tendons should not occur in the anchorage zones without taking special provisions to cover splitting forces and taking into account the bond behavior of bundles. Outside of the anchorage zone up to three tendons may be bundled.

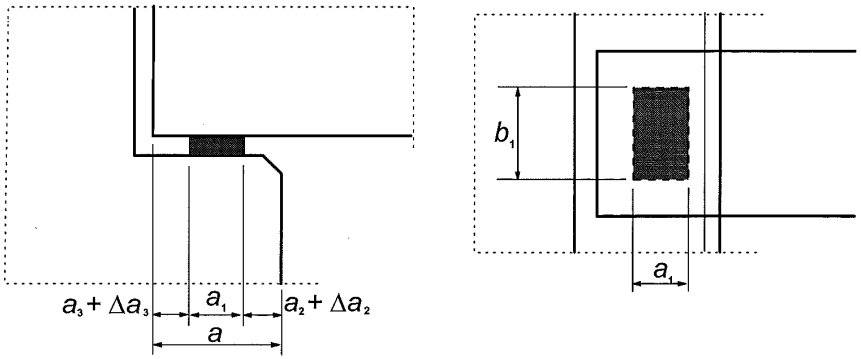


Figure 30.13-12: Definitions of bearing geometry.

30.13.7.3 Bearings

The nominal length of a simple bearing as shown in Figure 30.13-12 may be calculated as:

$$a = a_1 + a_2 + a_3 + \sqrt{(\Delta a_2^2 + \Delta a_3^2} \qquad (30.13\text{-}18)$$

where:

a_1 is the net bearing length with regard to the bearing stress, being:

$a_1 = F_{Ed}/(b_1 f_{Rd})$;

F_{Ed} is design value of support reaction;

b_t is net bearing width;

f_{Rd} is design value of bearing strength, with:

$f_{Rd} = 0.4 f_{cd}$ for dry connections;

$f_{Rd} = f_{bed} \leq 0.85 f_{cd}$;

where:

f_{cd} is lowest design strength between supporting and supported member;

f_{bed} is design strength of bedding material;

a_2 is distance between edge of bearing and end of supporting member (Figure 30.13-12);

a_3 is distance between most remote edge of bearing and end of supported member (Figure 30.13-12);

Δa_2 is allowance for tolerances for distance between supporting members (Figure 30.13-12);

Δa_3 is allowance for tolerances for the length of the supported member, with: $\Delta a_3 = l_n / 2500$, where l_n = length of supported member (Figure 30.13-12).

Minimum values for a_1, a_2, a_3 and Δa_2 are given in Tables 30.13-7 to 30.13-10.

Table 30.13-7: Minimum value of a_1 (Figure 30.13-12), in mm.

Relative bearing stress σ_{Ed} / f_{cd}	≤ 0.15	0.15–0.4	> 0.4
Line support (floors, roofs)	25	30	40
Ribbed floors and purlins	55	70	80
Concentrated supports	90	110	140

Table 30.13-8: Minimum distance a_2 (Figure 30.13-12), in mm.

Support material	σ_{Ed} / f_{cd}	≤ 0.15	0.15–0.4	> 0.4
Steel	line	0	0	10
	concentrated	5	10	15
Reinforced	line	5	10	15
Concrete ≥ C30	concentrated	10	15	20
Plain concrete	line	10	15	15
Reinf. Concrete < C30	conc.	20	25	25
Brickwork	line	10	15	—
	concentrated	20	25	—

Table 30.13-9: Minimum distance a_3 (Figure 27.13-12), in mm.

Detailing of reinforcement	**Support type**	
	Line	Concentrated
Continuous bars over support (restrained or not)	0	0
Straight bars, horizontal, close to end of member	5	15, but not less than end cover
Tendons or straight bars exposed at end of member		15
Vertical loop reinforcement	15	End cover + inner radius of bend

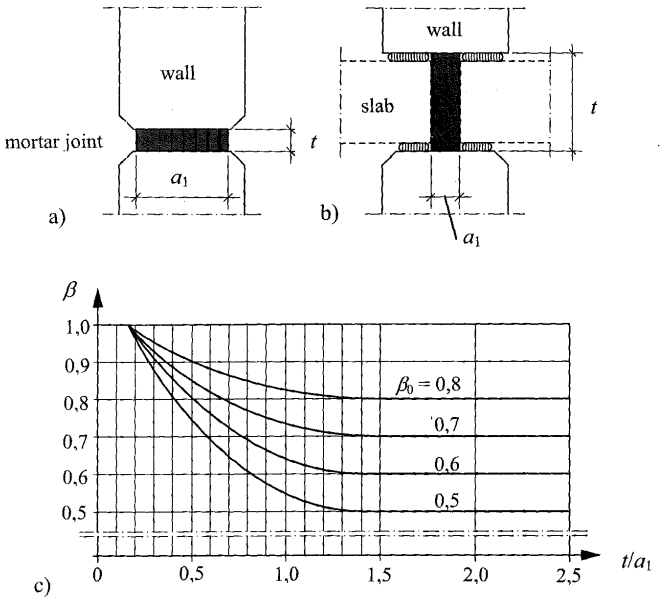


Figure 30.13-13: Capacity of mortar joint (fib Bulletin 43).

Table 30.13-10:Allowance Δa_2 (Figure 30.13-12), in mm.

Support material	Δa_2
Steel or precast concrete	$10 \le l/1200 \le 30$ mm
Brickwork or cast in situ	$15 \le l/1200 + 5 \le 40$ mm

30.13.7.4 Mortar joints

Joints filled with mortar are used as a connection between precast members transmitting compression forces. The joint bearing capacity is:

$$N_{Rd,j} = f_{cd,j} \cdot A_j = \beta \cdot f_{cd,w} \cdot a_1 \cdot l \qquad (30.13\text{-}19)$$

where:

$f_{cd,w}$ is design compressive strength of wall concrete;

$f_{cd,m}$ is design compressive strength of joint mortar;

$f_{cd,j}$ is design compressive strength of joint;

$\beta_0 = f_{cd,m}/f_{cd,w}$;

$\beta = f_{cd,j}/f_{cd,w}$.

The values β can be read from Figure 30.13-13.

The strength of the mortar should be at least 50% of the strength of the adjacent members.

30.13.7.5 Loop connections

Loop connections as exemplified in Figure 30.13-14 can be used to transfer tensile forces, bending moments and shear forces. Loop connections are used between solid slabs where continuity is demanded.

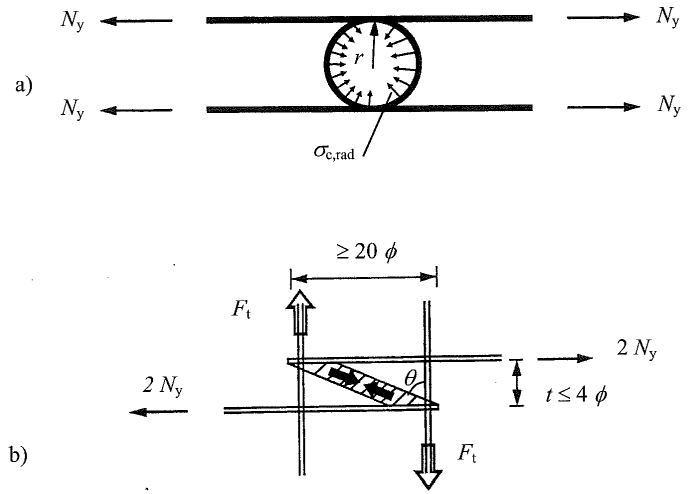
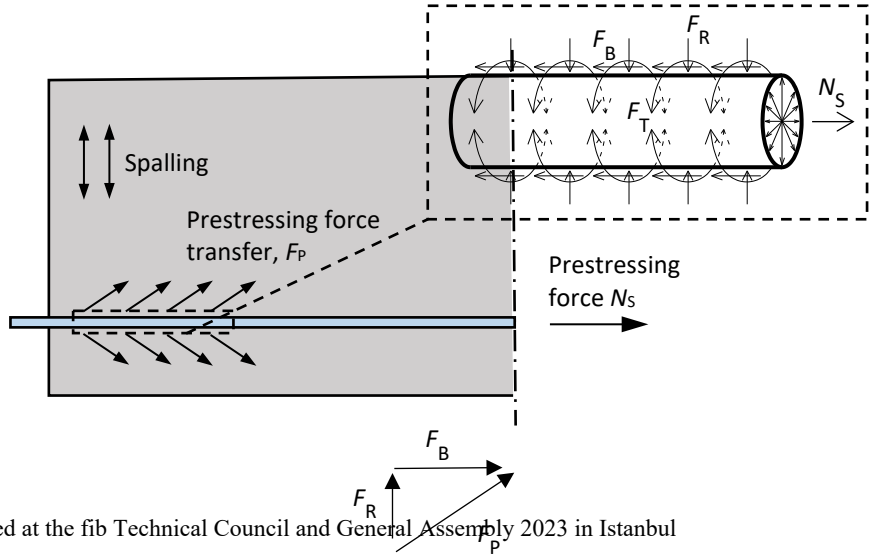


Figure 30.13-14: Loop connection (N_{yd} = yield force in longitudinal bar, F_t = force in transverse bar).



The radius of the loop should satisfy the following demands:

$$R \geq \frac{\pi \cdot \phi}{4} \frac{f_{yd}}{\sigma_{c,rad}} \geq 8f \qquad (30.13\text{-}20)$$

where:

$$\sigma_{c,rad} \leq f_{cd}\sqrt{b_i/\phi} \leq 3f_{cd} \qquad (30.13\text{-}21)$$

in which:

$b_i = 2 \cdot (c_e + \frac{\phi}{2})$ but not smaller than t (see Figure 30.13-14);

$\sigma_{c,rad}$ = radial compressive strength at inside bend of loop (Figure 30.13-14);

c_e = concrete cover between U-bar and edge of element;

ϕ = diameter of reinforcing bar.

30.13.7.6 Transverse stresses in the anchorage zone of prestressed tendons

30.13.7.6.1 General

In the case of prestressing with pretensioned tendons or wires, two ways of cracking can be distinguished:

– spalling can occur at the end of the member, especially in the case of I-beams with thin webs;

– splitting can occur along the transmission length as a result of excessive bursting force generated by spreading of the prestressing force radial component over the cross-section. Bursting force magnitude is a function of strands mechanical properties such as prestressing element geometry and Poisson effect.

Figure 30.13-15 shows the various mechanisms in the anchorage zone of a prestressed member.

F_P – Prestressing force
F_R – Radial (Bursting) force
F_B – Bond force
F_T – Tangential force*

Decomposed
Prestressing force

*Tangential force as a reaction to bursting force resulting into concrete splitting failure on the surface.

Figure 30.13-15:Actions in the anchorage zone of a member, prestressed with steel with direct bond.

Poisson effect causes prestressing element cross-section change due to its elongation / contraction.

30.13.7.6.2 Spalling

The spalling stresses can be read from Figure 30.13-16, based on elastic analysis. If $\sigma_{spl} < f_{ctd}$ no spalling reinforcement is necessary.

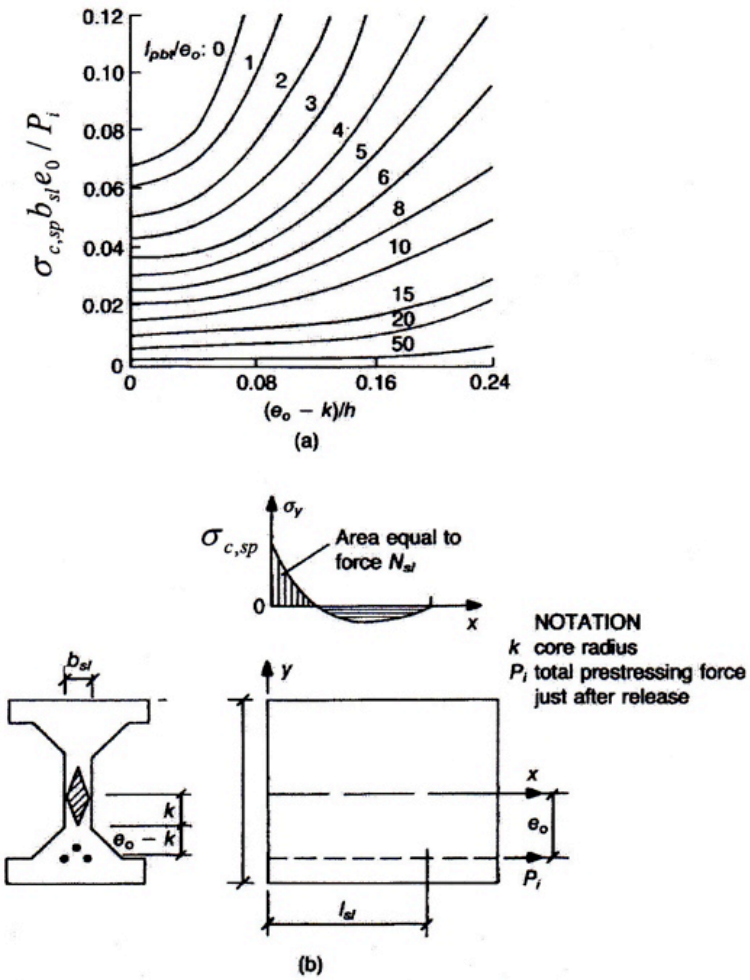


Figure 30.13-16: Maximum spalling stress as a function of eccentricity and transmission length (based on linear elastic analysis) for members with h < 400 mm.

30.13.7.6.3 Splitting

For the calculation of the bursting force causing splitting failure the symmetric prism analogy may be used (Figure 27.13-18). The calculation is based on a virtual

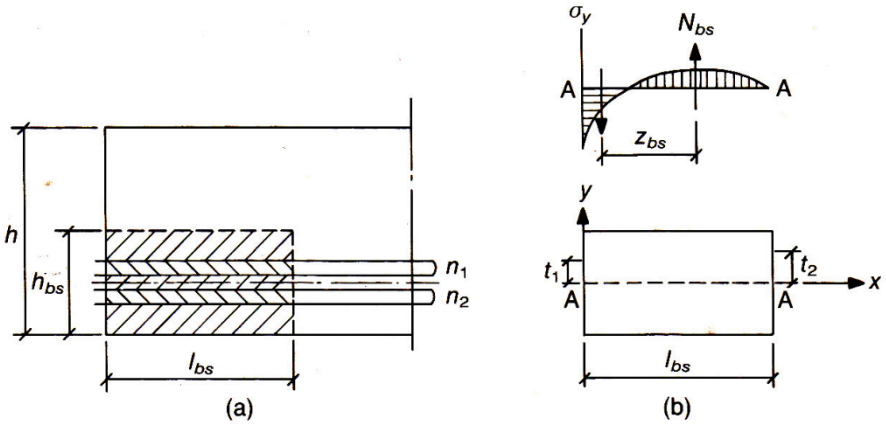


Figure 30.13-17:Calculation of the bursting force:(a) dimensions of the symmetrical prism; (b) moment equilibrium along section A-A.

prismatic element, defined in order to describe the bursting forces. The prism is shown in Figure 27.13-18(a) (shaded area). The length of the prism is:

$$l_{bs} = \sqrt{h_{bs}^2 + (0.6 l_{bpt})^2} < l_{bpt} \qquad (30.13\text{-}22)$$

30.13.7.6.4

where l_{bpt} is the transmission length, where now the 5% value is used: this is equal to 50% of the values given in Eq. (30.13-6a) or (30.13-6b).

The internal lever arm for the bursting force is:

$$z_{bs} = 0.5 l_{bs}$$

The bursting force N_{bs} follows from the moment equilibrium along section A-A (Figure 30.13-17):

$$N_{bs} = \frac{1/2(n_1+n_2)t_2 - n_1 t_1}{z_{bs}} \gamma_1 F_{sd} \qquad (30.13\text{-}23)$$

where:

t_1 is distance between the centroid of tendons above section A-A to the centroid of the prism;

t_2 is distance between the centroid of the concrete stress block above section A-A to the centroid of the prism;

n_1, n_2 are numbers of the tendons above and below section A-A, respectively;

F_{sd} is design prestressing force per tendon;

$\gamma_1 = 1.1$ being the supplementary safety factor against overstressing.

No reinforcement against splitting is necessary if the distance between the strands/wires (with diameter Ø) and the cover satisfies the minimum values given in Table 30.13-11:

Table 30.13-11:Minimum cover as a function of the clear spacing to resist the splitting stresses around pretensioned strands.

Concrete strength	Clear spacing	Cover
C20/25 to C50/60	≥ 3Ø = 2.5Ø	≥ 3Ø ≥ 4Ø
≥ C55/67	≥ 2.5Ø = 2Ø	≥ 2.5Ø ≥ 3Ø

For creating an envelope of allowable covers, interpolation between the clear spacings 2.5Ø–3.0Ø and 2.0Ø–2.5Ø is possible.

References

[30.13-1] *fib* Bulletin 40 — FRP reinforcement in RC structures, *fib*, 2007.

[30.13-2] Bank — L.C. Bank (2006), 'FRP Reinforcement Detailing', in Composites for Construction: Structural Design with FRP Materials, ISBN 9780471681267, DOI 10.1002/9780470121429

[30.13-3] fib Bulletin 90. — Design of Concrete Members Strengthened with Externally Applied Reinforcement. *fib*, 2019.

[30.13-4] Monney et. al. — Monney F., Fernández Ruiz M., Muttni A., "Design against splitting failures in reinforced concrete due to concentrated forces and minimum bend diameter of reinforcement", Engineering Structures (2021).

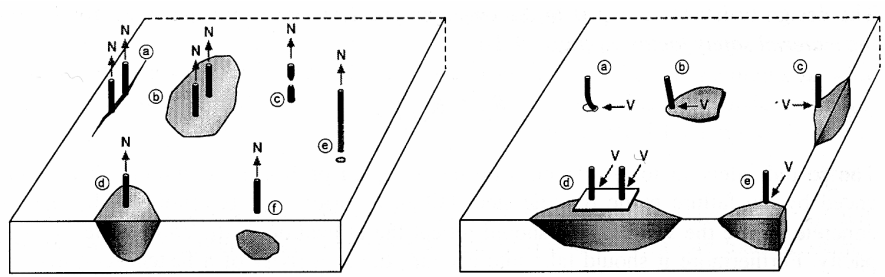


Figure 30.14-1: Failure modes of anchorages.

30.14 Evaluation of anchorages in concrete

30.14.1 General

General principles and classification of anchorages in concrete are given chapter 25. Further guidance is also provided in *fib* Bulletin 58 [30.14-1].

30.14.2 Strength evaluation

Anchorages should be verified for tension, shear and combined tension and shear loadings and all failure modes.

A summary of the failure modes of anchors under tension and shear loads is presented in Figure 30.14-1.

There are several types of failure modes which are exhibited by anchors when they are loaded to failure under

a) Tension loading

- steel fracture;
- concrete cone failure;
- pullout failure or pull-through failure of mechanical anchors;
- combined bond and concrete failure of bonded anchors concrete cone failure;
- concrete splitting;
- edge failure;

b) Shear loading

- steel failure;
- pryout failure;
- concrete edge failure.

Strength and failure mode of anchors are related to the steel and concrete strength (and the bonding mortar strength in case of bonded anchors), the depth of

embedment, the anchor's shaft and head diameter, the loading type, the loading direction, the edge distance and/or the anchor spacing.

Due to several causes tensile stresses in concrete can reach the concrete strength, so anchorages and connectors are often installed in cracked concrete regions. The influence of concrete cracking on the anchor performances depends on the type of anchor.

Only anchorages with a predictable behaviour in cracked concrete should be used; the anchor suitability has to be proven on an experimental basis by means of pre-qualification tests carried out according to relevant codes.

Characteristic resistance of anchors can be based on the computation or test evaluation of the steel tensile and shear resistance, the concrete breakout tensile and shear resistance, the concrete splitting resistance and the tensile pull-out resistance of the anchors.

The models should take into account factors affecting strength such as embedment depth, spacing and edge distance, depth of structural members as well as the presence or the absence of concrete cracking. Limits on edge-distance and anchor spacing in the design model should be consistent with the set of reference tests. The quality of installation can have a significant influence on the load-bearing capacity of anchorages. In case of unknown or sub-optimal execution quality, suitable safety calculations may need to be undertaken.

Interaction of tensile and shear loads should be considered in design, using interaction curves that result in a prediction of the strength in good agreement with results of comprehensive tests.

References

[30.14-1] *fib* Bulletin 58 Design of anchorages in concrete, *fib* Guide to good practice, ISBN 978-2-88394-098-7, July 2011[1]

[1] New Bulletin currently under preparation.

31 Evaluation of other aspects of social performance

31.1 Introduction

The goal of including this requirement consists of taking into account the satisfaction perceived for all those stakeholders involved from the construction phase to the service life of the structure. Better understanding of the connections between civil engineering and society can help to minimise their negative impacts and help to develop more socially sustainable structures (Josa and Aguado [31-1]).This requirement embraces several indicators that can lead to the evaluation of the structure. The social issues include, on the one hand, criteria linked with the social effects of construction and operation of structure (protection of cultural heritage, service quality improve, local unemployment decrease, availability of affordable dwelling), and on the other hand criteria regarding health, safety and comfort of occupants and workers on site.

Some social costs can also be monetarized and taken into account in the economic pillar of sustainability as well (chapter 33).

Some aspects to be taken into account are:

- Health and quality of built environment – health issues and quality of performance within construction, service life, maintenance, or dismantling;
- Safety and security of occupants and workers on site;
- Aesthetics, preservation of cultural heritage and visual impact;
- Impact on local community (positive and/or negative);
- Other.

31.2 Health and quality of built environment

Health and quality of built environment has a fundamental impact on how people perceive, function, and behave. It makes a direct contribution to community social

live and should be considered within all live cycle stages from design, construction, operation to the end of life of structure.

Concrete and concrete structures could positively contribute to specific aspects of social performance by following qualities:

- Acoustic comfort – concrete (due to high specific weight) can significantly contribute to air-born sound insulation of structures of buildings (concrete floors and/or walls separating different operational areas);
- Thermal comfort – concrete (due to high specific weight) can contribute to thermal stability of internal environment and consequently to energy savings associated with cooling and heating of buildings. This represents a direct link to reduction of carbon footprint;
- Maintainability - concrete surface produced in high quality can be easily maintained, cleaned and it could have a long durability;
- Flexibility – character of concrete technology enables significant design flexibility due to the possibility of forming almost any element shape limited only by structural reliability requirements;
- Healthiness - Concrete is not the source of toxic emissions or volatile organic compounds. In standard concrete are not used any hazardous chemicals.

Evaluation of these specific performance qualities should be done in direct cooperation with pertinent experts according to relevant standards during design of structure.

31.3 Safety and security

Evaluation of structural safety of concrete structures representing a key issue of structural design is described in detail in a chapter 30.

Security performance evaluation could be based on a risk analysis, where security risk could be generally expressed as a function of threats, vulnerabilities and potential impacts or expected loss. Nevertheless, application of risk analysis to the real problem is difficult due to a lack of relevant and meaningful variable data (e.g., cost of vulnerability, etc.).

A beautiful structure can only be achieved if in addition to efficient functioning the aesthetics are developed from the beginning as an essential part of the global structural concept. Owners and engineers have a responsibility and duty to contribute to the aesthetic aspect of a structure, at a reasonable cost.

Performance requirements for aesthetics:

- visual appearance of the structure: choice of shape and composition, selection of colours, textures, and materials;
- harmony of a structure and its environment;
- integration into the surroundings regarding historical context and cultural heritage.

Safety and security should be considered in all the life cycle of the structure (construction, operation, maintenance and dismantling).

31.4 Aesthetics and cultural heritage

Aesthetics is a branch of philosophy dealing with nature and expression of beauty. It deals with the perception that the users or other stakeholders have of the structure. Aesthetics of structures is determined by engineering and architectural concept considering surrounding environment including climate, client brief, regulations, budget etc. Close cooperation of structural engineer with architect from the very early stage of the conceptual design is thus essential and necessary for high quality (sustainable) design and construction of building, bridge, dam, or other type of infrastructure.

The evaluation of aesthetics is in general influenced by the subjective taste of individuals. It is usually a part of evaluation of jury within engineering and architectural competition.

Cultural heritage includes, inter alia, tangible culture such as buildings and other construction monuments. Protection and conservation of historical buildings and other constructions represent an essential and highly important part of preservation of cultural human history.

Evaluation of cultural heritage is subjected to legal protection of cultural property by number of international agreements and national laws, which must be considered and implemented.

31.5 Impact on local community

Impact on local community and neighbours could be positive or negative.

Negative impacts:

- Interferences associated to noise and particles pollution;
- Traffic disturbances during construction;

- Soil erosion impacts due to construction, including impacts to ground water level;
- Increased demand for water during construction and potential impact to water quality;
- Time delays or loss of activity due to maintenance, rehabilitation, repair activities.

Positive impacts:

- New construction can create economic advantages for the local community, incl. increase employment during construction and employment after start of new built infrastructure;
- Improving transportation networks provides economic benefits to local community properties. By reducing travel time, people in local communities can reduce fuel costs and vehicle depreciation;
- Increased attractivity of the area by improvement of infrastructure.

The evaluation of some above-mentioned aspects can be done by evaluation or measuring the pollution or the disturbance that is expected of perceived in a particular work site. It can also be evaluated by expert advice or by qualitative comparison of different possibilities.

References

[31-1] Josa, I., Aguado, A. Infrastructure and society: from a literature review to a conceptual framework, 2019. *Journal of Cleaner Production.*

32 Evaluation of environmental performance

32.1 General

The goal and scope of environmental impact evaluation shall be consistent with the intended application in the design, construction and operation processes of concrete structure. The evaluation and optimization model should be sufficiently well defined to ensure that the results of the study are compatible, relevant and sufficient to address the pre-defined goals.

Environmental problems such as the global warming, shortage of resources (energy, water, raw materials), waste management problem has accelerated the conversion of the current mass-production, mass-consumption, and mass-disposal type economic society into recycling-based “sustainable society”, see also chapter 39.

The construction industry has a significant impact on the state of the environment. For the implementation and operation of buildings, it consumes a significant amount of raw material and energy resources (most of which are non-renewable) and produces a decisive amount of waste and harmful emissions (CO2, SOX, NOX, etc.). The actual construction activity adversely affects the environment mainly by noise, dust, exhalations, road pollution, occupation of natural areas, damage to greenery and waste from construction activities. It is not possible to completely exclude all adverse effects of construction activities on the environment, but a necessary requirement must be to limit them to the extent strictly necessary, corresponding to the general global requirements of sustainable development of society.

In the period 1950 to 2020 global CO2 emissions increased 5.8 times and in the same period CO2 emissions from cement production increased 24.4 times (source data: Our World in Data).

As a result, recently the most important environmental focus is on low carbon technologies leading to carbon neutral solutions.

In the 6th IPCC report (published in 2022) is specified that achieving the global transition to a low-carbon, climate-resilient and sustainable world requires purposeful and increasingly coordinated planning and decisions at many scales of governance including local, subnational, national and global levels. A low-carbon transition will thus shift investment patterns and create new economic opportunities. This process should become essential concept of transition of construction industry including concrete industry.

However the greenwash type of technologies and systems should be prevented.

32.2 Objectives of evaluation of environmental performance

Due to the global character of the problem and complexity of relations among the elements of the analyzed system, environmental performance evaluation requires consideration of its multicriterial character. The use of multicriteria evaluation methodology and multicriteria optimization techniques respecting the significance of the system's interrelationships is thus essential and necessary.

The evaluation method should be capable to cover all relevant flows of material, energy and others, and should consider corresponding essential environmental criteria. However, simplification of the model is usually admitted.

32.3 Principles of evaluation of environmental performance evaluation

32.3.1 Introduction

The total environmental impact of product (i.e. concrete structure) should be considered throughout its whole life, from raw material acquisition, through production, use and disposal.

It is essential that the goal of optimization efforts should be to keep structural materials in the closed material cycle (the gray area) as long as possible. The high importance of maintenance and repair processes, which can increase the durability of concrete structure, is thus evident. Equally, the significance of renovation and recycling phases on the total environmental impact of concrete structure is considerable.

Life cycle approach allows for the choice of either a cradle-to-gate, cradle-to-grave or cradle-to-cradle life cycle (figure 32.3-2).

A characteristic life cycle of concrete structure with its typical material and energy flows and consequent environmental impacts is presented in the figure 32.3-1.

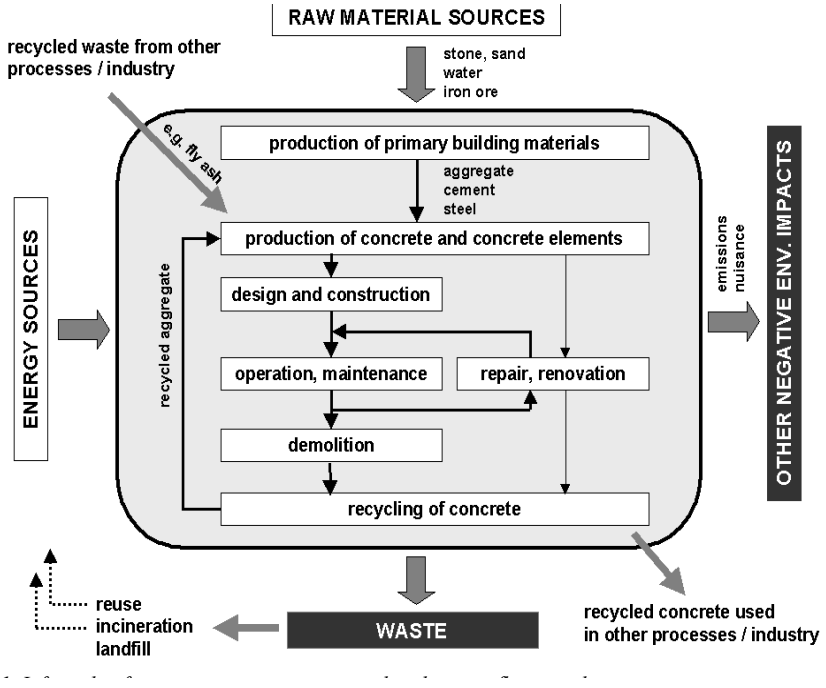


Figure 32.3-1: Life cycle of concrete structure – material and energy flows and consequent environmental impacts, fib Bulletin 71 [32-1].

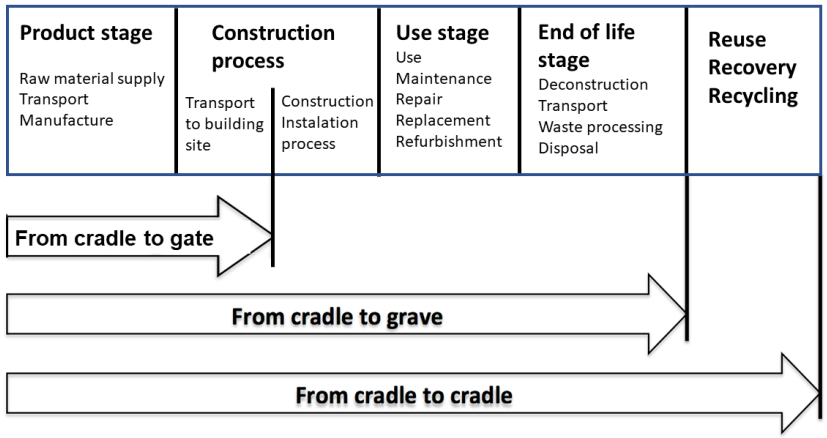


Figure 32.3-2: Different concepts of LCA of concrete structures.

ISO 13315-4 [32-2] provides the procedure for environmental design for concrete structures. Figure 32.3-3 shows the general procedure for environmental design of concrete structures. The environmental design of a concrete structure should be performed integrally with the structural. Durability and other aspects of designs. The procedure consists of environmental performance requirements expressed as multiple indicators. Examples of environmental indicators are provided as ISO 13315-6 [32-3] annex A.

32.3.2 Evaluation of environmental performance

The environmental performance of the entire structure can be expressed in two principal forms:

- environmental profile, which is composed from a set of values of different criteria;
- environmental impact, which is expressed by characteristic indicator(s) (the value can be a weighted sum of values of different criteria).

The environmental performance includes both positive and negative aspects of concrete and concrete structure to environment. Negative aspects include CO2 emission, waste emissions etc., while positive aspects include carbon capture at production stage and heat storage at use stage.

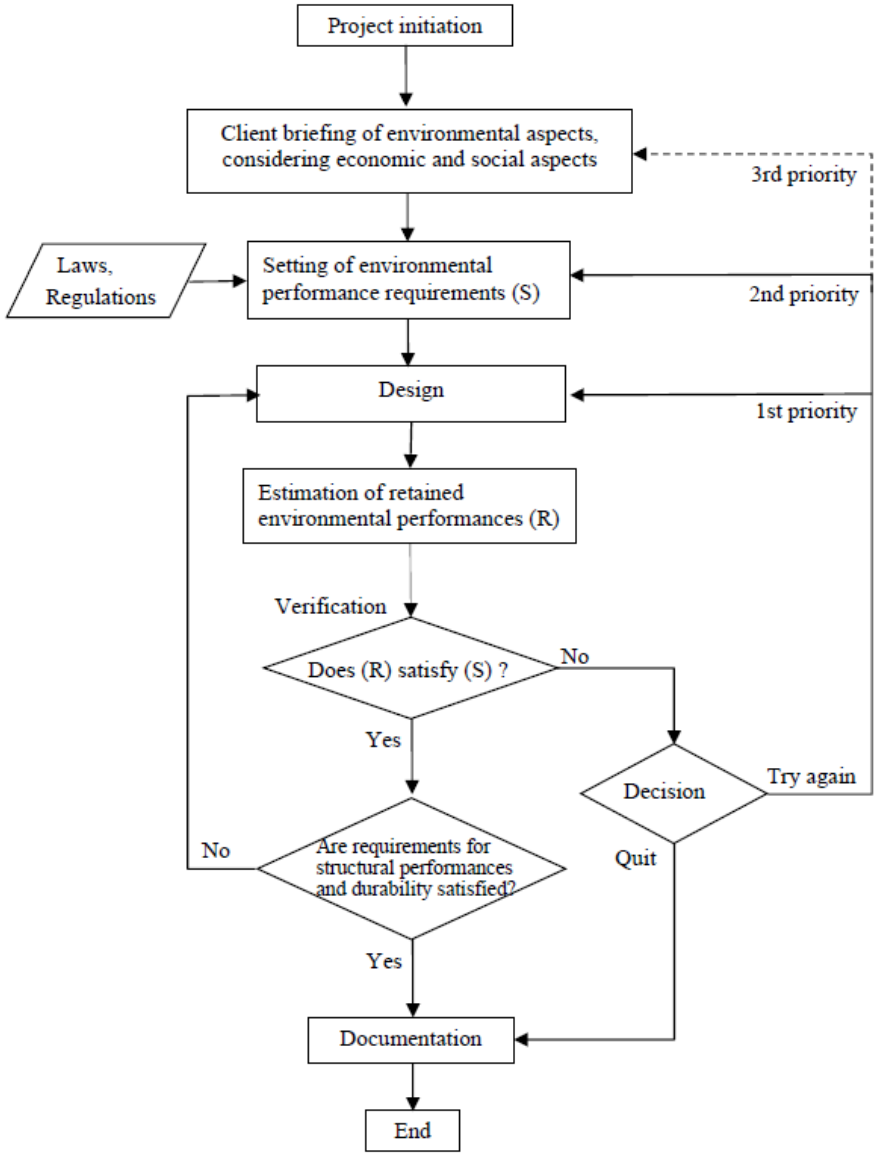


Figure 32.3-3: Procedure for environmental design of concrete structures according to ISO 13315-4 [32-2]

ISO 13315-4 [32-2] annex A provides impact categories to be considered in environmental management of concrete and concrete structures, and ISO 13315-6 [32-3] provides examples for positive aspects.

Taking into account the whole life cycle of the product, the environmental performance associated with particular criterion can be expressed as a sum of partial environmental performance E_i as follows:

$$E_{tot} = \sum E_i \qquad (32.3\text{-}1)$$

The value of environmental performance of the product and/or process can be expressed as an environmental cost – eco-cost or in normalized amounts of points.

For most cases of concrete structures, the general equation can be expressed in a more detailed form as:

$$E_{tot} = E_{ini} + E_{oper} + E_m + \sum E_{repair} + \sum E_{renov} + E_{demol} + E_{recycl} \qquad (32.3\text{-}2)$$

where:

E_{ini} represents the initial environmental performance covering production and construction phases defined as:

$$E_{ini} = E_{pbm} + E_{pc} + E_{constr} \qquad (32.3\text{-}3)$$

Particular environmental performance within life-cycle steps are:

- E_{oper} environmental performance associated with operation of the structure;
- E_m environmental performance associated with the maintenance;
- E_{repair} environmental performance associated with the repair of the failure;
- E_{renov} environmental performance associated with renovation;
- E_{demol} environmental performance associated with demolition;
- E_{recycl} environmental performance associated with reuse, recycling and waste disposal;
- E_{pbm} environmental performance associated with the production of primary building materials;
- E_{pc} environmental performance associated with the production of concrete and concrete elements;
- E_{constr} environmental performance associated with the design and construction of the structure.

Environmental performance E_i, related to a particular step of the life cycle should incorporate all environmental damages, which correspond to all essential environmental criteria:

$$E_i = \sum w_j \; Q_j \qquad (32.3\text{-}4)$$

where:

$\{w_j\} = (w_1 \ldots w_m)^T$ is the vector of weights representing importance of individual criteria;

m is the number of essential environmental criteria; and

$\{Q_j\} = (Q_1 \ldots Q_m)^T$ is the vector of embodied values of environmental criteria.

Considering the particular environmental criteria, the environmental performance in each phase of the life cycle can be written in the form:

$$E_i = w_1 Q_{CO2} + w_2 Q_{SO2} + \ldots + w_m Q_m \qquad (32.3\text{-}5)$$

where Q_{CO2},, Q_{SO2} and Q_m are total values of embodied CO_2, SO_2 , and embodied value of other environmental criteria, respectively.

This equation should be determined for every particular phase of the life cycle in order to analyze environmental performance of the structure within the entire life cycle.

In some cases the independent evaluation of environmental performance of single criterion could be important and useful:

$E_i = Q_{CO2}$ for Life Cycle CO_2 (LCCO2) evaluation;

$E_i = Q_{en}$ for Life Cycle Energy (LCE) evaluation.

Weighting is a very sensitive approach and if used it must be based on scientific backgrounds.

The weighting process represents critical step in the Life-Cycle Assessment process. The reliability of obtainable results is highly dependent on the quality of determination of weighting factors. The process of determination of weighting factors is very complex and should cover specific conditions, boundary limitations and preferences associated with the particular case. However, this process is very often subjective due to a variety of criteria with different characteristic features (the problem of “mixing apples and oranges“). Sensitivity analysis of the multicriteria problem is thus essential. The transparency of the criteria used is essential for the clarity and accountability of the results to all the stakeholders.

Weighting factors could be determined using different weighing approaches and methods such as:

EPS method – Environmental Priority Strategies in Product Design (developed by Institute for Environmental Research, Sweden). The model covers environmental, cost and other aspects. The weight of the particular aspect is based on the price that has to be paid by the society to prevent the corresponding environmental impact. The method has been used in the car industry.

Panel method - Expert-based determination of weighing factors. An expert makes professional judgement of weights among different environmental aspects. A single expert judgement represents rather subjective view. However, by the use of

higher number of experts (panel of experts) the subjectivity of determination of weights can be reduced.

NEL method – No-effect level method. The method is based on quantitative analysis of the relation between no-effect level and current level of particular environmental aspect. The difference between current environmental load and NEL level called sustainability indicator is used to determine weighting factors.

Combined Panel-NEL method. The combined method consists from following steps:

(1) criteria selection by a panel of experts;

(2) effect –criterion relation score using NEL method;

(3) criteria weighting – weighting factors determined by a panel of experts;

(4) calculation of environmental standards.

There is no single method for conducting LCA studies. Results from LCA studies may be useful inputs to different evaluation, optimization and other decision-making processes. The LCA considers the potential environmental impacts and aspects of a product system throughout its life cycle.

Cite literatures on above methodologies.

The comparative method assumes that the environmental performance during the lifecycle stages other than target of evaluation is the same between the target and reference. This assumption becomes a limitation for use of the method. The comparative evaluation can be applied only when reference material or reference activity can be appropriately defined.

32.3.3 Evaluation of environmental performance of a specific life cycle stage of the structure

The collection of inventory data for environmental evaluation of the entire concrete structure is generally difficult considering large number of stakeholders involved during the lifecycle of a concrete structure. ISO 13315 [32-4] series provide general procedure for environmental evaluation for each specific lifecycle stage of concrete and concrete structure.

Comparative evaluation procedure for environmental performance evaluation enables to mitigates above mentioned difficulty. In comparative evaluation, environmental performance of a specific lifecycle stage of a target structure or activity is evaluated by comparison with the reference structure or reference activity. Reference structure/activity for comparison should be appropriately set. The “reference structure” should have the same structural and durability performance as the target structure or the same existing structure prior to environmental

ISO 13315-6 [32-3] provides two methods for comparative evaluation, as follow:

- evaluation by difference between target and reference, where, environmental performance difference is defined as:

$$R^d = E^{\text{reference}} - E^{\text{target}} \tag{32.3-6}$$

- evaluation by rate between target and reference, where, environmental performance rate is defined as:

$$R^r = \frac{E^{\text{reference}} - E^{\text{target}}}{E^{\text{reference}}} \tag{32.3-7}$$

improvement. The “reference activity” should have the same purpose as the target activity.

ISO 13315-6 [32-3] provides comparative evaluation procedure for environmental performance evaluation and definitions for “target” and “reference” in comparative evaluation.

32.4 Life Cycle Assessment

32.4.1 Introduction

The evaluation of the environmental performance is performed using standard Life-cycle assessment (LCA) methodology. This methodology is defined in a set of ISO, CEN, ACI and other standards. This chapter presents basic principles of LCA.

32.4.2 General basis

The general methodology of LCA is defined in the International Standard ISO 14040 [32-5] and in the complementary International Standards ISO 14041 [32-6], ISO 14042 [32-7] and ISO 14043 [32-8] concerning various phases of LCA *fib* Bulletin 71 [32-1]. There is no single method for conducting LCA studies. Results from LCA studies may be useful inputs to different evaluation, optimization and other decision-making processes. The LCA technique includes phase of evaluating the potential environmental impacts associated with the whole life cycle of the product.

Life cycle assessment methodology specified in ISO 14040 [32-5] is an iterative assessment method. It includes several steps covering:

(i) definition of goal and scope;
(ii) inventory analysis;
(iii) impact assessment and
(iv) interpretation of results.

while these particular steps are in the state of mutual interaction *fib* Bulletin 28 [32-9]; see figure 32.4-1.

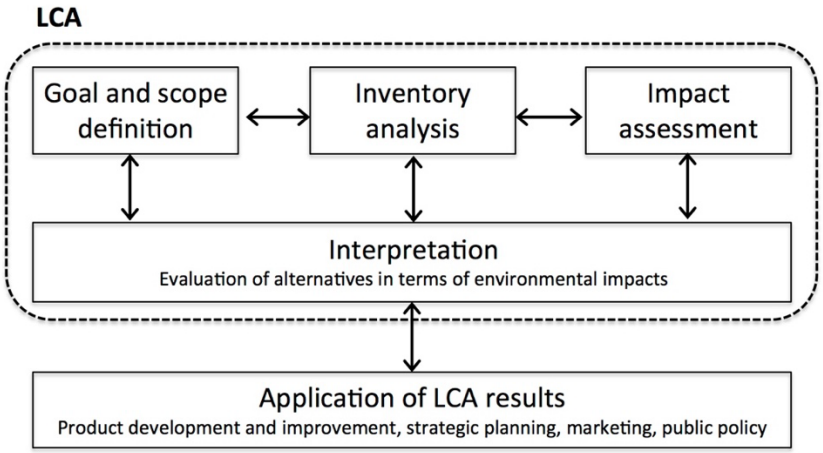


Figure 32.4-1: Relationships and interactions of the steps in a LCA, fib Bulletin 71 [32-1].

Goal and scope

The goal and scope of an LCA study must be clearly defined and consistent with its intended application. The scope must consider all relevant aspects and criteria and should be sufficiently well defined to ensure that the definition of the evaluation model and specification of the assessment data sets are compatible and sufficient to address the stated goal.

Inventory analysis

The inventory analysis (LCI – Life Cycle Inventory analysis) involves data collection and calculation procedures to quantify relevant material and energy inputs and outputs of a product system, e.g. concrete element, concrete structure, whole building or other civil engineering works, etc., through the whole life cycle.

Impact assessment

The target of the impact assessment phase (LCIA – Life Cycle Impact Assessment) is to examine the product system from an environmental point of view using impact categories and category indicators connected with the inventory analysis results.

Interpretation

The final phase of LCA is interpretation, in which the findings of either the inventory analysis or the impact assessment, or both, are combined (consistent with the defined goal and scope) in order to reach conclusions and recommendations.

The principal diagram of the LCA process flow and inherent subprocesses is shown in figure 32.4-2.

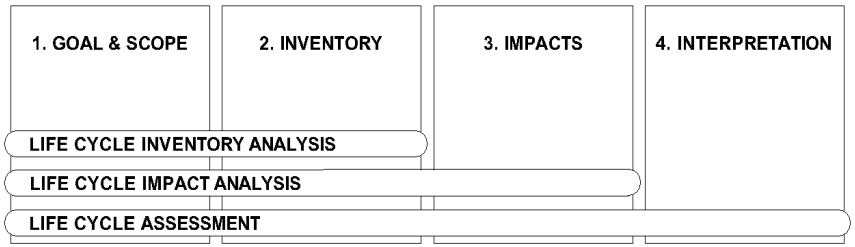


Figure 32.4-2: The LCA process flow and inherent subprocesses, fib Bulletin 71 [32-1].

32.4.3 Functional unit, Functional equivalent

In LCA methodology functional unit is a reference unit that is used to quantify the performance of a system. It is specified as a quantified performance of a product system. According to EN 15643-2 [32-10] functional equivalent represents quantified functional requirements and/or technical requirements for an assembled system for use as a basis for comparison.

32.4.4 Environmental performance / performance sustainability scenarios

The basis for the cost or environmental impact evaluation should be a condition assessment of the entire structure or structural elements, where the performance of the structure, deterioration mechanism, extent and future degradation rate are estimated. It is evident that higher initial cost and higher environmental impact (leading to higher performance quality) can at the end of entire life of structure result in lower total life cycle cost and total life cycle environmental impact. Appropriate settings of repair periods keeping or improving performance quality are essential for the extended life of the structure.

Technically feasible repair methods shall be set forward and priced both in relation to actual costs but also in relation to indirect costs, where the user inconvenience due to lower performance quality is priced depending on the duration of the repair works. This is typically important for repair of civil structures like bridges, roads, etc. The degradation rate is important, as an owner often has to prioritize between different repair works. Therefore the consequences of postponing a repair project have to be evaluated.

There are normally three different reasons for overruling the cost analysis as schematically shown above:

- safety issues, if action to ensure the overall safety of a structure is required;
- keeping of the service level (e.g. decongesting the traffic corridors on the bridge or road);
- capital preservation (allowing for a preventive strategy).

An important aspect in relation to LCC and LCA calculations is the consideration of interest rates. For LCC calculations, national applied interest rates can be used allowing for a calculation of net present value for a structure. In relation to LCA, it should be considered to what extent an interest rate makes sense.

32.5 Environmental Product Declaration

ISO 21930 [32-11] provides core product category rules (PCR) for all construction products and services including concrete components and related services. In accordance with ISO 21930 [32-11], ISO 13315-8 [32-12] provides principles, procedures and requirements for environmental labels and declaration for

concrete and concrete structures specifically. This standard provides rules to ensure that all Environmental Product Declarations (EPD) of construction products, services and processes are derived, verified and presented in a harmonised way. The overall goal of an EPD is to provide relevant, verified and comparable information about the environmental impact from construction products (e.g. concrete component like precast floor panel, precast column, etc.) and services.

The information in EPD is based on LCA, covering different life cycle stages and could be derived based on different system boundaries, see also figure 32.3-2:

- "Cradle to gate" LCA concept: Product stage – raw material supply (stone, gravel, sand, water…), transport, manufacturing and other associated processes influenced by the producer.
- "Cradle to gate with options" LCA concept: Product stage + selected further life cycle stages, e.g. end-of-life stage, recycling, etc.
- "Cradle to grave" LCA concept: Product stage + Construction process stage (transport, construction installation process) + Use stage (use, maintenance, repair, replacement, refurbishment) + End of life stage (deconstruction, demolition, transport, waste processing, disposal). Reuse, recovery and recycling potential may also be included in this concept.

32.6 EIA - Environmental Impact Assessment

Environmental Assessment is according to European legislation (https://ec.europa.eu) a procedure that ensures that the environmental implications of decisions are taken into account before the decisions are made. Environmental assessment can be undertaken for individual projects, such as a dam, motorway, airport or factory, on the basis of Directive 2011/92/EU (known as 'Environmental Impact Assessment' – **EIA** Directive) or for public plans or programmes on the basis of Directive 2001/42/EC (known as 'Strategic Environmental Assessment' – **SEA** Directive).

The common principle of both Directives is to ensure that plans, programmes and projects likely to have significant effects on the environment are made subject to an environmental assessment, prior to their approval or authorisation. Consultation with the public is a key feature of environmental assessment procedures.

References

[32-1] *fib* Bulletin 71 Integrated life cycle assessment of concrete structures. State of the art report – 2013

[32-2] ISO 13315-4:2017 Environmental management for concrete and concrete structures — Part 4: Environmental design of concrete structures

[32-3] ISO 13315-6:2019 Environmental management for concrete and concrete structures — Part 6: Use of concrete structures

[32-4] ISO 13315 Environmental management for concrete and concrete structures

[32-5] ISO 14040:2006 Environmental management – Life cycle assessment – Principles and framework.

[32-6] ISO 14041:1998 Environmental management – Life cycle assessment – Goal and scope definition and inventory analysis.

[32-7] ISO 14042:2000 Environmental management – Life cycle assessment – Life cycle impact assessment.

[32-8] ISO 14043:2000 Environmental management – Life cycle assessment – Life cycle interpretation.

[32-9] *fib* Bulletin 28 Environmental design. State of the art report – 2004

[32-10] EN 15643-2:2011 Sustainability of construction works - Assessment of buildings - Part 2: Framework for the assessment of environmental performance

[32-11] ISO 21930:2017 Sustainability in buildings and civil engineering works — Core rules for environmental product declarations of construction products and services

[32-12] ISO 13315-8:2019 Environmental management for concrete and concrete structures — Part 8: Environmental labels and declarations

33 Evaluation of economic performance

33.1 Introduction

The evaluation of the economic performance is normally done through a Life-Cycle Cost Analysis (LCCA)

LCCA is an approach for the assessment of the financial feasibility of projects. It is frequently based on Discount Cash flow analysis, which is commonly limited to Net Present Value analysis; however, other approaches can be used (e.g., real options). This chapter presents the basic principles, the definition of the terms included in LCCA, the cost categories and the methodology to perform LCCA.

LCCA is a useful tool for designers to develop, select and evaluate economically different alternatives of a project. It is also a useful tool for other stakeholders when considering the construction, maintenance, rehabilitation and dismantling ore reuse of structures.

Sometimes, the administration or other stakeholders impose analysis in a period shorter than the full LCCA and, thus, the evaluation of the economic performance should be carried out in the period as requested, even if in this ModelCode it is recommended to perform a the whole life (from design to dismantlement) LCCA.

Cost and risk are interdependant. Both should, in all cases, be considered together. Sometimes, the most economical solution from the point of view of the cost could also be that with the higher risk associated.

33.2 Cost categories

33.2.1 General

The two main cost categories are:

i) direct; and

ii) indirect costs.

In the literature, the term Life-Cycle Assessment (LCA) usually refers to the evaluation of a system that includes economic, social and environmental aspects. The focus of this chapter is on the economic aspects of the assessment.

The reason for the evaluation over a shorter period than the whole life (from design to dismantlement or reuse) LCCA is that, at present, the development of sustainability is leading to this ambition but in many cases is not feasible to perform it fully yet.

This chapter gives an overview of cost components which may be used to build LCC models / tools (see chapter 29.2.2) in order to assist decision-makers at all levels and within all phases (design, construction, maintenance, rehabilitation and

dismantling or reuse) in selecting the most cost-effective alternative from an array of applicable alternatives.

Direct costs should include also overhead, risk and benefit of the operation, which are indirect costs of the operation of construction.

p is a vector of parameters that describes the decision variables, which include aspects, such as the design criteria (geometry, material properties, external demands) and inspection and maintenance schedules.

It is important to distinguish between the end or service life, t_D and the time mission, t_m. The former is the time at which the system is taken out of service- i.e., actual life; while the time mission is the expected life of the system, which is a design parameter.

Direct costs may include also other costs such as those associated with the planning and pre-construction processes.

Eq. (33.2-2) is a discrete representation of all costs incurred throughout the life of the project. A continuous formulation may be possible if required for some parts or the entire equation.

Additional costs required for the system to operate as required may include:

- insurance coverage;
- quality assurance measurements;
- financial costs (e.g., finance charges such as loan interest payments);
- loss of business opportunity;
- direct and indirect losses in case of failure;
- loss of life.

The total cost of a project can then be computed as:

$$C_T(\boldsymbol{p}) = C_{T_d}(\boldsymbol{p}) + C_{T_i} \quad (33.2\text{-}1)$$

Where:

$C_{T_d}(\boldsymbol{p})$= total discounted direct costs; and

C_{T_i}= total discounted indirect costs;

$\boldsymbol{p}$ = vector parameter of system properties.

33.2.2 Direct Costs

33.2.2.1 Introduction

Direct costs are those directly borne by the owner during the entire life span of an asset. These are usually divided into four subcategories, namely:

i) design and construction costs,

ii) overhead, risk and benefit of the construction and operation;

iii) maintenance, inspection and operation costs, and

iv) end of service life costs.

For a fixed (deterministic) time history of interventions; and fixed end-of-service-life time, t_D, the total discounted direct cost, C_{T_d}, can be calculated as:

$$C_{T_d}(\boldsymbol{p}) = C_0(\boldsymbol{p}) + \sum_{n=1}^{N(t_D)} C_{m,n}(\boldsymbol{p})\delta(t_n) + C_D\delta(t_D) \quad (33.2\text{-}2)$$

where:

C_0	Design and construction costs (monetary unit);
$C_{m,n}$	Cost of the n^{th} preventive maintenance (monetary unit);
C_D	End-of-service-life costs –Decommissioning costs-(monetary unit);
$\delta(t)$	Discounting function;
t_n	time at which the n-th intervention occurs;
$N(t_D)$	the total number of interventions within time frame t_D;
t_D	end of service life of the system;

The discounting function $\delta(t)$ is calculated as $1/(1+r)^t$ or as $\exp(-rt)$ for small values of the discount rate r. The values of the discount rate vary depending upon the location of the project. Typical annual discounting rates in Europe and the US are between 2% and 4%; in middle-income countries, the rate varies between 5% and 12%. Larger values are only observed in other specific locations.

The design and construction costs are related with the processes of planning, designing and building the project. These costs vary depending on the nature of the project but commonly include materials cost, transportation and placement, cost of labour and equipment. These may also include the cost of risk transfer (insurance policies) and quality control strategies. Some of these costs are calculated directly (e.g., labour and material) but some are computed as a percentage of the total costs (e.g., insurance policy). The percentage and the value with respect to which that percentage is taken should be based on the market or on statistical data of the owner. All these costs are executed at the beginning of the life cycle; therefore, there is no need for discounting.

Additional terms can be added to Eq. (33.2-3), which are problem-dependent and dependent also on the level of approximation. These take into consideration unforeseen expenses at the time of design/construction which are normally calculated as a percentage of the direct constructions costs. The percentage and the value respect to which that percentage is taken should be based on the empirical/statistical data.

$\boldsymbol{p}$ vector parameter of system properties.

33.2.2.2 Design and construction costs

Design and construction costs are frequently computed as:

$$C_0(\boldsymbol{p}) = C_{0p} + C_{0f} + C_{0v}(\boldsymbol{p}) + C_{0e} \quad (33.2\text{-}3)$$

where:

C_{0p} Costs associated to the planning and design phases of the project.

C_{0f} Fixed component of construction costs.

$C_{0v}(\boldsymbol{p})$ Variable part of construction costs; which depends on the physical characteristics of the project $\boldsymbol{p}$;

C_{0e} Variable part of construction costs defined by other externalities.

33.2.2.3 Overhead, economic uncertainties and benefits of the construction and operation

Overheads can be considered the indirect costs of a company to perform a construction work.

There are several types of overheads that the companies will take into account. These overheads depend on the type of construction, the type of company and the country where the construction takes place.

Some type of overheads that are to be considered, but not limited to only these, are:

- Personnel and installations onsite that are necessary for the development of the work, like the site office, administration and management personnel.
- Installations and personnel of the central offices and management of the company.
- Costs associated with the risk of the operation and the finance of the construction.
- Costs associated with the financial needs for the operation of the company
- Assumed benefits for the construction and the operation of the construction company.
- Administrative permits for the construction of the works and general operation of the construction company.
- General taxes for the operation of the work site and the general taxes for the company. Taxes on the benefits of the company.

33.2.2.4 Operational, and inspection costs

Operational costs are the recurring costs required to keep the project functioning above a specific acceptable performance threshold defined to meet client's requirements. Operational costs are all investments made to guarantee the functionality of the project; for instance, in terms of safety for the user.

Inspections costs are related to the processes of data acquisition to decide whether an intervention is required or not. Inspections are frequently executed as a result of a predefined plan; however, extraordinary inspections may be required by an inspector, or if there are indications of malfunctioning.

33.2.2.5 Maintenance costs

While operational and inspection costs are usually well defined, maintenance costs are evaluated as follows:

$$C_m(\boldsymbol{p}) = C_{mf} + C_{mv}(\boldsymbol{p}) + C_{me} \quad (33.2\text{-}4)$$

Maintenance activities can be separated into preventive and corrective maintenance.

Planned periodic maintenance activities costs are modifications (interventions) made to the system to prevent the structure from failure. Preventive maintenance is

frequently the result of decay in a performance indicator related to a certain or various limit states. It is frequently executed in response to the output of an inspection result.

Unplanned reactive maintenance / repair activities are modifications (interventions) made to the system after failure -i.e. not compliance with a performance metrics. Corrective maintenance is frequently associated with the reconstruction of parts of a system after failure – defined in terms of some limit state. Corrective maintenance is also referred to as system rehabilitation.

In order to calculate periodic maintenance activities costs, maintenance scenarios that accurately describe the estimated required maintenance over the life cycle of an asset should identified. The European standard series EN 1504 [33-1] gives guidelines for the choice of repair materials and systems that are appropriate for rehabilitation and maintenance of concrete structures. European standard EN 1504 [33-1] describe the main points of rehabilitation of a damaged concrete structure:

- Assessment of the registered state of a concrete structure;
- Determination of the courses of damage;
- Determination of the objective of the rehabilitation of a damaged concrete structure;
- Choice of relevant principles for rehabilitation of a damaged concrete structure;
- Choice of methods for rehabilitation of a damaged concrete structure;
- Definition of the properties for repair materials and systems for rehabilitation of a damaged concrete structure or its members;
- Specification of requirements for the maintenance that should always follow rehabilitation of a damaged concrete structure or its members.

According to EN1504-9 [33-2], six different maintenance strategies are possible. Table 33.2-1 indicates the strategies that should be evaluated to decide upon the right maintenance for the structure. It should be emphasized that more than one strategy may be relevant and that each structural part should be given a separate evaluation. Selection of right maintenance strategy and repair method should at least be based on owners' requirements, as built documentation, evaluation of the bearing capacity, life-cycle cost analyses and a detailed condition assessment.

Table 33.2-1*: Maintenance strategies.*

where:

C_{mf} fix component of the intervention; it does not depend on the size of the intervention;

$C_{mv}(\boldsymbol{p})$ variable part of maintenance costs defined by the extent of the intervention;

C_{me} variable part of maintenance costs defined by other externalities.

The maintenance yearly costs through the life cycle of the system gives the total nominal maintenance costs. Since the maintenance costs occurs at different time periods over the lifespan of the system, the future cash flows should be discounted to establish a present value. The total discounted maintenance costs may be increased by a certain percentage that cover the *unassigned costs*, *indirect costs* and *unassigned object risks* (but not *engineering costs* and *other additional costs*), if that is the owner's practice.

The total maintenance costs for the object during its service life is therefore calculated by:

$$MC_{tot} = \sum_{t=0}^{T} \frac{C_{m.t}}{(1+r)^t} \times (1+\chi) \qquad (33.2\text{-}5)$$

where:

MC_{tot} the total maintenance costs during the life cycle of the system (monetary unit);

$C_{m,t}$ maintenance costs for year t (monetary unit);

t year in life cycle from 0 until end of life cycle T;

r the discount rate;

x an additional percentage to cover unassigned, indirect, engineering and other costs (this is related to the agency or contractor practice and should be determined individually).

No	**Options for protection and repair**
1	Do nothing for a certain time Postpone the repair work, but monitor the degradation process
2	Re-analysis of structural capacity, possibly leading to downgrading of the function of the concrete structure
3	Prevention or reduction of further deterioration, without improvement of the concrete structure
4	Improvement, strengthening or refurbishment of all or part of the concrete structure
5	Reconstruction of part or all of the concrete structure
6	Demolition of all or part of the concrete structure

Monetarized user costs are used as a surrogate measure of the total time lost by the users due to the unavailability of the functionality of the facility. There is not a

33.2.2.6 End of service life costs

The end-of-service-life costs, C_D, include costs of demolition and disposal minus the residual value or reuse. Normally, the end-of-service life costs will be calculated in the similar way that the initial construction costs such as material and labour costs associated with demolition.

Because the end of service costs take place at the end of the life cycle of the asset, the costs should be discounted to obtain net present value.

33.2.3 Indirect costs

33.2.3.1 Introduction

Indirect costs are those which society incurs as a result of the construction, operation, interventions or decommissioning of the project.

When evaluating the three pillars of sustainability sometimes it is possible to evaluate the environmental aspects and the social aspects in economic parameters. Normally both environmental and social aspects are much more than an economic

unique mathematical formulation for these costs and normally infrastructure owners have their own model for estimating user costs related to delays.

Although there is an important debate, in most cases environmental costs incurred during the life-cycle are not discounted as recommended by (Hellweg et al. [33-3]).

evaluation and they should be considered in other ways, like material consumption, energy, pollution on disturbances to society.

When considering some these aspects in an economic way the designer should not forget that they have also to be considered in the proper evaluation of environmental and social aspects of sustainability. The designer should also consider, when taking into account the global sustainability of a project, that maybe some of these aspects are considered in the economic pillar and in the other pillars at the same time.

Discounted indirect costs, mentioned in Eq. (33.2-1):

$$C_{T_i} = \sum_{u=1}^{N(t_D)} C_{i,u}\delta(t_u) \tag{33.2-6}$$

where:

$C_{i,u}$ indirect costs incurred at time u (monetary unit);

$\delta(t)$ discounting function;

$N(t_D)$ the total number of indirect costs recorded at time t_D;

t_D end of service life of the system.

Indirect costs, $C_{i,u}$, depends on the type and extent of the evaluation. Although these include a wide range of features; the most commons are:

i) user costs,

ii) societal costs, and

iii) environmental costs.

33.2.3.2 User costs

User costs are those incurred by the users of the system resulting from restrictions in the system's operation. In order to be included as part of the life-cycle cost assessment, these are commonly monetarized. User costs may include:

- Time delays or loss of activity due to maintenance, rehabilitation, repair activities;
- Costs of accidents due to poor safety management;

– Time delays or loss of activity due to saturation of the system (e.g., congestion);
– Loss or gain in property values by the construction or operation a nearby facility.

33.2.3.3 Societal costs

In a highly generalised formulation, societal costs are costs borne by society at large. For instance, the cost associated to the preservation of cultural values and the protection of their values, beliefs, arts, modes of perception, habits of thought and activity, in their natural and cultural conditions. Societal costs are difficult to value in monetary terms.

33.2.3.4 Environmental costs

Environmental costs can be divided into:

i) impact or damage to the project's surrounding area; and
ii) contribution to global environmental problems (e.g., climate change).

The former is related to the impact on, for example, surrounding water sources or flora and fauna. The latter includes the impact on climate change such as emissions to the environment.

The environmental costs are the costs caused by use of energy and resources during design, construction, operation, maintenance and end-of-life phase of the system and the accompanying emissions to the environment.

There is plentiful industrial life cycle inventory data on energy supply, resource extraction, material supply, chemicals, metals, waste management services, and transport services which can be translated into environmental performance indicators and eventually into costs.

For better understanding and comparison to economy category, the translation of environmental effects into costs is preferable. Using the 'revealed collective preference method' the environmental costs, C_E, can then be determined as:

$$C_E = \sum_{i=n}^{m} EE_i \times SP_i \qquad (33.2\text{-}7)$$

where:

EEi environmental effects for impact category *i* (kg of impact category equivalent (ICeq));

SPi the shadow price for environmental effect category *i* (m u/kg of ICeq).

The environmental effects per impact category can be determined as:

$$EE_i = \sum_{j=n}^{m} EE_{i,j} \times Mq_j \quad (33.2\text{-}8)$$

where:

$EE_{i,j}$ environmental effect for impact category *i* per kg of material *j* (kg ICeq/kg material);

M_{qj} material quantity per functional unit for material *j* (kg)

33.3 Methodology of LCCA

Despite the differences among existing LCC methodologies, the following steps can be identified as general in any method of LCC analysis:

- Definition of the project objective and scope;
- Establishing minimum performance requirements and restrictions;
- Identification of alternative options;
- Assumptions and parameters;
- Definition of the LCC model;
- Cost and time database;
- LCC Evaluation;
- LCC results and interpretation: identification of key cost drivers;
- LCC Sensitivity, uncertainty and risk analysis.

Definition of the project objective and scope

To be successful, an LCC study must have clear objectives, and it must define clearly what is and what is not included in the analysis. This is particularly important given that the analysis incorporates many aspects that are different in nature.

LCCA can be performed over a single element or a structure as a whole. In the latter, structure have to be clearly identified and defined. This is frequently carried out by establishing an inventory; this is a database where information associated with the assets is held, to provide accurate and up-to-date data to run LCC evaluation.

Identification of alternative options

The set of feasible project scenarios, whose life-cycle cost will be calculated, should be clearly defined. If, for example, the LCCA is to be carried out on an existing asset for future budgeting purposes, then there may be no need to identify alternative options. If, however, the analysis is being carried out to identify the best option from a range of potential options, then it is important that the appropriate options are identified and approved by the client prior to detailed cost and data being collected.

Assumptions and parameters

This includes choosing of the parameter to compare alternative options (normally Net Present Value of the structure), base time, study period (reflects investor's time horizon), general inflation rate, discount rate.

Once the reference framework has been defined, a strategy is needed to compare alternative options. Normally, the financial analysis is based on the expected Net Present Value (NPV). This analysis includes the selection of specific parameters such as the cost structure, the investor's time horizon and both inflation and discount rates.

In LCCA the parameter discount rate is of particular importance. It is used to convert cash flows occurring at different time to a common time to reflect the time value of money (see ISO 15686-5 [33-4]).

The parameter “time horizon” is the period of time over which life cycle cost is to be evaluated. The length of the analysis period is dependent on the nature of the project and the components taken into account.

The “life span” of a component is one of the key input parameters when performing LCC analysis. Following the Davis Langdon LCC Common Methodology Report (Langdon [33-5]), the life span of an asset can be defined in a number of ways.

Build LCC model

Different categories of LCC models are available depending on the degree of complexity.

Different levels and Equations related to the different levels as well as a summary table with advantages and disadvantages of each level may be found in XXXXBulletin.

Build cost data base and time data base

This includes the timing of all relevant costs, timing of each cost as it is expected to occur during the study period. In this step, the user identifies all relevant costs and timescales for each option identified above, in preparation for carrying out the analysis. The objectives are to make the best estimate for the value of each cost and identify the time related data for each cost.

The best LCCA alternative is obtained as the solution of an optimization problem with the objective to minimize an objective function with the required constraints.

The objective function can be written following equation (33.2-1).

LCC results and interpretation: identify key cost drivers

The output from the LCCA should cover the costs, the future interventions as well as the performance of the structure over the planned lifetime. Costs are forecasted either as deterministic or probabilistic values. When LCC analyses consider a breakdown of costs (e.g. agency costs, user costs, etc.), the output of such analyses provide also a breakdown of costs. Decisions at this level cannot be based only on the results of the model; they require an understanding of the context and the stakeholders interests. It will often be necessary for the LCC report to draw out the key findings of the analysis in clear terms.

Perform sensitivity analysis and uncertainty/risk analysis

Sensitivity analysis measures the impact on project outcomes of changing key input values about which there is uncertainty, typically:

- discount rate;
- future inflation assumptions;
- period of analysis;
- service life or maintenance, repair or replacement cycles;
- operational cost data.

References

[33-1] EN 1504. Products and systems for the protection and repair of concrete structures.

[33-2] EN 1504-9:2008 Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 9: General principles for the use of products and systems

[33-3] Hellweg, S., Hofstetter, T. B., Hungerbuhler, K. Discounting and the environment should current impacts be weighted differently than impacts harming future generations? The International Journal of Life Cycle Assessment, 8(1), 8–18, 2003.

[33-4] ISO 15686-5: 2017. Buildings and constructed assets – Service life planning – Part 5: Life-cycle costing.

[33-5] Langdon, D. Life Cycle Costing (LCC) as a Contribution to Sustainable Construction: A Common Methodology. Davis Langdon Management Consulting, 2007.

In case of an indicator or concept is susceptible to affect more than one pillar simultaneously (ex., emission taxes in the economic pillar and CO_2-eq emission in the environmental pillar), the decision-maker should notice that overlapping can occur if the results of each independent pillar are combined to assess sustainability from a global point of view.

When combining the three pillars the designer has to take into account that many indicators influence more than one performance. Some indicators may influence the performance of different pillars. In these situations, the stakeholders shall identify and establish those indicators that are representative while avoiding overlapping. One specific aspect can influence all performances, social, environmental and economic. It may also happen that one aspect influences two or just one.

Therefore, if an evaluation is done by combining the three pillars, it is required to consider a consistent multi-criteria procedure capable of guaranteeing transparency and, unambiguity of the combination and the results. and should be explicit from the beginning and should consider clearly the importance of each aspect of the three pillars of sustainability as desired by the administration or the final users. The validity and objectivity of the results are based upon those premises, which are the foundations of desired decisions to be made by administrations and users.

Economic cost is often the variable used for assessing sustainability, even for those indicators with different units.

Administrations, policy-makers, technical-scientific committees and other stakeholders have proposed conversion factors to be used for this purpose. These conversion factors can be modified and re-calibrated periodically and its magnitudes could vary according to different national strategies, economic situation, cultural aspects and sensitivities to environmental and social requirements.

34 Sustainability evaluation

34.1 Introduction

Sustainability evaluation embraces the assessment of the three pillars of sustainability, social, environmental and economic as a whole.

Each one of the pillars measures a different aspect of sustainability and the criteria and indicators that compose these pillars are generally not combinable. Therefore it is not possible to combine them in a direct way.

For the sustainability assessment of structures, stakeholders and decision-makers are the bodies that take the leading role in selecting the criteria and the relative importance of them that are used to organise the results derived from each pillar to estimate a global sustainability with a combination of the three pillars. Normally this combination is not explicit beforehand, and the combination of the factors follows different rules in each individual case.

The selection of the criteria is more important than the relative importance of each parameter in the comparison. The selection of criteria es specific for each type of structure and for each localisation.

Administrations normally have economic limitations but also value specific aspects or ambitions of the project. Normally they will give more preference to solutions that add more value to their requirements if they are in the economic range that is affordable to them. Administrations can be restricted by economic limitations but these should value other requirements and ambitions of the project by properly considering the other pillars.

A consistent multi-criteria decision-making approach oriented to assess the sustainability of structural elements and to assist stakeholders in the decision-making process could consist of the following steps:

1. To define the decision to be made (ex., the most sustainable structural typology for constructing a structure) and to stablish the technical (ex., span-length and loads) and particular requirements (ex., maximum construction-time) to be met;
2. To establish a decision-making diagram (or tree) that includes all those criteria and indicators representative of the three pillars of the sustainability and representative of the decision to be made;
3. Establish a system that allows normalizing the units of each of the indicators by considering an accepted approach for this purpose (ex., multi-attribute theory, value function concept);
4. To identify and design those alternatives (ex., precast prestressed concrete girders; posttensioned concrete slab, among others) that met those requirements established in point 1;
5. Assess the economic, environmental and social performance of each alternative by applying the model defined in points 2-3. A sustainability performance can be obtained for each alternative by combining the performance of the three pillars. This process must be made by means of an accepted approach that guarantees transparency and objectiveness;
6. To make the decision based on the results obtained in point 5.
7.

The *fib* Bulletin 88 [34-1] gathers several approaches for assessing the sustainability of precast concrete products and technologies that can be applied to any type of concrete element. Likewise, some real examples focused on the assessment of sustainability by using a multi-criteria decision-making model.

34.2 Evaluation

There are several available approaches oriented to assess sustainability by combining the three pillars. Some of these tools might dismiss the explicit consideration of the ISO regulations for the evaluation of the performance according to the three pillars.

It is recommended to be very careful with the use of tools that do not have explicit measurements linked to the performance of the three pillars of sustainability as described in the chapters of this document.

The evaluation of the three pillars should be performed as described in the following chapters:

- Social Performance. Chapter 31;
- Environmental Performance. Chapter 32;
- Economic Performance. Chapter 33;

Should the pillars be combined to derive a global sustainability that permits to make decisions, a transparent and consistent procedure has to be followed to establish objective performance for each of the three pillars and the combination of them.

It is important to clarify to the designers, the administrations and the final users that the way to combine the different aspects is as important as the evaluation of the performance of them individually and can have an important impact in the final solution.

References

[34-1] *fib* Bulletin 88. Sustainability of precast structures. State oft he art report. 2018.

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Guidance for the development of health and safety systems can be found in: ISO 45001 [35-2].

Guidance for the development of systems for environmental protection can be found in ISO 13315-1 [35-3], in ISO 14001 [35-4], Section 3 and Section 32 of this Model Code.

These assumptions correspond with the assumptions of ISO 22966 [35-1].

35 Execution management

35.1 General Principles

The execution of concrete works at the construction site shall comply with the requirements given in ISO 22966 [35-1] or a technical specification that fulfils the minimum requirements of this standard.

In the following subsections, key-elements from ISO 22966 [35-1] are given together with some amendments/ additions.

The “execution specification” is the total sum of documents required for the execution of the works as provided by the designer to the constructor.

It includes ISO 22966 [35-1] and the “project specification” prepared to supplement and qualify the requirements of ISO 22966 [35-1] for a particular project, as well as referring to mandatory local provisions relevant in the place of use.

Annex A of ISO 22966 [35-1] gives an overview of issues to be included in the “project specification” .

The execution of concrete works at the construction site shall comply with governing codes and regulations regarding health and safety. Additionally agreed requirements may apply.

The execution of concrete works at the construction site shall comply with governing codes and regulations regarding protection of the environment. Additionally agreed requirements may apply. Key aspects that should be considered within environmental performance are: (a) (re)use of resources, (b) use of energy, (c) use of land and water, (d) emissions to air, water and soil, (e) noise and vibration, (f) circularity/ waste production, (g) species and ecosystem, (h) landscape and history.

35.2 Assumptions

The following assumptions apply for this chapter:

This Model Code assumes that there is a site management which will take charge of the organization of the works and enable the correct and safe use of the equipment and machinery, the required quality of materials, the execution of a conforming structure and its safe use up to the delivery of the works.

This Model Code presupposes that the work is carried out with the necessary skill and adequate equipment and resources to perform the work in accordance with ISO 22966 [35-1] , the requirements of the execution specification, and formally required certificates and approvals.

This Model Code presupposes that Health, Safety and Environmental requirements for construction are specified.

35.3 Documentation

The execution specification is a document, as defined in ISO 22966 [35-1], that contains all drawings, technical data and requirements necessary for the execution (of any part) of the works.

Before the start of execution of any part of the works, the execution specification relevant to that part of the works shall be complete and available.

The following items should be included in the execution specification:

- a reference to ISO 22966 [35-1] and, if published, its national annex;
- a reference to other relevant international standards and national technical approvals;
- a reference to relevant national regulations and standards;
- a project specification giving information and requirements for the particular project prepared to supplement and qualify the requirements of the above listed documents;
- models/drawings and other technical documents needed for every phase of the execution;
- an interface document from design giving information to construction of aspects which do require special attention in aiming to assure health, safety and environmental care, as well as a global method statement as considered by design;
- a quality management plan which aims to prevent non conformities and to ensure that the works will perform as required;
- a safety plan which aims to prevent accidents to workers and the public during construction;

"As-built" documentation of the direct input parameters to the service life design models must include the design assumptions or, in case of non-conformities, give the applied corrective measures and consequences thereof for the service life design. '' It might also serve as a basis for the condition assessments of the structure during its service life. Such extracts of the "as-built" documentation is sometimes named the structure's "birth certificate".

Guidance for the development of systems for quality assurance and control can be found in ISO 9001 [35-5].

ISO 9001 [35-5] specifies 8 quality management principles: customer focus, leadership, involvement of people, process approach, systems approach to management, continual improvement, factual approach to decision making and mutually beneficial supplier relationships.

Guidance for the development of Risk Management Principles and Guidelines can be found in ISO 45001 [35-2].

Based on the risk analysis, it could be appropriate to specify inspections by members of the design team at designated hold points.

The three execution classes give the option to specify the required level of quality management based on the importance of the component/structure and the criticality of the execution for its ability to fulfil its function.

- an environmental protection plan which aims to prevent the adverse impact of construction on the environment;
- a data management plan which aims to assure the collection of relevant data for the birth certificate;
- a contingency plan which aims to avoid adverse effects of non-conformities.

In execution class 2 and 3 (see section 35.4), an inspection report for the materials and products brought to the construction site as well of the completed works is required.

35.4 Quality management

Execution management at the construction site must assure and control compliance of quality of the works with the contract documents, governing codes and regulations, and the execution specification.

Quality management forms an integrated part of the reliability differentiation for the project, see also section 11.3.1.

Quality management aims to prevent non- conformities and to ensure the works will perform as required.

Prior to the development of the quality management plan, a risk analysis and evaluation should be performed ,to identify the risks, required assurance measures and/ or mitigation measures. These measures should be included in the quality management plan. Upon completion of the works, an interface document should be prepared to support life cycle management (see chapter 5) during the service life of the works

Requirements for quality management are specified using one of the following three classes, for which the required strictness increases from class 1 to class 3:

Execution class 1 should only be used for structures where consequences in case of failure are small or negligible.

The execution classes comprise requirements for inspection and, depending on the relevant national annex or the execution specification, requirements for quality planning focusing on organizational measures and allocation of resources and personnel.

Supervision and inspection are parts of the quality management.

The types of inspection are normally differentiated in three categories:

– self-inspection by the operator;
– inspection in accordance with the procedures of the constructor;
– possible inspection by another company, that is an independent inspection.

If the inspection of the completed works reveals that the original design assumptions are not met, a partial or full re-design may be needed to get an overview of the possible consequences for the future performance of the structure. The new calculation should be supplemented with data from the inspection.

An example of such a situation may be if the inspection document mentions a too small concrete cover of the reinforcement. The actual cover should then:

– be implemented in a recalculation of the service life and form the basis for conclusions regarding possible rectifications;
– be judged to asses the ability of proper anchorage of the reinforcement.

Updates from the inspection should also result in updates of the “maintenance plan” and the “inspection plan” for the structure, if relevant.

– execution class 1;
– execution class 2;
– execution class 3.

The execution class may refer to the complete structure, to components of the structure or to certain materials/technologies used for the execution.

For all of them, the execution class being used shall be stated in the execution specification.

The needed inspection to perform a conformity evaluation of the completed work shall be carried out and the results documented.

The extent of inspections to be applied shall be according to national regulations and must be stated in the execution specification by the selection of the appropriate execution class.

Where inspection reveals a non-conformity, appropriate action shall be taken to ensure that the structure will perform as required

The following aspects shall be investigated in the listed order:

– the implications of the non-conformity on further execution and fitness for intended design purpose;
– the measures necessary to make the component acceptable;
– the necessity of rejection and replacement of the non-repairable component.

Before commencement of execution of any part of the works, management shall ensure that available workforce is appropriately trained, certified and has adequate experience to perform the tasks. If needed, additional assurance measures shall be taken to adequately counterbalance the potential risk from limited experience or training.

35.5 Health and Safety

35.5.1 Introduction

Guidance for the development of health and safety systems can be found in: ISO 45001 [35-2].

Table 35.5-1: Chapter reference for ISO 45001 and ISO 45002-2018

ISO 45002 - 2018	ISO 45001-2018	Overview
A 1	1	Specifies the requirements for an OH&S System. Summary of what the standard intends including outcomes.
A 4.1	4.1	Understanding the context of the organisation including internal / external factors that can be positive / negative
A 4.2	4.2	Understanding the needs / expectations of interested parties / workers
A 6.1	6.1	The requirements related to controlling risks and identifying opportunities.
A 8.2	8.2	Overview of requirements related to eliminating hazards and reducing risks.

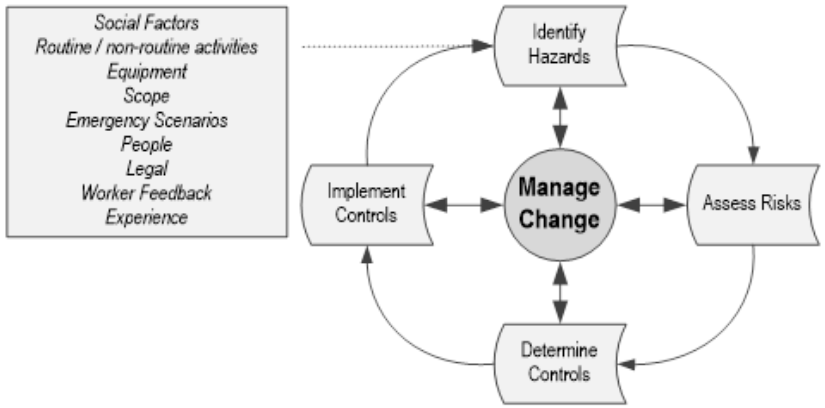

Figure: 35.5-1: Risk assessment process.

The purpose of the Health and Safety procedure is to ensure that hazards are identified, the associated risks / opportunities are assessed and the required control measures are implemented / monitored for all activities. It is also critical that social factors are assessed when identifying hazards – this will include bullying, leadership, hours worked, distance and time to travel to and from the workplace etc.

35.5.2 The Risk Assessment Process

Step 1 – Identify Hazards

The following should be considered when identifying hazards - i.e., those aspects of work (for example substances or equipment used, work processes, or work organization) which have the potential to cause harm:

- routine / non-routine activities;
- activities of all persons having access to the workplace;
- human behaviour, capabilities and other human factors (Social including workload / bullying / leadership / culture etc.;
- identified hazards originating outside the workplace capable of adversely affecting the health and safety of persons employed by the Contractor and sub- Contractors;
- hazards created in the vicinity of the workplace by work related activities under the control of the Contractor, including sub-Contractors;
- potential external interferences (access areas, etc.)
- infrastructure, equipment and material at the workplace whether provided by the Contractor or others;

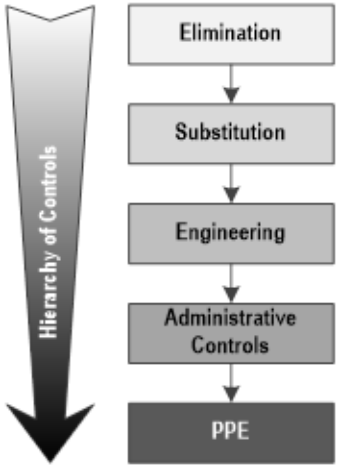

Figure: 35.5-2: Hierarchy of controls in risk assessment process.

– changes or proposed changes by Client, Contractor, or others involved.
– legal Requirements.

Step 2 – Assess Risks

Once the hazard is identified, the probability that the hazardous event may occur plus who may be exposed is assessed. A hazard may have more than one associated risk.

Step 3 – Determine Controls

This is the main element of the assessment where the hierarchy of controls should be considered. The section of the risk assessment should contain text detailing what is required to reduce the risk level to an acceptable level. The control measures should be described in a clear and unambiguous language. It should principally contain what actions the Project team are going to implement during the Construction Process. Control measures need to be communicated to all persons carrying out the activity with a focus on management / supervisors / engineers.

The approved controls shall be implemented for the identified hazard. All personnel, including workers / supervisors / project management are required to ensure the controls are in place and monitored.

Control measures should be realistic and achievable with inputs from those experienced in the activity being undertaken. The most important factor when determining controls is the hierarchy of risk management (see figure 35.5-2).

Following controls can be applied:

Elimination

Risks may be ‘eliminated’ by redesigning a job sequence and removing a hazardous step. This may take place at design stage, for either permanent or temporary works.

Substitution

Risks can be reduced through, for example, the ‘substitution’ of hazardous materials / substances for less hazardous materials / substances. This may take place at design stage for either permanent / temporary works.

Typical examples of engineering controls include barriers or guarding of machinery. The use of local exhaust ventilation in a workshop is another example of engineering controls. In construction, the use of engineering controls is a major factor in the development of a safe place of work. Much construction equipment has built in engineering controls such as automatic cutoff switches, interlocks etc.

Section 1 highlights the risks associated with the activity plus the details of consequences, i.e. – what can be the further risks.

Engineering

Engineering Controls are an effective means of managing risks. Engineering controls use physical measures to separate workers from hazards. This may take place at design stage for either permanent / temporary works.

Administrative Controls

Administrative Controls include the development of procedures, method statements, risk assessments etc. (Operational Controls).

Administrative controls also include training. Administrative controls can be effective as workers become more aware of hazards and more competent in carrying out their duties in a safe manner. Administrative controls may also include work scheduling. For example, reducing shift times may reduce workers exposure to hazards.

PPE (Personal Protective Equipment)

PPE is the final element in the hierarchy of risk management. PPE should be used where no 'reasonably practicable' alternative is available. PPE will include general personal safety equipment and RPE (Respiratory Protective Equipment).

Monitor and Review

The Project Manager and his supporting project team (construction managers, engineers, foremen, supervisors etc) shall monitor compliance with the planned arrangements and intervene where unsafe conditions / behaviours exist. In addition, the onsite HSS (Health, Safety and Security) team should provide feedback / improvement recommendations.

35.5.3 Risk Assessment Template

The Risk Assessment template, provides an appropriate tool for Risk Assessments, unless alternative requirements are required by the Client and / or legal requirements exist

Section 2 highlights who or what may be affected. Several options within this section may apply at the same time.

Section 3 assesses the risk rating prior to controls being in place. See section 6 below for further information.

Section 4 is the critical element of the risk assessment process. This section details the controls that shall be implemented. Where required, projects / areas may add to the control measures but are not permitted to remove any section from the approved list.

Section 5 highlights what are the permits required. One or more options may be ticked within this section. Permits not listed can be added as applicable.

Section 6 highlights who are the responsible and accountable persons for the implementation and supervision of the control measures that have been identified.

Section 7 details the internal / external training requirements associated with the risk assessment and essentially form part of the control measures included in section 4.

Section 8 details the Personal Protective Requirements

Section 9 is the assessment of the residual risk further to the control measures being implemented. Where the risk rating is not acceptable, additional controls may be required. Where the residual risk rating is 8 or above (Orange), the works / activity is not permitted to take place. In that case further review is required.

Section 10 provides a reference to additional risk assessments plus Project Specific Documents.

Each risk assessment is a ‘’stand-alone’’ document / record which has the following sections:

- Section 1 – Risks
- Section 2 – Who / What may be affected
- Section 3 – Initial Risk Rating
- Section 4 – Controls
- Section 5 – Required Permits
- Section 6 – Responsibilities and accountabilities
- Section 7 – Training Requirements
- Section 8 – PPE Requirements
- Section 9 – Residual Risk Rating (Further to the implementation of Controls)
- Section 10 –References (Including Risk Assessments)
- Section 11 - Opportunities

Section 11 provides the tool to identify / add opportunities further to the assessment, i.e. a circumstance or set of circumstances that can lead to an improvement in OH&S (Operational Health & Safety) Performance.

Severity X Likelihood – Risk Rating (RR)		
	1	Insignificant
	2 to 3	Medium Low
	4 to 6	Medium
	8 to 12	Medium High
	15 to 25	High

Likelihood of the incident occurring	Severity				
	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15
4	4	8	12	16	20
5	5	10	15	20	25

Figure 35.5-3: Matrix for severity – likelihood impact scoring.

35.5.4 Assessment of risk rating

When calculating the risk / impact both the likelihood and severity are considered. The matrix, figure 35.5-3, details how the assessment takes place. How to assess the Risk Rating is detailed in figure 35.5-4. Where the rating is rated at Medium High or High, additional controls are required to reduce the risk prior to the works taking place. Where the risk is rated as Medium or Medium Low, the works may proceed.

Figure 35.5-4:*Risk Rating Prior to mobilizing on site, a list of required risk assessments is compiled covering all potential hazards. The risk assessments should be discussed in detail at the Project Mobilization Meeting. During the development of method statements, the controls detailed within the risk assessments are the minimum mandatory requirements.*

Likelihood of Risk Occurring		Severity of Risk
Highly unlikely but maybe an occurrence in exceptional circumstances. It could happen but probably never will.	1	Causing minor injury (scratch etc)
Not expected but there's a slight possibility there may be an occurrence at some time.	2	Potentially causing injuries that may be a Category 4 (First Aid) or a Cat 6 (near miss)
The event might occur at some time, medium probability.	3	Causing significant injury (Stitches, Sprain, laceration with the potential for a Cat 2,3 or 5 incident
There is a strong possibility of occurrence	4	The potential to cause a lost time / serious injury and / or a high potential near miss.
Very likely, high chance of occurrence	5	The potential to cause a Cat 1 (Fatality / permanent disability) or an incident of greater damage than 50,000 euro

35.5.5 Method Statements

This section details Health and Safety requirements for management of method statements. A list of method statements are a topic for discussion in the Project premobilization safety meeting. This list is then included in the appendixes of the Specific Project Safety Plan.

For all Method Statements, a Risk Assessment is a mandatory requirement. It is permitted that additional controls may be added to risk assessments, however under no circumstances are Projects permitted to remove controls from the approved list of risk assessments.

If there is a slight deviation from the approved methodology, the method statement originator may introduce the method statement amendment template. The Method Statement originator is then required to ensure that the updated requirements are communicated / implemented.

Method Statements, in addition to client / legal requirements, should be drafted / approved under the following headings:

1) Introduction
    - Scope of Work
        - SIMOPS (Simultaneous Operations)
    - Anticipated Duration
    - Subcontractor Information if applicable
2) Reference Documents
    - Specifications
    - Operational Controls
    - Quality
    - Engineering
    - Etc.
3) Sequence of Work
    - Step by step overview of the activity(s) involved in the process
4) Plant / Equipment
    - List of ALL plant / equipment including static / mobile involved in the activity
5) Temporary Works
    - List of temporary works for the activity.
6) Responsibilities

- Clear list of who is involved in the activity plus their accountabilities / responsibilities including appointed persons such as;
  - Temporary works coordinator
  - Lifting Coordinator
  - Etc.

7) Occupational Safety
   - Risk assessment list
   - Training Requirements (Competencies)
   - Permits required
   - Emergency Response
8) QA/QC
9) Environmental
10) Sustainability
11) Appendixes
   - Risk Assessments
   - ITP
   - Activity Plan (Summary of the method statement with an emphasis on sequence of works / risks)
   - Other

Wherever possible, method statements will incorporate a full sequence of works.

35.6 Durability planning – execution aspects

35.6.1 Introduction

To achieve the specified design service life or extended service life objectives, it is necessary to incorporate durability considerations into the design, construction / execution processes and activities, along with those concerned with operational maintenance aspects, to avoid premature deterioration of concrete structures. This process is commonly referred to as *Durability planning*.

Durability verification (presented in a formal report) shall be provided prior to completion of execution / construction - refer Section 27.6.8. Compliance with the execution requirements shall be evaluated with respect to the project specifications

Under performance-based design, the objective is to create a concrete structure which has the ability to fulfil the performance requirements for the specified design service life at the required level of reliability.

The overall durability planning approach with its various facets and the associated component, parts helps deliver that objective - refer Section 27.6. The durability planning processes and inputs are described more fully in Bulletin XY: [35-6].

The durability planning inputs to achieve the design service life or extended service life are described in *fib* Bulletin XY [35-6]. Other contemporary guidance on durability planning also includes CIA Z7-01: *Durability Planning* [35-7].

The contractor tender stage Durability Assessment Report - the DAR: CT - is significantly influenced by the project contract delivery process and might include:

- Macro exposures allowed for (based on the contract document).
- Characteristics of micro environments experienced on the structures.
- Additional exposure information that will be gathered post- contract award and the potential implications for durability and project costs.
- Durability aspects of the owner's conceptual design (if issued by the owner) and any contractor proposed changes, plus any variations / enhancements of the SWTC (Scope of Works and Technical Criteria) durability requirements.
- Materials that will be used to achieve the required design service life.
- Approach to assessing optimum service life design and life-cycle costs.
- How the required condition and performance at the end of the design service life / any proposed extension to the design life will be achieved.
- Approach to assessing existing structures within the contract works.
- Potential materials related construction difficulties.
- How durability will be evaluated and the experience of those involved.
- How durability requirements will be implemented during construction and the in-use phase including provisions for monitoring performance.
- Cost to be allocated for durability management and input.

and drawings and the durability design intent as defined in the Durability Assessment Report: Detailed Design (DAR: DD) - refer Section 27.6.7.

Further details upon the overall durability planning approach, its various facets and the associated component parts are given in Section 27.6.

The following sections (35.6.2 and 35.6.3) present those aspects of the *Durability planning* activities which focus upon the execution / construction phase of seeking to ensure adequate durability is achieved for the concrete structure concerned.

35.6.2 Contractor project tender durability approach

The project tender Durability Assessment Report: Contractor Tender (DAR: CT) will provide the contractor's durability approach, which will be included as minimum requirements in the contract award documents.

The contractor's tender Durability Plan Report (DAR: CT) should demonstrate that the contractor understands the project durability requirements, exposure conditions, mechanisms of deterioration, any issues associated with the proposed construction materials, workmanship requirements and life-cycle management / maintenance activities needed after construction is completed.

The Durability Assessment Report: Contractor's Tender (DAR: CT) may include a Durability Assessment Report: Conceptual Design (DAR: CD) - refer Section 27.6.2 - where this is required for contractor early engagement delivery projects.

The execution / construction durability review requires ongoing expert durability input during construction in accordance with the Durability Assessment Report: Detailed Design (DAR: DD). For some tasks it may be appropriate to require that key personnel are named.

It is expected that the contractor would have an ongoing interaction / support relationship with the project durability expert / consultant.

Training might include workshops, tool box talks, etc. to ensure all staff have appropriate expertise to achieve the required construction quality.

Additional exposure assessment may be required, for example involving further ground/groundwater sampling (eg. chemical composition and sulfate reducing bacteria count), assessment of chloride ingress in adjacent structures, measurement of CO_2 levels, modelling of the influence of internal air CO_2 content / RH, etc. It may also include consideration of potential future changes in exposure conditions.

Site trials may be required to prove concrete mix stability, penetrability performance (eg. for chlorides, carbon dioxide), the development of heat of hydration and/or the ease of concrete placement and compaction.

35.6.3 Execution / construction durability approach

Perform reviews to ensure the durability design intent defined in the Durability Assessment Report: Detailed Design (DAR: DD) - refer section 27.6.7 - will be achieved by the contractor's execution / construction procedures for the project.

The issues which the contractor procedures should address include:

- A construction team with appropriate experience to achieve the specified durability requirements.
- Training requirements for the construction team to achieve the required construction quality.
- The means by which any additional exposure condition information required will be provided as the project is progressed after award.
- Procedures to assess whether materials provide acceptable durability.
- How to develop and implement a suitable quality management program to verify the durability criteria are achieved.
- How site trials will prove materials and construction methods meet the defined durability requirements, where these are not previously proven.
- The contractor's approach to:
    - Resolving durability issues arising out of construction difficulties.
    - Eradicating non-compliances which may affect durability.
    - Verifying that acceptable durability has been achieved prior to issue of a practical completion certificate for the works.
    - Implementing control and report inspection, monitoring and testing programs during construction to gather data to establish whether the measures required to deliver the durability design intent were achieved.

Figure 35.6-1 presents a flow chart of the processes intended to deliver durability during construction, this guidance is generally applicable to the construction process.

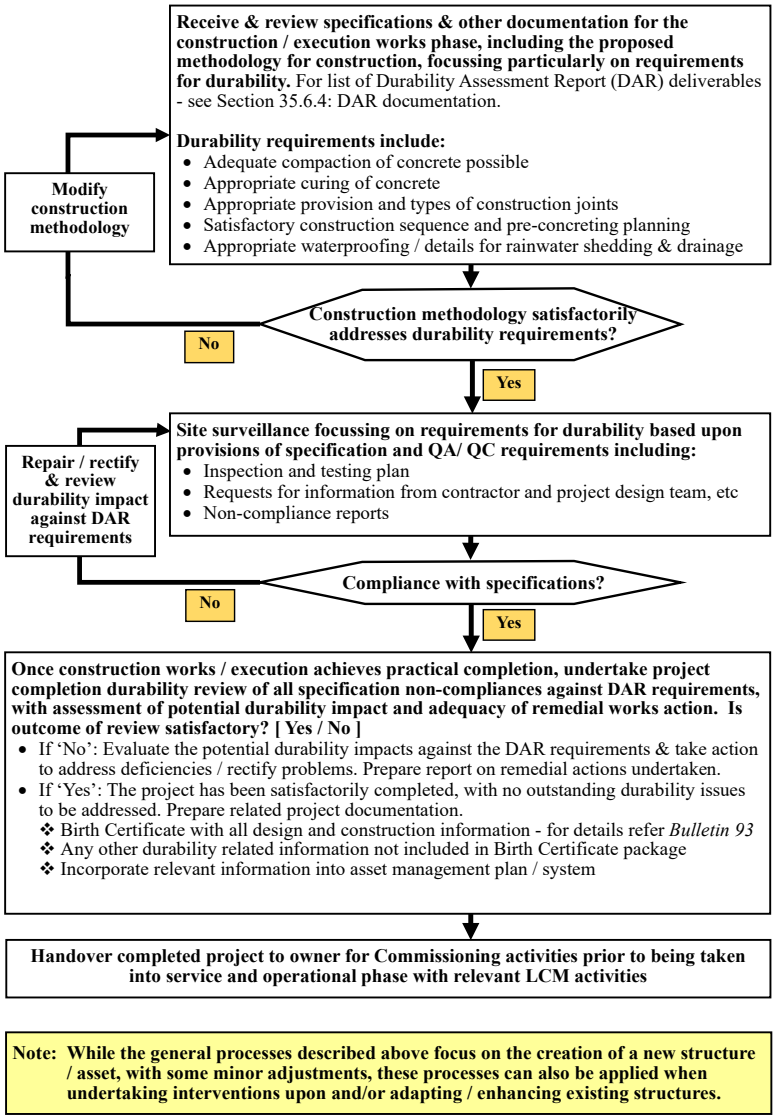

Figure 35.6-1: Durability process during execution / construction (after CIA Z7/01 [35-7], Fig 2.2)

The contractor's compliance shall be evaluated with respect to the requirements defined in the project specifications and drawings, which shall be consistent with the durability design intent defined in the DAR: DD.

Prior to completion of execution / construction a Durability Assessment Report: Verification Report (DAR: VR) is required to provide evidence that the durability design intent was achieved.

This verification will include a review of all non-compliances with the durability requirements defined on the drawings and in the specifications, plus the repairs undertaken to ensure that the design service life is achieved.

The Durability Assessment Report deliverables, which can include the following reports and associated supporting documentation, are also given in:

- *The Durability Assessment Report: Structure Owner's Brief (DAR: B)*
- *Durability Assessment Report: Scope of works & technical criteria SWTC (DAR: SWTC)*
- *Durability Assessment Report: Contractor Tender (DAR: CT)*
- *Durability Assessment Report: Conceptual Design (DAR: CD)*
- *Durability Assessment Report: Detailed Design (DAR: DD)*
- *Durability Assessment Report: Verification Report (DAR: VR)*

Further guidance on the preparation of the Durability assessment report is given in *fib* Bulletin XY [35-6]. Other contemporary guidance on durability planning includes CIA Z7-01 [35-7].

35.6.4 Durability assessment report (DAR) documentation

A sequence of durability assessment report (DAR) documentation deliverables shall be provided for a project to both formally define the durability requirements, starting from the structure owner's brief, and subsequently to provide evidence that the durability design intent defined in the DAR has been achieved.

A list of the durability assessment report (DAR) documentation deliverables expected for a work-flow sequence involving early contractor engagement is presented in *fib* Bulletin XY [35-6]. This is appropriate where the concrete structure owner has proactively issued the tender document project requirements which highlight durability issues that require the contractor's consideration and assessment during detailed design.

References

[35-1] ISO 22966:2009 — Execution of concrete structures.

[35-2] ISO 45001:2018. — Occupational health and safety requirements with guidance for use.

[35-3] ISO 13315-1:2012. — Environmental management for concrete and concrete structures.

[35-4] ISO 14001:2015. — Environmental management systems – Requirements with guidance for use.

[35-5] ISO 9001:2015. — Quality management systems.

[35-6] *fib* Bulletin XY — *Durability planning for concrete structures* [IN PREPARATION, COM8]

[35-7] CIA Z7/01 Durability Planning — A 'Recommended Practice Report' forming part of the CIA Concrete Durability Series, Concrete Institute of Australia (CIA), Sydney, Australia

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Prefabrications include cut reinforcement, cut and bent reinforcement and assemblies of these elements using ties, mechanical splices or welds manufactured off-site by a reinforcement fabricator and delivered to site or manufactured on the construction site.

Light surface rust, of a normal atmospheric nature, may be acceptable. However, if reinforcing steel is stored in an environment in which salt is present in the atmosphere – for example in a coastal zone – then rust formed may be considered deleterious

Passivation of the zinc coating can be achieved by storing the products outdoors for a sufficient period of time (4 weeks is thought to be a sufficient time) or by application of an appropriate passivation treatment.

36 Construction works

36.1 Reinforcing steel works

The following subsections apply to both prefabrications and site fabricated reinforcement. For principles reference is made to Chapter 7 and Chapter 35.

36.1.1 Transportation and storage

Reinforcing steel bars, coils, welded fabric and prefabrications should not be damaged (i.e. by mechanical damage and/or corrosion) during transportation and storage.

The surface condition of the reinforcing steel should be examined prior to use on site to ensure that it is free from mechanical damage (e.g. notches) and from loose rust and deleterious substances which may adversely affect the reinforcing steel, concrete or the bond between them.

When galvanized reinforcement is used, either the zinc coating should be sufficiently passive to avoid chemical reactions with the cement, or the concrete should be made with cement that has no detrimental effect on the bond to the galvanized reinforcement.

Where epoxy-coated reinforcement or other types of material with coatings (e.g. Zinc-coated) is used, care needs to be taken during storage and handling to avoid damage to the coatings. Even small areas of damage can result in preferential corrosion in localised areas.

Care should be taken in the storage of reinforcing steels of dissimilar materials to ensure separation to mitigate the risk of bi-metallic corrosion.

Marking steel grade and ductility class on the reinforcing steel bar is considered good practice since these marks are easy to check in practice and considered useful also during assessment of structures. For assessment then a clear database of marking used in the past should be available.

36.1.2 Identification

The reinforcing steel delivered on site shall be identified to steel grade, ductility class and manufacturer, using either the rolling marks on the steel when available or the accompanying documentation (e.g. Mill test certificate, conformity certificate or identification label) so that the reinforcing steel can be checked to determine whether it conforms to the appropriate standard referred to in the order. If the reinforcing steel is produced from coils or comes from straightening operations, this must be indicated in the documentation

Anchorage devices and couplers for mechanical splices shall be identified using the marks on the devices or on their packaging/documentation (e.g. Technical approval report) in order to check that they conform to the appropriate standard or approval referred to in the order.

If means of identification have been lost, additional acceptance tests on samples may be required.

36.1.3 Cutting, bending and straightening

Guidance can be found in BS 8666 [36-1]

Mechanical cutting is an appropriate method.

Cutting and bending of reinforcing steel shall be made in accordance with an accepted standard and shall conform to the execution specification.

Bending should be carried out by mechanical methods using equipment designed for the purpose and in one operation at a uniform rate with the aid of mandrels. The diameter of the mandrel used should be suitable for the actual type and diameter of reinforcement and may not be less than values that could induce bending fracture in the reinforcement or failure of the concrete inside the bend of the reinforcement.

Cutting procedures could be manual or automatic, but they shall not modify geometrical nor mechanical properties of the steel products.

Recommended values for the minimum permissible mandrel diameter for bending reinforcing steel are given in Table 36.1-1.

Table 36.1-1: Recommended minimum mandrel diameter for bending reinforcing bars.

Bar diameter (mm)	Recommended minimum mandrel diameter for bends, hooks and loops
Ø ≤ 16 mm	4Ø
16 mm < Ø ≤ 32 mm	7Ø
32 mm < Ø ≤ 40 mm	10Ø

The recommended minimum mandrel diameters for bends, hooks and loops given in Table 36.1-1 are minimum values for reinforcing bars in accordance with product standards, which specify mandrels for bending tests which are 3Ø, 6Ø and 7Ø, respectively. Larger minimum mandrel diameters in the order of 10Ø to 15Ø should be used if re-bending or straightening of bars is intended.

For the sake of good detailing, larger minimum mandrel diameters may be required – see also section 30.13.3.4.

Recommended values for the minimum permissible mandrel diameter for welded reinforcement or welded fabric bent after welding are given in Table 30.13-2.

Due to current production techniques applied heating of reinforcement steel might result in great loss of strength.

Bending of reinforcement steel at temperatures lower than –5°C should only be permitted if allowed by the execution specification and provided that the bending procedure conforms to accepted additional precautionary procedures.

Unless otherwise agreed, bending by heating should not be permitted.

When permitted by the execution specification, straightening of bent reinforcement steel should only be allowed if:

- special equipment to limit local stresses is used;
- appropriate straightening procedures are available;
- visual inspection of the straightened reinforcement is carried out for any cracks and other damage.

Reinforcement steel produced from coils will only be acceptable if an appropriate decoiling and straightening equipment is used and appropriate procedures in accordance with the equipment manufacturer's instructions are applied. The de-coiled and straightened bars shall meet the requirements for the reinforcement given

in chapter 15 or the relevant standards applicable, after straightening, and tested as specified in corresponding standards.

36.1.4 Welding

Welding of reinforcing bars shall be designed and detailed, and welds shall be specified in the execution specification.

Welding is permitted on reinforcing steel classified as weldable by the product specification unless otherwise specified in the execution specification.

Welding of reinforcing steel, and welding of reinforcing steel to structural steel, in load-bearing joints shall be performed as specified in the execution specification, and in accordance with ISO 17660-1 [36-2] unless otherwise specified.

In general, reinforcing steel should not be welded at or near bends of the bars. If welding is applied Section 30.13.3.4 gives guidance on the minimal mandrel diameter to be applied.

Welding of non load-bearing welds (tack-welding) performed according to ISO 17660-2 [36-3] is permitted unless otherwise specified in the execution specification.

Where a risk of fatigue exists for the concrete structure, welding of reinforcement should conform to special requirements. The requirements of ISO 17660 [36-2] + { 36-3] only apply to static loaded concrete structures.

36.1.5 Splices

Bars may be spliced by laps, couplers or welding in accordance with the execution specification. Splices should be located away from regions of high bar stress whenever possible and staggered (in different positions), if possible, for linear members (beams and columns) and planar structures (slabs, walls and shells).

Overlapping bars should be placed in contact; in beams and columns the laps should be tied. The laps should be enclosed with ties.

In general, the location of laps should be shown on construction drawings. If points of overlapping are not shown on the drawings, they should be evenly

Suitability for welding is defined in the appropriate product standards and is normally defined by limiting the carbon equivalent value of the steel (CEV).

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Further guidance on lap length and required transverse reinforcement are given in section 20.2.2.6.

It is assumed that the execution specification gives detailed information on the layout and spacing of reinforcement as well as any precautions to be taken in areas of congested reinforcement.

Acceptable installation tolerances are given in ISO 22966 [36-4], section 10. See also section 36.5.5 of this Model Code.

Concrete and cementitious spacers should have at least the same strength and should give at least the same corrosion protection as the concrete in the structure. Steel spacers should ensure that the correct concrete cover is achieved.

distributed and the longitudinal distance between two adjacent laps should not be less than 0.3 times the lap length.

Splices made with mechanical connecting devices should be carried out according to the manufacturer's instructions.

36.1.6 Assembly and placing of the reinforcement

The reinforcement should be placed according to the execution specification.

The reinforcement shall be fixed and secured so that its final position after placing and concreting lies within the tolerances given in the execution standard or specification.

The assembly of reinforcement may be done with tie wire or spot welding (see section 36.1.4).

The specified cover to the reinforcement should be ensured by the use of suitable chairs and spacers.

36.1.7 Construction documents – reinforcement

Reinforcing steel works should be defined with details and required quality of works in an execution specification containing as a minimum:

(i) construction drawings (shop drawings) giving all necessary information such as geometry of the structure and amount and position of reinforcing steel;

(ii) description of all products to be used with any requirement for the application of the products;

(iii) works description (method statements) including inspection requirements.

Actual construction works should be documented with sufficient detail and in sufficient number of construction records/certificates as specified in the construction specification.

36.2 Prestressing works

36.2.1 General

The prestressing works should be done by a specialised prestressing company. The qualification of the company and its personnel may be controlled via a standard such as CWA 14646 [36-5].

High amounts of energy are stored in stressed tendons. Sudden unexpected release of the tendons is therefore dangerous.

Ducts should be supported by different means to avoid any movement during concreting or compaction and ensure the specified geometry. Ducts can be tied to the rebar cage with wire. Tack welding is not recommended.

For principles reference is made to Chapter 7 and Chapter 35.

To ensure safety, reliability and durability of tendons it is necessary to provide:

- skilled supervisors and personnel, trained for the purpose and the prestressing system to be used;
- robust and safe equipment (regular maintenance and calibration) compatible with the prestressing system to be used.

Suitable safety precautions shall be taken on site during all prestressing works.

The engineer should provide detailed execution specifications to sufficiently specify the requirements for prestressing works to be done on site including at least the following:

- specifications for the tensile elements and the prestressing system to be used;
- drawings to specify the tendon geometry;
- minimum concrete strength for stage or final stressing of tendons;
- prestressing forces for each tendon and each tendon end, and sequence of stressing of tendons;
- permissible tolerances for prestressing force and tendon elongations;
- theoretical friction coefficients and wedge penetration considered in the prestressing design;
- requirement for any hold point during prestressing works needing approval of the engineer before continuation of works;
- requirements for any special testing to be performed (e.g. measurement of transmission, i.e. friction losses in tendon, or grouting tests);
- requirements for traceability of materials.

The company responsible for the installation of the prestressing system should provide information to site to define the specific methods for the installation of ducts and the prestressing works to be done including at least the following:

– site requirements for storage and handling of materials and equipment;
– shop drawings showing tendon layout (support heights), anchorage assembly, and grouting details (vent locations);
– method statements for duct installation and tendon installation, tensioning and grouting, adapted to the complexity of the site;
– theoretical tendon elongations and permissible tolerance for each tendon at specified stressing force;
– required quality control and reporting with optional hold points as specified by the engineer.

36.2.2 Packaging, transportation, storage and handling of materials and components

Materials and components should be packed such that they are protected from excessive moisture and/or dirt. Temporary corrosion protection may need to be provided on materials and components where required for the duration of storage and/or environmental conditions (e.g. humidity).

Materials and components should be protected from contact with any harmful substances.

Materials and components should be packed, transported, stored and handled such that they are not damaged and do not deteriorate (e.g. rust). Any damage or deterioration which is considered unacceptable should be repaired or the materials or components shall be rejected.

Certificates of materials received on site should be compared against the execution specifications. Materials not complying shall be rejected or placed in quarantine.

Materials should be used before their shelf life expires.

36.2.3 Prestressing works for post-tensioning tendons

36.2.3.1 Installation of tendons

Installation of the tendons and components should be in accordance with the shop drawings and method statements relevant for the project.

Tendon installation tolerances are provided in national or international standards; see for example ISO 22966 [36-4] or EN 13670 [36-6].

No particular temporary corrosion protection is required if tendons made of prestressing steel are provided with permanent corrosion protection within 2 and 6 weeks after stressing for aggressive and benign environments, respectively, or if the prestressing steel is supplied with a factory-applied corrosion protection.

If the above periods cannot be satisfied, temporary corrosion protection with a suitable water soluble oil may be provided. Only oils should be used which do not need to be removed before providing permanent corrosion protection (see *fib* Bulletin 20 [36-7]). Tendons should not be flushed with water.

Refer to the FIP State of the Art Report: "Tensioning of tendons: force-elongation relationship" [36-8].

In general, the tendon force is determined with the pressure in the stressing equipment. In exceptional cases, direct force measurement with load cells or similar may be done.

The placing of anchorages, coupling devices and ducts should be according to the specified geometry and tolerances. Care should be taken to avoid kinks in the tendon profile and damage to the components. Anchorages, coupling devices and ducts should be sealed against ingress of water and other foreign substances into the tendons during construction.

Tensile elements should be installed in the ducts as specified in the relevant procedures in the workshop (prefabricated tendons) or on site either before or after placing of the concrete.

Tensile elements should be protected from welding sparks.

The tensile elements should obtain temporary corrosion protection before installation if the expected time between installation/stressing and grouting of the duct exceeds acceptable time limits. Alternatively, the installed tensile elements may be protected by blowing of dry air or other equivalent methods. Temporary corrosion protection materials applied to the tendon should allow sufficient bond after grouting and should have no detrimental effect on the tensile elements or on the grouting material. Only products approved by the supplier of the prestressing system should be used.

Tendons, ducts and vents shall be suitably marked for identification and recording during stressing and grouting operations.

Tendon components should be inspected before placing of the concrete for compliance with the relevant project and system documents.

Traceability of tendon components should comply with the execution specification.

36.2.3.2 Tensioning operations

36.2.3.2.1 General

Stressing of tendons should be in accordance with the project specific method statements and detailed procedures.

In general, the calibration certificates should not be older than 6 months.

Typically post-tensioning tendons are stressed in the following stages:

- stressing from 0 to 100% of the specified stressing force in one stage for tendons with little tendon deviation and little tendon slack, and recording of the tendon elongations only at the 100% mark. Such tendons are typically found in building slabs as single strand tendons or tendons in flat ducts as well as bar tendons;
- stressing in stages of 25, 50, 75 and 100% of the specified stressing force for tendons with considerable tendon deviation and tendon slack, and recording of the tendon elongations at each stage. With this procedure, the effect of tendon slack may be deleted from the full force-elongation graph. Such tendons are typically found in bridge construction as multistrand/wire tendons in round ducts.

If the actual tendon force and/or elongations are below acceptable values, the following measures may be considered:

- if the available prestressing steels and prestressing system allow, to apply a higher stress at the stressing end(s) of the tendons. This stress should never exceed the value of 0.95 $f_{p0.1k}$;
- lubricate prestressing steel of tendons to reduce the friction losses;

36.2.3.2.2 Activities before stressing

- check concrete strength to have reached or exceeded the specified minimum strength for stressing;
- check that structure is free to move as may be needed during stressing;
- inspect exposed tendon ends and surfaces on which anchorages and stressing equipment will bear;
- check calibration certificates of stressing jacks to be recent.

36.2.3.2.3 Activities during stressing

- increase tendon force gradually and steadily in the stages indicated in the method statement;
- record tendon force (pressure) and tendon elongation at the stages specified in the method statement.

- compare specified force and elongation with measured and calculated values;

– install and stress additional internal tendons into provided spare anchorages and ducts or additional external tendons into provided spare anchorages and deviators as may have been specified in the execution specification.

Elongation measurements are only a rough indication for the friction losses during stressing. If action effects in structures are very sensitive to prestressing force deviations, the measurement of the force transmission with lift-off tests is recommended.

Values given for permissible deviations from expected elongations for deviated tendons are taken from ISO 22966 [36-4] and EN 13670 [36-6]. For short tendons the values might be challenging.

Tendon elongations should be calculated with the properties of the prestressing steel as given in the material certificates.

Where tendon tails are burned off, the anchorage devices should be protected from excessive heat development.

If during the stressing of post-tensioning tendons to the specified force, the actual elongation of the group of all tendons at a particular cross-section of the structure is not within ± 5 % of the calculated elongation, or of a single tendon in a group is not within ± 15 % of the calculated elongation, action shall be taken in accordance with the execution specification.

– maintain tendon tails to allow re-stressing; only cut tendon tails and grout after final approval.

36.2.3.2.4 Activities after stressing

– visual inspection of concrete and anchorages;
– supervision of temporary protection of tendons;
– cutting of tendon tails once the stressing records have been approved.

36.2.3.2.5 Records

– pressure in stressing jack;
– elongation of tendons with adjustment for seating in tendon anchorages, as applicable, and for effects of measuring procedure (e.g. elongation inside stressing jack, if applicable);
– any particular observations during tensioning;
– site stressing records to be transmitted to and approved by the design engineer in view of any necessary adaptation of above instructions.

For additional information, see the fib Bulletin 20 [36-7], and ISO standards 14824 parts 1 to 3 [36-9][36-10][36-11].

The main objectives of prestressing ducts grouting is to protect the tendons from corrosion and to provide and effective bonding between steel and concrete.

In hot or cold climates, only grouting materials should be used which have been confirmed to provide the specified performance and to remain sufficiently fluid over the expected period of time at the expected range of temperatures.

If measuring equipment (temperature gauges) is available, the 5°C value may be referred to the temperature of the structure in the vicinity of the tendon.

36.2.3.3 Grouting of prestressing ducts

36.2.3.3.1 General

Grouting of prestressing ducts should be in accordance with the project specific requirements and method statements. Grouting materials should be in accordance with section 16.2.2.3.

36.2.3.3.2 Activities before grouting

- equipment should be confirmed to be capable of adequately mixing and pumping the specified grouting material and should be operational;
- permanent supplies of water under pressure and of compressed air should be available;
- grouting material to be confirmed for use under conditions expected on site (temperature, time required for grouting operations, tendon profile) with suitability tests or similar;
- instructions for actions in case of incidents (e.g. fault during injection) and harmful climatic conditions (e.g. after and during periods with temperature lower than 5°C);
- materials batched (excess to allow for overflow);
- blowing of air through the ducts, or using other approved methods, to ensure that ducts are free of harmful material (e.g. water, ice);
- vents prepared and marked with identification;
- preparation of moulds and other tools for quality control tests for grouting material.

Grouting of a duct should be done without interruption. The speed at which the grouting material is pumped through the duct can vary from 5 to 15 m/min.

Provisional inspection ports in the anchorages are recommended to check complete filling of the anchorages behind the anchor head with grout.

In general, small voids which do not leave the prestressing steel unprotected do not need to be grouted.

To obtain a good seal of anchorage recesses, a preparatory treatment to the surface around the anchorages may be applied; in large recesses, connecting reinforcement should be provided.

36.2.3.3.3 Activities during grouting

- grouting to be carried out in a continuous operation and at the rate and in the sequence specified in the method statement;
- grouting should continue until the density of the grout flowing from the tendon ends and the vent openings is about the same as that of the injected grout; after closing the last vent, pressure may be held for about one minute to verify that there are no unacceptable leaks in the system.

36.2.3.3.4 Activities after grouting

- immediately after grouting, check all accessible tendon areas for complete grouting, for example by light tapping. If voids are suspected, pump grouting material again until voids are filled;
- all equipment to be thoroughly washed shortly after grouting, followed by thorough draining of pump, mixer and hoses;
- record all operations, measurements, tests and particular events and submit to the engineer;
- after setting of grouting material check all accessible tendon areas (caps, vents, inspection ports) for complete grouting. If unacceptable voids are found, top up with fresh grouting material, or with vacuum grouting as may be suitable.

36.2.3.3.5 Records

- records to be established on all operations, measures, tests, controls and particular events; to be transmitted to the design engineer.

36.2.3.3.6 Sealing of anchorages

Once the grouting material has set, all openings, grouting tubes and vents should be hermetically sealed and the anchorage recesses filled as specified in the execution

Tendon installation tolerances are provided in national or international standards – see for example ISO 22966 [36-4] or EN 13670 [36-6]. See also section 36.5.5 of this Model Code.

specification to prevent penetration of water and harmful products (e.g. de-icing agents).

36.2.4 Prestressing works for pretensioning tendons

36.2.4.1 Installation of tendons

Installation of the tendons and components such as hold-down devices at deviators should be in accordance with the shop drawings and method statements relevant for the project.

The placing of anchorages and tensile elements should be to the specified geometry and tolerances. Care should be taken to avoid kinks in the tendon profile. Installation should be in accordance with the relevant procedures.

Details of local debonding of tendons should be provided as specified in the relevant shop drawings.

Tendons should be protected from welding sparks.

The tensile elements should be suitably protected from adverse environmental conditions. The time between installation and stressing of the tendons and pouring of the concrete providing permanent corrosion protection should be limited in accordance with section 36.2.3.1.

Tendons shall be suitably marked for identification and recording during stressing and release operations.

Tendon components should be inspected before placing of the concrete for compliance with the relevant project and system documents.

Traceability of tendon components should comply with the execution specification.

36.2.4.2 Tensioning operations

36.2.4.2.1 General

Refer to the FIP State of the Art Report: "Tensioning of tendons: force-elongation relationship" [36-8].

In general, the tendon force is determined by the pressure in the stressing equipment. In exceptional cases, direct force measurement with load cells or similar may be done.

The calibration certificates should not be older than 6 months, in general.

If the actual tendon forces and/or elongations are below acceptable values, the following measures may be considered:

- if the available prestressing steels and prestressing system allow, to apply a higher stress at the stressing end(s) of the tendons. This stress should never exceed the value of 0.95 $f_{p0.1k}$;
- install and stress additional internal tendons as may have been specified in the execution specification.

Values given for permissible deviations from expected elongations for deviated tendons are taken from ISO 22966 [36-4] and EN 13670 [36-6]. For short tendons the values might be challenging.

- Tendon elongations should be calculated with the properties of the prestressing steel as given in the material certificates.

Stressing of tendons should be in accordance with the project specific method statements.

36.2.4.2.2 Activities before stressing

- inspect tendon ends and abutments on which anchorages and stressing equipment will bear;
- check calibration certificates of stressing jacks to be recent.

36.2.4.2.3 Activities during stressing

- stress tendons in the specified sequence gradually to the specified force and/or elongation;
- record pressure at jack or load cell and elongations of prestressing steel;
- compare measured and calculated values;
- take corrective actions if deviation between specified and measured values is greater than permitted tolerances.

If during the stressing of pre-tensioning tendons to the specified force, the actual elongation of the group of all tendons at a particular cross-section of the structure is not within ± 3 % of the calculated elongation, or of a single tendon is not within ± 5 % of the calculated elongation, action shall be taken in accordance with the execution specification.

Sudden release of the tendons may cause significant increase of the transmission length.

36.2.4.2.4 Release of tendons

- check that concrete strengths have reached or exceeded the specified minimum strength for release;
- release the tendons with the approved method and in the specified sequence;
- record the tendon draw-in as specified for the project;
- compare measured and expected draw-in;
- observe instructions if deviation is greater than expected.

36.2.4.2.5 Records

- pressure in stressing jack;
- elongations of tendons with adjustment for seating in tendon anchorages, as applicable, and for effects of measuring procedure (e.g. elongation inside stressing jack, if applicable);
- records of draw-in of tensile elements;
- records to be established on all operations, measures, tests and particular events; to be transmitted to the design engineer;
- any particular observations to be recorded during tensioning;
- site stressing records to be transmitted to and approved by the design engineer in view of any necessary adaptation of above instructions.

36.2.4.3 Sealing

All tendon ends, debonded tendon areas and any other penetration (e.g. hold-down devices at deviators) shall be hermetically sealed as specified for the project to prevent penetration of water and harmful products (e.g. de-icing agents).

36.2.5 Replacement of tendons

Any potential future replacement of internal unbonded or external tendons should be carefully planned and carried out by an experienced specialist contractor.

Particular procedures have to be specified and applied. When replacing the tendons by cutting individual elements, special procedures have to be applied to avoid the potential transfer of load from already cut individual prestressing elements to the remaining elements due to bond or clamping effects at deviation points which could overload the tendon and lead to an unintended sudden failure of the tendon.

Care should be taken to avoid unacceptably high unbalanced loads on deviators or diaphragms with tendon anchorages.

36.2.6 Construction documents – prestressing

ISO 22966 [36-4] may provide details of required construction documents.

Prestressing works should be defined with details and required quality of works in an execution specification containing as a minimum:

- construction drawings (shop drawings) giving all necessary information such as geometry of the structure, number and position of prestressing tendons and anchorages;
- description of all prestressing systems to be used with any requirement for the application of the systems;
- works description (method statements) including inspection requirements.

Alternatively, to assembly drawings, system documentation included in technical approvals giving the main dimensions of standard components may be considered as part of the as-built documentation.

Actual construction works should be documented with sufficient detail and in sufficient number of construction records/certificates as specified in the construction specification. Actual prestressing systems hardware should be documented with assembly drawings.

36.3 Non-metallic reinforcement

For the application of externally applied FRP reinforcement, reference is made to *fib* Bulletin 90 [36-12], chapter 10.

fib Bulletin 48 [36-14] gives guidance for the selection of falsework and formwork and aspects to be considered.

The actual upper sieve size "*D*" represents the sieve size of which at least 80% to 85% of the aggregate fraction, but not more than 99% of the sample pass.

For the application of internally applied FRP reinforcement, reference is made to the publications of ACI Committee 440, specifically ACI 440.1R-16 [36-13].

For additional principles reference is made to Chapter 7 and Chapter 35.

36.4 Falsework and formwork

For principles reference is made to Chapter 7 and Chapter 35.

Falsework and formwork including their supports and foundations should be designed and constructed so that they are:

- capable of resisting any foreseeable action to which they are submitted during the construction process, considering all construction stages; In prestressed concrete structures, the falsework must resist load distributions induced by the prestressing;
- stiff enough to ensure that the tolerances specified for the structure are satisfied and the integrity of the structural member is not affected;
- stable enough provide the visible surfaces the texture and uniformity required by the design.

The materials used for formwork should assure that the properties of the concrete and appearance of the concrete surface meet the requirements .

The contractor should have a complete design of the falsework, including Design basis, Calculations report, Specifications and drawings before starting the assembly of the falsework.

36.5 Concreting

36.5.1 Specification of concrete

Concrete and its specification shall comply with ISO 22965-1 [36-14].

The term d_{max} normally defines the "maximum nominal upper aggregate size" which is selected to ensure a proper casting taking into account the cover and free spacing between the reinforcement bars.

In principle, any value of D less than d_{max} satisfies a requirement related to d_{max}.

ISO 22965-1 [36-14] defines concrete as "…mixing cement, coarse and fine aggregate and water…".

Standards on aggregates may define coarse aggregates as aggregate with $D \geq 4$ mm.

Concrete with aggregates with actual upper size D in the range 4 – 16 mm might in some cases not support the design assumptions, for example concerning aggregate interlock, shear capacity, stiffness and fracture energy (see for instance effect of d_g on k_{dg} in section 30.1.3.2.1 affecting the shear resistance $V_{Rd,c}$).

ISO 22965-1 [36-14] therefore states in a note that in concrete for general purpose the coarse aggregate should normally have a D of at least 16 mm.

To ensure compatibility with the design assumptions, the acceptable range of D for the coarse aggregate, given with an upper and a lower value, should therefore be given in the execution specification, in particular if D smaller than 16 mm is allowed.

SCC mix design should comply with specific requirements in the fresh state depending on the type of application, and especially on:

- confinement conditions related to the concrete element geometry and the quantity, type and location of reinforcement, inserts and recesses;
- placing equipment (pump, truck-mixer, skip, etc.);
- placing methods (number of delivery points);
- finishing method.

Those requirements might be expressed and justified in terms of:

- flowability and grouting ability;
- viscosity (measure of the speed of flow);
- passing ability (flow without blocking);
- segregation stability.

The concrete specification should include requirements given in the execution specification and requirements related to the actual method of execution.

The actual upper and lower sieve size D of the aggregate to be used in the concrete should be within the range given in the execution specification.

36.5.2 Placing and compaction

The concrete shall be placed and compacted in order to ensure that all reinforcement and cast-in items are properly embedded and that the concrete achieves its intended strength and durability.

By the use of concrete described as self-compacting concrete (SCC), the compaction of the fluid concrete is achieved due to the effect of gravity. Working procedures for the actual cast should be established, based on the contractor's experience and/or pretesting, to enable the required compaction to be obtained.

Additional requirements to those given in ISO 22965-1 [36-14] if any, shall be agreed with the producer.

The following methods are suitable for curing used separately or in sequence:

- keeping the formwork in place;
- covering the concrete surface with vapour-proof sheets which are secured at the edges and joints to prevent draughts;
- placing of wet coverings on the surface and protection of these coverings against drying out;
- keeping the concrete surface visibly wet with suitable water;
- application of a curing compound of established suitability.

Other curing methods of equal effectiveness may be used.

At the time of publishing this Model Code, standardized test methods characterizing the properties of curing compound are not available.

The quality of the curing will influence the properties of the concrete cover. Good curing will realize the potential of the mixed concrete, while less good curing will not. Deemed-to-satisfy requirements for service life design normally end up with tabulated requirements to cover thickness and cover quality. The tradition differs from nation to nation on how the quality of curing is handled. In some national provisions a relatively high w/c ratio (proxy characteristic for the potential quality) is specified together with strict requirements for the curing regime. In other provisions lower w/c ratios are given with a more relaxed demand for the curing.

ISO 22966 [36-4] gives requirements for the construction operations involving structural precast elements from their reception at the site or in the case of site-manufactured elements from removal from the forms, until the completion of their installation and final acceptance.

36.5.3 Curing

Concrete in its early life should be cured and protected:

- to assure the desired moisture for continued hydration and development of strength
- to minimize plastic shrinkage;
- to ensure adequate surface strength;
- to ensure adequate surface zone durability;
- from harmful weather conditions;
- from freezing;
- from harmful vibration, impact or damage.

ISO 22966 [36-4] defines four “curing classes” for the duration of applied curing ranging from 12 hours to 70% of specified characteristic 28 days compressive strength.

The curing class to be applied should be stated in the execution specification.

36.5.4 Execution with precast concrete elements

Precast elements should be used as specified in the execution specification and the design coordination between them and the structural performance of the overall structure should be verified.

Handling (including transportation), storage and protection of the precast elements should be carried out in accordance with the execution specification.

Requirements for the placing and adjustment of the precast elements should be given in the erection specification.

The placing of any additional reinforcement and in-situ concreting for the completion of the structure shall conform to the relevant sections of ISO 22966.

Structural connections including other materials, for instance steel, should conform to the relevant standards for execution like international welding standards or the relevant standards for the application of grout in connecting ducts.

36.5.5 Geometrical tolerances

The completed structure shall be within the maximum allowable deviations to avoid detrimental effects in terms of:

- mechanical resistance and stability in transient and in service stages;
- service performance during the use of the construction works;
- placing compatibility for the erection of the structure and its non-structural components.

Deviations from the specified tolerance range should be handled in accordance with section 35.4. Small deviations which have no significant consequence on the performance of the finished structure may be ignored.

Any requirements for special tolerances should be identified in the execution specification and the following information should then be given:

- any amendments to the permitted deviations given in ISO 22966 [36-4];
- any further type of deviation to be controlled, together with defined parameters and permitted values;
- whether these special tolerances apply to all relevant components or to particular components which are identified;
- if the “box principle” should be applied, and what deviation is permitted.

ISO 22966 [36-4] provides two sets of permitted geometrical deviations. The requirements given in section 10 of ISO 22966 [36-4] are considered as achieving the design assumptions for the normally required level of safety, while those given in annex G of ISO 22966 [36-4] are assumed to have relevance for service performance.

The tolerances given in section 12.5.4.1.5.4 are slightly different with those given in ISO 22966, but may be considered as equivalent.

The box principle will require that all points of the structure are within the specified theoretical position with a margin in any direction corresponding to the permitted deviation.

The surface finish characteristics might be given in the client brief and be based on classifications applied in local regulations. Further guidance on surface characterisation can be found in:

- ACI 347.3R-13 [36-15];
- DBV Merkblatt Sichtbeton [36-16];
- CUR Aanbeveling 100 [36-17].

36.5.6 Aesthetics of concrete surfaces

The appearance of the finished concrete surfaces shall comply with the agreed specification.

References

[36-1] BS 8666:2020. Scheduling, dimensioning, bending and cutting of steel reinforcement for concrete. Specification.

[36-2] ISO 17660-1:2006. Welding – Welding of reinforcing steel – Part 1: Load-bearing welded joints.

[36-3] ISO 17660-2:2006 Welding – Welding of reinforcing steel – Part 2: Non load-bearing welded joints.

[36-4] ISO 22966:2009 Execution of concrete structures

[36-5] CWA 14646[1]. Requirements for the installation of post-tensioning kits for prestressing of structures and qualification of the specialist company and its personnel.

[36-6] EN 13670:2009. Execution of concrete structures.

[36-7] *fib* Bulletin 20. Grouting of tendons in prestressed concrete. 2002.

[36-8] FIP State of the Art Report. Tensioning of tendons: force-elongation relationship. 1986.

[36-9] ISO 14824-1:2012 Grout for prestressing tendons – Part 1: Basic requirements.

[36-10] ISO 14824-2:2012 Grout for prestressing tendons – Part 2: Grouting procedures.

[36-11] ISO 14824-3:2012 Grout for prestressing tendons - Part 3: Test Methods.

[36-12] *fib* Bulletin 90. Externally applied FRP reinforcement for concrete structures. 2019.

[36-13] ACI 440.1R-16. Guide for the Design and Construction of Structural Concrete Reinforced with Fiber-Reinforced Polymer (FRP) Bars. 2015.

[36-14] *fib* Bulletin 48. Formwork and falsework for heavy construction.

[36-15] ISO 22965-1:2007. Concrete – Part 1: Methods of specifying and guidance for the specifier.

[36-16] ACI 347.3R-13. Guide to formed concrete surfaces, 2014.

[36-17] Merkblatt Sichtbeton. Deutscher Beton- und Bautechnik-Verein, 2004.

[36-18] CUR Aanbeveling 100. Schoonbeton – specificatie, uitvoering en beoordeling van betonoppervlakken waaraan estetische eisen worden gesteld, (Aesthetic concrete - specification, execution and assessment of concrete surfaces subject to aesthetic requirements) [in Dutch]. 2004.

[1] – Due to be replaced by EN 17678: Installation of post-tensioned kits for prestressing of structures - Part 1: Competence of personnel

37 Execution of interventions

37.1 General

The term execution is used to refer to all the physical activities undertaken for the physical completion of the intervention works. These activities include procurement, scaffolding, falsework, formwork, reinforcement, concreting and curing, as well as the related inspection and documentation of those activities. The principles of execution of interventions are found in section 7.4.

The certainty that the desired outcomes will be achieved can be improved by "thinking intervention" long before the intervention process actually starts on site. In this way, those involved develop a process-focused approach which potentially leads not only to a reduction in problems on site and associated costs, but also to related improvements in health and safety for those involved, as well as potentially in subsequent additional (and unnecessary) environmental impacts, etc. The typical problems on site are related to space and time constraints from the service conditions of existing structures, which usually do not have to be considered for execution of new structures.

When making an intervention, the constraints relating to the space and time available for the execution activities need to be considered, along with the condition of the existing structure, as these can be quite different (i.e. more restricted) to those encountered with new structures. In order to achieve the target structural performance required after the intervention is made, it is important to have an execution plan appropriate to the selected intervention method(s).

Execution of interventions should be carried out to reduce disturbance to the surrounding environment and the service condition of the structure, and by considering climate- and environmental impact, cost, benefit and safety.

The common issues between execution of new structures and interventions are provided in chapter 35 and 36.

A detailed project specification for the execution of the intervention works should be developed clarifying and extending the general requirements of the ISO 22966 [37-1] for new structures and ISO 16311-4 [37-2] for intervention as necessary with project specific needs to meet the performance and durability requirements for the structure, together with the necessary supporting documents.

As part of the activities associated with selecting an appropriate intervention option, all execution processes of intervention should be effectively analysed with a view to identify significant potential problems and risks of non-conformance with the project requirements, including assessing the level of each risk involved.

37.2 Execution plan

The execution plan shall be prepared in advance considering not only the condition of the structure subject to the intervention but also the surrounding environment and the service conditions.

The execution plan should include not only the execution process but also the quality management plan and preparation of the existing structure.

Quality assurance activities: ensure that effective preventive actions are incorporated in relevant process control procedures at all stages of the definition and execution of the intervention so as to minimize the identified problems and/or reduce the level of risk to an acceptable value.

The process control procedures, and the preventive actions incorporated in them, should be documented so that the responsible operators and supervisors understand not only what is to be done during the course of each process, but also how the process has to be undertaken.

Quality control activities: ensure that verification procedures are prepared and implemented which define how, when, where and by whom each intervention related process will be monitored. The observations need to be recorded by appropriately qualified, experienced and independent staff, dedicated to surveillance activities, with adequate time and resources to conduct the required surveillance activities.

Appendix G of *fib Bulletin 44* [37-3] sets out steps for the development of supplementary documents to support the preparation of a project specification for the execution works for concrete structures, linking with the requirements of the ISO 22966 [37-1].

Further information on quality management during execution is also given in the Model Code chapter on life cycle management – see chapter 10.

37.3 Execution management

The performance of the structure after intervention is strongly affected by the quality of execution. Therefore, the execution of intervention should be properly planned and carried out under an approved quality management plan based on the selected intervention option(s) and the requirements set down in the project specification. These activities should be supported by appropriate quality assurance and quality control activities.

Appropriate verification tests should be carried out as part of the execution process for quality control purposes. A detailed record of the execution shall be maintained for future reference.

After execution, an initial inspection/survey shall be carried out to examine the structural performance after intervention and to collect necessary data for future performance assessment.

37.4 Preparation in existing structures

For remedial interventions such as coating and concrete patching, the materials for coating and patching should be bonded firmly to the substrate concrete surface. External bonding and overlaying as strengthening interventions should have strong/stiff enough interface between intervention materials and substrate concrete surface. For adding element, such as external cable and brace, as strengthening interventions, the added elements are connected mechanically with some device, such as anchor bolt. For all of such interventions, the quality of concrete at the

Generally materials/elements for interventions are connected with concrete in existing structures. The effectiveness of the load transfer mechanism (which will vary depending upon the nature of structural intervention being made and the form of mechanical connection involved) is a key consideration for satisfactory structural performance after intervention. In order to attain the target property, the proper preparation in the existing structure should be made in advance.

connection should be adequate. The preparation work to attain the required quality should be planned in advance and conducted in an appropriate way.

The preparation work includes cleaning the substrate concrete surface to remove laitance, dust, oil and other loose particles, and to create an appropriate surface texture (e.g. surface roughness).

For top surface overlaying and/or bottom surface (soffit) underlaying with cementitious material, reinforcing material usually in net/mesh form is attached firmly to the surface of the substrate concrete before placing the cementitious overlay/underlay material, in order to assure the precise location of reinforcing material and good bonding at the interface between substrate concrete and overlay/underlay.

In many intervention methods, such as overlaying/underlaying and jacketing with cementitious material, the cementitious material is placed in a layer, containing reinforcing materials, attached to surface of substrate concrete. Therefore, more care is necessary than in the case of execution of new structures, especially in respect of geometrical tolerances.

For overlaying/underlaying and jacketing, cementitious materials may be placed in a similar way as for new structures, but also by different techniques such as spraying and troweling.

Special attention should be paid to various issues for execution as follows:

– storage of intervention materials;

37.5 Reinforcing materials works

Ordinary steel reinforcement works, including cutting, bending, welding, assembling and jointing, may be the same as for those of execution of new structures. However, different reinforcing materials and materials with different details / special features need appropriate care and consideration for each particular case.

37.6 Cementitious materials works

Ordinary Portland cement concrete works, including placing, compaction, curing, precast concrete element execution and checking geometrical tolerance, may be the same as for those adopted in the execution of new structures. However, different cementitious materials and cementitious materials containing other materials, such as fibres and polymers, need special care for the special features and aspects of each particular case.

37.7 Other intervention works

There are various execution works for other interventions which differ from those employed for new structures as follows:

– execution speed;
– weather condition (temperature, moisture, etc) during execution (and in the subsequent curing period).

– Coating of protection materials and surface impregnation;
– Injecting materials in cracks/bandaging of cracks;
– Protection systems such as cathodic protection and corrosion inhibitors;
– External bonding of cementitious materials;
– External bonding of reinforcing materials such as steel and FRP in sheet/plate form;
– Mechanical connection of reinforcing elements such as braces and external cables;

For each method, every execution step should be conducted specifically to ensure the quality of intervention works.

37.8 Falsework and formwork

Falsework and formwork should be conducted considering restrictions in terms of time and space for execution since the existing structures are in service.

References

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[37-1] ISO 22966 ISO 22966:2009, Execution of concrete structures, ISO 2009

[37-2] ISO 16311-4:2014. Maintenance and repair of concrete structures – Part 4: Execution of repairs and prevention.

[37-3] *fib* Bulletin 44. Concrete structure management: Guide to ownership and good practice. 2008.

The available conservation strategies and respective tactics are classified into three categories as described in Section 8.2.1; namely, 'Proactive conservation', 'Reactive conservation' and 'No conservation'

For further details refer to:

– Chapter 5: Life-cycle management
– Chapter 8: Principles of conservation, and
– Chapter 27: Approach to design / Section 27.6: Durability planning

Information required for conservation of an existing structure / constructed asset is commonly derived from a variety of sources, such as the following:

i. Available design and construction documentation for the structure;
ii. Codes, standards and documented practice at the time of construction;
iii. Field investigations and structure measurements; and
iv. Destructive, in-situ and/or laboratory, and non-destructive tests.

For more information regarding the gathering of the required information and the specific tools and techniques associated with undertaking investigation, monitoring, assessment and interventions upon existing structures, see:

a. *fib* Bulletin 17 [38-1].
b. *fib* Bulletin 22: [38-2]
c. *fib* Bulletin 44 [38-3];
d. *fib* Bulletin 62 [38-4];

38 Conservation

38.1 Introduction

38.1.1 General

To allow effective and efficient life-cycle management (LCM) the durability design and the associated conservation processes shall be adequately planned, documented and managed through the life of the structure. In particular, the durability design and conservation planning shall be prepared taking account of the conservation categories (three are recognised - A, B and C) which have been selected for the concrete structure and/or its component parts – see Section 38.2.

To achieve the specified service life design objectives, it is necessary to incorporate durability considerations into the design, construction / execution processes and activities, along with operational maintenance aspects, to avoid premature deterioration of concrete structures. This process is commonly referred to as *Durability planning* – refer Sections 27.6 and 38.1.2.

Typically, three different conservation objectives are distinguished:

– conservation activities concerned with enabling a structure to meet its design service life, as envisaged at the time of design;
– conservation activities concerned with extending the planned service life of a structure; or
– enabling a structure to meet revised performance requirements (e.g. enhanced loading or changed functionality / needs).

An increase in the level of the performance requirements during the use stage may necessitate an intervention being made to increase the performance or possibly the durability of the structure / constructed asset.

An idealised representation of the life-cycle performance of a constructed asset / structure requiring a remedial intervention to meet its intended design service life is presented as Figure 8.3-1. This is complemented by the overview of the conservation management process presented in Figure 8.3-2, which defines the MC2020 life-cycle management and conservation process for concrete structures;

e. *fib* Bulletin 93 [38-5];
f. *fib* Bulletin ZZ [38-6]
g. *fib* Existing concrete structures [38-7];
h. ISO 13822 [38-8];
i. ISO 16311-1 [38-9];
j. ISO 16311-2 [38-10];
k. ISO 16311-3 [38-11];
l. ISO 16311-4 [38-12];
m. JCI 2014 [38-13];
n. JCSE 2007 [38-14];
o. CEN TS17440 [38-15];
p. Hajdin et al. [38-16].

along with the associated recording of life-cycle information. Both Figure 8.3-1 and 8.3-2 are contained in chapter 8: Principles of conservation.

In general, the life-cycle conservation process for a structure involves the following types and sequence of activities:

– *Condition survey and monitoring:* This involves gathering information on the current condition and performance – see section 38.2;

– *Identification of deterioration and damage mechanisms:* This involves identification of deterioration and damage mechanisms, plus evaluation of the predicted degree and rate of deterioration and damage – see section 38.3;

– *Performance evaluation (structural assessment):* This involves analysis of the available data to estimate current performance and to make a prognosis of future performance to support LCM of the asset, including deterioration effects. One important aspect may be to verify compliance with the original performance requirements in terms of the implied loads and usage conditions, involving evaluation of their possible influence on structural behaviour – see chapter 28 and section 38.4;

– *Decision-making for LCM activities including interventions:* This can be required for various reasons, including to enable a structure to:

  - meet its design service life, as envisaged at the time of design;
  - achieve an extension of its planned service life; or
  - meet revised performance requirements (e.g. enhanced loading or changed functionality / needs).

  These activities are concerned with establishing the implications of deterioration or damage and/or changed performance requirements upon the life-cycle management and conservation of an asset, evaluating potential management and conservation options, the selection of appropriate management and intervention option(s) and the timing of such works to address or remedy non-conformances with the performance requirements or achieve compliance with extended / enhanced performance requirements – see sections 38.5 and 38.6;

– *LCM activities including intervention works:* These activities are concerned with both LCM procedures employed as operational measures (refer section 38.6.1) but also physical intervention measures which are considered to be one of the range of LMC options. Section 38.6 is primarily concerned with physical intervention measures including matters relating to intervention

principles and methods; intervention strategies, methods and materials; the selection of intervention methods; condition survey for interventions; the design of interventions; the execution of interventions and conservation measures to be employed after interventions have been performed – see chapters 19, 37 and section 38.6;

– *Documenting LCM activities and conservation activities* – Recording - see section 38.7.

The wider context of these matters is considered in chapters 5 and 8, with the general approach taken to the assessment of existing structures being described in chapter 28.

Conservation activities are an element of the overall quality management processes applied to the structure / constructed asset. Those processes seek to ensure that its condition remains within satisfactory limits and that it continues to meet the defined performance requirements for the specified period of time – refer chapter 10.

Performance-based risk analysis procedures can be employed for the life-cycle management of constructed assets / infrastructure, enabling asset management activities and responsibilities to be clearly focused upon the performance of the 'system' under consideration – refer section 8.2.2.2.

The performance-based risk approach facilitates the adoption of performance-based service level agreements (SLA's) between the owner of the constructed asset / infrastructure and those responsible for its operational management, maintenance and undertaking remedial intervention or other forms of life-cycle management activities.

38.1.2 Preparatory activities – The role of durability planning

38.1.2.1 Introduction

To achieve the specified design service life or extended service life objectives, it is necessary to incorporate durability considerations into the design, construction / execution processes and activities, along with those concerned with operational maintenance aspects, to avoid premature deterioration of concrete structures. This process is commonly referred to as *Durability planning*.

Under performance-based design, the objective is to create a concrete structure which has the ability to fulfil the performance requirements for the specified design service life at the required level of reliability.

The durability planning approach is applicable to all materials and components used in the creation and life-cycle management of concrete structures.

The durability planning inputs required to achieve the design service life or extended service life are described in *fib* Bulletin XY [38-17]. Other contemporary

guidance on durability planning also includes CIA Z7-01: *Durability Planning* [38-18].

Durability planning is a formalised approach which applies durability knowledge during project design, execution / construction and operational maintenance activities to minimise the likelihood of premature deterioration.

Further details upon the overall durability planning approach, its various facets and the associated component parts are given in Section 27.6.

38.1.2.2 Life-cycle management activities - Durability planning

Life-cycle management / maintenance activities during use are an integral part of ensuring durability through the structure life-cycle. These shall include:

- A structure maintenance plan delivered by the contractor upon construction completion.
- Durability Assessment Reports for Detailed Design (DAR: DD, refer Section 27.6.7) and Verification Report (DAR: VR, refer Sections 27.6.8 & 35.6.3) to provide guidance on operational maintenance.
- Details of inspection audits and tests to be undertaken, plus the associated assessment criteria.
- Details of maintenance materials and methods to be adopted.
- LCM planning based on anticipated materials deterioration and costs to achieve acceptable performance predicted at appropriate intervals not exceeding 10 years.
- Anticipated remedial works required during maintenance periods.
- Minor repairs to be completed within scheduled maintenance periods.
- Keeping records of defects and repairs throughout structure life-cycle..

Figure 38.1-1 presents a flow chart of the processes delivering durability during operational maintenance activities, this guidance is generally applicable to life-cycle management / operational maintenance activities.

The operational maintenance provider shall be given the Durability Assessment Report: Detailed Design (DAR: DD, refer Section 27.6.7) and the Durability Assessment Report: Verification report (DAR: VR, refer Sections 27.6.8 & 35.6.3); plus associated documentation such as the Birth / Re-Birth Certificate / maintenance plan / concrete structure management plan.

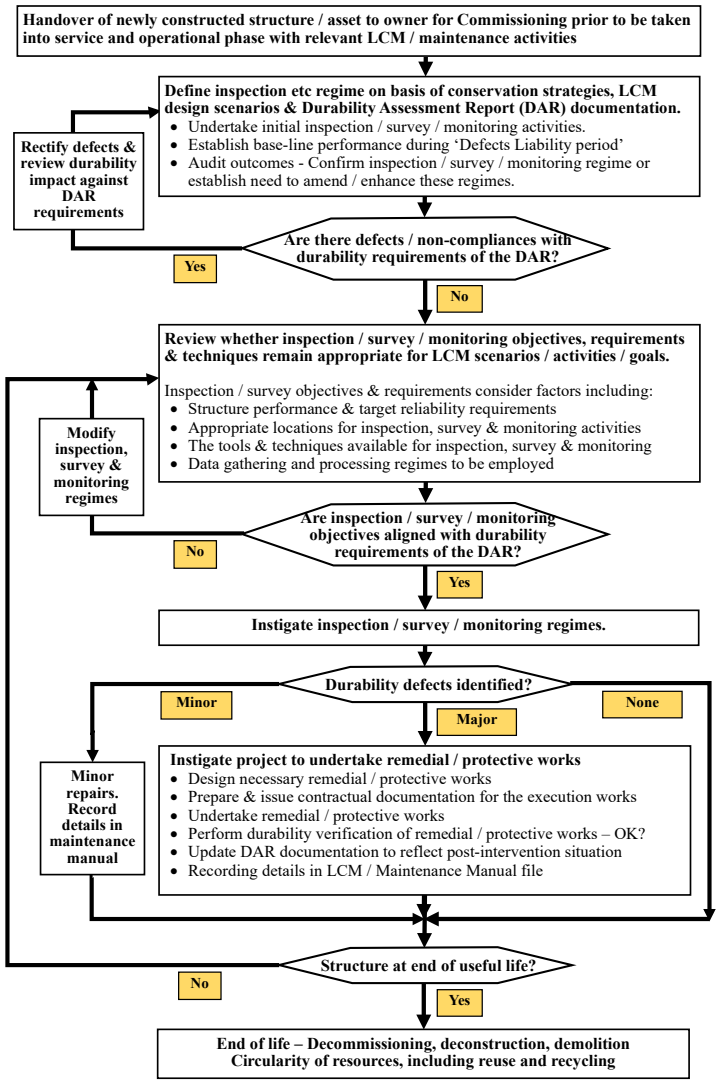


Figure 38.1-1: Durability process during LCM / design service life operational maintenance (after CIA Z7/01 [38-18], Fig 2.3)

The condition survey results are usually compared to specified requirements and standards for determining whether the behaviour of the item or activity is meeting those performance targets.

Identification and evaluation of active damage or deterioration processes is essential. Observable / measurable indications of such processes can provide a basis for anticipating the future development of damage / deterioration / change in behaviour of a structure / constructed asset, along with the associated effects upon its in-service performance.

Surveys may involve visual inspections, measurements, tests or some form of gauging applied to certain characteristics in regards to objects (e.g. the structure or structural components) or activity. Testing undertaken is usually non-invasive or non-destructive in nature. Monitoring often involves the automated recording of performance related data for the structure and possibly some degree of associated data processing. Strictly this does not need to be so, as there are a variety of means of gathering appropriate data.

Engineering evaluation and interpretation of the data obtained is central to the process of converting data into useful context related information, which may take the form of performance indicators, and their subsequent assessment.

Observations and data obtained from surveys, testing, monitoring and related activities can be used as the basis for Performance Indicators (PIs) providing an indication of the fitness for purpose of a structure / constructed asset or their component parts. This use of PIs is examined in Strauss et. al. [38-19].

It may be possible to gather indirect indications of performance through ancillary behaviours. For example, the performance of foundations might be inferred from the measurement of differential structural movements.

38.2 Condition survey and monitoring

38.2.1 Condition survey and monitoring activities

In this Model Code, condition survey and monitoring is understood to include all activities performed to gather information regarding the form and nature of the structure, current or potential deterioration mechanisms and/or change in performance of the structure or service conditions, which are needed to evaluate conformity with the design (data) for actions and/or material and/or product properties. As such, these could be expected to utilise a variety of procedures including structural health monitoring and associated techniques.

Accordingly, condition surveying is understood to mean an organized examination by a range of appropriate techniques and procedures undertaken on a periodic basis during the service life of the structure, while condition monitoring / structural health monitoring is understood to mean the process of observing such parameters on a quasi-continuous basis.

Significant changes in the condition of the structure, and/or the related data sets / performance indicators, may be indicative of a developing failure. However, such deductions can only be established through an adequate process of data assessment and evaluation, which enables an appropriate context-based engineering interpretation to be made of the indications obtained.

To gain the greatest benefit, the surveys, testing and condition / structural health monitoring activities shall be carried out from an early stage in the service life of the structure.

Regimes for surveys and/or monitoring must be devised taking account of the conservation strategies selected for the assets / structures or their component parts. Conservation planning recognises the following three conservation categories:

Category A - Proactive conservation measures;

Category B - Reactive conservation measures; and

Category C - No conservation measures practicable.

The associated potential approaches are described in Table 38.2-1.

Section 5.2 of Strauss et. al. [38-19] also gives consideration to correlations between damage / deterioration processes and the potentially observable indications of such processes which arise when they are active.

Proactive conservation requires more than just periodic visual inspections because environmental factors slowly and subtly change the internal conditions within the outer zone of the concrete (e.g. typically within the concrete providing the cover to the embedded reinforcing bars, such as by undertaking testing for the chloride ion content). As a result, concrete structures tend to experience changes in their condition which initially do not produce visible indications. Thus, these changes do not manifest themselves as observable deterioration. Accordingly, these changes cannot be recognized on the basis of visual observations alone.

It is only perhaps many years later that physical deterioration becomes sufficiently developed to be visibly evident (e.g. cracking and spalling of concrete arising from the corrosion of embedded reinforcing bars). As these changes and deterioration processes generally develop progressively from the surface of the concrete, and often over considerable periods of time, information on the progress of these actions is generally gathered using appropriate forms of non-destructive testing or sampling of the concrete. Appropriate data can also be obtained by suitable monitoring techniques.

Judgment criteria are further discussed in section 38.5 on condition evaluation and decision-making.

Table 38.2-1: Survey and monitoring regimes (CAL0 to CAL3) for different conservation strategies (A to C).

Conservation strategy	**Survey and monitoring regime (associated condition assessment level)**
Category A - Proactive conservation measures	Planned periodic and systematic survey / monitoring of parameters relevant to design, in particular to the deterioration processes which are critical for the verification of the condition states associated with durability. Applied to both structures and / or their structural elements. (condition assessment level: CAL3)
Category B - Reactive conservation measures	Planned periodic survey (eg. visual inspection by qualified personnel) applied to both structures and / or their structural elements. No systematic inspection, testing or monitoring. (condition assessment level: CAL2)
	Ad hoc survey and testing/investigation applied to both structures and / or their structural elements. No systematic inspection, testing or monitoring. (condition assessment level: CAL1)
Category C - No conservation measures	No direct survey, testing or monitoring applied to either structures or their structural elements. (condition assessment level: CAL0)

38.2.2 General flow of condition survey process

The general flow of the life-cycle conservation management process is shown in Figure 8.3-2. This is complemented by that in Figure 38.2-1, which presents a flow diagram illustrating the decision-making process for surveys/monitoring and interventions; with particular attention being given to circumstances where judgments need to be made.

When making surveys/monitoring or investigations of a deteriorated structure, consideration shall be given to whether the structure poses an immediate threat to the environment, its users, any third party, or the survey/investigation team. The consequences of some form of failure / collapse could potentially be more severe for a structure in a higher Consequence Class.

In such a situation, suitable emergency measures and temporary works may need to be taken immediately, in order to minimize risks and to attempt to avoid potential collapse and so on.

Depending upon the circumstances, it may be appropriate to take a number of immediate actions to minimize the risk of injury or worse. These measures might include closing the structure to further use, evacuating the structure and the surrounding area, setting up an exclusion zone to control access to the structure and the area which might be affected by any collapse and/or the release of stored/contained materials and shutting-off services such as piped gas which might be damaged and create secondary hazards.

Detailed investigations may be required once the structure has been "stabilised" and put into a "safe" condition.

Figure 8.3-1 illustrates the time dimension associated with the survey processes portrayed in Figure 38.2-1. Figure 8.3-1 shows an idealized representation of the life-cycle performance of a structure from the time of its construction up to the time when the first remedial intervention was undertaken so that the structure could meet its design service life. The various surveys/investigations portrayed in Figure 38.2-1 would correspond to the life-cycle monitoring activities (Stages 2 and 4) shown in Figure 8.3-1. The intervention activities portrayed in Figure 38.2-1 would correspond to Stage 3 in Figure 8.3-1.

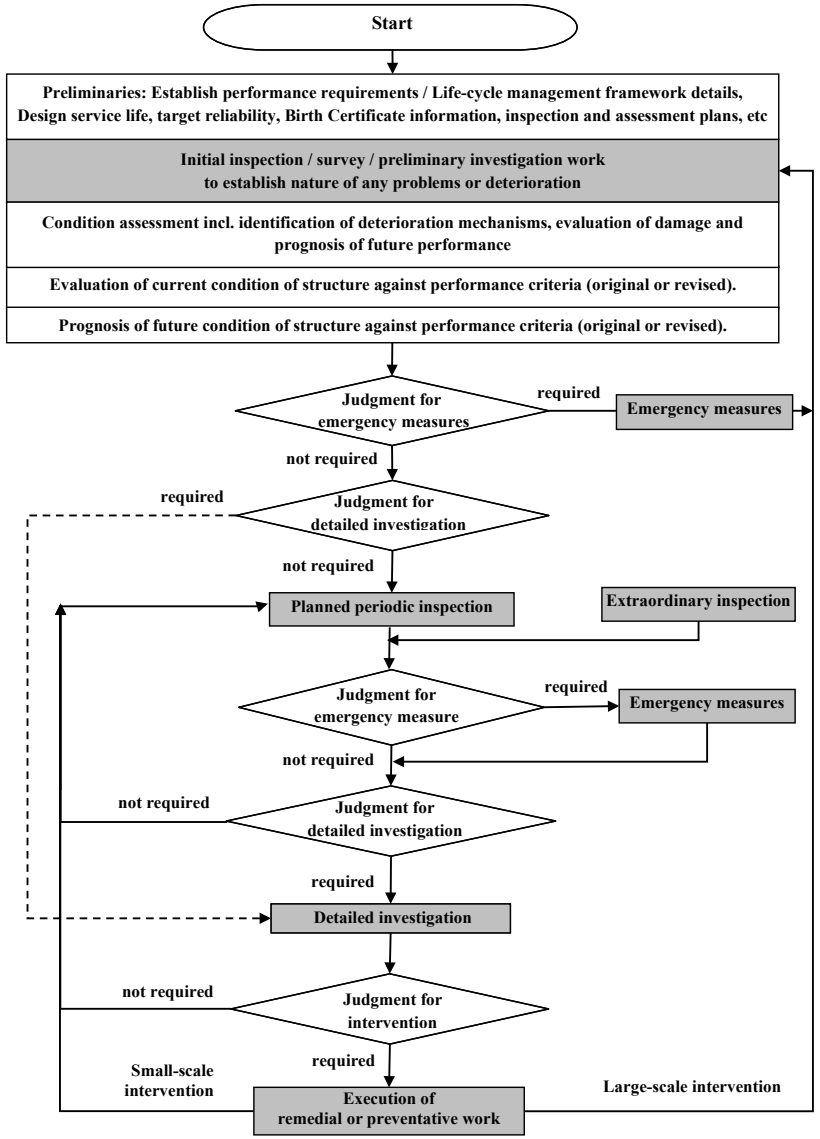


Figure 38.2-1: General flow of condition survey where a proactive conservation strategy implemented after construction.

Monitoring results shall contribute to this process and the associated outcomes e.g. if the monitoring data values exceed predetermined threshold value(s), an investigation would need to be performed.

Figure 1.1 of *fib* Bulletin 33 [38-20] presents illustrative hazard scenarios for prestressing steel in a 'typical' concrete box girder bridge, providing indications of potentially 'weak points' where water (possibly contaminated with chlorides and/or other substances) could gain access to the tendons, cause corrosion and thereby adversely affect the structural performance / safety of the bridge. Figure 1.1 of *fib* Bulletin 33 [38-20] illustrates how a risk-based approach to identifying hazards and potential 'weak' or vulnerable points, combined with an evaluation of the relevant potential failure modes and the resulting consequences, would assist in identifying the most appropriate locations for inspections, testing and monitoring activities.

The locations where a structure is most vulnerable to localised deterioration or damage may well also be the most difficult to inspect and to undertake testing and monitoring upon. Half-joints in bridges are a notorious example of a combination of a severe exposure environment, structural sensitivity to deterioration / damage and extremely difficult / limited access.

Vulnerable zones are those segments and/or elements of a structure in which damage or deterioration / poor durability can have the greatest adverse effect on safety and/or serviceability. The Strauss et. al. [38-19] describes some procedures for the determination of vulnerable zones in structures, along with associated performance indicators. These are illustrated by supporting examples.

Descriptions of the Environmental Classes adopted in MC2020 are provided in Section / Table xx.xx **ADD CROSS-REFERENCE**

38.2.3 Locations for surveys and monitoring activities

Locations where inspection, testing and condition monitoring activities are to be undertaken must be carefully selected so that the desired information about the deterioration of materials and/or structural performance can be obtained, keeping in mind factors such as:

- the environmental conditions, usage conditions and the nature of the threats / hazards posed;
- the likely mechanism(s) and rate of deterioration;
- locations of greatest hazard (eg. the locations subject to the most severe exposure conditions, potential deterioration / damage 'hot spots');
- locations where the structure is most vulnerable to deterioration or damage (eg. half-joints in bridges); and
- the conservation strategy and tactics, and the inspection testing and monitoring regimes defined at the time of design or re-design.

Consideration shall be given to employing a risk-based approach to identify the most appropriate locations for surveys, testing and monitoring activities – see adjacent discussion in the left-hand informative column.

While similar inspection, testing and monitoring activities may be employed at different locations, criteria for evaluating the results may differ where a structure is to meet either:

- its design service life, as envisaged at the time of design; or
- revised performance requirements, which might include an extension of service life, revised loading and/or other actions or changed functionality requirements.

To appreciate the potential diversity of the environmental conditions and the nature of the threats posed, it may be appropriate for environmental influences to be considered at different scales. These range from the macro-scale (affecting the overall structure), through the meso-scale (affecting, say, an individual element or component), down to the micro-scale (localised influences affecting a small part of a structure, e.g. a particular joint).

Information acquired by inspection, testing and monitoring the condition of the structure can greatly improve the accuracy of performance prediction by more accurately assessing the variability of the input parameters, which are typically assumed to be random variables. This can help reduce the uncertainty associated with the input parameters and aid characterization of their variability, as illustrated in Figure 38.2-2.

Further work is required to standardise inspection / monitoring procedures and to assessing the reliability of inspection / monitoring to be employed.

For more information regarding specific tools and techniques for inspection, testing and monitoring, see:

- *fib* Bulletin 17 [38-1];
- *fib* Bulletin 22 [38-2];
- *fib* Bulletin 59 [38-21];
- *fib* Bulletin 93 [38-5].

Figure 38.2-2 (c) illustrates how the estimated timing of required interventions may evolve during periodic inspection and assessment, which portrays a change (increase) in the estimated period required to the first intervention works. This is illustrated by the change (increase) in the mean of the estimated period to the first intervention works.

38.2.4 Tools and techniques for surveys and monitoring

Tools and techniques for inspection, testing and monitoring could involve a wide range of procedures. Typically, they are likely to include a combination of visual observations, material sampling and possibly selected non-destructive and non-invasive testing methods, as well as procedures including structural health monitoring and associated techniques.

Ideally the approaches chosen shall be appropriate to:

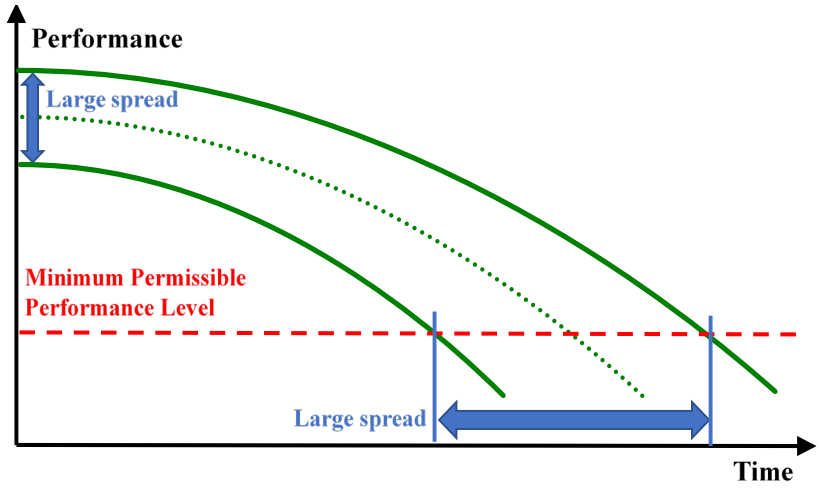
– the required reliability of the information to be obtained;

– the conservation strategy and tactics, and any inspection, testing or monitoring regimes defined at the time of design or re-design.

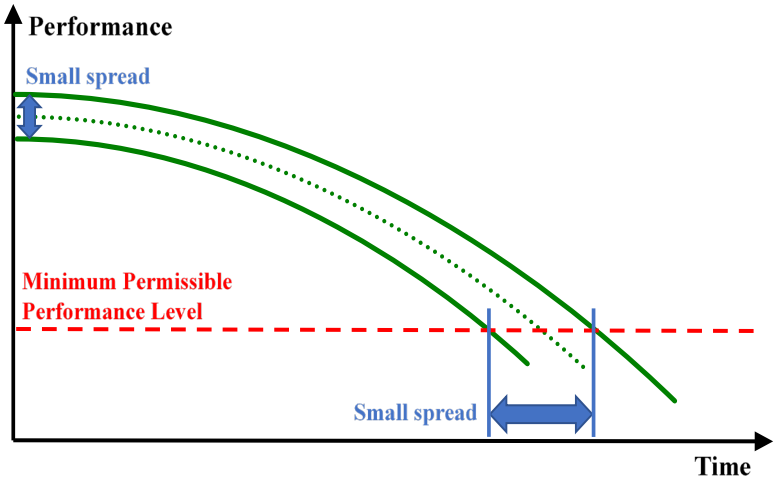
Figure 38.2-2 (a)-(d) illustrates how the accuracy of the data gathered, and the assessment made using it, can influence the estimated timing of required interventions, which can significantly change when the level of uncertainty in the data and the assessment outcome are taken into account. In turn, these factors may influence considerations made about the tools and techniques to be employed for inspection, testing and monitoring activities.

Higher levels of measured data accuracy generally reduce uncertainty. This greater precision, combined with shorter data acquisition intervals, will generally improve the accuracy of performance prediction made using the data obtained, which is also greatly influenced by the accuracy of the prediction models employed – see Figures 38.2-2 (a) and 38.2-2 (b), which provide greatly simplified representations of these influences.

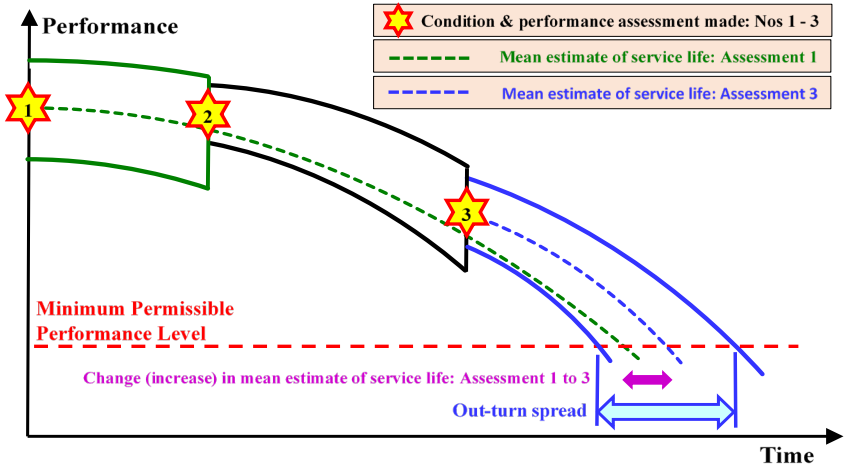
Figure 38.2-2 (c) seeks to portray the (greatly simplified) potential evolution of circumstances under a periodic inspection and assessment regime, with Figure 38.2-2 (d) seeking to portray the situation for continuous monitoring – SHM (this representation is also greatly simplified).



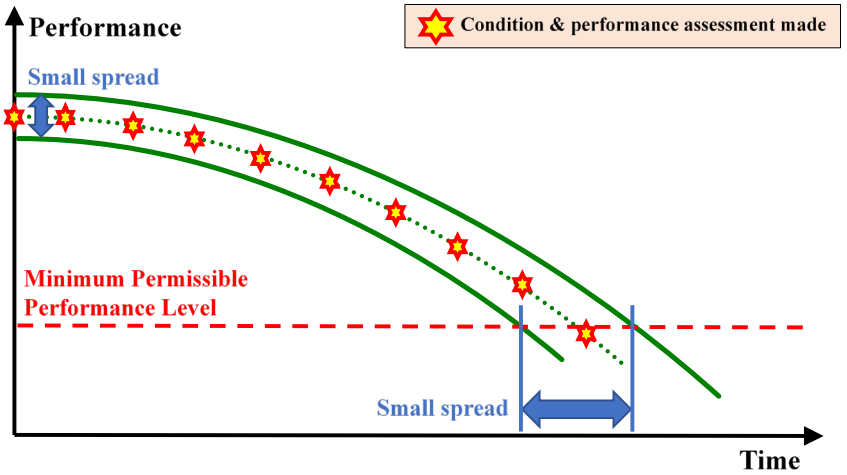
(a) Lower accuracy in test / inspection data and lower precision in performance modelling.



(b) Higher accuracy in test / inspection data and higher precision in performance modelling.



(c) Periodic inspection & assessment - Modest accuracy in monitoring data, extended intervals between data acquisition and modest precision in performance modelling.



(d) Continuous monitoring - SHM - Satisfactory accuracy in monitoring data, short intervals between data acquisition and higher precision in performance modelling.

Figure 38.2-2: Uncertainty in estimating the timing of making an intervention – Illustrating the conceptual influence of test / inspection data accuracy, changes in data acquisition intervals and the accuracy of the prediction models employed.

The process of structural assessment, particularly of a deteriorated structure, may involve a number of cycles of gathering and evaluating information; especially if these activities are taken forward using an incremental approach. Whilst this approach is aimed at minimising the investigation effort expended, the approach may not always represent good value when associated factors, such as the cost of gaining appropriate access to the relevant zones of the structure / constructed asset, are taken into account.

Considering the potentially high cost of gaining access, it may be more appropriate to take the opportunity to perform additional inspections / testing to gather a variety of complementary data and samples. The additional cost may be relatively small.

Further details about the information required to perform a structural assessment are given in chapter 28.

An inspection is a primarily visual examination, often at close range, of a structure or its components with the objective of gathering information about their form, current condition, service environment and general circumstances.

A survey is a process, often involving visual examination but which may utilize various forms of sampling and localized condition testing (e.g. measurement of depth of cover to reinforcement), whereby information is gathered about the form and current

38.2.5 Gathering data for LCM purposes

38.2.5.1 Introduction

The process of carrying out a structural assessment involves gathering information about the existing structure. This requires the identification, collection and the evaluation of information already existing about the structure concerned, coupled with an assessment of the nature and extent of any additional information required to complete the task. The types of information and understanding required include:

a. General information and history;

b. Required input data for defining structure, its condition, models etc ;

c. Categories of input data (geometry, construction details, materials etc);

d. Knowledge levels and implications thereof.

e. Representative values of material properties and any changes with time

In the case of a deteriorating concrete structure, the assessment will undoubtedly require some form of inspection and / or testing to define the mechanism(s) and severity of any structurally significant degradation, as well as to be able to make a prognosis about future condition.

It is often appropriate to limit the extent of structural assessment and associated investigation activities when the structure has clearly demonstrable and adequate reserves of strength (or of another property of interest) or where the use of more refined methods do not significantly improve the outcome.

38.2.5.2 Survey and monitoring regimes

Data may be gathered from the structure for condition assessment purposes by a variety of techniques which are used for undertaking inspections, measurements, testing and monitoring activities. As such, these could be expected to utilise a variety of procedures including structural health monitoring and associated techniques.

condition of a structure or its components. The testing undertaken is usually non-invasive or non-destructive in nature.

An investigation is a systematic process of inquiry undertaken to establish the specific details of a situation without restriction upon the methods used. For example, it may involve inspections, measurements, sampling and various forms of local condition and global response testing to establish the form and current condition of a structure. The process may also provide data for a prognosis of future condition and performance of the structure.

Structural and condition monitoring typically involves gathering data by a range of possible sensors, techniques and procedures to aid the life-cycle management of a structure. These activities typically involve the automated recording of performance data for the structure, with increasingly sophisticated procedures being employed in the associated data processing to extract relevant information / make the predictions required for LCM purposes. There is an ever-greater variety of techniques / sensor types for gathering data.

For more information regarding regimes for inspection, testing and monitoring, see:

- *fib* Bulletin 22 [38-2];
- *fib* Bulletin 44 [38-3];
- *fib* Bulletin 93 [38-5];
- *fib* Bulletin ZZ [38-6];

Initial survey shall include not only the inspection of external appearance of the structure, but shall address other “out of sight” parameters (e.g. cover to reinforcing bars) and workmanship issues. Data from the initial survey can be of use for subsequent routine/regular, detailed and extraordinary inspections/surveys.

An initial survey related to the development of a Birth Certificate for the structure shall be conducted as the first step in collecting information on the structure. While it is assumed that the design and construction of the asset / structure has been undertaken in accordance with the principles of quality and information management set down in

Survey and monitoring regimes can be classified on the methods to be used, their frequency, the planning and timing of inspections/surveys, as shown in Table 38.2-2.

Table 38.2-2: Gathering data for life-cycle management purposes for various conservation strategies

Gathering procedure (Survey and / or monitoring regime type)	**Proactive conservation measures**	**Reactive conservation measures**
Initial survey / monitoring	planned	planned
Routine (or regular) survey / monitoring	planned	planned or ad hoc
Detailed investigation, perhaps with survey / monitoring	planned	planned or ad hoc
Extraordinary investigation, perhaps with survey / monitoring	ad hoc	ad hoc

Initial survey / monitoring

An initial survey is carried out using appropriate tools and techniques in order to examine whether or not the structure is adequately constructed or whether previously specified preventative/remedial actions have been implemented satisfactorily.

It may also provide the means of collecting basic data about the as-built or repaired structure in order to initiate appropriate maintenance or preventative intervention works, or as part of a previously unplanned survey regime.

Three different types of survey which potentially could be required for the initial inspection, are detailed below:

chapter 10 of MC2020, the process of preparing a Birth Certificate shall find evidence to confirm this.

In the case of an existing structure requiring a remedial intervention, such a process would lead to the development of a re-Birth Certificate for the structure. This would be expected to link with the adoption of a proactive conservation strategy for the structure. Thus, such an initial survey shall be conducted as the first step in collecting appropriate information upon the structure.

For further details see *fib* Bulletin 93 [38-5].

The choice of specific tools/techniques to be used and the frequency of such inspections/surveys shall consider factors such as the likely mechanism(s) of deterioration, environmental conditions and the importance of the structure. It is anticipated that these issues would have been considered at the time of design or re-design, when an appropriate conservation strategy would be selected.

It is desirable that the structure or the components concerned be inspected at close range (arm's length). Some form of non-destructive testing, such as estimating the depth of carbonation or the penetration of chlorides, is likely to be required in order to understand the progress of initiation of deterioration. Not all survey regimes need to have the same objectives and coverage. Thus, the focus of particular surveys may differ and the scope and methods used could be expected to vary depending upon the nature of the structure or the component being inspected. Not all parts will need to be inspected at the same frequency or in the same degree of detail.

In some instances, it may be considered appropriate that some of the detailed survey / monitoring procedures described in the following paragraphs be repeated at an appropriate frequency as part of a regular survey / monitoring regime.

Section 7.5 of *fib* Bulletin 59 [38-21] provides guidance on determining / setting the timing of survey and monitoring activities, along with the use of Bayesian updating and probabilistic methods in the through-life management of concrete structures.

This type of activity is intended to obtain detailed data regarding the deterioration and performance degradation of the structure or the component(s) concerned. The approach will often be hypothesis driven on the basis of the deterioration/damage

a. Survey carried out prior to the in-service use of a newly constructed structure upon completion of construction and associated commissioning works (Birth Certificate);
b. The first survey of an existing structure which has been subjected to a major preventative or remedial intervention, or substantially renewed (re-Birth Certificate);
c. The first survey of an existing structure carried out for the purpose of an intervention made in the absence of previously collected data.

Routine (or regular) survey / monitoring

Follow-on routine (or regular) surveys shall be carried out periodically at intervals defined by the service life design to identify changing internal conditions within the concrete, indications of deterioration and their first observed occurrence.

These inspections/surveys must be carried out using appropriate tools and techniques to meet the objectives of the particular inspection.

In the absence of a specified survey programme, knowledge of the design service life (the period which the structure is expected to be used without major repair being necessary), coupled with information derived from the Birth Certificate (eg. the hazards / service environment, cover depth values, concrete properties, etc) typically allow an initial estimate to be made of the appropriate timing of the first round of inspections/surveys/tests.

The period to the next survey might be estimated considering the anticipated reduction in the reliability index (β-value) with time, with the survey being planned to take place well before the β-value of structural components fell below the specified target value (to allow time to take responsive actions in advance of breaching any critical performance requirements). The chosen timing and the target β-value would depend upon the limit state being considered - with a higher target value / more time being adopted for the ULS and with lesser values for SLS and CLS circumstances.

After an intervention, it may be possible to adopt a similar approach employing the re-Birth Certificate.

Detailed investigation

A detailed investigation must be carried out to obtain sufficient specific information concerning the deterioration of materials and/or structural performance to facilitate detailed evaluation or re-design of the structure or

mechanism(s) acting or suspected. These activities are likely to be guided by records of past inspections/surveys/testing regimes, as well as those of previous maintenance, preventative and remedial works interventions. This information would typically be reviewed in conjunction with the results of deterioration prediction/damage modelling.

The detailed quantitative data gathered might include characterization of the concrete surface, the quality of concrete, the depth of carbonation and chloride concentration in the concrete, the degree of corrosion of the reinforcing bars and/or similar parameters. Furthermore, the detailed inspection will need to include the data on the environmental conditions, such as temperature, humidity, airborne chlorides, loads and actions acting on the structure and so on.

The investigation will require use of appropriate tools and techniques, such as non-invasive and non-destructive testing and core drilling for material sampling, with the aim of locating and defining deterioration, damage and defects. Provision for close access to parts of the structure will be required.

Extraordinary investigations are usually carried out when the structure is subjected to some form of extreme event such as a natural disaster, perhaps involving an earthquake or typhoon, fire or collision involving vehicles or ships. Such an investigation is intended to accurately ascertain the state of the structure and judge if a remedial intervention is necessary. Extraordinary investigations may also be carried out after an accident associated with the deterioration of a structure, perhaps caused by a piece of spalled concrete falling causing injury to a passer-by.

An extraordinary survey/monitoring regime would usually use a similar methodology to routine or regular survey/monitoring. If further information upon the form or condition of a structure is judged to be necessary, a detailed investigation might then follow. Extraordinary investigations may also be carried out for a particular class of structure after an accident due to the deterioration of such a structure, as noted above. Such an investigation may be carried out for similar structures, where the same accident could occur.

Some degree of investigation / due diligence review shall be undertaken of the available documents to verify that they provide a satisfactory / acceptable (valid) representation of the structure under consideration.

components concerned. There are various circumstances when a detailed investigation may be required including when:

- signs of significant deterioration or a change in the performance level are observed during a routine/regular survey regime;
- a routine/regular survey regime is unable to provide the information required; or
- it is suspected that the structural integrity of the structure could have been adversely affected by the extent or severity of deterioration/damage.

More information is required to decide whether a major remedial intervention is necessary and upon its potential scope. In a detailed investigation the items for inspection and/or the locations where testing and/or measurements are to be made, must be appropriate both in number and the characteristics of the mechanism(s) of deterioration/damage acting or suspected.

Extraordinary or Special investigation

An Extraordinary investigation regime will usually be instigated after the structure has been subjected to an accidental load/action or some form of extreme event, such as that caused by an earthquake, storm, flood, fire, impact by a vehicle or ship, to assess the extent of the damage, the immediate safety of the structure and the need for a remedial intervention.

This type of investigation may also be referred to as a ‘Special’ investigation.

38.2.5.3 Study of documents

Much useful information about a building or structure is contained in the documents prepared during its design and construction (eg. design calculations, material specification, material delivery documentation, etc). Reports of earlier surveys and maintenance works, plus documents concerning subsequent

In addition to information about the specific structure in question, there is generally a large volume of supplementary explanatory information which relates to systems of construction, codes of practice, contemporary advice and guidance notes, textbooks and papers and associated matters such as patents. This type of information is most likely to be available from professional institutions and specialised sources such as research establishments, trade associations, libraries and manufacturers.

For hidden structural elements, such as foundations and reinforcements, old drawings may be the only practical source of information. Details of the exposure of the structure to natural hazard needs will need to be investigated and an assessment made on the basis of the most recent observations and data (e.g. update of magnitudes, re-assessment of site conditions, etc.).

Generally, the engineer will have to search a number of potential sources for information about an existing structure / constructed asset. Section 9.8.2.2 of *fib* Bulletin 62 [38-4] provides guidance on the potential sources of information about existing structures / constructed assets, as well as on the types of documents which may be of assistance.

Parameters which might be investigated, along with some corresponding investigation methods and damage / deterioration indications, include:

- Verification that the asset / structure being inspected is consistent with the available documentation for the design and any structural alterations / interventions made whilst in-service. This is achieved by checking site findings with available specifications, drawings and other documents;
- Structural integrity (eg. for hidden damage or inhomogeneity) using, for instance, impact echo testing;
- Material strength using tension and compression tests on material samples taken from the structure, plus on-site testing using sclerometer hardness test, rebound test hammer, pull-out tests, pull-off tests, etc.;
- Cross-sectional and longitudinal geometry changes (ie. effects of damage) due to overloading (eg. visible cracks, ruptures), other damage processes (eg. fatigue cracks) and from deterioration processes (eg. corrosion, spalling, delamination) using lasers, ultrasonic devices, slide gauges, electronic gauges, etc.;

modifications and repairs will provide an insight into the history, use, maintenance and problems encountered with the structure whilst in service.

Archive documents shall be examined since they may reveal assumptions made regarding loading situations and material properties.

Past and present loading codes can provide guidance on the nature and magnitude of loads which a structure might have been designed for. Information on environmental conditions / loadings may be available from design and / or re-design studies (possibly for adjacent structures), inspection and monitoring reports.

Resistance properties, such as material characteristic strength, geometric dimensions (e.g. cross-section size, bar diameters, cover depths) and associated structural parameters, can be obtained from codes, drawings and other design specifications (e.g. static calculations, geotechnical reports), from construction documents (e.g. material delivery documentation) and from reports of earlier surveys and maintenance / intervention works.

38.2.5.4 Methods for inspections and material testing

To reduce uncertainties about load and resistance parameters, site specific data shall be used within the assessment process whenever practicable. This data is obtained from site inspections, sampling and material testing. There are a large variety of methods which may be employed, starting with simple visual or photographic inspection procedures and extending to various high-end non-destructive/non-invasive imaging techniques.

Inspections are especially useful for the detection and investigation of deterioration/damage processes, such as corrosion and fatigue, and for detection of changes in the structural systems. Therefore, it is necessary to conduct inspections regularly/periodically.

Material tests are undertaken to determine the strength parameters (eg. compressive, tensile, shear strength) of the construction materials used. The tests which can be conducted on site can be semi-destructive or non-destructive in nature. Alternatively, material samples can be recovered from the structure and tested at an approved / certified laboratory.

The testing methods employed for ordinary condition surveys include those listed in:

- Parameters influencing the dead load and the superimposed dead load (eg. material densities, permanent equipment, unplanned variations / errors in structure geometry);
- Parameters influencing durability / length of service life of the structure (eg. environmental conditions, carbonation depth, chloride content of concrete, achieved concrete cover depth, location of zones of low concrete cover depth) using pH-test, phenolphthalein indicator test, quantitative chloride analysis of samples, etc.; and
- Serviceability (eg. surface crack widths, excessive deflection and/or deformed profile, excessive vibration).

Structural health monitoring (SHM)

SHM has become increasingly affordable for owners and responsible authorities as a long-term monitoring technique for the continuous observation and evaluation of structural integrity. SHM involves the continuous or periodic measurement of time-variant parameters such as displacement, strains and stresses, crack width (as a proxy for damage) and vibration characteristics; with the aim of detecting changes in structural properties. In some installations, alarms can be triggered when condition states are reached or exceeded.

The objectives for a SHM system commonly include:

1. Detecting the existence of damage to the structure;
2. Locating where the damage has occurred;
3. Identifying the types of damage experienced; and
4. Quantifying the severity of the damage.

SHM is being applied to newly constructed structures as a tool for life-time monitoring. In these circumstances the technique can estimate the structural properties of the original structure in an undamaged state. This allows future data to be compared with this benchmark, facilitating the direct detection of changes in the structural properties, which makes the prediction of future performance easier.

- Table 30.11-xx: *Types of non-destructive testing methods (NDT);* and
- Table 30.11-xx: *Types of semi-destructive testing methods (SDT).*

However, it is possible for these testing / inspection methods to also be employed as part of a monitoring procedure when applied as components of a periodic / repetative survey regime.

For further details on fatigue life verification and testing procedures see section 30.2.2.

38.2.5.5 Structural monitoring and performance testing

Monitoring / inspection / testing regimes may be differentiated on the basis of the measurement of load related effects as follows:

- Load-independent procedures: Periodic inspections employing visual examination, combined with non-destructive testing (NDT) and semi-destructive testing (SDT) techniques.
- Load-dependent procedures: Observation of the structural and other responses of a constructed asset to loads and environmental effects by continuous monitoring using installed technical measuring equipment.

Further details are given in Section 30.11 ??

Amongst other benefits, such monitoring / inspection / testing regimes can be helpful in facilitating:

- identification/verification of actual structural performance parameters;
- identification of loads and environmental actions;
- monitoring of the structural reaction to loads and other influences;
- recognition of active degradation/damage processes;
- detection of structural defects/damage; and
- documentation of historical performance and the prediction of its remaining service lifetime.

It is important to recognise that measured data obtained by a structural monitoring system does not necessarily display all aspects of the actual behaviour being experienced by a structure (i.e. reality). Accordingly, care needs to be taken with the selection, positioning and orientation of the sensors forming the monitoring system, as this will influence the sensitivity of that monitoring system to the behaviours and effects being experienced by the structure. Care is also required to ensure the correct handling and processing of the recorded / measured data.

For further details upon SHM refer to CIRIA Report C788 [38-22]. Ye et al [38-23] discuss the state-of-the-art in objective performance and condition evaluation of constructed systems in the context of technologies for the condition and performance evaluation of highway bridges.

It shall be recognised that environmental conditions, especially temperature, can have a large influence on the static and dynamic measurands. These influences must be considered when evaluating the structural properties.

When system identification is applied periodically or even permanently, the time variance of structural properties by the deterioration processes or other damage causing events, can be identified and monitored. The structural model can then be updated using the new measurement data taken.

Structural identification seeks a mechanistic and quantitative description of a structural system, typically in the context of a physics-based finite element model, which is now commonly referred to as being a digital twin.

The best practices for structural identification were summarised as a six-step analysis-experiment-decision integration cycle by the ASCE Technical Committee on Structural identification of constructed systems (ASCE [38-24]) as follows:

1. Observation and conceptualisation;
2. A-priori modelling;
3. Controlled experimentation;
4. Processing and interpretation of data;

In situations where the measured data obtained from the monitoring system does not bring the expected results or does not provide an adequate understanding / clarity of the structural behaviour(s) being experienced, consideration shall be given to undertaking performance testing.

Performance testing entails the measurement of the static and/or dynamic behaviour of the structure, which provides information about actual structural properties. The measured values can then be compared with the required structural properties. The measurement can be performed once, periodically or on an on-going basis (i.e. continuously).

Caution shall be exercised when interpreting the collected data. Structural modelling can be a powerful tool to assist with the interpretation of the measured data and help understand the behaviour(s) of a structure.

38.2.5.6 System identification by static and dynamic response measurements

If dimensions and material properties of the real structure cannot be estimated by direct measurement or testing (e.g. due to inaccessibility, hidden damage, etc), structural properties such as stiffness of structural members and joints, flexibility of hinges or bearing conditions can be estimated by system identification testing procedures. These procedures are also an efficient tool for damage detection and monitoring for damage evaluation (quantification).

With this structural identification procedure static properties like displacements (e.g. deflections, inclinations) developed under defined loads, as well as dynamic properties such as natural frequencies and mode shapes, are measured upon the actual structure. The structural system model is then refined, so that the model reflects the same behaviour characteristics as those observed for the actual structure.

5. Model calibration and parameter identification; and
6. Utilisation of the calibrated model (i.e. Digital-twin for simulations and decision-making).

Controlled proof-load testing (refer section 30.11.??) can be especially useful in demonstrating the location and impacts of damage and evaluating the remaining load capacity. However, such an approach can only be considered for the most critical cases due to its high cost and disruption to the operation of the highway. Operational monitoring was shown as a sufficient and much cheaper alternative for structural identification, permitting the development of a 3D digital twin of the test bridge, which proved critical in identifying the root causes of its deficiencies and formulating meaningful interventions.

Stiffening effects produced by non-structural floor screeds, for example, can be considerable. Tests (Moss 1993 and Moss 1994 [38-25][38-26]) have shown that a four-fold increase in stiffness was achieved when a bonded screed was applied to some forms of floor construction. However, the associated increase in the load capacity of such floors was much more modest, being only some 20% - 30%.

Laboratory testing (Moss 1993, Moss 1994 and Moss 1995 [38-25][38-26][38-27]) demonstrated that these mechanisms tended to breakdown as the ultimate load condition is reached. Thus, whilst these mechanisms added significantly to the observed stiffness of the tested floors at certain stages of applied loading, they cannot be relied upon to be present when the overall failure condition for the floor is reached.

Accordingly, it is unwise to extrapolate load test results beyond the load level applied to a structure during a proof load test.

The same proviso also applies when considering extremely good past behaviour of (say) a floor or roof structure. Excellent past behaviour might give false (excessive) confidence that the structure would exhibit an enhanced (higher load) capacity when the ultimate load condition was reached.

38.2.5.7 **Proof load tests**

The application of defined loads to a structure to verify its load carrying capacity is a powerful tool for evaluating existing structures.

Different types of proof load tests are employed depending on the limit state which is being investigated. For the serviceability limit state, the load effects are measured after application of the proof load. There will be the verification of compliance with the limit state if the measured response does not exceed the magnitude defined for the limit state.

In the case of proof loading at the ultimate limit state, the fact that the structure (or elements of the structure) does not fail during the test verifies that the limit state was not exceeded.

Composite action between structural and non-structural components may mean that the structural components appear to perform much better than if they were tested in isolation. Load sharing occurring between individual structural components forming a larger element can produce similar results, floors formed from individual precast concrete beam of plank components are a particular case. Such influences may need to be taken into account when load testing regimes are designed and when interpreting the results of load tests.

Non-structural elements often have a significant influence upon the observed stiffness and strength of both individual structural members and the overall structure under normal and accidental load situations, especially where damage has been caused and alternative load paths are mobilised. Some types of internal partitions have the ability to provide (alternative) secondary load paths in the event of damage to the main structure of a building.

There are a number of basic considerations when defining a monitoring system including, for example, the:

- Nature of effects / the range of parameters to be monitored;
- Frequency of inspections/measurements;
- The types of elements which may be monitored and/or the situations or circumstances which might arise; and
- The various means by which the data can be transmitted.

Considering the above factors in turn, in respect of the nature of effects / range of parameters to be monitored, the types of occurrences might include:

- Physical parameters (both static and dynamic);
- Chemical processes involving both concrete deterioration and corrosion of reinforcement and prestressing components; and
- Biological phenomena; plus
- Synergetic / coupled or complex mechanisms.

The above listed range of parameters to be monitored is only indicative – section 30.11.?? considers a different grouping of possible types of monitoring, listing some types of monitoring systems which might be employed for a bridge. These are concerned with the monitoring of (a) Actions; (b) Reactions; (c) Structural performance and (d) Structural health.

The frequency of inspection / measurement options includes:

- Continuous observation with a defined frequency of measurement;
- Periodic observations with a predefined schedule / program; and
- Irregular observation / measurement regime, triggered when specified circumstances / conditions occur.

The types of elements which may require monitoring and/or the situations or circumstances which might arise, such as:

- Different structural components (superstructure, towers, substructures, piles, forces in tendons, cables etc) and sensitive structural details;
- Critical changes in deteriorating or damaged structural components;

38.2.5.8 Monitoring implementation and operation

Whilst the circumstances relating to individual monitoring schemes will be different, there are often common aspects in the process of defining, implementing and operating a monitoring scheme. In broad terms these might be as set down below. Several iterations and a degree of compromise may be required before a final scheme can be defined which satisfies the various parties involved:

a. Define the problem or combination of effects which need to be addressed and over what period, plus associated data processes.

b. Undertake initial (analytical) studies needed to establish an adequate understanding of the actions, response and behaviours of structure.

c. Obtain an initial indication of the budget required / available for installation and subsequent operation of the monitoring system.

d. Define operational requirements and constraints upon the functioning of the monitoring scheme / reporting and response processes.

e. Define requirements for sensitivity, long-term stability and reliability of the instrumentation and allied interpretative processes.

f. Select appropriate methodologies for monitoring. Evaluate if they will be able to provide timely data / information and be cost effective.

g. Define types and appropriate numbers of sensors, the degree of redundancy, etc for the various parts of monitoring system.

h. Undertake detailed design of monitoring scheme and the associated data processing regime / interpretative / reporting processes.

i. Make estimate of expected cost of installing and operating monitoring system over the expected life of the monitoring scheme.

j. Receive approval and proceed with scheme.

k. Review the performance of the monitoring system and subsequent information handling / reporting; modify as necessary.

It is also important for any monitoring scheme designed to provide warning of sudden distress to be both active (ie. operational at the time the distress arises) and to provide sufficient time for appropriate responsive action to be taken.

- Accessibility for installation / maintenance of the monitoring system.

The various means by which the sensors / instruments can be powered and data can be transmitted / made available for stakeholder review:

- Wired architecture (eg. cables and data loggers, fibre optic links, 'industrial' digital bus systems);
- Wireless operation;
- Combination of wired architecture and wireless operation; and
- Internet enabled / hosted platform for monitoring system

Adequate time and resources need to be allowed for the installation and commissioning of a monitoring system, including all aspects of setting up the data processing, presentation and interpretation (and related analytical modelling studies) components. In many instances the effort, and hence the cost, associated with the latter tasks is greatly underestimated.

Similarly, consideration must be given at the outset to the operation and management of the monitoring scheme. The resources required to support an intensive monitoring programme can be significant. Lack of resources at this stage may greatly reduce the effectiveness of the monitoring system. Experience shows that the costs associated with the operation and management of a long-term monitoring scheme may be considerable and can be substantially more than those of its initial installation.

Further discussion of the deployment of instrumentation for in-service monitoring of structures is given in [Matthews [38-28]].

The following illustrates some of the fundamental issues which will need to be resolved during the design and development of a monitoring scheme.

i. Is the structure experiencing deterioration / damage / ongoing external disturbance? How quickly are these effects progressing?
ii. How critical are these effects to the functionality & safety of the structure? How vulnerable is the structure?
iii. Could the structure transition into a critical state (ULS / SLS)?
iv. What to measure (ie. which parameters)?
v. How to measure (ie. selection of sensor types & specific instruments)?
vi. Where to measure (ie. locations on structure)?
vii. When to measure (eg. duration and frequency)/
viii. How to process measured data (ie. collection, transfer, analysis and associated analytical studies / modelling to assist data interpretation)?
ix. When to process and review the measured data?
x. How / when to make monitoring outcomes available / report findings?
xi. Who needs to know what and how quickly?
xii. What subsequent life-cycle management activities or interventions are to be undertaken, if any, in response to particular circumstances / recorded monitoring outcomes?

38.2.5.9 Knowledge Levels

The process of carrying out a structural assessment involves gathering information about the existing structure. Part of these activities is the evaluation of the information which already exists about the structure in order to establish the nature and extent of any additional information which is required to complete the task.

Information and data available about or gathered for the purpose of making an assessment of an existing structure may be classified in accordance with its completeness / the contribution it is able to make to the structural assessment process, with this being referred to as the Knowledge Level achieved for the particular parameters concerned.

The amount and quality of the collected information in each category is expressed through three distinct knowledge levels (KL) relating to:

The information collected is classified into three categories, as follows:

i. KLG: knowledge level for Geometry;
ii. Geometry of the structure;
iii. The Construction details;

ii. KLD: knowledge level for the Construction details;

iii. KLM: knowledge level for the Material properties (for each material).

The rating values of the KLs may differ in different zones of the structure. Individual values for each category of information shall be determined for the different areas of the structure.

The titles adopted for three knowledge rating values varies with the different categories of information being considered, for example in regard to:

- The original design documents, the three categories are: Not available, Incomplete set available, and a complete set available.
- Information upon materials, the three categories are: Limited, Extended and Comprehensive.

Section X.xx.xx gives further details on Knowledge Levels and their use.

This process would be expected to draw upon information obtained for and detailed in the Birth Certificate, and potentially also in the re-Birth Certificate, assuming that these to be available. For further details on these topics see *fib* Bulletin 93 [38-5].

Deterioration is a worsening of condition with time, or a progressive reduction in the ability of a structure or its components to perform some aspect of their intended function. Typically, these changes will be driven by chemical, mechanical or physical processes or agents, or a combination thereof.

Damage is a physical disruption or change in the condition of a structure or its components, brought about by external actions and influences, such that some aspect of either the current or future functionality of the structure or its components will be impaired. Continuous structural and condition monitoring can be an effective tool for characterising the development of damage due to an on-going process (eg. fatigue).

iii. The various Materials used in the structure.

Three knowledge rating values are to be used to represent the assessed quantity and quality of the collected information for each KL:

i. Minimum knowledge: A minimum quantity / quality of information;

ii. Average knowledge: An average quantity / quality of information;

iii. High knowledge: A high quantity / quality of information.

38.3 Deterioration mechanisms and damage identification

38.3.1 Introduction

The deterioration mechanism, present deterioration level and deterioration rate of materials and/or structural performance must be determined using appropriate models on the basis of the information obtained during inspection, testing and monitoring activities, from the design and construction records, information upon previous interventions and the environmental conditions.

The presence, severity and extent of existing damage must also be identified and recorded during the condition survey activities. It is also necessary to establish the cause of such damage and, if the damage is attributed to an on-going process (eg. cracking associated with fatigue inducing actions), to make a prognosis of its future development.

As activities required for conservation of a concrete structure to meet its design service life must be planned when designing the structure, this process would be expected to lead to the identification of deterioration mechanisms which might potentially adversely affect the structure or its components.

The processes of evaluation and re-design required to extend the design service life of a concrete structure or to accommodate revised performance requirements would also be expected to lead to the identification of similar adverse effects upon the performance of the structure or its components.

The following list illustrates some potential deterioration mechanisms, but this list shall not be considered to include all possible mechanisms:

- carbonation induced corrosion of reinforcement;
- chloride induced corrosion of reinforcement;
- corrosion induced cracking and spalling of concrete;
- scaling, surface spalling or internal damage (disintegration) due to freezing-thawing with or without de-icing-agents;
- alkali silica reaction (ASR) induced cracking and deformation (expansion);
- cracking, deformation (expansion, swelling) or dissolution due to chemical attack (sulfate attack, magnesium attack, acid attack, chloride attack etc.);
- fatigue induced cracking or internal damage;
- abrasion and / or erosion damage;
- sustained load induced cracking or internal damage (load / force may arise from external or internal sources of restraint; and
- induced displacement (thermal movements, fire induced displacements)

Among the above-mentioned deterioration mechanisms, corrosion and corrosion induced effects are the most common in many countries. Deterioration involving the combined effects of a number of the above mechanisms is also commonly observed.

Extreme events (eg. fire, accidental and malicious actions) can rapidly create severe and extensive damage, possibly threatening the stability and safety of a structure, potentially even causing its failure and collapse.

38.3.2 **Identification of deterioration mechanisms and damage**

It is necessary to identify the deterioration and/or damage mechanisms which are affecting, or might potentially affect, the structure or its components.

These must be identified from consideration of factors such as the environment to which the structure is exposed, variations in the exposure conditions, previous condition and performance data, if available, and any sign of deterioration that may be observed or detected.

Damage may arise from external actions and influences, such as imposed loadings, as well as the effects of induced and / or restrained deformations.

A rapid assessment of the safety and stability of a structure could be required following damage caused by an extreme event. Such an assessment would require a rapid estimate to be made of the severity and extent of the damage caused.

Table 38.3-1 lists six classes of defects affecting constructed assets and an associated classification of the defects in terms of the type of the defect (i.e. multiple types – see column 2) and the category of defect (i.e. multiple categories – see column 3).

Table 38.3-2 lists some degradation mechanisms which may affect constructed assets and an associated classification of the effect of the deterioration into six classes of defect.

Table 38.3-2 also provides an indication of the effect of the various degradation mechanisms, differentiating between the primary effects of the respective degradation mechanisms (identified as a Basic degradation mechanism) and secondary effects (identified as an Additional degradation mechanism). The illustrations provided are based upon circumstances likely to apply to concrete bridges, other constructed assets (eg. sewers) exposed to different environmental conditions could exhibit appreciably different relationships.

In terms of seeking to achieve damage identification at a global level, some insights and understanding might be gained from undertaking dynamic / ambient vibration testing and proof load response tests. These procedures are discussed in Section 38.2.5.6: *System identification by static and dynamic response measurements* and Section 38.2.5.7: *Proof load tests.* Refer to section 30.X.XX ?? for further information.

Table 38.3-1: *Some defects affecting constructed assets - Classification of defects.*

Class of defect	Type of defect	Category of defect
Column 1	Column 2	Column 3
Deformation	Incorrect geometry of constructed element	Incorrect shape / dimensions of concrete
		Invalid arrangement of reinforcement
		Invalid arrangement of prestressing tendons
	Change of the geometry of the element axis	Excessive elastic deformation
		Permanent deformation
	Change of the geometry along the element length	Excessive elastic deformation
		Permanent deformation
Destruction of material	Change of the chemical characteristics	Change of concrete characteristics
		Change of reinforcing steel characteristics
		Change of prestressing steel characteristics
		Change of protective layer characteristics
	Change of the physical characteristics	Change of concrete characteristics
		Change of reinforcing steel characteristics
		Change of prestressing steel characteristics
		Change of protective layer characteristics
Loss of material	Loss of structural material	Loss of concrete
		Loss of reinforcing steel
		Loss of prestressing steel
	Loss of material from protective layer	Loss of material of concrete protection
		Loss of protection of reinforcing steel
		Loss of protection of prestressing steel
Discontinuity	Crack	Crack of concrete
		Crack of reinforcing steel
		Crack of prestressing steel
		Crack of protective layer
	Fracture	Fracture of concrete
		Fracture of reinforcing steel
		Fracture of prestressing steel
		Fracture of protective layer
Contamination	Inorganic	Aggressive
		Neutral
	Organic	Aggressive
		Neutral
Displacement	Incorrect translation/linear displacement	Excessive movement
		Restricted movement
	Incorrect rotation	Excessive movement
		Restricted movement

Table 38.3-2: *Some degradation mechanisms and classes of defect which may affect constructed assets (eg. concrete bridges).*

Degradation mechanisms		Class of defect					
		Deformation	Destruction	Loss of material	Discontinuity	Contamination	Displacement
Col1	Column2	Col3	Col4	Col5	Col6	Col7	Col8
Physical	Accumulation of inorganic material		□	□		■	■
	Cyclic freeze-thaw action		■	■	□	□	
	Erosion		□	■			
	Crystallisation		■	□	□		
	Extreme temperatures	□	□	□	□	□	■
	Creep	□					■
	Relaxation	□			□		
	Shrinkage	□	□	□	■		■
	Overloading	■	□	■	■		■
	Fatigue		■		□		
	Geotechnical condition change	■		□	■		■
Chemical	Carbonation		■				
	Corrosion	□	■	■	□	□	
	Action of aggressive compounds		■	□		□	
	Chemical dissolution/leaching		■	□		■	
	Reactions between component materials	□	■	■	□		
Biological	Accumulation of organic material/debris		□	□		■	□
	Activity of microbes		□	□		■	
	Activity of plants	□	□	□	□	■	□
	Activity of animals		□			□	
Legend: ■ – Basic degradation mechanism, □ – Additional degradation mechanism							

Consideration shall be given whether any of the factors which influence the development of deterioration and damage might evolve or change with time (ie. are not constant).

38.3.3 Factors influencing deterioration and damage

Influencing factors may potentially be classified as follows:

More than one factor may influence the development and progress of deterioration arising in a structure/member. In such circumstances the potential combined (synergetic) effects of the deterioration factors shall be considered.

This information would be used to make a prognosis of the behaviour of the structure under the envisaged environmental and loading conditions over the remainder of its design or extended service life, considering the performance requirements that need to be met.

Assessment is a fundamental activity for managing existing structures, being used to establish the need for intervention works (eg. maintenance, repair, upgrading works) and / or appropriate associated LCM / operational actions (eg. restrictions on usage).

The objectives of the performance evaluation (assessment) of an existing structure will typically be agreed in consultation with the client / stakeholders (eg. the owner, the relevant authorities, insurance companies, etc.) and be based upon defined performance levels. These might typically entail it's:

- safety performance level;
- continued functional performance for specified hazards; and
- any special performance requirements defined by the client.

The process of performance evaluation is expected to be able to draw upon information from the design and construction process and / or as detailed in the Birth Certificate, and potentially also in the re-Birth Certificate documents, assuming these to be available.

– Actions upon the structure (external factors) such as temperature, humidity and any other characteristics of the environment, including physical loading and imposed or restrained deformations; and
– Factors related to the nature of the structure (internal factors) such as designed parameters (e.g. material characteristics, concrete cover for the reinforcing bars, geometry of sections etc.) and quality management during construction.

38.3.4 Determination of deterioration/damage level & rate

The deterioration / damage level and the rate of change of material properties and/or structural performance must be determined from the results of the surveys and/or monitoring carried out and by using appropriate models for the mechanism(s) of deterioration.

38.4 Performance evaluation (Structural assessment)

The adequacy of an existing asset / structure for continued service is established by comparing its structural capabilities (eg load capacity, ability to resist / accommodate actions applied to the structure) with the current and/or future performance requirements, such as a defined set of loadings or actions (original or revised) and the environmental circumstances.

Performance evaluation may be undertaken to verify the adequacy of an asset / structure (or its component parts):

a) To achieve its design service life, as envisaged at the time of design;
b) To achieve an extension of the planned length of service life or meet revised performance requirements (eg. revised loading, actions or some other aspect of the required functionality).

An increase in the level of the performance requirements during the use stage of an asset / structure may necessitate an intervention being made to increase the performance / durability of the asset / structure.

It is assumed that the survey and monitoring processes outlined above (refer section 38.2) will be able to furnish sufficient information to permit decisions to be made in respect of options for the management of the structure. If there is not sufficient information it may be necessary to initiate a detailed investigation (refer Figure 28.4-1 and section 38.2.5).

The structural assessment will need to take account of the conservation strategy and tactics defined at the time of design or re-design of the structure (refer section 8.2), which are derived from the Life-cycle management (LCM) strategy for a building / constructed asset (refer chapter 5) which in turn is informed by the sustainability objectives for the asset (refer chapter 3).

Thus, the Life-cycle management (LCM) strategy for a building / constructed asset are derived from the objectives defined under the three pillars of sustainability; namely those of social responsibility, environmental quality and economic efficiency. The LCM strategy shall contribute to achieving an acceptable balance between these factors.

Life-cycle management activities: These might include:

- Managing hazards and/or hazard scenarios;
- Monitoring of in-service performance and structural utilisation levels to manage the safety, condition and/or serviceability of the structure or parts thereof;
- Performing re-analysis and/or re-design evaluations of the structure, as well as making prognoses about future behaviour and performance;
- Instigating a change of functional use of the structure or parts thereof;
- When appropriate, imposing usage limitations or other forms functional restriction upon the structure or parts thereof; and

It may be necessary for a performance evaluation to consider the effects of any current and / or future deterioration / damage upon structural performance:

a) To consider the effects of any current deterioration / damage upon the current adequacy of the asset / structure; and / or

b) To make a prognosis of the future effects of any deterioration upon the adequacy of the asset / structure and / or the length of time that its performance is anticipated to remain adequate.

An overall evaluation of the extent of any material and/or structural performance reduction due to deterioration / damage must be based on:

– records in the design file and the as-built-documentation, see chapter 10;

– results of survey and monitoring as outlined above; and

– assessment of the nature, severity, significance and rate of deterioration.

The approach to be taken for the performance assessment of existing structures is described in chapter 28: *Approach to Assessment*. Further details on approaches to performance evaluation / structural assessment are given in ISO 13822 [38-8].

38.5 Decision-making for LCM activities and interventions

38.5.1 General

Decision-making for LCM activities and interventions can be required for various reasons including to enable a structure / constructed asset to:

– meet its design service life, as envisaged at the time of design;

– achieve an extension of its planned service life; or

– meet revised performance requirements (e.g. enhanced loading / action effects or changed functionality / needs).

These activities are concerned with:

- When appropriate, ending the operational use of the structure or parts thereof.

Physical intervention works options: These might include works for maintenance, prevention / avoidance of damage or deterioration, repair / remediation, upgrading / strengthening and possibly for adaptation / reconstruction – see Table 38.6-1 (a) – (b) for further details.

Decision-making can be made either on a proactive (ie. preventative intervention) or reactive (i.e. remedial intervention) basis, depending upon the underlying philosophy adopted for the management of the structure. It may be sufficient to re-analyse/re-design/evaluate the structure to address a performance non-conformity regarding the limit states associated with durability or other parameters. Consideration may also need to be given to changed (increased or extended) performance requirements.

In addition to direct physical intervention measures, such as employing repair / upgrading activities, operational control measures including usage restrictions, targeted monitoring and the possibility of termination of use, may also be considered as potential control options for the effective LCM of a structure / constructed asset.

A higher level of deterioration/damage in a structure (ie. more severe) typically results in a higher cost for the necessary remedial works, as well as a more restricted choice of intervention methods (ie. more restricted / fewer appropriate options). The threshold levels for deterioration in structural performance need to be set to achieve economically and technically feasible interventions.

– Establishing the implications of deterioration or damage and/or changed performance requirements upon the life-cycle management and conservation of an asset;

– Evaluating potential management and conservation options;

– The selection of appropriate management and intervention option(s) and the timing of such works to address or remedy non-conformances with the performance requirements or achieve compliance with extended / enhanced performance requirements

These processes involve a comparison of the assessed capacity (or other appropriate performance or response metric / proxy) of the existing structure, or parts thereof, with the specified performance requirements.

Life-cycle management activities and/or physical interventions works may take a variety of forms – see details in the adjacent informative column entry.

38.5.2 Threshold levels for deterioration in structural performance

The structural performance is evaluated and verified for the defined performance requirement by following the structural analysis and verification procedures provided in MC2020 Parts V: Input data for Materials, VI: Input data for Interfaces and VII: Design and assessment using the information related to materials and structural configurations, which have been obtained through the condition survey and monitoring processes.

Depending on how the outcomes of the above evaluations compare with a minimum required performance level(s), a decision must be taken on how these findings and prognoses impact upon the current and any proposed future way of managing the structure or its component parts. It may be necessary to undertake appropriate preventative (proactive) and/or remedial (reactive) interventions.

The level set shall take account of the required performance, such as the load carrying capacity to achieve the required structural safety, as well as serviceability, repairability and other requirements, such as appearance.

In setting these values, account would need to be taken of the conservation strategy and tactics defined at the time of design and/or re-design. The

If the results of a survey raise doubts as to whether the safety and serviceability of the structure is acceptable, a detailed investigation shall be undertaken. The in-depth investigation shall include all measures that may reduce uncertainties, such as testing of material properties, checking of dimensions, stiffness measurements, etc.

During the period to the next detailed evaluation (assessment) of the condition and performance of the structure – ie. during the review period – appropriate LCM activities would need to be performed. These could take the form of regular or periodic visual inspections and/or other appropriate forms of survey / monitoring of the structure – refer section 38.2.5.2.

damage/deterioration level for preventive intervention is set lower than that typically associated with reactive remedial interventions.

38.5.3 Judgement criteria

If the defects/damage/changes obviously put structural safety and/or asset value in immediate jeopardy, emergency measures shall be undertaken immediately before detailed investigation is undertaken.

If the detailed investigation and evaluation indicates that the structural or some other form of performance does not satisfy the defined performance requirements, or may not do so at some future time during the design service life, it may be appropriate to instigate suitable life-cycle management (LCM) activities and/or undertake appropriate and timely intervention(s) to rectify the situation – refer section 38.6.

The outcomes of the detailed evaluation will typically allow a length of time to be estimated when the structure is expected to satisfy the defined performance requirements at an acceptable reliability level. This information will typically provide a rational basis for a 'review period' to be defined, which would be the length of time before further a detailed assessment of the condition and performance of the structure was expected to be performed.

38.6 LCM activities including intervention works

While life-cycle management of a concrete structure will typically be provided through the following two linked approaches, section 38.6 is primarily concerned with the activities for executing physical interventions:

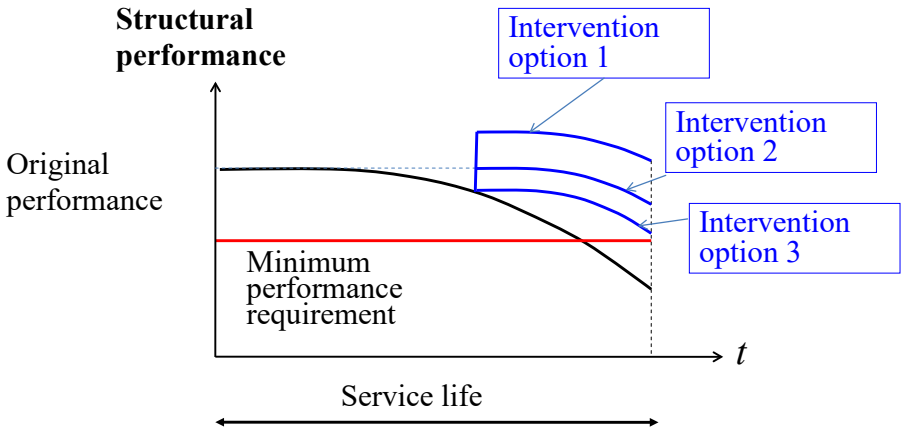
a) Life cycle management activities; and

b) Physical interventions for maintenance, repair / remediation, upgrading / strengthening and possibly by adaptation / reconstruction works.

c)

38.6.1 Introduction / General

Figure 38.6-1 shows schematically the performance of alternative intervention options. The evaluations made in this process need to consider a range of factors such as the importance of structure, the significance of non-compliance with the structure's performance requirement, and an extension of service life in future.



Note:
Option 1 is to upgrade the performance and to lessen the deterioration rate.
Option 2 is to restore the performance and to lessen the deterioration rate.
Option 3 is to lessen the deterioration rate only.

Figure 38.6-1: Schematic for intervention option selection.

Note to Figure 38.6-1: This shows only one intervention being made. In reality several interventions are likely to be made during the service life of a structure.

Planned interventions are typical for a proactive conservation strategy. A reactive conservation strategy may require both planned and unplanned interventions. For planned interventions, consideration shall be given to not only the timing of the first intervention but also the timing of the planned sequence of intervention cycles.

In situations where conservation measures are not feasible, should significant damage / deterioration to the structure be detected, a potentially difficult and expensive reactive remedial intervention could be required.

Undertaking interventions is one facet of the overall strategy for the life-cycle management of a structure. The evaluation of possible options for making an intervention necessarily involves consideration of the available time and space required to make the intervention, as well as its cost and environmental and societal impacts.

Thus, when reviewing potential intervention methods, it is necessary to give consideration to the requirements for the execution of the interventions, as well as to the relevant health and safety, environmental protection and fire regulations.

In future, cost may need to include issues such as the carbon burden and other environmental impacts associated with the structure and the proposed works, as well as their monetary cost (i.e. their construction-cost and their whole-life cost).

Sufficient diagnostic tests must be carried out before design to ensure that the intervention option selected is appropriate to the circumstances and future performance requirements.

Once the most appropriate preventative or remedial intervention option(s) has been selected it is still necessary to get the work carried out. To do this, the owner will usually need the support of a team of technical advisers, depending upon the expertise held by the owner. Together they shall be able to address the various technical and process matters that may arise. Intervention activities need particular knowledge and experience. This group shall be concerned with the performance of the concrete structure after intervention. In essence what is required is an integrated team approach, wherever the necessary expertise actually resides.

In many cases interventions need subsequent (planned) interventions since the effectiveness of interventions has a design service life. Therefore, interventions shall be carried out in a manner which makes subsequent interventions feasible and acceptable from the technical, social, economic and environmental perspectives.

Potential interventions can be classified, based on their planning, as follows:

– *Planned interventions:* These include maintenance, preventative remedial and upgrading interventions which were planned at the time of design of a new structure. Such interventions may take various form, including repairs to and upgrading of an existing structure;

– *Unplanned interventions:* These include remedial or strengthening interventions which are not planned at the time of design but become

For more information regarding interventions, see *fib* Bulletin 17 [38-1], *fib* Bulletin 44 [38-3], *fib* Bulletin 102 [38-29] and to *fib* Bulletin 103 [38-30].

Section 38.6 assumes that the reasons and mechanisms associated with the development of deterioration / damage have been established for the particular case, which is by having conducted an appropriate investigation of the causes and evaluation of the implications of the deterioration / damage - Refer chapters 5 and 8, together with earlier sections of chapter 38.

An increase in the level of the performance requirements during the use stage may necessitate an intervention being made to increase the performance / durability of the constructed asset / structure to satisfy the new requirements.

ISO 16311 Parts 1 to 4 [38-9][38-10][38-11][38-12] provides an overall framework for undertaking the condition survey, investigation & assessment of existing concrete structures, evaluation of the adequacy of their structural performance, together with the design/selection and execution of interventions.

necessary due to unexpected damage or revision of performance requirements.

Appropriate steps must also be taken to ensure the use of suitable materials, methods and quality management. A plan for the execution of the intervention including details of the materials to be used, the method of undertaking the proposed works and quality control tests to ensure quality of the work, must be drawn up before the work commences; see chapters 17 and 33.

A detailed record of the intervention work, along with other relevant information, must be maintained for future reference; see section 38.7.

38.6.2 Basis for intervention works

As noted in earlier parts of this chapter, during the in-service stage of the life of a structure, periodic survey / monitoring / diagnosis shall be conducted to confirm the ongoing satisfactory performance of the structure. A prognosis of subsequent performance shall be made as required. Not only physical measures including maintenance, repair and upgrading works, but also operational measures including use restrictions or the possibility of termination of use, shall be considered as elements of the LCM of the constructed asset / structure.

The following approach is commonly used for the selection and implementation of protection / repair / strengthening intervention options within the overall LCM framework:

1. Choose an option to be adopted for the future LCM of the structure.
2. Adopt a principle for the intervention upon the structure.
3. Where an intervention is to be undertaken, choose an appropriate intervention method.
4. Where an intervention is to be undertaken, choose a product or system to be employed for the works.
5. Define the future survey / monitoring & maintenance requirements.

Table 38.6-1A: Intervention principles & methods (after ISO 16311-3 [38-11].)

Principle / Remedy	Examples of intervention methods based upon the Principle
1. Protection against ingress	• Undertake hydrophobic impregnation • Undertake impregnation • Apply coating • Surface bandaging of cracks • Fill / inject cracks in concrete • Change cracks into joints • Provide external envelope / over-cladding • Apply membrane
2. Moisture control	• Undertake hydrophobic impregnation • Undertake impregnation • Apply coating • Provide external envelope / cladding • Apply electrochemical treatments (dehumidification)
3. Concrete restoration	• Localised patch-repairs – hand applied mortar • Recast members with concrete or mortar • Add concrete overlay • Apply sprayed concrete (shotcrete) • Replace entire members
4. Increased physical resistance	• Apply coating • Undertake impregnation • Add mortar or concrete • Apply membrane
5. Increased resistance to chemical attack	• Apply coating and / or membrane • Undertake impregnation • Add mortar or concrete

Notes to Table 38.6-1:

1 Refer to ISO 16311, Part 3 Table 1 for further details of the cited intervention methods. Note the differences in numbering of intervention methods in ISO 16311.

*2 Refer fib Bulletin ** for an overview of the intervention methods.*

LCM Options for managing a concrete structure which does not or is predicted not to meet performance requirement include the following six technical steps:

a) Do nothing for a certain time while monitoring the structure.

b) Re-analyse structural capacity, possibly leading to a reduction in the accepted functional capability of the structure.

c) Make intervention to prevent or reduce further deterioration/damage.

d) Make intervention to improve / strengthen / protect / refurbish all or part of the structure.

e) Make intervention - Reconstruct all or part of the structure.

f) Make intervention - Demolish all or part of the structure.

g)

38.6.3 Intervention principles and methods

Principles are the physical concepts underlying the methods of structure protection, repair and structural enhancement. These relate to the nature of the deterioration/defect which exists. They are grouped under the three headings given below. It is necessary to select a Principle appropriate to the Intervention Option chosen (see section 38.6.2 above):

***Defects in Concrete*:**

1. Protection against ingress
2. Moisture control
3. Concrete restoration
4. Increase physical resistance
5. Resistance to chemicals

***Reinforcement Corrosion*:**

6. Preserve or restore passivity
7. Increase resistivity
8. Cathodic control
9. Cathodic protection
10. Control of anodic areas

***Structural Repair and Enhancement*:**

11. Structural repair / strengthening / upgrading
12. Adding new systems / devices to control structural response

Table 38.6-1B:. Intervention principles & methods (after ISO 16311: Part 3 [38-11])
- Continued.

Principle / Remedy	Examples of intervention methods based upon the Principle
6. Preserve or restore passivity	• Increase cover to reinforcement with additional mortar or concrete (preservation only) or apply coating • Replace contaminated or carbonated concrete • Electrochemical re-alkalisation of carbonated concrete • Re-alkalisation of carbonated concrete by diffusion • Undertake electrochemical chloride extraction • Apply membrane (for preserving passivity only)
7. Increase resistivity	• Undertake hydrophobic impregnation • Undertake impregnation • Apply coating
8. Cathodic control	• Limit oxygen availability at cathode by saturation or application of surface coating
9. Cathodic protection / prevention	• Apply electrical current to provide protective electrochemical potential
10. Control of anodic areas	• Apply active coating to the reinforcement • Apply barrier coating to the reinforcement • Apply corrosion inhibitors onto or into the concrete • Install discrete galvanic anodes
11. Structural strengthening	• Adding or replacing embedded or external reinforcing bars • Adding reinforcement anchored in pre-formed / drilled holes • Bonding plate reinforcement • Adding mortar or concrete • Injecting cracks, voids, or interstices • Filling cracks, voids, or interstices • Prestressing - (post tensioning) or FRP strengthening
12. Addition of new systems / devices to control response	• Base-isolation devices • Energy-absorption devices • Other new systems / devices

The application of methods to situations not shown in Table 38.6-1 shall not be taken to mean that such a method or application is necessarily unsatisfactory in those

Maintenance, protection, repair and strengthening interventions are based on chemical, electrochemical or physical principles / remedies that can be used to prevent or stabilise the deterioration of concrete, or corrosion of the steel or other embedded metals, or to strengthen the concrete structure.

Methods are the means of achieving the objective of the chosen Principle, which is the concrete protection, repair and structural enhancement method to be used for the intervention.

A number of different approaches are included under the broad banner of concrete protection and repair intervention options. The protection actions include the application of some form of surface protection or impregnation, electrochemical re-alkalisation, electrochemical chloride extraction and cathodic prevention / protection. Repair actions include concrete restoration, surface protection or impregnations, overlays or other forms of barrier, as well as cathodic protection. It will be noted that some techniques can be utilised as both a protection and as a repair, depending upon circumstances.

Table 38.6-1 contains examples of intervention methods for protection, repair, strengthening and structural enhancement which apply the intervention principles.

It is recommended that only intervention methods which comply with the principles shall be selected, taking into account any possible undesirable consequences of applying a particular method or combination of methods under the specific conditions of the individual intervention.

situations. The use of methods which are not listed in Table 38.6-1, may be satisfactory in appropriate circumstances.

fib Bulletins 102 & 103 [38-29] + [38-30] provides an overview of a number of the intervention methods which are applied to actual structures. Details include some newly developed intervention methods, including intervention methods employing more than one remedial principle / remedy, cathodic protection and structural strengthening utilising FRP carbon fibre laminates.

38.6.4 Intervention strategies, methods and materials

38.6.4.1 Maintenance interventions

This term is commonly applied in the context of building fabric components with a limited life (e.g. coatings, surface finishes), components associated with water management and rainwater run-off, items where regular intervention is required to maintain their effective operation by removing any build-up of detritus (e.g. gutters, gullies, drains, etc.), to overcome the effects of ageing or fatigue upon materials (paints, coatings, sealants, movement joints, etc.), to maintain an acceptable appearance by cleaning of surfaces, etc.

Maintenance interventions involve various recurrent activities performed to either prevent or correct the effects of minor deterioration, degradation or mechanical wear of the structure or its components in order to keep their future performance at the level anticipated by the designer.

38.6.4.2 Preventative interventions

There are various methods and materials used for undertaking preventative interventions, including the following (see intervention Option 3 in Figure 38.6-1):

Methods include:

- cathodic prevention;
- chloride extraction;
- re-alkalization;
- electrochemical dehumidification;
- surface coating, surface barriers, sealants and so on;

Preventative interventions are a proactive conservation activity concerned with applying some form of treatment or taking action to avoid a change in a material property (e.g. such as that caused by the influence of carbonation or chlorides) adversely affecting the ability of the structure, or parts thereof, to meet the required performance levels because of deterioration.

In particular, proactive maintenance interventions seek to utilize actions which will result in an extension of the period of initiation, prior to the onset of the propagation of deterioration, or slow down the further development of deterioration.

– overlays, wearing courses and such like.

Materials include:

– various materials available.

Preventative interventions generally have a lower life-cycle cost and reduced environmental impacts than remedial interventions.

There are various methods and materials for undertaking repairs including the following (see intervention Option 2 in Figure 38.6-1):

Methods include:

– restoration of affected concrete section (patch and sprayed repairs);
– restoration of reinforcement provision (replacement of reinforcing bars, addition of supplementary reinforcement);
– surface coating, surface barriers, overlays, sealants and such like;
– the preventative intervention methods listed in section 38.6.4.2 to lessen the deterioration rate (see intervention Option 3 in Figure 38.6-1);
– the strengthening / upgrading intervention methods listed in section 38.6.4.4 to restore degraded mechanical performance (see intervention Option 1 in Figure 38.6-1).

Materials include:

– various materials available including cast and sprayed concretes and mortars, reinforced concrete (using various metallic and non-metallic reinforcements eg. carbon steel, stainless steel, carbon fibre), fibre reinforced concrete, polymer concrete, proprietary repair mortars.

There are various methods and materials used for making strengthening or upgrading interventions including the following (see intervention Option 1 in Figure 38.6-1):

The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended. It is implied that the treatment or action will be taken prior to deterioration and damage becoming apparent/visible on the structure, for example cracking or spalling of concrete.

38.6.4.3 Remedial interventions

Remedial interventions are undertaken to reinstate to an acceptable level the current and future performance of a structure or its components which are defective, deteriorated, degraded or damaged in some way so their performance level is below that anticipated by the designer; generally without restriction upon the materials or methods employed.

Remedial interventions are typically required after a change in a material property (e.g. such as that caused by the influence of carbonation or chlorides) has adversely affected the ability of the structure, or parts thereof, to meet the required performance levels because of deterioration (e.g. corrosion of embedded reinforcing bars). The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended.

Remedial interventions might be instigated for the purposes of repair, rehabilitation, remediation, restoration or retrofitting of the structure concerned, or of elements thereof.

38.6.4.4 Strengthening and upgrading interventions

Strengthening or upgrading interventions are undertaken to improve the strength or some other performance property of a structure or structural elements (e.g. stiffness) above the level anticipated by the designer.

Methods include:

- external bonding (plate or sheet bonding, over or under-laying, jacketing, etc.);
- introduction of external prestressing and addition of tension cables;
- addition of a member (girder, brace, support, etc.);
- increase in member cross-sectional size (area);
- replacement of member.

Materials include:

- concrete (reinforced concrete (using various metallic and non-metallic reinforcements eg. carbon steel, stainless steel, carbon fibre), prestressed concrete, fibre reinforced concrete, polymer concrete, etc.);
- steel (plate, cable, bars, etc.);
- non-metallic materials (plate, sheet, cable, etc.).

There are various methods and materials for undertaking these types of works which could include all the methods and techniques used in original construction. There is a wide range of materials available including cast and sprayed concretes, reinforced concrete, fibre reinforced concrete and polymer concrete.

In response to the developing climate emergency, amongst other potential courses of action, upgrading existing buildings and other constructed assets for extended or alternative use may offer a more carbon efficient alternative to demolition and construction of a new replacement facility. This approach, which requires a structural assessment to be made of the assets concerned, supports the aim that engineering design and life-cycle management processes move towards achieving a net zero carbon life-cycle.

The revised performance requirements shall be defined at the outset of the re-design process. Accordingly, after these works the structure or structural elements would be expected to meet the revised performance requirements.

38.6.4.5 Reconstruction and replacement

Rebuild, reconstruction and replacement activities are intended to provide a new structure or structural components to replace the original damaged, defective or deteriorated entity, generally without restriction upon the materials or methods employed.

Commonly, this type of intervention would be undertaken to restore the performance level of the structure or specific structural components to the level anticipated by the designer. In this case, these activities would be deemed to be a remedial intervention. However, in other circumstances these works might provide the opportunity to revise the performance requirements to a higher level (i.e. above the level anticipated by the designer). In this case, these activities might be deemed to be a strengthening or upgrading intervention.

Thus, in some instances this approach may be considered to be an extreme form of remedial intervention, which involves the creation of a new structure or structural component to replace the original damaged, defective or deteriorated entity after its destruction or demolition. These activities require careful planning and management in order not to compromise the safety and/or performance of other parts of the structure while the works are carried out.

This approach is only valid on the basis that the additional information gathered is able to contribute directly to the judgments to be made on the performance of the structure, including safety considerations, and the remaining service life.

For example, where shear or brittle failure of an element was the main concern, it would not be appropriate to rely on the monitoring of flexural deflections because it is unlikely that this parameter would give adequate indication of distress or developing failure prior to catastrophic collapse.

In some situations, the aesthetic appearance of a structure or building may be paramount and it may be necessary to assign this parameter a particularly high importance and to ensure that this aspect is always maintained at a satisfactory standard.

While the introduction of usage restrictions is effectively recognising a reduction in the achieved performance of the structure, it might also be looked upon as a decision to downgrade its required performance.

Plans for further action must be drawn up on the basis of the known or suspected extent of deterioration, taking account of what further information is required in order to make a satisfactory assessment of its current condition and safety. Additional investigations may be required. Once the immediate threat and concerns have been addressed, consideration shall turn to aspects of the remaining service life (estimated and design), actual and desired structural performance and functionality, along with conservation strategy, tactics and any previous interventions undertaken.

See also flowchart and observations made in section 38.2.2.

38.6.4.6 Other LCM activities and measures

Intensified survey / monitoring

Intensified survey refers to inspections which are carried out by suitably experienced personnel and by increasing one or more of: frequency of survey, the number of surveyed items and the number of locations surveyed.

It is not appropriate to rely on an intensified survey or monitoring if the data or information gathered does not contribute directly to the decision-making procedure associated with the safety management of the structure.

Usage restriction

Usage restriction is a structure management strategy which imposes some restriction on usage (e.g. restriction on the maximum traffic load or speed, maximum imposed floor load in a building, etc.), to ensure that the maximum effects of the specified actions on the structure are reduced.

The extent of restriction must be determined depending on the level of deterioration observed or suspected and a prognosis for the (near) future.

Monitoring and/or other measures, such as detailed investigations, may also be necessary in addition to imposing the restriction.

Emergency measures

When a deteriorated structure poses an immediate threat to the environment, its users or any third party, suitable emergency action and temporary works must be implemented immediately. Detailed investigations may be required once the structure has been "stabilised" and put into a temporarily "safe" condition.

Dismantling and removal of a severely deteriorated structure is generally the last option. It must be carried out only after taking into account of all relevant factors including hazards to the environment, public safety and disposal of the debris. Details are provided in MC2020 chapter 39: Circularity, dismantlement and reuse. Special considerations may be required for prestressed concrete structures.

See also flowchart and observations made in section 38.2.2.

While the performance requirements for the structure will relate to functionality, there can be many other facets which may be relevant. Performance requirements typically include focus on topics such as safety, usability/serviceability, durability, user and third-party interests, appearance, restorability, and the like. A potential performance-based approach to undertaking interventions upon existing concrete structures is presented in CONREPNET [38-31].

In order to perform an appropriate intervention design /selection, it is necessary to understand many things, including:

- the mechanism(s) responsible for the deterioration/damage;
- the degree of performance deterioration of structure which has occurred;
- the design concept adopted at the time of construction;
- the remaining (design) service life required after the intervention;
- the target performance level required for the structure after the intervention has been made (related to the importance of structure);
- the actions on and environmental conditions affecting the structure, both those experienced now and those expected to occur in the future;
- the conservation strategies which can be employed;
- other more general requirements (eg aesthetics) or constraints (eg. access, work site space limitations, sequencing of works, etc); and

Dismantling and removal

Dismantling and removal of a severely deteriorated structure requires careful planning and evaluation before embarking upon the process, as there are a wide range of potential complications which may arise.

38.6.5 Selection of intervention methods

38.6.5.1 General

Selection of an intervention method, including the materials to be used, must consider the reasons for undertaking the intervention, the conditions under which the execution will have to be carried out and the conservation strategy to be adopted after the intervention is made, so that the purpose of the intervention can be achieved in the most reliable way. Ease of conservation after intervention must also be taken into account.

The steps involved will typically include:

- performance requirements review, work analysis and work statement development;
- intervention design: goals and requirements/acceptance criteria;
- intervention design: specification development;
- execution of the selected intervention option(s); and
- monitoring of the effectiveness of the intervention work.

The selection of the optimum intervention method can operate at a number of different levels, from making decisions upon the treatment of individual members up to that of the overall structure. Thus, the selection perspective may be on the basis of a particular element, a zone or the overall structure. Thus, different preventative or remedial intervention methods could potentially be used in conjunction to meet the various objectives and performance goals.

- the life-cycle cost (LCC) and life-cycle environmental performance (as evaluated by LCA) consequences of the potential intervention options.

See chapter 26 *Conceptual design*, in regard to the design concept adopted at the time of design and construction

See section 8.2 *Conservation strategies and tactics*, in regard to the relevant conservation strategies.

See section 8.3 *Conservation management*, in regard to the appropriate conservation categories (ie. the conservation classification).

See section 38.2 *Condition survey and monitoring*, in regard to survey / monitoring regimes which can be employed.

See section 38.3 *Deterioration mechanisms and damage identification*, in regard to the active mechanism(s) of the deterioration and damage.

The process of selecting the most appropriate intervention option for the given circumstances shall include:

– confirmation of the conservation strategy and tactics to be applied to the asset concerned, and whether there is any intentional differentiation of the performance standards across the asset (e.g. a higher standard of aesthetic appearance may be required on facades which are in public view, or a higher performance standard may be required on some facades to minimize the risk of spalled concrete falling into public zones);

– confirmation of appropriate indicators of life-cycle performance;

– establishing the threshold levels for the performance indicators to be used in the design and evaluation of the intervention, together with the subsequent post-intervention monitoring; and

– identification of intervention methods appropriate to the desired result (e.g. the proposed improvement in the condition of the structure), which may or may not involve restoration back to the initial condition of the structure or fulfilment of the design requirements – see Figure 38.6-1.

The process of choosing an intervention option shall include:

The performance requirements to be achieved after an intervention has been made shall be established considering the importance of structure, the level of structural reliability demanded, the conservation strategies which can be employed, the remaining design service life, the deterioration / damage mechanism(s) and the degree of performance deterioration which has occurred to the structure.

This subsection of the MC is relevant to structures whose conservation classification is either Category A (proactive conservation strategy) or Category B (reactive conservation strategy).

A condition survey/investigation shall be undertaken to establish the mechanism(s) of deterioration/damage or how initial defect(s) were caused.

The target performance level shall be determined before selecting an appropriate intervention method(s), to ensure that the structure will retain the performance required after the intervention for the design service life.

38.6.5.2 Selection of intervention methods

Principles addressing defects in concrete and reinforcement corrosion

Principles / remedies 1 to 5 and 11 in Table 38.6-1 address defects in the concrete or concrete structures that can be caused by the following actions, separately or in combination:

a) mechanical: e.g. impact, overloading, movement caused by settlement, vibration, seismic actions, fatigue and blast;

b) chemical and biological: e.g. sulfate attack, alkali-aggregate reaction;

c) physical: e.g. freeze–thaw action, fire, thermal cracking, moisture movement, salt crystallization and erosion.

Principles / remedies 6 to 11 in Table 38.6-1 address the effects of reinforcement corrosion caused by:

e) physical loss of the protective concrete cover;

f) chemical loss of alkaline pH in the protective concrete cover as a result of reaction with atmospheric carbon dioxide (carbonation);

– defining the nature of the acceptance testing and verifications required;
– confirming the means for monitoring life-cycle performance post-intervention; and
– benchmarking or comparison of the available intervention methods to establish the most appropriate (best) method for the particular circumstances and requirements.

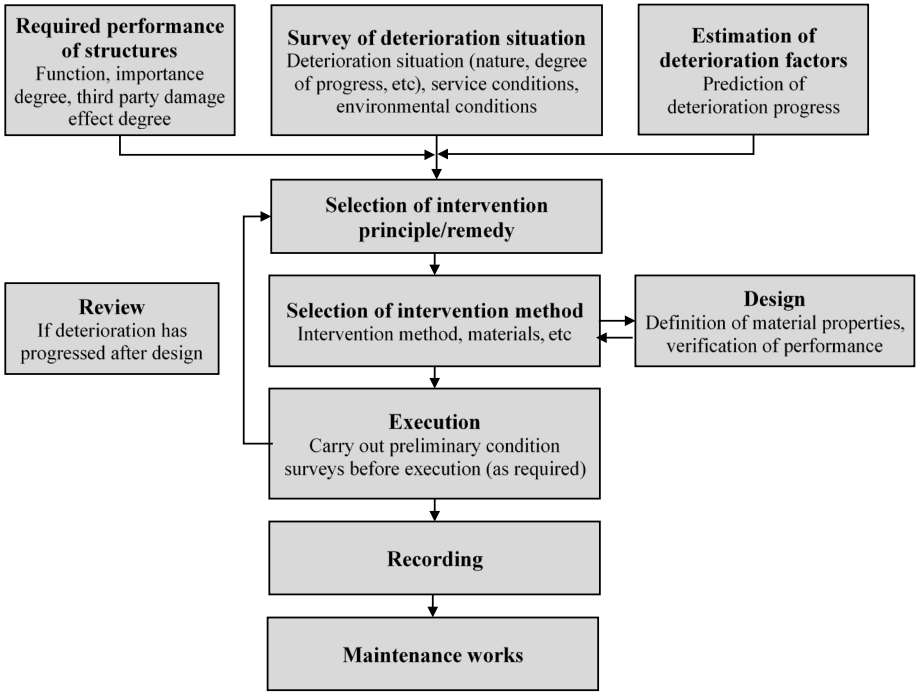


Figure 38.6-2: Flow chart for intervention process.

g) contamination of the protective concrete cover with corrosive agents (usually chloride ions) incorporated in the concrete when it was mixed or which have penetrated into the concrete from the environment;
h) stray electrical currents conducted or induced in the reinforcement from neighbouring electrical installations;
i) stress corrosion cracking of prestressed members;
j) galvanic corrosion (e.g. dissimilar metals, differential environments).

Where there is existing corrosion of reinforcement or a danger that corrosion will occur in the future, one or more of Principles / remedies of corrosion repair and prevention shall be selected.

In addition, the concrete itself can be repaired, where necessary, using Principles / remedies 1 to 5 and 11.

Figure 38.6-2 presents the flow chart for the intervention process, which includes steps for the '*Selection of intervention principle/remedy*' and then for the '*Selection of intervention method*'.

Guidance is provided for the selection of intervention methods for the following forms of deterioration in tables 38.6-2 to -8.

- Chloride-induced corrosion of carbon steel reinforcement
- Carbonation-induced corrosion of carbon steel reinforcement
- Frost damage
- Alkali silica reaction
- Thermal / shrinkage cracking
- Fatigue; and
- Seismic damage.

Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration.

Tables 38.6-2 to -8 are to be considered as being provided solely on an 'Informative' basis.

Table 38.6-2: *Suggested intervention methods for **chloride-induced corrosion of carbon steel reinforcement** in concrete – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration.*

Stage 1: No change in appearance: **Chloride ion concentration at depth of reinforcing bar below corrosion threshold**	**Stage 2: No change in appearance:** **Chloride ion concentration at depth of reinforcing bar above corrosion threshold**	**Stage 3: Concrete exhibiting cracking, spalling and rust staining**	**Stage 4: Rebar corrosion having caused a reduction in load capacity (strength) of affected members**

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Before initiation of rebar corrosion: No change in external appearance of concrete members	Moisture control	Keeping concrete dry
		Dehumidify concrete
	Protection against ingress	Apply surface impregnation
	Protection against ingress	Apply surface coating
	Preserve / restore passivity	Chloride extraction
	Prevention of rebar corrosion	Use cathodic protection

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Early-stage propagation of rebar corrosion: Rebar corrosion underway, but **no cracking** of concrete	Moisture control	Keeping concrete dry ®
		Dehumidify concrete®
	Protection against ingress	Apply surface impregnation®
	Protection against ingress	Apply surface coating ®
	Preserve / restore passivity	Make patch repair
		Chloride extraction
	Prevention of rebar corrosion	Use cathodic protection

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Mid- stage propagation of rebar corrosion: Rebar corrosion underway, **cracking of concrete** has developed	Moisture control	Keeping concrete dry ®
		Dehumidify concrete®
	Protection against ingress	Apply surface impregnation®
	Protection against ingress	Apply surface coating ®
	Preserve / restore passivity	Make patch repair
		Chloride extraction
	Prevention of rebar corrosion	Use cathodic protection
		Apply corrosion inhibitors
	Prevention of delamination	Use anchorage reinforcement, Jacketing Φ
	Concrete restoration	Make patch repair

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Advanced / late-stage propagation of rebar corrosion: Rebar corrosion underway, cracking of concrete and with **loss of structural strength** of affected concrete members	Moisture control	Keeping concrete dry Ω
	Protection against ingress	Apply surface impregnation Ω
	Protection against ingress	Apply surface coating Ω
	Preserve / restore passivity	Make patch repair Ω
		Chloride extraction Ω
	Prevention of rebar corrosion	Use cathodic protection Ω
		Apply corrosion inhibitors Ω
	Prevention of delamination	Use anchorage reinforcement, Jacketing Φ
Scaling, delamination	Concrete restoration	Make patch repair
Severe rebar corrosion	Rebar reinstatement	Rebar replacement
Inadequate structural capacity	Structural strengthening	Strengthening, rebuilding

Key: ® If a patch repair is made, implement after that activity completed. Ω Repair shall be conducted after cracks have been properly repaired (eg. by crack injection or patch repair).
Φ Anti-crack reinforcement (mesh or bar), which is placed in the jacketing concrete, is not for structural strengthening, but primarily for prevention of cracking / delamination of cover concrete, etc

*Table 38.6-3: Suggested intervention methods for **carbonation-induced corrosion of carbon steel reinforcement** in concrete – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration.*

Stage 1: No change in appearance: **Carbonation front at a depth smaller than the concrete cover to outer reinforcing bars**	**Stage 2: No change in appearance:** **Carbonation front at a depth greater than the concrete cover to outer reinforcing bars**	**Stage 3: Concrete exhibiting cracking, spalling and rust staining**	**Stage 4: Rebar corrosion having caused a reduction in load capacity (strength) of affected members**

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Before initiation of rebar corrosion: No change in external appearance of concrete members	Moisture control	Keep concrete dry
		Dehumidify concrete
	Protection against ingress	Apply surface impregnation
	Protection against ingress	Apply surface coating
	Preserve / restore passivity	Re-alkalisation
	Prevention of rebar corrosion	Use cathodic protection

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Early-stage propagation of rebar corrosion: Rebar corrosion underway, but **no cracking** of concrete	Moisture control	Keep concrete dry®
		Dehumidify concrete ®
	Protection against ingress	Apply surface impregnation®
	Protection against ingress	Apply surface coating ®
	Preserve / restore passivity	Make patch repair
		Re-alkalisation
	Prevention of rebar corrosion	Use cathodic protection

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Mid- stage propagation of rebar corrosion: Rebar corrosion underway, **cracking of concrete** has developed	Moisture control	Keep concrete dry®
		Dehumidify concrete ®
	Protection against ingress	Apply surface impregnation®
	Protection against ingress	Apply surface coating ®
	Preserve / restore passivity	Make patch repair
		Re-alkalisation
	Prevention of rebar corrosion	Use cathodic protection
		Apply corrosion inhibitors
	Prevention of delamination	Use anchorage reinforcement, Jacketing Φ
	Concrete restoration	Make patch repair

Deterioration / distress phenomena	**Intervention principle / remedy**	**Intervention method examples**
Advanced / late-stage propagation of rebar corrosion: Rebar corrosion underway, cracking of concrete and with **loss of structural strength** of affected concrete members	Moisture control	Keep concrete dry Ω
	Protection against ingress	Apply surface impregnation Ω
	Protection against ingress	Apply surface coating Ω
	Preserve / restore passivity	Make patch repair Ω
		Re-alkalisation Ω
	Prevention of rebar corrosion	Use cathodic protection Ω
		Apply corrosion inhibitors Ω
	Prevention of delamination	Use anchorage reinforcement, Jacketing Φ
Scaling, delamination	Concrete restoration	Make patch repair
Severe rebar corrosion	Rebar reinstatement	Rebar replacement
Inadequate structural capacity	Structural strengthening	Structural strengthening, rebuilding

Key: ® If a patch repair is made, implement after that activity completed. Ω Repair shall be conducted after cracks have been properly repaired (eg. by crack injection or patch repair).
Φ Anti-crack reinforcement (mesh or bar), which is placed in the jacketing concrete, is not for structural strengthening, but primarily for prevention of cracking / delamination of cover concrete, etc

*Table 38.6-4: Suggested intervention methods for **frost damage to concrete** – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration – Illustrations of some possibilities.*

Deterioration stage	Intervention principle / remedy	Intervention method examples
Stage 1: No change in appearance	Moisture control / Keep concrete dry	Apply surface coating
Stage 2: Small change in appearance - Some surface deterioration	Moisture control / Keep concrete dry	Apply surface coating / impregnation
Stage 3: Exposure of aggregate or cracking / spalling of surface / rust staining	Concrete restoration	Make patch repair
Stage 4: Cover concrete spalling, reinforcing bar exposure and corrosion, deterioration & concern about reduction of the load-carrying capacity	Structural strengthening	Reinforcing bar replacement
	Structural strengthening / reconstruction	Strengthening / reconstruction of elements

Table 38.6-5: *Suggested intervention methods for **alkali-silica reaction (ASR) damage to concrete** – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration – Illustrations of some possibilities.*

Deterioration stage	Intervention principle / remedy	Intervention method examples
Stage 1: Expansion - No observable change in appearance / concrete cracking	Moisture control / Keep concrete dry	Apply surface coating
Stage 2: Observable cracking - Fine surface cracking / surface runs of ASR gel	Moisture control / Keep concrete dry	Apply surface coating Undertake crack filling / crack injection
Stage 3: Further development of cracks / more extensive cracking, reinforcing steel corrosion may occur	Increased physical resistance - Prevention of peeling-off of surface concrete zone	Provide anti-crack rebar in anchorage zones Undertake jacketing
Stage 4: Further development of cracks - width and density of cracks increase / concrete deterioration, potential reinforcing bar exposure and corrosion - concerns about reduction of the load-carrying capacity	Concrete restoration	Make patch repair
	Structural strengthening	Reinforcing bar replacement

*Table 38.6-6: Suggested intervention methods for **thermally and shrinkage-induced cracks in concrete** – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration – Illustrations of some possibilities.*

Deterioration stage	Intervention principle / remedy	Intervention method examples
Stage 1: Surface crack widths of less than 0.2mm (which typically require no repair)	Protection against ingress of water / salts (In environment with high chloride level)	Undertake surface impregnation
Stage 2: Surface crack width 0.2mm to 1.0mm	Provide protection against ingress	Apply surface coating
Stage 3: Surface crack width greater than 1.0mm	Moisture control / Keep concrete dry	Undertake crack filling / crack injection, etc

*Table 38.6-7: Suggested intervention methods for **fatigue damage to concrete** (eg. concrete slabs forming road bridge decks) – Illustrations of some possibilities – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration.*

Deterioration stage	Intervention principle / remedy	Intervention method examples
Stage 1: Cracks occur transversely to the main direction of span of the concrete slab main girder	Moisture control / Keep concrete dry to eliminate the adverse influence of water	Introduce top-surface waterproofing coating or membrane [N2]
Stage 2: Development of longitudinal cracks parallel to the main direction of span of the concrete slab, plus development of a two-dimensional map cracking pattern	Moisture control / Keep concrete dry to eliminate the adverse influence of water	Introduce top-surface waterproofing coating or membrane [N2]
	Control of crack opening	Introduce bonded fibre reinforced polymer (FRP) sheet
	Installing strengthening / additional member /material at extreme tension fibre	Steel-plate or FRP bonding Bottom-surface (soffit) underlay / top-surface overlay [N3/N4]. Introduce supplementary support girders
	Improving shear stiffness of compression zone of cross-section	Introduce top-surface overlay
Stage 3: Further development of cracks, load-carrying capacity of slab declines rapidly with growth / increase in severity of cracking	Moisture control / Keep concrete dry to eliminate the adverse influence of water	Introduce top-surface waterproofing coating or membrane [N2]
	Installing strengthening / additional members at extreme tension fibre	Steel plate or FRP bonding; Add supplementary support girders Bottom-surface (soffit) underlay / top-surface overlay [N3/N4].
	Improving shear stiffness of compression zone of cross-section	Introduce top-surface overlay [N3]
Stage 4: Further development of cracks – Deep penetration of cracks into slab such that the structural continuity of the concrete slab is lost	Structural strengthening / reconstruction - Improve shear stiffness of compression zone of cross-section	Remove / replace affected concrete slabs - either totally or partially - or (possibly) overlay the top surface of slab [N5]

Notes:

N1. In an environment where de-icing agents (eg. road salts) are applied, the degree of deterioration must be carefully evaluated considering the influence of the deterioration caused by the combined action of fatigue effects and chloride-induced corrosion damage, frost damage and / or ASR, existing and in the future, where such effects are present.

N2. Implemented in conjunction with other preventative maintenance interventions, such as actions to keep the structure as dry as possible (as moisture significantly enhances fatigue damage) / minimise ingress of aggressive substances, keep coatings, membranes, gulleys and drainage pipes in good order, discharge rainwater from structure, etc.

N3 Top-surface overlaying is a method where the thickness of the structural element associated with the top surface of the existing concrete members (eg. deck slab) is increased by adding cementitious materials, which will commonly be reinforced. The technique enhances the performance (e.g. strength, stiffness) of the existing concrete structure and is applicable to highway bridge decks, etc. The overlay concrete may comprise UHPC to take advantage of its high strength, whilst minimising the additional weight added to the structure. Top-surface overlaying / introduction of additional tensile capacity may be necessary at continuous slab / girder support locations.

N4 A bottom-surface (soffit) underlay augments the performance of the bottom surface member (eg soffit slab) by adding cementitious material and tensile reinforcement – refer N3.

N5. Implemented with other appropriate remedial / preventative interventions, such as the installation of a top-surface waterproof membrane / Note N2 example actions.

Table 38.6-8: *Suggested intervention methods for **damage induced by seismic actions** (to lower section of concrete piers) – Intervention method selection depends upon the objectives of the intervention and the degree of existing deterioration – Illustrations of some possibilities.*

Deterioration stage	Intervention principle / remedy	Intervention method examples
Stage 1: No or minor damage, no appreciable reduction (<5%) in the load-carrying capacity in the short-term [N1]	Restoration of stiffness and load-bearing capacity	Undertake resin injection Make patch repairs
Stage 2: Moderate damage, modest degradation effects & decrease in the load-carrying capacity (<15%)	Restoration of stiffness and load-bearing capacity	Undertake resin injection Make patch repairs
	Improvement in flexural strength	Undertake steel plate jacketing / RC jacketing (not anchored)
	Improvement in toughness	Construct continuous fibre-reinforced jacket
Stage 3: Major damage, significant degradation effects & decrease in the load-carrying capacity (<30%)	Restoration of stiffness and load-bearing capacity	Undertake resin injection Make patch repairs
	Improvement in flexural strength	Undertake steel plate jacketing / RC jacketing (not anchored)
	Improvement in toughness	Construct continuous fibre-reinforced jacket
	Replacement / reconstruction of structural element	Removal / reconstruction of structural element
Stage 4: Collapse or partial-collapse (ie. no reliable residual resistance capacity)	Replacement / reconstruction of structural elements or the total structure	Removal / reconstruction of structural elements or the total structure

Notes:

N1. Once some reduction (even a minor reduction) in the load-carrying capacity has been established, a seismic retrofit will generally be undertaken, since even minor seismic damage (Stage 1) indicates that the current seismic performance is unlikely to be satisfactory.

N2. Interventions for abutments are based on the details given above for piers.

N3. In the case of damage to parapet or wing walls, such as cracking, this is treated as being equivalent to Stage 1 deterioration.

N4. The restoration of stiffness and load-bearing capacity is a necessary preparation for seismic strengthening works for Stages 1, 2 and 3.

N5. The seismic strengthening is necessary to prevent damage under future seismic events – applicable to Stages 1, 2 and 3.

38.6.5.3 Important considerations influencing selection of intervention methods

When selecting an intervention method, consideration shall be given to the state of deterioration of the structure, along with matters such as the application conditions, execution conditions, costs, environmental impacts etc relating to the various possible execution methods.

The important considerations are summarised in tables 38-6-9 to -14 for the following forms of deterioration:

- Chloride-induced corrosion of carbon steel reinforcement
- Carbonation-induced corrosion of carbon steel reinforcement
- Frost damage
- Alkali silica reaction
- Thermal and moisture shrinkage cracking
- Fatigue; and
- Seismic damage.

Tables 38.6-9 to -14 below are provided solely on an 'Informative' basis.

Table 38.6-9: Factors affecting intervention method selection for chloride-induced / carbonation-induced corrosion of carbon steel reinforcement in concrete.

Deterioration stages >>> **Intervention methods ∨∨∨**	**Stage 1: No change in appearance** **(Chloride ion concentration at depth of reinforcing bar is below corrosion threshold)**	**Stage 2: No change in appearance** **(Chloride ion concentration at depth of reinforcing bar is above corrosion threshold)**	**Stage 3: Concrete exhibiting cracking, spalling and rust staining**	**Stage 4: Rebar corrosion having caused a reduction in load capacity (strength) of affected members**
Moisture control	Preventative measures shall be implemented to keep concrete dry (eg introduce coating / membrane)	Same as left	Same as left.	Same as left.
Surface coating	• Necessary to confirm surface coating will be effective. • Avoid moisture accumulation below the coating (eg employ water vapour permeable coating) • Periodic re-coating is required, as the coating material deteriorates, to prevent water and chloride penetration.	• Same as left • Surface coating will be ineffective against previous chloride ingress into the concrete. • Consider the effect of internal diffusion of ingressed chlorides after coating application.	• Same as left • If a patch repair is to be carried out, surface coating shall be undertaken after that is work is completed and cured.	Consideration shall be given to reconstruction as other interventions will only provide temporary relief / a slowing of deterioration / damage.
Surface impregnation	• Necessary to confirm that water penetration into the concrete can be suppressed by surface impregnation. • Avoid moisture collection below impregnated surface. • Less effective as a barrier than surface coating. • Performance depends on impregnation material used. • There is little empirical data on durability benefits. • Surface impregnation may reduce the adhesion of subsequent patch repair / surface coating treatments.	• Same as left • Surface impregnation ineffective against previous chloride ingress into the concrete. • After impregnation application, consider the effect of internal diffusion of high levels of previous chloride ingress into the concrete.	• Same as left • If patch repair is to be carried out, surface impregnation will be undertaken after that work is completed and cured.	
Patch repair	Not applicable	• Necessary to check remaining load-capacity after concrete substrate removal. • Where third-party damage or injury is possible (eg. to a member of the public), such as might arise from spalled concrete, an intervention to prevent spalling is required.	Same as left	
Chloride extraction	• It is necessary to consider the cost. • There is the possibility of the occurrence of ASR.	Same as left	Same as left	
Cathodic protection	It is necessary to consider the maintenance of the cathodic protection system equipment.	Same as left	Same as left	
Anchoring reinforcement, jacketing	Not applicable	Not applicable	Where third-party damage or injury is possible, an intervention to prevent spalling are required.	
Rebar replacement	Not applicable	Not applicable	Not applicable	
Strengthening, reconstructing	Not applicable	Not applicable	Not applicable	

Notes: *Interventions applied for carbonation-induced corrosion are essentially the same as those for chloride-induced damage, except that chloride extraction and cathodic protection would not be used. However, as carbonation-induced corrosion progresses slowly, a structure might be managed by 'doing-nothing and monitoring for any adverse changes'.*

*Table 38.6-10: Factors affecting the selection of intervention methods for **frost damage to concrete** – Illustrations of some possibilities.*

Deterioration stages >>> **Intervention methods** VVV	**Stage 1: No change in appearance**	**Stage 2: Small change in appearance - Some surface deterioration**	**Stage 3: Concrete exhibiting exposure of aggregate and / or cracking / spalling of surface / rust staining**	**Stage 4: Cover spalling, rebar exposure and corrosion, deterioration / reduction in load capacity (strength) of affected members**
Moisture control	Preventative measures shall be implemented to keep concrete dry (eg introduce coating / membrane)	• Same as left	As deterioration progresses, more water tends to accumulate within the concrete, exacerbating deterioration	Consideration shall be given to reconstruction as other interventions will only provide temporary relief / a slowing of deterioration.
Surface coating	Not applicable	• Water ingress, including from the rear or joints, trapped within concrete may result in frost damage from freezing of internal water. • Periodic re-coating is required, as the coating material deteriorates, to prevent water penetration and resulting frost damage.	• Same as left • If patch repair is carried out, surface coating will be undertaken after that is work is completed and cured	
Surface impregnation	Not applicable	• Less effective as a barrier than surface coating. • Water ingress, including from the rear or joints, trapped within concrete may result in frost damage from freezing of internal water • There can be a difference in performance, depending on the material used for impregnation. • There is little empirical data on durability benefits available. • Surface impregnation may reduce the adhesion of subsequent patch repair / surface coating treatments.	• Same as left • If patch repair is carried out, surface impregnation will be undertaken after that is work is completed and cured.	
Crack injection	Not applicable	• Crack injection may be carried out in the cracked zone. However, loose concrete shall be removed and a patch repair carried out.	Not applicable	
Patch repair	Not applicable	• In situations where deterioration may result in damage or injury to a third-party (eg. to a member of the public), such as might arise from spalled concrete, an intervention to prevent spalling is required.	• Necessary to check remaining load-capacity after removal of damaged concrete substrate. • Where a hazard or injury to a third-party is anticipated, a preventative intervention is required.	
Rebar replacement	Not applicable	Not applicable	Not applicable	
Strengthening / reconstruction	Not applicable	Not applicable	Not applicable	

*Table 38.6-11: Factors affecting the selection of intervention methods for **alkali-silica reaction (ASR) damage to concrete** – Illustrations of some possibilities.*

Deterioration stages **Intervention methods**	**Stage 1: No change in appearance** **Expansion, but no observable change in appearance or concrete cracking**	**Stage 2: Observable cracking - Fine surface cracking / surface runs of ASR gel**	**Stage 3: Further growth of cracks / more extensive cracking, reinforcing steel corrosion may occur**	**Stage 4: Concern about residual load capacity (strength) due to extent of concrete cracking / rebar corrosion in members**
Moisture control	Preventative measures shall be implemented to keep concrete dry (eg introduce coating / membrane)	Same as left	Same as left	Same as left
Surface coating	• May be employed as a preventative measure • Water ingress, including from the rear or joints, trapped within concrete may result in ASR. • There can be a difference in performance, depending on the material used for impregnation. • There is little empirical data on durability benefits. • Surface impregnation may reduce the adhesion of subsequent patch repair	• Water ingress, including that from the rear or joints, trapped within concrete may result in ASR. • Periodic re-coating is required, as the coating material deteriorates, to prevent water penetration which promotes ASR related expansion.	• Same as left • Employed in combination with crack injection	Same as left
Surface impregnation	• As above. • Less effective as a barrier than a surface coating • Surface impregnation may reduce the adhesion of subsequent patch repair / surface coating treatments	Same as left	• Same as left • Possible use with crack injection	Same as left
Bandaging of surface crack/ crack filling	Not applicable	Always use deformable filler material, unless ASR related expansion has been shown to have effectively ceased.	• Same as left • Used in combination with crack injection	Same as left
Crack injection	Not applicable	Always use deformable filler material, unless ASR related expansion has been shown to have effectively ceased.	Same as left	Same as left
Anchoring reinforcement, jacketing	Not applicable	Not applicable	Typically, the effect of jacketing is small unless the jacket completely envelopes the full circumference of the member	Same as left
Patch repair	Not applicable	Not applicable	Not applicable	The potential effects of residual ASR expansion shall be considered.
Rebar replacement	Not applicable	Not applicable	Not applicable	If ASR related expansion has not ceased, interventions will be of only temporary benefit. Reconstruction shall be considered.
Strengthening, reconstructing	Not applicable	Not applicable	Not applicable	

*Table 38.6-12: Factors affecting the selection of intervention methods for **thermal- and shrinkage-induced cracks in concrete** – Illustrations of some possibilities.*

Deterioration stages / **Intervention methods**	**Stage 1: Surface crack widths of less than 0.2mm, which typically require no repair**	**Stage 2: Surface crack width 0.2mm to 1.0mm**	**Stage 3: Surface crack width greater than 1.0mm**
Surface impregnation	• There can be a difference in performance, depending on the material used to make the impregnation. • There is little empirical data on durability benefits. • Surface impregnation may reduce the adhesion of subsequent patch repair / surface coating treatments.	Not applicable	Not applicable
Crack injection	Not applicable	If crack width growth has not effectively ceased, use deformable filler material.	When the cracks are wide, it may be difficult to achieve complete / satisfactory crack injection.
Bandaging of surface crack/ crack filling	Not applicable	Sufficient protection / an adequate surface barrier may not be achieved if extensive / severe concrete damage / cracking has been caused by chloride-induced / carbonation-induced corrosion of carbon steel reinforcement in concrete.	Same as left

*Table 38.6-13: Factors affecting the selection of intervention methods for **fatigue damage to concrete** (eg concrete slabs forming road bridge decks) – Illustrations of some possibilities.*

Deterioration stages **Intervention methods**	**Stage 1: Cracks occur transversely to the main direction of span of the concrete slab**	**Stage 2: Development of longitudinal cracks parallel to the main direction of span of the concrete slab, plus development of a two-dimensional map cracking pattern**	**Stage 3: Further development of cracks, load-carrying capacity of slab declines rapidly with growth / increase in severity of cracking**	**Stage 4: Further development of cracks– Deep penetration of cracks into slab such that the structural continuity of the concrete slab is lost**
Bridge deck waterproof membrane	• It is necessary to carefully examine the complete surface water drainage arrangements / system to avoid problems that allow moisture / water accumulation in the concrete, especially below the deck (top surface) waterproofing membrane. • The installation of the deck waterproofing membrane must be undertaken with great care to avoid problems. The number of membrane joints shall be minimised, as they and the edges of the treated zone form weak points. • Unless the deterioration is at a very early / initiation stage, it is desirable to undertake all necessary concrete and reinforcement repairs prior to applying a deck waterproofing membrane.			Not applicable as a single intervention at this deterioration stage. Consider reconstruction, as other interventions will only provide a temporary improvement.
Introduce bonded steel plate / FRP sheet strengthening	Not applicable	• The strengthening materials must be bonded to the existing members / substrate with adhesive • It is necessary to ensure that the strengthening system material (eg. steel plate / FRP sheet) does not exhibit a decrease in strength (weaker zones) at joints and corners. • The surface of the concrete must be prepared to a satisfactory standard to ensure an adequate bond between the concrete substrate surface and the strengthening material.		As above
Top-surface overlay	Not applicable	• In order to achieve the desired repair / strengthening effects, it is necessary to remove damaged parts / deteriorated materials and to properly prepare the concrete substrate interface. • Ensure adequate bond between the top-surface overlay and the substrate / existing members. • When laying an asphalt pavement over (above) a top-surface overlay, instal a waterproof layer on (above) the top-surface overlay to ensure the deck is effectively waterproofed. Pay particular attention to the detailing and arrangements for drainage of water from the deck slab, especially where drainage pipes pass through the slab.		As above
Bottom-surface (soffit) underlay See Notes N3/N4 - Table 38.6-7	Not applicable	• Ensure bond between the substrate / existing member and the bottom-surface (soffit) underlay. • Avoid moisture accumulation above the bottom-surface (soffit) underlay.	Not applicable	As above
Introduction of supplementary support beam(s)	Not applicable	The construction of supplementary support beam(s) is essentially the same as that of new construction. It is also important to keep concrete dry to avoid adverse influences of water – see above.	Same as left	As above
Remove / replace affected concrete slabs - either totally or partially or the upper layer of existing slab, as appropriate	Not applicable	Not applicable	Not applicable	• Partial reconstruction is often used for severe localised deterioration of a slab. • Upper layer replacement is undertaken after removing the deteriorated portion of the slab, especially when there is the influence of water, by using a material having a quality equal to or higher than that forming the existing slab.

*Table 38.6-14: Factors affecting the selection of intervention methods for **damage induced by seismic actions** – Illustrations of some possibilities.*

Deterioration stages **Intervention methods**	**Stage 1: Minor damage, no significant reduction in the load-carrying capacity in the short term**	**Stage 2: Moderate damage, modest effects and decrease in the load-carrying capacity**	**Stage 3: Major damage, significant effects and decrease in the load-carrying capacity**	**Stage 4: Collapsed or partially-collapsed**
Injection method of resin-based materials and cross-sectional repair	• When the amount of cross-sectional loss is large, it is preferable to use a material having a large Young's modulus, such as resin concrete containing coarse aggregate, in order to minimise the decrease in rigidity after repair. • When the amount of cross-sectional loss is large, it is advisable to consider the prepacked construction method in which the formwork is assembled, aggregates are put in advance, and resin is injected to bind them.	Not applicable	Not applicable	Not applicable
Steel plate jacketing	• In the case of flexural strengthening, anchor the steel plate to the member joint or foundation. • It is necessary to employ anti-corrosion / protection measures for the steel plate.			Not applicable
RC jacketing	• In order to reliably improve the flexural strength of columns and piers, the axial reinforcing bars shall be effectively anchored into the concrete foundation / footing construction – This may be a difficult operation to perform. • The increase of dead load due to jacketing will be substantial (large).			Not applicable
FRP jacketing				Not applicable
Prestressing	Strengthening effect can also be expected when prestressing is used in combination with RC jacketing.	Not applicable	Not applicable	Not applicable
Additional wall / extension method	Not applicable	Resin or mechanical anchorages may be used to fix the reinforcing bars into the substrate / the existing concrete member.		Not applicable
Removal / reconstruction of structural element	Not applicable	Not applicable	• If the works required to anchor the vertical reinforcing bars necessary to strengthen a bridge pier / column would decrease the strength of the footing, the footing would also be thickened as appropriate. • If the pier weight increases significantly, it may be necessary to take account of the increased load upon the foundation.	

38.6.6 Condition survey for interventions

The general approach to undertaking a (diagnostic) condition survey of an existing structure is presented in section 38.2. This section 38.6.6 is concerned with performing a condition survey before an intervention is made. For intervention design and execution, it is necessary to understand the condition of the parts where an intervention is to be made, the properties of the materials in the existing structure, the actions on and environmental conditions affecting the existing structure (both now or as expected to be experienced in the future) and, if any change in condition has occurred, its type, spatial distribution, etc shall be estimated.

Also, if the intervention history and the design and execution records are available, they shall also be taken into account when selecting suitable intervention methods.

Design documents for the existing structure can provide useful information for evaluating both the residual performance of the existing structure before intervention, and to estimate its performance after intervention.

In addition, some materials used for interventions are adversely affected by environment influences, such as temperature. Therefore, it is necessary to understand what the site environment conditions will in the period when the execution works will take place. There may also be limited space on the work site, which may introduce constraints upon the storage/delivery of materials and equipment. Such factors will need to be taken into account when the works execution plan is formulated.

In obtaining the expected effect of intervention, integrity between the repaired or retrofitted parts and existing structure is important. Ensuring integrity requires taking necessary measures based on the understanding of degradation such as carbonation of the surface of the existing structure, damage such as cracks, sprinkling or leakage of water, etc.

Degradation of a concrete structure after intervention progresses at a different rate depending on the type of cause and degree of degradation. Therefore, before intervention is performed for a deteriorating concrete structure, it is necessary to understand the cause(s), degree and extent of the degradation which has occurred.

The condition of an existing structure where interventions are planned shall be investigated to obtain the information necessary for intervention design and planning the execution activities. This condition survey is intended to aid selection of appropriate intervention methods, but also to establish the quantity of the various types of work and the extent of the chosen intervention methods.

While this condition survey will typically involve the review of existing documentary evidence (such as earlier diagnostic condition surveys), the survey / testing activities will involve on-site-investigations.

The review of existing documentary evidence could include:

- Specifications and design standards used;
- Design and re-design documents;
- Previous investigation reports, records and histories of repairs etc;
- Records from construction / execution of any previous interventions;
- Records / reports on in-service changes in the usage of structure or exposure environments or revised performance requirements, etc;
- Verification of required performance;
- Maintenance works records; and
- Reports on climatic / environmental issues / geographical conditions.

The on-site-surveys / investigations could need to estimate parameters such as material properties, degradation, damage and defects in the existing structure.

The above inputs will be used to facilitate the following:

- An evaluation of the residual performance of the structure;
- Structural integrity between the existing parts and the retrofitted parts;
- Adequate structural performance after intervention; and
- Prediction of durability / degradation after intervention.

The parts of the structure where interventions have been made (i.e. the retrofitted parts) may be stiffer and/or heavier as result of the intervention.

For the structure to meet the specified performance requirements after intervention, appropriate structural details and methods of bonding and anchoring reinforcing materials shall be selected to ensure structural integrity between the existing parts and the retrofitted elements.

Also, the load-share and stress distribution between the existing parts and retrofitted parts shall be considered for the circumstances before, during and after intervention. The load-bearing capacity and stiffness (rigidity) of strengthened members shall be estimated considering to the structural properties of these members, the intervention method and so forth.

For the performance verification of the structure after intervention, the necessary design assumptions for the evaluation of material properties and effects of actions shall be set appropriately, taking into account the current condition of the existing structure, along with the condition in which the structure is expected to be placed during the remaining service life.

When establishing the design assumption, it may be possible to take account of the as-built execution records and/or measure the actual dimensions of existing members.

Generally, verification of the safety and serviceability of a structure after an intervention has been made is based on the assumption that the elements of the original structure and the interventions effectively act together as a single structural entity (ie. act integrally) to resist external forces and actions. When deciding upon structural details for an intervention, it is necessary to adopt a bonding method which ensures that adequate structural integrity is achieved.

38.6.7 Design of interventions

38.6.7.1 General

In intervention design a rational structural concept must be developed, together with associated structural details, so that after intervention the structure fulfils the required performance for the remaining service life.

The conceptual design phase is a critical part of this process. The concept phase is where the identified requirements are examined, potential solutions satisfying these are defined, evaluated, refined and a suitable structural concept chosen for further detailed development. Many iterations of the process are commonly required to refine the solution concepts to accord with the functional requirements and comply with the sustainability/financial/other constraints which exist.

The procedures described in section 38.6.7 follow a common approach for all types of physical intervention methods. Each particular intervention method may need a specific verification procedure.

The specific verification procedure can follow the verification procedures provided in chapter 30, but shall consider the differences described in section 38.6.7. Appropriate materials must be chosen for the selected intervention method. The relevant information for intervention materials is provided in chapter 19.

38.6.7.2 Structural plan and details

An appropriate method of bonding supplementary reinforcing materials must be selected to ensure that the structure achieves the satisfactory level of integrity after the intervention has been made.

The type of reinforcing material and the structural properties of the retrofitting parts need to be selected carefully, taking into account the differences in load-bearing capacity and rigidity between the existing parts and the retrofitting parts of the structure, to ensure that the overall structure has the required load-bearing capacity and stiffness (rigidity).

If the rigidity ratio of the existing parts and retrofitting parts is large, care needs to be exercised because an adequate strengthening effect may not be obtained depending on the performance of the bonding method. It is important to give full consideration (in advance) as to how to make the existing parts, the bonding parts and retrofitting parts resistant to each load level, respectively. Note that there is the need to fully understand the reinforcement arrangement of the existing parts in the study of the existing structure, including whether round bars are used.

In the structure after intervention, the permanent action (dead load) of the existing parts is assumed to be borne by only those existing parts. All the variable actions (live loading), plus the permanent action (dead load) of the retrofitting parts, are assumed to be borne by the existing parts integrated with those retrofitting parts. It is important to set the rigidity ratio between the existing parts and retrofitted parts appropriately, in order to control the resistance mechanism and the load-sharing ratio between these two types of member.

Care needs to be taken to prevent the existing parts of the structure (where no intervention has been made) from being damaged or the overall integrity of the structure being impaired by time-dependent deformation of reinforcing materials and/or bonding materials, such as creep occurring in reinforcing materials or adhesive, due to the stress from a permanent action. Also, there are cases where the existing parts of the structure restrict the shrinkage (change in length) of the retrofitting parts (where an intervention has been made). Accordingly, a thorough consideration of indirect actions is also necessary.

The structural analysis method employed shall be based upon the modelling of actions and structural responses. Where possible, a structural analysis method shall be selected which provides the required verification indices, such as bending moment for safety and crack width for serviceability as a direct output. If the chosen structural analysis method does not provide this output, the response values from the

The reinforcing material used, the preparation of the material surface and the structural properties provided by the retrofitting parts of the structure must be carefully selected, in order that, after intervention, the structure achieves the required load-bearing capacity and an appropriate stiffness (rigidity).

The effects of stress redistribution upon the load-bearing behaviour must be considered in the structural analysis of the overall structure and its component parts (both those which are original and those subject to an intervention).

38.6.7.3 Structural analysis

A structural analysis procedure shall be used which can adequately evaluate the chosen verification indices such as bending moment for safety and crack width for serviceability requirements.

The verification process does not necessarily require that the same structural model and analysis theory be used for all limit states. A structural model and an analysis theory appropriate to each individual limit state may be used. Also, it is

structural analysis will have to be subsequently converted into values for the respective verification indices.

Since the stiffness of a repaired or strengthened structure changes due to such factors as the occurrence of cracks, the non-linearity of the materials of the members needs to be taken into account. In general, the mechanical model for materials mentioned in MC2020 may be used to represent material non-linearity.

The use of cross-sectional analysis techniques to verify the ultimate limit state (failure) of repaired and/or strengthened structural members assumes that the integrity of the existing and retrofitting parts is assured (ie. that the elements of the original structure and the interventions effectively act together as a single structural entity – see comments on bond in the following section).

The possible failure modes of the repaired and/or strengthened structure include peeling failure in the repaired or strengthened members, in addition to bending and shear failure modes. Here, it is desirable that the structural analysis for the repaired and/or strengthened structure involves considering the influence of the peeling and delamination failure modes in retrofitted members.

In regard to potential fatigue failures, there may be a need to take account of the different "condition states" of the existing materials and any new materials introduced as part of an intervention scheme. The existing 'old' reinforcement might have little residual fatigue life, being able to bear only very few more load cycles. However, the new additional reinforcement, which may have to be installed to strengthen the structure, could provide a considerable extension of the fatigue life of the structure / the respective elements. The 'existing' and new materials shall be considered as separate entities – refer to section 38.2.5.4 for further discussion of this topic.

The durability of structures after the intervention has two issues. The first one is the durability of the existing part (original element) and the second one is that of the retrofitting part (intervention element). If further degradation occurs in the existing part, the intervention method may need to be changed to another method for better containment of the degradation mechanism or the intervention may need to be conducted again to improve the quality of intervention work. The long-term

necessary to adopt a safety margin appropriate to the reliability of analysis theory used to calculate the response values.

As for structural members, the influence of non-linearity shall be taken into account according to its influence upon the (type of) response being evaluated. A structural member may be regarded as acting linearly if it is evident that the non-linearity of the member does not affect the verification indices, such as sectional force, or that conservative and rational evaluation results are obtained.

Structural analysis for the verification of safety for:

- fatigue failures in repaired or strengthened structures; or
- cross-sectional failure (such as flexure and shear failure) of a structure or elements thereof;

shall take account of the influence of the peeling and delamination of reinforcing materials (or other forms of deterioration) in existing and retrofitted members.

38.6.7.4 Verification of durability

It is necessary to verify that, after intervention, the structure will achieve the required durability for the design (remaining) service life. This process shall consider both the original structure and those elements where interventions have been performed, seeking to ensure that all components are working together

efficiency of the intervention may not be assured due to the limited available information on its long-term performance. Therefore, regular survey / monitoring / maintenance after the intervention is necessary.

The bond between the original structure (ie. the substrate) and the materials applied during the intervention is affected by many factors including external mechanical actions (forces, displacements, vibration etc), indirect actions on cement-based materials and bonding materials (time-dependent deformation, such as shrinkage and creep), degradation of cement-based materials (freezing and thawing, alkali-aggregate reaction etc), problems with the construction method (inappropriate breaking out of original concrete, surface preparation or priming) and incompatibility between cement-based materials and bonding materials. The presence of moisture interface between the substrate and the materials applied during the intervention can have an adverse impact on the bond achieved / the longer-term integrity of the intervention. Understanding the reasons why moisture may be present at this interface is important. Appropriate maintenance activities can minimise moisture related problems.

Therefore, in intervention design, it is of great importance to eliminate the factors which adversely affect the integrity of the structure after intervention. Careful examination of the designed details and construction method are required. It is important to select cement-based materials and bonding materials having appropriate levels of strength, durability and water-tightness for the intended application, as well as a satisfactory long material 'open time' – a parameter indicating how long the material is workable for (thus an indicator of buildability). The careful selection and application of a bonding aid to the interface can improve the strength of the bond between the substrate and cementitious materials applied during the intervention.

Pull-off strength testing is one measure used to verify the strength/integrity of the bond achieved between the substrate and the materials applied during the intervention. However, additional measures are required for other performance characteristics, such as bond durability/longevity - with the suitability of parameters such as fracture energy and elongation capacity being investigated.

The durability verification methods in section 30.6 are for concrete structures with embedded steel reinforcement. Generally, the intervention component is situated on the outside of the original element. In addition, the materials used in the intervention component may be different to the conventional concrete and steel reinforcement used in the original member. Therefore, the condition limit states need to be verified for both the intervention component and the remaining parts of the effectively (i.e. structural integrity has not been compromised) and that there are no adverse interactions.

If survey/monitoring reveals that deterioration of an intervention is occurring or that it may not be durable for the service life specified for the intervention design, the intervention design shall be reviewed. If deterioration is occurring, the cause and mechanism(s) involved shall be established, so that another intervention can be planned and/or appropriate LCM activities instigated.

To achieve the desired performance characteristics after intervention, the integrity of the structure after intervention needs to be assured (i.e. that the elements of the original structure and the interventions effectively act together as a single structural entity). A critical aspect of this is that a satisfactory bond is achieved between the original structure (i.e. the substrate) and the materials applied during the intervention (e.g. cementitious materials, polymers, coatings etc), so there is no debonding, delamination, peeling etc.

The durability verification methods are provided in section 30.6 as an evaluation of the condition limit states (CLS) associated with the durability of structures after interventions have been made.

original element, which are exposed to the external environment. Separate condition limit states may be necessary for the intervention component and the original element, as well as for the interface/connection between these two parts.

The ultimate limit state of structures after intervention can be verified by a similar method to that used for structures without an intervention. However, in addition, there are also the ultimate limit states relating to the interface/connection between the existing part (original element) and the retrofitting part (intervention element), such as debonding and localised connection failure.

The verification of structural safety for static loading, non-static loading and seismic action can be conducted according to the methods provided in sections 30.1, 30.2 and 30.3, respectively. For the case of interventions with external bonded non-metallic reinforcement, sections 30.3 and 30.8 shall be referred to. However, the scope of the verification methods provided in chapter 30 are constrained to the scope specified for the individual verification methods. The scope may include the applicable range of material properties for the concrete, reinforcement and other materials employed. For interventions, the material-related conditions can be different to those for ordinary concrete structures. The ultimate limit states relating to the interface between the existing part and externally bonded non-metallic reinforcement and externally-bonded cementitious material can be verified on the basis of sections 30.1.30.6 and 30.8, respectively. However, no verification method for an interface / connection is provided in chapter 30 for other intervention methods.

After intervention the functionality of structure, and parts thereof, shall sustain sufficient functions suitable for their purpose of use, such as deflection and ‘comfort’ of the users, for the design service life.

The integrity of the structure after intervention needs to be assured (ie. that the elements of the original structure and the interventions effectively act together as a single structural entity).

38.6.7.5 Verification of structural safety

In principle, the verification of structural safety after intervention will generally involve checking that neither the ultimate limit state condition nor the fatigue failure limit state, using member cross-sectional analysis or other suitable means, will be reached during the design (remaining) service life.

During the structural verifications, the initial stress states of both the existing cross-sections and materials (possibly reduced by damage, deterioration or the intervention methods) and the new cross-sections and materials, must be considered.

The verification method for structural safety is provided in sections 30.1, 30.2, 30.3 and 30.8 as the evaluation of structural safety (ULS) for structures with interventions.

38.6.7.6 Verification of serviceability

It is necessary to verify that, after intervention, the structure achieves the required serviceability throughout the design (remaining) service life.

The serviceability verification shall be performed by checking that after intervention none of the members of the structure reach the serviceability limit state of under the design actions.

During the structural verifications, the initial stress states of both the existing cross-sections and materials (possibly reduced by damage, deterioration or during

The serviceability limit states of structures after intervention can be verified by a similar method to that used for structures where no intervention has been performed, refer section 30.5. For an intervention made using externally bonded non-metallic reinforcement, the method provided in section 30.8.5 can be applied.

The scope of the verification method in section 30.5 is appropriate to the range of material properties for the concrete, reinforcement and other materials employed. For interventions, the material-related parameters can be different to those employed for ordinary concrete structures.

When formulating a post-intervention conservation plan, it is necessary to:

- Determine the period of maintenance based on the design service life of the structure, as considered during the design of the intervention.
- Consider the actual characteristics of the methods and materials used to make the intervention. In particular, attention shall be paid to the members and material properties deemed to be vulnerable to change with time / interaction with the environment (ie. durability implications).
- Formulate the conservation plan early in the planning and design phases for the intervention.
- Verify the adequacy of the conservation plan by collecting information about the evolving condition of the structure after intervention. If the revised degradation prediction based on survey results is appreciably different to the prediction made during design, it is important to assess the implications for the conservation plan and identify any remedial actions necessary.

The post-intervention conservation plan depends on the long-term performance of the intervention method. There are various intervention methods available, however reliable information on long-term performance may not be available at the the implementation of the chosen intervention methods) and the new cross-sections and materials, must be considered.

The verification method of serviceability is provided in sections 30.5 and 30.8.5 as the evaluation of serviceability (SLS) for structures with interventions.

38.6.8 Execution of interventions

After the design of the intervention has been completed, execution of intervention works shall be conducted in accordance with chapter 37.

38.6.9 Conservation after interventions

After intervention, the method of condition survey, the intervention method used, plus design and execution details, shall be recorded.

Conservation of the structure after intervention shall be planned taking account of the intervention method(s) employed.

The post-intervention conservation plan shall be made according to the estimated long-term performance of the intervention method.

time that the intervention is carried out. Therefore, regular survey / monitoring / maintenance after the intervention is necessary

The information relating to quality control and assurance measures undertaken during execution of intervention shall be documented in the Service Life File for the structure, together with other documents relating to conservation activities after interventions.

This process would be expected to draw upon information contained in the *Birth Certificate* document.

There is the expectation that these activities would draw upon/contribute to the preparation of a *re-Birth Certificate.*

For further details see *fib* Bulletin 93 [38-5].

The conservation records shall be kept for the life of the structure. It is also important that an appropriate way is established to transfer the knowledge and understanding of the structure from the present generation of engineers/persons with responsibility for the management of the structure to the following generations who will assume responsibility for its through-life (life-cycle) management.

38.7 Recording

To allow effective and efficient life cycle management, the conservation records must be kept and preserved for future reference in the service-life file – refer Figure 8.3-2. Conservation records must be included in the life cycle file. The conservation records for a structure must be preserved while it remains in service/in existence.

Details of recording during conservation are given in MC2020 section 10.6, along with other information regarding quality management.

References

[38-1] *fib* Bulletin 17 – Management, maintenance and strengthening of concrete structures, *fib* Technical Report, (fib, 2002).

[38-2] *fib* Bulletin 22. – Monitoring and safety evaluation of existing concrete structures, *fib* State-of-the-art Report, 2003

[38-3] *fib* Bulletin 44. – Concrete structure management: Guide to ownership and good practice, *fib* 2008

[38-4] *fib* Bulletin 62. – Through-life Care and Management of Concrete Structures - Assessment, Protection, Repair & Strengthening, Volume 5 of the *fib* Structural Concrete Textbook on behaviour, design and performance (2nd edition), published as *fib* Bulletin 62, January 2012

[38-5] *fib* Bulletin 93. – Birth Certificate and Through-Life Management Documentation, *fib* Technical Report, *fib* 2020

[38-6] *fib* Bulletin ZZ - Draft — Existing concrete structure life-management, testing and structural health monitoring, State-of-the-art Report [Draft by TG3.3 – In preparation, 2022]

[38-7] *fib* Structural Concrete 2020 — Existing concrete structures: Structural Health monitoring and testing for condition assessment, *Special Theme (15 papers) in Structural Concrete journal,* 2020, 21(4): ISSN 1464 – 4177 (print version).

[38-8] ISO 13822:2010. — Bases for design of structures – Assessment of existing structures, ISO 2010.

[38-9] ISO 16311-1:2014 — Maintenance and repair of concrete structures – Part 1: General principles., ISO 2014

[38-10] ISO 16311-2:2014. — Maintenance and repair of concrete structures – Part 2: Assessment of existing concrete structures. ISO 2014

[38-11] ISO 16311-3:2014. — Maintenance and repair of concrete structures – Part 3: Design of repair and prevention. ISO 2014

[38-12] ISO 16311-4:2014. — Maintenance and repair of concrete structures – Part 4: Execution of repairs and prevention. ISO 2014

[38-13] JCI 2014 — JCI Guidelines for assessment of existing concrete structures, Japan Concrete Institute, 2014.

[38-14] JCSE 2007 — JCSE, Guidelines for Concrete No 17, Standard Specifications for Concrete Structures - Maintenance, Japan Society of Civil Engineers (JSCE, 2007 – English version).
Note: The 2007 document is the latest English language version, 2010 version in Japanese.

[38-15] CEN TS 17440:2020. — CEN TS17440, Assessment and Retrofitting of Existing Structures, CEN, July 2020

[38-16] Hajdin et. al. — Hajdin R, Kušar M, Mašović S, Linneberg P, Amado J and Tanasić N, Working Group WG3 Technical Report: Establishment of a Quality Control Plan, September 2018, ISBN: 978-86-7518-200-9. An output of COST ACTION TU1406: Quality specifications for roadway bridges, standardization at a European level, European Co-operation in Science & Technology (COST), Cost TU1406 WG3 Report

[38-17] *fib* Bulletin XY — Durability planning for concrete structures, *fib* Technical Report, *fib* xxxx [In preparation by COM8]

[38-18] CIA Z7-01 — CIA Z7-01. Durability Planning, which is a Recommended Practice Report forming part of the CIA Concrete Durability Series, Concrete Institute of Australia, 2014.

[38-19] Strauss et. al. — Strauss, A., Madic Ivankovic, A., Matos, J., Casas, J, COST Action TU1406 Working Group WG1 Technical Report: Performance indicators for Roadway Bridges of COST Action TU1406, July 2016, ISBN 978-3-900932-41-1, Cost TU1406 WG1 Report

[38-20] *fib* Bulletin 33. — Durability of post-tensioning tendons. *fib* Recommendations 2005.

[38-21] *fib* Bulletin 59. — Condition control and assessment of reinforced concrete structures exposed to corrosive environments (carbonation/chlorides). *fib* State-of-the-art Report, 2011.

[38-22] CIRIA Report C788. — Sparkes P and Webb J: Structural health monitoring in civil engineering. CIRIA Report C788, Construction Industry Research and Information Association, London, 2020.

[38-23] Ye et. al. — Ye S, Lai X, Bartoli B and Aktan E: Technology for condition and performance evaluation of highway bridges. Journal of Civil Structural Health Monitoring, published online at: https://doi.org/10.1007/s13349-020-00403-6.

[38-24] ASCE — ASCE Technical Committee on Structural Identification (2013), Structural identification of constructed systems: approaches, methods, and technologies for effective practice of structural identification.
In: Çatbas FN, Kijewski-Correa T, Aktan AE (eds). ASCE. pp 1–248, 2013, published online at: https://doi.org/10.1061/9780784411971

[38-25] Moss 1993 Moss, RM., Load testing of beam and block floors. Structures and Buildings, Proceedings of the Institution of Civil Engineers, London, May 1993, Volume 99, p211-223.

[38-26] Moss 1994 Moss, RM., Assessment of stiffness increase due to non-structural screeds. The Structural Engineer, Institution of Structural Engineers, London, Volume 72, No 14, 19 July 1994, p221 - 228.

[38-27] Moss 1995 Moss, RM., Load testing of hollow plank concrete floors. The Structural Engineer, Institution of Structural Engineers, London, Volume 73, No 10, 16 May 1995, p161-168.

[38-28] Matthews, S. Deployment of instrumentation for in-service monitoring of structures; The Structural Engineer, Vol 78, No 13, 4 July 2000, pp28-32.

[38-29] *fib* Bulletin 102 Guide for protection and repair of concrete structures, *fib* Guide to Good Practice, March 2022

[38-30] *fib* Bulletin 103 Guide for strengthening of concrete structures, *fib* Guide to Good Practice, May 2022

[38-31] CONREPNET Matthews et al: A Performance Based Approach: Concepts and application for the use of performance concepts to achieve durable remediation of concrete structures, CONREPNET Project final publishable report, IHS-BRE Press, Report EP77, 2007

POSSIBLY ADD:

Draft ISO Standards: ISO 210301**:** Structural intervention upon existing concrete structures using cementitious materials" (4 Parts) – Part 1: General principles; Part 2: Top-surface overlaying; Part 3: Bottom-surface underlaying; Part 4: Jacketing. Need to review these documents to see if their wording has been sufficiently improved by further editing work to make them suitable references.

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Specific provisions for implementation of circularity principles in life cycle of the structures are still under development.

39 Circularity and Dismantlement

39.1 General provisions for circularity

For general principles of implementation of circularity approach for concrete structures reference is made to Chapter 9.1.

In general, provisions relevant for new and existing structures related to sustainability in conceptual design approach, structural design approach, construction and through-life management apply.

In order to enable circular solutions for concrete structures a variety of specific aspects related to circularity require particular attention at the relevant phases of the project development. These aspects concern, but are not limited, to:

- availability of reliable as-built data for existing structures intended to form part of a circular solution
- assessment techniques for the determination of geometry and cross sectional composition and assessment of properties for construction materials obtained via dismantlement,
- implementation of structural performance evaluation framework (incl. safety, serviceability, durability and robustness evaluation concepts) for non-traditional circular solutions, ,
- consideration of time dependent material and structural performance in circular applications,
- detailing aspects for non-traditional circular solutions, ,
- constructability and dismantlement aspects of concepts subject to circularity,
- integral evaluation of environmental impact of circular solutions reflecting minimization of the need for new materials. ,

39.2 General provisions for dismantlement

39.2.1 General

For general principles of dismantlement of concrete structures reference is made to section 9.2.

During dismantlement all requirements as presented in this Model Code for the construction of new structures with respect to hazards and risks, health, safety, environment provisions, quality management and execution management do apply. This chapter 39.2 addresses additional requirements specifically related to dismantlement.

39.2.2 Hazards and risks in dismantlement

In general, analysis of hazards and risks in dismantlement should address the following three categories of hazards (see chapter 11):

- Category 1: natural hazards and unintentional human-caused hazards;
- Category 2: malevolence and vandalism;
- Category 3: negligence and human errors:

In general, in the planning stage of dismantlement consideration should be given to define measures for handling relevant hazards during execution of the dismantlement and eliminating associated risks. In particular,

- in every stage of the dismantlement the stability of the remaining part of the structure, including the eventual effect of dynamic loads shall be guaranteed,
- in every stage of the dismantlement the health and safety of workers and others should be assured
- in every stage of the dismantlement negative impact on environment shall be avoided.

The inspection of a building in the preparation stage of dismantlement can be carried out according to Figure 39.2-1.

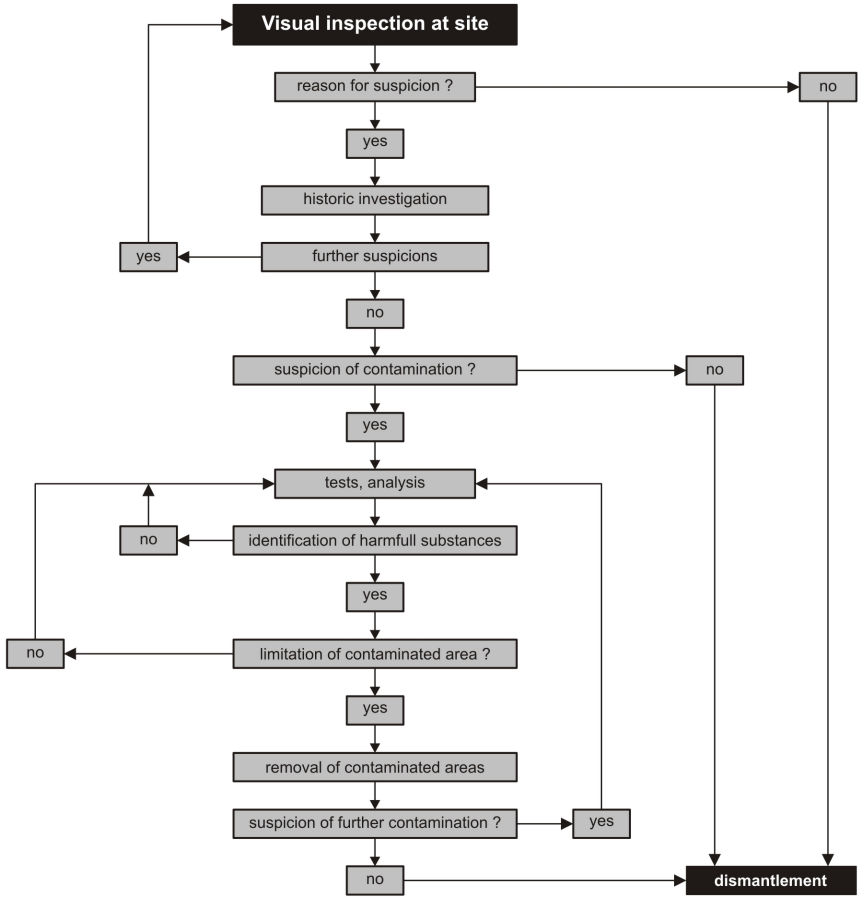

Figure 39.2-1: The inspection of a building in the preparation stage of dismantlement (Toppel [39-1]).

The inspection of a structure in the preparation stage of dismantlement shall be carried out followed by the appropriate evaluation of treats and identification of the measures to deal with the recognised threats.

Particular consideration should be given to the use of parts of the structure (eventually already in the partially dismantled stage) for supporting demolition tools or equipment for dismantlement. Especially the use of rigorous demolition techniques, for example breakdown by explosion, deserves thorough analysis and experience.

The structural system (framework, stability by stiffening walls or cores, mixed systems) should have a significant influence on the choice of the dismantlement procedure.

The materials used have to be the subject of inventory. This especially applies to materials which can be health-damaging or can disturb the environment (types, masses, integration in the structure).

The history of the building has to be regarded (previous loading history, previous damage, for instance by fire or earthquake, and the previous interventions and method of repair).

39.2.3 Structural behaviour during dismantlement

39.2.3.1 General considerations

In every stage of the dismantlement stability of the remaining part of the structure, including the eventual effect of dynamic loads shall be guaranteed.

39.2.3.2 Structural analysis for dismantlement

Dismantlement is a transient situation. For the phase of dismantlement, basically the same safety requirements apply as for the construction stage. By means of a structural analysis, unstable situations during dismantlement should be prevented. Intermediate stages have to be considered and may require temporary measures for stabilization.

In structural analysis for dismantlement, the following aspects should be regarded:

- function and size of the structure,
- structural system including stability provisions,
- structural materials used,
- age and possible deterioration,
- previous repair or structural alteration.

Harmful substances are substances, which have a negative effect on human health or environment, can be of organic or inorganic origin.

Disturbing substances are components of demolished material which can negatively influence the properties in case of re-use (gypsum, isolation material, foam, wood).

39.2.4 Health, safety and environment provisions

39.2.4.1 Health and safety aspects

A plan should be made to deal with the following labour health and safety aspects:

- threats for the work environment: this aspect regards protection against dynamic effects, flying material fractions and dust. This can be reached by providing various types of barriers and moistening.
- threats of injury to persons due to: falling parts, splitters and larger scale collapse. This can be avoided by the right sequence of dismantlement and a reliable structural analysis of all intermediate stages of dismantlement. Moreover, reliable scaffolding has to be used and information on the presence and position of electric ducts should be supplied.

39.2.4.2 Environmental protection

Investigation of potential contamination should be conducted followed by the appropriate evaluation of treats and identification of the measures to deal with the recognised threats.

The structure should be investigated with regard to the eventual presence of harmful substances.

Moreover the presence of disturbing substances should be traced.

39.2.5 Disposal concept

The plan for on-site or off-site waste management incl. material disposal should, as a minimum, regard the following aspects:

- volume of waste and quality;
- local spatial conditions;
- possibilities of regional re-use;
- possibilities for regional disposal.

The strategy should be to optimize the circularity of materials through re-use or recycling.

39.2.6 Planning of dismantlement

39.2.6.1 Consideration

In general, in the planning stage of dismantlement aspects to be considered include:

- consequence class of the structure and other technical data;
- site conditions;
- time constraints;
- hazards and threats during dismantlement (see section 35.2.2);
- stability during dismantlement (see section 35.2.3);
- health and safety aspects (see section 35.2.4.1);
- environmental protection aspects (see section 35.2.4.2);
- disposal concept (see section 35.2.5)
- dismantlement sequence and method set-up options.

Dismantling of major components may be done in stages with the sequence and method that guarantee maintaining the required structural safety level during all stages of deconstruction. In particular, deconstruction of the large-scale components (e.g. frames, decks, flooring or roofing) may be done in stages, and include more detailed separation of members on site or off site.

The process of removing building parts in often conducted in the reverse order to construction and deconstruction of objects usually starts with the removal of

39.2.6.2 Quality management and reporting

Reference is made to the requirements given in chapter 7 and chapter 31.

The preparation phase should end with a number of conclusions reported in preparation report, including:

- choice of dismantlement technique;
- measures for coping with harmful substances and disturbing material components before and during dismantlement;
- definition of necessary safety provisions;
- choice for waste disposal;
- definition of necessary capacities for transport, intermediate storage and further processing.

At least for structures in the consequence classes 2 and 3, a detailed plan for dismantlement is required, based on structural analysis for dismantlement (see section 35.2.3.2).

39.2.7 Execution of dismantlement

39.2.7.1 Considerations

Prior to the execution of dismantlement appropriate measures should be taken to deal with:

- dismantlement sequence and method set-up
    - deconstruction sequence and method set-up for structural components;
    - deconstruction sequence and method set-up for non-structural components;

(interior and exterior) non-structural element (e.g. doors, windows, linings, cladding, roofing etc.) followed by the removal of structural components and (if applicable) foundations. Multi-storey buildings are typically deconstructed floor by floor, beginning with the roof and upper floor.

Issues that may be considered include for instance (but not only):

- safe and appropriate storage conditions of the reclaimed materials and diligent transportation off to reduce damage and contamination that could devalue the materials;
- identification and labelling of the reclaimed materials as they are removed, for unproblematic future use;
- appropriate storage conditions to protect the reclaimed material from damage, mishandling, theft, vandalism and fire.

Hazardous materials (e.g. asbestos) should only be removed by approved contractors.

Hazards related to health and safety issues may include e.g. noise, dust, water, contamination, vibration and shocks, and release of dangerous substances. Measures should be taken to minimize the risks of such hazards in order not to violate health and safety demands during deconstruction.

- on-site or off-site waste management incl.:
  - waste separation and storage;
  - handling of recyclable and reusable materials;
  - handling of non-recyclable and non-reusable materials;
- health and safety issues.

In every stage of the stability of the remaining part of the structure, including the eventual effect of dynamic loads shall be considered.

References

[39-1] Toppel — Toppel, C. O. , Technische und ökonomische Bewertung verschiedener Abbruch-verfahren im Industriebau, Selbstverlag, 2003.